



Waveserver 5

User Guide

OS 2.2

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CHAPTER 1

About this document

Waveserver 5 User Guide, 323-3001-100, provides detailed descriptions of the features and operations for Waveserver 5, OS 2.2 and subsequent maintenance releases.

This section covers the following topics:

- [Documentation Roadmap](#)

Publication History

Issue 2

Added support for new QSFP28 pluggable, 160-9433-900: 100GBASE-LR1, SMF, 1310nm, 10km QSFP28 at various sections in this guide.

For more detail, refer to [Pluggables and specifications](#).

Issue 1

Issue 1 is the first release of this document for Waveserver 5, OS 2.2.

New in this release

Waveserver 5 introduces the following hardware and software features and enhancements in OS 2.2:

- [Support for new WL5e 2x MOTR 12x modules](#)
- [4x100GE-ZR client xcvr support on WL5e 2x MOTR 12x module](#)
- [ITS support for OTL4.4 Clients](#)
- [Support for a new pluggable](#)
- [Connection Validation](#)
- [Support for independent service profile per line in the same module](#)
- [New transmission modes and LOM](#)
- [Streaming of link data collection information](#)
- [Flexible client port configuration in mixed-otn-ethernet service profile](#)
- [Support for FEC Uncorrected Seconds parameter in the new ptp show statistics command](#)

- [Untimed-Read-And-Clear](#)
- [Web User Interface enhancements](#)
- [Security enhancements to certificate authorities](#)
- [New ordering codes](#)
- [Waveserver 5, OS 2.2 upgrade paths](#)
- [Waveserver 5, OS 2.2 licensing](#)

Support for new WL5e 2x MOTR 12x modules

Waveserver 5, OS 2.2 introduces the following new service modules:

- Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module (186-3120-900)
- Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module (186-3121-900)

For more detail, refer to [Module descriptions, equipping rules, and specifications \(Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD\)](#).

4x100GE-ZR client xcvr support on WL5e 2x MOTR 12x module

The newly introduced WL5e 2x MOTR 12x module allows user to provision 400ZR QSFP-DD pluggables on the client port in the 4x100GE-ZR mode. See [Service profiles for WL5e 2x MOTR 12x modules](#) for details.

ITS support for OTL4.4 Clients

Added ITS support on OTL4.4 clients in race track mode. Refer to [Integrated test set](#) section for detail.

Support for a new pluggable

A new pluggable 400GBASE-LR4, SMF, 10Km, QSFP-DD (160-9603-900) is supported in this release. For details see [Ordering](#).

Connection Validation

Connection validation allows you to validate fiber inter-connections between line-side Waveserver transponder PTPs and the Photonic CMD / WSS. Connection validation implements an optical loopback test at the WSS, which verifies that the transponder is properly connected and functional. For more information, refer to [Connection validation](#).

Support for independent service profile per line in the same module

This release supports separate or independent service profiles on each standalone line port of the same module. For example, you can now provision mixed-otn-ethernet service profile one line and mixed-ethernet on the other line.

New transmission modes and LOM

Waveserver 5, OS 2.2 introduces the following transmission modes and link optimization modes:

- 95-600-O (Fast-Tracking)
- 95-400-O (Fast-Tracking)
- 95-650-E (SNR-Tolerant)
- 93.3-600-E (SNR-Tolerant)
- 93.3-400-E (SNR-Tolerant)
- 90-600-E (Standard, SNR-Tolerant)
- 90-400-E (SNR-Tolerant)
- 85-600-E (Standard)
- 85-400-E (Standard, SNR-Tolerant)
- 85-300-E (Standard)
- 82-400-E (SNR-Tolerant)

Note: The WL5e 2x MOTR 16x modules supports only 95 and 91.6 Gbaud. However, the WL5e 2x MOTR 12x module supports all Gbauds.

Streaming of link data collection information

OS 2.2 supports gNMI streaming telemetry of WL5e link data. A new Ciena YANG model, `ciena-waveserver-link-data-telemetry`, is used for subscription and streaming of link data. For more information, refer to [“Streaming of link data collection information”](#).

Flexible client port configuration in mixed-otn-ethernet service profile

This release allows a flexible configuration of 100G ports (8, 9, 16, and 17) to support both 100GE or OTL4.4 protocols when configuring 400GE clients with the mixed-otn-ethernet service profile at 500G & 600G line rates. Earlier to this release, the ports 8 and 16 are fixed to support OTU4 protocol and the ports 9 and 17 are fixed to support 100GE protocol only.

Support for FEC Uncorrected Seconds parameter in the new ptp show statistics command

A new cli command `ptp show ptp <slot/port> statistics` is introduced in this release with all existing ptp instantaneous parameters and a new FEC Uncorrected Seconds parameter. See [ptp show ptp <slot/port> statistics](#) for detail.

Untimed-Read-And-Clear

This feature enables the user to clear the untimed bin after a get (read) request.

Web User Interface enhancements

The Waveserver 5 Web User Interface (Web UI) is updated in this release to have a consistent look and feel with the user interfaces of other Ciena products. For details, see [Waveserver 5 Web User Interface](#).

Security enhancements to certificate authorities

You are now required to specify a name (`cert-name`) for certificate authorities when you install them. The certificate name parameter can be used to display certificate information and to uninstall the certificate. If necessary, you can rename certificate authorities after you install them.

New ordering codes

The following table lists new ordering codes for OS 2.2 that apply to Waveserver 5.

Table 1 New ordering codes in this release

Description	Order code
Modules	
Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module	186-3120-900
Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module	186-3121-900
Pluggables	
400G-LR4-10/4x100GE, SMF, 10km QSFP-DD	160-9603-900
100GBASE-LR1, SMF, 1310nm, 10km QSFP28	160-9433-900
Software Licenses	
Release 2.2 Base Licenses	S86-LIC-WOSBASE020200 S86-LIC-WOSUPG020200

Waveserver 5, OS 2.2 upgrade paths

You can upgrade to Waveserver 5 OS 2.2 in a one-step upgrade from Release OS 2.0.2, 2.0.3, 2.0.4, 2.1, 2.1.1, and 2.1.2.

Waveserver 5, OS 2.2 licensing

The following software license is available for OS 2.2. For more information, see [Ordering](#).

- Waveserver OS Release 2.2.0 Base Software License (S86-LIC-WOSBASE020200)
- Waveserver OS Release 2.2.0 Software Upgrade License (S86-LIC-WOSUPG020200)

Documentation Roadmap

The following table provides information about documentation supporting Waveserver 5 for OS 2.2.

Table 2 Documentation roadmap

Documents	Description
<i>Waveserver 5 User Guide</i> , 323-3001-100	Includes product description, ordering, software licensing, hardware and software information.
<i>Waveserver 5 Installation Guide</i> , 323-3001-200	Includes hardware description and installation information.

CHAPTER 2

Release history

This section lists the new hardware and software features introduced for each Waveserver 5 OS software release.

OS 2.1

OS 2.1 introduces the following hardware and software features:

New hardware

The following hardware is added in OS 2.1:

Support for Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900)

The main purpose of Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900) is to map three groups of 4x100GE clients to three QSFP-DD DCO 400ZR line-side plugs. The client ports can be comprised of 4 x QSFP28 (100G). It performs the role of a transponder and its total data throughput is 1.2 Tbs.

Support for WL5n 400ZR Type 3.0 QSFP-DD Transceiver (180-3530-900)

The WL5n 400ZR Type 3.0 QSFP-DD Module (180-3530-900) is a multi-rate, single carrier C-band coherent transceiver, capable of transmitting and receiving a 400GE or 4x100GE client payload. This line port plug is used in Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900).

Support for 400GE, QSFP-DD - QSFP-DD Active Optical Cables

The 400GE, QSFP-DD - QSFP-DD Active Optical Cables (AOCs) are offered in the following five different lengths:

- 400GE, 3m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-003)
- 400GE, 5m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-005)
- 400GE, 10m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-010)
- 400GE, 20m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-020)
- 400GE, 100m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-100)

Each 400GE Active Optical Cable (AOC) has a QSFP-DD plug at each end and is used to interconnect two QSFP-DD ports. The use of 400GE Active Optical Cable (AOC) is an alternative to using two optical 400GE QSFP-DD transceivers and an optical patch cord.

New software

The following software features are added in OS 2.1:

Streaming telemetry enhancements

Streaming telemetry is a method of network monitoring that allows you to continuously stream data from network devices with efficient, incremental updates. Telemetry is sent based on subscriptions. Streaming telemetry is expanded in this release to include support on NETCONF and the CLI. In addition, gNMI support is expanded to include Ciena YANG model paths.

NETCONF supports the basic notification delivery mechanisms defined in RFC 5277, that is, the <create-subscription> operation. The <create-subscription> operation is used to set up session-specific filtering and optional buffered notification delivery parameters.

Declarative configurations

Declarative configuration allows you to send a full device configuration to Waveserver 5 and then determines the differences between its currently running configuration and the candidate configuration and makes only the minimum number of changes required to achieve the new configuration. Declarative configuration is expanded in this release to include support on NETCONF and the CLI.

Candidate datastores

A candidate datastore is used to hold configuration data that can be manipulated without impacting the device's current configuration. The candidate configuration is a full configuration data set that serves as a workplace for creating and manipulating configuration data. Candidate datastores are expanded in this release to include CLI support.

DHCP relay agent

The DHCP relay agent feature allows a remote Waveserver node, which is hosted on different subnet, to communicate and obtain the IP address from the DHCP server on a different subnet. This feature is useful for connecting remote nodes that does not have their own DHCP server.

Additional YANG model support

This release supports additional Ciena YANG and OpenConfig YANG models. All models are supported on REST, NETCONF, and gNMI.

TACACS+ AAA support for gNMI/gNOI, REST, and NETCONF

This release supports TACACS+ authentication, authorization, and accounting for gNMI/gNOI, REST, and NETCONF operations.

gNOI factory reset

This release supports the gNOI:system:FactoryReset command to reset the shelf to the factory default.

gRPC library version

The gNMI specification outlines the protocol buffer definitions and services used by the gRPC server to perform management actions, or retrieve and/or modify configuration on Waveserver 5. Waveserver 5 uses gRPC library 1.34.0, protobuf v3.5.0, and gNMI proto v0.7.0.

SPLI with RLS

This release supports Service and Photonic Layer Interoperability (SPLI) with the RLS Photonics. SPLI communicates DWDM channel provisioning information between a Waveserver 5 transponder or line module and the adjacent RLS CMD. For Waveserver 5, SPLI facilitates tighter integration of its line ports with the RLS Photonics to reduce OAM costs associated with wavelength configuration and maintenance. Modem provisioning is sent from the Photonic line SPLI (RLS) to the service layer SPLI (Waveserver 5) after the transmitter type is discovered by RLS.

Session timeouts

This release supports a global inactivity timeout setting and an HTTPS inactivity timeout setting. If the global setting is enabled, any management interface session that has been inactive for the duration of the global timeout period will be closed. If the HTTPS setting is enabled, any management interface that uses HTTPS (Web user interface, REST, and WebSocket) will be closed if the session has been inactive for the duration of the HTTPS timeout period. The HTTPS timeout setting takes precedence over the global timeout setting.

New transmission modes and LOM

Waveserver 5 introduces the following transmission modes and link optimization modes in this release:

- 95-800-O (SNR-Tolerant)
- 95-650-O (Standard)
- 95-350-O (Non-Linear)
- 95-700-E (Standard)
- 95-650-E (Standard)
- 95-600-E (Standard, SNR-Tolerant)
- 95-350-E (Standard, Non-Linear, SNR-Tolerant)

- 95-300-E (Standard, Non-Linear)
- 91.6-700-E (Standard)
- 91.6-550-E (SNR-Tolerant)
- 91.6-500-E (Standard, SNR-Tolerant)
- 91.6-450-E (SNR-Tolerant)
- 91.6-400-E (SNR-Tolerant)
- 91.6-300-E (Standard, Non-Linear)

ZTP process improvements

The ZTP process is improved in this release of Waveserver 5:

- ZTP in Progress Alarm: This alarm is raised as soon as the ZTP process starts. The ZTP in Progress alarm clears once commissioning is complete and the ZTP process stops. See [ZTP In Progress](#) for detail.
- NTP server configuration: NTP server configuration is now optional before starting the ZTP process.
- No system restart: ZTP process does not perform a system restart after activating the configuration.

OS 2.0.3

OS 2.0.3 introduces the following hardware and software features:

New hardware

The following hardware is added in OS 2.0.3:

Support for new AC Power Cords

- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, black (NTK955GF)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, black (NTK955GG)
- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, white (NTK955GH)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, white (NTK955GJ)

Support for new QSFP-28 and QSFP-DD pluggables

- 160-9600-900: 400G-FR4, SMF, 2km QSFP-DD
- 160-9431-900: 100G-FR, SMF, 1310nm, 2km QSFP28

Support for three new L-Band service modules

- Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFPDD Module (186-3110-901).

- Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module (186-3111-901).
- Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3112-901).

New software

The following software features are added in OS 2.0.3:

Border Gateway Protocol (BGP) Peering

The BGP peering feature allows you to have two DCN ports (DCN-1 and DCN-2 or DCN-3 and DCN-4) running BGP on the Waveserver 5. Each DCN port is connected to a router and the BGP protocol decides which router the packets should be sent to. If a failure occurs locally or remotely in the network, BGP protects against these failures.

GCC0 comms over different subnet

OTUCn GCC0 is now supported over different subnets. Now, local and remote Waveserver 5 chassis can be on same or different IPv4 or IPv6 subnets.

Control Processor (CP) replacement - auto-recovery

This feature enables automatic backup of the software load, system configuration, certificate files, and licenses files on the access panel. Upon a CP replacement, if the AP contains the backup information for the chassis, the new CP can auto recover from the backup information available on the AP. Configuration data will automatically be backed-up each time you use CLI command `configuration save`.

Zero Touch Provisioning

Zero Touch Provisioning (ZTP) simplifies the initial commissioning of the Waveserver 5 and extends the automatic provisioning capabilities of the Waveserver 5.

ZTP feature allows you to retrieve its provisioning information from an xFTP server and then provision the Waveserver 5 automatically.

New transmission modes and LOM

Waveserver 5 introduces the following transmission modes and link optimization modes in this release:

- 95-300-O (Standard, Non-Linear, SNR-Tolerant)
- 95-350-O (Standard, SNR-Tolerant)
- 95-400-O (Non-Linear)
- 95-600-O (SNR-Tolerant)
- 95-700-O (Standard)

- 95-400-E (Standard, Non-Linear, SNR-Tolerant)
- 95-450-E (Standard, Non-Linear)
- 95-500-E (Standard)
- 95-550-E (SNR-Tolerant)
- 91.6-400-E (Standard)

400GE Clients and ITS

New 400GE clients are added which can be configured using the `xcvr set xcvr` command and their corresponding ITS are added to verify connection settings.

400GE Service Profile

The 400GE service profile supports 400GE client ports over QSFP-DD pluggables.

Mixed-Ethernet Service Profile

The Mixed-Ethernet service profile supports 100GE/400GE client ports over QSFP28 / QSFP-DD pluggables.

Real time stats in the `ptp show` command

Following instantaneous statistics related parameters are added to the `ptp show ptp` command.

- Pre-FEC BER
- Q-Factor
- ESNR
- PDL
- SNR External
- Cycle Slip Indicator

OpenConfig support over NetConf and gNMI

OpenConfig is a collaborative effort by network operators to develop a common set of APIs for a dynamic, programmable method for configuring and managing multi-vendor networks. These vendor-neutral data models allow you to configure and manage a network, and define the configuration and operational state of network devices for common network protocols or services. The data models are written in YANG. For details, see [OpenConfig](#), [NETCONF](#), [gRPC/gNMI](#).

OS 2.0.2

OS 2.0.2 introduces the following hardware and software features:

New hardware

The following hardware is added in OS 2.0.2:

- AC Power Cord, Saf-D-Grid to C20, 12 AWG, 250V, 2m, white (NTK955GC)
- AC Power Cord, Saf-D-Grid to C14, 14 AWG, 250V, 2m, black (NTK955GD)
- AC Power Cord, Saf-D-Grid to CEE7/7, 12 AWG, 250V, 2m, black (NTK955GE)

New software

The following software features are added in OS 2.0.2:

OTL4.4 client ports

OTL4.4 clients on QSFP28 pluggables and Mixed-OTN-Ethernet service profiles are introduced for Waveserver 5, OS 2.0.2. The Mixed-OTN-Ethernet service profile supports a mix of OTL4.4 and 100GE client ports over QSFP28 pluggables.

Mated terminal line configurations

Mated terminal line configurations are now supported where the line ports (ports 1 and 2) work together. If the module is set to mated mode, you can split the last supported client port on the primary line (port 1) in half. The 100GE and Mixed-OTN-Ethernet service profiles both support mated terminal line configurations.

Transmission modes

Waveserver 5, OS 2.0.2 introduces the following transmission modes:

- x00G transmission modes: 95-500-O, 95-600-O, 91.6-600-E, and 91.6-800-E
- x50G transmission modes: 95-250-O, 95-450-O, and 95-550-O

Dispersion scan

A Q-based dispersion scan can be performed on the line ports. This feature automatically sweeps pre-compensation values to optimize the Q-factor. The scan results provide the pre-compensation value and associated Q. You can use the scan results to set the PTP parameters for optimal quality.

OS 2.0.1

OS 2.0.1 adds the following new hardware:

New hardware

- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901)

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- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901)

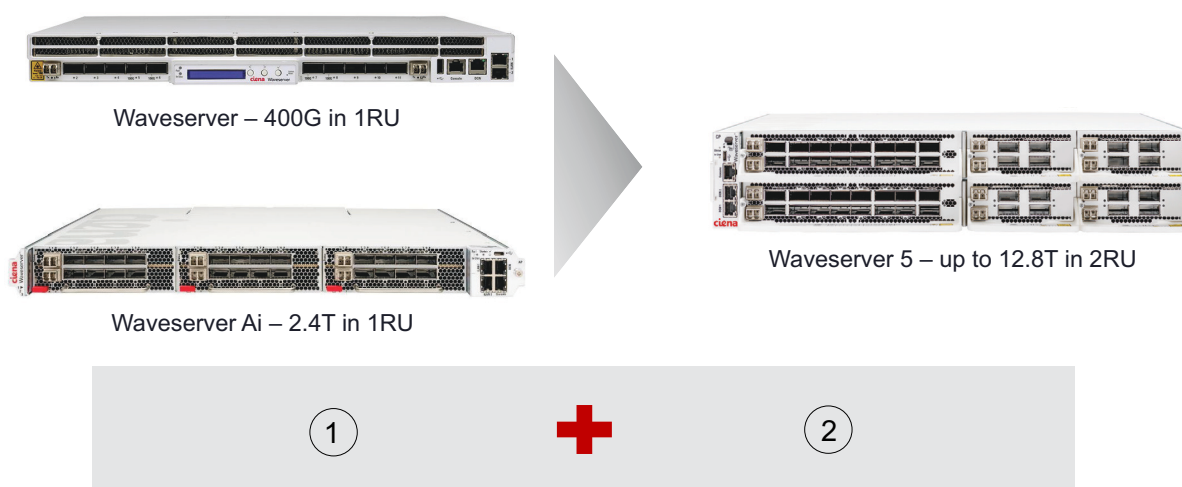
CHAPTER 3

Overview

The Waveserver 5 provides high-speed, point-to-point connectivity for Data Center Interconnect (DCI) applications in a scalable and small footprint solution.

The Waveserver 5 is an extension of Waveserver family as shown in [Figure 1](#).

Figure 1 Waveserver family value extension to Waveserver 5



Legend

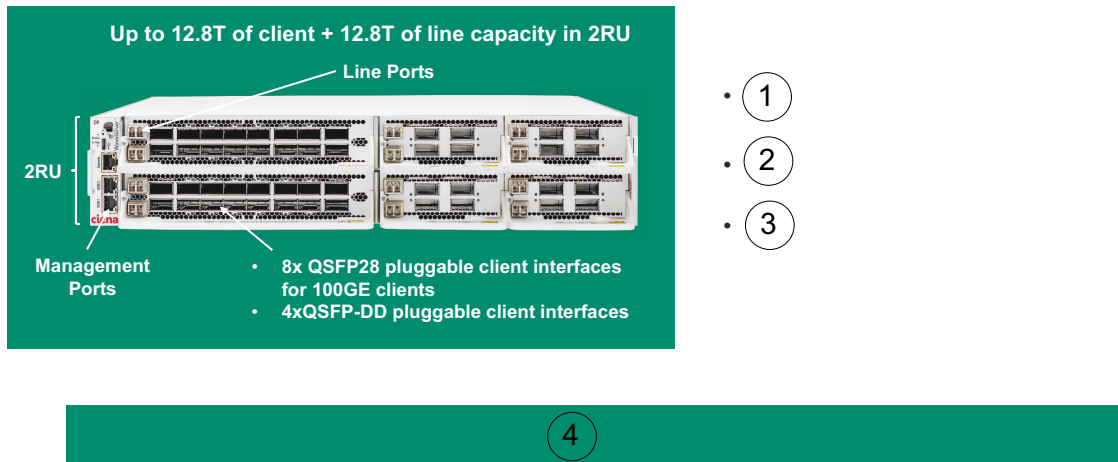
- | | | | |
|---|---------------------------------|---|--------------------------------------|
| 1 | Transforming DCI Economics | 2 | Simple to Deploy |
| | Unique, industry-leading | | Simple server-like operational model |
| | capacity and power efficiencies | | Single platform for any distance |

Waveserver 5 features:

- a mix of 100GbE and 400GbE services across any distance, from metro to long-haul and subsea, by leveraging Ciena's Waveserver 5 Extreme and Nano coherent technology innovation.

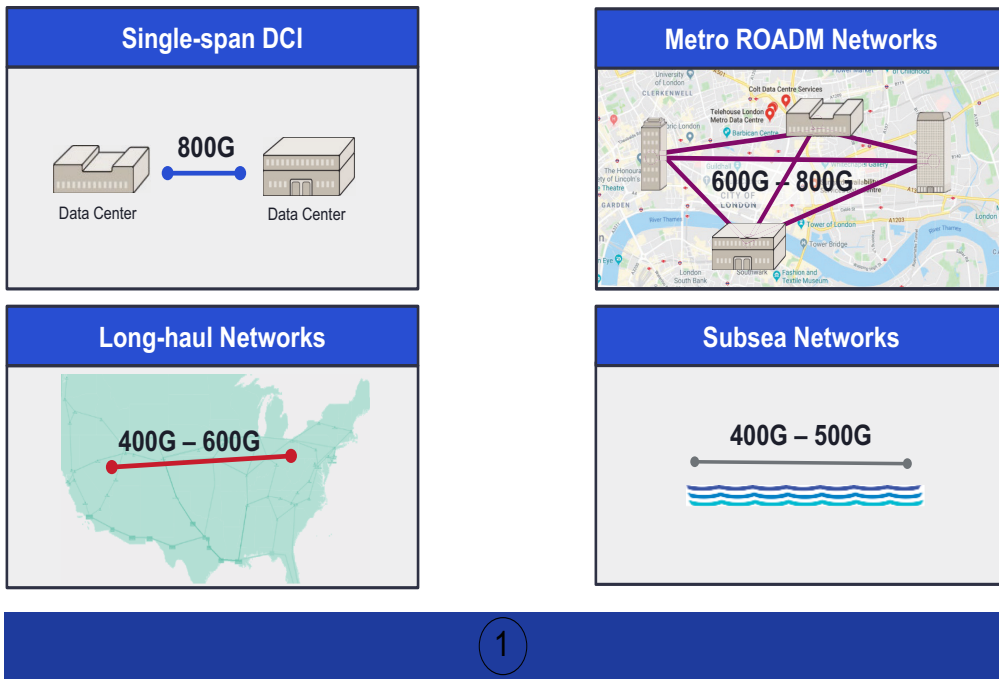
- matching system capacity to available network margin for any network application, including single-span DCI, metro ROADM networks, long-haul routes, and subsea links. By enabling higher line rates at any distance, Waveserver 5 improves spectral efficiency and increases overall fiber capacity up to 38.4 Tb/s. It can also operate across fixed-grid networks and third-party line systems. By removing constraints on scalability, Waveserver 5 satisfies the capacity requirements of the most demanding data services and applications.
- high-growth applications in an ultra-dense, compact platform.
- highly programmable, enabling rapid scalability with tunable coherent optics that help providers adapt to changing bandwidth requirements.
- higher capacity, with up to 800 Gb/s per wavelength.
- highest service density with up to 12.8T of capacity in two rack-units.
 - reduces power utilization and cost/bit
 - sets a new industry benchmark for high-performance DCI capacity and platform density
 - small stackable 2RU chassis designed for future growth
- up to four double slot Service Modules (SMs).
- field replaceable Service Modules.
- redundant and in-service field replaceable Power Supply Units (PSUs) and fan modules.
- in-service field replaceable Control Processor (CP) and Access Panel (AP).
- automatic and simplified provisioning, open Application Programming Interfaces (APIs).
- Service Photonic Layer Interoperability (SPLI) interworking with 6500 and RLS.

The Waveserver 5 raises the bar for high capacity and density in a small form-factor device as shown in [Figure 2](#).

Figure 2 Waveserver 5 capacity and density**Legend**

- | | | | |
|---|--|---|---|
| 1 | Enables rapid capacity growth with support for single carrier 800G for DCI and 400G for long-haul applications | 3 | Easy to install and operate |
| 2 | New industry benchmark for density and capacity in a small form-factor device | 4 | Ideal for applications using large traffic flows with 100GE and 400GE |

The Waveserver 5 applications extend from metro to subsea as shown in [Figure 3](#).

Figure 3 Waveserver 5 applications**Legend**

- 1 Waveserver 5 suited for all applications requiring highest spectral efficiency

The Waveserver 5 supports the following northbound interfaces (NBI):

- Command Line Interface (CLI), a text-based command interface used to configure and view system information.
- Representational state transfer application programming interface (REST API). REST systems typically communicate over the Secure Hypertext Transfer Protocol (HTTPS) using the same HTTP methods that web browsers use to retrieve and send data to remote servers. You can use REST to configure and view system information.
- general Remote Procedure Call (gRPC)/gRPC Network Management Interface (gNMI), a communication protocol that allows application developers to use software and scripts to easily interact with and monitor Waveserver 5.
- NETCONF, an XML-based network management protocol specification developed by the Internet Engineering Task Force (IETF). You can use NETCONF to install, manipulate, and delete network device configurations, and to provision and activate network services.
- Simple Network Management Protocol (SNMP). You can use SNMP Get operations to retrieve system information.

YANG modules define the configuration data, state data, and the capabilities available on a Waveserver 5 device. Waveserver 5 supports Ciena YANG modules and OpenConfig YANG models. All models are supported on REST, NETCONF, and gNMI. As OpenConfig does not provide full configuration support, OpenConfig users can supplement operation requests with native Ciena YANG models as required.

CHAPTER 4

Chassis descriptions and technical specifications

This section contains the following topics:

- [Overview](#)
 - [Chassis \(186-3001-900\)](#)
 - [Control Processor Subsystem \(186-3010-900\)](#)
 - [Access Panel \(186-3020-900\)](#)
 - [Power and fan modules \(186-3040-900/186-3041-900/186-3042-900\)](#)
 - [Filler module \(186-3031-900\)](#)
 - [Supported upright spacing depths](#)

Overview

This section provides information on chassis descriptions and technical specifications.

Chassis (186-3001-900)

The Waveserver 5 chassis is a 2RU chassis that carries up to 6.4 Tb/s with four double-width service modules. The Waveserver 5 chassis supports up to four WaveLogic 5e and WaveLogic 5n modules. Both C-Band and L-Band modules are available in different GBauds. For details, refer to appropriate section of this guide.

Field replaceable units (FRUs) are available at the front and rear of the Waveserver 5 chassis.

FRUs found at the front of the chassis include;

- Control Processor (in-service field replaceable)
- Four double-width service modules

FRUs found at the rear of the chassis include;

- Access panel (in-service field replaceable)

- Redundant power supply units with redundant fan units (in-service field replaceable)

See the following figures for the front and rear panel views of the chassis.

Figure 4 Front panel view - Waveserver 5 chassis with a Control Processor and four double width service modules installed

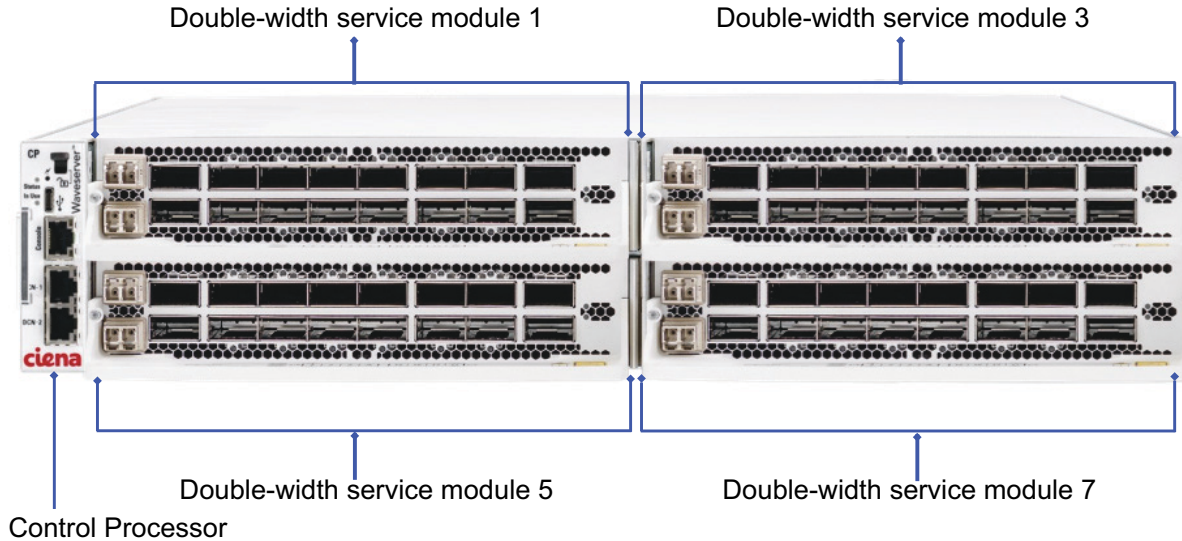


Figure 5 Waveserver 5 chassis with AC PSU, fan, and access panel - rear panel view

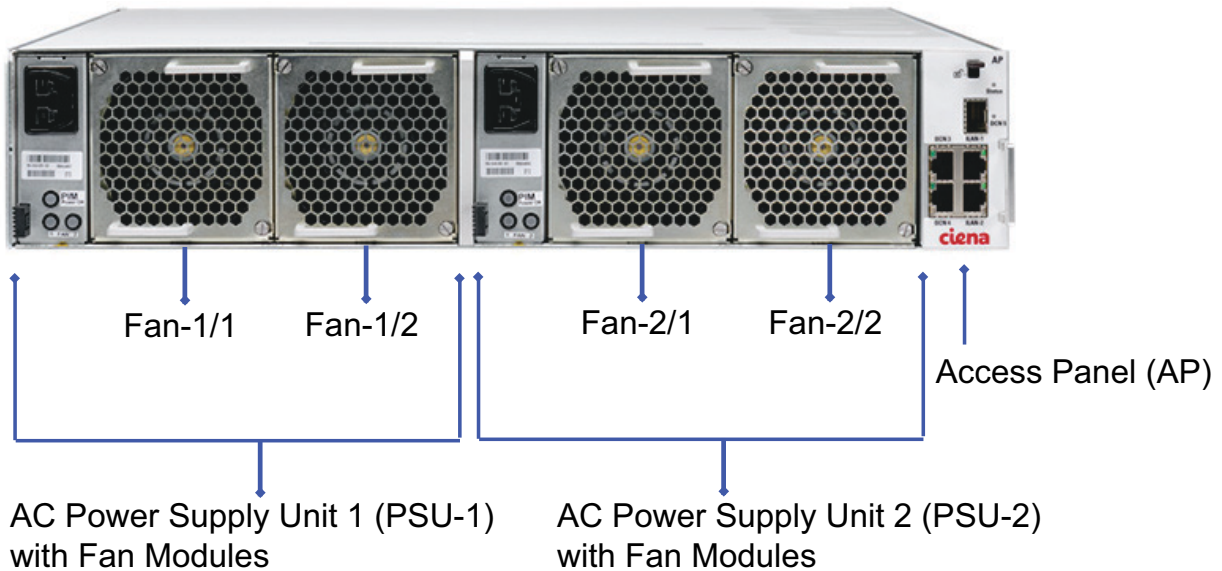
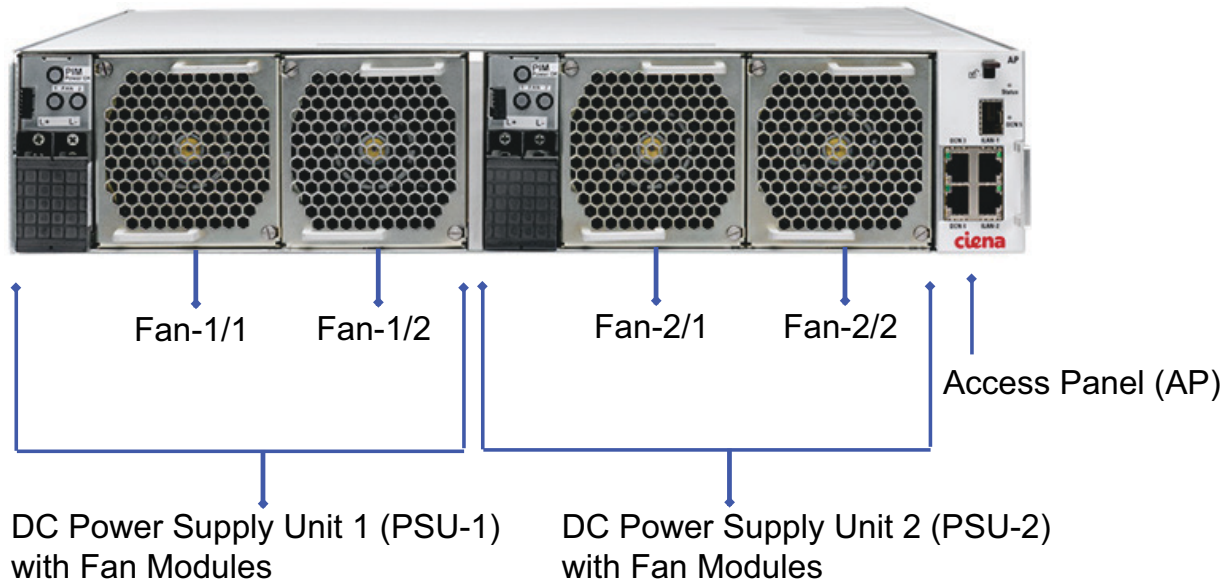


Figure 6 Waveserver 5 chassis with DC PSU, fan, and access panel - rear panel view

Lamp flash is supported on the chassis, or a traffic module, or a port, where you can flash a LED remotely for a specific amount of time. Use CLI command `system lamps set` to enable lamp flash for the whole chassis. Lamp flash cannot be done for a Control Processor or an access panel but Control Processor and access panel LEDs are flashed when chassis is flashed.

The following table defines each slot in the Waveserver 5 chassis.

Table 3 Slotting in the Waveserver 5 chassis

Slot #	# of slots	Components	Reference
1, 3, 5, and 7	4	Service Modules (SM)	• for WL5e 2x MOTR 16x modules, see “Module descriptions, equipping rules, and specifications (Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD)” • for WL5e 2x MOTR 12x module modules, see “Module descriptions, equipping rules, and specifications (Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD)” • for WL5n DCO 3xQSFP-DD 12x modules, see “Module descriptions, equipping rules, and specifications (Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD)”
—	1	Control Processor Subsystem (CPS)	Control Processor Subsystem (186-3010-900)
—	1	Access Panel (AP)	Access Panel (186-3020-900)

Slot #	# of slots	Components	Reference
—	2	Power Supply Unit (PSU) with Fan Modules	Power and fan modules (186-3040-900/186-3041-900/186-3042-900)

Technical specifications

The following table provides technical specifications for the Waveserver 5 chassis.

Table 4 Technical specifications for Waveserver 5 chassis

Specification		Value
Dimensions (H x W x D) (mm)		88.45 x 479.8 x 544.87 Note: H: Height is 2 rack units (88.9 mm). W: Includes flanges for mounting fiber managers. Behind these flanges, width is 444 mm. D: Does not include fiber managers, FRU handles, or power cables.
Weight (kg)	Empty	9.158
	Fully equipped	28.952
Power Consumption (W)	Chassis with common equipment (two power and fan modules, CP, and AP), output power (not including efficiency)	• Typical: 290 W • Max: 400 W
	Fully equipped chassis including four service modules, input power without pluggables (including efficiency, dual PSU)	Typical 25 °C: 1700 W
	Power Budget (single PSU to cover a PSU failure)	• 40 °C: 2250 W • 45 °C: 2300 W

Control Processor Subsystem (186-3010-900)

The Control Processor (CP) controls the chassis and it is an in-service field replaceable unit (FRU) found at the front of the Waveserver 5 chassis. The Waveserver 5 service modules do not require the CP to be present, in order to maintain their transport capabilities.

Note: During a CP replacement procedure, the chassis will not be accessible, as all Operation Administration and Maintenance (OAM) functions reside on the CP.

The Control Processor supports these features:

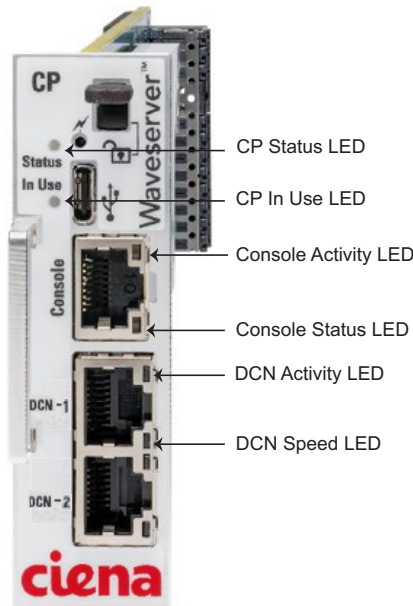
- One RJ45 universal serial console port with auto-baud detection
- DCN-1 and DCN-2 ports (10/100/1000Base-T)

Note: DCN ports on the CP support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

- One USB-C console port with auto-baud detection
- One recessed reset button (warm, cold, RTFD)
- Provides redundancy for the software load, system configuration, certificate files, and licenses files. These files are saved on both the access panel and the control processor.

See the following figure.

Figure 7 Control Processor



LEDs

The following LED definitions exist for the Control Processor (CP):

- In-Use LED
 - Red (solid): Indicates common equipment failure.
 - Green (solid): Indicates Waveserver 5 chassis is operational.
- Status LED
 - Red (solid): Indicates CP failure.

- Green (flashing): Indicates that the software is initializing.
- Green (solid): Indicates that the Waveserver 5 CP is operational.
- Console Status LED
 - Green (solid): Indicates connected and recognized.
 - Yellow (solid): Indicates unrecognized.
- Console Activity LED
 - Green: Indicates baud detected.
 - Yellow: Indicates baud not detected.
- DCN-1 and DCN-2 Speed LEDs
 - Green (1 flash per sequence): 10 Mbps
 - Green (2 flashes per sequence): 100 Mbps
 - Green (3 flashes per sequence): 1000 Mbps
- DCN-1 and DCN-2 Activity LEDs
 - Green (flashing): Indicates that the DCN-1 or DCN-2 port is operating normally.

Technical specifications

The following table provides technical specifications for the control processor.

Table 5 Technical specifications for Control Processor (186-3010-900)

Specification	Value
Dimensions (H x W x D) (mm)	87.7 x 24.5 x 344.38
Weight (kg)	0.421

Alarms

For a complete list of alarms and alarm clearing procedures, see [Chapter 28, “Alarms”](#).

Access Panel (186-3020-900)

The Access Panel (AP) provides rear connections and Ethernet comms to the chassis. It is an in-service field replaceable unit (FRU) in the rear of the Waveserver 5 chassis. In the absence of the CP, the AP remains fully operational with the exception that the chassis is inaccessible, as the CP is required for authentication.

The Access Panel supports these features:

- Two ILAN (ILAN-1 and ILAN-2) ports (1000Base-T and 2500Base-T)
- Two DCN (DCN-3, DCN-4) ports (10/100/1000Base-T)

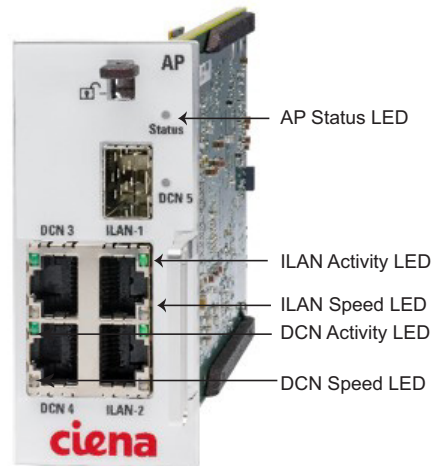
Note 1: DCN SFP+ (1G/10G) (DCN-5) port is not supported for this release.

Note 2: DCN and ILAN ports on the AP support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

- Provides redundancy for the software load, system configuration, certificate files, and licenses files. These files are saved on both the access panel and the control processor.

See the following figure for an example of the AP.

Figure 8 Access Panel



LEDs

The following LED definitions exist for the access panel (AP):

- Status LED
 - Red (solid): Indicates a hardware failure on the AP.
 - Green (solid): Indicates that the AP is operational.
- RJ45 DCN-3, DCN-4, ILAN-1, and ILAN-2 Speed LEDs
 - Green (1 flash per sequence): 10 Mbps
 - Green (2 flashes per sequence): 100 Mbps
 - Green (3 flashes per sequence): 1000 Mbps
 - Yellow (solid): 2500 Mbps

Note: DCN-3 supports up to 1000 Mbps and ILAN-1/ILAN-2 supports up to 2500 Mbps.

- RJ45 DCN-3, ILAN-1, and ILAN-2 Activity LEDs

- Green (flashing): Indicates that the DCN-3, ILAN-1 or ILAN-2 RJ45 port is operating normally.

Technical specifications

The following table provides technical specifications for the access panel.

Table 6 Technical specifications for Access Panel (186-3020-900)

Specification	Value
Dimensions (H x W x D) (mm)	87.7 x 38.14 x 221.54
Weight (kg)	0.242

Alarms

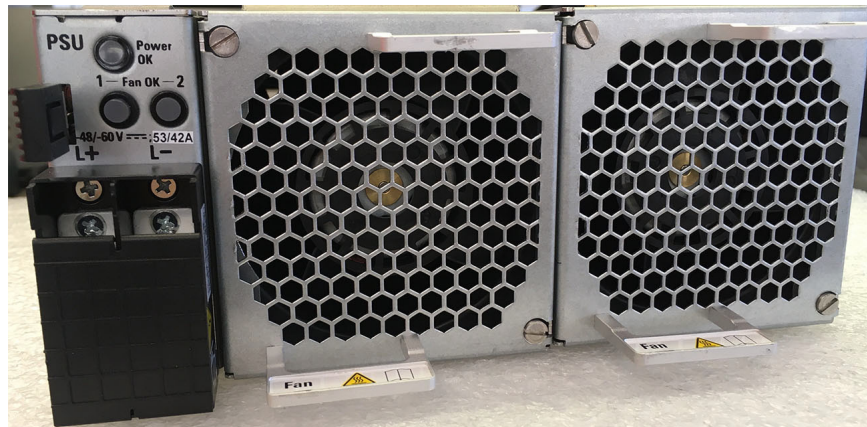
For a complete list of alarms and alarm clearing procedures, see [Chapter 28, “Alarms”](#).

Power and fan modules (186-3040-900/186-3041-900/186-3042-900)

The power and fan modules provide power and cooling to the shelf. Two shelf cooling fan modules are integrated with each power module. There are two power and fan modules per chassis for redundancy and to allow for in-service replacement. The power and fan modules are in-service field replaceable units (FRUs) installed at the rear of the Waveserver 5 chassis. See the following figure for an example of a DC power and fan module.

Note: The fan serial number is not listed in the CLI outputs of the `chassis show` commands for FAN-1/1, FAN-1/2, FAN-2/1, and FAN- 2/2. Refer to the fan label for the serial number.

Figure 9 Power and fan module (DC module shown)



The power and fan module supports the following features:

- Each fan module consists of a dual rotor fan.

- Waveserver 5 draws air from intake vents at the front of the chassis and exhausts it through vents at the rear.
- The Waveserver 5 operates normally if one of the two rotors fails within the fan module.
- Supports module variants;
 - 100-277VAC / 240VDC power modules
 - -48V DC power modules

Note: If there is a hardware issue with one of the power supplies, the two fans within that power supply will still operate by using the redundant power supply.

For information on power specifications, see [Power and grounding specifications](#).

LEDs

The following LED definitions exist for the power and fan module:

- Power OK LED
 - Off: Indicates that the incoming voltage source is below the minimum power module turn-on specification or outside the specified operating range. An “Off” condition can also indicate a damaged or faulty power supply module.
 - Red (flashing): Indicates a warning condition in the power module.
 - Red (solid): Indicates a shutdown fault condition in the power module.
 - Green (flashing): Indicates a missing fan module.
 - Green (solid): Indicates the power module is operating normally.

The following LED definitions exist for the fan module:

- Fan OK 1/2 LED
 - Off: Indicates that the fan module is not present or fan power is not present.
 - Red (flashing): Indicates the fan is operating at above the set speed threshold for longer than 20 seconds.
 - Red (solid): Indicates the fan is operating at below the set speed threshold for longer than 20 seconds.
 - Green (flashing): Indicates a missing fan module.
 - Green (solid): Indicates the fan is operating at the expected speed.

Technical specifications

The following table provides technical specifications for the power and fan modules.

Table 7 Technical specifications for power and fan modules (186-3040-900/186-3041-900/186-3042-900)

Description	Specification	Value
AC power and fan module (186-3040-900) (Note 1)	Dimensions (H x W x D) (mm)	87.8 x 200.29 x 220.51
	Weight (kg)	3.426
DC power and fan module (186-3041-900) (Note 2)	Dimensions (H x W x D) (mm)	87.8 x 200.29 x 220.51
	Weight (kg)	2.891
Fan module (186-3042-900) (Note 3)	Dimensions (H x W x D) (mm)	87.88 x 83.29 x 145.822
	Weight (kg)	0.661
Note 1: AC power and fan module (186-3040-900) is shipped with two fan modules equipped (186-3042-900). Note 2: DC power and fan module (186-3041-900) is shipped with two fan modules equipped (186-3042-900). Note 3: Fan module (186-3042-900) is separately orderable for replacement only.		

The following tables provide technical specifications for the 100-277Vac/240Vdc AC power and fan module (186-3040-900) and the -48Vdc DC power and fan module (186-3041-900).

Table 8 Waveserver 5 100-277 Vac/240Vdc AC power and fan module specifications (186-3040-900)

Characteristic	Nominal	Min	Max
AC input voltage operating range (3 kW maximum output power)	200 - 277 Vac	180 Vac	305 Vac
AC input voltage operating range (2 kW maximum output power)	100 - 200 Vac	90 Vac	180 Vac
DC input voltage operating range (3 kW maximum output power)	240 Vdc	180 Vdc	300 Vdc
Output voltage	12 Vdc	11.8 Vdc	12.18 Vdc
Output power	—	—	3 kW/ 2 kW
AC input frequency	50/60 Hz	47 Hz	63 Hz
AC or DC fuse/breaker protection (per module)	—	—	25

Characteristic	Nominal	Min	Max
Note 1: All Waveserver 5 shelf assemblies are suitable for installation as part of the Common Bonding Network (CBN) or an Isolated Bonding Network (IBN) or both, or an European Telecommunications Standards Institute (ETSI) Mesh Bonding Network. The Waveserver 5 supports the connection of the battery return (for the DC power module) in an isolated DC return (DC-I) or common DC return (DC-C) configuration. The DC power module return lead, L+, must be earthed (connected to ground) at some point within the building, consistent with the applicable power infrastructure (DC-C or DC-I) to satisfy safety requirements. Note 2: All AC inputs must be equipped with an external surge protection device (SPD not supplied) to meet the Telcordia requirements for lightning criteria for equipment interfacing with AC power port(s). Note 3: All Waveserver 5 shelf assemblies are suitable for installation in Network Telecommunication Facilities and locations where the United States National Electrical Code (NEC) applies.			

Table 9 Waveserver 5 –48Vdc DC power and fan module specifications (186-3041-900)

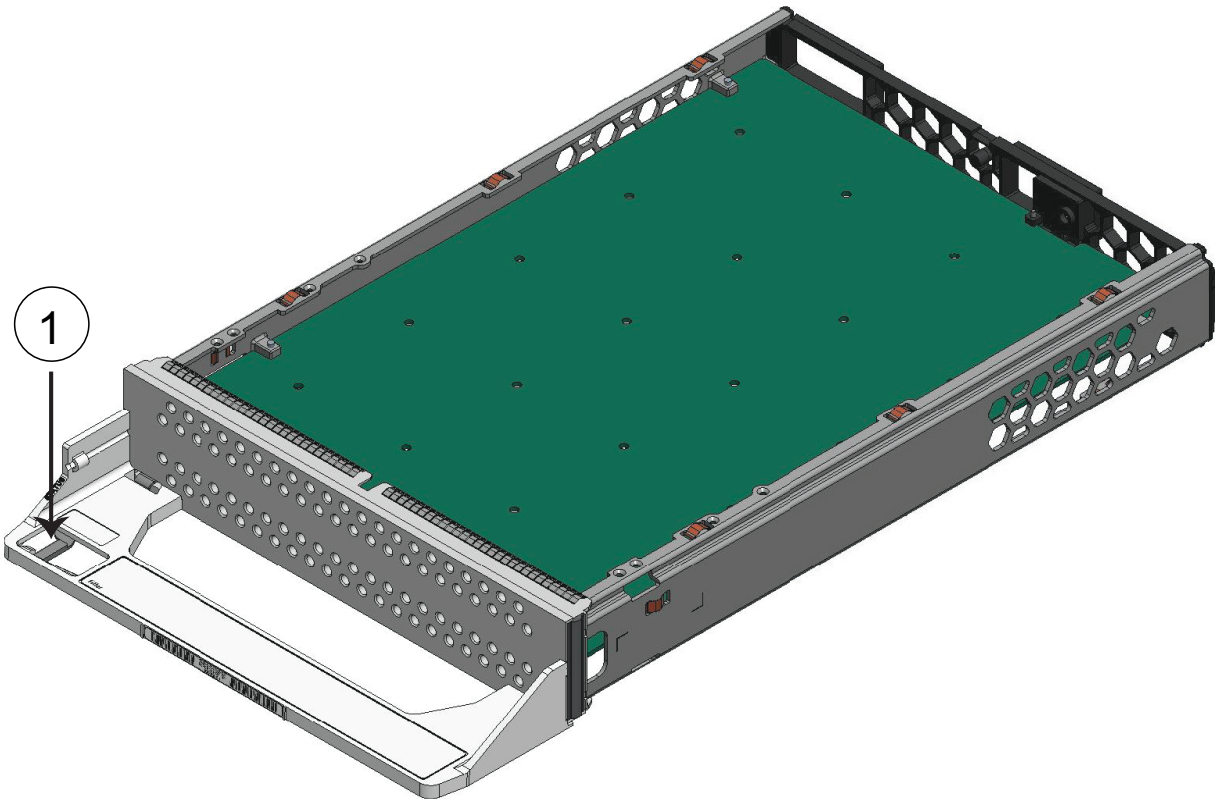
Characteristic	Nominal	Min	Max
DC input voltage operating range	–48Vdc/–60 Vdc	–40 Vdc	–72 Vdc
Output	12 Vdc	11.8 Vdc	12.18 Vdc
Output power	–	–	2.3 kW
DC fuse/breaker protection (per module)	–	–	60 A
Note 1: All Waveserver 5 shelf assemblies are suitable for installation as part of the Common Bonding Network (CBN) or an Isolated Bonding Network (IBN) or both, or an ETSI Mesh Bonding Network. The Waveserver 5 supports the connection of the battery return (for the DC power module) in an isolated DC return (DC-I) or common DC return (DC-C) configuration. The DC power module return lead, L+, must be earthed (connected to ground) at some point within the building, consistent with the applicable power infrastructure (DC-C or DC-I) to satisfy safety requirements. Note 2: All AC inputs must be equipped with an external surge protection device (SPD not supplied) to meet the Telcordia requirements for lightning criteria for equipment interfacing with AC power port(s). Note 3: All Waveserver 5 shelf assemblies are suitable for installation in Network Telecommunication Facilities and locations where the United States National Electrical Code (NEC) applies.			

Alarms

For a complete list of alarms and alarm clearing procedures, see [Chapter 28, “Alarms”](#).

Filler module (186-3031-900)

A double slot filler module (186-3031-900) must be installed in empty slots on the Waveserver 5 chassis. The filler module ensures that there is sufficient cooling of the chassis. An alarm is raised when an empty traffic slot has no service module or filler module. See the following figure.

Figure 10 Filler module**Legend**

1 Module latch

Technical specifications

The following table provides technical specifications for the filler modules.

Table 10 Technical specifications for filler module (186-3031-900)

Specification	Value
Dimensions (H x W x D) (mm)	43.16 x 204.532 x 363.1
Weight (kg)	0.423

Alarms

For a complete list of alarms and alarm clearing procedures, see [Chapter 28, "Alarms"](#).

Supported upright spacing depths

This table provides information on supported upright spacing depths. Review this table prior to ordering bracket and installation kits.

Table 11 Bracket kit selection - supported upright frame depths (without ground bracket)

Rack type (Note 1, Note 2)			Upright spacing				
19" (Electronic Industries Association) EIA	23" EIA	ETSI	75mm - 438mm (3" - 17.25")	438mm - 583mm (17.25" - 24")	583mm - 914mm (24" - 36")	914mm - 1067mm (36" - 42")	1067mm - 1200mm (42" - 48")
X			186-1300-900	186-1300-900	186-1306-900	Not supported	Not supported
	X		186-1301-900	186-1301-900	186-1307-900	N/A	Not supported
		X	186-1302-900	N/A	N/A	N/A	N/A

Note 1: Dimension 438mm (17.25") varies depending on the setback of the front bracket, the location of the ground bracket installation on the frame, the cable exit direction, and the type of ground lug used. This dimension assumes a 5" setback and cables exiting away from the rack (recommended).

Note 2: If you are using a 100mm (approximately 4") setback, the maximum upright spacing is 18.3". For a flush mount setback, the maximum upright spacing is 19".

Table 12 Bracket kit selection - supported upright frame depths (with ground bracket)

Rack type (Note 1, Note 2)			Upright spacing				
19" EIA	23" EIA	ETSI	75mm - 438mm (3" - 17.25")	438mm - 583mm (17.25" - 24")	583mm - 914mm (24" - 36")	914mm - 1067mm (36" - 42")	1067mm - 1200mm (42" - 48")
X			186-1300-900	Not supported	186-1306-900	Not supported	Not supported
	X		186-1301-900	Not supported	186-1307-900	N/A	Not supported
		X	186-1302-900	N/A	N/A	N/A	N/A

Note 1: Dimension 438mm (17.25") varies depending on the setback of the front bracket, the location of the ground bracket installation on the frame, the cable exit direction, and the type of ground lug used. This dimension assumes a 5" setback and cables exiting away from the rack (recommended).

Note 2: If you are using a 100mm (approximately 4") setback, the maximum upright spacing is 18.3". For a flush mount setback, the maximum upright spacing is 19".

CHAPTER 5

Module descriptions, equipping rules, and specifications (Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD)

This section contains the following topic:

- [Overview](#)
 - [Service modules \(186-3100-900/186-3100-901/186-3110-901/186-3101-900/186-3101-901/186-3111-901/186-3102-900/186-3102-901/186-3112-901\)](#)

Overview

Service modules are the traffic carrying modules on the Waveserver 5. The Waveserver 5 chassis supports up to four Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD modules. Both C-Band and L-Band modules are available, in either 95GBaud or 91.6GBaud.

Service modules (186-3100-900/186-3100-901/186-3110-901/186-3101-900/186-3101-901/186-3111-901/186-3102-900/186-3102-901/186-3112-901)

The following service modules are supported:

- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-900)
- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901)
- Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFP-DD Module (186-3110-901)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-900)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901)
- Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module (186-3111-901)

- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-900)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901)
- Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3112-901)

Note 1: For Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-900), Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-900), and Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-900), only the following traffic configurations are supported:

- Unmated lines
- Unprotected traffic
- Any x00-E transmission mode with a mix of 100GE/400GE clients
- Any x00-O transmission mode up to 600G with a mix of 100GE/OTL4.4 client rates

Note 2: The Premium variants offer better reach performance than Enhanced variants for applications that require 600G of line capacity and above (link-engineering allowed).

Note 3: For the documentation purpose, a simplified name WL5e 2x MOTR 16x module is used to represent this type of service modules at other places in this document.

See the following figures.

Figure 11 Service modules (186-3100-900/186-3100-901/186-3110-901/186-3101-900/186-3101-901/
186-3111-901/186-3102-900/186-3102-901/186-3112-901)

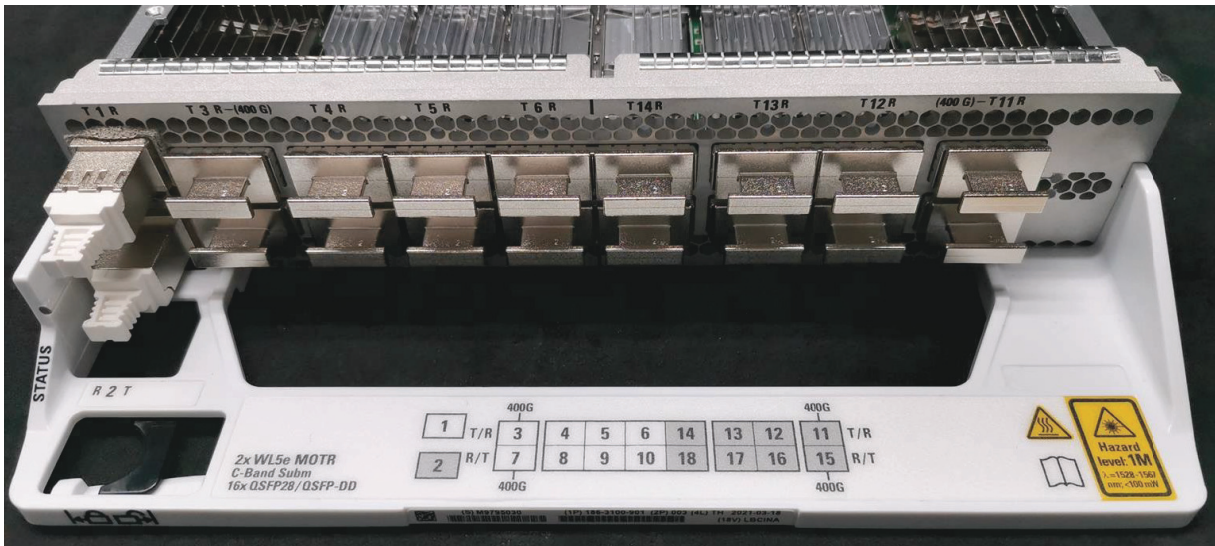


Figure 12 Service module client and line ports

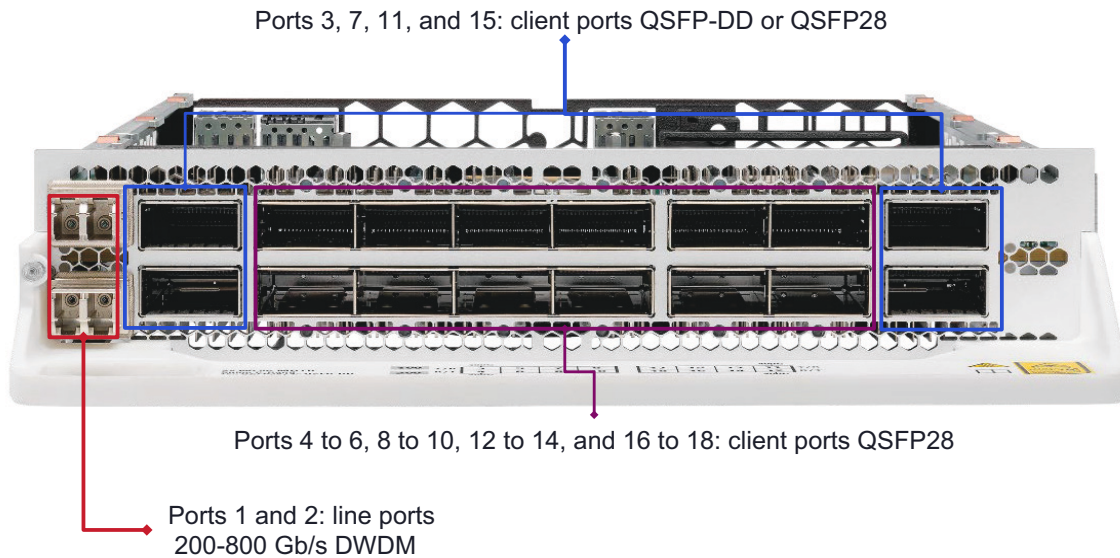
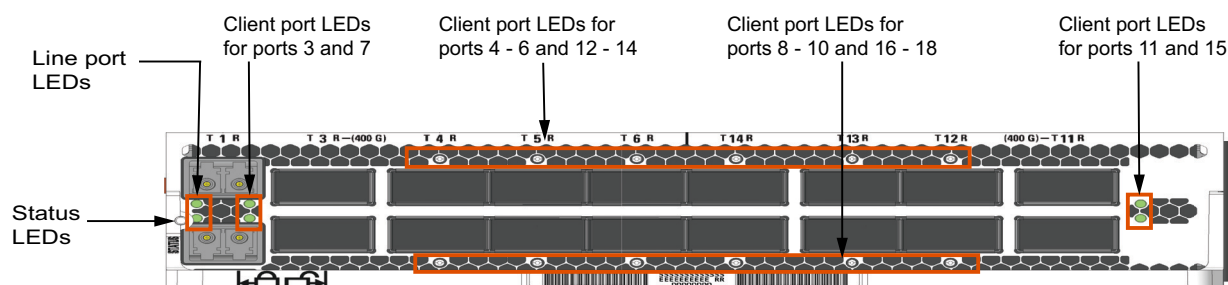


Figure 13 Service module LEDs



LEDs

The following LED definitions exist for the service modules:

- Status LED
 - Red (solid): Indicates there is a hardware fault on the service module.
 - Red (flashing): Indicates the service module is in low level boot.
 - Green (solid): Indicates the service module is operational.
 - Green (flashing): Indicates the software on the service module is initializing.
- Line port LED
 - Green (solid): Indicates line optics are operational.
 - Yellow/red bi-color (solid): Yellow indicates a Rx loss of signal on the line port and red indicates a line optics failure.
- Client port LED
 - Green (solid): Indicates the client pluggable is operational.
 - Yellow/red bi-color (solid): Yellow indicates a Rx loss of signal on the client port and red indicates a client pluggable failure.
 - Off: Indicates the client pluggable is not present or not operational.

Supported features

The service modules support these features:

- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-900) is used for submarine C-Band applications.
- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901) is used for submarine C-Band applications.
- Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFP-DD Module (186-3110-901) is used for submarine L-Band applications.
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-900) is used for terrestrial (premium) C-Band applications.

- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901) is used for terrestrial (premium) C-Band applications.
- Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module (186-3111-901) is used for terrestrial (premium) L-Band applications.
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-900) is used for terrestrial (enhanced) C-Band applications.
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901) is used for terrestrial (enhanced) C-Band applications.
- Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3112-901) is used for terrestrial (enhanced) L-Band applications.
- Up to four double-width service modules are supported in a Waveserver 5 chassis.
- 2 line ports (2 x 800 Gb/s) for up to 1.6 Tb/s
- 16 client ports
 - 4 x bilingual QSFP-DD/QSFP28 client ports (ports 3, 7, 11, and 15)
 - 12 x QSFP28 client ports (ports 4 to 6, 8 to 10, 12 to 14, and 16 to 18)
- Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD line transmission modes

The following table provides information on Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD line transmission modes.

Table 13 Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD line transmission modes

GBaud	x00G transmission modes	x50G transmission modes
95	— 95-200-O — 95-300-O — 95-300-E — 95-400-O — 95-400-E — 95-500-O — 95-500-E — 95-600-O — 95-600-E — 95-700-O — 95-700-E — 95-800-O — 95-800-E	— 95-250-O — 95-350-O — 95-350-E — 95-450-O — 95-450-E — 95-550-O — 95-550-E — 95-650-O — 95-650-E
91.6	— 91.6-300-E — 91.6-400-E — 91.6-500-E — 91.6-600-E — 91.6-700-E — 91.6-800-E	— 91.6-450-E — 91.6-550-E

- Auto-provisioning of client port when a client pluggable is inserted
- Supports Ethernet and OTN clients
 - 100GBASE-R4 (AOC, SR4, PSM4, CWDM4, LR4, 4WDM)
 - 100GBASE-R (DR1, FR1)
 - 400GE (AOC, FR4, LR4)
 - OTL4.4
- Client OPS protection support for
 - 100GE and OTU4 clients
 - all xxx-xxxx-901 variants of Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD modules as well as Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD encryption modules when 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 pluggable module (160-9402-900) is used.

- The traffic outage time will be less than 50ms typical on client ports using OPS in a point-to-point configuration, when no hold-off time is applied, measured at the OPS hardware connected to Waveserver5.
- Conditioning-type of the client ports that are attached to the OPS hardware should be set to laser-off.
- A Laser off active alarm is introduced which is triggered when the system intentionally shuts down the Tx laser.
- The conditioning hold off timer is only used on Ethernet ports when a line fault is detected. Conditioning hold off timer will not be used on OTU4 ports or on upstream client fault.
- Trunk OPS protection support for
 - unmated line
 - 100GE clients with the following holdoff time:
 - Ethernet Conditioning w/ 100ms holdoff for WL5e 2x800G 16xQSFP28/QSFP-DD non-encryption modules
 - Ethernet Conditioning w/ 200ms holdoff for WL5e 2x800G 16xQSFP28/QSFP-DD encryption modules
 - None Conditioning
 - Laser Off Conditioning w/ 100ms holdoff for WL5e 2x800G 16xQSFP28/QSFP-DD non-encryption modules
 - Laser Off Conditioning w/ 200ms holdoff for WL5e 2x800G 16xQSFP28/QSFP-DD encryption modules
 - 400GE clients with the following holdoff time:
 - Ethernet Conditioning w/ 100ms holdoff for WL5e 2x800G 16xQSFP28/QSFP-DD non-encryption modules
 - None Conditioning
 - Laser Off Conditioning w/ 100ms holdoff for WL5e 2x800G 16xQSFP28/QSFP-DD non-encryption modules
 - OTL4.4 clients (OTN Conditioning (holdoff N/A))
- Lamp flash is supported on the service modules where you can flash a LED to a slot or port for a specified amount of time. Use CLI command `system lamps set` to enable lamp flash for all ports on a specific service module or on a specific port.
- Terminal (Tx) and facility (Rx) loopbacks on client and line ports
- Integrated Test Set (ITS) in racetrack mode for OTL4.4/100GE/400GE clients
- Supported on 6500 photonic and RLS configurations

Service module port types, port number and supported pluggables

The following table provides information on service module port types and supported pluggables.

Table 14 Service module port types, port number and supported pluggables

Service module port type	Port number	Supported pluggable types (See Pluggables and specifications)
Line port	1 and 2	—
Client Port	3, 7, 11, and 15	QSFP28, QSFP-DD
	4 to 6	QSFP28
	8 to 10	
	12 to 14	
	16 to 18	

Technical specifications

This section provides information on the service module and line port technical specifications.

Table 15 Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD service module specifications

Specification	Value
Dimensions (H x W x D) (mm)	43.16 x 204.532 x 363.1
Weight (kg)	2.27
Power Consumption, output power without pluggables	• Typical: 330 W • Max: 360 W

Table 16 Line port specifications

Item	Specification
Transmitters connector type	LC
Line rate	See Supported line port transmission mode, baud, mapping, rate and client protocols .
Transmitter tunable frequency spacing	Flexgrid: Step-tunable with 100 MHz step size
Transmitter tunable wavelength and frequency range	C-Band: 191.2500 to 196.1250 THz with tuning precision 0.1 GHz (100 MHz or 0.0001 THz) for C-band variants (186-3100-900/186-3100-901/186-3101-900/186-3101-901/186-3102-900/186-3102-901) L-Band: 186.0875 to 190.9562 THz with tuning precision 0.1 GHz (100 MHz or 0.0001 THz) for L-band variants (186-3110-901/186-3111-901/186-3112-901)

Item	Specification
Transmitter central frequency accuracy	+/-1.5 GHz
Transmit output power tuning range	-9 dBm to +4 dBm
Tx power accuracy	+/-0.6 dB
Receiver connector type	LC
Receiver type	Balanced PIN Coherent Receiver
Per-Channel back-to-back sensitivity (minimum EOL), colored application (Note1)	<550 Gb/s: -25.0 dBm 550 Gb/s: -23.5 dBm 600 Gb/s: -22.5 dBm 650 Gb/s: -21.0 dBm 700 Gb/s: -19.5 dBm 800 Gb/s: -15 dBm
Input Power Range (per channel power)	-7 dBm to +5 dBm (Note 3)
Receiver power low	-22.0 dBm for <= 600G line rates -18.5 dBm for > 600G line rates (Note 5)
Receiver power high	12 dBm
Receiver channel power low	-18.5 dBm for all line rates (Note 5)
Receiver channel power high	9 dBm
Receiver Overload	+9 dBm (per channel) +14 dBm (total power) (Note 4)
Receiver Damage level	+17 dBm
Rx power monitor accuracy	+/-2 dB (total power)
Polarization Mode Dispersion (PMD) tolerance	<800 Gb/s: 50 ps mean, 150 ps peak 800 Gb/s: 30 ps mean, 90 ps peak

44 Module descriptions, equipping rules, and specifications (Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD)

Item	Specification
Polarization Dependent Loss (PDL) tolerance (peak)	200 Gb/s: 9.0 dB 250 Gb/s: 8.7 dB 300 Gb/s: 8.4 dB 350 Gb/s: 8.4 dB 400 Gb/s: 8.4 dB 450 Gb/s: 8.1 dB 500 Gb/s: 7.8 dB 550 Gb/s: 6.9 dB 600 Gb/s: 6.0 dB 650 Gb/s: 5.4 dB 700 Gb/s: 4.5 dB 800 Gb/s: 4.5 dB
ROADM support	Up to 28 (link dependent)
Chromatic dispersion tolerance	- For Enhanced and Premium variants, see “Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD Enhanced and Premium variants”. - For Submarine variants, see “Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD Submarine variants”.
Reach (Note 2)	Application and reach are managed by using the reach class licenses, modem-class and reach limit parameter on the line port facility. This is a licensed parameter, see Licenses and associated enforcement modes .
Note 1: Back-to-back sensitivity is defined as the Rx power for which the FEC BER threshold is not exceeded in Tx-to-Rx fiber loopback through fiber patch cords <10m total length and a variable optical attenuator (VOA). Note 2: Maximum supported reach is subject to appropriate link engineering analysis. Note 3: Across all transmission modes. No power related OSNR penalty when operating in this range. Recommended target Rx input power = -7 dBm unless another value is recommended for a specific modem deployment by Ciena link budget analysis. Note 4: Overload power level is beyond the range of specified receiver operation and subject to significant measurement inaccuracy. Note 5: The Rx channel power low alarm and/or the Rx power low alarm may be raised spuriously for a loss-limited unamplified span application. The power thresholds for these alarms are not programmable, therefore cannot be set to a lower power level to avoid the alarms.	

Table 17 Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 16xQSFP28/ QSFP-DD Enhanced and Premium variants

Parameter	C-Band (Note)				L-Band (Note)			
Gbaud	95 Gbaud		91.6 Gbaud		95 Gbaud		91.6 Gbaud	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Total dispersion compensation range	-45,300	120,000	-48,700	120,000	-42,900	120,000	-46,100	120,000
Tx dispersion compensation range	-5,700	80,400	-6,100	77,400	-5,400	82,500	-5,800	79,700
Rx dispersion compensation range (nominal)	-39,600	39,600	-42,600	42,600	-37,500	37,500	-40,300	40,300
Note: The supported Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD Enhanced and Premium variants are: - Enhanced variants: - C-Band: 186-3102-900/186-3102-901 - L-Band: 186-3112-901 - Premium variants: - C-Band: 186-3101-900/186-3101-901 - L-band: 186-3111-901								

Table 18 Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 16xQSFP28/ QSFP-DD Submarine variants

Parameter	C-Band (Note)				L-Band (Note)			
Gbaud	95 Gbaud		91.6 Gbaud		95 Gbaud		91.6 Gbaud	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Total dispersion compensation range	-45,300	469,200	-48,700	504,600	-42,900	444,200	-46,100	477,700
Tx dispersion compensation range	-5,700	429,600	-6,100	462,000	-5,400	406,700	-5,800	437,400
Rx dispersion compensation range (nominal)	-39,600	39,600	-42,600	42,600	-37,500	37,500	-40,300	40,300

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Parameter	C-Band (Note)	L-Band (Note)
Note: The supported Waveserver WL5e 2x800G 16xQSFP28/QSFP-DD Submarine variants are: • C-Band: 186-3100-900/186-3100-901 • L-band: 186-3110-901		

Latency

The following table shows latency values for the service modules.

Table 19 Service module (186-3100-900/186-3100-901/186-3110-901/186-3101-900/186-3101-901/186-3111-901/186-3102-900/186-3102-901/186-3112-901) latency specifications

Client XCVR Mode	Client FEC	Line XCVR mode (transmission mode)	Latency, typical (us) (Note 1, Note 2)
100GE	• no FEC • KR4 FEC • KP4 FEC	• x00G-E unmated • x00G-O unmated	36 for line rates <= 500G 27 for line rates >= 600G
		• x50G-E mated • x50G-O mated	47
OTL4.4	no FEC	x00G-O unmated	35 for line rates <= 500G 27 for line rates >= 600G
		x50G-O mated	47
	GFEC	x00G-O unmated	36 for line rates <= 500G 28 for line rates >= 600G
		x50G-O mated	47
400GE	KP4 FEC	x00G-E unmated	• 36 for line rates <= 500G • 27 for line rates >= 600G
		x00G-E mated	47

Note 1: The latency reported in this table is the latency of the entire datapath, including the line modem, client functions, and the client FEC.

Note 2: Preliminary values for transmission modes 95-500-O, 95-600-O, 95-600-E, 95-700-O,95-700-E, 91.6-500-E,91.6-600-E, 91.6-800-E, 95-250-O, 95-450-O, 95-450-E, 95-550-O, 95-550-E, 91.6-450E, 91.6-550E, 95-650-O, and 95-650-E.

Performance Monitoring

For information on performance monitoring, see [Chapter 27, “Performance Monitoring”](#).

Alarms

For a complete list of alarms and alarm clearing procedures, see [Chapter 28, “Alarms”](#).

Equipping

The Waveserver 5 chassis supports up to 4 double-width service modules. The double-width service modules are only permitted in slots 1, 3, 5, and 7 in the Waveserver 5 chassis.

Provisioning

For information on provisioning the service modules, see [Service module port types, port number and supported pluggables](#), [Module attributes](#) and [Line ports and client ports in WL5e 2x MOTR 16x modules](#).

CHAPTER 6

Module descriptions, equipping rules, and specifications (Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD)

This section contains the following topic:

- [Overview](#)
 - [Service modules \(186-3120-900/186-3121-900\)](#)

ATTENTION

Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD modules (186-3120-900/186-3121-900) do not support Datapath encryption.

Overview

Service modules are the traffic carrying modules on the Waveserver 5. The Waveserver 5 chassis supports up to four Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD modules. This module is available in 95GBaud, 93.3GBaud, 91.6GBaud, 90GBaud, 85GBaud, or 82GBaud.

Service modules (186-3120-900/186-3121-900)

The following service modules are supported:

- Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module (186-3120-900)
- Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module (186-3121-900)

Note: For the documentation purpose, a simplified name WL5e 2x MOTR 12x module is used to represent this type of service modules at other places in this document.

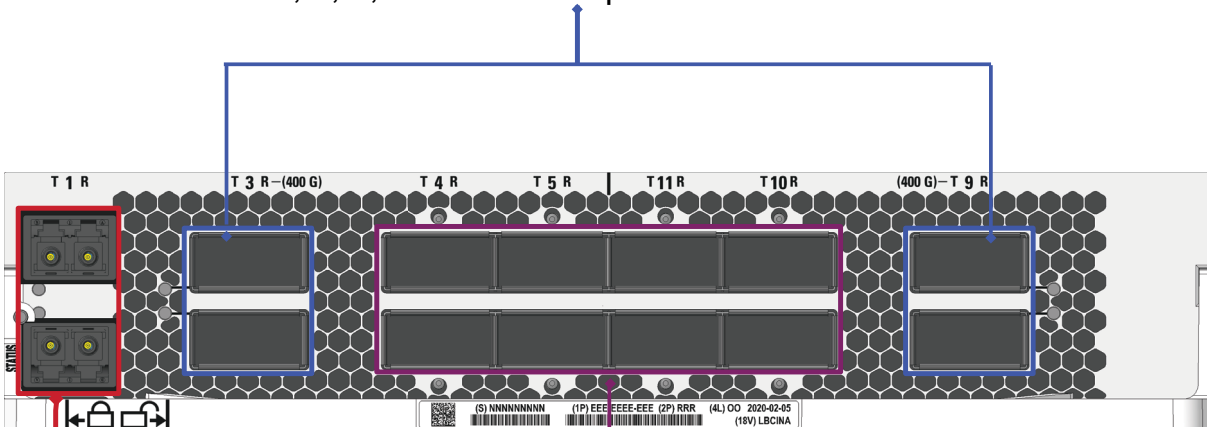
See the following figures.

Figure 14 Service modules (186-3120-900/186-3121-900)



Figure 15 Service module client and line ports

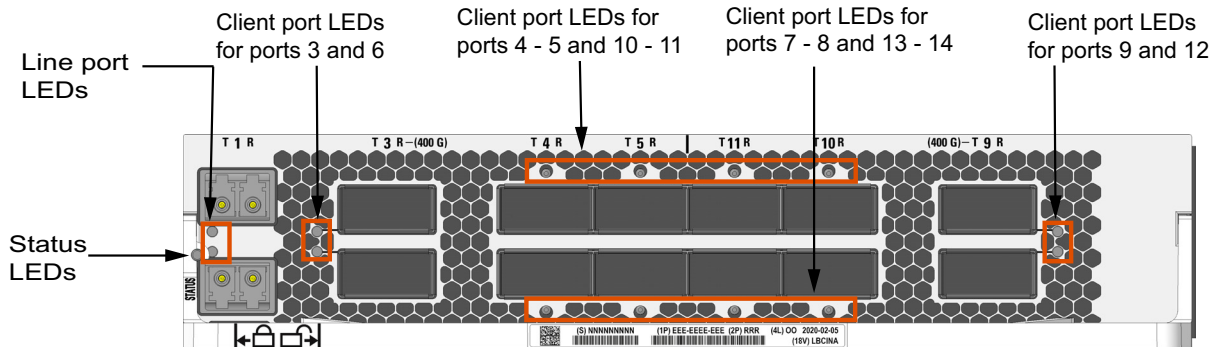
Ports 3, 6, 9, and 12: client ports QSFP-DD or QSFP28



Ports 4-5, 7-8, 10-11, and 13-14: client ports QSFP28

Ports 1 and 2: line ports 200-800 Gb/s DWDM

Figure 16 Service module LEDs



LEDs

The following LED definitions exist for the service modules:

- Status LED
 - Red (solid): Indicates there is a hardware fault on the service module.
 - Red (flashing): Indicates the service module is in low level boot.
 - Green (solid): Indicates the service module is operational.
 - Green (flashing): Indicates the software on the service module is initializing.
- Line port LED
 - Green (solid): Indicates line optics are operational.
 - Yellow/red bi-color (solid): Yellow indicates a Rx loss of signal on the line port and red indicates a line optics failure.
- Client port LED
 - Green (solid): Indicates the client pluggable is operational.
 - Yellow/red bi-color (solid): Yellow indicates a Rx loss of signal on the client port and red indicates a client pluggable failure.
 - Off: Indicates the client pluggable is not present or not operational.

Supported features

The service modules support these features:

- Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module (186-3120-900) is used for submarine C-Band applications.
- Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module (186-3121-900) is used for terrestrial (premium) C-Band applications.
- Up to four double-width service modules are supported in a Waveserver 5 chassis.

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- 2 line ports (2 x 800 Gb/s) for up to 1.6 Tb/s (ports 1 and 2)
- 12 client ports
 - 4 x bilingual QSFP-DD/QSFP28 client ports (ports 3, 6, 9, and 12)
 - 8 x QSFP28 client ports (ports 4-5, 7-8, 10-11, and 13- 14)
- Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD line transmission modes

The following table provides information on Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD line transmission modes.

Table 20 Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD line transmission modes

GBaud	x00G transmission modes	x50G transmission modes
95	— 95-200-O — 95-300-O — 95-300-E — 95-400-O — 95-400-E — 95-500-O — 95-500-E — 95-600-O — 95-600-E — 95-700-O — 95-700-E — 95-800-O — 95-800-E	— 95-250-O — 95-350-O — 95-350-E — 95-450-O — 95-450-E — 95-550-O — 95-550-E — 95-650-O — 95-650-E
93.3	— 93.3-400-E — 93.3-600-E	N/A
91.6	— 91.6-300-E — 91.6-400-E — 91.6-500-E — 91.6-600-E — 91.6-700-E — 91.6-800-E	— 91.6-450-E — 91.6-550-E
90	— 90-400-E — 90-600-E	N/A

GBaud	x00G transmission modes	x50G transmission modes
85	— 85-300-E — 85-400-E — 85-600-E	N/A
82	— 82-400-E	N/A

- Auto-provisioning of client port when a client pluggable is inserted
- Supports Ethernet and OTN clients
 - 100GBASE-R4 (AOC, SR4, PSM4, CWDM4, LR4, 4WDM)
 - 100GBASE-R (DR, FR)
 - 400GE (AOC, FR4, LR4)
 - 4x100GE (ZR)
 - OTL4.4
- Trunk OPS protection support for
 - unmated line

Note: Line OPS is not supported for 4x100GE-ZR client.

 - 100GE/400GE clients with the following holdoff time:
 - Ethernet Conditioning w/ 100ms holdoff
 - None Conditioning
 - OTL4.4 clients (OTN Conditioning (holdoff N/A))
- The conditioning hold off timer is only used on Ethernet ports when a line fault is detected. Conditioning hold off timer will not be used on OTU4 ports or on upstream client fault.
- Lamp flash is supported on the service modules where you can flash a LED to a slot or port for a specified amount of time. Use CLI command `system lamps set` to enable lamp flash for all ports on a specific service module or on a specific port.
- Terminal (Tx) and facility (Rx) loopbacks on client and line ports
- Integrated Test Set (ITS) in racetrack mode for OTL4.4, 100GE, and 400GE clients
- Supported on 6500 photonic and RLS configurations

Service module port types, port number and supported pluggables

The following table provides information on service module port types and supported pluggables.

Table 21 Service module port types, port number and supported pluggables

Service module port type	Port number	Supported pluggable types (See Pluggables and specifications)
Line port	1 and 2	—
Client Port	3, 6, 9, and 12	QSFP28, QSFP-DD
	4-5	QSFP28
	7-8	
	10-11	
	13-14	

Technical specifications

This section provides information on the service module and line port technical specifications.

Table 22 Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD service module specifications

Specification	Value
Dimensions (H x W x D) (mm)	43.16 x 204.532 x 363.1
Weight (kg)	2.27
Power Consumption, output power without pluggables	• Typical: 330W • Max: 360W

Table 23 Line port specifications

Item	Specification
Transmitters connector type	LC
Line rate	See Supported line port transmission mode, baud, mapping, rate and client protocols .
Transmitter tunable frequency spacing	Flexgrid: Step-tunable with 100 MHz step size
Transmitter tunable wavelength and frequency range	C-Band: 191.2500 to 196.1250 THz with tuning precision 0.1 GHz (100 MHz or 0.0001 THz) for C-band variants (186-3120-900/186-3121-900)
Transmitter central frequency accuracy	+/-1.5 GHz
Transmit output power tuning range	-9 dBm to +4 dBm
Tx power accuracy	+/-0.6 dB
Receiver connector type	LC
Receiver type	Balanced PIN Coherent Receiver

Item	Specification
Per-Channel back-to-back sensitivity (minimum EOL), colored application (Note 1)	<550 Gb/s: -25.0 dBm 550 Gb/s: -23.5 dBm 600 Gb/s: -22.5 dBm 650 Gb/s: -21.0 dBm 700 Gb/s: -19.5 dBm 800 Gb/s: -15 dBm
Input Power Range (per channel power)	-7 dBm to +5 dBm (Note 3)
Receiver power low	• -22.0 dBm for ≤ 600G line rates • -18.5 dBm for > 600G line rates (Note 5)
Receiver power high	12 dBm
Receiver channel power low	• -24.0 dBm for ≤ 600G line rates • -20.5 dBm for > 600G line rates (Note 5)
Receiver channel power high	9 dBm
Receiver Overload	+9 dBm (per channel) +14 dBm (total power) (Note 4)
Receiver Damage level	+17 dBm
Rx power monitor accuracy	+/-2 dB (total power)
Polarization Mode Dispersion (PMD) tolerance	<800 Gb/s: 50 ps mean, 150 ps peak 800 Gb/s: 30 ps mean, 90 ps peak

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Item	Specification
Polarization Dependent Loss (PDL) tolerance (peak)	200 Gb/s: 9.0 dB 250 Gb/s: 8.7 dB 300 Gb/s: 8.4 dB 350 Gb/s: 8.4 dB 400 Gb/s: 8.4 dB 450 Gb/s: 8.1 dB 500 Gb/s: 7.8 dB 550 Gb/s: 6.9 dB 600 Gb/s: 6.0 dB 650 Gb/s: 5.4 dB 700 Gb/s: 4.5 dB 800 Gb/s: 4.5 dB
ROADM support	Up to 28 (link dependent)
Chromatic dispersion tolerance	• For Premium variant, see “Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD Premium variant”. • For Submarine variant, see “Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD Submarine variant”.
Reach (Note 2)	Application and reach are managed by using the reach class licenses, modem-class and reach limit parameter on the line port facility. This is a licensed parameter, see Licenses and associated enforcement modes .
Note 1: Back-to-back sensitivity is defined as the Rx power for which the FEC BER threshold is not exceeded in Tx-to-Rx fiber loopback through fiber patch cords <10m total length and a variable optical attenuator (VOA). Note 2: Maximum supported reach is subject to appropriate link engineering analysis. Note 3: Across all transmission modes. No power related OSNR penalty when operating in this range. Recommended target Rx input power = -7 dBm unless another value is recommended for a specific modem deployment by Ciena link budget analysis. Note 4: Overload power level is beyond the range of specified receiver operation and subject to significant measurement inaccuracy. Note 5: The Rx channel power low alarm and/or the Rx power low alarm may be raised spuriously for a loss-limited unamplified span application. The power thresholds for these alarms are not programmable, therefore cannot be set to a lower power level to avoid the alarms.	

Table 24 Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 12xQSFP28/ QSFP-DD Premium variant

Parameter	C-Band (Note)					
Gbaud	95 Gbaud	93.3 Gbaud	91.6 Gbaud	90 Gbaud	85 Gbaud	82 Gbaud
	Min./Max.	Min./Max.	Min./Max.	Min./Max.	Min./Max.	Min./Max.
Total dispersion compensation range	-45,300/120,000	-46,900/120,000	-48,700/120,000	-50,400/120,000	-56,500/120,000	-60,800/120,000
Tx dispersion compensation range	-5,700/80,400	-5,900/79,000	-6,100/77,400	-6,300/75,900	-7,100/70,600	-7,600/66,800
Rx dispersion compensation range (nominal)	-39,600/39,600	-41,000/41,000	-42,600/42,600	-44,100/44,100	-49,400/49,400	-53,200/53,200
Note: The supported Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD Premium variant is • C-Band: 186-3121-900						

Table 25 Chromatic dispersion tolerance values (ps/nm) for Waveserver WL5e 2x800G 12xQSFP28/ QSFP-DD Submarine variant

Parameter	C-Band (Note)					
Gbaud	95 Gbaud	93.3 Gbaud	91.6 Gbaud	90 Gbaud	85 Gbaud	82 Gbaud
	Min./Max.	Min./Max.	Min./Max.	Min./Max.	Min./Max.	Min./Max.
Total dispersion compensation range	-45,300/469,200	-46,900/486,400	-48,700/504,600	-50,400/523,100	-56,500/586,000	-60,800/630,300
Tx dispersion compensation range	-5,700/429,600	-5,900/445,400	-6,100/462,000	-6,300/479,000	-7,100/536,600	-7,600/577,100
Rx dispersion compensation range (nominal)	-39,600/39,600	-41,000/41,000	-42,600/42,600	-44,100/44,100	-49,400/49,400	-53,200/53,200
Note: The supported Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD Submarine variant is: • C-Band: 186-3120-900						

Latency

The following table shows latency values for the service modules.

Table 26 Service module (186-3120-900/186-3121-900) latency specifications

Client XCVR Mode	Client FEC	Line XCVR mode (transmission mode)	Latency, typical (us) (Note)
100GE	• no FEC • KR4 FEC • KP4 FEC	• x00G-E unmated • x00G-O unmated	36 for line rates <= 500G 27 for line rates >= 600G
		• x50G-E mated • x50G-O mated	47
OTL4.4	no FEC	x00G-O unmated	35 for line rates <= 500G 27 for line rates >= 600G
		x50G-O mated	47
	GFEC	x00G-O unmated	36 for line rates <= 500G 28 for line rates >= 600G
		x50G-O mated	47
400GE	KP4 FEC	x00G-E unmated	• 36 for line rates <= 500G • 27 for line rates >= 600G
		x00G-E mated	47
4x100GE-ZR	CFEC	• x00G-E unmated • x00G-O unmated	44 for line rates <= 500G 36 for line rates >= 600G

Note: The latency reported in this table is the latency of the entire datapath, including the line modem, client functions, and the client FEC.

Performance Monitoring

For information on performance monitoring, see [Chapter 27, “Performance Monitoring”](#).

Alarms

For a complete list of alarms and alarm clearing procedures, see [Chapter 28, “Alarms”](#).

Equipping

The Waveserver 5 chassis supports up to 4 double-width service modules. The double-width service modules are only permitted in slots 1, 3, 5, and 7 in the Waveserver 5 chassis.

Provisioning

For information on provisioning the service modules, see [Service modules \(186-3120-900/186-3121-900\)](#) and [Line ports and client ports in WL5e 2x MOTR 12x modules](#).

CHAPTER 7

Module descriptions, equipping rules, and specifications (Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD)

This section contains the following topic:

- [Overview](#)
 - [Service module \(186-3302-900\)](#)

Overview

Service modules are the traffic carrying modules on the Waveserver 5. The Waveserver 5 chassis supports up to four Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD service modules.

Service module (186-3302-900)

The following service module is supported:

- Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900)

Note: For the documentation purpose, a simplified name WL5n DCO 3xQSFP-DD 12x module is used to represent this type of service module at other places in this document.

See the following figures.

60 Module descriptions, equipping rules, and specifications (Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD)

Figure 17 Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD service module (186-3302-900)

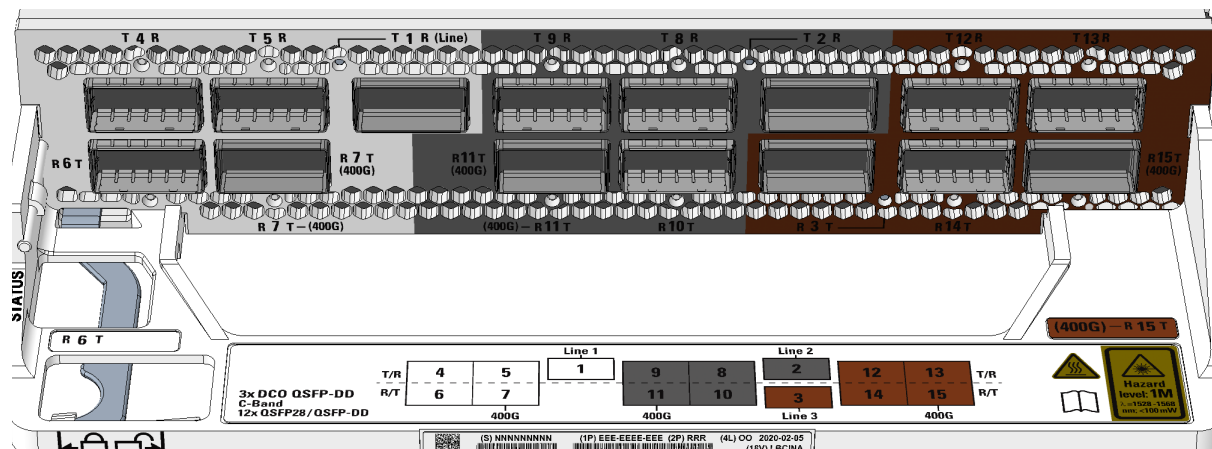


Figure 18 Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD service module client and line ports

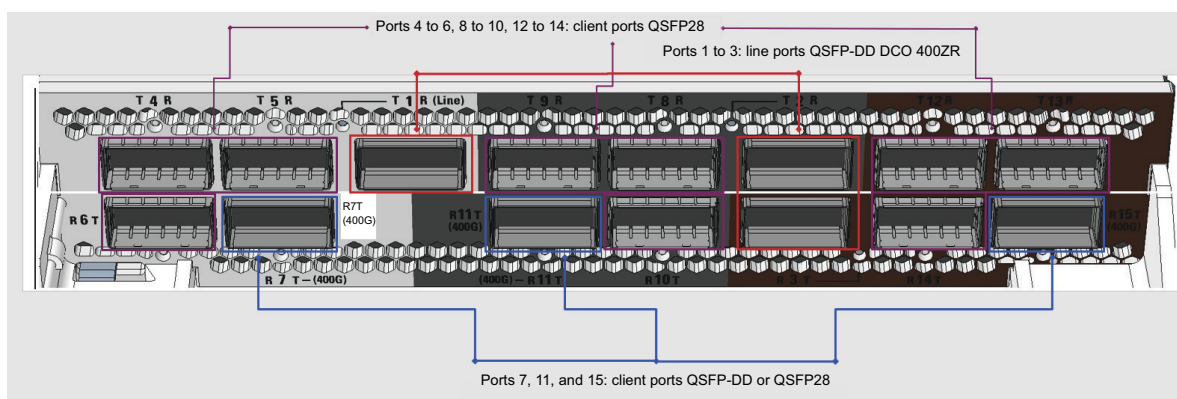
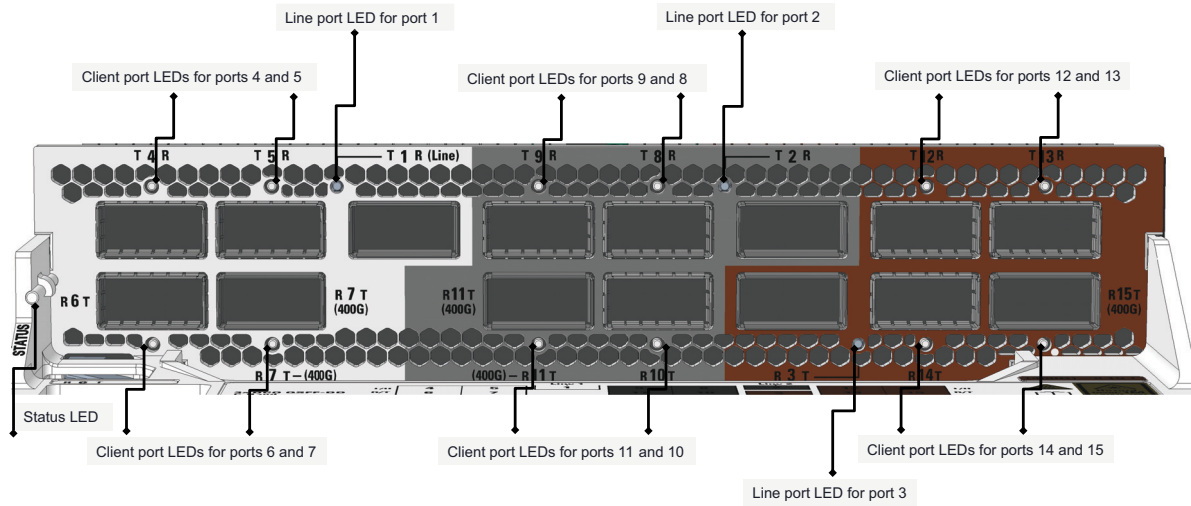


Figure 19 Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD service module LEDs



LEDs

The following LED definitions exist for the service modules:

- Status LED
 - Red (solid): Indicates there is a hardware fault on the service module.
 - Red (flashing): Indicates the service module is in low level boot.
 - Green (solid): Indicates the service module is operational.
 - Green (flashing): Indicates the software on the service module is initializing.
- Line port LED
 - Green (solid): Indicates line optics are operational.
 - Yellow/red bi-color (solid): Yellow indicates a Rx loss of signal on the line port and red indicates a line optics failure.
- Client port LED
 - Green (solid): Indicates the client pluggable is operational.
 - Yellow/red bi-color (solid): Yellow indicates a Rx loss of signal on the client port and red indicates a client pluggable failure.
 - Off: Indicates the client pluggable is not present or not operational.

Supported features

The service module supports these features:

- Up to four double-width service modules are supported in a Waveserver 5 chassis for a total of 4.8Tb/s capacity.
- 3 QSFP-DD 400ZR line ports (3 x 400 Gb/s) for up to 1.2 Tb/s.

- 12 client ports
 - 3 x bilingual QSFP-DD/QSFP28 client ports (ports 7, 11, and 15)
Note: Only QSFP28 is supported in this release.
 - 9 x QSFP28 client ports (ports 4 to 6, 8 to 10, and 12 to 14)
- Mapping three groups of 4x100GE clients to three QSFP-DD DCO 400ZR line-side plugs as shown below. The client ports can be comprised of 4 x QSFP28 (100G). It performs the role of a transponder. The module's total data throughput is 1.2 Tbs.
 - Group 1: 4 x QSFP28 client ports (ports 4, 5, 6, and 7) mapped to line port 1 (QSFP-DD DCO 400ZR).
 - Group 2: 4 x QSFP28 client ports (ports 8, 9, 10, and 11) mapped to line port 2 (QSFP-DD DCO 400ZR).
 - Group 3: 4 x QSFP28 client ports (ports 12, 13, 14, and 15) mapped to line port 3 (QSFP-DD DCO 400ZR).

Note: Mapping 1 x QSFP-DD (4x100G) to a line port (QSFP-DD DCO 400ZR) is not supported in this release. This release only supports mapping 4 x QSFP28 (100G) to a line port (QSFP-DD DCO 400ZR).
- Auto-provisioning of line and client ports when a client or line pluggable is inserted
- 400ZR line rate at 60 GBaud 16QAM
 - 100GE client protocol carried in a 400ZR line
- Supports Ethernet clients
 - 100GBASE-R4
 - 100GBASE-R (DR, FR)
- Line transmission modes
 - 4x100GE-ZR
- Lamp flash is supported on the service modules where you can flash a LED to a slot or port for a specified amount of time. Use CLI command `system lamps set` to enable lamp flash for all ports on a specific service module or on a specific port.
- Terminal (Tx) and facility (Rx) loopbacks on client and line ports

Service module port types, port number and supported pluggables

The following table provides information on service module port types and supported pluggables.

Table 27 Service module port types, port number and supported pluggables

Service module port type	Port number	Supported pluggable types (See Pluggables and specifications)
Line port	1, 2, and 3	WL5n 400ZR Type 3.0 QSFP-DD
Client Port	7, 11, and 15	QSFP28, QSFP-DD Note: The QSFP-DD is currently not supported in this release.
	4 to 6 8 to 10 12 to 14	QSFP28

Technical specifications

This section provides information on the service module and line port technical specifications.

Table 28 Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD service module and line port specifications

Specification	Value
Dimensions (H x W x D) (mm)	43.16 x 204.532 x 363.1
Weight (kg)	1.38
Power Consumption, output power without pluggables	• Typical: 80 W • Max: 83 W
Line port specifications	See “180-3530-900: WL5n 400ZR Type 3.0 QSFP-DD” on page 91

Latency

The following table shows latency values for the service modules.

Table 29 Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD service module (186-3302-900) latency specifications

Transmission mode	Client XCVR Mode	Client FEC	Latency, typical (us) (Note)
4x100GE-ZR	100GE	None FEC KR4 (NRZ plugs) KP4 (PAM4 plugs)	9
Note: The latency reported in this table is the latency of the entire datapath, including the line modem, client functions, and the client FEC.			

Performance Monitoring

For information on performance monitoring, see [Chapter 27, “Performance Monitoring”](#).

Alarms

For a complete list of alarms and alarm clearing procedures, see [Chapter 28, “Alarms”](#).

Equipping

The Waveserver 5 chassis supports up to 4 double-width service modules. The double-width service modules are only permitted in slots 1, 3, 5, and 7 in the Waveserver 5 chassis.

Provisioning

For information on provisioning the service modules, see [Service module port types, port number and supported pluggables](#), [Module attributes](#) and [Line ports and client ports in WL5n DCO 3xQSFP-DD 12x modules](#).

CHAPTER 8

Pluggables and specifications

This section covers the following topics:

- [Overview](#)
 - [QSFP28](#)
 - [QSFP-DD](#)
- [Datasheets](#)

Overview

QSFP28 and QSFP-DD pluggables are used on the client ports on the Waveserver 5.

QSFP28

The Enhanced quad (4-channel) small form-factor pluggable 28 (QSFP28) have similar form-factor as the conventional QSFP+ pluggables, but they have higher data rates in comparison to QSFP+ and also support 4x25G+ optical channels. The QSFP28 pluggables are used as client interfaces in the service modules.

QSFP-DD

The QSFP-DD pluggables have a similar form-factor as the standard QSFP28 pluggables (but different overall dimension as QSFP28 can be inserted into a QSFP-DD cage but not vice versa) but at four times the data rate compared to QSFP28. QSFP-DD pluggables support a variety of optical interfaces for 400Gb/s transport, including:

- 400G-FR4
- 400G-ZR
- 400G-LR4

The QSFP-DD are used as client interfaces in the service modules except 400GBASE-ZR QSFP-DD which can be used as line interfaces in the service modules where applicable.

Datasheets

This section provides technical specifications for Waveserver 5 pluggables.

- [160-9400-900: 100GBASE-SR4, 4x25G, MMF, 850 nm, 100m, MPO QSFP28 specifications](#)
- [160-9401-900: 100GBASE-LR4, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications](#)
- [160-9402-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications](#)
- [160-9403-900: 100GE CWDM4, 4x25G, CWDM, SMF, 1310nm, 2 km QSFP28 specifications](#)
- [160-9404-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 40km QSFP28 specifications](#)
- [160-9405-900: 100GE 4WDM-40, 4x25G, SMF, 1310nm, 40km QSFP28 specifications](#)
- [160-9406-900: 100GE PSM4, 4x25G, SMF, 1310nm, 500m, MPO QSFP28 specifications](#)
- [160-9431-900: 100G-FR, SMF, 1310nm, 2km QSFP28 specifications](#)
- [160-9433-900: 100GBASE-LR1, SMF, 1310nm, 10km QSFP28 specifications](#)
- [160-9460-xxx: 103.1G-112.28G, QSFP28-QSFP28 Active Optical Cables](#)
- [160-9660-xxx: 400GE, QSFP-DD - QSFP-DD Active Optical Cables](#)
- [160-9600-900: 400G-FR4, SMF, 2km QSFP-DD](#)
- [180-3530-900: WL5n 400ZR Type 3.0 QSFP-DD](#)
- [160-9603-900: 400G-LR4-10/4x100GE, SMF, 10km QSFP-DD](#)

QSFP28 and QSFP-DD pluggable specifications

This section provides datasheets and support information for QSFP28 and QSFP-DD pluggables.

160-9400-900: 100GBASE-SR4, 4x25G, MMF, 850 nm, 100m, MPO QSFP28 specifications

The following table provides information on 160-9400-900: 100GBASE-SR4, 4x25G, MMF, 850 nm, 100m, MPO QSFP28 specifications.

Table 30 160-9400-900: 100GBASE-SR4, 4x25G, MMF, 850 nm, 100m, MPO QSFP28 specifications

Specifications for 160-9400-900	Value
PEC (Part Number)	160-9400-900
CLEI	WOTRDP0FAA

Specifications for 160-9400-900	Value
Classification	100GBASE-SR4
Fiber	Multi-mode fiber (MMF)
Pluggable Form-Factor	QSFP28
Application	100G Ethernet
Standards	• QSFP28 MSA Compliant • IEEE 802.3 100GBASE-SR4
Transmitter connector type A single connector is used for both Tx and Rx on this QSFP28.	MPO12
Transmitter nominal wavelength	840 nm to 860 nm
Transmit average output power	-8.4 dBm to 2.4 dBm
Launch power in OMA minus TDEC (min)	-7.3 dBm
Transmitter and dispersion eye closure (TDEC), each lane (max)	4.3 dB
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Transmitter power high alarm threshold	5.4 dBm
Transmitter power low alarm threshold	-11.4 dBm
Receiver connector type A single connector is used for both Tx and Rx on this QSFP28.	MPO12
Receiver wavelength range	840 nm to 860 nm
Average Rx power range Average power is informative and not the principal indicator of signal strength. A received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-10.3 dBm to 2.4 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 5E-5	-5.2 dBm
Receiver overload	2.4 dBm
Receiver damage level	3.4 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB

Specifications for 160-9400-900	Value
Receiver power high alarm threshold	2.4 dBm
Receiver power low alarm threshold	-10.3 dBm
Maximum channel insertion loss	1.9 dB, OM4 fiber 1.8 dB, OM3 fiber
Nominal reach each standard	100 m, OM4 fiber 70 m, OM3 fiber
Typical power consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	2
Power budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	3
Supported Rates	103.125G (100GE)

160-9401-900: 100GBASE-LR4, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications

The following table provides information on 160-9401-900: 100GBASE-LR4, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications.

Table 31 160-9401-900: 100GBASE-LR4, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications

Specifications for 160-9401-900	Value
PEC (Part Number)	160-9401-900
CLEI	WOTRD2ZFAA
Classification	100GBASE-LR4
Fiber	Single-mode fiber (SMF)
Pluggable Form-Factor	QSFP28
Application	100G Ethernet
Standards	QSFP28 MSA Compliant IEEE 802.3 100GBASE-LR4
Transmitter connector type	LC

Specifications for 160-9401-900	Value
Transmitter wavelength range	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm
Transmit average output power, each lane	-4.3 dBm to 4.5 dBm
Launch power in OMA minus TDP, each lane (min)	-2.3 dBm
Transmitter dispersion penalty (TDP), each lane (max)	2.2 dB
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Transmitter power high alarm threshold	6.5 dBm
Transmitter power low alarm threshold	-6.3 dBm
Receiver connector type	LC
Receiver wavelength range	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm
Average Rx power range Average power is informative and not the principal indicator of signal strength. A received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-10.6 dBm to 4.5 dBm
Receiver sensitivity (OMA), each lane (max) at 1 E-12 BER For Ethernet application, receiver sensitivity is informative.	-8.6 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 1E-12	-6.8 dBm
Receiver overload	4.5 dBm
Receiver damage level	5.5 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Receiver power high alarm threshold	4.5 dBm
Receiver power low alarm threshold	-10.6 dBm

Specifications for 160-9401-900	Value
Maximum channel insertion loss	6.3 dB
Nominal reach each standard	10 km
Typical Power consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	3
Power budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	4
Supported Rates	103.125G (100GE)

160-9402-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications

The following table provides information on 160-9402-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications.

Table 32 160-9402-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 10 km QSFP28 specifications

Specifications for 160-9402-900	Value	
Classification	100GBASE-LR4	OTU-4 4L1-9D1F
PEC (Part Number)	160-9402-900	
CLEI	WOTRD20FAA	
Fiber	Single-mode fiber (SMF)	
Pluggable Form-Factor	QSFP28	
Application	100G Ethernet applications	OTU4 applications
Standards	QSFP28 MSA Compliant, IEEE 802.3 100GBASE-LR4	ITU-T G.959.1 4L1-9D1F
Transmitter connector type	LC	
Transmitter nominal wavelength	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm	

Specifications for 160-9402-900	Value	
Classification	100GBASE-LR4	OTU-4 4L1-9D1F
Transmit average output power (each lane) Average power for 100GBASE-LR4 is informative and not the principal indicator of signal strength.	-4.3 dBm to 4.5 dBm	-0.6 to 4.0 dBm
Maximum optical path penalty	-	1.5 dB
Transmitter dispersion penalty (TDP), each lane (max)	2.2 dB	-
Optical Modulation Amplitude (OMA), each lane	-1.3 dBm to 4.5 dBm	-
Transmitter launch power minimum in OMA minus TDP (each lane)	-2.3 dBm	-
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB	
Transmitter power high alarm threshold	6.5 dBm	6 dBm
Transmitter power low alarm threshold	-6.3 dBm	-3.6 dBm
Receiver connector type	LC	
Receiver wavelength range	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm	
Average Rx power range (each lane) Average power for 100GBASE-LR4 is informative and not the principal indicator of signal strength. Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance. For OTU-4, specified at a BER of 10^{-6} (pre-FEC), per ITU-T G.supp39.	-10.6 dBm to 4.5 dBm	-10.3 dBm to 4 dBm
Receiver overload	4.5 dBm	4 dBm
Receiver damage level	5.5 dBm	
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB	

Specifications for 160-9402-900	Value	
Classification	100GBASE-LR4	OTU-4 4L1-9D1F
Receiver sensitivity (OMA), each lane (max) at 1 E-12 BER For Ethernet application, receiver sensitivity is informative.	-8.6 dBm	-
Receiver sensitivity (max), each lane	-	-8.8 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 1E-12, each lane	-6.8 dBm	-
Receiver power high alarm threshold	4.5 dBm	4 dBm
Receiver power low alarm threshold	-10.6 dBm	-10.3 dBm
Maximum channel insertion loss	6.3 dB	
Nominal reach each standard	10 km	
Typical Power Consumption (W) The typical power consumption values are based on operation at an ambient temperature of 25°C (+/-3°C) and voltage of 54 V dc (+/-2.5 V) or in the full operational voltage range in the case of AC-powered equipment. For practical purposes, the rounded typical power consumption of an equipment can be used as the equipment heat dissipation when calculating facilities thermal loads (an estimate of the long-term heat release of the item in a system).	3	
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V) or in the full operational voltage range in the case of AC-powered equipment. These rounded power values must be used in sizing feeders and estimating theoretical maximum power draw.	4 (max)	
Supported Rates	103.125G (100GE)	111.810G (OTU4)

160-9403-900: 100GE CWDM4, 4x25G, CWDM, SMF, 1310nm, 2 km QSFP28 specifications

The following table provides information on 160-9403-900: 100GE CWDM4, 4x25G, CWDM, SMF, 1310nm, 2 km QSFP28 specifications.

Table 33 160-9403-900: 100GE CWDM4, 4x25G, CWDM, SMF, 1310nm, 2 km QSFP28 specifications

Specifications for 160-9403-900	Value
PEC (Part Number)	160-9403-900
CLEI	WOTRD4MFAA
Classification	100GE CWDM4
Fiber	Single mode fiber (SMF)
Pluggable Form-Factor	QSFP28
Application	100G Ethernet
Standards	QSFP28 MSA Compliant 100G CWDM4 MSA
Transmitter Connector type	LC
Transmitter wavelength range	1271 nm, 1291 nm, 1311 nm, 1331 nm
Transmit average output power, each lane	–6.5 dBm to 2.5 dBm
Launch power in OMA minus TDP, each lane (min)	–5.0 dBm
Transmitter dispersion penalty (TDP), each lane (max)	3.0 dB
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Transmitter power high alarm threshold	5.5 dBm
Transmitter power low alarm threshold	-9.5 dBm
Receiver Connector type	LC
Receiver Wavelength range	1264.5 nm to 1277.5 nm 1284.5 nm to 1297.5 nm 1304.5 nm to 1317.5 nm 1324.5 nm to 1337.5 nm
Average Rx power range Average power is informative and not the principal indicator of signal strength. A received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	–11.5 dBm to 2.5 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 1E-12	–7.3 dBm

Specifications for 160-9403-900	Value
Receiver overload	2.5 dBm
Receiver damage level	3.5 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Receiver power high alarm threshold	2.5 dBm
Receiver power low alarm threshold	-11.5 dBm
Maximum channel insertion loss	5.0 dB
Nominal reach each standard	2 km
Typical Power Consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	3
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	4
Supported Rates	103.125G (100GE)

160-9404-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 40km QSFP28 specifications

The following table provides information on 160-9404-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 40km QSFP28 specifications.

Table 34 160-9404-900: 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 40km QSFP28 specifications

Specifications for 160-9404-900	Value	
Classification	4WDM-40	OTU-4 4L1-9D1F
PEC (Part Number) All parameters are calculated for each individual optical lane (and not for the total of all optical lanes supported by the QSFP28).	160-9404-900	
CLEI	WOTRERBFAA	
Fiber	Single-mode fiber (SMF)	
Pluggable Form-Factor	QSFP28	

Specifications for 160-9404-900	Value	
Classification	4WDM-40	OTU-4 4L1-9D1F
Application	100G Ethernet applications	OTU4 applications
Standards	4WDM-40 MSA	ITU-T G.959.1 4L1-9D1F
Transmitter Connector type	LC	
Transmitter wavelength range	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm	
Transmit average output power, each lane	-2.5 dBm to 6.5 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.	0.6 dBm to 5.1 dBm
Maximum optical path penalty	—	1.5 dB
Launch power in OMA minus TDP, each lane (min)	-0.5 dBm	—
Transmitter dispersion penalty (TDP), each lane (max)	3 dB	—
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB	
Transmitter power high alarm threshold	8 dBm	8 dBm
Transmitter power low alarm threshold	-4.5 dBm	-3.6 dBm
Receiver Connector type	LC	
Receiver Wavelength range	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm	

Specifications for 160-9404-900	Value	
Classification	4WDM-40	OTU-4 4L1-9D1F
Average Rx power range Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance. For OTU-4, specified at a BER of 10^{-6} (pre-FEC), per ITU-T G.sup39.	-20.5 dBm to -3.5 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.	-18.9 dBm to -4.9 dBm
Receiver sensitivity (OMA), each lane (max) at 5E-5 BER	-18.5 dBm	—
Receiver sensitivity (OMA), each lane (max) at 1E-12 BER	-15 dBm	—
Stressed receiver sensitivity in OMA (max), BER not worse than 5E-5	-16 dBm	—
Receiver sensitivity (max), each lane	See above Receiver sensitivity (OMA) values	-18.9 dBm back-to-back
Receiver overload	-3.5 dBm per lane	-4.9 dBm
Receiver damage level	-2.5 dBm	
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB	
Receiver power high alarm threshold	-3.5 dBm	-4.9 dBm
Receiver power low alarm threshold	-20.5 dBm	-18.9 dBm

Specifications for 160-9404-900	Value	
Classification	4WDM-40	OTU-4 4L1-9D1F
Maximum channel insertion loss	• 10 dB to 18 dB using Clause 91 FEC per IEEE802.3 For operation with compliant 4WDM-40 MSA transceivers. • 6.4 dB to 12.4 dB using FEC off (BER $\leq 1E-12$) For operation with compliant 100GBASE-ER4 transceivers. The receiver of the connected 100GBASE-ER4 module must have a minimum of 2 dB attenuation to prevent overload from the 4WDM-40 transmitter.	• 10 to 18 dB when interoperating with another 4L1-9D1F TX • 8 to 13 dB when interoperating with legacy 4L1-9C1F TX It is recommended to consider a minimum of 1 dB attenuation at the 4L1-9C1F RX when connected to the 4L1-9D1F TX to prevent overload.
Minimum channel insertion loss	10 dB	
Nominal reach each standard	• 40 km For operation with compliant 4WDM-40 MSA transceivers. • 25 km For operation with compliant 100GBASE-ER4 transceivers. The receiver of the connected 100GBASE-ER4 module must have a minimum of 2 dB attenuation to prevent overload from the 4WDM-40 transmitter.	• 40 km with 4L1-9D1F • 27 km with 4L1-9C1F
Typical Power Consumption (W) The typical power consumption values are based on operation at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V) or in the full operational voltage range in the case of AC-powered equipment. For practical purposes, the rounded typical power consumption of an equipment can be used as the equipment heat dissipation when calculating facilities thermal loads (an estimate of the long term heat release of the item in a system).	3	3

Specifications for 160-9404-900	Value	
Classification	4WDM-40	OTU-4 4L1-9D1F
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V) or in the full operational voltage range in the case of AC-powered equipment. These rounded power values must be used in sizing feeders and estimating theoretical maximum power draw.	4	4
Supported Rates	103.125G (100GE)	111.810G (OTU4)

160-9405-900: 100GE 4WDM-40, 4x25G, SMF, 1310nm, 40km QSFP28 specifications

The following table provides information on 160-9405-900: 100GE 4WDM-40, 4x25G, SMF, 1310nm, 40km QSFP28 specifications.

Table 35 160-9405-900: 100GE 4WDM-40, 4x25G, SMF, 1310nm, 40km QSFP28 specifications

Specifications for 160-9405-900	Value
PEC (Part Number)	160-9405-900
CLEI	WOTREKZFAA
Classification	4WDM-40 MSA
Fiber	Single-mode fiber (SMF)
Pluggable Form-Factor	QSFP28
Application	100G Ethernet applications
Standards	4WDM-40 MSA
Transmitter Connector type	LC
Transmitter wavelength range	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm
Transmit average output power, each lane Average power is informative and not the principal indicator of signal strength.	-2.5 dBm to 6.5 dBm
Launch power in OMA minus TDP, each lane (min)	-0.5 dBm

Specifications for 160-9405-900	Value
Transmitter dispersion penalty (TDP), each lane (max)	3 dB
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Transmitter power high alarm threshold	8 dBm
Transmitter power low alarm threshold	-4.5 dBm
Receiver Connector type	LC
Receiver Wavelength range	1294.53 nm to 1296.59 nm 1299.02 nm to 1301.09 nm 1303.54 nm to 1305.63 nm 1308.09 nm to 1310.19 nm
Average Rx power range Average power is informative and not the principal indicator of signal strength. A received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-20.5 dBm to -3.5 dBm
Receiver sensitivity (OMA), each lane (max) at 5 E-5 BER	-18.5 dBm
Receiver sensitivity (OMA), each lane (max) at 1 E-12 BER	-15 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 5E-5	-16 dBm
Receiver overload	-3.5 dBm per lane
Receiver damage level	-2.5 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Receiver power high alarm threshold	-3.5 dBm
Receiver power low alarm threshold	-20.5 dBm
Maximum channel insertion loss	• 10 dB to 18 dB using Clause 91 FEC per IEEE802.3 (Note 1) • 6.4 dB to 12.4 dB using FEC off (BER <= 1E-12) (Note 2)
Nominal reach per MSA	• 40 km (Note 1) • 25 km (Note 2)

Specifications for 160-9405-900	Value
Typical Power Consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	3
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	4
Supported Rates	103.125G (100GE)
Note 1: For operation with compliant 4WDM-40 MSA transceivers. Note 2: For operation with compliant 100GBASE-ER4 transceivers. The receiver of the connected 100GBASE-ER4 module must have a minimum of 2 dB attenuation to prevent overload from the 4WDM-40 transmitter.	

160-9406-900: 100GE PSM4, 4x25G, SMF, 1310nm, 500m, MPO QSFP28 specifications

The following table provides information on 160-9406-900: 100GE PSM4, 4x25G, SMF, 1310nm, 500m, MPO QSFP28 specifications.

Table 36 160-9406-900: 100GE PSM4, 4x25G, SMF, 1310nm, 500m, MPO QSFP28 specifications

Specifications for 160-9406-900	Value
PEC (Part Number)	160-9406-900
CLEI	WOTRERWFAA
Classification	100GBASE-PSM4
Fiber	Single mode fiber (SMF)
Pluggable Form-Factor	QSFP28
Application	100G Ethernet
Standards	100G PSM4 Specification Version 2.0 MSA Compliant
Transmitter connector type A single connector is used for both Tx and Rx on this QSFP28.	MPO12
Transmitter nominal wavelength	1295 nm to 1325 nm
Transmit average output power, each lane Average power for Ethernet applications is informative and not the principal indicator of signal strength.	-9.4 dBm to 2.0 dBm

Specifications for 160-9406-900	Value
Transmitter dispersion penalty (TDP), each lane (max)	2.9 dB
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Transmitter power high alarm threshold	5 dBm
Transmitter power low alarm threshold	-12.4 dBm
Receiver connector type A single connector is used for both Tx and Rx on this QSFP28.	MPO12
Receiver wavelength range	1295 nm to 1325 nm
Average Rx power range Average power for Ethernet applications is informative and not the principal indicator of signal strength. Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-12.66 dBm to 2.0 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 5E-5 $Rx\ sens_stressed \leq MAX \left(-9.29 + \frac{(\lambda - 1310)^2}{100}, -8.79 \right) \text{dBm}$ where Rx sens_stressed is the stressed receiver sensitivity (OMA) of each receive lane l is center wavelength (in nm).	-7.04 dBm OMA
Receiver overload	+2.0 dBm Avg +22 dBm OMA
Receiver damage level	3.0 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Receiver power high alarm threshold	2 dBm
Receiver power low alarm threshold	-12.7 dBm
Maximum channel insertion loss	3.3 dB
Nominal reach each standard	500 m
Typical power consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	3

Specifications for 160-9406-900	Value
Power budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	4
Supported Rates	103.125G (100GE)

160-9431-900: 100G-FR, SMF, 1310nm, 2km QSFP28 specifications

The following table provides information on 160-9431-900: 100G-FR, SMF, 1310nm, 2km QSFP28 specifications.

Table 37 160-9431-900: 100G-FR, SMF, 1310nm, 2km QSFP28 specifications

Specifications for 160-9431-900	Value
PEC (Part Number)	160-9431-900
CLEI	WOTREUZFAA
Classification	100G-FR
Fiber	Single mode fiber (SMF)
Pluggable Form-Factor	QSFP28
Application	100G Ethernet
Standards	100G Lambda MSA
Transmitter connector type	LC
Transmitter nominal wavelength	1304.5 nm to 1317.5 nm
Transmit average output power Average power for Ethernet applications is informative and not the principal indicator of signal strength.	-2.4 dBm to 4 dBm
Transmitter dispersion penalty eye closure for PAM4 (TDECQ), (max)	3.4 dB
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Transmitter power high alarm threshold	7.0 dBm
Transmitter power low alarm threshold	-6.1 dBm
Receiver connector type	LC
Receiver wavelength range	1304.5 nm to 1317.5 nm

Specifications for 160-9431-900	Value
Average Rx power range Average power for Ethernet applications is informative and not the principal indicator of signal strength. Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-6.4 dBm to 4.5 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 2.4E-4	-2.5 dBm
Receiver overload (max)	4.5 dBm
Receiver damage level (min)	5.5 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB
Receiver power high alarm threshold	4.0 dBm
Receiver power low alarm threshold	-7.1 dBm
Maximum channel insertion loss	4.0 dB
Nominal reach	2 km
Typical power consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	4
Power budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	4.5 (FEC enabled mode) 4 (FEC bypass mode)
Supported Rates	106.25G (100GE)
Pluggable optical FEC mode	Enabled by default (FEC requirement is KP4)

160-9433-900: 100GBASE-LR1, SMF, 1310nm, 10km QSFP28 specifications

The following table provides information on 160-9433-900: 100GBASE-LR1, SMF, 1310nm, 10km QSFP28 specifications.

Table 38 160-9433-900: 100GBASE-LR1, SMF, 1310nm, 10km QSFP28 specifications

Specifications for 160-9433-900	Value
PEC (Part Number)	160-9433-900

Specifications for 160-9433-900	Value
CLEI	WOTREY2FAA
Classification	100G-LR1
Fiber	Single mode fiber (SMF)
Pluggable Form-Factor	QSFP28
Application	100G Ethernet
Standards	IEEE Std 802.3 100GBASE-LR1
Transmitter connector type	LC
Transmitter nominal wavelength	1304.5 nm to 1317.5 nm
Transmit average output power Average power for Ethernet applications is informative and not the principal indicator of signal strength.	-1.9 dBm to 4.8 dBm
Transmitter dispersion penalty eye closure for PAM4 (TDECQ), (max)	3.4 dB
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Transmitter power high alarm threshold	4.8 dBm
Transmitter power low alarm threshold	-1.9 dBm
Receiver connector type	LC
Receiver wavelength range	1304.5 nm to 1317.5 nm
Average Rx power range Average power for Ethernet applications is informative and not the principal indicator of signal strength. Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-8.2 dBm to 4.8 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 2.4E-4	-4.1 dBm
Receiver overload (max)	4.8 dBm
Receiver damage level (min)	5.8 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Receiver power high alarm threshold	4.8 dBm
Receiver power low alarm threshold	-8.2 dBm

Specifications for 160-9433-900	Value
Maximum channel insertion loss	6.3 dB
Nominal reach	10 km
Typical power consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	4
Power budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	4.5 (FEC enabled mode) 4 (FEC bypass mode)
Supported Rates	106.25G (100GE)
Pluggable optical FEC mode	Enabled by default (FEC requirement is KP4)

160-9460-xxx: 103.1G-112.28G, QSFP28-QSFP28 Active Optical Cables

The following Active Optical Cables (AOCs) are supported:

- 103.1G-112.28G, 3m, QSFP28-QSFP28 Active Optical Cable (160-9460-003)
- 103.1G-112.28G, 5m, QSFP28-QSFP28 Active Optical Cable (160-9460-005)
- 103.1G-112.28G, 7m, QSFP28-QSFP28 Active Optical Cable (160-9460-007)
- 103.1G-112.28G, 10m, QSFP28-QSFP28 Active Optical Cable (160-9460-010)
- 103.1G-112.28G, 15m, QSFP28-QSFP28 Active Optical Cable (160-9460-015)
- 103.1G-112.28G, 20m, QSFP28-QSFP28 Active Optical Cable (160-9460-020)
- 103.1G-112.28G, 50m, QSFP28-QSFP28 Active Optical Cable (160-9460-050)
- 103.1G-112.28G, 100m, QSFP28-QSFP28 Active Optical Cable (160-9460-100)

The following table provides information on 160-9460-xxx: 103.1G-112.28G, QSFP28-QSFP28 Active Optical Cables.

Table 39 160-9460-xxx: 103.1G-112.28G, QSFP28-QSFP28 Active Optical Cables specifications

Specifications for 160-9460-xxx	Value
PEC	• 160-9460-003 • 160-9460-005 • 160-9460-007 • 160-9460-010 • 160-9460-015 • 160-9460-020 • 160-9460-050 • 160-9460-100
CLEI	• WOPQAE7FAA for 160-9460-003 • WOPQAE8FAA for 160-9460-005 • WOPQAE9FAA for 160-9460-007 • WOPQAFHFAA for 160-9460-010 • WOPQAFJFAA for 160-9460-015 • WOPQAFAFAA for 160-9460-020 • WOPQAFNFAA for 160-9460-050 • WOPQAFBFAA for 160-9460-100
Integrated Cable	Four pairs of multimode fiber, in a multi-fiber optical cable
Length	• 3m for 160-9460-003 • 5m for 160-9460-005 • 7m for 160-9460-007 • 10m for 160-9460-010 • 15m for 160-9460-015 • 20m for 160-9460-020 • 50m for 160-9460-050 • 100m for 160-9460-100
Pluggable Form-Factor	QSFP28 (both ends)
Application	100G Ethernet, OTU4, and OTUCn applications
Standards	QSFP28 MSA Compliant IEEE 802.3 100GBASE-SR4, ITU-T G.959.1 4L1-9D1F
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 3 dB

Specifications for 160-9460-xxx	Value
Receiver power high alarm threshold	7 dBm
Receiver power low alarm threshold	-7 dBm
Typical Power Consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V). This is the typical power for only one end of the AOC cable as equipped in a QSFP28 port and the total typical power including both plug ends is 4W.	2
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw. This is the budget power for only one end of the AOC cable as equipped in a QSFP28 port and the total budget power including both plug ends is 4W.	2.1
Supported Rates	• 103.125G (100GE) • 111.810G (OTU4)

160-9660-xxx: 400GE, QSFP-DD - QSFP-DD Active Optical Cables

The following Active Optical Cables (AOCs) are supported:

- 400GE, 3m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-003)
- 400GE, 5m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-005)
- 400GE, 10m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-010)
- 400GE, 20m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-020)
- 400GE, 100m, QSFP-DD - QSFP-DD Active Optical Cable (160-9660-100)

The following table provides information on 160-9660-xxx: 400GE, QSFP-DD - QSFP-DD Active Optical Cables.

Table 40 160-9660-xxx: 400GE, QSFP-DD - QSFP-DD Active Optical Cables specifications

Specifications for 160-9660-xxx	Value
PEC	• 160-9660-003 • 160-9660-005 • 160-9660-010 • 160-9660-020 • 160-9660-100
CLEI	• WOPQAFRFAA for 160-9660-003 • WOPQAFSFAA for 160-9660-005 • WOPQAFTHAA for 160-9660-010 • WOPQAFUFAA for 160-9660-020 • WOPQAFVFAA for 160-9660-100
Integrated Cable	Eight pairs of multimode fiber, in a multi-fiber optical cable
Length	• 3m for 160-9660-003 • 5m for 160-9660-005 • 10m for 160-9660-010 • 20m for 160-9660-020 • 100m for 160-9660-100
Pluggable Form-Factor	PQSFP (both ends)
Application	400G Ethernet applications
Standards	• QSFP-DD MSA Compliant • QSFP-DD Common Management Interface Specification (CMIS), Rev 4.0, May 8, 2019 • 400GAUI-8, specified by IEEE802.3-2018, Annex 120E
Tx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Transmitter power high alarm threshold	4.0 dBm (per lane)
Transmitter power low alarm threshold	-6.5 dBm (per lane)
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Receiver power high alarm threshold	4.0 dBm (per lane)

Specifications for 160-9660-xxx	Value
Receiver power low alarm threshold	-8.4 dBm (per lane)
Typical Power Consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V). This is the typical power for only one end of the AOC cable as equipped in a QSFP-DD port and the total typical power including both plug ends is 18W.	9
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw. This is the budget power for only one end of the AOC cable as equipped in a QSFP-DD port and the total budget power including both plug ends is 18W.	9
Supported Rates	425G (400GE)

160-9600-900: 400G-FR4, SMF, 2km QSFP-DD

The following table provides information on 160-9600-900: 400G-FR4, SMF, 2km QSFP-DD specifications.

Table 41 160-9600-900: 400G-FR4, SMF, 2km QSFP-DD specifications

Specifications for 160-9600-900	Value
PEC (Part Number)	160-9600-900
CLEI	WMOTC02FAA
Classification	400GBASE-FR4
Fiber	Single-mode fiber (SMF, dual LC/PC connector)
Pluggable Form-Factor	QSFP-DD
Application	400G Ethernet applications
Standards	400G-FR4
Transmitter Connector type	LC
Transmitter nominal wavelength	1271 nm, 1291 nm, 1311 nm, and 1331 nm

Specifications for 160-9600-900	Value
Transmit average output power, each lane	-3.3 dBm to 3.5 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.
Launch power in OMA minus TDP (min)	-1.7 dB (for extinction ratios ≥ 4.5 dB)
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Transmitter power high alarm threshold	3.5 dBm
Transmitter power low alarm threshold	-3.3 dBm
Receiver Connector type	LC
Receiver Wavelength range	1264.5 nm to 1277.5 nm 1284.5 nm to 1297.5 nm 1304.5 nm to 1317.5 nm 1324.5 nm to 1337.5 nm
Average Rx power range (each lane) Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-7.3 dBm to 3.5 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.
Receiver sensitivity in OMA per lane (max) Receiver sensitivity is informative.	-4.6 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 2.4E-4	-2.6 dBm
Receiver overload	3.5 dBm per lane for average power 3.7 dBm per lane for the maximum Rx input power (OMA)
Receiver damage level	4.5 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Receiver power high alarm threshold	3.5 dBm
Receiver power low alarm threshold	-7.3 dBm
Maximum channel insertion loss	4 dB

Specifications for 160-9600-900	Value
Nominal reach each standard	2 km
Typical Power Consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	10
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	12
Supported Rates	425G (400GE)

180-3530-900: WL5n 400ZR Type 3.0 QSFP-DD

The following table provides information on 180-3530-900: WL5n 400ZR Type 3.0 QSFP-DD specifications.

Table 42 180-3530-900: WL5n 400ZR Type 3.0 QSFP-DD specifications

Specifications for 180-3530-900	Value
PEC (Part Number)	180-3530-900
CLEI	WMOTC08FAA
Classification	400GBASE-ZR
Fiber	Single-mode fiber (SMF, dual LC/PC connector)
Pluggable Form-Factor	QSFP-DD
Application	100G and 400G Ethernet applications
Standards	• QSFP-DD MSA Compliant • IEEE802.3cd-2018 and IEEE802.3cm-2020 • QSFP-DD Common Management Interface Specification (CMIS), Rev 4.0, May 8, 2019 • 400GAUI-8 • 4x100GAUI-2
Transmitter Connector type	LC
Transmitter tunable frequency spacing	Flexgrid: Step-tunable with 100 MHz step size

Specifications for 180-3530-900	Value
Transmitter tunable frequency range	C-Band: 191.250 to 196.125 THz with tuning precision 0.1 GHz (100 MHz or 0.0001 THz)
Transmit average output power	-13.0 dBm to -9.0 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 1 dB
Transmitter power high alarm threshold	-5.0 dBm
Transmitter power low alarm threshold	-17.0 dBm
Receiver Connector type	LC
Receiver Wavelength range	1528.77 nm to 1567.54 nm
Average Rx power range Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-12.0 dBm to 0 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.
Stressed receiver sensitivity in OMA (max), BER not worse than 2.4E-4	-18.0 dBm
Receiver overload	9 dBm for average power 14 dBm for the maximum Rx input power (OMA)
Receiver damage level	14 dBm
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Maximum channel insertion loss	-22 dB
Nominal reach each standard	Up to 120 km
Receiver power high alarm threshold	5.0 dBm
Receiver power low alarm threshold	-16.0 dBm
Typical Power Consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	15

Specifications for 180-3530-900	Value
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	15
Supported Rates	425G (400ZR) (60 GBaud DP-16QAM)
Pluggable optical FEC mode	KP4

160-9603-900: 400G-LR4-10/4x100GE, SMF, 10km QSFP-DD

The following table provides information on 160-9603-900: 400G-LR4-10/4x100GE, SMF, 10km QSFP-DD specifications.

Table 43 160-9603-900: 400G-LR4-10/4x100GE, SMF, 10km QSFP-DD specifications

Specifications for 160-9603-900	Value
PEC (Part Number)	160-9603-900
CLEI	WMOTC06FAA
Classification	400G-LR4-10/4x 100G-LR1 Note: Even though the 160-9603-900 QSFP-DD pluggable can support 4x100G-LR1 clients, there is no hosting module currently available to support 4x100G-LR1 clients with 160-9603-900 QSFP-DD pluggables.
Fiber	Single-mode fiber (SMF, dual LC/PC connector)
Pluggable Form-Factor	QSFP-DD
Application	• 400G Ethernet applications • 4x100G applications, point to point Note: Even though the 160-9603-900 QSFP-DD pluggable can support 4x100G point to point applications, there is no hosting module currently available to support 4x100G point to point applications with 160-9603-900 QSFP-DD pluggables.
Standards	100GLambda MSA Group
Transmitter Connector type	LC
Transmitter nominal wavelength	1271 nm, 1291 nm, 1311 nm, and 1331 nm

Specifications for 160-9603-900	Value
Transmit average output power, each lane	-2.7 dBm to 5.1 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.
Outer OMA (OMAxouter) per optical lane (max)	4.4 dBm
Outer OMA (OMAxouter) per optical lane (min)	• 0.3 dBm (for TDECQ < 1.4 dB) • -1.1 + TDECQ (for 1.4 ≤ TDECQ ≤ 3.9 dB)
Tx average power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Transmitter power high alarm threshold	7.1 dBm
Transmitter power low alarm threshold	-4.7 dBm
Receiver Connector type	LC
Receiver Wavelength range	• 1264.5 nm to 1277.5 nm • 1284.5 nm to 1297.5 nm • 1304.5 nm to 1317.5 nm • 1324.5 nm to 1337.5 nm
Average Rx power range (each lane) Received power below the minimum average Rx power cannot be compliant; however, a received power above the value listed does not ensure compliance.	-9.0 dBm to 5.1 dBm The minimum average power for Ethernet applications is informative and not the principal indicator of signal strength.
Receiver sensitivity in OMA per lane (max)	-6.8 dBm
Stressed receiver sensitivity in OMA (max), BER not worse than 2.4E-4	-4.3 dBm
Receiver overload in OMA	5.1 dBm for average power
Receiver damage level	6.1 dBm for average power
Rx power monitor accuracy Power monitor accuracy stated applies to nominal operating power range.	+/- 2 dB
Receiver power high alarm threshold	5.1 dBm
Receiver power low alarm threshold	-9.0 dBm
Maximum channel insertion loss	6.3 dB

Specifications for 160-9603-900	Value
Nominal reach each standard	10 km
Typical Power Consumption (W) The typical power consumption values are based on operation in a system operating at an ambient temperature of 25°C (+/-3°C) and voltage of 54 Vdc (+/-2.5 V).	10
Power Budget (W) The power budget values are based on the maximum power consumption in a system operating at an ambient temperature range from 5°C to 40°C at a voltage of 40 Vdc (+/-2.5 V). These values must be used in sizing feeders and estimating theoretical maximum power draw.	12
Supported Rates	• 425G (400GE) • 106.25G (4x100GE)

CHAPTER 9

Operating specifications and standards compliance

This section covers the following topics:

- [Overview](#)
 - [Safety Compliance](#)
 - [Environmental and electromagnetic specifications](#)
 - [Power and grounding specifications](#)

Overview

This section provides information on operating specification and standards compliance.

Safety Compliance

Waveserver 5 meets or exceeds the following standards:

- Full Restriction of Hazardous Substances (RoHS) compliant
- CE compliant
- IEC 60825-1 Edition 3 compliant
- IEC 60825-2 Edition 3.2 compliant as (maximum) Hazard Level 1M
- CAN/CSA-C22.2 No. 62368-1-14
- NRTL certified to UL 62368-1 Edition 2.0
- CB certified to IEC 62368-1 Edition 2.0
- EN 62368-1:2014

Environmental and electromagnetic specifications

The following table provides the operating environmental and electromagnetic specifications for the Waveserver 5 chassis.

Table 44 Operating environmental and electromagnetic specifications for WS5 system

Specification	Description
Air flow	Air intake: Front of chassis Air exhaust: Rear of chassis
Operating temperature	-5°C to 45°C (23°F to 113°F)
Heat dissipation	1000W typical configuration dissipates 1kW/hr or equivalently 1000 x 3.412 BTU/hr Note: Be aware of replacing working pluggable modules with elevated case temperature which is expected at higher ambient temperatures of over 25°C. These plugs such as WL5n 400ZR Type 3.0 QSFP-DD (180-3530-900) are indicated with “hot surface” symbol, so it is recommended to handle the plug only by its pull tab until it cools down.
Operating humidity (Telcordia GR-63 CORE, Issue 5 (NEBS))	Ambient relative humidity between 5% and 85%
Transportation/handling	Telcordia GR-63 CORE, Issue 5 (NEBS)
Storage	Telcordia GR-63 CORE, Issue 5 (NEBS) Storage temperature: -40°C to 70°C (-40°F to 158°F) Storage humidity: 93% relative humidity ETSI EN 300 019-1.1 (Class 1.2)
Office vibration	Telcordia GR-63 CORE, Issue 5 (NEBS) (in 4-post racks only)
Earthquake/seismic	Telcordia GR-63 CORE, Issue 5 (NEBS) (in 4-post racks only)
Fire resistance and flame spread	Telcordia GR-63 CORE, Issue 5 (NEBS)
Network Equipment-Building System (NEBS) Requirements for Telecommunications Data Center Equipment and Spaces	GR-3160-CORE (Compliant as per GR-63-CORE and GR-1089-CORE)

Specification	Description
Electromagnetic compatibility:	• FCC Part 15 (Class A) • UL Std No. 62368-1, 2nd Ed • CISPR 32 Edition 2 (Class A) • CISPR 35 Edition 1 • ICES-003 (Industry Canada), Iss.6 • ETSI EN 300 386 V1.6.1 (Class A, Other than telecommunications centres) • British Telecom GS07 (Iss.14) • VCCI V-32: 2016 (Class A) • Telecordia GR-1089-CORE (Comply with applicable criteria for DC, AC and Type 4 ports) • KS C 9832:2019 - Electromagnetic interference prevention test for multimedia equipment Note: This device has been evaluated for the purpose of use in a work environment. If it is used in a home environment, there is a risk of interference with radio waves. 이 기기는 업무용 환경에서 사용할 목적으로 적합성평가를 받은 기기로서 가정용 환경에서 사용하는 경우 전파간섭의 우려가 있습니다.  R-R-Cgv-Waveserver5 • KS C 9835:2019 - Electromagnetic wave immunity test for multimedia devices
	Note 1: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at user's own expense. Note 2: LAN interfaces (e.g. DCN, ILAN) deploying rates up to 1000BASE-T may use Cat 5E unshielded twisted pair (UTP) cable. LAN interfaces deploying rates in excess of 1000BASE-T must use Cat 6 (or better) shielded twisted pair (STP).

Power and grounding specifications

The following sections outline the power and grounding specifications for Waveserver 5, including:

- [DC input voltage ranges](#)
- [AC input voltage ranges](#)

The Waveserver 5 is suitable for installation as part of the Common Bonding Network (CBN) or an Isolated Bonding Network (IBN) or both, or an ETSI Mesh Bonding Network.

The Waveserver 5 supports the connection of the battery return (for the DC power module) in an isolated DC return (DC-I) or common DC return (DC-C) configuration.

ATTENTION

The DC power module return lead, L+, must be earthed (connected to ground) at some point within the building consistent with the applicable power infrastructure (DC-C or DC-I) to satisfy safety requirements.

All AC inputs to the Waveserver 5 must be equipped with an external surge protection device (SPD not supplied) to meet the Telcordia criteria for lightning criteria for equipment interfacing with AC power port(s).

The Waveserver 5 is suitable for installation in Network Telecommunication Facilities and locations where the United States National Electrical Code (NEC) applies.

The AC/240Vdc power module meets international specifications. The AC/240Vdc power module is also compliant with the National Industry Standards for Telecommunications of the People's Republic of China (YD/T2378 - 2011).

Note: An external main disconnect must be provided on all power feeds connected to the equipment.



DANGER

Risk of electrical shock and equipment damage

Grounding is mandatory to satisfy local electrical codes/regulations for the safe use of the equipment.

Ground the bay/cabinet to the common building network (CBN), isolated bonding network (IBN) or ETSI Mesh Bonding Network ground/protective earth.

Correct grounding of the equipment bay/cabinet is mandatory in accordance to electrical safety standards and mitigates the electrostatic discharge (ESD) and operational concerns described in the "Consequences of incorrect grounding" section of *Waveserver 5 installation*, 323-3001-200.

Also, ground every shelf and peripheral tray as described in the related procedure.

DC input voltage ranges

The operational input voltage range for Waveserver 5 in -48 DC configurations is -40 Vdc to -72 Vdc with a start-up voltage requirement of -44 +/- 1 Vdc (measured at the power input terminals). Waveserver 5 is not damaged when an input voltage of correct polarity, with a value between 0 and 45 Vdc (maximum start-up voltage) is applied for any period of time. The system is not damaged by a reversed polarity voltage supply, for example, 0 to +72 Vdc (the system does not function with a reversed polarity voltage supply).

Table 45 -48 Vdc input power specifications

Specification	Description/value (shelf equipped with -48/-60Vdc power modules)
Redundant -48 Vdc power input description	The -48 Vdc A and B redundant power inputs to the Waveserver 5 are connected to separate A and B power supply units.
-48 Vdc power connectors	See the applicable power lug. Recommended lugs are listed in the "Recommended lugs for DC Waveserver 5 installations" table in <i>Waveserver 5 Installation</i> , 323-3001-200.
Operational input voltage range	-40 to -72 Vdc
Maximum power capacity of one PSU	2300 W over the input voltage range of -40 to -72 Vdc
Low voltage alarm	-41 Vdc
Turn-on voltage	43 - 45 Vdc
Turn-off voltage	37.5 - 40 Vdc
Input feed current rating	67A Internally protected by a 80A fuse. The breaker or fuse that limit the current on the A/B feeds supplying power to the Waveserver 5 must be sized according to local electrical safety regulations.
In-rush current	11A typical at 52 Vdc input Note: Meets ETSI EN 300 132-2 V2.4.6 criteria.

Specification	Description/value (shelf equipped with -48/-60Vdc power modules)
Note 1: A Waveserver 5 has all of the A power connections on one power module and all the B power connections on the other power module. All A/B power feeds must be connected and working for alarm-free operation. Each A and B power input is converted to isolated and regulated 12V by the corresponding power module and applied to the shelf backplane, where they are connected together as a common 12V bus. Typically, the difference in power drawn between the A and B power input terminals for total input power greater than 650W is up to 10% of the total power. In the event that either the A or B power source is disconnected, all power is drawn on the remaining feed. DC power cables and source circuits must be rated to handle 67A (the maximum input current draw of the power supply unit). Note 2: During a low voltage event where voltage falls below -40 Vdc, modules will power down automatically when both the A and B feeds to the shelf are within 37.5 to 40Vdc. Note 3: If the voltage on one or more feeds is not at least -41 Vdc, a “Power Supply Feed Failed” alarm will be raised. If neither the A nor B power feed meets the minimum operating voltage, the modules will shut down automatically in the -37.5 Vdc to -40 Vdc range. Note 4: Waveserver 5 automatically recovers without any manual intervention when the input voltage on either the A or B power input returns to a value exceeding the start-up voltage. During a recover from a low voltage battery condition, modules recover between -43 Vdc and -45 Vdc.	

AC input voltage ranges

The operational input voltage range for Waveserver 5 is 100 Vac to 277 Vac. See the following table for details.

Table 46 AC input power specifications

Specification	Description/value
AC power input configuration and redundancy	Redundant A and B feeds
AC power connectors	See the applicable AC power cable. Refer to Ordering and AC Power Cables for supported AC power cables.
Input voltage range	Nominal: 100-277 Vac (90 to 305 Vac) Nominal: 240 Vdc (180 to 300 Vdc)
Turn-on voltage	81 - 89 Vac, 155 - 175 Vdc
Turn-off voltage	70-78 Vac, 150 - 170 Vdc
Maximum power capacity per PSU	3000W for 200-277 Vac and 240 Vdc 2000W for 120 Vac 1900W for 100-110 Vac
Input frequency range	Minimum: 47 Hz Maximum: 63 Hz
Power Factor Correction (PFC)	Minimum: 0.95

Specification	Description/value
Total Harmonic Distortion (THD)	Maximum: 0.995 Note: Complies with IEC/EN 61000-3-2: 2014.
Current rating per AC feed	18A max at 200-277 Vac input, 3000W output 18A max at 240 Vdc input, 3000W output 20A max at 100 Vac input, 1550W output 20A max at 110 Vac input, 1700W output 20A max at 120 Vac input, 2000W output 22A max at 110 Vac input, 1900W output 25A max at 100 Vac input, 1900W output Internally protected by a 30A fuse. The breaker or fuse that limit the current on the A/B feeds supplying power to the Waveserver 5 must be sized according to local electrical safety regulations.
In-rush current per AC feed	50 Apk at 305 Vac or 300 Vdc with 25 °C cold start
Holdup time	12 ms minimum
Note 1: It is recommended that the two feeds are supplied from separate AC circuits to maintain feed redundancy. Each of the A and B AC power feeds is hot-pluggable. A Waveserver 5 has all of the A power connections on one power module and all the B power connections on the other power module. All A/B power feeds must be connected and working for alarm-free operation. Each A and B power input is converted to isolated and regulated 12V by the corresponding power module and applied to the shelf backplane, where they are connected together as a common 12V bus. Typically, the difference in power draw between the A and B power input terminals for total input power greater than 750W is up to 10% of the total power. In the event that either the A or B power source is disconnected, all power is drawn on the remaining feed. AC power cables and source circuits must be rated to handle the maximum input current draw of the power supply unit as listed in the above table under “Current rating per AC feed”. Note 2: The AC voltage and phase supplied to each AC input can be different as long as each feed complies with the specified voltage and frequency ranges. AC power converter will shut down and enter a fail-safe mode below 90 Vac but will recover automatically following an under-voltage condition. If both AC feeds are under-voltage, a shelf supplied with two feeds will restart on recovery of the first feed. Other than correcting the source of the AC power failure, no manual intervention is required. During a low voltage event where voltage falls below 90 Vac or 180 Vdc, modules will power down automatically when both the A and B feeds to the shelf reach 74 +/- 4 Vac or 160 +/- 10 Vdc. Note 3: If the voltage on one or more feeds is not at least 80 Vac or 170 Vdc, a “Power Supply Feed Failed” alarm will be raised. If neither the A nor B power feed meets the minimum operating voltage, the modules will shut down automatically in the 70 to 78 Vac or 150 to 170 Vdc range. Note 4: Waveserver 5 automatically recovers without any manual intervention when the input voltage on either the A or B power input returns to a value exceeding the start-up voltage. During a recover from a low voltage battery condition, modules recover in the 81 to 89 Vac or 155 to 175 Vdc range.	

AC Power Cables

This section provides information on Waveserver 5 AC power cables. You must order AC power cables from Ciena prior to the installation, see [Ordering](#).

The AC power modules are equipped with Saf-D-Grid connectors. Supported cables are:

- AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2m (NTK955GA)
- AC Power Cord, Saf-D-Grid to Open, 12AWG, 250V, 3m (NTK955GB)
- AC Power Cord, Saf-D-Grid to C20, 12 AWG, 250V, 2m, white (NTK955GC)
- AC Power Cord, Saf-D-Grid to C14, 14 AWG, 250V, 2m, black (NTK955GD)
- AC Power Cord, Saf-D-Grid to CEE7/7, 12 AWG, 250V, 2m, black (NTK955GE)
- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, black (NTK955GF)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, black (NTK955GG)
- AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, white (NTK955GH)
- AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, white (NTK955GJ)

The following figures show examples of the AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2m (NTK955GA) and the AC Power Cord, Saf-D-Grid to Open, 12AWG, 250V, 3m (NTK955GB) AC power cables.

Figure 20 AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2m (NTK955GA)



Figure 21 AC Power Cord, Saf-D-Grid to Open, 12AWG, 250V, 3m (NTK955GB)



CHAPTER 10

Waveserver 5 Web User Interface

The Waveserver 5 Web User Interface (Web UI) provides access to system tools and resources through the Waveserver 5 homepage.

The Web UI consists of a sidebar navigation menu and a main window that displays the content of the menu selection.

Note: You can access the Waveserver 5 homepage immediately after Waveserver 5 powers on and initializes. Use an Ethernet cable to connect the Waveserver 5 DCN port to a terminal, and enter the Waveserver 5 IP address in a Web browser.

Overview

To access Waveserver 5 through the Web UI, type the URL of the Waveserver 5 in a web browser, in this format:

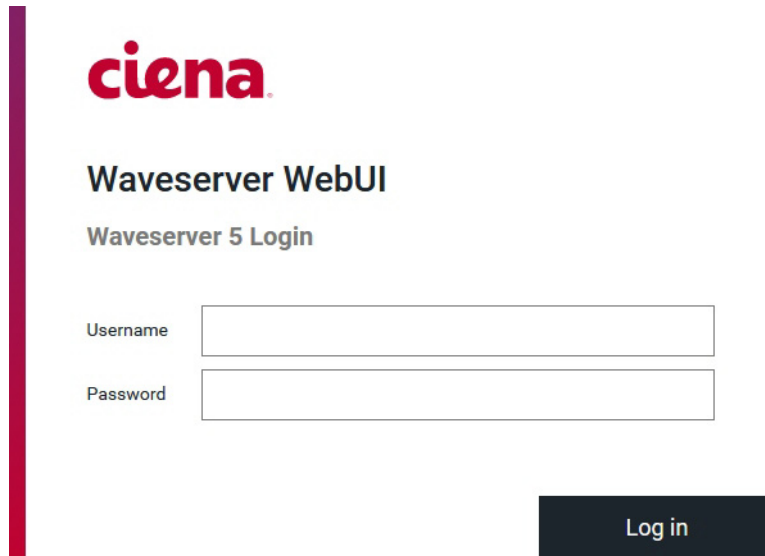
```
https://<Waveserver 5 IP address>:443/
```

where, 443 is the secure HTTP port number.

Waveserver 5 uses a default self-signed browser security certificate. You can access Waveserver 5 by adding the self-signed security certificate to your browser, or by uploading and activating an RSA certificate on Waveserver 5 to replace the default self-signed browser security certificate. See [Installing a new SSL certificate](#). You can also re-direct the HTTPS server to use an RSA or ECC X.509 certificate enrolled through PKIX. See [Public Key Infrastructure \(PKI\) and X.509](#).

Login page

The Waveserver 5 Web UI login page prompts you for Waveserver 5 local user credentials: User name and password.

Figure 22 Waveserver 5 Web UI loginThe image shows the Waveserver 5 Web UI login page. It features the Ciena logo at the top, followed by the text "Waveserver WebUI" and "Waveserver 5 Login". Below this, there are two input fields: "Username" and "Password". At the bottom right, there is a dark blue "Log in" button. A thick vertical red bar is on the left side of the page.

ciena

Waveserver WebUI

Waveserver 5 Login

Username

Password

Log in

To log in to the Waveserver 5, you must have an account with at least limited user access privileges. Your user login and password are authenticated. If authentication fails, you have a limited number of login attempts. If you exceed the maximum number of login attempts, your user account is locked out.

Web UI sessions

Once your credentials are authenticated and you are logged in, Waveserver 5 creates and tracks a user session for the Web UI client.

Web UI and RESTCONF sessions both use the common interface type REST. You can view and retrieve information about the active Web UI and REST sessions using the four main interfaces (CLI, gNMI, NETCONF, REST) and their associated commands. The table at the end of this section lists the interfaces and commands.

Waveserver 5 supports a total of ten concurrent user sessions. The ten concurrent sessions can be through a single interface (Web UI, REST, or WebSocket) sessions or distributed across the three interfaces. Once the maximum number of concurrent sessions is reached, Waveserver 5 will reject any attempt to establish a new session until one of the active sessions is closed.

There are two inactivity timers associated with the HTTPS server that affect the duration of a Web UI session:

- **global-inactivity-timer** — This timer controls session inactivity timeouts for all management interfaces. This timer takes a value between 1 and 1500 minutes. When enabled, all management interfaces are subject to the assigned timeout value. When disabled, Web UI sessions never time out.

- **HTTPS inactivity timeout** — This timer only applies to HTTPS sessions. This timer takes a value between 1 and 1500 minutes. The HTTPS inactivity timeout value takes precedence over the global-inactivity-timer for Web UI and RESTCONF sessions. As a result, sessions through Web UI and sessions through other management systems can have different timeout values. If the global-inactivity-timer is disabled, it takes precedence and Web UI sessions never time out.

The table at the end of this section lists the interfaces and commands to configure these timeout values.

There are two ways to close a Web UI session: manually (user session kill), or through a system event (client disconnect, disabling the Waveserver 5 HTTPS server, rebooting the Waveserver 5, or a software upgrade.) The following table lists the interfaces and commands to end a Web UI session.

Table 47 Waveserver 5 Web UI settings

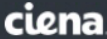
Setting	Interface	Command	Mechanism
Session management	CLI	user who	
	gNMI	gNMI.GET	waveserver-aaa/user-management/active-users
	NETCONF	get-config	waveserver-aaa/user-management/active-users
	REST	get	waveserver-aaa/user-management/active-users
Inactivity timer / timeout values	CLI	system server set global-inactivity-timer <on/off> system server set global-inactivity-timeout <1..1500min> system server https set inactivity-timeout <1..1500min>	
	gNMI	gNMI.SET	waveserver-system/server-config waveserver-system/server-config/https
	NETCONF	edit-config	waveserver-system/server-config waveserver-system/server-config/https
	REST	PATCH / PUT	waveserver-system/server-config waveserver-system/server-config/https

Setting	Interface	Command	Mechanism
Closing Web UI session	CLI	user kill session <session-id> user kill all-but-me	
	gNMI	gNMI.RPC	operations/waveserver-aaa-kill-user
	NETCONF	RPC	operations/waveserver-aaa-kill-user
	RESTCONF	POST	operations/waveserver-aaa-kill-user

System Data

System Data displays information about the Waveserver 5 that you are logged into.

Figure 23 Waveserver 5 homepage (System Data window)

 Waveserver 5 System Data Basic Event Stream Advanced Event Stream (beta) REST Interface File Transfer Model Files User Guide	Waveserver-5 waveserver-2.2.0.199 System Data Log Out	
	Host Name	Waveserver-5
	Chassis Type	Waveserver 5
	Time/date	Thu Nov 11 16:45:06 UTC 2021
	System uptime	up 47 minutes
	Booted from	zone A WARM
	Running load	waveserver-2.2.0.199

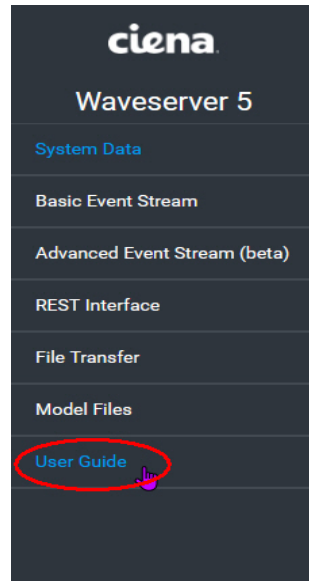
The System Data window displays these details:

- Host name
- Chassis Type: Waveserver 5
- Time/date
- System uptime
- Booted from
- Running load

User Guide

User Guide is a link that opens a separate browser window and displays the HTML5 version of the Waveserver 5 User Guide.

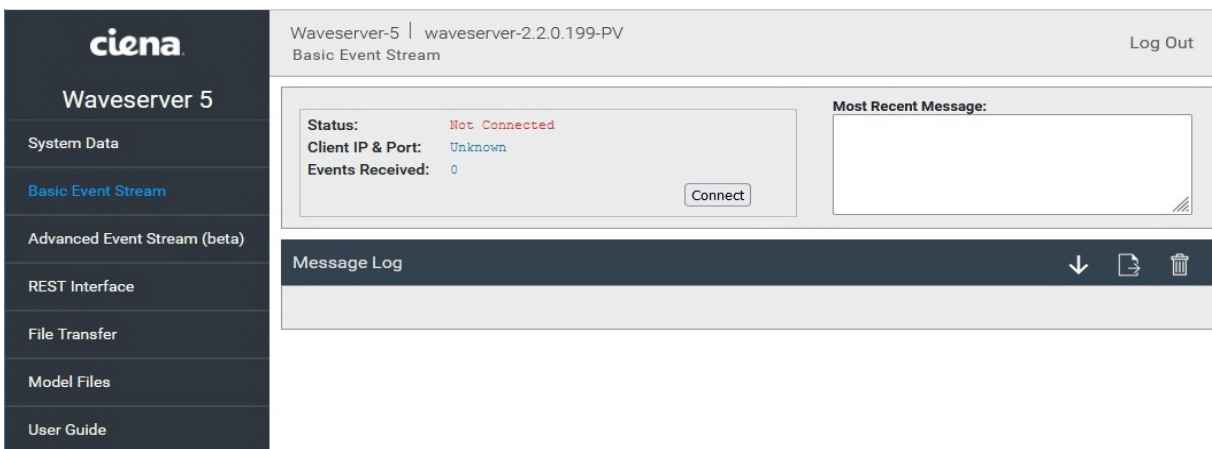
Figure 24 User Guide menu item



Basic Event Stream

Basic Event Stream enables use of the WebSocket protocol, as defined in RFC 6455, to listen for Waveserver 5 autonomous messages.

Figure 25 Basic Event Stream window



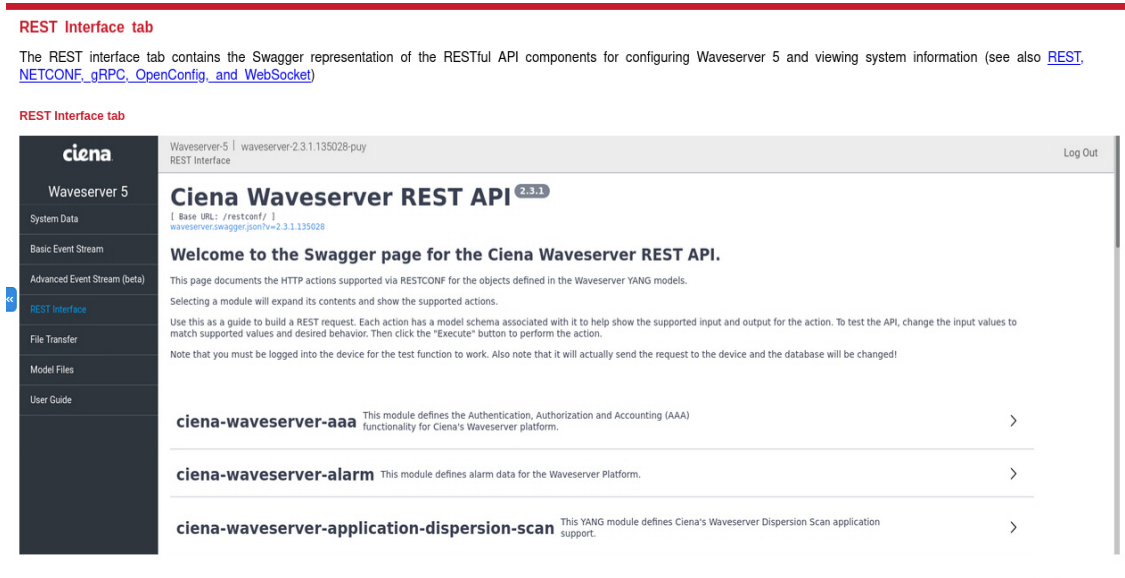
Advanced Event Stream

Note: Advanced Event Stream is a beta feature that is not supported in the current release.

REST Interface

The REST interface menu item contains links to the Swagger representation of the RESTful API components for configuring Waveserver 5 and viewing system information (see also [REST](#), [NETCONF](#), [gRPC](#), [OpenConfig](#), and [WebSocket](#)).

Figure 26 REST Interface window



Model Files

Model Files contains links to display and download a tar file containing the raw YANG files. To view individual Yang models in the browser, click one of the links on the homepage.

Figure 27 Model File window

	Waveserver-5 waveserver-2.2.0.199-PV Model Files Log Out	
Waveserver 5	Download Ciena YANG module files	
System Data	Click here to download an archive of all raw Ciena YANG files	
Basic Event Stream	Download OpenConfig YANG module files	
Advanced Event Stream (beta)	Click here to download an archive of all raw OpenConfig YANG files	
REST Interface	SNMP MIB Files	
File Transfer	Click here to download an archive of all SNMP MIB files.	
Model Files	Browse Ciena YANG module files	Browse OpenConfig YANG module files
User Guide	ciena-common-version.html ciena-common-yang.html ciena-pro-telemetry.html ciena-waveserver-aaa.html ciena-waveserver-alarm.html ciena-waveserver-application-dispersion-scan.html ciena-waveserver-application-its.html ciena-waveserver-application-link-data-collection.html ciena-waveserver-application-otdr.html ciena-waveserver-chassis.html ciena-waveserver-configuration.html ciena-waveserver-encryption.html ciena-waveserver-interfaces.html ciena-waveserver-license.html ciena-waveserver-link-data-telemetry.html ciena-waveserver-ldp.html ciena-waveserver-logging.html ciena-waveserver-module.html ciena-waveserver-ndp.html ciena-waveserver-pkix.html ciena-waveserver-pm-tca.html ciena-waveserver-pm.html ciena-waveserver-port.html ciena-waveserver-protection.html ciena-waveserver-ptp-cmd.html ciena-waveserver-ptp-modem.html ciena-waveserver-ptp.html ciena-waveserver-snmp.html ciena-waveserver-software.html ciena-waveserver-spli.html ciena-waveserver-statistics.html ciena-waveserver-system.html ciena-waveserver-topology.html ciena-waveserver-typedefs.html ciena-waveserver-xcvt-modem.html ciena-waveserver-xcvt-pluggable.html ciena-waveserver-xcvt.html ietf-inet-types.html ietf-yang-types.html yuma-ncx.html	openconfig-if-ip.html openconfig-local-routing.html openconfig-platform-transceiver.html openconfig-platform.html openconfig-system.html openconfig-terminal-device.html

Table 48 Waveserver 5 homepage resources

Resource	Description
Model files	Click the link to download a tar file containing the raw YANG files. To view individual Yang models in the browser, click one of the following links on the homepage. • ciena-common-version.html • ciena-common-yang.html • ciena-pro-telemetry.html • ciena-waveserver-aaa.html • ciena-waveserver-alarm.html • ciena-waveserver-application-dispersion-scan.html • ciena-waveserver-application-its.html • ciena-waveserver-application-link-data-collection.html • ciena-waveserver-application-otdr.html • ciena-waveserver-chassis.html • ciena-waveserver-configuration.html • ciena-waveserver-encryption.html • ciena-waveserver-interfaces.html • ciena-waveserver-license.html • ciena-waveserver-lldp.html • ciena-waveserver-logging.html

Resource	Description
Model files	• ciena-waveserver-module.html • ciena-waveserver-ndp.html • ciena-waveserver-pkix.html • ciena-waveserver-pm-tca.html • ciena-waveserver-pm.html • ciena-waveserver-port.html • ciena-waveserver-protection.html • ciena-waveserver-ptp-cmd.html • ciena-waveserver-ptp.html • ciena-waveserver-ptp-modem.html • ciena-waveserver-snmp.html • ciena-waveserver-software.html • ciena-waveserver-spli.html • ciena-waveserver-statistics.html • ciena-waveserver-system.html • ciena-waveserver-topology.html • ciena-waveserver-typedefs.html • ciena-waveserver-xcvr-modem.html • ciena-waveserver-xcvr-pluggable.html • ciena-waveserver-xcvr.html • ietf-inet-types.html • ietf-yang-types.html • yuma-ncx.html
Openconfig Yang Model files	• openconfig-if-ip.html • openconfig-local-routing.html • openconfig-platform-transceiver.html • openconfig-platform.html • openconfig-system.html • openconfig-terminal-device.html

File Transfer

File Transfer allows you to upload and download files such as software and license files between Waveserver 5 and the local machine using Secure Hypertext Transfer Protocol (HTTPS). Using the Waveserver 5 homepage to transfer files eliminates the need for an xFTP server.

Figure 28 File Transfer window

ciena

Waveserver 5

System Data

Basic Event Stream

Advanced Event Stream (beta)

REST Interface

File Transfer

Model Files

User Guide

WS5-0126 | waveserver-2.2.0.241

File Transfer

Log Out

File Download

To download a file from a directory, firstly make sure the file is in /tmp/public, then select the file name below and click **Download**.

File Name:

Select a file

Download

File Upload

1. Choose the file you would like to upload

2. Click **Upload**

Note: Uploaded file can be found in the /tmp/public directory on Waveserver

Browse...

No files selected.

Upload

CHAPTER 11

Ordering

This section provides ordering information for the Waveserver 5.

Ordering information

The following tables list Waveserver 5 components and software that can be ordered from Ciena.

- [Pick-to-Order \(PTO\) hardware and software kits](#)
- [Bracket and installation kits](#)
- [Common Equipment, PSUs with Fan Modules, PSU AC Cords, and Filler Modules](#)
- [Service Modules](#)
- [QSFP28 and QSFP-DD pluggable transceivers](#)
- [High-speed pluggable interconnect cable assemblies](#)
- [MPO-MPO patch cables for 100G SR pluggables](#)
- [MPO-LC MMF 4x10G SR breakout cables](#)
- [MPO-LC SMF 4x10G LR breakout cables](#)
- [LC fiber tool kit](#)
- [Software licenses - Base and Add-ons](#)

Pick-to-Order (PTO) hardware and software kits

The following table provides ordering information on pick-to-order hardware and software kits.

Table 49 Pick-to-Order hardware and software kits

Ordering code (Note)	PTO hardware and software kit description	Code	Description	Qty
K86-3500-900	Waveserver 5 Chassis w/ AC Power Modules, Ctrl, Fan Modules	186-3001-900	Waveserver 5 Chassis	1
		186-3010-900	Waveserver 5 Control Processor Subsystem	1
		186-3020-900	Waveserver 5 Access Panel	1
		186-3040-900	Waveserver 5 100-277Vac/ 240Vdc Power and Fan Module	2
		186-3051-900	Waveserver 5 Shipping Brackets and Packaging	1
K86-3501-900	Waveserver 5 Chassis w/ DC Power Modules, Ctrl, Fan Modules	186-3001-900	Waveserver 5 Chassis	1
		186-3010-900	Waveserver 5 Control Processor Subsystem	1
		186-3020-900	Waveserver 5 Access Panel	1
		186-3041-900	Waveserver 5 -48Vdc Power and Fan Module	2
		186-3051-900	Waveserver 5 Shipping Brackets and Packaging	1
Note: 186-3031-900 (Waveserver 5 Double Slot Filler) is not included in the PTO kits and must be ordered, if required, to provide proper cooling since all empty slots require a slot filler.				

Bracket and installation kits

This section provides information on ordering bracket and installation kits.

Before ordering bracket and installation kits, ensure that you have the following information:

- The frame type used; 19-in EIA, 23-in EIA, or ETSI.
- Whether a ground bracket is required. A ground bracket is required for all DC Waveserver 5 installations, and for AC Waveserver 5 installations that require GR-1089-CORE compliance. For AC installations that do not require GR-1089-CORE compliance, the ground bracket is recommended but not required.
- The frame upright (post) spacing. For information on supported upright spacing depths with and without the ground bracket installation, refer to [Supported upright spacing depths](#).

The following table provides ordering information on bracket and installation kits.

Table 50 Bracket and installation kits

Ordering Code (Note)	Bracket Kit Name	Description
186-1300-900	Waveserver 19-inch Bracket Kit	Contains parts for installing Waveserver in a 2-post or 4-post 19-in equipment frame.
186-1301-900	Waveserver 23-inch Bracket Kit	Contains parts for installing Waveserver in a 2-post or 4-post 23-in equipment frame.
186-1302-900	Waveserver ETSI Bracket Kit	Contains parts for installing Waveserver in an ETSI equipment frame.
186-1303-900	Waveserver 19-inch, 23-inch, and ETSI Bracket Kits	Contains all contents of Waveserver 19-inch, 23-inch, and ETSI bracket kits (186-1300-900, 186-1301-900, and 186-1302-900, respectively).
186-1306-900	Waveserver Data Center Rack Kit, 19-in Wide x 24 - 36-in Deep	Contains parts for installing Waveserver in a 19-inch wide, 24 - 36-inch deep data center frame.
186-1307-900	Waveserver Data Center Rack Kit, 23-in Wide x 24 - 36-in Deep	Contains parts for installing Waveserver in a 23-inch wide, 24 - 36-inch deep data center frame.
186-1310-900	Waveserver Metric Installation Kit	Optional fastener kits for installing a Waveserver 5 chassis into frames and cabinets.
186-1311-900	Waveserver Imperial Installation Kit	
186-1312-900	Waveserver Metric and Imperial Installation Kit	
Note: Refer to “Site planning and preparation” in <i>Waveserver 5 Installation</i> , 323-3001-200 for lists and diagrams of bracket kit contents.		

Common Equipment, PSUs with Fan Modules, PSU AC Cords, and Filler Modules

The following table provides ordering information on common equipment, PSUs with fan modules, and filler modules.

Table 51 Common equipment, PSUs with fan module, PSU AC cords and filler modules

Ordering code	Description
186-3001-900	Waveserver 5 Chassis
186-3010-900	Waveserver 5 Control Processor Subsystem

Ordering code	Description
186-3020-900	Waveserver 5 Access Panel
186-3040-900	Waveserver 5 100-277Vac/240Vdc Power and Fan Module
186-3041-900	Waveserver 5 -48Vdc Power and Fan Module
186-3042-900	Waveserver 5 Fan Module
NTK955GA	AC Power Cord, Saf-D-Grid to C20, 12AWG, 250V, 2m For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.
NTK955GB	AC Power Cord, Saf-D-Grid to Open, 12AWG, 250V, 3m For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation. Order this AC power cable if you want to terminate the AC cable with a connector of your choice.
NTK955GC	AC Power Cord, Saf-D-Grid to C20, 12 AWG, 250V, 2m, white For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.
NTK955GD	AC Power Cord, Saf-D-Grid to C14, 14 AWG, 250V, 2m, black For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.
NTK955GE	AC Power Cord, Saf-D-Grid to CEE7/7, 12 AWG, 250V, 2m, black For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.
NTK955GF	AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, black For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.
NTK955GG	AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, black For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.

Ordering code	Description
NTK955GH	AC Power Cord, Saf-D-Grid, left routed to C20, 12 AWG, 250V, 2m, white For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.
NTK955GJ	AC Power Cord, Saf-D-Grid, right routed to C20, 12 AWG, 250V, 2m, white For more information, refer to AC Power Cables . Note: Ciena AC power cables must be ordered prior to the Waveserver 5 installation.
186-3031-900	Waveserver 5 Double Slot Filler

Service Modules

The following table provides ordering information on service modules.

Table 52 Service Modules

Ordering code	Description
186-3100-900	Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module
186-3100-901	Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module
186-3110-901	Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFP-DD Module
186-3101-900	Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module
186-3101-901	Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module
186-3111-901	Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module
186-3102-900	Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module
186-3102-901	Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module
186-3112-901	Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module
186-3120-900	Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module
186-3121-900	Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module
186-3302-900	Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module

QSFP28 and QSFP-DD pluggable transceivers

The following table provides ordering information on the QSFP28 and QSFP-DD pluggable transceivers.

Table 53 QSFP28 and QSFP-DD pluggable transceivers

Ordering code	Description	Application
160-9400-900	100GBASE-SR4, 4x25G, MMF, 850NM, 100M, MPO QSFP28	100GE MMF Ribbon (NTTC95AA – AD, NTTC95AG – AH). Refer to 40GE/100GE MMF Ribbon NTTC95AA - NTTC95AH and MPO-MPO patch cables for 100G SR pluggables . FEC is required when you are using an SR4 pluggable transceiver at 103.125G (100GE).
160-9401-900	100GBASE-LR4, 4x25G, WDM, SMF, 1310NM, 10KM QSFP28	100GE SMF. Refer to LC-LC SM Simplex/Duplex patch cords .
160-9402-900	103.1G-111.8G, 4x25G, WDM, SMF, 1310NM, 10 KM QSFP28	Intended for interoperability with 6500 100G and 200G OCI OTU4 clients.
160-9403-900	100GE CWDM4, 4x25G, SMF, 1310NM, 2KM QSFP28	100GE SMF. Refer to LC-LC SM Simplex/Duplex patch cords .
160-9404-900	103.1G-111.8G, 4x25G, WDM, SMF, 1310NM, 40KM QSFP28	100GE SMF and OTU4
160-9405-900	100GE 4WDM-40, 4x25G, SMF, 1310nm, 40km QSFP28	100GE SMF
160-9406-900	100GE PSM4, 4x25G, SMF, 1310nm, 500m, MPO QSFP28	100GE SMF
160-9431-900	100G-FR, SMF, 1310nm, 2km QSFP28	100GE SMF
160-9433-900	100GBASE-LR1, SMF, 1310nm, 10km QSFP28	100GE SMF
160-9600-900	400G-FR4, SMF, 2KM QSFP-DD	400GE SMF
160-9603-900	400G-LR4-10/4x100GE, SMF, 10km QSFP-DD	400GE SMF
180-3530-900	WL5n 400ZR Type 3.0 QSFP-DD	400G SMF

High-speed pluggable interconnect cable assemblies

The following table provides ordering information on high-speed pluggable interconnect cable assemblies.

Table 54 High-speed pluggable interconnect cable assemblies

Ordering code (Note)	Description	Application
400GE Active Optical Cables (AOCs)		
160-9660-003	400GE, 3m, QSFP-DD - QSFP-DD Active Optical Cable	Intended for connecting terminal client ports to subtending equipment. FEC is required at all rates when you are using an AOC.
160-9660-005	400GE, 5m, QSFP-DD - QSFP-DD Active Optical Cable	
160-9660-010	400GE, 10m, QSFP-DD - QSFP-DD Active Optical Cable	
160-9660-020	400GE, 20m, QSFP-DD - QSFP-DD Active Optical Cable	
160-9660-100	400GE, 100m, QSFP-DD - QSFP-DD Active Optical Cable	
100GE Active Optical Cables (AOCs)		
160-9460-003	103.1G-112.28G, 3M, QSFP28-QSFP28 Active Optical Cable	Intended for connecting terminal client ports to subtending equipment. FEC is required at all rates when you are using an AOC.
160-9460-005	103.1G-112.28G, 5M, QSFP28-QSFP28 Active Optical Cable	
160-9460-007	103.1G-112.28G, 7M, QSFP28-QSFP28 Active Optical Cable	
160-9460-010	103.1G-112.28G, 10M, QSFP28-QSFP28 Active Optical Cable	
160-9460-015	103.1G-112.28G, 15M, QSFP28-QSFP28 Active Optical Cable	
160-9460-020	103.1G-112.28G, 20M, QSFP28-QSFP28 Active Optical Cable	
160-9460-050	103.1G-112.28G, 50M, QSFP28-QSFP28 Active Optical Cable	
160-9460-100	103.1G-112.28G, 100M, QSFP28-QSFP28 Active Optical Cable	
Note: Since Active Optical Cables are available in a limited number of fixed lengths, their use requires the ability to manage optical fiber slack. This must be taken into consideration during the planning and ordering process. Depending on the number of optical fibers and their lengths, fiber management trays or upright (vertical) fiber managers may be required.		

Cabling

This section provides figures of cables used with Waveserver 5.

Figure 29 LC-LC SM Simplex/Duplex patch cords



Figure 30 40GE/100GE MMF Ribbon NTTC95AA - NTTC95AH



Figure 31 4 x 10GE MMF Ribbon Breakout NTTC95BA - NTTC95BK



Figure 32 4 x 10GE SMF Ribbon Breakout NTTC97BA - NTTC97BH

MPO-MPO patch cables for 100G SR pluggables

The following table provides ordering information on MPO-MPO patch cables for 40G and 100G SR pluggables.

Table 55 MPO-MPO patch cables for 40 and 100G SR pluggables

Ordering Code	Description
NTTC95AA	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM3 Fiber, Formable Boot, TX-RX, 5 Meter
NTTC95AB	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM3 Fiber, Formable Boot, TX-RX, 10 Meter
NTTC95AC	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM3 Fiber, Formable Boot, TX-RX, 25 Meter
NTTC95AD	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM3 Fiber, Formable Boot, TX-RX, 50 Meter
NTTC95AE	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM3 Fiber, Formable Boot, TX-RX, 100 Meter
NTTC95AF	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM4 Fiber, Formable Boot, TX-RX, 150 Meter
NTTC95AG	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM3 Fiber, Formable Boot, Tx-Rx, 3 Meter
NTTC95AH	FPC, MPO(F)-MPO(F), 3.0D, 12F, MM, OM3 Fiber, Formable Boot, Tx-Rx, 7 Meter

MPO-LC MMF 4x10G SR breakout cables

The following table provides ordering information on MPO-LC MMF 4x10G SR breakout cables.

Table 56 MPO-LC MMF 4x10G SR breakout cables

Ordering Code	Description
NTTC95BA	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 1M
NTTC95BB	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 2M
NTTC95BC	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 3M
NTTC95BD	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 5M
NTTC95BE	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 10M
NTTC95BF	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 15M
NTTC95BG	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 20M
NTTC95BH	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 30M
NTTC95BJ	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 50M
NTTC95BK	FPC, BREAKOUT MPO(F) 3.0D, 12F, MM, OM3 BIF, Formable Boot - 4 Duplex LC - 100M

MPO-LC SMF 4x10G LR breakout cables

The following table provides ordering information on MPO-LC SMF 4x10G LR breakout cables.

Table 57 MPO-LC SMF 4x10G LR breakout cables

Ordering Code	Description
NTTC97BA	FPC, BREAKOUT MPO(F) 3.0D, 12F, SM, BIF, Formable Boot - 4 Duplex LC - 1M
NTTC97BB	FPC, BREAKOUT MPO(F) 3.0D, 12F, SM, BIF, Formable Boot - 4 Duplex LC - 3M
NTTC97BD	FPC, BREAKOUT MPO(F) 3.0D, 12F, SM, BIF, Formable Boot - 4 Duplex LC - 5M
NTTC97BE	FPC, BREAKOUT MPO(F) 3.0D, 12F, SM, BIF, Formable Boot - 4 Duplex LC - 7M
NTTC97BF	FPC, BREAKOUT MPO(F) 3.0D, 12F, SM, BIF, Formable Boot - 4 Duplex LC - 10M
NTTC97BG	FPC, BREAKOUT MPO(F) 3.0D, 12F, SM, BIF, Formable Boot - 4 Duplex LC - 15M
NTTC97BH	FPC, BREAKOUT MPO(F) 3.0D, 12F, SM, BIF, Formable Boot - 4 Duplex LC - 30M

LC fiber tool kit

The following table provides the ordering information on the LC fiber tool kit.

Table 58 LC fiber tool kit

Ordering Code	Description
174-0099-900	Kit, LC Fiber Tool This tool is used to assist in installing and removing LC fibers or removing pluggables when the space is restricted. When not in use, this tool can be stored in its container. The LC fiber tool kit is optional. Order one or two (if a backup is required) for each Waveserver 5 chassis.

Console port (RJ-45 or USB-C) to terminal (PC) adaptors

The following table provides information on the adaptors required for commissioning the Waveserver 5. These adaptors allow you to connect locally to the RJ-45 or USB-C console port of the Waveserver 5. For additional information, refer to [Using console port or USB-C port for management access during commissioning](#).

Note: These are user supplied and must be available prior to commissioning.

Table 59 Console port (RJ-45 or USB-C) to terminal (PC) adaptors

Adaptor	Description
RJ45 to USB adaptor	Used to connect to the RJ-45 console port on the Waveserver 5.
RJ45 to DB9 adaptor	Used to connect to the RJ-45 console port on the Waveserver 5.
USB-C to USB adaptor	Used to connect to the USB-C console port on the Waveserver 5.

Software licenses - Base and Add-ons

The following tables provide ordering information on software base licenses and add-on licenses.

- [Software licenses - Base](#)
- [Software licenses - Add-ons](#)

Table 60 Software licenses - Base

License Type	Ordering Code	Description
Release 2.2 Base Licenses	S86-LIC-WOSBASE020200	Waveserver OS Release 2.2 Base Software License. Enables the use of Waveserver OS Release 2.2 system software. Purchase one license per Waveserver 5.
	S86-LIC-WOSUPG020200	Waveserver OS Release 2.2 Upgrade License. Enables the use of the Waveserver OS Release 2.2 system software. This license is an upgrade for the base system software. Purchase one license per Waveserver 5.

Table 61 Software licenses - Add-ons

License Type	Ordering code	Description
API License	S86-LIC-WOSAPIS	Waveserver OS Application API Add-On License. Enables the use of northbound interfaces. Purchase one license per Waveserver 5. Software license S86-LIC-WOSAPIS is required when accessing the Waveserver 5 using API (gRPC, NETCONF, REST and WebSocket), when it does not involve MCP. This license is not required when managing Waveserver 5 using MCP and when direct API access is not required. This license is not required for SNMP or CLI.
Foreign Client Plug License	S86-LIC-WOSFRGNCLPLUG	Waveserver OS Foreign Client Plug Add-On License. Ciena-branded pluggable optics are supported by Ciena. This license enables the use of third-party client pluggable optics. Purchase one license per module.
Foreign Line Plug License	S86-LIC-WOSFRGNLNPLUG	Waveserver OS Foreign Line Plug Add-On License. Ciena-branded pluggable optics are supported by Ciena. This license enables the use of third-party coherent line pluggable optics. Purchase one license per module.

License Type	Ordering code	Description
ITS License	S86-LIC-WOSITS	Waveserver OS Application ITS Add-On License. Purchase one license per WL5e module. The ITS Add-On License is released when the module is deleted using the <code>module delete slot</code> command.
Modem Licenses (Terrestrial)	S86-LIC-WL5ETERADD100G	WaveLogic 5e Terrestrial Incremental Capacity License (per 100G)
	S86-LIC-WL5ETER250KM	WaveLogic 5e Terrestrial Reach License (up to 250km). Purchase one license per modem.
	S86-LIC-WL5ETER500KM	WaveLogic 5e Terrestrial Reach License (up to 500km). Purchase one license per modem.
	S86-LIC-WL5ETER1000KM	WaveLogic 5e Terrestrial Reach License (up to 1000km). Purchase one license per modem.
	S86-LIC-WL5ETER2000KM	WaveLogic 5e Terrestrial Reach License (up to 2000km). Purchase one license per modem.
	S86-LIC-WL5ETER3000KM	WaveLogic 5e Terrestrial Reach License (up to 3000km). Purchase one license per modem.
	S86-LIC-WL5ETER4000KM	WaveLogic 5e Terrestrial Reach License (up to 4000km). Purchase one license per modem.
	S86-LIC-WL5ETER5000KM	WaveLogic 5e Terrestrial Reach License (up to 5000km). Purchase one license per modem.
	S86-LIC-WL5ETER8000KM	WaveLogic 5e Terrestrial Reach License (up to 8000km). Purchase one license per modem.
Modem Licenses (Terrestrial)	S86-LIC-WL5NTERADD100G	Waveserver Transponder Connection Bandwidth License (per 100G)
Modem Licenses (Submarine)	S86-LIC-WL5ESUBADD100G	WaveLogic 5e Submarine Incremental Capacity License (per 100G)
	S86-LIC-WL5ESUB3500KM	WaveLogic 5e Submarine Reach License (up to 3500km). Purchase one license per modem.
	S86-LIC-WL5ESUB9000KM	WaveLogic 5e Submarine Reach License (up to 9000km). Purchase one license per modem.
	S86-LIC-WL5ESUB14000KM	WaveLogic 5e Submarine Reach License (up to 14000km). Purchase one license per modem.

CHAPTER 12

Licensing

This section contains the following topics:

- [Overview](#)
 - [Licensing process](#)
 - [External License Server](#)
 - [Manual installation of licenses](#)
 - [License enforcement mode](#)
 - [Terrestrial Reach Limit](#)
 - [Licensing considerations](#)
 - [Commissioning mode](#)
 - [Listing licenses in arrears](#)
 - [Licensing related alarms](#)
- [Procedure list](#)

Overview

Ciena's product licensing model establishes a single process for ordering, managing, and deploying Ciena-licensed features. Ciena makes licenses available to you through the License Portal interface. After you obtain licenses from the portal, you can deploy the licenses to each Waveserver 5 in your network using two different methods:

- through an external license server
- through local installation of licenses on each Waveserver 5 manually

You can use these methods individually, or use a combination of both methods to distribute licenses in your network.

A Waveserver 5 software license contains all licenses (entitlements) for purchased features in a single file. Once you install and activate the first license on a Waveserver 5, each subsequent software license you purchase for that device contains all previous software licenses plus the new software licenses.

For a list of Waveserver 5 software licenses, see [Licenses and associated enforcement modes](#).

Note: If your Waveserver 5 is running a software release that is different from the release specified on the software license, see [Upgrading Waveserver 5 when the release number of the software on the device does not match the release number on your software license](#).

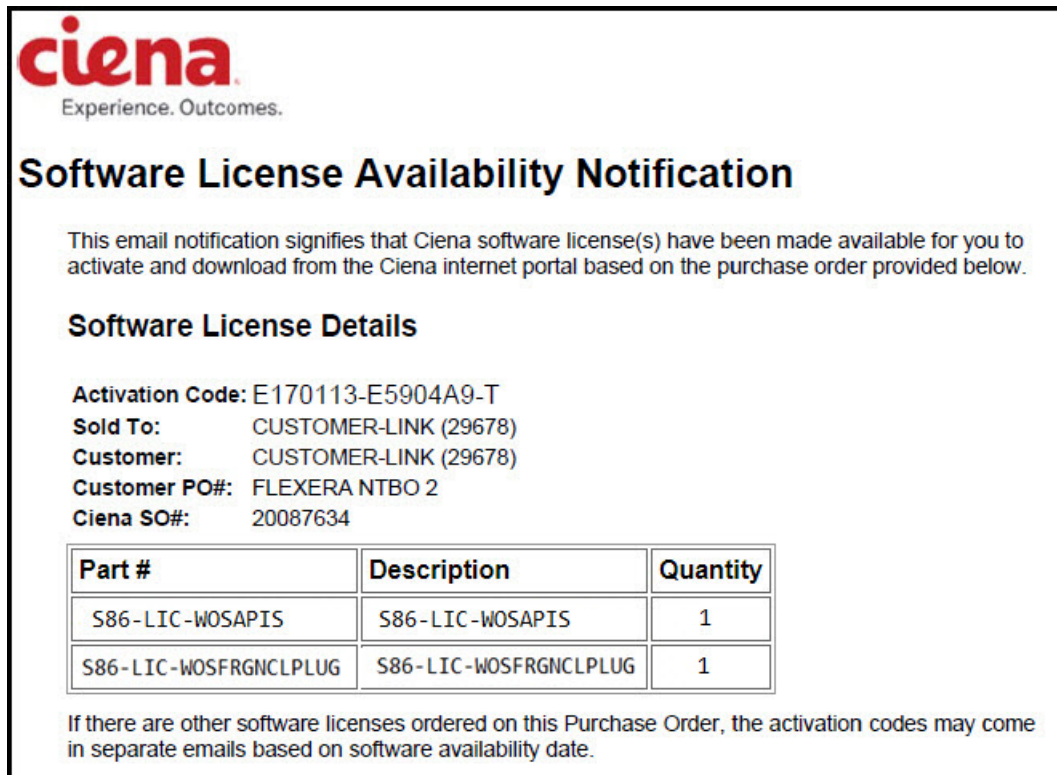
Licensing process

The Ciena licensing process is built on the Flexera Software licensing architecture. At the center of this process is the License Portal, which is a dedicated interface within the Ciena customer portal that you use to obtain, manage, and activate licenses.

Ordering and license generation

When you order a license, Ciena generates the licenses and makes them available to you through the License Portal. The date of license generation aligns with the shipping date on the purchase order. Ciena sends you a Software License Availability Notification email. This email contains license activation codes that allow you to obtain software licenses from the License Portal. See the following figure.

Figure 33 Sample Software License Notification email



Registering devices and obtaining licenses

To obtain software licenses from the License Portal, you must first register the device where you intend to store or install the licenses. The tasks to register a device and obtain licenses vary depending on whether you will distribute licenses using an external license server (see [External License Server](#)) or you will install licenses on each Waveserver 5 manually (see [Manual installation of licenses](#)).

External License Server

The external license server application manages, distributes, and tracks license keys. The license server is available from the Ciena customer portal. Waveserver 5 supports an external license server with and without redundancy.

Note: Contact Ciena if you manually installed licenses on each local Waveserver 5 and you would like to transition to using an external license server for distributing licenses.

License server host requirements

The license server runs on a standalone Linux host deployed within the Waveserver 5 network. The table lists the Linux host requirements.

Table 62 External license server requirements

Platform component	Minimum requirement
Hardware	A Linux-capable host with at least: • CPU: — 2 vCPU x 2 GHz for small deployments (fewer than 100 NEs) — 4 vCPU x 2 GHz for large deployments (100 or more NEs) • RAM: 4 GB for small deployments (fewer than 100 NEs); otherwise, 16 GB is recommended • Storage: 150 GB free disk space

Platform component	Minimum requirement
Operating system	One of these Linux distributions: • Red Hat 7.4+ or later • Oracle Linux 7.4+ or later • CentOS 7.4+ or later Minimum OS package set: Infrastructure Server Ciena has tested and supports deploying the License Server in the following hypervisor environments: • VMWare ESXi • OpenStack

Registering the license server host

Register the license server host on the License Portal once only, the first time you obtain licenses from the License Portal to store on the server. The table lists the tasks to register the license server host and obtain Waveserver 5 software licenses using an external license server.

Table 63 Registering an external license server and obtaining licenses

Task	Procedure
• Generate a Registration Request File on the License Server Host. • Register the License Server Host on the License Portal. • Download licenses from the License Portal to the PC. • Transfer licenses to the external license server host and activate the licenses.	These tasks are part of the procedure: Obtaining software licenses: external license servers
Provision Waveserver 5 access to the license server host	Provisioning access to an external license server
Provision license-server sync-interval	Provisioning the license server sync-interval

License server sync-interval

Each Waveserver periodically synchronizes its license holdings with the license inventory on the license server, if one is configured. Synchronizing the licenses ensures the Waveserver has the feature licenses it requires and clears any license violation alarms if the required licenses are available on the license server.

The interval for performing the synchronization is controlled by the `sync-interval` parameter. The default interval is 15 minutes. You can provision `sync-interval` to a value between 15 and 1440 minutes (24 hours).

Increasing the sync-interval reduces the number of times the Waveserver syncs its installed licenses with the license server. For most deployments, you do not need to change the value of `sync-interval`. However, the number and mix of products accessing the license server and the value of `sync-interval` can affect license server performance.

You can also force a license-server synchronization at any time by running the command: `license client server sync-now`

For information on provisioning access to the license server, see [Procedure 7, “Provisioning access to an external license server”](#). For information on setting the license-server sync-interval, see [Procedure 8, “Provisioning the license server sync-interval”](#).

High Availability (HA) license server

Waveserver 5 supports a High Availability (HA) license server configuration consisting of two license servers: a primary (active) license server and a secondary (backup) license server that work together. The HA license server configuration provides a simple failover mechanism that ensures licenses are available if the primary license server goes out of service or cannot be reached by Waveserver 5 devices in the network.

Note: Both the primary and secondary license servers must be behind the same proxy. The High Availability (HA) license server configuration does not support geo-redundancy.

Manual installation of licenses

The following table lists the tasks to obtain and install Waveserver 5 software licenses manually.

Table 64 Registering a Waveserver 5 and obtaining and manually installing licenses

Task	Procedure
Create or acquire a Registration ID for the Waveserver 5	Setting the Registration ID on Waveserver 5
• Generate a Registration Request File on the Waveserver 5. • Register the Waveserver 5 on the License Portal. • Download licenses from the License Portal to the PC.	These tasks are part of the procedure: Obtaining software licenses: manual installation
Transfer licenses to the Waveserver 5 and activate licenses	Installing and activating a software license

License summary information

Software license files that you obtain from Ciena contain all purchased licenses in a single file. To obtain software licenses you must register the Waveserver 5 with the licensing portal, using the device Registration ID. Once you register Waveserver 5, you can activate licenses on it.

To add software licenses after you install an initial license, you must activate a new license file that contains the original software licenses and the new software licenses you want to add.

Registering Waveserver 5 for licensing

To register a Waveserver 5 unit, you must set the Registration ID manually. The Registration ID is a user-defined string. For details, see [Setting the Registration ID on Waveserver 5](#).

License filename

Each software license filename contains the Waveserver 5 Registration ID. The format of the software license filename is:

ciena-COMPANY-X_001-201624111150.lic where COMPANY-X_001 is the Registration ID.

Registration ID and Waveserver 5 chassis replacements

There are two licensing options available for Waveserver 5 chassis replacements, depending on whether you have access to the chassis Registration ID.

You can look up the Registration ID on the Ciena portal, verify the Registration ID if you have access to the license file, or retrieve the Registration ID using the CLI command `license client show`. See [Retrieving or looking up the Registration ID](#).

If you have access to the Registration ID for the Waveserver 5 chassis you are replacing, you can:

- 1 Set the Registration ID on the replacement Waveserver 5 to match the Waveserver 5 you are replacing.
- 2 Obtain the software license from your local PC or download a license from the License Portal.
- 3 Install and activate the existing license on the replacement Waveserver 5.

See [Setting the Registration ID on Waveserver 5](#) and [Installing and activating a software license](#).

If you cannot access the Registration ID for the Waveserver 5 chassis that you are replacing, you must:

- 1 Set the Registration ID on the new Waveserver 5.

- 2 Obtain a software license.
- 3 Install and activate the software license.

See [Setting the Registration ID on Waveserver 5](#), [Obtaining software licenses: manual installation](#) and [Installing and activating a software license](#).

License enforcement mode

This section provides information on license enforcement modes. Currently, Waveserver 5 supports a single enforcement mode:

- Pre-authorized - All licenses with the Pre-authorized enforcement mode allow you to provision the associated feature without a valid license. If you provision a feature without a valid license, the system raises a License Violation alarm, which prevents you from performing software upgrades.

Licenses and associated enforcement modes

The following table provides a list of licenses and their enforcement modes.

Table 65 Base and Add-on licenses and associated enforcement modes

License	Enforcement Mode	Description of license
S86-LIC-WOSBASEXXXXXX	Pre-authorized	Software license, where XXXXXX is the current release number.
S86-LIC-WOSAPIS	Pre-authorized	API Add-On License.
S86-LIC-WOSFRGNCLPLUG	Pre-authorized	Waveserver OS Foreign Client Plug Add-On License.
S86-LIC-WOSFRGNLNPLUG	Pre-authorized	Waveserver OS Foreign Line Plug Add-On License.
S86-LIC-WOSITS	Pre-authorized	Waveserver OS Application Integrated Test Set (ITS) Add-On License. One ITS Add-On License is required per module. The ITS Add-On License is released when the module is deleted using the <code>module delete slot</code> command.

Table 66 WaveLogic 5e Add-on licenses and associated enforcement modes

License	Enforcement Mode	Description of license
S86-LIC-WL5ETERADD100G	Pre-authorized	WaveLogic 5e Terrestrial Incremental Capacity License (per 100G).
S86-LIC-WL5ETER250KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 250km. Purchase one license per modem.

License	Enforcement Mode	Description of license
S86-LIC-WL5ETER500KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 500km. Purchase one license per modem.
S86-LIC-WL5ETER1000KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 1000km. Purchase one license per modem.
S86-LIC-WL5ETER2000KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 2000km. Purchase one license per modem.
S86-LIC-WL5ETER3000KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 3000km. Purchase one license per modem.
S86-LIC-WL5ETER4000KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 4000km.
S86-LIC-WL5ETER5000KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 5000km.
S86-LIC-WL5ETER8000KM	Pre-authorized	WaveLogic 5e Terrestrial Reach License up to 8000km
S86-LIC-WL5ESUBADD100G	Pre-authorized	WaveLogic 5e Submarine Incremental Capacity License (per 100G).
S86-LIC-WL5ESUB3500KM	Pre-authorized	WaveLogic 5e Submarine Reach License up to 3500km.
S86-LIC-WL5ESUB9000KM	Pre-authorized	WaveLogic 5e Submarine Reach License up to 9000km.
S86-LIC-WL5ESUB14000KM	Pre-authorized	WaveLogic 5e Submarine Reach License up to 14000km.

Table 67 Waveserver Transponder Connection Bandwidth license

License	Enforcement Mode	Description of license
S86-LIC-WL5NTERADD100G	Pre-authorized	Waveserver Transponder Connection Bandwidth License (per 100G).

Terrestrial Reach Limit

The default modem reach limit is now a provisionable, system-level parameter: `terrestrial-reach-limit`. The default value is 1000km, but you can assign it any one of the following values: 250, 500, 2000, 3000, 4000 or 5000 km. Use this command to assign a ptp terrestrial reach limit in kilometers:

```
system defaults set terrestrial-reach-limit <250, 500,  
1000, 2000, 3000, 4000, 5000, 8000>
```

The ability to modify the reach to the desired limit ensures that modules will acquire the correct reach licenses when the modules are inserted.

Note: This parameter (`terrestrial-reach-limit`) only applies to newly provisioned modems and does not affect existing services. There is no equivalent system level parameter for the submarine modem class.

For information on other system level defaults, see [Commissioning](#).

Licensing considerations

These considerations apply to module and pluggable licenses:

- When a module is replaced, the licenses associated with that module are re-used automatically on the replacement module.
- A pluggable automatically acquires a capacity license when you insert or configure the pluggable on the client or line port of the modem. The type of license that is acquired is based on the provisioned modem class.

For example, when you insert a supported pluggable in the client port of a Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD Module, the pluggable automatically acquires a WaveLogic 5e Capacity License, either for terrestrial applications (S86-LIC-WL5ETERADD100G) or submarine applications (S86-LIC-WL5ESUBADD100G) depending on the provisioned modem-class on the associated line port.

Similarly, when you insert or configure pluggables in both the client ports and associated line port of a Waveserver 5 DCO 12xQSFP28/QSFP-DD Module and a connection is established, the pluggable automatically acquires a WaveLogic 5n Capacity License (S86-LIC-WL5NTERADD100G).

However, when you insert a 400G-ZR pluggable in the client port of a Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD Module, the pluggable acquires a WaveLogic 5e Capacity License either for terrestrial applications (S86-LIC-WL5ETERADD100G) or submarine applications (S86-LIC-WL5ESUBADD100G) depending on the provisioned modem-class on the associated line port.

Commissioning mode

Commissioning mode is a temporary state that allows you to commission a Waveserver without having the required licenses for the features you deploy.

On first boot after a factory-reset, the Waveserver license system starts up in commissioning mode. The operational-state of the license client is reported as commissioning in CLI, REST, NETCONF, and gNMI when in commissioning mode.

Waveserver remains in commissioning mode until one of the following events occurs:

- you activate a license file of entitlements
- a license server is configured
- 90 days have elapsed since Waveserver entered commissioning mode

While Waveserver is in commissioning mode, the Entitlement Enforcement Disabled alarm is active (For details on this alarm, see [Procedure 247, “Entitlement Enforcement Disabled”](#)) and all other licensing alarms are blocked. You can perform software upgrades while Waveserver is in commissioning mode regardless of whether licensed features are provisioned.

Once Waveserver exits commissioning mode, the Entitlement Enforcement Disabled alarm clears, and the operational-state of the license client changes to up. Waveserver will now raise a License Missing alarm for any licensed feature that you provision without having the required license. Waveserver will also block software upgrades if a License Missing alarm is active.

Listing licenses in arrears

Waveserver provides a command to list licenses in arrears. This capability can assist you in identifying missing licenses that are causing the system to raise License Missing alarms. A license is in arrears when a provisioned feature requires the license, but insufficient license entitlements are available to satisfy the provisioning.

When a license is in arrears and Waveserver raises License Missing alarms, it also prevents you from performing system software upgrades. The ability to list licenses in arrears lets you immediately identify missing licenses and clarify how many entitlements are required for the current provisioning. For more information, see [Procedure 9, “Listing licenses in arrears”](#).

Licensing related alarms

This section provides a list of licensing related alarms. For more information on these alarms, see [Alarms](#).

- Entitlement Enforcement Disabled
- External License Server Comms Failure
- External License Server Time Out of Sync
- License Missing

- License Expiration Warning

Procedure list

This section provides the following procedures:

- [Procedure 1, "Retrieving license information"](#)
- [Procedure 2, "Setting the Registration ID on Waveserver 5"](#)
- [Procedure 3, "Retrieving or looking up the Registration ID"](#)
- [Procedure 4, "Obtaining software licenses: external license servers"](#)
- [Procedure 5, "Obtaining software licenses: manual installation"](#)
- [Procedure 6, "Installing and activating a software license"](#)
- [Procedure 7, "Provisioning access to an external license server"](#)
- [Procedure 8, "Provisioning the license server sync-interval"](#)
- [Procedure 9, "Listing licenses in arrears"](#)
- [Procedure 10, "Removing a license file"](#)
- [Procedure 11, "Releasing a Reach or Add-On license"](#)

Procedure 1 Retrieving license information

Use this procedure to retrieve information about licenses installed on Waveserver 5.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Overview

For a description of the parameters returned by the operations in this procedure, see the tables at the end of this procedure:

- [License feature parameter descriptions](#)
- [Pluggable transceiver \(XCVR\) license parameter descriptions](#)
- [License client state parameter descriptions](#)

Steps

- | | | |
|----------|--|--|
| 1 | <p>If you are retrieving</p> <hr/> <p>status information for a Waveserver 5 license</p> <p>an inventory of licenses installed on Waveserver 5</p> <p>license information for a pluggable transceiver</p> <p>state information for the Waveserver 5 license client</p> | <p>Then go to</p> <hr/> <p>step 2</p> <p>step 3</p> <p>step 4</p> <p>step 5</p> |
|----------|--|--|
- 2** To retrieve status information of a Waveserver 5 license:

```
license client show license <license name | all>
```

This command displays the license feature information for all licenses or for the specified license. For a description of the license parameters returned, see [License feature parameter descriptions](#).
 - 3** To retrieve inventory information for software licenses stored on Waveserver 5:

```
license client show inventory [all] [served] [expired] [checked-out] [available] [preauth]
```

This command displays a table listing all licenses or all licenses that match the specified criteria.
 - 4** To retrieve license information for pluggable transceivers:

```
xcvr show xcvr <slot>/<xcvr>
```

This command displays three tables. The XCVR license information is located in the Licensed Features section of the XCVR Information table, at the top of the output. For a description of the license parameters returned, see [Pluggable transceiver \(XCVR\) license parameter descriptions](#).

5 To retrieve state information for the Waveserver 5 license client:

```
license client show
```

This command displays the license client state information for the device. For a description of the license client state parameters returned, see [License client state parameter descriptions](#).

Table 68 License feature parameter descriptions

Parameter	Description	Possible values
Name	The Waveserver 5 license name.	Alphanumeric
Description	The descriptive name for the Waveserver 5 license.	Alphanumeric
Version	The license version number.	Numeric
Status	The current status of the license, as follows: • Valid — Indicates that the license does not have any problems associated with it and can be used. • Invalid — The license is invalid and cannot be used. This typically indicates that there is a registration-id mismatch between the license and the Waveserver 5 chassis. • Expired — The license is expired. Served licenses expire when communication with an external license server has been lost for 7 days or more.	• Valid • Invalid • Expired
Issuer Name	The issuer of the license.	• Ciena
Issued Date	The date the license was issued.	Alphanumeric
Start Date	The date at which the license becomes valid.	Alphanumeric
Type	The license category. License types include Pre-Auth and Served. • Pre-Auth — Indicates that the license has a Pre-Authorized enforcement mode. See License enforcement mode. • Served — Indicates that the license is permanent and served from a local file or from an external license server.	• PreAuth • Served
Host ID	The host-id of the Waveserver 5.	Alphanumeric
Count	The total number of licenses available on the system.	Numeric
Checked out Count	The number of licenses currently in use.	Numeric
Expiry Date	The date the license expires.	Alphanumeric
Time Remaining	The time remaining before the license expires.	Numeric

Parameter	Description	Possible values
Notice	Indicates that the license is for Waveserver 5.	• Waveserver OS License • WaveLogic 5e License • WaveLogic 5n License • Waveserver Ai License
Source	Indicates the location from which Waveserver 5 is acquiring the license. Possible values are as follows: • PreAuthorized — sourced from PreAuth licenses • Local — sourced from a local file • Server — sourced from an external server	• PreAuthorized • Local • Server

Table 69 Pluggable transceiver (XCVR) license parameter descriptions

Parameter	Description	Possible values
In-Use	Indicates whether the pluggable transceiver license is in use or not. • Yes — Indicates that the license is installed and acquired from the source. • No — Indicates that the license is not installed, or the license is installed but not acquired from the source.	• Yes • No
Availability	The license availability status: • Missing — Indicates that the license is not installed. • Available — Indicates that the license is installed, but not acquired from the source. • Held — Indicates that the license is installed and acquired from the source.	• Missing • Available • Held
Type	The license category. License types include Pre-Auth and Served. • Pre-Auth — Indicates that the license has a Pre-Authorized enforcement mode. See License enforcement mode. • Served — Indicates that the license is permanent and served from a local file or from an external license server.	• Pre-Auth • Served

Parameter	Description	Possible values
Status	The current status of the license, as follows: • Valid — Indicates that the license does not have any problems associated with it and can be used. • Invalid — The license is invalid and cannot be used. This typically indicates that there is a registration-id mismatch between the license and the Waveserver 5 chassis. • Expired — The license is expired. Served licenses expire when communication with an external license server has been lost for 7 days or more.	• Valid • Invalid • Expired

Table 70 License client state parameter descriptions

Parameter	Description	Possible values
Admin State	Administrative state of the license service.	• Enabled • Disabled
Oper State	Operational state of the license service.	• Down • Up • Commissioning
Compliance State	Whether the license client is compliant.	• Compliant • Not-compliant
Registration ID	The license client registration ID.	Alphanumeric
Number of License Servers	The number of license servers.	Numeric

Procedure 2 Setting the Registration ID on Waveserver 5

Use this procedure to set the Registration ID on a Waveserver 5.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

You require a Registration ID to generate a Registration Request File. You use the Registration Request File to obtain a Waveserver 5 software license.

The Registration ID is a string that you create. The Registration ID must conform to these rules:

- can be a maximum of 64 characters
- can contain these characters: a-z, A-Z, 0-9, "-", "_"
- cannot contain spaces

Steps

- 1 To set the Registration ID on a Waveserver 5, use the command:

```
license client set registration-id <Registration ID>
```

For example:

```
license client set registration-id COMPANY-X_001
```

Note: To restore the license client Registration ID to the default setting, use the command:

```
license client unset registration-id
```

You have completed this procedure.

Procedure 3 Retrieving or looking up the Registration ID

Use this procedure to retrieve the Registration ID for a Waveserver 5 using the CLI, or look up the Registration ID on the Ciena portal.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Overview

If you are replacing a Waveserver 5 chassis, you can re-use the Registration ID and then re-activate the existing software license from the chassis you are replacing.

Note: You can also obtain the Waveserver 5 Registration ID from the license filename, if it is available. For more information, see [License filename](#).

Steps

1	If you are	Then go to
	able to log in to the Waveserver 5	step 2
	unable to log in to the Waveserver 5 but you have access to the Ciena portal and you know the Waveserver 5 Device Name	step 3

Retrieving the Registration ID from Waveserver 5

- 2 To retrieve the Waveserver 5 Registration ID:

```
license client show
```

Looking up the Registration ID on the Ciena portal

- 3 Log in to the Ciena portal: (<https://my.ciena.com>).
- 4 Select the Software tab in the Customer Service Center window.
- 5 Click **Software Licenses**.
- 6 Select the **Device Management** tab, and locate the Waveserver 5 Device Name.
- 7 Select **View** from the Function drop-down list and click **Go**.

Figure 34 Device Management tab — Waveserver 5 device name

Device Management

License Management

Name

Find

Clear

Add Device

NAME	NOTES	PRODUCTS INSTALLED	TYPE	FUNCTION	Export
Ciena-MTL-Lab	Waveserver	View Details	CLIENT	View	Go

The Registration ID is displayed as Device ID.

Figure 35 Waveserver 5 Device ID display

NAME	CREATED ON	NOTES
Ciena-MTL-Lab	31 Jan 2017	Waveserver
DEVICE ID	LAST MODIFIED ON	
Ciena-MTL-Lab	31 Jan 2017	
DEVICE TYPE	LAST MODIFIED BY	
CLIENT	ciena24@ciena.com	

Download License

Product

Find

Clear

Add Query

Procedure 4 Obtaining software licenses: external license servers

Use this procedure to obtain Waveserver 5 software licenses and to install them on an external license server.

Requirements

- You installed, powered-up, and commissioned the Waveserver 5.
- You installed an external license server to make licenses available to Waveserver 5 nodes in your network. See [External License Server](#).
- You received an email from Ciena, which contains the license activation codes, and links to the Ciena portal and reference materials for license activation. For an example of the Software License Availability email, see [Ordering and license generation](#).

Overview

When you initially power up Waveserver 5, it starts in commissioning mode and it raises the Entitlement Enforcement Disabled alarm. Commissioning mode lets you install and commission the Waveserver 5 as is, then apply the required licenses within 90 days. While in commissioning mode, upgrades and downgrades are not blocked, so you can install the correct software release without a license. Once you install a license, the system exits commissioning mode, the Entitlement Enforcement Disabled alarm clears, and the normal licensing enforcement rules and procedures take effect. After 90 days, if no license is applied, the system automatically exits commissioning mode, the Entitlement Enforcement Disabled alarm clears, and the normal licensing enforcement rules and procedures take effect.

Note: During multi-step upgrades, a software license missing alarm is raised after each intermediate upgrade is completed. To clear the alarm and to be able to perform the next upgrade, you must install and activate the missing software license.

In specific cases, you may not be required to activate a license:

- Return merchandise authorization (RMA) Waveserver 5 — RMA
Waveserver 5 chassis ship with the manufacturing baseline software load. As a result, you may need to upgrade the software load. You must have a software license to change the software load and apply the software license before you upgrade; or if you do not have the software license to change the software load, you must follow [Procedure 106, “Performing a one-step upgrade when the software release on the Waveserver 5 does not match the software license”](#).

Steps

Generating the Registration Request File

- 1 Generate the Registration Request File on the workstation that hosts the external license server software. On the workstation, do the following:
 - Change to the License Server Home directory.

```
cd /opt/ciena/flexnetls
```

 - Generate the Registration Request File.

```
./getRegistrationID
```

The license server software generates a Registration Request File. The format of the file name is:

```
<host_name>_actid.bin
```

where <host_name> is the name of the workstation that hosts the license server software.
- 2 Transfer the generated Registration Request File from the external license server host to your local PC.

Registering the license server host on the License Portal

- 3 Register the license server host on the License Portal.
 - Log into the portal (<https://my.ciena.com>) from the PC where you transferred the Registration Request File.
 - Select the Software tab in the Customer Service Center window.
 - Click **Software Licenses**.
 - Click **Add Device** in the Device Management tab.

Figure 36 Device Management tab — Add device

Device Management | **License Management**

Name

NAME	NOTES	PRODUCTS INSTALLED	TYPE	FUNCTION
You currently have no devices registered. You must register a device in order to apply licenses. How to register a device.				

[Export](#)

- 4 Provide information about the license server host in the **Add Device** window.

Figure 37 Add Device window

NAME

NOTES

Notes to help with search and/or maintenance of the device.

REGISTRATION REQUEST FILE

RFF.bin

- Name — the license server host name.
- Notes — optional information about the license server host.
- Registration Request File — click **Choose File**. Navigate to the file you generated in [step 1](#) to pair the license server host to the License Portal.
- Click **Save**.

The Device Management tab displays the license server host.

Figure 38 Device Management tab — license server host

Device Management **License Management**

Name

[Export](#)

NAME ↑	NOTES	PRODUCTS INSTALLED	TYPE	FUNCTION
Ciena-MTL-Lab	Waveserver	View Details	CLIENT	-- Select -- Go

Downloading licenses from the License Portal

- 5 Select the License Management tab.
 - Type the license activation code (from the Software License Availability Notification email) into the Activation Code field.
 - Click **Find**.

Figure 39 License Management tab — Find function.

Device Management | License Management

ACTIVATION CODE(S)

E111111-E111111 Find Clear [I don't know my Activation Code.](#)

Activate License for Selected

Results for: E111111-E111111

0 selected Export

☐ SELECT ALL	PRODUCT	PART #	PO #	AVAILABLE	TOTAL	USE	FUNCTION
☐	S86-LIC-WOSAPIS	S86-LIC-WOSAPIS	CIE-0000000	7	7	0	Activate Go
☐	S86-LIC-WOSFRGNCLPLUG	S86-LIC-WOSFRGNCLPLUG	CIE-0000000	9	9	0	Activate Go

The License Management tab displays licenses associated with the activation code.

- 6 Download licenses from the License Portal to your PC.
 - Click the checkbox next to the licenses to download.
 - Specify the number of licenses to activate in the Use column.
 - Click **Activate License for Selected**.

- 7

If you	Then go to
did not previously select a device	step 8
previously selected a device	step 9

- 8 Select a device in the Add Selected Device window.

Figure 40 Add Selected Device window

Browse Downloads | Software Alerts | Software Licenses | Services, Lifecycle & Support | Key Codes | Help

Licenses for Ciena Technical Support

Select a Device to Apply License.

Name Find Clear

Add License to Selected Device

NAME	NOTES	PRODUCTS INSTALLED	TYPE
☐ CienaLab	Ciena	BluePlanet_MCP_Base	SERVER
☐ Test		View Details	SERVER

A confirmation window displays the licenses you selected. Go to [step 10](#).

- 9 When the selected device appears in the Device management window, select **Add License** in the Function field list box and click **Go**.

Figure 41 Device Management window — Add License

Device Management **License Management**

Name **Find** **Clear**

Add Device

NAME ↑	NOTES	PRODUCTS INSTALLED	TYPE	FUNCTION
MTL-LAB-1		None	CLIENT	Add License Go

[Export](#)

10 Verify your license selections and click **Commit**.

Figure 42 Commit license function

You have selected the following 2 license(s) for **Ciena**

PRODUCT(S)	PART	TOTAL
S86-LIC-WOSAPIS	S86-LIC-WOSAPIS	1
S86-LIC-WOSFRGNCLPLUG	S86-LIC-WOSFRGNCLPLUG	1

Cancel **Commit**

The license file downloads to your local PC.

The license file name is in the format:

ciena-<host_name>-<timestamp>.lic

where:

- <host_name>— is the ID of the external license server host.
- <timestamp>— is the date and time the portal generated the license file.

Transferring license files to the external license server host

- 11** Transfer the license from the PC to the license server host.
- Log in to the license server host, and change directory to the license server home directory:
- ```
cd /opt/ciena/flexnetls
```
- Transfer the license file from your PC (where you downloaded it from the License Portal) to the license server home directory on the license server host.

- 12** Activate the license file.

On the license server host type:

```
./loadLicense ciena-<host_name>-<timestamp>.lic
```

where:

- `<host_name>`— is the ID of the external license server host.
- `<timestamp>`— is the date and time the portal generated the license file.

- 13** Confirm the licenses were installed on the license server host:

```
./queryLicenses
```

The server responds with a list of installed licenses.

```
} -bash-4.2$ sudo ./queryLicenses
```

| Features              | Expiry    | Purchased | Used |
|-----------------------|-----------|-----------|------|
| S86-LIC-WL5ETER250KM  | permanent | 5000      | 0    |
| S86-LIC-WL5ETER2000KM | permanent | 100       | 0    |

- 14**
- | If you are                                | Then                                                                      |
|-------------------------------------------|---------------------------------------------------------------------------|
| obtaining multiple software licenses      | repeat <a href="#">step 5</a> to <a href="#">step 13</a> for each license |
| have finished obtaining software licenses | go to <a href="#">step 15</a>                                             |
- 15** Perform [Procedure 7](#), “Provisioning access to an external license server”.

---

## Procedure 5 Obtaining software licenses: manual installation

---

Use this procedure to obtain software licenses to install licenses on Waveserver 5 manually.

### Requirements

- You have installed, powered-up, and commissioned the Waveserver 5.
- You received an email from Ciena containing the license activation codes. The email includes a link to the Ciena portal and a link to download reference materials for license activation. For an example of the Software License Availability email, see [Ordering and license generation](#).

### Overview

When you initially power up Waveserver 5, it starts in commissioning mode and it raises the Entitlement Enforcement Disabled alarm. Commissioning mode lets you install and commission Waveserver 5 as is, then apply the required licenses within 90 days. While in commissioning mode, upgrades and downgrades are not blocked, so you can install the correct software release without a license. Once you install a license, the system exits commissioning mode, the Entitlement Enforcement Disabled alarm clears, and the normal licensing enforcement rules and procedures take effect. After 90 days, if no license is applied, the system automatically exits commissioning mode, the Entitlement Enforcement Disabled alarm clears, and the normal licensing enforcement rules and procedures take effect.

**Note:** During multi-step upgrades, a software license missing alarm is raised after each intermediate upgrade is completed. To clear the alarm and to be able to perform the next upgrade, you must install and activate the missing software license.

In specific cases, you may not be required to activate a license:

- Return merchandise authorization (RMA) Waveserver 5 — RMA Waveserver 5 chassis ship with the manufacturing baseline software load. As a result, you may need to upgrade the software load. You must have a software license to change the software load and apply the software license before you upgrade; or if you do not have the software license to change the software load, you must follow [Procedure 106, “Performing a one-step upgrade when the software release on the Waveserver 5 does not match the software license”](#).

### Steps

- 1 If you have not done so already, perform [Procedure 2, “Setting the Registration ID on Waveserver 5”](#).



**Generating a Registration Request File**

- 2 Generate the Registration Request File.

Type the following command:

```
license file generate request
```

The system responds with the location and name of the Registration Request File. For example:

```
Generated the registration request file: /license/
request/MTL-LAB-5_registration.bin
```

The Registration Request File filename contains the Registration ID, in the format: <registration\_id>-registration.bin.

- 3 Make note of the system response, or retrieve the path and name of the registration request using this CLI command:

```
license file list requests
```

- 4 Transfer the generated request file from the Waveserver 5 to your local PC or server. Type the CLI command using the appropriate protocol (xftp, sftp, or scp), for example:

```
system xftp putfile ... <path/filename>
```

For example:

```
system xftp putfile sftp-server <ip address> login-id
<login id> password <password> local-filename /license/
request/MTL-LAB-5_registration.bin
```

**Registering Waveserver 5 on the License Portal**

- 5 Register the Waveserver 5 on the License Portal.
  - Log into the portal (<https://my.ciena.com>).
  - Select the Software tab in the Customer Service Center window.
  - Click **Software Licenses**.
  - Click **Add Device** in the Device Management tab.

**Figure 43** Device Management tab — Add Device

- 6 Provide details about the Waveserver 5 in the **Add Device** window.

**Figure 44** Add Device window

**NAME**

Ciena-MTL-Lab

**NOTES**

Waveserver

**REGISTRATION REQUEST FILE**

Choose File Ciena-MTL-La...tration.bin

Cancel Save

- Name — type in the Waveserver 5 Registration ID.
- Notes — optionally, add descriptive information about the Waveserver 5.
- Registration Request File — click **Choose File**. Select the file you downloaded in [step 2](#) to pair the device to the portal for license activation.
- Click **Save**.

The Device Management tab displays the registered Waveserver 5.

**Figure 45** Device Management tab — registered Waveserver 5

Device Management License Management

Name  Find Clear

Add Device

Export

| NAME ↑        | NOTES      | PRODUCTS INSTALLED           | TYPE   | FUNCTION        |
|---------------|------------|------------------------------|--------|-----------------|
| Ciena-MTL-Lab | Waveserver | <a href="#">View Details</a> | CLIENT | -- Select -- Go |

- 7 Add licenses for the registered Waveserver 5.
  - Select Add License in the Function field list box.
  - Click **Go**.

The Activate License window (License Management tab) displays.
- 8 Select the License Management tab.
  - Type the activation code (the Software License Availability Notification email) into the Activation Code search box.
  - Click **Find**.

The License Management tab displays the available licenses.

***Downloading a license from the License Portal***

- 9 Apply a license to the registered Waveserver 5.
  - Select the licenses in the Select column.
  - Specify the quantity of licenses to apply in the Use column.
  - Select **Activate** in the Function field list box.
  - Click **Go**.

**Figure 46** License Management tab

The screenshot shows the 'License Management' tab. At the top, there's a search bar for 'ACTIVATION CODE(S)' with the value 'E111111-E111111' and buttons for 'Find' and 'Clear'. A link 'I don't know my Activation Code.' is also present. Below the search bar is a button 'Activate License for Selected'. The results section shows 'Results for: E111111-E111111' and '0 selected'. A table lists available licenses with columns: SELECT ALL, PRODUCT, PART #, PO #, AVAILABLE, TOTAL, USE, and FUNCTION. Two licenses are listed: 'S86-LIC-WOSAPIS' and 'S86-LIC-WOSFRGNCLPLUG'. Each row has a checkbox, a 'Go' button, and a dropdown menu set to 'Activate'.

| SELECT ALL               | PRODUCT               | PART #                | PO #        | AVAILABLE | TOTAL | USE | FUNCTION |
|--------------------------|-----------------------|-----------------------|-------------|-----------|-------|-----|----------|
| <input type="checkbox"/> | S86-LIC-WOSAPIS       | S86-LIC-WOSAPIS       | CIE-0000000 | 7         | 7     | 0   | Activate |
| <input type="checkbox"/> | S86-LIC-WOSFRGNCLPLUG | S86-LIC-WOSFRGNCLPLUG | CIE-0000000 | 9         | 9     | 0   | Activate |

- 10 A confirmation window displays the licenses you selected.
 

Verify your license selections.

Click **Commit** to obtain the license for the registered Waveserver 5.

**Figure 47** License selection confirmation

You have selected the following 2 license(s) for **Ciena**

| PRODUCT(S)            | PART                  | TOTAL |
|-----------------------|-----------------------|-------|
| S86-LIC-WOSAPIS       | S86-LIC-WOSAPIS       | 1     |
| S86-LIC-WOSFRGNCLPLUG | S86-LIC-WOSFRGNCLPLUG | 1     |

- The license file downloads to your local PC.
- |           |                                           |                                                                           |
|-----------|-------------------------------------------|---------------------------------------------------------------------------|
| <b>11</b> | <b>If</b> you are                         | <b>Then</b>                                                               |
|           | obtaining multiple software licenses      | repeat <a href="#">step 8</a> to <a href="#">step 10</a> for each license |
|           | have finished obtaining software licenses | go to <a href="#">step 12</a>                                             |
- 12** You have completed this procedure. Perform [Procedure 6, “Installing and activating a software license”](#) to move the downloaded license to a server or device in your network.

## Procedure 6 Installing and activating a software license

Use this procedure to download and install a purchased (permanent) software license.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You obtained the latest software license file and it is available on a remote server (scp or sftp) or terminal. To obtain a software license, see [Obtaining software licenses: manual installation](#). To access a default remote server, the server must be configured on Waveserver 5, see [Managing and configuring default servers](#).
- Verify the currently installed software licenses using the command:  
`license client show inventory served`
- You have set the Registration ID on the Waveserver 5, see [Procedure 2, “Setting the Registration ID on Waveserver 5”](#).

**Note:** During multi-step upgrades, a software license missing alarm is raised after each intermediate upgrade is completed. To clear the alarm and to be able to perform the next upgrade, you must install and activate the missing software license.

### Steps

- 1 Download the software file from the License portal:  
**Note:** Download the file that includes the same registration ID that you set in [Procedure 2, “Setting the Registration ID on Waveserver 5”](#).  

```
license file download [default-server] [default-sftp-server] [default-scp-server] [sftp-server <IP address or host name>] [scp-server <IP address or host name>] [login-id <String>] [password <String>] [url <string> login-id <String[1..32]> password <Password String[1..128]> secret <String[1..256]>] [filename <full directory path and filename>]
```

**Note 1:** For SFTP and SCP license servers, ensure the key exchange SSH algorithms match on the Waveserver 5 and the license server.  
**Note 2:** You must specify [login-id] and [password] for an SCP or SFTP server.  
 For example, `license file download sftp-server 10.111.112.113 login-id admin password admin filename Directory/Path/ciena-Waveserver-COMPANY-X_001-20162411-1150.lic`  
 or

```
license file download scp-server 10.111.112.113 login-id
admin password admin filename Directory/Path/Ciena-
Waveserver-M1234567-20162110-1027.lic
```

**Note 3:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

```
For example, license file download url sftp://
[2222:111:dddd:5444:aced::293]/waveserver/license/ciena-
license.lic login-id waveserver password *****
```

The message `license file downloaded` appears when the download completes.

- 2 Optionally, confirm that the downloaded software license file is available on Waveserver 5:

```
license file list downloaded
```

**Note:** Ensure that the license is stored in and activated from the `/var/opt/license/download` location. Do not activate the license file from the `/tmp/public` location.

- 3 Activate the software license:

```
license client activate <filename>
```

For example,

```
license client activate filename ciena-COMPANY-X_001-
201624111150.lic
```

- 4 Optionally, confirm that the activated license is available on Waveserver 5:

```
license client show inventory all
```

- 5 Optionally, confirm that the license was successfully activated:

```
license client show inventory served
```

| 6 | If you are                                                 | Then                                        |
|---|------------------------------------------------------------|---------------------------------------------|
|   | installing and activating multiple software licenses       | repeat the above procedure for each license |
|   | not installing and activating additional software licenses | the procedure is complete                   |

## Procedure 7 Provisioning access to an external license server

---

Use this procedure to enable the Waveserver 5 access to an external license server.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You must have information about the external license server, including the hostname and IP address.

### Steps

- 1 Enable the Waveserver 5 to access the external license server.  

```
license client server set hostname <ip_addr> port <port>
protocol <http | https> sync-interval <15..1440>
```

Where

  - `<ip_addr>` is the IP address of the external license server.
  - `<port>` is the external license server port. Values: 7071 (for https) or 7072 (for http). Default: 7071.
  - `protocol` sets the protocol Waveserver uses to communicate with the license server. Values: http or https. Default value: https.
  - `sync-interval` is the time interval (in minutes) between each sync of licenses on the Waveserver with the licenses on the license server. Values: 15 min to 1440 min. Default: 15 min.
- 2 Verify that Waveserver 5 has access to the license server. Type the command:  

```
license client server show
```

You have completed this procedure.

## Procedure 8 Provisioning the license server sync-interval

Use this procedure to configure the license server sync-interval value.

### Requirements

- This procedure assumes you have already configured access to the external license server. If you have not, see [Procedure 7, “Provisioning access to an external license server”](#).
- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You must have information about the external license server, including the hostname and IP address.

### Overview

A Waveserver node that is configured to acquire licenses from a license server, also regularly synchronizes its license holdings with the license inventory on the license server. The interval for synchronizing licenses is controlled by the `sync-interval` parameter.

### Steps

- 1 Display the current license-server access settings.  

```
license client server show
```

The system displays the license-server access settings.

**Figure 48** License server access settings

```
WS5-0126# license client server show
```

| LICENSE SERVER SUMMARY      |       |
|-----------------------------|-------|
| Parameter                   | Value |
| Number of License Server(s) | 0     |
| Hostname                    |       |
| Port                        | 0     |
| Protocol                    | HTTPS |
| Sync Interval (minutes)     | 15    |

### Setting and unsetting the sync-interval

- 2 To set the license server sync-interval, run the following command:  

```
license client server set sync-interval <15..1440>
```

Where the value of sync-interval is in the range from 15 minutes (the default) to 1440 minutes (24 hours).



- 3 To return the value of sync-interval to the default value (15 minutes), run the following command:

```
license client server unset sync-interval
```

***Forcing the license server synchronization***

- 4 You can force a license-server synchronization at any time by running the following command:

```
license client server sync-now
```

You have completed this procedure.

---

## Procedure 9 Listing licenses in arrears

---

Use this procedure to list licenses in arrears.

### Requirements

- This procedure assumes you have already configured access to the external license server. If you have not, see [Procedure 7, “Provisioning access to an external license server”](#).
- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You must have information about the external license server, including the hostname and IP address.

### Overview

The ability to list licenses in arrears can assist you in identifying missing licenses that are causing the system to raise License Missing alarms. The ability to list licenses in arrears also lets you immediately identify missing licenses and clarify how many entitlements are required for the current provisioning.

### Steps

- 1 To list the licenses in arrears, run the following command:

```
license client show arrears
```

The system displays a table listing all the licenses in arrears. The following table describes the information provided for each license in arrears.

**Table 71** Licenses in arrears parameter descriptions

| Parameter       | Description                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature         | The name of the license in arrears.                                                                                                                              |
| Description     | A description of the license.                                                                                                                                    |
| Installed-count | The total number of entitlements installed for this license by a license file or license server. This is the total number allocated to this specific Waveserver. |
| Check-out-count | The number of entitlements that were provisioned for this license. If this number is larger than the installed count, then this license is in arrears.           |
| Arrear-count    | The number of entitlements that are in arrears for this license.                                                                                                 |

## Procedure 10 Removing a license file

Use this procedure to remove a license file from the Waveserver 5 license client.

### Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

### Steps

- 1 To display a list of the license file names:

```
license file list
```

For example:

```
WS5_0212*# license file list
Processed:
/var/opt/license/processed/ciena-WS5_2_0_0-202002110916.lic 11073 Wed Mar 4 15:18:43 2020

Downloads:
ciena-WS5_partial-202002191456.lic 6756 Mon Mar 9 14:59:07 2020

Requests:
/var/opt/license/request/WS5_2_0_0_registration.bin 397 Mon Mar 9 14:59:13 2020
```

- 2 Note the filenames of the license files you want to remove.
- 3 To delete a license file from the Waveserver 5 license client, use the command:  

```
license file delete file <filename>
```

Where <filename> is the name of the license file you want to remove.

For example:

```
WS5_0212*# license file delete filename /var/opt/license/processed/ciena-WS5_2_0_0-202002110916.lic
Deleted license file.
WS5_0212*# license file delete ciena-WS5_partial-202002191456.lic
Deleted license file.
WS5_0212*# license file delete filename /var/opt/license/request/WS5_2_0_0_registration.bin
Deleted license file.
```

## Procedure 11 Releasing a Reach or Add-On license

Use this procedure to release a license so that it can be reused.

### Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

You can release a Reach license from a line port, or a capacity Add-On license from a client port.

See [Ordering](#) for a list of supported Reach and Add-On licenses.

### Steps

| 1 | If you are releasing                  | Then go to             |
|---|---------------------------------------|------------------------|
|   | a Reach license from a line port      | <a href="#">step 2</a> |
|   | an Add-On license from client port(s) | <a href="#">step 5</a> |
|   | an ITS Add-On license                 | <a href="#">step 8</a> |

#### Releasing a reach license

- 2 Display the number of available reach licenses:  

```
license client show license <license>
```

**Note:** Count identifies the number of licenses available on the system. Checked out Count identifies the number of licenses in use.
- 3 Disable and delete the module where the license is used:  

```
module disable slot <slotNumber>
```

```
module delete slot <slotNumber>
```
- 4 Verify that the number of checked out reach licenses on Waveserver 5 has decreased:  

```
license client show license <license>
```

**Note:** Typically, deleting a module releases two reach licenses, as there are two line ports per module.

*You have completed this procedure.*

#### Releasing a capacity add on license

- 5 Display the number of available capacity add-on licenses:  

```
license client show license <license>
```

**Note:** The parameter Count identifies the number of licenses available on the system. Checked out Count identifies the number of licenses in use.

- 6 Disable and delete the XCVR that is assigned the capacity add-on license you want to release:

```
xcvr disable xcvr <slot>/<xcvr>
```

```
xcvr delete xcvr <slot>/<xcvr>
```

- 7 Verify that the number of checked-out add-on licenses has decreased:

```
license client show license <license>
```

**Note:** The number of licenses released depends on the number of capacity licenses allocated to the XCVR. For a 100G XCVR, one 100G capacity license is released. For a 400G XCVR, four 100G capacity licenses are released.

*You have completed this procedure.*

#### **Releasing an ITS Add-On license**

- 8 Display the number of available ITS Add-On licenses:

```
license client show license <license>
```

**Note:** The parameter Count identifies the number of licenses available on the system. Checked out Count identifies the number of licenses in use.

- 9 Disable and delete the module where the license is used:

```
module disable slot <slotNumber>
```

```
module delete slot <slotNumber>
```

- 10 Verify that the number of checked out ITS licenses on Waveserver 5 has decreased:

```
license client show license <license>
```

*You have completed this procedure.*

---

## CHAPTER 13

# Comms

---

This section contains the following topics:

- [Overview](#)
  - [IPv4 and IPv6 support](#)
  - [DHCP and IP addressing](#)
  - [Domain Name System \(DNS\)](#)
  - [Console port and USB-C port](#)
  - [DCN ports](#)
  - [Direct DCN connection](#)
  - [DCN redundancy and Virtual Router Redundancy Protocol \(VRRP\)](#)
  - [Border Gateway Protocol \(BGP\) peering](#)
  - [ILAN ports](#)
  - [ILAN linear configuration](#)
  - [ILAN ring configuration and loop mitigation](#)
  - [Remote Group Management](#)
  - [Neighbor Discovery Protocol \(NDP\)](#)
  - [Link Layer Discovery Protocol](#)
    - [LLDP on DCN port](#)
    - [LLDP on client port](#)
  - [Network Time Protocol \(NTP\)](#)
  - [xFTP, SCP, and HTTPS servers](#)
  - [Protocols and associated open ports on the Waveserver 5](#)
- [Procedure list](#)

### Overview

This section provides information on Waveserver 5 comms.

### IPv4 and IPv6 support

Waveserver 5 supports both IPv4 and IPv6 protocols, however, only DHCPv4 is supported for DHCP.

### DHCP and IP addressing

By default, a DHCPv4 client is enabled on Waveserver 5. If DHCPv4 servers are configured on the management network, the DHCPv4 client automatically communicates with a server and is assigned an IPv4 IP address and gateway, which is then configured on the local management interface. This will overwrite a manually provisioned (static) IPv4 IP address and gateway. Disable the DHCPv4 client to prevent overwriting.

**Note:** When you use SPLI, Ciena recommends you to manually assign a static IPv4 address to the Waveserver 5, instead allowing DHCP to assign the IPv4 address. This prevents the need to re-provision the 6500 SPLI table to reestablish SPLI communication with Waveserver 5 if the DHCP assigns a different IPv4 address to the Waveserver 5.

### Automatic Private IP Addressing (APIPA)

Automatic Private IP Addressing (APIPA) is a client-side, network process that autoconfigures IPv4 (link-local) addresses. A link-local address is valid only for communications within the network segment (link) or the broadcast domain to which the host is connected.

APIPA allows a device to choose an IPv4 address from the pool of 65,534 Class B network IPv4 addresses in the address range 169.254.0.0/16. The Waveserver 5 then verifies that the IP address is not being used by another device on the network.

While Waveserver 5 uses the randomly generated IP address from APIPA, it continues to send out DHCP discover broadcasts to locate a DHCP server on the network. When a DHCP server becomes available, the APIPA address is replaced with one from the DHCP server.

### Domain Name System (DNS)

Waveserver 5 supports the DNS client on the Waveserver 5. DNS settings can be either manually provisioned, or they can be pushed from the DHCP server to the DNS client on Waveserver 5 using DHCP Options 6 (Domain Server) and 15 (Domain Name) values.

### Console port and USB-C port

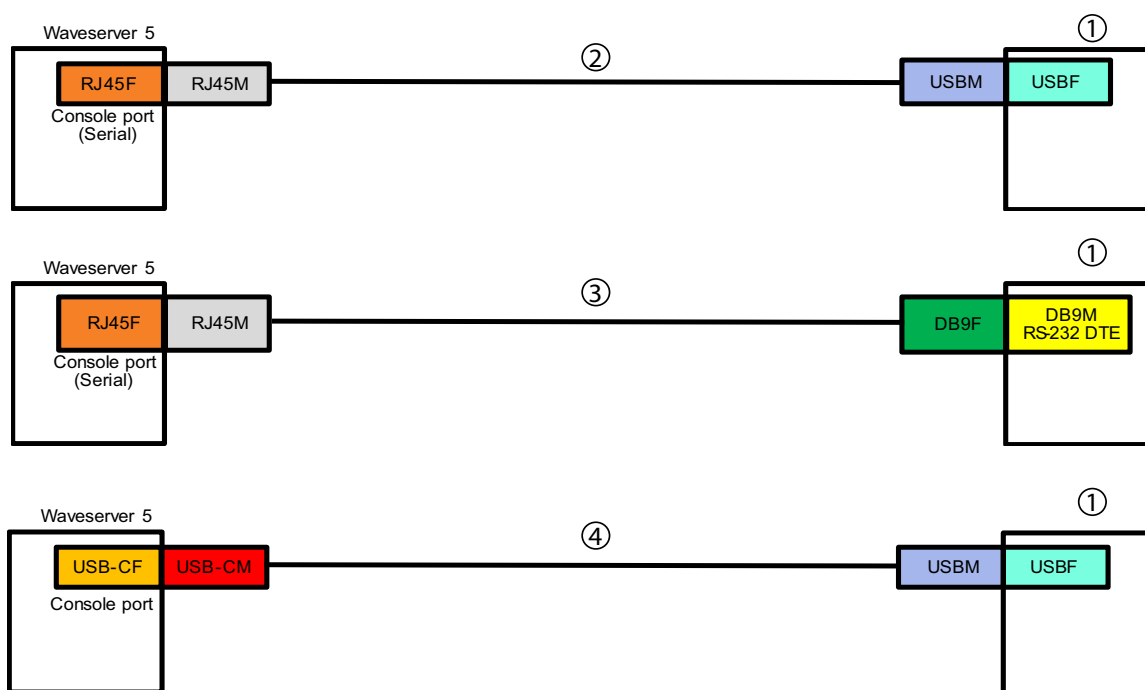
The console and USB-C ports on Waveserver 5 enable serial communications between Waveserver 5 and a terminal (PC, laptop) for management, set up, and troubleshooting purposes. These ports are found on the faceplate of the control processor. Waveserver 5 supports DB9-to-RJ45 cables for the console port, or USB-C-to-USB-C or USB-C-to-USB-A

cables for the USB-C port. You can use the console or USB-C ports to provision a static IPv4 or IPv6 address and gateway for the management network interface. For details, see [Procedure 12, “Setting up management access using the console or USB port”](#).

The default connection speed (baud) is 115200. Waveserver 5 has an auto-detect mechanism enabled by default. See [Procedure 13, “Changing the console port baud”](#).

As an example, when connecting a terminal server (RJ45) to the Waveserver 5 console port, a straight Ethernet cable can be used. Once the straight cable is connected, the console port detects the serial connection pin-out and adjusts the baud accordingly, as shown in the following figure:

**Figure 49** Using console port or USB-C port for management access during commissioning



#### Legend

- |   |                     |   |                     |
|---|---------------------|---|---------------------|
| 1 | Terminal (e.g. PC)  | 3 | RJ45 to DB9 adaptor |
| 2 | RJ45 to USB adaptor | 4 | USB - C to USB      |



The following security features are available on the Waveserver 5 when using the console port:

- The default super user account (su) initially has no password until you provision a password. Press any character string or the Enter key to access the system. Ciena recommends that you set a password for the default user account the first time you log in.
- There is a global session inactivity timeout while the user is logged in using the console port.
- The Linux root user can log in to the Waveserver 5 console port via the root user id and password.
- Any user, local or from remote authentication, can log in to the Waveserver 5 console port.
- All commands executed via the console port are logged to standard Waveserver 5 logs such as command logs, syslog, or security logs.

Waveserver 5 supports console protocols YOST DTE/DCE, Cyclades, and EIA/TIA-561. Waveserver 5 supports the following bauds: 9600, 19200, 38400, 57600, 115200, 230400, 460800, and 921600.

## DCN ports

Data Communications Network (DCN) ports provide access to the management network.

DCN-1 and DCN-3 ports are used to provide DCN access to the management network. DCN-1 is a RJ-45 (Ethernet) port located on the faceplate of the control processor and supports 10, 100 and 1000 Base-T. DCN-3 is a RJ-45 (Ethernet) port located on the faceplate of the access panel and supports 10, 100 and 1000 Base-T. Only one of these ports (DCN-1 or DCN-3) can be enabled at a time.

DCN-2 is an RJ-45 (Ethernet) port located on the faceplate of the control processor and supports 10, 100 and 1000 Base-T. DCN-2 supports wayside-channel mode only. For more information on SLTE comms, see [Comms channel for 6500-based SLTE configurations](#).

**Note 1:** DCN ports support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

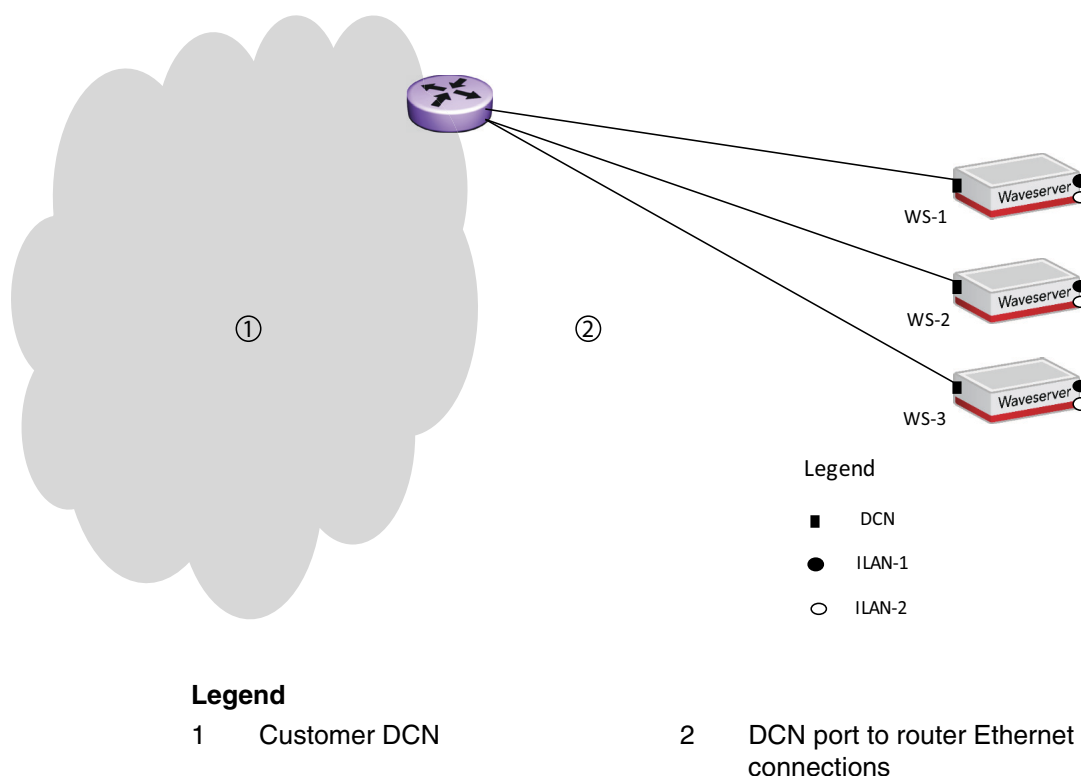
**Note 2:** DCN-5 port is not supported for this release.

**Note 3:** A disconnected DCN port has unreliable operational status and you should disable its administrative state using CLI command `chassis port disable port`. When the DCN port is administratively disabled, the default port speed of 1G will be shown.

### Direct DCN connection

In this configuration, each Waveserver 5 is connected directly by Ethernet cable from the DCN port on the Waveserver 5 to the DCN/router, as in the following figure. A failure in any one connection to the DCN/router affects one Waveserver 5 only. For this configuration, each Waveserver 5 is provisioned with its own “local host” management IP, subnet mask and default gateways.

**Figure 50** Direct connection to DCN



**Note:** It is recommended to enable Auto Negotiation (AN) on the network side when connecting the Waveserver 5 DCN port to a Router port to avoid CRC errors. If Auto-negotiation is not enabled, it may result in a mismatch Speed / Duplex leading to packet fragments or CRC errors.

### DCN redundancy and Virtual Router Redundancy Protocol (VRRP)

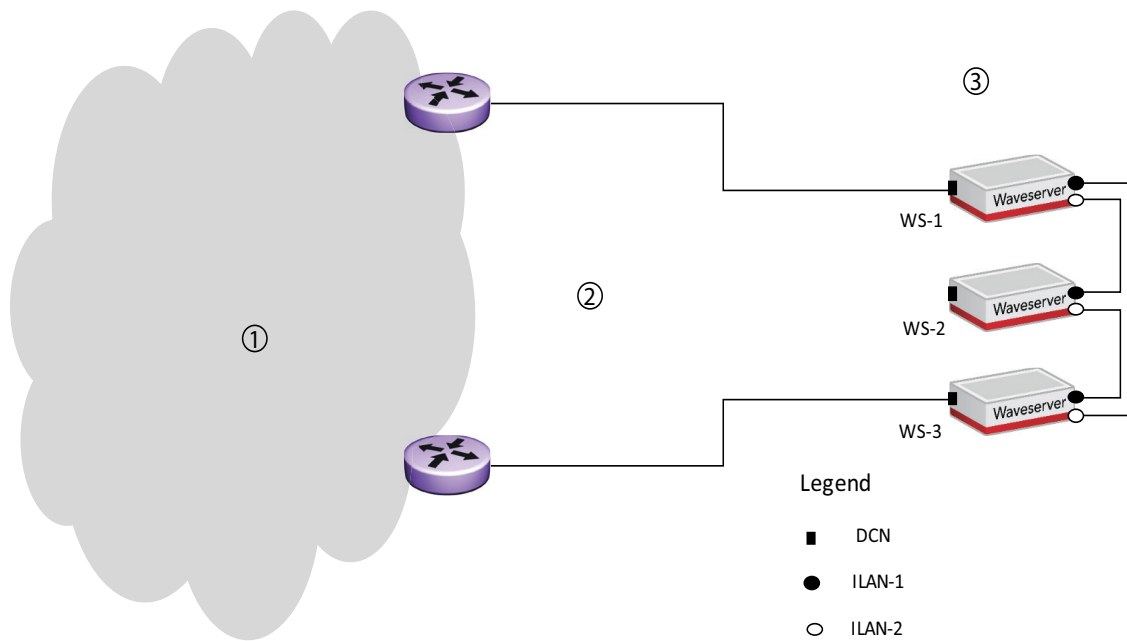
DCN redundancy on the Waveserver 5 can be achieved by using the Virtual Router Redundancy Protocol (VRRP). VRRP is a networking protocol designed to eliminate a single point of failure, in a static default routed environment. VRRP always runs on an external network and no provisioning is required on the Waveserver 5. VRRP runs an election protocol that dynamically assigns responsibility for a virtual router to one of the VRRP routers on the LAN. The VRRP router controlling the IP address associated

with the virtual router is the master and it forwards packets sent to this IP address. Should the master fail, the election process selects the backup router to take over.

Redundant routers (master and backup routers) running VRRP can be used to provide DCN connectivity protection against a cable disconnection, router port failure or router failure.

For DCN redundancy with a Waveserver 5 group, connectivity to the Waveserver 5 group must use the DCN ports on two different Waveserver 5 chassis, as shown in the following figure:

**Figure 51** DCN redundancy with Waveserver 5 group



**Note 1:** WS-2 achieves connectivity to the Redundant Routers (running VRRP) by bridging over the ILAN connections.

**Note 2:** When using VRRP, care must be taken externally in the sub-tending routed network with a Waveserver 5 group to avoid asymmetric TCP connections.

When the master router fails and the backup router takes over, all existing SSH sessions are lost. The user must re-establish SSH connections.

---

## Border Gateway Protocol (BGP) peering

The BGP peering feature allows you to have two DCN ports (DCN-1 and DCN-2 or DCN-3 and DCN-4) running BGP on the Waveserver 5. Each DCN port is connected to a router and the BGP protocol decides which router the packets should be sent to. If a failure occurs locally or remotely in the network, BGP protects against local and remote failures in the network.

BGP is an external gateway protocol to connect different networks. It is normally used between Internet Service Providers (ISPs) or between an Enterprise and ISP. BGP is built for reliability, scalability, and control.

BGP uses the concept of autonomous systems (AS). An AS is a group of networks under a common administrator. The Internet Assigned Numbers Authority (IANA) assigns the AS numbers.

Within the autonomous system, the Waveserver 5 nodes are meshed together using Interior Gateway Protocol (IGP) to provide maximum protection and scalability. The different autonomous systems can then be connected using Exterior Gateway Protocol (EGP).

Adjacent neighbors are called BGP peers and must be statically configured. BGP peers perform these functions:

- Exchange routing updates.
- Sends periodic keep alive messages to each other.
- Tells its peer autonomous system, which autonomous systems it can reach.
- Manages a table with all the routes to each network and sends the information to its neighbor autonomous system.
- Provides information to its own autonomous system.

BGP is a path-vector protocol. BGP is responsible for having a list of autonomous systems in the path to a specific network. With this information, BGP can find the best route to a specific network. For the BGP loop prevention mechanism, it makes use of the autonomous system numbers. When a routing update leaves an autonomous system, that autonomous system's number is prepended to the list of autonomous systems that have handled this update. When an autonomous system receives an update, it verifies its own autonomous system number in the list. If its own autonomous system is in the list, the update is discarded for loop prevention.

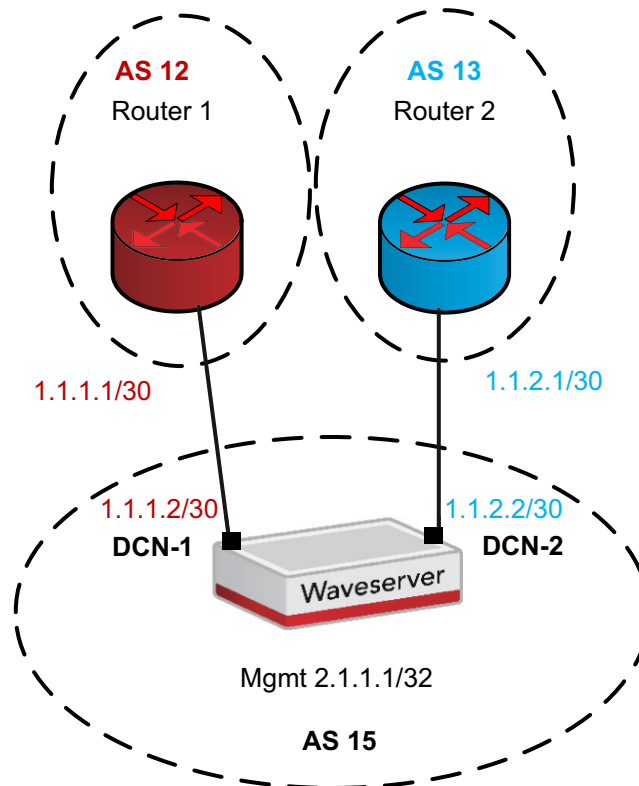
An autonomous system (AS) is a type of network architecture that includes a collection of routes, whose prefixes and routing policies are under common administrative control, for example an Internet Service Provider or a corporate

network. An autonomous system controls a collection of connected routing prefixes representing a range of IP addresses. The AS determines the routing policy inside the network.

Each BGP peer exchanges routing information with neighboring peers in the form of network prefix announcements. This means that an AS knows the network prefix of another AS without being directly connected. BGP analyzes the information and sets one of its peers as the next stop for forwarding packets to a specific destination. Each peer manages a table with all of the routes that it knows for each network and then propagates that information to its neighboring AS. The Waveserver 5 supports either IPv4 or IPv6 address and therefore, propagates either IPv4 or IPv6 routes to its neighboring AS. This allows an AS to collect all of the routing information from its neighboring autonomous systems and advertises the information further. The system being advertised may not be a BGP system, and may be running a different protocol, such as OSPF.

Different Waveserver 5 nodes must be in a different autonomous system from the routers as shown in the following figure:

**Figure 52** Waveserver 5 and routers in different autonomous systems



### Considerations

The following considerations apply for the BGP peering feature:

- DHCP should not be used. IP addresses must be assigned statically.
- ZTP should not be used, as it depends on DHCP.
- This feature is supported on DCN ports.
- The chassis mode for the required port running BGP peering must have its mode set to redundant-bgp using CLI command `chassis port set port <dcn-n>`.
- Serial port access must be available during the transition to BGP.
- BGP cannot be used in conjunction with the [Remote Group Management](#) feature.
- Waveserver 5 support either IPv4 or IPv6 type of IP address to be assigned to the peer.
- Both peers must have the same type of IP address, either IPv4 or IPv6.
- For local interface, the user IPv6 address is advertised to the peers when IPv6 address are used. Otherwise local IPv4 address is advertised.
- The external devices can access the Waveserver 5 using the local IPv4 or IPv6 address.

To configure BGP peering, see [Procedure 14, “Configuring BGP peering”](#).

### Provisionable BGP and BGP peer attributes

The following table provides information on BGP provisionable attributes. To set BGP provisionable attributes, use CLI command `interface protocol bgp set`.

**Table 72** BGP provisionable attributes

| Attribute         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Values                                   |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| autonomous-system | Sets the autonomous system for the local chassis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1..4294967295                            |
| keepalive         | Sets the keep alive timer. The keep alive timer is the time in which BGP sends keep alive messages to its neighbor. The keep alive timer should match the keep alive timer for the peer routers, but if the values do not match, BGP negotiates to the lower value between the BGP peers. If the keep alive is changed after the BGP connection has been established, the new value does not take effect until CLI command <code>interface protocol bgp clear</code> is performed. This CLI command resets the BGP connection. | 1..21845 seconds<br>30 seconds (default) |

| Attribute | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Values                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| holdtime  | Sets the hold-off timer. The hold-off timer is the time that must pass without receiving a keep alive message from its peer, prior to BGP reporting its neighbor to be down. The hold-off timer should be at least three times the neighbors keep alive time. The hold-off timer should match the hold-off timer for the peer routers, but if the values do not match, BGP negotiates to the lower value between the BGP peers. If the holdtime is changed after the BGP connection has been established, the new value does not take effect until CLI command <code>interface protocol bgp clear</code> is performed. This CLI command resets the BGP connection. | 3..65535 seconds<br>90 seconds (default)                |
| peer      | Sets the provisionable attributes for BGP peer-1 and/or peer-2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | See <a href="#">BGP peer provisionable attributes</a> . |

The following table provides information on BGP peer provisionable attributes. To set BGP peer provisionable attributes, use CLI command `interface protocol bgp set peer <peer-1 | peer-2>`.

**Table 73** BGP peer provisionable attributes

| Attribute         | Description                                                                                                      | Values        |
|-------------------|------------------------------------------------------------------------------------------------------------------|---------------|
| autonomous-system | Sets the BGP peer's autonomous system.                                                                           | 1..4294967295 |
| ip                | Sets the BGP peer's IP address for the BGP interface.<br><br>Waveserver 5 supports both IPv4 and IPv6 addresses. | IP Address    |
| interface         | Sets the peer's BGP interface.                                                                                   | dcn-n         |

### Operational BGP attributes

The following tables provide information on BGP, peer, and route summary information.

To view BGP summary information, use CLI command `interface protocol bgp show`.

**Table 74** BGP summary information

| Attribute         | Description                   |
|-------------------|-------------------------------|
| Operational State | The operational state of BGP. |

| Attribute         | Description                                           |
|-------------------|-------------------------------------------------------|
| Autonomous System | The autonomous system number assigned to the chassis. |
| Hold Time         | The holdoff timer value in seconds.                   |
| Keep Alive Time   | The keep alive timer value in seconds.                |
| Peer1 Ip          | The IP address for the peer1.                         |
| Peer1 AS          | The autonomous system number for the peer1.           |
| Peer2 Ip          | The IP address for the peer2.                         |
| Peer2 As          | The autonomous system number for the peer2.           |
| Network           | The network identifier.                               |

To view summary information for the peers, use CLI command `interface protocol bgp show peer-summary`.

**Table 75** Peer summary information

| Attribute | Description                                                |
|-----------|------------------------------------------------------------|
| Peer IP   | The IP address for the peer.                               |
| Peer AS   | The assigned autonomous system number for the peer.        |
| MsgRecv   | The number of BGP messages received.                       |
| MsgSend   | The number of BGP messages sent.                           |
| Uptime    | The duration the peer has been in an “Up” or “Down” state. |



| Attribute | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| State     | <p>The state of the peer. Possible states include:</p> <ul style="list-style-type: none"> <li>• Idle - BGP is starting and trying to initiate a TCP connection to the BGP peer, and is listening for a new connection from a peer router.</li> <li>• Connect - BGP is initiating the TCP connection.</li> <li>• Active - BGP is starting a new 3-way TCP handshake. If the connection is established, an “open” message is sent, the holdoff timer is set to 4 minutes, and the state changes to OpenSent.</li> <li>• OpenSent - an “open” message has been sent from the originating router and is waiting for an “open” message from the other router. After the originating router receives the “open” message from the other router, both messages are checked for errors.</li> <li>• OpenConfirm - BGP is waiting for a keep alive or notification message. Once the neighbor’s keep alive message is received, the state changes to Established.</li> <li>• Established - the connection is working. BGP session is established and BGP neighbors exchange routes using Update messages.</li> </ul> |

To view peer data information, use CLI command `interface protocol bgp show peer <peer-1 | peer-2>`.

**Table 76** Peer data information

| Attribute     | Description                                                        |
|---------------|--------------------------------------------------------------------|
| Peer Address  | The IP address of the remote peer.                                 |
| Peer Route Id | The route identifier of the remote peer.                           |
| Local Address | The local address for the interface that is connected to the peer. |
| Up Time       | The “Up” time since the last failure.                              |

| Attribute                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Peer State                  | <p>The state of the peer. Possible states include:</p> <ul style="list-style-type: none"> <li>• Idle - BGP is starting and trying to initiate a TCP connection to the BGP peer, and is listening for a new connection from a peer router.</li> <li>• Connect - BGP is initiating the TCP connection.</li> <li>• Active - BGP is starting a new 3-way TCP handshake. If the connection is established, an “open” message is sent, the holdoff timer is set to 4 minutes, and the state changes to OpenSent.</li> <li>• OpenSent - an “open” message has been sent from the originating router and is waiting for an “open message from the other router. After the originating router receives the “open” message from the other router, both messages are checked for errors.</li> <li>• OpenConfirm - BGP is waiting for a keep alive or notification message. Once the neighbor’s keep alive message is received, the state changes to Established.</li> <li>• Established - the connection is working. BGP session is established and BGP neighbors exchange routes using Update messages.</li> </ul> |
| Last Read Time              | The last time a BGP packet was received from the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Peer Keep Alive             | The keep alive timer setting for the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Peer Hold Time              | The hold off timer setting for the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Received Refresh Count      | The number of refresh packets received from the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Send Refresh Count          | The number of refresh packets sent to the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Total Received Msg Count    | The total number of BGP packets received from the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Total Send Msg Count        | The total number of BGP packets sent to the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Advertising Time Interval   | The minimum time delay between updates for a neighbor session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Accepted Prefix Count       | The number of prefixes received from the peer that were accepted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Announced Prefix Count      | The number of prefixes advertised to the peer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Received Routes from Peer 1 | A list of received routes from Peer 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Advertised Routes to Peer 1 | A list of advertised routes to Peer 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

To view the addresses advertised to BGP and the connectivity of those addresses, use CLI command `interface protocol bgp show runtime-routes`.

**Table 77** Runtime Routes attribute information

| Attributes | Description                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prefix     | Indicates the reachable route network.                                                                                                                                                          |
| Gateway    | Indicates the IP address of the peer interface to be used to get to the prefix. If the route is local to the Waveserver 5 or is not in use, the Gateway is blank.                               |
| Interface  | The interface used to reach the connection.                                                                                                                                                     |
| Up Time    | The “Up” time since the last failure.                                                                                                                                                           |
| Type       | The type of connection. <ul style="list-style-type: none"> <li>• connected - Waveserver 5 is locally connected</li> <li>• bgp - Waveserver 5 is connected using the network over BGP</li> </ul> |

### ILAN ports

The ILAN-1 and ILAN-2 ports are used to interconnect co-located Waveserver 5 chassis. They are RJ-45 (Ethernet) ports that are located on the faceplate of the access panel. These ports support 1000, and 2500Base-T.

**Note 1:** ILAN ports support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

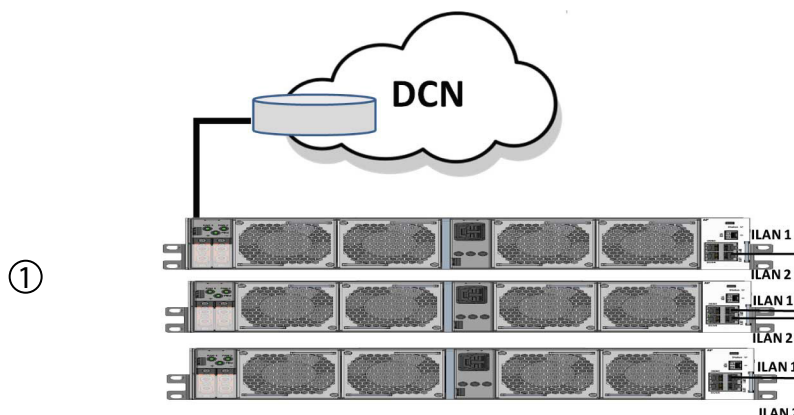
**Note 2:** A disconnected ILAN port has unreliable operational status and you should disable its administrative state using CLI command `chassis port disable port`. When the ILAN port is administratively disabled, the default port speed of 1G is shown.

### ILAN linear configuration

In this configuration, one Waveserver 5 is connected directly by Ethernet cable to the DCN/router. The remaining Waveserver 5 chassis are interconnected in a linear series by ILAN port to the Waveserver 5 directly connected to the DCN/router. For this configuration, each Waveserver 5 is provisioned with its own “local host” management IP, subnet mask and default gateways (if IPv4 is used). If IPv6 is used, then only the chassis with the direct DCN connection requires a default gateway address to be configured. Use an Ethernet cable to connect the ILAN 2 port on one Waveserver 5 to the ILAN1 port on the next Waveserver 5 in the series. If the connection between the one Waveserver 5 and the DCN/router fails, Waveserver 5 DCN communication is lost to that Waveserver 5 and to all Waveserver 5 chassis connected to it by

ILAN. If the connection between any two ILAN interconnected Waveserver 5 chassis fails, communication is lost to all ILAN interconnected Waveserver 5 chassis after the point of the failure. Refer to the following figure.

**Figure 53** ILAN linear configuration for DCN access

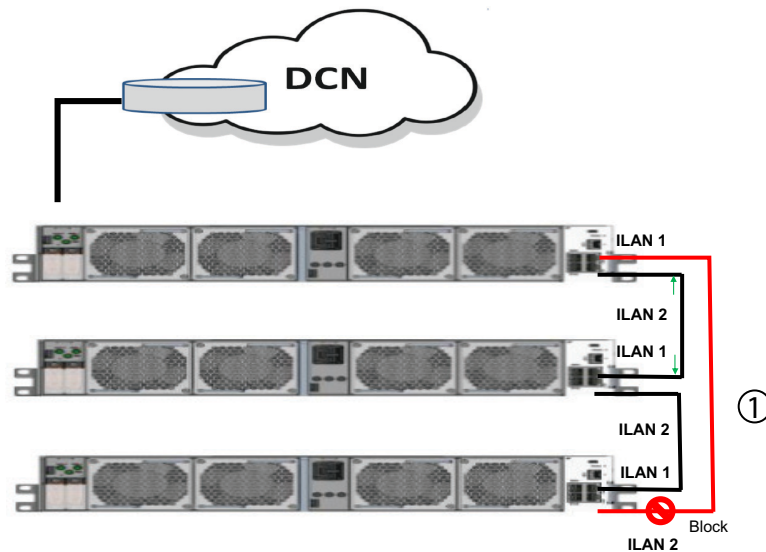


#### Legend

- 1 DCN port to router Ethernet connection

### ILAN ring configuration and loop mitigation

The DCN connectivity from one **Waveserver 5** can be extended to a group of **Waveserver 5** chassis using the ILAN ports on the faceplate of the **Waveserver 5**. Each chassis has two ILAN ports, which allows a group of **Waveserver 5** chassis to be connected in an ILAN ring. Only one **Waveserver 5** should be connected by Ethernet cable to the DCN/router (unless VRRP is being used on external routers, refer to the [DCN redundancy and Virtual Router Redundancy Protocol \(VRRP\)](#) for more detail). This setup creates a dual path to every chassis, which provides DCN connectivity protection in the case of an ILAN cable failing or being removed. To avoid traffic loops, Waveserver 5 supports the loop mitigation feature. This feature provides protection for a single failure, but multiple breaks in the ring causes chassis isolation. No user provisioning is required, once the Waveserver 5 chassis have been physically connected in an ILAN ring, the feature becomes active. The ILAN ring supports up to 32 Waveserver 5 nodes.

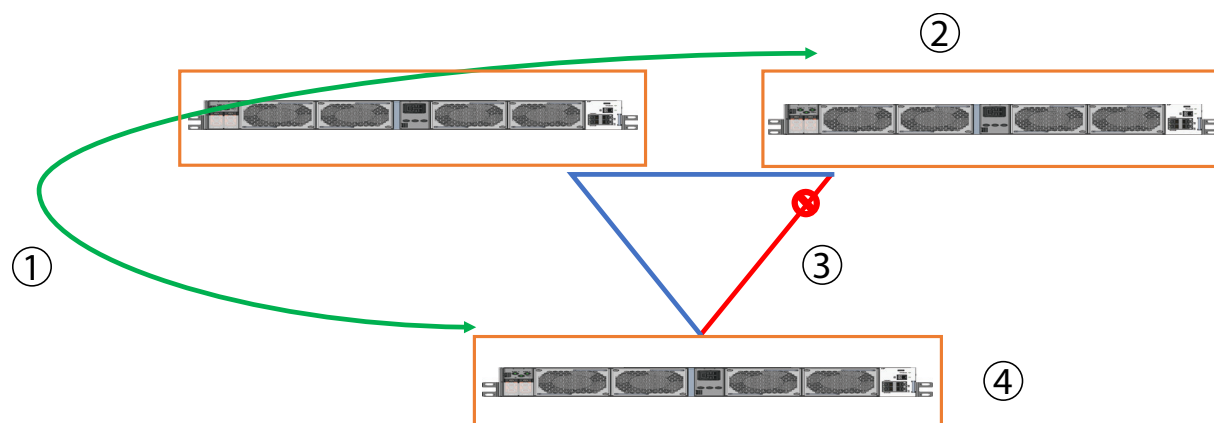
**Figure 54** ILAN ring configuration and loop mitigation**Legend**

- 1 ILAN Blocked Link

**Note:** If you are migrating from a direct DCN connection configuration to an ILAN configuration, prior to the migration, ensure DCN ports are disconnected or disabled prior to the ILAN ports being connected. If VRRP is being used, there are two chassis in the ILAN chain that are connected to the external DCN using their DCN port.

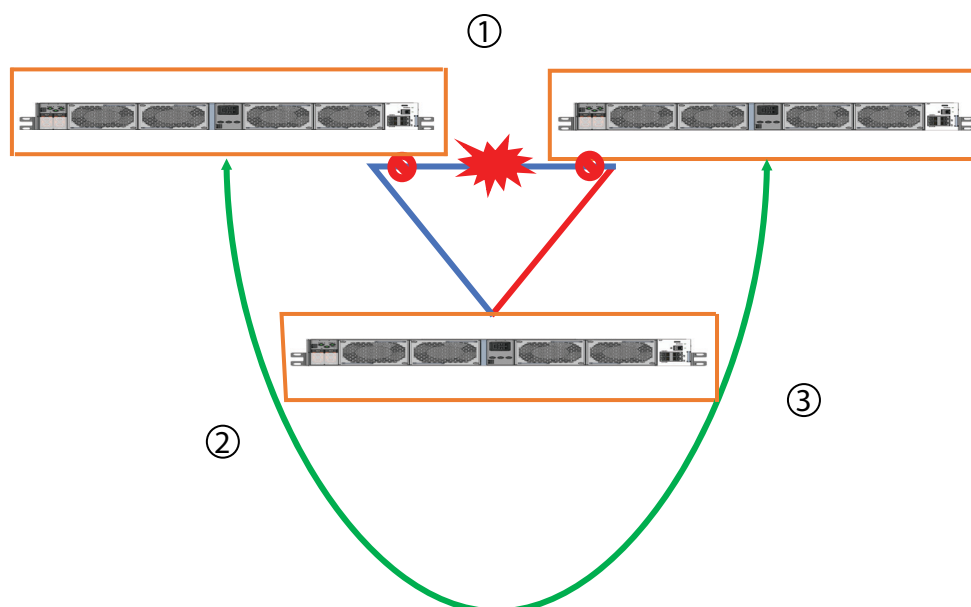
**ILAN loop avoidance and ring protection process**

Waveserver 5 avoids ILAN traffic loops by guaranteeing traffic always has a block on at least one of the ring links, this block prevents traffic from flowing through. Refer to the following figure for the ILAN ring in a normal state:

**Figure 55** ILAN ring in normal state**Legend**

- |   |                   |   |              |
|---|-------------------|---|--------------|
| 1 | ILAN traffic flow | 3 | Blocked port |
| 2 | Block owner       | 4 | Protect Link |

The protect link blocks traffic forwarding until a failure is detected elsewhere in the ring. When a failure is detected, the original traffic block is removed to allow traffic to continue to flow around the ring. Refer to the following diagram for ILAN ring in a protection state:

**Figure 56** ILAN ring in protection state

**Legend**

- |   |                   |   |                        |
|---|-------------------|---|------------------------|
| 1 | Link failure      | 3 | Protect link unblocked |
| 2 | ILAN traffic flow |   |                        |

**Remote Group Management**

The Remote Group Management feature provides a means to access a remote group of Waveserver 5 nodes in a network, where DCN access is not available. A group of Waveserver 5 nodes, can be one or more Waveserver 5 nodes connected together via ILAN. This feature uses General Communication Channels (GCCs) for routing management traffic between local DCNs and remote Waveserver 5 groups.

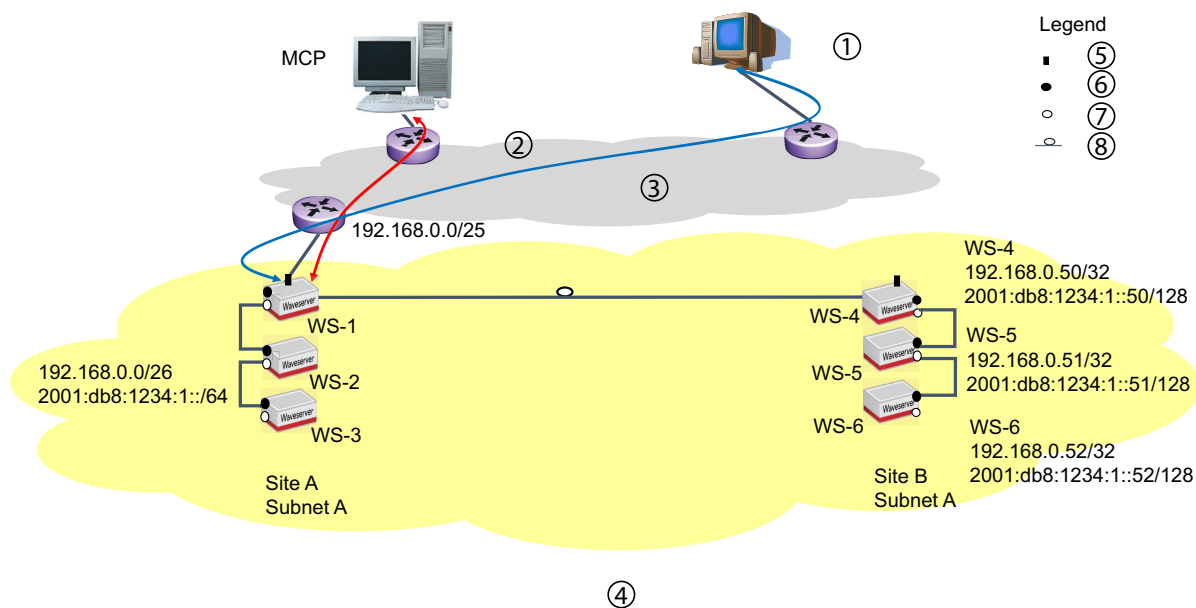
This feature can also use the DHCP relay agent option, which is enabled by default, to communicate and use dynamic IP addressing to support DHCP on remotely managed groups within the same subnet using the local DCN. The Waveserver 5 has the inter-shelf routing interfaces (GCC and ILAN) as the relay agent trusted interfaces by default. The IP address obtained by interface-shelf routing interfaces uses the 32-bit mask and ignores the default route. The IP address obtained by the management interface applies the netmask and default route. However, the DHCP relay options have the following limitations:

- The DHCP Relay does not support IPv6.
- The DHCP option Relay agent info (option82) is disabled.
- The mix of static IP with a different DHCP IP subnet is not supported.

The OTUCn GCC0 channels are also available on each modem to provide point-to-point links, where link end-points have routing interface bindings. Routing interfaces are auto-created for all OTUCn GCC0 line interfaces and are automatically bound to the associated line port. Routing interfaces on the service module line ports are not tied to the service module's administrative or operational states. Auto-created routing interfaces cannot be deleted but they can be administratively disabled if the user wants to suppress alarms or prevent the interface from being used. To disable remote management interfaces using an OTUCn GCC, use CLI commands `interface disable interface if0-<slot>-<port>`. For this release, OTUCn GCC0 is supported using the same subnet and different subnets.

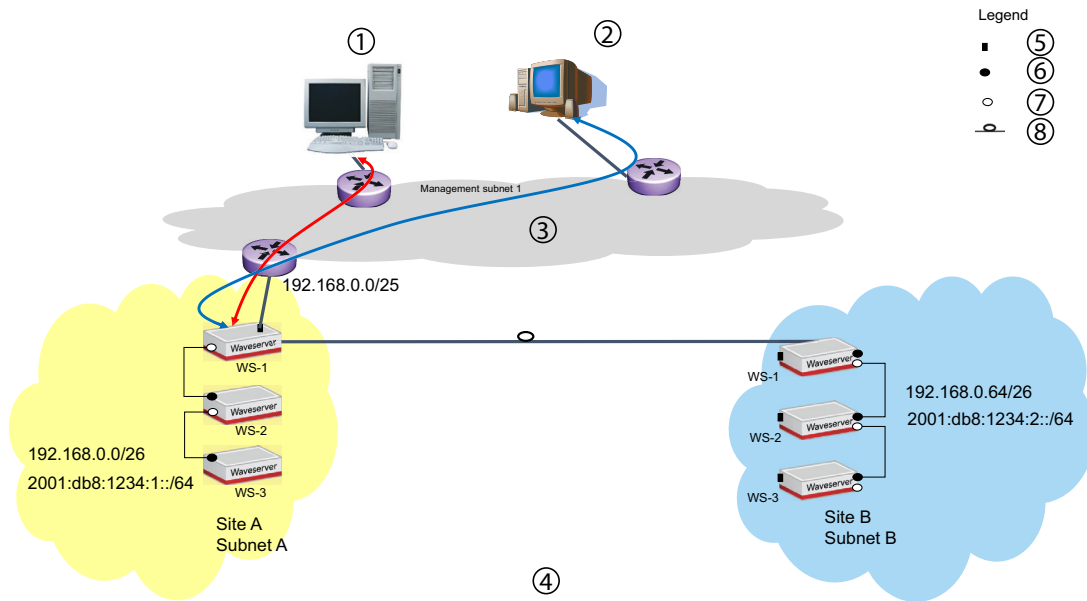
**Note:** For mated line configuration, GCC is only supported on line 1.

Refer to the following figure for remote management access via Line OTUCn GCC0 Channels:

**Figure 57** Remote management via line OTUCn GCC0 channels using same subnet**Legend**

- |   |                                                             |   |           |
|---|-------------------------------------------------------------|---|-----------|
| 1 | Network Admin                                               | 5 | DCN       |
| 2 | Management subnet 1                                         | 6 | ILAN-1    |
| 3 | External DCN                                                | 7 | ILAN-2    |
| 4 | Routed interfaces bound to line ports (one per OTUCn GCC0). | 8 | GCC comms |



**Figure 58** Remote management via line OTUCn GCC0 channels using different subnet**Legend**

|   |                                                             |   |           |
|---|-------------------------------------------------------------|---|-----------|
| 1 | BluePlanet                                                  | 5 | DCN       |
| 2 | Network Admin                                               | 6 | ILAN-1    |
| 3 | External DCN                                                | 7 | ILAN-2    |
| 4 | Routed interfaces bound to line ports (one per OTUCn GCC0). | 8 | GCC comms |

The following considerations and configuration details apply for remote management via line OTUCn GCC0 channels:

- Supports multiple remote management paths using the line ports for network redundancy.
- Supports one remote management interface per line port and up to 8 (double slot service module) remote management interfaces per WaveServer 5 chassis.
- The remote management interface is supported from any local interface to any remote interface.
- Line interfaces are automatically created and they are enabled by default. One line interface is automatically selected to be “active” and the remaining redundant interfaces are available for automatic “activation”. If the “active” management interface becomes unavailable, one of the redundant interfaces takes over.

- Local and remote Waveserver 5 chassis can be on the same or different IPv4 or IPv6 subnets.
- If the range of local and remote subnets are contiguous, a single summary static route can be installed on the external router, with the DCN port connection. If route summarization is not desired or possible, separate static network routes must be installed on the external router. The static network route is required if multiple subnets are used.
- Prior to setting up Remote Group Management,
  - Ensure that there is DCN connectivity for the node that is directly connected to the DCN/Router. For information on how to provision IPv4 and IPv6, refer to [Procedure 12, “Setting up management access using the console or USB port”](#) for more detail.
  - Ensure that the line ports are administratively enabled and operationally up. Refer to the [Commissioning](#) chapter for more detail.
- For IPv6, if Remote Management interfaces are enabled, the IPv6 default gateway must be configured, on the Waveserver 5, in the local group which is directly connected to the DCN network.
- Dynamic IPv4 addressing is supported on remotely managed groups within the same subnet when DHCP relay is enabled.
- For IPv6, stateless address auto-configuration (via SLAAC), from network prefix advertisement messages, is not supported.
- Remote group for same subnet IP addressing must be setup as /32 for IPv4, and /128 for IPv6. For information on how to provision IPv4 and IPv6, refer to the [Procedure 12, “Setting up management access using the console or USB port”](#).
- All Waveserver 5 nodes in a remote group have no default gateway specified. The default route is automatically learned, which specifies the desired Remote Management interface.

### Neighbor Discovery Protocol (NDP)

The Neighbor Discovery Protocol (NDP) is a standard protocol used on the DCN and ILAN ports to discover IPv6 neighbors that are on the same link. The NDP messages communicate information about a device to its immediate neighbors.

### Link Layer Discovery Protocol

Waveserver 5 supports Link Layer Discovery Protocol (LLDP) on both the client port and DCN ports configured for management traffic. LLDP on the client port enables discovery of client-side, far-end equipment on the network, as defined in IEEE 802.1ab (802.1AB-2005 recommended). LLDP on a DCN port provides the ability to discover the management router(s) connected to the DCN port. LLDP can be disabled/enabled on the chassis or on the port.

LLDP is a link-constrained, Layer 2 (L2) protocol that allows discovery of the physical-link topology of the network. LLDP agents on the network exchange LLDP messages with adjacent LLDP agents. These messages communicate information about a device and its immediate neighbors (see [LLDP TLVs on the client port](#)). You can use this information to validate connections and determine the last LLDP message exchanged before a link failure occurred.

Waveserver 5 LLDP updates the database every 30 seconds or less. There can be some delay between the incoming LLDP packet arrival and the information update.

### **LLDP on DCN port**

Waveserver 5 supports the Link Layer Discovery Protocol over a DCN port (DCN-1 and DCN-3 ports). LLDP peering on the DCN port provides the ability to discover the management router(s) connected to the DCN port.

Waveserver 5 supports the transmit and receive capability on the DCN port. With this capability, Waveserver 5 and the router are able to discover each other and exchange information about each other.

LLDP peering on a DCN port can be configured using CLI and NBI. Retrievals can be performed using CLI, NBI and SNMP. You can configure SNMP notifications for LLDP peering on a DCN port, see [Procedure 35, "Configuring SNMP notifications for LLDP peering on a DCN port"](#).

**Note:** Waveserver 5 only provides peering to the directly connected sub-tending equipment; Waveserver 5 only transmits and receives the "nearest bridge" LLDPDUs.

### **LLDP operating modes on a DCN port**

For LLDP peering on the DCN, there are two operating modes: disabled and tx-rx. In disabled mode, there is no monitoring or capturing of the LLDP packets. In tx-rx mode, incoming LLDP packets from the DCN port are captured by the Waveserver 5 and LLDP packets are generated by Waveserver 5, then sent back to the DCN port. Using the tx-rx mode allows the management port to participate in LLDP sessions.

**Note:** The tx-rx mode is not supported on Waveserver 5 client ports.

### **LLDP TLVs for LLDP peering on DCN ports**

The LLDPDU transmitted by Waveserver 5 includes the following mandatory TLVs (Type-Length-Values) defined in IEEE 802.1AB-2009.

**Table 78** Mandatory LLDP TLVs for LLDP peering on DCN

| Name               | Description                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Chassis ID         | Identifies the chassis containing the IEEE 802 LAN station associated with the transmitting LLDP agent.                                 |
| Port ID            | Identifies the port component for the MSAP (MAC server access point) identifier associated with the transmitting LLDP agent.            |
| Time to Live (TTL) | Indicates the number of seconds the recipient LLDP agent is to regard the information associated with this MSAP identifies to be valid. |

Waveserver 5 collects the following optional TLVs from the inbound LLDPDU. Also, Waveserver 5 includes these optional TLVs in the outbound LLDPDU if they are provisioned by the user.

**Table 79** Optional LLDP TLVs for LLDP peering on DCN

| Name               | Description                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Management Address | Identifies a network address associated with the local LLDP agent which can be used to reach the agent on the port identified in the Port ID TLV.                                                          |
| System Name        | Allows network management to advertise the system's assigned name. The value of the System Name TLV can be changed using CLI command <code>system set host-name &lt;String[2..63]&gt;</code> or using NBI. |
| System Description | Allows network management to advertise the system's description.                                                                                                                                           |
| Port Description   | Allows network management to advertise the IEEE 802 LAN station's port description.                                                                                                                        |

### LLDP statistics for LLDP peering on DCN port

Waveserver 5 collects statistics for the LLDP packets received and transmitted on the DCN port. The table below provides details on the statistics supported.

**Table 80** LLDP statistics supported for LLDP peering on DCN

| Name               | Description                                                                 |
|--------------------|-----------------------------------------------------------------------------|
| Out TotalPkt       | Number of LLDP frames successfully transmitted to the Management port.      |
| In TotalPkt        | Number of LLDP frames received from the Management port.                    |
| InErrPkt Discarded | Number of LLDP frames discarded due to errors.                              |
| Unknown Tlv        | The number of LLDP TLVs received on the given port that are not recognized. |

| Name         | Description                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------|
| AgedOutTotal | The number of age-outs that occurred on a given port because the information timeliness interval has expired. |
| PDULngthErr  | The number of LLDPDU length restriction errors detected.                                                      |

### LLDP on client port

On the 100GE client ports, Waveserver 5 supports the mode of operation LLDP snooping. You can manually enable or disable LLDP on each Waveserver 5 client port.

LLDP snooping listens for LLDP messages received on the client port, and then tunnels them transparently through the Waveserver 5 network to the far end client egress. As information communicated by the LLDP messages accumulates, you can retrieve it through the CLI, NBI, SNMP or WebSocket management application.

LLDP messages are sent out at a defined interval and whenever LLDP data changes. When a Waveserver 5 client port receives changed data from a neighboring device, an SNMP notification is generated. Upon receipt of this notification, the management application polls the LLDP MIB objects to determine what information has changed.

To view the generated traps, turn the LLDP port mode on using CLI command `lldp set port <slot>/<port> mode snoop`.

By default, LLDP snooping is enabled on all client ports. When LLDP snooping is disabled on a client port, or when snooping is enabled on a client port, but the client port is disabled, LLDP messages do not accumulate, and no information can be retrieved.

### LLDP TLVs on the client port

As explained earlier, an LLDP message is a packet that consists of an L2 source, an L2 destination, an Ethertype field, and four or more Type Length and Value fields (TLVs). The LLDP standard specifies nine common TLV fields, four of which are mandatory (see [Mandatory TLVs on the client port](#)).

The remaining fields are optional TLVs that carry information about sender information. These can be excluded from LLDP messages. See [Optional TLVs supported on the client port](#) for the list of optional TLVs that Waveserver 5 supports.

**Table 81** Mandatory TLVs on the client port

| TLV                | Description                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Time To Live (TTL) | Indicates the number of seconds that the recipient LLDP agent is to regard the information associated with this LLDPDU as valid. |
| Port ID            | Identifies the port component that transmits the TLV.                                                                            |
| Chassis ID         | Identifies the chassis.                                                                                                          |
| End of LLDPDU      | Defines the end of the LLDPDU with all 0 values.                                                                                 |

**Table 82** Optional TLVs supported on the client port

| TLV                 | Description                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Management Address  | Identifies the address associated with the local LLDP agent that may be used to reach higher layer entities to assist discovery by network management. |
| System Capabilities | Identifies the primary functions of the system and whether they are enabled.                                                                           |
| System Description  | Alphanumeric string describing the system                                                                                                              |
| System Name         | Alphanumeric string specifying the system name                                                                                                         |
| Port Description    | Alphanumeric string describing the port                                                                                                                |

### LLDP statistics on the client port

The following table list the LLDP port-based statistics that Waveserver 5 supports.

**Table 83** LLDP port statistics on the client port

| Statistic                                                                                                                                                     | Description ( <b>Note</b> )                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| OutTotalPkt                                                                                                                                                   | The number of LLDP frames transmitted by this LLDP agent on the indicated port.                                      |
| InTotalPkt                                                                                                                                                    | The number of valid LLDP frames received by this LLDP agent on the indicated port, while this LLDP agent is enabled. |
| InErrPktDiscarded                                                                                                                                             | The number of LLDP frames received by this LLDP agent on the indicated port, and then discarded for any reason.      |
| UnknownTlv                                                                                                                                                    | The number of LLDP TLVs received on the given port that are not recognized.                                          |
| AgedOutTotal                                                                                                                                                  | The number of age-outs that occurred on a given port because the information timeliness interval has expired.        |
| PDULngthErr                                                                                                                                                   | The number of LLDPDU length restriction errors detected.                                                             |
| <b>Note:</b> As described in 802.1AB IEEE Standard for Local and Metropolitan Area Networks: <i>Station and Media Access Control Connectivity Discovery</i> . |                                                                                                                      |

### LLDP bridge statistics

The following table list the LLDP bridge-based (global) statistics that Waveserver 5 supports.

**Table 84** LLDP bridge statistics

| Statistic                                                                                                                                                     | Description (Note)                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inserts                                                                                                                                                       | The number of times the complete set of information advertised by a neighbor has been inserted into the LLDP neighbor database.                                                        |
| Deletes                                                                                                                                                       | The number of times the complete set of information advertised by a neighbor has been deleted into the LLDP neighbor database.                                                         |
| Drops                                                                                                                                                         | The number of times the complete set of information advertised by a neighbor could not be entered into LLDP neighbor database because of insufficient resources.                       |
| AgeOuts                                                                                                                                                       | The number of times the complete set of information advertised by a neighbor has been deleted from the LLDP neighbor database because the information timeliness interval has expired. |
| <b>Note:</b> As described in 802.1AB IEEE Standard for Local and Metropolitan Area Networks: <i>Station and Media Access Control Connectivity Discovery</i> . |                                                                                                                                                                                        |

### Network Time Protocol (NTP)

Waveserver 5 can automatically obtain the system time when it is connected to the DCN, and a DHCP server is configured on the DCN and the network time protocol (NTP) option is set.

If a DHCP server is not configured on the DCN, or if you are logging in to Waveserver 5 through the Console port, set the system date and time manually (see [Procedure 21, “Manually setting the system clock”](#)) or by configuring the NTP client (see [Procedure 22, “Configuring NTP MD5 authentication”](#) and [Procedure 20, “Configuring the NTP client”](#)).

Waveserver 5 also supports MD5 authentication for the NTP client, which prevents unauthorized access by an intruder masquerading as an NTP server. Authentication requires a keyed-MD5 private key, that is encrypted on both the server and client used to identify the NTP time server. For more information, see [Procedure 22, “Configuring NTP MD5 authentication”](#).

### Network Time Protocol version 4 (NTPv4) Autokey authentication

Waveserver 5 supports the Autokey authentication scheme. This allows the Waveserver 5 to verify that the configured NTP server is a known and trusted server. NTPv4 authentication is used to perform an integrity check of the received NTP packet and verify the data received was not changed during

transmission. To enable the NTPv4 Autokey for the Waveserver 5, which acts as a NTP client, a host key pair and a self-signed certificate is created and maintained on the Waveserver 5.

The Autokey message digest algorithm is configured to SHA1 on the Waveserver 5 and the NTP server must also use SHA1 for Autokey exchanges.

The NTPv4 Autokey feature can be disabled or enabled using the CLI command `ntp autokey-auth <disable | enable>`. The certificate used for the NTPv4 Autokey feature is valid for one year and should be updated periodically using CLI command `ntp autokey-auth generate`. If the NTPv4 Autokey feature is enabled, you can select the Autokey method per NTP server using the CLI command `ntp client add server <IP address or host name> autokey`. Up to 10 NTP client servers can be added for NTPv4 Autokey authentication. If the server was previously added, you can turn Autokey on or off using the CLI command `ntp client set server autokey`. You can display the NTPv4 Autokey authentication summary using CLI command `ntp autokey-auth show`.

Waveserver 5 supports NTPv4 management using CLI, NETCONF, REST, SNMP (read only), and WebSocket (event notifications).

To configure NTPv4 Autokey authentication, see [Procedure 23, “Configuring NTPv4 Autokey authentication”](#).

## **xFTP, SCP, and HTTPS servers**

This section describes how to configure access to a default remote xFTP (FTP, TFTP, SFTP, or SCP) server from Waveserver 5. You can use a default server in operations such as backing up and restoring configuration files or saving system state files.

This section also describes how to transfer files using Secure Copy Protocol.

Additional information is provided on how to enable and disable Secure Hypertext Transfer Protocol (HTTPS), HTTPS server mutual authentication, and HTTPS server OCSP.

### **Transferring files using the Secure Copy Protocol**

Waveserver 5 provides Secure Copy Protocol (SCP) support to enable the transfer of files between devices over an SSH connection.

File transfer using SCP facilitates the following tasks:

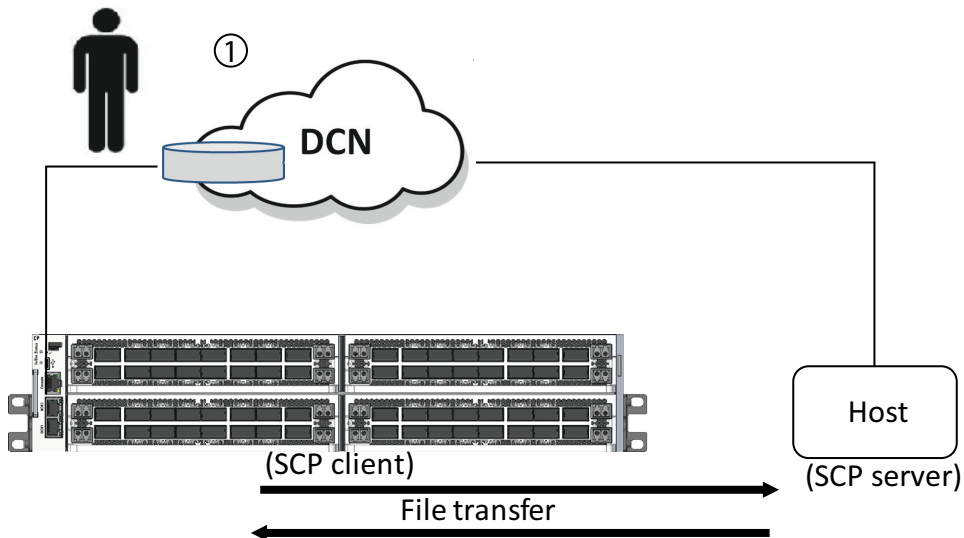
- Transferring a debug log file from Waveserver 5 to a remote server (see [Procedure 25, “Transferring a debug log file”](#))



- Transferring a backup configuration file from Waveserver 5 to a remote server (see [Procedure 118, “Backing up the configuration datastores”](#))
- Transferring a configuration file from a remote server to Waveserver 5 so that the configuration settings it specifies can be restored on Waveserver 5 (see [Procedure 120, “Restoring backed-up configuration datastores”](#))

The following figure shows two examples of the file-transfer scenarios that can be performed using SCP. For each example, Waveserver 5 is connected to the data communications network (DCN) using management configuration option described in this chapter.

**Figure 59** SCP examples—Waveserver 5 requests file transfer from Waveserver 5 to server or from server to Waveserver 5



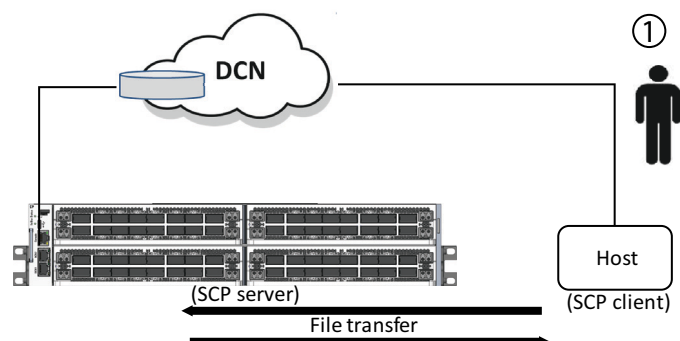
#### Legend

- 1 User issues SCP request on Waveserver

#### Waveserver 5 as an SCP server

Waveserver 5 can also act as an SCP server. It can respond to SCP requests from the host when port 20022 is used. The following figure shows two examples of the file transfer scenario when Waveserver 5 acts as an SCP server.

**Figure 60** SCP examples - Request is issued on Host and file(s) is transferred from Host to Waveserver 5 or Waveserver 5 to Host



### Legend

- 1 User issues SCP request on Host

### Secure Hypertext Transfer Protocol (HTTPS)

Secure Hypertext Transfer Protocol (HTTPS) is enabled by default. If you are concerned with security, you can turn off the HTTPS server using CLI command `system server https disable`. Rest API, Websocket, and the Waveserver 5 homepage require the HTTPS server to be enabled to create an HTTPS connection. For more details on the configuration of HTTPS, see [Procedure 28, “Configuring the HTTPS server”](#).

**Note:** Waveserver 5 requires you to login again if the HTTPS session is inactive for a period of time that exceeds the HTTPS inactivity timeout setting. Depending on the browser you are using, Waveserver 5 automatically renews your session using the credentials stored in the browser cache and a login window is not displayed. To confirm that your session was automatically renewed, view the login timestamps in the command log.

### HTTPS server mutual authentication

Waveserver 5 supports mutual authentication when the Waveserver 5 acts as a TLS server; for example, when the user connects to Waveserver 5 with HTTPS. By turning on the mutual authentication of the Apache HTTPS server and using the client certificate, the HTTPS server can confirm the identity of an HTTPS client, and validate the revocation status of the client certificate against an OCSP responder.

When mutual authentication is enabled, certain conditions must be met to pass the client certificate verification:

- A previously created client certificate must have been imported and installed on the HTTPS client.

- The CA certificate for the client certificate must have been imported and installed on Waveserver 5.
- The certificate sent by the client has had its signature verified against the CA certificate.

### HTTPS server OCSP

When the client accesses the Waveserver 5's HTTPS server, Waveserver 5 uses OCSP to verify the revocation status of the client certificate. If the client certificate is revoked, access to the Waveserver 5 HTTPS is denied. For more details on OCSP, see [Online Certificate Status Protocol \(OCSP\)](#).

To configure HTTPS server mutual authentication and OCSP, see [Procedure 29, "Configuring HTTPS server mutual authentication and OCSP"](#).

### Protocols and associated open ports on the Waveserver 5

The following table provides information on the protocols and associated open ports available on the Waveserver 5.

**Table 85** Protocols and associated open port

| Protocol                  | Port           | Additional Information                                                                          |
|---------------------------|----------------|-------------------------------------------------------------------------------------------------|
| SSF / SFTP                | TCP port 22    | Provisionable from 22 to 65535                                                                  |
| HTTPS                     | TCP port 443   | --                                                                                              |
| NETCONF                   | TCP port 830   | --                                                                                              |
| gRPC                      | TCP port 10161 | --                                                                                              |
| SCP                       | TCP port 20022 | --                                                                                              |
| SPLI                      | TCP port 20040 | --                                                                                              |
| NTP broadcast / multicast | UDP port 123   | UDP client port 123 is opened when the NTP client mode is set to either broadcast or multicast. |
| SNMP                      | UDP port 161   | By default UDP port 161 is closed, SNMP must be enabled to open it.                             |

### Procedure list

This section provides the following procedures:

- [Procedure 12, "Setting up management access using the console or USB port"](#)
- [Procedure 13, "Changing the console port baud"](#)

- Procedure 14, “Configuring BGP peering”
- Procedure 15, “Clearing or removing BGP peering”
- Procedure 16, “Enabling and disabling interface ports”
- Procedure 17, “Enabling and disabling General Communication Channel (GCC) for remote management interface”
- Procedure 18, “Configuring the DHCP client”
- Procedure 19, “Configuring the DNS client”
- Procedure 20, “Configuring the NTP client”
- Procedure 21, “Manually setting the system clock”
- Procedure 22, “Configuring NTP MD5 authentication”
- Procedure 23, “Configuring NTPv4 Autokey authentication”
- Procedure 24, “Managing and configuring default servers”
- Procedure 25, “Transferring a debug log file”
- Procedure 26, “Transferring a file using Waveserver 5 as an SCP server”
- Procedure 27, “Enabling and disabling the HTTPS server”
- Procedure 28, “Configuring the HTTPS server”
- Procedure 29, “Configuring HTTPS server mutual authentication and OCSP”
- Procedure 30, “Configuring LLDP on the client port”
- Procedure 31, “Retrieving LLDP information on the client port”
- Procedure 32, “Retrieving Waveserver 5 TLV data on the client port”
- Procedure 33, “Retrieving or clearing LLDP statistics on the client port”
- Procedure 34, “Configuring LLDP on a DCN port”
- Procedure 35, “Configuring SNMP notifications for LLDP peering on a DCN port”
- Procedure 36, “Retrieving information for LLDP on a DCN port”
- Procedure 37, “Retrieving or clearing LLDP statistics on a DCN port”

---

## Procedure 12 Setting up management access using the console or USB port

---

Use this procedure to connect to Waveserver 5 using the console or USB-C port and acquire or provision an IPv4 or IPv6 address and gateway for the management network interface. This procedure also describes the steps to connect Waveserver 5 to the management network and log in.

### Requirements

- If you are assigning a static IPv6 address and gateway, the terminal supports IPv6.
- One of the following cables are available for local access:
  - A USB-C-to-USB-C cable or a USB-C-to-USB-A cable (dependent on compatibility with your PC or laptop)
  - A DB9-to-RJ45 or USB-to-RJ45 (straight connection) console cable
- An Ethernet cable is available.
- The management network supports IPv4 or IPv4 + IPv6.
- Waveserver 5 line ports are not physically connected to photonic system equipment, and transceivers are not installed in client ports.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

- A terminal emulator (for example, PuTTY) is open on the terminal.
- If you are using a Macintosh computer or MacBook to connect to Waveserver 5, Ciena recommends installing Minicom for serial port setup. Minicom is available at the Web site:  
<http://macappstore.org/minicom/>
- Enable Auto Negotiation (AN) on the network side when connecting the Waveserver 5 DCN port to a Router port to avoid CRC errors.

### Overview

**Note 1:** The DCN and ILAN interfaces described in this procedure are intended for intra-building use only. In customer premises locations where compliance to Telcordia GR-1089-CORE intra-building lightning is required, install shielded twisted pair (STP) wire with shield terminations crimped into the RJ-45 connectors for connection at both ends.

**Note 2:** DCN and ILAN ports support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

**Note 3:** External adapters are not required except when the Waveserver 5 is connected to a complete non-standard proprietary console port.

## Steps

- 1 Connect Waveserver 5 to the management network:

| If                                                                   | Then                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| You are setting up a group of interconnected Waveserver 5 chassis    | Connect the first Waveserver 5 chassis in the group to the management network (router). Use an Ethernet cable to connect the Waveserver 5 DCN port to the router.<br><br>Then, use an Ethernet cable to connect the ILAN 2 port on the Waveserver 5 to the ILAN 1 port on the next Waveserver 5 chassis in the group. See, <a href="#">ILAN linear configuration</a> and <a href="#">ILAN ring configuration and loop mitigation</a> . |
| You are connecting a Waveserver 5 directly to the management network | Use an Ethernet cable to connect the DCN port on the Waveserver 5 to the management network (router). See, <a href="#">Direct DCN connection</a> .                                                                                                                                                                                                                                                                                     |

**Note:** Group refers to a set of interconnected Waveserver 5 units that are connected to the same photonic system.

- 2 Connect the terminal to Waveserver 5.

| If you are                                                                                                           | Then go to              |
|----------------------------------------------------------------------------------------------------------------------|-------------------------|
| connecting the terminal (PC or laptop) to the Waveserver 5 USB-C port using a USB-C-to-USB-C or USB-C-to-USB-A cable | <a href="#">step 3</a>  |
| connecting the terminal (Macintosh or MacBook) to the Waveserver 5 USB-C port using a USB-C-to-USB-A cable           | <a href="#">step 6</a>  |
| connecting the terminal to the Waveserver 5 console port using a console port cable                                  | <a href="#">step 13</a> |

- 3 Connect a USB-C cable to the Waveserver 5 USB-C port and then to the PC or laptop. For additional information, see, [Using console port or USB-C port for management access during commissioning](#).

The PC detects the cable and automatically installs any required drivers.

**Note:** If the drivers do not automatically install, download and install CP210x USB to UART Bridge Virtual COM Port (VCP) from the website:

<https://www.silabs.com/products/development-tools/software/usb-to-uart-bridge-vcp-drivers>

- 4 Open the Device Manager on your PC or laptop and under the Ports (COM & LPT) section, determine the COM port used to connect to Waveserver 5.
- 5 With the terminal emulator, configure the connected terminal with the following settings:
  - Serial line: COM port noted in [step 4](#).
  - Port: a supported baud (the USB-C port supports bauds up to 921600)
  - Connection type: SerialGo to [step 15](#).
- 6 Connect a USB-C to USB-A cable to the Waveserver 5 USB-C port and then to the Macintosh computer or MacBook. See, [Using console port or USB-C port for management access during commissioning](#).

The Macintosh computer or MacBook detects the cable and automatically installs any required drivers.

**Note:** If the drivers do not automatically install, download and install CP210x USB to UART Bridge Virtual COM Port (VCP) for Macintosh from the website: <https://www.silabs.com/products/development-tools/software/usb-to-uart-bridge-vcp-drivers>
- 7 Restart the Macintosh computer or MacBook.
- 8 Verify the presence of the USB cable by searching the /dev/cu folder.
- 9 Open a terminal window and connect to Waveserver 5. Ciena recommends using Minicom for the connection.
- 10 If you are using Minicom, in the terminal window, type `minicom -s`.
- 11 Use the arrow keys to scroll to `Serial port setup` and press **Enter** to select it.
- 12 Configure the connected terminal with the following settings;
  - Serial Device: USB device noted in [step 8](#)
  - Bps/Par/Bits: 9600 8N1
  - Hardware Flow Control: No
  - Software Flow Control: NoGo to [step 15](#).
- 13 Connect a console port cable to the Waveserver 5 console port and then to the PC or laptop. For additional information, refer to [Using console port or USB-C port for management access during commissioning](#).

If using a USB to RJ45 console cable or adapter, ensure the associated drivers are installed.

- 14 With the terminal emulator, configure the connected terminal with the following RS-232 settings:
- Speed: Default value
  - Data bits: 8
  - Stop bits: 1
  - Parity: None
  - Flow control: None
- 15 Press any key for auto-baud detection.
- Note 1:** You can change the baud by disconnecting the cable, changing the baud on the terminal, and then reconnecting the cable. The RJ-45 port supports bauds up to 921600 and the default baud is 115200.
- Note 2:** Auto-baud detection supports these rates: 9600, 19200, 38400, 57600, 115200, 230400, 460800, and 921600.
- 16 At the login prompt, enter a username and a password to access the Waveserver 5. Log in as `su`. The default super user (su) account initially has no password; press **Enter** for system access.
- 17 Note the following:

| If                                                                                                                             | Then                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| The Waveserver 5 DHCPv4 client successfully acquires an IP address and gateway from a DHCPv4 service on the management network | With the terminal emulator, retrieve the provisioned IP address:<br><code>interface show</code><br>Go to <a href="#">step 19</a>           |
| The DHCPv4 client cannot acquire an IP address and gateway from a DHCPv4 service on the management network                     | Check for and resolve issues with the DHCPv4 services on the management network, and reattempt to acquire a DHCPv4 IP address and gateway. |
| Waveserver 5 successfully acquires an IPv6 SLAAC IP address and gateway from the management network                            | With the terminal emulator, retrieve the provisioned IP address:<br><code>interface show</code><br>Go to <a href="#">step 19</a>           |



| If                                                                                                                                                                                                                                                                                                         | Then                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waveserver 5 cannot acquire an IPv6 SLAAC IP address and gateway from the management network                                                                                                                                                                                                               | Check for and resolve issues with the router configuration, and reattempt to acquire an IPv6 SLAAC IP address and gateway.                                                                                 |
| You are assigning a static IPv4 IP address and the management network that Waveserver 5 will be connected to might support DHCPv4 services, which will automatically assign an IPv4 IP address that will overwrite the static IP address you are provisioning                                              | With the terminal emulator, disable the DHCPv4 client on Waveserver 5:<br><br>dhcp client disable                                                                                                          |
| You are assigning a static IPv6 IP address and the management network that Waveserver 5 will be connected to might support SLAAC, which will automatically assign an IPv6 IP address. Multiple IPv6 addresses can exist on a system and the SLAAC IPv6 address will not overwrite the static IPv6 address. | With the terminal emulator, prevent Waveserver 5 from accepting an auto-assigned SLAAC IP address:<br><br>interface ipv6-stack set<br>accept-router-advertisement<br>off<br><br>interface ipv6-stack flush |

- 18 Assign the IP address and gateway:

| Type | Use the commands                                                            |
|------|-----------------------------------------------------------------------------|
| IPv4 | interface set interface local ip <IP address / mask> [gateway <IP address>] |
|      | interface set gateway <IP address>                                          |
| IPv6 | interface add interface local ip <IP address / mask>                        |
|      | interface add gateway <IPv6 address>                                        |

- 19 With the terminal emulator, log in to Waveserver 5 using an SSH connection with the provisioned IP address. Enter the user name (`su`) of the default user account. The default super user (`su`) account initially has no password; press Enter for system access.
- Note:** Ciena strongly recommends that you set a password for the default user account as soon as possible after you log in to Waveserver 5 for the first time. The “Credentials Warning” alarm will activate to remind you to do this. For more information, refer to [User account types](#) and [Credentials Warning](#).
- 20 Disable unused communications interfaces on Waveserver 5 (see figures [ILAN linear configuration for DCN access](#) and [Direct connection to DCN](#)):

| If                                                                                                                                                                                          | Then                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| The Waveserver 5 is the first in a group and the Waveserver 5 will be connected using ILAN to another Waveserver 5.                                                                         | Disable any unused ILAN port:<br><pre>chassis port disable port &lt;ilan-1   ilan-2&gt;</pre>                                                          |
| The Waveserver 5 is the first in a group and the Waveserver 5 will not be connected using ILAN to another Waveserver 5.                                                                     | Disabled the ILAN 1 and ILAN 2 ports:<br><pre>chassis port disable port ilan-1 chassis port disable port ilan-2</pre>                                  |
| The Waveserver 5 is being added to an existing group with access to the management network                                                                                                  | Disable a DCN port:<br><pre>chassis port disable port &lt;dcn-n&gt;</pre>                                                                              |
| The Waveserver 5 is part of an existing linear group with access to the management network, no other Waveserver 5 will be connected to it and there is no intention to close the ILAN ring. | Disable the DCN port and the unused ILAN port:<br><pre>chassis port disable port &lt;dcn-n&gt; chassis port disable port &lt;ilan-1   ilan-2&gt;</pre> |
| <b>21</b>                                                                                                                                                                                   | Save the provisioned settings to the configuration file:<br><pre>configuration save</pre>                                                              |

## Procedure 13 Changing the console port baud

---

Use this procedure to configure the console port baud.

### Overview

By default, Waveserver 5 uses a speed (baud) of 115200 for console port connections. To support different configurations, the Waveserver 5 has an auto-baud detection mechanism which is invoked on all Waveserver 5 units.

Auto-baud detection supports these rates: 9600, 19200, 38400, 57600, 115200, 230400, 460800, and 921600.

### Steps

- 1 Connect a terminal (PC, laptop) to the serial console port.
- 2 Configure the connected terminal with the following RS-232 settings:  
Character size = 8  
Parity = None  
Stop bits = 1  
Flow control = None
- 3 Press any key for auto-baud detection.  
**Note:** You can change the baud by disconnecting the cable, changing the baud on the terminal, and then reconnecting the cable. The RJ-45 port supports bauds up to 921600 and the default baud is 115200.
- 4 A login prompt appears when the auto-baud detection is complete, enter a username and a password to access the Waveserver 5.  
**Note:** If you are changing the baud, you may need to start a new session for text to appear correctly.

## Procedure 14 Configuring BGP peering

Use this procedure to configure BGP peering.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- Clustering must be disabled using CLI command `system set cluster-enrollment disabled`.
- Management IP must be statically configured and DHCP must be disabled.
- Enable Auto Negotiation (AN) on the network side while you connect the Waveserver 5 DCN port to a Router port to avoid CRC errors.

### Steps

- 1 Repurpose the DCN port:  

```
chassis port set port <dcn-n> mode redundant-bgp
```

**Note 1:** This validates that clustering is disabled.

**Note 2:** In this command, if you set the port as DCN-1 then the associated redundant port would be DCN-2. Similarly, if you set the port as DCN-3, the associated redundant port would be DCN-4.
- 2 Enable BGP:  

```
interface protocol bgp enable
```
- 3 Set the autonomous number:  

```
interface protocol bgp set autonomous-system <NUMBER: 1..4294967295>
```
- 4 Set the BGP attributes:  

```
interface protocol bgp set [autonomous-system <NUMBER: 1..4294967295>] [keepalive <NUMBER: 1..21845>] [holdtime <NUMBER: 3..65535>]
```

**Note:** This CLI command validates that the interface has a mode of redundant-bgp. If the interface has a mode of redundant-bgp, the logical interface is created for the peer.
- 5 Set the BGP peer attributes:  

```
interface protocol bgp set peer <peer-1 | peer-2> autonomous-system <NUMBER: 1..4294967295> ip <IP Address>
```
- 6 Set the logical interfaces for the DCN ports:  

```
interface set interface <peer> ip <IP Address with Mask> gateway <IP Address of gateway>
```

Or

```
interface add interface <peer> ip <IPv6 Address with Mask>
gateway <IPv6 Address of gateway>
```

- 7** Change the management interface's IP address, if required:

```
interface set interface local ip <IP Address with Mask>
```

Or

```
interface add interface local ip <IPv6 Address with Mask>
```

- 8** Display the BGP summary information:

```
interface protocol bgp show [peer-summary] [runtime-
routes] [peer <peer-1 | peer-2>]
```

- 9** Save the provisioned settings to the configuration file:

```
configuration save
```

## Procedure 15 Clearing or removing BGP peering

Use this procedure to clear or remove BGP peering.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You must be logged in to Waveserver 5 with serial port access.

**Note:** You can also remove BGP by performing a return to factory defaults, see [Procedure 71, “Performing a Reset To Factory Defaults \(RTFD\)”](#).

### Steps

| 1 | If you want to                                                                                                                                                                                                                                                                                                                                                                                                         | Then go to             |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|   | perform a soft clear on the BGP connection                                                                                                                                                                                                                                                                                                                                                                             | <a href="#">step 2</a> |
|   | perform a hard clear on the BGP connection                                                                                                                                                                                                                                                                                                                                                                             | <a href="#">step 3</a> |
|   | remove the BGP peering                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">step 4</a> |
| 2 | Perform a soft clear on the BGP connection:<br><pre>interface protocol bgp clear soft</pre> <p><b>Note:</b> Using CLI command <code>interface protocol bgp clear soft</code>, updates the advertised network information passed between the BGP nodes and leaves the BGP connection established.</p> <p>The procedure is complete.</p>                                                                                 |                        |
| 3 | Perform a hard clear on the BGP connection:<br><pre>interface protocol bgp clear</pre> <p><b>Note:</b> Using CLI command <code>interface protocol bgp clear</code>, tears down the BGP connection and allows the BGP connection to re-establish. Any updated provisioning for BGP (for example, provisioning updates made to the keepalive or holdtime) is applied at this time.</p> <p>The procedure is complete.</p> |                        |
| 4 | Disable DCN-n ports:<br><pre>chassis port disable port dcn-n</pre>                                                                                                                                                                                                                                                                                                                                                     |                        |
| 5 | To disable and clear BGP data:<br><pre>interface protocol bgp disable</pre> <p><b>Note:</b> Using CLI command <code>interface protocol bgp disable</code>, removes all BGP provisioning and tears down the BGP connection.</p>                                                                                                                                                                                         |                        |
| 6 | Re-enable DCN-n port to be management ports:<br><pre>chassis port set port dcn-n mode management</pre>                                                                                                                                                                                                                                                                                                                 |                        |

- 7 Set the management interface IP address and gateway, if you previously set the local management IP address during the BGP configuration:  

```
interface set interface local ip <Ip Address with Mask>
gateway <IP Address>
```
- 8 Set the router to the non-BGP settings.
- 9 Re-enable the DCN port.  

```
chassis port enable port dcn-n
```
- 10 Save the provisioned settings to the configuration file:  

```
configuration save
```

## Procedure 16 Enabling and disabling interface ports

Use this procedure to enable and disable interface ports. The interface port must be enabled prior to enabling a remote management interface.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Steps

|   |                                                                                                                                                                            |                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1 | <b>If you want to</b>                                                                                                                                                      | <b>Then go to</b>      |
|   | disable an interface port                                                                                                                                                  | <a href="#">step 2</a> |
|   | enable an interface port                                                                                                                                                   | <a href="#">step 3</a> |
| 2 | To disable an interface port.<br><code>interface port disable ifp0-&lt;slot&gt;-&lt;port&gt;</code><br>Go to <a href="#">step 4</a> .                                      |                        |
| 3 | To enable an interface port.<br><code>interface port enable ifp0-&lt;slot&gt;-&lt;port&gt;</code>                                                                          |                        |
| 4 | Optionally display summary information for all the interface ports or a specific interface port.<br><code>interface port show [port ifp0-&lt;slot&gt;-&lt;port&gt;]</code> |                        |
| 5 | Save the provisioned settings to the configuration file.<br><code>configuration save</code>                                                                                |                        |



## Procedure 17 Enabling and disabling General Communication Channel (GCC) for remote management interface

Use this procedure to enable and disable General Communication Channels (GCCs) on the Waveserver 5 line ports for a remote management interface.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You have enabled the interface port in which you want to enable GCC remote management. See [Procedure 16, “Enabling and disabling interface ports”](#).
- Ensure that line ports are up; if not refer to [Procedure 85, “Operating loopback test on Waveserver 5”](#) or [Procedure 235, “Loss of Adjacency”](#).
- Corresponding interface and interface ports are enabled. It is enabled, by default. `interface if0-<slot>-<port>`.
- Verify DCN connectivity is disabled on the remote nodes.

### Steps

- 1 Provision IP addresses according to the provision rules for both far-end and near-end group devices. On the far-end, you can choose to use the same subnet IP addresses or different subnet IP addresses.

**Note:** If the range of local and remote subnets are contiguous, a single summary static route can be installed on the external router, with the DCN port connection. If route summarization is not desired or possible, separate static network routes must be installed on the external router.

- 2 Verify that local interface port is able to verify the neighbor router id.

```
interface show interface if0-<slot>-<port>
```

The neighbor is the remote node on the far-end of the line port.

- 3 Verify that remote Waveserver 5 is able to see the default route.

```
interface route show
```

- 4 Verify that remote node can be pinged from and connected to the nodes outside the subnet under these circumstances.

- 5 Save the provisioned settings to the configuration file.

```
configuration save
```

- | 6 | If you want to                                               | Then go to             |
|---|--------------------------------------------------------------|------------------------|
|   | disable GCC on the line port for remote management interface | <a href="#">step 7</a> |
|   | enable GCC on the line port for remote management interface  | <a href="#">step 8</a> |
- 7** To disable GCC on the line port for a remote management interface.  
`interface disable interface if0-<slot>-<port>`  
 Go to [step 9](#).
- 8** To enable GCC on the line port for a remote management interface.  
`interface enable interface if0-<slot>-<port>`
- 9** Optionally display summary information for the remote management interface.
- ```

interface show interface if0-<slot>-<port>
+-----+-----+
| Parameter | Value |
+-----+-----+
| Name | if0-1-1 |
| Admin State | Enabled |
| Oper State | Down |
| Interface Port | ifp0-1-1 |
| Metric | 12 |
| IPv4 Neighbor router ID | |
| IPv4 Operational State | Down |
+-----+-----+

```
- 10** Save the provisioned settings to the configuration file.
`configuration save`

Procedure 18 Configuring the DHCP client

Use this procedure to configure the DHCP client on Waveserver 5. The DHCP client is enabled by default.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- If you want to assign Waveserver 5 to a subnet on a DHCP server configured on the management network, use the Waveserver 5 MAC address as the UID. To retrieve the MAC address:


```
chassis show mac
```
- Do any of the following:
 - Disable or enable the DHCP client or renew the lease on the DHCP client:


```
dhcp client <disable | enable>
dhcp client lease renew
```

Note: Waveserver 5 does not clear the IP address when you disconnect it from the network. To clear the IP address and obtain a new one from the DHCP server, disable the DHCP client, then re-enable it. You must perform this operation through an IPv6 or console connection.
 - Set writable attributes and corresponding options of the DHCP client:


```
dhcp client set {[interface-type <active | remote>]
[discovery-interval <Seconds: 1...60>] [requested-lease-
time <Seconds: 45...4294967294]}

dhcp client options set {[boot-file <on | off>] [dns <on
| off>] [domain-name <on | off>] [host-name <on | off>]
[lease-time <on | off>] [log-server <on | off>] [ntp <on
| off>] [router <on | off>] [subnet <on | off>][time-
offset <on | off>]}
```
 - Retrieve summary information for the DHCP client:


```
dhcp client show [options] [state]
```
- Save the provisioned settings to the configuration file:


```
configuration save
```

Procedure 19 Configuring the DNS client

Use this procedure to configure the DNS client on Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Assign a server to or remove an assigned server from the DNS client:
`dns client <add | remove> server <IP address>`
- 2 Set the domain name of the DNS client:
`dns client set domain-name <Domain Name String: 1...63>`
- 3 Perform any of the following tasks:
 - Disable or enable the DNS client or a server assigned to the DNS client:
`dns client disable [server <IP address>]`
`dns client enable [server <IP address>]`
 - Restore the domain name of the DNS client to the configuration default:
`dns client unset domain-name`
 - Set the IP address for a DNS server:
`dns client set server <IP address> priority <Number: 1...3>`
 - Resolve the host name or IP address of the DNS client:
`dns client resolve ip <IP address or host name>`
 - Retrieve summary information for the DNS client:
`dns client show [domain-name <Domain Name String: 1...63>] [servers]`
- 4 Save the provisioned settings to the configuration file:
`configuration save`

Procedure 20 Configuring the NTP client

Use this procedure to configure the NTP client on Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Assign a server to the NTP client:

```
ntp client add server <IP address or host name: 1...63>
[key-id <authentication key ID>][autokey]
```

Note: To use the `key-id` attribute, the NTP MD5 authentication key must have been added to the NTP server. See [Procedure 22, “Configuring NTP MD5 authentication”](#).

- 2 Set one or more writable attributes for the NTP client:

```
ntp client set {[mode <polling | broadcast | multicast>]
[polling-interval <Seconds: 16, 32, 64, 128, 256, 512,
1024, 2048, 4096, 8192, 16384, 32768, 65536>]}
```

- 3 Do any of the following:

- Set the NTP MD5 authentication key for a server assigned to the NTP client:

```
ntp client set server <IP address or host name> key-id
<MD5 authentication key>
```

Note: To use this command, the NTP MD5 authentication key must have been added to the NTP server. See [Procedure 22, “Configuring NTP MD5 authentication”](#).

- Set the NTP Autokey authentication to a pre-added server to be on or off.

```
ntp client set server <IP address or hostname> autokey <on
| off>
```

- Disable or enable a server assigned to the NTP client:

```
ntp client <disable | enable> <IP address or host name>
```

- Retrieve summary information about the NTP client:

```
ntp client show [oper] [state] [keys]
```

- Retrieve summary information for a server assigned to the NTP client:

```
ntp client show server <IP address or host name>
```

- 4 Save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 21 Manually setting the system clock

Use this procedure to manually set the Waveserver 5 system clock.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin user privileges.
- The NTP client must be disabled using CLI command `ntp client disable`. The NTP client is enabled by default.

Steps

- 1 Set the system date:
`system set date <Date: yyyy-mm-dd | yy-mm-dd | mm-dd>`
Note: After setting the date, wait one to two minutes before setting the system time.
- 2 Set the system time, a time offset, and a timestamp:
`system set time <Time: hh:mm:ss | hh:mm> time-offset <-43200..50400> timestamp <local|UTC>`
Note: The `time-offset` attribute specifies the offset in seconds from the specified timestamp, that is either local time or UTC (Coordinated Universal Time).
- 3 Confirm the system time and date:
`system show time`
- 4 Save the provisioned setting to the configuration file:
`configuration save`

Procedure 22 Configuring NTP MD5 authentication

Waveserver 5 supports RSA Message Digest 5 (MD5) authentication for the NTP client. Use this procedure to configure NTP MD5 authentication for Waveserver 5 on the NTP server.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Waveserver 5 supports RSA Message Digest 5 (MD5) authentication for the NTP client.

Note: MD5 authentication is not supported for NTP servers configured with DHCP option 42. When the NTP servers are configured with DHCP, any user-configured NTP settings are overridden.

Steps

- Enable or disable MD5 authentication:

```
ntp md5-auth <disable | enable>
```

Note: The default NTP MD5 authentication state is disabled.
- Do one of the following to add an MD5 authentication key to the NTP server:
 - Manually add an MD5 authentication key:

```
ntp md5-auth add {key-id <NTP MD5 key object number:1...65534> md5 <MD5 auth key string: 2...31>}
```
 - Add a pre-encrypted NTP MD5 authentication key to the NTP server:

```
ntp md5-auth secret {key-id <NTP MD5 key object number:1...65534> md5 <Encrypted MD5 auth key string: 4...62>}
```
 - Optionally, import MD5 authentication keys from a file to the NTP server:

```
ntp md5-auth import <filename>
```

Note: For information about accessing the NTP server (for example, a remote server), see [xFTP, SCP, and HTTPS servers](#).
- Show summary information for NTP MD5 authentication:

```
ntp md5-auth show
```
- Remove an MD5 authentication key from the NTP server:

```
ntp md5-auth remove {key-id <NTP MD5 key object number:1...65534>}
```
- Save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 23 Configuring NTPv4 Autokey authentication

Use this procedure to configure NTPv4 Autokey authentication on the Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- Autokey has been configured on the NTP server side and uses SHA1 digest algorithms.
- You know the IP address or host name for the NTP server.

Steps

- | If you want to | Then go to |
|--|------------------------|
| disable the global administrative state for NTP Autokey authentication | step 2 |
| enable the global administrative state for NTP Autokey authentication | step 3 |
- Set the global administrative state for NTP Autokey authentication to disabled:

```
ntp autokey-auth disable
```

Go to [step 8](#).
- Set the global administrative state for NTPv4 Autokey authentication to enabled:

```
ntp autokey-auth enable
```
- Generate a certificate for NTPv4 Autokey authentication:

```
ntp autokey-auth generate
```
- The NTPv4 Autokey authentication can be selected per server:

If you would like to enable NTPv4 Autokey on a	Then go to
newly created NTP server	step 6
existing NTP server	step 7

Note: Up to 10 NTP client servers can be added for NTPv4 Autokey authentication.
- Enable NTPv4 Autokey authentication on a newly created NTP server:

```
ntp client add server <IP address or host name> autokey
```

Go to [step 8](#).
- Enable NTPv4 Autokey authentication on an existing NTP server:

```
ntp client set server <IP address or host name> autokey on
```


- 8 Save the provisioned settings to the configuration file:
`configuration save`

Procedure 24 Managing and configuring default servers

Use this procedure to manage and configure default servers.

Requirements

- To manage or configure a default server, you must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- Set the global default xFTP server mode:

```
system xftp set mode <ftp | sftp | scp | tftp>
```

Note: Setting the default xFTP server mode enables the use of the default server for backup and restore operations.
- Configure a default remote server (FTP, TFTP, SFTP, or SCP):

Note: When you use the code in the following examples to configure a remote server, specify the password or the secret, but not both.

 - To configure a default remote FTP server:

```
system xftp set ftp-server <IP address or host name>
login-id <String: 1...32> password <String: 1...128>
secret <String: 1...256>
```
 - To configure a default remote SFTP server:

```
system xftp set sftp-server <IP address or host name>
login-id <String: 1...32> password <String: 1...128>
secret <String: 1...256>
```
 - To configure a default remote SCP server:

```
system xftp set scp-server <IP address or host name>
login-id <String: 1...32> password <String: 1...128>
secret <String: 1...256>
```
 - To configure a default remote TFTP server:

```
system xftp set tftp-server <IP address or host name>
```
- Do any of the following:
 - To remove a remote server as the default for the specified server type:

```
system xftp unset <ftp-server | tftp-server | sftp-server
| scp-server>
```
 - To retrieve summary information for all configured default xFTP servers:

```
system xftp show
```
- To save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 25 Transferring a debug log file

Use this procedure to create a debug log file and transfer it from Waveserver 5 to a remote server.

Requirements

- You must be logged in to Waveserver 5 using an account with super user privileges to create and transfer the log file from Waveserver 5 to a remote host.
- You must have name, IP address, user name, and password of the remote server.
- To access the default remote server, the server must be configured on the system (see [Procedure 24, “Managing and configuring default servers”](#)).

Steps

Remote host is available and debug log file can be transferred instantly

- 1 Create the debug log file and transfer it from Waveserver 5 to a remote host:

```
system state-dump file-name <String> [default-server |
default-sftp-server | default-scp-server | sftp-server
<IP address or host name> login-id <String: 1..32>
password <String: 1..128> secret <String: 1..256> | scp-
server <IP address or host name> login-id <String: 1..32>
password <String: 1..128> secret <String: 1..256>]
```

Note: The file-name string supports alphanumeric characters (space not supported) including these special characters: ~ @ % ^ * _ = + { } [] : ? . However, some strings are allowed as valid filenames due to your management platform system's OS restrictions. For example, if your management platform system has Windows OS then do not use the ? (question mark) character in the file-name string, although it is supported by the Waveserver 5 and allowed on a Linux system.

Remote host is not available

- 2 Create a debug log file locally on the Waveserver 5 node.

```
system state-dump file-name <String>
```

Note: The state dump process is an asynchronous activity that keeps running in the background without interrupting any other activity or action on the Waveserver 5 node.

Remote host is available again

- 3 Check state dump process status and also the debug log file location:

```
system show state-dump
```

*The typical location of the debug log file is /var/opt/stateDump/
<filename>.txz*

- 4 If the state dump process is complete, transfer the generated debug log file from the Waveserver 5 node to your local PC or server. Use the following CLI command with appropriate protocol (xftp, sftp, or scp):

```
system xftp putfile ... <path/filename>
```

For example:

```
system xftp putfile sftp-server <ip address> login-id  
<login id> password <password> local-filename /var/opt/  
stateDump/sample_debug_log.txz
```

Procedure 26 Transferring a file using Waveserver 5 as an SCP server

Waveserver 5 can respond to SCP requests from the host when port 20022 is used. This procedure shows an example of Waveserver 5 as an SCP server.

Overview

Note: In the SCP examples below, the command-line option `-P` (which specifies the port) is case sensitive. You must type an uppercase `-P` to specify the port.

Steps

- | | | |
|---|--|--|
| 1 | <p>If you want to</p> <hr/> <p>Transfer a file from Waveserver 5 to Host</p> <p>Transfer a file from Host to Waveserver 5</p> | <p>Then go to</p> <hr/> <p>step 2</p> <p>step 3</p> |
|---|--|--|
- 2 To transfer a file from Waveserver 5 to a Linux machine using Waveserver 5 as an SCP server, issue the following command on a Linux machine outside of Waveserver 5:
- ```
scp -P 20022 user@IP:remotepath <local_file>
```
- For example: The file named `my_info` is transferred by user1 from the remote host whose IP address is 123.123.123.123 and filepath is `/var/www/public` to the local folder with the filepath `/home/records/`
- ```
scp -P 20022 user1@123.123.123.123:/var/www/public/my_info /home/records/
```
- 3 To transfer a file from a Linux machine to Waveserver 5 using Waveserver 5 as an SCP server, run the following command on a Linux machine outside of Waveserver 5:
- ```
scp -P 20022 <local_file> user@IP:remotepath
```
- For example: The file named `my_info` is transferred by user 1 from the local folder with the filepath `/home/records/` to the remote host whose IP address is 123.123.123.123 and filepath is `/var/www/public`.
- ```
scp -P 20022 /home/records/my_info user1@123.123.123.123:/var/www/public/
```

Procedure 27 Enabling and disabling the HTTPS server

Use this procedure to enable or disable the HTTPS server.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- Note:** Rest API and the homepage communicate with Waveserver 5 over a secure hypertext transfer protocol (HTTPS), therefore HTTPS must be enabled for these applications.

Steps

- 1

Optionally, retrieve the current administrative state of the servers:
`system server https show`
- 2

If you want to	Then go to
disable the HTTPS server	step 3
enable the HTTPS server	step 4
- 3

To disable the HTTPS server, use the command:
`system server https disable`
Go to [step 5](#).
- 4

To enable the HTTPS server, use the command:
`system server https enable`
- 5

Optionally retrieve the current administrative state of the servers to confirm the setting:
`system server https show`
- 6

Save the provisioned settings to the configuration file:
`configuration save`

Procedure 28 Configuring the HTTPS server

Use this procedure to configure the HTTPS server settings. You can use this procedure to set the HTTPS server certificate name, the HTTPS session inactivity timeout duration, and the trusted DNS name to be used when checking the peer's certificate.

Requirements

- You must be logged in to Waveserver 5 using an account with super user access privileges.

Steps

- 1 Enable the HTTPS server, see [Procedure 27, “Enabling and disabling the HTTPS server”](#).
- 2 To configure HTTPS server settings, use the CLI command:

```
system server https set [cert-name] [inactivity-timeout] [trusted-dns]
```
- 3 To set the HTTPS server certificate name, which assigns an installed RSA or ECC certificate to the server, use the command:

```
system server https set cert-name <certificate name>
```
- 4 To set the HTTPS server inactivity-timeout period, use the command:

```
system server https set inactivity-timeout <MINUTES: 1..1500>
```
- 5 To set the trusted DNS name, use the command:

```
system server https set trusted-dns <a fully qualified domain name that can accept a leading wildcard period>
```
- 6 To enable or disable the web file transfer feature on the Waveserver 5 homepage:

```
system server https web-ui-file-transfer <enable |disable>
```

Note: You can display the administrative state of the web file transfer feature using the CLI command `system server https web-ui-file-transfer show`.
- 7 To revert the HTTPS server attributes to default settings, use the command:

```
system server https unset [cert-name] [inactivity-timeout] [trusted-dns]
```

Procedure 29 Configuring HTTPS server mutual authentication and OCSP

Use this procedure to configure HTTPS server mutual authentication and OCSP.

Requirements

- You must be logged in to Waveserver 5 using an account with super user access privileges.
- You have the server IP or host name and filename for the CA certificate files.
- A valid X.509 certificate has been installed on the HTTPS client, for example on your web browser.
- You have set the trusted DNS, if required. See [Procedure 28, “Configuring the HTTPS server”](#).

Steps

- 1 To install a CA certificate:

```
certificates authorities install {default-ftp-server |
default-sftp-server | default-tftp-server | default-
server | default-scp-server | sftp-server <IP address or
host name> login-id <String [1..32]> password <String
[1..128] secret <String [1..128] | tftp-server <IP
address or host name> | scp-server <IP address or host
name> login-id <String [1..32]> password <String [1..128]
secret <String [1..128] | ftp-server <IP address or host
name> login-id <String [1..32]> password <String [1..128]
secret <String [1..128] | url <String [1..128]>} filename
<String [1..127]>
```

For example:

```
certificates authorities install tftp-server 1.2.3.4
filename ca.pem
```

- | | | |
|---|--|---|
| 2 | If you want to
disable the HTTPS server mutual authentication
enable the HTTPS server mutual authentication | Then go to
step 3
step 4 |
| 3 | To disable the HTTPS server mutual authentication, use the command:
<pre>system server https mutual-authentication disable</pre> Go to step 5 . | |
| 4 | To enable the HTTPS server mutual authentication, use these commands:
<pre>system server https enable system server https mutual-authentication enable</pre> | |

5	If you want to	Then go to
	disable HTTPS server OCSP verification	step 6
	enable HTTPS server OCSP verification	step 7
6	To disable HTTPS OCSP verification: <pre>system server https ocsf disable</pre> Go to step 12 .	
7	To enable HTTPS OCSP verification, use the following CLI commands: <pre>system server https enable</pre> <pre>system server https ocsf enable</pre>	
8	Set the default OCSP responder: <pre>system server https ocsf set default-responder <OCSP responder URL[1..127]></pre> For example: <pre>system server https ocsf set default-responder http://10.10.10.10:8442</pre> or <pre>system server https ocsf set default-responder http://ocsp.mydomain.com:8442</pre>	
	Note 1: When using an IPv6 address with the default-responder <OCSP responder URL[1..127]>, the IPv6 address must be in square brackets as defined in RFC 3986 (https://tools.ietf.org/html/rfc3986). For example, <code>system server https ocsf set default-responder http://[2222:111:dddd:5444:aced::293]</code>	
	Note 2: The default value for the [default-responder] attribute is blank.	
9	Set the OCSP responder preference: <pre>system server https ocsf set responder-preference <aia default responder></pre> Note: The default value for the [responder-preference] attribute is aia.	
10	Set whether you would like the OCSP responders to contain nonce: <pre>system server https ocsf set nonce <on off></pre> Note: The default value for the [nonce] attribute is on.	
11	Optionally, retrieve HTTPS server settings: <pre>system server https show</pre>	
12	Save the provisioned settings to the configuration file: <pre>configuration save</pre>	

Procedure 30 Configuring LLDP on the client port

Use this procedure to configure LLDP on the client port.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Use the following commands to configure LLDP.
 - To globally disable or enable LLDP:
`lldp <disable | enable>`
Note: LLDP is enabled by default.
 - To set the LLDP mode for a client port:
`lldp set port <slot>/<port> mode <snoop | disable>`
Note: The default LLDP mode is snoop.
 - To save the provisioned LLDP settings to the configuration file:
`configuration save`

Procedure 31 Retrieving LLDP information on the client port

Use this procedure to retrieve data (as shown in the following example) for LLDP-enabled network devices (neighbors) connected to Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Steps

- 1 Use the following commands to retrieve data for LLDP-enabled network devices connected to Waveserver 5.
 - To view global LLDP configuration settings:
`lldp show configuration`
 - To view LLDP configuration settings for one or more ports:
`lldp show port <slot>/<port> configuration`
 - To retrieve data for an LLDP-enabled network device:
`lldp show neighbors`

Procedure 32 Retrieving Waveserver 5 TLV data on the client port

Use this procedure to retrieve data for Waveserver 5 (local bridge) TLVs.

Requirements

- You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Overview

See the tables [LLDP bridge statistics](#) and [LLDP port statistics on the client port](#).

Steps

- 1 Use the following command to retrieve Waveserver 5 TLV data.
`lldp show`

Procedure 33 Retrieving or clearing LLDP statistics on the client port

Use this procedure to retrieve or clear LLDP bridge and/or port statistics.

Requirements

- You must be logged in to Waveserver 5 using an account with at least limited access privileges to retrieve the LLDP statistics on the client port.
- You must be logged in to Waveserver 5 using an account with at least admin access privileges to clear the LLDP statistics on the client port.

Steps

- 1 Use the following commands to retrieve and/or clear LLDP bridge and/or port statistics.
 - To retrieve all LLDP bridge and port statistics:
`lldp show statistics`
 - To retrieve LLDP port statistics for one or more ports:
`lldp show port <slot>/<port> statistics`
 - To clear all LLDP statistics (bridge and port):
`lldp clear statistics`
 - To clear LLDP statistics for one or more ports:
`lldp clear <slot>/<port> statistics`

Procedure 34 Configuring LLDP on a DCN port

Use this procedure to configure LLDP on the Waveserver 5 DCN port.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Note: The default LLDP mode for the DCN-n port is disabled.

Steps

1	If you want to	Then go to
	disable LLDP peering on the DCN port	step 2
	enable LLDP peering on the DCN port	step 4
2	To globally disable LLDP.	
	<code>lldp disable</code>	
3	To disable LLDP mode for the DCN port.	
	<code>lldp set interface-port ifp0-dcn-n mode disable</code>	
	Go to step 6 .	
4	To globally enable LLDP.	
	<code>lldp enable</code>	
5	To set the LLDP mode for the DCN port.	
	<code>lldp set interface-port ifp0-dcn-n mode tx-rx</code>	
6	To save the provisioned settings to the configuration file.	
	<code>configuration save</code>	

Procedure 35 Configuring SNMP notifications for LLDP peering on a DCN port

Use this procedure to configure SNMP notifications for LLDP peering on the DCN port.

Requirements

- You must be logged in to the Waveserver 5 using an account with at least admin access privileges.

Note: The default for LLDP notifications on the DCN-n port is off.

Steps

- 1 **If** you want to set SNMP notifications for the DCN port **Then** go to
to be

off	step 2
on	step 3
- 2 To set SNMP notifications for the DCN port to be off.
`lldp set interface-port ifp0-dcn-n notification off`
Go to [step 4](#).
- 3 To set SNMP notifications for the DCN port to be on.
`lldp set interface-port ifp0-dcn-n notification on`
- 4 To save the provisioned settings to the configuration file.
`configuration save`

Procedure 36 Retrieving information for LLDP on a DCN port

Use this procedure to retrieve LLDP information for LLDP peering on the DCN port.

Requirements

- You must be logged in to the Waveserver 5 using an account with at least limited access privileges.

Steps

- 1 Use the following commands to retrieve LLDP information for LLDP peering on the DCN port.
 - To display summary information about the LLDP peering on the DCN port:

```
lldp show interface-port ifp0-dcn-n
```
 - To display the configuration information about the LLDP peering on the DCN port:

```
lldp show interface-port ifp0-dcn-n configuration
```
 - To display the DCN port's remote neighbor:

```
lldp show interface-port ifp0-dcn-n neighbors
```

Procedure 37 Retrieving or clearing LLDP statistics on a DCN port

Use this procedure for retrieving or clearing statistics for LLDP peering on a DCN port.

Requirements

- You must be logged in to the Waveserver 5 using an account with at least limited access privileges to retrieve the LLDP statistics on a DCN port.
- You must be logged in to the Waveserver 5 using an account with at least admin access privileges to clear the LLDP statistics on a DCN port.

Steps

- 1 Use the following commands to retrieve or clear statistics for LLDP peering on a DCN port.
 - To retrieve statistics for LLDP peering on a DCN port:
`lldp show interface-port ifp0-dcn-n statistics`
 - To clear LLDP statistics for the DCN port:
`lldp clear interface-port ifp0-dcn-n statistics`

CHAPTER 14

User accounts, user authentication and security

This section contains the following topics:

- [Overview](#)
 - [User account types](#)
 - [User authentication](#)
 - [Local user account authentication \(default\)](#)
 - [RADIUS](#)
 - [RADSec](#)
 - [TACACS+](#)
 - [Access Control List](#)
 - [User lockout](#)
 - [Public Key Infrastructure \(PKI\) and X.509](#)
 - [X.509 certificates](#)
 - [SSH algorithms selection](#)
 - [TLS algorithms selection](#)
 - [Reset to Factory Defaults \(RTFD\)](#)
- [Session management](#)
- [Procedure list](#)

Overview

This section provides information on user accounts, user authentication, and security features available on Waveserver 5.

User account types

Waveserver 5 supports user account types: super user, admin and limited. This section provides details about the functionality of the user account types and the associated access privileges.

Table 86 User account types

Account	Capabilities	Limitations
Super user (See Note 1)	• Create, delete, modify, display user accounts. See Creating local user accounts. • Change passwords on any user account. • Change the Linux root account password. See Changing the Linux root account password. • Upload SSH server keys and SSL certificates • Configure and manage RADSec, RADIUS, TACACS+, and HTTPS. • Manage PKIX X.509v3 certificates. • Can provision up to 10 additional CLI SSH sessions, for a total of 11 CLI SSH sessions that can be active simultaneously. (The default is 5 simultaneously active CLI SSH sessions.) You can also open 10 NETCONF SSH super user sessions simultaneously with the CLI super user sessions. See Note 2.	• Cannot view or execute commands from the command-log for which you do not have execute privileges. • Cannot exceed 11 simultaneous active SSH sessions for a super user.
Admin user	• Manage system state changes, modify system configuration, and execute commands. • Change own account password. • View all user accounts at the admin user or limited user privilege level. • Manage Syslog, Syslog TLS, NTP, and NTPv4 Autokey authentication. • Can provision up to 9 additional CLI SSH sessions, for a total of 10 CLI SSH sessions that can be active simultaneously. (The default is 5 simultaneously active SSH sessions.) You can also open 10 NETCONF SSH admin user sessions simultaneously with the CLI admin user sessions. See Note 2.	• Cannot view or execute commands from the command-log for which you do not have execute privileges. • Cannot change the password of other user accounts. • Cannot create, edit, or delete information in other user accounts. • Cannot access Linux system security files, Linux password and shadow files, or the SSH and SSL server keys. • Cannot exceed 10 simultaneous active SSH sessions for an admin user.

Account	Capabilities	Limitations
Limited user	• System monitoring and gathering information about the system configuration and performance. • Change own account password. See Note 3. • Can provision up to 9 additional CLI SSH sessions, for a total of 10 CLI SSH sessions that can be active simultaneously. (The default is 5 simultaneously active SSH sessions.) You can also open 10 NETCONF SSH limited user sessions simultaneously with the CLI limited user sessions. See Note 2.	• Cannot view or execute commands from the command-log for which you do not have execute privileges. • Cannot change the password of other user accounts. • Cannot create, edit, or delete information in other user accounts. • Cannot access Linux system security files, Linux password and shadow files, or the SSH and SSL server keys. • Cannot exceed 10 simultaneous active SSH sessions for a limited user.

Note 1: The CLI prompt for sessions started by someone with super user privileges contains the hash or pound character (for example: `Waveserver#`).

Note 2: Waveserver 5 supports a total maximum of 21 simultaneous SSH login sessions. This count includes access using SSH (CLI and NETCONF interfaces), but does not include user login using the Waveserver 5 console port. For a procedure to change this limit for individual accounts, see [Setting login limits and minimum password lengths on user accounts](#).

Note 3: A limited user cannot save the configuration file, therefore a password change by a limited user will revert to the old password if there is a cold restart. To make a password change permanent, limited users should contact an admin user or a super user to save the configuration.

User authentication

Waveserver 5 supports the following methods for authenticating users:

- [Local user account authentication \(default\)](#)
- [RADIUS](#)
- [RADSec](#)
- [TACACS+](#)

You can configure Waveserver to use RADIUS, RADSec, and TACACS+ services that are either already available on the management network or configured directly on the device. RADIUS, RADSec and TACACS+ support IPv4 and IPv6.

Local user account authentication (default)

Local user account authentication uses a username and password as the default method for logging into Waveserver 5. A username and password is managed locally at each Waveserver 5.

RADIUS

Remote Authentication Dial-In User Service (RADIUS) is a client/server system used to secure networks against unauthorized remote access. When authenticating a user, Waveserver 5 sends authentication requests to a RADIUS server or servers, which keep track of all user authentication and service access information. A RADIUS server returns authentication results to Waveserver 5, and the user is either allowed or denied access based on this information.

RADIUS authentication is enabled by default on Waveserver 5, which can support up to eight RADIUS servers. You must specify a priority for each RADIUS server that you add to the Waveserver configuration. The priority value for a server is in the range of 1 to 8, with 1 being the highest priority.

You cannot assign the same priority value to two RADIUS servers of the same type in the network. You do not need to assign priority values in consecutive order. For example, in a network with two RADIUS authentication servers, you can assign one RADIUS server priority 2, and the backup RADIUS server priority 8, which leaves priority values 1 and 3 to 7 available to be assigned to other RADIUS servers.

For more information, see [Procedure 47, “Configuring RADIUS authentication”](#).

RADIUS Vendor-Specific Attributes (VSA)

The Internet Engineering Task Force (IETF) draft standard specifies a method for communicating vendor-specific information between the network access device and the RADIUS server. By using attribute 26, the vendor-specific attribute, vendors can support their own extended attributes otherwise not suitable for general use.

The Ciena RADIUS client supports attribute 26, whose elements can be used to define the user privilege level assigned to the user being authenticated:

- vendor-id (1271 = Ciena)
- vendor-type (10 = User Privilege Level)
- vendor-len (6)

- vendor-data (4-byte integer from 1 to 3; see the following table)

Table 87 VSA mappings

User group	Value
super	3
admin	2
limited	1

Note: Use `dictionary.ciena` as the filename of the dictionary file. Ciena does not supply a dictionary file. You must add entries to the dictionary file for your specific RADIUS or TACACS+ servers.

RADSec

RADSec provides a transport profile for RADIUS using Transport Layer Security (TLS) over Transmission Control Protocol (TCP). Using the RADSec protocol enables dynamic trust between RADIUS servers.

There are two types of authentication with RADSec: end user authentication and TLS authentication.

End user authentication

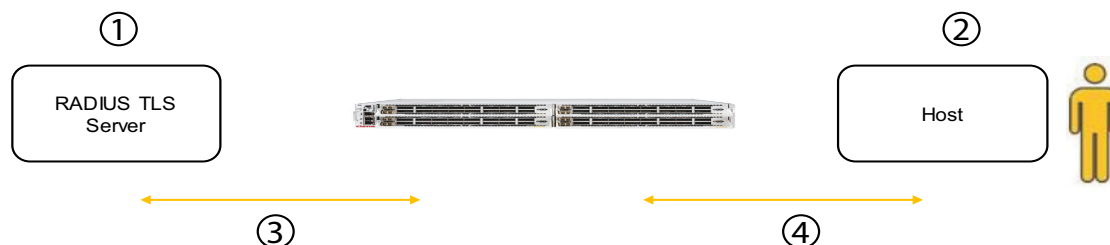
End user authentication is used for logging into a Waveserver 5. It verifies the identity of a user attempting to gain access to the management functionalities.

TLS authentication

TLS connection is established between the Waveserver 5's RADIUS client and external RADSec server. The TLS authentication verifies the peer identity and the trust relationship using X.509v3 certificates. For more details on TLS and X.509v3 certificates, see [TLS algorithms selection](#) and [X.509 certificate for TLS](#).

End user authentication with RADIUS TLS server

A TLS connection is initiated from Waveserver 5 to the RADSec server. Waveserver 5 and the RADSec server then perform TLS authentication and establish a TLS connection. Once the TLS connection is established, RADIUS messages are exchanged through the TLS connection for end user authentication, as shown in the following figure.

Figure 61 End user authentication with RADSec server**Legend**

- | | | | |
|---|-------------------------|---|--------------------|
| 1 | End User Authentication | 3 | TLS Authentication |
| 2 | User login | 4 | Console port, SSH |

RADSec provisioning rules and considerations

The following RADSec provisioning rules and considerations apply:

- Device certificates with private keys and signing CA certificates must be configured on both the Waveserver 5 and RADSec server.
- Waveserver 5 supports a single device certificate and private key for all RADSec operations.
- Up to 8 RADSec servers can be configured on the Waveserver 5.
- The IP configuration of the RADSec Server must be configured on the Waveserver 5. You must specify a priority for a new RADSec server that you add to the Waveserver configuration. The priority value for a server is in the range of 1 to 8, with 1 being the highest priority. You cannot assign the same priority value to two RADSec servers in the network. You do not need to assign priority values in consecutive order. For example, in a network with two RADSec servers, you can assign priority value 2 to one server and priority value 8 to the other server, which leaves the unused priority values available to be assigned to future RADSec servers.
- For RADSec connections, TLS mutual authentication with X.509v3 certificates using PKIX trust models are supported as per RFC6614. Each device is identified by the tuple (the certificate issuer and the serial number of the presented client certificate).
- RADSec connections use TLS 1.2 and support IPv4 and IPv6 addresses. Waveserver 5 also supports resolution of the RADSec server host name to use either IPv4 and IPv6 addresses.
- If the RADSec server host name has not been resolved, the RADSec server's operational state is disabled. If the RADSec server's administrative state is enabled and the server host name has been resolved, the RADSec server's operational state is enabled. To display the RADSec server's operational state, use CLI command `radsec show`.

- If the check-ip-host option is enabled, Waveserver 5 checks the host name that was configured when the server was added to RADSec against the Subject Alternative Name (SAN) on the certificate that is presented by the server during the TLS connection process. compares the literal HostName string to the SAN and does not expand it to a Fully Qualified Domain Name (FQDN) if an IP address or a Partially Qualified Domain Name (PQDN) was used to add the server.
- If the server HostName string does not match the SAN, you can provision a Trusted DNS Name using the command `radsec set server trusted-dns`. Waveserver 5 then checks the Trusted DNS Name against the SAN.
- As per RFC6614, when connecting to a RADSec server, the RADIUS shared secret (key) is fixed to “radsec”.
- RADSec only uses the enabled TLS algorithms. To display enabled TLS algorithms, use the CLI command `ssl algorithm show`. The highest priority for TLS cipher suite negotiation is `TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384`. For more details on TLS algorithms, see [TLS algorithms selection](#).
- You can display the RADSec server statistics using CLI command `radsec show statistics`.
- All successful and unsuccessful TLS authentications with a RADSec server are logged in syslog.
- Waveserver 5 supports RADSec management through CLI, NETCONF, REST, and WebSocket (event notifications).

RADSec provisionable attributes

The following table shows the configurable RADSec and RADSec OCSP attributes on Waveserver 5. For more information on available RADSec commands, see [RADSec commands](#).

Table 88 RADSec provisionable attributes

Attribute (Note)	Description	Values	Default
Certificate name	Certificate to use.	String	blank
Check fingerprint	Determines whether the fingerprint of the peer's certificate is checked.	• on • off	off
Check IP host	Determines whether the peer's specified IP address or hostname is checked against the Subject Alternative Names (SAN) or Common Names (CN) field of its certificate.	• on • off	on

Attribute (Note)	Description	Values	Default
Fingerprint	A fingerprint of the peer's certificate 95 characters long. For example, 00:11:22:33:44:55:66:77:88:99:AA:BB:CC:DD:EE:FF:A1:B1:C1:D1:E1:F1:1A:1B:1C:1D:1E:1F:2A:2B:2C:2D.	32 byte SHA-256 fingerprint	Blank
Port	RADSec port.	Number: 1..65535	2083
Priority	Priority of the RADSec server.	Number: 1..8 Note: 1 is the highest priority. You must assign a priority. You cannot assign the same priority value to two RADSec servers in the network. You do not need to assign priority values in consecutive order.	1
Search method	Search method for RADSec server.	- cached - priority	priority
Server	RADSec server IP address or host name.	IPv4/IPv6 IP address or host name	blank
Timeout	Determines the time allowed for a RADSec connection to be made.	Seconds: 1..30	6
Trusted DNS	If the check-ip-host is on and the trusted DNS is configured, Waveserver 5 checks whether the peer's domain is in the SAN or CN is in the Trusted DNS Name in the configuration.	String Note: Must be a Fully Qualified Domain Name (FQDN) that can accept a leading wildcard period.	blank
Admin State	Sets the administrative state (enabled or disabled) for the RADSec server Online Certificate Status Protocol (OCSP). OCSP performs real-time certificate status checks when validating a RADSec server's X.509 certificate.	- enabled - disabled	disabled

Attribute (Note)	Description	Values	Default
Default Responder	Sets the default responder for the RADSec server Online Certificate Status Protocol (OCSP). URL of the default OCSP responder starting with http://	- String: 1..255 - blank	blank
Nonce	Determines whether nonce is sent with an OCSP request.	- on - off	on
Responder Preference	Sets the OCSP responder to try and stop after the first aia/default responder.	- default-responder - aia	aia
Note: Fingerprint, Priority, Port and Trusted-DNS attributes are assigned per RADSec server. The other attributes within this table are shared globally by all RADSec servers.			

To configure RADSec, see [Configuring RADSec authentication](#).

TACACS+

Terminal Access Controller Access Control System Plus (TACACS+) is a set of security protocols that controls and monitors access to Waveserver nodes in your network. TACACS+ separates the functions of authentication, authorization, and accounting (AAA) into individual services that can run on separate hosts. For example, you can designate servers in your network to provide a single AAA service, or you can designate global servers that provide one or more AAA services. When you deploy TACACS+, the authentication service is globally enabled by default.

TACACS+ uses a client-server model to provide AAA services to Waveserver 5 requests. Communication between Waveserver 5 and TACACS+ is made secure by a shared key to encrypt information.

Waveserver 5 supports TACACS+ AAA services on CLI, OpenConfig, and gRPC (gNMI/gNOI) commands. You can configure TACACS+ servers using CLI, Ciena YANG (ciena-waveserver-aaa), or OpenConfig YANG (openconfig-aaa-tacacs). These YANG models are available on the Waveserver 5 home page, either to work with directly using a REST API (Swagger) or to download and use with a third-party RESTconf client tool. For details on how to access the interfaces available on the Waveserver 5 home page, see [Waveserver 5 Web User Interface](#). For details on the OpenConfig and gRPC (gNMI/gNOI) northbound interfaces, see [REST](#), [NETCONF](#), [gRPC](#), [OpenConfig](#), and [WebSocket](#)

TACACS+ servers: unique settings and global settings

You can configure global attributes for TACACS+ servers, including: a non-encrypted shared key (key) used to generate a shared secret between Waveserver nodes and TACACS+ servers; an encrypted, shared-secret key (secret) to ensure secure transmission of TACACS+ packets between Waveserver nodes and TACACS+ servers; and, a timeout value that specifies how long a Waveserver node waits for a response from a TACACS+ server. You can now configure these attributes (key, secret, and timeout) with unique values at the individual server level, instead of globally, for additional security. For details on configuring the servers, see [Procedure 49, “Configuring TACACS+”](#).

The following table describes the TACACS+ AAA services.

Note: TACACS+ is not compatible with TACACS and XTACACS protocols.

Table 89 Summary of TACACS+ services

Service	Description
Authentication	Controls log-in access to Waveserver 5. • Authentication is globally enabled by default. • Authentication services must be enabled to use authorization or accounting services. • Authentication fails if any of these conditions are true: — TACACS+ or TACACS+ authentication is globally disabled. — Waveserver 5 is not configured to use TACACS+ to authenticate the login source (remote, all, serial). — The TACACS+ key is not configured or configured incorrectly. — The TACACS+ TCP port number is configured incorrectly. — The user account is not recognized by the authentication server. See Configuring TACACS+ and Configuring TACACS+ authentication servers.
Authorization	Controls access to commands based on user or group access privileges specified on the TACACS+ server. (See Privilege levels and User account types.) • Authorization service is disabled by default. • When authorization is enabled and TACACS+ servers are configured: — TACACS+ performs authorization only if user-authentication is done by TACACS+. — Waveserver 5 authorizes each attempt to execute a command with the server. Note: Authorization is done on user sessions that start after you enable authorization. To enable authorization on sessions that were active before you started the service, you must log out and log in again using TACACS+ authentication. See Configuring TACACS+ and Configuring TACACS+ authorization servers.

Service	Description
Accounting	Records user actions performed on Waveserver 5 for security audits or billing. - Accounting service is disabled by default. - When accounting is enabled and TACACS+ servers are configured, TACACS+ performs accounting only if user-authentication is done by TACACS+. See Configuring TACACS+ and Configuring TACACS+ accounting servers.
Note: For a description of the CLI commands to configure TACACS+, see TACACS+ commands .	

TACACS+ servers

You can designate up to eight servers for each AAA service and up to eight global servers. Services are performed by the corresponding TACACS+ component on an AAA server. If AAA servers are not configured or are inaccessible, the global server list is used.

Server priority is configurable (1= highest) and determines the order that clients contact TACACS+ servers. You must specify a priority for a new TACACS+ server that you add to the Waveserver configuration. The priority value for a server is in the range of 1 to 8.

You cannot assign the same priority value to two TACACS+ servers of the same type in the network. You do not need to assign priority values in consecutive order. For example, in a network with two TACACS+ servers, you can assign one server priority 2, and the backup TACACS+ server priority 8, which leaves priority values 1 and 3 to 7 available to be assigned to other servers. You can search for servers by priority or cached value (last accessed). The AAA services also maintain statistics that you can retrieve or clear.

To add a TACACS+ server to Waveserver 5, specify:

- the IP address and TCP port number (default port number is 49)
- an unencrypted shared key value
- a timeout value
- user privilege level
- priority

See [Configuring TACACS+](#) and [Configuring the global servers list](#).

Authorization and accounting of non-CLI commands

TACACS+ authorization and accounting applies to commands issued in the REST, NETCONF, gNMI, and gNOI interfaces. For commands issued in the CLI, the full validated command is used for TACACS+ authorization and accounting. No TACACS+ authorization or accounting is done for WebUI HTTP requests, WebSocket, and SNMP interfaces.

TACACS+ authorization of a user request on a non-CLI interface first authorizes the requested operation. You can configure regular expressions to perform additional authorization based on the data itself.

TACACS+ accounting of a user request on a non-CLI interface records the requested operation only, not the data the request contains.

When authorization and accounting are enabled, the NETCONF, REST, gNMI, and gNOI user commands listed in this table are sent to a TACACS+ server for authorization and accounting. They are also displayed in the local command-log.

Table 90 Non-CLI commands sent to TACACS+ servers for authorization and accounting

Interface	Commands	
NETCONF	NETCONF.edit-config NETCONF.get-config NETCONF.get NETCONF.copy-config NETCONF.delete-config NETCONF.lock NETCONF.unlock NETCONF.close-session NETCONF.kill-session NETCONF.validate	NETCONF.commit NETCONF.discard-changes NETCONF.cancel-commit NETCONF.get-schema NETCONF.establish-subscription NETCONF.delete-subscription NETCONF.kill-subscription NETCONF.resync-subscription NETCONF.rpc.<rpc-name> where <rpc-name> is a supported RPC defined in a YANG module (e.g., waveserver-configuration-save)
gNMI	gNMI.Set gNMI.Get gNMI.Subscribe gNMI.Capabilities	
gNOI	gNOI.System.Reboot gNOI.System.RebootStatus gNOI.System.Time gNOI.System.SetPackage gNOI.File.Get	gNOI.File.TransferToRemote gNOI.File.Put gNOI.File.Stat gNOI.File.Remove gNOI.FactoryReset.Start
REST	REST.GET <uri> REST.PUT <uri> REST.PATCH <uri> REST.DELETE <uri> REST.HEAD <uri>	REST.OPTIONS <uri> REST.POST <uri> - where <uri> is a data target (i.e. /waveserver-aaa/user), not an rpc target REST.POST.<rpc-name> - where <rpc-name> is a supported RPC defined in a YANG module (e.g., waveserver-configuration-save)

Privilege levels

The privilege level of a TACACS-authenticated user is derived by mapping the TACACS+ privilege level to the supported Waveserver 5 user level mapping.

Table 91 TACACS+ privilege mapping to default CLI user groups

TACACS+ Privilege	TACACS+ Numerical Levels	Default TACACS+ Numerical Levels	CLI User Group
Super	2-15	10	super
Admin	1-13	2	admin
Read	0-1	—	limited
Note: The value assigned to the Super privilege must be greater than the value assigned to the Admin privilege.			

Access Control List

The Access Control List (ACL) feature adds filtering to ingress and egress traffic on a physical interface for both IPv4 and IPv6 traffic. This functionality adds an additional layer of security and lowers the risk of unauthorized access. The filtering rules determine whether incoming or outgoing DCN traffic is allowed or dropped based on a combination of factors, including: destination address and port; source address; protocol; and, interface.

Currently, ACL filtering rules are supported on externally facing DCN ports only. To apply ACL rules to a remote Waveserver 5 node or a node that is connected by ILAN, you must apply the ACL rules to all head-end nodes with a DCN connection.

ACL filtering rules are applied to DCN traffic regardless of whether the Waveserver is connected to the DCN through:

- a direct Ethernet-cable connection
- an ILAN-cable connection to another Waveserver 5 node that is connected to the DCN
- an overhead communication channel

Note 1: In BGP mode, create ACL rules for both the if0-peer-1 and if0-peer-2 interface instead of the local interface.

Note 2: When you create ACL rules on a Waveserver 5 node that is configured as a Gateway Network Element (GNE) and the rules specify an ingress ACL destination address of 0.0.0.0 (apply to all), those ACL rules can also be applied to subtending nodes managed by the GNE through ILAN. To limit the scope of the ACL rules to the GNE, set the ingress ACL destination address to the IP address of the GNE.

After recovery from a cold or warm boot, ACL rules are immediately applied to Waveserver interfaces, with no window of opportunity for non-matched packets to get through the data communications firewall.

Incoming traffic is processed based on the Waveserver 5 default firewall setting. You can override that behavior by adding your own user-defined ACL rules. By default, when traffic does not match any ACL rule, the traffic passes through. To drop packets that do not match any ACL rule, you must add a “drop-all” rule to the end of the list of ACL rules. For examples, see the last rule in each configuration scenario in [Waveserver 5 default firewall setting](#).

Note: An ACL rule only affects the chassis on which it is provisioned. It does not affect management traffic to other nodes.

Each Waveserver 5 supports an ACL that allows you to provision up to 100 IPv4 filter rules and 100 IPv6 filter rules to allow or drop in-bound IP packets from a specified:

- egress / ingress
- protocol
- source IP/mask
- source port
- destination IP/mask
- destination port
- interface name

The syntax of the `interface acl` command is shown below:

```
interface acl <add | delete> <egress | ingress> <ipv4 | ipv6> action <allow | drop> [destination-address <String>] [interface-name <String>] [protocol <tcp | udp | icmp | ipv6-icmp | all | 0 to 255>] [sequence-id <1..50>] [source-address <String>]
```

The following table describes the parameters and values that can be specified with the `interface acl` command for creating ACL rules.

Table 92 Options for creating ACL rules

Parameter	Details	Requirement	Values	Default
[rule type]		mandatory	egress ingress	no default
[stack]		mandatory	ipv4 ipv6	no default
action		mandatory	allow drop	no default

Parameter	Details	Requirement	Values	Default
destination-address	ip address/mask	optional	IPv4: • CIDR format (x.x.x.x/x) • mask is optional and has a range of 0-32 • when mask is not specified, the rule is applied to a single IP address • to specify any/all IPv4 addresses, use 0.0.0.0/0 IPv6: • CIDR format (specified in hexadecimal and using 16-bit values between colons) • mask is optional and has a range of 0-128 • when mask is not specified, the rule is applied to a single IP address • to specify any/all IPv6 addresses, use ::/0	IPv4: 0.0.0.0/0 IPv6: 0.0.0.0.0.0.0.0/0
interface-name		optional	Name of the logical management interface	no default
protocol	protocol type	optional	• decimal: 0-255 (Note) • TCP • UDP • ICMP • IPv6-ICMP • all (all protocols, not just TCP, UDP, and ICMP)	all
sequence-id	integer	mandatory	Integer in the range 1 to 50, inclusive. Specifies the order in which the rule is applied. You do not need to assign sequence IDs in consecutive order. For example, the first ACL rule can be sequence ID 2 and the second ACL rule can be sequence ID 9. The value of sequence-id must be unique; you cannot assign the same sequence-id to more than one ACL rule.	The next available ID number.

Parameter	Details	Requirement	Values	Default
source-address	ip address/mask	mandatory	IPv4: • CIDR format (x.x.x.x/x) • mask is optional and has a range of 0-32 • when mask is not specified, the rule is applied to a single IP address • to specify any/all IPv4 addresses, use 0.0.0.0/0 IPv6: • CIDR format (specified in hexadecimal and using 16-bit values between colons) • mask is optional and has a range of 0-128 • when mask is not specified, the rule is applied to a single IP address • to specify any/all IPv6 addresses, use ::/0	no default
Note: Commonly used protocols are: 1-IPv4 ICMP, 58-IPv6 ICMP, 6-TCP, and 17-UDP				

For a full list of TCP and UDP port numbers, see <http://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml>.

For example parameter configurations, see [Waveserver 5 ACL example scenarios](#).

Rules are maintained in the order they are created and take effect immediately after creation. When you create a new ACL rule, it is added to the end of the list of existing ACL rules. To change the list order, see [Modifying an ACL list](#).

Packet matching starts from the first ACL rule. When a matching rule is found, all remaining rules are ignored. Waveserver then processes the packet according to the action defined in the matching rule (allow or drop). In the case where an IP datagram is fragmented, the allow/drop action is applied to the first and all subsequent fragmented packets.

Waveserver 5 default firewall setting

User-defined ACL rules have a higher priority than the Waveserver 5 default firewall setting. If the last ACL rule drops all unmatched packets, it can cause communication problems on internal features that require specific port(s) to be open on the firewall.

Waveserver 5 maintains ACL user settings in the configuration file. If you would like to back up and restore these settings, refer to [Configuration management](#).

The following two tables list the Waveserver 5 services that need to be considered when configuring ACL rules.

Note 1: The SSH server and the SFTP client are independent. If you are using ACL and a non-default port for the SSH server, you must add separate ACL rules for the non-default SSH server port and services related to the default port 22, such as the SFTP or SCP client.

Note 2: To enable traceroute to work over UDP, you must create an ACL rule to do one of the following:

- allow all incoming UDP traffic from a specific device IP address.
- specify each individual UDP port (UDP ports 33434 to 33523) on the Waveserver to be opened.
- allow all incoming traffic from all devices from a specific subnet. To change the default firewall settings by creating an ACL rule, see [Creating an ACL rule](#).

Table 93 Waveserver 5 service considerations for configuring ACL rules for incoming requests

Service provided (for incoming requests)	Protocol	Service Port (Note)	Notes
ICMP	ICMP	N/A	
CLI over SSH and SFTP	TCP	22	Always enabled. Port 22 is the default, but you can configure other ports.
SNMP	UDP	161	Disabled by default, can be enabled See snmp <disable enable>
HTTPS	TCP	443	Enabled by default, can be disabled See system server https <enable disable>
NETCONF over SSH	TCP	830	Enabled by default, can be disabled See system server netconf <enable disable>
SCP	TCP	20022	Enabled by default, can be disabled by ACL rule.
SPLI over SSH	TCP	20040	Enabled by default, can be disabled See spli <disable enable>

Service provided (for incoming requests)	Protocol	Service Port (Note)	Notes
Traceroute over UDP	UDP	33434 to 33523	Waveserver default firewall settings block traceroute over UDP. To enable traceroute to work over UDP, you must create an ACL rule to do one of the following: • allow all incoming UDP traffic from a specific device IP address. • specify each individual UDP port (UDP ports 33434 to 33523) on the Waveserver to be opened. • allow all incoming traffic from all devices from a specific subnet. To change the default firewall settings by creating an ACL rule, see “Creating an ACL rule” on page 317.
Note: The service ports listed in this table are the default service ports. You can provision some services to use a service port other than the default.			

Table 94 Waveserver 5 service considerations for configuring ACL rules for outgoing requests and incoming responses

Service needed (for outgoing requests, incoming responses)	Protocol	Service Port (Note)	Notes
ICMP	ICMP	N/A	
FTP	TCP	21	FTP client on Waveserver for file transfer. Waveserver 5 uses passive FTP mode to send requests to TCP port 21 on the FTP server for the FTP command channel. A dynamic TCP port number is used for the FTP data channel. Ciena recommends you add an ACL rule to allow all incoming TCP traffic from the FTP server if FTP is required. Secured File Transfer Protocol (SFTP) is recommended for file transfer.
SSH/SFTP	TCP	22	Waveserver SSH client connects to other boxes and SFTP clients on Waveserver for file transfer. Port 22 is the default, but you can configure other ports.
SCP	TCP	22	SCP client uses SSH for file transfer. Port 22 is the default, but you can configure other ports.
TACACS+	TCP	49	TACACS+ client on Waveserver. Port 49 is the default but you can configure other ports.
DNS	UDP	53	DNS client on Waveserver to resolve DNS name.

Service needed (for outgoing requests, incoming responses)	Protocol	Service Port (Note)	Notes
DHCP	UDP	67/68	Client uses 68, server uses 67. The Waveserver DHCP client implementation uses raw sockets that bypass firewall screening. To block DHCP, use the command: dhcp client disable.
TFTP	UDP	69	TFTP client on Waveserver for file transfer. Similar to FTP, the TFTP file transfer uses dynamic port numbers. To allow or drop TFTP traffic, the recommended ACL rule will allow or drop the entire TFTP server for individual source-IP addresses.
NTP	UDP	123	NTP client requests NTP time from server.
Syslog	UDP	514	Syslog client sends logs to syslog server, outgoing one way. Port 514 is the default, but you can configure a different port.
TLS Syslog	TCP	6514	Syslog client sends logs to tls syslog server, outgoing one way. Port 6514 is the default, but you can configure a different port.
RADIUS	UDP	1812	RADIUS client on Waveserver. Port 1812 is the default, but you can configure other ports.
RADSec	TCP	2083	RADSec client on Waveserver. Port 2083 is the default, but you can configure other ports
Note: The service ports listed in this table are the default service ports. You can provision some services to use a service port other than the default.			

ACL provisioning is completed using the interface acl commands;

- `interface acl show [stack]`, see [Listing all ACL rules](#)
- `interface acl add [stack]`, see [Creating an ACL rule](#)
- `interface acl delete [stack]`, see [Deleting ACL rules](#)

Waveserver 5 ACL example scenarios

The examples below use IPv4, but the same settings can be used for IPv6, using a source-address of `::/0` instead of `0.0.0.0/0`.

Configuration 1: Only allow CLI over SSH from certain subnet and DNS clients

```
interface acl add ingress ipv4 source-address <ip/mask>
protocol tcp destination-port 22 sequence-id 1 action
allow
```

```
interface acl add ingress ipv4 source-address 0.0.0.0/0
protocol udp source-port 53 sequence-id 2 action allow
```

```
interface acl add ingress ipv4 source-address 0.0.0.0/0
sequence-id 3 action drop
```

Configuration 2: Only allow CLI from one IP and drop everything else

```
interface acl add ingress ipv4 source-address <ip>
protocol tcp destination-port 22 sequence-id 1 action
allow
```

```
interface acl add ingress ipv4 source-address 0.0.0.0/0
sequence-id 2 action drop
```

Configuration 3: Only allow CLI from one IP, ICMP and DNS/DHCP/SSH/SFTP responses from a subnet and then drop everything else

```
interface acl add ingress ipv4 source-address <ip>
protocol tcp destination-port 22 sequence-id 10 action
allow
```

```
interface acl add ingress ipv4 source-address 0.0.0.0/0
protocol icmp sequence-id 11 action allow
```

```
interface acl add ingress ipv4 source-address <ip/mask>
protocol udp source-port 53 sequence-id 12 action allow
```

```
interface acl add ingress ipv4 source-address <ip/mask>
protocol udp source-port 67 sequence-id 13 action allow
```

```
interface acl add ingress ipv4 source-address <ip/mask>
protocol tcp source-port 22 sequence-id 14 action allow
```

```
interface acl add ingress ipv4 source-address 0.0.0.0/0
sequence-id 15 action drop
```

Configuration 4: Only allow CLI from one IP, RADIUS client, and then drop everything else

```
interface acl add ingress ipv4 source-address <ip>
protocol tcp destination-port 22 sequence-id 20 action
allow
```

```
interface acl add ingress ipv4 source-address 0.0.0.0/0
protocol udp source-port 1812 sequence-id 21 action allow
```

```
interface acl add ingress ipv4 source-address 0.0.0.0/0
sequence-id 22 action drop
```

Configuration 5: Do not allow CLI to one IP and allow everything else

```
interface acl add egress ipv4 source-address 0.0.0.0/0
destination-address <ip> protocol all sequence-id 1
action drop
```

User lockout

The user lockout feature disables a user account when it reaches a specified number of unsuccessful login attempts. The user lockout feature can be used to disable a user account under the following conditions:

- the user account exists in the local database
- it is a remote login attempt (not a local console port login)
- the number of login failures due to an incorrect password exceeds the max-login-attempt threshold

Note 1: If a user with super user access privileges is using the console port, the user lockout feature does not apply. For example, the super user can login at any time, using the console port and the correct password.

Note 2: Performing a warm or cold restart on Waveserver 5 clears the account lockout status.

The user is locked out of the Waveserver 5 for a specified duration (lockout interval) when the maximum number of unsuccessful login attempts (max-login-attempt) is reached. Any login attempt when the lockout interval is active renews the lockout interval timer for that account. Users are not provided with any indication that their account is locked out.

After the lockout interval expires, Waveserver 5 allows the user one attempt to login with a correct password. If the password is incorrect, the user is locked out again for the lockout interval time. After a successful login, the failed-login-attempt counter is reset to 0 and the account is set to an unlocked state.

The maximum number of login attempts for a user and the lockout interval can be configured. To set the maximum number of login attempts and to define a lockout interval, use the CLI command `user set [max-login-attempt <Number: 2..10>] [lockout-interval <Number: <1..30>]`. Increasing or decreasing the max-login-attempt immediately sets a new threshold for the failed-login attempt counter. Increasing or decreasing the lockout-interval or system time may cause the lockout-interval timer to expire. See the following table.

Table 95 User security parameters

Parameter	Description	Values
max-login-attempt	The maximum number of login attempts before the user is locked out of the Waveserver 5.	2 to 10, 5 (default)
lockout-interval (minutes)	The duration of the lockout time, in minutes.	1 to 30, 10 (default)

To display the status of locked out accounts, use CLI command `user show`. See the following table.

Table 96 User lockout status

Parameter	Description
User Name	User account that is currently locked out.
Failures	The number of failed login attempts. Failures are reset to 0, once the user is able to successfully login or after a Waveserver 5 restart.
Latest Failure	The date and time in which the last failed login attempt occurred.

The user lockout functionality is only applicable to user accounts that exist in the local database. The user lockout does not apply to external user accounts, such as RADIUS, RADSec, or TACACS accounts. If there is a user account with the same name as an external user account, the Waveserver 5 will enforce the lockout functionality, even if the login is externally authenticated.

The Waveserver 5 homepage can be accessed by entering `https://<Waveserver IP address>:443/` in to a web browser. If a user attempts to access the Waveserver 5 homepage and they have reached the maximum number of login attempts, that user account will be locked out.

Installing and naming certificate authorities

You are now required to specify a name for the certificate authorities (CA) when you install them. Use the certificate name parameter (`cert-name`) of the `certificates authorities` commands to assign a name to the CA. You can refer to the CA using the `cert-name` parameter to retrieve and display information about the CA and to uninstall the CA. After you install the CA, you can rename it by assigning a new value to the `cert-name` parameter.

Public Key Infrastructure (PKI) and X.509

A public-key infrastructure (PKI) is a set of hardware, software, people, and policies required to create, manage, distribute, use, store, and revoke digital certificates. Waveserver 5 supports Public Key Infrastructure with Public-Key Cryptography Standards #12 format V3 (PKCS#12 format V3).

Waveserver 5 PKI uses asymmetric cryptography where a private and a public key is used. The private and public keys are mathematically linked such that messages encrypted with either of these keys can be decrypted with the other.

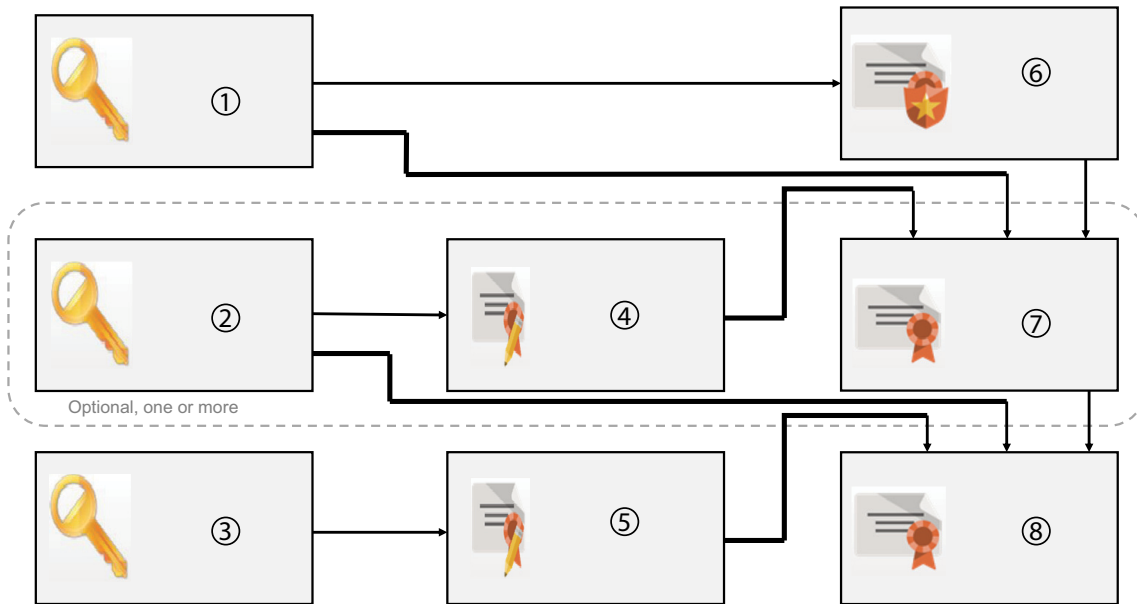
The public key can be used to create messages that can only be decrypted by the owner of the private key, and the private key owner is the only one that can create messages that can be decrypted with the public key. This can be used to verify the identity of the private key owner, as well as communicate securely with the private key owner.

Waveserver 5 allows you to install (enroll) trusted CA and Entity certificates using PKIX. Waveserver 5 verifies the trusted CA certificates when the remote peer presents its certificate, for example using a TLS connection. You can assign an installed Entity certificate to one or multiple applications, including RADSec, TLS syslog, HTTPS server, or gRPC server using mutual authentication.

X.509 is an ITU-T standard for encapsulating public keys in certificates and providing standards for the certification path validation algorithms. Certificates contain additional information such as identifying the owner of the associated private key, restrictions on use, and a digital signature.

Certificates are signed with a private key. They can be self-signed or signed by a more trusted private key owner using their private key.

The figure below illustrates an example relationship between private keys and device certificates, as well as certificate signing requests used to create certificates signed by a higher level certificate/key owner.

Figure 62 Private key and certificate associations**Legend**

1	Root Private Key (Passphrase Protected)	5	Certificate Signing Request
2	Intermediate Private Key (Passphrase Protected)	6	Root Certificate (Self Signed)
3	Device Private Key (Passphrase Protected)	7	Intermediate Certificate
4	Certificate Signing Request	8	Device Certificate

Certificates are validated by checking the certificate signature with the public key contained in the issuer's certificate, which is stored in a local CA certificate cache. Validity dates and allowed certificate uses may be checked as well.

Using PKI and X.509

This section includes information on installing trusted CA certificates and installing the device certificate and private keys.

Installing trusted CA certificates

Waveserver 5 can retrieve and install trusted CA certificates from a xFTP (TFTP, FTP, SFTP, and SCP) server. Up to 16 Privacy Enhanced Mail (PEM) encoded trusted CA certificates can be installed by a user with super access privileges. Each CA (root) and subCA (intermediate) certificate required in a

trusted chain should be enrolled with a separate PEM file. These certificates are used to verify a device certificate. Trusted CA certificates are maintained in non-volatile memory. See [Installing a trusted CA certificate](#).

Installing the device certificate and private keys

There are two scenarios in which the user can install the device certificate and private keys:

- 1 The certificate and private/public key pair are created completely with an external CA.
- 2 The private/public key pair is created within Waveserver 5 and a Certificate Signing Request (CSR) is used to complete the certificate creation process.

When the device certificate is created with an external CA, the certificate is downloaded to the Waveserver 5 in a bundled single PKCS#12 file. PKCS#12 is the standard format for combining private keys, device certificates, and optionally intermediate certificates. Waveserver 5 allows you to install up to 16 PKCS#12 files containing both a device certificate and a private key. PKCS#12 provides passphrase protection on the private keys. If required, external tools such as OpenSSL provide a mechanism to convert other formats of bundled keys and certificates into PKCS#12 format prior to installing them on the Waveserver 5. To install device certificates and private keys, certificate files can be retrieved from an xFTP server and installed for a specific application and/or port. Users must have super user access privileges to manage device certificate and private key files. Device certificate and private key files may additionally include one or more intermediate certificates used in the signing path of the device certificate. See [Installing an X.509 certificate and private key](#).

Alternatively, the private/public key pair can be created on the Waveserver 5. Once the private/public key pair and CSR are created by Waveserver 5, the user must have the CSR signed by a CA. The signed response must be enrolled back to Waveserver 5 to complete the device certificate creation process. See [Creating a private key and installing a device certificate](#).

The device certificate and private key files are stored in non-volatile memory. The device certificate must meet the following criteria:

- The Basic Constraints setting is CA:FALSE
- If the certificate will be assigned for HTTPS server use, set the Key Usage Extension field to TLS Server Authentication
- If the certificate will be assigned for RADSec client or TLS Syslog client use, set the Key Usage Extension field to TLS Client Authentication

X.509 certificates

An X.509 certificate is a digital certificate that is used to verify that a public key belongs to a certain owner. The X.509 certificate contains owner information, a public key, and a signature to prove that the certificate has been approved by a higher level certificate authority. Additional information contained in the certificate includes restrictions of use, for example validity date range.

The signed public device certificate and private device key/passphrase must be installed on the client before a client can prove its identity to a server. Also, the root CA certificate that signed the client's certificate must also be installed on the server. Intermediate CA certificates can be enrolled as a separate PEM file on the server or bundled with the device certificate on the client. During authentication, the client provides its device certificate to the server and uses its own private key to prove ownership of that certificate. The server verifies the certificate's signature against a chain of certificates back to the root certificate it holds and trusts.

Authentication can be done in the opposite direction provided that the server has its own signed device certificate and private key. The client must also have a copy of the root certificate that signed it. This can be the same root certificate that signed the certificate on the client device or a different root certificate.

When creating a CA (root) or SubCA (intermediate) certificate for use with Waveserver 5, the signing certificate must have:

- the Basic Constraint extension;
- the CA flag in Basic Constraint set to True;
- the Key Usage field present;
- Certificate Signing in the Key Usage field set;

Waveserver 5 supports RSA and ECC based certificates types in PKCS#12 format V3 (X.509v3).

RSA-based certificates

Waveserver 5 supports RSA key types:

- RSA2048
- RSA3072
- RSA4096

RSA2048 is supported only if the `supportRSA2048` option is enabled using the `certificates set support-RSA-2048 enabled` command. This option is disabled on Waveserver 5 after an RTFD operation or on Waveserver 5 nodes with new system software installations.

Three signing algorithms for RSA based certificates are supported:

- RSA with SHA256
- RSA with SHA384
- RSA with SHA512

Elliptic Curve Cryptography (ECC) based certificates

Waveserver 5 supports the following elliptic curves key types:

- SECP256R1
- SECP384R1
- SECP521R1

Three signing algorithms for elliptic curves based certificates are supported:

- ECDSA with SHA256
- ECDSA with SHA384
- ECDSA with SHA512

SSL/TLS Server Certificates

Waveserver 5's SSL/TLS server includes client and server authentication and data encryption for application-specific access for REST and SWAGGER.

The SSL/TLS server has a default self-signed RSA certificate that is generated on the first boot of the Waveserver 5. To change this default certificate, you can import externally-generated RSA security certificates into the SSL/TLS server or regenerate a new self-signed certificate. See [Installing a new SSL certificate](#) and [Regenerating the default SSL server RSA key](#).

You can assign a device RSA or ECC certificate that was installed through PKIX for use by the HTTPS server.

Certificate display

The following basic information from an imported certificate can be queried:

- Subject Common Name
- Issuer Common Name
- Valid From
- Valid To
- Signature Algorithm
- Serial Number
- Extended Key Usage

Certificate signing request support

Waveserver 5 supports the generation of private/public key pair and Certificate Signing Request (CSR) based on the user's CSR configuration file. The Waveserver 5 imports the user's CSR configuration and exports the CSR by xFTP (TFTP, FTP, SFTP and SCP).

Waveserver 5 allows the user to generate a well-formed Certificate Signing Request (CSR). The CSR includes a mandatory auto-generated public key and a mandatory Common Name from the CSR configuration file. The CSR must be signed by a CA and then be imported back to Waveserver 5 in order to complete the device certificate creation process. The private key is maintained within Waveserver 5 and is not part of the CSR.

The following asymmetric cryptographic key types for CSR generation are supported:

- RSA2048
- RSA3072
- RSA4096
- SECP256R1
- SECP384R1
- SECP521R1

You can optionally enter the following information in the CSR:

- Company Name or Organization (O)
- Department or Organization Unit (OU)
- Country (C)
- State (ST)
- Location (L)
- Email (E)

Certificate validation and expiry management

The Waveserver 5 TLS client checks the validity of the TLS sever certificate before establishing a TLS connection with the TLS server. The Waveserver 5 TLS server also checks the validity of the TLS client certificate prior to making a TLS connection with the TLS client. Certificate validity is determined based on Reference ID verification, certificate path, basic constraints extension, ExtendedKeyUsage field, certificate expiry date, and the certificate revocation status.

Waveserver 5 rejects a certificate if it does not include the ExtendedKeyUsage field. The presented certificate's ExtendedKeyUsage field is dependent on the use case.

- For TLS server presented certificates, Waveserver 5 verifies that the X.509v3 ExtendedKeyUsage field is set to TLS Web Server Authentication.
- For TLS client presented certificates, Waveserver 5 verifies that the X.509v3 ExtendedKeyUsage field is set to TLS Web Client Authentication.
- For OCSP certificates presented for OCSP responses, Waveserver 5 verifies that the X.509v3 ExtendedKeyUsage field is set to OCSP Signing.

As per RFC6125, Waveserver 5 verifies the certificate fields against the locally configured peer DNS name or IP address (Subject Name Authorization).

All invalid certificates are rejected by Waveserver 5. If you attempt to activate an invalid certificate, the activation fails. If a presented certificate from a remote device is invalid, the service and connection will not be established.

Managing certificates for HTTPS servers

Waveserver 5 ships with a default self-signed RSA certificate. This certificate can be changed by regenerating and then activating it using the commands `ssl server rsa regenerate`. The default certificate is valid for REST and WebSocket sessions. See [Regenerating the default SSL server RSA key](#) and [Activating the default SSL server RSA certificate](#) for details.

You can install and set an RSA or ECC certificate using the command `certificates entity install` and then `system server https set cert-name` (the Ciena-recommended method). The certificate is valid for RESTCONF and WebSocket applications. See [Installing a new SSL certificate](#).

You can determine which certificate is being used by the HTTPS server using the CLI command `system server https show`.

Online Certificate Status Protocol (OCSP)

Online Certificate Status Protocol (OCSP) is used for obtaining the revocation status of an X.509v3 digital certificate, defined in RFC6960. The messages communicated via OCSP are communicated over HTTP and based on the HTTP Post/Get operation.

When a peer provides a certificate for validation, the certificate is checked against the locally stored CA certificates. If these checks succeed and OCSP is set up on the Waveserver 5, an HTTP request is sent to an external OCSP responder. The responder validates the certificate against its certificate revocation status. The responder then responds with a status of "good",

“revoked”, or “unknown”. If the responder is inaccessible or cannot definitively report the status as “good” or “revoked”, requests to other OCSP responders may be initiated.

The OCSP response is signed and the signature is checked again using the locally stored CA certificates on the Waveserver 5. The OCSP request and reply may include a nonce to prevent a replay attack.

Waveserver 5 supports the Delegated Trust Model and the Trusted Responder Model. In the Delegated Trust Model, a trusted CA (or intermediate) certificate signs a certificate intending to be used for signing the OCSP response. When Waveserver 5 receives an OCSP response it verifies that certificate against an installed, trusted CA certificate. The Trusted Responder Model is also supported, where a self-signed CA certificate is used to sign the OCSP response. In this case, the self-signed CA certificate must be installed onto the Waveserver 5 to verify its validity.

OCSP responder URLs may be provided in the certificate being validated. One or more OCSP URLs are listed in the certificate’s Authority Information Access (AIA) section. A default responder may be configured for the case in which the certificate does not contain any OCSP responder URLs or alternatively, in the case where the responders do not provide a definitive response. Additionally, you can specify a preferred order in which responders are attempted, AIA or default responder first.

Below is an example of text output from a certificate with OCSP responder.

Waveserver 5 supports OCSP with HTTPS server, RADSec, and TLS syslog. For more information, see [HTTPS server OCSP](#), [RADSec](#) and [Logs](#).

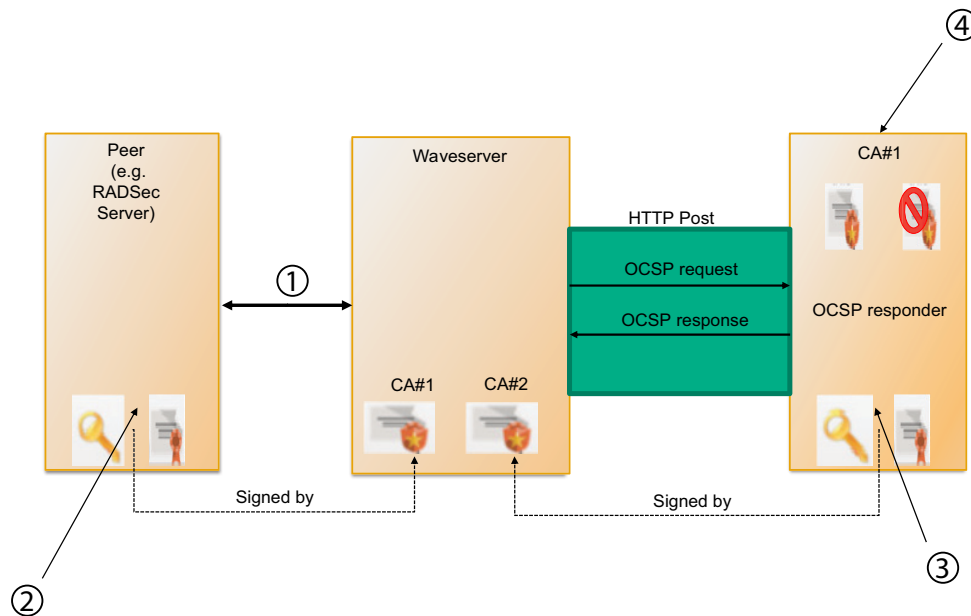
Figure 63 Example — text output from certificate with OCSP responder

```
Authority Information Access:
```

```
OCSP URI:  http://myCA.com/ocsp
```

```
OCSP URI:  http://1.2.3.4:2560
```

The figure below illustrates the OCSP operation.

Figure 64 OCSP operation**Legend**

- | | | | |
|---|---|---|--|
| 1 | Authenticating Peers | 3 | Private key and cert used to sign the OCSP response |
| 2 | Cert may contain OCSP responder's URL(s) inside the authorityInfoAccess section | 4 | CA that signed Peer 1's certificate and associated CRL |

SSH algorithms selection

For a Waveserver 5 secure shell (SSH) server and client, the encryption, key exchange, message authentication, and host key algorithms are limited to the allowed algorithms list. These algorithms are applied to both the SSH server and SSH client.

OpenSSH is used for the Waveserver 5 SSH login server and when the Waveserver 5 is connecting to a remote SSH server. For supported OpenSSH SSH algorithms, see [Encryption algorithms](#), [Key exchange algorithms](#), [Message Authentication Code algorithms](#), and [Host key algorithms](#).

OpenSSH encryption, key exchange, message authentication code, and host key algorithms

This section provides information on the OpenSSH algorithms, including:

- [Encryption algorithms](#)
- [Key exchange algorithms](#)

- [Message Authentication Code algorithms](#)
- [Host key algorithms](#)

Encryption algorithms

Waveserver 5 uses symmetric algorithms for encrypting the transferred data. The encryption algorithms can be disabled or enabled using the CLI command `ssh algorithm <disable | enable> encryption <encryption_algorithm>`. The administrative states (disabled or enabled) for the encryption algorithms can be displayed using the CLI command `ssh algorithm show encryption`. Provisioning for encryption algorithms is supported through CLI, RESTCONF, and NETCONF. See the following table for a list of supported encryption algorithms and their priorities.

Table 97 Supported encryption algorithm and priorities

Encryption algorithm	Priority
AES-256-GCM	1 (enabled by default)
AES-256-CTR	2 (enabled by default)
AES-128-GCM	3 (disabled by default)
AES-128-CTR	4 (disabled by default)

Key exchange algorithms

Waveserver 5 SSH uses symmetric keys to encrypt the entire connection. The secret key is created through a process known as a key exchange algorithm in which the client and server contribute towards establishing the key. The key exchange algorithm can be disabled or enabled using the CLI command `ssh algorithm <disable | enable> kex`. The administrative states (disabled or enabled) for the key exchange algorithms can be displayed using CLI command `ssh algorithm show kex`. Provisioning for the key exchange algorithms is supported through CLI, RESTCONF, and NETCONF. See the following table for a list of supported key exchange algorithms and their priorities.

Table 98 Supported key exchange algorithms and priorities

Key exchange algorithm	Priority
ECDH-SHA2-NISTP521	1 (enabled by default)
ECDH-SHA2-NISTP384	2 (enabled by default)
ECDH-SHA2-NISTP256	3 (enabled by default)
DIFFIE-HELLMAN-GROUP14-SHA1	4 (enabled by default)

Message Authentication Code algorithms

Waveserver 5 SSH uses hash-based message authentication codes (HMAC) for data integrity to ensure that the received message text is intact and has not been modified. The message authentication code algorithm can be disabled or enabled using the CLI command `ssh algorithm <disable | enable> mac`. The administrative states (disabled or enabled) of the message authentication code algorithms can be displayed using the CLI command `ssh algorithm show mac`. Provisioning for message authentication code algorithms is supported through CLI, RESTCONF, and NETCONF. See the following table for a list of supported message authentication code algorithms and their priorities.

Table 99 Supported message authentication code algorithm and priorities

Message authentication code algorithm	Priority
HMAC-SHA2-512	1 (enabled by default)
HMAC-SHA2-256	2 (enabled by default)
AEAD-AES-256-GCM	Implicitly enabled when the AES-256-GCM encryption algorithm is enabled.
AEAD-AES-128-GCM	Implicitly enabled when the AES-128-GCM encryption algorithm is enabled.

Host key algorithms

Waveserver 5 SSH accepts public key authentication. The public key authentication algorithms can be disabled or enabled using the CLI command `ssh algorithm <disable | enable> public-key-authentication`. The administrative states (disabled or enabled) for the public key authentication algorithms can be displayed using the CLI command `ssh algorithm show public-key-authentication`. Provisioning for the public key authentication algorithms is supported through CLI, RESTCONF, and NETCONF.

Waveserver 5 supports the following public key authentication algorithms, which are enabled by default:

- SSH-RSA
- ECDSA-SHA2-NISTP256
- ECDSA-SHA2-NISTP384
- ECDSA-SHA2-NISTP521

The Waveserver 5 SSH server host key has both a public component and a private component. The public key is saved on each SSH client and used by the client to decrypt an authentication message sent from the server when a connection is taking place. For more information on private and public keys, see [Public Key Infrastructure \(PKI\) and X.509](#).

You can delete the host key using the CLI command `ssh server key delete`. You can also regenerate the host key using the CLI command `ssh server key generate [key-type <host_key_type>] [force]`.

You can activate a pre-downloaded encryption host key pair using the CLI command `ssh server key activate host-key <String[1..64]> passphrase <Password String[1..128]>`.

The following table lists the supported host key types.

Table 100 Supported host key types

Key type	Key size	OpenSSH Key name
RSA2048	2048 bits	SSH-RSA
RSA3072	3072 bits	SSH-RSA
RSA4096	4096 bits	SSH-RSA
ECDSA256	256 bits	ECDSA-SHA2-NISTP256
ECDSA384	384 bits	ECDSA-SHA2-NISTP384
ECDSA521	521 bits	ECDSA-SHA2-NISTP521

You can specify whether RSA2048 host keys are supported on Waveserver 5 using the CLI command `certificates set support-RSA-2048 <disabled | enabled>`. If enabled, Waveserver 5 can generate/activate a 2048 bit RSA key and use it as the host key for the SSH server. Also, when enabled a 2048 bit RSA key can be used for public key authentication. If disabled, 2048 bit RSA keys are not supported. To display the administrative state for the support-RSA-2048 setting, use the CLI command `certificates show`.

TLS algorithms selection

Transport Layer Security (TLS) is a cryptographic protocol designed to create secure network connections. It is used by applications to create secure TLS connections to external devices.

Waveserver 5 supports TLS 1.2 (RFC 5246) over IPv4 and IPv6. Waveserver 5 can act as a TLS server, for example when you connect to Waveserver 5 with HTTPS. It can also act as a TLS client, for example when the Waveserver 5 connects to an external RADSec or syslog server using TLS.

Waveserver 5 has a system-level TLS cipher suite list that is created and applied to all services that use TLS. TLS cipher suites are named combinations of the authentication, encryption, message authentication code (MAC), and key exchange algorithms used to negotiate the security settings for TLS. The cipher suite list is provided by the Waveserver 5 TLS service during the SSL (Secure Sockets Layer) handshake and a cipher suite is selected from the list based on there being a match with the remote end.

To add or remove a TLS cipher suite to or from the list, use the CLI command `ssl algorithm <enable | disable> cipher-suite <TLS_cipher_suite>`. You can display the administrative states (disabled or enabled) of the TLS cipher suites using the CLI command `ssl algorithm show`. Provisioning of TLS cipher suites is supported through CLI, RESTCONF, and NETCONF.

See the following table for a list of supported TLS cipher suites and their default values.

Table 101 TLS cipher suites supported on the Waveserver 5 and default administrative states

TLS cipher suite	Default administrative state
TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	Enabled (highest priority in cipher suite negotiation)
TLS_DHE_RSA_WITH_AES_256_CBC_SHA256	Enabled
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384	Enabled
TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384	Enabled
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384	Enabled
TLS_RSA_WITH_AES_256_CBC_SHA256	Enabled
TLS_RSA_WITH_AES_256_GCM_SHA384	Enabled
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256	Disabled
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384	Disabled
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256	Disabled
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256	Disabled
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256	Disabled

TLS cipher suite	Default administrative state
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	Disabled
TLS_RSA_WITH_AES_128_CBC_SHA256	Disabled

HTTPS Mutual Authentication

Mutual authentication refers to the process of a client and server authenticating one another. Mutual authentication ensures that the client and server are performing communication to the appropriate parties. Waveserver 5 supports mutual authentication using X.509v3 certificates. For more information, see [X.509 certificates](#).

Mutual Authentication is performed when the Waveserver 5 acts as a TLS server, for example, when you connect to the Waveserver 5 with HTTPS. With the support of the mutual authentication feature, the TLS client validates the TLS server's digital certificate. The TLS client also presents its own digital certificate for validation by the TLS server.

X.509 certificate for TLS

Waveserver 5 supports the X.509 certificate authentication for TLS connections. The enrollment of the Waveserver 5 certificate (passphrase protected PKCS#12 format V3 file) is generated using the external PKI infrastructure. Waveserver 5 also supports the generation of a certificate signing request which can be signed by an external PKI infrastructure.

Waveserver 5 allows each TLS service (RADSec, TLS Syslog, and HTTPS) to be configured with its own certificate. Once a certificate is configured for RADSec, TLS Syslog, or for an HTTPS server, the same certificate is used for that specific TLS service. For example, when a RADSec certificate is configured, that certificate will be used for all RADSec server connection authentications. Similarly, once a certificate is configured for TLS Syslog, that certificate will be used for all TLS Syslog collector server connection authentications. Once a certificate is configured for an HTTPS server, that certificate will be used for all HTTPS connection authentications.

For more information on X.509 certificates, see [X.509 certificates](#).

Reset to Factory Defaults (RTFD)

The Reset to Factory Defaults feature securely and permanently removes configuration data from Waveserver 5's non-volatile storage. You can initiate an RTFD operation through the CLI or directly on the hardware by pressing a recessed reset button on the control processor.

**CAUTION**

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

RTFD removes the following information:

- Configuration data
- Executable files
- User files
- Library files
- Licenses
- Logs (for example command logs and syslogs)
- IP addresses
- Host names
- User names
- Passwords
- Certificates, for example, trusted certificates and device certificates
- All software releases except the current running software load, which is reinstalled

During RTFD, default passwords are reset and all information is permanently removed. This information is not recoverable. After a RTFD, Waveserver 5 checks the contents of the memory card to ensure that the information is erased.

Session management

A client connection to the Waveserver 5 is established when you (or a client application) attempt to connect to Waveserver 5 from a remote device. Once Waveserver 5 authenticates you or the client application, Waveserver 5 creates a user session.

User sessions

A user session allows you or a client application to interact with the Waveserver 5. A user session is uniquely identified by the client connection; the username; the IP address of the client host; and, the management interface or protocol used. A user session tracks and records the requests and modifications made by the client.

Waveserver 5 permits visibility and management of user sessions across all connections:

- SSH-based connections covering CLI, NETCONF, SFTP, and SCP connections
- Webserver-based HTTP/TLS connections for REST WebSocket and WebUI access
- gRPC-based connections with gRPC channels carrying gNMI or gNOI RPC requests
- Console port access
- SNMP Agent based activity

To terminate sessions in any of these management interfaces, you require super level privileges.

User session limits

Each management interface is limited to a total of 10 user sessions across all three privilege levels (super, admin, limited). The CLI management interface reserves an extra, eleventh user session for super privilege level emergency access. The following table lists the user session limits by management interface.

Table 102 User session limits per management interface

Protocol	Server	Interface	User session limit
SSH	SSH	CLI SCP SFTP	10 + 1
SSH	NETCONF	NETCONF	10
HTTP 1.1 / TLS	Webserver	REST WebSocket WebUI	10
HTTP 2.0 / TLS	gRPC	gNMI gNOI	10

User session lifespan

When you establish a client connection to a Waveserver 5 from a remote device, Waveserver 5 authenticates your credentials and then allocates a user session. In general, the lifespan of the user session starts after authenticating the credentials until the connection is closed or terminated.

Table 103 User session lifespan by connection type

Connection type	User session termination
CLI NETCONF SFTP SCP	Terminates when the client connection closes.
Serial console port	Terminates when the authenticated user logs out.
Webserver	Terminates when the lifespan of the authentication token allocated by the Webserver expires. By default, the lifespan of the token is 10 minutes, but this value is configurable at the system level by a user with super level privileges. When the token expires, the associated user session terminates and the HTTPS client must re-authenticate to allocate a new token and begin a new user session. Although the WebSocket interface is handled by Webserver, The life span of a WebSocket user session is not determined by the lifespan of an authentication token. Once the WebSocket client connection is established and the user is authenticated, the WebSocket user session remains open until the WebSocket connection is closed.
gRPC	A user session is allocated with each RPC, and terminates after it executes. A client connection opens a gRPC channel to the Waveserver to send it gNMI or gNOI RPCs. Each RPC contains user credentials in the overhead. When Waveserver authenticates the credentials, it allocates a user session that exists until the RPC completes execution, after which the user session terminates. A gRPC channel is not directly associated with a user session, so the gRPC channel does not automatically close after a gNMI or gNOI RPC executes and the user session closes.

Session inactivity timeouts

Waveserver 5 terminates client connections and associated user sessions that have been inactive longer than the value specified by an inactivity timer.

The following table describes the Waveserver 5 inactivity timers.

Table 104 Waveserver 5 inactivity timers

Inactivity timer	Description
Global inactivity timer (global-inactivity-timer)	This inactivity timer applies to all management interfaces: CLI, REST, NETCONF gRPC, WebUI, WebSocket, SCP, and SFTP. When enabled, all management interfaces are monitored for inactivity, and are subject to the global inactivity timer value. When disabled, Waveserver 5 does not monitor management interfaces for inactivity and sessions never time out. Value: 1 to 1500 minutes. Default: 10 minutes
Webserver inactivity timer	This timer only applies to Webserver sessions (REST, WebSocket, WebUI), and it takes precedence over the global inactivity timer for Webserver sessions. As a result, Webserver sessions and sessions through other management interfaces can have different inactivity timeout values. When the global inactivity timer is disabled, it takes precedence and Webserver sessions never time out. Value: 1 to 1500 minutes. Default: 10 minutes.

Session monitoring and management

You can retrieve and view a list of active user sessions by running the following command:

```
user who
```

This command returns a list of active users that contains the following information

- user name
- Idle time
- Session ID
- Interface type
- Client IP address

Terminating active user sessions

A user with super level privileges can terminate any active user session listed in the list of active users, by running the following command:

```
user kill session <session_id>
```

A super level user can terminate active user sessions in from CLI, REST, or NETCONF. Currently, gNMI and gNOI do not support this capability.

User session statistics

You can retrieve and view user session statistics for each of the connection types: Console, SSH, NETCONF, Webserver, and gRPC. Run the following command:

```
user session show statistics
```

This command returns the following information for each session type:

- Max sessions — The high-water mark for the number of simultaneous sessions.
- Active sessions — The current number of active sessions.
- Closed sessions — The number of sessions previously terminated.

Procedure list

This section provides the following procedures:

- [Procedure 38, “Creating local user accounts”](#)
- [Procedure 39, “Configuring local user accounts”](#)
- [Procedure 40, “Deleting local user accounts”](#)
- [Procedure 41, “Setting login limits and minimum password lengths on user accounts”](#)
- [Procedure 42, “Displaying user account information”](#)
- [Procedure 43, “Configuring user lockout”](#)
- [Procedure 44, “Changing the Linux root account password”](#)
- [Procedure 45, “Configuring SSH shell type and configuring diag user sudo access”](#)
- [Procedure 46, “Configuring user authentication methods”](#)
- [Procedure 47, “Configuring RADIUS authentication”](#)
- [Procedure 48, “Configuring RADSec authentication”](#)
- [Procedure 49, “Configuring TACACS+”](#)
- [Procedure 50, “Configuring TACACS+ authentication servers”](#)
- [Procedure 51, “Configuring TACACS+ authorization servers”](#)
- [Procedure 52, “Configuring TACACS+ accounting servers”](#)
- [Procedure 53, “Configuring the global servers list”](#)
- [Procedure 54, “Generating the SSH server key”](#)
- [Procedure 55, “Activating an SSH server key”](#)
- [Procedure 56, “Installing an SSH user key”](#)
- [Procedure 57, “Managing SSH server clients”](#)
- [Procedure 58, “Configuring SSH server attributes”](#)
- [Procedure 59, “Configuring SSH algorithms for encryption, key exchange, message authentication code, and public key authentication”](#)
- [Procedure 60, “Configuring TLS algorithms”](#)

- [Procedure 61, “Installing a new SSL certificate”](#)
- [Procedure 62, “Activating the default SSL server RSA certificate”](#)
- [Procedure 63, “Regenerating the default SSL server RSA key”](#)
- [Procedure 64, “Listing all ACL rules”](#)
- [Procedure 65, “Creating an ACL rule”](#)
- [Procedure 66, “Deleting ACL rules”](#)
- [Procedure 67, “Modifying an ACL list”](#)
- [Procedure 68, “Installing a trusted CA certificate”](#)
- [Procedure 69, “Installing an X.509 certificate and private key”](#)
- [Procedure 70, “Creating a private key and installing a device certificate”](#)
- [Procedure 71, “Performing a Reset To Factory Defaults \(RTFD\)”](#)

Procedure 38 Creating local user accounts

Use this procedure to create local user accounts on Waveserver 5.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Overview

Waveserver 5 supports a maximum of 50 provisioned user accounts. The maximum number of active sessions for each management interface on the Waveserver 5 is 10. For details on sessions and session management, see [User session limits](#).

Steps

- 1 To create a local user account, use the command:

```
user create user <String: 1...64> access-level <limited |
admin | super> [password <String: 1...128>] [secret
<1...256>]
```

The following restrictions apply to the user string:

- The user string must not contain only numbers; for example, 583247.
- The user string must not start with a number.
- The user string must start with an alpha character or an underscore (_).
- The user string must not contain the characters ? # \$ & ' () + / : ; < > = @ [] ^ ` ~ { } ! ! % * , or a space.

Note: The user string (that is, the username) cannot be greater than 42 characters if the user is going to log in to Waveserver 5 using the console port. The user name can be longer than 42 characters for remote login using the supported NBIs over SSH.

The password string may contain special characters ~ ! % ^ & * () - _ : = + { } [] | \ ; ' < > . , @ # ? / \$

The default minimum password length is 8 characters. You can change the minimum length using CLI command: `user set min-password-length <NUMBER: 1..128>`.

ATTENTION

The default minimum password length of 8 characters is compliant with Federal Information Processing Standards (FIPS) rules for password strength.

Although you can change the minimum password length to less than 8 characters when you create a user account with an initial password, Ciena recommends that you set the minimum password length to 8 characters to ensure FIPS compliance.

When you subsequently change the password, the minimum default of 8 characters is enforced, which is the same behavior before this change. If you previously set the minimum-password-length to less than 8 characters, the default minimum-password-length resets to 8 characters when you upgrade. Future requests to create or modify a user password to less than the minimum-password-length will be rejected.

To create or enter the password, you can also use the equivalent ASCII code. The following table provides information on the supported password characters and their equivalent ASCII codes.

Table 105 Supported password characters and equivalent ASCII codes

Supported password character	ASCII code	Supported password character	ASCII code	Supported password character	ASCII code	Supported password character	ASCII code
!	!	#	#	\$	$	%	%
&	&	'	'	(	(	)	)
*	*	+	+	,	,	-	-
.	.	/	/	0	0	1	1
2	2	3	3	4	4	5	5
6	6	7	7	8	8	9	9
:	:	;	;	<	<	=	=
>	>	?	?	@	@	A	A
B	B	C	C	D	D	E	E
F	F	G	G	H	H	I	I
J	J	K	K	L	L	M	M
N	N	O	O	P	P	Q	Q
R	R	S	S	T	T	U	U
V	V	W	W	X	X	Y	Y

Supported password character	ASCII code	Supported password character	ASCII code	Supported password character	ASCII code	Supported password character	ASCII code
Z	Z	[	[	\	\	]	]
^	^	_	_	a	a	b	b
c	c	d	d	e	e	f	f
g	g	h	h	i	i	j	j
k	k	l	l	m	m	n	n
o	o	p	p	q	q	r	r
s	s	t	t	u	u	v	v
w	w	x	x	y	y	z	z
{	{		|	}	}	~	~

2 Save the provisioned settings to the configuration file:

```
configuration save
```

You have completed this procedure.

Procedure 39 Configuring local user accounts

Use this procedure to set attributes, such as passwords and access level for local user accounts on Waveserver 5. You can change the password of your own account regardless of the level of account you have.

Requirements

To change the accounts of other users, you must be logged in to Waveserver 5 using an account with super user access privileges.

Steps

- 1 To configure a local user account, use the command:

```
user set user <local user name> access-level <limited |  
admin | super> password <String: 1...128> [secret  
<1...256>]
```

The password string may contain special characters ~ ! % ^ & * () - _ : = + { } [] | \ ; ' < > . , @ # ? / \$

For information on supported password characters and equivalent ASCII codes, see [Creating local user accounts](#).

- 2 Save the provisioned settings to the configuration file:

```
configuration save
```

You have completed this procedure.

Procedure 40 Deleting local user accounts

Use this procedure to delete local user accounts on Waveserver 5.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Guidelines

The Linux root account cannot be deleted.

Steps

- 1 To delete a local user account, use the command:
`user delete user <String: 1...64>`
- 2 Save the provisioned settings to the configuration file:
`configuration save`
You have completed this procedure.

Procedure 41 Setting login limits and minimum password lengths on user accounts

Use this procedure to set limits for simultaneous user account sessions on Waveserver 5.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Steps

- 1 To set limits for simultaneous logins for one or more account types, use the command:

```
user set [lockout-interval <Number: 1..30>] [max-login-attempt <Number:2..10>] [min-password-length <Number: 1..128>]
```
- 2 Save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 42 Displaying user account information

Use this procedure to show account information such as account name, privilege, and account status on Waveserver 5.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited user access privileges.

Steps

- To retrieve summary information for user accounts, use the command:

```
user show
```

See [Table 106, “User session parameters”](#) for a description of the information returned by this command.

You have completed this procedure.

Table 106 User session parameters

Parameter	Values	Description
Username	string	The user name for the user session.
Privilege	• limited • admin • super • diag	The privilege level of the user.
User Security Parameters		
max-login-attempt	integer	The maximum allowable number of failed attempts to log in to the network element.
lockout-interval (in minutes)	integer	The length of time the user is prevented from attempting to log in to the system, after exceeding the maximum number of unsuccessful login attempts.
min-password-length	integer	The minimum length of a user password.
User lockout status		
Username	string	The username of the locked out user.
Failures	integer	The number of failed attempts to log in.
Latest failure	Ddd Mmm dd hh:mm:ss yyyy	The timestamp of the latest failed attempt to log in by the locked-out user.

Procedure 43 Configuring user logout

Use this procedure to set the user security parameters.

Requirements

- You must be logged in to Waveserver 5 using an account with at least super user access privileges.
- At least one user must exist.

Steps

- 1 Optionally, display the user security parameters:
`user show`
- 2 To set user security parameters:
`user set [max-login-attempt <Number: 2..10>] [lockout-interval <Number: <1..30>]`
- 3 Optionally, confirm the user security parameter settings:
`user show`
- 4 Save the provisioned settings to the configuration file:
`configuration save`

Procedure 44 Changing the Linux root account password

With a super user account, you can change the password of the Linux root account. This root account cannot be deleted.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Guidelines

Ciena strongly recommends that you set a password for the default super user account and/or the root user account as soon as possible after you log in to Waveserver 5 for the first time. The “Credentials Warning” alarm will appear to remind you to do this. For more information, see [User account types](#) and [Credentials Warning](#).

Steps

- 1 To configure a password for the Linux root account, use the command:

```
system environment root set password <String:1..128>
```

Procedure 45 Configuring SSH shell type and configuring diag user sudo access

With a super user account, you can configure the SSH diag user shell type to be CLI or system (Linux). You can also configure the diag user to have sudo access.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Guidelines

Ciena recommends that you do not create a diag user account that can configure or monitor Waveserver 5 operations as it can represent a security policy risk.

Steps

- | 1 | If you want to | Then go to |
|---|---|------------------------|
| | configure the SSH diag user shell type | step 2 |
| | restore the SSH diag user shell type to the default value | step 3 |
| | configure diag user sudo access | step 4 |
| | disable the diagnostics shell | step 5 |
- 2 To configure the SSH diag user shell type:
`system environment diag set shell <cli | system>`
 Go to [step 6](#).
 - 3 To restore the SSH diag user shell type to the default value:
`system environment diag unset shell`
Note: The default SSH diag user shell type is CLI.
 Go to [step 6](#).
 - 4 To configure diag user sudo access:
`system environment diag set sudo <on | off>`
 - 5 To disable diagnostics shell access:
`system environment diag disable diag-shell`
Note: Disabling access to the diagnostics shell limits diag and root level access to the system. When you execute this command, the system performs a chassis warm-restart.
 - 6 Optionally, confirm the SSH diag user shell type and diag user sudo access:
`system environment show`

----- SYSTEM ENVIRONMENT SETTINGS -----	
Parameter	Value
Diag User Shell Type	cli
Diag User Sudo Allow	off
Diag Shell Admin State	enabled

7 To save the provisioned settings to the configuration file:
configuration save

Procedure 46 Configuring user authentication methods

Use this procedure to manage Waveserver 5 user authentication methods.

Requirements

You are logged in to Waveserver 5 using an account with super user access privileges.

Steps

- 1 Perform the required task:

If you want to	Then
Set the order of authentication methods to be used	<pre>user auth set order <Value or a comma-separated list of values (value,value,value,value): local radius radsec tacacs></pre> Note: Set local authentication as a backup method to ensure access to Waveserver 5 in the event that RADIUS and/or TACACS+ services are unavailable. See Example 1 below.
Set the authentication method for a login source	<pre>user auth set method <local radius radsec tacacs> scope <all serial remote></pre> Note 1: You can set an authentication method for each available login source. See Example 2 below. Note 2: For an authentication scope (all, serial, or remote), if no authentication method is configured, the authentication method defaults to local.
Display summary and statistics information for authentication services	<pre>user auth show</pre>
Clear authentication statistics for all authentication services or only the specified service	<pre>user auth clear [provider <local radius radsec tacacs>]</pre>

- 2 Save the provisioned settings to the configuration file:

```
configuration save
```

You have completed this procedure.

Example 2

For local logins using the Console port:

```
user auth set method local scope serial
```

For remote logins using SSH over the data communications network:

```
user auth set method radius scope remote
```

```
user auth set method tacacs scope remote
```


Procedure 47 Configuring RADIUS authentication

Use this procedure to configure RADIUS authentication on Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with super user access privileges.
- You know the IP address and UDP port number for each RADIUS server to be added.

Steps

- 1 Specify global RADIUS settings:

```
radius set {[key <Password String:8...64>]} [retries
<Number:0...3>] [secret <String:16...128>] [timeout
<Seconds:1...30>] [search-method <cached | priority>]}
```

Note: The string specified for the [key] attribute must match the shared key value that is located on the Radius server.

- 2 Add a RADIUS server to the Waveserver 5 system, specifying a priority for the server:

```
radius add server <IP address or host name> [priority
<Number:1...8>] [udp-port <Number:1...65535>]
```

Note 1: You must specify a priority for each new RADIUS server you add. The priority value is in the range of 1 to 8, with 1 being the highest priority. You cannot assign the same priority value to more than one RADIUS server. You do not need to assign priority values in consecutive order.

Note 2: Repeat [step 2](#) for each RADIUS server you want to add. You can add up to eight RADIUS servers.

Note 3: The default value for the [udp-port] attribute is 1812.

- 3 Set the order of available authentication providers:

```
user auth set order radius [,local][,radsec][,tacacs]
```

Note: Set TACACS+ and/or local authentication as a backup provider to ensure access to the Waveserver 5 in the event that RADIUS services are unavailable.

- 4 Set RADIUS as the authentication method for all connections over SSH:

```
user auth set method radius scope remote
```

- 5 Save the provisioned settings to the configuration file:

```
configuration save
```

- 6 To view summary or statistics information for all configured RADIUS servers:

```
radius show [statistics]
```

You have completed this procedure.

Procedure 48 Configuring RADSec authentication

Use this procedure to configure RADSec authentication on Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with super user access privileges.
- You have the server IP address or host name and the filename for the X.509v3 certificate files.
- You know the IP address or host name and port number for each RADSec server to be added.
- A valid X.509v3 certificate has been installed on the RADSec server.

Steps

- 1 Install the trusted CA certificate for the corresponding RADSec server, see [Procedure 68, "Installing a trusted CA certificate"](#).
- 2 Install the X.509 certificate for RADSec:

```
certificates entity install cert-name <String: [1..32]>
cert-passphrase <cert-passphrase <Password
String[1..128]> filename <String [1..127]> {default-ftp-
server | default-sftp-server | default-tftp-server |
default-server | default-scp-server | sftp-server <IP
address or host name> login-id <String [1..32]> password
<String [1..128] | tftp-server <IP address or host name>
| scp-server <IP address or host name> login-id <String
[1..32]> password <String [1..128] | ftp-server <IP
address or host name> login-id <String [1..32]> password
<String [1..128]>} [certificate-only]
```

For example:

```
certificates entity install cert-name radseccert ftp-
server 10.120.99.194 filename radseccert.p12 cert-
passphrase *****
```
- 3 Optionally, display the installed certificate:

```
certificates entity show [cert-name <Certificate name>]
```
- 4 Specify the global RADSec settings:

```
radsec set [cert-name <String: [1..32]>] [timeout
<Seconds: 1..30>] [check-fingerprint <on | off>] [check-
ip-host <on | off>] [search-method <cached | priority>]]
```

Note: For the [cert-name] attribute, use the cert-name used from [step 2](#).

- 5 Add a RADSec server to the Waveserver 5 system, specifying a priority for the server:

```
radsec add server <IP address or host name> [fingerprint
<a SHA256 certificate fingerprint[95]>] [priority
<Number:1..8>] [port <Number:1..65535>] [trusted-dns <a
fully qualified domain name that can accept a leading
wildcard period>]
```

Note 1: Repeat [step 5](#) for each RADSec server you want to add. You can add up to eight RADSec servers.

Note 2: The value of the [priority] attribute is in the range 1 to 8, with 1 being the highest priority. You must specify a priority for a RADSec server that you add to the Waveserver configuration. You cannot assign the same priority value to two servers of the same type in the network. You do not need to assign priority values in consecutive order.

Note 3: The default value for the [port] attribute is 2083.

- 6 OSCP can optionally be enabled to perform real time certificate status checks when validating a RADSec server's X.509 certificate.

If you want to	Then go to
disable RADSec with OSCP	step 7
enable RADSec with OSCP	step 8

- 7 Disable RADSec with OSCP:

```
radsec oosp disable
```

Go to [step 12](#).

- 8 Enable RADSec with OSCP:

```
radsec oosp enable
```

- 9 Set the default OSCP responder:

```
radsec oosp set default-responder <String: [1..255]>
```

Note: The default value for the [default-responder] attribute is blank.

- 10 Set the OSCP responder preference.

```
radsec oosp set responder-preference <aia | default-
responder>
```

Note: The default value for the [responder-preference] attribute is aia.

- 11 Set whether you want the OSCP responders to contain nonce:

```
radsec oosp set nonce <on | off>
```

Note: The default value for the [nonce] attribute is on.

- 12 Set the order of available authentication providers:

```
user auth set order radsec [,radius][,local][,tacacs]
```

Note: Set RADIUS, TACACS+ and/or local authentication as a backup provider to ensure access to the Waveserver 5 in the event that RADSec services are unavailable.

- 13** Set RADSec as the authentication method for all connections over SSH:
`user auth set method radsec scope remote`
- Note:** You can also set the scope to “serial” to set the authentication method for Console port logins, or to “all” to set the authentication method for all types of connections.
- 14** Optionally, view RADSec summary or statistics information for all configured RADSec servers:
`radsec show [statistics]`
- 15** Save the provisioned settings to the configuration file:
`configuration save`

Procedure 49 Configuring TACACS+

Use this procedure to configure TACACS+ on Waveserver 5.

Requirements

- You are logged in to Waveserver 5 using an account with super user access privileges.
- You know the IP address and TCP port number for each TACACS+ server to be added.

Steps

- Ensure that TACACS+ is globally enabled:
`tacacs enable`
- Add each required TACACS+ server type to the system:

To	Use the command
Add a TACACS+ authentication server	<code>tacacs authentication add server <IP address or host name> priority <Number: 1...8> {[tcp-port <Number: 1...65535>][timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}</code>
Add a TACACS+ authorization server	<code>tacacs authorization add server <IP address or host name> priority <Number: 1...8> {[tcp-port <Number: 1...65535>][timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}</code>
Add a TACACS+ accounting server	<code>tacacs authorization add server <IP address or host name> priority <Number: 1...8> {[tcp-port <Number: 1...65535>][timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}</code>
Add a server to the global server list	<code>tacacs add server <IP address or host name> priority <Number:1...8> {[tcp-port <Number:1...65535>][timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}</code>

Note 1: If you do not add an authentication server, an authorization server, or an accounting server, the corresponding function can be performed by a global server (see [TACACS+ servers](#)) if configured.

Note 2: The default value of the [tcp-port] attribute for each server type is 49. Ensure that the TCP port to be used is not already reserved.

Note 3: You can add up to eight servers of each type.

Note 4: You must specify a priority for each new TACACS+ server you add. The priority value is in the range of 1 to 8, with 1 being the highest priority. You cannot assign the same priority value to more than one of the same type of TACACS+ server in the network. You do not need to assign priority values in consecutive order.

Note 5: The string specified for the [key] attribute must match the shared key value located on the TACACS+ server.

3 Set global attributes for TACACS+:

```
tacacs set {[key <Password string>] [keyminlen <Number:
2...64>] [secret <String>] [timeout <Number of
seconds:1...30>] [privilege-level-admin <Number:
1...13>] [privilege-level-super <Number: 2...15>]
[global-servers <on | off>] [search-method <priority |
cached>]}
```

Note 1: If your application must use a global server for any AAA function, ensure that the [global-servers] attribute is set to on.

Note 2: The value specified for the [privilege-level-super] attribute must be higher than the value specified for the [privilege-level-admin] attribute.

Note 3: The string specified for the [key] attribute must match the shared key value that is located on the TACACS+ server.

4 Set the order of available authentication providers:

```
user auth set order tacacs [,radius] [,radsec] [,local]
```

Note: Set radius and/or local authentication as a backup provider to ensure access to the Waveserver 5 in the event that TACACS+ services are unavailable.

5 Do one of the following:

- Set TACACS+ as the authentication method for all connections over SSH:

```
user auth set method tacacs scope remote
```

- Set TACACS+ as the authentication method for all connections including console:

```
user auth set method tacacs scope all
```

- 6 If required, set the administrative state of TACACS+ message logging:
`tacacs syslog <disable | enable>`
Note: If you enable TACACS+ message logging, ensure the following:
 - Message logging is configured and enabled on Waveserver 5 (see [Procedure 135, “Configuring message logging \(syslog\)”](#)).
 - TACACS+ authorization is enabled (see [step 8](#) in this procedure).
- 7 Ensure that TACACS+ authentication is enabled:
`tacacs authentication enable`
Note: If enabled, authorization and accounting services are operational only while authentication is enabled.
- 8 If required, enable TACACS+ authorization:
`tacacs authorization enable`
Note: Enabling authorization does not make it operational for active user sessions started before it was enabled. To use authorization, each user must log out of their session and then log in again using TACACS+ authentication.
- 9 If required, enable TACACS+ accounting:
`tacacs accounting enable`
- 10 Save the provisioned settings to the configuration file:
`configuration save`
- 11 If required, view TACACS+ summary or statistics information:
`tacacs show [statistics]`
- 12 Complete the following:
 - Go to [Procedure 50, “Configuring TACACS+ authentication servers”](#).
 - Go to [Procedure 51, “Configuring TACACS+ authorization servers”](#).
 - Go to [Procedure 52, “Configuring TACACS+ accounting servers”](#).
 - Go to [Procedure 53, “Configuring the global servers list”](#).
 - Close the connection to Waveserver 5.

Procedure 50 Configuring TACACS+ authentication servers

Use this procedure to configure TACACS+ authentication servers.

Requirements

- You are logged in to Waveserver 5 using an account with super user access privileges.
- You know the IP address and TCP port number for each TACACS+ authentication server to be added.

Steps

- 1 Perform the required task:

If you want to	Then use the command
Add a TACACS+ authentication server Note: You can add up to eight authentication servers.	<pre>tacacs authentication add server <IP address or host name> {[priority <Number: 1...8>] [tcp- port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}</pre> Note: The default value of the [tcp-port] attribute is 49. Ensure that the TCP port to be used is not already reserved.
Set the administrative state of an authentication server	<pre>tacacs authentication <disable enable> server <IP address or host name></pre>
Retrieve summary information or statistics for configured authentication servers	<pre>tacacs authentication show [statistics]</pre>
Clear statistics for the specified authentication server	<pre>tacacs authentication clear server <IP address or host name> statistics</pre>

Note: You must specify a priority for each new TACACS+ server you add. The priority value is in the range of 1 to 8, with 1 being the highest priority. You cannot assign the same priority value to more than one of the same type of TACACS+ server in the network. You do not need to assign priority values in consecutive order.

- 2 Save the provisioned settings to the configuration file:

```
configuration save
```

You have completed this procedure.

Procedure 51 Configuring TACACS+ authorization servers

Use this procedure to configure TACACS+ authorization servers.

Requirements

- You are logged in to Waveserver 5 using an account with super user access privileges.
- You know the IP address and TCP port number for each TACACS+ authorization server to be added.

Steps

- 1 Perform the required task:

If you want to	Then use the command
Add a TACACS+ authorization server Note: You can add up to eight authorization servers.	<code>tacacs authorization add server <IP address or host name> {[priority <Number: 1...8>] [tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}</code> Note: The default value of the [tcp-port] attribute is 49. Ensure that the TCP port to be used is not already reserved.
Set the administrative state of an authorization server	<code>tacacs authorization <disable enable> server <IP address or host name></code>
Retrieve summary information or statistics for configured authorization servers	<code>tacacs authorization show [statistics]</code>
Clear statistics for the specified authorization server	<code>tacacs authorization clear server <IP address or host name> statistics</code>

Note: You must specify a priority for a new TACACS+ server that you add to the Waveserver configuration. The priority value for a server is in the range of 1 to 8, with 1 being the highest priority. You cannot assign the same priority value to more than one of the same type of TACACS+ server in the network. You do not need to assign priority values in consecutive order.

- 2 Save the provisioned settings to the configuration file:

```
configuration save
```

You have completed this procedure.

Procedure 52 Configuring TACACS+ accounting servers

Use this procedure to configure TACACS+ accounting servers.

Requirements

- You are logged in to Waveserver 5 using an account with super user access privileges.
- You know the IP address and TCP port number for each TACACS+ accounting server to be added.

Steps

- 1 Perform the required task:

If you want to	Then use the command
Add a TACACS+ accounting server Note: You can add up to eight accounting servers.	<pre>tacacs accounting add server <IP address or host name> priority <Number:1...8> {[tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}</pre> Note: The default value of the [tcp-port] attribute is 49. Ensure that the TCP port to be used is not already reserved.
Set the administrative state of an accounting server	<pre>tacacs accounting <disable enable> server <IP address or host name></pre>
Set the status of command and/or session logging for TACACS+ accounting globally	<pre>tacacs accounting set [command <on off>] [session <on off>]</pre>
Retrieve summary information or statistics for configured accounting servers	<pre>tacacs accounting show [statistics]</pre>
Clear statistics for the specified accounting server	<pre>tacacs accounting clear server <IP address or host name> statistics</pre> Note: You must specify a priority for each new TACACS+ server you add. The priority value is in the range of 1 to 8, with 1 being the highest priority. You cannot assign the same priority value to more than one of the same type of TACACS+ server in the network. You do not need to assign priority values in consecutive order.
2 Save the provisioned settings to the configuration file:	<pre>configuration save</pre> You have completed this procedure.

Procedure 53 Configuring the global servers list

Use this procedure to configure TACACS+ global servers.

Requirements

- You are logged in to Waveserver 5 using an account with super user access privileges.
- You know the IP address and TCP port number for each server to be added to the global server list.

Steps

- 1 Perform the required task:

If you want to	Then use the command
Add a server to the global server list Note: You can add up to eight global servers.	<code>tacacs add server <IP address or host name> {[priority <Number:1..8>] [tcp-port <Number:1..65535>] [timeout <Seconds: 1..30>] [key <Password String: 2..64>] [secret <String: 4..128>]}</code> Note: The default value of the [tcp-port] attribute is 49. Ensure that the TCP port to be used is not already reserved.
Set the administrative state of a global server	<code>tacacs <disable enable> server <IP address or host name></code>
Retrieve statistics for configured global servers	<code>tacacs show [statistics]</code>
Clear statistics for the specified global server	<code>tacacs clear server <IP address or host name> statistics</code>

Note: You must specify a priority for a TACACS+ server that you add to the Waveserver configuration. The priority value for a server is in the range of 1 to 8, with 1 being the highest priority. You cannot assign the same priority value to two servers of the same type in the network. You do not need to assign priority values in consecutive order.

- 2 Save the provisioned settings to the configuration file:

```
configuration save
```

You have completed this procedure.

Procedure 54 Generating the SSH server key

Use this procedure to generate the host key pair for the SSH login server. Waveserver 5 also allows you to export the public key to an xFTP server during the key generation. The public key file name is appended with the chassis MAC address, for example 00238af49809-rsa4096.pub.

Requirements

You must be logged in to Waveserver 5 using an account with at least super user access privileges.

Guidelines

Waveserver 5 supports the OpenSSH format only for private authentication keys.

Steps

- 1 Generate the host key pair for the SSH login server and export the public key to an xftp server, if required:

```
ssh server key generate [key-type <rsa2048 | rsa3072
|rsa4096 | ecdsa256 | ecdsa384 | ecdsa521>] [default-
tftp-server | default-ftp-server | default-server |
default-sftp-server | default-scp-server | sftp-server |
tftp-server | ftp-server | scp-server][force]
```

Note 1: If the key has already been generated, the command fails and the message `ERROR: Key already exists` appears. You can delete the existing key (`ssh server key delete`), then generate a new key or you can use the `ssh server key generate [key-type <rsa2048 | rsa3072 |rsa4096 | ecdsa256 | ecdsa384 | ecdsa521>]`

Note 2: You can use the `[force]` option which will overwrite the existing key, then generate a new key.

Note 3: The `[key-type]` attribute is optional, it will default to “rsa3072” if a `[key-type]` is not entered.

- 2 If required, retrieve SSH server key information:

```
ssh server show key
```

- 3 Save the configuration file:

```
configuration save
```

Procedure 55 Activating an SSH server key

Use this procedure to activate an SSH server key. Waveserver 5 supports an OpenSSH format private keys. The supported key types are RSA2048, RSA3072, RSA4096, ECDSA256, ECDSA384, and ECDSA521.

Requirements

You must be logged in to Waveserver 5 using an account with at least super user access privileges.

Guidelines

For Waveserver 5 to support RSA2048 keys, the `certificates set support-RSA-2048` attribute must be enabled. To confirm `support-RSA-2048` is enabled, use the CLI command `certificates show`.

Steps

- 1 Transfer the private key file onto the Waveserver 5 using any of the existing file transfer commands.

For more information on how to transfer files, see [Transferring files using the Secure Copy Protocol](#) and [System commands](#) (System xftp commands).
- 2 Activate the key:


```
ssh server key activate host-key <String [1..64]>  
passphrase <Password String [1..128]>
```
- 3 Save the configuration file:


```
configuration save
```

Procedure 56 Installing an SSH user key

Use this procedure to install an SSH user key on Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with at least super user access privileges.
- You have generated public and private SSH keys using an external tool.
- The SSH user key file must be available on a remote server.

Guidelines

This SSH server key installation procedure is used for local user authentication, because the user name associated with the account must match the user name string of the specified key file.

Steps

- 1 Install the SSH user key file on Waveserver 5:

```
ssh server key install user <String: 1...64> <default-tftp-server | default-ftp-server | default-server | default-sftp-server | default-scp-server | sftp-server | tftp-server | ftp-server | scp-server <IP address or host name> login-id <String> password <String> secret <String>
```

Note 1: You must specify [login-id], [password], and [secret] for an FTP or SFTP server.

Note 2: When uploading the user name string of the specified key file, disregard the file extension.
- 2 Optionally, retrieve SSH server key information to confirm that the user key file is installed.

```
ssh server show key
```

Figure 65 Confirming SSH user key file installation

```
Waveserver5_0132# ssh server show key
```

SSH SERVER KEY		
Key Status	Generated	
Key Type	rsa3072	
Key Fingerprint SHA256	1112222111AAAx00AAA/555/nnnnnnnnVVVVVVVVVV	

SSH PUBLIC KEY AUTHENTICATION USERS		
Username	Key Type	Key Status
user1		installed note: shell user account not created

- 3 Create the shell user account to be associated with the user key. See [Procedure 38, "Creating local user accounts"](#).

Note: The user name of the account must match the user name string of the specified key file.

- 4 Optionally, retrieve SSH server key information to confirm that the key file has been assigned to a shell user account:

```
ssh server show key
```

Figure 66 Confirming SSH server key shell user account

```
Waveserver5_0132# ssh server show key
```

```
+----- SSH SERVER KEY -----+
| Key Status          | Generated          |
| Key Type            | rsa3072            |
| Key Fingerprint SHA256 | 1112222111AAAAx00AAA/555/nnnnnnnnVVVVVVVVVV |
+-----+

+----- SSH PUBLIC KEY AUTHENTICATION USERS -----+
| Username            | Key Type           | Key Status          |
+-----+-----+-----+
| user1               |                    | installed           | note: shell user account not created
+-----+-----+-----+
```

- 5 Save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 57 Managing SSH server clients

Use this procedure to manage SSH server clients.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Use the following commands to manage SSH server clients.
 - To add a valid client IP address to, or remove a valid IP client from the list of addresses from which a connection is accepted:

```
ssh server <add | remove> client <IP address>
```

Note: By default, all IP addresses are accepted. However, after one client IP address is added, only that IP address, and IP addresses subsequently added, are allowed to connect.
 - Retrieve client summary information for the SSH server:

```
ssh server show
```
 - To save the provisioned settings to the configuration file:

```
configuration save
```


Procedure 58 Configuring SSH server attributes

Use this procedure to configure SSH server attributes.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Use the following commands to configure SSH server attributes.
 - To configure writable SSH server settings:

```
ssh server set {[authentication-retries <Number: 1...3>]
[listener-port <Number: 22...65535>]}
```

Note: The default value for the attribute [authentication-retries] is 3. The default value for the [port] attribute is 22.
 - To restore modified SSH server settings to their default values:

```
ssh server unset {[authentication-retries] [listener-
port]}
```
 - To retrieve client summary information for the SSH server:

```
ssh server show
```
 - To save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 59 Configuring SSH algorithms for encryption, key exchange, message authentication code, and public key authentication

Use this procedure to configure the encryption, key exchange, message authentication code, or host key algorithms.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Steps

1	If you want to configure the	Then go to
	encryption algorithms	step 2
	key exchange algorithms	step 4
	message authentication code algorithms	step 6
	public key authentication algorithms	step 8

Configuring the encryption algorithms

- 2 Set the administrative state (disabled or enabled) for the encryption algorithm:

```
ssh algorithm <disable | enable> encryption <aes256-gcm | aes256-ctr | aes128-gcm | aes128-ctr>
```
- 3 Display the encryption algorithm settings (optional):

```
ssh algorithm show encryption
```

Go to [step 10](#).

Configuring the key exchange algorithms

- 4 Set the administrative state (disabled or enabled) for the key exchange algorithm:

```
ssh algorithm <disable | enable> kex <ecdh-sha2-nistp521 | ecdh-sha2-nistp384 | ecdh-sha2-nistp256 | diffie-hellman-group14-sha1>
```
- 5 Display the key exchange algorithm settings (optional):

```
ssh algorithm show kex
```

Go to [step 10](#).

Configuring the message authentication algorithms

- 6 Set the administrative state (disabled or enabled) for the message authentication algorithm:

```
ssh algorithm <disable | enable> mac <hmac-sha2-512 | hmac-sha2-256>
```

- 7 Display the message authentication algorithm settings (optional):

```
ssh algorithm show mac
```

Go to [step 10](#).

Configuring the public key authentication algorithms

- 8 Set the administrative state (disabled or enabled) for the public key authentication algorithm:

```
ssh algorithm <disable | enable> public-key-  
authentication <ssh-rsa | ecdsa-sha2-nistp521 | ecdsa-  
sha2-nistp384 | ecdsa-sha2-nistp256>
```

- 9 Display the public key authentication algorithm settings (optional):

```
ssh algorithm show public-key-authentication
```

- 10 Save the provisioned settings to the configuration file.

```
configuration save
```

Procedure 60 Configuring TLS algorithms

Use this procedure to configure Transport Layer Security (TLS) algorithms.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Steps

- 1 Set the administrative state of the desired TLS algorithm:

```
ssl algorithm <enable |disable> cipher-suite
<TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 |
TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384 |
TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 |
TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384 |
TLS_DHE_RSA_WITH_AES_256_CBC_SHA256 |
TLS_RSA_WITH_AES_256_GCM_SHA384 |
TLS_RSA_WITH_AES_256_CBC_SHA256 |
TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 |
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 |
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 |
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 |
TLS_DHE_RSA_WITH_AES_256_GCM_SHA384 |
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256 |
TLS_RSA_WITH_AES_128_CBC_SHA256>
```
- 2 Optionally display the administrative states for the TLS algorithms:

```
ssl algorithm show
```
- 3 Save the provisioned settings to the configuration file.

```
configuration save
```

Procedure 61 Installing a new SSL certificate

Use this procedure to activate a pre-downloaded RSA or ECC certificate on Waveserver 5's HTTPS server. The key and certificate must be bundled into PKCS#12 format V3 (X.509v3). This procedure is only applicable to HTTPS (REST and WebSocket sessions).

Requirements

- You must be logged in to Waveserver 5 using an account with super user access privileges.
- The server must be configured on Waveserver 5. See [Procedure 24, "Managing and configuring default servers"](#).

Note 1: Performing this procedure does not affect the default SSL RSA key and certificate file.

Note 2: This procedure forcefully closes any WebSocket connection and may interrupt REST operations. Ciena recommends that you avoid performing any REST operations while activating a new SSL server key.

Steps

- 1 Install the X.509 certificate on the Waveserver 5:

```
certificates entity install cert-name <String: [1..32]>
{default-ftp-server | default-sftp-server | default-tftp-
server | default-server | default-scp-server | sftp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128] | tftp-server <IP address or
host name> | scp-server <IP address or host name> login-
id <String [1..32]> password <String [1..128] | ftp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128]>} filename <String [1..127]>
cert-passphrase <String 1..128> [certificate-only]
```

For example:

```
certificates entity install cert-name cert tftp-server
1.2.3.4 cert.pem cert-passphrase
```
- 2 Set the certificate name for HTTPS:

```
system server https set cert-name <cert_name>
```
- 3 Verify the certificate name:

```
system server https show
```
- 4 Optionally display the certificate:

```
certificates entity show
```

Procedure 62 Activating the default SSL server RSA certificate

Waveserver 5 ships with a default, self-signed certificate. You can regenerate and replace this certificate. Use this procedure to activate a pre-downloaded RSA certificate on the Waveserver 5 HTTPS server. The key and certificate must be bundled into PKCS#12 format and only RSA format can be used.

Requirements

- You must be logged in to Waveserver 5 using an account with super user access privileges.
- The server must be configured on Waveserver 5.

Note 1: This procedure forcefully closes any WebSocket connection and may interrupt REST operations. Ciena recommends that you avoid performing any REST operations while regenerating an SSL server key.

Note 2: If an X.509 certificate has already been assigned to the HTTPS server using the [Installing a new SSL certificate](#) procedure, any changes to the default SSL server RSA key and certificate file will not affect the certificate used by HTTPS server, for example, the default is used only when the cert-name is not set.

Overview

The following externally generated RSA certificate types are supported:

- Key length: 2048, 4096
- Supported Ciphers: AES128, AES192, AES256, 3DES
- Hash: sha256, sha384, sha512

Note: This certificate can be used for HTTPS, REST, and WebSocket sessions.

Steps

- 1 Upload the following to Waveserver 5: An RSA key and certificate that is bundled into a PKCS#12 format. Transfer the RSA key from a remote SFTP or SCP server:

```
system xftp getfile local-filename <String: full
directory path and filename> [remote-filename <String: up
to 127 characters>] <default-server | default-sftp-server
| default-scpserver| sftp-server <IP address or host
name> login-id <String> password <String> secret <String>
| scp-server <IP address or host name> login-id <String>
password <String> secret <String>
```

Note: The attributes [login-id] and [password] must be specified.

2 Type the command:

```
ssl server rsa activate p12-file path <String[1..64]>  
password <Password string[1..128]>
```

The `path` is the full file path to the PKCS#12 file you uploaded in [step 1](#).

The `password` is the PKCS#12 password. If there is no password, you can omit the `password` parameter.

The system will respond:

```
WARNING: Successful command will trigger a httpd server  
restart.
```

```
Enter [Y/y] to continue or [N/n] to abort
```

3 Enter `y`.

The system will respond:

```
INFO: New certificate imported, restarting httpd server.  
It may take several minutes for httpd server to restart.
```

When the `activate` command has completed successfully, the `httpd` server will restart and any WebSocket connection will close. During this time, no REST operations can be performed. This activation operation may take several minutes to complete.

4 When the activation command is complete, reconnect the WebSocket connection, if required.**5** Verify the certificate name:

```
system server https show
```

Procedure 63 Regenerating the default SSL server RSA key

Waveserver 5 ships with a default, self-signed certificate that you can regenerate and activate. Use this procedure to delete an existing default RSA key and certificate file and regenerate a new default RSA self-signed key and certificate file. You can use the RSA certificate on the HTTPS server.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Note 1: This procedure forcefully closes any WebSocket connection and may interrupt REST operations. It is recommended that you avoid performing any REST operations while regenerating an SSL server key.

Note 2: If an X.509 certificate has already been assigned to the HTTPS server using the [Installing a new SSL certificate](#) procedure, any changes to the default SSL server RSA key and certificate file will not affect the certificate used by HTTPS server, for example, the default is used only when the cert-name is not set.

Steps

- 1 To delete the existing default key and certificate file and regenerate the new self-signed key and certificate file:

```
ssl server rsa regenerate
```

The system will respond:

```
WARNING: Successful command will trigger an httpd server
restart.
```

```
Enter [Y/y] to continue or [N/n] to abort
```

- 2 Enter y.

The system will respond:

```
INFO: New self-signed certificate generated, restarting
httpd server... It may take several minutes for httpd
server to restart.
```

When the regenerate command has completed successfully, the Apache server will restart and any WebSocket connection will close. During this time, no REST operations can be performed. This regeneration operation may take several minutes to complete.

- 3 When the regeneration command is complete, reconnect to your previous connection (HTTPS, WebSocket, or REST).

Note: You may need to remove the self-signed browser certificate to re-establish your connection.

Procedure 64 Listing all ACL rules

Use this procedure to generate a list of all existing ACL rules. The ACL rules will be displayed in the same order as they were created, which is the same order that they are used in packet matching.

Requirements

- You must be logged in to Waveserver using an account with at least limited access privileges.
- At least one ACL rule must exist.

Steps

- 1 In the command line, enter the command:

```
interface acl show [ipv4 | ipv6]
```

The existing acl rules are displayed:

Access Control List					
ID	Interface Name	Protocol	Source IP address	Destination IP address	Action
IPv4 Ingress					
1	local	ALL	10.134.161.251	0.0.0.0/0	allow
2	local	UDP	47.134.116.63/22	0.0.0.0/0	allow
IPv4 Egress					
4	local	all	0.0.0.0/0	192.168.2.21/0	drop

Procedure 65 Creating an ACL rule

Use this procedure to create a new Access Control List (ACL) rule. You can create up to 100 IPv4 filter rules and 100 IPv6 filter rules.

Requirements

You must be logged in to Waveserver using an account with at least super user access privileges.

Overview

The ACL feature adds filtering to ingress and egress traffic on a physical interface for both IPv4 and IPv6 traffic. The filtering rules determine whether incoming/outgoing DCN traffic is allowed or dropped based on a combination of factors, including:

- egress / ingress
- destination IP address/mask
- destination port
- source IP address/mask
- source port
- interface name
- protocol

Note 1: When you create ACL rules on a Waveserver node that is configured as a Gateway Network Element (GNE) and the rules specify an ingress ACL destination address of 0.0.0.0 (apply to all), those ACL rules may also be applied to subtending nodes managed by the GNE through ILAN. To limit the scope of the ACL rules to the GNE, set the ingress ACL destination address to the IP address of the GNE.

Note 2: In BGP mode, create ACL rules for both the if0-peer-1 and if0-peer-2 interface instead of the local interface.

Refer to the table [Options for creating ACL rules](#) for parameter value options. Rules are maintained in the same order as they were created by the user. A new ACL rule is appended to the end of the existing ACL rules.

Note: Misconfiguring the ACL rules could result in blocking of network access using SSH. If this happens, use the console port to correct the rules to unblock the network access.

Steps

- 1 Use the following command to create a new ACL rule:

```
interface acl <add | delete> <egress | ingress> <ipv4 |  
ipv6> action <allow | drop> destination-address <ip  
address/mask> destination-port <1 to 65535 | all>  
interface-name <string> protocol <tcp | udp | icmp | ipv6-  
icmp | all | 0 to 255> sequence-id <1 to 50> source-  
address <ip address/mask> source-port <1 to 65535 | all>
```

For details on the command parameters, see [Options for creating ACL rules](#).

Note 1: `source-address` must be specified and is any valid IP address/mask combination.

Note 2: `destination-address` defaults to zeros (0.0.0.0/0 for IPv4 and 0:0:0:0:0:0:0:0/0 for IPv6) if left blank, but if specified, can be any valid IP address/mask combination.

Procedure 66 Deleting ACL rules

Use this procedure to delete an ACL rule or all user defined ACL rules.

Requirements

You must be logged in to Waveserver using an account with at least super user access privileges.

Steps

- | | | | | | | | |
|------------------------------|---|------------------------------|-------------|----------------------------|------------------------------|---------------------|------------------------------|
| 1 | <table border="0"> <tr> <td style="padding-right: 20px;">If you want to delete</td> <td>Then</td> </tr> <tr> <td style="border-top: 1px solid black; padding-top: 5px;">all user defined ACL rules</td> <td style="border-top: 1px solid black; padding-top: 5px;">go to step 2</td> </tr> <tr> <td style="padding-top: 5px;">a specific ACL rule</td> <td style="padding-top: 5px;">go to step 4</td> </tr> </table> | If you want to delete | Then | all user defined ACL rules | go to step 2 | a specific ACL rule | go to step 4 |
| If you want to delete | Then | | | | | | |
| all user defined ACL rules | go to step 2 | | | | | | |
| a specific ACL rule | go to step 4 | | | | | | |
| 2 | To delete all ACL rules, enter:
<pre>interface acl delete <egress ingress> <ipv4 ipv6> all</pre> | | | | | | |
| 3 | Go to step 7 . | | | | | | |
| 4 | To find the values of the parameters for the ACL rule you want to delete, enter the command:
<pre>interface acl show <egress ingress> <ipv4 ipv6></pre> | | | | | | |
| 5 | From the resulting display, note the ID (sequence-id) of the rule you want to delete. | | | | | | |
| 6 | Enter the following command, with the appropriate parameters:
<pre>interface acl delete <egress ingress> <ipv4 ipv6>
sequence-id <1 to 50></pre> | | | | | | |
| 7 | Optionally, confirm that the ACL rule has been deleted:
<pre>interface acl <egress ingress> <ipv4 ipv6> show</pre> | | | | | | |

Procedure 67 Modifying an ACL list

Because ACL rules are maintained in the same order that you create them, a new ACL rule will be appended to the end of the existing ACL rules. Use this procedure to modify the ACL list.

Requirements

- You must be logged in to Waveserver using an account with at least admin user access privileges.
- At least one ACL rule must exist.

Steps

- 1 Display the list of ACL rules in the configuration file:
`config show`
ACL rules are listed in the INTERFACE MANAGER CONFIG: section of the configuration file.
- 2 Copy the INTERFACE MANAGER CONFIG section of the configuration file to a local file on your PC.
- 3 Modify, rearrange, add, or delete ACL rules in the local file.
- 4 Optionally, save the modified list to your local PC.
- 5 Delete the existing ACL list:
`interface acl <egress | ingress> <ipv4 | ipv6> delete all`
- 6 Cut and paste the modified list of ACL rules to Waveserver.
- 7 Optionally, confirm that the modified ACL rules are in place:
`interface acl <egress | ingress> <ipv4 | ipv6> show`

Procedure 68 Installing a trusted CA certificate

Use this procedure to install a trusted CA certificate from an external xFTP server onto the Waveserver 5. CA certificates are used to validate peer certificates.

Requirements

You must be logged in to Waveserver 5 using an account with at least super user access privileges.

Steps

- 1 To install a trusted CA certificate.

```
certificates authorities install cert-name
<user_assigned_Certificate_name> {default-ftp-server |
default-sftp-server | default-tftp-server | default-
server | default-scp-server | sftp-server <IP address or
host name> login-id <String [1..32]> password <String
[1..128] | tftp-server <IP address or host name> | scp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128] | ftp-server <IP address or host
name> login-id <String [1..32]> password <String
[1..128]>} filename <String [1..127]>
```

For example:

```
certificates authorities install cert-name myCAcert tftp-
server 1.2.3.4 filename ca.pem
```

- 2 Optionally display the summary information for the installed CA certificate.

```
certificates authorities show cert-name
<user_assigned_Certificate_name> [ca-cert-hash <CA
certificate hash[1..8]>]
```

Note 1: For RSA key types, the hash fields contain the hash values of the modulus field of the public key and the private key. You can use these values to verify that the certificate data matches the private key on disk. Use the following commands to obtain these values:

```
openssl x509 -noout -modulus -in certName.pem | openssl md5
openssl rsa -noout -modulus -in privKey.pem | openssl md5
```

Note 2: For ECDSA key types, the hash fields contain the hash value of the public key from the certificate and from the private key. The ECDSA key type does not have the modulus field. Use the following commands to obtain these values:

```
openssl x509 -noout -pubkey -in certName.pem | openssl md5
openssl pkey -pubout -in privateKey.pem | openssl md5
```

- 3 To rename the CA certificate after you have installed it:

```
certificates authorities rename current-cert-name  
<current_Certificate_name> new-cert-name  
<new_Certificate_name>
```

Procedure 69 Installing an X.509 certificate and private key

Use this procedure to install an X.509v3 certificate and private key generated by an external CA onto the Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with at least super user access privileges.
- You must have moved the PKCS#12 file generated by the external CA to the xFTP server, which is accessible by the Waveserver 5.

Steps

- 1 To install a X.509 certificate and private key.

```
certificates entity install cert-name <String: [1..32]>
{default-ftp-server | default-sftp-server | default-
tftpserver | default-server | default-scp-server |
sftpserver <IP address or host name> login-id <String
[1..32]> password <String [1..128] | tftp-server <IP
address or host name> | scp-server <IP address or host
name> loginid <String [1..32]> password <String [1..128]
| ftpserver <IP address or host name> login-id <String
[1..32]> password <String [1..128]>} filename <String
[1..127]> cert-passphrase
```

For example:

```
certificates entity install cert-name test1 tftp-server
1.2.3.4 filename test.pem.p12 cert-passphrase *****
```
- 2 Optionally display the summary information for the X.509 device certificate and private key.

```
certificates entity show [cert-name <Certificate name>]
```


Procedure 70 Creating a private key and installing a device certificate

Use this procedure to create a private key and install a device certificate for use by RADSec, Syslog over TLS, HTTPS connection from a web browser to access Waveserver 5 securely.

Requirements

- You must be logged in to Waveserver 5 using an account with at least super user access privileges.
- Access to a certificate authority or OpenSSL to sign the X.509v3 certificate.

Overview

The device certificate configuration file (.cnf file) should contain all of the arguments required by the OpenSSL command `openssl req` for CSR generation. If these arguments are missing from the file, and you are using CLI, OpenSSL prompts you to enter them, even if `[req] prompt= no` is present in the configuration file. However, if you are using a northbound interface to generate a CSR, ensure that all fields in the .cnf file are entered and correct, since OpenSSL cannot prompt you to enter missing fields.

Steps

- 1 On the external xFTP (FTP, SFTP, SCP, or TFTP) server, create a device certificate configuration file (.cnf file). The configuration file must contain the configuration for the certificate.

The device certificate configuration file fields for CLI are shown below:

```
[ req ]
prompt = no distinguished_name = req_distinguished_name
x509_extensions=v3_req
[ req_distinguished_name ]
```

C = Country

ST = State

L = Locality

O = organization

OU = Engineering

CN = Common name emailAddress = test@none.invalid

The device certificate configuration file fields for RESTCONF and NETCONF are shown below;

```
{{[ req ]prompt = no}}
{{distinguished_name = req_distinguished_name}}
{{[ req_distinguished_name ]}}
```

```

{{C = GB}}
{{ST = Test State}}
{{L = Test Locality}}
{{O = Organization Name}}
{{OU = Organization Unit Name}}
{{CN = Common Name}}
{{emailAddress = test@none.invalid}}

```

- 2 On the Waveserver 5, create a private key, download the certificate configuration file from the xFTP server, and generate a certificate signing request (CSR).

```

certificates entity csr generate cert-name <String:
[1..32]> key-type <rsa2048 | rsa3072 | rsa4096 | secp256r1
| secp384r1 | secp521r1 | use-existing> filename <String
[1..127]>] {default-ftp-server | default-sftp-server |
default-tftp-server | default-server | default-scp-server
| sftp-server <IP address or host name> login-id <String
[1..32]> password <String [1..128] | tftp-server <IP
address or host name> | scp-server <IP address or host
name> login-id <String [1..32]> password <String [1..128]
| ftp-server <IP address or host name> login-id <String
[1..32]> password <String [1..128]}}

```

For example:

```

certificates entity csr generate cert-name test1 tftp-
server 1.2.3.4 key-type rsa4096 filename test1.cnf

```

Note: The Waveserver 5 automatically uploads the test1.cnf.csr to the tftp server. This file should not exist on the server, otherwise the upload will fail.

- 3 Sign the certificate using a Certificate Authority or an OpenSSL tool.
- 4 Copy the signed certificate to the xFTP server.
- 5 On the Waveserver 5, install the device certificate.

```

certificates entity install cert-name <String: [1..32]>
{default-ftp-server | default-sftp-server | default-tftp-
server | default-server | default-scp-server | sftp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128] | tftp-server <IP address or
host name> | scp-server <IP address or host name> login-
id <String [1..32]> password <String [1..128] | ftp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128]}} filename <String [1..127]>
cert-passphrase [certificate-only]

```

For example:

```

certificates entity install cert-name test1 tftp-server
1.2.3.4 filename test.pem cert-passphrase certificate-
only

```

Note 1: The cert-name must match the cert-name specified in [step 2](#).

Note 2: If the signed certificate copied to the xFTP server in [step 4](#) has an embedded certificate chain, you must first install the RootCA's certificate into Waveserver 5 using the procedure, [Installing a trusted CA certificate](#).

- 6 Optionally display the summary information for the private key and device certificate.

```
certificates entity show [cert-name <String: 1..32>]
```

Procedure 71 Performing a Reset To Factory Defaults (RTFD)

Use this procedure to perform a reset to factory default settings (RTFD) using a software command or the reset button. This operation performs a secure erase that removes all critical and sensitive security parameters along with all other configuration data.

Requirements

- You are aware of the impact of a reset-to-factory-defaults action and have taken appropriate precautions. A reset-to-factory-defaults is traffic impacting.
- You have access to a PC with a console connection to the Waveserver 5.
- For a hardware reset, you must have a 1.3 mm (1/16-in) hex key or another small, blunt object to access the pinhole reset button.

Overview

Using the `configuration reset-to-factory-defaults` CLI command causes a cold restart of Waveserver 5. However, using the reset button to reset the chassis to factory defaults causes Waveserver 5 to be left in a ready to power down state. If you then want to return the Waveserver 5 to service, perform a power cycle after executing the hardware reset.

A reset-to-factory-defaults operation clears the existing configuration and licenses but does not delete the software load.

The `configuration reset-to-factory-defaults` command deletes user-configured information such as the hostname, IP address, configuration file, logs, user accounts, and passwords (except for the default su account). It does not reinstall the software load. After the RTFD operation and the resulting chassis-level cold restart, an existing AP backup is replaced with the currently committed software library and configuration data.



CAUTION

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

After the RTFD operation and the resulting chassis-level cold restart, an existing Access Panel backup is replaced with the currently committed software library and configuration data. For details, see [Auto configuration recovery](#).

Steps

- 1 If a backup configuration file is not available, create and save a backup configuration file. See [Configuration management](#).



WARNING

Performing a reset-to-factory defaults on Waveserver 5 is service-affecting. Before you perform the next step, ensure that configuration settings are saved and a backup configuration file is available. For more information, see [Configuration management](#).

- 2 **If** you want to perform a reset-to-factory-defaults action using

the software command

[step 3](#)

the reset button

[step 4](#)

- 3 To perform a software reset-to-factory-defaults action, use the command:

```
configuration reset-to-factory-defaults
```

A confirmation message appears. Type Y to proceed or N to cancel.

Waveserver 5 automatically performs a cold restart.

You have completed this procedure.

- 4 To perform a hardware reset-to-factory-defaults operation using a 1.3 mm (1/16 inch) hex key or another small, blunt object, press and hold the pinhole reset button until the “Status” LED flashes green and the “In Use” LED is yellow.

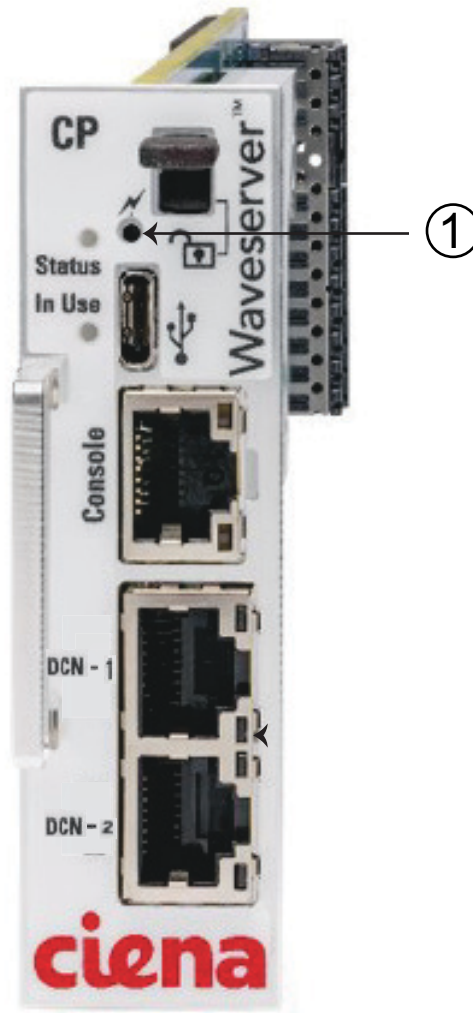


CAUTION

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena’s ability to isolate a failure event.

It may take some time (up to 40 seconds) before the “Status” LED flashes green and the “In Use” LED is yellow. During that time, the LEDs behave as follows:

- the “Status” LED flashes red and the “In Use” LED turns off
- the “Status” LED continues to flash red for a period of time
- the “Status” LED and “In Use” LED both flash green
- the “Status” LED flashes green and the “In Use” LED turns yellow. At this point you may release the pinhole reset button.

Figure 67 Control Processor pinhole reset button**Legend**

- 1 Pinhole reset button

Note: You must continue to press and hold the pinhole reset button until the “Status” LED flashes green and the “In Use” LED is yellow.

- 5 Release the reset button when the “Status” LED flashes green and the “In Use” LED is yellow.

- 6 When the RTFD completes, the “Status” and “In Service” LEDs are not illuminated, which indicates the Waveserver 5 is in a ready to power down state.

Note: Perform a power cycle only after the RTFD completes and the “Status” and “In Service” LEDs are no longer illuminated. When the RTFD completes and the “Status” and “In Service” LEDs are not illuminated you can perform a power cycle to return the Waveserver 5 to service. If you perform a power cycle before the RTFD completes, you risk deleting the software load.

CHAPTER 15

Equipment Management

This section contains the following topics:

- [Overview](#)
 - [Chassis attributes](#)
 - [Module attributes](#)
 - [Slot numbering](#)

Overview

This section provides information on equipment management.

Chassis attributes

The chassis has provisionable and operational attributes. The chassis also provides statistical data for analyzing and managing various components of the chassis.

Chassis provisionable attributes

The table below provides the chassis provisionable attributes when using CLI command `chassis <access-panel | fan | port | power | control-module>`.

Table 107 Chassis provisionable attributes

Component Name	Attribute		Description
	Name	Value	
chassis access-panel	state	• disable • enable	Administratively disables or enables the access panel.
chassis fan	state	• disable • enable	Administratively disables or enables the fan.

Component Name	Attribute		Description
	Name	Value	
chassis port	state	• disable • enable	Administratively disables or enables the management port.
	mode	• management • wayside-channel • redundant-bgp	Set the management port to management, wayside-channel, or redundant-bgp mode.
chassis power	state	• disable • enable	Administratively disables or enables the power supply.
chassis control-module	state	• disable • enable	Administratively disables or enables the Control Module (CM).

Chassis operational attributes

The following tables provide information on chassis operational attributes when using CLI command `chassis show`.

Table 108 Chassis capabilities operational attributes

Parameter	Description	Values
Chassis Model	Chassis model	Waveserver 5 Chassis
Chassis Type	Chassis type	3
No. of Control Modules	Number of control modules in the chassis	1
Control Module Type	Indicates the control module type	Field Replaceable
No. of Access Panel Modules	Number of access panel modules	1
Access Panel Module Type	Indicates the access panel module type	Field Replaceable
No. of Fan Modules	Number of fan modules in the chassis	4
Fan Module Type	Indicates the fan module type	Field Replaceable
No. of Power Supply Modules	Number of power supply modules in the chassis	2
Power Supply Module Type	Indicates the power supply module type	Field Replaceable
No. of Line Modules	Number of line modules in the chassis	8
Line Module Type	Indicates the line module type	Field Replaceable

Parameter	Description	Values
Air Filter	Indicates if there is an air filter	No
Air Filter Type	Indicates the air filter type	N/A
Air Filter Active	Indicates if the air filter is active	No
DC Power	Indicates if there is DC power	Yes
Redundant Power	Indicates if there is redundant power	Yes

Table 109 Chassis MAC partitions

Attribute	Description	Value
Partition	Indicates the partition type	• Chassis • Local Mgmt • Reserved
Base MAC	Indicates the base MAC	base MAC
Block Size	Indicates the block size	block size

Table 110 Chassis restart show

Attribute	Description	Value
Unit	Indicates the unit that was restarted	• Chassis • CM-1 • 1 to 8
Type	Indicates the restart type	• Warm • Cold • Unknown
Reason	Indicates the reason due to which the system was restarted	• User • Unknown • PowerOn • System
Last Restart Time	Indicates the last time the unit was restarted	yyyy-mm-ddThh:mm:ss+timezone

Table 111 Secure erase state

Parameter	Description	Value
Validation	Indicates if the RTFD was successful	• PASSED • FAILED

Parameter	Description	Value
Type	Indicates the type of secure erase performed	• RTFD - User request • RTFD - Push button • RTFD - Halt • RTFD - Recovery menu
Date	Indicates the date of the last secure erase	yyyy-mm-ddThh:mm:ss+timezone For example, 2021-04-30T18:20:05+00:00

Chassis hardware components utilization statistics

Waveserver 5 introduces enhanced temperature reporting for hardware components of the chassis and statistics for the chassis control-module (for example, Control Processor) CPU, memory, and disk utilization. Each statistic is sampled every 60-seconds. An average, minimum, and maximum values are calculated over the last 60-minutes of samples. The following tables provide information on attributes that are provided for each new statistic being gathered.

Table 112 Temperature and utilization statistic information

Statistic Information	Description
Instant	The value of the last sample value of the component for statistic use.
Average	The average value calculated over “interval” seconds. The “interval” value is 60 seconds by default.
Minimum	The minimum value calculated over the last 60 minutes.
Maximum	The maximum value calculated over the last 60 minutes.
Interval	The interval value used for collecting sample of statistical value and calculating average, minimum, and maximum values.

Temperature statistics for CP, AP, Chassis and individual service modules

Waveserver 5 introduces temperature statistic reporting for the following components:

- Chassis – air temperature at the intake of the chassis
- Control Module – module temperature
- Access Panel – module temperature
- Service Module – module temperature in one of the service slots of the chassis

All temperature readings are reported in Celsius. You can use the following commands to retrieve temperature readings of these components:

Table 113 Waveserver 5 CLI commands to retrieve temperature statistic data

Component Name	CLI Command
Chassis	<code>chassis show statistics</code>
Control Processor	<code>chassis control-module show statistics</code>
Access Panel	<code>chassis access-panel show statistics</code>
Service Module	<code>module show slot <slot> statistics</code>

CPU, memory, and disk utilization statistics for CP

Waveserver 5 introduces utilization statistics for the following components of the control module:

- CPU Utilization
- Disk Utilization
- Memory Utilization

All utilization statistics are reported in percentage (%). You can use the following command to retrieve utilization statistics:

Table 114 Waveserver 5 CLI commands to retrieve CPU, disk and memory utilization statistics

Component Name	CLI Command
CPU utilization (%)	<code>chassis control-module show statistics</code>
Disk utilization (%)	
Memory utilization (%)	

Module attributes

The Waveserver 5 allows you to set and view all attributes of the service module.

Module provisionable attributes

The following table provides information on module provisionable attributes when using CLI command `module set`.

Table 115 Module provisionable attributes

Attribute	Description	Values
Slot	The slot name for the module.	Slot Name
Label	The label for the module.	String[1..128]

Attribute	Description	Values
Mode	The mode for the module.	• none • mated

Module operational attributes

The following table provides information on module operational attributes when using CLI command `module show`.

Table 116 Module operational attributes

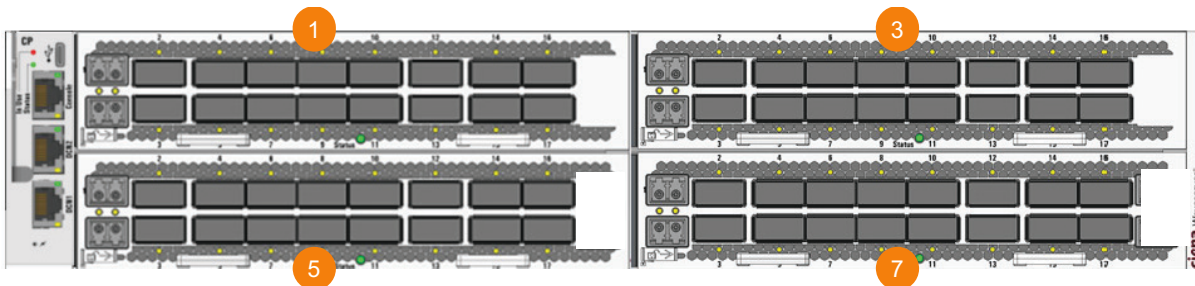
Admin Configuration / Operational Status	Attribute	Description	Values
Admin Configuration	Type	The type of module.	• MOTR-400-2-16 • MOTR-400-2-12 • MOTR-DCO-400-3-12 • Blank
	State	The administrative state of the module.	• Enabled • Disabled

Admin Configuration / Operational Status	Attribute	Description	Values
Operational Status	Type	The type of module.	• MOTR-400-2-16 • MOTR-400-2-12 • MOTR-DCO-400-3-12 • Filler
	State	The operational state of the module. Unequipped is displayed when there is no service module present. Equipped is displayed when the service module is present and the Admin Configuration is Enabled. Down is displayed when the service module is present and the Admin Configuration is Disabled. Occupied is displayed in the even slot, when there is a double-width service module present in the odd slot.	• Unequipped • Equipped • Down • Up • Occupied • Deprovisioned
	Last Restart	The last restart date and time of the module.	yyyy-mm-ddThh:mm:ss

Slot numbering

This section provides information on slot numbering on the Waveserver 5 chassis. See the following figure for slot numbering for double slot modules.

Figure 68 Slot numbering for double slot modules



CHAPTER 16

Facilities

This section provides information on the facilities available on the service modules supported by Waveserver 5. You can refer to the following topics to get information about facilities available on these modules:

- [Facilities for WL5e 2x MOTR 16x module](#)
- [Facilities for WL5e 2x MOTR 12x module](#)
- [Facilities for WL5n DCO 3xQSFP-DD 12x module](#)

Facilities for WL5e 2x MOTR 16x module

This section provides information on the facilities available on WL5e 2x MOTR 16x type service modules:

Module attributes

The following table defines provisionable module attributes:

Table 117 Provisionable module attributes

Attribute	Setting	Default Value	Description
Admin State	Enabled, Disabled	Enabled	The administrative state of the module.
Type	• MOTR-400-2-16	automatically detected by the system	Defines the module type.
Mode	• none • mated	none	Defines the module mode in which module is used.
Label	String (1...128)	—	Sets a user-defined description of the module for the ease of identification purpose in a network infrastructure.

You can also display module status using the `module show slot <slot>` command:

Table 118 Module status attributes

Attribute	Possible values	Description
Operational State	• Up • Down	Defines the current operational state of the module.
Model	• Waveserver WL5e 2x800G 16x100G/4x400G MOTR	Defines the model of the module.
Width	double	Defines the module width in a system.
Last Restart Type	• Warm • Cold	Defines the last restart type of the module.
Last Restart	yyyy-mmddThh:mm:ss+timezone	Indicates the last time the module was restarted.
Uptime	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the module has been in its current state.

You can also display module statistics, which is related to module temperature, using the `module show slot <slot> statistics` command:

Table 119 Module statistics

Attribute	Possible values	Description
Temperature, Celsius	—	Indicates the temperature statistics of the module: • Instant • Minimum • Maximum • Average • Interval

Line Transceiver (XCVR) attributes

The following table defines provisionable transceiver attributes for transceiver 1 and 2.

Table 120 Provisionable transceiver attributes for embedded coherent optics

Attribute	Setting	Default Value	Description
Admin state	Enabled, Disabled	Enabled	The administrative state of the line port.

Attribute	Setting	Default Value	Description
Mode	• 95-200-O • 95-250-O • 95-300-O • 95-350-O • 95-400-O • 95-450-O • 95-500-O • 95-550-O • 95-600-O • 95-650-O • 95-700-O • 95-800-O • 95-300-E • 95-350-E • 95-400-E • 95-450-E • 95-500-E • 95-550-E • 95-600-E • 95-650-E • 95-700-E • 95-800-E • 91.6-300-E • 91.6-400-E • 91.6-450-E • 91.6-500-E • 91.6-550-E • 91.6-600-E • 91.6-700-E • 91.6-800-E	95-400-O	Defines the transmission mode used by the modem transceiver.
Service Profile	• 100GE • 400GE • mixed-ethernet • mixed-otn-ethernet	100GE	Defines the service profile applied to the line port.

You can display the provisioned XCVR attributes in the Transceiver-Status table using the CLI command `xcvr show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Line PTP attributes

This section provides information on PTP attributes for PTP 1 and 2.

Table 121 Provisionable PTP attributes for embedded coherent optics

Attribute	Settings	Default value	Description
Admin state	• Enabled • Disabled	Enabled	Sets the administrative state of the port.

Attribute	Settings	Default value	Description
Channel contention detection avoidance	• Enabled • Disabled	Disabled	Sets the status (enabled or disabled) of the Channel Contention Detection and Avoidance feature. This feature is triggered by new wavelength/frequency provisioning and detects whether the provisioned wavelength/frequency is already used. If it is, channel contention is declared. Line port Tx is held in a low power state (<-35 dBm) and a Tx Channel Contention alarm is raised.
Expected-restoration-time	0 - 255 (seconds)	10	Sets the maximum restoration time (in seconds).
Forward-error-correction	Enabled	Enabled	Sets the status of forward error correction.
Frequency (GHz)	191250.0..196125.0 GHz (C-Band) 186087.5...190956.2 GHz (L-Band)	0 - for both bands	Sets the frequency the modem transceiver uses.
HCCS-threshold	-1.00..4.00 dB (N.MM)	0	High Correction Count Second (HCCS) for the line ports.
Laser centering mode	• Enabled • Disabled • Holdover	Disabled	Sets the laser-centering mode. • Enabled: Laser centering algorithm enabled. • Disabled: Laser centering algorithm disabled. • Holdover: A holdover setting for troubleshooting purposes.
Laser centering range	Laser Centering Range in GHz (N.M): 0.0 .. 3.0	2.5	Sets the laser-centering range.
Laser shutdown mode	• Normal • Accelerated	Accelerated	Sets the laser shutdown mode.
Laser tuning mode	• Normal • Accelerated • Fast	Normal	Sets the laser-tuning mode. • Normal: includes a self-calibration test to optimize performance. This results in a longer tuning time than Accelerated tuning mode. • Accelerated: performs a partial self-calibration test. As a result, there can be a small degradation in performance. • Fast: omits the self-calibration test. As a result, there can be a small degradation in performance.

Attribute	Settings	Default value	Description
Line system type	• Coloured • Colourless • Contentionless • Coherent Select (CS)-Coloured • Coherent Select (CS)- Colourless	Coloured	Sets the optical channel line system type.
Link optimization mode	• Standard • Non-linear • SNR-Tolerant • Fast-Tracking	Standard	Sets the performance optimization mode.
Modem class	• terrestrial • submarine	Default is based on module variant.	The modem type.
Prefec-sd-threshold	Prefec-sd-threshold in dB (N.MM): -1.00 - 3.50 (dB)	0.5 dB	Sets the threshold level for the Pre-FEC Signal Degrade alarm (in dB). The threshold attribute setting represents the margin in dB units from the 0 dB reference point. The 0 dB reference point represents an operating point where the system may see significant post FEC errors and become unstable. To retrieve the Signal Degrade values, use the command <code>port show port <slot>/<port></code> .
Prefec-sf-threshold	Prefec-sf-threshold in dB (N.MM): -1.00 - 1.00	0 dB	Sets the threshold level for the Pre-FEC Signal Fail alarm (in dB). The threshold attribute setting represents the margin in dB units from the 0 dB reference point. The 0 dB reference point represents an operating point where the system may see significant post FEC errors and become unstable. To retrieve the Signal Fail values, use the command <code>port show port <slot>/<port></code> .

Attribute	Settings	Default value	Description
Reach limit	• 250 • 500 • 1000 • 2000 • 2100 • 3000 • 3500 • 4000 • 5000 • 8000 • 9000 • 14000	• Terrestrial - 1000 • Submarine - 3500	Maximum reach of the modem.
SPLI	• Enabled • Disabled	Enabled	Sets the administrative state of SPLI.

Attribute	Settings	Default value	Description
Tx dispersion compensation mode	• Automatic (default) • Fixed • Search	Automatic	Determines how dispersion compensation for the transmitter is managed. In Automatic mode, the user must provision the Tx dispersion path value based on a determined link engineering value from One Planner, or based on the value determined by search mode. For more information on search mode, see details below. The Tx dispersion precompensation value is not used in automatic mode. In Fixed mode, the user uses the dispersion scan tool to measure the Tx dispersion precompensation setting required for the link. The Tx dispersion precompensation value can then be provisioned based on the results from the dispersion scan tool. For more information on the dispersion scan tool, see Dispersion scan. The Tx dispersion path value is not used in manual mode. In Search mode, the algorithm slowly scans for the best Tx dispersion precompensation value in the absence of any fiber information. This mode should only be used for new line ports (no running traffic) and can be useful when no link information is available. After the search completes, the user can retrieve the path dispersion using CLI command <code>application dispersion-scan show ptp <slot/port></code>, provision the Tx dispersion path value based on the scan results, and then switch the Tx dispersion compensation mode back from search to automatic. The Tx-dispersion-compensation-mode can be automatic, fixed or search, and must be set based on the application; protected/unprotected terrestrial or submarine. For the required application settings, see Line port PTP parameters - chromatic dispersion compensation details.
Tx dispersion path value	-89000..1044500	76000	Sets the Tx dispersion value for the path that is used to manage Tx dispersion compensation. The system default value for Tx dispersion path value is 76000 ps/nm and can accommodate links up to 4000 km.

Attribute	Settings	Default value	Description
Tx dispersion precompensation value	-44500..1000000	38000	Sets the Tx precompensation value that is used to manage Tx dispersion compensation. The Tx-dispersion-compensation-mode is automatic, fixed or search, and must be set based on the application; protected/unprotected terrestrial or submarine. For the required application settings, see Line port PTP parameters - chromatic dispersion compensation details .
Tx dispersion search mode maximum	-89000..1044500	0	Sets the maximum value for tx dispersion search mode.
Tx dispersion search mode minimum	-89000..1044500	0	Sets the minimum value for tx dispersion search mode.
Tx power	-9 dBm to 4 dBm	-9 dBm	Sets the line transmit optical power. See Technical specifications .
Tx power reduction state	• Enabled • Disabled	Disabled	Sets the status (enabled or disabled) of the Tx Power Reduction feature. • Enabled: forces the Tx power to be held in the low power state. • Disabled: releases the Tx power from being held in the low power state (only if there is no channel conflict detected when Channel Contention and Avoidance is enabled).
Tx state	• Enabled • Disabled	Enabled	Sets the laser state of the transmitter.

The following table describes the Line PTP Tx Dispersion Compensation Mode settings and the other parameters that must be set for specific applications.

In Automatic mode, an algorithm looks for the best Tx pre-compensation value. The Tx pre-compensation value adjusts automatically following a path change. In Fixed mode, the Tx pre-compensation value is manually set. The Tx pre-compensation value does not adjust automatically following a path change.

Note: The settings apply to all module variants.

Table 122 Line port PTP parameters - chromatic dispersion compensation details

Application	Line PTP Tx Dispersion Compensation Mode setting specified by OnePlanner	Line PTP Tx Dispersion Compensation Mode setting	Line PTP Tx Dispersion path-value	Line PTP Tx Dispersion precompensation-value
Unprotected terrestrial links with path dispersion	Automatic	Automatic	Set to the value provided by OnePlanner (“Receiver Dispersion Max”). (see Note)	Not required
Protected terrestrial links	Automatic	Automatic	Set to the value provided by OnePlanner (“Receiver Dispersion Max”). (see Note)	Not required
Protected terrestrial links	Fixed	Fixed	Not required	Set to the value provided by OnePlanner.
Protected terrestrial links	Not applicable	Fixed	Not required	Set the value to the average of the two extremes of the set of protection paths.
Submarine links	Not applicable	Fixed	Not required	Set to the value provided by OPNet.
Note: If no path dispersion value is provided by OnePlanner (for example for a deployment on a non-Ciena line system) the actual path dispersion reported by the modem should be used as the provisioned path dispersion value instead of the default value of 76000 ps/nm.				

Transmission Mode and Link Optimization Mode combinations

The following table lists the Link Optimization Mode settings supported for each transmission mode.

Table 123 Transmission Mode and Link Optimization Mode combinations

Transmission Mode	Link Optimization Mode (Note)
95-200-O	Non-linear
95-250-O	Non-linear
95-300-O	Standard, SNR-Tolerant, Non-linear
95-350-O	Standard, SNR-Tolerant, Non-linear

Transmission Mode	Link Optimization Mode (Note)
95-400-O	Standard, SNR-Tolerant, Non-linear, Fast-Tracking
95-450-O	Standard, SNR-Tolerant, Non-linear
95-500-O	Standard, SNR-Tolerant, Non-linear
95-550-O	Standard, SNR-Tolerant
95-600-O	Standard, SNR-Tolerant, Fast-Tracking
95-650-O	Standard
95-700-O	Standard
95-800-O	SNR-Tolerant
95-300-E	Standard, Non-linear
95-350-E	Standard, SNR-Tolerant, Non-linear
95-400-E	Standard, SNR-Tolerant, Non-linear
95-450-E	Standard, SNR-Tolerant, Non-linear
95-500-E	Standard, SNR-Tolerant
95-550-E	Standard, SNR-Tolerant
95-600-E	Standard, SNR-Tolerant
95-650-E	Standard, SNR-Tolerant
95-700-E	Standard
95-800-E	Standard, SNR-Tolerant
91.6-300-E	Standard, Non-linear
91.6-400-E	Standard, SNR-Tolerant
91.6-450-E	SNR-Tolerant
91.6-500-E	Standard, SNR-Tolerant
91.6-550-E	SNR-Tolerant
91.6-600-E	Standard
91.6-700-E	Standard
91.6-800-E	Standard, SNR-Tolerant

Transmission Mode	Link Optimization Mode (Note)
Note: When Link Optimization Mode is set to Non-linear, modem performance is optimized for compensated submarine lines with higher non-linear noise. When Link Optimization Mode is set to SNR-tolerant, the modem is optimized for performance on uncompensated submarine lines or for 800Gb/s terrestrial lines. When Link Optimization Mode is set to Standard, modem performance is optimized for typical terrestrial applications. The Link Optimization Mode used should be that specified by the Ciena link budget tools or analysis.	

Display Line PTP attributes

You can display the provisioned PTP attributes in the PTP Status Summary table using the CLI command `ptp show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Line Port attributes

The following table defines provisionable port attributes.

Table 124 Provisionable port attributes for embedded coherent optics

Attribute	Settings	Default Value	Description
Conditioning Type	OTN	OTN	The type of conditioning signal is always OTN.
Admin state	- Enabled - Disabled	Enabled	Sets the administrative state of the port.
Label	String (1...128)		Sets a user-defined description for the port.
Loopback	- Disabled - Rx - Tx	Disabled	Operates loopback on a port.
OTU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the OTU signal degrade threshold.
ODU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the ODU signal degrade threshold.
Mismatch mode	operator-only, sapi, dapi, sapi-and-dapi	operator-only	Specifies the string, which is used by the system to determine whether a mismatch is present. Note: Applicable to both section and path trace.

Attribute	Settings	Default Value	Description
Mismatch fail mode	none, alarm-only	none	Specifies the action to take when a string mismatch is detected. • alarm-only: the system raises an alarm. • none: no action is taken. Note: Applicable to both section and path trace.
tx-sapi	String (1...15)		Specifies the SAPI string to transmit to the far-end node. Note: Applicable to both section and path trace.
tx-dapi	String (1...15)		Specifies the DAPI string to transmit to the far-end node. Note: Applicable to both section and path trace.
tx-oper	String (1...32)		Sets value of the transmit operator string. Note: Applicable to both section and path trace.
tx-oper-mode	automatic, manual	automatic	Sets the transmit operator mode. • automatic: sets by the system. • manual: as specified in the tx-oper attribute. Note: Applicable to both section and path trace.
exp-sapi	String (1...15)		Specifies the SAPI string expected from the far-end node. Note: Applicable to both section and path trace.
exp-dapi	String (1...15)		Specifies the DAPI string expected from the far-end node. Note: Applicable to both section and path trace.
exp-oper	String (1...32)		Sets value of the expected transmit operator string. Note: Applicable to both section and path trace.

You can display the provisioned port attributes in the Port Table using the CLI command `port show`. The following table defines the possible port attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 125 Port status attributes for embedded coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Label	—	Name of the port, displayed as <slot>/<port> or <slot>/<port>.<port section>
	State	Enabled, Disabled	The administrative state of the port.
	Type	OTUCn, OTUCn_Flex	The port type.
	Rate	OTUC2 - OTUC16	The port rate (speed).
	Interface Type	I-NNI	The type of interface used by the port.
	Loopback	Tx, Rx, Disabled	The administrative state of loopback testing on the port.
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the port: • Up — the port is operating normally. • Down — the port is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the port is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the port.
	Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the port has been in its current state.
	ODU Termination Mode	Terminated, Passthrough	Sets the termination mode for the port that is configured by the system.

Line Channel attributes

The following table defines provisionable channel attributes.

Table 126 Provisionable channel attributes for embedded coherent optics

Attribute	Settings	Default value	Description
Label	String (1...128)		Sets a user-defined description of the channel.
admin-state	enabled, disabled	enabled	The administrative state of the channel.
ODU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the ODU signal degrade threshold. Tjjs attribute is not applicable for ET channel types.
Mismatch mode	operator-only, sapi, dapi, sapi-and-dapi	operator-only	Specifies the string, which is used by the system to determine whether a mismatch is present. Note: Applicable to the path trace only.
Mismatch fail mode	none, alarm-only	none	Specifies the action to take when a string mismatch is detected. - alarm-only: the system raises an alarm. - none: no action is taken. Note: Applicable to the path trace only.
tx-sapi	String (1...15)		Specifies the SAPI string to transmit to the far-end node. Note: Applicable to the path trace only.
tx-dapi	String (1...15)		Specifies the DAPI string to transmit to the far-end node. Note: Applicable to the path trace only.
tx-oper	String (1...32)		Sets value of the transmit operator string. Note: Applicable to the path trace only.
tx-oper-mode	automatic, manual	automatic	Sets the transmit operator mode. - automatic: sets by the system. - manual: as specified in the tx-oper attribute. Note: Applicable to the path trace only.
exp-sapi	String (1...15)		Specifies the SAPI string expected from the far-end node. Note: Applicable to the path trace only.
exp-dapi	String (1...15)		Specifies the DAPI string expected from the far-end node. Note: Applicable to the path trace only.

Attribute	Settings	Default value	Description
exp-oper	String (1...32)		Sets value of the expected transmit operator string. Note: Applicable to the path trace only.

You can display the provisioned channel attributes in the Channel table using the CLI command `channel show`. The following table defines the possible channel attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 127 Channel status attributes for embedded coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Name	—	Name of the channel, displayed as <slot>/<port>.<port section>
	Admin State	Enabled, Disabled	The administrative state of the channel.
	Type	ODUk, ET	The channel type.
	Rate	ODU4, ET	The channel rate (speed).
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the channel: • Up — the channel is operating normally. • Down — the channel is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the channel is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the channel.
	State cli Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the channel has been in its current state.

You can also display the connection information between the line port channel and their corresponding client port using the `connection show` command.

Table 128 Connection information

Attribute	Possible Values	Description
"From" Facility name	Port-<slot/port>	Client port name.
Bandwidth	• 100G • 400G	Defines the required bandwidth of the client port. For 100GE, OTL4.4 clients it is 100G and for 400GE clients it is 400G.
"To" Facility name	• Channel-<slot/port.channel> • Port-<slot/port>	For terminal applications, defines the channel name of the line port to which the client port is connected. For regen application, defines the line port name to which the client port is connected.

Client Transceiver (XCVR) attributes

The following table defines provisionable transceiver attributes for transceiver ports 3 to 18 (the client side ports).

Table 129 Provisionable transceiver attributes for pluggable non-coherent optics

Attribute	Description
Mode (Client)	The mode used by the client port. • 100GE • 400GE • OTL4.4

You can display the provisioned client XCVR attributes in the Transceiver-Status table using the CLI command `xcvr show`. For more information about displaying entity attributes, see [Procedure 82, "Retrieving Waveserver 5 entity data"](#).

Client PTP attributes

The following table defines provisionable ptp attributes for a client port.

Table 130 Provisionable Client PTP attributes pluggable non-coherent optics

Attribute	Possible Values	Default Value	Description
Forward Error Correction	Enabled Disabled	• 100GE -- Note 2, Note 3 • OTL4.4 -- Note 2, Note 3 • 400GE -- Enabled	Sets the status of forward error correction. Note 1: The FEC is always <i>enabled</i> for the 100G-FR (160-9431-900/160-9433-900) plug. Note 2: For 100GE and OTL4.4 client ports, if the plug is available while provisioning the xcvr (automatic or manual) the system selects the recommended FEC value, by default, based on the plug type. Note 3: For 100GE and OTL4.4 client ports, if the plug is not available and xcvr is provisioned manually, the default FEC is <i>disabled</i>. Note 4: For 400GE and FOIC1.4 client ports, the FEC is always <i>enabled</i>.
Tx state	Enabled Disabled	Enabled	Sets the laser state of the transmitter.

Display Client PTP attributes

You can display the provisioned PTP attributes in the PTP Status Summary table using the CLI command `ptp show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Client Port attributes

The following table defines provisionable client port attributes.

Table 131 Provisionable client port attributes for pluggable non-coherent optics

Attribute	Settings	Default Value	Description
Conditioning Holdoff	From 0 ms to 100 ms in 10 ms increments; From 100 ms to 1000 ms in 100 ms increments	10ms	Sets the conditioning hold-off timer.

Attribute	Settings	Default Value	Description
Conditioning Type	Ethernet None Laser-off OTN	• OTL4.4 -- OTN • 100GE, 400GE -- Ethernet	Sets the type of conditioning signal to transmit to the client subtending equipment when a failure is detected on the line port or upstream remote client port. The conditioning type choices are dependent on the client protocol. For Ethernet (100GE), the options are none, laser-off, and Ethernet (default) and for OTN (OTL4.4), the conditioning type should be OTN (default). • Ethernet: Ethernet primitives (Local Fault and Remote Fault). • None: No signal conditioning applied. • Laser-off: shut-down the laser at the local chassis.
Label	String (1...128)		Sets a user-defined description for the port.
Loopback	Disabled Rx Tx	Disabled	Operates loopback on a port.
OTU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the OTU signal degrade threshold.
ODU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the ODU signal degrade threshold.
Mismatch mode	operator-only, sapi, dapi, sapi-and-dapi	operator-only	Specifies the string, which is used by the system to determine whether a mismatch is present. Note: Applicable to the section trace.
Mismatch fail mode	none, alarm-only	none	Specifies the action to take when a string mismatch is detected. • alarm-only: the system raises an alarm. • none: no action is taken. Note: Applicable to the section trace.
tx-sapi	String (1...15)		Specifies the SAPI string to transmit to the far-end node. Note: Applicable to the section trace.

Attribute	Settings	Default Value	Description
tx-dapi	String (1...15)		Specifies the DAPI string to transmit to the far-end node. Note: Applicable to the section trace.
tx-oper	String (1...32)		Sets value of the transmit operator string. Note: Applicable to the section trace.
tx-oper-mode	automatic, manual	automatic	Sets the transmit operator mode. • automatic: sets by the system. • manual: as specified in the tx-oper attribute. Note: Applicable to the section trace.
exp-sapi	String (1...15)		Specifies the SAPI string expected from the far-end node. Note: Applicable to the section trace.
exp-dapi	String (1...15)		Specifies the DAPI string expected from the far-end node. Note: Applicable to the section trace.
exp-oper	String (1...32)		Sets value of the expected transmit operator string. Note: Applicable to the section trace.

You can display the provisioned client port attributes in the Port Table using the CLI command `port show`. The following table defines the possible port attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 132 Client Port attributes for pluggable non-coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Name	—	Name of the port, displayed as <slot>/<port> or <slot>/<port>.<port section>
	State	Enabled, Disabled	The administrative state of the port.
	Type	OTUCn, OTUCn_Flex, OTUk, Ethernet	The port type.
	Rate	100GE, 400GE, OTU4, OTUC2-OTUC16	The port rate (speed).
	Interface Type	UNI, I-NNI, E-NNI	The type of interface used by the port.
	Loopback	Tx, Rx, disabled	The administrative state of loopback testing on the port.
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the port: • Up — the port is operating normally • Down — the port is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the port is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the port.
	Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the port has been in its current state.

Facilities for WL5e 2x MOTR 12x module

This section provides information on the facilities available on WL5e 2x MOTR 12x type modules.

Module attributes

The following table defines provisionable module attributes.

Table 133 Provisionable module attributes

Attribute	Setting	Default Value	Description
Admin State	Enabled, Disabled	Enabled	The administrative state of the module.
Type	MOTR-400-2-12	MOTR-400-2-12	Defines the module type.
Mode	• none • mated	none	Defines the module mode in which module is used.
Label	String (1...128)	—	Sets a user-defined description of the module for the ease of identification purpose in a network infrastructure.

You can also display module status using the `module show slot <slot>` command:

Table 134 Module status attributes

Attribute	Possible values	Description
Operational State	• Up • Down	Defines the current operational state of the module.
Model	Waveserver WL5e 2x800G 12x100G/4x400G MOTR	Defines the model of the module.
Width	double	Defines the module width in a system.
Last Restart Type	• Warm • Cold	Defines the last restart type of the module.
Last Restart	yyyy-mmddThh:mm:ss+timezone	Indicates the last time the module was restarted.
Uptime	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the module has been in its current state.

You can also display module statistics, which is related to module temperature, using the `module show slot <slot> statistics` command:

Table 135 Module statistics

Attribute	Possible values	Description
Temperature, Celsius	—	Indicates the temperature statistics of the module: • Instant • Minimum • Maximum • Average • Interval

Line Transceiver (XCVR) attributes

The following table defines provisionable transceiver attributes for transceiver 1 and 2.

Table 136 Provisionable transceiver attributes for embedded coherent optics

Attribute	Setting	Default Value	Description
Mode	• 95-200-O • 95-250-O • 95-300-O • 95-350-O • 95-400-O • 95-450-O • 95-500-O • 95-550-O • 95-600-O • 95-650-O • 95-700-O • 95-800-O • 82-400-E • 85-300-E • 85-400-E • 85-600-E • 90-400-E • 90-600-E • 93.3-400-E • 93.3-600-E • 95-300-E • 95-350-E • 95-400-E • 95-450-E • 95-500-E • 95-550-E • 95-600-E • 95-650-E • 95-700-E • 95-800-E • 91.6-300-E • 91.6-400-E • 91.6-450-E • 91.6-500-E • 91.6-550-E • 91.6-600-E • 91.6-700-E • 91.6-800-E	95-400-O	Defines the transmission mode used by the modem transceiver.
Service Profile	• 100GE • 400GE • 4x100GE • mixed-ethernet • mixed-otn-ethernet	100GE	Defines the service profile applied to the line port.
admin-state	enabled, disabled	enabled	The administrative state of the line port.

You can display the provisioned XCVR attributes in the Transceiver-Status table using the CLI command `xcvr show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Line PTP attributes

This section provides information on PTP attributes for PTP 1 and 2.

Table 137 Provisionable PTP attributes for embedded coherent optics

Attribute	Settings	Default value	Description
Channel contention detection avoidance	Enabled Disabled	Disabled	Sets the status of the Channel Contention Detection and Avoidance feature. This feature is triggered by new wavelength/frequency provisioning and detects whether the provisioned wavelength/frequency is already used. If it is, channel contention is declared. Line port Tx is held in a low power state (<-35 dBm) and a Tx Channel Contention alarm is raised.
Expected-restoration-time	0 - 255 (seconds)	10	Sets the maximum restoration time (in seconds).
Forward-error-correction	Enabled	Enabled	Sets the status of forward error correction.
Frequency (GHz)	191250.0..196125.0 GHz (C-Band) 186087.5...190956.2 GHz (L-Band)	0 - for both bands	Sets the frequency the modem transceiver uses.
HCCS-threshold	-1.00..4.00 dB (N.MM)	0	High Correction Count Second (HCCS) for the line ports.
Laser centering mode	• Enabled • Disabled • Holdover	Disabled	Sets the laser-centering mode. • Enabled: Laser centering algorithm enabled. • Disabled: Laser centering algorithm disabled. • Holdover: A holdover setting for troubleshooting purposes.
Laser centering range	Laser Centering Range in GHz (N.M): 0.0 .. 3.0	2.5	Sets the laser-centering range.
Laser shutdown mode	• Normal • Accelerated	Accelerated	Sets the laser shutdown mode.

Attribute	Settings	Default value	Description
Laser tuning mode	• Normal • Accelerated • Fast	Normal	Sets the laser-tuning mode. • Normal: includes a self-calibration test to optimize performance. This results in a longer tuning time than Accelerated tuning mode. • Accelerated: performs a partial self-calibration test. As a result, there can be a small degradation in performance. • Fast: omits the self-calibration test. As a result, there can be a small degradation in performance.
Line system type	• Coloured • Colourless • Contentionless • Coherent Select (CS)-Coloured • Coherent Select (CS)- Colourless	Coloured	Sets the optical channel line system type.
Link optimization mode	• Standard • Non-linear • SNR-Tolerant • Fast-Tracking	Standard	Sets the performance optimization mode.
Modem class	• terrestrial • submarine	Default is based on module variant.	The modem type.
Prefec-sd-threshold	Prefec-sd-threshold in dB (N.MM): -1.00 - 3.50 (dB)	0.5 dB	Sets the threshold level for the Pre-FEC Signal Degrade alarm (in dB). The threshold attribute setting represents the margin in dB units from the 0 dB reference point. The 0 dB reference point represents an operating point where the system may see significant post FEC errors and become unstable. To retrieve the Signal Degrade values, use the command <code>port show port <slot>/<port></code>.

Attribute	Settings	Default value	Description
Prefec-sf-threshold	Prefec-sf-threshold in dB (N.MM): -1.00 - 1.00	0 dB	Sets the threshold level for the Pre-FEC Signal Fail alarm (in dB). The threshold attribute setting represents the margin in dB units from the 0 dB reference point. The 0 dB reference point represents an operating point where the system may see significant post FEC errors and become unstable. To retrieve the Signal Fail values, use the command <code>port show port <slot>/<port></code> .
Reach limit	• 250 • 500 • 1000 • 2000 • 2100 • 3000 • 3500 • 4000 • 5000 • 8000 • 9000 • 14000	• Terrestrial - 1000 • Submarine - 3500	Maximum reach of the modem.
SPLI	• Enabled • Disabled	Enabled	Sets the administrative state of SPLI.
admin-state	• Enabled • Disabled	Enabled	The administrative state of the ptp.

Attribute	Settings	Default value	Description
Tx dispersion compensation mode	• Automatic • Fixed • Search	Automatic	Determines how dispersion compensation for the transmitter is managed. In Automatic mode, the user must provision the Tx dispersion path value based on a determined link engineering value from OnePlanner, or based on the value determined by search mode. For more information on search mode, see details below. The Tx dispersion precompensation value is not used in automatic mode. In Fixed mode, the user uses the dispersion scan tool to measure the Tx dispersion precompensation setting required for the link. The Tx dispersion precompensation value can then be provisioned based on the results from the dispersion scan tool. For more information on the dispersion scan tool, see Dispersion scan. The Tx dispersion path value is not used in manual mode. In Search mode, the algorithm slowly scans for the best Tx dispersion precompensation value in the absence of any fiber information. This mode should only be used for new line ports (no running traffic) and can be useful when no link information is available. After the search completes, the user can retrieve the path dispersion using CLI command <code>application dispersion-scan show ptp <slot/port></code>, provision the Tx dispersion path value based on the scan results, and then switch the Tx dispersion compensation mode back from search to automatic. The Tx-dispersion-compensation-mode can be automatic, fixed or search, and must be set based on the application; protected/unprotectedterrestrial or submarine. For the required application settings, see Line port PTP parameters - chromatic dispersion compensation details.
Tx dispersion path value	-89000..1044500	76000	Sets the Tx dispersion value for the path that is used to manage Tx dispersion compensation. The system default value for Tx dispersion path value is 76000 ps/nm and can accommodate links up to 4000 km.

Attribute	Settings	Default value	Description
Tx dispersion precompensation value	-44500..1000000	38000	Sets the Tx precompensation value that is used to manage Tx dispersion compensation. The Tx-dispersion-compensation-mode is automatic, fixed or search, and must be set based on the application; protected/unprotected terrestrial or submarine. For the required application settings, see Line port PTP parameters - chromatic dispersion compensation details .
Tx dispersion search mode maximum	-89000..1044500	0	Sets the maximum value for tx dispersion search mode.
Tx dispersion search mode minimum	-89000..1044500	0	Sets the minimum value for tx dispersion search mode.
Tx power	-9 dBm to 4 dBm	-9 dBm	Sets the line transmit optical power. See Technical specifications .
Tx power reduction state	• Enabled • Disabled	Disabled	Sets the status (enabled or disabled) of the Tx Power Reduction feature. • Enabled: forces the Tx power to be held in the low power state • Disabled: releases the Tx power from being held in the low power state (only if there is no channel conflict detected when Channel Contention and Avoidance is enabled)
Tx state	• Enabled • Disabled	Enabled	Sets the laser state of the transmitter.

The following table describes the Line PTP Tx Dispersion Compensation Mode settings and the other parameters that must be set for specific applications.

In Automatic mode, an algorithm looks for the best Tx pre-compensation value. The Tx pre-compensation value adjusts automatically following a path change. In Fixed mode, the Tx pre-compensation value is manually set. The Tx pre-compensation value does not adjust automatically following a path change.

Note: The settings apply to all module variants.

Table 138 Line port PTP parameters - chromatic dispersion compensation details

Application	Line PTP Tx Dispersion Compensation Mode setting specified by OnePlanner	Line PTP Tx Dispersion Compensation Mode setting	Line PTP Tx Dispersion path-value	Line PTP Tx Dispersion precompensation-value
Unprotected terrestrial links with path dispersion	Automatic	Automatic	Set to the value provided by OnePlanner ("Receiver Dispersion Max"). (see Note)	Not required
Protected terrestrial links	Automatic	Automatic	Set to the value provided by OnePlanner ("Receiver Dispersion Max"). (see Note)	Not required
Protected terrestrial links	Fixed	Fixed	Not required	Set to the value provided by OnePlanner
Protected terrestrial links	Not available	Fixed	Not required	Set the value to the average of the two extremes of the set of protection paths.
Submarine links	Not applicable	Fixed	Not required	Set to the value provided by OPNet.
Note: If no path dispersion value is provided by OnePlanner (for example for a deployment on a non-Ciena line system) the actual path dispersion reported by the modem should be used as the provisioned path dispersion value instead of the default value of 76000 ps/nm.				

Transmission Mode and Link Optimization Mode combinations

The following table lists the Link Optimization Mode settings supported for each transmission mode.

Table 139 Transmission Mode and Link Optimization Mode combinations

Transmission Mode	Link Optimization Mode (Note)
95-200-O	Non-linear
95-250-O	Non-linear

Transmission Mode	Link Optimization Mode (Note)
95-300-O	Standard, SNR-Tolerant, Non-linear
95-350-O	Standard, SNR-Tolerant, Non-linear
95-400-O	Standard, SNR-Tolerant, Non-linear, Fast-Tracking
95-450-O	Standard, SNR-Tolerant, Non-linear
95-500-O	Standard, SNR-Tolerant, Non-linear
95-550-O	Standard, SNR-Tolerant
95-600-O	Standard, SNR-Tolerant, Fast-Tracking
95-650-O	Standard
95-700-O	Standard
95-800-O	SNR-Tolerant
95-300-E	Standard, Non-linear
95-350-E	Standard, SNR-Tolerant, Non-linear
95-400-E	Standard, SNR-Tolerant, Non-linear
95-450-E	Standard, SNR-Tolerant, Non-linear
95-500-E	Standard, SNR-Tolerant
95-550-E	Standard, SNR-Tolerant
95-600-E	Standard, SNR-Tolerant
95-650-E	Standard, SNR-Tolerant
95-700-E	Standard
95-800-E	Standard, SNR-Tolerant
93.3-600-E	SNR-Tolerant
93.3-400-E	SNR-Tolerant
91.6-300-E	Standard, Non-linear
91.6-400-E	Standard, SNR-Tolerant
91.6-450-E	SNR-Tolerant
91.6-500-E	Standard, SNR-Tolerant
91.6-550-E	SNR-Tolerant
91.6-600-E	Standard

Transmission Mode	Link Optimization Mode (Note)
91.6-700-E	Standard
91.6-800-E	Standard, SNR-Tolerant
90-400-E	SNR-Tolerant
90-600-E	Standard, SNR-Tolerant
85-300-E	Standard
85-400-E	Standard, SNR-Tolerant
85-600-E	Standard
82-400-E	SNR-Tolerant
Note: When Link Optimization Mode is set to Non-linear, modem performance is optimized for compensated submarine lines with higher non-linear noise. When Link Optimization Mode is set to SNR-tolerant, the modem is optimized for performance on uncompensated submarine lines or for 800Gb/s terrestrial lines. When Link Optimization Mode is set to Standard, modem performance is optimized for typical terrestrial applications. The Link Optimization Mode used should be that specified by the Ciena link budget tools or analysis.	

You can display the provisioned PTP attributes in the PTP Status Summary table using the CLI command `ptp show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Line Port attributes

The following table defines provisionable port attributes.

Table 140 Provisionable port attributes for embedded coherent optics

Attribute	Settings	Default Value	Description
Conditioning Type	OTN	OTN	The type of conditioning signal is always OTN.
Admin state	- Enabled - Disabled	Enabled	Sets the administrative state of the port.
Label	String (1...128)	-	Sets a user-defined description for the port.
Loopback	- Disabled - Rx - Tx	Disabled	Operates loopback on a port.

Attribute	Settings	Default Value	Description
OTU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the OTU signal degrade threshold.
ODU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the ODU signal degrade threshold.
Mismatch mode	operator-only, sapi, dapi, sapi-and-dapi	operator-only	Specifies the string, which is used by the system to determine whether a mismatch is present. Note: Applicable to both section and path trace.
Mismatch fail mode	none, alarm-only	none	Specifies the action to take when a string mismatch is detected. - alarm-only: the system raises an alarm. - none: no action is taken Note: Applicable to both section and path trace.
tx-sapi	String (1...15)		Specifies the SAPI string to transmit to the far-end node. Note: Applicable to both section and path trace.
tx-dapi	String (1...15)		Specifies the DAPI string to transmit to the far-end node. Note: Applicable to both section and path trace.
tx-oper	String (1...32)		Sets value of the transmit operator string. Note: Applicable to both section and path trace.
tx-oper-mode	automatic, manual	automatic	Sets the transmit operator mode. - automatic: sets by the system. - manual: as specified in the tx-oper attribute. Note: Applicable to both section and path trace.
exp-sapi	String (1...15)		Specifies the SAPI string expected from the far-end node. Note: Applicable to both section and path trace.

Attribute	Settings	Default Value	Description
exp-dapi	String (1...15)		Specifies the DAPI string expected from the far-end node. Note: Applicable to both section and path trace.
exp-oper	String (1...32)		Sets value of the expected transmit operator string. Note: Applicable to both section and path trace.

You can display the provisioned port attributes in the Port Table using the CLI command `port show`. The following table defines the possible port attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 141 Port status attributes for embedded coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Label	—	Name of the port, displayed as <slot>/<port> or <slot>/<port>.<port section>
	State	Enabled, Disabled	The administrative state of the port.
	Type	OTUCn, OTUCn_Flex	The port type.
	Rate	OTUC2 - OTUC16	The port rate (speed).
	Interface Type	I-NNI	The type of interface used by the port.
	Loopback	Tx, Rx, Disabled	The administrative state of loopback testing on the port.

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the port: • Up — the port is operating normally. • Down — the port is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the port is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the port.
	Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the port has been in its current state.
	ODU Termination Mode	Terminated, Passthrough	Sets the termination mode for the port that is configured by the system.

Line Channel attributes

The following table defines provisionable channel attributes.

Table 142 Provisionable channel attributes for embedded coherent optics

Attribute	Settings	Default value	Description
Label	String (1...128)		Sets a user-defined description of the channel.
admin-state	enabled, disabled	enabled	The administrative state of the channel.
ODU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the ODU signal degrade threshold. This attribute is not applicable for ET channel types.
Mismatch mode	operator-only, sapi, dapi, sapi-and-dapi	operator-only	Specifies the string, which is used by the system to determine whether a mismatch is present. Note: Applicable to the path trace only.

Attribute	Settings	Default value	Description
Mismatch fail mode	none, alarm-only	none	Specifies the action to take when a string mismatch is detected. • alarm-only: the system raises an alarm. • none: no action is taken Note: Applicable to the path trace only.
tx-sapi	String (1...15)		Specifies the SAPI string to transmit to the far-end node. Note: Applicable to the path trace only.
tx-dapi	String (1...15)		Specifies the DAPI string to transmit to the far-end node. Note: Applicable to the path trace only.
tx-oper	String (1...32)		Sets value of the transmit operator string. Note: Applicable to the path trace only.
tx-oper-mode	automatic, manual	automatic	Sets the transmit operator mode. • automatic: sets by the system. • manual: as specified in the tx-oper attribute. Note: Applicable to the path trace only.
exp-sapi	String (1...15)		Specifies the SAPI string expected from the far-end node. Note: Applicable to the path trace only.
exp-dapi	String (1...15)		Specifies the DAPI string expected from the far-end node. Note: Applicable to the path trace only.
exp-oper	String (1...32)		Sets value of the expected transmit operator string. Note: Applicable to the path trace only.

You can display the provisioned channel attributes in the Channel table using the CLI command `channel show`. The following table defines the possible channel attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 143 Channel status attributes for embedded coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Name	—	Name of the channel, displayed as <slot>/<port>.<port section>
	Admin State	Enabled, Disabled	The administrative state of the channel.
	Type	ODUk, ET	The channel type.
	Rate	ODU4, ET	The channel rate (speed).
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the channel: • Up — the channel is operating normally. • Down — the channel is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the channel is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the channel.
	State cli Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the channel has been in its current state.

You can also display the connection information between the line port channel and their corresponding client port using the `connection show` command.

Table 144 Connection information

Attribute	Possible Values	Description
“From” Facility name	Port-<slot/port>	Client port name.
Bandwidth	• 100G • 400G	Defines the required bandwidth of the client port. For 100GE, OTL4.4 clients it is 100G and for 400GE clients it is 400G.

Attribute	Possible Values	Description
"To" Facility name	Channel-<slot/port.channel>	For terminal applications, defines the channel name of the line port to which the client port is connected.

Client Transceiver (XCVR) attributes

The following table defines provisionable transceiver attributes for transceiver 3 to 14 (the client side ports).

Table 145 Provisionable transceiver attributes for pluggable non-coherent optics

Attribute	Settings	Default Value	Description
Mode (Client)	• 100GE • 400GE • OTL4.4	QSFP28 - 100GE QSFPDD-400GE	The mode used by the client port. • 100GE • 400GE • OTL4.4

Table 146 Provisionable transceiver attributes for pluggable coherent optics

Attribute	settings	Default value	Description
Mode	4x100GE-ZR	4x100GE-ZR	The mode supported by the client ports 3, 6, 9, and 12.

You can display the provisioned client XCVR attributes in the Transceiver-Status table using the CLI command `xcvr show`. For more information about displaying entity attributes, see [Procedure 82, "Retrieving Waveserver 5 entity data"](#).

Client PTP attributes

The following table defines provisionable ptp attributes for a client port.

Table 147 Provisionable Client PTP attributes pluggable non-coherent optics

Attribute	Possible Values	Default Value	Description
Forward Error Correction	Enabled Disabled	100GE - Note 2, Note 3 - OTL4.4 - Note 2, Note 3 400GE - Enabled	Sets the status of forward error correction. Note 1: The FEC is always <i>enabled</i> for the 100G-FR (160-9431-900/160-9433-900) plug. Note 2: For 100GE and OTL4.4 client ports, if the plug is available while provisioning the xcvr (automatic or manual) the system selects the recommended FEC value, by default, based on the plug type. Note 3: For 100GE and OTL4.4 client ports, if the plug is not available and xcvr is provisioned manually, the default FEC is <i>disabled</i>. Note 4: For 400GE client ports, the FEC is always <i>enabled</i>.
Tx state	Enabled Disabled	Enabled	Sets the laser state of the transmitter.

Table 148 Provisionable Client PTP attributes for coherent optics

Attribute	Settings	Default value	Description
Frequency (GHz)	Frequency in GHz	0	Sets the frequency the pluggable transceiver uses.
Tx state	Enabled Disabled	Enabled	Sets the laser state of the transmitter.
Tx power	Transceiver Tx Power	-9	Sets the line transmit optical power. See Technical specifications .
Tx power reduction state	Enabled Disabled	Disabled	Sets the status of the Tx Power Reduction feature. - Enabled: forces the Tx power to be held in the low power state - Disabled: releases the Tx power from being held in the low power state
Forward-error-correction	Enabled	Enabled	Sets the status of forward error correction.

Display Client PTP attributes

You can display the provisioned PTP attributes in the PTP Status Summary table using the CLI command `ptp show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Client Port attributes

The following table defines provisionable client port attributes.

Table 149 Provisionable client port attributes for pluggable non-coherent optics

Attribute	Settings	Default Value	Description
Conditioning Holdoff	From 0 ms to 100 ms in 10 ms increments; From 100 ms to 1000 ms in 100 ms increments	10ms	Sets the conditioning hold-off timer.
Conditioning Type	Ethernet OTN	• OTL4.4 -- OTN • 100GE, 400GE -- Ethernet	Sets the type of conditioning signal to transmit to the client subtending equipment when a failure is detected on the line port or upstream remote client port. The conditioning type choices are dependent on the client protocol. For Ethernet (100GE and 400GE), the option is Ethernet (default) and for OTN (OTL4.4), the conditioning type should be OTN (default). • Ethernet: Ethernet primitives (Local Fault and Remote Fault)
Label	String (1...128)		Sets a user-defined description for the port.
Loopback	Disabled Rx Tx	Disabled	Operates loopback on a port.
OTU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the OTU signal degrade threshold.
ODU SD Threshold	BER String: XE-YY, for example, 1E-06, 5E-78, 0E-00	1E-09	Sets the ODU signal degrade threshold.
Mismatch mode	operator-only, sapi, dapi, sapi-and-dapi	operator-only	Specifies the string, which is used by the system to determine whether a mismatch is present. Note: Applicable to the section trace.

Attribute	Settings	Default Value	Description
Mismatch fail mode	none, alarm-only	none	Specifies the action to take when a string mismatch is detected. • alarm-only: the system raises an alarm. • none: no action is taken Note: Applicable to the section trace.
tx-sapi	String (1...15)		Specifies the SAPI string to transmit to the far-end node. Note: Applicable to the section trace.
tx-dapi	String (1...15)		Specifies the DAPI string to transmit to the far-end node. Note: Applicable to the section trace.
tx-oper	String (1...32)		Sets value of the transmit operator string. Note: Applicable to the section trace.
tx-oper-mode	automatic, manual	automatic	Sets the transmit operator mode. • automatic: sets by the system. • manual: as specified in the tx-oper attribute. Note: Applicable to the section trace.
exp-sapi	String (1...15)		Specifies the SAPI string expected from the far-end node. Note: Applicable to the section trace.
exp-dapi	String (1...15)		Specifies the DAPI string expected from the far-end node. Note: Applicable to the section trace.
exp-oper	String (1...32)		Sets value of the expected transmit operator string. Note: Applicable to the section trace.

You can display the provisioned client port attributes in the Port Table using the CLI command `port show`. The following table defines the possible port attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 150 Client Port attributes for pluggable non-coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Name	—	Name of the port, displayed as <slot>/<port> or <slot>/<port>.<port section>
	State	Enabled, Disabled	The administrative state of the port.
	Type	OTUCn, OTUCn_Flex, OTUk, Ethernet	The port type.
	Rate	100GE, 400GE, OTU4, OTUC2-OTUC16	The port rate (speed).
	Interface Type	UNI, I-NNI, E-NNI	The type of interface used by the port.
	Loopback	Tx, Rx, disabled	The administrative state of loopback testing on the port.
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the port: • Up — the port is operating normally. • Down — the port is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the port is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the port.
	Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the port has been in its current state.

Facilities for WL5n DCO 3xQSFP-DD 12x module

This section provides information on the facilities available on WL5n DCO 3xQSFP-DD 12x type service modules.

Module attributes

The following table defines provisionable module attributes.

Table 151 Provisionable module attributes

Attribute	Setting	Default Value	Description
Admin State	Enabled, Disabled	Enabled	The administrative state of the module.
Type	MOTR-DCO-400-3-12	MOTR-DCO-400-3-12	Defines the module type.
Mode	none	none	Defines the module mode in which module is used.
Label	String (1...128)	—	Sets a user-defined description of the module for the ease of identification purpose in a network infrastructure.

You can also display module status using the `module show slot <slot>` command:

Table 152 Module status attributes

Attribute	Possible values	Description
Operational State	• Up • Down	Defines the current operational state of the module.
Model	Waveserver DCO 3x400G-QSFP-DD 12x100G/3x400G MOTR	Defines the model of the module.
Width	double	Defines the module width in a system.
Last Restart Type	• Warm • Cold	Defines the last restart type of the module.
Last Restart	yyyy-mmddThh:mm:ss+timezone	Indicates the last time the module was restarted.
Uptime	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the module has been in its current state.

You can also display module statistics, which is related to module temperature, using the `module show slot <slot> statistics` command:

Table 153 Module statistics

Attribute	Possible values	Description
Temperature, Celsius	—	Indicates the temperature statistics of the module: • Instant • Minimum • Maximum • Average • Interval

Line Transceiver (XCVR) attributes

The following table defines provisionable transceiver attributes for transceiver 1, 2, and 3.

Table 154 Provisionable transceiver attributes for pluggable coherent optics

Attribute	Setting	Default Value	Description
Mode	4x100GE-ZR	4x100GE-ZR	Defines the transmission mode used by the pluggable transceiver.
Service Profile	mixed-ethernet	mixed-ethernet	Defines the service profile applied to the line port.
Admin state	Enabled, Disabled	Enabled	Defines the administrative state of the line port.

You can display the provisioned XCVR attributes in the Transceiver-Status table using the CLI command `xcvr show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Line PTP attributes

This section provides information on PTP attributes for PTP 1, 2, and 3.

Table 155 Provisionable line PTP attributes for pluggable coherent optics

Attribute	Settings	Default value	Description
Frequency (GHz)	Frequency in GHz	0	Sets the frequency the pluggable transceiver uses.
Tx state	Enabled Disabled	Enabled	Sets the laser state of the transmitter.
Tx power	Transceiver Tx Power	-9	Sets the line transmit optical power. See Technical specifications .

Attribute	Settings	Default value	Description
Tx power reduction state	Enabled Disabled	Disabled	Sets the status of the Tx Power Reduction feature. • Enabled: forces the Tx power to be held in the low power state • Disabled: releases the Tx power from being held in the low power state
Forward-error-correction	Enabled	Enabled	Sets the status of forward error correction.

You can display the provisioned PTP attributes in the PTP Status Summary table using the CLI command `ptp show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

You can also display the connection information between the line ports and their corresponding client ports using the `connection show` command.

Table 156 Connection information

Attribute	Possible Values	Description
“From” Facility name	Port-<slot/port>	Client port name.
Bandwidth	100G	Defines the required bandwidth of the client port. For 100GE, clients it is 100G.
“To” Facility name	Port-<slot/port>	For terminal applications, defines the line port name to which the client port is connected.

Client Transceiver (XCVR) attributes

The following table defines provisionable transceiver attributes for transceiver 4 to 15 (the client side ports).

Table 157 Provisionable transceiver attributes for pluggable non-coherent optics

Attribute	Settings	Default Value	Description
Mode (Client)	100GE	100GE	The mode used by the client port.

You can display the provisioned client XCVR attributes in the Transceiver-Status table using the CLI command `xcvr show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Client PTP attributes

The following table defines provisionable ptp attributes for a client port.

Table 158 Provisionable Client PTP attributes pluggable non-coherent optics

Attribute	Possible Values	Default Value	Description
Forward Error Correction	Enabled Disabled	100GE -- Note 2, Note 3	Sets the status of forward error correction. Note 1: The FEC is always <i>enabled</i> for the 100G-FR (160-9431-900/160-9433-900) plug. Note 2: If the plug is available while provisioning the xcvr (automatic or manual) the system selects the recommended FEC value, by default, based on the plug type. Note 3: If the plug is not available and xcvr is provisioned manually, the default FEC is <i>disabled</i>.
Tx state	Enabled Disabled	Enabled	Sets the laser state of the transmitter.

You can display the provisioned PTP attributes in the PTP Status Summary table using the CLI command `ptp show`. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Client Port attributes

The following table defines provisionable client port attributes.

Table 159 Provisionable client port attributes for pluggable non-coherent optics

Attribute	Settings	Default Value	Description
Conditioning Holdoff	• 0 ms • 10ms • 50ms • 100ms • 200ms • 300ms • 500ms • 1000ms	10ms	Sets the conditioning hold-off timer. This is only supported when using the Ciena's WL5n pluggable in the line port.

Attribute	Settings	Default Value	Description
Conditioning Type	Ethernet	Ethernet	Sets the type of conditioning signal to transmit to the client subtending equipment when a failure is detected on the line port or upstream remote client port. The conditioning type choices are dependent on the client protocol. Ethernet: Ethernet primitives (Local Fault and Remote Fault)
Label	String (1...128)		Sets a user-defined description for the port.
Loopback	Disabled Rx Tx	Disabled	Operates loopback on a port.

You can display the provisioned client port attributes in the Port Table using the CLI command `port show`. The following table defines the possible port attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 160 Client Port attributes for pluggable non-coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Name	—	Name of the port, displayed as <slot>/<port> or <slot>/<port>.<port section>
	State	Enabled, Disabled	The administrative state of the port.
	Type	Ethernet	The port type.
	Rate	100GE	The port rate (speed).
	Interface Type	UNI	The type of interface used by the port.
	Loopback	Tx, Rx, disabled	The administrative state of loopback testing on the port.

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the port: • Up — the port is operating normally. • Down — the port is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the port is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the port.
	Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the port has been in its current state.

Line Port attributes

The following table defines provisionable port attributes.

Table 161 Provisionable port attributes for pluggable coherent optics

Attribute	Settings	Default Value	Description
Conditioning Type	Ethernet	Ethernet	The type of conditioning signal.
Admin state	• Enabled • Disabled	Enabled	Sets the administrative state of the port.
Label	String (1...128)		Sets a user-defined description for the port.
Loopback	Tx, Rx, disabled	Disabled	Operates loopback on a port.

You can display the provisioned port attributes in the Port Table using the CLI command `port show`. The following table defines the possible port attributes and their values. For more information about displaying entity attributes, see [Procedure 82, “Retrieving Waveserver 5 entity data”](#).

Table 162 Port status attributes for pluggable coherent optics

Admin Configuration / Operational Status	Attribute	Possible Values	Description
Admin Configuration	Label	—	Name of the port, displayed as <slot>/<port> or <slot>/<port>.<port section>
	State	Enabled, Disabled	The administrative state of the port.
	Type	Ethernet	The port type.
	Rate	100GE	The port rate (speed).
	Interface Type	I-NNI	The type of interface used by the port.
	Loopback	Tx, Rx, Disabled	The administrative state of loopback testing on the port.
Operational Status	State	Up, Down, LLD, LoopTx, LoopRx, Flt	Indicates the current state of the port: • Up — the port is operating normally. • Down — the port is not operational. It is either experiencing an error, does not have fibers connected, or is disabled. • LLD — indicates a “LowerLevelDown” condition where the parent of the port is not functional. • LoopTx — Tx loopback is active. • LoopRx — Rx loopback is active. • Flt — a fault exists on the port.
	Duration	dd/hh/mm/ss	Indicates the length of time in days/hours/minutes/seconds that the port has been in its current state.

CHAPTER 17

Commissioning

This section contains the following topics:

- [System attributes](#)
- [Date and time](#)
- [Licensing](#)
- [Comms](#)
- [Waveserver 5 WL5e 2x800G 16xQSFP28/QSFP-DD service module details](#)
 - [Line ports and client ports in WL5e 2x MOTR 16x modules](#)
 - [Engineering rules for mated line configurations for WL5e 2x MOTR 16x modules](#)
 - [Standalone and mated line terminals for WL5e 2x MOTR 16x modules](#)
 - [Service profiles for WL5e 2x MOTR 16x modules in terminal applications](#)
- [Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD type service module details](#)
 - [Line ports and client ports in WL5e 2x MOTR 12x modules](#)
 - [Engineering rules for mated line configurations for WL5e 2x MOTR 12x module](#)
 - [Standalone and mated line terminals for WL5e 2x MOTR 12x modules](#)
 - [Service profiles for WL5e 2x MOTR 12x modules](#)
- [Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD module details](#)
 - [Line ports and client ports in WL5n DCO 3xQSFP-DD 12x modules](#)
 - [Engineering rules for mated line configuration for in WL5n DCO 3xQSFP-DD 12x modules](#)
 - [Standalone terminals for in WL5n DCO 3xQSFP-DD 12x modules](#)
 - [Service profiles for WL5n DCO 3xQSFP-DD 12x modules](#)

- [Commission third-party XCVR pluggables](#)
- [Procedure list](#)

The following sections provide information on commissioning or provisioning the Waveserver 5.

System attributes

System attributes let you specify information that uniquely identifies each Waveserver 5 in your network.

CLI commands and attributes

Table 163 Provisionable system attributes

Attributes under	Attribute	Description
system defaults set	Conditioning type	Specifies whether the client sends an LF or idles when there is a line fault.
	Conditioning holdoff	Specifies the holdoff timer before the client port conditions LF when it detects a line fault.
	Terrestrial reach limit	Sets the terrestrial reach limit as one of the following: • 250 • 500 • 1000 • 2000 • 2100 • 3000 • 4000 • 5000 • 8000
system environment	Environment settings	• Diag: Changes the SSH diag user shell type to CLI or system (Linux). • Root: Changes the Linux root account password.
system lamps set	Lamps settings	• Admin State: Sets the administrative state for the lamp settings. • Target: Sets the target (slot, port or chassis) for the lamp settings. • Timeout: Sets the timeout for the lamp settings. Note: A zero-minute timeout setting results in permanently flashing LEDs until <code>system lamp</code> is disabled or the system is restarted.

Attributes under	Attribute	Description
system server	Server settings	• GRPC: Sets the GRPC server settings. • HTTPS: Sets the HTTPS server settings. • NETCONF: Sets the NETCONF server settings. • SFTP: Sets the SFTP server settings. • SCP: Sets the SCP server settings. • Global inactivity timer: Sets the global inactivity timer to be on or off. If on, automatically ends inactive user sessions after a period (global inactivity timeout) of inactivity. • Global inactivity timeout: If the global inactivity timer is on, this setting provides the time of inactivity, in which the user session shuts down automatically.
system shell set	Shell settings	• Login authentication message: The login authentication message. • Login banner file: The login banner file. • Welcome banner file: The welcome banner file. • Global more: Sets the global more to be on or off. If set to off, this disables the “more” paging on the shell session. • Global more lines: The number of lines to display before presenting the option to display more lines for the current session.
system sleep	Sleep settings	• For: Suspends interactive process for the specified duration in the current session. • Message: Message displayed while the Waveserver 5 is in sleep mode. Note: These settings apply to the current session only.

Attributes under	Attribute	Description
system state-dump	State dump settings	Dumps the system state to a file that can be saved to a specified location. You must have the super user privileges to perform this activity. • Default server: Sets the default server for state-dump. • Default SFTP server: Sets the default SFTP server for state-dump. • Default SCP server: Sets the default SCP server for state-dump. • SFTP server: Sets the SFTP server for state-dump. • SCP server: Sets the SCP server for state-dump. • Filename: Filename for the state-dump. Note: The filename string supports alphanumeric characters (space not supported) including these special characters: ~ @ % ^ * _ = + { } [] : ?. However, some strings may not be allowed as valid filenames due to your management platform system's OS restrictions. For example, if your management platform system has Windows OS then you cannot use ? (question mark) character in the file-name string, although it is supported by the Waveserver 5 and allowed on a Linux system. Note: This is a one-time activity and you have to execute this command every time as required.
system xftp set	xFTP settings	• Mode: Sets the global default xFTP mode. • FTP server: To configure a default remote FTP server. • TFTPserver: To configure a default TFTP server. • SFTP server: To configure a default SFTP server. • SCP server: To configure a default SCP server.

Attributes under	Attribute	Description
system set	Time settings	• Date: The date assigned to the system. • Time: The time assigned to the system. • Timestamp: Specified as either local or UTC.
	Time offset	The time-offset specifies the time offset from UTC (Coordinated Universal Time). The timestamp specifies whether to display the UTC time or the local time, where the local time is UTC + offset.
	Host name	The host name of the Waveserver 5 system.
	Site information	• Site ID: The ID (integer) assigned to the data center where Waveserver 5 is located. Note: The site ID must be unique to each data center in your network. • Site name: The name assigned to the site. • Site description: The description of the site. • Site location: The geographic co-ordinates (site-longitude, site-latitude) of the site and/or street address (site-address) information. Note: Although only the site ID value is required, Ciena recommends that you also provision the site name, description, and location values.
	Group information	• Group ID: The ID (integer) of the group to which Waveserver 5 is assigned. Note: The group ID must be unique to each group at a site. • Group name: The name assigned to the group. • Group description: The description of the group. Note: Although only the group ID value is required, Ciena recommends that you also provision the group name and description values.
	Member information	• Member ID: Sets the member ID. • Member name: Sets the member name. • Member description: Sets the member description. • Member frame ID: Sets the member frame ID. • Member rack unit number: Sets the member rack unit number.
	Network information	• Network ID: Sets the system network ID. • Network name: Sets the system network name. • Network description: Sets the system network description.

Date and time

The Waveserver 5 can automatically obtain the system time when it is connected to the DCN and the network time protocol (NTP) is enabled. For additional information, see [Network Time Protocol \(NTP\)](#). The system date and time can also be manually provisioned.

CLI commands and attributes

The NTP is enabled by default. However, to provision the NTP manually use the following CLI command:

```
ntp client enable <IP address or host name>
```

To provision the system date and time manually, disable the NTP and then set date and time using the following CLI commands:

```
ntp client disable <IP address or host name>
```

```
system set date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>
```

```
system set time <time: hh:mm:ss or hh:mm>
```

Licensing

If licenses are required, the licenses need to be ordered, registered, activated, and transferred. For more information on licensing, see [Licensing](#).

Comms

To commission the shelf, at a minimum, the following needs to be provisioned:

- DCN port (if connected directly to the external DCN)
- ILAN ports (if connected directly to another Waveserver 5)

For additional information on comms, see [Comms](#).

CLI commands and attributes

To provision the DCN port, use CLI commands:

```
interface set interface local ip <IP address/mask>  
[gateway <IP address>]
```

```
interface set gateway <IP address>
```

To provision the ILAN port, use CLI commands:

```
interface create interface <if0-ilan-1 | if0-ilan-2>
```

```
interface enable interface <if0-ilan-1 | ifp0-ilan-2>
```

Waveserver 5 WL5e 2x800G 16xQSFP28/QSFP-DD service module details

The WL5e 2x MOTR 16x service modules are double-slot wide modules that have the following properties:

- 2 x WL5e modem line interfaces
- 4 x bilingual QSFP-DD / QSFP-28 client interfaces
- 12 x QSFP-28 client interfaces

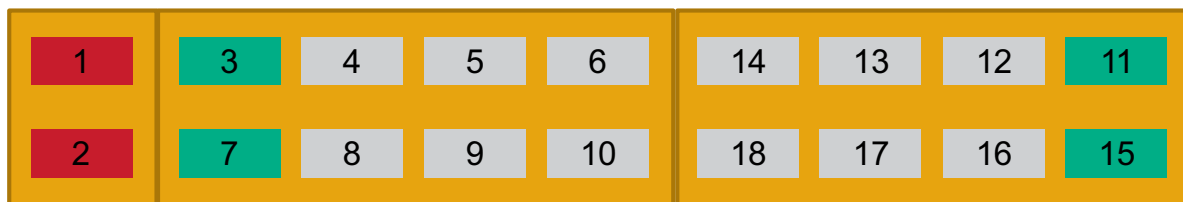
This section outline the following details about the WL5e type service modules:

- [Line ports and client ports in WL5e 2x MOTR 16x modules](#)
- [Engineering rules for mated line configurations for WL5e 2x MOTR 16x modules](#)
- [Standalone and mated line terminals for WL5e 2x MOTR 16x modules](#)
- [Service profiles for WL5e 2x MOTR 16x modules in terminal applications](#)

Line ports and client ports in WL5e 2x MOTR 16x modules

The service modules have line and client ports which allow traffic to be carried end-to-end. This section provides information on the line and client ports. See the following figure on service module line and client port numbering:

Figure 69 Service module - line and client port numbering



Legend

- 1 Red - Ports 1 and 2: line ports
 - 2 Green - Ports 3, 7, 11, and 15: Bilingual QSFP-DD / QSFP28 client ports
 - 3 Grey - Ports 4 – 6, 8 – 10, 12 – 14, and 16 – 18: QSFP28 client ports
- Client ports 3 to 10 use line port 1, client ports 11 to 18 use line port 2

Line ports (WL5e 2x MOTR 16x Modules)

The following table lists the supported line port types, baud, bit rate, mapping and supported client protocols.

Table 164 Supported line port transmission mode, baud, mapping, rate and client protocols

Line XCVR Mode (Note)	Baud or Symbol Rate (GBaud)	Bit Rate (Gbps)	Supported client protocols
95-200-O	95	200	100GE, 400GE, OTL4.4
95-250-O	95	250	100GE, OTL4.4
95-300-O	95	300	100GE, OTL4.4
95-350-O	95	350	100GE, OTL4.4
95-400-O	95	400	100GE, 400GE, OTL4.4
95-450-O	95	450	100GE, OTL4.4
95-500-O	95	500	100GE, 400GE, OTL4.4
95-550-O	95	550	100GE, OTL4.4
95-600-O	95	600	100GE, 400GE, OTL4.4
95-650-O	95	650	100GE
95-700-O	95	700	100GE, OTL4.4
95-800-O	95	800	100GE, 400GE, OTL4.4
95-300-E	95	300	100GE
95-350-E	95	350	100GE
95-400-E	95	400	100GE, 400GE
95-450-E	95	450	100GE
95-500-E	95	500	100GE, 400GE
95-550-E	95	550	100GE
95-600-E	95	600	100GE, 400GE
95-650-E	95	650	100GE
95-700-E	95	700	100GE
95-800-E	95	800	100GE, 400GE
91.6-300-E	91.6	300	100GE
91.6-400-E	91.6	400	100GE, 400GE
91.6-450-E	91.6	450	100GE
91.6-500-E	91.6	500	100GE, 400GE
91.6-550-E	91.6	550	100GE

Line XCVR Mode (Note)	Baud or Symbol Rate (GBaud)	Bit Rate (Gbps)	Supported client protocols
91.6-600-E	91.6	600	100GE, 400GE
91.6-700-E	91.6	700	100GE
91.6-800-E	91.6	800	100GE, 400GE
Note: When the line port transmission mode has Ethernet mapping, all OTN related configuration, operational data, PM, and alarms are not applicable to the line channel object.			

Client ports (WL5e 2x MOTR 16x Modules)

The following table provides information on the client port types supported.

Table 165 Client port summary

Client XCVR Mode (Note)	Protocol	FEC	Supported Line Mapping	Supported Pluggable (See Pluggables and specifications)	Ports
100GE	100GE	- None - KR4 (NRZ pluggables) - KP4 (PAM4 pluggables)	OTN Ethernet	QSFP28 (including AOC)	3 to 18
400GE	400GE	KP4	OTN Ethernet	QSFP-DD	3, 7, 11, and 15
OTL4.4	OTU4	- None - Generic Forward Error Correction (GFEC)	OTN	QSFP28 (including AOC)	4 to 6, 8 to 10, 12 to 14, 16 to 18
Note: 100GE KR4 - 802.3 CL-91 RS (528, 514) over 4 lanes 100GE KP4 - 802.3 CL-91 RS (544, 514) over 4 or 2 lanes 400GE KP4 - 802.3 CL-119 RS (544, 514) over 8 lanes OTL4.4 GFEC-G.709 Annex A RS (255, 239)					

Engineering rules for mated line configurations for WL5e 2x MOTR 16x modules

When using mated line configurations using service modules, the allowable skew budget dictates the maximum spectral separation between the two mated wavelengths.

Consider the following engineering rules to minimize skew between the paired signals when a service module has both line ports set to the same transmission mode and there is an odd total number of client interfaces provisioned.

Note: For information on setting line port frequencies, see [Line PTP attributes](#).

- The mated line configuration is supported only on the following 901 modules:
 - Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901)
 - Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901)
 - Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901)
 - Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFP-DD Module (186-3110-901)
 - Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module (186-3111-901)
 - Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3112-901)
- Mated x50G lines can be used to carry 100GE and OTL4.4 clients.
- Mated x00G lines can be used to carry 400GE clients as 2x200G and 2x600G.
- For uncompensated C-band and L-band, a 1.2 Terahertz difference in frequency between the line ports is permitted without fiber distance restrictions (below 25,000 km).
- For C-band and L-band, if the fiber distance does not exceed 6000km, and in compensated submarine links, the difference in frequencies between the two line ports can be unlimited within the band.
- Line port 1 deskews the paired 50G signals. The skew tolerance is a minimum of +/- 7 μ s.
- Ensure that the wavelength pairs take the same photonic path and have similar length patch cords.

Standalone and mated line terminals for WL5e 2x MOTR 16x modules

The service modules support both standalone and mated line terminals.

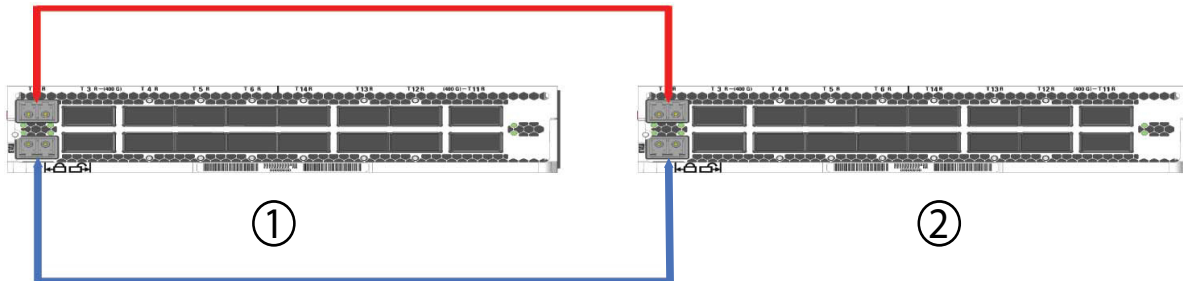
Standalone line terminals are supported on service modules, where the line ports (1 and 2) on a service module work independently of each other.

Standalone line terminals have the following considerations:

- Provisionable at the module level using CLI command `module set mode none slot <slot>`.
- Line port odu-termination-mode is set to terminate automatically by the system.
- The line XCVR mate is set to none by the system.
- Supports client traffic up to 800G.
- Last 50G bandwidth for x50G mode is not used.

See the following figure.

Figure 70 Standalone line terminal



Legend

1 Terminal mode = "none"

2 Terminal mode = "none"

Mated line terminals are supported on the service modules, where the line ports (1 and 2) work together. If the module is set to mated mode, you can split the last supported client port on the primary line (port 1) in half.

Mated line terminals have the following considerations:

- Provisionable at the module level using CLI command `module set mode mated slot <slot>`. This is a traffic impacting operation.
- Line port odu-termination-mode is set to terminate automatically by the system.
- The line XCVR mate indicates a `<slot>/<port>`.
- Only line XCVRs of the same slot can be mated, and line 1 is always the primary.

- Supports x50G and x100G transmission modes.
- A logical line port 101 is automatically created as a child of the two parent PTPs (on the same module) that are mated.

See the following figure.

Figure 71 Mated line terminal

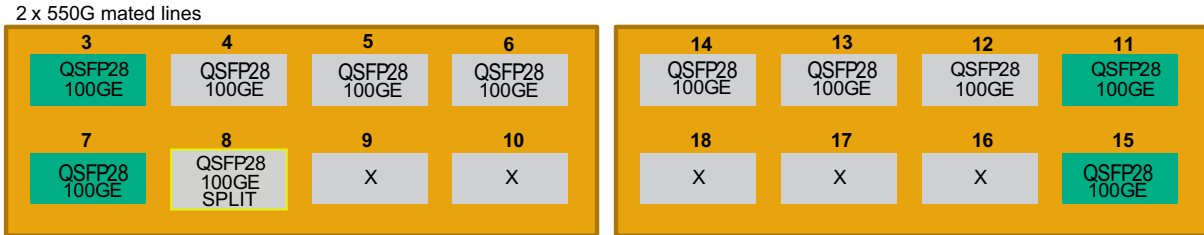


Legend

- 1 Terminal mode = "mated" 2 Terminal mode = "mated"

The following figure provides an example of a 2x550G mated line using the 100GE service profile, where the client port 8 is the split port. For information on the 100GE service, see [100GE service profile](#).

Figure 72 2 x 550G mated lines



Service profiles for WL5e 2x MOTR 16x modules in terminal applications

This section provides information on the service profiles available in terminal applications. Each module supports one service profile. You can provision different service profile on each line of the service module.

Waveserver 5 supports these service profiles:

- [100GE service profile](#)
- [400GE service profile](#)
- [Mixed-Ethernet service profile](#)
- [Mixed-OTN-Ethernet service profile](#)

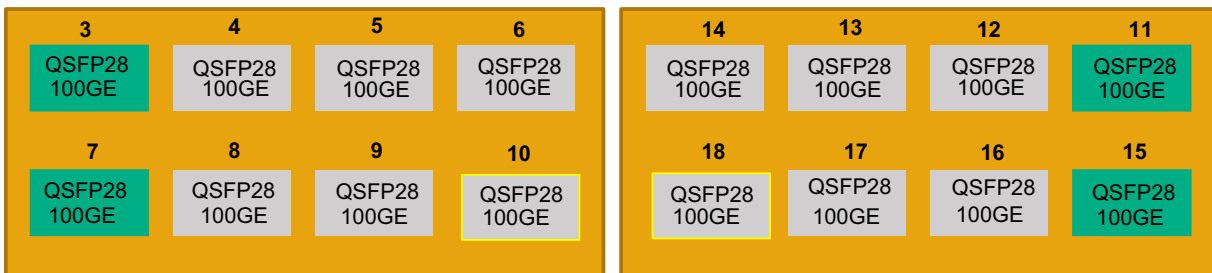
Note: If you change the service profile it impacts the traffic running on that module. Also, the service profile change is only permitted when the existing client provisioning is valid for the new service profile; any invalid client provisioning with respect to the new profile must be deleted prior to changing to a new service profile.

100GE service profile

The 100GE service profile supports 100GE client ports over QSFP28 pluggables.

The following figure shows the supported transceiver modes and pluggables for client ports using the 100GE service profile.

Figure 73 100GE service profile - transceiver modes and pluggables for client ports



Note: Client ports 10 and 18 are not usable when mated.

100GE service profile provisioning considerations

The following 100GE service profile provisioning considerations apply:

- The line XCVR service profile must be set to 100GE, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile 100GE`.
- Ports 3 to 18 support QSFP28 pluggables in 100GE mode.
- Unsupported client XCVR modes and pluggable types are blocked.
- Supports standalone line terminal and mated line terminal configurations.
 - Standalone line terminal
 - All clients are provisionable.
 - At 200G, client ports 3, 4, 11, and 12 traffic is carried over the line signal.
 - At 300G, client ports 3, 4, 5, 11, 12, and 13 traffic is carried over the line signal.
 - At 400G, client ports 3, 4, 5, 6, 11, 12, 13, and 14 traffic is carried over the line signal.

- At 500G, client ports 3, 4, 5, 6, 7, 11, 12, 13, 14, and 15 traffic is carried over the line signal.
- At 600G, client ports 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, and 16 traffic is carried over the line signal.
- At 700G, client ports 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, and 17 traffic is carried over the line signal.
- At 800G, all client ports traffic is carried over the line signal.
- Mated line terminal
 - All clients are provisionable.
 - Ports 10 and 18 are not usable, the Waveserver 5 disables the administrative and connection state of the port.
 - At 2x250G, client ports 3, 4, 5, 11, and 12 traffic is carried over the line signal.
 - At 2x350G, client ports 3, 4, 5, 11, and 12 traffic is carried over the line signal.
 - At 2x450G, client ports 3, 4, 5, 6, 7, 11, 12, 13, and 14 traffic is carried over the line signal.
 - At 2x550G, client ports 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, and 15 traffic is carried over the line signal.
 - At 2x650G, client ports 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15 and 16 traffic is carried over the line signal.

For additional information on standalone and mated line terminals, see [Standalone and mated line terminals for WL5e 2x MOTR 16x modules](#).

- When the port connection state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
 - ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
 - Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
 - Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

400GE service profile

The 400GE service profile supports 400GE client ports over QSFP-DD pluggables.

The following figure shows the supported transceiver modes and pluggables for the client ports using the 400GE service profile.

Figure 74 400GE service profile - transceiver modes and pluggable for client ports

3	4	5	6	14	13	12	11
QSFP-DD 400GE	X	X	X	X	X	X	QSFP-DD 400GE
7	8	9	10	18	17	16	15
QSFP-DD 400GE	X	X	X	X	X	X	QSFP-DD 400GE

400GE service profile provisioning considerations

The following 400GE service Profile provisioning considerations apply:

- The line XCVR service-profile must be set to 400GE, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile 400GE`.
- Ports 3, 7, 11, and 15 support QSFP-DD pluggables in 400GE mode.
- Unsupported client XCVR modes and pluggable types are blocked.
- Provisioning is blocked for ports 4 to 6, 8 to 10, 12 to 14, and 16 to 18.
- Supports standalone line terminal and mated line terminal configurations.
 - Standalone line terminal configuration
 - at 800G, client ports 3, 7, 11, and 15 traffic is carried over the line signal.
 - at 400G, client ports 3 and 11 traffic is carried over the line signal.
 - Mated line terminal configuration
 - at 2x600G, client ports 3, 7, and 11 traffic is carried over the line signal.
 - at 2x200G, client 3 traffic is carried over the line signal.
- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
 - ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).

- Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
- Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

Mixed-Ethernet service profile

The Mixed-Ethernet service profile supports 100GE / 400GE client ports over QSFP28 / QSFP-DD pluggables.

The following figure shows the supported transceiver modes on the client ports using the Mixed-Ethernet service profile.

Figure 75 Mixed-Ethernet service profile - transceiver modes and pluggables for client ports

3 QSFP28 / QSFP-DD 100GE / 400GE	4 QSFP28 100GE	5 QSFP28 100GE	6 QSFP28 100GE	14 QSFP28 100GE	13 QSFP28 100GE	12 QSFP28 100GE	11 QSFP28 / QSFP-DD 100GE / 400GE
7 QSFP28 / QSFP-DD 100GE / 400GE	8 QSFP28 100GE	9 QSFP28 100GE	10 QSFP28 100GE	18 QSFP28 100GE	17 QSFP28 100GE	16 QSFP28 100GE	15 QSFP28 / QSFP-DD 100GE / 400GE

Mixed-Ethernet service profile provisioning considerations

The following Mixed-Ethernet service profile provisioning considerations apply:

- The line XCVR service profile must be set to mixed-ethernet using the following CLI command


```
xcvr set xcvr <slot>/<port> service-profile mixed-ethernet
```
- Ports 3, 7, 11, and 15 support QSFP28 and QSFP-DD pluggables.
 - QSFP28 pluggables in 100GE mode
 - QSFP-DD pluggables in 400GE mode
- Ports 4 to 6, 8 to 10, 12 to 14, and 16 to 18 support QSFP28 pluggables in 100GE mode.
- QSFP28 and QSFP-DD clients are mutually exclusive. For example, when port 3 is provisioned as 400GE, provisioning for ports 4, 5, and 6 are blocked.
- When port 3 is provisioned as 400GE then ports 9 and 10 are not usable in 100GE mode.
- When port 11 is provisioned as 400GE then ports 17 and 18 are not usable in 100GE mode

- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
 - ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
 - Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
 - Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).
- Traffic on all clients for the same associated line port could be impacted when creating or deleting any client XCVR provisioning.
 - Client ports 3 to 10 use line port 1 and client ports 11 to 18 use line port 2.
 - Client XCVR provisioning can be done manually or automatically when a supported pluggable is inserted in the unprovisioned port.
- It is recommended to provision 400GE clients before 100GE clients to avoid traffic interruption on existing 100GE clients.
- Mated line configuration is not supported in Mixed-Ethernet service profile.
- Supports standalone line terminal configurations as follows:
 - **8 x 100GE per line**
 - all clients are provisionable.
 - at 200G, client ports 3, 4, 11, and 12 traffic is carried over the line signal.
 - at 300G, client ports 3, 4, 5, 11, 12, and 13 traffic is carried over the line signal.
 - at 400G, client ports 3, 4, 5, 6, 11, 12, 13, and 14 traffic is carried over the line signal.
 - at 500G, client ports 3, 4, 5, 6, 7, 11, 12, 13, 14, and 15 traffic is carried over the line signal.
 - at 600G, client ports 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, and 16 traffic is carried over the line signal.
 - at 700G, client ports 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, and 17 traffic is carried over the line signal.
 - at 800G, all client ports traffic is carried over the line signal

— 4 x 100GE + 1 x 400GE per line

- client ports 8, 9, 10 and 16, 17, 18 are blocked.
- client ports 7 and 15 are provisioned in 400G mode.
- at 200G, client ports 3, 4, 11, and 12 traffic is carried over the line signal.
- at 300G, client ports 3, 4, 5, 11, 12, and 13 traffic is carried over the line signal.
- at 400G, client ports 3, 4, 5, 6, 11, 12, 13, and 14 traffic is carried over the line signal.
- at 800G, client ports 3, 4, 5, 6, 7, 11, 12, 13, 14, and 15 traffic is carried over the line signal.

— 1 x 400GE + 2 x 100GE per line

- client ports 4, 5, 6 and 12, 13, 14 are blocked.
- client ports 9, 10 and 17, 18 are always disabled.
- client ports 3 and 11 are provisioned in 400G mode.
- at 400G, client ports 3 and 11 traffic is carried over the line signal.
- at 500G, client ports 3, 7, 11, and 15 traffic is carried over the line signal.
- at 600G, client ports 3, 7, 8, 11, 15, and 16 traffic is carried over the line signal.

— 2 x 400GE per line

- client ports 4, 5, 6, 8, 9, 10 and 12, 13, 14, 16, 17, 18 are blocked.
- client ports 3, 7 and 11, 15 are provisioned in 400G mode.
- at 800G, client ports 3, 7, 11, and 15 traffic is carried over the line signal.
- at 400G, client ports 3 and 11 traffic is carried over the line signal.

ATTENTION**Risk of traffic interruption**

Adding or deleting a 400GE client port by manually provisioning or inserting a QSFP-DD pluggable when there is an existing client port sharing the same line is service-affecting. If mixed 400GE and 100GE clients will be deployed on the same line (mixed-ethernet profile), it is recommended to provision 400GE clients before provisioning the 100GE clients.

Mixed-OTN-Ethernet service profile

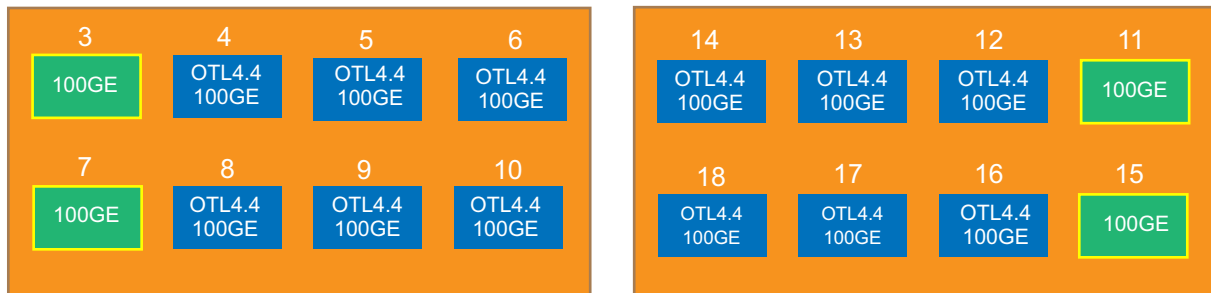
The Mixed-OTN-Ethernet service profile supports a mix of 100GE, 400GE, and OTL4.4 client ports over QSFP28 and QSFP-DD pluggable.

However, there are certain limitations when a port is provisioned in 400GE mode. The following two sections show the supported transceiver modes on the client ports using the Mixed-OTN-Ethernet service profile without and with 400GE mode.

- [Mixed-OTN-Ethernet service profile - without 400GE](#)
- [Mixed-OTN-Ethernet service profile - with 400GE](#)

Mixed-OTN-Ethernet service profile - without 400GE

Figure 76 Mixed-OTN-Ethernet service profile - Transceiver modes 100GE and OTL4.4



Note: Ports 10 and 18 cannot be used in mated line configuration.

The following Mixed-OTN-Ethernet service profile provisioning considerations apply:

- The line XCVR mode must be set to a mode that is OTN Transcoded, where the max line rate 800G.
- The line XCVR service-profile must be set to mixed-otn-ethernet, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile mixed-otn-ethernet`.
- Ports 4 to 6, 8 to 10, 12 to 14 and 16 to 18 support QSFP28 pluggables in OTL4.4 or 100GE mode.
- Port 3, 7, 11, and 15 support QSFP28 pluggables in 100GE mode.
- Port 3 can only be provisioned as 100GE when port 4 is 100GE and port 5 is OTL4.4
- Port 7 can only be provisioned as 100GE when port 8 is 100GE and port 9 is OTL4.4
- Port 11 can only be provisioned as 100GE when port 12 is 100GE and port 13 is OTL4.4

- Port 15 can only be provisioned as 100GE when port 16 is 100GE and port 17 is OTL4.4
- Port 10 and 18 are not usable in mated line configuration. The port admin state and the connection state remains disabled.
- Unsupported client xcvr modes and pluggable types are blocked.
- Supports standalone line terminal and mated line terminal configurations.
 - Standalone line terminal
 - At 200G, client ports 4, 5, 12, and 13 traffic are carried over the line signal.
 - At 400G, client ports 4, 5, 6, 8, 12, 13, 14, and 16 traffic are carried over the line signal.
 - At 500G, client ports 4, 5, 6, 8, 9, 12, 13, 14, 16, and 17 traffic are carried over the line signal.
 - At 600G, all client ports are carried over the line signal, except ports 3, 7, 11, and 15.
 - At 700G, all client ports are carried over the line signal, except ports 7 and 15.
 - At 800G, all client ports are carried over the line signal.
 - Mated line terminal
 - Client ports 10 and 18 cannot be used.
 - At 2x250G, client ports 4, 5, 6, 12, and 13 traffic are carried over the line signal.
 - At 2x350G, client ports 4, 5, 6, 8, 12, 13, and 14 traffic are carried over the line signal.
 - At 2x450G, client ports 4, 5, 6, 8, 9, 12, 13, 14, and 16 traffic are carried over the line signal.
 - Mated line configuration is supported only when no 400GE client is provisioned.

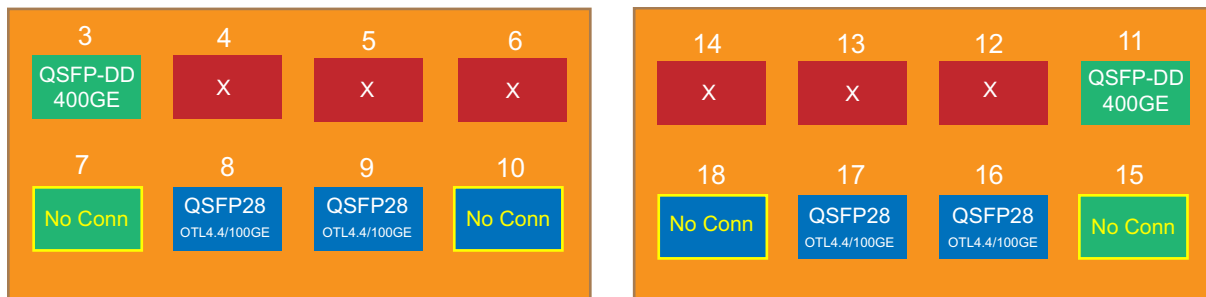
For additional information on standalone and mated line terminals, see [Standalone and mated line terminals for WL5e 2x MOTR 16x modules](#).

- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
 - ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).

- Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
- Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

Mixed-OTN-Ethernet service profile - with 400GE

Figure 77 Mixed-OTN-Ethernet service profile - Transceiver modes 400GE, 100GE and OTL4.4



The following consideration applies in Mixed-OTN-Ethernet with 400GE client:

- The line XCVR mode must be set to a mode that is OTN Transcoded, where the max line rate 600G.
- The line XCVR service-profile must be set to mixed-otn-ethernet, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile mixed-otn-ethernet`.
- Port 3 and 7 support QSFP-DD pluggables in 400GE mode.
- Ports 4, 5, 6 and 12, 13, 14 are blocked.
- Port 7, 10, 15 and 18 are always disabled. The port admin state and the connection state remains disabled.
- Mated line Configuration is not supported.
- Supports following standalone line terminal configurations.
 - At 400G, client ports 3 and 11 traffic are carried over the line signal.
 - At 500G, client ports 3, 8, 11, and 16 traffic are carried over the line signal.
 - At 600G, client ports 3, 8, 9, 11, 16, and 17 are carried over the line signal.
- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).

- ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
- Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
- Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

ATTENTION**Risk of traffic interruption**

Adding or deleting a 400GE client port by manually provisioning or inserting a QSFP-DD pluggable when there is an existing client port sharing the same line is service-affecting. If mixed 400GE and 100GE clients will be deployed on the same line (mixed-otn-ethernet profile). It is recommended to provision 400GE clients before provisioning the 100GE/OTL4.4 clients.

Service Profile Migration

After comparing the service profiles, you can use this information to migrate a service profile into another one. Typically, there are two type of migration can happen:

- [100GE and 400GE into Mixed-Ethernet service profile](#)
- [100GE, 400GE, and Mixed-Ethernet into Mixed-OTN-Ethernet service profile](#)

The following sections provide a set of guidelines to migrate a service profile into another one.

100GE and 400GE into Mixed-Ethernet service profile

- When migrating from 100GE or 400GE service profile to mixed-ethernet profile, re-provisioning of clients is not required.
- Before migrating from mixed-ethernet profile to 100GE or 400GE profile, client not supported in destination profile needs to be deleted.
- There is traffic impact to all clients when changing the service profile.

100GE, 400GE, and Mixed-Ethernet into Mixed-OTN-Ethernet service profile

- Before migrating from 100GE service profile to mixed-otn-ethernet profile, line xcvr mode has to be modified to -O mode (like 95-400-O), and 100GE provisioning in client ports 5, 9, 13, and 17 have to be deleted.
- Before migrating from 400GE service profile to mixed-otn-ethernet profile, line xcvr mode has to be modified to -O mode (like 95-400-O), and 400GE provisioning other than client ports 3, 11 have to be deleted.

- Before migrating from mixed-ethernet profile to mixed-otn-ethernet profile, line xcvr mode has to be modified to -O mode (like 95-400-O), 400GE provisioning other than client port 3, 11 have to be deleted, and 100GE provisioning on client ports 9, 17 have to be deleted.
- Before migrating from mixed-otn-ethernet profile to 100GE, 400GE, or mixed-ethernet profile, client not supported in destination profile needs to be deleted.
- There is traffic impact on all clients when changing the service profile.

Service profile comparisons

The following table provides information on the differences between the service profiles.

Table 166 Service profile comparisons

Service profile	Clients	Application	Line mapping	Pluggable (Note 1)	Mated lines
100GE	100GE	Terminal	Ethernet, OTN	QSFP28	Yes
400GE	400GE	Terminal	Ethernet, OTN	QSFP-DD	Yes
Mixed-Ethernet	100GE / 400GE (Note 2)	Terminal	Ethernet, OTN	QSFP28, QSFP-DD	No
Mixed-OTN-Ethernet	100GE / 400GE / OTL4.4 (Note2)	Terminal	OTN	QSFP28, QSFP-DD	Yes (Note 3)

Note 1: For more information on pluggables, see [Pluggables and specifications](#).

Note 2: Traffic hits on all clients sharing the same line when adding or deleting 400GE clients.

Note 3: Mated line configuration is supported only when no 400GE client is provisioned.

Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD type service module details

The Waveserver WL5e 2x800G 12xQSFP28/QSFP-DD type service modules are double-slot wide modules has following properties:

- 2 x WL5e modem line interfaces
- 4 x bilingual QSFP-DD / QSFP-28 client interfaces
- 8 x QSFP-28 client interfaces

This section outline the following details about the WL5e 2x MOTR 12x modules:

- [Line ports and client ports in WL5e 2x MOTR 12x modules](#)
- [Engineering rules for mated line configurations for WL5e 2x MOTR 12x module](#)

- Standalone and mated line terminals for WL5e 2x MOTR 12x modules
- Service profiles for WL5e 2x MOTR 12x modules

Line ports and client ports in WL5e 2x MOTR 12x modules

The service modules have line and client ports which allow traffic to be carried end-to-end. This section provides information on the line and client ports. See the following figure on service module line and client port numbering:

Figure 78 Service module - line and client port numbering



Line ports (WL5e 2x MOTR 12x modules)

The following table lists the supported line port types, baud, bit rate, mapping and supported client protocols.

Table 167 Supported line port transmission mode, baud, mapping, rate and client protocols

Line XCVR Mode (Note)	Baud or Symbol Rate (GBaud)	Bit Rate (Gbps)	Supported client protocols
95-200-O	95	200	100GE, 400GE, OTL4.4
95-250-O	95	250	100GE, OTL4.4
95-300-O	95	300	100GE, OTL4.4
95-350-O	95	350	100GE, OTL4.4
95-400-O	95	400	100GE, 400GE, OTL4.4
95-450-O	95	450	100GE, OTL4.4
95-500-O	95	500	100GE, 400GE, OTL4.4
95-550-O	95	550	100GE, OTL4.4
95-600-O	95	600	100GE, 400GE, OTL4.4
95-650-O	95	650	100GE
95-700-O	95	700	100GE, OTL4.4
95-800-O	95	800	100GE, 400GE, OTL4.4
82-400-E	82	400	100GE
85-300-E	85	300	100GE

Line XCVR Mode (Note)	Baud or Symbol Rate (GBaud)	Bit Rate (Gbps)	Supported client protocols
85-400-E	85	400	100GE
85-600-E	85	600	100GE
90-400-E	90	400	100GE
90-600-E	90	600	100GE
93.3-400-E	93.3	400	100GE
93.3-600-E	93.3	600	100GE
95-300-E	95	300	100GE
95-350-E	95	350	100GE
95-400-E	95	400	100GE, 400GE
95-450-E	95	450	100GE
95-500-E	95	500	100GE, 400GE
95-550-E	95	550	100GE
95-600-E	95	600	100GE, 400GE
95-650-E	95	650	100GE
95-700-E	95	700	100GE
95-800-E	95	800	100GE, 400GE
91.6-300-E	91.6	300	100GE
91.6-400-E	91.6	400	100GE, 400GE
91.6-450-E	91.6	450	100GE
91.6-600-E	91.6	600	100GE, 400GE
91.6-800-E	91.6	800	100GE, 400GE
Note: When the line port transmission mode has Ethernet mapping, all OTN related configuration, operational data, PM, and alarms are not applicable to the line channel object.			

Client ports (WL5e 2x MOTR 12x modules)

The following table provides information on the client port types supported.

Table 168 Client port summary

ClientXCVR Mode (Note)	Protocol	FEC	Supported Line Mapping	Supported Pluggable (See Pluggables and specifications)	Ports
100GE	100GE	- None - KR4 (NRZ pluggables) - KP4 (PAM4 pluggables)	OTN Ethernet	QSFP28 (including AOC)	3 to 14
400GE	400GE	KP4	OTN Ethernet	QSFP-DD	3, 6, 9, and 12
4x100GE-ZR	100GE	CFEC	OTN Ethernet	400ZR QSFP-DD	3, 6, 9, and 12
OTL4.4	OTU4	- None - GFEC	OTN	QSFP28 (including AOC)	3 to 14
Note: 100GE KR4 - 802.3 CL-91 RS (528, 514) over 4 lanes 100GE KP4 - 802.3 CL-91 RS (544, 514) over 4 or 2 lanes 400GE KP4 - 802.3 CL-119 RS (544, 514) over 8 lanes OTL4.4 GFEC-G.709 Annex A RS (255, 239)					

Engineering rules for mated line configurations for WL5e 2x MOTR 12x module

When using mated line configurations using service modules, the allowable skew budget dictates the maximum spectral separation between the two mated wavelengths.

Consider the following engineering rules to minimize skew between the paired signals when a service module has both line ports set to the same transmission mode and there is an odd total number of client interfaces provisioned.

Note: For information on setting line port frequencies, see [Line PTP attributes](#).

- The mated line configuration is supported on all variants of this module.
- Mated x50G lines can be used to carry 100GE and OTL4.4 clients.
- Mated x00G lines can be used to carry 400GE or 4X100GE-ZR clients as 2x200G and 2x600G.

- For uncompensated C-band and L-band, a 1.2 Terahertz difference in frequency between the line ports is permitted without fiber distance restrictions (below 25,000 km).
- For C-band, if the fiber distance does not exceed 6000km, and in compensated submarine links, the difference in frequencies between the two line ports can be unlimited within the band.
- Line port 1 deskews the paired 50G signals. The skew tolerance is a minimum of +/- 7 μ s.
- Ensure that the wavelength pairs take the same photonic path and have similar length patchcords.

Standalone and mated line terminals for WL5e 2x MOTR 12x modules

The service modules support both standalone and mated line terminals.

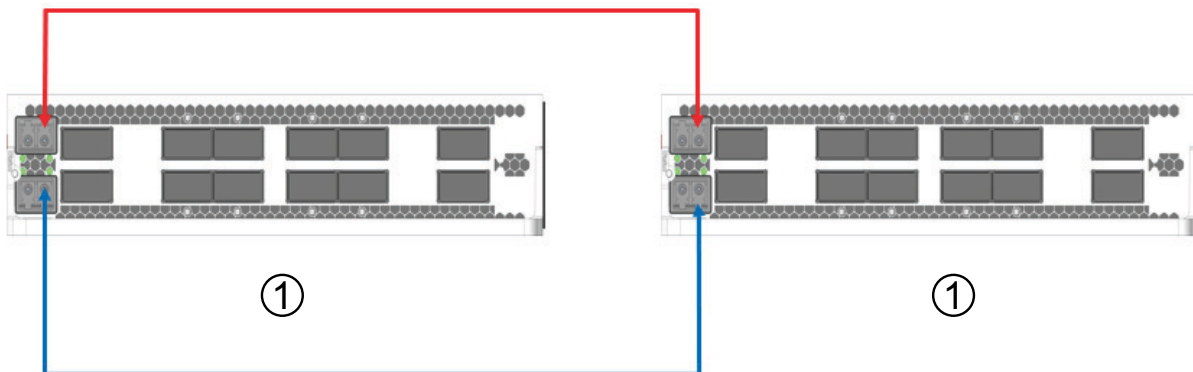
Standalone line terminals are supported on service modules, where the line ports (ports 1 and 2) on a service module work independently of each other.

Standalone line terminals have the following considerations:

- Provisionable at the module level using CLI command `module set mode none slot <slot>`.
- Line port odu-termination-mode is set to terminate automatically by the system.
- The line XCVR mate is set to none by the system.
- Supports client traffic up to 800G.
- Last 50G bandwidth for x50G mode is not used.

See the following figure.

Figure 79 Standalone line terminal



Legend

1 Terminal mode = "none"

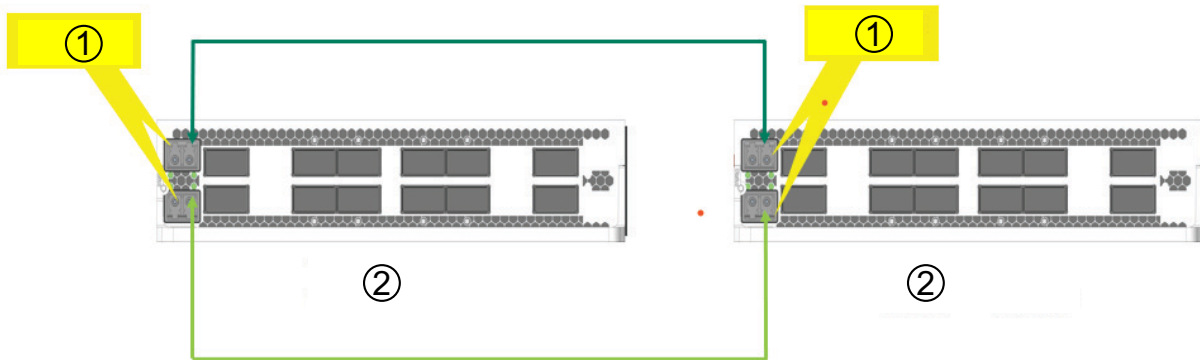
Mated line terminals are supported on the service modules, where the line ports (1 and 2) work together. If the module is set to mated mode, you can split the last supported client port on the primary line (port 1) in half.

Mated line terminals have the following considerations:

- Provisionable at the module level using CLI command `module set mode mated slot <slot>`. This is a traffic impacting operation.
- Line port odu-termination-mode is set to terminate automatically by the system.
- The line XCVR mate indicates a <slot>/<port>.
- Only line XCVRs of the same slot can be mated, and line 1 is always the primary.
- Supports x50G and x100G transmission modes.
- A logical line port 101 is automatically created as a child of the two parent PTPs (on the same module) that are mated.

See the following figure.

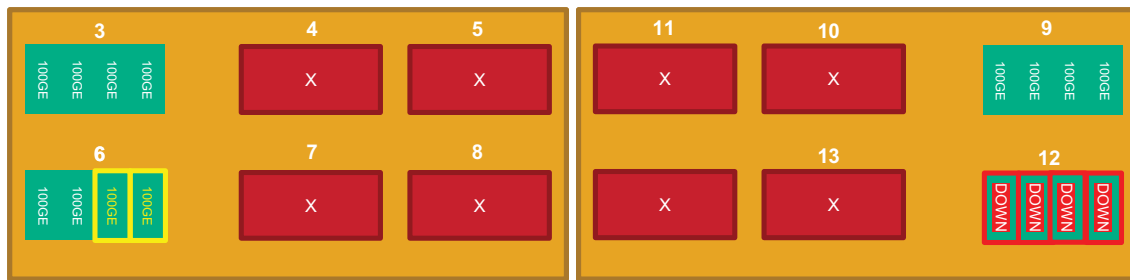
Figure 80 Mated line terminal

**Legend**

1 Line1 mated to Line2

2 Terminal mode = mated

The following figure provides an example of a 2x600G mated line using the 4x100GE service profile, where the client port 6 is the split port. For more information see [4x100GE service profile](#).

Figure 81 2 x 600G mated lines

Service profiles for WL5e 2x MOTR 12x modules

This section provides information on the service profiles available. You can provision different service profile on each line of the service module.

Waveserver 5 supports these service profiles:

- [100GE service profile](#)
- [400GE service profile](#)
- [4x100GE service profile](#)
- [Mixed-Ethernet service profile](#)
- [Mixed-OTN-Ethernet service profile](#)

Note: If you change the service profile it impacts the traffic running on that module. Also, the service-profile change is only permitted when the existing client provisioning is valid for the new service profile; any invalid client provisioning with respect to the new profile must be deleted prior to changing to a new service-profile.

100GE service profile

The 100GE service profile supports 100GE client ports over QSFP28 pluggables.

The following figure shows the supported transceiver modes and pluggables for client ports using the 100GE service profile.

Figure 82 100GE service profile - transceiver modes and pluggables for client ports

100GE service profile provisioning considerations

The following 100GE service profile provisioning considerations apply:

- The line XCVR service-profile must be set to 100GE, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile 100GE`.
- Ports 3 to 14 support QSFP28 pluggables in 100GE mode.
- Unsupported client XCVR modes and pluggable types are blocked.
- Supports standalone line terminal and mated line terminal configurations.

— Standalone line terminal

- All clients are provisionable.
- At 200G, client ports 3, 4, 9, and 10 traffic is carried over the line signal.
- At 300G, client ports 3, 4, 5, 9, 10, and 11 traffic is carried over the line signal.
- At 400G, client ports 3, 4, 5, 6, 9, 10, 11, and 12 traffic is carried over the line signal.
- At 500G, client ports 3, 4, 5, 6, 7, 9, 10, 11, 12, and 13 traffic is carried over the line signal.
- At 600G, all client ports traffic is carried over the line signal.

— Mated line terminal

- All clients are provisionable.
- At 2x250G, client ports 3, 4, 5, 9, and 10 traffic is carried over the line signal.
- At 2x350G, client ports 3, 4, 5, 6, 9, 10, and 11 traffic is carried over the line signal.
- At 2x450G, client ports 3, 4, 5, 6, 7, 9, 10, 11, and 12 traffic is carried over the line signal.
- At 2x550G, client ports 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, and 13 traffic is carried over the line signal.

For additional information on standalone and mated line terminals, see [Standalone and mated line terminals for WL5e 2x MOTR 12x modules](#).

- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
- Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).

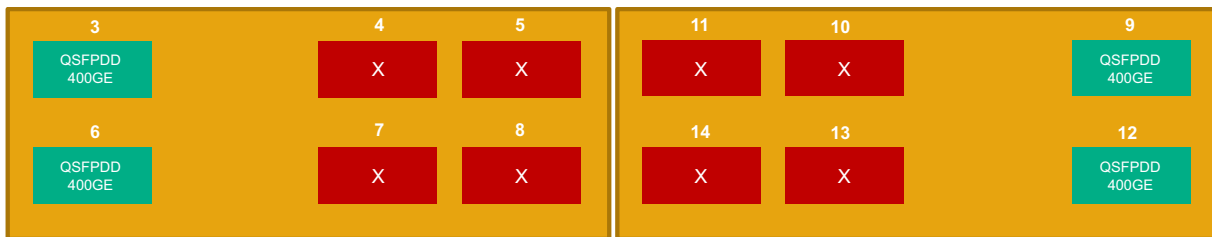
- ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
- Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
- Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

400GE service profile

The 400GE service profile supports 400GE client ports over QSFP-DD pluggables.

The following figure shows the supported transceiver modes and pluggables for the client ports using the 400GE service profile.

Figure 83 400GE service profile - transceiver modes and pluggable for client ports



400GE service profile provisioning considerations

The following 400GE service Profile provisioning considerations apply:

- The line XCVR service-profile must be set to 400GE, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile 400GE`.
- Ports 3, 6, 9, and 12 support QSFP-DD pluggables in 400GE mode.
- Unsupported client XCVR modes and pluggable types are blocked.
- Provisioning is blocked for ports 4, 5, 7, 8, 10, 11, 13, and 14.
- Supports standalone line terminal and mated line terminal configurations.
 - Standalone line terminal configuration
 - at 800G, client ports 3, 6, 9, and 12 traffic is carried over the line signal.
 - at 400G, client ports 3 and 9 traffic is carried over the line signal.
 - Mated line terminal configuration
 - at 2x600G, client ports 3, 6, and 9 traffic is carried over the line signal.
 - at 2x200G, client 3 traffic is carried over the line signal.

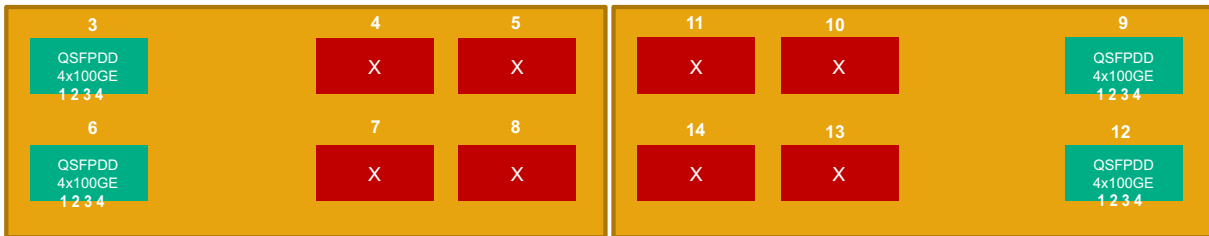
- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
 - ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
 - Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
 - Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

4x100GE service profile

The 4x100GE service profile supports 4x100GE-ZR client ports over 400ZR QSFP-DD pluggables.

The following figure shows the supported transceiver modes and pluggables for the client ports using the 4x100GE service profile.

Figure 84 4x100GE service profile - transceiver modes and pluggables for client ports



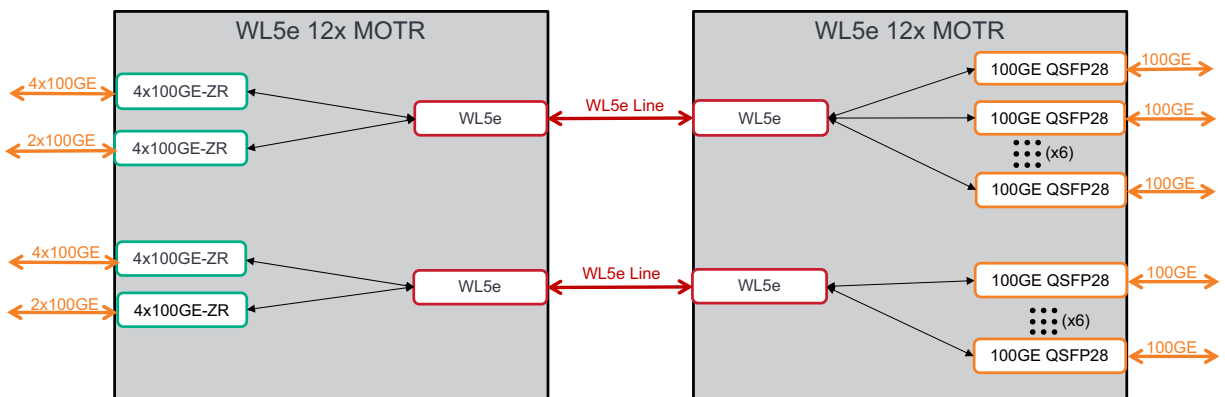
4x100GE service profile provisioning considerations

The following 4x100GE service Profile provisioning considerations apply:

- The line XCVR service-profile must be set to 4x100GE, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile 4x100GE`.
- Ports 3, 6, 9, and 12 support QSFP-DD pluggables in 4X100GE-ZR mode.
- Unsupported client XCVR modes and pluggable types are blocked.
- Provisioning is blocked for ports 4, 5, 7, 8, 10, 11, 13, and 14.
- Supports standalone line terminal and mated line terminal configurations.
 - Standalone line terminal configuration
 - All clients are provisionable.

- At 200G, client ports 3.1, 3.2, 9.1, and 9.2 traffic is carried over the line signal.
 - At 300G, client ports 3.1, 3.2, 3.3, 9.1, 9.2, and 9.3 traffic is carried over the line signal.
 - At 400G, client ports 3.1, 3.2, 3.3, 3.4, 9.1, 9.2, 9.3, and 9.4 traffic is carried over the line signal.
 - At 500G, client ports 3.1, 3.2, 3.3, 3.4, 6.1, 9.1, 9.2, 9.3, 9.4, and 12.1 traffic is carried over the line signal.
 - At 600G, client ports 3.1, 3.2, 3.3, 3.4, 6.1, 6.2, 9.1, 9.2, 9.3, 9.4, 12.1, and 12.2 traffic is carried over the line signal.
 - At 700G, client ports 3.1, 3.2, 3.3, 3.4, 6.1, 6.2, 6.3, 9.1, 9.2, 9.3, 9.4, 12.1, 12.2, and 12.3 traffic is carried over the line signal.
 - At 800G, all client ports traffic is carried over the line signal.
- Mated line terminal configuration
- Client port 12.3 and 12.4 are not usable.
 - at 2x600G, client ports 3.1, 3.2, 3.3, 3.4, 6.1, 6.2, 6.3, 6.4, 9.1, 9.2, 9.3, and 9.4 traffic is carried over the line signal.
 - at 2x200G, client 3.1, 3.2, 3.3, and 3.4 traffic is carried over the line signal.
- For 4x100-ZR to 100GE QSFP28 hand-off based applications the 12xMOTR module carrying the 100GE traffic is also need to be configured as 4x100GE profile. See the following figure for detail.

Figure 85 Application example – 4x100GE-ZR to 100GE Clients over WL5e Line



- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.

- Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
- ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
- Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
- Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

Mixed-Ethernet service profile

The Mixed-Ethernet service profile supports 100GE / 400GE client ports over QSFP28 / QSFP-DD pluggables.

The following figure shows the supported transceiver modes on the client ports using the Mixed-Ethernet service profile.

Figure 86 Mixed-Ethernet service profile - transceiver modes and pluggables for client ports



Mixed-Ethernet service profile provisioning considerations

The following Mixed-Ethernet service profile provisioning considerations apply:

- The line XCVR service-profile must be set to mixed-ethernet, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile mixed-ethernet`.
- Ports 3, 6, 9, and 12 support QSFP28 and QSFP-DD pluggables.
 - QSFP28 pluggables in 100GE mode
 - QSFP-DD pluggables in 400GE mode
- Ports 4, 5, 7, 8, 10, 11, 13, and 14 support QSFP28 pluggables in 100GE mode.
- When port 3 and port 6 are provisioned as 400GE then ports 4, 5, 7, and 8 are blocked.
- When port 3 is provisioned as 400GE and ports 4, 5, 7, and 8 are provisioned as 100GE then port 6 is blocked.

- When port 6 is provisioned as 400GE and ports 4, 5, 7, and 8 are provisioned as 100GE then port 3 is blocked.
- When port 9 and port 12 are provisioned as 400GE then ports 10, 11, 13, and 14 are blocked.
- When port 9 is provisioned as 400GE and ports 10, 11, 13, and 14 are provisioned as 100GE then port 12 is blocked.
- When port 12 is provisioned as 400GE and ports 10, 11, 13, and 14 are provisioned as 100GE then port 9 is blocked.
- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
 - ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
 - Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
 - Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).
- Traffic on all clients for the same associated line port could be impacted when creating or deleting any client XCVR provisioning.
 - Client ports 3 to 8 use line port 1 and client ports 9 to 14 use line port 2.
 - Client XCVR provisioning can be done manually or automatically when a supported pluggable is inserted in the unprovisioned port.
- It is recommended to provision 400GE clients before 100GE clients to avoid traffic interruption on existing 100GE clients.
- Mated line configuration is not supported in Mixed-Ethernet service profile.
- Supports standalone line terminal configurations as follows:
 - **6 x 100GE per line**
 - all clients are provisionable.
 - at 200G, client ports 3, 4, 9, and 10 traffic is carried over the line signal.
 - at 300G, client ports 3, 4, 5, 9, 10, and 11 traffic is carried over the line signal.
 - at 400G, client ports 3, 4, 5, 6, 9, 10, 11, and 12 traffic is carried over the line signal.

- at 500G, client ports 3, 4, 5, 6, 7, 9, 10, 11, 12, and 13 traffic is carried over the line signal.
- at 600G, all client ports traffic is carried over the line signal
- **4 x 100GE + 1 x 400GE per line**
 - client ports 3 and 9 are blocked.
 - client ports 6 and 12 are provisioned in 400G mode.
 - at 400G, client ports 6 and 12 traffic is carried over the line signal.
 - at 500G, client ports 4, 6, 10, and 12 traffic is carried over the line signal.
 - at 600G, client ports 4, 5, 6, 10, 11, and 12 traffic is carried over the line signal.
 - at 700G, client ports 4, 5, 6, 7, 10, 11, 12, and 13 traffic is carried over the line signal.
 - at 800G, client ports 4, 5, 6, 7, 8, 10, 11, 12, 13, and 14 traffic is carried over the line signal.
- **1 x 400GE + 4 x 100GE per line**
 - client ports 6 and 12 are blocked.
 - client ports 3 and 9 are provisioned in 400G mode.
 - at 400G, client ports 3 and 9 traffic is carried over the line signal.
 - at 500G, client ports 3, 4, 9, and 10 traffic is carried over the line signal.
 - at 600G, client ports 3, 4, 5, 9, 10, and 11 traffic is carried over the line signal.
 - at 700G, client ports 3, 4, 5, 7, 9, 10, 11, and 13 traffic is carried over the line signal.
 - at 800G, client ports 3, 4, 5, 7, 8, 9, 10, 11, 13, and 14 traffic is carried over the line signal.
- **2 x 400GE per line**
 - client ports 4, 5, 7, 8 and 10, 11, 13, 14 are blocked.
 - client ports 3, 6 and 9, 12 are provisioned in 400G mode.
 - at 800G, client ports 3, 6, 8, and 12 traffic is carried over the line signal.
 - at 400G, client ports 3 and 9 traffic is carried over the line signal.

ATTENTION**Risk of traffic interruption**

Adding or deleting a 400GE client port by manually provisioning or inserting a QSFP-DD pluggable when there is an existing client port sharing the same line is service-affecting. If mixed 400GE and 100GE clients are deployed on the same line (mixed-ethernet profile), it is recommended to provision 400GE clients before provisioning the 100GE clients.

Mixed-OTN-Ethernet service profile

The Mixed-OTN-Ethernet service profile supports a mix of 100GE, 400GE, and OTL4.4 client ports over QSFP28 and QSFP-DD pluggable.

However, there are certain limitations when a port is provisioned in 400GE mode. The following two sections show the supported transceiver modes on the client ports using the Mixed-OTN-Ethernet service profile without and with 400GE mode.

- [Mixed-OTN-Ethernet service profile - without 400GE](#)
- [Mixed-OTN-Ethernet service profile - with 400GE](#)

Mixed-OTN-Ethernet service profile - without 400GE**Figure 87** Mixed-OTN-Ethernet service profile - Transceiver modes 100GE and OTL4.4

Note: Ports 8 and 14 are not usable in mated configuration.

The following Mixed-OTN-Ethernet service profile provisioning considerations apply:

- The line XCVR mode must be set to a mode that is OTN Transcoded, where the max line rate 800G.
- The line XCVR service-profile must be set to mixed-otn-ethernet, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile mixed-otn-ethernet`.
- All ports support QSFP28 pluggables in OTL4.4 or 100GE mode.
- Port 8 and 14 are not usable in mated line configuration. The port admin state and the connection state remains disabled.

- Unsupported client xcvr modes and pluggable types are blocked.
- Supports standalone line terminal and mated line terminal configurations.
 - Standalone line terminal
 - At 200GE, client ports 3, 4, 9, and 10 traffic are carried over the line signal.
 - At 300GE, client ports 3, 4, 5, 9, 10, and 11 traffic are carried over the line signal.
 - At 400GE, client ports 3, 4, 5, 6, 9, 10 11, and 12 traffic are carried over the line signal.
 - At 500GE, client ports 3, 4, 5, 6, 7, 9, 10 11, 12, and 13 traffic are carried over the line signal.
 - At 600GE, all client ports traffic are carried over the line signal.
 - Mated line terminal
 - Client ports 8 and 14 cannot be used.
 - At 2x250G, client ports 3, 4, 5, 9, and 10 traffic are carried over the line signal.
 - At 2x350G, client ports 3, 4, 5, 6, 9, 10, and 11 traffic are carried over the line signal.
 - At 2x450G, client ports 3, 4, 5, 6, 7, 9, 10, 11 and 12 traffic are carried over the line signal.
 - Mated line configuration is supported only when no 400GE client is provisioned.

For additional information on standalone and mated line terminals, see [Standalone and mated line terminals for WL5e 2x MOTR 12x modules](#).

- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).
 - ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
 - Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
 - Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

Mixed-OTN-Ethernet service profile - with 400GE

Figure 88 Mixed-OTN-Ethernet service profile - Transceiver modes 400GE, 100GE and OTL4.4



The following consideration applies in Mixed-OTN-Ethernet with 400GE client:

- The line XCVR mode must be set to a mode that is OTN Transcoded, where the max line rate 800G.
- The line XCVR service-profile must be set to mixed-otn-ethernet, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile mixed-otn-ethernet`.
- Port 3 and 9 support QSFP-DD pluggables in 400GE mode.
- Ports 6 and 12 are blocked.
- Port 4, 5, 7, 8, 10, 11, 13, and 14 support QSFP28 pluggables in 100GE or OTL4.4 mode.
- Port 8 and 14 support QSFP28 pluggables in 100GE mode only.
- Mated line Configuration is not supported.
- Supports following standalone line terminal configurations.
 - At 400G, client ports 3 and 9 traffic are carried over the line signal.
 - At 500G, client ports 3, 4, 9, and 10 traffic are carried over the line signal.
 - At 600G, client ports 3, 4, 5, 9, 10, and 11 traffic are carried over the line signal.
 - At 700G, client ports 3, 4, 5, 7, 9, 10, 11, and 13 are carried over the line signal.
 - At 800G, all client ports (except 6 and 12) traffic are carried over the line signal.
- When the port connection-state is disabled, the considerations below apply. To confirm if the connection-state is disabled, use CLI command `port show port <slot>/<port>` and check the connection peers state.
 - Laser shutdown conditioning is applied. For additional information on signal conditioning, see [Signal conditioning](#).

- ITS functionality is unavailable. For additional information on ITS, see [Integrated test set](#).
- Client port loopback functionality is unavailable. For additional information on loopbacks, see [Loopback testing](#).
- Client port alarms and PMs are unavailable. For additional information on alarms and PMs, see [Alarms](#) and [Performance Monitoring](#).

ATTENTION

Risk of traffic interruption

Adding or deleting a 400GE client port by manually provisioning or inserting a QSFP-DD pluggable when there is an existing client port sharing the same line is service-affecting. If mixed 400GE and 100GE clients will be deployed on the same line (mixed-otn-ethernet profile). It is recommended to provision 400GE clients before provisioning the 100GE/OTL4.4 clients.

Service profile comparisons

The following table provides information on the differences between the service profiles.

Table 169 Service profile comparisons

Service profile	Clients	Application	Line mapping	Pluggable (Note 1)	Mated lines
100GE	100GE	Terminal	Ethernet, OTN	QSFP28	Yes
400GE	400GE	Terminal	Ethernet, OTN	QSFP-DD	Yes
4x100GE	4x100GE	Terminal	Ethernet, OTN	QSFP-DD	Yes
Mixed Ethernet	100GE / 400GE (Note 2)	Terminal	Ethernet, OTN	QSFP28, QSFP-DD	No
Mixed-OTN-Ethernet	100GE / 400GE / OTL4.4 (Note 2)	Terminal	OTN	QSFP28, QSFP-DD	Yes (Note 3)

Note 1: For more information on pluggables, see [Pluggables and specifications](#).

Note 2: Traffic hits on all clients sharing the same line when adding or deleting 400GE and 4x100GE clients.

Note 3: Mated line configuration is supported only when no 400GE client is provisioned.

Service Profile Migration

After comparing the service profiles, you can use this information to migrate a service profile into another one. Typically, there are two type of migration can happen:

- [100GE and 400GE into Mixed-Ethernet service profile](#)

- [100GE, 400GE, and Mixed-Ethernet into Mixed-OTN-Ethernet service profile](#)

The following sections provide a set of guidelines to migrate a service profile into another one.

100GE and 400GE into Mixed-Ethernet service profile

- When migrating from 100GE or 400GE service profile to mixed-ethernet profile, re-provisioning of clients is not required.
- Before migrating from mixed-ethernet profile to 100GE or 400GE profile, client not supported in destination profile needs to be deleted.
- There is traffic impact to all clients when changing the service profile.

100GE, 400GE, and Mixed-Ethernet into Mixed-OTN-Ethernet service profile

- Before migrating from 100GE service profile to mixed-otn-ethernet profile, line xcvr mode has to be modified to -O mode (like 95-400-O), and 100GE provisioning on client ports can remain as it is.
- Before migrating from 400GE service profile to mixed-otn-ethernet profile, line xcvr mode has to be modified to -O mode (like 95-400-O), and 400GE provisioning other than client ports 3, and 9 have to be deleted.
- Before migrating from mixed-ethernet profile to mixed-otn-ethernet profile, line xcvr mode has to be modified to -O mode (like 95-400-O), 400GE provisioning other than client port 3 and 9 have to be deleted, and 100GE provisioning on client ports can remain as it is.
- Before migrating from mixed-otn-ethernet profile to 100GE, 400GE, or mixed-ethernet profile, client not supported in destination profile needs to be deleted.
- There is traffic impact on all clients when changing the service profile.

Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD module details

The Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD (WL5n DCO 3xQSFP-DD 12x) modules are double-slot wide modules has following properties:

- 3 x 400ZR QSFP-DD coherent line interfaces
- 3 x bilingual QSFP-DD / QSFP-28 client interfaces
- 9 x QSFP-28 client interfaces

This section outline the following details about the WL5n DCO 3xQSFP-DD 12x modules:

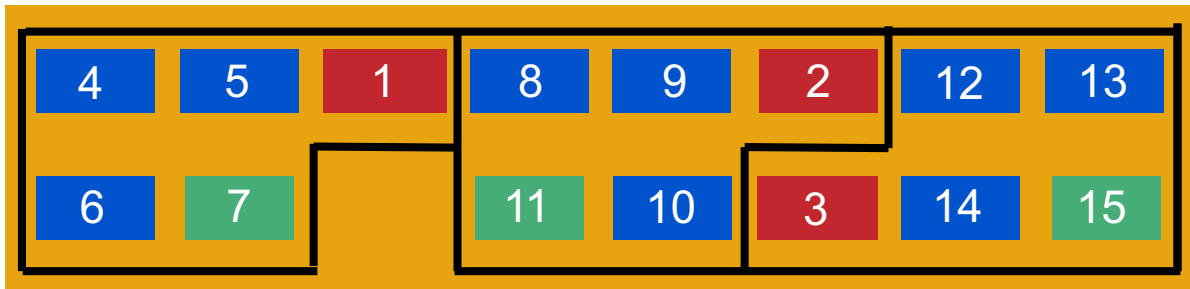
- [Line ports and client ports in WL5n DCO 3xQSFP-DD 12x modules](#)

- [Engineering rules for mated line configuration for in WL5n DCO 3xQSFP-DD 12x modules](#)
- [Standalone terminals for in WL5n DCO 3xQSFP-DD 12x modules](#)
- [Service profiles for WL5n DCO 3xQSFP-DD 12x modules](#)

Line ports and client ports in WL5n DCO 3xQSFP-DD 12x modules

The service modules have line and client ports which allow traffic to be carried end-to-end. This section provides information on the line and client ports. See the following figure on service module line and client port numbering.

Figure 89 Service module - line and client port numbering



Legend

- 1 Red - Ports 1, 2, and 3: 400ZR QSFP-DD line ports
 - 2 Green - Ports 7, 11, and 15: Bilingual QSFP-DD/QSFP28 client ports
 - 3 Blue - Ports 4-6, 8-10, and 12-14: QSFP28 client ports
- Client ports 4-7 use line port 1, client ports 8-11 use line port 2, and client ports 12-15 use line port 3

Line ports in WL5n DCO 3xQSFP-DD 12x modules

The following tables lists the supported line port types, baud, bit rate, mapping and supported client protocols.

Table 170 Supported line port transmission mode, baud, mapping, rate and client protocols

Line XCVR mode	Baud or Symbol Rate (GBaud)	Bit Rate (Gbps)	Supported client protocols
4x100GE-ZR	60	400	100GE

Client ports in WL5n DCO 3xQSFP-DD 12x modules

The following table provides information on the client port types supported.

Table 171 Client port summary

Client XCVR Mode	Protocol	FEC	Supported Line Mapping	Supported Pluggable (See Pluggables and specifications)	Ports
100GE	100GE	- None - KR4 (NRZ pluggables) - KP4 (PAM4 pluggables)	Ethernet	QSFP28 (including AOC)	4 to 15

Engineering rules for mated line configuration for in WL5n DCO 3xQSFP-DD 12x modules

Mated line configuration is NOT supported on WL5n type service module.

Standalone terminals for in WL5n DCO 3xQSFP-DD 12x modules

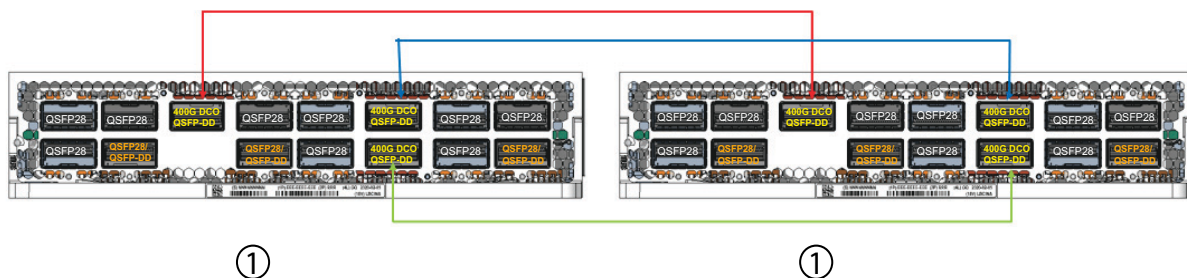
The service modules support only standalone line terminals, where the line ports (1, 2, and 3) on a service module work independently of each other.

Note: Mate line terminals are not supported on WL5n DCO 3xQSFP-DD 12x module.

Standalone line terminals have the following considerations:

- Provisionable at the module level using CLI command `module set mode none slot <slot>`.
- The line XCVR mate is set to none by the system.
- Supports client traffic up to 400G.

See the following figure.

Figure 90 Standalone line terminal

Legend

- 1 Terminal mode = "None"

Service profiles for WL5n DCO 3xQSFP-DD 12x modules

This section provides information on the service profiles available. Each service module supports one service profile.

The WL5n DCO 3xQSFP-DD 12x modules support the following service profile:

- [Mixed-ethernet service profile](#)

Mixed-ethernet service profile

The mixed-ethernet service profile supports 100GE client ports over QSFP28 pluggables.

The following figure shows the supported transceiver modes on the client ports using the mixed-ethernet service profile.

Figure 91 Mixed-Ethernet service profile - transceiver modes and pluggables for client ports



Mixed-Ethernet service profile provisioning considerations

The following mixed-ethernet service profile provisioning considerations apply:

- The line XCVR service-profile must be set to mixed-ethernet, to set the service profile, use CLI command `xcvr set xcvr <slot>/<port> service-profile mixed-ethernet`.
- The line ports 1, 2, and 3 supports QSFP-DD ZR plug in 4x100GE-ZR mode only.
- All client ports 4 to 7, 8 to 11, and 12 to 15 can support only QSFP28 pluggables in 100GE mode.
- Independent traffic configuration between all 3 port groups.

Commission third-party XCVR pluggables

Waveserver 5 supports some third-party XCVR plugs and to provision them the `xcvr profile` command is used. Mostly, at time the pluggable is inserted in the service module it gets auto-provisioned similar to Ciena branded plugs. This `xcvr profile` command allows you to configure power threshold values for third-party XCVR plugs. Each XCVR profile is

represented by a unique profile ID, which represents a specific third-party XCVR pluggable. There are following vendor-specific parameters used to distinguish each pluggable:

- vendor name
- vendor OUI
- part number
- part revision number

Note 1: The `xcvr profile` command is used ONLY for third-party plugs.

Note 2: The use of `xcvr profile` command is completely optional and is required only when you need to override the Rx/Tx power alarm thresholds of the pluggable.

For each XCVR profile, the user can define the TX/Rx power high/low warning/alarm thresholds and Tx optical power monitor override. The profile-id must be an integer value between 1 and 255, each XCVR profile must have a unique profile-id, and each XCVR pluggable can only have one XCVR profile defined.

For third-party pluggable, there are default receive and transmit power thresholds recommended by the vendor and stored in the EEPROM (read-only memory) of the pluggable. By default, XCVR pluggable use the power thresholds stored in the EEPROM. If you want to change the default power thresholds, you must consider that setting the power thresholds too high or too low, may result in no alarms being raised or generating false alarms. For example, Rx Power Low alarm may not indicate a loss of signal. A loss of signal is indicated by additional faults such as high bit error rate, receive loss of clock, loss of alignment marker, and so on.

once an XCVR profile is assigned to a third-party pluggable and power thresholds are reached, the corresponding alarms are raised on the Waveserver 5. These alarms are raised when power thresholds are reached:

- Rx Power High warning alarm
- Rx Power High alarm
- Rx Power Low warning alarm
- Rx Power Low alarm
- Tx Power High warning alarm
- Tx Power High alarm
- Tx Power Low warning alarm
- Tx Power Low alarm

Note: The warning alarms are raised only when they are configured using the XCVR profile. If an XCVR profile is not assigned to a third-party pluggable then no warning alarm is raised. However, warning alarm threshold, which is readable from EEPROM of the pluggable, and warning alarm flags indicating if the thresholds are crossed, are visible using the `ptp show ptp status` command.

The following table provides information on the provisionable attributes for an XCVR profile. To set XCVR profile attributes, use CLI command `xcvr profile set`.

Table 172 Provisionable attributes for XCVR profiles

Attribute	Description	Values
profile-id	Profile ID for the profile, each profile is identified with a unique ID.	NUMBER: 1..255
description	Allows the user to provide a description of the XCVR profile.	String[1..255], blank (default)
vendor-name	Vendor name for the profile.	String[1..16]
vendor-oui	Vendor Organizationally Unique Identifier (OUI) for the profile.	OUI of the vendor (xx:yy:zz)
vendor-partnum	Vendor part number or product name for the profile.	String[1..16]
vendor-revision	Vendor revision number for the profile.	String[1..2]
rx-power-high-alarm-threshold	The Rx power high threshold for when the Rx Power High alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
rx-power-high-warning-threshold	The Rx power high warning threshold for when the Rx Power High warning alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
rx-power-low-alarm-threshold	The Rx power low threshold for when the Rx Power Low alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
rx-power-low-warning-threshold	The Rx power low warning threshold for when the Rx Power Low warning alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
tx-optical-power-monitor-override	If the tx-optical-power-monitor is off, the pluggable's EEPROM is used for power monitoring. If on, the EEPROM settings are ignored for power monitoring.	• on • off

Attribute	Description	Values
tx-power-high-alarm-threshold	The Tx power high alarm threshold for when the Tx Power High alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
tx-power-high-warning-threshold	The Tx power high warning threshold for when the Tx Power High warning alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
tx-power-low-alarm-threshold	The Tx power low alarm threshold for when the Tx Power Low alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
tx-power-low-warning-threshold	The Tx power low warning threshold for when the Tx Power Low warning alarm is raised.	Power Threshold in dBm (NN.MM): -40.00 .. 8.16 (Note)
Note: The default value is read from the pluggable's EEPROM.		

A syslog is generated when a XCVR power threshold or optical power monitor override is modified. Waveserver 5 validates the XCVR power threshold to ensure the values that are set, are appropriate. For example, the high power alarm threshold value must be greater than the high power warning threshold. Also, the high power warning threshold must be greater than the low power warning threshold and the low power warning threshold must be greater than low power alarm threshold.

For information on configuring power thresholds for third-party pluggables, see [Configuring power thresholds for third-party pluggables](#).

Procedure list

This section provides the following procedures:

- [Procedure 72, "Provisioning Waveserver 5"](#)
- [Procedure 73, "Manually provisioning slots and modules"](#)
- [Procedure 74, "Automatically provisioning slots and modules"](#)
- [Procedure 75, "Provisioning line ports for WL5e 2x MOTR 16x module"](#)
- [Procedure 76, "Provisioning line ports for WL5e 2x MOTR 12x modules"](#)
- [Procedure 77, "Provisioning line ports for WL5n DCO 3xQSFP-DD 12x modules"](#)
- [Procedure 78, "Provisioning client ports"](#)
- [Procedure 79, "Changing the line XCVR mode"](#)
- [Procedure 80, "Confirming traffic"](#)
- [Procedure 81, "Configuring power thresholds for third-party pluggables"](#)

- [Procedure 82, “Retrieving Waveserver 5 entity data”](#)

Procedure 72 Provisioning Waveserver 5

Use this procedure to provision the Waveserver 5 at either end of a network. Provision the same attribute values on each Waveserver 5 in a system.

Requirements

- You have reviewed [System attributes](#) and have access to the information you need to complete this procedure.
- Waveserver 5 is connected to the management network, and has an IP address assigned to it. See [Comms](#).
- You have checked the software load running on Waveserver 5 using the CLI command `software show`. If Waveserver 5 is not running the load you require, upgrade the software load. See [Out-of-service software upgrades](#).
- If you have a **Waveserver 5** running a different release than the software license you have, perform procedure [Performing a one-step upgrade when the software release on the Waveserver 5 does not match the software license](#). To confirm the software release on the Waveserver 5, use CLI command `software show`. To confirm the software license you have, check the Software License Availability email you received from Ciena. For an example of the Software License Availability email, see [Ordering and license generation](#).
- The required licenses are installed on Waveserver 5. The Waveserver 5 is already configured in commissioning mode which is valid for 90-day period. The 90-day period for each license begins when the system time is set. See [Licensing](#).
- Waveserver 5 line ports are not physically connected to another or to photonic system equipment, and transceivers are not installed in client ports.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

**CAUTION****Risk of traffic interruption**

Adding or deleting a 400GE client port by manually provisioning or inserting a QSFP-DD pluggable when there is an existing 100GE client port sharing the same line is service-affecting. If mixed 400GE and 100GE clients will be deployed on the same line (mixed-ethernet profile), you must pre-provision 400GE clients first before provisioning the 100GE clients.

- An SSH client is installed on the terminal.

Steps

- 1 Start the SSH client on the terminal, and log in to Waveserver 5 using an account with at least admin access privileges.

Note: If the following warning message appears: “WARNING The system could not load the default or backup configuration files”, contact Ciena support.

Setting the system-level attributes

- 2 Set the system conditioning behavior.

```
system defaults set conditioning-type <laser-off | none |
protocol-specific>
```

```
system defaults set conditioning-holdoff <0ms - 1000ms>
```

Note: For a 100GE client port, the default conditioning type is Ethernet and the default conditioning holdoff is 10 ms and for an OTL4.4 client port, the default conditioning type is OTN and conditioning holdoff is not supported.

Specify the conditioning-holdoff as follows:

- From 0 ms to 100 ms in 10 ms increments
- From 100 ms to 1000 ms in 100 ms increments

- 3 Specify the site ID, name, description, and location of the data center where Waveserver 5 is installed:

```
system set site-id <Number: 1...65535> [site-name
<String: 1...31>] [site-description <String: 1...64>]
[site-latitude <Number: -90...90>] [site-longitude
<Number: -180...180>] [site-address <String: 1...128>]
```

Note 1: The [site-id] attribute value must be unique to the data center.

Note 2: The site name, description, and address strings cannot contain the characters: , ! ? % * # @ &. A string that contains one or more spaces must be enclosed in quotation marks; for example, “Main Street Data Center”.

- 4 Specify the ID, name, and description of the group to which Waveserver 5 is assigned:

```
system set group-id <Number: 1...99> [group-name <String:
1...32>] [group-description <String: 1...64>]
```

Note 1: The group-id value must be unique to the site.

Note 2: Ciena recommends using the format <Site Name>-< Group ID> to set the group-name attribute.

Note 3: The group name and description strings cannot contain the characters: , ! ? % * # @ &. Also, a string that contains one or more spaces must be enclosed in quotation marks.

5 Set the system host name:

```
system set host-name <String: 2...63>
```

Note 1: The host-name has the following rules:

-Must be between 2 to 63 characters. Host-name strings that exceed 63 characters raise a "String is too long" warning message.

-Must contain only alphanumeric characters, hyphens, periods, and underscores. Hyphens are not supported as the first or last character. Periods are not supported as the first character. Using unsupported characters raises an "Invalid Hostname" warning message.

-The default host name is "Waveserver-5".

Note 2: If you assigned a DNS host name to Waveserver 5, you can set it as the system host name.

Note 3: You can also assign the chassis host name to Waveserver 5 to immediately identify the type of chassis.

Provisioning the slots and modules

6 Determine the module state by retrieving module data.

If you want to	Then go to
provision and enable a slot for a module type before module installation	Manually provisioning slots and modules
Install a module and let the system provision it automatically	Automatically provisioning slots and modules

Provisioning line ports

7 Determine the module type and select the appropriate option to provision the line ports:

If you want to	Then go to
provision line ports for a WL5e 2x MOTR 16x module	Provisioning line ports for WL5e 2x MOTR 16x module
provision line ports for a WL5e 2x MOTR 12x module	Provisioning line ports for WL5e 2x MOTR 12x modules
provision line ports for a WL5n DCO 3x QSFP-DD 12x module	Provisioning line ports for WL5n DCO 3xQSFP-DD 12x modules

Setting the client port attributes

- 8 To configure the client ports go the [Procedure 78, "Provisioning client ports"](#)

Saving the configuration

- 9 Save the provisioned settings to the configuration file:
`configuration save`
- 10 Provision the Waveserver 5 at the far-end of the network.
- 11 Go to [Confirming traffic](#).

Procedure 73 Manually provisioning slots and modules

Use this procedure to manually provision slots and modules before installing actual module to the Waveserver 5 chassis. This type of provisioning is required when you override the default installation properties of the module as per your requirement so that as and when the module is available it can be plugged in to the chassis and make it ready for managing the traffic.

Requirements

- You have reviewed [System attributes](#) and have access to the information you need to complete this procedure.
- Waveserver 5 is connected to the management network, and has an IP address assigned to it. See [Comms](#).
- You have checked the software load running on Waveserver 5 using the CLI command `software show`. If Waveserver 5 is not running the load you require, upgrade the software load. See [Out-of-service software upgrades](#).
- You must be logged in to the CLI using an account with at least limited access privileges.

Overview

This procedure is useful while commissioning the Waveserver 5 node for the first time and also when you need to add a new module to already provisioned Waveserver 5 node.

Steps

- 1 Determine the empty slot and the module state by retrieving module data.
`module show`
- 2 Provision and enable the slot for a module:
`module create slot <Number: 1..8> type <motr-400-2-16 | motr-400-2-12 | motr-dco-400-3-12>`
Note 1: Only odd slots are used in the current release.
Note 2: Use `motr-400-2-16` to provision WL5e 2x MOTR 16x module, use `motr-400-2-12` to provision WL5e 2x MOTR 12x module, and `motr-dco-400-3-12` to provision WL5n DCO 3xQSFP-DD 12x module.
- 3 Assign a label to the slot, if required:
`module set slot <slot> label <String[1..128]>`
Note: You can remove the description assigned to the slot using the command: `module unset slot <slot> label`.
- 4 Assign a mode to the slot, if required:
`module set slot <slot> mode <none | mated>`

Now you can insert the provisioned module type in the selected slot as and when required. If you insert a different module then the [Equipment Mismatch](#) alarm is raised.

- 5 Save the provisioned settings to the configuration file:
`configuration save`
- 6 If required, return to the main procedure [Provisioning Waveserver 5](#) to continue provisioning the Waveserver 5.

Procedure 74 Automatically provisioning slots and modules

Use this procedure to provision module automatically by the Waveserver 5 system software at time of installing the actual module in the empty slot of the Waveserver 5 chassis. This type of provisioning is required when a module is installed with their default installation properties.

Requirements

- You have reviewed [System attributes](#) and have access to the information you need to complete this procedure.
- Waveserver 5 is connected to the management network, and has an IP address assigned to it. See [Comms](#).
- You have checked the software load running on Waveserver 5 using the CLI command `software show`. If Waveserver 5 is not running the load you require, upgrade the software load. See [Out-of-service software upgrades](#).
- You must be logged in to the CLI using an account with at least limited access privileges.

Overview

This procedure is useful while commissioning the Waveserver 5 node for the first time and also when you need to add a new module to already provisioned Waveserver 5 node.

Steps

- 1 Determine the empty slot and the module state by retrieving module data.
`module show`
- 2 Holding the module by its handle, carefully insert it into the appropriate slot until the holding clip engages.
- 3 The system automatically provisions the module, which is inserted in the empty slot, and assigns the following default values:

Table 173

Parameter name	WL5e 2x MOTR 16x module	WL5e 2x MOTR 12x module	WL5n DCO 3xQSFP-DD 12x module
	Default value	Default value	Default value
Line XCVR mode	95-400- O	95-400- O	4x100GE-ZR
LOM	Standard	Standard	Standard
Service profile	100GE	100GE	mixed-ethernet

- 4 Determine the module state by retrieving module data.
`module show`
- 5 Assign a label to the slot, if required:
`module set slot <slot> label <String[1..128]>`
Note: You can remove the description assigned to the slot using the command: `module unset slot <slot> label`.
- 6 Assign a mode to the slot:
`module set slot <slot> mode <none | mated>`
- 7 Save the provisioned settings to the configuration file:
`configuration save`
- 8 If required, return to the main procedure [Provisioning Waveserver 5](#) to continue provisioning the Waveserver 5.

Procedure 75 Provisioning line ports for WL5e 2x MOTR 16x module

Use this procedure to provision line ports for the WL5e 2x MOTR 16x modules on the Waveserver 5 on either end of the system.

Requirements

- You have reviewed the Waveserver 5 [System attributes](#) and have access to the information you need to complete this procedure.
- The required licenses are installed on Waveserver 5 or it is running in the commissioning mode. See [Ordering](#) and [Retrieving license information](#).
- The following equipment is available:
 - Fiber-optic cables to connect transceivers to subtending equipment. See [Ordering](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

- An SSH client is installed on the terminal.

Overview

This procedure is useful in the following situations:

- The line ports are not provisioned automatically by the software.
- Re-provision of the existing xcvr is required.
- The line xcvr needs to support a different service profile other than the default that is 100GE.
- A new service module is added to the already provisioned Waveserver 5 to fulfill the service expansion needs.

This procedure assumes the following:

- For a Waveserver 5 connected directly to another, the other might not be commissioned.

Note: For information about client entity variables used in CLI commands, see [Waveserver 5 entities in CLI commands](#).

Steps

- 1 Set the service profile for both line ports.

```
xcvr set xcvr <slot>/<xcvr 1..2> service-profile <100GE | 400GE | mixed-otn-ethernet | mixed-ethernet>
```

Note: Setting the service profile impacts traffic.
- 2

If you are	Then go to
not using SPLI functionality	step 3
using SPLI functionality	step 14

Note: When using SPLI with a service module in mated mode, SPLI blocks provisioning to the service module until the transmission mode and link optimization mode provisioned are the same between the line ports. If there is a mismatch between the line ports, the “Adjacency Mismatch” alarm is raised on the 6500. If the user changes the module mode from mated to none on the Waveserver 5, the SPLI provisioning will be automatically unblocked and applied on the service module.
- 3 If SPLI functionality is not used, disable the SPLI using the following command:

```
spli disable
```
- 4 Configure the link optimization mode of the XCVR.

```
ptp set ptp <slot/port> link-optimization-mode <Standard | Non-linear | SNR-Tolerant | Fast-Tracking>
```

The default link optimization mode is Standard.
- 5 Set the transmission mode for both line ports.

```
xcvr set xcvr <slot>/<xcvr 1..2> mode <95-200-O | 95-250-O | 95-300-O | 95-350-O | 95-400-O | 95-450-O | 95-500-O | 95-550-O | 95-600-O | 95-650-O | 95-700-O | 95-800-O | 95-300-E | 95-350-E | 95-400-E | 95-450-E | 95-500-E | 95-550-E | 95-600-E | 95-650-E | 95-700-E | 95-800-E | 91.6-300-E | 91.6-400-E | 91.6-450-E | 91.6-500-E | 91.6-550-E | 91.6-600-E | 91.6-700-E | 91.6-800-E>
```

Note 1: Setting the transmission mode impacts traffic.

Note 2: The modes listed above are supported for this release.

Note 3: The default transmission mode for an XCVR is 95-400-O.

Note 4: When using the mixed-otn-ethernet service-profile, provisioning -E transmission modes are blocked.

6 Configure the transmitter frequency:

```
ptp set ptp <slot>/<ptp> frequency <Frequency in
GHz (N.M): C-Band: 191250.0..196125.0> | <Frequency in
GHz (N.M): L-Band: 186087.5..190956.2>
```

7 Set the transmitter output power:

```
ptp set ptp <slot/port> tx-power <-9.0..4.0 dBm>
```

8 Set the line-system type, if required. By default, the line-system type is coloured.

```
ptp set ptp <slot>/<ptp> line-system-type <coloured |
colourless | contentionless | cs-coloured | cs-colourless>
```

9 Set the channel contention detection avoidance, if required:

```
ptp set ptp <slot>/<ptp> channel-contention-detection-
avoidance <enabled | disabled>
```

10 Set the tx dispersion compensation mode, if required:

```
ptp set ptp <slot>/<ptp> tx-dispersion-compensation-mode
<automatic | fixed | search>
```

The Tx-dispersion-compensation-mode is set based on the application; protected/unprotected terrestrial or submarine. For the required application settings, see [Line port PTP parameters - chromatic dispersion compensation details](#).

11 If the tx dispersion compensation mode is **Then go to**

automatic	step 12
fixed	step 13
search	step 14

12 Set the Tx dispersion path value according to the application you are configuring.

```
ptp set ptp <slot>/<ptp> tx-dispersion-path-value
<number>
```

Note: See [Line port PTP parameters - chromatic dispersion compensation details](#) for configuration details.

Go to [step 14](#).

13 Set the Tx dispersion pre-compensation value according to the application you are configuring:

```
ptp set ptp <slot>/<ptp> tx-dispersion-precompensation-
value <number>
```

- 14 Set the channel attributes and enable channel, if required.

```
channel set channel <channelList> [label <String[1..31]>
[odu-sd-threshold <BER String: XE-YY>]
Note: To clear these settings, use CLI command channel unset
channel <slot>/<port>.<channel> [label] [odu-sd-
threshold].
channel enable channel <channel List>
```
- 15 Confirm channel settings, if required:

```
channel show channel <channel List>
```
- 16 Retrieve PTP, XCVR and port data to confirm that the provisioned settings are correct:

```
ptp show ptp <slot>/<ptp>
xcvr show xcvr <slot>/<ptp>
port show port <slot>/<port>
```
- 17 Save the provisioned settings to the configuration file:

```
configuration save
```
- 18 If required, return to the main procedure [Provisioning Waveserver 5](#) to continue provisioning the Waveserver 5.

Procedure 76 Provisioning line ports for WL5e 2x MOTR 12x modules

Use this procedure to provision line ports for the WL5e 2x MOTR 12x modules on the Waveserver 5 on either end of the system.

Requirements

- You have reviewed the Waveserver 5 [System attributes](#) and have access to the information you need to complete this procedure.
- The required licenses are installed on Waveserver 5 or it is running in the commissioning mode. See [Ordering](#) and [Retrieving license information](#).
- The following equipment is available:
 - Fiber-optic cables to connect transceivers to subtending equipment. See [Ordering](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

- An SSH client is installed on the terminal.

Overview

This procedure is useful in the following situations:

- The line ports are not provisioned automatically by the software.
- Re-provision of the existing xcvr is required.
- The line xcvr needs to support a different service profile other than the default that is 100GE.
- A new service module is added to the already provisioned Waveserver 5 to fulfill the service expansion needs.

This procedure assumes the following:

- For a Waveserver 5 connected directly to another, the other might not be commissioned.

Note: For information about client entity variables used in CLI commands, see [Waveserver 5 entities in CLI commands](#).

Steps

- 1 Set the service profile for both line ports.

```
xcvr set xcvr <slot>/<xcvr 1..2> service-profile <100GE |
400GE | 4x100GE | mixed-otn-ethernet | mixed-ethernet>
```

Note: Setting the service profile impacts traffic.

- | | | |
|---|---|--|
| 2 | If you are
not using SPLI functionality
using SPLI functionality | Then go to
step 3
step 14 |
|---|---|--|

Note: When using SPLI with a service module in mated mode, SPLI blocks provisioning to the service module until the transmission mode and link optimization mode provisioned are the same between the line ports. If there is a mismatch between the line ports, the “Adjacency Mismatch” alarm is raised on the 6500. If the user changes the module mode from mated to none on the Waveserver 5, the SPLI provisioning will be automatically unblocked and applied on the service module.

- 3 If SPLI functionality is not used, disable the SPLI using the following command:

```
spli disable
```

- 4 Configure the link optimization mode of the XCVR.

```
ptp set ptp <slot/port> link-optimization-mode <Standard
| Non-linear | SNR-Tolerant | Fast-Tracking>
```

The default link optimization mode is Standard.

- 5 Set the transmission mode for both line ports.

```
xcvr set xcvr <slot>/<xcvr 1..2> mode <95-200-O | 95-250-
O | 95-300-O | 95-350-O | 95-400-O | 95-450-O | 95-500-O
| 95-550-O | 95-600-O | 95-650-O | 95-700-O | 95-800-O |
95-300-E | 95-350-E | 95-400-E | 95-450-E | 95-500-E | 95-
550-E | 95-600-E | 95-650-E | 95-700-E | 95-800-E | 93.3-
600-E | 93.3-400-E | 91.6-300-E | 91.6-400-E | 91.6-450-
E | 91.6-500-E | 91.6-550-E | 91.6-600-E | 91.6-700-E |
91.6-800-E> | 90-600-E | 90-400-E | 85-600-E | 85-400-E
| 85-300-E | 82-400-E
```

Note 1: Setting the transmission mode impacts traffic.

Note 2: The modes listed above are supported for this release.

Note 3: The default transmission mode for an XCVR is 95-400-O.

Note 4: When using the mixed-otn-ethernet service-profile, provisioning -E transmission modes are blocked.

6 Configure the transmitter frequency:

```
ptp set ptp <slot>/<ptp> frequency <Frequency in
GHz (N.M): C-Band: 191250.0..196125.0> | <Frequency in
GHz (N.M): L-Band: 186087.5..190956.2>
```

7 Set the transmitter output power:

```
ptp set ptp <slot/port> tx-power <-9.0..4.0 dBm>
```

8 Set the line-system type, if required. By default, the line-system type is coloured.

```
ptp set ptp <slot>/<ptp> line-system-type <coloured |
colourless | contentionless | cs-coloured | cs-colourless>
```

9 Set the channel contention detection avoidance, if required:

```
ptp set ptp <slot>/<ptp> channel-contention-detection-
avoidance <enabled | disabled>
```

10 Set the tx dispersion compensation mode, if required:

```
ptp set ptp <slot>/<ptp> tx-dispersion-compensation-mode
<automatic | fixed | search>
```

The Tx-dispersion-compensation-mode is set based on the application; protected/unprotected terrestrial or submarine. For the required application settings, see [Line port PTP parameters - chromatic dispersion compensation details](#).

11 If the tx dispersion compensation mode is **Then** go to

automatic	step 12
fixed	step 13
search	step 14

12 Set the Tx dispersion path value according to the application you are configuring.

```
ptp set ptp <slot>/<ptp> tx-dispersion-path-value
<number>
```

Note: See [Line port PTP parameters - chromatic dispersion compensation details](#) for configuration details.

Go to [step 14](#).

13 Set the Tx dispersion pre-compensation value according to the application you are configuring:

```
ptp set ptp <slot>/<ptp> tx-dispersion-precompensation-
value <number>
```

- 14 Set the channel attributes and enable channel, if required.

```
channel set channel <channelList> [label <String[1..31]>
[odu-sd-threshold <BER String: XE-YY>]
Note: To clear these settings, use CLI command channel unset
channel <slot>/<port>.<channel> [label] [odu-sd-
threshold].
channel enable channel <channel List>
```
- 15 Confirm channel settings, if required:

```
channel show channel <channel List>
```
- 16 Retrieve PTP, XCVR and port data to confirm that the provisioned settings are correct:

```
ptp show ptp <slot>/<ptp>
xcvr show xcvr <slot>/<ptp>
port show port <slot>/<port>
```
- 17 Save the provisioned settings to the configuration file:

```
configuration save
```
- 18 If required, return to the main procedure [Provisioning Waveserver 5](#) to continue provisioning the Waveserver 5.

Procedure 77 Provisioning line ports for WL5n DCO 3xQSFP-DD 12x modules

Use this procedure to provision line ports for the WL5n DCO 3xQSFP-DD 12x modules on the Waveserver 5 on either end of the system.

Requirements

- You have reviewed the Waveserver 5 [System attributes](#) and have access to the information you need to complete this procedure.
- The required licenses are installed on Waveserver 5 or it is running in the commissioning mode. See [Ordering](#) and [Retrieving license information](#).
- The following equipment is available:
 - Fiber-optic cables to connect transceivers to subtending equipment. See [Ordering](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

- An SSH client is installed on the terminal.

Overview

This procedure is useful in the following situations:

- The line ports are not provisioned automatically by the software.
- The existing line xcvr is deleted and re-provision of the xcvr is required.
- The line xcvr need to support a different service profile other than the default that is 100GE.
- A new service module is added to the already provisioned Waveserver 5 to fulfill the service expansion needs.

This procedure assumes the following:

- For a Waveserver 5 connected directly to another, the other might not be commissioned.

Note: For information about client entity variables used in CLI commands, see [Waveserver 5 entities in CLI commands](#).

Steps

- 1 Set the service profile for all three line ports.

```
xcvr set xcvr <slot>/<xcvr 1..3> service-profile <mixed-ethernet>
```

Note 1: Setting the service profile impacts traffic.

Note 2: You cannot use SPLI functionality to provision WL5n DCO 3xQSFP-DD 12x modules.
- 2 Set the transmission mode for all 3 line ports.

```
xcvr set xcvr <slot>/<xcvr 1..3> mode <4x100GE-ZR>
```

Note 1: Setting the transmission mode impacts traffic.

Note 2: The modes listed above are supported for this release.
- 3 Configure the transmitter frequency:

```
ptp set ptp <slot>/<ptp> frequency <Frequency in GHz (N.M)>
```
- 4 Set the transmitter output power:

```
ptp set ptp <slot>/<port> tx-power <N.M dBm>
```
- 5 Retrieve PTP, XCVR and port data to confirm that the provisioned settings are correct:

```
ptp show ptp <slot>/<ptp>  
xcvr show xcvr <slot>/<ptp>  
port show port <slot>/<port>
```
- 6 Save the provisioned settings to the configuration file:

```
configuration save
```
- 7 If required, return to the main procedure [Provisioning Waveserver 5](#) to continue provisioning the Waveserver 5.

Procedure 78 Provisioning client ports

Use this procedure to provision client ports on the Waveserver 5 at either end of a system.

Requirements

- You have reviewed the Waveserver 5 [System attributes](#) and have access to the information you need to complete this procedure.
- The required licenses are installed on Waveserver 5 or it is running in the commissioning mode. See [Ordering](#) and [Retrieving license information](#).
- The following equipment is available:
 - Fiber-optic cables to connect pluggable transceivers to subtending equipment. See [Ordering](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

- An SSH client is installed on the terminal.

Overview

For detail on various client ports, see [Commissioning](#).

This procedure is useful in the following situations:

- The client ports are not provisioned automatically by the software. For example, the service profile of the line port did not allow provisioning of that client xcvr.
- The existing xcvr is deleted and re-provision of the xcvr is required.
- The client xcvr supports two different line transcoding (Ethernet and OTN) at 100GE and you need to toggle between default transcoding as configured by the system upon insertion of the plug with the other supported transcoding.

This procedure assumes the following:

- For a Waveserver 5 connected directly to another, the other might not be commissioned.

Note: For information about client entity variables used in CLI commands, see [Waveserver 5 entities in CLI commands](#).

This procedure is applicable for following service modules:

- WL5e 2x MOTR 16x modules
- WL5e 2x MOTR 12x modules
- WL5n DCO 3xQSFP-DD 12x modules

Steps

- 1 Start the SSH client on the terminal, and log in to Waveserver 5 using an account that has at least admin privileges.
- 2

If the transceiver is available and you want to provision it instantly you want to pre-provision the transceiver before physical installation	Then go to step 3 step 6
--	---
- 3 Clean the optical surfaces of the transceivers, and install transceivers in supported client ports (see [Client ports \(WL5e 2x MOTR 16x Modules\)](#) and [Client ports in WL5n DCO 3xQSFP-DD 12x modules](#)). See also *"Installing or removing a pluggable transceiver"* in *Waveserver 5 Installation, 323-3001-200*.
- 4 Retrieve the existing client provisioning:


```
xcvr show
xcvr show xcvr <slot>/<xcvr>
ptp show ptp <slot>/<ptp>
```

Note: You must create the 400GE client before creating any OTL4.4 or 100GE client when using the mixed-otn-ethernet service-profile. Similarly, you must delete the OTL4.4 or 100GE clients before deleting the 400GE client.
- 5 If required, change the client xcvr mode:


```
xcvr set xcvr <slot>/<xcvr> mode <100GE | 400GE | 4x100GE-ZR | OTL4.4>
```

Note 1: The above command is used for overriding the existing client xcvr mode, which is either configured by the system, or to provision the client xcvr that is not provisioned before.

Note 2: If you want to configure handoff between the ZR plugs installed on the client port of WL5e 2x MOTR 12x modules and line port of the WL5n DCO 3xQSFP-DD 12x module use the 4x100GE-ZR mode.

- Go to [step 8](#).
- 6 Retrieve the Waveserver 5 chassis inventory information to verify that the port is empty (no xcvr is already installed on that port) and the port is not yet provisioned.

```
chassis inventory show
```

Note: After pre-provisioning the port, if an incompatible pluggable is inserted then the [Provisioning Mismatch](#) alarm is raised.
 - 7 Set the client xcvr mode on the port, you need to pre-provision

```
xcvr set xcvr <slot>/<xcvr> mode <100GE | 400GE | 4x100GE-ZR | OTL4.4>
```

Note: If you want to configure handoff between the ZR plugs installed on the client port of WL5e 2x MOTR 12x modules and line port of the WL5n DCO 3xQSFP-DD 12xmodule use the 4x100GE-ZR mode.
 - 8 Configure the transmitter frequency:

```
ptp set ptp <slot>/<ptp> frequency <Frequency in GHz (N.M): C-Band: 191250.0..196125.0> | <Frequency in GHz (N.M): L-Band: 186087.5..190956.2>
```

Note: Ensure that the transmitter frequency must be same between the ZR plugs installed on the client port of WL5e 2x MOTR 12x modules and line port of the WL5n DCO 3xQSFP-DD 12xmodule, if you want to configure handoff between them.
 - 9 Optionally, retrieve the transceiver information to verify the configuration:

```
xcvr show
```

```
xcvr show xcvr <slot>/<xcvr>
```
 - 10 Save the provisioned settings to the configuration file:

```
configuration save
```
 - 11 Clean the ends of the fiber-optic cable, and connect the client ports to the subtending equipment. Verify connectivity.

Note: Use the command `alarm show` to retrieve alarms that indicate connectivity issues and, if required, see [Alarms](#) for information about clearing these alarms.
 - 12 Exit the SSH client, or go to [Confirming traffic](#).
 - 13 If the Waveserver 5 at the other end of the system is not already commissioned, ensure that the following procedures are performed to commission it:
 - [Provisioning Waveserver 5](#)
 - [Provisioning client ports](#)

Procedure 79 Changing the line XCVR mode

Use this procedure to change the line XCVR mode (transmission mode) after a Waveserver 5 has been commissioned.

Requirements

- A Waveserver 5 Base Software License and the appropriate Waveserver 5 Modem Licenses (See [Software licenses - Add-ons](#)).
- The Waveserver 5 is configured in commissioning mode to upgrade and provision the Waveserver 5 without a license for up to 90 days. This 90 days period begins for each license when the Waveserver 5 system time is set. See [Licensing](#).
- You have reviewed and have access to the information you need to complete this procedure.
- The following equipment is available:
 - Fiber-optic cables to connect pluggable transceivers to subtending equipment.



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

- An SSH client is installed on the terminal.

Overview

In Waveserver 5 the service modules 186-3100-900, 186-3101-900, and 186-3102-900 supports only x00G transmission modes. For all other service modules that support both x50G and x00G transmission modes the following rules apply when changing the transmission mode:

- Both line ports can be set to different x00G modes.

- Service modules do not support a combination of x00G and x50G transmission modes. For example, you cannot provision a x00G transmission mode on one line port and a x50 transmission mode on the other line port on the same module.
- Before changing the line port mode from x00G to x50G, the ODU Termination Mode for both line ports must have the same setting (either Terminal or Passthrough), the Port Loopback setting must be disabled, and the module mode must be set to mated.
- If you change one line port to a x50G transmission mode, the other line port is automatically changed to the same mode.
- Configuring a line port to a x50G mode causes an automatic creation of a new line logical port named OTUCn-Flex and deletes line Port 1 and 2. Waveserver 5 designates the OTUCn-Flex port as Port 101 and it represents a single OTUCn service or signal striped across two physical line interfaces.

Steps

- 1 Start the SSH client on the terminal, and log in to Waveserver 5 using an account with at least admin access privileges.
- 2 Use the following command to view the supported line xcvr modes and ptp link optimization modes:

`xcvr show xcvr <slot>/<port>`
- 3 Use the following commands to set the transceiver mode for each modem transceiver:

Note: Disable SPLI on the affected line port, if required, before setting the transceiver mode.

`xcvr set xcvr <slot>/<xcvr 1..2> mode <95-200-O | 95-250-O | 95-300-O | 95-350-O | 95-400-O | 95-450-O | 95-500-O | 95-550-O | 95-600-O | 95-650-O | 95-700-O | 95-800-O | 95-300-E | 95-350-E | 95-400-E | 95-450-E | 95-500-E | 95-550-E | 95-600-E | 95-650-E | 95-700-E | 95-800-E | 93.3-600-E | 93.3-400-E | 91.6-300-E | 91.6-400-E | 91.6-450-E | 91.6-500-E | 91.6-550-E | 91.6-600-E | 91.6-700-E | 91.6-800-E | 90-600-E | 90-400-E | 85-600-E | 85-400-E | 85-300-E | 82-400-E`
- 4 Optionally, retrieve transceiver data:

`xcvr show`
- 5 If the adj-tx provisioning pointing to the Waveserver 5 was removed from the 6500, re-add it:

`adjacencies adjacency set ADJ-<shelf>-<slotNumber>-<port> prov-fe-addr "<WS Name>" prov-addr-form NODENAME-SLT-PRT`

- 6 Save the provisioned settings to the configuration file:
`configuration save`
- 7 Clean the ends of the fiber-optic cable, and connect the client ports to the subtending equipment. Verify connectivity.
Note 1: Use the command `alarm show` to retrieve alarms that indicate connectivity issues and, if required, see [Alarms](#) for information about clearing these alarms.
Note 2: See “*Site planning and preparation*” in *Waveserver 5 Installation*, 323-3001-200 for information about routing and securing fiber-optic cables.
- 8 If the transmission mode on the Waveserver 5 at the other end of the system is not already changed, repeat this procedure and then go to [Confirming traffic](#).

Procedure 80 Confirming traffic

Use this procedure to confirm that traffic is being transported between the local and remote Waveserver 5 units in a system.

Requirements

- The Waveserver 5 at each end of the system is connected to the management network and fully commissioned.
- The Waveserver 5 units in the system are either connected by a commissioned photonic system or connected directly to each other.
- An SSH client is installed on the terminal.

Steps

- 1 Start the SSH client on the terminal, and log in to the Waveserver 5 node using its IP address.
- 2 Clear the port statistics:

```
pm clear port <port>
```

Note: You can clear all port statistics using the command `port clear all`.
- 3 Retrieve and note the port statistics:

```
pm show port <port>
```

Non-zero statistics indicate that traffic is being transmitted and/or received on the port.
- 4 Close the terminal session, and then log in to the Waveserver 5 at the other end of the system using its IP address.
- 5 Repeat [step 2](#) and [step 3](#).

Non-zero statistics indicate that traffic is being transmitted and/or received on the port.

Procedure 81 Configuring power thresholds for third-party pluggables

Use this procedure to configure power thresholds for third-party pluggables.

Requirements

- You must be logged in to the CLI using an account with at least admin access privileges.
- You have installed a third-party pluggable.

Steps

- | If you want to | Then go to |
|---|------------------------|
| create an XCVR profile based on an inserted third-party pluggable | step 2 |
| create an XCVR profile with specific vendor details | step 3 |
| apply XCVR profiles to the matching XCVR pluggables | step 4 |
| set the XCVR profile parameters to default values | step 7 |
| delete an XCVR profile | step 8 |
- Create an XCVR profile based on an inserted third-party pluggable:

```
xcvr profile create profile-id <NUMBER: 1..255> from-xcvr <slot>/<xcvr>
```

Note: This above form of the command is used when a plug must be inserted. All parameter values are used from the inserted plug only and you cannot set any other parameters.

Go to [step 4](#).
- Create an XCVR profile with specific vendor details:

```
xcvr profile create profile-id <NUMBER: 1..255> vendor-revision <String[1..2]> vendor-partnum <String[1..16]> vendor-name <String[1..16]> vendor-oui <Organization Unique Identifier (xx:yy:zz)> [description <String[1..255]>] [rx-power-high-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-low-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-high-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-low-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-high-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-low-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-high-warning-threshold <Power Threshold
```

```
in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-low-warning-
threshold <Power Threshold in dBm (NN.MM): -40.00 ..
8.16>] [tx-optical-power-monitor-override <off | on>]
```

Note: The above form of command is used to create an xcvr profile without inserting the plug. In this case, the vendor-name, vendor-oui, vendor-partnum, and vendor-rev parameters are mandatory.

Go to [step 4](#).

- 4 Apply an XCVR profile with a certain profile ID to all the matching third-party XCVR pluggables:

```
xcvr profile apply profile-id <XCVR Profile Index:
1..255>
```

Optionally, use the “apply all” CLI command to modify the power threshold attributes for all the third party XCVR pluggables that have a matching XCVR profile (if an XCVR profile exists).

```
xcvr profile apply all
```

- 5 Set XCVR profile attributes:

```
xcvr profile set [vendor-name<String[1..16]>] [vendor-
oui <Organization Unique Identifier (xx:yy:zz)>] [vendor-
revision <String[1..2]>] [vendor-partnum
<String[1..16]>] [description <String[1..255]>] [rx-
power-high-alarm-threshold <Power Threshold in dBm
(NN.MM): -40.00 .. 8.16>] [rx-power-low-alarm-threshold
<Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-
power-high-warning-threshold <Power Threshold in dBm
(NN.MM): -40.00 .. 8.16>] [rx-power-low-warning-threshold
<Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-
power-high-alarm-threshold <Power Threshold in dBm
(NN.MM): -40.00 .. 8.16>] [tx-power-low-alarm-threshold
<Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-
power-high-warning-threshold <Power Threshold in dBm
(NN.MM): -40.00 .. 8.16>] [tx-power-low-warning-threshold
<Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-
optical-power-monitor-override <off | on>]
```

To update the power thresholds repeat [step 4](#), then go to the next step.

- 6 Display summary information for all the XCVR profiles or a specific XCVR profile:

```
xcvr profile show [profile-id <XCVR Profile Index:
1..255>]
```

Go to [step 9](#).

- 7 To set the XCVR profile parameters to default values:

```
xcvr profile unset profile-id <Number 1 ..255> [rx-power-
high-alarm-threshold] [rx-power-low-alarm-threshold] [rx-
power-high-warning-threshold] [rx-power-low-warning-
threshold] [tx-power-high-alarm-threshold] [tx-power-
```

```
low-alarm-threshold] [tx-power-high-warning-threshold]  
[tx-power-low-warning-threshold] [tx-optical-power-  
monitor-override]
```

Go to [step 9](#).

- 8** To delete an XCVR profile:

```
xcvr profile delete profile-id <NUMBER: 1..255>
```

Note: You can delete an xcvr profile even when it is applied on a currently provisioned third party plug. However, deleting a profile does not impact the provisioned plug.

- 9** Save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 82 Retrieving Waveserver 5 entity data

Use the commands in this procedure to retrieve summary information for a Waveserver 5 entity.

Requirements

- You must be logged in to the CLI using an account with at least limited access privileges.
- For a client entity, a transceiver must be present at the client interface.

Steps

1	If you want to retrieve	Then go to
	summary information for all modules	step 2
	summary information for a specific module	step 3
	summary information for all transceivers	step 4
	summary information for a specific transceiver	step 5
	summary information for all physical termination points (PTPs)	step 6
	summary information for a specific PTP	step 7
	summary information for all ports	step 8
	summary information for a specific port	step 9
	summary connection information	step 10
	summary connection information for a port and/or slot	step 11
2	To retrieve summary information for all modules: <code>module show</code>	
3	To retrieve summary information for a specific module: <code>module show slot <slotNumber></code>	
4	To retrieve summary information for all transceivers: <code>xcvr show</code>	
5	To retrieve summary information for a specific transceiver: <code>xcvr show xcvr <slot>/<xcvr> [diagnostics] [status] [vendor]</code>	
6	To retrieve summary information for all physical termination points (PTPs): <code>ptp show</code>	
7	To retrieve summary information for a specific PTP: <code>ptp show ptp <slot>/<ptp> [status] [ranges]</code>	
8	To retrieve summary information for all ports: <code>port show [ethernet-config] [status] {summary}</code>	

- 9 To retrieve summary information for a specific port:
`port show port <slot>/<port>`
- 10 To retrieve summary connection information:
`connection show`
- 11 To retrieve summary connection information for the port and/or slot:
`connection show [port <slot>/<port>] [slot <slot>]`

CHAPTER 18

Zero Touch Provisioning

This section contains the following topics:

- [Overview](#)
- [ZTP prerequisites and options](#)
- [Zero Touch Provisioning sequence](#)
- [Additional details for DHCP Option 61](#)
- [Boot file](#)
- [Generic and custom provisioning](#)
- [ZTP status and parameters](#)
- [Procedure list](#)

Overview

Zero Touch Provisioning (ZTP) simplifies the initial commissioning of Waveserver 5 and extends the automatic provisioning capabilities. The ZTP feature allows the Waveserver 5 to retrieve its provisioning from an xFTP server and subsequently provision itself automatically.

Note: Zero Touch Provisioning is intended to be used for greenfield deployments.

ZTP is enabled by default. When you connect Waveserver 5 to the DCN (through the DCN port or through an ILAN connection), the DHCP client running on Waveserver 5 sends a request to the DHCP server. The DHCP server acknowledges the request by returning information about the network (such as IP address, NTP server) and the boot file. The boot file is an XML file containing the names and locations of files needed to commission the Waveserver 5; the license file, the configuration file, and the software load, or a subset of these items depending on how you configure ZTP.

Although ZTP is enabled by default, ZTP does not start automatically unless you specify values for the DHCP options for TFTP Server Name (option 66) and Boot File Name (option 67). See [Supported DHCP options](#). If you do not configure values for these DHCP options, the DHCP client acquires an IP

address and does not attempt to start the ZTP process until values are assigned for DHCP options 66 and 67, and the DHCP lease is renewed, either automatically or manually. You can manually perform a DHCP lease renewal to start the ZTP process.

ZTP processes the boot file, and applies the specified files to the Waveserver 5. If necessary, ZTP upgrades the software load and reboots the Waveserver 5. When ZTP finishes commissioning a Waveserver 5, ZTP disables itself automatically.

ZTP prerequisites and options

To use ZTP, you require:

- A DHCP server that provides DHCP options to support ZTP. You must configure the DHCP options. See [Supported DHCP options](#).
- An FTP-type server that supports one or more of the FTP, TFTP, HTTP, HTTPS, SCP, or SFTP protocols. The DHCP and FTP servers can reside on the same device.
- Detailed network design information for your network.

In addition, you can optionally use these items with ZTP:

- A Waveserver 5 software license file made available on a networked device (for details, see [Licensing](#)).
- A Waveserver 5 software load made available on a networked device.
- A Waveserver 5 configuration file made available on a networked device.

Zero Touch Provisioning sequence

The following sequence of events illustrates the ZTP sequence:

- The Waveserver 5 is connected to the DCN.
- The Waveserver 5 DHCP client sends "DHCP discover" messages at approximately 30-second intervals onto the DCN.
- One or more DHCP servers reply to the discover messages by sending a "DHCP offer" message to the requesting DHCP client.
 - The DHCP client accepts an offer from one DHCP server, establishing a session between the DHCP client and server. The discover message (and subsequent messages) contain the Waveserver 5 serial number as a transaction ID to identify the DHCP client. Once the DHCP client accepts an offer message, it stops sending discovery messages.
- The DHCP client sends a "DHCP request" message to the DHCP server, and waits for a "DHCP acknowledge" ("ACK") message from the server.

- The DHCP server responds to the request by sending an acknowledgment ("ACK") message to the DHCP client, the name of the boot file and details of the server where the boot file is located. Then, the DHCP client will publish the boot file name and server details to the ZTP service.
 - The "ACK" message contains configuration information, called options. The DHCP client processes the contents of the "ACK" message. For details, see [Supported DHCP options](#).
- Note:** The boot file is an XML file that contains the names and locations of files required by ZTP to provision and commission Waveserver 5. For details, see [Boot file](#).
- ZTP service downloads the boot file from the server, which is provided by the DHCP client, and processes the contents of the boot file.
 - If there is a problem downloading or parsing the boot file, the DHCP client renews the lease; that is, it requests a new "ACK" message from the DHCP server.

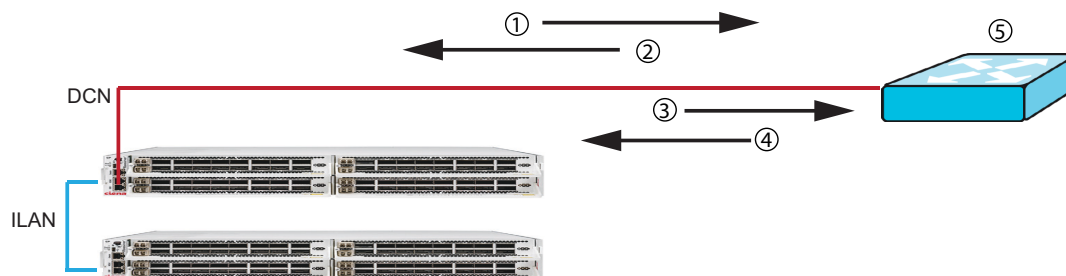
The boot file specified in DHCP option 67 is the boot file that is downloaded. If DHCP option 67 specifies a directory the following sequence of tasks occurs:

- The file with the name that matches the Waveserver 5 serial number (for example, M1234567.xml) is downloaded. If this fails then;
- The file with the name that matches the chassis/DCN MAC address (for example, 0102030a0b0c.xml) is downloaded. If this fails then;
- The file with the name that matches the platform name (for example, Waveserver5.xml) is downloaded.
- If this also fails, download the file with the name that matches platform name Waveserver5.xml.

Supported DHCP options

The following figure shows the ZTP starting sequence, which occurs when you connect the Waveserver 5 to the DCN.

Figure 92 ZTP sequence



Legend

- | | | | |
|---|---------------|---|------------------|
| 1 | DHCP discover | 4 | DHCP acknowledge |
| 2 | DHCP offer | 5 | DHCP Server |
| 3 | DHCP request | | |

The table below lists the supported DHCP options for ZTP.

Table 174 Supported DHCP options

DHCP option	Description
IP Address Subnet Mask (option 1) Router (Gateway) (option 3)	These options configure the interface used by the DHCP client, to allow the system to communicate with services on the network, such as TFTP.
Vendor Class (option 60) Client Identifier (option 61) TFTP Server Name (option 66) Boot File Name (option 67)	These options specify the TFTP server and the boot file to download from it. The Vendor Class (option 60) contains the unique identifier 1271-00013000-001. The Client Identifier (option 61) contains the Waveserver 5 serial number DHCP uses to identify the device. (The MAC address is also sent by default). If the ACK message does not specify the TFTP Server Name (option 66) or Boot File Name (option 67), the DHCP client acquires an IP address and stops ZTP processing until the DHCP lease is renewed. When the lease is renewed, the DHCP server sends another ACK message, which may contain these options. This approach is backward-compatible with current DHCP behavior in case you want to use DHCP to acquire IP addresses, but do not want to use ZTP. If the ACK message contains both the TFTP Server Name and Boot File Name options, the DHCP client downloads the boot file from the TFTP server.
Time offset (option 2) Domain Name server (option 6) Log server (option 7) Host name (option 12) Domain name (option 15) Network Time Protocol Servers (option 42)	Other supported options which can be retrieved from the DHCP server.

Additional details for DHCP Option 61

The DHCP client ID option 61 includes the Waveserver 5 serial number. To avoid an IP address change during an upgrade, Ciena recommends adding the following host static address allocation entry to the DHCP server:

Figure 93 Host static address allocation

```

host ws10 {
    hardware ethernet aa:bb:cc:dd:ee:ff;          /* <- Specify the Local Mgmt MAC address
    fixed-address      11.22.33.44;                /* <- Assign the desired fixed IP address
}

```

You can retrieve the Local Mgmt MAC address using the `chassis show mac` command, as shown below.

Figure 94 Local Mgmt MAC address display

```

ws# chassis show mac

+----- CHASSIS MAC PARTITIONS -----+
| Partition          | Base MAC          | Block Size |
+-----+-----+-----+
| Chassis            | 22:88:55:bb:a6:10 | 16         |
+-----+-----+-----+
| Local Mgmt         | 22:88:55:bb:a6:10 | 16         |
| Reserved           | 22:88:55:bb:a6:11 | 15         |
+-----+-----+-----+

```

Boot file

The boot file is an XML file containing the names and locations of files you require to provision and commission Waveserver 5. The boot file contains attributes that point to these files:

- Configuration file (configFilePath)
- License file (licenseFile)
- Software load (packagePath)
- Script file (scriptFile)
- Login banner (loginBanner)
- Welcome banner (welcomeBanner)

The following figure shows the contents of a sample boot file.

Figure 95 Sample boot file contents

```

<XmlCienaCommandFile>
  <XmlCmdPlatformClass name="waveserver5"
    registrationId="Waveserver_R2.0.2"
    licenseFile="sftp://user:password@10.1.2.3/
ciena-WaveserverAi_R1-3-1234567890.lic"
    welcomeBanner="ftp://user:password@10.1.2.3/
welcomeBanner.txt"
    loginBanner="scp://user:password@10.1.2.3/
loginBanner.txt"
    scriptFile="http://10.1.2.3/myScriptFile"
    configFilePath="https://user:password@10.1.2.3/
myConfigFile"
    configFileRule="activate"
    ztpTimeout=45
    packagePath="http://10.1.2.3"
    operation="upgrade"
    version="waveserver-2.0.2.66">
  </XmlCmdPlatformClass>
</XmlCienaCommandFile>

```

Boot file attributes

When you assign a path to a boot file attribute, the path is always relative to the xFTP server root directory:

- tftpboot directory for TFTP
- home directory for FTP/SFTP/SCP
- document root directory for HTTP/HTTPs web server

You can assign a URL to a boot file attribute, using the formats shown in the following table (user ID and password are optional).

Table 175 URL types to assign to boot file attributes

URL type (Note)	Example
With uid and password	option bootfile-name "http://user:password@machineID.company.com/test.xml";
Without uid and password	option bootfile-name "https://machineID.company.com/test.xml";
IP address instead of hostname	option bootfile-name "http://<ip_address>/test.xml";
Note: To use a hostname instead of an IP address in the URL, you must have already configured the DNS either manually or using DHCP option 6.	

You can also specify multiple <XmlCmdPlatformClass> attributes in the same boot file for different platforms, as shown in the following figure.

Figure 96 Multiple <XmlCmdPlatformClass> attributes in boot file

```

<XmlCienaCommandFile>
  <XmlCmdPlatformClass name="waveserver"
    ...
  </XmlCmdPlatformClass>
  <XmlCmdPlatformClass name="waveserver5"
    ...
  </XmlCmdPlatformClass>
</XmlCienaCommandFile>

```

Boot file processing

The boot file can list the attribute name-value pairs in any order. ZTP always processes the attributes in this order:

- 1 ztpTimeout
- 2 registrationId
- 3 licenseFile
- 4 loginBanner
- 5 welcomeBanner
- 6 configFilePath and configFileRule
- 7 scriptFile
- 8 packagePath, operation, and version

ZTP Timeout attribute

By default, if ZTP fails to download or process a file, ZTP fails and then waits for a forced DHCP lease renewal (30 seconds) before a retry. You can modify this timeout value by configuring the optional ztpTimeout attribute in the boot file. The valid ZTP Timeout range is 30 to 3600 seconds. The ZTP timeout value is persisted, even after completing the ZTP process.

License file and Registration ID attributes

The licenseFile attribute specifies the path and name of the Waveserver 5 software license file.

Note: To use the Waveserver 5 in commissioning mode instead of applying a license during ZTP, remove the licenseFile attribute from the boot file.

To obtain a **Waveserver 5** software license, you must register the **Waveserver 5** with the licensing portal using the device Registration ID. (See [Licensing](#) for details.) License file names are determined by the Waveserver 5 device type and the device Registration ID.

The Registration ID is a string that you create and set on the Waveserver 5. For details, see [Setting the Registration ID on Waveserver 5](#).

The license filename format is: `ciena-COMPANY-X_001-201624111150.lic`, where `COMPANY-X_001` is the Registration ID.

Table 176 License file attribute

Attribute	Description
licenseFile	Specifies the location and name of the license file. You must also specify the registrationId attribute. You can also use a URL as the licenseFile attribute, and the ZTP feature will use the specified protocol and server to download the file.
registrationId	Specifies the license passphrase (that is, Registration Id). Note: If you are activating the configuration file using the Configuration file attributes then you must set the Registration ID in that configuration file too. While activating the configuration file it causes the license-manager to reset its configuration parameters and also sets the registration Id. If the registration Id is not specified in the configuration file it sets none as the registration Id.

You can set Waveserver 5 to use an external license server for ZTP by removing licensing names and values from the boot file and adding the CLI commands for setting up the external license server to the configuration file. See, [Provisioning access to an external license server](#) for more information.

Login banner and welcome banner attributes

The loginBanner and welcomeBanner attributes specify the paths and names of the login and welcome banner files, respectively.

If you specify these attributes and also apply a configuration file (using the configFilePath attribute) with operation="activate", you must also assign values to the login-banner-file and welcome-banner-file properties in the configuration file using the CLI commands:

```
system shell set login-banner-file <filename>
```

```
system shell set welcome-banner-file <filename>
```

If you apply a configuration file with operation="augmentAndSave", you do not need to specify the login-banner-file and welcome-banner-file properties; the login and welcome banner settings are automatically saved.

If you assign a URL to loginBanner or welcomeBanner, Waveserver 5 uses the specified protocol to download the banner file. Otherwise, the banner file is downloaded from the TFTP server (DHCP option 66).

Table 177 Login banner and welcome banner attribute

Attribute	Description
loginBanner	Specifies the location and name of the login banner file.
welcomeBanner	Specifies the location and name of the welcome banner file.

scriptFile attribute

The scriptFile attribute specifies a script file (bash script or python script) to run. You can also use a URL for the attribute, and Waveserver 5 will use the specified protocol and server to download the script file.

In this release, script based CLI session is not supported. If the user is using an old script that uses such commands, then the script will fail. For example, a python script will fail that uses a command like `os.system("woscli <<< \"system set host-name WSOS_0111_Test\"")`. You can either use a REST interface in your script or use the configuration file attribute to achieve the same.

Configuration file attributes

The configFilePath attribute specifies the path and name of the configuration file. The configuration file is optional. If you do not want to apply a Waveserver 5 configuration file, remove the configFilePath attribute from the boot file.

Table 178 Configuration file attribute

Attribute	Description
configFilePath	Specifies the location and name of the configuration file. When you include the configFilePath attribute in the boot file, the specified configuration file is downloaded to <code>/var/opt/config</code> and applied based on the criteria specified in the configFileRule. Note: Ensure that the Registration ID is also specified in the configuration file, which is same as specified in the License file attribute because it causes the license-manager to reset the configuration parameters and also sets the registration Id. If the registration Id is not specified in the configuration file it sets none as the registration Id.
configFileRule	An optional attribute that specifies how to apply the configuration. It has two possible values (Note 1, Note 2, Note 3): activate — Default value. The configFilePath replaces the existing configuration and it does not require a reboot to load the new configuration. Select the “activate” value for new Waveserver 5 installations or when performing a return to factory defaults operation. augmentAndSave — The configFilePath adds to the existing configuration and saves the result. This does not cause a reboot. Select the “augmentAndSave” value for in-service Waveserver 5 chassis, for example when you are re-activating ZTP.

Attribute	Description
Note 1: When configFileRule = " activate", the configuration file specified by the configFilePath attribute in the boot file is applied. The loginBanner and welcomeBanner attributes of that configuration file must point to the downloaded welcome and login banner files you want applied. Otherwise, the downloaded banner files are not applied. When configFileRule = " augmentAndSave", the banner files are applied without having to specify them in the configuration file pointed to by configFilePath. Note 2: ZTP tries to apply the boot file until it succeeds or you intervene. Any errors in the configuration file or in the ZTP process (such as installing a license file that does not exist), stops and then restarts ZTP after a defined interval. You must ensure that ZTP completes successfully, or that you resolve any issues preventing it from completing (for example, a missing or incorrect boot file or license files). Note 3: Use caution when you include statements in the configuration file that disable ZTP or DHCP. If the ZTP process fails, the commands may still be applied which can affect or stop the ZTP process. Only use these commands when you are certain the boot file has no errors and will complete successfully the first time you run it.	

Software package load file attributes

The packagePath attribute specifies the path of the Waveserver 5 software load file. If you assign a URL to the packagePath attribute, Waveserver 5 uses the specified protocol to download the software load file. Otherwise, the software load file is downloaded from the TFTP server (DHCP option 66).

Note: The software load file name is specified by the version attribute.

The optional "operation" attribute currently supports only one operation: "upgrade". If you assign any other value to the operation attribute, it triggers "Error Parsing Boot File" error.

The "version" attribute specifies the release to upgrade to. A Waveserver 5 software load consists of several RPM files, packaged in a tar.gz file. ZTP compares the versions of the currently installed RPMs to the value assigned to the "version" attribute, and determines whether to upgrade the software load. If an upgrade is required, ZTP downloads the new software load file and upgrades the node. The upgrade process initiates a reboot to complete the upgrade.

Table 179 Software load attribute

Attribute	Description
packagePath	Specifies the path where the file containing the software load is located.
operation	This optional attribute specifies the action to take on the software load. Currently supports only one operation: " upgrade". Any other value triggers the " Error Parsing Boot File" error.

Attribute	Description
version=<version-string>	Specifies the Waveserver 5 software version you want to install and run. The value of <version_string> is based on the name of the software load file. For example, if the file name is <code>waveserver-rel-2.1.0.15.tar.gz</code>, then assign <version-string> the value: <code>waveserver-rel-2.1.0.15.tar.gz</code> or <code>waveserver-2.1.0.15</code>. The value assigned to version is compared to the version of the currently installed software load. If the versions match, no further action is taken. If they do not match, the system upgrades the software load.

Generic and custom provisioning

ZTP supports two provisioning methods:

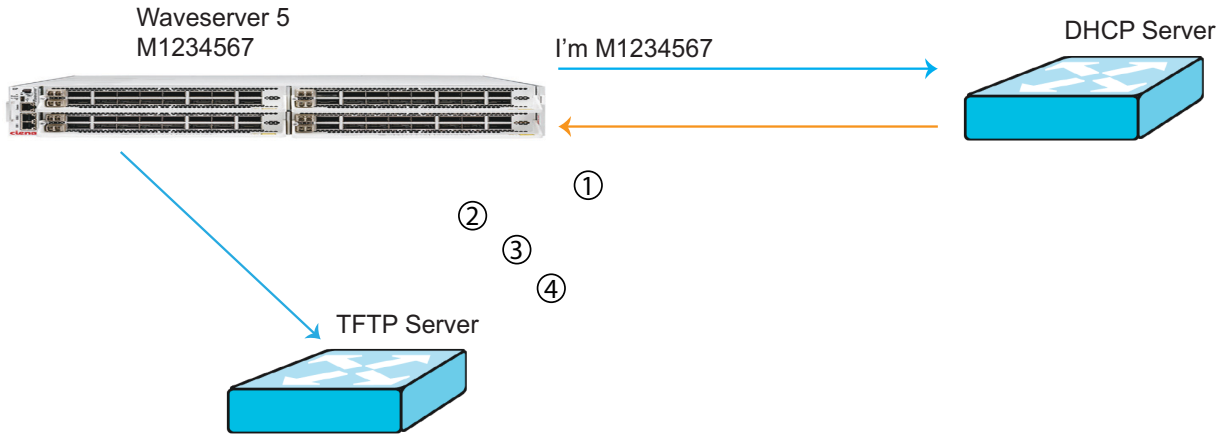
- [Zero-touch \(generic provisioning\)](#)
- [Low-touch \(custom provisioning\)](#)

For greater flexibility, you can configure the DHCP server to use a combination of both approaches.

Zero-touch (generic provisioning)

Generic provisioning ensures that provisioning is the same on each Waveserver 5. For generic provisioning, you configure the DHCP server to return the same boot file to every provisioning request from a Waveserver 5 DHCP client. When you specify a specific filename in DHCP option 67, Waveserver 5 downloads the file. However, if you specify a directory instead of a filename, Waveserver 5 attempts to download the file with a filename containing the serial number. If this fails, Waveserver 5 then attempts to download the file with a filename containing the MAC address. If this also fails, Waveserver 5 downloads the file with a filename that contains the platform name (`Waveserver-5.xml`). If this also fails, download the file with the name that matches platform name `Waveserver5.xml`.

Figure 97 Zero-touch (generic provisioning)



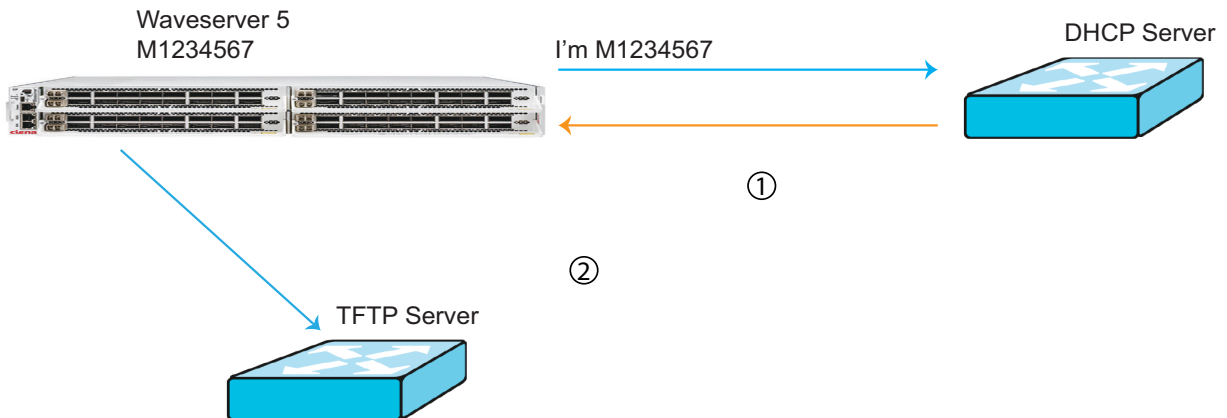
Legend

- | | | | |
|---|---|---|---|
| 1 | All nodes get your file from directory/waveserverfiles at the TFTP server | 3 | If not, give me file with filename = Local Management MAC address |
| 2 | Give me file with filename = serial number | 4 | If not, give me file with filename = platform name Waveserver-5.xml TFTP Server |

Low-touch (custom provisioning)

For custom provisioning, configure the DHCP server to return a different boot file to provisioning requests from DHCP clients, based on the unique Waveserver 5 serial number contained in the "DHCP request" message. The following graphic describes the custom-provisioning process.

Figure 98 Low-touch (custom provisioning)



Legend

- | | | | |
|---|---|---|---|
| 1 | Go get file M1234567 from TFTP server under directory / waveserverfiles | 2 | Give me file /waveserverfiles/ M1234567 |
|---|---|---|---|

ZTP status and parameters

The ZTP status and parameters can be displayed using the CLI command `configuration ztp show`. See the following table:

Table 180 ZTP status and parameters

Parameter	Description
Admin state	Shows whether ZTP is enabled or disabled.

Parameter	Description
Oper State	Displays status of the current ZTP step. After ZTP commissions the node, the status is Completed. Any other status can indicate that commissioning is still in progress or an error occurred. The value of Last ZTP Failure indicates whether a previous failure caused ZTP to restart. Possible values of Oper State are: • Idle — Not running • Started — ZTP received a DHCP request. • Restarting — ZTP failed and is starting over. • Restarting (waiting for DHCP lease renew) — ZTP failed, has restarted, and is waiting for a DHCP lease renewal to complete. • Downloading Boot File — ZTP is downloading the boot file. • Parsing Boot File — ZTP downloaded the boot file, and is parsing the XML content. • Processing License Registration ID — Setting the license Registration ID. This is done when a Full-rate (400G) license is applied so the license file is against a string (Registration ID) and not the Waveserver 5 serial number. • Downloading Licenses — Downloading the license file. • Installing Licenses — Installing the license file. • Downloading Script File — Downloading the script file. • Applying Script File — Executing the script file. • Processing ZTP Timeout — Processing ZTP timeout attribute in the boot file. • Upgrading Software — Performing the software upgrade. • Failed (ZTP or DHCP is disabled or SW rollback) — ZTP has failed and cannot proceed further without a manual intervention because there was a software rollback or ZTP and/or DHCP got disabled. • Downloading Config File — Downloading the configuration file. • Applying Config File — Applying the configuration file rule. Currently, two rules are supported: activate and augmentAndSave. • Downloading Login Banner — Downloading and installing the login banner. • Downloading Welcome Banner — Downloading and installing the welcome banner to print when the Waveserver 5 boots. • Processing Software Package — Downloading/processing the software load. • Requesting Reboot — Requesting a system reboot to finish the ZTP process. • Completed — ZTP successfully commissioned the Waveserver 5 and is disabled.

Parameter	Description
Last ZTP Failure	Displays the cause of the last failure ZTP encountered. When ZTP encounters a failure, it renews the lease and restarts. Either fix the problem in the configuration file or disable ZTP. Because ZTP continues to retry, the current state could be running a normal step. This field indicates the previous failure. Possible values are: • Blank — If this field is blank, no error was detected. • Boot File Download Failed — Unable to download the boot file. Ensure the name and path to the boot file on the specified server matches the information specified by the DHCP server. • Error Parsing Boot File — The boot file is formatted incorrectly or has invalid attribute values. Correct the XML syntax of the boot file. ZTP periodically retries to download the boot file and process it. • Error Setting License Registration ID — Cannot set the license Registration ID. The ID string may be too long (more than 64 characters) or contain unsupported characters. • Error Downloading Licenses — Failed to download the license file. Make sure the license file is present with the correct name on the specified server. • Error Installing Licenses — Failed to apply the license file. The file is corrupt or invalid. • Error Installing Licenses: Host Id mismatch — Failed to apply the license file. The license file Registration ID does not match the Registration ID in the boot file. • Error Downloading/Installing Login Banner — Failed to download or apply the login banner. Ensure the name and location of the login banner file in the boot file is correct. • Error Downloading/Installing Welcome Banner — Failed to download or apply the welcome banner. Ensure the name and location of the welcome banner file in the boot file is correct. • Config File Download Failed — Failed to download the configuration file. Ensure the name and location of the configuration file in the boot file is correct. • Software Package Upgrade Failed — Cannot upgrade the software load package, which indicates the software load is corrupt. Contact Ciena support. • Software Download Failed — Cannot download the software load. Ensure the name and location of the software load in the boot file is correct. • No SW upgrade performed. Correct load is installed — The specified software load was already installed, so no software load upgrade was performed. Not an error; informational only. • Error Processing ZTP Timeout — There was an error processing ZTP timeout attribute in the boot file (for example, the timeout value was out of range). • Script File Download Failed — Failed to download the script file. Ensure the name and location of the script file in the boot file is correct. • Executing Script File Failed — Failed to execute the script file specified in the boot file. • Error Applying Config File — Failed to apply the configuration file specified in the boot file.

Parameter	Description
Last ZTP Command File	The name of the most recently processed command file (boot file).
Last ZTP Config File	The name of the most recently applied configuration file.
ZTP Timeout (seconds)	ZTP timeout value received in the boot file. The default is 30 seconds.
TFTP Server	TFTP server received in DHCP Option 66.

Procedure list

This section provides the following procedure:

- [Procedure 83, “Configuring a Waveserver 5 using Zero Touch Provisioning”](#)

Procedure 83 Configuring a Waveserver 5 using Zero Touch Provisioning

Use this procedure to configure a Waveserver 5 using Zero Touch Provisioning.

Requirements

- Waveserver 5 is connected to the DCN (through the DCN port or through an ILAN connection).
- If the Waveserver 5 has been previously logged into, verify that ZTP is enabled using the CLI command `configuration ztp show`. To enable ZTP, use the CLI command `configuration ztp enable`.
- The following information is available:
 - DHCP server and configuration attributes
 - boot file and configuration file information

Overview

Note 1: ZTP is enabled by default on the Waveserver 5.

Note 2: ZTP is intended to be used for greenfield deployments.

Steps

- 1 Download and install a DHCP server on your PC.
- 2 Configure the DHCP server and update the appropriate configurable attributes, for example:
 - Listening Port = 192.168.0.1
 - IP Range = 192.168.0.2 to 192.168.0.4
 - Global Option
 - NTP server = 192.168.0.1
 - TFTP server name = 192.168.0.1
 - Bootfile Option = waveserver/bootfile/

DHCP server configuration example:

 - TFTPServerName="192.168.0.1"
 - BootFileOption=waveserver/bootfile/

For more information, see [Supported DHCP options](#).
- 3 Restart the DHCP server to apply the changes.
- 4 Download and install a TFTP server.
- 5 Create a `waveserver/bootfile` folder under the root directory of TFTP server.
- 6 Create a `waveserver/configs` folder under the root directory of TFTP server.

7 Create a configuration file, for example "WS_Aug29.txt" and place the file in the `waveserver/configs` folder under the TFTP server's root directory.

8 Create a boot file and place it in the `waveserver/bootfile` folder under the TFTP server's root directory. For more information, see [Boot file](#).

Example:

```
<XmlCienaCommandFile>
<XmlCmdPlatformClass name="waveserver5"
registrationId="MTL-LAB-1"
configFilePath="waveserver/configs/WS_Aug29.txt"
configFileRule="activate">

</XmlCmdPlatformClass>
</XmlCienaCommandFile>
```

9 Connect the Waveserver 5 to the DHCP server (running on your PC). Wait a few minutes.

10 Attempt to log in to the Waveserver 5 remotely.

11	If you are	Then
	able to log in to the Waveserver 5 remotely	go to step 16
	unable to log in to the Waveserver 5 remotely	go to step 12

12 Connect a console port to the Waveserver 5. See [Setting up management access using the console or USB port](#).

13 Confirm the ZTP feature is enabled using the CLI command `configuration ztp show`.

Figure 99 Zero touch provisioning state summary

```
> configuration ztp show
+----- ZERO-TOUCH PROVISIONING STATE -----+
| Parameter                               | Value                               |
+-----+-----+-----+-----+-----+
| Admin State                             | Enabled                             |
| Oper State                              | Idle                                |
| Last ZTP Failure                         |                                     |
| ZTP Timeout (seconds)                   | 30                                  |
+-----+-----+-----+-----+-----+
| Last ZTP Command File                   |                                     |
| Last ZTP Config File                    |                                     |
| Tftp Server                             |                                     |
+-----+-----+-----+-----+-----+
```

14	If the ZTP feature is	Then
	disabled	go to step 15
	enabled	go to step 18

- 15 Enable the ZTP feature:
`configuration ztp enable`
- 16 Using CLI command `configuration ztp show`, verify the following on the Waveserver 5:
 - The Waveserver 5 Oper State is “Completed”, indicating that the configuration was successfully completed using the ZTP feature with the configuration file created in [step 7](#).
 - The ZTP Admin State is “Disabled”, indicating that ZTP was automatically disabled after the successful configuration using ZTP.
- 17

If the Waveserver 5 provisioning was	Then
not completed via ZTP using the configuration file	go to step 18
completed via ZTP using the configuration file	go to step 19
- 18 Contact your next level of support or your Ciena support group.
- 19

If the Waveserver 5 ZTP feature has	Then
been automatically disabled	the procedure is complete
not been automatically disabled	step 20
- 20 Disable the ZTP feature on the Waveserver 5 using the CLI command
`configuration ztp disable`

CHAPTER 19

Signal conditioning

This section contains the following topics:

- [Overview](#)
- [Procedure list](#)

Overview

Waveserver 5 supports link state forwarding and signal conditioning.

The `Link State Forwarding` attribute is applied at the service level and specifies whether fault messages are exchanged between the Waveserver 5 chassis in a system in the event of a fault at one end. Link State Forwarding is used as a mechanism to propagate port link states from the upstream chassis to the downstream chassis.

The `Conditioning Type` attribute applies only to client ports. It determines the conditioning signal that is transmitted towards the subtending Ethernet equipment, in the event that there is a line fault or far end client fault.

For client port conditioning, Waveserver 5 supports user provisionable Link State Holdoff for Ethernet clients, and it is applied in the event of a line fault:

- from 0 ms to 100 ms in 10 ms increments
- from 100 ms to 1000 ms in 100 ms increments

You can provision the conditioning type globally for each chassis as `laser-off`, `none`, or `protocol-specific`.

Procedure list

This section provides the following procedures:

- [Procedure 84, “Configuring signal conditioning”](#)

Procedure 84 Configuring signal conditioning

Use this procedure to configure signal conditioning.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Note: Once the default values are set, the new ports are created with the system default values.

Steps

- Use any of the following CLI commands to configure signal conditioning:
 - To set the port conditioning type for one or more ports:


```
port set port <portList> conditioning-type <ethernet | none | laser-off | otn>
```

Note: The default conditioning type is `ethernet`.
 - To set the port conditioning for all ports in the chassis:


```
system defaults set conditioning-type <laser-off | none | protocol-specific>
```
 - To set the port conditioning-holdoff timer for one or more ports:


```
port set port <portList> conditioning-holdoff <0 to 1000 in increments of 100>
```
 - To set the port conditioning-holdoff timer for all existing ports in the chassis:


```
system defaults set conditioning-holdoff <0 to 100ms in 10ms increments; 200ms to 1000ms in 100ms increments>
```

Note: The default conditioning-holdoff is 10 ms.
 - To restore conditioning type to the factory default setting for all ports in the chassis:


```
system defaults unset conditioning-type
```
 - To restore conditioning-holdoff timer to the factory default setting for all ports in the chassis:


```
system defaults unset conditioning-holdoff
```
 - To display system defaults for conditioning type and conditioning-holdoff:


```
system defaults show
```

CHAPTER 20

Test tools

This section contains the following topics:

- [Overview](#)
 - [Loopback testing](#)
 - [Integrated test set](#)
 - [Dispersion scan](#)
 - [Link data collector](#)
- [Procedure list](#)

Overview

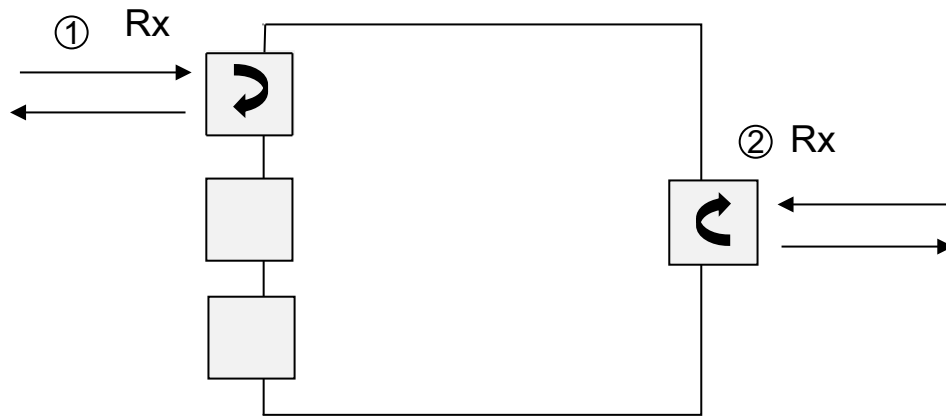
This section provides information on tools that can be used on Waveserver 5.

Loopback testing

Waveserver 5 supports loopback testing. Loopback tests (Rx (facility) and Tx (terminal)) enable you to test the performance of ports or identify connectivity issues between two chassis. Client and line ports support Rx and Tx loopback testing.

Note: Loopbacks must be disabled prior to upgrades.

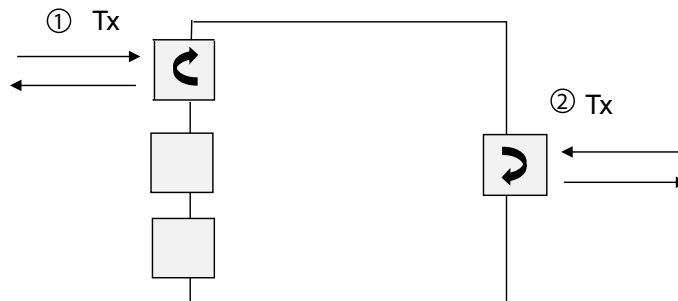
- The figure below shows a Rx loopback test, where it forwards the ingress traffic from the Rx port directly to the Tx port (toward the fiber).

Figure 100 Rx loopback - client or line port**Legend**

1 Client RX

2 Line RX

- The figure below shows a Tx loopback test, where it forwards egress traffic from the Tx port directly to the Rx port (back toward the Waveserver 5).

Figure 101 Tx loopback - client or line port**Legend**

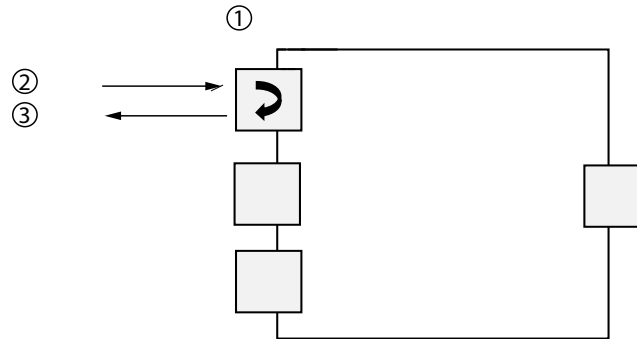
1 Client TX

2 Line TX

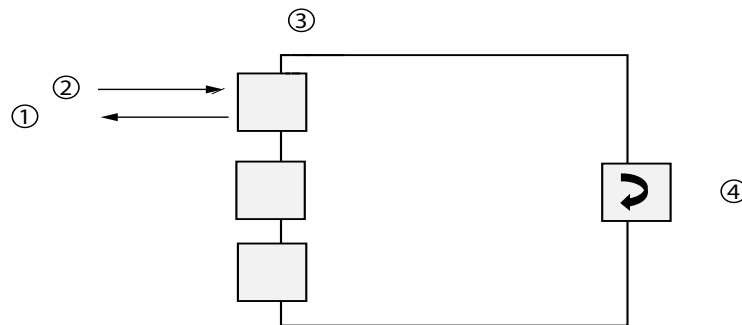
Note 1: You cannot perform loopback tests on client and line ports at the same time, however, all loopback tests are supported individually.

Note 2: If a remote management interface is enabled on a line port, Tx and Rx loopbacks are blocked.

- You can test the port performance on an individual Waveserver 5 by operating a Rx loopback test on the client port or by operating a Tx loopback test on the line port. See the following figures:

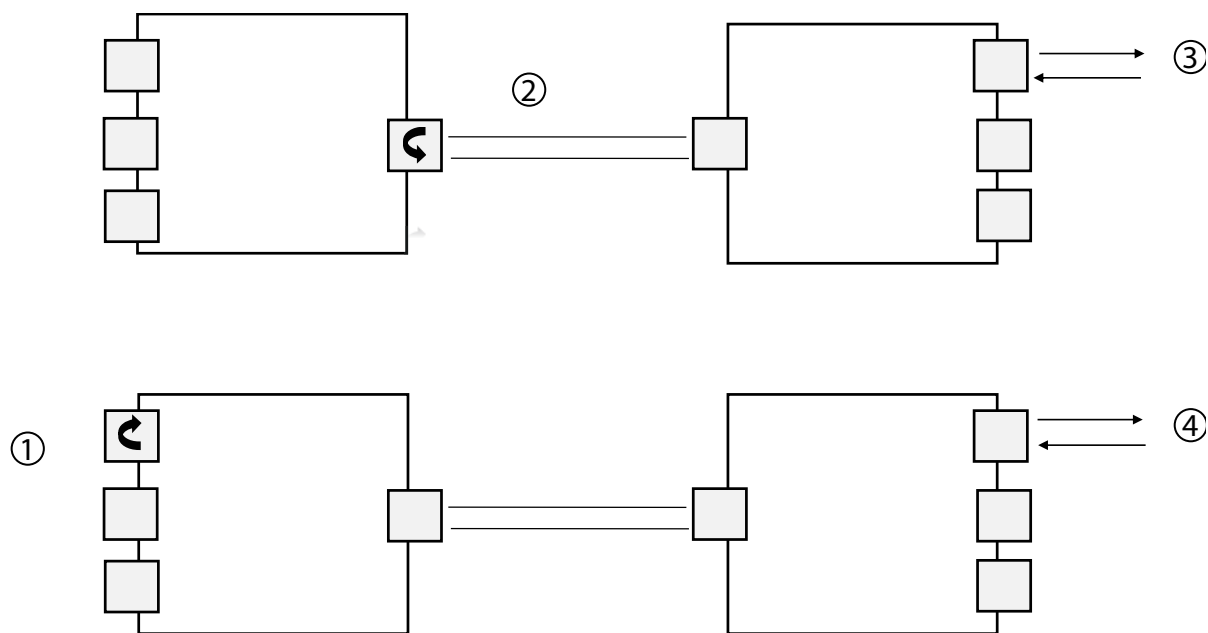
Figure 102 Client Rx Loopback**Legend**

- | | | | |
|---|----------------------------------|---|----------------------------------|
| 1 | Client Rx Loopback | 3 | Test set or Subtending Equipment |
| 2 | Test set or Subtending Equipment | | |

Figure 103 Line Tx loopback**Legend**

- | | | | |
|---|----------------------------------|---|------------------|
| 1 | Test set or Subtending Equipment | 3 | Client |
| 2 | Test set or Subtending Equipment | 4 | Line Tx Loopback |

- You can identify connectivity issues between two Waveserver 5s by operating Rx loopbacks on the line port or Tx loopbacks on the client port. See the following figures:

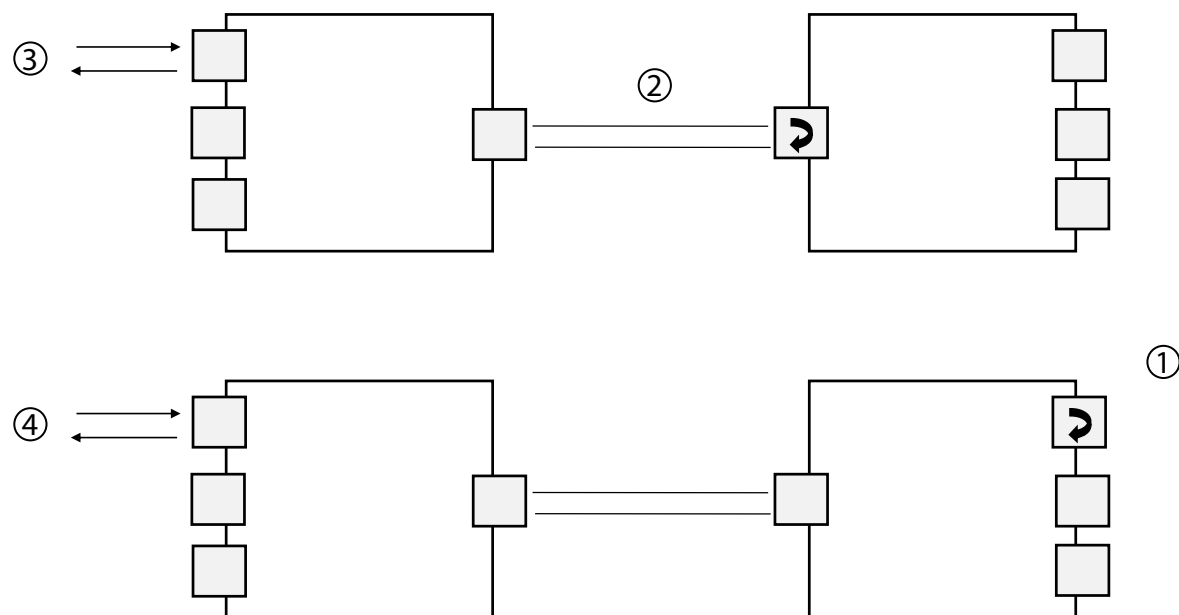
Figure 104 Example 1 - Line Rx Loopback and Client Tx Loopback between Waveserver 5s**Legend**

1 Client Tx Loopback

2 Line Rx Loopback

3 Test set or Subtending Equipment

4 Test set or Subtending Equipment

Figure 105 Example 2 - Line Rx Loopback and Client Tx Loopback between Waveserver 5s

Legend

1	Client Tx Loopback	3	Test set or Subtending Equipment
2	Line Rx Loopback	4	Test set or Subtending Equipment

Integrated test set

The Integrated Test Set (ITS) feature alleviates the need for expensive test sets to validate 100GE, OTL4.4 (100G), and 400GE circuits in your networks. Waveserver 5 supports testing on Ethernet client ports.

The ITS can generate traffic patterns and monitor them, allowing you to test traffic continuity between various traffic path endpoints. The port on which you use the ITS must be disabled and must remain out of service while the test is running.

Waveserver 5 supports the ITS in race track mode only. When testing in a race track configuration, you must enable the Integrated Test Set generators before completing the race track circuit by applying Tx (terminal) or fiber loopbacks to avoid the creation of timing loops. Rx (facility) and Tx (terminal) loopbacks can be provisioned while the ITS is provisioned.

The test pattern for 100GE consists of scrambled idles as defined in IEEE 802.3ba, clauses 82.2.1 and 82.2.10.

Note 1: If a remote management interface is enabled on a line port, a Tx loopback is blocked.

Note 2: ITS must be disabled prior to upgrades.

Note 3: ITS does not survive module restarts (warm or cold).

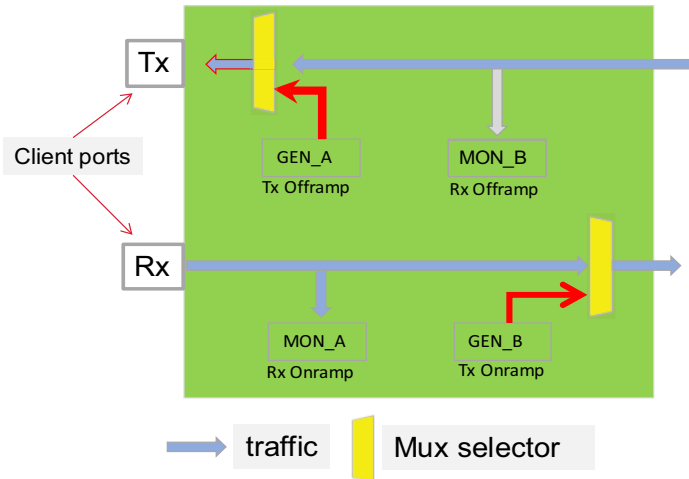
Table 181 Supported ITS application modes and provisionable parameters

Client XCVR Mode	Mode	Generation Tx	Monitor RX	BER threshold	Duration
100GE	Race Track	100GE 802.3ba Ethernet Test Pattern	100GE 802.3ba Ethernet Test Pattern	0 (default) 0, 1E-16 to 1E-5	Continuous (default) dd-hh-mm-ss
400GE	Race Track	4000GE 802.3ba Ethernet Test Pattern	400GE 802.3ba Ethernet Test Pattern	0 (default) 0, 1E-16 to 1E-5	Continuous (default) dd-hh-mm-ss
100G (OTL4.4)	Race Track	OPU4 PRBS31	OPU4 PRBS31	0 (default) 0, 1E-16 to 1E-5	Continuous (default) dd-hh-mm-ss

See the following graphic that illustrates the ITS reference model:

Figure 106 Integrated Test Set reference model

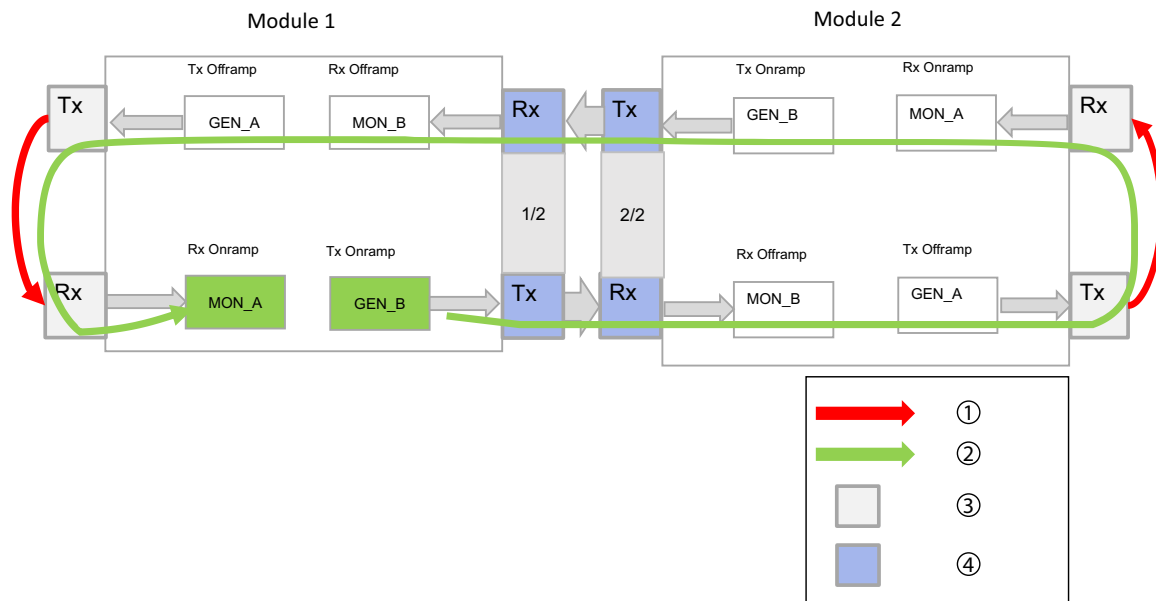
- GEN_A: Test Signal is generated towards the client (off-ramp direction)
- MON_A: Test Signal is monitored from the client (on-ramp direction)
- MON_B: Test Signal is monitored from the line (off-ramp direction).
- GEN_B: Test Signal is generated towards the line (on-ramp direction)



Note: Waveserver 5 supports race track mode, in which only Gen_B and MON_A points apply.

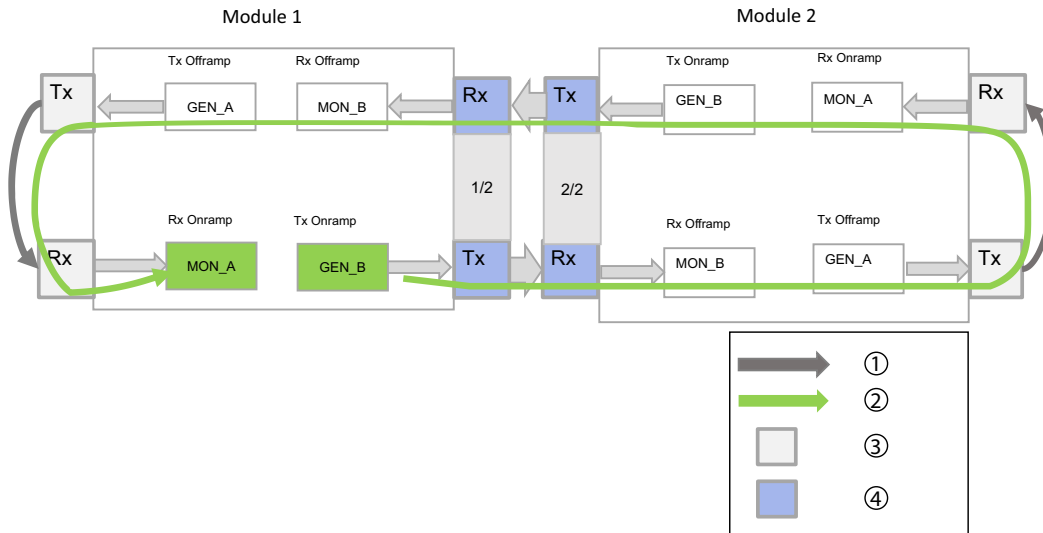
Waveserver 5 supports three ITS race track mode configurations:

- Race track mode with fiber loopbacks at both ends: In this mode, you must install fiber loopbacks on both the near-end and far-end client ports and start the test in race track mode. This mode tests the entire traffic path, including ingress and egress optical transceivers. For more information, see [Performing a test with the Integrated Test Set in race track mode - fiber loopbacks at both ends](#).

Figure 107 Race track mode with fiber loopbacks at both ends**Legend**

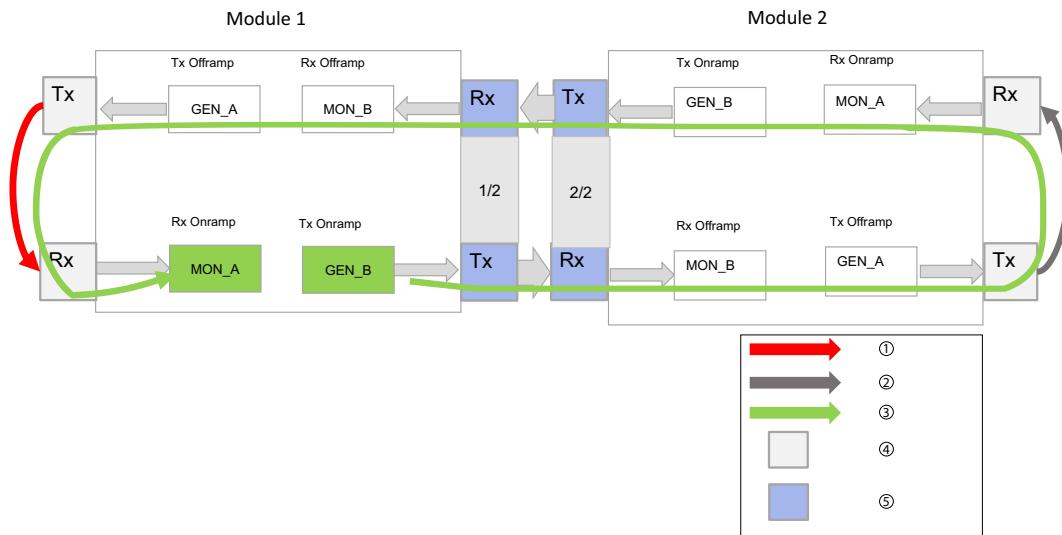
- | | | | |
|---|----------------|---|-------------|
| 1 | Fiber loopback | 3 | Client Port |
| 2 | Test Signal | 4 | Line Port |

- Race track mode with Tx (terminal) software loopbacks at both ends: This mode tests the traffic path, excluding ingress and egress optical transceivers. You can execute this test set without the presence of client pluggables. For more information, see [Performing a test with the Integrated Test Set in race track mode - Tx software loopbacks at both ends](#).

Figure 108 Race track mode with Tx software loopbacks at both ends**Legend**

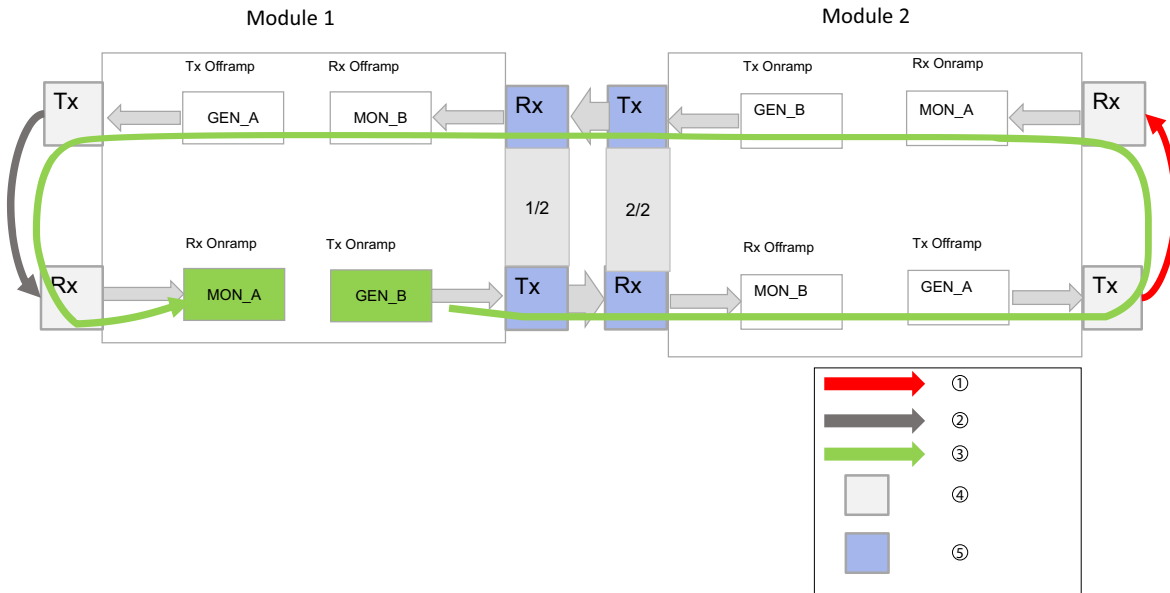
1	Tx SW loopback	3	Client Port
2	Test Signal	4	Line Port

- Race track mode with a Tx fiber loopback at the near-end and a Tx software loopback at the far-end; or conversely Tx software loopback at the near-end and Tx fiber loopback at the far-end: This mode tests the traffic path, excluding ingress and egress optical transceivers at the end where the software Tx loopback is applied. On the end where there is a Tx fiber loopback, the ingress and egress optical transceivers are being tested. For more information, see [Performing a test with the Integrated Test Set in race track mode - fiber loopback \(near-end\) and software \(far-end\) loopbacks](#).

Figure 109 Race track mode with Tx fiber (near-end) and software (far-end) loopbacks**Legend**

- | | | | |
|---|----------------|---|-------------|
| 1 | Fiber loopback | 4 | Client Port |
| 2 | Tx SW loopback | 5 | Line Port |
| 3 | Test Signal | | |

Figure 110 Race track mode with Tx software (near-end) and fiber (far-end) loopbacks



Legend

- | | | | |
|---|----------------|---|-------------|
| 1 | Fiber loopback | 4 | Client Port |
| 2 | Tx SW loopback | 5 | Line Port |
| 3 | Test Signal | | |

ITS Status, Report, and Logs

After performing ITS on supported client ports of the Waveserver 5 you can view the ITS result to take necessary actions. The ITS results are available in following formats:

- ITS Status: displays a summary of ITS session with ITS status and other important parameters.
- ITS Report: displays a detailed view of the ITS session containing card details, plug details, test mode, test status, Rx/Tx test patterns and other relevant parameters.
- ITS log summary: displays the events recorded during the ITS session with a time-stamp.

The following table defines the ITS report parameters and their corresponding descriptions:

Table 182 ITS report parameters

Parameter	Description
Test Report Time	ITS report time when it is generated.
Card Serial Number	Displays card details.
Card Part Number	
Plug Serial Number	Displays plug details.
Plug Vendor Name	
Plug Ciena Item Number	
Plug Ciena Revision	
Monitor Status	Overall monitoring status of the ITS session. It depends on Rx Pattern Sync Status, Estimated BER, and Rx Pattern Sync BER Threshold values: • Pass: when all of the following conditions are true: — Test completed — Rx Pattern Sync Status = Green & Estimated BER $\leq$ Rx Pattern Sync BER Threshold • Fail: any one of the following conditions is true: — Test in progress/completed & Rx Pattern Sync Status = Yellow/Red — Estimated BER > Rx Pattern Sync BER Threshold • In Progress: when the following condition is true: — Test is in progress and Rx Pattern Sync Status is Green
Rx Pattern Sync Status	Status of the Rx pattern sync: • Green: no defect reported during the session. • Red: a defect is currently detected. • Yellow: a defect is reported but has cleared during the session.
Status Fail Reason	Reason for failed ITS session: It remain blank when ITS status is PASS. • None: ITS status is PASS. • Pattern Sync Lost: the Rx Pattern Sync Status is YELLOW or RED. • BER Exceeded: BER threshold exceeded. • Warm Restart: module warm restarted.
Payload Errors	Payload error reported during the session.

Parameter	Description
Estimated BER	Total estimated BER during the session.
Rx Pattern Sync BER Threshold	Provisioned threshold value of the Rx pattern sync BER.
Test Mode	Race Track is the only supported mode.
Tx Test Pattern	Tx test pattern type.
Rx Test Pattern	Rx test pattern type.
Test Start Time	ITS session start time stamp.
Test Stop Time	ITS session end time stamp.
Error Free Duration	ITS session time when no error is occurred.
Elapsed Time	Total time of the ITS session in dd-hh-mm-ss format.

Dispersion scan

An ESNR-based dispersion scan can be performed on the line ports. This feature automatically sweeps pre-compensation values to optimize the ESNR-factor. The scan results provide the pre-compensation value and associated ESNR. You can use the scan results to set the PTP parameters for optimal quality. For information on performing an ESNR-based dispersion scan, see [Performing a ESNR-based pre-compensation dispersion scan and retrieving scan results](#).

Link data collector

The optical link performance data for a line port can be retrieved for a line port. This feature allows you to retrieve the PMD, PDL, BER, Rx Power, and other information/parameters for an optical link for a selected interval. You can then transfer the data records file to a specified location using an SCP server. For a complete list of parameters, see [Link Data parameters](#). For information on retrieving and uploading link performance data for the line port, see [Retrieving and uploading link performance data for a line port](#).

Table 183 Link Data parameters

Parameter	Description
Time and Date	Displays the date and time in the format yyyy-mm-dd-hh-mm-ss, indicating the year-month-day-hour-minute-second for the data.
Elapsed Time ms	Displays the elapsed time (ms).
Total Pwr mdBm	Displays the total power (mdBm).
Chan Pwr mdBm	Displays the channel power (mdBm).

Parameter	Description
FE Tx CD Pre-Comp ps/nm	Displays the far end transmit chromatic dispersion pre-compensation (ps/nm). Dispersion is the phenomenon in which the phase velocity of a wave depends on its frequency. Chromatic is used to distinguish this type of dispersion from other types, which are relevant for optical fibers.
Rx CD Comp ps/nm	Displays the receive chromatic dispersion compensation (ps/nm).
BER	Displays the Bit Error Rate (decimal).
Fast Ber	Displays the fast BER (decimal).
FER	Displays the Frame Error Rate.
Uncorrected Blocks	Displays the number of Uncorrected Blocks.
Cycle Slip	Displays the Cycle Slip. A cycle slip is a jump of an integer number of cycles in the carrier phase observable, caused by the loss of lock of a receiver phase lock loop.
PM Tick Count	Displays the performance monitoring tick count.
FreqOffset MHz	Displays the frequency offset (MHz). Frequency offset is an intentional shift of the frequency to reduce interference with other transmitters.
MERx dB	Displays the Modulation Error Ratio (MER) for value X polarization (dB). MER is the ratio between the root mean square (RMS) power of the reference constellation points and the RMS error (defined as the Euclidean distance between the reference and received points).
MERy dB	Displays the Modulation Error Ratio for value Y polarization (dB).
SPM dB	Displays the Self Phase Modulation (SPM) (dB). SPM is a nonlinear optical effect of light-matter interaction. Note: SPM data is not supported in the current release.
TNLE dB	Displays the Total NonLinear Noise (TNLE) (dB). Note: TNLE data is not supported in the current release.
ESNR dB	Displays the Effective Signal to Noise Ratio (ESNR) (dB). The Effective Signal to Noise Ratio (ESNR) is the equivalent SNR on the received symbols at the input to the decision circuit given the observed pre-FEC BER. The ESNR contains contributions from noise sources that are internal to the modem as well as external sources from the line and tracks any eye closure (distortion) present as an effective noise.
OSNR dB	Displays the Optical Signal-to-Noise Ratio (OSNR) in decibels.
SNRExt_dB	Displays the external Signal-to-Noise Ratio (ESNR) in decibels.
SNRAse_dB	Displays the Signal-to-Noise Ratio for Amplified Spontaneous Emission (SNR_ASE) in decibels.

Parameter	Description
PDLAvg_dB	Displays the Polarization Dependent Loss (PDL) (dB). PDL is a measure of the peak-to-peak difference in transmission of an optical component or system with respect to all possible states of polarization.
PDLPeak_dB	Displays the Polarization Dependent Loss (PDL) (dB). PDL is a highest level of transmission of an optical component or system with respect to all possible states of polarization.
DGDAvg	Displays the average Differential Group Delay (DGD).
DGDPeak_ps	Displays the maximum Differential Group Delay (DGD).
SOPs1f0	Displays the State of Polarization (SOP) for the first element of Stokes vector (s1) corresponding to subchannel 0.
SOPs2f0	Displays the State of Polarization (SOP) for the second element of Stokes vector (s2) corresponding to subchannel 0.
SOPs3f0	Displays the State of Polarization (SOP) for the third element of Stokes vector (s3) corresponding to subchannel 0.
SOPs1f1	Displays the State of Polarization (SOP) for the first element of Stokes vector (s1) corresponding to subchannel 1.
SOPs2f1	Displays the State of Polarization (SOP) for the second element of Stokes vector (s2) corresponding to subchannel 1.
SOPs3f1	Displays the State of Polarization (SOP) for the third element of Stokes vector (s3) corresponding to subchannel 1.
SOPs1f2	Displays the State of Polarization (SOP) for the first element of Stokes vector (s1) corresponding to subchannel 2.
SOPs2f2	Displays the State of Polarization (SOP) for the second element of Stokes vector (s2) corresponding to subchannel 2.
SOPs3f2	Displays the State of Polarization (SOP) for the third element of Stokes vector (s3) corresponding to subchannel 2.
SOPs1f3	Displays the State of Polarization (SOP) for the first element of Stokes vector (s1) corresponding to subchannel 3.
SOPs2f3	Displays the State of Polarization (SOP) for the second element of Stokes vector (s2) corresponding to subchannel 3.
SOPs3f3	Displays the State of Polarization (SOP) for the third element of Stokes vector (s3) corresponding to subchannel 3.

Procedure list

This section provides the following procedure:

- [Procedure 85, “Operating loopback test on Waveserver 5”](#)

- Procedure 86, “Performing a test with the Integrated Test Set in race track mode - fiber loopbacks at both ends”
- Procedure 87, “Performing a test with the Integrated Test Set in race track mode - Tx software loopbacks at both ends”
- Procedure 88, “Performing a test with the Integrated Test Set in race track mode - fiber loopback (near-end) and software (far-end) loopbacks”
- Procedure 89, “Retrieving and saving the test report and logs after performing a test using the Integrated Test Set”
- Procedure 90, “Performing a ESNR-based pre-compensation dispersion scan and retrieving scan results”
- Procedure 91, “Retrieving and uploading link performance data for a line port”

Procedure 85 Operating loopback test on Waveserver 5

Use this procedure to operate a loopback test on a line or client port.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- Waveserver 5 and the test set or subtending equipment are connected correctly for the loopback test being performed.

See previous figures, [Client Rx Loopback, Example 1 - Line Rx Loopback and Client Tx Loopback between Waveserver 5s](#) and [Example 2 - Line Rx Loopback and Client Tx Loopback between Waveserver 5s](#).

Overview

Note: If a remote management interface is enabled on a line port, Tx and Rx loopbacks are blocked.

Steps

- 1 Disable the port to be tested:

```
port disable <slot>/<port>
```

Note: Disabling a port affects end-to-end traffic on that port.
- 2 Operate the loopback test on the port:

```
port set port <slot>/<port> loopback <disabled | Rx | Tx>
```

Note: You can operate only one type of loopback test (Rx or Tx) at a time on any one port.
- 3 Optionally, retrieve summary information for the port to confirm that the loopback is operating:

```
port show port <slot>/<port>
```
- 4 Retrieve performance monitoring port statistics to view transmission quality:

```
pm show port <slot>/<port>
```

Note: Frame size and percentage of bandwidth utilization indicate port performance and connectivity issues.
- 5 When required, end the loopback test on the port and set the administrative state of the port to enabled to restore end-to-end traffic on the port:

```
port unset port <slot>/<port> loopback  
port enable port <slot>/<port>
```

Note: Setting the tx-state (laser on/off) at the PTP layer of the same port, which is independently configured, will disrupt the loopback test.

Procedure 86 Performing a test with the Integrated Test Set in race track mode - fiber loopbacks at both ends

Use this procedure to perform a test using the Integrated Test Set (ITS) in race track mode by installing fiber loopbacks at both the near-end and far-end client ports. This mode tests the entire traffic path, including ingress and egress optical transceivers.

Requirements

- You must be logged in to the CLI using an account with at least admin access privileges.
- You must have an Integrated Test Set (ITS) add-on license. One Integrated Test Set (ITS) add-on license is required per module.
- Pluggable transceivers must be installed at the client interface.
- Fiber-optic loopback test cables are available.



CAUTION

Risk of damage to the chassis, pluggable modules, and transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable modules or transceivers.

Overview

For an example illustration of the test configuration, see [Race track mode with fiber loopbacks at both ends](#).

The following procedure uses the example of slot 1, client port 9 as the ITS source.

Note: Fiber loopbacks or Tx loopbacks on both ends of the client ports are only supported on 100GE, OTL4.4 (100G), and 400GE ports for the ITS application.

Steps

Enabling ITS on near-end module

- 1 Disable the near-end client port:

```
port disable port <slot/port>
```

 For Example:

```
port disable port 1/9
```

- 2 Enable the ITS on the near-end client port:
`application its enable port <slot/port>`

For example:

`application its enable port 1/9`

Note: ITS must be enabled before applying the loopbacks to avoid timing loops.

Installing fiber loopbacks (near-end and far-end)

- 3 At the far-end module, install a fiber loopback on the client port that will be used as the source for the test. For example, fiber loopback is installed on slot 2 port 9.
- 4 At the near-end module, install a fiber loopback on the client port that will be used as the source for the test. For example, fiber loopback is installed on slot 1 port 9.

Configuring test parameters and running the test

- 5 Set the test duration and monitoring threshold on the near-end client port:
`application its set port <slot/port> [duration <number/time format N[yMwdhms] [ber-threshold <NUMBER: 0 or 5 to 16>]`

For example:

`application its set port 1/9 duration 0h0m3s ber-threshold 5`

- 6 Start the test on near-end client port:
`application its start port <slot/port>`

For example:

`application its start port 1/9`

Note: You can stop the test using the command: `application its stop port <slot/port>`.

- 7 To view the status of the test, display the integrated test set report on the near-end client port:

`application its show port <slot/port> status`

- 8 When the test is complete, retrieve the ITS logs on the near-end client port:
`application its show port <slot/port> log-summary`

For example:

`application its show port 1/9 log-summary`

For more information, see [Retrieving and saving the test report and logs after performing a test using the Integrated Test Set](#).

- 9 If you have completed testing, do the following:

- a.** Stop the test on near-end client port.

```
application its stop port <slot/port>
```

For example:

```
application its stop port 1/9
```

- b.** Uninstall the fiber loopbacks from the near-end and far-end client ports.

- c.** Disable the ITS on the near-end client port:

```
application its disable port <slot/port>
```

- d.** Enable the near-end client port:

```
port enable port <slot/port>
```

Procedure 87 Performing a test with the Integrated Test Set in race track mode - Tx software loopbacks at both ends

Use this procedure to perform a test using the Integrated Test Set (ITS) in race track mode by configuring Tx (terminal) software loopbacks at both the near-end and far-end client ports. This mode tests the traffic path, excluding ingress and egress optical transceivers.

Requirements

- You must be logged in to the CLI using an account with at least admin access privileges.

Note 1: It is not necessary for pluggable transceivers to be installed at the client interface before running the test set.

Note 2: If a remote management interface is enabled on a line port, a Tx loopback is blocked.

- You must have an Integrated Test Set (ITS) add-on license. One Integrated Test Set (ITS) add-on license is required per module.

Overview

For an example illustration of the test configuration, see [Race track mode with Tx software loopbacks at both ends](#).

The following procedure uses the example of slot 1 client port 9 as the ITS source.

Note: Fiber loopbacks or Tx loopbacks on both ends of the client ports are only supported on 100GE, OTL4.4 (100G), and 400GE ports for the ITS application.

Steps

Enabling ITS on near-end module

- 1 Disable the near-end client port:

```
port disable port <slot/port>
```

For Example:

```
port disable port 1/9
```
- 2 Enable the ITS on the near-end client port:
Note: GEN_B and MON_A are automatically enabled.

```
application its enable port <slot/port>
```

For example:

```
application its enable port 1/9
```

Note: ITS must be enabled before applying the loopbacks to avoid timing loops.

Provisioning software loopbacks (near-end and far-end)

- 3 At the far-end module, perform the following actions on the client port that you are using for the test:

- a. Set the client rate:

```
xcvr set xcvr <slot/port> mode <mode>
```

For example:

```
xcvr set xcvr 2/9 mode 100GE
```

- b. Disable the client port:

```
port disable port <slot/port>
```

For example:

```
port disable port 2/9
```

- c. Set a tx loopback on the client port:

```
port set port <slot/port> loopback Tx
```

For example:

```
port set port 2/9 loopback Tx
```

- 4 At the near-end module, perform the following actions on the client port you are using as the source for the test:

- a. Set the client rate:

```
xcvr set xcvr <slot/port> mode <mode>
```

For example:

```
xcvr set xcvr 1/9 mode 100GE
```

- b. Provision a tx loopback on the client port:

```
port set port <slot/port> loopback Tx
```

For example:

```
port set port 1/9 loopback Tx
```

Configuring test parameters and running the test

- 5 Set the test duration and monitoring threshold on the near-end client port:

```
application its set port <slot/port> [duration <number/
time format N[yMwdhms] [ber-threshold <NUMBER: 0 or 5 to
16>]
```

For example:

```
application its set port 1/9 duration 0h0m3s ber-
threshold 5
```

- 6 Start the test on the near-end client port:

```
application its start port <slot/port>
```


For example:

```
application its start port 1/9
```

Note: You can stop the test using the command: `application its stop port <slot/port>`.

- 7 To view the status of the test, display the integrated test set report on the near-end client port:

```
application its show port <slot/port> status
```

- 8 When the test is complete, retrieve the ITS logs on the near-end client port:

```
application its show port <slot/port> log-summary
```

For Example:

```
application its show port 1/9 log-summary
```

For more information, see [Retrieving and saving the test report and logs after performing a test using the Integrated Test Set](#).

- 9 If the testing is complete, perform the following steps:

- a. Stop the test on near-end client port.

```
application its stop port <slot/port>
```

For example:

```
application its stop port 1/9
```

- b. Remove the Tx loopback from the near-end and far-end client port:

```
port unset port <slot/port> loopback
```

- c. Disable the ITS on the near-end client port:

```
application its disable port <slot/port>
```

- d. Enable the client port that were previously disabled:

```
port enable port <slot/port>
```

For example:

```
port enable port 2/9
```

Procedure 88 Performing a test with the Integrated Test Set in race track mode - fiber loopback (near-end) and software (far-end) loopbacks

Use this procedure to perform a test using the Integrated Test Set (ITS) in race track mode by configuring a Tx (terminal) software loopback at the far-end client port and installing a fiber loopback on the near-end client port. This mode tests the traffic path, excluding ingress and egress optical transceivers at the end where the software Tx loopback is applied.

Requirements

- You must be logged in to the CLI using an account with at least admin access privileges.
- You must have an Integrated Test Set (ITS) add-on license. One Integrated Test Set (ITS) add-on license is required per module.
- A fiber-optic loopback test cable is available.

Note: If a remote management interface is enabled on a line port, a Tx loopback is blocked.



CAUTION

Risk of damage to the chassis, pluggable modules, and transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable modules or transceivers.

Overview

For an example illustration of the test configuration, see [Race track mode with Tx fiber \(near-end\) and software \(far-end\) loopbacks](#).

The following procedure uses the example of slot 1 client port 9 as the ITS source.

Note: Fiber loopbacks or Tx loopbacks on both ends of the client ports are only supported on 100GE, OTL4.4 (100G), and 400GE ports for the ITS application.

Steps

Enabling ITS on near-end module

- 1 Disable the near-end client port:

```
port disable port <slot/port>
```

For Example:

```
port disable port 1/9
```

- 2 Enable the ITS on the near-end client port.

Note: GEN_B and MON_A are automatically enabled.

```
application its enable port <slot/port>
```

For example:

```
application its enable port 1/9
```

Note: ITS must be enabled before applying the loopbacks to avoid timing loops.

Provisioning the software loopback (far-end)

- 3 At the far-end module, perform the following actions on the client port that you are using for the test:

- a. Set the client rate:

```
xcvr set xcvr <slot/port> mode <mode>
```

For example:

```
xcvr set xcvr 2/9 mode 100GE
```

- b. Disable the client port:

```
port disable port <slot/port>
```

For example:

```
port disable port 2/9
```

- c. Provision a tx loopback on the client port:

```
port set port <slot/port> loopback Tx
```

For example:

```
port set port 2/9 loopback Tx
```

Installing the fiber-optic loopback test cable (near-end)

- 4 On the near-end module, install a fiber loopback on the client port to be used in the test. For example, the fiber loopback is installed on slot 1 port 9.

Configuring test parameters and running the test

- 5 Set the test duration and monitoring threshold on the near-end client port:

```
application its set port <slot/port> [duration <number/  
time format N[yMwdhms]][ber-threshold <NUMBER: 0 or 5 to  
16>]
```

For example:

```
application its set port 1/9 duration 0h0m3s ber-  
threshold 5
```

- 6 Start the test on the near-end client port:

```
application its start port <slot/port>
```

For example:

```
application its start port 1/9
```

Note: You can stop the test using the command: `application its stop port <slot/port>`.

- 7 To view the status of the test on the near-end client port:

```
application its show port <slot/port> status
```

- 8 When the test is complete, retrieve the ITS logs on the near-end client port:

```
application its show port <slot/port> log-summary
```

For example:

```
application its show port 1/9 log-summary
```

For more information, see [Retrieving and saving the test report and logs after performing a test using the Integrated Test Set](#).

- 9 If you have completed testing, do the following:

- a. Stop the test on near-end client port.

```
application its stop port <slot/port>
```

For example:

```
application its stop port 1/9
```

- b. Uninstall the fiber loopback on the near-end client port.

- c. Remove the Tx loopback from the far-end client port:

```
port unset port <slot/port> loopback
```

- d. Disable the ITS on the near-end client port:

```
application its disable port <slot/port>
```

- e. Enable the client port that were previously disabled:

```
port enable port <slot/port>
```

For example:

```
port enable port 2/9
```

Procedure 89 Retrieving and saving the test report and logs after performing a test using the Integrated Test Set

Use the commands in this procedure to retrieve and save a report and logs after performing a test using the Integrated Test Set (ITS). You can also use this procedure to show the status of a test port, or upload the test report to a server.

Requirements

- You must be logged in to the CLI using an account with at least limited access privileges.

Overview

Note: The Error Free Duration (seconds) parameter indicates the number of error-free seconds between the start of the test and the error. The value stops incrementing after an error occurs.

The report is uploaded as a zip file that contains a CSV file of the report and a CSV file of the log summary. An example of the zip file format is: 186-1600-900_NNTMRT000TNJ_port_3_20180709_2006ws53005.zip

Where

- 186-3100-900 is the module part number
- NNTMRT000TNJ is the module serial number
- port_3 is the port number
- 20180709 is the test date
- ws53005 is the suffix

The event log records the following events:

- START (Start monitoring port under test)
- PAYLOAD_ERR (Payload Error Detected)
- PAYLOAD_SYNC (True: Payload Pattern Sync lost, False: Payload Pattern Sync regained)
- STOP (Stop monitoring port under test)

Note 1: When uploading the results file to the remote server, if the remote-path or url ends with a "/", the Waveserver 5 assumes the target path is a directory and it appends the results file with a ".zip" extension. If the remote-path or url does not end with a "/", the Waveserver 5 assumes the target path is a file and it does not append the results file with a ".zip" extension. In this case, the user must add the ".zip" extension when specifying the remote-path or url.

Note 2: The results file is stored on the Waveserver 5 in “.zip” format. When uploading to remote server, ensure the same “.zip” extension is specified.

Steps

- 1 Use the following commands to display the ITS status of a port, and to save a report and logs after performing a test.

- To display the status of a port:

```
application its show port <slot/port> status
```

Figure 111 ITS port status

ITS STATUS	
Parameter	Value
Monitor Status	FAIL
Sync Status	RED
Fail Reason	Pattern Sync Lost
Payload Errors	0
Estimated BER (Mantissa)	0
Estimated BER (Exponent)	+00
Error Free Duration (seconds)	0
Elapsed time (seconds)	458

- To generate and display the ITS report for a port:

```
application its show port <slot/port> report
```

Figure 112 ITS report for a port

ITS REPORT	
Parameter	Value
Test Report Time	2022-03-15 2:45:54
Card Serial Number	M975B0BE
Card Part Number	186-3101-901
Plug Serial Number	EXLGJP71EY010363
Plug Vendor Name	Ciena
Plug Ciena Item Number	160-9401-900
Plug Ciena Revision	001
Monitor Status	FAIL
Rx Pattern Sync Status	RED
Status Fail Reason	Pattern Sync Lost
Payload Errors	0
Estimated BER	0E-00
Rx Pattern Sync BER Threshold	0E-00
Test Mode	Race Track
Tx Test Pattern	100GE 802.3ba Ethernet
Rx Test Pattern	100GE 802.3ba Ethernet
Test Start Time	2022-03-08 4:53:02
Test Stop Time	2022-03-08 5:00:40
Error Free Duration	00-00-00-00
Elapsed Time	00-00-07-38

- To generate and display the ITS logs for a port:

application its show port <slot/port> log-summary

Figure 113 ITS logs port summary

LOG SUMMARY								
IDX	UNIT	DESCRIPTION	Start Time		End Time		Value	Duration
0	CONFIG	START	2022-03-08	4:53:02	2022-03-08	4:53:02	TRUE	00-00-00-00
1	MONA	PAYLOAD_SYNC	2022-03-08	4:53:02	-	-	TRUE	00-00-00-00
2	MONA	PAYLOAD_SYNC	2022-03-08	4:53:02	2022-03-08	5:00:40	TRUE	00-00-30-30
3	CONFIG	STOP	2022-03-08	5:00:40	2022-03-08	5:00:40	TRUE	00-00-00-00

- To retrieve summary information for all ITS logs, including port name, test mode, duration, and BER threshold:

application its show

Figure 114 ITS logs summary information

-----ITS CONFIG-----			
Port Name	Test Mode	Duration(s)	BER Threshold
2/3	Enabled	0	0E-00

- To upload the ITS report and event log:

```
application its upload-detailed-logs port <portList>
{scp-server remote-path <String[1..127]> suffix
<String[1..127]> login-id <String[1..32]> password
<Password String[1..128]> | url <String> login-id
<String[1..32]> password <Password String[1..128]>}
```

For example:

```
application its upload-detailed-logs port 2/3 url scp://
10.20.30.40:/tmp/itslogs.zip login-id user1 password
*****
```

or

```
application its upload-detailed-logs port 2/3 scp-server
10.20.30.40 remote-path /home/waveserver/ suffix ws53005
login-id user1 password *****
```

Note: When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

For example, application its upload-detailed-logs port 2/3
url scp://user@[2222:111:dddd:5444:aced::293]:/tmp/
itslogs.zip password *****

Procedure 90 Performing a ESNR-based pre-compensation dispersion scan and retrieving scan results

Use this procedure to perform a ESNR-based dispersion scan on the Waveserver 5 line ports. The ESNR-based pre-compensation dispersion scan feature automatically sweeps pre-compensation values to optimize ESNR factor. The scan results provide the pre-compensation value and associated ESNR. You can use the scan results to set the PTP parameters for optimal quality. For uncompensated (positive dispersion) submarine systems with more than 40000ps/nm of total dispersion, the tool should not be used to set the dispersion as the required setting will be outside of the scan range.

Requirements

- You must be logged in to the CLI using an account with at least admin access privileges.
- The PTP must be administratively disabled.

Steps

Running the scan

- 1 Select the line port from which you want to run the pre-compensation dispersion scan and the scan range:

```
application dispersion-scan start ptp <slot/port> from
<ps/nm (Range: -40000 - 40000)>
```

```
application dispersion-scan start ptp <slot/port> to <ps/
nm (Range: -40000 - 40000)>
```

Note 1: Default values for the submarine PTP modem class are:

From = -5000 ps/nm, To = 5000 ps/nm.

Note 2: Default values for the terrestrial PTP modem class are:

From = 0 ps/nm, To = 40000 ps/nm.

- 2 Configure the scanning step size:

```
application dispersion-scan start ptp <slot/port> step
<ps/nm (Range: 100 - 1000 in the increments of 100)>
```

Note 1: Default value for the submarine PTP modem class is Step = 200 ps/nm.

Note 2: Default value for the terrestrial PTP modem class is Step = 1000 ps/nm.

- 3 Configure the scan dwell time:

```
application dispersion-scan start ptp <slot/port> dwell
<SECONDS (Range: 1 - 100 in the increments of 1)>
```

Note: Default value for the submarine and terrestrial PTP modem class is Dwell time = 5 seconds.

- The scan can take up to five minutes to complete.
- 4 To confirm that the scan is running, execute the following command and view scan status.
- ```
application dispersion-scan show
```
- The status is "in-progress" and changes to "completed" when the scan successfully completes.*
- 5 If you want to cancel the scan, run the following command:
- ```
application dispersion-scan stop ptp <slot/port>
```

Viewing scan results

- 6 To view the scan results, run the command:
- ```
application dispersion-scan show ptp <slot/port>
```

## Procedure 91 Retrieving and uploading link performance data for a line port

Use this procedure to retrieve optical link performance data for a line port.

### Requirements

- You must be logged in to the CLI using an account with at least admin access privileges.

### Overview

The link data collection feature allows you to retrieve the PMD, PDL, BER, Rx Power, and other information/parameters for an optical link for a selected interval. You can then transfer the data records file to a specified location using an SCP server. For a complete list of parameters, see [Link Data parameters](#).

**Note:** The results file is stored on the Waveserver 5 in “.csv” format. When uploading to remote server, ensure the same “.csv” extension is specified.

### Steps

#### Retrieving link data

- 1 Select the line port for which you want to retrieve link data, the data collection interval, and the number of records you want to retrieve:

```
application link-data-collector start ptp <slot/port>
interval <MILLISECONDS (Range: 10..60000)> numrecords
<NumRecords count (Range: 100 - 50000)>
```

- 2 To display the status of the data collection, type the command:

```
application link-data-collector show ptp <slot/port>
```

- 3 To view the status of the collection and the input parameters, type the command `application link-data-collector show ptp <slot/port>`.

The status displays “initiated” when the link collection is in process and “completed” when the collection is finished. The status is also “completed” when the collection is canceled. An “internal-error” status indicates that signaling to the module failed.

**Note:** You can cancel the link data collection using the command `application link-data-collector stop ptp <slot/port>`. Canceling the collection generates a “completed” status.

When the data collection is complete, data is saved to a .csv file on the control processor in the folder: `/var/opt/LinkDataCollector`. The file name includes the slot and port number for easy identification.

**Uploading the link data file**

- 4 To transfer the .csv file to a specified location using an SCP, FTP, SFTP, or TFTP server, run the command:

```
application link-data-collector upload ptp <slot/port>
{default-ftp-server | default-scp-server | default-server
| default-sftp-server | default-tftp-server | scp-server
<IP address (w/optional port) or host name> file-name
<String> login-id <String[1..32]> password <Password
String[1..128]> | url <String> login-id <String[1..32]>
password <Password String[1..128]> secret
<String[1..32]>}
```

For example:

```
application link-data-collector upload ptp 1/1 url scp:/
/user@10.20.30.40:/tmp/linkperformancedata.csv password

```

or

```
application link-data-collector upload ptp 1/1 url scp:/
/10.20.30.40:/tmp/linkperformancedata.csv login-id user1
password *****
```

or

```
application link-data-collector upload ptp 1/1 scp-server
10.20.30.40 file-name linkperformancedata.csv login-id
user1 password *****
```

**Note 1:** Repeat the command to transfer multiple files, however, only one file is transferred at a time.

**Note 2:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

The password can contain special characters:

```
~!^&*()-_:=+{ }[]\|'; '<>.,
```

For example,

```
application link-data-collector upload ptp 1/1 url scp:/
/[2222:111:dddd:5444:aced::293]/tmp/
linkperfromancedata.csv login-id user1 password *****
```

**Note 3:** If the remote server information is not provided in the URL then the file is transferred from the /var/opt/LinkDataCollector folder on the control processor to the default server.

- 5 Double-click the .csv file to open the file in MS Excel. If you do not have MS Excel on your computer, you can also open the file using Notepad; however, the text display is less legible.
- 6 Refer to the [Link Data parameters](#) table for complete list of parameters.



---

## CHAPTER 21

# Interworking

---

This section contains the following topics:

- [Overview](#)
  - [Photonic line systems](#)
  - [Protection configurations](#)
  - [Comms channel for 6500-based SLTE configurations](#)
  - [Waveserver WL5e interworking with 6500 WL5e](#)
  - [SPLI with 6500](#)
  - [SPLI with RLS](#)
  - [Connection validation](#)
- [Procedure list](#)

### Overview

This section provides information on Waveserver 5 interworking with the 6500 or RLS.

#### Photonic line systems

Waveserver 5 operating at 95 GBaud interworks with these photonic line systems:

- CMD42 ROADM or Thin Terminal
- 1x9 CDA ROADM
- 1x20 CDA ROADM
- 1x9 CD ROADM
- 1x20 CD ROADM
- 1x20 CDC ROADM
- 1x5 RLA CDA ROADM
- 1x5 RLA CD ROADM
- CCMD12

- RLS CDC ROADM
- RLS CDA ROADM
- RLS 1x12 RLA + CMD48
- 6500 fixed and flexible grid ROADM based photonic line systems as Ciena native wavelengths

Waveserver 5 operating at 91.6 GBaud interworks with these photonic line systems:

- Passive photonic line system with CMD48 and MLA3
- Passive photonic line system with CMD48 and MLA3/ESAM
- Passive photonic line system with CMD48 and RLS TPS + DLM

Waveserver 5 interworks with these submarine photonic line systems:

- Flexible Grid 1x9 GMD10 SLTE
- Flexible Grid 1x9 CD SLTE

### Protection configurations

Waveserver 5 supports the following protection configurations:

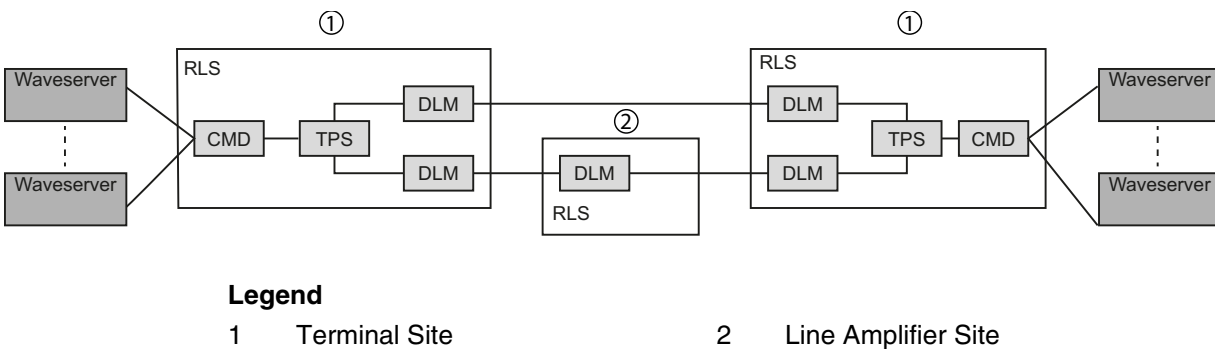
- [Point-to-point protected configurations with RLS](#)
- [Link trunk protection configurations with the 6500 S-Series](#)
- [Client OPS with 6500 S-Series](#)

#### Point-to-point protected configurations with RLS

Waveserver 5 supports point-to-point protected configurations with RLS. Point-to-point protected configurations on RLS use a combination of CMD, TPS, and DLM modules. For more information on RLS point-to-point protected configurations, refer to the *RLS Planning Guide*.

The following figure shows an example configuration.

**Figure 115** Point-to-point protected configuration with RLS



### Link trunk protection configurations with the 6500 S-Series

Waveserver 5 supports link trunk protection configurations with the 6500 S-Series. Link trunk protection configurations on 6500 S-Series use a combination of OPS, CMD, and line amplifier circuit packs. However, the trunk-ops is supported for the following configurations:

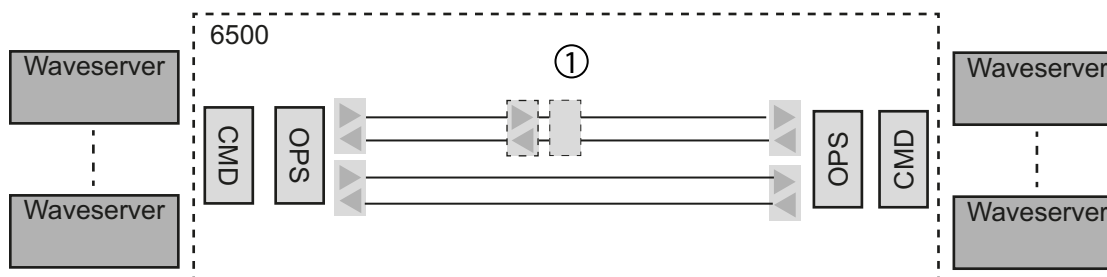
- Unmated line configurations
- 100GE/400GE clients (Ethernet Conditioning w/ 200ms holdoff, None Conditioning w/ 200ms holdoff, and Laser Off Conditioning w/ 200ms holdoff)
- OTL4.4 clients (OTN Conditioning (holdoff N/A))

**Note:** Line OPS switch times are typically < 50ms

For more information on 6500 protection configurations, refer to *6500 Protection Switching*, 323-1851-315.

The following figure shows an example configuration.

**Figure 116** Link trunk protection configuration with 6500 S-Series



#### Legend

- 1 Optional Line Amplifiers

### Client OPS with 6500 S-Series

Waveserver 5 supports client-layer optical protection switch (OPS) configurations with 6500 S-Series OPS circuit packs. Client OPS is supported on 100GE and OTU4 clients of all WL5e 2x MOTR 16x and WL5e 2x MOTR 12x modules

The client OPS feature allows the Waveserver 5 to shutoff a client Tx laser within a given timeframe in order to allow the attached 6500 OPS circuit pack to react.

The following provisioning considerations apply:

- The typical switch time is less than 50 ms when no conditioning hold-off timer is applied.



- The conditioning hold-off timer is only used on 100GE ports when a line fault is detected. The hold-off timer is not used on OTU4 ports or on upstream client faults.
- The Conditioning Type parameter of client ports attached to the OPS circuit pack should be set to laser-off. The system raises a “Laser off active” alarm when the system intentionally shuts down the Tx laser.
- The only pluggable that can be connected to the OPS circuit pack when used in the client-layer optical path protection configuration is the QSFP28 (160-9402-900).

### **Waveserver 5 and 6500 interconnect model and 6500 port preferences**

The Waveserver 5 and 6500 interconnect model illustrates how to connect Waveserver 5 to the 6500, when you want to manage Waveserver 5 through the 6500.

For the Waveserver 5 and 6500 interconnect model:

- 6500 Gateway Network Element (GNE) configurations with IP static route (non-redundant) and OSPF (redundant) configurations are supported, private IP (NAT) is not supported. For additional information on 6500 GNE setups, see *6500 Data Communications Planning and User Guide*, 323-1851-101.
- Auto-negotiation must be enabled and an OSPF passive circuit must be provisioned on the 6500 port that connects to the Waveserver 5.
- The 6500 management port must be numbered. IP subnets must be large enough to accommodate the 6500 management port and all the subtending Waveserver 5 nodes.
- When there are multiple Waveserver 5 nodes subtending a 6500, a single Waveserver 5 DCN port must be connected to the 6500 COLAN/ILAN, with the remaining Waveserver 5 nodes connected in an ILAN daisy-chain (either linear or in a ring configuration).

### **DCN connectivity at all sites**

This section provides an example of a configuration where each site has DCN access. As a result, Waveserver 5 management traffic generally flows through the COLAN port of the local 6500 or RLS GNE.

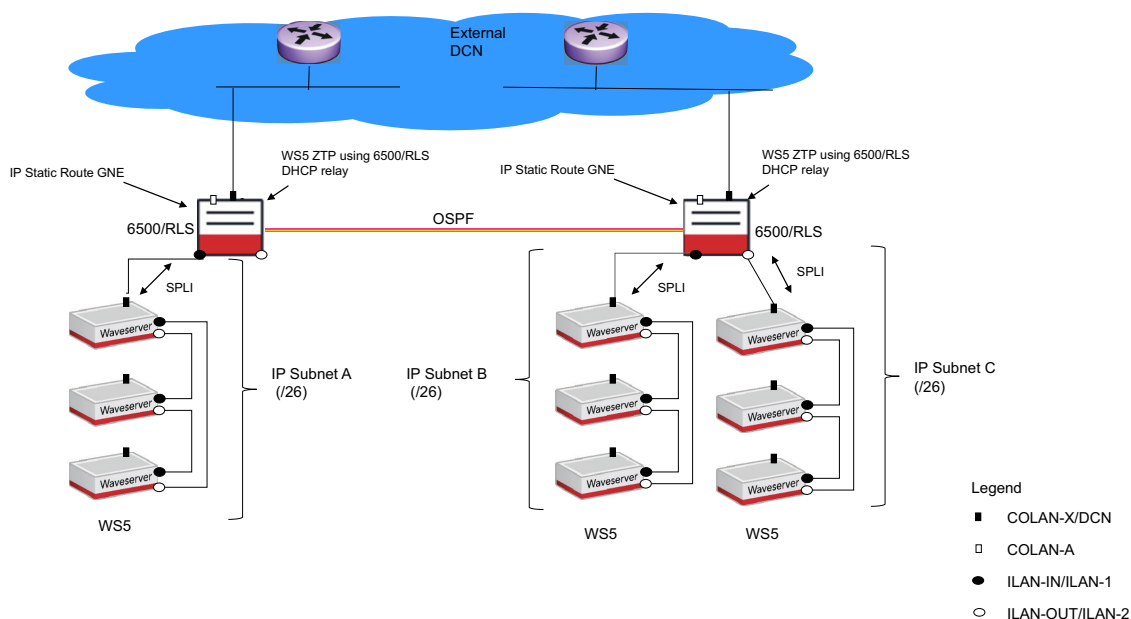
In this example, there is DCN access and a single 6500 or RLS chassis at each site. It is assumed that there is only a single connection to the external DCN available at each site. Therefore, the configuration requires Waveserver 5 to subtend the 6500 or RLS. For 6500, the shelf type is assumed to be the 14-slot, where COLAN-A is not recommended, thus the Waveserver 5 nodes are connected to the 6500 ILAN ports.

These notes apply:

- An IP static route is provisioned for the non-redundant 6500 or RLS GNE.
- OSPF is used within the 6500 or RLS network only.
- A /26 subnet is allocated for the Waveserver 5 nodes to allow for future growth.
- OSPF passive circuits are required on the 6500 or RLS ports to the Waveserver 5.
- Waveserver 5 ZTP is supported using 6500 or RLS DHCP relay (plain mode).

See the following figure.

**Figure 117** DCN connectivity at all sites

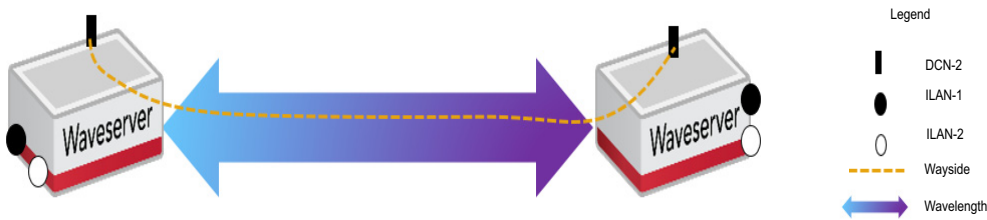


### Comms channel for 6500-based SLTE configurations

Waveserver 5 supports a wayside channel for use in submarine applications when interworking with a 6500 as the photonic line system. In submarine applications, due to distance, no OSC is available for photonic control messaging, and since Waveserver 5 provides the wavelengths, 6500 GCC is also not available. In this situation, the Waveserver 5 wayside channel can be used to provide a channel for 6500-to-6500 communications across the submarine link that are required by the photonic applications.

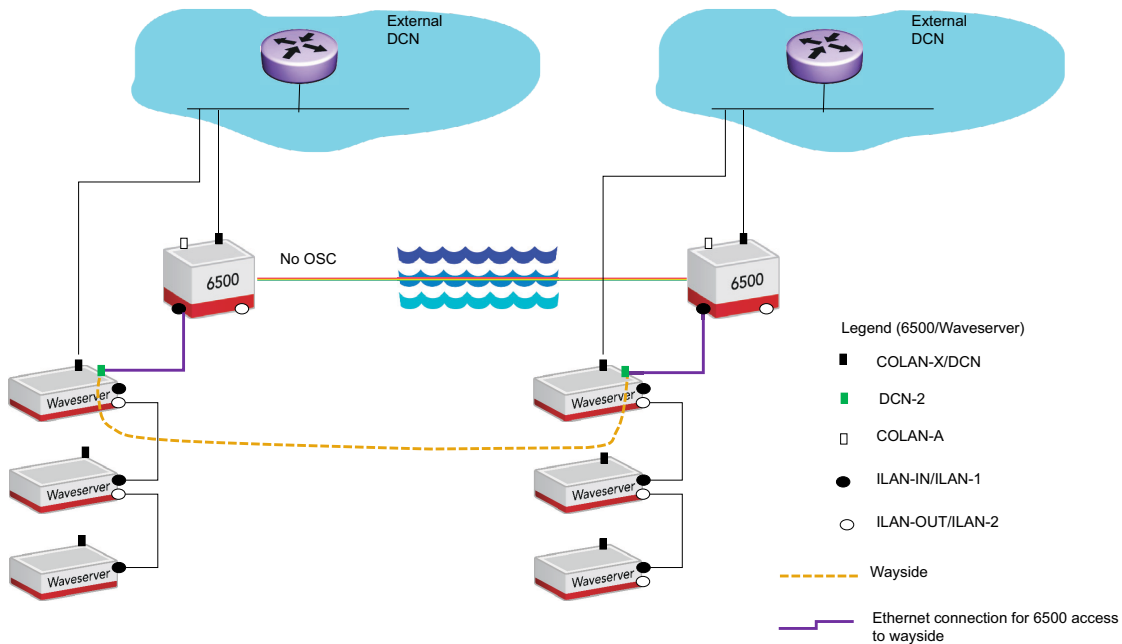
For an end-to-end wayside channel, the DCN-2 port is used as a wayside access port on a pair of Waveserver 5 chassis (one at each end of the submarine link). To provision this, the DCN-2 port must be administratively enabled using CLI command `chassis port enable port dcn-2`. See the following figure.

**Figure 118** Waveserver 5 wayside access via DCN port



The DCN-2 port on a Waveserver 5 chassis is used for a wayside channel. This means that the chassis must have access to the external DCN via DCN-1, DCN-3 or ILAN ports. See the following figure for an example of a typical application.

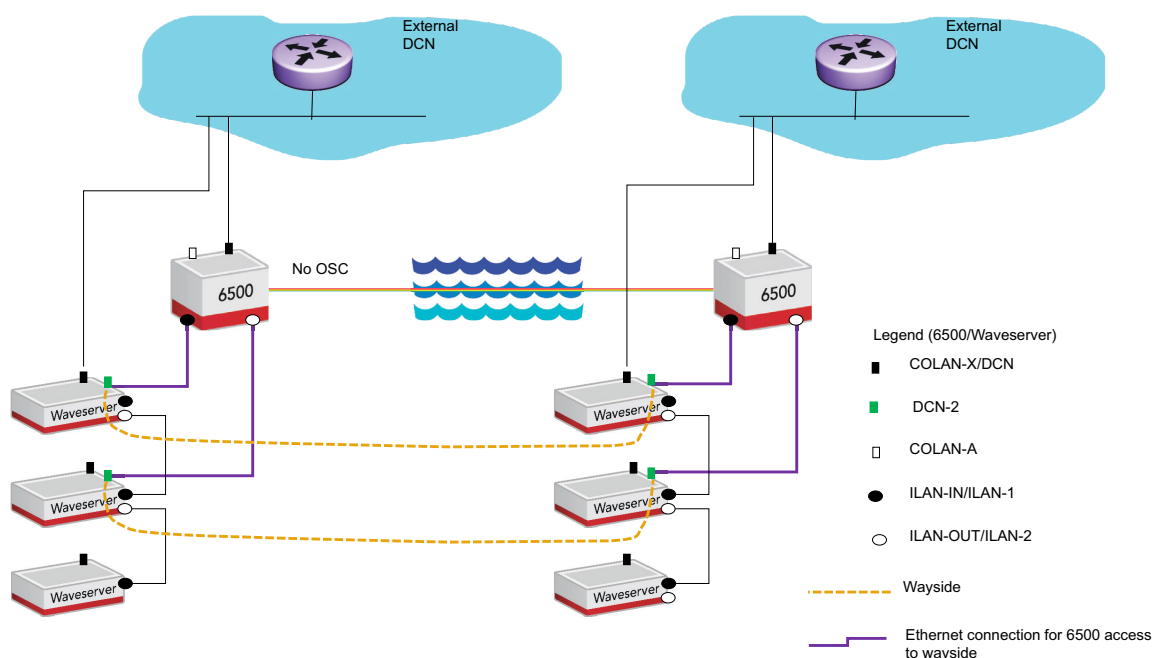
**Figure 119** Waveserver 5 wayside for 6500 SLTE communications



As shown in the figure above, logically the 6500 nodes are using an ILAN-to-ILAN connection, and with OSPF configured on that connection, the 6500 photonic control software components are able to communicate and therefore manage the optical link.

You can create more than one wayside channel by using additional Waveserver 5 pairs. This provides improved resiliency for certain failures; see the following figure for an example.

**Figure 120** Multiple Waveserver 5 wayside channels used for 6500 SLTE communications

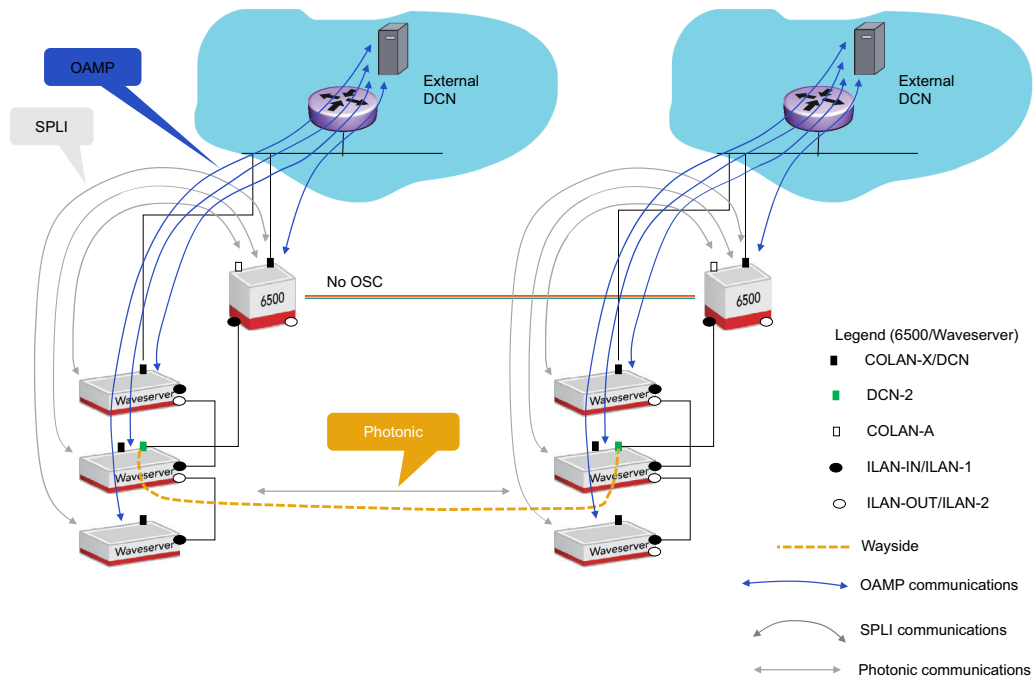


### Wayside channel guidelines

The following section provides both general and more detailed guidelines for implementing a wayside channel.

- The wayside channel is supported on Waveserver 5.
- The wayside channel operates at Layer 2. When connecting to the wayside channel, take care to prevent loops. For example, do not connect the same two wayside channels into a Layer 2 switch at each end of the submarine link.
- Ensure that the Waveserver 5 chassis to be used for the wayside channel has the MOTR module installed in slot 1. Only port 1/1 carries the wayside channel.
- To configure a comms channel for 6500-based SLTE configurations, DCN-2 must be administratively enabled using CLI command `chassis port enable port dcn-2`.

- DCN-2 supports wayside-channel mode only.
- Ciena recommends using a 6500 ILAN port to connect to the Waveserver 5 DCN-2 port for wayside access. A 6500 COLAN port can also be used, however COLAN-A port is not supported on 6500 14-slot shelves with SP redundancy for this application. A COLAN-A port on a 6500 14-slot shelf with a single SP is supported for this application.
- The 6500 ILAN or COLAN port connected to the Waveserver 5 DCN-2 port must be:
  - Set to AUTO mode (auto-negotiate). The ports auto-negotiate to 100BT full-duplex.
  - Set as a numbered interface. In routable IP configurations, the IP addresses must be routable. A /30 subnet is recommended.
  - Configured with OSPF with opaque LSAs enabled. Note that if a COLAN port is used, opaque LSAs are disabled by default.
- The configuration should be engineered so that north/south OAMP traffic does not normally traverse the wayside channel, since the channel bandwidth is intended for 6500 photonics use.
- Use of the Waveserver 5 wayside channel does not restrict or affect the use of a 6500 GNE configuration, however the type of configuration can affect local SPLI communications. For SPLI communication between 6500 and Waveserver 5, the 6500 shelf-IP address must be visible to the Waveserver 5; therefore some GNE configurations, such as Private-IP, cannot be used. The following figure shows the various communications flows, including Waveserver 5 to 6500 SPLI.

**Figure 121** Communication flows

For information on how to configure 6500-based SLTE configurations using Waveserver 5 wayside channel, see [Configuring 6500-based SLTE configurations using Waveserver 5 wayside channel](#).

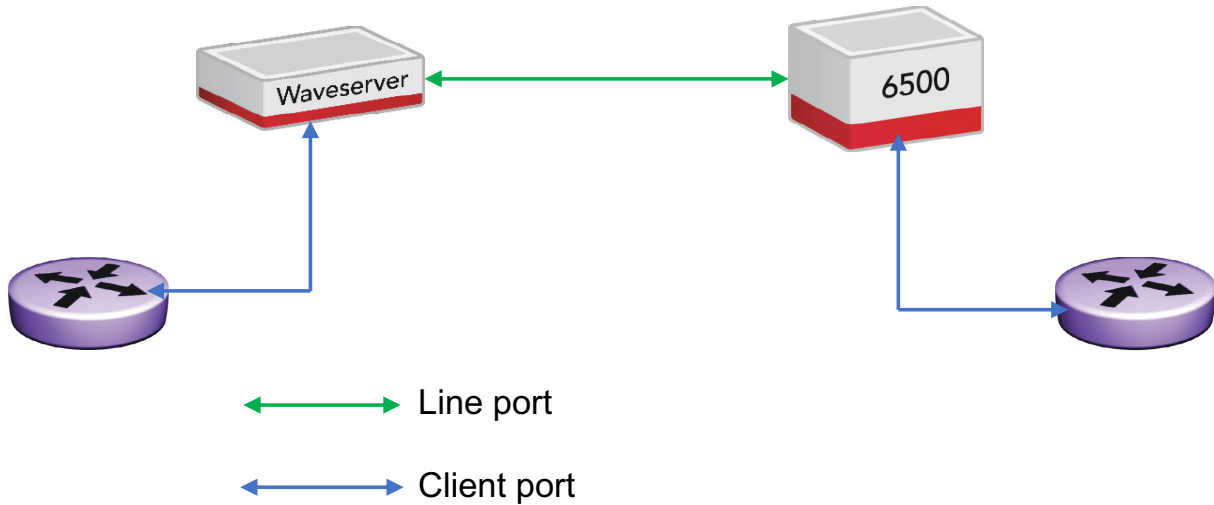
### Waveserver WL5e interworking with 6500 WL5e

This section outlines interworking between the:

- Waveserver WL5e module and the 6500 S-Series WL5e MOTR circuit pack as a transponder
- Waveserver WL5e module and the 6500 T-Series WL5e USSM as a transponder

See the following.

**Figure 122** Waveserver WL5e interworking with 6500 WL5e as a transponder



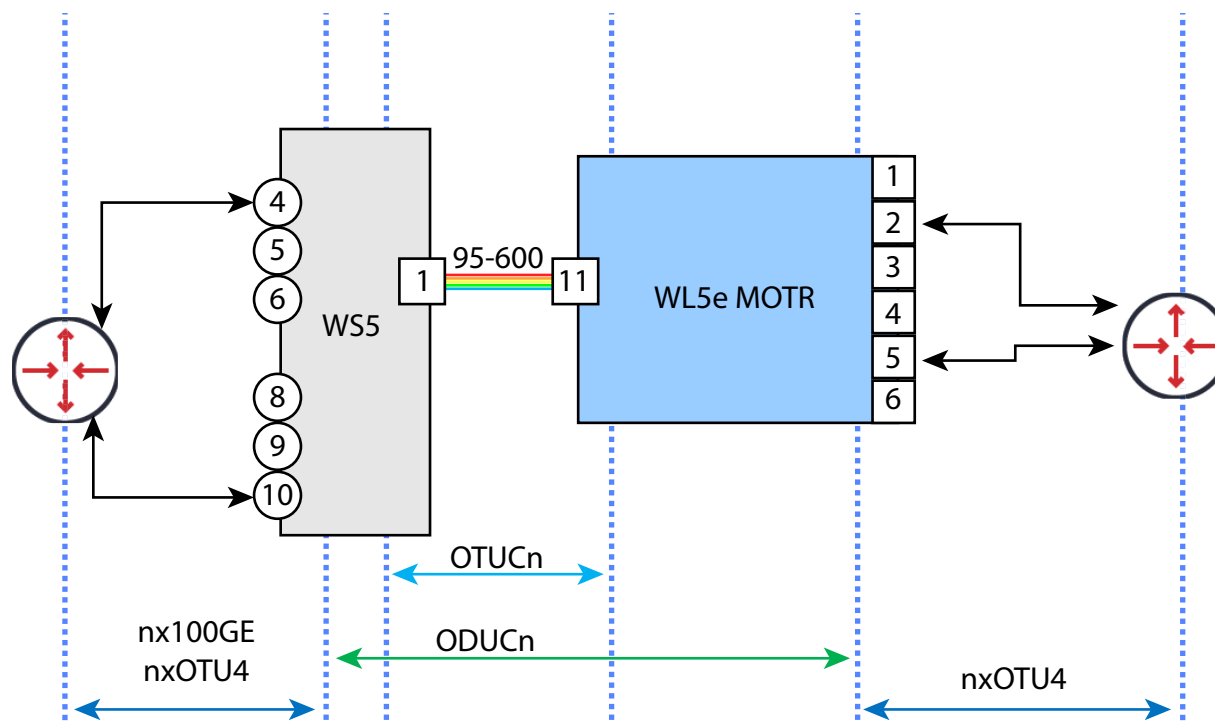
### Waveserver WL5e module and 6500 S-Series WL5e MOTR interworking

The following figure shows the Waveserver WL5e interworking with 6500 WL5e MOTR circuit pack as a transponder.

The configuration supports the following equipment:

- 6500 WL5e MOTR (1xQSFP-DD/5xQSFP28) circuit pack (NTK540BD)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD module (186-3102-901)
- 100GBASE-LR4, 4x25G, WDM, SMF, 1310nm, 10km QSFP28 pluggable (160-9401-900) for 100GE services
- 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 10km QSFP28 pluggable (160-9402-900) for OTU4 services

**Figure 123** Example of Waveserver WL5e interworking with 6500 WL5e MOTR circuit pack as a transponder



**Note:** Although the above figure shows 95-600G, 95-400G interworking is also supported.

The following mapping applies to the above configuration. The mapping is fixed, that is, cannot be provisioned.

- WS5 port 4 maps to 6500 WL5e MOTR port 6
- WS5 port 5 maps to 6500 WL5e MOTR port 5
- WS5 port 6 maps to 6500 WL5e MOTR port 4
- WS5 port 8 maps to 6500 WL5e MOTR port 3
- WS5 port 9 maps to 6500 WL5e MOTR port 2
- WS5 port 10 maps to 6500 WL5e MOTR port 1

The following interworking considerations apply:

- The configuration supports 95GBAUD\_600G\_O and 95GBAUD\_400G\_O transmission modes using Profile 1 or Profile 3 on the 6500 WL5e MOTR. Profile 1 is the default setting.
- Profile 1 supports up to six 100GE/OTU4 client facilities. 400GE client facilities are not supported.



- Profile 3 supports up to two 100GE/OTU4 client facilities and a 400GE client facility.
  - If a 400GE client facility is not required, 100GE facilities are supported in ports 1 to 6 and/or OTU4 facilities are supported in ports 1, 2, 4, 5, and 6.
- On the Waveserver WL5e module, you must use the Mixed-OTN-Ethernet service profile.
- The OTN Tributary Port (TP) values provisioned on the 6500 WL5e MOTR must be the same as the value used with the WS5. For example, use:
  - TP value = 2 for Tributary Slots (TS) = 1-20
  - TP value = 12 for TS = 21-40
  - TP value = 22 for TS = 41-60
  - and so on
- Waveserver WL5e does not support provisioning of "Tx Physical Layer ID" and "Tx Group ID" parameters for PTP facilities. On 6500, these parameters must be kept at their default values. Otherwise, traffic is not carried.
- The configuration supports all client traffic types, alarms, signal conditioning, performance monitoring, loopbacks, Integrated Test Set (ITS), and Trail Trace Identifier (TTI).
- MCP considerations
  - You can use MCP to add, delete and visualize the Photonic sub-network connections (SNC) or to create adjacencies for the SNC between the Waveserver WL5e and 6500 WL5e USSM.
  - You can use MCP to up-shift and down-shift the rate of the transmission mode for the line interface.
  - MCP does not support provisioning of transport client services for this configuration. Provisioning of these services must be performed using 6500 Site Manager and the Waveserver CLI.
  - Refer to MCP documentation for information on MCP provisioning.

### **Waveserver WL5e interworking with 6500 T-Series WL5e USSM as a transponder**

The following figure shows the Waveserver WL5e interworking with the 6500 WL5e USSM as a transponder for 95-400G interworking.

The configuration supports the following equipment:

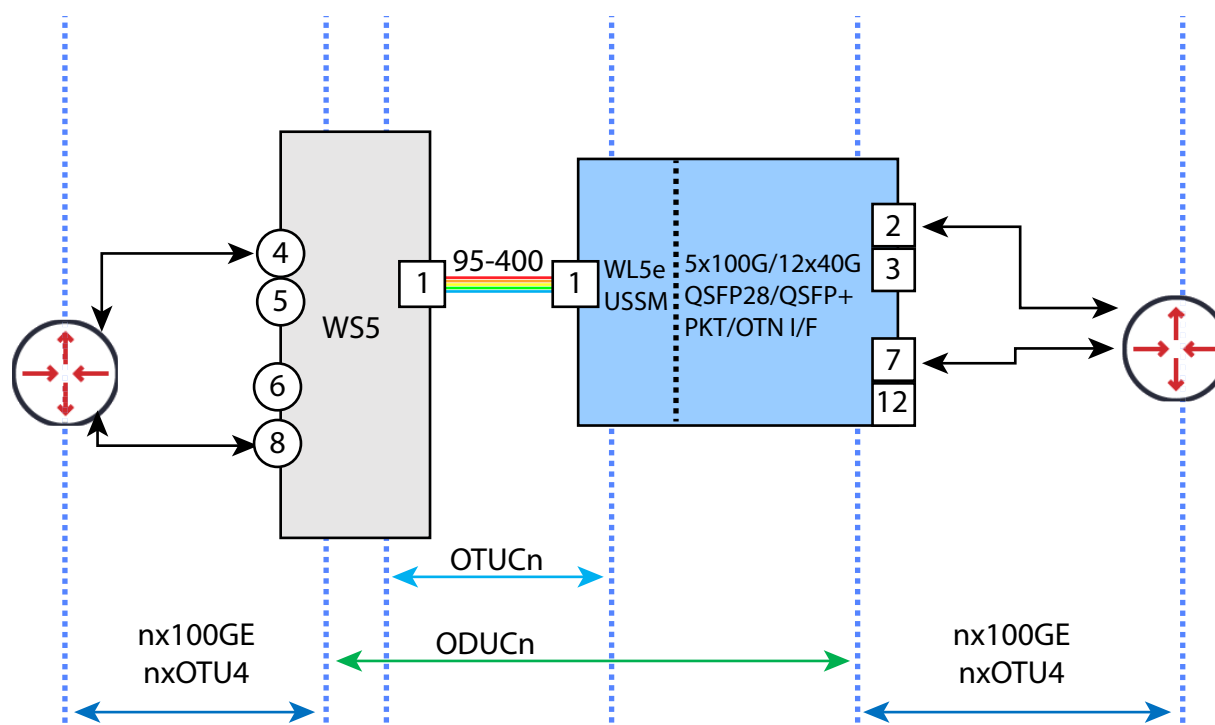
- 6500 WL5e USSMs
  - 1xWaveLogic 5e Enhanced C-Band Subslot Module (174-0662-900)

— 1xWaveLogic 5e Premium C-Band Subslot Module (174-0663-900)

The WL5e USSM must be housed in the 500G 2xUSS/2xQSFP28 module (NTK762VA).

- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD module (186-3102-901)
- 100GBASE-LR4, 4x25G, WDM, SMF, 1310nm, 10km QSFP28 pluggable (160-9401-900) for 100GE services
- 103.1G-111.8G, 4x25G, WDM, SMF, 1310nm, 10km QSFP28 pluggable (160-9402-900) for OTU4 services

**Figure 124** Example of Waveserver WL5e interworking with 6500 WL5e USSM as a transponder for 95-400G interworking



The following interworking considerations apply:

- The configuration supports 95GBAUD\_400G\_O transmission modes.
- The configuration supports 100GE and OTU4 client protocols.
- On the Waveserver WL5e module, you must use the Mixed-OTN-Ethernet service profile.
- On the 6500 WL5e USSM, use Profile1 as the tributary slot profile. Profile1 supports a mix of 100GE and OTU4 clients.
- The OTN Tributary Port (TP) values provisioned on the 6500 WL5e USSM must be the same as the value used with the WS5. For example, use:

- TP value = 2 for Tributary Slots (TS) = 1-20
- TP value = 12 for TS = 21-40
- TP value = 22 for TS = 41-60
- and so on
- The above figure shows 100GE and OTU4 client services dropped at the local 6500 T-Series node but it is also supported to carry these services on the OTN Control Plane network (if applicable) and have them dropped at remote OTN Control Plane nodes.
- Waveserver WL5e does not support provisioning of "Tx Physical Layer ID" and "Tx Group ID" parameters for PTP facilities. On 6500, these parameters must be kept at their default values. Otherwise, traffic is not carried.
- The configuration supports all client traffic types, alarms, signal conditioning, performance monitoring, loopbacks, and Trail Trace Identifier (TTI).
  - Integrated Test Set (ITS) sessions using Network Mode for 100GE ports on the 6500 T-Series are not supported for this configuration. If required, use race track mode on the Waveserver WL5e node and loopbacks at the end points to establish an ITS session.
  - GCC comms are not supported between the Waveserver WL5e and the 6500 WL5e USSM.
  - TCM provisioning is not supported between the Waveserver WL5e and the 6500 WL5e USSM.
- MCP considerations
  - You can use MCP to visualize the Photonic sub-network connections (SNC) between the Waveserver WL5e and 6500 WL5e USSM. However, you cannot use MCP to add or delete the Photonic SNC or to create adjacencies for the SNC.
  - You can use MCP to up-shift and down-shift the rate of the transmission mode for the line interface.
  - MCP does not support provisioning of transport client services. Provisioning of these services must be performed using 6500 Site Manager and the Waveserver CLI.
  - Refer to MCP documentation for information on MCP provisioning.

### **SPLI with 6500**

Waveserver 5 supports Service Photonic Layer Interoperability (SPLI) with a 6500 photonic line system. SPLI communicates DWDM channel provisioning information between a device (for example a transponder or line module) and the adjacent 6500 photonics line system. For Waveserver 5, SPLI facilitates

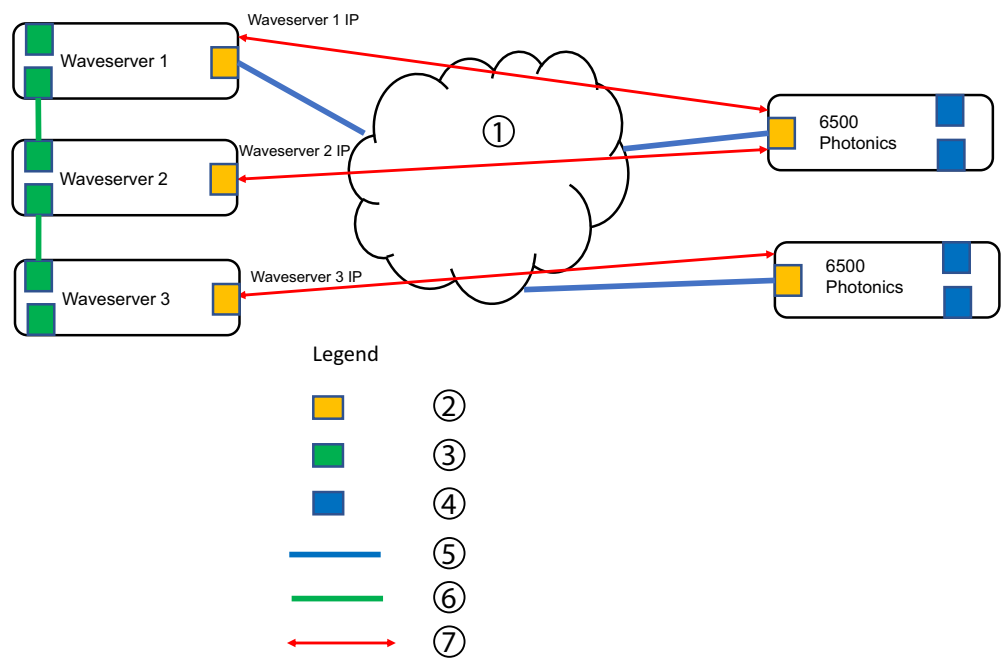
tighter integration of its line ports with the 6500 Release 12.6 (and higher) photonics control functions to reduce OAM costs associated with wavelength configuration and maintenance. Modem provisioning is sent from the Photonic line SPLI (6500) to the service layer SPLI ( Waveserver 5) after the transmitter type is discovered by 6500. SPLI is enabled by default on the line ports of the Waveserver 5.

**Note 1:** The 6500 must be running Release 12.6 or higher.

**Note 2:** Only IPv4-based SPLI is supported for this release.

To support communications in a SPLI-enabled system, you must connect Waveserver 5 to the DCN either through the ILAN (see [ILAN linear configuration](#) and [ILAN ring configuration and loop mitigation](#)) or directly ([Direct DCN connection](#)). See the following figures.

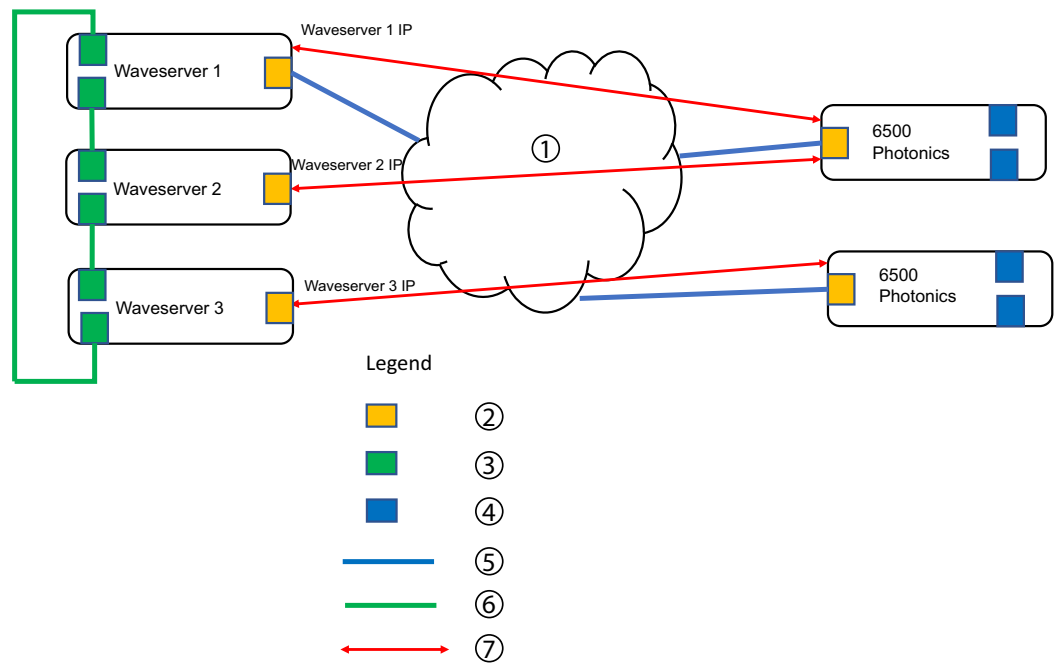
**Figure 125** SPLI communications between Waveserver 5 and 6500; interconnected Waveserver 5 ILAN linear — DCN connection



**Legend**

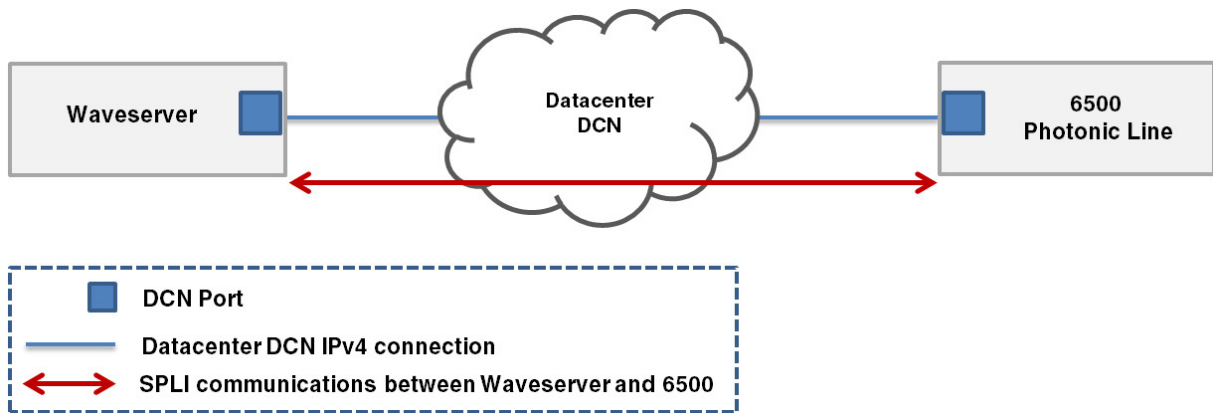
- |   |                       |   |                                        |
|---|-----------------------|---|----------------------------------------|
| 1 | Datacenter DCN        | 5 | Datacenter DCN Connection (IPv4)       |
| 2 | DCN port              | 6 | Waveserver ILAN daisy chain connection |
| 3 | Waveserver ILAN ports | 7 | SPLI Communication using 6500 DCN port |
| 4 | 6500 ILAN ports       |   |                                        |

**Figure 126** SPLI communications between Waveserver 5 and 6500; interconnected Waveserver 5 ILAN ring — DCN connection



| Legend |                                        |
|--------|----------------------------------------|
| 1      | Datacenter DCN                         |
| 2      | DCN port                               |
| 3      | Waveserver ILAN ports                  |
| 4      | 6500 ILAN ports                        |
| 5      | Datacenter DCN Connection (IPv4)       |
| 6      | Waveserver ILAN daisy chain connection |
| 7      | SPLI Communication using 6500 DCN port |

**Figure 127** SPLI communications between Waveserver 5 and 6500; direct — DCN connection



SPLI connections between Waveserver 5 and 6500 use Secure Socket Shell (SSH) over TCP. The designated SPLI port is port 20040.

**Note:** On 6500, SSH must be provisioned to use the HMAC-SHA2 algorithm.

For information about provisioning SPLI communications on 6500, see *6500 Administration and Security*, 323-1851-301. For information about provisioning SSH on 6500, see *6500 Data Communications Planning and User Guide*, 323-1851-101.

### Waveserver 5 and 6500 SPLI information sharing

When the discovery process completes, the Waveserver 5 shares the following information with the 6500:

- Transmitter type parameters:
  - Modulation scheme/transmission mode of modem
  - Platform type
  - Modem type
- Ciena product code of the Waveserver 5
- Modem Output
- Line System Configuration - SPLI Management
- Transmitter Wavelength (nm)
- Tx Tuned
- Tx Power (dBm)
- Tx Power Reduction State
- Frequency (GHz)
- Frequency Resolution (GHz)
- Laser Control - Centering Mode
- Laser Control - Centering Range (GHz)
- Tx Minimum Spectral Width (GHz)
- Tx Signal Bandwidth 10dB (GHz)
- Tx Signal Bandwidth 3dB (GHz)
- Link Optimization Mode
- Tx Pre-compensation Mode
- Tx Dispersion Compensation Management - Tx Path Dispersion (ps/nm)
- Tx Dispersion Compensation Management - Tx Precompensation (ps/nm)

In a SPLI-enabled system, the 6500 communicates DWDM channel provisioning information to the Waveserver 5 it is matched with. The following attributes are under the SPLI control software:

- Actual Tx Power (dBm)
- ITU Center Wavelength (nm)
- OCH Line System Type
- Tx Power Reduced State
- Laser Centering
- Center Frequency (THz)
- Provisioned Transmission Mode
- Allocated NMC Spectral Width (GHz)
- Discovered Path Dispersion (ps/nm)
- Discovered Line Optimization Mode
- Discovered Transmitter Dispersion Mode
- Discovered Transmitter Dispersion Value (ps/nm)
- Expected Restoration Time (seconds)

#### Waveserver 5 and 6500 SPLI comms requirements

You must provision an IPv4 IP address on the Waveserver 5 (see [Setting up management access using the console or USB port](#)).

On the 6500, these comms rules apply:

- The 6500 must be DCN routable (public) and its IPv4 IP address must be visible to the Waveserver 5. Private configurations are not supported, such as Private-IP and Redundant NAT. For more information on supported GNE configurations with SPLI, see [Supported GNE configurations for Waveserver 5 SPLI](#).
- The COLAN-X port is normally used, but not required. The 6500 ILAN port can be used as a DCN port, provided that the shelf-IP is DCN-routable.

**Table 184** Supported GNE configurations for Waveserver 5 SPLI

| GNE configuration              | Supported for Waveserver 5 SPLI |
|--------------------------------|---------------------------------|
| Individual DCN drops           | Yes                             |
| Statically routed              | Yes                             |
| Proxy ARP (single, redundant)  | Yes                             |
| OSPF (single, redundant)       | Yes                             |
| Private IP (single, redundant) | No                              |

| GNE configuration | Supported for Waveserver 5 SPLI |
|-------------------|---------------------------------|
| Redundant NAT     | No                              |

### SPLI setting engineering considerations

The following SPLI setting engineering considerations apply:

- The MTU is 1500 bytes.
- Number of IP hops < max TTL (90)
- 100 Base-T Full Duplex is recommended on the 6500. Waveserver 5 supports up to 1000 Base-T.
- If external firewalls exist, TCP port 20040 must be allowed.
- The maximum number of Waveserver 5s communicating with a single 6500 is 100.

**Note:** When using SPLI, Ciena recommends that you manually assign a static IPv4 address to the Waveserver 5, instead of letting DHCP assign the IPv4 address automatically. Doing so prevents the need to re-provision the 6500 SPLI Table to reestablish SPLI communication with Waveserver 5, in the case where DHCP automatically provisions it with an IPv4 address.

### SPLI database

After SPLI adjacencies have been created on 6500, you can display the SPLI database on the Waveserver 5 using CLI command `spli database show`.

### SPLI and service modules in mated mode

When using SPLI with a service module in mated mode, SPLI blocks provisioning to the service module until the transmission mode and link optimization mode provisioned are the same between the line ports. If there is a mismatch between the line ports, the “Adjacency Mismatch” alarm is raised on the 6500. If the user changes the module mode from mated to none on the Waveserver 5, the SPLI provisioning will be automatically unblocked and applied on the service module.

### Fast tuning support

6500 (Release 12.4 and higher) introduces L0 Control Plane Mesh Restoration Speed Improvements to enable faster traffic capacity changes and quicker L0 Control Plane Mesh Restoration times in comparison to previous releases. For this feature, channel capacity modes “Mode-1”, “Mode-2” and “Mode-3” are introduced. When a new channel is created on the 6500 (Release 12.4 or higher) with channel capacity mode “Mode-1”, the ADJ-TX Tuning mode will be set to ‘Accelerated Tuning’ and pushed to the modem (**Waveserver 5**) via SPLI. Similarly, when a new channel is



created with channel capacity modes “Mode-2” or “Mode-3”, the ADJ-TX Tuning mode will be set to 'Fast Tuning' and pushed to the modem (Waveserver 5) via SPLI.

### SPLI with RLS

Waveserver 5 supports Service and Photonic Layer Interoperability (SPLI) with the RLS Photonics. SPLI communicates DWDM channel provisioning information between a Waveserver 5 transponder or line module and the adjacent RLS CMD. For Waveserver 5, SPLI facilitates tighter integration of its line ports with the RLS Photonics to reduce OAM costs associated with wavelength configuration and maintenance. Modem provisioning is sent from the Photonic line SPLI (RLS) to the service layer SPLI (Waveserver 5) after the transmitter type is discovered by RLS. SPLI is enabled by default on RLS.

To support SPLI between RLS and Waveserver 5:

- RLS must be running Release 2.3 or higher
- Waveserver 5 must be running Release 2.1 or higher

**Note 1:** On lower releases, you can use the Wave-Line Synchronizer application within MCP to synchronize the settings between interconnected modems and Photonic lines. Refer to MCP documentation for details.

**Note 2:** SPLI is enabled by default on RLS starting with Release 2.3. However, if Waveserver 5 was connected to RLS prior to RLS Release 2.3, and if managed by Wave-Line Synchronizer, the connection will continue to be managed by Wave-Line Synchronizer after RLS upgrades to Release 2.3.

Only IPv4-based SPLI is supported for this release.

To establish a connection:

- On RLS, the Waveserver 5 platform must be provisioned as a neighbor of the RLS shelf. The platform type must also be provisioned to indicate waveserver.
- On RLS, the far-end address of the connected transponder port must be provisioned. SPLI uses the far-end address and neighbor settings to establish a match. On Waveserver 5, discovered far end addresses to RLS show the shelf number as 0.
- On RLS, provision the link from CMD to the Waveserver. Ensure that the “to-label” and “from-label” point to the Waveserver transponder. This must match the name of the Waveserver in the neighbour list. In addition to the node name, the “to-label” and “from-label” must contain the port information in the format “NODENAME-SL-PRT” format.

Once a match is established, SPLI autoprovisions certain parameters on the transponder and certain transponder discovery parameters on the RLS. You can display the SPLI database on the Waveserver 5 using the CLI command `spli database show`.

SPLI connections between Waveserver 5 and RLS use Secure Socket Shell (SSH) over TCP. The designated SPLI port is port 20040.

**Note:** On RLS, SSH must be provisioned to use the HMAC-SHA2 algorithm.

### Tuning mode support

On RLS, the `tx-proxy` tuning mode parameter is set to fast by default. Since Waveserver does not support fast tuning mode, you must set the RLS tx-proxy tuning mode parameter to accelerated. Use the following command from the RLS CLI: `set slots config tx-proxy <name> parent-port <Value> tuning-mode accelerated`.

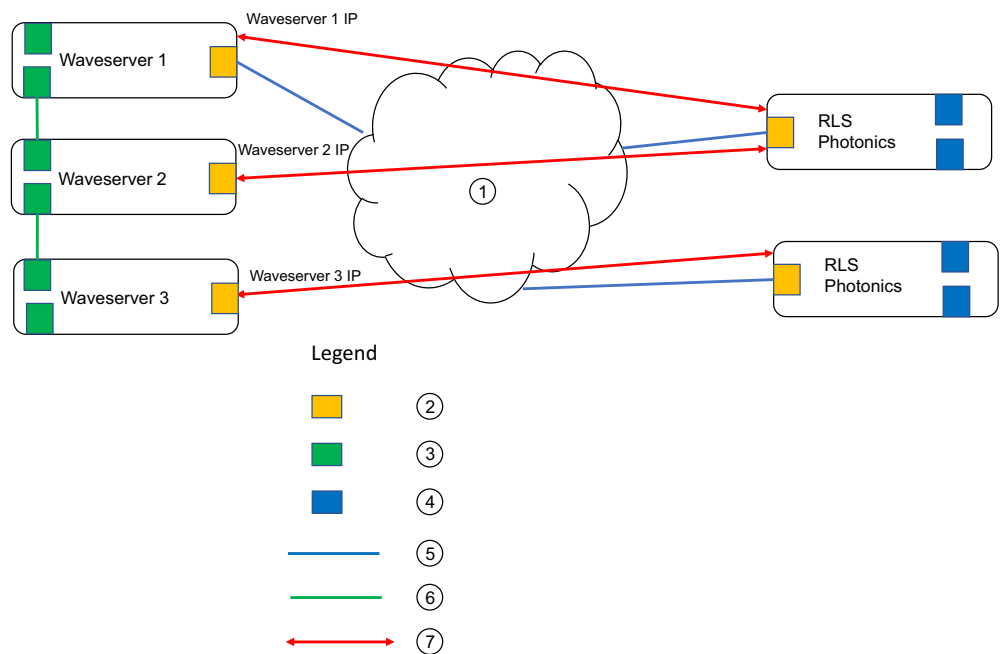
RLS raises a data mismatch alarm when SPLI is managing a connection and the tuning mode parameters do not match.

### SPLI communications between Waveserver 5 and RLS

To support communications in a SPLI-enabled system, you must provision an IPv4 IP address on the Waveserver 5 and connect the Waveserver 5 to the DCN either through the ILAN or directly. See the following figures.

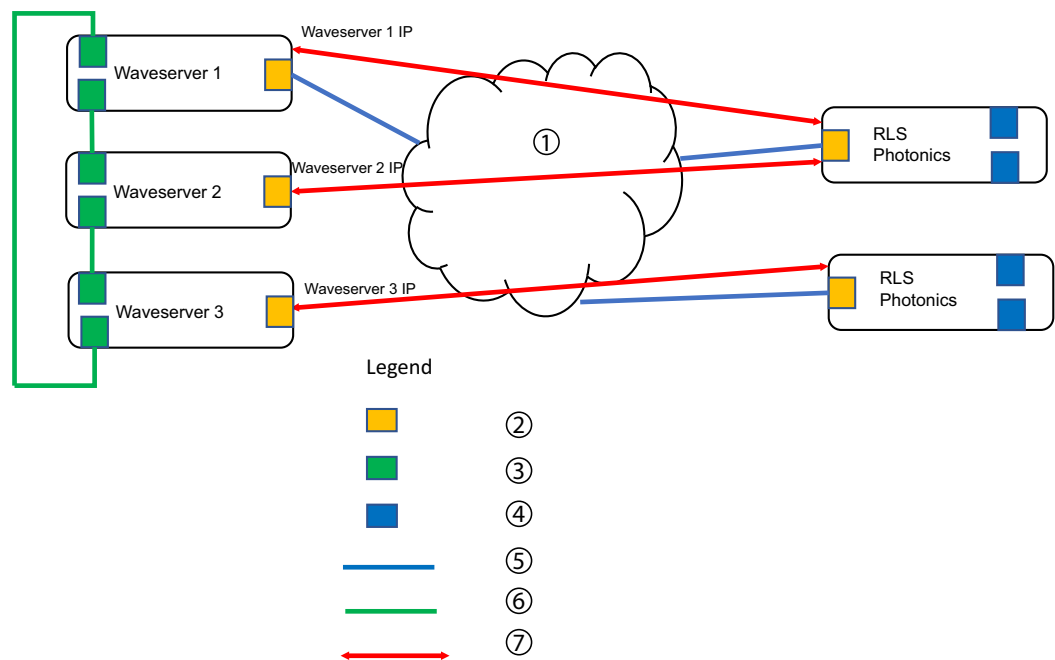
When using SPLI, Ciena recommends that you manually assign a static IPv4 address to the Waveserver 5, instead of letting DHCP assign the IPv4 address automatically. Doing so prevents the need to re-provision the RLS SPLI table to reestablish SPLI communication with Waveserver 5, in the case where DHCP automatically provisions it with an IPv4 address.

Figure 128 SPLI communications between Waveserver 5 and RLS: ILAN linear DCN connection



| Legend |                       |   |                                        |
|--------|-----------------------|---|----------------------------------------|
| 1      | Datacenter DCN        | 5 | Datacenter DCN Connection (IPv4)       |
| 2      | DCN port              | 6 | Waveserver ILAN daisy chain connection |
| 3      | Waveserver ILAN ports | 7 | SPLI Communication using RLS DCN port  |
| 4      | RLS ILAN ports        |   |                                        |

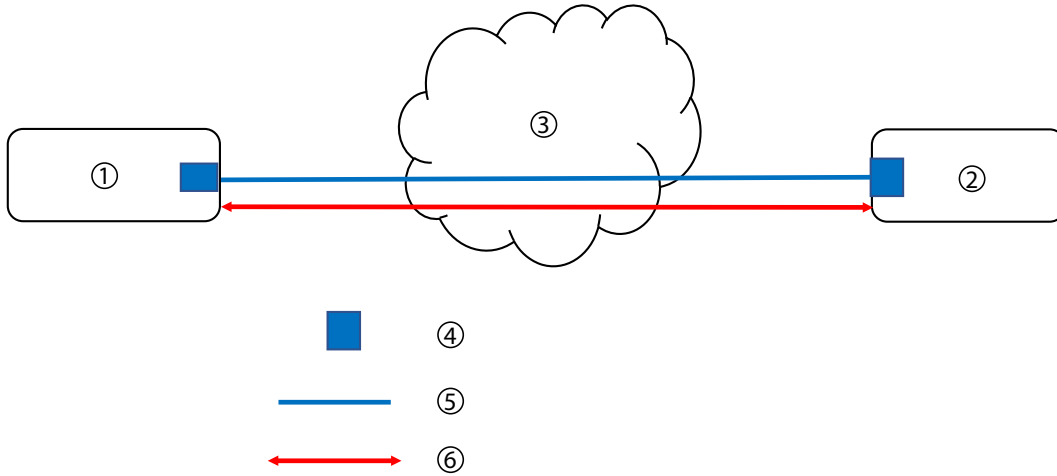
Figure 129 SPLI communications between Waveserver 5 and RLS: ILAN ring DCN connection



- Legend**

|   |                       |   |                                        |
|---|-----------------------|---|----------------------------------------|
| 1 | Datacenter DCN        | 5 | Datacenter DCN Connection (IPv4)       |
| 2 | DCN port              | 6 | Waveserver ILAN daisy chain connection |
| 3 | Waveserver ILAN ports | 7 | SPLI Communication using RLS DCN port  |
| 4 | RLS ILAN ports        |   |                                        |

**Figure 130** SPLI communications between Waveserver 5 and RLS: direct DCN connection



**Legend**

|   |                |   |                                               |
|---|----------------|---|-----------------------------------------------|
| 1 | Waveserver     | 4 | DCN Port                                      |
| 2 | RLS Photonics  | 5 | Datacenter DCN Connection (IPv4)              |
| 3 | Datacenter DCN | 6 | SPLI Communication between Waveserver and RLS |

**Waveserver 5 and RLS SPLI information sharing**

When the discovery process completes, the Waveserver 5 and RLS share the following information.

**Transponder information discovery parameters**

- Discovered Tx Power (`total-target-tx-power`)
- Discovered Frequency (`center-freq`)
- Discovered Transmission Mode (`transmission-mode`)
- Discovered Link Optimization Mode (`link-optimization-mode`)
- Discovered Path Dispersion (`path-chromatic-dispersion`)
- Discovered Tx Blanking (`blanking`)
- Discovered Laser Centering Mode (`laser-centering-mode`)
- Discovered Tuning Mode (`tuning-mode`)
- Discovered CCDA (`ccda-enabled`)
- Discovered Tuning State (`tuning-state`)
- Discovered Minimum Spectrum Width (`tx-minimum-spectral-width`)

- OCH\_DOWN (och-fault)
- OCH\_DEGRADE (och-degrade)

### Photonics parameters

- Provisioned Tx Power (total-target-tx-power)
- Frequency (center-freq)
- Transmission Mode (transmission-mode)

**Note:** The transmission mode on the SPLI enabled Waveserver PTPs expects the provisioning to be in the following text format: <BaudRate>GBAUD\_<BitRate>\_<Mapping>. The decimal section of the BaudRate must be indicated by the separator 'P'. For example, "95-800-O" transmission mode expects the SPLI port provisioning from the Photonics side to be "95GBAUD\_800G\_O". "91.6-200-E" transmission mode expects the SPLI port provisioning from the Photonics side to be "91P6GBAUD\_200G\_E". "93.3-400-E" expects "93P3GBAUD\_400G\_E".

- Link Optimization Mode (link-optimization-mode)
- Path Dispersion (path-chromatic-dispersion)
- Tx Tuning Mode (tuning-mode)
- Tx Blanking (blanking)
- Laser Centering Mode (laser-centering-mode)
- Channel Contention Detection and Avoidance (CCDA) (ccda-enabled)
- Tx Tuning Units/Wavelength Spacing (tuning-units)
- Tuning grid (tuning-grid)
- Expected Restoration Time/Rx Acquisition Hold Off (rx-acquisition-hold-off)
- Line System Type (not configurable)
- Dispersion Mode (not configurable)

### Connection validation

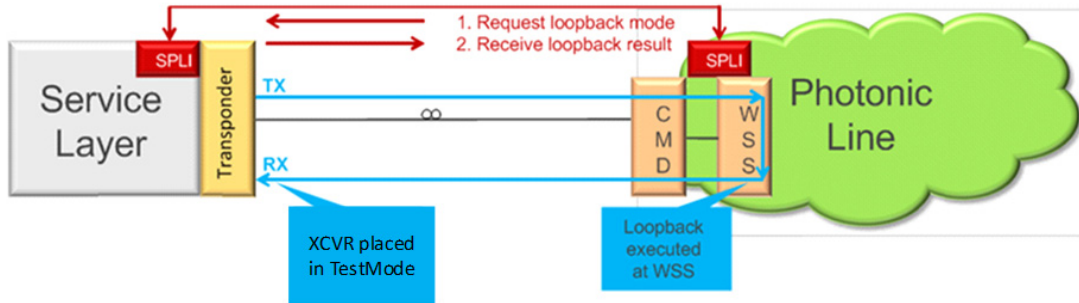
Connection validation allows you to validate fiber inter-connections between line-side Waveserver transponder PTPs and the Photonic CMD / WSS. Connection validation implements an optical loopback test at the WSS, which verifies that the transponder is properly connected and functional. The test loops the signal back to the originating transponder, allowing connectivity validation and optical power measurements on the line.

Connection validation is supported between Waveserver WL5e service modules and the 6500 Photonic line. SPLI must be enabled.

Connection validation is user-initiated from the 6500 Photonic line, using an optical loopback test. The optical loopback test signals Waveserver to initiate validation over the SPLI interface. For information on initiating connection validation from the 6500 Photonic line, refer to *6500 Configuration - Provisioning and Operating*, 323-1851-310.

Refer to the following figure.

**Figure 131** Connection validation



#### CAUTION

Connection validation is service-affecting. Line-side traffic is down for the duration of the operation. The local receiver does not receive valid traffic. All connected client ports are operationally down and conditioned appropriately.

## Procedure list

This section provides the following procedures:

- [Procedure 92, “Configuring 6500-based SLTE configurations using Waveserver 5 wayside channel”](#)
- [Procedure 93, “Enabling or disabling SPLI”](#)
- [Procedure 94, “Adding Waveserver 5 information to the 6500 SPLI list”](#)
- [Procedure 95, “Adding Waveserver 5 information to the RLS SPLI list”](#)
- [Procedure 96, “Retrieving SPLI information”](#)
- [Procedure 97, “Managing the SPLI database”](#)
- [Procedure 98, “Configuring client-layer OPS protection”](#)

## Procedure 92 Configuring 6500-based SLTE configurations using Waveserver 5 wayside channel

Use this procedure to set up Waveserver 5 and 6500 for 6500-based SLTE configurations using the Waveserver 5 wayside channel.

### Requirements

- You must be logged into all applicable Waveserver 5 CLI sessions.
- You must be logged into all applicable 6500 Site Manager sessions.
- You must have a LAN cable available.
- You are using Waveserver 5 DCN-2 for wayside-channel mode.

The Waveserver 5 wayside channel allows 6500 SLTE shelves on either side of the submarine link to communicate for the purposes of managing that link. For more information, see [Comms channel for 6500-based SLTE configurations](#).

**Note:** Waveserver 5 uses single-slice GCC mode to allow interoperability between different service profiles.

### Steps

#### Configuring Waveserver 5 DCN ports

- 1 Administratively enable the DCN-2 ports:  
`chassis port enable port dcn-2`
- 2 Verify that the Waveserver 5 DCN ports are administratively enabled:  
`chassis port show`

#### Configuring the 6500 ILAN/COLAN ports

- 3 In Site Manager select the required network element in the navigation tree, then select **Comms Setting Management** from the **Configuration** menu.
- 4 Select the **Interfaces** tab and select the 6500 shelf number corresponding to the shelf that will be connected to the Waveserver 5, then select **LAN** as the Interface Type.
- 5 Click the **Add** button. Select the appropriate 6500 LAN port (ILAN/COLAN) in the Name field. Ensure that the Configuration is set to Automatic and click **OK**. For more information, see *6500 Data Communications Planning and User Guide*, 323-1851-101.
- 6 Select **IP** from the **Interface** Type drop-down list, then click **Add** and select the appropriate 6500 LAN port (ILAN/COLAN) from the list. Ensure that the IP address is routable to the DCN. Ciena recommends that the IP address have a /30 subnet. The IP address must be unique and must not overlap with existing COLAN or Shelf IP addresses.



- 7 Add the 6500 LAN port to the OSPF area, by selecting the **Routers** tab and selecting OSPF Circuit from the **Router Type** drop-down list.
- 8 Click **Add** and select the appropriate 6500 LAN port.
- 9 In the Add OSPF Circuit Parameters dialog box, set the required Network Area. Set other OSPF parameters, such as Cost and Authentication Type, appropriately. Ensure that Opaque Link State Advertisement is set to **On**. Click **OK**. See *6500 Data Communications Planning and User Guide*, 323-1851-101 and *6500 Configuration - Control Plane*, 323-1851-330 for details.
- 10 Complete [step 3](#) to [step 9](#) on both the near-end and far-end 6500 nodes. Once completed, go to next step.

**Connecting the LAN cables**

- 11 At both the near-end and far-end sites, connect a LAN cable between the designated 6500 ILAN or COLAN port and the Waveserver 5 DCN port to be used for wayside access. Verify that the link negotiates to 100BT Full-duplex.

**Verifying traffic**

- 12 In Site Manager, open the IP Routing table and verify that the 6500 NE is receiving communications through the specified LAN port.

**Configuring OSRP**

- 13 Set the OSRP to use out-of-band communications. See *6500 Data Communications Planning and User Guide*, 323-1851-101 and *6500 Configuration - Control Plane*, 323-1851-330 for details.

## Procedure 93 Enabling or disabling SPLI

Use this procedure to set the global SPLI flag, to set the SPLI flag of the modem transceiver (PTP), or to retrieve a summary of SPLI global, remote shelf, and local port parameters.

**Requirements**

- To disable or enable SPLI, you must be logged in to Waveserver 5 using an account with at least admin access privileges.

**Overview**

The SPLI feature is enabled on a port only when both of these conditions exist:

- The global SPLI flag is enabled
- The port SPLI flag is enabled

**Note:** If you disable SPLI globally and then re-enable it, you must re-enable each Waveserver 5 modem transceiver (PTP) individually to establish a SPLI match.

**Steps**

| 1                                                                | <table><tr><th>If you want to</th><th>Then go to</th></tr><tr><td>set the global SPLI flag on Waveserver 5</td><td><a href="#">step 2</a></td></tr><tr><td>enable or disable SPLI on a Waveserver 5 modem transceiver (PTP)</td><td><a href="#">step 3</a></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | If you want to | Then go to | set the global SPLI flag on Waveserver 5 | <a href="#">step 2</a> | enable or disable SPLI on a Waveserver 5 modem transceiver (PTP) | <a href="#">step 3</a> |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|------------------------------------------|------------------------|------------------------------------------------------------------|------------------------|
| If you want to                                                   | Then go to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                |            |                                          |                        |                                                                  |                        |
| set the global SPLI flag on Waveserver 5                         | <a href="#">step 2</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |            |                                          |                        |                                                                  |                        |
| enable or disable SPLI on a Waveserver 5 modem transceiver (PTP) | <a href="#">step 3</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |            |                                          |                        |                                                                  |                        |
| 2                                                                | <p>To set the global SPLI flag on Waveserver 5, enter the command:</p> <pre>spli &lt;disable   enable&gt;</pre> <p>The global SPLI flag is enabled by default. Disabling the global SPLI flag on the Waveserver 5 will disable SPLI control by the matching Photonic line. This operation will also automatically disable the modem transceiver (PTP) SPLI flag for ports 1 and 2. Enabling or disabling the global SPLI flag is not service-affecting. Ciena recommends that SPLI be disabled during commissioning if your data center application does not require this functionality. For information, see <a href="#">Provisioning Waveserver 5</a>.</p> <p>Go to <a href="#">step 4</a>.</p> <p><b>Note:</b> Enabling or disabling the global SPLI flag on Waveserver 5 triggers a reapplication of the firewall rules, resulting in all users being temporarily blocked. As a result, subsequent commands may time out or take longer to complete, or sessions may be dropped and require re-connection. If TACACS+ is enabled, Waveserver 5 may generate a "waiting for accounting reply" message.</p> |                |            |                                          |                        |                                                                  |                        |
| 3                                                                | <p>To enable or disable SPLI on a Waveserver 5 modem transceiver (PTP):</p> <pre>ptp set ptp &lt;slot&gt;/&lt;ptp&gt; spli &lt;disabled   enabled&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |            |                                          |                        |                                                                  |                        |

SPLI is enabled by default on each modem transceiver. Disabling or enabling SPLI on a modem transceiver is not service affecting. Disabling SPLI on a modem transceiver disables SPLI control for that modem transceiver only.

- 4 To save provisioned SPLI settings to the configuration file:  
`configuration save`

## Procedure 94 Adding Waveserver 5 information to the 6500 SPLI list

Use this procedure to add a Waveserver 5 node name to the 6500 SPLI list and provision the Waveserver 5 line port as Far End Address in 6500's ADJ facility.

### Requirements

- Have the IPv4 address(es) of the Waveserver 5 node(s).
- If you are using Site Manager, have Site Manager software installed on your craft computer for this 6500 software load release.
- Use an account with a UPC level of 4 or higher.
- On the Waveserver 5, SPLI must be enabled globally and on the PTP connecting to the 6500. See [Enabling or disabling SPLI](#).
- For more information on Waveserver 5 and SPLI information sharing, see [Waveserver 5 and 6500 SPLI information sharing](#).
- This procedure refers to CLI commands. For more information on CLI command syntax, refer to *6500 CLI Reference*, 323-1851-193.

**Note:** The Waveserver 5 node name will be added on the 6500 NE to which the Waveserver 5 line port is connected. That is, in a TIDc configuration, if the Waveserver 5 node is connected to the CMD on the member shelf, the Waveserver 5 node name will be added to the member shelf instead of the primary shelf.

### Steps

- 1 Log into the 6500 NE using Site Manager or use the CLI, as applicable.
- 2 Add the IPv4 address(es) of the Waveserver 5 node name(s) to the SPLI list of the 6500 node as follows:
  - If you are using Site Manager, refer to the procedure on adding SPLI entries in *6500 Administration and Security*, 323-1851-301.
  - If you are using the CLI, enter:
 

```
adjacencies spli create SHELF-<shelf number> fend-ip-addr
<FEIPADD> spli-id <TID>
```

 where
    - FEIPADD is the IPv4 address of the Waveserver 5 node being added to the 6500 SPLI list. Surround the IP address with double-quotes.

- TID is a string defined by the user to represent the Waveserver 5 node. Surround the string with double-quotes. It is recommended that the Waveserver 5's member-name string be used as TID.

**Note:** DHCP is not recommended to be used on Waveserver 5 when SPLI is used as the Waveserver 5 could get assigned a different IPv4 address through DHCP. When that occurs, the user would have to return to this step to manually update the IPv4 address provisioned in the 6500 SPLI table.

**3** Provision the Waveserver 5 line port as Far End Address in 6500's ADJ facility as follows.

- If you are using Site Manager, refer to 6500 *Commissioning and Testing*, 323-1851-221, and *Configuration - Provisioning and Operating*, 323-1851-310 for more information.

**Note:** Refer to the following for information on the supported Waveserver 5 far end addresses that can be entered in Site Manager far end address builder dialog.

- If you are using the CLI, enter:

```
adjacencies adjacency set <AID> prov-addr-form <prov-addr-form> prov-fe-addr <prov-fe-addr>
```

where

- <AID> is the CMD adjacency in the form ADJ-SH-SL-PORT for the CMD port connected to the Waveserver 5.

- <prov-addr-form> defines the format of the far end address. For example, 'NODENAME-SL-PRT' for Waveserver 5 node name-Slot-Port.

- <prov-fe-addr> is a string defined by the user to represent the far end address of the Waveserver 5 in the form TID-Slot-Port.

For Waveserver 5, the Slot can be '1', '3', '5' or '7' and the Port is either '1' or '2'

For example, slot-port is '1-1', '1-2', '3-1', '3-2', '5-1', '5-2', '7-1', '7-2'.

If following recommendations, the TID is the Waveserver 5's member-name string. Surround the string with double-quotes.

For example, if connecting slot 3 port 2 of a Waveserver 5 with a TID of 'WSTEST' to port 26 of the 6500 CMD in shelf 2 slot 83, you would enter:

```
adjacencies adjacency set ADJ-3-83-26 prov-addr-form
NODENAME-SL-PRT prov-fe-addr "WSTEST-3-2"
```

---

## Procedure 95 Adding Waveserver 5 information to the RLS SPLI list

---

Use this procedure to add a Waveserver 5 node name to the RLS SPLI list and to provision the Waveserver 5 line port as a far end address on the RLS CMD Tx adjacency facility.

### Requirements

- Have the IPv4 addresses of the Waveserver 5 nodes.
- Globally enable SPLI on the Waveserver 5 and on the PTP connecting to the RLS. See [Enabling or disabling SPLI](#).
- Log into the RLS CLI. Refer to the *6500 RLS User Guide*, 323-2051-100.

### Steps

- 1 From the RLS CLI, add Waveserver 5 node names to the RLS SPLI neighbor list using the following command:
  - `set config-neighbours <remote-node> ip-address <remote-ip-addr> platform-type ciena-waveserver`  
`<remote-node>` is the Waveserver 5 node name  
`<remote-ip-addr>` is the IPv4 address of the Waveserver 5 node  
For example:  
`set config-neighbours HANWS5-A ip-address 1.2.3.4 platform-type ciena-waveserver`
- 2 Verify the neighbor list using the following command: `show config-neighbours`

- 3** From the RLS CLI, provision two half-links (one link provisioned with only a 'To' port and another link provisioned with only a 'From' port) between the Waveserver 5 line port and the RLS CMD using the following commands:

- ```
create link <from_link_name> from "slots <slot#>
config port <switch-out port#>" to-label <text>
create link <to_link_name> to "slots <slot#> config
port <switch-in port#> from-label <text>
```

<from_link_name> is a unique user-assigned name for the from-link
<to_link_name> is a unique user-assigned name for the to-link
<slot#> identifies the RLS slot
<port#> identifies the RLS port
<text> identifies the far-end Waveserver 5 node, slot, and port

For example, to create links between Waveserver node HANWS5-A on slot 5 / port 1 and an RLS node with a CMD in slot 7, enter:

```
create link CCMD7p29 from "slots 7 config port 29" to-
label HANWS5-A-5-1
create link CCMD7p30 to "slots 7 config port 30" from-
label HANWS5-A-5-1
```

- 4** Verify the half-links using the following commands:

- ```
show link <from_link_name>
show link <to_link_name>
```

<from\_link\_name> is the name for the from-link  
<to\_link\_name> is the name for the to-link

---

## Procedure 96 Retrieving SPLI information

---

Use this procedure to retrieve SPLI adjacency information for the ports and PTPs on a Waveserver 5.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least limited access privileges.
- SPLI entries are shown when the Waveserver 5 is provisioned at 6500's SPLI ID Table.

### Steps

- 1 To retrieve SPLI summary information:  
`spli show`
- 2 To retrieve SPLI port adjacency information:  
`spli adjacency show`
- 3 To retrieve SPLI adjacency status information for a modem transceiver PTP:  
`spli adjacency show ptp <slot>/<ptp>`



## Procedure 97 Managing the SPLI database

---

Use this procedure to delete all unreliable entries from the SPLI database on the Waveserver 5 or to retrieve SPLI database summary information. You can also use this procedure to clear “Adjacency Discovery Unreliable” alarms resulting from changes to the photonic network or to Waveserver 5 configurations.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges to delete unreliable entries from the SPLI database.
- You must be logged in to Waveserver 5 using an account with at least limited access privileges to retrieve SPLI database summary information.

### Overview

For more information about the SPLI database, see [SPLI database](#).

### Steps

- 1 Display SPLI database summary information:  
`spli database show`
- 2 To retrieve SPLI database information for a PTP:  
`spli database show ptp <slot>/<ptp>`
- 3 Delete all unreliable entries from the SPLI database:  
`spli database delete all-unreliable`
- 4 Save provisioned settings:  
`configuration save`

## Procedure 98 Configuring client-layer OPS protection

Use this procedure to configure client-layer optical protection switch (OPS) protection.

### Requirements

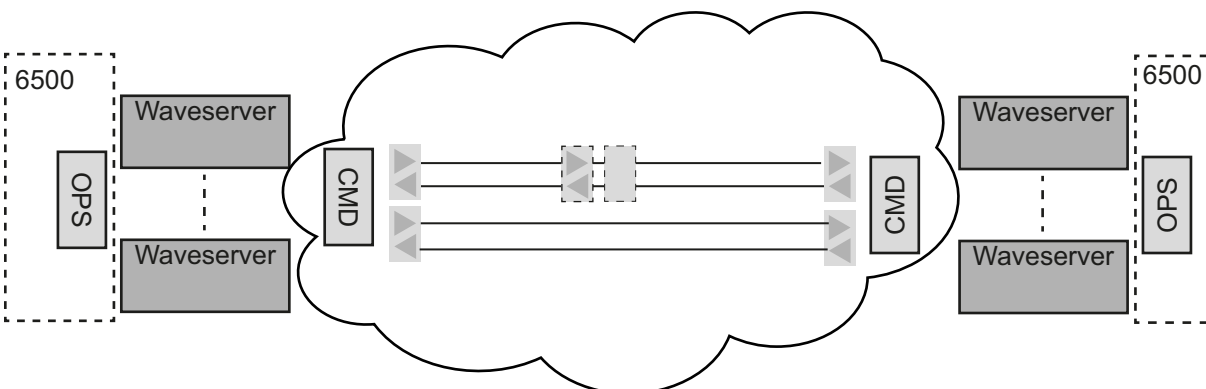
- To configure client-layer OPS protection, you must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

Waveserver 5 supports client-layer OPS configurations with 6500 S-Series OPS circuit packs. Client OPS is supported on 100GE and OTU4 clients of WL5e 2x MOTR 16x and WL5e 2x MOTR 12x modules.

Refer to the following figure.

**Figure 132** Client-layer OPS protection configuration with 6500 S-Series



**Note 1:** The only pluggable that can be connected to the OPS circuit pack when used in the client-layer optical path protection configuration is the QSFP28 (160-9402-900).

**Note 2:** The modules connected to the OPS circuit pack can be equipped in the same chassis or in a separate chassis.

### Steps

- On each Waveserver client port connected to the 6500 OPS, set the client conditioning type parameter to laser-off using the following command, where <String> refers to the port name:  

```
port set port <String> conditioning-type laser-off
```
- For 100GE clients or OTU4 clients, set the conditioning holdoff timer parameter to 0 ms using the following command, where <String> refers to the port name.  

```
port set port <String> conditioning-holdoff 0ms
```



---

## CHAPTER 22

# Trail Trace Identifier

---

This section contains the following topics:

- [Overview](#)
  - [TTI provisioning considerations](#)
- [Procedure list](#)

### Overview

The Waveserver 5 CLI allows direct manipulation of the trail-trace identifier (TTI) strings of the OTN overhead. The TTI is information that is part of the OTUk frame and is used to communicate device identification information. The transmitted and received information can then be used to build a network topology at the far end by comparing the transmitted and expected information.

You can provision TTI properties for a port path, port section, or a channel.

The TTI string has three components:

- SAPI (Source Access Point Identifier; the source address)
- DAPI (Destination Access Point Identifier; the destination address)
- OPER (Operator specified)

Each TTI component has three parts:

- Transmit string (tx) — the string you send to the far-end chassis.
- Receive string (rx) — the string you receive from the far-end chassis.
- Expected string (exp) — the string you expect to receive from the far-end chassis.

A Trace Identifier Mismatch alarm is raised when the transmitted information does not match the expected received information. You can enable the detection of a mismatch for each of the three TTI string components (SAPI, DAPI, or Operation), enable the mismatch detection alarm to be raised when a mismatch is detected, and specify a response. You can select the "squench traffic" option to stop the flow of traffic if Waveserver 5 detects a mismatch.

**Note 1:** Automatically generated TTI information is used by MCP for network topology discovery and display if Waveserver 5 chassis are not connected using a 6500 Photonics system.

**Note 2:** Channel ODU TTI is not supported on Ethernet Transcoded (ET) line transmission modes.

### TTI provisioning considerations

Waveserver 5 supports multiple layers of TTI on the line side port, the line side channel, and the OTL4.4 client port.

You can provision the TTI attributes on the following layers:

- Line side port: port trace section = OTUCn SM (Section Monitoring) TTI
- Line side port: port trace path = ODUCn PM (Path Monitoring) TTI
- Line side channel: channel trace = ODUk PM TTI
- OTL4.4 client port: port trace section = OTUk SM TTI
- OTL4.4 client port: port trace path = ODUk PM TTI

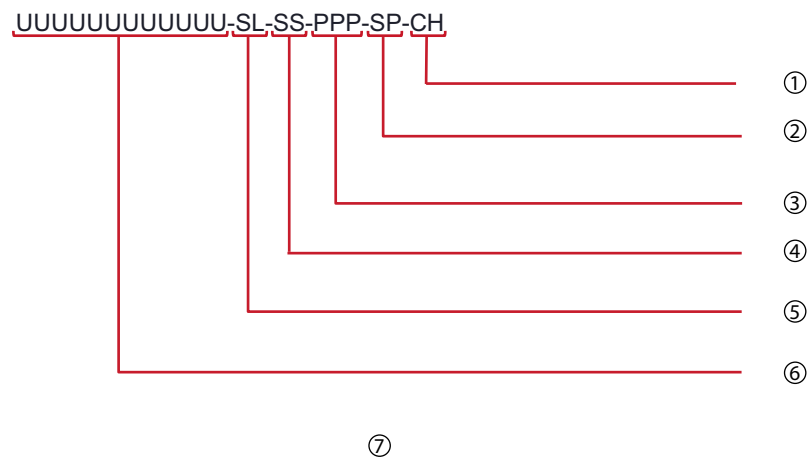
The layer being configured (transparent or terminated) defines the TTI strings that can be provisioned as follows:

- Line side port in terminated mode: OTUCn SM TTI and ODUCn PM TTI are provisionable
- Line side channel paired with a 100GE client: ODUk PM TTI is provisionable.
- Line side channel paired with an OTL4.4 client: For the ODUk PM TTI you can retrieve Rx values only (ODUk is transparent in this mode).
- OTL4.4 client port: OTUk SM TTI is provisionable. For ODUk PM TTI you can retrieve Rx values only (ODUk is transparent in this mode).

### TTI String format

TTI automatically generates a TX-OPER string in a certain format. See the following figure.

Figure 133 TTI TX-OPER string format



**Legend**

- |   |                |   |                                                                                                         |
|---|----------------|---|---------------------------------------------------------------------------------------------------------|
| 1 | Channel Number | 5 | Slot Number                                                                                             |
| 2 | Subport Number | 6 | Universally Unique ID – MAC*                                                                            |
| 3 | Port Number    | 7 | *Note : MAC used is the CP Local Management MAC (as seen when using the "chassis show mac" cli command) |
| 4 | Subslot Number |   |                                                                                                         |

**Procedure list**

This section provides the following procedure:

- [Procedure 99, “Assigning, displaying, and clearing TTI settings”](#)

## Procedure 99 Assigning, displaying, and clearing TTI settings

Use the commands in this procedure to assign values to TTI properties.

### Requirements

- You must be logged in to the CLI using an account with at least admin access privileges.

### Overview

In this procedure you can:

- assign values to TTI string components (SAPI, DAPI, and OPER) for a port section and path
- assign values to TTI string components (SAPI, DAPI, and OPER) for a channel
- set responses when a mismatch is detected between expected and actual TTI string values for the port path, port section, and channel
- clear values assigned to TTI string components (SAPI, DAPI, and OPER)
- display current TTI values

### Steps

- Select your step:

| If you want to                                     | Then go to              |
|----------------------------------------------------|-------------------------|
| assign values to TTI properties for a port path    | <a href="#">step 2</a>  |
| assign values to TTI properties for a port section | <a href="#">step 8</a>  |
| assign values to TTI properties for a channel      | <a href="#">step 14</a> |
| clear TTI settings                                 | <a href="#">step 20</a> |
| display TTI settings                               | <a href="#">step 21</a> |

### *Assigning values to TTI properties for a port path*

- To set the TTI transmit string (SAPI, DAPI, or OPER) for a port path:
  - For SAPI:
 

```
port trace path set port <slot/port> tx-sapi <0-15 char string>
```
  - For DAPI:
 

```
port trace path set port <slot/port> tx-dapi <0-15 char string>
```
  - For OPER:
 

```
port trace path set port <slot/port> tx-oper <0-32 char string>
```

- 3 To set the TTI expected string (SAPI, DAPI, or OPER) for a port path:
  - For SAPI:
 

```
port trace path set port <slot/port> exp-sapi <0-15 char string>
```
  - For DAPI:
 

```
port trace path set port <slot/port> exp-dapi <0-15 char string>
```
  - For OPER:
 

```
port trace path set port <slot/port> exp-oper <0-32 char string>
```
- 4 Set the mismatch mode for the port path. This indicates the strings to compare to determine whether there is a mismatch. The options are SAPI, DAPI, SAPI and DAPI, or Operator only. Default is operator-only.
 

```
port trace path set port <slot/port> mismatch-mode <sapi | dapi | sapi-and-dapi | operator-only>
```
- 5 Set the mismatch failure mode for the port path. This specifies the action when a mismatch is detected. Default is alarm-off.
 

```
port trace path set port <slot/port> mismatch-fail-mode <none | alarm-on | squelch-traffic>
```
- 6 Set the transmit operator mode for the port path. This specifies the topology string and how it is sent. Default is automatic.
 

```
port trace path set <slot/port> tx-oper-mode <automatic | manual>
```
- 7 Set the rx-to-exp string. This sets the string that you expect to receive.
 

```
port trace path set port <slot/port> rx-to-exp <sapi | dapi | oper>
```

***Assigning values to TTI properties for a port section***

- 8 To set the TTI transmit string (SAPI, DAPI, or OPER) for a port section:
  - For SAPI:
 

```
port trace section set port <slot/port> tx-sapi <0-15 char string>
```
  - For DAPI:
 

```
port trace section set port <slot/port> tx-dapi <0-15 char string>
```
  - For OPER:
 

```
port trace section set port <slot/port> tx-oper <0-32 char string>
```



- 9 To set the TTI expected string (SAPI, DAPI, or OPER) for a port section:
- For SAPI:  

```
port trace section set port <slot/port> exp-sapi <0-15 char string>
```
  - For DAPI:  

```
port trace section set port <slot/port> exp-dapi <0-15 char string>
```
  - For OPER:  

```
port trace section set port <slot/port> exp-oper <0-32 char string>
```
- 10 Set the mismatch mode for the port section. This specifies the strings to compare to determine whether there is a mismatch. The options are SAPI, DAPI, SAPI and DAPI, or Operator only. Default is operator-only.
- ```
port trace section set port <slot/port> mismatch-mode <sapi | dapi | sapi-and-dapi | operator-only>
```
- 11 Set the mismatch failure mode for the port section, which specifies the action when a mismatch of the string specified in mismatch mode is detected. Default is alarm-off.
- ```
port trace section set port <slot/port> mismatch-fail-mode <none | alarm-only | squelch-traffic>
```
- 12 Set the transmit operator mode for the port section, which determines the topology string and how it is sent. Default is automatic.
- ```
port trace section set port <slot/port> tx-oper-mode <automatic | manual>
```
- 13 Set the rx-to-exp string. This sets the string that you expect to receive.
- ```
port trace section set port <slot/port> rx-to-exp <sapi | dapi | operator>
```

***Assigning values to TTI properties for a channel***

- 14 To set the TTI transmit string (SAPI, DAPI, or OPER) for a channel:
- For SAPI:  

```
channel trace set channel <slot>/<port>.<channel> tx-sapi <String:0..15 char string>
```
  - For DAPI:  

```
channel trace set channel <slot>/<port>.<channel> tx-dapi <0-15 char string>
```
  - For OPER:  

```
channel trace set channel <slot>/<port>.<channel> tx-oper <0-32 char string>
```

- 15** To set the TTI expected string (SAPI, DAPI, or OPER) for a channel:
- For SAPI:  

```
channel trace set channel <slot>/<port>.<channel> exp-sapi <0-15 char string>
```
  - For DAPI:  

```
channel trace set channel <slot>/<port>.<channel> exp-dapi <0-15 char string>
```
  - For OPER:  

```
channel trace set channel <slot>/<port>.<channel> exp-oper <0-32 char string>
```
- 16** Set the mismatch mode for a channel, which determines which strings to compare to determine whether there is a mismatch. The options are SAPI, DAPI, SAPI and DAPI, or Operator only. Default is operator-only.
- ```
channel trace set channel <slot>/<port>.<channel> mismatch-mode <sapi | dapi | sapi-and-dapi | operator-only>
```
- 17** Set the mismatch failure mode for a channel, which specifies the action to take when a mismatch of the string specified in mismatch mode is detected. Default is alarm-off.
- ```
channel trace set channel <slot>/<port>.<channel> mismatch-fail-mode <none | alarm-only | squelch-traffic>
```
- 18** Set the transmit operator mode for a channel, which determines the topology string and how it is sent. Default is automatic.
- ```
channel trace set channel <slot>/<port>.<channel> tx-oper-mode <automatic | manual>
```
- 19** Set the rx-to-exp string. This sets the string that you expect to receive.
- ```
channel trace set channel <slot>/<port>.<channel> rx-to-exp <sapi | dapi | oper>
```

### ***Clearing TTI settings***

- 20** To clear the transmit, expected, and mismatch strings, and return them to their default settings, do any of the following:
- port trace path unset port <slot/port> <tx-sapi | exp-sapi | tx-dapi | exp-dapi | tx-oper | exp-oper | tx-oper-mode | mismatch-fail-mode | mismatch-mode>
  - port trace section unset port <slot/port> <tx-sapi | exp-sapi | tx-dapi | exp-dapi | tx-oper | exp-oper | tx-oper-mode | mismatch-fail-mode | mismatch-mode>
  - channel trace unset channel <slot>/<port>.<channel> <tx-sapi | exp-sapi | tx-dapi | exp-dapi | tx-oper | exp-oper | mismatch-mode | mismatch-fail-mode | tx-oper-mode> port ptp trace unset ptp <port> tx-dapi

**Displaying TTI settings**

- 21 To display the TTI values, use any of the following commands:
- `port trace path show port <slot/port>`
  - `port trace section show port <slot/port>`
  - `channel trace show channel <slot>/<port>.<channel>`

**Table 185** Description of TTI properties

| TTI property                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mismatch mode               | Specifies the TTI string to use to determine whether a mismatch is present. The options are SAPI, DAPI, SAPI and DAPI, or Operator. You cannot look for mismatch on all three strings simultaneously.                                                                                                                                                                                                                                                                                                 |
| Mismatch Failure mode       | Specifies the action when a mismatch is detected. Possible actions are: do nothing, raise an alarm on that port or channel, or squelch traffic.                                                                                                                                                                                                                                                                                                                                                       |
| Operator Tx String          | Specifies the operator string (OPER) to transmit to the far-end chassis. Length: 0 to 32 characters.                                                                                                                                                                                                                                                                                                                                                                                                  |
| Operator Tx String Actual   | The actual operator string transmitted to the far-end chassis. If the value of Operator Tx Mode is “automatic”, then the value of the actual operator string is overridden by the default topology string. This allows you to control which string is actually transmitted. Topology information is carried in this string. Length: 0 to 32 characters.                                                                                                                                               |
| Operator Tx Mode            | Determines which OPER string is sent to the far-end chassis. When the value is “automatic” the default topology string is transmitted, regardless of the user-assigned value of the transmit operator string. You can still change the Operator TX string value, but the change only affects the stored value stored and not the actual string that is transmitted on the TTI Operator overhead bytes.<br><br>When the value is “manual”, the value assigned to the Operator Tx field is transmitted. |
| Operator Expected Rx String | Specifies the operator string (OPER) we expect to receive from the far-end chassis. This string is compared with OPER string that is actually received to determine whether there is a mismatch. It can be 0 to 32 characters.                                                                                                                                                                                                                                                                        |
| Operator Rx String          | The operator string (OPER) received from the far-end chassis. This string contains topology information. Length: 0 to 32 characters.                                                                                                                                                                                                                                                                                                                                                                  |
| SAPI Tx String              | Specifies the source access point identifier (SAPI) string to transmit to the far-end chassis. Length: 0 to 15 characters.                                                                                                                                                                                                                                                                                                                                                                            |
| SAPI Rx String              | The SAPI string received from the far-end chassis. Length: 0 to 15 characters.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| SAPI Expected Rx string     | The SAPI string expected from the far-end chassis. This string is compared to the SAPI string actually received from the far-end chassis to determine whether there is a mismatch. Length: 0 to 15 characters.                                                                                                                                                                                                                                                                                        |
| DAPI Tx String              | Specifies the destination access point identifier (DAPI) string to transmit to the far end. Length: 0 to 15 characters.                                                                                                                                                                                                                                                                                                                                                                               |

| TTI property            | Description                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DAPI Rx String          | The DAPI string received from the far end. Length: 0 to 15 characters.                                                                                                                                         |
| DAPI Expected Rx string | The DAPI string expected from the far-end chassis. This string is compared to the DAPI string actually received from the far-end chassis to determine whether there is a mismatch. Length: 0 to 15 characters. |



---

## CHAPTER 23

# Out-of-service software upgrades

---

This section contains the following topics:

- [Overview](#)
  - [Upgrade considerations](#)
  - [Upgrading Waveserver 5 when the release number of the software on the device does not match the release number on your software license](#)
- [Procedure list](#)

### Overview

This section provides information on upgrading chassis that are not in-service. See *Software Upgrade Procedure for Waveserver OS*, 323-3001-900, for detailed procedures for in-service upgrades. The procedures in this section are intended for upgrading Waveserver 5 chassis that are not in-service.

### Upgrade considerations

These upgrade considerations apply:

- Waveserver 5 cannot be upgraded if the License Missing alarm is raised.
- To avoid post-upgrade connectivity issues due to unsupported cryptographic SSH or SSL algorithms, compare the SSH or SSL algorithms supported in your network with those supported by Waveserver 5. For a list of supported algorithms, see [SSH algorithms selection](#) and [TLS algorithms selection](#).
- To conserve disk space, Ciena recommends using the `delete-from-load` command whenever possible to delete the previous software load after the upgrade successfully completes. To remove unused software loads, see [Deleting unused software loads](#).

### Upgrading Waveserver 5 when the release number of the software on the device does not match the release number on your software license

This section describes workarounds for upgrading the software release on Waveserver 5 in the case where the release number of the software on the device does not match your software license.

**The software release number is 2.1 or greater**

This section provides details on upgrading the software release on the Waveserver 5, when the software release number on the Waveserver 5 is Release 2.1 or greater and does not match your software license. To confirm the software release on the Waveserver 5, use CLI command `software show`. To confirm the software license you have, check the Software License Availability email you received from Ciena. For an example of the Software License Availability email, see [“Ordering and license generation”](#).

If you are upgrading and the license operational state is not “commissioning”, the following steps apply:

- 1 Perform an RTFD on the Waveserver 5.

**CAUTION**

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena’s ability to fully investigate a module or network failure event.

- 2 After the RTFD completes, check the license status and verify the license operational state is: “commissioning”.
  - a. If the license operational state is “commissioning”, continue to the next step.
  - b. If the license operational state is not “commissioning”, then go to [step 1](#).
- 3 Upgrade to the current release.

**The software release number is less than 2.1**

This section provides details on upgrading the software release on the Waveserver 5, when the software release number on the Waveserver 5 is less than Release 2.1 does not match your software license. To confirm the software release on the Waveserver 5, use CLI command `software show`. To confirm the software license you have, check the Software License Availability email you received from Ciena. For an example of the Software License Availability email, see [Ordering and license generation](#).

If you are upgrading and the system clock has been set, the following steps apply:

- 1 Disable the NTP configuration on the DHCP server, if it is configured for the Waveserver 5 you are upgrading.
- 2 Perform a RTFD on the Waveserver 5.

**CAUTION**

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

- 3 After the RTFD completes, check if the "System Clock Not Set" alarm is raised.
  - a. If "System Clock Not Set" alarm is raised, then go to [step 4](#).
  - b. If "System Clock Not Set" alarm is cleared, then go to [step 1](#).
- 4 Upgrade to the current release.
- 5 Enable the NTP configuration on the DHCP server, if it was disabled in step 1.

## Procedure list

This section provides the following procedures:

- [Procedure 100, "Performing a local check for file integrity and performing a remote check for file accessibility"](#)
- [Procedure 101, "Deleting unused software loads"](#)
- [Procedure 102, "Performing a one-step system software upgrade"](#)
- [Procedure 103, "Performing a multi-step system software upgrade"](#)
- [Procedure 104, "Performing a one-step system software downgrade"](#)
- [Procedure 105, "Performing a multi-step system software downgrade"](#)
- [Procedure 106, "Performing a one-step upgrade when the software release on the Waveserver 5 does not match the software license"](#)
- [Procedure 107, "Canceling a software upgrade"](#)
- [Procedure 108, "Upgrading pluggable transceiver firmware"](#)



## Procedure 100 Performing a local check for file integrity and performing a remote check for file accessibility

Use this procedure to perform a local check for file integrity and a remote check for file accessibility. For a local software version, Waveserver 5 checks the file integrity. For a remote software version, Waveserver 5 checks the file extension and verifies that the URL can be reached.

### Requirements

You have downloaded the software load to either the Waveserver 5 or to the remote server. The software load consists of .rpm and signature files.

### Steps

- 1 Use one of the following commands to check the software version on the local processor for errors or to check the default software server (remote) for file accessibility issues:

```
software check local version <version>
```

Or

```
software check remote <url> login-id <username> password
<password>
```

Where <version> is the software release and load; for example:  
waveserver-2.2.0.xyz-GA.tar.gz

**Note:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

The password may contain special characters:

```
~ ! ^ & * () - _ : = + { } [] | \ ; ' < > . ,
```

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

For example:

```
software check remote url sftp://
ciena@[2222:111:dddd:5444:aced::293]/262/waveserver-
2.2.0.10-GA.tar.gz password ad#min
```

---

## Procedure 101 Deleting unused software loads

---

Use this procedure to delete unnecessary software loads from Waveserver 5. Ciena recommends keeping only the current running software load.

### Steps

- 1 Display the list of software loads saved in the Waveserver 5 memory:  
`software show installed`
- 2 Determine the software loads that are no longer required. Ciena recommends keeping only the current running software load and deleting all prior software versions. To determine the current running system load use the CLI command:  
`software show`
- 3 For each software load that you want to remove, use the CLI command:  
`software delete version <String: 1..31>`  
You have completed this procedure.

## Procedure 102 Performing a one-step system software upgrade

This procedure describes the use of a single CLI command (`software install url`) that performs an immediate download, activation, and installation of upgrade system software (one-step system software upgrade) from the specified URL. The software load is automatically committed after activation completes and Waveserver 5 performs a warm restart.

### Requirements

#### ATTENTION

See *Software Upgrade Procedure for Waveserver OS*, 323-3001-900, for information on in-service upgrades.



#### CAUTION

##### Upgrade remotely managed nodes first

Upgrade nodes managed via remote comms before you upgrade nodes with direct DCN access. After the remote node is upgraded, the remote comms is temporarily unavailable until the software load on the local node is also upgraded.

- The upgrade software has been downloaded from the Ciena portal (<https://my.ciena.com>) and saved to the designated remote server, and you have recorded the IP address or host name of the server and the URL, as well as the user name and password of the server.

**Note:** Waveserver 5 supports IPv4 and IPv6 on SFTP, and SCP servers.

- The target upgrade software file ends with `tar.gz`
- You are logged in to Waveserver 5 using an account with at least admin privileges.
- Sufficient memory is available in the Waveserver 5 memory to support the new software load. To ensure that there is sufficient memory space, delete any unnecessary software files, see [Deleting unused software loads](#).
- Save a copy of the running (active) configuration to the default load file. This ensures that the existing configuration settings are applied when you commit the upgrade software load.

### Steps

- 1 Save the running (active) configuration to ensure these configuration settings are applied when you commit the upgrade software load. Run this command:  

```
configuration save
```

The configuration is saved under the filename assigned to default-save-filename.

- 2** Ensure the filenames assigned to default-save-filename and default-load-filename (the configuration loaded after a restart) are the same. Run this command:

```
configuration list
```

The system displays the filenames assigned to default-save-filename and default-load-filename.

| If default-save-filename and default-load-filename | Then go to             |
|----------------------------------------------------|------------------------|
| are the same                                       | <a href="#">step 4</a> |
| are different                                      | <a href="#">step 3</a> |

- 3** Ensure the filename assigned to default-save-filename is also assigned to default-load-filename. Run this command:

```
configuration set default-load-filename <value-of-
default-save-filename>
```

where

|                                  |                                                   |
|----------------------------------|---------------------------------------------------|
| <value-of-default-save-filename> | is the filename assigned to default-save-filename |
|----------------------------------|---------------------------------------------------|

### Performing the software upgrade

- 4** Use the following command to download, activate, and install the upgrade software from the server:

```
software install url <url> login-id <userid> password
<password> [delete-from-load]
```

Use the format:

```
software install url sftp://10.10.10.10/path/waveserver-
2.2.0-xyz.tar.gz login-id userid password my_password
```

Waveserver 5 initiates a warm restart.

**Note 1:** If the software release file was manually downloaded to Waveserver 5, use the syntax: software install url file://<absolute path to release file>. For example, software install url file:///tmp/users/diag/waveserver-2.2.0.10-GA.tar.gz

**Note 2:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

The password may contain special characters:

```
~ ! ^ & * () - _ : = + { } [] \ ; ' < > . ,
```

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

For example:

```
software install url sftp://
ciena@[2222:111:dddd:5444:aced::293]/262/waveserver-
2.2.0.10-GA.tar.gz password ad#min
```

**Note 3:** The “password” keyword is not supported in the NETCONF interface. If you are using NETCONF to perform the software upgrade, use the format `userid:password`. For example, in the following NETCONF command, `su:su` represents the userid and password:

```
<software:waveserver-software-install
xmlns:software="urn:ciena:params:xml:ns:yang:cienaws:cie
na-waveserver-software"
xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
<software:url>sftp://su:su@10.10.10.10/path/waveserver-
2.2.0.10-GA.tar.gz</software:url> </software:waveserver-
software-install>
```

**Note 4:** To conserve disk space, Ciena recommends using the `delete-from-load` option to automatically delete the previous software load.

### **Retrieving the upgrade status**

- 5 Retrieve the software upgrade status and verify the Active Release Version:

```
software show upgrade-status
```

The software upgrade process can take several minutes to complete.

The upgrade is complete when the Upgrade State is “Idle” and the Last Upgrade Operation shows that the software release version is committed.

**Note:** When the system software upgrade procedure is complete, a software license missing alarm is raised. To clear the alarm, install and activate the software license. For details on license installation, see [Licensing](#).

### **Deleting the previous software release (optional)**

- 6 Delete the previous software version (optional):

```
software delete version <string: 1 to 31>
```

You have completed this procedure.

---

## Procedure 103 Performing a multi-step system software upgrade

---

Use this procedure to manually download the upgrade software and then activate and commit the software only when required.

### Requirements

#### ATTENTION

See *Software Upgrade Procedure for Waveserver OS*, 323-3001-900, for detailed instructions on in-service system software upgrades.



#### CAUTION

##### Upgrade remotely managed nodes first

Upgrade nodes managed via remote comms before you upgrade nodes with direct DCN access. After the remote node is upgraded, the remote comms is temporarily unavailable until the software load on the local node is also upgraded.

- The upgrade software has been downloaded from the Ciena portal (<https://my.ciena.com>) and saved to the designated remote server. You must have the IP address and host name of the server as well as the user name and password.

**Note 1:** To access the software server where the upgrade software is located, you must have the IP address or host name of the server, and the user name and password if the server is an SFTP or SCP server.

**Note 2:** Waveserver 5 supports IPv4 and IPv6 on SFTP and SCP servers.

- You are logged in to Waveserver 5 using an account with at least admin privileges.
- Sufficient memory is available in the Waveserver 5 memory to support the new software load. To ensure that there is sufficient memory space, delete any unnecessary software files, see [Deleting unused software loads](#).
- Save a copy of the running (active) configuration to the default load file. This ensures that the existing configuration settings are applied when you commit the upgrade software load.

**Note:** During multi-step upgrades, a software license missing alarm is raised after each intermediate upgrade is completed. To clear the alarm and to be able to perform the next upgrade, you must install and activate the missing software license. For details on license installation, see [Licensing](#).

## Overview

**Note:** You can use the `software activate <version> auto-commit` command to automatically commit the software.

## Steps

- 1 Save the running (active) configuration to ensure these configuration settings are applied when you commit the upgrade software load. Run this command:

```
configuration save
```

The configuration is saved under the filename assigned to `default-save-filename`.

- 2 Ensure the filenames assigned to `default-save-filename` and `default-load-filename` (the configuration loaded after a restart) are the same. Run this command:

```
configuration list
```

The system displays the filenames assigned to `default-save-filename` and `default-load-filename`.

| If default-save-filename and default-load-filename | Then go to             |
|----------------------------------------------------|------------------------|
| are the same                                       | <a href="#">step 4</a> |
| are different                                      | <a href="#">step 3</a> |

- 3 Ensure the filename assigned to `default-save-filename` is also assigned to `default-load-filename`. Run this command:

```
configuration set default-load-filename <value-of-
default-save-filename>
```

where

|                                                     |                                                                |
|-----------------------------------------------------|----------------------------------------------------------------|
| <code>&lt;value-of-default-save-filename&gt;</code> | is the filename assigned to <code>default-save-filename</code> |
|-----------------------------------------------------|----------------------------------------------------------------|

## Downloading the software version to Waveserver 5

- 4 Download the software upgrade file:  

```
software download url <url> login-id <String> password
<String>
```

For example,

```
software download url sftp://10.10.10.10/path/
waveserver-2.2.0.10-GA.tar.gz login-id userid password
my_password
```

**Note 1:** The target software upgrade file should end with `.tar.gz`

**Note 2:** You must specify the absolute path for the remote server.

**Note 3:** The password may contain special characters:

```
~ ! ^ & * () - _ : = + { } [] | \ ; ' < > . ,
```

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

For example, using a password admin?

```
software download url sftp://
[2222:111:dddd:5444:aced::293]/262/waveserver-2.2.0.10-
GA.tar.gz login-id waveserver password admin?
```

**Note 4:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

```
For example, software download url sftp://
[2222:111:dddd:5444:aced::293]/262/waveserver-
2.2.0.10.tar.gz login-id waveserver password ****
```

5 Software download can take several minutes to complete. To view the status of this process, use the command `software show upgrade-status`.

| 6 | If you want to                                                | Then go to                                   |
|---|---------------------------------------------------------------|----------------------------------------------|
|   | activate the new software release                             | <a href="#">step 7</a>                       |
|   | cancel the upgrade and remain at the current software release | <a href="#">Canceling a software upgrade</a> |

### Activating the software version

7 Activate the upgrade software:

```
software activate version <version> [auto-commit]
[delete-from-load]
```

**Note:** You can use the `auto-commit` option to automatically commit the software upgrade. (If you choose not to commit the software upgrade automatically, you must commit it manually.) You can also use the `delete-from-load` option to delete the previous software load after the upgrade software is committed.

Waveserver 5 initiates a warm restart. This process can take several minutes to complete.

| 8 | If you want to                                                | Then go to                                   |
|---|---------------------------------------------------------------|----------------------------------------------|
|   | commit the new software release manually                      | <a href="#">step 9</a>                       |
|   | cancel the upgrade and remain at the current software release | <a href="#">Canceling a software upgrade</a> |

### Manually committing the software

9 Verify that the system is ready for the software commit.

```
software show upgrade status
```

The system returns a table of Upgrade Status Information.



You can manually commit the software upgrade when the value of `Last Upgrade Operation` is `Ready for Manual Commit`.

**10** Manually commit the upgrade software:

```
software commit [delete-from-load]
```

**Note:** The `delete-from-load` option automatically deletes the previous software load after the upgrade software is committed.

The software upgrade process can take several minutes to complete. The process is complete when the Upgrade State is “Commit Complete”.

**Note:** When the system software upgrade procedure is complete, a software license missing alarm is raised. To clear the alarm, install and activate the software license. For details on license installation, see [Licensing](#).

**11** To retrieve the software upgrade status and confirm the software version, use the command:

```
software show upgrade-status
```

***Deleting the previous software release (optional)***

**12** If you did not use the `[delete-from-load]` option when committing the upgrade software, use this command to delete the previous software version:

```
software delete version <string: 1..31>
```

You have completed this procedure.

---

## Procedure 104 Performing a one-step system software downgrade

---

Use this procedure to downgrade Waveserver 5 to a previous software release.

### Requirements

- The downgrade software has been downloaded from the Ciena portal (<https://my.ciena.com>) and saved to the designated remote server, and you have recorded the IP address or host name of the server and the URL, as well as the user name and password of the server.

**Note:** Waveserver 5 supports IPv4 and IPv6 on SFTP, and SCP servers.

- The target downgrade software file ends with `tar.gz`
- You are logged in to Waveserver 5 using an account with at least admin privileges.
- Sufficient memory is available in the Waveserver 5 memory to support the new software load. To ensure that there is sufficient memory space, delete any unnecessary software files, see [Deleting unused software loads](#).
- Disable configuration mismatch detection to allow to downgrade to Release 2.0.x to run to completion.

### Overview

#### ATTENTION

This procedure is intended for an out-of-service Waveserver 5 only.

### Steps

#### *Disabling configuration mismatch detection*

- 1 Disable configuration mismatch detection:

```
system auto-config-recovery config-mismatch-detection
disable
```

**Note:** Disabling configuration mismatch detection allows the software downgrade to complete. Release 2.0.x cannot use the configuration database used in Release 2.1 and later releases. If you do not disable configuration mismatch detection, differences in the configuration databases cause the system to raise the Configuration Database Mismatch alarm when the CP attempts to restart with Release 2.0.x. When the system raises this alarm, it stops the software downgrade.

**Performing the software downgrade**

- 2 Use the following command to download, activate, and install the downgrade software from the server:

```
software install url <url> login-id <userid> password
<password> [delete-from-load]
```

Use the format:

```
software install url sftp://10.10.10.10/path/waveserver-
2.1.0-xyz.tar.gz login-id userid password my_password
```

Waveserver 5 initiates a warm restart.

**Note 1:** If the software release file was manually downloaded to Waveserver 5, use the syntax: `software install url file://<absolute path to release file>`. For example, `software install url file:///tmp/users/diag/waveserver-2.1.0.10-GA.tar.gz`

**Note 2:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

The password may contain special characters:

~ ! ^ & \* ( ) - \_ : = + { } [ ] \ ; ' < > . ,

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

For example:

```
software install url sftp://
ciena@[2222:111:dddd:5444:aced::293]/262/waveserver-
2.1.0.10-GA.tar.gz password ad#min
```

**Note 3:** The “password” keyword is not supported in the NETCONF interface. If you are using NETCONF to perform the software upgrade, use the format `userid:password`. For example, in the following NETCONF command, `su:su` represents the userid and password:

```
<software:waveserver-software-install
xmlns:software="urn:ciena:params:xml:ns:yang:cienaws:cie
na-waveserver-software"
xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
<software:url>sftp://su:su@10.10.10.10/path/waveserver-
2.1.0.10-GA.tar.gz</software:url> </software:waveserver-
software-install>
```

**Note 4:** To conserve disk space, Ciena recommends using the `delete-from-load` option to automatically delete the previous software load.

**Note 5:** If the Configuration Database Mismatch alarm is raised after the Release 2.0.x software is installed, you must override the alarm to complete the downgrade. To override the alarm run this command: `system auto-config-recovery override-config-mismatch`

### **Retrieving the downgrade status**

- 3 Retrieve the status of the software downgrade and verify the Active Release Version:

```
software show upgrade-status
```

The software downgrade process can take several minutes to complete.

The downgrade process is complete when the Upgrade State is “Idle” and the Last Upgrade Operation shows that the software release version is committed.

**Note:** When the system software downgrade procedure is complete, a software license missing alarm is raised. To clear the alarm, install and activate the software license. For details on license installation, see [Licensing](#).

### **Deleting the previous software release (optional)**

- 4 Delete the previous software version (optional):

```
software delete version <string: 1 to 31>
```

- 5 Restore the factory-set configuration values:

```
configuration reset-to-factory-defaults
```



#### **CAUTION**

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena’s ability to isolate a failure event.

In the confirmation message, verify that you want to proceed.

**Note:** The RTFD triggers a cold restart of Waveserver 5.

## Procedure 105 Performing a multi-step system software downgrade

---

Use this procedure to perform a manual downgrade of the Waveserver 5 system software to a previous software release. In this procedure, you manually download the previous software release, then activate and commit the software only when required.

### Requirements

- The downgrade software has been downloaded from the Ciena portal (<https://my.ciena.com>) and saved to the designated remote server. You must have the IP address and host name of the server as well as the user name and password.

**Note 1:** To access the software server where the upgrade software is located, you must have the IP address or host name of the server, along with a user name and password if the server is an SFTP or SCP server.

**Note 2:** Waveserver 5 supports IPv4 and IPv6 on SFTP and SCP servers.

- You are logged in to Waveserver 5 using an account with at least admin privileges.
- Sufficient memory is available in the Waveserver 5 memory to support the new software load. To ensure that there is sufficient memory space, delete any unnecessary software files, see [Deleting unused software loads](#).
- Disable configuration mismatch detection to allow a downgrade to Release 2.0.x to run to completion.

**Note:** During multi-step downgrades, a software license missing alarm is raised after each intermediate downgrade is completed. To clear the alarm and to be able to perform the next downgrade, you must install and activate the missing software license. For details on license installation, see [Licensing](#).

### Overview

**Note:** You can use the `software activate <version> auto-commit` command to automatically commit the software.

#### ATTENTION

This procedure is service-affecting and causes a cold-restart of the Waveserver 5.

## Steps

### *Disabling configuration mismatch detection*

- 1 Disable configuration mismatch detection:  

```
system auto-config-recovery config-mismatch-detection
disable
```

**Note:** Disabling configuration mismatch detection allows the software downgrade to complete. Release 2.0.x cannot use the Release 2.1 configuration database. If you do not disable configuration mismatch detection, differences in the configuration databases cause the system to raise the Configuration Database Mismatch alarm when the CP attempts to restart with Release 2.0.x. When the system raises this alarm, it stops the software downgrade.

### *Downloading the software version to Waveserver 5*

- 2 Download the software downgrade file:  

```
software download url <url> login-id <String> password
<String>
```

For example,

```
software download url sftp://10.10.10.10/path/
waveserver-2.1.0.10-GA.tar.gz login-id userid password
my_password
```

**Note 1:** The target software downgrade file should end with `.tar.gz`

**Note 2:** You must specify the absolute path for the remote server.

**Note 3:** The password may contain special characters:

`~!^&*()-_:=+{ } [ ] | \ ; ' < > . ,`

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

For example, using a password admin?

```
software download url sftp://
[2222:111:dddd:5444:aced::293]/262/waveserver-2.1.0.10-
GA.tar.gz login-id waveserver password admin?
```

**Note 4:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

```
For example, software download url sftp://
[2222:111:dddd:5444:aced::293]/262/waveserver-
2.1.0.10.tar.gz login-id waveserver password ****
```

- 3 Software download can take several minutes to complete. To view the status of this process, use the command `software show upgrade-status`.

| 4 | If you want to                                                  | Then go to                                   |
|---|-----------------------------------------------------------------|----------------------------------------------|
|   | activate the downgrade software release                         | <a href="#">step 5</a>                       |
|   | cancel the downgrade and remain at the current software release | <a href="#">Canceling a software upgrade</a> |

### **Activating the software version**

- 5 Activate the downgrade software:
- ```
software activate version <version> [auto-commit]
[delete-from-load]
```
- Note:** You can use the `auto-commit` option to automatically commit the software upgrade. (If you choose not to commit the software upgrade automatically, you must commit it manually.) You can also use the `delete-from-load` option to delete the previous software load after the downgrade software is committed.
- Waveserver 5 initiates a warm restart. This process can take several minutes to complete.

6	If you want to	Then go to
	commit the downgrade software release manually	step 7
	cancel the downgrade and remain at the current software release	Canceling a software upgrade

Manually committing the software

- 7 Verify that the system is ready for the software commit.
- ```
software show upgrade status
```
- The system returns a table of Upgrade Status Information.
- You can manually commit the software upgrade when the value of `Last Upgrade Operation` is `Ready for Manual Commit`.
- 8 Manually commit the software:
- ```
software commit [delete-from-load]
```
- The `delete-from-load` option automatically deletes the previous software load after the upgrade software is committed.
- The software upgrade process can take several minutes to complete. The process is complete when the Upgrade State is “Commit Complete”.
- Note 1:** If the Configuration Database Mismatch alarm is raised after the Release 2.0.x software is installed, you must override the alarm to complete the downgrade. To override the alarm run this command: `system auto-config-recovery override-config-mismatch`

Note 2: When the system software downgrade procedure is complete, a software license missing alarm is raised. To clear the alarm, install and activate the software license. For details on license installation, see [Licensing](#).

- 9 To retrieve the software upgrade status and confirm the software version, use the command:

```
software show upgrade-status
```

Deleting the previous software release (optional)

- 10 If you did not use the [delete-from-load] option when committing the software, use the following command to delete the previous software version:

```
software delete version <string: 1..31>
```

- 11 Restore the factory-set configuration values:

```
configuration reset-to-factory-defaults
```



CAUTION

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

In the confirmation message, verify that you want to proceed.

Note: The RTFD triggers a cold restart of Waveserver 5.

Procedure 106 Performing a one-step upgrade when the software release on the Waveserver 5 does not match the software license

Use this procedure to perform a one-step upgrade when the software release on the Waveserver 5 does not match the software license you have.

Requirements

ATTENTION

See *Software Upgrade Procedure for Waveserver OS*, 323-3001-900, for information on in-service upgrades.

- You have confirmed the software release on the Waveserver 5 does not match the software license you have. To confirm the software release on the Waveserver 5, use CLI command `software show`. To confirm the software license you have, check the Software License Availability Notification email you received from Ciena. For an example of the sample software license availability email, see [Ordering and license generation](#).
- The upgrade software has been downloaded from the Ciena portal (<https://my.ciena.com>) and saved to the designated remote server, and you have recorded the IP address or host name of the server and the URL, as well as the user name and password of the server.

Note: Waveserver 5 supports IPv4 and IPv6 on SFTP, and SCP servers.

- The target upgrade software file ends with `tar.gz`
- Sufficient memory is available in the Waveserver 5 memory to support the new software load. To ensure there is sufficient memory, delete any unnecessary software files. For details, see [Deleting unused software loads](#).
- You are logged in to Waveserver 5 with at least admin privileges.

Steps

- 1 Disable the NTP configuration on the DHCP server, if it is configured for the Waveserver 5 you are upgrading or downgrading.
- 2 Perform a reset to factory defaults.

**CAUTION**

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

```
configuration reset-to-factory-defaults
```

A confirmation message displays. Type `y` to confirm that you want to perform a Secure Erase. This operation takes approximately five minutes to complete and includes a full disk validation. Waveserver 5 will restart with a new software installation and factory default configuration.

- 3 Check to see that Waveserver 5 information was properly erased using CLI command `chassis show secure-erase`. If Waveserver 5 was not properly erased, repeat the previous step, otherwise proceed to the next step. Contact your next level of support or your Ciena support, if you have performed the previous step twice and you are unable to successfully perform a reset to factory defaults.

```
ws5_0134# chassis show secure-erase
```

----- SECURE ERASE STATE -----	
Parameter	Value

Last Secure Erase	
Validation	PASSED
Type	RTFD - User Request
Date	Sun Mar 22 17:42:52 UTC 2020

Performing the software upgrade

- 4 Download, activate, and install the upgrade software from the server.

```
software install url <url> login-id <userid> password
<password> [delete-from-load]
```

Use the format:

```
software install url sftp://10.10.10.10/path/waveserver-
2.2.0.xyz-GA.tar.gz login-id userid password my_password
```

Waveserver 5 initiates a warm restart.

Note 1: If the software release file was manually downloaded to Waveserver 5, use the syntax: `software install url file://<absolute path to release file>`. For example, `software install url file:///tmp/users/diag/waveserver-2.2.0.10-GA.tar.gz`

Note 2: When using an IPv6 address with the `url` parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

The password may contain special characters:

~ ! ^ & * () - _ : = + { } [] | \ ; ' < > . ,

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

(use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

For example:

```
software install url sftp://
ciena@[2222:111:dddd:5444:aced::293]/262/waveserver-
2.2.0.10-GA.tar.gz password ad#min
```

Note 3: The “password” keyword is not supported in the CLI interface. If you are using CLI to perform the software upgrade, use the format `userid:password`. For example, in the following CLI command, `su:su` represents the userid and password:

```
software install url sftp://su:su@10.10.10.10/path/
waveserver-2.2.0.10-GA.tar.gz
```

Note 4: To conserve disk space, Ciena recommends using the `delete-from-load` option to automatically delete the previous software load.

Retrieving the upgrade status

- 5 Retrieve the software upgrade status and verify that the Active Release Version is correct.

```
software show upgrade-status
```

The software upgrade process can take several minutes to complete. The upgrade is complete when the Upgrade State is “Idle” and the Last Upgrade Operation shows that the software release version is committed.

Note: When the system software upgrade or downgrade procedure is complete, a software license missing alarm is raised. To clear the alarm, install and activate the software license. For details on license installation, see [Licensing](#).

Deleting the previous software release (optional)

- 6 Delete the previous software version (optional).

```
software delete version <string: 1 to 31>
```

- 7 Enable the NTP configuration on the DHCP server if it was previously disabled in [step 1](#).

Procedure 107 Canceling a software upgrade

Use this procedure to cancel an initiated upgrade and return Waveserver 5 to its pre-upgrade state.

Requirements

- The Waveserver 5 is currently activating the new software load.
- You are logged in to Waveserver 5 using an account with at least admin privileges.
- You have a configuration file saved remotely on a server that supports SFTP or SCP.



CAUTION

Canceling the software upgrade causes a service-affecting cold restart of Waveserver 5

Canceling the software upgrade also requires a reset-to-factory-defaults operation and a configuration restore to return to the previous system state. After performing a reset-to-factory-defaults operation, Waveserver 5 must be recommissioned with an IP address and gateway, or the DCN interface must be re-provisioned.

Steps

- 1 Cancel the software upgrade:

```
software cancel
```

Waveserver 5 starts rebooting.
- 2 When the reboot is complete, perform a reset-to-factory-defaults operation to prevent any invalid configuration data from being carried over.

```
configuration reset-to-factory-defaults
```



CAUTION

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

- 3 Type `y` to confirm the reset-to-factory-defaults operation.
- 4 Contact your service personnel to recommission Waveserver 5 with an IP address and gateway, or login to the chassis through a console port connection and re-provision the DCN interface.

- 5 The configuration file is saved remotely on a server that supports SFTP or SCP, use the following CLI command:

```
configuration install filename [default-server | default-  
sftp-server | default-scp-server | sftp-server <IP  
address or host name> login-id <String> {password  
<String> secret <String>} | scp-server <IP address or host  
name> login-id <String> {password <String> secret  
<String>}]
```

Note 1: Ensure that you specify the full path to the file in the command.

Note 2: You must specify [login-id] and [password] for SFTP and SCP servers.

Note 3: The CLI command `configuration install`, copies the configuration file from its storage location to the `~ config` directory. It also sets the configuration file as the default load file. To verify the default load file, use CLI command `configuration list`.

- 6 Restore the configuration file by using `reset-to-user-config`:

```
configuration reset-to-user-config filename <filename>  
[revert-timeout <Duration: number/time format  
N[yMwdhms]>]
```

Note 1: Restoring the configuration file can take several minutes.

Note 2: For [revert-timeout], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. If the specified time elapses before the installation is successfully completed, the configuration reverts to the currently running configuration file.

Procedure 108 Upgrading pluggable transceiver firmware

Use this procedure to upgrade the firmware load on 400ZR pluggable transceivers.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You have transferred the firmware upgrade load to the Waveserver 5. Make sure the file access permission on the firmware upgrade load is readable by everyone.

Overview

Note: Contact your Ciena representative for information about how to obtain the upgrade firmware loads and how to upgrade firmware loads on Ciena pluggable transceivers.

ATTENTION

You cannot upgrade the firmware load on the 400ZR pluggable transceivers at the same time as you perform a state-dump on the 400ZR pluggable transceivers.

Steps

- 1 Start the firmware upgrade:

```
xcvr upgrade xcvr <slot>/<port> file <filename> [restart
<cold | warm>]
```

Note: The optional `restart` argument lets you specify the type of restart that occurs after the firmware upgrade completes. If you do not specify the `restart` argument, a warm restart (a hitless upgrade) occurs by default.

For example:

```
> xcvr upgrade xcvr 5/1 file /home/cws/signed_image_w15n-
mcs1_1.0.134.muf restart warm
```

- 2 Verify the progress of the upgrade:

```
xcvr show xcvr <slot>/<port> upgrade-status
```

For example:

```
> xcvr show xcvr 5/1 upgrade-status
```

The system displays the upgrade status in a table. Continue to run the `xcvr show xcvr` command until the upgrade state is Idle and the Upgrade Operation is Upgrade is Completed, as shown below.

```
> xcvr show xcvr 3/1 upgrade-status
```

-----UPGRADE STATUS INFORMATION-----	
Parameter	Value
Active Release Version	1.0.135 BankB Committed
Upgrade State	Idle
Upgrade Operation	Upgrade is Completed

CHAPTER 24

Configuration management

Waveserver 5 configuration settings are maintained in a set of three datastores, which stores this information in binary format. Together, the three datastores make up the configuration database, which is used to create, maintain, and apply configuration settings to the Waveserver 5 device.

This section contains the following topics:

- [Declarative configuration](#)
- [Datastore model](#)
- [Procedure list](#)

Declarative configuration

Waveserver 5 implements the configuration datastore model defined in Internet Engineering Task Force (IETF) RFC 6241. The IETF document describes a framework for configuring network devices based on configuration datastores: a running configuration datastore that holds the currently active configuration settings of a network device; a startup configuration datastore (which can differ in content from the running datastore) that is loaded by the network device when it boots; and, a candidate configuration datastore that holds uncommitted configuration settings that can work on without affecting the current (running) configuration of the network device.

This framework also describes the behavior and interaction between the three configuration datastores, and how users can interact and work with the datastores. The capabilities of the configuration datastore model are accessible to Waveserver users from currently supported NBIs: CLI, REST, NETCONF, gRPC/gNMI, and OpenConfig.

The configuration datastore framework enables Waveserver 5 to implement a declarative configuration scheme in which you request the desired final-state of the configuration, rather than execute individual commands to achieve the desired configuration. You do this by modifying the candidate configuration datastore. Waveserver 5 determines whether the candidate configuration is valid. If the modified candidate configuration is valid, Waveserver identifies

the differences between the running configuration and the candidate configuration and only makes the changes needed to attain that configuration state. If the modified candidate configuration is invalid, no changes are made. Waveserver 5 rejects the candidate configuration and identifies the invalid modifications so you can correct them.

Datastore model

In previous releases, the Waveserver 5 configuration file was a set of CLI commands contained in a single text file (startup config). At system startup, the startup config was loaded into memory and became the active, running configuration. Although you could modify the running configuration, unless you saved it (which overwrote the startup config file) your changes were lost on a cold reboot.

The Waveserver 5 configuration settings are now maintained in three datastores. Each datastore contains a complete Waveserver 5 configuration. The datastores maintain the configuration information in binary format. The three datastores are:

- Running datastore
- Startup datastore
- Candidate datastore

Running datastore

The running datastore contains the currently active configuration settings.

When you first bring up the Waveserver 5 system during commissioning or when you perform a cold restart, the configuration settings in the startup datastore are loaded into the running datastore and become the active configuration. A warm restart of the Control Module (CM) or the Waveserver 5 chassis retains the existing system and traffic configurations by loading the running datastore configuration settings on initialization.

You can modify the active configuration in the running datastore any time during or after commissioning. Changes you make to the active configuration take effect in real-time, but do not persist. To persist the running datastore configuration settings, use the CLI command: `configuration save`. This command saves the configuration in the running datastore to the startup datastore.

Note: A cold restart loads the last saved configuration in the startup datastore and overwrites the active configuration in the running datastore. If you made changes to the active configuration but did not save them, those changes are lost when the cold restart occurs. Unsaved changes to the active configuration are not lost during a warm restart.

When Waveserver 5 automatically creates a new system configuration it does so in the running datastore and then saves the contents of the running datastore to the startup datastore. Events that cause Waveserver 5 to create a new system configuration include: the detection of a new module in a deprovisioned slot; detection of a new pluggable in an unconfigured XCVR interface; or, the arrival of new Layer-0 Control Plane configuration data through SPLI.

If you do not specify a datastore when performing configuration operations, the running datastore is the default target. You can use any of the supported management interfaces to modify the running datastore: CLI, REST, NETCONF, gNMI/gNOI, SNMP, and WebSocket. Unlike the other management interfaces, gNOI and gNMI operations can only target the running datastore. When you use gNMI/gNOI to make changes to the running datastore, the changes are automatically saved to the startup datastore. SNMP MIBs reflect the current running configuration and operational state of the system. The running datastore receives output from telemetry subscriptions, including WebSocket publications that occur when a WebSocket client connects to the Waveserver 5.

Startup datastore

The startup datastore contains the most recently saved running datastore configuration settings.

Initially, the startup datastore contains the Waveserver 5 default configuration settings. The default configuration settings can be used for most Waveserver 5 applications.

The first time you bring up the Waveserver 5 system during commissioning and each time you perform a cold restart, the configuration in the startup datastore is loaded into the running datastore and becomes the active configuration.

If the startup datastore was deleted or does not exist, the running datastore contains the default initial state of the system, which may include modules and facilities that were automatically provisioned by the Waveserver 5 on discovery during initialization.

Candidate datastore

The candidate datastore contains a configuration that you can develop, modify, and validate in real time without affecting the running datastore or disrupting traffic.

The candidate datastore initially contains the system default configuration, which is equivalent to the configuration the system applies to a chassis containing no modules after a `reset-to-factory-defaults` operation.

You can seed the candidate datastore with the contents of the running datastore or startup datastore, or with an external configuration that you import. You can also modify the initial contents of the candidate datastore manually using CLI, NETCONF, or REST operations. You cannot modify the candidate datastore using gNMI/gNOI, SNMP, or WebSocket clients.

The primary use of the candidate datastore is to create and validate a configuration that is intended for future use. Creating and validating the candidate datastore does not affect the running datastore or disrupt traffic. Changes to the candidate datastore are validated as they are made to ensure the candidate datastore always contains a valid configuration.

Use the candidate datastore to create and validate a configuration for one or more Waveserver 5 nodes in a network before you commit the new configuration, so you can synchronize configuration changes across the network. When you commit the candidate datastore to the running datastore, the system determines the differences between the running datastore configuration and the candidate datastore configuration and only applies the changes. Configuration settings in the running datastore that are not in the candidate datastore are deleted. To commit the candidate datastore to the running datastore use the `configuration commit` command.

Table 186 Datastore summary

Datastore	Contents	Persistent	Interface access	Telemetry
Running	Configuration settings Operational state	Yes	CLI, REST, NETCONF, gNMI/gNOI, SNMP, WebSocket	Yes
Startup	Configuration settings	Yes	CLI, REST, NETCONF	No
Candidate	Configuration settings	Yes	CLI, REST, NETCONF	No
For examples of working with datastores using REST, NETCONF, and gNMI/gNOI commands, see REST , NETCONF , gRPC , OpenConfig , and WebSocket .				

Saving configurations

As you make changes to the active configuration, you can make those changes persist by saving the running datastore to the startup datastore using the `configuration save` command.

You can also save the configuration settings in a specific datastore to a file. Ciena recommends that you create a backup configuration file that can be restored and installed as needed on an existing or newly installed and commissioned Waveserver 5.

Working with datastores

This section describes new and changed CLI commands you can use to perform operations on datastores.

Datastore operations

The Waveserver 5 CLI includes a set of configuration commands for working with datastores.

You can perform operations on datastores using CLI, REST, NETCONF, and gNMI/gNOI commands. The following table describes CLI commands that perform datastore operations. For information on working with datastores using the other management interfaces, see [REST](#), [NETCONF](#), [gRPC](#), [OpenConfig](#), and [WebSocket](#).

Table 187 Datastore CLI commands

CLI Datastore command	Description
<code>cli candidate <disable enable></code>	Enables or disables candidate mode. When you enable candidate mode and make a configuration change during a CLI session, the change is only made to the candidate datastore for that CLI session. Other, concurrent CLI sessions that reference the running datastore are not affected.
<code>configuration datastore lock <startup candidate running></code>	Locks the specified datastore, preventing any changes to it. Typically, you lock the running datastore to prevent changes to the configuration settings. You can also lock the candidate datastore to limit access to it. When a lock exists for a particular datastore only the NBI user session that created the lock is allowed to update that datastore.
<code>configuration datastore unlock <startup candidate running> [force]</code>	Releases a previously locked datastore. A lock can only be unlocked by the user who created the lock, unless you use the <code>force</code> option. The <code>force</code> option forces the unlock operation on the specified datastore. You require super user privileges to use the <code>force</code> option.
<code>configuration datastore show datastore <startup candidate running> configuration datastore show differences source <startup candidate running> destination <startup candidate running></code>	Lists the contents of the specified datastore. The datastore show differences command compares the contents of the two specified datastores then outputs a combined list of the datastore contents indicating which configuration settings commands were added or removed.

CLI Datastore command	Description
configuration datastore copy source <startup candidate running> destination <startup candidate running>	Copies the contents of the specified source datastore to the specified destination datastore. Note 1: Use <code>configuration save</code> (not the copy command) to copy the running datastore to the startup datastore. Note 2: Use <code>configuration commit</code> (not the copy command) to copy the contents of the candidate datastore to the running datastore.
configuration datastore delete <startup candidate running>	Deletes the contents of the specified datastore.
configuration save	Saves the configuration settings in the running datastore to the startup datastore. For variations on this command, see Changed CLI commands for configuration operations .

Changed CLI commands for configuration operations

The datastore model maintains configuration information in binary format. The datastores and their contents are not directly accessible to users as the text files containing CLI configuration commands were in previous releases.

As a result, some existing CLI commands have changed to accommodate the change in how configurations are created, stored, and manipulated. The following table summarizes these changes.

Table 188 Changed CLI commands

CLI command	Description
configuration backup filename <filename>	Creates a full backup archive of the configuration datastores in binary format. The backup archive is stored in the local config directory. Use the backup archive in a restore operation.

CLI command	Description
<pre>configuration check filename <filename> [verbose] [hide-progress] [default- server] [default-tftp-server] [default- ftp-server] [default-sftp-server] [default-scp-server] [tftp-server <ip- addr-or-hostname>] [ftp-server <ip-addr- or-hostname> login-id <username> password <password> secret <secret>] [sftp-server <ip-addr-or-hostname> login-id <username> password <password> secret <secret>] [scp-server <ip-addr- or-hostname> login-id <username> password <password> secret <secret>] [url <string>]</pre>	Configuration check is only applicable to text-based configuration files that contain CLI commands.
<pre>configuration copy filename <filename> to filename <filename></pre>	Makes a copy of a file in the system config directory. Applies to both CLI files and backup archive files.
<pre>configuration install filename <filename> [verbose] [hide-progress] [default-server] [default-tftp-server] [default-ftp-server] [default-sftp- server] [default-scp-server] [tftp- server <ip-addr-or-hostname>] [ftp- server <ip-addr-or-hostname> login-id <username> password <password> secret <secret>] [sftp-server <ip-addr-or- hostname> login-id <username> password <password> secret <secret>] [scp-server <ip-addr-or-hostname> login-id <username> password <password> secret <secret>] [url <string>]</pre>	Writes a configuration file in its native binary database format or CLI text format from a remote server location to the \config directory on the local node.
<pre>configuration list</pre>	This command lists all files in the config directory, with a timestamp of the last file modification.
<pre>configuration rename filename <original_name> to filename <new_name></pre>	Renames files in the config directory.
<pre>configuration reset-to-defaults</pre>	Reset the configuration file to the system default settings.

CLI command	Description
<code>configuration reset-to-factory-defaults</code>	Performs a full reset to factory defaults (RTFD) operation on the configuration and system files. This operation deletes user configuration information such as: hostname, IP address, configuration files, logs, user accounts, and passwords. Note: Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.
<code>configuration reset-to-user-config filename <filename> [destination <datastore>] [revert-timeout] [force]</code>	Replaces the running configuration with the set of CLI commands in the specified text-based configuration file. If you do not specify a datastore this command operates on the running configuration similar to legacy behavior. If you do specify a datastore, it must be the candidate or running datastore. See The <code>[revert-timeout]</code> parameter sets a time limit for executing the command. If the specified time elapses before the <code>reset-to-user-config</code> command successfully completes, the configuration reverts to the currently running configuration (running datastore); that is, the system state before the <code>reset-to-user-config</code> operation started. The <code>[force]</code> option is used to bypass the service-affecting confirmation prompt that requires you to answer yes or no. You require super user privileges to use the <code>reset-to-user-config</code> command.

CLI command	Description
<pre>configuration save configuration save filename <filename> configuration save filename <filename> datastore <startup candidate running></pre>	By itself, the <code>configuration save</code> command saves the running datastore to the startup datastore. If you specify a filename only, the command saves the running datastore as a file of CLI commands, under the specified filename, in the local config directory. If you specify a filename and a datastore, the command saves the specified datastore as a file of CLI commands, under the specified filename, in the local config directory.
<pre>configuration show configuration show filename <filename> configuration show status configuration show differences-from-saved</pre>	By itself, <code>configuration show</code> displays the contents of the running datastore. If you specify <code>filename <filename></code>, the command shows the contents of that file. You can only use this command on a configuration file that was save as a text file containing CLI commands. The command <code>configuration show status</code> displays the status of the three datastores: running, startup, and candidate. The <code>differences-from-saved</code> argument displays a summary of the differences between the running datastore and the startup datastore. That is, it shows the changes made to the running datastore since the last time it was saved.

Procedure list

This section provides the following procedures:

- [Procedure 109, “Displaying and comparing datastore contents”](#)
- [Procedure 110, “Retrieving the status of the datastores”](#)
- [Procedure 111, “Saving a datastore configuration”](#)
- [Procedure 112, “Copying a datastore”](#)
- [Procedure 113, “Locking and unlocking datastores”](#)
- [Procedure 114, “Validating a datastore”](#)
- [Procedure 115, “Committing the candidate datastore”](#)
- [Procedure 116, “Enabling and disabling candidate mode”](#)
- [Procedure 117, “Deleting a datastore”](#)
- [Procedure 118, “Backing up the configuration datastores”](#)

- [Procedure 119, “Saving a configuration datastore as a set of CLI commands in a text file”](#)
- [Procedure 120, “Restoring backed-up configuration datastores”](#)
- [Procedure 121, “Applying a text file of CLI commands to a configuration datastore”](#)
- [Procedure 122, “Resetting the Waveserver 5 configuration”](#)

Procedure 109 Displaying and comparing datastore contents

Use this procedure to display the contents of a datastore and to compare the contents of two datastores.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Steps

- 1 To display the contents of the running datastore, run the command:

```
configuration show
```

This command lists the contents of the running datastore.

- 2 To display the contents of a datastore, run the command:

```
configuration datastore show datastore <startup |  
candidate | running>
```

For example:

```
configuration datastore show datastore startup
```

This command lists the contents of the startup datastore.

- 3 To compare the contents of two datastores, run the command:

```
configuration datastore show differences source <startup  
| candidate | running> destination <startup | candidate |  
running>
```

For example:

```
configuration datastore show differences source running  
destination startup
```

The system outputs a list of the differences between the two datastores indicating which settings were added (+) and deleted (-).

For example:

```
# config datastore show differences source running destination startup  
--- /tmp/tempSourceConfig  
+++ /tmp/tempDestinationConfig  
@@ -37,7 +37,6 @@  
chassis set slot 6 power-state automatic  
chassis set slot 7 power-state automatic  
chassis set slot 8 power-state automatic  
-command-log enable dhcp client disable  
dhcp client set discovery-interval 30  
dhcp client set interface-type active
```

In this example, `command-log enable` is in the running datastore but is not in the startup datastore, as indicated by the minus sign preceding the statement.

Procedure 110 Retrieving the status of the datastores

Use this procedure to retrieve the status of the datastores.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Steps

- 1 To retrieve and display the status of the three datastores, enter this command:

```
configuration show status
```

The system displays the status of each datastore, which includes the state, the lock status, and the session ID of the lock holder, if it is locked. For example:

CONFIGURATION STATUS			
Datastore	Status	Lock Status	Lock Holder
Running	Unsaved	Locked	3
Startup	Ready	Unlocked	
Candidate	Uncommitted	Locked	3

The following table describes the datastore statuses.

Table 189 Datastore status

	Status	Description
Running datastore	Default	The running datastore contains system initial state configuration settings with no user-applied changes.
	Saved	The running datastore configuration was saved to the startup datastore. No changes were made to the running datastore configuration settings since it was saved. The configuration data in the running datastore and the startup datastore should be the same.
	Unsaved	The running datastore configuration was changed, but was not saved. That is, the changes to the running datastore have not been saved to the startup datastore.
Startup datastore	Default	The startup datastore contains system initial state configuration settings with no user-applied changes.
	Empty	The startup datastore has not been created yet or has been deleted.
	Ready	The startup datastore exists and is ready to be used on the next control-module or chassis cold restart.

	Status	Description
Candidate datastore	Default	The candidate datastore contains system initial state configuration settings with no user applied changes.
	Committed	The candidate datastore has been committed to the running datastore. No changes were made to the candidate datastore configuration settings since it was committed.
	Uncommitted	The candidate datastore configuration was changed, but the changes were not committed to the running datastore.
Lock status	Locked	The datastore has been locked. Configuration changes from other user sessions are blocked. Only the session holding the lock can modify the datastore.
	Unlocked	The datastore is not locked and can be modified.
Lock Holder	<session-id>	The <session-id> is the ID of the session that locked the datastore. Only that session can make configuration changes while the datastore is locked.

Procedure 111 Saving a datastore configuration

Use this procedure to save the Waveserver 5 configuration in the running datastore.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Saving the active configuration in the running datastore:

- writes the contents of the running datastore to the startup datastore.
- persists the active configuration in the running datastore. Any unsaved modifications to the running datastore are saved.

Note: The running datastore is also automatically saved to the startup datastore when the system automatically provisions a new configuration state.

Steps

- | | |
|---|--|
| 1 | <div style="display: flex; justify-content: space-between;"> <div> <p>If you want to save</p> <hr/> <p>the running datastore to the startup datastore</p> <p>the running datastore to a text file as CLI commands</p> <p>any datastore to a text file as CLI commands</p> </div> <div> <p>Then go to</p> <hr/> <p>step 2</p> <p>step 3</p> <p>step 4</p> </div> </div> |
| 2 | <p>To save the active configuration settings in the running datastore to the startup datastore, run this command:</p> <pre>configuration save</pre> <p>Go to step 5.</p> |
| 3 | <p>To save the configuration settings in the running datastore as a list of CLI commands in a text file, run this command:</p> <pre>configuration save filename <filename></pre> <p>For example, to save the running datastore to a CLI-based file called runningDS run this command:</p> <pre>configuration save filename runningDS</pre> <p>Note: Waveserver 5 supports filenames containing alphanumeric characters, hyphens, and underscores. Waveserver 5 allows the following special characters in filenames: ~ ' @ # \$ () + = { } [] . <> Spaces are not supported.</p> <p>The file is saved to the local config directory.</p> <p>Go to step 5.</p> |

- 4 To save the configuration settings in any datastore as a list of CLI commands in a text file, run this command:

```
configuration save filename <filename> datastore <running  
| startup | candidate>
```

For example, to save the candidate datastore to a CLI-based file called cand06-15.

```
configuration save filename cand06-15 datastore candidate
```

Note: Waveserver 5 supports filenames containing alphanumeric characters, hyphens, and underscores. Waveserver 5 allows the following special characters in filenames: ~ ' @ # \$ () + = { } [] . <> Spaces are not supported.

The file is saved to the local config directory.

- 5 You have completed this procedure.

Procedure 112 Copying a datastore

Use this procedure to copy the configuration settings from one datastore to another.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

The copy operation is a root-level replace. That is, the system determines the differences between the source and destination datastores and merges the differences into the destination by creating, deleting, and modifying the configuration state to make it identical to the source.

The datastore copy operation affects the current system and hardware configuration if the destination datastore is the running datastore.

Copying is not supported between all source and destination datastores. The following table summarizes all possible datastore copy operations.

Table 190 Supported source-destination datastore copy operations

Source datastore	Destination datastore	Supported	Notes
Running	Candidate	Yes	Populates the candidate datastore with the configuration settings from the running datastore. You can modify the contents of the candidate datastore and commit the changes in the future.
	Startup	No	Use <code>configuration save</code> instead, which writes the contents of the running datastore to the startup datastore.
Startup	Running	Yes	Replaces the current running datastore configuration with the startup datastore configuration. Do this when the running datastore contains unsaved configuration changes that you want to delete. This method lets you revert to the startup datastore without performing a cold restart.
	Candidate	Yes	Populates the candidate datastore with the configuration settings from the startup datastore. You can modify the contents of the candidate datastore and commit the changes in the future.

Source datastore	Destination datastore	Supported	Notes
Candidate	Running	No	Use the datastore commit operation.
	Startup	No	Commit the candidate datastore to the running datastore first; then save the running datastore to write to contents to the startup datastore.

Steps

- 1 To copy the contents of one datastore to another datastore, run this command:

```
configuration datastore copy source <startup | candidate  
| running> destination <startup | candidate | running>
```

Procedure 113 Locking and unlocking datastores

Use this procedure to lock and unlock a datastore.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

When you lock a datastore, you are given exclusive access to the datastore to modify the configuration settings it contains. The datastore lock is assigned to your user session, and only lasts for the life of the session. Only you have access to a datastore that you have locked. Until you unlock the datastore, other users (user sessions) are denied access to the locked datastore and cannot modify the datastore contents. Other users can query a locked datastore.

When the running datastore is locked, Waveserver 5 can still auto-provision the system and make configuration changes in the running datastore. For example, when a module is inserted into a slot that is not provisioned, the new module instance and the dependent child objects are still created even when the running datastore is locked. If the startup datastore is locked, the system does not automatically save the running datastore to the startup datastore as normally occurs for system auto-provisioning.

A locked datastore also prevents other users (user sessions) from performing the actions listed in the following table.

Table 191 Operations blocked by a locked datastore

Operation	Notes
commit	A commit operation is blocked if the running datastore is blocked.
datastore copy	A copy operation is blocked if the destination datastore is locked.
reset-to-factory-defaults	Blocked if at least one datastore is locked.
reset-to-user-config augment	These actions are blocked if the target datastore is blocked.
restore	Restoring a system configuration from a backup is blocked.
save	A save operation is blocked if the startup datastore is locked.

A locked datastore is unlocked when:

- an unlock operation is performed in the user session that holds the lock.
- the user session that holds the lock closes, expires, or terminates.
- a super level user forces the datastore to unlock.

Steps

- 1 To lock a datastore, run the command:

```
configuration datastore lock <startup | candidate |  
running>
```

For example, to lock the candidate datastore to make modifications to it:

```
configuration datastore lock candidate
```

- 2 To unlock a datastore, run the command:

```
configuration datastore unlock <startup | candidate |  
running> [force]
```

For example, to unlock the candidate datastore after making modifications to it:

```
configuration datastore unlock candidate
```

To force an unlock on a locked datastore, you must have super level privileges. To force an unlock in the previous example:

```
configuration datastore unlock candidate force
```

Procedure 114 Validating a datastore

Use this procedure to validate the configuration settings in a datastore.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

You can use this procedure to validate the contents of each of the three datastores.

Steps

- 1 To validate a datastore, run the command:

```
configuration datastore validate <startup | candidate | running>
```

For example, to validate the contents of the startup datastore:

```
configuration datastore validate startup
```

Procedure 115 Committing the candidate datastore

Use this procedure to commit the configuration settings in the candidate datastore to the running datastore.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

When you commit the candidate datastore to the running datastore, the process performs a root-level replace. That is, the system determines the differences between the running datastore configuration and the candidate datastore configuration and only applies the changes. Configuration settings in the running datastore that are not included in the candidate datastore are deleted.

Steps

- 1 To commit the candidate datastore to the running datastore, run the command:

```
configuration commit [revert-timeout <N[yMwdhms]>]
```

where:

[revert-timeout] — sets a timer for execution of the commit command. If the specified time elapses before the commit successfully completes, the commit operation is canceled. Format: to specify 1 hour, 10 minutes and 3 seconds for example, the format is 1h10m3s.

Note: If you issue a commit command with a timeout value, then issue a second commit command with a second revert-timeout value while the timer of the first command is still running, the timer is updated to the second revert-timeout value. If you issue a third commit command with no revert-timeout value while the timer of the second command is still running, the timer is canceled and the commit becomes permanent. You can also cancel the revert-timeout timer by issuing the `configuration cancel-revert` command, which also causes the commit to become permanent.

Procedure 116 Enabling and disabling candidate mode

Use this procedure to enable or disable candidate mode.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Overview

When you enable candidate mode and make a configuration change in a CLI session, the change is only made to the candidate datastore for that CLI session. Other, concurrent CLI sessions that reference the running datastore are not affected.

Steps

- 1 To enable candidate mode, run this command:
`cli candidate enable`
- 2 To disable candidate mode, run this command:
`cli candidate disable`

Procedure 117 Deleting a datastore

Use this procedure to delete a datastore.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Deleting a datastore removes the current contents of the datastore, and repopulates it with the initial Waveserver 5 system configuration.

Deleting the running datastore is handled differently because it contains the current system and hardware configuration, including the traffic configuration. Deleting the running datastore causes a root-level replace of the current running datastore configuration with the initial system configuration, changing only the configuration settings that are different.

The following table describes operations you can perform as an alternative to deleting the running datastore.

Table 192 Alternatives to deleting the running datastore

Alternative operation	Description
Commit the candidate datastore containing only default/initial configuration-to-factory-defaults	Performs a root-level replace with the Waveserver initial state from the candidate datastore.
Set running to <no-config> at the root ("/") with a replace operation	Performs an explicit root-level replace to an empty configuration that has the same effect as deleting all current configuration and re-initializing to the initial system configuration.
Control-module cold restart when startup datastore contains only default/initial configuration	On control-module cold restart the running datastore is replaced by the startup datastore initializing with the initial Waveserver configuration. Any existing traffic will be deleted and reconfigured with default values which may be different than the previous configuration.

Note: Deleting any or all datastores is not equivalent to performing a reset-to-factory-default operation. Deleting a datastore removes the contents of the datastore and re-initializes the datastore contents. A reset-

to-factory-defaults deletes all system configurations before rebuilding the configuration from the committed software load. For details, see [Reset to Factory Defaults \(RTFD\)](#).



CAUTION

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

Steps

- 1 To delete a datastore, run the command:

```
configuration datastore delete <startup | candidate | running>
```

For example, to delete the contents of the startup datastore:

```
configuration datastore delete startup
```

Procedure 118 Backing up the configuration datastores

Use this procedure to create a full backup of the configuration datastores in native binary format. You can back up the configuration datastores to the local Waveserver 5 config directory or to a remote server.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- To access a default remote server, the server must be configured on Waveserver 5, see [Managing and configuring default servers](#).

Overview

Note 1: You can back up more than one configuration and store the backed-up files on the local Waveserver 5 or a remote server. The backup creates an archive of the datastores (running, startup, candidate) in native binary format, and logs and system files, and stores the archive in the `/config` directory. Use the `configuration restore` command to restore the backed-up configuration files.

Note 2: To ensure that the running and startup configuration datastores are the same when you back up the datastores, perform a `configuration save` operation just before you perform a full backup of the datastores.

Note 3: Waveserver 5 supports filenames containing alphanumeric characters, hyphens, and underscores. Waveserver 5 allows the following special characters in filenames: `~ ' @ # $ ( ) + = { } [ ] . <>` Spaces are not supported.

ATTENTION

Do not back up Integrated Test Set (ITS) or loopbacks to a configuration file, or restore them from a configuration file. Before you backup a configuration file, make sure that ITS is disabled and that loopbacks are not active.

Steps

- 1 To backup the configuration datastores (running, startup, and candidate) to a file that is stored on the local device, run this CLI command:

```
configuration backup filename <filename>
```

where:

`<filename>` — is the name of the file containing the backed-up configuration. The backed-up configuration is stored in the local config directory: `/var/opt/config`

- 2 Verify that the configuration datastores were backed up to the file you specified. At the system command prompt run this command:

```
configuration list
```
- 3 If you want to save the backed up configuration datastores to a remote SFTP or SCP server, verify the directory path and filename of the backed-up configuration and run this command:

```
system xftp putfile local-filename <String: full  
directory path and filename> [remote-filename <String: up  
to 127 characters>] <default-server default-sftp-server |  
default-scp-server | sftp-server <IP address or host  
name> login-id <String> password <String> secret <String>  
| scp-server <IP address or host name> login-id <String>  
password <String> secret <String>
```

For example:

```
system xftp putfile local-filename /home/su/config/  
startup-config remote-filename /tmp/startup-  
config.ws5_01 sftp-server 10.1.2.3 login-id root password  
root
```

Note 1: The server must be configured on Waveserver 5. See [Managing and configuring default servers](#).

Note 2: You must specify [login-id] and [password] for SFTP and SCP servers.

Note 3: The local-filename is the path and filename of the backed-up configuration.
- 4 To restore a backed-up configuration datastore, see [Procedure 120, "Restoring backed-up configuration datastores"](#).

Procedure 119 Saving a configuration datastore as a set of CLI commands in a text file

Use this procedure to save a configuration datastore as a set of CLI commands in a text file.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- To access a default remote server, the server must be configured on Waveserver 5, see [Managing and configuring default servers](#).

Overview

In previous releases, the Waveserver 5 configuration file was a set of CLI commands contained in a single text file (startup config). At system startup, the startup config was loaded into memory to become the active, running configuration.

As of Waveserver 5 Release 2.1 configuration settings are maintained in binary format in a database that consists of three datastores. Each datastore contains a complete Waveserver 5 configuration. For details see [“Datastore model” on page 596](#). The three datastores are:

- Running datastore
- Startup datastore
- Candidate datastore

Note 1: You can save more than one configuration datastore and store the saved configuration file on the local Waveserver 5 or a remote server.

Note 2: Waveserver 5 supports filenames containing alphanumeric characters, hyphens, and underscores. Waveserver 5 allows the following special characters in filenames: ~ ' @ # \$ () + = { } [] . <> Spaces are not supported.

ATTENTION

Do not back up Integrated Test Set (ITS) or loopbacks to a configuration file, or restore them from a configuration file. Before you backup a configuration file, make sure that ITS is disabled and that loopbacks are not active.

Steps

- 1 To backup a configuration datastore (running, startup, or candidate) as a set of CLI commands in a text file that is stored on the local device, run this CLI command:

```
configuration save filename <filename> datastore <running  
| startup | candidate>
```

where:

<filename> — is the name of the file where the datastore is saved as a set of CLI commands in a text file. The saved configuration is stored in the local config directory: `/var/opt/config`

Note: If you do not specify a datastore, by default this command saves the running datastore to <filename>. For example:

```
configuration save filename <filename>
```

- 2 Verify that the configuration datastore was saved to the file you specified. At the system command prompt run this command to list the contents of the config directory:

```
configuration list
```

- 3 If you want to save the CLI-based text file to a remote SFTP or SCP server, verify the directory path and filename of the saved configuration, and run this command:

```
system xftp putfile local-filename <String: full  
directory path and filename> [remote-filename <String: up  
to 127 characters> <default-server default-sftp-server |  
default-scp-server | sftp-server <IP address or host  
name> login-id <String> password <String> secret <String>  
| scp-server <IP address or host name> login-id <String>  
password <String> secret <String>
```

For example:

```
system xftp putfile local-filename /home/su/config/  
startup-config remote-filename /tmp/startup-  
config.ws5_01 sftp-server 10.1.2.3 login-id root password  
root
```

Note 1: The server must be configured on Waveserver 5. See [Managing and configuring default servers](#).

Note 2: You must specify `login-id` and `password` for SFTP and SCP servers.

Note 3: The `local-filename` is the path and filename of the backed-up configuration.

- 4 To apply a configuration that you saved as a set of CLI commands in a text file, you must use the `reset-to-user-config` command. See [Procedure 121, “Applying a text file of CLI commands to a configuration datastore”](#).

Procedure 120 Restoring backed-up configuration datastores

Use this procedure to restore configuration datastores that were backed up to a file.

Requirements

- You must be logged in to Waveserver 5 using an account with admin or super user access privileges. Admin users can only restore configuration files that contain commands an admin user has privileges to execute. If the configuration file contains commands requiring super user access privileges, the admin user will not be able to restore the configuration file.
- To access a default remote server, the server must be configured on Waveserver 5. See [Managing and configuring default servers](#).
- The system time must be set (either manually or via the NTP server) prior to restoring a configuration file.

Overview

ATTENTION

Do not restore a configuration with Integrated Test Set (ITS) or loopbacks
Before you backup a configuration, make sure that ITS is disabled and that loopbacks are not active.

Note: When the restore operation is performed the CM is restarted.

Steps

- 1

If the backed-up configuration datastore is on a remote server that supports SFTP or SCP	Then go to step 2
the local node	step 3
- 2

If the backed-up datastore is saved on a remote server that supports SFTP or SCP, use the following command to copy the backed-up datastore to the local node:

```
configuration install filename [default-server | default-sftp-server | default-scp-server | sftp-server <IP address or host name> login-id <String> {password <String> secret <String>} | scp-server <IP address or host name> login-id <String> {password <String> secret <String>}]
```

Note 1: Ensure that you specify the full path to the file in the command.

Note 2: You must specify `login-id` and `password` for SFTP and SCP servers.

Note 3: The CLI command `configuration install` copies the backed-up configuration file from its storage location to the config directory on the local node. Once the configuration file is on the local node you must restore the file. To verify that the backed-up configuration file is the default load file, use CLI command `configuration list`.

- 3 To restore a backed-up configuration datastore stored on the local device, run this CLI command:

```
configuration restore filename <filename>
```

where:

<filename> — is the name of the file containing the backed-up configuration. By default, configuration files are backed up to the local config directory: `/var/opt/config`

Note: When the restore operation is performed the CM is restarted.

- 4 Optionally, log in to Waveserver 5, and retrieve the summary list of configurations to confirm that the specified configuration has been restored:

```
configuration show
```

Procedure 121 Applying a text file of CLI commands to a configuration datastore

Use this procedure to apply a configuration that was saved as a set of CLI commands in a text file.

Requirements

- You must be logged in to Waveserver 5 using an account with admin or super user access privileges. Admin users can only apply configuration files that contain commands an admin user has privileges to execute. If the configuration file contains commands requiring super user access privileges, the admin user will not be able to apply the configuration file.
- To access a default remote server, the server must be configured on Waveserver 5. See [Managing and configuring default servers](#).
- The system time must be set (either manually or via the NTP server) prior to restoring a configuration file.

Overview

In previous releases, the Waveserver 5 configuration file was a set of CLI commands contained in a single text file (startup config). At system startup, the startup config was loaded into memory to become the active, running configuration.

As of Waveserver 5 Release 2.1 configuration settings are maintained in binary format in a database that consists of three datastores. Each datastore contains a complete Waveserver 5 configuration. For details see [“Datastore model” on page 596](#). The three datastores are:

- Running datastore
- Startup datastore
- Candidate datastore

ATTENTION

Do not restore a configuration with Integrated Test Set (ITS) or loopbacks
Before you backup a configuration, make sure that ITS is disabled and that loopbacks are not active.

Steps

1	If the configuration text file is saved	Then go to
	on a remote server that supports SFTP or SCP	step 2
	on the local node	step 3

- 2 If the file is saved on a remote server that supports SFTP or SCP, you must first install the saved configuration text file on the local node.

Note: The `configuration install` command writes a copy of the specified file in its native database format or CLI text format to the `\config` directory on the local node.

Run the following CLI command:

```
configuration install filename [default-server | default-  
sftp-server | default-scp-server | sftp-server <IP  
address or host name> login-id <String> {password  
<String> secret <String>} | scp-server <IP address or host  
name> login-id <String> {password <String> secret  
<String>}]
```

Note 1: Ensure that you specify the full path to the file in the command.

Note 2: You must specify [login-id] and [password] for SFTP and SCP servers.

- 3 Apply the configuration file by using `reset-to-user-config`:

```
configuration reset-to-user-config filename <filename>  
[revert-timeout <N[yMwdhms]>]
```

where:

<filename> — is the name of the text configuration file containing the set of CLI commands. The saved configuration is stored in the local config directory: `/var/opt/config`

Note 1: Restoring the backed-up configuration can take several minutes.

Note 2: For the [revert-timeout] parameter, to specify 1 hour, 10 minutes and 3 seconds, the format is: `1h10m3s`. If the specified time elapses before the installation successfully completes, the configuration reverts to the currently running configuration (running datastore); that is, the system state before the `reset-to-user-config` operation started. Cancel the revert timer to make the configuration permanent with the command. To review the status of the revert timer, use the command `configuration show status`.

The prompt does not return until the restore of the backed-up configuration completes.

Procedure 122 Resetting the Waveserver 5 configuration

Use this procedure to reset the Waveserver 5 configuration.

Requirements

You must be logged in to Waveserver 5 using an account with super user access privileges.

Overview

Use this procedure to reset the Waveserver 5 configuration. There are three reset commands:

Table 193 Configuration reset commands

Command	Description
<code>configuration reset-to-defaults</code>	Resets the configuration file to the system default settings.
<code>configuration reset-to-user-config filename <filename> datastore <datastore> revert-timeout <N[yMwdhms]></code>	Replaces the running configuration with the specified configuration file. If you do not specify a datastore this command operates on the running configuration similar to legacy behavior. If you do specify a datastore, it must be the candidate or running datastore.
<code>configuration reset-to-factory-defaults</code>	This operation performs a full reset to factory defaults (RTFD) operation on the configuration and system files. It deletes user configuration information such as: hostname, IP address, configuration files, logs, user accounts, and passwords. This command is not covered in this procedure. See Procedure 71, "Performing a Reset To Factory Defaults (RTFD)". Note: Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

Steps

- | | | |
|---|---|---|
| 1 | If you want to reset the configuration file to the system default settings
to the user configuration file | Then go to
step 2
step 3 |
| 2 | To reset the active configuration settings in the running datastore to the system default settings, run this command:

<code>configuration reset-to-defaults</code>
Go to step 4 . | |
| 3 | To reset the configuration settings in the running datastore with the user configuration file, run this command:

<code>configuration reset-to-user-config filename <filename>
datastore <running candidate> revert-timeout
<N[yMwdhms]></code> | |



CAUTION

Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to isolate a failure event.

For example, to replace the running datastore configuration settings with a configuration file config_01, run this command:

```
configuration reset-to-user-config filename config_01  
datastore running
```

Note: The revert-timeout argument specifies a time limit on the datastore replacement after which the replaced datastore reverts to its original state. To specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. If the specified time elapses before the `reset-to-user-config` is successfully completed, the configuration reverts to the previously running configuration (running datastore).

- 4 You have completed this procedure.

CHAPTER 25

SNMP

This section contains the following topics:

- [Overview](#)
 - [Proprietary and standard MIBs](#)
 - [ENTITY-MIB](#)
 - [Physical entity overview](#)
 - [Index schemes](#)
 - [PM-MIB](#)
 - [SNMPv2c](#)
 - [SNMPv3](#)
 - [SNMP traps](#)
 - [SNMP security considerations](#)
- [SNMP MIB files and external yang interfaces](#)
- [Object Identifiers \(OID\)](#)
 - [Finding the entity index associated with an ifIndex](#)
- [Procedure list](#)

Overview

This section describes Simple Network Management Protocol (SNMP) and how to configure it.

ATTENTION

Waveserver 5 performance issues can occur when multiple northbound interfaces (NBIs) are used simultaneously on Waveserver 5.

The use of multiple primary NBIs is not supported on Waveserver 5. Use one instance of a single primary NBI (REST, NETCONF or SNMP) to extract data from Waveserver 5. If you are running a secondary interface for light monitoring, for example an SNMP trap monitor or WebSocket, do not duplicate the data management activity on the secondary interface. Avoid overlapping requests for the same data over two interfaces on the same Waveserver 5.

Avoid bulk data retrievals on Waveserver 5. Retrieve the specific data whenever possible. Waveserver 5 APIs allow you to specify individual data elements or group data.

For more information, see [General NBI guidelines](#).

Waveserver 5 supports SNMP versions v1, v2c, and v3. SNMPv2c and SNMPv3 use the same Management Information Base (MIB) definitions and primarily differ in authentication method. SNMPv2c uses plain-text community strings for authentication and SNMPv3 supports user-based authentication and encryption. Both authentication methods are supported by Waveserver 5 simultaneously and may be separately configured.

SNMP supports both IPv4 and IPv6.

SNMPv2c authentication (community strings) is enabled by default. To configure SNMPv2c authentication, see [Configuring SNMP community mapping](#).

SNMPv3 authentication (user-based) is disabled by default. See [Configuring SNMPv3 users](#), [Displaying SNMP users](#) and [Creating and attaching an SNMPv3 user to an SNMPv3 access entry group](#) to set up SNMPv3 authentication.

Note 1: A timeout value of 10 seconds is recommended when performing SNMP operations.

Note 2: Waveserver 5 supports SNMPv2c and SNMPv3 for sending alarm traps. Link-defect traps are also supported for Ethernet and management ports.

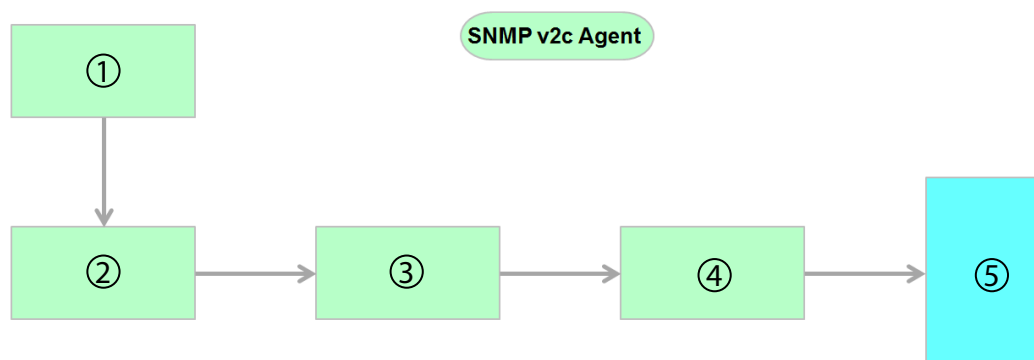
Note 3: Set operations are not supported using SNMP.

The following table lists the components in the SNMP architecture.

Table 194 Components in the SNMP architecture

Component	Description
Message dispatcher	The message dispatcher is responsible for sending and receiving SNMP messages. When an SNMP request message is received, the Dispatcher determines the SNMP version and then passes the message to the appropriate Message Processing Model (SNMPv2c or SNMPv3). The dispatcher is also responsible for dispatching SNMP response messages to the NMS interface application.
Community authentication	SNMPv2c supports a community-based security model. This model provides limited security based upon the configured community name. No encryption or user authentication is provided. All SNMP messages between Agent and manager contain “Community Name” in the plain text.
User-based authentication	SNMPv3 supports a user-based security model. This model requires users to provide a secret key for authentication and privacy. The supported authentication protocols are Keyed Hashing for Message Authentication Message Digest 5 (HMAC-MD5) and HMAC Secure Hash Algorithm 1 (SHA-1) algorithms.
View-based access control	SNMPv3 provides additional security by limiting access to specific managed objects.
Protocol PDU operations	The Message Processing Subsystem handles the Protocol PDU operations of preparing messages for sending and extracting data from received messages.
MIB (Management Information Base)	The system software supports standard MIBs that store configuration, operational state, and statistics information in the form of managed objects. See Proprietary and standard MIBs for a list of available and supported MIBs.

The following figure illustrates the SNMPv2c agent architecture.

Figure 134 SNMPv2c agent engine architecture

Legend

1	NMS Interface	4	Protocol PDU Operations
2	Message Dispatcher	5	MIB Handlers
3	Community Authentication		

Proprietary and standard MIBs

The system software supports standard and proprietary MIBs that store configuration, operational state, and statistics information in the form of managed objects. Each managed object is identified by an Object Identifier (OID) and exchanges information between the MIBs and the system software modules through an Application Programming Interface (API). The MIBs also define SNMP traps and notifications.

Note 1: Waveserver 5 supports RFC-4293, including IPv6 functionality.

Note 2: Set requests are not supported.

The Waveserver 5 homepage contains a compressed archive of the following proprietary MIB files which can be downloaded and extracted:

- CIENA-WS-MIB.my
- CIENA-WS-NOTIFICATION-MIB.my
- CIENA-WS-PLATFORM-ALARM-MIB.my
- CIENA-WS-PLATFORM-PM-MIB.my
- CIENA-WS-PLATFORM-TYPEDEFS-MIB.my

Note 1: A timeout value of 10 seconds is recommended when performing SNMP operations.

Note 2: Use SNMPv2 Get Bulk instead of SNMP Get to optimize performance when retrieving large table views while browsing MIBs.

For a detailed list of the proprietary MIB OIDs, see [Proprietary MIB OIDs](#).

The following standard MIB files are supported and are available online:

- ENTITY-MIB.my
- IF-MIB.my
 - ifAdminStatus
 - ifAlias
 - ifDescr
 - ifHighSpeed
 - ifInOctets
 - ifLastChange
 - ifMtu

- ifOperStatus
- ifOutOctets
- ifPhysAddress
- ifSpeed
- ifType
- IP-MIB.my
 - ipAddrEntry
 - ipAdEntIfIndex
 - IPadEntNetMask
- LLDP-MIB.my (see [Note 1](#) and [Note 2](#))
- LLDP-V2-MIB.my
- LLDP-V2-TC-MIB.my
- LLDP-EXT-DOT3-MIB.my
- LLDP-EXT-DOT3-V2-MIB.my
- RFC1213-MIB.my
- SNMPv2-CONF.my
- SNMPv2-MIB.my
- SNMPv2-SMI.my
- SNMPv2-TC.my
- SNMP-USER-BASED-SM-MIB.my

Note 1: The LLDP MIB uses Standard 802.1AB-2005.

Note 2: As of OS 2.1, Waveserver uses the open-source Linux LLDP daemon, which does not support the OID String attribute of the management address of the neighbor nodes.

ENTITY-MIB

ENTITY-MIB (Version 4) describes logical and physical entities that are managed by a single SNMP agent. ENTITY-MIB (Version 4) is defined by RFC-6933. Waveserver 5 implements a subset of the features specified in RFC-6933, specifically the Waveserver 5 physical entities and their hierarchy.

Standards compliance

The ENTITY-MIB agent specifies five tables, one stand-alone field, and one trap. Waveserver 5 implements the subset of ENTITY-MIB elements listed in the following table.

Table 195 ENTITY-MIB components

Name	Type	Implementation and compliance
entPhysicalTable	Table	Implemented. Partial compliance.
entPhysicalContainsTable	Table	Implemented. Fully compliant
entLogicalTable	Table	Not implemented
entLPMappingTable	Table	
entAliasMappingTable	Table	
entLastChangeTime	Field	
entConfigChange	Trap	

entPhysicalTable object

The entPhysicalTable object maintains a brief inventory of Waveserver 5 physical components, which can include:

- Chassis
- Access Panel
- Control Processor
- Service Modules
- Filler Modules
- Power Supply Units
- Cooling Fan Units

The following table lists the fields of the entPhysicalTable object implemented in Waveserver 5 in compliance with RFC-6933 standards.

Table 196 entPhysicalTable object fields

Field Name	Availability	Compliant
entPhysicalIndex	All	Yes
entPhysicalDescr	All	Partial
entPhysicalVendorType	None	No
entPhysicalContainedIn	All	Yes
entPhysicalClass	All	Yes
entPhysicalParentRelPos	All	Partial
entPhysicalName	All	Yes
entPhysicalHardwareRev	All	Yes

Field Name	Availability	Compliant
entPhysicalFirmwareRev	None	Yes
entPhysicalSoftwareRev	Chassis	Yes
entPhysicalSerialNum	All	Partial
entPhysicalMfgName	All	Yes
entPhysicalModelName	All	Yes
entPhysicalAlias	None	No
entPhysicalAssetID	None	No
entPhysicalIsFRU	All	Yes
entPhysicalMfgDate	All	Yes
entPhysicalUris	None	No
entPhysicalUUID	None	No

entPhysicalContainsTable object

The entPhysicalContainsTable object describes the parent-child relationships of physical components.

The following table lists the two fields in the entPhysicalContainsTable object (entPhysicalIndex and entPhysicalContainedIn) implemented in Waveserver 5 in compliance with RFC-6933 standards.

Table 197 entPhysicalContainsTable object fields

Field name	Availability	Compliant
entPhysicalIndex	All	YES
entPhysicalChildIndex	All	YES

These two fields indicate the parent-child relationship by showing which items are 'inserted' and which are 'contained in'. These fields have the same values in both the entPhysicalContainsTable object and the entPhysicalTable object.

The entPhysicalContainsTable object uses these fields to find children from the parent. The entPhysicalTable object uses these fields to find parents from the children. The entPhysicalContainsTable object does not provide a parent-child relationship for the Chassis as a child, because it has no parent entity.

Physical entity overview

ENTITY-MIB provides an inventory of the physical elements present in the Waveserver 5. Physical elements include, the chassis, control processor, power modules and fan modules, line and client ports, and access panel.

Considerations

Currently, you cannot disable ENTITY-MIB without affecting all other MIB agents. If you do a full SNMP walk, ENTITY-MIB and the tables EntPhysicalEntry and EntPhysicalContainsEntry are visible.

To display values from ENTITY-MIB in a MIB manager, you must include the MIBs listed in the following table.

Table 198 MIBs required to display values from ENTITY-MIB

MIB Name	Version	Source RFC	Purpose
ENTITY-MIB	4	RFC-6933	Field names and values
UUID-TC-MIB	—	RFC-6933	Field values
IANA-ENTITY-MIB	—	RFC-6933	Field values
SNMP-FRAMEWORK-MIB	—	RFC-3411	Field values
SNMPv2-CONF	—	RFC-2580	Compilation only
SNMPv2-TC	—	RFC-2579	Field values
SNMPv2-SMI	—	RFC-2578	Field values and OIDs

MIBs that are not specified in RFC-6933 are used for nearly all SNMP v2c agents and are likely already available to SNMP users.

Index schemes

The numbering schemes are used to uniquely identify an interface or piece of equipment. The ifIndex numbering scheme is used in the IF-MIB for SNMP Get operations and is also used to identify objects in the CIENA-WS-PLATFORM-PM-MIB. The following figure shows a graphical representation of the ifIndex numbering scheme.

Figure 135 ifIndex format

① (1 bit)	② (2 bits)	③ (5 bits)	④ (3 bits)	⑤ (12 bits)	⑥ (4 bits)	⑦ (5 bits)
0	xx	xxxxx	xxx	xxxxxxxxxxxx	xxxx	xxxxx

Legend

1	Sign Bit	5	Port
2	Version	6	Sub-port
3	Slot	7	Layer
4	Sub-slot		

The ifIndex format is based on a signed 32-bit integer, which is divided into the following fields:

- Sign bit: Not used and always set to 0.
- Version bits: Not used and always set to 0.
- Slot number: Identifies the slot number. Values are:
 - 00001 — Slot 1
 - 00010 — Slot 2
 - 00011 — Slot 3
 - 00100 — Slot 4
 - 00101 — Slot 5
 - 00110 — Slot 6
 - 00111 — Slot 7
 - 01000 — Slot 8
- Sub-Slot number: Not used and always set to 000.
- Port number: Identifies the port number within the module. Any value from 1 to 18. For logical ports, values are 101, 201, 202, and 301.
- Sub-Port number: Identifies the sub-port (channel) number. It is set to 0 for Ports that do not have any sub layers. For example, XCVR and PTP.
- Layer: Indicates the Port (and Sub-Port, if applicable). Values are:
 - 0 — Physical Slot. For example, XCVR.
 - 1 — WaveLogic Layer (physical layer)
 - 2 — PTP - optical power (physical layer)
 - 3 — Ethernet (logical layer)
 - 4 — OTUk (logical layer)
 - 5 — ODUk (logical layer)
 - 6 — OTUCn (logical layer)
 - 7 — ODUCn (logical layer)
 - 8 — Not Used/Reserved
 - 9 — Photonics (physical layer)

Waveserver 5 also uses a non-formatted ifIndex scheme to define management interfaces. This scheme uses only one-based numbers that are based on slot, port, sub-port, and type. The first ifIndex that uses this scheme is 16777728, which is slot 1, subSlot 0, port 1, subPort 0, type 0 (an XCVR installed in port 1 in chassis slot 1).

ifIndex 1 - 5, and 7 corresponds to the following management interfaces:

- ifIndex 1 - Console
- ifIndex 2 - ILAN-1
- ifIndex 3 - ILAN-2
- ifIndex 4 - DCN-1
- ifIndex 5 - Local
- ifIndex 6 - DCN-2
- ifIndex 7 - DCN-3

Interface Name (ifName)

The ifName is the read-only text name for the interface, which is assigned by the Waveserver 5.

Table 199 Interface and an example of ifName

Interface	ifName
Console	Console
ILAN-1	ILAN-1
ILAN-2	ILAN-2
DCN-1	DCN-1
DCN-2	DCN-2
DCN-3	DCN-3
Local	Local
Line XCVR	XCVR-<slot>/<1 2>
PTP	PTP-<slot>/<ptp>-WaveLogic PTP-<slot>/<ptp>
QSFP28 pluggable	XCVR-<slot>/<xcvr>
OTUCn or OTUCn_flex	Port-<slot>/<port>-OTUCn
ODU Channel	CHANNEL-<slot>/<port>.<channel>-ODUK
Client Ethernet port	PORT-<slot>/<port>

Interface Descriptor (ifDesc)

The ifDesc is the read-only text description of the interfaces, which is assigned by the Waveserver 5. The following table provides information on the interface and the values that may be returned for this field.

Table 200 Interfaces and ifDesc examples

Interface	ifDesc examples (Note)
Console port	Ciena Waveserver-5, Rx.x.x: Console management port
ILAN-1 port	Ciena Waveserver-5, Rx.x.x: ILAN-1 management port
ILAN-2 port	Ciena Waveserver-5, Rx.x.x: ILAN-2 management port
DCN-1 port	Ciena Waveserver-5, Rx.x.x: DCN-1 management port
DCN-3 port	Ciena Waveserver-5, Rx.x.x: DCN-3 management port
Local port	Ciena Waveserver-5, Rx.x.x: Local chassis management interface
Line XCVR	Ciena Waveserver-5, Rx.x.x: slot 1, sub-slot 0, port 1, sub-port 0, XCVR layer(0)
Line Port	Ciena Waveserver-5, Rx.x.x: slot 1, sub-slot 0, port 1, sub-port 0, WaveLogic layer(1)
Line PTP	Ciena Waveserver-5, Rx.x.x: slot 1, sub-slot 0, port 1, sub-port 0, PTP layer(2)
Line OTUCn	Ciena Waveserver-5, Rx.x.x: slot 1, sub-slot 0, port 1, sub-port 0, OTUCn layer(6)
Line ODUCn	Ciena Waveserver-5, Rx.x.x: slot 1, sub-slot 0, port 1, sub-port 0, ODUCn layer(7)
Line ODUk	Ciena Waveserver-5, Rx.x.x slot 1, sub-slot 0, port 1, sub-port 1, ODUk layer(5) Ciena Waveserver-5, Rx.x.x slot 1, sub-slot 0, port 1, sub-port 2, ODUk layer(5)
Client XCVR	Ciena Waveserver-5, Rx.x.x: slot 1, sub-slot 0, port 3, sub-port 0, XCVR layer(0)
Client Ethernet Port	Ciena Waveserver-5, Rx.x.x: slot 1, sub-slot 0, port 3, sub-port 0, Ethernet layer(3)
Note: Rx.x.x represents the Waveserver 5 Release number.	

PM-MIB

PM-MIB provides performance monitoring values reported by the Waveserver 5. This includes the PM TCA threshold information, such as when the PM TCA has been crossed, the TCA administrative state for each PM instance, global TCA administrative state and global TCA notification. SNMP supports sending traps for the TCA notifications that are of type alarm and summary alarm.

Note: SNMP traps are not sent for TCA notifications of type Event and summary event.

PM-MIB and ifName

The PM-MIB uses different ifNames and indexes for the PM statistics collected on the same line. The following table lists the interfaces and associated ifNames and the PM statistics collected. For a detailed list of the indexes, see [ifIndex values mapped to Interface type](#).

Table 201 Interface and ifName and PM statistics

Interface	ifName	PM statistics collected
Line XCVR	XCVR- slot / port	
PTP	PTP- slot / ptp -WaveLogic	Modem PM statistics
	PTP- slot / ptp	Optical Power PM statistics
	PTP- slot / ptp -PHOTONICS	Photonics PM statistics
OTUCn or OTUCn_flex	PORT- slot / port -OTUCn	OTN PM statistics
ODUCn	PORT- slot / port -ODUCn	ODU PM statistics
ODU Channel	CHANNEL- slot / port . channel -ODUK	ODU PM statistics
QSFP28 or QSFP+ pluggable	XCVR- slot / xcvr	
Client Ethernet port	PORT- slot / port	Ethernet PM statistics

SNMPv2c

SNMPv2c is defined by RFC1901, RFC3416 (SNMPv2), RFC2578 (SMIv2) and RFC2579. The following table lists the operations supported by SNMPv2c in Waveserver 5.

Table 202 Operations supported by SNMPv2c

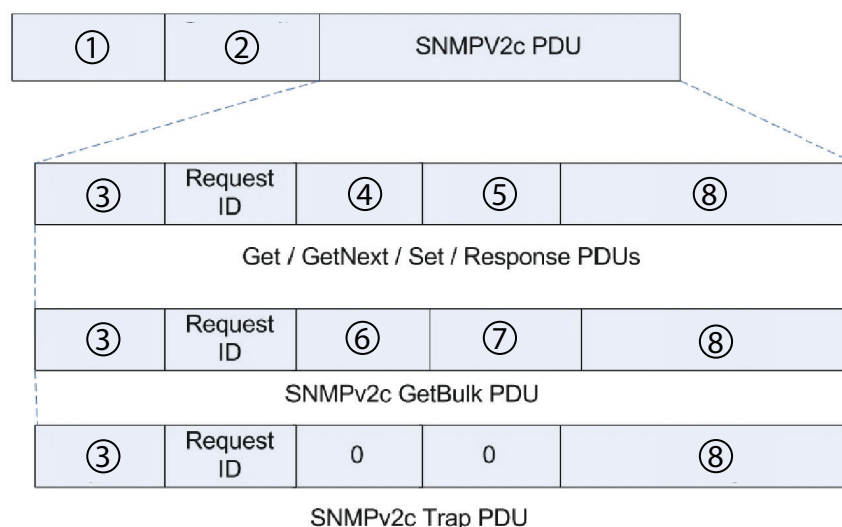
Operation	Description
Get Request	manager requests the current value of a specified managed object from the agent.

Operation	Description
Get Next Request	manager requests the current value of the next managed object in the MIB from the agent.
Get Response	agent responds to a Get-Request or Get-Next-Request with the value of the managed object.
Trap	agent notifies manager of system event.
Get Bulk (RFC3616)	for efficient retrieval of large amounts of data.

This version defines an SNMPv2 Trap Protocol Description Unit (PDU) format and supports data types based on SMIPv2, including 64-bit counters. The SNMPv2c message and PDU format is shown in the following figure.

Note: SNMP set requests are not supported in the current release.

Figure 136 SNMPv2c message and PDU format



Legend

1	Version	5	Error Index
2	Community Name	6	Non Repeaters
3	Request Type	7	Max Repetition
4	Error Status	8	Variable Bindings

SNMPv2c default table entries

The following tables list supported SNMPv2c default table entries.

Table 203 Default SNMP user table entries

User name	Auth protocol	Priv protocol
Public	noAuth	noPriv
Private	noAuth	noPriv

Table 204 Default SNMP security map table entries

User name	Sec model	Group name
Public	v2c	Public
Private	v2c	Private

Table 205 Default SNMP VACM access table entries

Group name	Sec model	Sec level	Read view	Write view	Notify view
Public	v2c	noAuth	V12cView	—	V12cView
Private	v2c	noAuth	V12cView	V12cView	V12cView

Table 206 Default SNMP VACM view tree table entries

View tree name	Sub tree	Type
V12cView	iso	include
V12cView	snmpResearch	exclude

Table 207 Default SNMP target table entries

Target name	IP address	Param name	Tags	UDP port	Retry count	Retry timeout
Anywhere	0.0.0.0	None	AnywhereTag	0	0	0
anywhereIpv6	::/	None	AnywhereTag	0	0	0

Table 208 Default SNMP community map table entries

Community index (Note)	Community name	Sec name	Transport tag
t0000000	Public	Public	AnywhereTag
t0000001	Private	Private	AnywhereTag
Note: These default community indices are system-reserved and cannot be modified. To create a new SNMP community string, see Configuring a custom SNMP community string .			

Table 209 Default SNMP notification table entries

Notification name	Notification tag	Notification type
Traps	TrapTag	Trap

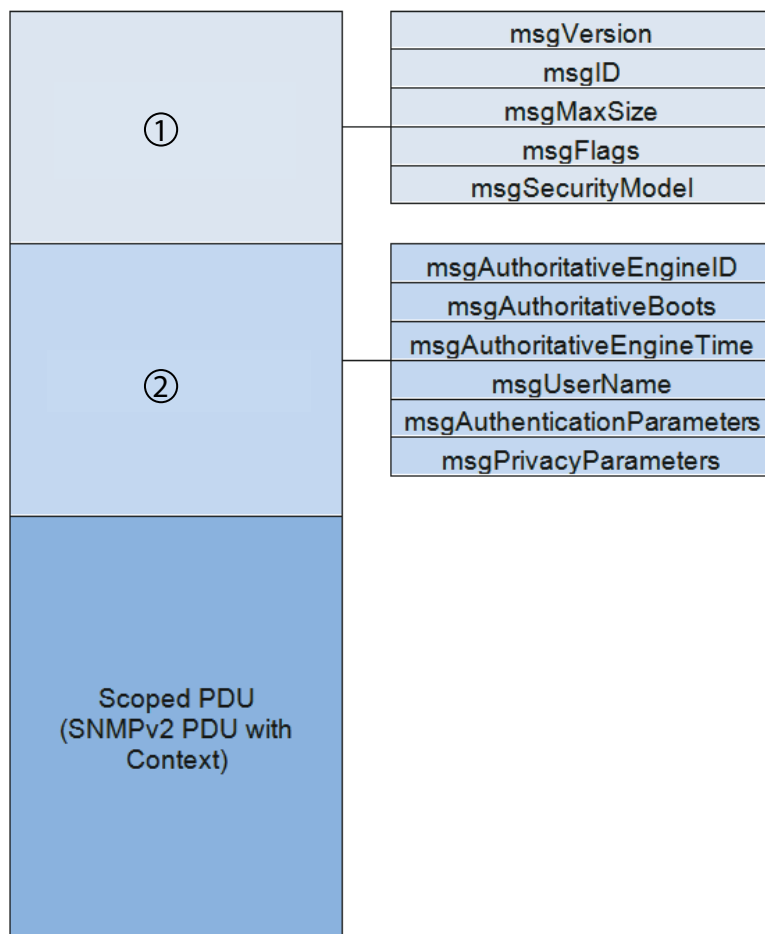
SNMPv3

Waveserver 5 SNMPv3 supports enhanced security features such as payload hashing and PDU encryption. Supported hashing algorithms are MD5 and SHA, and supported encryption algorithms are DES, 3DES, and AES.

Note: SNMPv3 uses the same PDU format and supports all operations that are defined in SNMPv2c, but it is not considered a stand-alone protocol. Instead, it employs an extensible framework that uses SNMPv2c information that is encapsulated in the SNMPv3 packet format.

Packet format

The SNMPv3 packet format includes a message header, security parameters, and PDU information as shown in the following figure.

Figure 137 SNMPv3 packet format**Legend**

1 Message Header

2 Security Parameters

Supported SNMPv3 packets

The following table lists the supported SNMPv3 fields.

Table 210 SNMPv3 packets

Packet name	Description
<code>msgVersion</code>	Displays the SNMP version of the packet (for example, 2 or 3).
<code>msgID</code>	Coordinates request and response messages between manager and agent. A <code>msgID</code> in a response must be the same as the <code>msgID</code> received in the request.
<code>msgMaxSize</code>	Indicates the maximum message size that the sender can support.

Packet name	Description
msgFlags	Indicates how to process the message. Contains the following flags: • reportableFlag • authFlag • privFlag
msgSecurityModel	Contains the security model that was used for security processing upon message reception.
msgSecurityParameters	Contains the security model specific data. This data is defined and used only by the security model specified by msgSecurityModel.
scopedPDU	Contains the normal PDU and information for identifying the administratively unique context for processing the PDU.

Supported concepts

The following table lists the supported SNMPv3 concepts.

Table 211 SNMPv3 concepts

Concept	Description
snmpEngineID	A unique identifier for an SNMP engine within an administrative domain. The snmpEngineID is computed using the chassis MAC address and an administratively assigned string.
snmpEngineBoots	Counts the number of times an SNMP engine has rebooted or reinitialized since the snmpEngineID was last set. This counter is initially set to zero.
snmpEngineTime	Specifies the number of seconds since the snmpEngineBoots counter was last incremented. This counter is initially set to zero. If the snmpEngineTime reaches its maximum (2147483647), the snmpEngineBoots is incremented by one and the snmpEngineTime starts counting again from zero.
snmpSecurityLevel	Contains three security levels that rank from highest to lowest, respectively: • authPriv • authNoPriv • noAuthNoPriv
Authoritative SNMP Engine	Protects against message replay, message delay, and message redirection.
Agent Discovery	Used to first determine the snmpEngineID of the agent, and then determines the snmpEngineBoots and snmpEngineTime.

Configuring SNMPv3

This section provides information on configuring SNMPv3.

Creating and attaching an SNMPv3 user to an SNMPv3 access entry group

You can create and attach an SNMPv3 user to an SNMPv3 access entry group, see [Creating and attaching an SNMPv3 user to an SNMPv3 access entry group](#).

Configuring notification entries

You can create, modify, delete, and display entries in the *snmpNotifTable* specified in *SNMP-NOTIFICATION-MIB.my*. The *snmpNotifTable* contains a default entry tag called the Trap Tag that is used to qualify entries in the *snmpTargetTable* for SNMPv1 traps. See [Configuring notification entries](#).

The following table lists the supported SNMPv3 table entries.

Table 212 Supported SNMPv3 table entries

Table name	Description
usmUserTable	Specifies the creation of new users who are allowed to access SNMP supported devices. This table is only used when an ingress SNMP packet utilizes SNMPv3. usmUserTable includes the following information: • usmUserEngineID • usmUserName • usmUserAuthProtocol • usmUserPrivProtocol
vacmSecurityToGroupTable	Attaches a security model to a group that is defined in usmUserTable. This table does not, however, validate whether a user ID specified in the table exists in usmUserTable. vacmSecurityToGroupTable includes the following information: • vacmSecurityModel • vacmSecurityName • vacmGroupName

Table name	Description
vacmAccessTable	Specifies either a read view, write view, or notify view access policy for the group specified in vacmSecurityToGroupTable. vacmAccessTable includes the following information: • vacmGroupName • vacmAccessSecurityModel • vacmAccessSecurityLevel • vacmAccessReadViewName • vacmAccessWriteViewName • vacmAccessNotifyViewName vacmAccessSecurityLevel contains three security levels that rank from highest to lowest, respectively: • authPriv • authNoPriv • noAuthNoPriv
vacmViewTreeFamilyTable	Specifies the following three views that are used in vacmAccessTable: • vacmViewTreeFamilyViewName • vacmViewTreeFamilySubtree • vacmViewTreeFamilyType
snmpTargetAddrTable	Specifies the transport addresses that can either be used when generating SNMP traps or allow SNMP requests that come from specified IP addresses. snmpTargetAddrTable includes the following information: • snmpTargetAddrName • snmpTargetAddrTAddress • snmpTargetAddrTagList • snmpTargetAddrParams
snmpTargetParamTable	Specifies additional information about the snmpTargetAddrTable. This additional information includes what security model, security level, and trap type is to be used when generating SNMP traps. snmpTargetParamTable includes the following information: • snmpTargetParamsName • snmpTargetParamsSecurityModel • snmpTargetParamsSecurityName • snmpTargetParamsSecurityLevel

Table name	Description
snmpCommunityTable	Used to map earlier SNMPv1 and SNMPv2c communities, notifications, and configurations into the SNMPv3 framework. snmpCommunityTable includes the following information: • snmpCommunityIndex • snmpCommunityName • snmpCommunitySecurityName • snmpCommunityTransportTag

The following table lists the maximum number of SNMPv3 table entries that are supported.

Table 213 SNMPv3 maximum number of table entries

Table name	Maximum entries supported
usmUserTable	10
vacmSecurityToGroupTable	15
vacmAccessTable	15
vacmViewTreeFamilyTable	20
snmpTargetAddrTable	17
snmpTargetParamTable	15
snmpCommunityTable	18
snmpNotifyTable	17

SNMP traps

Waveserver 5 supports these SNMP traps:

- [Alarm SNMP traps](#)
- [Standard SNMP traps](#)
 - [coldStart and warmStart traps](#)
 - [linkDown and linkUp traps](#)

Note: The authenticationFailure trap is supported for a SNMP user login failure. No authenticationFailure trap is generated when the Waveserver 5 user login fails.

Alarm SNMP traps

Alarm SNMP traps are generated when an alarm is raised or cleared on the Waveserver 5. See the following example.

Example

```
sysUpTimeInstance = Timeticks: (414392) 1:09:03.92
SNMPv2-MIB::snmpTrapOID.0 = OID: SNMPv2-SMI::enterprises.1271.3.2.11
SNMPv2-SMI::enterprises.1271.3.2.11.11 = Gauge32: 1
SNMPv2-SMI::enterprises.1271.3.2.11.10 = INTEGER: 2
SNMPv2-SMI::enterprises.1271.3.2.11.9 = STRING: "Link Down"
SNMPv2-SMI::enterprises.1271.3.2.11.8 = STRING: "ILAN-2"
SNMPv2-SMI::enterprises.1271.3.2.11.7 = Gauge32: 4
SNMPv2-SMI::enterprises.1271.3.2.11.6 = Gauge32: 109
SNMPv2-SMI::enterprises.1271.3.2.11.5 = STRING: "Tue Apr 14 13:34:40 2020"
SNMPv2-SMI::enterprises.1271.3.2.11.4 = Gauge32: 22
SNMPv2-SMI::enterprises.1271.3.2.11.3 = Gauge32: 0
SNMPv2-SMI::enterprises.1271.3.2.11.2 = Gauge32: 0
SNMPv2-SMI::enterprises.1271.3.2.11.1 = Gauge32: 0
```

Standard SNMP traps

Waveserver 5 supports standard SNMP traps: [coldStart](#) and [warmStart](#) traps and [linkDown](#) and [linkUp](#) traps.

coldStart and warmStart traps

When the Waveserver 5 chassis restarts, a coldStart or warmStart trap is sent. There are no traps sent for module restarts. See the following example.

Example

```
2019-05-06 17:37:31 <UNKNOWN> [UDP: [10.92.12.179]:38686->[10.92.22.62]:162]:
sysUpTime.0 = 0:0:03:46.76      snmpTrapOID.0 = warmStart      snmpTrapEnterprise.0 = waveserver
2019-05-06 17:49:00 <UNKNOWN> [UDP: [10.92.12.179]:59541->[10.92.22.62]:162]:
sysUpTime.0 = 0:0:03:46.26      snmpTrapOID.0 = coldStart      snmpTrapEnterprise.0 = waveserver
```

linkDown and linkUp traps

When the management (DCN, ILAN ports) and/or client Ethernet port's operational state changes, a linkDown or linkUp trap is sent. See the following example of a linkDown trap.

Example

```
2019-05-06 17:47:21 <UNKNOWN> [UDP: [10.92.12.173]:55657->[10.92.22.62]:162]:
sysUpTime.0 = 0:0:04:42.00      snmpTrapOID.0 = linkDown      ifIndex.33557027 = 33557027
ifAdminStatus.33557027 = up      ifOperStatus.33557027 = down      snmpTrapEnterprise.0 = waveserver
```

SNMP security considerations

To reduce the risk of a distributed denial of service (DDoS) attack involving a GetBulk request against a remote SNMP v2c server, Ciena recommends that you disable the SNMP server on a remote host if you are not using the SNMP service. (See [Procedure 123, "Enabling and disabling SNMP"](#) for details on disabling an SNMP server.)

If you do use the SNMP server on a remote host, Ciena recommends that you configure address filtering on the SNMP v2c server to allow messages only from trusted subnets. See [Procedure 134, "Configuring address filtering"](#) for details on configuring address filtering on the SNMP v2 server.

For additional security, use SNMP v3 with `priv-protocol` and `auth-protocols` enabled.

SNMP MIB files and external yang interfaces

The SNMP MIB files available with Waveserver 5 were generated from the external yang files that are also available with Waveserver.

Each module's OID starts from CIENA-WS-PLATFORM-CONFIGURATION (1.3.6.1.4.1.1271.3.5). The following table lists the associated MIB OID for the listed yang module.

Table 214 Yang modules and associated OIDs under WsPlatformConfig

Yang Module	OID under WsPlatformConfig
ciena-waveserver-aaa	29
ciena-waveserver-alarm	4
ciena-waveserver-application-its	7
ciena-waveserver-chassis	6
ciena-waveserver-configuration	19
ciena-waveserver-encryption	23
ciena-waveserver-interfaces	31
ciena-waveserver-license	25
ciena-waveserver-lldp	43
ciena-waveserver-logging	32
ciena-waveserver-module	5
ciena-waveserver-ndp	40
ciena-waveserver-pkix	34
ciena-waveserver-pm-tca	42
ciena-waveserver-pm	22
ciena-waveserver-port	7
ciena-waveserver-protection	8
ciena-waveserver-ptp-cmd	8
ciena-waveserver-ptp-modem	8
ciena-waveserver-ptp	8
ciena-waveserver-snmp	9
ciena-waveserver-software	14
ciena-waveserver-spli	35

Yang Module	OID under WsPlatformConfig
ciena-waveserver-statistics	39
ciena-waveserver-system	12
ciena-waveserver-topology	36
ciena-waveserver-typedefs	41
ciena-waveserver-xcvr-modem	10
ciena-waveserver-xcvr-pluggable	10
ciena-waveserver-xcvr	10

Object Identifiers (OID)

This section covers the following topics:

- [Proprietary MIB OIDs](#)
- [Finding the entity index associated with an ifIndex](#)

Proprietary MIB OIDs

The following tables list OIDs for the following Ciena Management Information Bases (MIBs):

- [CIENA-WS-NOTIFICATION-MIB.my](#)
- [CIENA-WS-PLATFORM-ALARM-MIB.my](#)
- [CIENA-WS-PLATFORM-PM-MIB.my](#)

Note: CIENA-WS-NOTIFICATION-MIB contains the definitions for the wsAlarmNotification and wsLinkStateAlarmNotification traps.

Table 215 CIENA-WS-NOTIFICATION-MIB.my

Indicator	OID	Description
wsAlarmNotificationSiteId	1.3.6.1.4.1.1271.3.2.11.1	The system Site ID.
wsAlarmNotificationGroupId	1.3.6.1.4.1.1271.3.2.11.2	The system Group ID.
wsAlarmNotificationMemberId	1.3.6.1.4.1.1271.3.2.11.3	The system Member ID.
wsAlarmNotificationInstanceId	1.3.6.1.4.1.1271.3.2.11.4	The alarm Instance ID.
wsAlarmNotificationDateAndTime	1.3.6.1.4.1.1271.3.2.11.5	The local date and time when alarm was raised.
wsAlarmNotificationTableId	1.3.6.1.4.1.1271.3.2.11.6	The alarm table ID.
wsAlarmNotificationSeverity	1.3.6.1.4.1.1271.3.2.11.7	The alarm severity.

Indicator	OID	Description
wsAlarmNotificationInstance	1.3.6.1.4.1.1271.3.2.11.8	The object instance the alarm is raised against.
wsAlarmNotificationDescription	1.3.6.1.4.1.1271.3.2.11.9	The alarm description.
wsAlarmNotificationActiveStatus	1.3.6.1.4.1.1271.3.2.11.10	The alarm active status.
wsAlarmNotificationEntityType	1.3.6.1.4.1.1271.3.2.11.11	The alarm source entity type if applicable.
wsLinkStateAlarmNotificationSiteId	1.3.6.1.4.1.1271.3.2.12.1	The line fault site ID.
wsLinkStateAlarmNotificationGroupId	1.3.6.1.4.1.1271.3.2.12.2	The system Group ID.
wsLinkStateAlarmNotificationMemberId	1.3.6.1.4.1.1271.3.2.12.3	The system Member ID.
wsLinkStateAlarmNotificationInstanceId	1.3.6.1.4.1.1271.3.2.12.4	The line fault instance ID.
wsLinkStateAlarmNotificationDateAndTime	1.3.6.1.4.1.1271.3.2.12.5	The local date and time when alarm was raised.
wsLinkStateAlarmNotificationSeverity	1.3.6.1.4.1.1271.3.2.12.7	The alarm severity.
wsLinkStateAlarmNotificationInstance	1.3.6.1.4.1.1271.3.2.12.8	The object instance the alarm is raised against.
wsLinkStateAlarmNotificationDescription	1.3.6.1.4.1.1271.3.2.12.9	The alarm description.
wsLinkStateAlarmNotificationPtpRxLos	1.3.6.1.4.1.1271.3.2.12.10.1	PTP alarms for the link.
wsLinkStateAlarmNotificationPtpRxLol	1.3.6.1.4.1.1271.3.2.12.10.2	PTP alarms for the link.
wsLinkStateAlarmNotificationPtpTxLos	1.3.6.1.4.1.1271.3.2.12.10.3	PTP alarms for the link.
wsLinkStateAlarmNotificationPtpTxLol	1.3.6.1.4.1.1271.3.2.12.10.4	PTP alarms for the link.
wsLinkStateAlarmNotificationEthPcshighber	1.3.6.1.4.1.1271.3.2.12.11.1	Ethernet PCS high bit error rate.
wsLinkStateAlarmNotificationEthPcs loam	1.3.6.1.4.1.1271.3.2.12.11.2	Ethernet PCS loss of alignment marker.
wsLinkStateAlarmNotificationEthPcs lobl	1.3.6.1.4.1.1271.3.2.12.11.3	Ethernet PCS loss of block lock.
wsLinkStateAlarmNotificationEthRs link down	1.3.6.1.4.1.1271.3.2.12.11.4	Ethernet RS link down.
wsLinkStateAlarmNotificationEthRs lf	1.3.6.1.4.1.1271.3.2.12.11.5	Ethernet RS local fault.
wsLinkStateAlarmNotificationEthRs rf	1.3.6.1.4.1.1271.3.2.12.11.6	Ethernet RS remote fault.
wsLinkStateAlarmNotificationEth fec loss sync	1.3.6.1.4.1.1271.3.2.12.11.7	Ethernet FEC loss of alignment marker.

Indicator	OID	Description
wsLinkStateAlarmNotificationEth pma sool	1.3.6.1.4.1.1271.3.2.12.11.8	Ethernet PMA serdes out of lock.
wsLinkStateAlarmNotificationOtuLoc	1.3.6.1.4.1.1271.3.2.12.12.1	OTU loss of clock.
wsLinkStateAlarmNotificationOtuFreqOor	1.3.6.1.4.1.1271.3.2.12.12.2	OTU frequency out of range.
wsLinkStateAlarmNotificationOtuLof	1.3.6.1.4.1.1271.3.2.12.12.3	OTU loss of frame.
wsLinkStateAlarmNotificationOtuOreFecSf	1.3.6.1.4.1.1271.3.2.12.12.4	OTU pre-FEC signal fail.
wsLinkStateAlarmNotificationOtuOreFecSd	1.3.6.1.4.1.1271.3.2.12.12.5	OTU pre-FEC signal degrade.
wsLinkStateAlarmNotificationOtuLom	1.3.6.1.4.1.1271.3.2.12.12.6	OTU loss of multi-frame.
wsLinkStateAlarmNotificationOtuBdi	1.3.6.1.4.1.1271.3.2.12.12.7	OTU backwards defect indicator.
wsLinkStateAlarmNotificationOtuTtiMismatch	1.3.6.1.4.1.1271.3.2.12.12.8	OTU TTIMismatch.
wsLinkStateAlarmNotificationOduOci	1.3.6.1.4.1.1271.3.2.12.13.1	ODU open connection indication.
wsLinkStateAlarmNotificationOduAis	1.3.6.1.4.1.1271.3.2.12.13.2	ODU alarm indicating signal.
wsLinkStateAlarmNotificationOduLck	1.3.6.1.4.1.1271.3.2.12.13.3	ODU locked.
wsLinkStateAlarmNotificationOduSf	1.3.6.1.4.1.1271.3.2.12.13.4	ODU signal fail.
wsLinkStateAlarmNotificationOduSd	1.3.6.1.4.1.1271.3.2.12.13.5	ODU signal degrade.
wsLinkStateAlarmNotificationOduTtiMismatch	1.3.6.1.4.1.1271.3.2.12.13.6	ODU TTI mismatch.
wsLinkStateAlarmNotificationOduBdi	1.3.6.1.4.1.1271.3.2.12.13.7	ODU backwards defect indicator.
wsLinkStateAlarmNotificationOduPtiMismatch	1.3.6.1.4.1.1271.3.2.12.13.8	ODU payload type indicator mismatch.
wsLinkStateAlarmNotificationOduFeClientSf	1.3.6.1.4.1.1271.3.2.12.13.9	ODU far-end client signal fail.
wsLinkStateAlarmNotificationOduSkewOor	1.3.6.1.4.1.1271.3.2.12.13.10	ODU skew out of range.
wsLinkStateAlarmNotificationEntityType	1.3.6.1.4.1.1271.3.2.12.14	The alarm source entity type if applicable.

Table 216 CIENA-WS-PLATFORM-ALARM-MIB.my

Indicator	OID	Description
activeAlarmInstanceid	1.3.6.1.4.1.1271.3.5.4.3.1.1	The alarm Instance ID
activeAlarmAcknowledged	1.3.6.1.4.1.1271.3.5.4.3.1.2	Alarm has been acknowledged.
activeAlarmTableid	1.3.6.1.4.1.1271.3.5.4.3.1.3	The alarm table ID.
activeAlarmSeverity	1.3.6.1.4.1.1271.3.5.4.3.1.4	The alarm severity.
activeAlarmLocalDateTime	1.3.6.1.4.1.1271.3.5.4.3.1.5	The local date and time when alarm was raised.
activeAlarmInstance	1.3.6.1.4.1.1271.3.5.4.3.1.6	The object instance the alarm is raised against.
activeAlarmDescription	1.3.6.1.4.1.1271.3.5.4.3.1.7	The alarm description.
activeAlarmIntermittent	1.3.6.1.4.1.1271.3.5.4.3.1.8	Alarm condition is intermittent.
activeAlarmSiteIdentifier	1.3.6.1.4.1.1271.3.5.4.3.1.9	An integer to uniquely identify the site where this Waveserver is located.
activeAlarmGroupIdentifier	1.3.6.1.4.1.1271.3.5.4.3.1.10	An integer to uniquely identify a group of Waveservers within a site.
activeAlarmMemberIdentifier	1.3.6.1.4.1.1271.3.5.4.3.1.11	An integer to uniquely identify a Waveserver chassis within a group of Waveservers.
historyAlarmId	1.3.6.1.4.1.1271.3.5.4.4.1.1	The alarm history ID
historyAlarmReason	1.3.6.1.4.1.1271.3.5.4.4.1.2	The reason for the alarm condition.
historyAlarmInstanceid	1.3.6.1.4.1.1271.3.5.4.4.1.3	The alarm instance ID.
historyAlarmTableid	1.3.6.1.4.1.1271.3.5.4.4.1.4	The alarm table ID.
historyAlarmSeverity	1.3.6.1.4.1.1271.3.5.4.4.1.5	The alarm severity.
historyAlarmLocalDateTime	1.3.6.1.4.1.1271.3.5.4.4.1.6	The local date and time when alarm was raised.
historyAlarmInstance	1.3.6.1.4.1.1271.3.5.4.4.1.7	The object instance the alarm is raised against.
historyAlarmDescription	1.3.6.1.4.1.1271.3.5.4.4.1.8	The alarm description.

Indicator	OID	Description
historyAlarmSiteIdentifier	1.3.6.1.4.1.1271.3.5.4.4.1.9	An integer to uniquely identify the site where this Waveserver is located.
historyAlarmGroupIdentifier	1.3.6.1.4.1.1271.3.5.4.4.1.10	An integer to uniquely identify a group of Waveservers within a site.
historyAlarmMemberIdentifier	1.3.6.1.4.1.1271.3.5.4.4.1.11	An integer to uniquely identify a Waveserver chassis within a group of Waveservers.
definedAlarmId	1.3.6.1.4.1.1271.3.5.4.5.1.1	The alarm table ID.
definedAlarmTableId	1.3.6.1.4.1.1271.3.5.4.5.1.2	The alarm table ID.
definedAlarmEnabled	1.3.6.1.4.1.1271.3.5.4.5.1.3	Alarm is enabled.
definedAlarmActive	1.3.6.1.4.1.1271.3.5.4.5.1.4	Alarm is active.
definedAlarmThreshold	1.3.6.1.4.1.1271.3.5.4.5.1.5	The alarm threshold.
definedAlarmCap	1.3.6.1.4.1.1271.3.5.4.5.1.6	The alarm cap.
definedAlarmSeverity	1.3.6.1.4.1.1271.3.5.4.5.1.7	The alarm severity.
definedAlarmInstance	1.3.6.1.4.1.1271.3.5.4.5.1.8	The object instance the alarm is raised against.
definedAlarmDescription	1.3.6.1.4.1.1271.3.5.4.5.1.9	The alarm description.
alarmStatisticsIndex	1.3.6.1.4.1.1271.3.5.4.6.1.1	The alarm statistics table index.
alarmStatisticsActive	1.3.6.1.4.1.1271.3.5.4.6.1.2	There is alarm active for the alarm type
alarmStatisticsDisabled	1.3.6.1.4.1.1271.3.5.4.6.1.3	There is alarm disabled for the alarm type
alarmStatisticsCount	1.3.6.1.4.1.1271.3.5.4.6.1.4	The count of active alarms for the alarm type
alarmStatisticsCumulative	1.3.6.1.4.1.1271.3.5.4.6.1.5	The cumulative count of active alarms for the alarm type
alarmStatisticsType	1.3.6.1.4.1.1271.3.5.4.6.1.6	The alarm type.

The following table lists the Waveserver PM MIB indicators.

- For the pmOtu, pmOdu, pmOpticalPower, and pmGcm indicators listed in the following table, see [Table 221, “PM indexes for OTU, ODU, Optical, GCM entities”](#) for the corresponding PM indexes.
- For the pmModem and pmEthernet indicators listed in the following table, see [Table 222, “PM indexes for Ethernet entities”](#) for the corresponding PM indexes.

Table 217 CIENA-WS-PLATFORM-PM-MIB.my

Indicator	OID	Description
pmGlobalConfigTableSnmpKey	1.3.6.1.4.1.1271.3.5.22.3.1.1	Unique key for pmGlobalConfig
pmAdminState	1.3.6.1.4.1.1271.3.5.22.3.1.2	Global admin state.
pmTcaAdminState	1.3.6.1.4.1.1271.3.5.22.3.1.3	Global TCA admin state.
pmTcaNotificationType	1.3.6.1.4.1.1271.3.5.22.3.1.4	Global TCA Notification Type
pmEthernet15MinMonType	1.3.6.1.4.1.1271.3.5.22.4.1.1	Second Index for pmEthernetCurrent15minStatsTable. This is the PM counter Type
pmEthernet15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.4.1.2	Ciena PM Monitor Type description.
pmEthernet15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.4.1.3	Ciena PM If Index description
pmEthernet15MinMonValue	1.3.6.1.4.1.1271.3.5.22.4.1.4	Ciena 15 min monitored value.
pmEthernet15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.4.1.5	Invalid flag value for the MonType
pmEthernet15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.4.1.6	Flag indicating if PM Mon Value is supported or not
pmEthernet15MinAdminState	1.3.6.1.4.1.1271.3.5.22.4.1.7	Profile admin state.
pmEthernet15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.4.1.8	Start date and time.
pmEthernet15MinThValue	1.3.6.1.4.1.1271.3.5.22.4.1.9	Threshold value.
pmEthernet15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.4.1.10	Flag to show if TCA is supported or TCA is active
pmEthernet15MinTcaAdminState	1.3.6.1.4.1.1271.3.5.22.4.1.11	TCA Profile admin state.

Indicator	OID	Description
pmEthernet15MinTcaProfileId	1.3.6.1.4.1.1271.3.5.22.4.1.12	TCA Profile Id
pmEthernet15MinTcaProfileName	1.3.6.1.4.1.1271.3.5.22.4.1.13	TCA Profile Name
pmEthernet24HrMonType	1.3.6.1.4.1.1271.3.5.22.5.1.1	Second Index for pmEthernetCurrent24Hr StatsTable. This is the PM counter Type
pmEthernet24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.5.1.2	Ciena PM Monitor Type description.
pmEthernet24HrIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.5.1.3	Ciena PM If Index description
pmEthernet24HrMonValue	1.3.6.1.4.1.1271.3.5.22.5.1.4	Ciena 24Hr monitored value.
pmEthernet24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.5.1.5	Invalid flag value for the MonType
pmEthernet24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.5.1.6	Flag indicating if PM Mon Value is supported or not
pmEthernet24HrAdminState	1.3.6.1.4.1.1271.3.5.22.5.1.7	Profile admin state.
pmEthernet24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.5.1.8	Start date and time.
pmEthernet24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.5.1.9	Threshold value.
pmEthernet24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.5.1.10	Flag to show if TCA is supported or TCA is active
pmEthernet24HrTcaAdminState	1.3.6.1.4.1.1271.3.5.22.5.1.11	TCA Profile admin state.
pmEthernet24HrTcaProfileId	1.3.6.1.4.1.1271.3.5.22.5.1.12	TCA Profile Id
pmEthernet24HrTcaProfileName	1.3.6.1.4.1.1271.3.5.22.5.1.13	TCA Profile Name
pmEthernetUntimedMonType	1.3.6.1.4.1.1271.3.5.22.6.1.1	Second Index for pmEthernetUntimedStatsTable. This is the PM counter Type
pmEthernetUntimedMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.6.1.2	Ciena PM Monitor Type description.
pmEthernetUntimedIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.6.1.3	Ciena PM If Index description

Indicator	OID	Description
pmEthernetUntimedMonValue	1.3.6.1.4.1.1271.3.5.22.6.1.4	Ciena Untimed monitored value.
pmEthernetUntimedMonIdf	1.3.6.1.4.1.1271.3.5.22.6.1.5	Invalid flag value for the MonType
pmEthernetUntimedMonSupported	1.3.6.1.4.1.1271.3.5.22.6.1.6	Flag indicating if PM Mon Value is supported or not
pmEthernetUntimedAdminState	1.3.6.1.4.1.1271.3.5.22.6.1.7	Profile admin state.
pmEthernetUntimedMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.6.1.8	Start date and time.
pmEthernetUntimedthvalue	1.3.6.1.4.1.1271.3.5.22.6.1.9	Threshold value.
pmEthernetUntimedTcaActive	1.3.6.1.4.1.1271.3.5.22.6.1.10	Flag to show if TCA is supported or TCA is active
pmEthernetUntimedTcaAdminState	1.3.6.1.4.1.1271.3.5.22.6.1.11	TCA Profile admin state.
pmEthernetUntimedTcaProfileId	1.3.6.1.4.1.1271.3.5.22.6.1.12	TCA Profile Id
pmEthernetUntimedTcaProfileName	1.3.6.1.4.1.1271.3.5.22.6.1.13	TCA Profile Name
pmEthernetHistory15MinBinIndex	1.3.6.1.4.1.1271.3.5.22.7.1.1	Second index for pmEthernetHistory15minStatsTable .
pmEthernetHistory15MinMonType	1.3.6.1.4.1.1271.3.5.22.7.1.2	Third index for pmEthernetHistory15minStatsTable. This is the PM counter Type
pmEthernetHistory15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.7.1.3	Ciena PM Monitor Type description.
pmEthernetHistory15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.7.1.4	Ciena PM If Index description
pmEthernetHistory15MinMonValue	1.3.6.1.4.1.1271.3.5.22.7.1.5	Ciena History 15 min monitored value.
pmEthernetHistory15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.7.1.6	Invalid flag value for the MonType
pmEthernetHistory15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.7.1.7	Flag indicating if PM Mon Value is supported or not
pmEthernetHistory15MinAdminState	1.3.6.1.4.1.1271.3.5.22.7.1.8	Profile admin state.

Indicator	OID	Description
pmEthernetHistory15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.7.1.9	Start date and time.
pmEthernetHistory15Minthvalue	1.3.6.1.4.1.1271.3.5.22.7.1.10	Threshold value.
pmEthernetHistory15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.7.1.11	Flag to show if TCA is supported or TCA is active
pmEthernetHistory24HrMonType	1.3.6.1.4.1.1271.3.5.22.8.1.1	Second Index for pmEthernetUntimedStatsTable. This is the PM counter Type
pmEthernetHistory24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.8.1.2	Ciena PM Monitor Type description.
pmEthernetHistory24HrIndexDescr	1.3.6.1.4.1.1271.3.5.22.8.1.3	Ciena PM If Index description
pmEthernetHistory24HrMonValue	1.3.6.1.4.1.1271.3.5.22.8.1.4	Ciena 24Hr monitored value.
pmEthernetHistory24HrMonidf	1.3.6.1.4.1.1271.3.5.22.8.1.5	Invalid flag value for the MonType
pmEthernetHistory24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.8.1.6	Flag indicating if PM Mon Value is supported or not
pmEthernetHistory24HrAdminState	1.3.6.1.4.1.1271.3.5.22.8.1.7	Profile admin state.
pmEthernetHistory24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.8.1.8	Start date and time.
pmEthernetHistory24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.8.1.9	Threshold value.
pmEthernetHistory24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.8.1.10	Flag to show if TCA is supported or TCA is active
pmModem15MinMonType	1.3.6.1.4.1.1271.3.5.22.9.1.1	Second Index for pmModemCurrent15minStatsTable. This is the PM counter Type
pmModem15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.9.1.2	Ciena PM Monitor Type description.
pmModem15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.9.1.3	Ciena PM If Index description

Indicator	OID	Description
pmModem15MinMonValue	1.3.6.1.4.1.1271.3.5.22.9.1.4	Ciena 15 min monitored value.
pmModem15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.9.1.5	Invalid flag value for the MonType
pmModem15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.9.1.6	Flag indicating if PM Mon Value is supported or not
pmModem15MinAdminState	1.3.6.1.4.1.1271.3.5.22.9.1.7	Profile admin state.
pmModem15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.9.1.8	Start date and time.
pmModem15Minthvalue	1.3.6.1.4.1.1271.3.5.22.9.1.9	Threshold value.
pmModem15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.9.1.10	Flag to show if TCA is supported or TCA is active
pmModem15MinTcaAdminState	1.3.6.1.4.1.1271.3.5.22.9.1.11	TCA Profile admin state.
pmModem15MinTcaProfileId	1.3.6.1.4.1.1271.3.5.22.9.1.12	TCA Profile Id
pmModem15MinTcaProfileName	1.3.6.1.4.1.1271.3.5.22.9.1.13	TCA Profile Name
pmModem24HrMonType	1.3.6.1.4.1.1271.3.5.22.10.1.1	Second Index for pmModemCurrent24HrStatsTable. This is the PM counter Type
pmModem24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.10.1.2	Ciena PM Monitor Type description.
pmModem24HrIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.10.1.3	Ciena PM If Index description
pmModem24HrMonValue	1.3.6.1.4.1.1271.3.5.22.10.1.4	Ciena 24Hr monitored value.
pmModem24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.10.1.5	Invalid flag value for the MonType
pmModem24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.10.1.6	Flag indicating if PM Mon Value is supported or not
pmModem24HrAdminState	1.3.6.1.4.1.1271.3.5.22.10.1.7	Profile admin state.
pmModem24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.10.1.8	Start date and time.
pmModem24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.10.1.9	Threshold value.

Indicator	OID	Description
pmModem24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.10.1.10	Flag to show if TCA is supported or TCA is active
pmModem24HrTcaAdminState	1.3.6.1.4.1.1271.3.5.22.10.1.11	TCA Profile admin state.
pmModem24HrTcaProfileId	1.3.6.1.4.1.1271.3.5.22.10.1.12	TCA Profile Id
pmModem24HrTcaProfileName	1.3.6.1.4.1.1271.3.5.22.10.1.13	TCA Profile Name
pmModemUntimedMonType	1.3.6.1.4.1.1271.3.5.22.11.1.1	Second Index for pmModemUntimedStats Table. This is the PM counter Type
pmModemUntimedMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.11.1.2	Ciena PM Monitor Type description.
pmModemUntimedIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.11.1.3	Ciena PM If Index description
pmModemUntimedMonValue	1.3.6.1.4.1.1271.3.5.22.11.1.4	Ciena Untimed monitored value.
pmModemUntimedMonIdf	1.3.6.1.4.1.1271.3.5.22.11.1.5	Invalid flag value for the MonType
pmModemUntimedMonSupported	1.3.6.1.4.1.1271.3.5.22.11.1.6	Flag indicating if PM Mon Value is supported or not
pmModemUntimedAdminState	1.3.6.1.4.1.1271.3.5.22.11.1.7	Profile admin state.
pmModemUntimedMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.11.1.8	Start date and time.
pmModemUntimedthvalue	1.3.6.1.4.1.1271.3.5.22.11.1.9	Threshold value.
pmModemUntimedTcaActive	1.3.6.1.4.1.1271.3.5.22.11.1.10	Flag to show if TCA is supported or TCA is active
pmModemUntimedTcaAdminState	1.3.6.1.4.1.1271.3.5.22.11.1.11	TCA Profile admin state.
pmModemUntimedTcaProfileId	1.3.6.1.4.1.1271.3.5.22.11.1.12	TCA Profile Id
pmModemUntimedTcaProfileName	1.3.6.1.4.1.1271.3.5.22.11.1.13	TCA Profile Name
pmModemHistory15MinBinIndex	1.3.6.1.4.1.1271.3.5.22.12.1.1	Second index for pmModemHistory15min StatsTable .

Indicator	OID	Description
pmModemHistory15MinMonType	1.3.6.1.4.1.1271.3.5.22.12.1.2	Third Index for pmModemHistory15min StatsTable. This is the PM counter Type
pmModemHistory15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.12.1.3	Ciena PM Monitor Type description.
pmModemHistory15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.12.1.4	Ciena PM If Index description
pmModemHistory15MinMonValue	1.3.6.1.4.1.1271.3.5.22.12.1.5	Ciena History 15 min monitored value.
pmModemHistory15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.12.1.6	Invalid flag value for the MonType
pmModemHistory15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.12.1.7	Flag indicating if PM Mon Value is supported or not
pmModemHistory15MinAdminState	1.3.6.1.4.1.1271.3.5.22.12.1.8	Profile admin state.
pmModemHistory15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.12.1.9	Start date and time.
pmModemHistory15MinThvalue	1.3.6.1.4.1.1271.3.5.22.12.1.10	Threshold value.
pmModemHistory15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.12.1.11	Flag to show if TCA is supported or TCA is active
pmModemHistory24HrMonType	1.3.6.1.4.1.1271.3.5.22.13.1.1	Second Index for pmModemUntimedStats Table. This is the PM counter Type
pmModemHistory24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.13.1.2	Ciena PM Monitor Type description.
pmModemHistory24HrIndexDescr	1.3.6.1.4.1.1271.3.5.22.13.1.3	Ciena PM If Index description
pmModemHistory24HrMonValue	1.3.6.1.4.1.1271.3.5.22.13.1.4	Ciena History 24Hr monitored value.
pmModemHistory24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.13.1.5	Invalid flag value for the MonType
pmModemHistory24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.13.1.6	Flag indicating if PM Mon Value is supported or not

Indicator	OID	Description
pmModemHistory24HrAdminState	1.3.6.1.4.1.1271.3.5.22.13.1.7	Profile admin state.
pmModemHistory24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.13.1.8	Start date and time.
pmModemHistory24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.13.1.9	Threshold value.
pmModemHistory24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.13.1.10	Flag to show if TCA is supported or TCA is active.
pmOtu15MinMonType	1.3.6.1.4.1.1271.3.5.22.14.1.1	Second Index for pmOtuCurrent15minStatsTable. This is the PM counter Type
pmOtu15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.14.1.2	Ciena PM Monitor Type description.
pmOtu15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.14.1.3	Ciena PM If Index description
pmOtu15MinMonValue	1.3.6.1.4.1.1271.3.5.22.14.1.4	Ciena 15 min monitored value.
pmOtu15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.14.1.5	Invalid flag value for the MonType
pmOtu15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.14.1.6	Flag indicating if PM Mon Value is supported or not
pmOtu15MinAdminState	1.3.6.1.4.1.1271.3.5.22.14.1.7	Profile admin state.
pmOtu15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.14.1.8	Start date and time.
pmOtu15Minthvalue	1.3.6.1.4.1.1271.3.5.22.14.1.9	Threshold value.
pmOtu15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.14.1.10	Flag to show if TCA is supported or TCA is active
pmOtu15MinTcaAdminState	1.3.6.1.4.1.1271.3.5.22.14.1.11	TCA Profile admin state.
pmOtu15MinTcaProfileId	1.3.6.1.4.1.1271.3.5.22.14.1.12	TCA Profile Id
pmOtu15MinTcaProfileName	1.3.6.1.4.1.1271.3.5.22.14.1.13	TCA Profile Name
pmOtu24HrMonType	1.3.6.1.4.1.1271.3.5.22.15.1.1	Second Index for pmOtuCurrent24HrStatsTable. This is the PM counter Type

Indicator	OID	Description
pmOtu24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.15.1.2	Ciena PM Monitor Type description.
pmOtu24HrIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.15.1.3	Ciena PM If Index description
pmOtu24HrMonValue	1.3.6.1.4.1.1271.3.5.22.15.1.4	Ciena 24Hr monitored value.
pmOtu24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.15.1.5	Invalid flag value for the MonType
pmOtu24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.15.1.6	Flag indicating if PM Mon Value is supported or not
pmOtu24HrAdminState	1.3.6.1.4.1.1271.3.5.22.15.1.7	Profile admin state.
pmOtu24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.15.1.8	Start date and time.
pmOtu24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.15.1.9	Threshold value.
pmOtu24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.15.1.10	Flag to show if TCA is supported or TCA is active
pmOtu24HrTcaAdminState	1.3.6.1.4.1.1271.3.5.22.15.1.11	TCA Profile admin state.
pmOtu24HrTcaProfileId	1.3.6.1.4.1.1271.3.5.22.15.1.12	TCA Profile Id
pmOtu24HrTcaProfileName	1.3.6.1.4.1.1271.3.5.22.15.1.13	TCA Profile Name
pmOtuUntimedMonType	1.3.6.1.4.1.1271.3.5.22.16.1.1	Second Index for pmOtuUntimedStatsTable. This is the PM counter Type
pmOtuUntimedMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.16.1.2	Ciena PM Monitor Type description.
pmOtuUntimedIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.16.1.3	Ciena PM If Index description
pmOtuUntimedMonValue	1.3.6.1.4.1.1271.3.5.22.16.1.4	Ciena Untimed monitored value.
pmOtuUntimedMonIdf	1.3.6.1.4.1.1271.3.5.22.16.1.5	Invalid flag value for the MonType
pmOtuUntimedMonSupported	1.3.6.1.4.1.1271.3.5.22.16.1.6	Flag indicating if PM Mon Value is supported or not

Indicator	OID	Description
pmOtuUntimedAdminState	1.3.6.1.4.1.1271.3.5.22.16.1.7	Profile admin state.
pmOtuUntimedMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.16.1.8	Start date and time.
pmOtuUntimedThvalue	1.3.6.1.4.1.1271.3.5.22.16.1.9	Threshold value.
pmOtuUntimedTcaActive	1.3.6.1.4.1.1271.3.5.22.16.1.10	Flag to show if TCA is supported or TCA is active
pmOtuUntimedTcaAdminState	1.3.6.1.4.1.1271.3.5.22.16.1.11	TCA Profile admin state.
pmOtuUntimedTcaProfileId	1.3.6.1.4.1.1271.3.5.22.16.1.12	TCA Profile Id
pmOtuUntimedTcaProfileName	1.3.6.1.4.1.1271.3.5.22.16.1.13	TCA Profile Name
pmOtuHistory15MinBinIndex	1.3.6.1.4.1.1271.3.5.22.17.1.1	Index for pmOtuHistory15minStatsTable .
pmOtuHistory15MinMonType	1.3.6.1.4.1.1271.3.5.22.17.1.2	Third Index for pmOtuUntimedStatsTable. This is the PM counter Type
pmOtuHistory15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.17.1.3	Ciena PM Monitor Type description.
pmOtuHistory15MinIndexDescr	1.3.6.1.4.1.1271.3.5.22.17.1.4	Ciena PM If Index description
pmOtuHistory15MinMonValue	1.3.6.1.4.1.1271.3.5.22.17.1.5	Ciena History 15 min monitored value.
pmOtuHistory15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.17.1.6	Invalid flag value for the MonType
pmOtuHistory15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.17.1.7	Flag indicating if PM Mon Value is supported or not
pmOtuHistory15MinAdminState	1.3.6.1.4.1.1271.3.5.22.17.1.8	Profile admin state.
pmOtuHistory15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.17.1.9	Start date and time.
pmOtuHistory15MinThvalue	1.3.6.1.4.1.1271.3.5.22.17.1.10	Threshold value.
pmOtuHistory15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.17.1.11	Flag to show if TCA is supported or TCA is active

Indicator	OID	Description
pmOtuHistory24HrMonType	1.3.6.1.4.1.1271.3.5.22.18.1.1	Second Index for pmOtuUntimedStatsTable. This is the PM counter Type
pmOtuHistory24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.18.1.2	Ciena PM Monitor Type description.
pmOtuHistory24HrIndexDescr	1.3.6.1.4.1.1271.3.5.22.18.1.3	Ciena PM If Index description
pmOtuHistory24HrMonValue	1.3.6.1.4.1.1271.3.5.22.18.1.4	Ciena History 24Hr monitored value.
pmOtuHistory24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.18.1.5	Invalid flag value for the MonType
pmOtuHistory24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.18.1.6	Flag indicating if PM Mon Value is supported or not
pmOtuHistory24HrAdminState	1.3.6.1.4.1.1271.3.5.22.18.1.7	Profile admin state.
pmOtuHistory24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.18.1.8	Start date and time.
pmOtuHistory24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.18.1.9	Threshold value.
pmOtuHistory24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.18.1.10	Flag to show if TCA is supported or TCA is active
pmOdu15MinMonType	1.3.6.1.4.1.1271.3.5.22.19.1.1	Second Index for pmOduCurrent15minStatsTable. This is the PM counter Type
pmOdu15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.19.1.2	Ciena PM Monitor Type description.
pmOdu15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.19.1.3	Ciena PM If Index description
pmOdu15MinMonValue	1.3.6.1.4.1.1271.3.5.22.19.1.4	Ciena 15 min monitored value.
pmOdu15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.19.1.5	Invalid flag value for the MonType
pmOdu15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.19.1.6	Flag indicating if PM Mon Value is supported or not
pmOdu15MinAdminState	1.3.6.1.4.1.1271.3.5.22.19.1.7	Profile admin state.

Indicator	OID	Description
pmOdu15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.19.1.8	Start date and time.
pmOdu15Minthvalue	1.3.6.1.4.1.1271.3.5.22.19.1.9	Threshold value.
pmOdu15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.19.1.10	Flag to show if TCA is supported or TCA is active
pmOdu15MinTcaAdminState	1.3.6.1.4.1.1271.3.5.22.19.1.11	TCA Profile admin state.
pmOdu15MinTcaProfileId	1.3.6.1.4.1.1271.3.5.22.19.1.12	TCA Profile Id
pmOdu15MinTcaProfileName	1.3.6.1.4.1.1271.3.5.22.19.1.13	TCA Profile Name
pmOdu24HrMonType	1.3.6.1.4.1.1271.3.5.22.20.1.1	Second Index for pmOduCurrent24HrStatsTable. This is the PM counter Type
pmOdu24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.20.1.2	Ciena PM Monitor Type description.
pmOdu24HrIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.20.1.3	Ciena PM If Index description
pmOdu24HrMonValue	1.3.6.1.4.1.1271.3.5.22.20.1.4	Ciena 24Hr monitored value.
pmOdu24HrMonidf	1.3.6.1.4.1.1271.3.5.22.20.1.5	Invalid flag value for the MonType
pmOdu24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.20.1.6	Flag indicating if PM Mon Value is supported or not
pmOdu24HrAdminState	1.3.6.1.4.1.1271.3.5.22.20.1.7	Profile admin state.
pmOdu24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.20.1.8	Start date and time.
pmOdu24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.20.1.9	Threshold value.
pmOdu24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.20.1.10	Flag to show if TCA is supported or TCA is active
pmOdu24HrTcaAdminState	1.3.6.1.4.1.1271.3.5.22.20.1.11	TCA Profile admin state.
pmOdu24HrTcaProfileId	1.3.6.1.4.1.1271.3.5.22.20.1.12	TCA Profile Id
pmOdu24HrTcaProfileName	1.3.6.1.4.1.1271.3.5.22.20.1.13	TCA Profile Name

Indicator	OID	Description
pmOduUntimedMonType	1.3.6.1.4.1.1271.3.5.22.21.1.1	Second Index for pmOduUntimedStatsTable. This is the PM counter Type
pmOduUntimedMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.21.1.2	Ciena PM Monitor Type description.
pmOduUntimedIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.21.1.3	Ciena PM If Index description
pmOduUntimedMonValue	1.3.6.1.4.1.1271.3.5.22.21.1.4	Ciena Untimed monitored value.
pmOduUntimedMonIdf	1.3.6.1.4.1.1271.3.5.22.21.1.5	Invalid flag value for the MonType
pmOduUntimedMonSupported	1.3.6.1.4.1.1271.3.5.22.21.1.6	Flag indicating if PM Mon Value is supported or not
pmOduUntimedAdminState	1.3.6.1.4.1.1271.3.5.22.21.1.7	Profile admin state.
pmOduUntimedMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.21.1.8	Start date and time.
pmOduUntimedthvalue	1.3.6.1.4.1.1271.3.5.22.21.1.9	Threshold value.
pmOduUntimedTcaActive	1.3.6.1.4.1.1271.3.5.22.21.1.10	Flag to show if TCA is supported or TCA is active
pmOduUntimedTcaAdminState	1.3.6.1.4.1.1271.3.5.22.21.1.11	TCA Profile admin state.
pmOduUntimedTcaProfileId	1.3.6.1.4.1.1271.3.5.22.21.1.12	TCA Profile Id
pmOduUntimedTcaProfileName	1.3.6.1.4.1.1271.3.5.22.21.1.13	TCA Profile Name
pmOduHistory15MinBinIndex	1.3.6.1.4.1.1271.3.5.22.22.1.1	Index for pmOduHistory15minStatsTable .
pmOduHistory15MinMonType	1.3.6.1.4.1.1271.3.5.22.22.1.2	Third Index for pmOduHistory15minStatsTable. This is the PM counter Type
pmOduHistory15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.22.1.3	Ciena PM Monitor Type description.
pmOduHistory15MinIndexDescr	1.3.6.1.4.1.1271.3.5.22.22.1.4	Ciena PM If Index description

Indicator	OID	Description
pmOduHistory15MinMonValue	1.3.6.1.4.1.1271.3.5.22.22.1.5	Ciena History 15 min monitored value.
pmOduHistory15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.22.1.6	Invalid flag value for the MonType
pmOduHistory15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.22.1.7	Flag indicating if PM Mon Value is supported or not
pmOduHistory15MinAdminState	1.3.6.1.4.1.1271.3.5.22.22.1.8	Profile admin state.
pmOduHistory15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.22.1.9	Start date and time.
pmOduHistory15MinThvalue	1.3.6.1.4.1.1271.3.5.22.22.1.10	Threshold value.
pmOduHistory15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.22.1.11	Flag to show if TCA is supported or TCA is active
pmOduHistory24HrMonType	1.3.6.1.4.1.1271.3.5.22.23.1.1	Second Index for pmOdu24HrStatsTable. This is the PM counter Type
pmOduHistory24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.23.1.2	Ciena PM Monitor Type description.
pmOduHistory24HrIndexDescr	1.3.6.1.4.1.1271.3.5.22.23.1.3	Ciena PM If Index description
pmOduHistory24HrMonValue	1.3.6.1.4.1.1271.3.5.22.23.1.4	Ciena History 24Hr monitored value.
pmOduHistory24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.23.1.5	Invalid flag value for the MonType
pmOduHistory24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.23.1.6	Flag indicating if PM Mon Value is supported or not
pmOduHistory24HrAdminState	1.3.6.1.4.1.1271.3.5.22.23.1.7	Profile admin state.
pmOduHistory24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.23.1.8	Start date and time.
pmOduHistory24HrThvalue	1.3.6.1.4.1.1271.3.5.22.23.1.9	Threshold value.
pmOduHistory24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.23.1.10	Flag to show if TCA is supported or TCA is active

Indicator	OID	Description
pmOpticalPower15MinLaneIndex	1.3.6.1.4.1.1271.3.5.22.24.1.1	Second Index for pmOpticalPowerCurrent 15minStatsTable.
pmOpticalPower15MinMonType	1.3.6.1.4.1.1271.3.5.22.24.1.2	Third Index for pmOpticalPowerCurrent 15minStatsTable. This is the PM counter Type
pmOpticalPower15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.24.1.3	Ciena PM Monitor Type description.
pmOpticalPower15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.24.1.4	Ciena PM If Index description
pmOpticalPower15MinMonValue	1.3.6.1.4.1.1271.3.5.22.24.1.5	Ciena 15 min monitored value.
pmOpticalPower15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.24.1.6	Invalid flag value for the MonType
pmOpticalPower15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.24.1.7	Flag indicating if PM Mon Value is supported or not
pmOpticalPower15MinAdminState	1.3.6.1.4.1.1271.3.5.22.24.1.8	Profile admin state.
pmOpticalPower15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.24.1.9	Start date and time.
pmOpticalPower15Minthvalue	1.3.6.1.4.1.1271.3.5.22.24.1.10	Threshold value.
pmOpticalPower15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.24.1.11	Flag to show if TCA is supported or TCA is active
pmOpticalPower15MinTcaAdminState	1.3.6.1.4.1.1271.3.5.22.24.1.12	TCA Profile admin state.
pmOpticalPower15MinTcaProfileId	1.3.6.1.4.1.1271.3.5.22.24.1.13	TCA Profile Id
pmOpticalPower15MinBaseLineValue	1.3.6.1.4.1.1271.3.5.22.24.1.14	BaseLine value for threshold.
pmOpticalPower15MinTcaProfileName	1.3.6.1.4.1.1271.3.5.22.24.1.15	TCA Profile Name
pmOpticalPower24HrLaneIndex	1.3.6.1.4.1.1271.3.5.22.25.1.1	Second Index for pmOpticalPowerCurrent 24HrStatsTable.
pmOpticalPower24HrMonType	1.3.6.1.4.1.1271.3.5.22.25.1.2	Third Index for pmOpticalPowerCurrent 24HrStatsTable. This is the PM counter Type

Indicator	OID	Description
pmOpticalPower24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.25.1.3	Ciena PM Monitor Type description.
pmOpticalPower24HrIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.25.1.4	Ciena PM If Index description
pmOpticalPower24HrMonValue	1.3.6.1.4.1.1271.3.5.22.25.1.5	Ciena 24Hr monitored value.
pmOpticalPower24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.25.1.6	Invalid flag value for the MonType
pmOpticalPower24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.25.1.7	Flag indicating if PM Mon Value is supported or not
pmOpticalPower24HrAdminState	1.3.6.1.4.1.1271.3.5.22.25.1.8	Profile admin state.
pmOpticalPower24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.25.1.9	Start date and time.
pmOpticalPower24HrThvalue	1.3.6.1.4.1.1271.3.5.22.25.1.10	Threshold value.
pmOpticalPower24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.25.1.11	Flag to show if TCA is supported or TCA is active
pmOpticalPower24HrTcaAdminState	1.3.6.1.4.1.1271.3.5.22.25.1.12	TCA Profile admin state.
pmOpticalPower24HrTcaProfileId	1.3.6.1.4.1.1271.3.5.22.25.1.13	TCA Profile Id
pmOpticalPower24HrBaseLineValue	1.3.6.1.4.1.1271.3.5.22.25.1.14	BaseLine value for threshold.
pmOpticalPower24HrTcaProfileName	1.3.6.1.4.1.1271.3.5.22.25.1.15	TCA Profile Name
pmOpticalPowerUntimedLaneIndex	1.3.6.1.4.1.1271.3.5.22.26.1.1	Second Index for pmOpticalPowerUntimedStatsTable.
pmOpticalPowerUntimedMonType	1.3.6.1.4.1.1271.3.5.22.26.1.2	Third Index for pmOpticalPowerUntimedStatsTable. This is the PM counter Type
pmOpticalPowerUntimedMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.26.1.3	Ciena PM Monitor Type description.
pmOpticalPowerUntimedIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.26.1.4	Ciena PM If Index description
pmOpticalPowerUntimedMonValue	1.3.6.1.4.1.1271.3.5.22.26.1.5	Ciena untimed monitored value.

Indicator	OID	Description
pmOpticalPowerUntimedMonIdf	1.3.6.1.4.1.1271.3.5.22.26.1.6	Invalid flag value for the MonType
pmOpticalPowerUntimedMonSupported	1.3.6.1.4.1.1271.3.5.22.26.1.7	Flag indicating if PM Mon Value is supported or not
pmOpticalPowerUntimedAdminState	1.3.6.1.4.1.1271.3.5.22.26.1.8	Profile admin state.
pmOpticalPowerUntimedMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.26.1.9	Start date and time.
pmOpticalPowerUntimedThvalue	1.3.6.1.4.1.1271.3.5.22.26.1.10	Threshold value.
pmOpticalPowerUntimedTcaActive	1.3.6.1.4.1.1271.3.5.22.26.1.11	Flag to show if TCA is supported or TCA is active
pmOpticalPowerUntimedTcaAdminState	1.3.6.1.4.1.1271.3.5.22.26.1.12	TCA Profile admin state.
pmOpticalPowerUntimedTcaProfileId	1.3.6.1.4.1.1271.3.5.22.26.1.13	TCA Profile Id
pmOpticalPowerUntimedBaseLineValue	1.3.6.1.4.1.1271.3.5.22.26.1.14	BaseLine value for threshold.
pmOpticalPowerUntimedTcaProfileName	1.3.6.1.4.1.1271.3.5.22.26.1.15	TCA Profile Name
pmOpticalPowerHistory15MinBinIndex	1.3.6.1.4.1.1271.3.5.22.27.1.1	Index for pmOpticalPowerHistory15minStatsTable .
pmOpticalPowerHistory15MinLaneIndex	1.3.6.1.4.1.1271.3.5.22.27.1.2	Index for pmOpticalPowerHistory15minStatsTable .
pmOpticalPowerHistory15MinMonType	1.3.6.1.4.1.1271.3.5.22.27.1.3	Third Index for pmOpticalPowerUntimedStatsTable. This is the PM counter Type
pmOpticalPowerHistory15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.27.1.4	Ciena PM Monitor Type description.
pmOpticalPowerHistory15MinIndexDescr	1.3.6.1.4.1.1271.3.5.22.27.1.5	Ciena PM If Index description
pmOpticalPowerHistory15MinMonValue	1.3.6.1.4.1.1271.3.5.22.27.1.6	Ciena History 15 min monitored value.
pmOpticalPowerHistory15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.27.1.7	Invalid flag value for the MonType

Indicator	OID	Description
pmOpticalPowerHistory15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.27.1.8	Flag indicating if PM Mon Value is supported or not
pmOpticalPowerHistory15MinAdminState	1.3.6.1.4.1.1271.3.5.22.27.1.9	Profile admin state.
pmOpticalPowerHistory15MinMonStartTime	1.3.6.1.4.1.1271.3.5.22.27.1.10	Start date and time.
pmOpticalPowerHistory15MinThvalue	1.3.6.1.4.1.1271.3.5.22.27.1.11	Threshold value.
pmOpticalPowerHistory15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.27.1.12	Flag to show if TCA is supported or TCA is active
pmOpticalPowerHistory24HrLanelIndex	1.3.6.1.4.1.1271.3.5.22.28.1.1	Index for pmOpticalPowerHistory24HrStatsTable .
pmOpticalPowerHistory24HrMonType	1.3.6.1.4.1.1271.3.5.22.28.1.2	Second Index for pmOpticalPowerUntimedStatsTable. This is the PM counter Type
pmOpticalPowerHistory24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.28.1.3	Ciena PM Monitor Type description.
pmOpticalPowerHistory24HrIndexDescr	1.3.6.1.4.1.1271.3.5.22.28.1.4	Ciena PM If Index description
pmOpticalPowerHistory24HrMonValue	1.3.6.1.4.1.1271.3.5.22.28.1.5	Ciena 15 min monitored value.
pmOpticalPowerHistory24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.28.1.6	Invalid flag value for the MonType
pmOpticalPowerHistory24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.28.1.7	Flag indicating if PM Mon Value is supported or not
pmOpticalPowerHistory24HrAdminState	1.3.6.1.4.1.1271.3.5.22.28.1.8	Profile admin state.
pmOpticalPowerHistory24HrMonStartTime	1.3.6.1.4.1.1271.3.5.22.28.1.9	Start date and time.
pmOpticalPowerHistory24HrThvalue	1.3.6.1.4.1.1271.3.5.22.28.1.10	Threshold value.
pmOpticalPowerHistory24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.28.1.11	Flag to show if TCA is supported or TCA is active

Indicator	OID	Description
pmGcm15MinMonType	1.3.6.1.4.1.1271.3.5.22.29.1.1	Second Index for pmGcmCurrent15minStatsTable. This is the PM counter Type
pmGcm15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.29.1.2	Ciena PM Monitor Type description.
pmGcm15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.29.1.3	Ciena PM If Index description
pmGcm15MinMonValue	1.3.6.1.4.1.1271.3.5.22.29.1.4	Ciena 15 min monitored value.
pmGcm15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.29.1.5	Invalid flag value for the MonType
pmGcm15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.29.1.6	Flag indicating if PM Mon Value is supported or not
pmGcm15MinAdminState	1.3.6.1.4.1.1271.3.5.22.29.1.7	Profile admin state.
pmGcm15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.29.1.8	Start date and time.
pmGcm15MinThvalue	1.3.6.1.4.1.1271.3.5.22.29.1.9	Threshold value.
pmGcm15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.29.1.10	Flag to show if TCA is supported or TCA is active
pmGcm15MinTcaAdminState	1.3.6.1.4.1.1271.3.5.22.29.1.11	TCA Profile admin state.
pmGcm15MinTcaProfileId	1.3.6.1.4.1.1271.3.5.22.29.1.12	TCA Profile Id
pmGcm15MinTcaProfileName	1.3.6.1.4.1.1271.3.5.22.29.1.13	TCA Profile Name
pmGcm24HrMonType	1.3.6.1.4.1.1271.3.5.22.30.1.1	Second Index for pmGcmCurrent24HrStatsTable. This is the PM counter Type
pmGcm24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.30.1.2	Ciena PM Monitor Type description.
pmGcm24HrIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.30.1.3	Ciena PM If Index description
pmGcm24HrMonValue	1.3.6.1.4.1.1271.3.5.22.30.1.4	Ciena 24Hr monitored value.
pmGcm24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.30.1.5	Invalid flag value for the MonType

Indicator	OID	Description
pmGcm24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.30.1.6	Flag indicating if PM Mon Value is supported or not
pmGcm24HrAdminState	1.3.6.1.4.1.1271.3.5.22.30.1.7	Profile admin state.
pmGcm24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.30.1.8	Start date and time.
pmGcm24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.30.1.9	Threshold value.
pmGcm24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.30.1.10	Flag to show if TCA is supported or TCA is active
pmGcm24HrTcaAdminState	1.3.6.1.4.1.1271.3.5.22.30.1.11	TCA Profile admin state.
pmGcm24HrTcaProfileId	1.3.6.1.4.1.1271.3.5.22.30.1.12	TCA Profile Id
pmGcm24HrTcaProfileName	1.3.6.1.4.1.1271.3.5.22.30.1.13	TCA Profile Name
pmGcmUntimedMonType	1.3.6.1.4.1.1271.3.5.22.31.1.1	Second Index for pmGcmUntimedStatsTable. This is the PM counter Type
pmGcmUntimedMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.31.1.2	Ciena PM Monitor Type description.
pmGcmUntimedIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.31.1.3	Ciena PM If Index description
pmGcmUntimedMonValue	1.3.6.1.4.1.1271.3.5.22.31.1.4	Ciena Untimed monitored value.
pmGcmUntimedMonIdf	1.3.6.1.4.1.1271.3.5.22.31.1.5	Invalid flag value for the MonType
pmGcmUntimedMonSupported	1.3.6.1.4.1.1271.3.5.22.31.1.6	Flag indicating if PM Mon Value is supported or not
pmGcmUntimedAdminState	1.3.6.1.4.1.1271.3.5.22.31.1.7	Profile admin state.
pmGcmUntimedMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.31.1.8	Start date and time.
pmGcmUntimedthvalue	1.3.6.1.4.1.1271.3.5.22.31.1.9	Threshold value.
pmGcmUntimedTcaActive	1.3.6.1.4.1.1271.3.5.22.31.1.10	Flag to show if TCA is supported or TCA is active
pmGcmUntimedTcaAdminState	1.3.6.1.4.1.1271.3.5.22.31.1.11	TCA Profile admin state.
pmGcmUntimedTcaProfileId	1.3.6.1.4.1.1271.3.5.22.31.1.12	TCA Profile Id

Indicator	OID	Description
pmGcmUntimedTcaProfileName	1.3.6.1.4.1.1271.3.5.22.31.1.13	TCA Profile Name
pmGcmHistory15MinBinIndex	1.3.6.1.4.1.1271.3.5.22.32.1.1	Index for pmGcmHistory15minStatsTable .
pmGcmHistory15MinMonType	1.3.6.1.4.1.1271.3.5.22.32.1.2	Third Index for pmGcmHistory15minStatsTable. This is the PM counter Type
pmGcmHistory15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.32.1.3	Ciena PM Monitor Type description.
pmGcmHistory15MinIndexDescr	1.3.6.1.4.1.1271.3.5.22.32.1.4	Ciena PM If Index description
pmGcmHistory15MinMonValue	1.3.6.1.4.1.1271.3.5.22.32.1.5	Ciena History 15 min monitored value.
pmGcmHistory15MinMonIdf	1.3.6.1.4.1.1271.3.5.22.32.1.6	Invalid flag value for the MonType
pmGcmHistory15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.32.1.7	Flag indicating if PM Mon Value is supported or not
pmGcmHistory15MinAdminState	1.3.6.1.4.1.1271.3.5.22.32.1.8	Profile admin state.
pmGcmHistory15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.32.1.9	Start date and time.
pmGcmHistory15Minthvalue	1.3.6.1.4.1.1271.3.5.22.32.1.10	Threshold value.
pmGcmHistory15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.32.1.11	Flag to show if TCA is supported or TCA is active
pmGcmHistory24HrMonType	1.3.6.1.4.1.1271.3.5.22.33.1.1	Second Index for pmGcm24HrStatsTable. This is the PM counter Type
pmGcmHistory24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.33.1.2	Ciena PM Monitor Type description.
pmGcmHistory24HrIndexDescr	1.3.6.1.4.1.1271.3.5.22.33.1.3	Ciena PM If Index description
pmGcmHistory24HrMonValue	1.3.6.1.4.1.1271.3.5.22.33.1.4	Ciena History 24Hr monitored value.
pmGcmHistory24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.33.1.5	Invalid flag value for the MonType

Indicator	OID	Description
pmGcmHistory24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.33.1.6	Flag indicating if PM Mon Value is supported or not
pmGcmHistory24HrAdminState	1.3.6.1.4.1.1271.3.5.22.33.1.7	Profile admin state.
pmGcmHistory24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.33.1.8	Start date and time.
pmGcmHistory24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.33.1.9	Threshold value.
pmGcmHistory24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.33.1.10	Flag to show if TCA is supported or TCA is active
pmPhotonics15MinMonType	1.3.6.1.4.1.1271.3.5.22.34.1.1	Second Index for pmPhotonicsCurrent15 minStatsTable. This is the PM counter Type
pmPhotonics15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.34.1.2	Ciena PM Monitor Type description.
pmPhotonics15MinIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.34.1.3	Ciena PM If Index description
pmPhotonics15MinMonValue	1.3.6.1.4.1.1271.3.5.22.34.1.4	Ciena 15 min monitored value.
pmPhotonics15MinMonidf	1.3.6.1.4.1.1271.3.5.22.34.1.5	Invalid flag value for the MonType
pmPhotonics15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.34.1.6	Flag indicating if PM Mon Value is supported or not
pmPhotonics15MinAdminState	1.3.6.1.4.1.1271.3.5.22.34.1.7	Profile admin state.
pmPhotonics15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.34.1.8	Start date and time.
pmPhotonics15Minthvalue	1.3.6.1.4.1.1271.3.5.22.34.1.9	Threshold value.
pmPhotonics15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.34.1.10	Flag to show if TCA is supported or TCA is active
pmPhotonics15MinTcaAdminState	1.3.6.1.4.1.1271.3.5.22.34.1.11	TCA Profile admin state.
pmPhotonics15MinTcaProfileId	1.3.6.1.4.1.1271.3.5.22.34.1.12	TCA Profile Id
pmPhotonics15MinBaseLineValue	1.3.6.1.4.1.1271.3.5.22.34.1.13	BaseLine value for threshold.
pmPhotonics15MinTcaProfileName	1.3.6.1.4.1.1271.3.5.22.34.1.14	TCA Profile Name

Indicator	OID	Description
pmPhotonics24HrMonType	1.3.6.1.4.1.1271.3.5.22.35.1.1	Second Index for pmPhotonicsCurrent24HrStatsTable. This is the PM counter Type
pmPhotonics24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.35.1.2	Ciena PM Monitor Type description.
pmPhotonics24HrIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.35.1.3	Ciena PM If Index description
pmPhotonics24HrMonValue	1.3.6.1.4.1.1271.3.5.22.35.1.4	Ciena 24Hr monitored value.
pmPhotonics24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.35.1.5	Invalid flag value for the MonType
pmPhotonics24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.35.1.6	Flag indicating if PM Mon Value is supported or not
pmPhotonics24HrAdminState	1.3.6.1.4.1.1271.3.5.22.35.1.7	Profile admin state.
pmPhotonics24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.35.1.8	Start date and time.
pmPhotonics24HrThvalue	1.3.6.1.4.1.1271.3.5.22.35.1.9	Threshold value.
pmPhotonics24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.35.1.10	Flag to show if TCA is supported or TCA is active
pmPhotonics24HrTcaAdminState	1.3.6.1.4.1.1271.3.5.22.35.1.11	TCA Profile admin state.
pmPhotonics24HrTcaProfileId	1.3.6.1.4.1.1271.3.5.22.35.1.12	TCA Profile Id
pmPhotonics24HrBaseLineValue	1.3.6.1.4.1.1271.3.5.22.35.1.13	BaseLine value for threshold.
pmPhotonics24HrTcaProfileName	1.3.6.1.4.1.1271.3.5.22.35.1.14	TCA Profile Name
pmPhotonicsUntimedMonType	1.3.6.1.4.1.1271.3.5.22.36.1.1	Second Index for pmPhotonicsUntimedStatsTable. This is the PM counter Type
pmPhotonicsUntimedMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.36.1.2	Ciena PM Monitor Type description.
pmPhotonicsUntimedIfIndexDescr	1.3.6.1.4.1.1271.3.5.22.36.1.3	Ciena PM If Index description
pmPhotonicsUntimedMonValue	1.3.6.1.4.1.1271.3.5.22.36.1.4	Ciena untimed monitored value.

Indicator	OID	Description
pmPhotonicsUntimedMonidf	1.3.6.1.4.1.1271.3.5.22.36.1.5	Invalid flag value for the MonType
pmPhotonicsUntimedMonSupported	1.3.6.1.4.1.1271.3.5.22.36.1.6	Flag indicating if PM Mon Value is supported or not
pmPhotonicsUntimedAdminState	1.3.6.1.4.1.1271.3.5.22.36.1.7	Profile admin state.
pmPhotonicsUntimedMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.36.1.8	Start date and time.
pmPhotonicsUntimedthvalue	1.3.6.1.4.1.1271.3.5.22.36.1.9	Threshold value.
pmPhotonicsUntimedTcaActive	1.3.6.1.4.1.1271.3.5.22.36.1.10	Flag to show if TCA is supported or TCA is active
pmPhotonicsUntimedTcaAdminState	1.3.6.1.4.1.1271.3.5.22.36.1.11	TCA Profile admin state.
pmPhotonicsUntimedTcaProfileId	1.3.6.1.4.1.1271.3.5.22.36.1.12	TCA Profile Id
pmPhotonicsUntimedBaseLineValue	1.3.6.1.4.1.1271.3.5.22.36.1.13	BaseLine value for threshold.
pmPhotonicsUntimedTcaProfileName	1.3.6.1.4.1.1271.3.5.22.36.1.14	TCA Profile Name
pmPhotonicsHistory15MinBinIndex	1.3.6.1.4.1.1271.3.5.22.37.1.1	Second Index for pmPhotonicsHistory15m inStatsTable.
pmPhotonicsHistory15MinMonType	1.3.6.1.4.1.1271.3.5.22.37.1.2	Third Index for pmPhotonicsUntimedStatsTable. This is the PM counter Type
pmPhotonicsHistory15MinMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.37.1.3	Ciena PM Monitor Type description.
pmPhotonicsHistory15MinIndexDescr	1.3.6.1.4.1.1271.3.5.22.37.1.4	Ciena PM If Index description
pmPhotonicsHistory15MinMonValue	1.3.6.1.4.1.1271.3.5.22.37.1.5	Ciena History 15 min monitored value.
pmPhotonicsHistory15MinMonidf	1.3.6.1.4.1.1271.3.5.22.37.1.6	Invalid flag value for the MonType
pmPhotonicsHistory15MinMonSupported	1.3.6.1.4.1.1271.3.5.22.37.1.7	Flag indicating if PM Mon Value is supported or not
pmPhotonicsHistory15MinAdminState	1.3.6.1.4.1.1271.3.5.22.37.1.8	Profile admin state.

Indicator	OID	Description
pmPhotonicsHistory15MinMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.37.1.9	Start date and time.
pmPhotonicsHistory15Minthvalue	1.3.6.1.4.1.1271.3.5.22.37.1.10	Threshold value.
pmPhotonicsHistory15MinTcaActive	1.3.6.1.4.1.1271.3.5.22.37.1.11	Flag to show if TCA is supported or TCA is active
pmPhotonicsHistory24HrMonType	1.3.6.1.4.1.1271.3.5.22.38.1.1	Second Index for pmPhotonicsUntimedStatsTable. This is the PM counter Type
pmPhotonicsHistory24HrMonTypeDescr	1.3.6.1.4.1.1271.3.5.22.38.1.2	Ciena PM Monitor Type description.
pmPhotonicsHistory24HrIndexDescr	1.3.6.1.4.1.1271.3.5.22.38.1.3	Ciena PM If Index description
pmPhotonicsHistory24HrMonValue	1.3.6.1.4.1.1271.3.5.22.38.1.4	Ciena 15 min monitored value.
pmPhotonicsHistory24HrMonIdf	1.3.6.1.4.1.1271.3.5.22.38.1.5	Invalid flag value for the MonType
pmPhotonicsHistory24HrMonSupported	1.3.6.1.4.1.1271.3.5.22.38.1.6	Flag indicating if PM Mon Value is supported or not
pmPhotonicsHistory24HrAdminState	1.3.6.1.4.1.1271.3.5.22.38.1.7	Profile admin state.
pmPhotonicsHistory24HrMonStartDateTime	1.3.6.1.4.1.1271.3.5.22.38.1.8	Start date and time.
pmPhotonicsHistory24Hrthvalue	1.3.6.1.4.1.1271.3.5.22.38.1.9	Threshold value.
pmPhotonicsHistory24HrTcaActive	1.3.6.1.4.1.1271.3.5.22.38.1.10	Flag to show if TCA is supported or TCA is active

Finding the entity index associated with an ifIndex

This section describes a method of finding the entity index associated with an interface index (ifIndex). When you work with SNMP on Waveserver platforms you can do this manually using a MIB browser.

This method assumes the following:

- you have downloaded the Waveserver5 SNMP MIBs
- you are using a MIB browser, such as MG-SOFT MIB Browser.

- you have compiled the MIBs using your MIB browser.
- you have enabled SNMP on the Waveserver5
- you have connected to the Waveserver5 with your MIB browser using the IP address of the Waveserver5 and SNMPv1 with public community.

To find the ifIndex associated with the entity index, do the following:

- Perform a walk on the interface description (ifDescr) under ifTable to find ifIndex.
- Perform a walk on entity description (entPhysicalDescr) under entityMIB to find entity Index.

Example: Finding an OID

Find the OID for pmOtuUntimedMonValue for OTU PM entity OtuFeES on interface PORT-1/2-OTUCn. The following table lists the location and component parts of the complete OID.

Table	Component	Value
Table 217, "CIENA-WS-PLATFORM-PM-MIB.my"	MIB indicator: pmOtuUntimedMonValue	OID: 1.3.6.1.4.1.1271.3.5.22.16.1.4
Table 218, "ifIndex values mapped to Interface type"	Interface: PORT-1/2-OTUCn	ifIndex value: 16778246
Table 221, "PM indexes for OTU, ODU, Optical, GCM entities"	OTU entity: OtuFeES	PM index: 5

The complete OID is:

1.3.6.1.4.1.1271.3.5.22.16.1.4.16778246.5

Find the OID for pmOpticalPower15MinMonValue for Optical PM entity rxAverage on lane 1 of interface PTP-1/5. The following table lists the location and component parts of the complete OID.

Table	Component	Value
Table 217, "CIENA-WS-PLATFORM-PM-MIB.my"	MIB indicator: pmOpticalPower15MinMonValue	OID: 1.3.6.1.4.1.1271.3.5.22.24.1.5
Table 218, "ifIndex values mapped to Interface type"	Interface: Lane 1 of PTP-1/5	ifIndex value: 16779778
Table 221, "PM indexes for OTU, ODU, Optical, GCM entities"	Optical entity: rxAverage	PM index: 1.2

The complete OID is:

1.3.6.1.4.1.1271.3.5.22.24.1.5.16779778.1.2

Find the OID for serial number (entPhysicalSerialNum) of the Access Panel Module. The following table lists the location and component parts of the complete OID.

Table	Component	Value
Table 219, "EntityMIB entPhysicalEntry OIDs by information type"	MIB indicator: entPhysicalSerialNum	OID: 1.3.6.1.2.1.47.1.1.1.1.11
Table 220, "Entity MIB indexes for components"	Entity (chassis common equipment): Access Panel Instance	Index value: 111

The complete OID is:

1.3.6.1.2.1.47.1.1.1.1.11.111

OID indexes

The following tables show the relationship between the interfaces and the SNMP indexes.

- [ifIndex values mapped to Interface type](#)
- [EntityMIB entPhysicalEntry OIDs by information type](#)
- [Entity MIB indexes for components](#)
- [PM indexes for OTU, ODU, Optical, GCM entities](#)
- [PM indexes for Ethernet entities](#)
- [IF-MIB mib-2 elements](#)

Table 218 ifIndex values mapped to Interface type

Interface	ifIndex
Console-1	1
ILAN-1	2
ILAN-2	3
DCN-1	4
Local	5
DCN-2	6
DCN-3	7
DCN-4	8

Interface	ifIndex
DCN-5	9
Console-2	10
XCVR-1/1	16777728
PTP-1/1-WaveLogic	16777729
PTP-1/1	16777730
PORT-1/1-OTUCn	16777734
PORT-1/1-ODUCn	16777735
CHANNEL-1/1.1-ODUk	16777771
CHANNEL-1/1.2-ODUk	16777803
CHANNEL-1/1.3-ODUk	16777835
CHANNEL-1/1.4-ODUk	16777867
CHANNEL-1/1.5-ODUk	16777899
CHANNEL-1/1.6-ODUk	16777931
CHANNEL-1/1.7-ODUk	16777963
CHANNEL-1/1.8-ODUk	16777995
XCVR-1/2	16778240
PTP-1/2-WaveLogic	16778241
PTP-1/2	16778242
PORT-1/2-OTUCn	16778246
PORT-1/2-ODUCn	16778247
CHANNEL-1/2.1-ODUk	16778283
CHANNEL-1/2.2-ODUk	16778315
CHANNEL-1/2.3-ODUk	16778347
CHANNEL-1/2.4-ODUk	16778379
CHANNEL-1/2.5-ODUk	16778411
CHANNEL-1/2.6-ODUk	16778443
CHANNEL-1/2.7-ODUk	16778475
CHANNEL-1/2.8-ODUk	16778507
XCVR-1/3	16778752

Interface	ifIndex
PTP-1/3	16778754
PORT-1/3	16778755
XCVR-1/4	16779264
PTP-1/4	16779266
PORT-1/4	16779267
XCVR-1/5	16779776
PTP-1/5	16779778
PORT-1/5	16779779
XCVR-1/6	16780288
PTP-1/6	1678029
PORT-1/6	16780291
XCVR-1/7	16780800
PTP-1/7	16780802
PORT-1/7	16780803
XCVR-1/8	16781312
PTP-1/8	16781314
PORT-1/8	16781315
XCVR-1/9	16781824
PTP-1/9	16781826
PORT-1/9	16781827
XCVR-1/10	16782336
PTP-1/10	16782338
PORT-1/10	16782339
XCVR-1/11	16782848
PTP-1/11	16782851
PORT-1/11	16782852
XCVR-1/12	16783360
PTP-1/12	16783362
PORT-1/12	16783363

Interface	ifIndex
XCVR-1/13	16783372
PTP-1/13	16783374
PORT-1/13	16783375
XCVR-1/14	16784384
PTP-1/14	16784386
PORT-1/14	16743887
XCVR-1/15	16784896
PTP-1/15	16784298
PORT-1/15	16784899
XCVR-1/16	16785408
PTP-1/16	16785410
PORT-1/16	16785411
XCVR-1/17	16785920
PTP-1/17	16785922
PORT-1/17	16785923
XCVR-1/18	16786432
PTP-1/18	16786434
PORT-1/18	16786435
XCVR-3/1	50332160
XCVR-5/1	83886592
XCVR-7/1	117441024

The following table lists the OIDs for the physical entry information types (Path: iso(1).org(3).dod(6).internet(1).mgmt(2).mib-2(1).entityMIB(47).entityMIBObjects(1).entityPhysical(1).entPhysicalTable(1).entPhysicalEntry(1)).

To retrieve the physical description of a specific module use the OID of the Physical Description (1.3.6.1.2.1.47.1.1.1.1.2) along with the ID from the description. For example, the complete OID to get the hardware description of the module in slot 1 would be: 1.3.6.1.2.1.47.1.1.1.1.2.61

Table 219 EntityMIB entPhysicalEntry OIDs by information type

Information	OID
Physical Description (entPhysicalDescr)	1.3.6.1.2.1.47.1.1.1.1.2
Class (entPhysicalClass)	1.3.6.1.2.1.47.1.1.1.1.5
Name (entPhysicalName)	1.3.6.1.2.1.47.1.1.1.1.7
Hardware Rev (entPhysicalHardwareRev)	1.3.6.1.2.1.47.1.1.1.1.8
Firmware Rev (entPhysicalFirmwareRev)	1.3.6.1.2.1.47.1.1.1.1.9
Software Rev (entPhysicalSoftwareRev)	1.3.6.1.2.1.47.1.1.1.1.10
Serial Number (entPhysicalSerialNum)	1.3.6.1.2.1.47.1.1.1.1.11
Mfg Name (entPhysicalMfgName)	1.3.6.1.2.1.47.1.1.1.1.12
Model (entPhysicalModelName)	1.3.6.1.2.1.47.1.1.1.1.13
FRU (entPhysicalIsFRU)	1.3.6.1.2.1.47.1.1.1.1.16
Mfg Date (entPhysicalMfgDate)	1.3.6.1.2.1.47.1.1.1.1.17

The following table lists the Entity MIB indexes for entities. (Path: iso(1).org(3).dod(6).internet(1).mgmt(2).mib-2(1).entityMIB(47)).

Table 220 Entity MIB indexes for components

Component	Index	Physical Description
Chassis Instance	1	Waveserver 5 Chassis
Chassis common equipment slots		
Control Module Slot	2	Control Module Slot 1
Access Panel Slot	11	Access Panel Slot 1
Power Supply Slot 1	21	Power Supply Slot 1
Power Supply Slot 2	22	Power Supply Slot 2
Fan Slot 1/1	31	Fan Slot 1/1
Fan Slot 1/2	32	Fan Slot 1/2
Fan Slot 2/1	36	Fan Slot 1/3
Fan Slot 2/2	37	Fan Slot 1/4
Service Module Slot 1	61	Module Slot 1
Service Module Slot 2	62	Module Slot 2
Service Module Slot 3	63	Module Slot 3

Component	Index	Physical Description
Service Module Slot 4	64	Module Slot 4
Service Module Slot 5	65	Module Slot 5
Service Module Slot 6	66	Module Slot 6
Service Module Slot 7	67	Module Slot 7
Service Module Slot 8	68	Module Slot 8
Service Module Instance		
Service Modules from slot 1 to 8	161-176	Increments continuously for chassis with more slots (e.g., 10-slot chassis uses 161-170, 16-slot chassis uses 161-176)
Chassis common equipment instance		
Control Module Instance	101	Waveserver 5 Control Processor Module
Access Panel Instance	111	Waveserver 5 Access Panel
Power Supply Instance	121	Waveserver 5 DC Power and Fan Module
Power Supply Instance	122	Waveserver 5 DC Power and Fan Module
Fan Instance	131	Waveserver 5 Fan Module
Fan Instance	132	Waveserver 5 Fan Module
Fan Instance	136	Waveserver 5 Fan Module
Fan Instance	137	Waveserver 5 Fan Module
Chassis management port instances		
Console Port Instance	201	Management Port Console RJ-45
Console Port Instance	202	Management Port Console USB-C
ILAN Port Instance	211	Management Port ILAN-1
ILAN Port Instance	212	Management Port ILAN-2
DCN Port Instance	221	Management Port DCN-1
DCN Port Instance	222	Management Port DCN-2
DCN Port Instance	223	Management Port DCN-3
DCN Port Instance	224	Management Port DCN-4
DCN Port Instance	225	Management Port DCN-5
Slot 1 Transceivers & Pluggables		

Component	Index	Physical Description
XCVR slots for slot 1, port 1-18	1001-18	XCVR slot from 1-18
XCVR instances for slot 1, port 1-18	1501-18	Pluggable in xcvr slots 1-18
Slot 3 Transceivers & Pluggables		
XCVR slots for slot 3, port 1-18	3001-18	XCVR slot from 1-18
XCVR instances for slot 3, port 1-18	3501-18	Pluggable in xcvr slots 1-18
Slot 5 Transceivers & Pluggables		
XCVR slots for slot 5, port 1-18	5001-18	XCVR slot from 1-18
XCVR instances for slot 5, port 1-18	5501-18	Pluggable in xcvr slots 1-18
Slot 7 Transceivers & Pluggables		
XCVR slots for slot 7, port 1-18	7001-18	XCVR slot from 1-18
XCVR instances for slot 7, port 1-18	7501-18	Pluggable in xcvr slots 1-18

The following table lists the Performance Monitoring (PM) indexes for OTU, ODU Optical and GCM data. (Path: iso(1).org(3).dod(6).internet(1).private(4).enterprises(1).ciena(1271).waveserver(3).cienaWsPlatformConfig(5).cienaWsPlatformPmMIB(22))

Table 221 PM indexes for OTU, ODU, Optical, GCM entities

OTU	ODU	Optical (see Note)	GCM
0 - OtuBBE	0 - OduBBE	n.0 - rxMinimum	0 - gcmErrorCounts
1 - OtuES	1 - OduES	n.1 - rxMaximum	1 - gcmSES
2 - OtuSES	2 - OduSES	n.2 - rxAverage	2 - gcmUAS
3 - OtuUAS	3 - OduUAS	n.3 - txMinimum	3 - gcmES
4 - OtuFeBBE	4 - OduFeBBE	n.4 - txMaximum	4 - gcmSESFE
5 - OtuFeES	5 - OduFeES	n.5 - txAverage	5 - gcmESFE
6 - OtuFeSES	6 - OduFeSES	n.6 - chanRxMinimum	
7 - OtuFeUAS	7 - OduFeUAS	n.7 - chanRxMaximum	
		n.8 - chanRxAverage	
		9 - aggregateRxMinimum 10 - aggregateRxMaximum 11 - aggregateRxAverage	

OTU	ODU	Optical (see Note)	GCM
		12 - aggregateTxMinimum 13 - aggregateTxMaximum 14 - aggregateTxAverage	
Note: 'n' represents the lane number. Possible values: 1, 2, 3, or 4.			

The following table lists the Performance Monitoring (PM) indexes for Ethernet data. (Path: iso(1).org(3).dod(6).internet(1).private(4).enterprises(1).ciena(1271).waveserver(3).cienaWsPlatformConfig(5).cienaWsPlatformPmMIB(22))

Table 222 PM indexes for Ethernet entities

Ethernet (only on client port)		
0 - rxBytes	22 - rxMinLinkUtilization	44 - tx1024To1518OctsPkts
1 - rxPkts	23 - rxMaxLinkUtilization	45 - tx1519ToJumboOctsPkts
2 - rxCrcErrorPkts	24 - rxBlockErrorCount	46 - txJumboOctsPkts
3 - rxMcastPkts	25 - rxPcsLaneBipErrorCount	47 - txBytesPerSec
4 - rxBcastPkts	26 - rxFrameErrorRatio	48 - txFramesPerSec
5 - rxUndersizePkts	27 - txBytes	49 - txAverageLinkUtilization
6 - rxOversizePkts	28 - txPkts	50 - txBlockErrorCount
7 - rxFragmentPkts	29 - txExcessiveDeferredPkts	51 - txPcsLaneBipErrorCount
8 - rxJabberPkts	30 - txUnderRunPkts	52 - txMinLinkUtilization
9 - rxLenOutOfRangePkts	31 - txCrcErrorPkts	53 - txMaxLinkUtilization
10 - rxPausePkts	32 - txLenCheckErrorPkts	54 - txFrameErrorRatio
11 - rx64OctsPkts	33 - txLenOutOfRangePkts	55 - pcsES
12 - rx65To127OctsPkts	34 - txPausePkts	56 - pcsSES
13 - rx128To255OctsPkts	35 - txGiantPkts	57 - pcsUAS
14 - rx256To511Octs	36 - txMcastPkts	58 - syncHeaderErrorCount
15 - rx512To1023OctsPkts	37 - txBcastPkts	59 - fecCorrCodewordCount
16 - rx1024To1518OctsPkts	38 - txPacketsDropCountSummary	60 - fecUncorrCodewordCount
17 - rx1519ToJumboOctsPkts	39 - tx64OctsPkts	61 - fecErrorCount lane 0
18 - rxJumboOctsPkts	40 - tx65To127OctsPkts	62 - fecErrorCount lane 1

Ethernet (only on client port)		
19 - rxBytesPerSec	41 - tx128To255OctsPkts	63 - fecErrorCount lane 2
20 - rxFramesPerSec	42 - tx256To511OctsPkts	64 - fecErrorCount lane 3
21 - rxAverageLinkUtilization	43 - tx512To1023OctsPkts	

The following table lists the Performance Monitoring (PM) indexes for Modem data. (Path: iso(1).org(3).dod(6).internet(1).private(4).enterprises(1).ciena(1271).waveserver(3).cienaWsPlatformConfig(5).cienaWsPlatformPmMIB(22))

Table 223 PM indexes for Modem entities

Modem entity indexes			
0 - bitErrorRate	10 - pdlMaximum	20 - cdMin	30 - snrAseMin
1 - berMaximum	11 - pdlAverage	21 - qStdev	31 - snrTnleAvg
2 - qFactor	12 - esnrAvg	22 - snrExtAvg	32 - snrTnleMax
3 - qMinimum	13 - esnrMax	23 - snrExtMax	33 - snrTnleMin
4 - qMaximum	14 - esnrMin	24 - snrExtMin	34 - cfoAvg
5 - fecUncorrectedSecs	15 - osnrAvg	25 - csiAvg	35 - cfoMax
6 - unCorrectedBlockCount	16 - osnrMax	26 - csiMax	36 - cfoMin
7 - highCorrectionCountSeconds	17 - osnrMin	27 - esnrDelta	37 - evmAvg
8 - dgdMaximum	18 - cdAvg	28 - snrAseAvg	38 - evmMax
9 - dgdAverage	19 - cdMax	29 - snrAseMax	39 - evmMin

The following table lists the Performance Monitoring (PM) indexes for Photonic data. (Path: iso(1).org(3).dod(6).internet(1).private(4).enterprises(1).ciena(1271).waveserver(3).cienaWsPlatformConfig(5).cienaWsPlatformPmMIB(22))

Table 224 PM indexes for Photonic entities

Photonic entity indexes			
0 - returnLossMinimum	23 - txEMRDshortMin	46 - txCLlong	69 - rxCRshortMax
1 - returnLossMaximum	24 - txCLshortMax	47 - txCLlongMin	70 - rxCRshort
2 - returnLossAverage	25 - txCLshort	48 - txCRlongMax	71 - rxCRshortMin
3 - rxSpanLossMinimum	26 - txCLshortMin	49 - txCRlong	72 - rxECLongMax

Photonic entity indexes			
4 - rxSpanLossMaximum	27 - txCRshortMax	50 - txCRLongMin	73 - rxEClong
5 - rxSpanLossAverage	28 - txCRshort	51 - rxECshortMax	74 - rxEClongMin
6 - txSpanLossMinimum	29 - txCRshortMin	52 - rxECshort	75 - rxEMLlongMax
7 - txSpanLossMaximum	30 - txEClongMax	53 - rxECshortMin	76 - rxEMLlong
8 - txSpanLossAverage	31 - txEClong	54 - rxEMLshortMax	77 - rxEMLlongMin
9 - txECshortMax	32 - txEClongMin	55 - rxEMLshort	78 - rxEMLDlongMax
10 - txECshort	33 - txEMLlongMax	56 - rxEMLshortMin	79 - rxEMLDlong
11 - txECshortMin	34 - txEMLlong	57 - rxEMLDshortMax	80 - rxEMLDlongMin
12 - txEMLshortMax	35 - txEMLlongMin	58 - rxEMLDshort	81 - rxEMRlongMax
13 - txEMLshort	36 - txEMLDlongMax	59 - rxEMLDshortMin	82 - rxEMRlong
14 - txEMLshortMin	37 - txEMLDlong	60 - rxEMRshortMax	83 - rxEMRlongMin
15 - txEMLDshortMax	38 - txEMLDlongMin	61 - rxEMRshort	84 - rxEMRDlongMax
16 - txEMLDshort	39 - txEMRlongMax	62 - rxEMRshortMin	85 - rxEMRDlong
17 - txEMLDshortMin	40 - txEMRlong	63 - rxEMRDshortMax	86 - rxEMRDlongMin
18 - txEMRshortMax	41 - txEMRlongMin	64 - rxEMRDshort	87 - rxCLlongMax
19 - txEMRshort	42 - txEMRDlongMax	65 - rxEMRDshortMin	88 - rxCLlong
20 - txEMRshortMin	43 - txEMRDlong	66 - rxCLshortMax	89 - rxCLlongMin
21 - txEMRDshortMax	44 - txEMRDlongMin	67 - rxCLshort	90 - rxCRLongMax
22 - txEMRDshort	45 - txCLlongMax	68 - rxCLshortMin	91 - rxCRLong
			92 - rxCRLongMin

The following table lists the OIDs for two object types:

- System (Path: iso(1).org(3).dod(6).internet(1).mgmt(2).mib-2(1).system(1).sysDescr(1))
- Interface (Path: iso(1).org(3).dod(6).internet(1).mgmt(2).mib-2(1).interfaces(2).ifTable(2).ifEntry(1))

Table 225 IF-MIB mib-2 elements

Information	OID
System	
System Info (sysDescr)	1.3.6.1.2.1.1.1

Information	OID
System Object ID (sysObjectID)	1.3.6.1.2.1.1.2
System uptime (sysUpTime)	1.3.6.1.2.1.1.3
System Contact (sysContact)	1.3.6.1.2.1.1.4
System Name (sysName)	1.3.6.1.2.1.1.5
System Location (sysLocation)	1.3.6.1.2.1.1.6
System Services (sysServices)	1.3.6.1.2.1.1.7
System Last Change (sysORChange)	1.3.6.1.2.1.1.8
Interface	
Interface MTU (ifMtu)	1.3.6.1.2.1.2.2.1.4
Interface Speed (ifSpeed)	1.3.6.1.2.1.2.2.1.5
Interface Address (ifPhysAddress)	1.3.6.1.2.1.2.2.1.6
Interface Admin Status (ifAdminStatus)	1.3.6.1.2.1.2.2.1.7
Interface Operational Status (ifOperStatus)	1.3.6.1.2.1.2.2.1.8
Interface Last state Change (ifLastChange)	1.3.6.1.2.1.2.2.1.9
Interface In Octet (ifInOctets)	1.3.6.1.2.1.2.2.1.10
Interface Out Octet (ifOutOctets)	1.3.6.1.2.1.2.2.1.16
Interface Alias	1.3.6.1.2.1.2.2.1.18

Procedure list

This section provides the following procedures:

- [Procedure 123, “Enabling and disabling SNMP”](#)
- [Procedure 124, “Configuring a device to send SNMPv2c traps”](#)
- [Procedure 125, “Configuring a device to send SNMPv3 traps”](#)
- [Procedure 126, “Setting contact and location attributes”](#)
- [Procedure 127, “Configuring SNMPv3 users”](#)
- [Procedure 128, “Displaying SNMP users”](#)
- [Procedure 129, “Configuring SNMP community mapping”](#)
- [Procedure 130, “Creating and attaching an SNMPv3 user to an SNMPv3 access entry group”](#)
- [Procedure 131, “Configuring a custom SNMP community string”](#)

- [Procedure 132, “Configuring notification entries”](#)
- [Procedure 133, “Configuring SNMP source address checking”](#)
- [Procedure 134, “Configuring address filtering”](#)

Procedure 123 Enabling and disabling SNMP

Use this procedure to enable or disable SNMP on Waveserver 5. When the operational status is disabled, attempts to connect to the SNMP agent will timeout. SNMP is disabled by default.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

1	If you want to	Then go to
	enable SNMP	step 2
	disable SNMP	step 3

2 Enable SNMP status:

```
snmp enable
```

Go to [step 4](#).

3 Disable SNMP status:

```
snmp disable
```

4 Confirm the SNMP status:

```
snmp show
```

```
WS_0208*# snmp show
+-----+-----+-----+-----+-----+-----+
| Parameter | Value |
+-----+-----+-----+-----+-----+-----+
| Operational status | Enabled |
| System Contact | Customer Support, Ciena |
| System Location | Not Specified |
| SnmpEngineBoots | 70 |
| SnmpEngineTime | 194642 |
| SnmpEngineMaxMessageSize | 2048 |
| SnmpEngineID | 80:00:0a:ff:55:22:88:55:cc:66:80:00 |
| Invalid Messages | 0 |
| Not In Time Windows | 0 |
| Unknown Security Models | 0 |
| Unknown PDU Handlers | 0 |
| Unsupported Security Levels | 0 |
| Unknown User Names | 0 |
| Unknown Engine IDs | 208 |
| All Port Trap State | Enabled |
| Std Link Trap State | Enabled |
+-----+-----+-----+-----+-----+-----+
```

5 Save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 124 Configuring a device to send SNMPv2c traps

Configure a device to send SNMPv2c traps when you want to enable the device to send SNMP version 2 (v2) message format traps. SNMPv2 traps are defined in the MIB files using the "NOTIFICATION-TYPE" macro.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

You can also configure the network element to send both versions of the trap at the same time by performing the procedure for both SNMP versions.

Steps

- 1 Create an entry in the snmpTargetAddrTable.

```
snmp create target <String: 1...32> {addr <IP address
(with optional mask) or host name>} {param-name <String:
1...32>} [tag <String: 1...32>] [udp-port <Number:
0...65535>] [transport-domain <snmp-udp | v4-udp | v6-
udp>] [retry-count <Number: 0...255>] [retry-timeout
<Centi-seconds: 0...65535>]
```

For example,

```
snmp create target informTarget param-name informParamv2c
addr 10.32.41.117
```

- 2 Create a target parameter entry and security name:

```
snmp create target-param <String: 1...32> {sec-name
<String>} sec-model v2c
```

For example,

```
snmp create target-param informParamv2c sec-name public
sec-model v2c
```

Validating the configuration

- 3 Display the target configuration:

```
snmp show target
```

```
WS_0208*# snmp show target
```

TargetName	IPAddr_w_Mask/Host	ParamName	Tags	Udp Port	Transport	Retry Count	Retry Timeout
					Domain		
InformTarget1	10.111.222.300/32	InformParamv3	TrapTag	162	snmp-udp	3	1500
informTarget	10.111.222.300/32	informParamv2c	TrapTag	162	snmp-udp	3	1500

4 Display the target parameter configuration:

```
snmp show target-params
```

```
WS_0208*# snmp show target-params
```

TargetParamName	SecurityName	Sec Model	Sec Level	MP Model
InformParamv3	public	v3	noAuth	3
informParamv2c	public	v2c	noAuth	1

Procedure 125 Configuring a device to send SNMPv3 traps

Configure a device to send SNMPv3 traps when you want to enable the device to send SNMPv3 message format traps.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

You can also configure the network element to send both versions of the trap at the same time by performing the procedure for both SNMP versions.

Steps

- 1 Create a secure SNMPv3 user.

```
snmp create user <String: 1...64> {auth-protocol
<noAuth|md5|sha>} [auth-password <String: 1...64>] [priv-
protocol <aes-128|des|3des|noPriv>] [priv-password
<String: 1...64>] [auth-secret <String: 1...32>] [priv-
secret <String: 1...32>] [engine-id <SNMP
EngineID:XX:XX... (5-32 octets)>]
```

For example:

```
snmp create user ciena auth-protocol sha auth-password
password priv-protocol des priv-password password
```

Note: A matching SNMPv3 user must be created on the SNMP client.
- 2 Attach the user created in the previous step to the group.

```
snmp security-to-group attach user <String: 1...64> sec-
model v3 {group <String: 1...32>}
```

For example:

```
snmp security-to-group attach user ciena sec-model v3
group cienav3Group
```
- 3 Create SNMP view tree.

```
snmp create viewtree <String: 1...32> sub-tree <String:
1...64> [mask <Octet String>] {type <include | exclude>}
```

For example:

```
snmp create viewtree cienaAll sub-tree iso type include
```
- 4 Create an access policy for the specified group name:

```
snmp create access-entry <String: 1...32> sec-model v3
{sec-level <noAuth|authNoPriv|authWithPriv>} {read-view
<SNMP View Tree:1...32>} [write-view <SNMP View Tree:
1...32>] [notify-view <SNMP View Tree: 1...32>]
```

For example:

```
snmp create access-entry cienav3Group sec-model v3 sec-
level authWithPriv read-view cienav3 notify-view
cienav3
```

5 Create SNMP notify entry.

```
snmp create notify <String: 1...32> {notify-tag <String:
1...32>} [notify-type <trap>]
```

For example:

```
snmp create notify notify notify-tag notifyTag notify-
type trap
```

6 Create an entry in the snmpTargetAddrTable.

```
snmp create target <String: 1...32> {addr <IP address
(with optional mask) or host name>} {param-name <String:
1...32>} [tag <String: 1...32>] [udp-port <Number:
0...65535>] [transport domain <snmp-udp | v4-udp | v6-
udp>] [retry-count <Number: 0...255>] [retry-timeout
<Centi-seconds: 0...65535>]
```

For example:

```
snmp create target trapTarget addr 10.1.2.3/24 param-name
trapParam tag notifyTag
```

7 Create a target parameter entry and security name:

```
snmp create target-param <String: 1...32> {sec-name
<String>} sec-model v3
```

For example:

```
snmp create target-param trapParam sec-name cienav3 sec-
model v3 sec-level authWithPriv
```

Validating the configuration

8 Display the target configuration:

```
snmp show target
```

The following figure shows an example output for the `snmp show target` command.

Figure 138 Example — snmp show target command output

TargetName	IPAddr_w_Mask/Host	ParamName	Tags	Udp Port	Transport Domain	Retry Count	Retry Timeout
Informtarget1	10.1.2.3/32	informParamv3	TrapTag	162	snmp-udp	3	1500
anywhere	0.0.0.0/0	none	anywhereTag	0	snmp-udp	0	0
anywhereIpv6	::/0	none	anywhereTag	0	v6-udp	0	0
informtarget	10.1.2.3/32	informParamv2c	TrapTag	162	snmp-udp	3	1500

9 Display the target parameter configuration:

```
snmp show target-params
```

The following figure shows an example output for the `snmp show target-params` command.

Figure 139 Example — snmp show target-params command output

TargetParamName	SecurityName	Sec Model	Sec Level	MP Model
InformParamv3	public	v3	noAuth	3
informParamv2c	public	v2c	noAuth	1

- 10** Display users:
 snmp show users

Procedure 126 Setting contact and location attributes

You can set contact and location attributes. By default, all traps are enabled.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Set SNMP contact and location attributes:

```
snmp set contact <String: 1...255> [location <String: 1...255>]
```

For example,

```
snmp set contact "Customer Support, Ciena" location "Spokane Valley, USA"
```

Procedure 127 Configuring SNMPv3 users

You can create and delete SNMPv3 users. Create at least one SNMPv3 user for an SNMP engine to send and receive SNMPv3 messages. This procedure describes how to configure a User-based Security Model (USM) for SNMPv3.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Note: SNMPv3 is active if a user is provisioned (`snmp show users`), an SNMPv3 access-entry is provisioned (`snmp show access-entry`), and there is a mapping between the SNMPv3 access-entry and a user (`snmp show security-to-group`).

Steps

Creating an SNMP user

- 1 To create an SNMP user:

```
snmp create user <String: 1...64> {auth-protocol
<noAuth|md5|sha>} [auth-password <String: 1...64>] [priv-
protocol <aes-128|des|3des|noPriv>] [priv-password
<String: 1...64>] [auth-secret <String: 1...32>] [priv-
secret <String: 1...32>] [engine-id <SNMP
EngineID:XX:XX... (5-32 octets)>]
```

For example,

```
snmp create user user1 auth-protocol md5 priv-protocol
noPriv auth-password mypassword
```

Deleting an SNMP user

- 2 To delete an SNMP user:

```
snmp delete user <String: 1...64> [engine-id <SNMP
EngineID:XX:XX... (5-32 octets)>]
```

For example,

```
snmp delete user user1
```

Procedure 128 Displaying SNMP users

You can display SNMP users and their associated protocols and engine IDs.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Steps

- 1 To display SNMP users:

```
snmp show users
```

The following figure shows a sample output for the `snmp show users` command.

Figure 140 Example — `snmp show users` output

UserName	AuthProt	PrivProt	Engine ID
private	noAuth	noPriv	80:00:04:ff:05:20:dd:ee:bb:aa:00:00
public	noAuth	noPriv	80:00:04:ff:05:20:dd:ee:bb:aa:00:00

Procedure 129 Configuring SNMP community mapping

You can configure SNMP community mapping.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Note: SNMPv2c is active if a community is provisioned (`snmp show community-map`), an SNMPv2c access-entry is provisioned (`snmp show access-entry`), and a provisioned user is connected to the SNMPv2c access-entry.

Steps

- 1 Create an entry in the `snmpCommunityTable`:

```
snmp create community-index <String: 1...32> {community
<String: 1...64>} {sec-name <String: 1...32>}
```

For example: `snmp create community-index comm1 community mycommunity sec-name user1`

- 2 Validate that the new entry was created:

```
snmp show community-map
```

The following figure shows an example of the command output.

Figure 141 Example — `snmp show community-map` output

CommunityIndex	CommunityName	SecName	TransportTag	ContextName
comm1	mycommunity	user1		
t0000000	public	public	anywhereTag	
t0000001	private	private	anywhereTag	

- 3 Create an entry in the `vacmSecurityToGroupTable`:

```
snmp security-to-group attach user <String: 1...64> {sec-
model <v1|v2c|v3>} {group <String: 1...32>}
```

For example: `snmp security-to-group attach user user1 sec-model v2c group group1`

- 4 Validate that the new entry was created:

```
snmp show security-to-group
```

The following figure shows an example of the command output.

Figure 142 Example — snmp show security-to-group command output

UserName	SecModel	GroupName
user1	v2c	group1
public	v2c	public
private	v2c	private

5 Create an entry in the vacmAccessTable.

```
snmp create access-entry <String: 1...32> {sec-model
<v1|v2c|v3>} {sec level <noAuth|authNoPriv|
|authWithPriv>} {read-view <SNMP View Tree: 1...32>}
[write-view <SNMP View Tree: 1...32>] [notify-view <SNMP
View Tree: 1...32>]
```

For example: `snmp create access-entry group1 sec-model v2c sec-level noAuth read-view viewtree1`

6 Validate that the new entry was created:

```
snmp show access-entry
```

The following figure shows an example of the command output.

Figure 143 Example — snmp show access-entry command output

Group Name	Sec Sec Mod Level	Read View Write View Notify View
private	v1 noAu	V12cView
		V12cView
		V12cView
private	v2c noAu	V12cView
		V12cView
		V12cView
public	v1 noAu	V12cView
		V12cView
		V12cView
public	v2c noAu	V12cView
		V12cView
		V12cView

7 Create an entry in the vacmViewTreeFamilyTable:

```
snmp create viewtree <String: 1...32> {sub-tree <String:
1...64>} {type <include|exclude>}
```

For example: `snmp create viewtree viewtree1 sub-tree mgmt type include`

8 Validate that the new entry was created:

```
snmp show viewtree
```

The following figure shows an example of the command output.

Figure 144 Example — snmp show viewtree command output

```
+-----+
|                                     |
|                               vacmViewTreeFamilyTable |
|-----+
| ViewName  : V12cView              |
| Subtree   : iso                   |
| Type      : include               |
| Mask      : -                     |
|-----+
| ViewName  : V12cView              |
| Subtree   : snmpResearch          |
| Type      : exclude              |
| Mask      : -                     |
|-----+
```

Procedure 130 Creating and attaching an SNMPv3 user to an SNMPv3 access entry group

You can create and attach an SNMPv3 user to an SNMPv3 access entry group.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Create an SNMPv3 user with an authentication protocol, privacy protocol, authentication password, and privacy password:


```
snmp create user <String: 1...64> {auth-protocol
<noAuth|md5|sha>} [auth-password <String: 1...64>] [priv-
protocol <aes-128|des|3des|noPriv>] [priv-password
<String: 1...64>] [auth-secret <String: 1...32>] [priv-
secret <String: 1...32>] [engine-id <SNMP
EngineID:XX:XX... (5-32 octets)>]
```

Note: Users inherit the permissions of the group that they are attached to.

For example,

```
snmp create user SNMPv3_User auth-protocol md5 priv-
protocol des auth-password SNMPv3_password1 priv-password
SNMPv3_password2
```
- 2 Validate that the new entry was created:


```
snmp show users
```
- 3 Create a view tree entry in the vacmViewTreeFamilyTable with a sub-tree that can be viewed:


```
snmp create viewtree <String: 1...32> {sub-tree <String:
1...64>} {type <include | exclude>}
```

For example,

```
snmp create viewtree SNMPv3_view sub-tree mgmt type
include
```
- 4 Validate that the new entry was created.


```
snmp show viewtree
```
- 5 Create an access policy in the vacmAccessTable with a v3 security model.


```
snmp create access-entry <String: 1...32> sec-model v3
{sec-level <noAuth|authNoPriv|authWithPriv>} {read-view
<SNMP View Tree: 1...32>} [write-view <SNMP View Tree:
1...32>] [notify-view <SNMP View Tree: 1...32>]
```

For example,

```
snmp create access-entry SNMPv3_Group sec-model v3 sec-
level authWithPriv read-view V3View write-view V3View
notify-view V3View
```

- 6** Validate that the new entry was created:

```
snmp show access-entry
```

- 7** Attach the SNMPv3 user to an SNMPv3 access group using the `vacmSecurityToGroupTable`:

```
snmp security-to-group attach user <String: 1...64> sec-
model v3 {group <String: 1...32>}
```

For example,

```
snmp security-to-group attach user SNMPv3_User sec-model
v3 group SNMPv3_Group
```

- 8** Validate that the new entry was created:

```
snmp show security-to-group
```

The following figure shows an example output for the `snmp show security-to-group` command.

Figure 145 Example — `snmp show security-to-group` command output

UserName	SecModel	GroupName
public	v2c	public
private	v2c	private
SNMPv3_User	v3	SNMPv3_Group

Procedure 131 Configuring a custom SNMP community string

You can set up a custom SNMP community string with a unique sec-user or you can delete a custom SNMP community string with the unique sec-user.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Note: Community strings inherit the permissions of the user that they are attached to.

Steps

- | | | |
|---|--|--|
| 1 | <p>If you want to</p> <hr/> <p>setup a custom SNMP community string with a unique sec-user</p> <p>delete a custom SNMP community string with the unique sec-user</p> | <p>Then go to</p> <hr/> <p>step 2</p> <p>step 7</p> |
| 2 | <p>Enable snmp:</p> <pre>snmp enable</pre> | |
| 3 | <p>Create a user:</p> <pre>snmp create user <String: 1...64> {auth-protocol <noAuth md5 sha>} [authpassword <String: 1...64>] [priv-protocol <aes-128 des 3des noPriv>] [privpassword <String: 1...64>] [auth-secret <String: 1...40>] [priv-secret <String: 1...64>] [engine-id <SNMP EngineID:XX:XX... (5-32 octets)>]</pre> <p>For example,</p> <pre>snmp create user user1 auth-protocol noAuth</pre> | |
| 4 | <p>Create a security group for the user:</p> <pre>snmp create access-entry <String: 1...32> {sec-model <v1 v2c v3>} {seclevel <noAuth authNoPriv authWithPriv>} {read-view <SNMP View Tree:1...32>} [write-view <SNMP View Tree: 1...32>] [notify-view <SNMP View Tree: 1...32>]</pre> <p>For example,</p> <pre>snmp create access-entry laboratory sec-model v2c sec-level noAuth read-view V12cView</pre> | |

- 5 **Attach the user to the group:**

```
snmp security-to-group attach user <String: 1...64> {sec-model <v1 | v2c | v3>} {group <String: 1...32>}
```

For example,

```
snmp security-to-group attach user user1 sec-model v2c group laboratory
```
- 6 **Make a community string for that user:**

```
snmp create community-index <String: 1...32> {community <String: 1...64>} {sec-name <String: 1...32>}
```

For example,

```
snmp create community-index t0000002 community laserlab sec-name user1
```

Note: The community string accepts all numbers and special characters except “.”.

Go to [step 11](#).
- 7 **Delete the community string for the user you want to delete:**

```
snmp delete community-index <String[1..32]>
```

For example,

```
snmp delete community-index t0000002
```
- 8 **Detach the user from the group:**

```
snmp security-to-group detach user <String[1..64]> sec-model <v1 | v2c | v3>
```

For example,

```
snmp security-to-group detach user user1 sec-model v2c
```
- 9 **Delete the security group for the user:**

```
snmp delete access-entry <String: 1...32> {sec-model <v1 | v2c | v3>} {sec-level <noAuth | authNoPriv | authWithPriv>}
```

For example,

```
snmp delete access-entry laboratory sec-model v2c sec-level noAuth
```
- 10 **Delete the user:**

```
snmp delete user <String: 1...64> [engine-id <SNMP EngineID:XX:XX... (5-32 octets)>]
```

For example,

```
snmp delete user user1
```
- 11 **Save the provisioned settings to the configuration file:**

```
configuration save
```

Procedure 132 Configuring notification entries

You can create, modify, delete, and display notification entries.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

Creating notification entries

- 1 Create notification entries:

```
snmp create notify <String: 1...32> {notify-tag <String: 1...32>} [notify-type <trap>]
```

For example, `snmp create notify myNotifyName notify-tag myNotifyTag notify-type trap`

Modifying notification entries

- 2 Modify notification entries:

```
snmp set notify <String: 1...32> {notify-tag <String: 1...32>} [notify-type <trap>]
```

Deleting notification entries

- 3 Delete notification entries.

```
snmp delete notify <String[32]>
```

For example, `snmp delete notify myNotifyName`

Displaying a list of notification entries

- 4 Display a list of notification entries.

```
snmp show notify
```

The following figure shows an example output for the `snmp show notify` command.

Figure 146 Example — `snmp show notify` command output

Notification Name	Notification Tag	Notification Type
Traps	TrapTag	trap
myNotifyName	myNotifyTag	inform

Procedure 133 Configuring SNMP source address checking

Use this procedure to configure SNMP source address checking to reject incoming SNMPv2c messages unless they meet certain criteria.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Configure source address checking to reject incoming SNMPv2c messages unless they meet the following criteria:

- Community string matches a configured entry in the community map table.
- Transport tag has a corresponding entry in the target table.
- Subnet-mask indicates bit 0 or 1 and matches the corresponding bit in the “addr” value.

Source address checking can be configured to match:

- A specific source address. If the target IP address in the SNMP target table is a valid IP address and the transport-tag in the SNMP community table matches the tag in the SNMP target table, then SNMP packets from only this specific IP address are accepted.
- A specific source address in a subnet. If the target address in the SNMP target table is a valid IP address, the desired subnet mask is set to a valid subnet mask other than the default, and the transport-tag in the SNMP community table matches the tag in the SNMP target table, then only SNMP packets from IP addresses in the specified subnet are accepted.
- Any source address. If the target address in SNMP target table is 0.0.0.0 and subnet mask is 0.0.0.0 for IPv4, and the transport-tag in SNMP community table matches tag in the SNMP target table, then SNMP packets from any IP are accepted.

Steps

- 1 Create an SNMP community mapping:


```
snmp create community-index <String: 1...32> {community
<String: 1...64>} {sec-name <String: 1...32>} [transport-
tag <String: 1...32>]
```
- 2 Create an SNMP target entry:


```
snmp create target <String: 1...32> {addr <IP address
(with optional mask) or host name>} {param-name <String:
1...32>} [tag <String: 1...32>] [udp-port <Number:
```

```
0...65535>] [transport domain <snmp-udp|v4-udp|v6-udp>]  
[retry-count <Number: 0...255>] [retry-timeout <Centi-  
seconds: 0...65535>]
```

Example

In this example, the incoming source address must match 192.147.142.35 (as set in the target table) or the incoming SNMP request will be rejected. The subnet-mask is 255.255.255.255 and all the bits have a value of “1.”

```
snmp create community-index target1 community comm1 sec-name user1 transport-  
tag TrapTag  
snmp create target TRAPS addr 192.147.142.35/32 param-name TRAPS tag TrapTag
```

Example

The following example shows source address checking configured to match any source address. The target address in the SNMP target table is 0.0.0.0, the subnet mask is 0.0.0.0, and the transport-tag in the SNMP community table matches the tag in the SNMP target table.

```
snmp create community-index target1 community comm1 sec-name user1 transport-  
tag TrapTag  
snmp create target TRAPS addr 0.0.0.0/0 param-name TRAPS tag TrapTag
```

Procedure 134 Configuring address filtering

Use this procedure to configure SNMP address filtering to reject incoming SNMP v2c messages unless they meet certain criteria.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

If you use the SNMP v2c server on a remote host, Ciena recommends that you configure address filtering on the SNMP server to allow messages only from trusted subnets.

Steps

- 1 Delete the anywhere targets.
For example:

```
snmp delete target anywhere
```


or

```
snmp delete target anywhereIpv6
```
- 2 Create targets that only allow specific source IP subnets.

```
snmp create target <String: 1...32> {addr <IP address  
(with optional mask) or host name>} {param-name <String:  
1...32>} [tag <String: 1...32>]
```


For example:

```
snmp create target snmpMgmtUsers00 addr 192.168.99.40  
param-name none tag anywhereTag
```
- 3 Test the access to the device from specified networks.
For example:

```
snmpget -v 2c -c public xxx system.sysUpTime.0
```
- 4 Delete public and private community strings.

```
snmp delete community-index <String>
```


For example:

```
snmp delete community-index t0000001
```
- 5 Create a new community string.

```
snmp create community-index <String: 1...32> {community  
<String: 1...64>} {sec-name <String: 1...32>} [transport-  
tag <String: 1...32>]
```


For example:

```
snmp create community-index comm3 community mycommunity33  
sec-name user3
```

- 6** Configure access entry, group security, and the view tree for the community string you created in the previous step.

```
snmp create access-entry <String> sec-model <v1|v2c|v3>  
sec-level <noAuth|authNoPriv|authWithPriv> read-view  
<Name>
```

```
snmp security-to-group attach user <String> sec-model  
<v1|v2c|v3> group <String>
```

```
snmp create viewtree <String> sub-tree <String> type  
<include|exclude>
```

For example:

```
snmp create access-entry group3 sec-model v2 sec-level  
noAuth read-view viewtree3
```

```
snmp security-to-group attach user user3 sec-model v2  
group group3
```

```
snmp create viewtree viewtree3 sub-tree mgmt type include
```

- 7** Test the access with new community string and SNMP IP filter.

```
snmpget -v 2c -c mycommunity33 xxx system.sysUpTime.0
```

CHAPTER 26

Logs

This section contains the following topics:

- [Overview](#)
 - [Syslog global and per-collector configurable attributes](#)
 - [TLS syslog global and per-collector configurable attributes](#)
 - [Syslog/TLS syslog provisioning and considerations](#)
- [Procedure list](#)

Overview

This section describes the message logging (syslog) and configuring message logging on the Waveserver 5.

Syslog is a protocol that allows the Waveserver 5 to automatically send events to a syslog server. Waveserver 5 syslog messages conform to RFC 5424 standard for event log messaging.

A syslog message is sent from the Waveserver 5 to the syslog server:

- when configuration changes are made, for example, when a feature is enabled or disabled.
- when alarms are raised or cleared.
- when there is a removal and insertion of equipment.
- when there are thermal state changes.
- when there is a software download in progress.
- when there are operational state changes.
- when there are hardware checks.
- when cryptographic keys are generated, enrolled, changed, or removed.
- when system software is upgraded (both successful and failed upgrade attempts).
- when passwords are reset.

- for every user login/logout/lockout occurrence and sessions that unexpectedly terminate. These syslog messages include the source IP address. For failed user authentications, the user name is not included in the syslog message.
- for discontinuous changes of time.
- when establishing or removing a TLS connection to an external server.

Syslog messages sent from the Waveserver 5 to the syslog server contain the following information:

- date and time
- type of event
- subject identity
- origin of request
- outcome (success or failure)
- feature specific data
- user account (for password resets)

Waveserver 5 supports syslog where plain text User Datagram Protocol (UDP) connections are used with an external syslog server. Waveserver 5 also supports syslog over Transport Layer Security (TLS) over Transport Control Protocol (TCP), where the syslog payload is wrapped into a TLS stream. TLS delivers syslog messages from source to destination through a secure encrypted tunnel, where both the sender and receiver identify themselves using X.509 certificates. For more details on TLS, see [TLS algorithms selection](#). For more details on X.509 certifications, see [X.509 certificate for TLS](#).

These topics are covered:

- [Syslog global and per-collector configurable attributes](#)
- [TLS syslog global and per-collector configurable attributes](#)
- [Syslog/TLS syslog provisioning and considerations](#)

Syslog global and per-collector configurable attributes

You can configure syslog global and per-collector attributes.

To set the syslog global collector attributes, use this CLI command:

```
syslog set [custom-prefix <String: 1..15>] [facility  
<Number: 0..23>] [udp-port <Number: 1..65535>] [severity  
<emergency | alert | error | warning | notice | info |  
debug | all>]
```

To set the syslog per-collector attributes, use this CLI command:

```
syslog set collector <IP address or host name> [custom-  
prefix <String: 1..15>] [facility <Number: 0..23>] [udp-  
port <Number: 1..65535>]
```

See the following table for a list of syslog global and per-collector attributes.

Table 226 Syslog global and per-collector configurable attributes

Attribute	Description	Value	Default
Custom prefix	This string is included as a prefix in the syslog message.	String: 1..15	Blank
Facility	Defines the facility code that will be sent in each syslog message for this collector.	The RFC 5424 codes are a number from 0 to 23, as follows: • 0 - kernel messages • 1 - user-level messages • 2 - mail system • 3 - system daemons • 4 - security/authorization messages • 5 - messages generated internally by syslog • 6 - line printer subsystem • 7 - network news subsystem • 8 - UUCP subsystem • 9 - clock daemon • 10 - security/authorization messages • 11 - FTP daemon • 12 - NTP subsystem • 13 - log audit • 14 - log alert • 15 - clock daemon • 16 to 23 - local0 to local7	3
Udp-port	The udp port on the collector that will be used for the syslog connection.	Number: 1..65535	514

Attribute	Description	Value	Default
Severity	A list of severities to be sent to the collector.	The RFC 5424 codes are a number from 0 to 7, as follows: • 0 - Emergency: system is unusable • 1 - Alert: action must be taken immediately • 2 - Critical: critical conditions • 3 - Error: error conditions • 4 - Warning: warning conditions • 5 - Notice: normal but significant condition • 6 - Informational: informational messages • 7 - Debug: debug-level messages	Blank

TLS syslog global and per-collector configurable attributes

You can configure TLS syslog global and per-collector attributes.

To set the TLS syslog global attributes, use this CLI command:

```
syslog tls set [cert-name <certificate name>] [check-fingerprint <on | off>] [check-ip-host <on | off>]
[custom-prefix <String: 1..15>] [facility <Number: 0..23>] [port <Number: 1..65535>] [severity <emergency | alert | error | warning | notice | info | debug | all>]
[timeout <Number: 1..30>]
```

To set the TLS syslog per-collector attributes, use this CLI command:

```
syslog tls set collector <IP address or host name>
[custom-prefix <String: 1..15>] [facility <Number: 0..23>] [fingerprint <fingerprint>] [port <Number: 1..65535>] [severity <emergency | alert | error | warning | notice | info | debug | all>] [trusted-dns <trusteddns>]
```

See the following table for a list of TLS syslog global and per-collector attributes.

Table 227 TLS syslog global and per-collector configurable attributes

Attribute	Description	Value	Default
Certificate name	Certificate to use.	String	Blank
Check fingerprint	Determines if the fingerprint of the peer's certificate is checked.	• on • off	off

Attribute	Description	Value	Default
Check IP host	Determines whether the peer's specified IP or hostname is checked against the Subject Alternative Names (SAN) or Common Names (CN) field of its certificate.	• on • off	on
Custom prefix	This string is included as a prefix in the TLS syslog message.	String: 1..15	Blank
Facility	Defines the facility code that will be sent in each TLS syslog message for this collector.	The RFC 5424 codes are a number from 0 to 23, as follows: • 0 - kernel messages • 1 - user-level messages • 2 - mail system • 3 - system daemons • 4 - security/authorization messages • 5 - messages generated internally by syslog • 6 - line printer subsystem • 7 - network news subsystem • 8 - UUCP subsystem • 9 - clock daemon • 10 - security/authorization messages • 11 - FTP daemon • 12 - NTP subsystem • 13 - log audit • 14 - log alert • 15 - clock daemon • 16 to 23 - local0 to local7	3
Fingerprint	A fingerprint of the peer's certificate 95 characters long. For example, 00:11:22:33:44:55:66:77:88:99:A A:BB:CC:DD:EE:FF:A1:B1:C1:D 1:E1:F1:1A:1B:1C:1D:1E:1F:2A: 2B:2C:2D.	32 byte SHA-256 fingerprint	Blank

Attribute	Description	Value	Default
Port	The port on the collector that will be used for the TLS syslog connection.	Number: 1..65535	6514
Severity	A list of severities to be sent to the TLS syslog collector.	The RFC 5424 codes are a number from 0 to 7, as follows: • 0 - Emergency: system is unusable • 1 - Alert: action must be taken immediately • 2 - Critical: critical conditions • 3 - Error: error conditions • 4 - Warning: warning conditions • 5 - Notice: normal but significant condition • 6 - Informational: informational messages • 7 - Debug: debug-level messages	emergency, alert
Timeout	Determines the time allowed for a TLS syslog connection to be made.	Number: 1..30	6
Trusted DNS	If the check-ip-host is on and the trusted DNS is configured, Waveserver 5 checks whether the peer's domain in the SAN or CN is in the Trusted DNS Name in the configuration.	String Note: Must be a Fully Qualified Domain Name (FQDN), no wildcards(*) or quotes("").	Blank

Syslog/TLS syslog provisioning and considerations

The following syslog/TLS syslog provisioning and considerations apply:

- Waveserver 5 syslog/TLS syslog messages conform to RFC5424.
- Up to 16 syslog collectors can be provisioned.
- Up to 4 TLS syslog collectors can be provisioned.

- Device certificates with private keys and signing CA certificates must be configured on both the TLS syslog server and Waveserver 5. Waveserver 5 supports a single device certificate and private key for all TLS syslog operations. See [Creating a private key and installing a device certificate](#) and [Installing a trusted CA certificate](#).

When you install a trusted CA certificate, you are required to assign a name to the certificate using the `cert-name` parameter. When you configure the TLS syslog, you also assign the name of the CA certificate to the TLS syslog certificate name. If you later uninstall the CA certificate, it unsets the TLS syslog certificate name.

- If the “check-ip-host” option is enabled, Waveserver 5 checks the host name set on the TLS Syslog server against the Subject Alternative Name (SAN) on the certificate. Waveserver 5 compares the literal HostName string to the SAN and does not expand it to a Fully Qualified Domain Name (FQDN) if an IP address or a Partially Qualified Domain Name (PQDN) was used to add the server.
- If the server HostName string does not match the SAN, you can provision a Trusted DNS Name using the command `syslog tls set collector trusted-dns`. Waveserver 5 then checks the Trusted DNS Name against the SAN.
- Syslog/TLS syslog clients support IPv4/IPv6 addresses and host names for collectors. Syslog/TLS syslog also supports IPv4/IPv6 message formats.
- Syslog/TLS syslog collector configurable attributes can be set globally (for all collectors) or per collector for custom prefix, facility, port, and severity list. For more details on the configurable attributes, see tables [Syslog global and per-collector configurable attributes](#) and [TLS syslog global and per-collector configurable attributes](#).
- The system will only transmit syslog/TLS syslog messages with the syslog severity values that match the user-configured severity list.
- Any messages sent to a collector for a specific severity will be sent to a secure collector of the same configured severity.
- As per RFC5425, Waveserver 5 supports syslog over TLS in the role of the syslog originator and TLS client.
- TLS Syslog supports Online Certificate Status Protocol (OCSP).
- If a TLS syslog collector fails, any of the enabled authentication methods of the connection will be terminated and the failure reason is logged for that collector in the security log.
- The system transmits TLS syslog messages as SYSLOG_FRAMES with a leading message length.

- For the TLS syslog connection, Waveserver 5 supports TLS mutual authentication with X.509 certificates using PKIX trust models, as per RFC6614. Each device is uniquely identified by the tuple: the certificate issuer and the serial number of presented client certificate.
- The TLS syslog connection uses TLS 1.2.
- Re-authentication of the certificate is performed by the Waveserver 5 once every hour while a TLS connection is open.
- When re-authentication has been run 24 times and OCSP is enabled, Waveserver 5 automatically runs OCSP.
- If OCSP is enabled, TLS syslog uses OCSP to verify an incoming certificate before a connection is made.
- If verification of a certificate fails, TLS syslog releases all connections.
- The operational state of each collector is enabled if the global admin state, global operational state, and collector admin state are enabled; also the IP address must be resolved and a certificate must have been previously installed.
- The user can specify the IP address or host name for a syslog/TLS syslog collector. If the user specifies a host name, the host name is resolved automatically. If the host name is unresolvable, the operational status of the collector will be disabled and the IP address will show UNRESOLVED.
- Waveserver 5 supports syslog TLS management through CLI, NETCONF, REST, SNMP (read only), and WebSocket (event notifications).

Procedure list

This section provides the following procedures:

- [Procedure 135, “Configuring message logging \(syslog\)”](#)
- [Procedure 136, “Configuring TLS syslog”](#)

Procedure 135 Configuring message logging (syslog)

Use this procedure to configure message logging (syslog) on Waveserver 5.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 Set the administrative state of syslog message logging:
`syslog <disable | enable>`
- 2 Create a collector for syslog severity values:
`syslog create collector <IP address or host name> [custom-prefix <String: 1..15>] [facility <Number: 0..23>] [udp-port <Number: 1...65535>] [severity <emergency | alert | error | warning | notice | info | debug | all>]`
- 3 Add an available collector for syslog severity values:
`syslog add collector <IP address or host name> severity <emergency | alert | error | warning | notice | info | debug | all>`

4 Do any of the following:

- Change writable attributes for one or all collectors in the system:

```
syslog set [collector <IP address or host name>] {[custom-
prefix <String: 1...15>] [facility <Number: 0...23>]
[udp-port <Number: 1...65535>] [severity <all or String
list: [...,...] emergency, alert, error, warning, notice,
info, debug>]}
```

- Delete a collector for syslog severity values:

```
syslog delete collector <IP address or host name>
```

- Remove a collector for syslog severity values:

```
syslog remove collector <IP address or host name> severity
<emergency | alert | error | warning | notice | info |
debug | all>
```

- Add a value to or remove a value from the list of severity values that can egress to one or all collectors:

```
syslog add severity <emergency | alert | error | warning
| notice | info | debug | all>
```

- Disable or enable a collector:

```
syslog <disable | enable> collector <IP address or host
name>
```

- Restore modified attributes for one or all collectors in the system to their default configuration values:

```
syslog unset [collector <IP address or host name>]
{[custom-prefix] [facility] [udp-port] [severity]}
```

- Retrieve summary information for the syslog server:

```
syslog show [default-attributes] [severity-mapping]
```

- Retrieve summary information for one or all collectors in the system:

```
syslog show collector <IP address or host name>
```

Note: An asterisk in the Default column indicates a parameter that is using global default settings. Any modifications to the global default settings will also change the syslog collector parameters that set to default values.

5 Save the provisioned settings to the configuration file:

```
configuration save
```

You have completed this procedure.

Procedure 136 Configuring TLS syslog

Use this procedure to configure syslog over TLS.

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.
- You have the server IP or host name and filename for the X.509 certificate files.
- Device certificate with private key and CA certificate has been installed on the TLS syslog server.
- You know the IP address or host name for the TLS syslog server.

Steps

- 1 Create a private key and install a device certificate. See [Creating a private key and installing a device certificate](#).
- 2 Install a trusted CA certificate:

```
certificates authorities install cert-name
<user_assigned_Certificate_name> {default-ftp-server |
default-sftp-server | default-tftp-server | default-
server | default-scp-server | sftp-server <IP address or
host name> login-id <String [1..32]> password <String
[1..128] | tftp-server <IP address or host name> | scp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128] | ftp-server <IP address or host
name> login-id <String [1..32]> password <String
[1..128]>} filename <String [1..127]>
```

For example:

```
certificates authorities install cert-name tlssyslogcert
tftp-server 1.2.3.4 filename ca.pem
```
- 3 Set the TLS syslog certificate name to the certificate you installed in [step 2](#):

```
syslog tls set cert-name <String: [1..32]>
```

For example:

```
syslog tls set cert-name tlssyslogcert
```

Note: If you later uninstall the CA certificate you installed in [step 2](#), it will also unset the TLS syslog certificate name.
- 4 Set the global administrative state of TLS syslog message logging:

```
syslog tls <disable | enable>
```

5 Create a collector for TLS syslog with the desired attributes:

```
syslog tls create collector <IP address or host name>  
[custom-prefix <String: 1...15>] [fingerprint  
<fingerprint>] [facility <Number: 0..24>] [port <Number:  
1..65535>] [severity <emergency | alert | error | warning  
| notice | info | debug | all>] [trusted-dns <trusteddns>]
```

6 Do any of the following:

- **Change writable attributes for all of the TLS syslog collectors in the system:**

```
syslog tls set [check-fingerprint <on | off>] [check-ip-host <on | off>] [custom-prefix <String: 1..15>] [facility <Number: 0..23>] [port <Number: 1..65535>] [severity <emergency | alert | error | warning | notice | info | debug | all>] [timeout <Number: 1..30>]
```

- **Change writable attributes for one of the TLS syslog collectors in the system:**

```
syslog tls set collector <IP address or host name> [custom-prefix <String: 1..15>] [facility <Number: 0..23>] [fingerprint <fingerprint>] [port <Number: 1..65535>] [severity <emergency | alert | error | warning | notice | info | debug | all>] [trusted-dns <trusteddns>]
```

- **Delete a collector for TLS syslog:**

```
syslog tls delete collector <IP address or host name>
```

- **Remove a TLS syslog collector severity:**

```
syslog tls remove collector <IP address or host name> severity <emergency | alert | error | warning | notice | info | debug | all>
```

- **Add a TLS syslog collector severity:**

```
syslog tls add collector <IP address or host name> severity <emergency | alert | error | warning | notice | info | debug | all>
```

- **Disable or enable a TLS syslog collector:**

```
syslog tls <disable | enable> collector <IP address or host name>
```

- **Restore modified attributes for all TLS syslog collectors in the system to their default configuration values:**

```
syslog tls unset [collector <IP address or host name>] [cert-name] [check-fingerprint] [check-ip-host] [custom-prefix] [facility] [port] [severity] [timeout]
```

- **Restore modified attributes for one of the TLS syslog collectors in the system to their default configuration values:**

```
syslog tls unset collector <IP address or host name> [custom-prefix] [facility] [fingerprint] [port] [severity] [trusted-dns]
```

- **Retrieve summary information for one or all collectors in the system:**

```
syslog tls show [collector <IP address or host name>] [statistics]
```

7	If you want to	Then go to
	disable TLS syslog OCSP verification	step 8
	enable TLS syslog OCSP verification and configure parameters	step 9
8	To disable TLS syslog OCSP verification: <code>syslog tls ocsf disable</code> Go to step 14 .	
9	To enable TLS syslog OCSP verification: <code>syslog tls ocsf enable</code>	
10	To set the default OCSP responder:	
	<code>syslog tls ocsf set default-responder <String: 1..255></code>	
11	To set the OCSP responder preference:	
	<code>syslog tls ocsf set responder-preference <aia default responder></code>	
12	To set whether you would like the OCSP responders to contain nonce:	
	<code>syslog tls ocsf set nonce <on off></code>	
13	Optionally, retrieve TLS syslog OCSP settings:	
	<code>syslog tls ocsf show</code>	
14	Save the provisioned settings to the configuration file:	
	<code>configuration save</code>	

CHAPTER 27

Performance Monitoring

This section contains the following topics:

- [Overview](#)
 - [Performance monitoring profiles](#)
 - [Performance monitoring threshold crossing alerts](#)
 - [Untimed read and clear](#)
 - [High Correction Count Second \(HCCS\), Pre-FEC Signal Fail threshold, and Pre-FEC Signal Degrade threshold](#)
- [Procedure list](#)

Overview

This section provides details on Performance Monitoring (PM) on the Waveserver 5.

Waveserver 5 also supports [Configurable HCCS threshold](#).

Note: When the line port transmission mode has Ethernet mapping, all OTN related configuration, operational data, PM, and alarms are not applicable to the line channel object.

Performance monitoring profiles

This section provides information on the performance monitoring statistics available for each performance monitoring profile type.

Waveserver 5 supports these performance monitoring profiles and statistics:

- [Optical power profile statistics](#)
- [Modem profile statistics](#)
- [OTU profile statistics](#)
- [ODU profile statistics](#)
- [Ethernet profile statistics](#)

Table 228 Optical power profile statistics

Statistic	Description
Rx Optical Power Min (per lane)	The minimum Rx optical power reading for the client and line PTP.
Rx Optical Power Max (per lane)	The maximum Rx optical power reading for the client and line PTP.
Rx Optical Power Avg (per lane)	The average Rx optical power for the client and line PTP.
Tx Optical Power Min (per lane)	The minimum Tx optical power for the client and line PTP.
Tx Optical Power Max (per lane)	The maximum Tx optical power for the client and line PTP.
Tx Optical Power Avg (per lane)	The average Tx optical power for the client and line PTP.
Rx Channel Power Min	The minimum channel power for line PTP. Channel power only applies to line PTP when it is set to "colourless" mode.
Rx Channel Power Max	The maximum channel power for line PTP. Channel power only applies to line PTP when it is set to "colourless" mode.
Rx Channel Power Avg	The average channel power for line PTP. Channel power only applies to line PTP when it is set to "colourless" mode.

Table 229 Modem profile statistics

Statistic	Description	Supported PTP facility type
Pre-FEC BER	Bit Error Rate before Forward Error Correction	WL5e, 400ZR, WL5n
Pre-FEC BER Max	Maximum Bit Error Rate before Forward Error Correction	WL5e, 400ZR, WL5n
Q-factor	Q-PM representation of FEC Error Count. (Note)	WL5e
Q-factor Max	Q-PM representation of FEC Error Count (maximum). (Note)	WL5e
Q-factor Min	Q-PM representation of FEC Error Count (minimum). (Note)	WL5e
Q-factor Std Dev	Q-PM representation of FEC Error Count (standard deviation). (Note)	WL5e
FEC Uncorrected Count Second	The number of uncorrected FEC errors in one second.	WL5e, 400ZR, WL5n
FEC Uncorrected Block Count	The number of uncorrected FEC errors	400ZR, WL5n

Statistic	Description	Supported PTP facility type
High Correction Count Second	The number of FEC corrections in one second. The HCCS value is incremented when there is one or more FEC error, or when the corrected FEC count crosses the HCCS threshold.	WL5e, 400ZR, WL5n
DGD Average	Average Differential Group Delay	WL5e, 400ZR, WL5n
DGD Maximum	Maximum Differential Group Delay	WL5e, 400ZR, WL5n
PDL Average	Average Polarization Dependent Loss	WL5e, 400ZR, WL5n
PDL Maximum	Maximum Polarization Dependent Loss	WL5e, 400ZR, WL5n
ESNR Avg	Average Effective Signal To Noise Ratio	WL5e, 400ZR, WL5n
ESNR Max	Maximum Effective Signal To Noise Ratio	WL5e, 400ZR, WL5n
ESNR Min	Minimum Effective Signal To Noise Ratio	WL5e, 400ZR, WL5n
ESNR Delta	Effective Signal To Noise Ratio Delta	WL5e
SNR-ASE Avg	Average Signal To Noise Ratio Amplified Spontaneous Emission	WL5e
SNR-ASE Max	Maximum Signal To Noise Ratio Amplified Spontaneous Emission	WL5e
SNR-ASE Min	Minimum Signal To Noise Ratio Amplified Spontaneous Emission	WL5e
SNR-TNLE Avg	Average Signal To Noise Ratio Total Non Linear Noise Estimate	WL5e
SNR-TNLE Max	Maximum Signal To Noise Ratio Total Non Linear Noise Estimate	WL5e
SNR-TNLE Min	Minimum Signal To Noise Ratio Total Non Linear Noise Estimate	WL5e
SNR External Avg	Average Signal To Noise Ratio External	WL5e
SNR External Max	Maximum Signal To Noise Ratio External	WL5e

Statistic	Description	Supported PTP facility type
SNR External Min	Minimum Signal To Noise Ratio External	WL5e
Cycle Slip Indicator Avg	Average Carrier Slip Indicator	WL5e
Cycle Slip Indicator Max	Maximum Carrier Slip Indicator	WL5e
Chromatic Dispersion Avg	Average dispersion compensation	WL5e, 400ZR, WL5n
Chromatic Dispersion Min	Minimum dispersion compensation	WL5e, 400ZR, WL5n
Chromatic Dispersion Max	Maximum dispersion compensation	WL5e, 400ZR, WL5n
CFO Average	Average Carrier Frequency Offset	400ZR, WL5n
CFO Maximum	Maximum Carrier Frequency Offset	400ZR, WL5n
CFO Minimum	Minimum Carrier Frequency Offset	400ZR, WL5n
EVM Average	Average Error Vector Magnitude	400ZR, WL5n
EVM Maximum	Maximum Error Vector Magnitude	400ZR, WL5n
EVM Minimum	Minimum Error Vector Magnitude	400ZR, WL5n
Note: Q parameter values are calculated using the formula: $Q[\text{dBQ}] = 20 \times \log_{10}(Q[\text{linear}])$, where $Q(\text{linear})$ is calculated as $Q(\text{linear}) = \sqrt{2} \times \text{ERFCINV}(1 - 2 \times \text{PreFECBER})$.		

Table 230 OTU profile statistics

Statistic	Description
OTU BBE	OTU Background Block Errors
OTU ES	OTU Errored Seconds
OTU SES	OTU Severely Errored Seconds
OTU UAS	OTU Unavailable Seconds
OTU BBE-FE	OTU Far-End Background Block Errors
OTU ES-FE	OTU Far-End Errored Seconds
OTU SES-FE	OTU Far-End Severely Errored Seconds
OTU UAS-FE	OTU Far-End Unavailable Seconds

Table 231 ODU profile statistics

Statistic	Description
ODU BBE	ODU Background Block Errors

Statistic	Description
ODU ES	The number of Rx errored seconds at the ODU layer. An errored second is a second in which there was a traffic-impacting defect experienced at the ODU layer.
ODU SES	The number of Rx severely errored seconds at the ODU layer. A severely errored second is a second in which more than the configured number of errors have occurred on the ODU layer.
ODU UAS	The number of Rx unavailable seconds that have occurred on the ODU layer. When any of the ODU defects (AIS, OCI, LCK) are detected for 10 seconds or more, these seconds are counted as ODU UAS.
ODU BBE-FE	ODU far-end Background Block Errors
ODU ES-FE	ODU far-end Errored Seconds
ODU SES-FE	ODU far-end Severely Errored Seconds
ODU UAS-FE	ODU far-end Unavailable Seconds

Table 232 Ethernet profile statistics

Statistic	Description
(Rx/Tx) Bytes	Number of bytes received/transmitted including good and bad packets. This includes FCS bytes, but excludes framing bits.
(Rx/Tx) Packets	Number of packets received/transmitted, including all unicast, multicast, broadcast, MAC control, and bad packets.
(Rx/Tx) CRC Errored Packets	Number of packets received/transmitted that contained an FCS error and were between 64 bytes and up to the configured maximum frame size. Rx CRC Errored Packets are not discarded, they are counted and pass through the system transparently.
(Rx/Tx) Multicast Packets	Number of good multicast packets received/transmitted that were greater than 63 bytes in length, up to the configured maximum frame size.
(Rx/Tx) Broadcast Packets	Number of good broadcast packets received/transmitted that were greater than 63 bytes in length, up to the configured maximum frame size.
(Rx/Tx) 64 Octet Packets	Number of packets (including bad packets) received/ transmitted that were 64 bytes in length (excluding framing bits but including FCS bytes).
(Rx/Tx) 65 to 127 Octet Packets	Number of good and bad packets received/ transmitted that were between 65 and 127 bytes in length (excluding framing bits but including FCS bytes).

Statistic	Description
(Rx/Tx) 128 to 255 Octet Packets	Number of good and bad packets received/ transmitted that were between 128 and 255 bytes in length inclusive (excluding framing bits but including FCS bytes).
(Rx/Tx) 256 to 511 Octet Packets	Number of good and bad packets received/ transmitted that were between 256 and 511 bytes in length inclusive (excluding framing bits but including FCS bytes).
(Rx/Tx) 512 to 1023 Octet Packets	Number of good and bad packets received/transmitted that were between 512 and 1023 bytes in length inclusive (excluding framing bits but including FCS bytes).
(Rx/Tx) 1024 to 1518 Octet Packets	Number of good and bad packets received/transmitted that were between 1024 and 1518 bytes in length inclusive (excluding framing bits but including FCS bytes).
(Rx/Tx) 1519 to Jumbo Octet Packet	Number of good and bad packets received/transmitted that were 1519 bytes and greater in length (excluding framing bits but including FCS bytes).
(Rx/Tx) Jumbo Octet Packet	Number of good and bad packets received/transmitted that were greater than maximum transmission unit size.
(Rx/Tx) Average Link Utilization (%)	The average percentage link utilization (receive/transmit) for the specified port. When a new average link utilization sample is calculated, the number is averaged into the running average.
(Rx/Tx) Min Link Utilization (%)	Minimum percentage link utilization received/transmitted.
(Rx/Tx) Max Link Utilization (%)	Maximum percentage link utilization received/transmitted.
(Rx/Tx) Frame Error Ratio	The ratio of FCS errored frames/total frames received/transmitted.
Rx Sync Header Error Count	The sync header error count received.
(Rx/Tx) Block Error Count	The number of errored blocks that have been received/ transmitted on a given port.
(Rx/Tx) PCS Lane BIP Error Count	The measure of the sum of bit interleaved parity errors over all lanes received/transmitted at the given port.
PCS-ES Errored Seconds - PCS	The number of Rx errored seconds at the PCS layer. An errored second is a second in which there was a traffic-impacting defect experienced at the PCS layer.
PCS-SES Severely Errored Seconds - PCS	The number of Rx severely errored seconds at the PCS layer. A severely errored second is a second in which more than the configured number of errors have occurred on the PCS layer.

Statistic	Description
PCS-UAS Unavailable Seconds - PCS	The number of Rx unavailable seconds that have occurred on the MAC layer. When any of the PCS defects (LOBL, HIBER, LOAM, LF, RF) are detected for 10 seconds or more, these seconds are counted as PCS-UAS.
Rx FEC Corrected Codeword Count	The total number of corrected FEC errors for the modem. (Note)
Rx FEC Uncorrected Codeword Count	The total number of uncorrected FEC errors for the modem. A block can have multiple errors. (Note)
Rx FEC Symbol Error Count	The total number of FEC symbol errors for the modem. (Note)
Note: Not supported for 400ZR and WL5n ports.	

Performance monitoring threshold crossing alerts

Performance monitoring threshold crossing alerts (PM TCAs) notify you when an error count crosses a specified threshold. When the specified threshold is reached, a PM TCA is raised. You can configure TCAs as events and/or alarms.

PM thresholds have the following attributes:

- User configurable.
- Supported on all module types that support PMs.
- Supported for 15-min, current 24-hour, and untimed bins.
- Enabled by default.
- Support Performance Monitoring counters. See [TCA profiles](#).
- Can be enabled or disabled on the:
 - Chassis level
 - Module level
 - PTP level
 - Port level
 - Channel level
 - PM instance level by specifying the interface (port/ptp/channel) and the profileType
- TCA admin states persist and recover after Waveserver 5 restarts.

- Reset to Factory Defaults resets all TCA setting to their default settings.
- PM TCA information is included in system state dump.

Optical power PM TCAs

For the Optical power PM TCA, the Tx and Rx optical power baselines must be captured by a user-initiated action that captures the current optical power value readings. It is recommended that you capture the baseline when all modules and traffic are in a working, stable condition. If a baseline is set for dark fiber or when the channel has no light, any TCAs raised are deemed non-service affecting and may be ignored when attention is required. Capturing a baseline when a module is restarting may also cause false TCAs to be generated. A baseline may be triggered for the chassis, slot or PTP. To capture the baseline, use CLI command `pm tca baseline set {chassis | slot <NUMBER: 1..8> | ptp <slot>/<ptp> [direction <Rx | Tx>]}`. The low and high TCA thresholds are calculated from the baseline as follows:

- Low TCA threshold = baseline - (offset value in dB)
- High TCA threshold = baseline + (offset value in dB)

Note: The offset value in dB is user provisionable; the default value is 3 dB.

Optical power baselines are stored and recovered after restarts. A low TCA alarm is raised when a value is less than or equal to the low TCA threshold. A high TCA alarm is raised when a value is greater than or equal to the high TCA threshold. Optical power PM TCAs are only raised and cleared when the user sets a baseline.

TCA profiles

Waveserver 5 supports up to 40 custom TCA profiles. These custom TCA profiles can be modified or deleted. A TCA profile is attached to a PTP, Port or Channel. The Waveserver 5 also has default TCA profiles which cannot be modified or deleted. By default, when a PM instance is created, it uses the corresponding default TCA profile for that PM profile type. All default and custom TCA profiles are persisted and recover after a restart.

Waveserver 5 initializes with the following default TCA profiles:

- Three default TCA profiles for [PMBasicOtuProfile](#):
Auto-OTU-15-min, Auto-OTU-24-hr, and Auto-OTU-untimed
- Three default TCA profiles for [PMBasicOduProfile](#): Auto-ODU-15-min, Auto-ODU-24-hr, and Auto-ODU-untimed
- Three default TCA profiles for [PMEthernetProfile](#): Auto-Ethernet-15-min, Auto-Ethernet-24-hr, and Auto-Ethernet-untimed

- Three default TCA profiles for [PMOpticalPowerProfile](#): Auto-OpticalPower-15-min, Auto-OpticalPower-24-hr, and Auto-OpticalPoweruntimed
- Three default TCA profiles for [PMModemProfile](#): Auto-Modem-15-min, Auto-Modem-24-hr, and Auto-Modem-untimed
- Three default TCA profiles for [Photonics Profile](#): Auto-Photonics-15-min, Auto-Photonics-24-hr, and Auto-Photonics-untimed

The following tables provide information on counters you can monitor and customize in the TCA profiles. All thresholds have a minimum value of 0.

- [PMBasicOtuProfile](#)
- [PMBasicOduProfile](#)
- [PMEthernetProfile](#)
- [PMOpticalPowerProfile](#)
- [PMModemProfile](#)
- [Photonics Profile](#)

PMBasicOtuProfile

The following table provides information on counters you can monitor and customize for the PMBasicOtuProfile.

Table 233 PMBasicOtuProfile counters - minimum, default and maximum values

Monitor type (Note)	15-min		24-hr		untimed	
	Default	Maximum	Default	Maximum	Default	Maximum
OTU BBE	1772	38700	4430	215913600	0	4294967295
OTU ES	5	899	15	86399	0	86399
OTU SES	2	899	4	86399	0	86399
OTU UAS	10	899	10	86399	0	86399
OTU BBE-FE	1772	38700	4430	215913600	0	4294967295
OTU ES-FE	5	899	15	86399	0	86399
OTU SES-FE	2	899	4	86399	0	86399
OTU UAS-FE	10	899	10	86399	0	86399

Note: OTU thresholds have a minimum value of 0.

PMBasicOduProfile

The following table provides information on counters you can monitor and customize for the PMBasicOduProfile.

Table 234 PMBasicOduProfile counters - minimum, default and maximum values

Monitor type (Note)	15-min		24-hr		untimed	
	Default	Maximum	Default	Maximum	Default	Maximum
ODU BBE	1772	38700	4430	215913600	0	4294967295
ODU ES	5	899	15	86399	0	86399
ODU SES	2	899	4	86399	0	86399
ODU UAS	10	899	10	86399	0	86399
ODU BBE-FE	1772	38700	4430	215913600	0	4294967295
ODU ES-FE	5	899	15	86399	0	86399
ODU SES-FE	2	899	4	86399	0	86399
ODU UAS-FE	10	899	10	86399	0	86399
Note: ODU thresholds have a minimum value of 0.						

PMEthernetProfile

The following table provides information on counters you can monitor and customize for the PMEthernetProfile.

Table 235 PMEthernetProfile counters - minimum, default and maximum values

Monitor type (Note)	15-min		24-hr		untimed	
	Default	Maximum	Default	Maximum	Default	Maximum
Rx CRC Errored Packets	0	13392900	0	1285718400	0	4294967295
Tx CRC Errored Packets	0	13392900	0	1285718400	0	4294967295
Rx Frame Error Ratio	1.00E-07	1.00E-05	1.00E-07	1.00E-05	0	1.00E-05
Rx Average Link Utilization	90	100	90	100	0	100
Tx Frame Error Ratio	1.00E-07	1.00E-05	1.00E-07	1.00E-05	0	1.00E-05
Tx Average Link Utilization	90	100	90	100	0	100
PCS ES	60	899	600	86399	0	86399

Monitor type (Note)	15-min		24-hr		untimed	
	Default	Maximum	Default	Maximum	Default	Maximum
PCS SES	3	899	7	86399	0	86399
PCS UAS	10	899	10	86399	0	86399

Note: Ethernet thresholds have a minimum value of 0.

PMOpticalPowerProfile

The following table provides information on counters you can monitor and customize for the PMOpticalPowerProfile. For the PMOpticalPowerProfile, a baseline must be captured for PM TCA notifications. For additional information, see [Optical power PM TCAs](#).

Table 236 PMOpticalPowerProfile counters - minimum and default values

Monitor type (Note)	15-min		24-hr		untimed	
	Default		Default		Default	
Rx Power Min	Rx baseline - 3 dBm		Rx baseline - 3 dBm		Rx baseline - 3 dBm	
Rx Power Max	Rx baseline + 3 dBm		Rx baseline + 3 dBm		Rx baseline + 3 dBm	
Tx Power Min	Tx baseline - 3 dBm		Tx baseline - 3 dBm		Tx baseline - 3 dBm	
Tx Power Max	Tx baseline + 3 dBm		Tx baseline + 3 dBm		Tx baseline + 3 dBm	
Rx Channel Power Min	Rx channel baseline - 3 dBm		Rx channel baseline - 3 dBm		Rx channel baseline - 3 dBm	
Rx Channel Power Max	Rx channel baseline + 3 dBm		Rx channel baseline + 3 dBm		Rx channel baseline + 3 dBm	

Note: Optical power offset defaults to 3 dBm and is user provisionable with a range of 0 to 40 dBm.

PMModemProfile

The following table provides information on counters you can monitor and customize for the PMModemProfile.

Table 237 PMModemProfile counters - minimum, default and maximum values

Monitor type (Note)	15-min		24-hr		untimed	
	Default	Maximum	Default	Maximum	Default	Maximum
Q-Factor (in dbQ)	0	2.15	0	2.15	0	2.15

Monitor type (Note)	15-min		24-hr		untimed	
	Default	Maximum	Default	Maximum	Default	Maximum
DGD Average	0	250	0	250	0	250
PDL Average	0	250	0	250	0	250
ESNR Avg	0	50	0	50	0	50
SNR External Avg	0	50	0	50	0	50
Chromatic Dispersion Avg	0	2000000	0	2000000	0	2000000
High Correction Count Second	2	899	4	86399	0	86399
Cycle Slip Indicator Avg	0	1000000000	0	1000000000	0	1000000000
ESNR Delta	0	50	0	50	0	50
SNR-ASE Avg	0	50	0	50	0	50
SNR-TNLE Avg	0	50	0	50	0	50
CFO Avg	0	32768	0	32768	0	32768
EVM Avg	0	100	0	100	0	100
Note: Modem thresholds have a minimum value of 0.						

Photonics Profile

The following table provides information on counters you can monitor and customize for the Photonics Profile.

Table 238 Photonics Profile counters - minimum and default

Monitor type (Note)	15-min Default	24-hr Default	untimed Default
Return Loss Min	baseline – 3 dB	baseline – 3 dB	baseline – 3 dB
Return Loss Max	baseline + 3 dB	baseline + 3 dB	baseline + 3 dB
Rx Span Loss Min	baseline – 3 dB	baseline – 3 dB	baseline – 3 dB
Rx Span Loss Max	baseline + 3 dB	baseline + 3 dB	baseline + 3 dB
Tx Span Loss Min	baseline – 3 dB	baseline – 3 dB	baseline – 3 dB
Tx Span Loss Max	baseline + 3 dB	baseline + 3 dB	baseline + 3 dB

Monitor type (Note)	15-min Default	24-hr Default	untimed Default
Note: The baseline offset is configurable. The default offset value is 3 dB. The range of possible offset values is 0 dB to 40 dB.			

TCA notifications

The user has the ability to provision the TCA notifications to be received. Only one notification type can be set at any time. The notification type can be set to any of the following:

- event - Generates an event and a syslog entry.
- alarm - Generates an alarm, event and syslog entry.
- summary event - Generates a summary event and syslog entry.
- summary alarm - Generates a summary alarm, event, and a syslog entry.

Note: The default TCA notification type is summary event.

TCA notification types are persistent and recover after restarts. There are event and alarm definitions for each PM counter. TCA alarms are reported and cleared against specific interfaces (PTP/Port/Channel) and bins (15-min/24-hr/untimed) that raised the TCA. For example PORT-1/3-15m or PORT-1/3-24h. TCA events and summary events are always generated even if the associated interface is administratively disabled, but TCA alarms are suppressed if the associated interface is administratively disabled. Any PM TCA configuration change generates a config event and syslog entry.

TCA alarms are not raised under the following conditions:

- Module/XCVR is missing.
- If a XCVR is disabled, PTP, Port, Channel TCA alarms are not raised.
- If a PTP is disabled, PTP, Port, Channel TCA alarms are not raised.
- If a Port is disabled, Port and Channel TCA alarms are not raised.
- If a Channel is disabled, Channel TCA alarms are not raised.

TCA alarms are cleared when:

- TCA is administratively disabled either globally or per PM instance.
- Current 15-min or 24-hr bin period ends.
- Baseline is cleared or current count is 0, as a result of a pm clear request.
- TCA thresholds are reconfigured, the original TCA clears and a new TCA is raised if a threshold is crossed.
- The TCA notification type is changed.

- A new TCA profile is attached to the PM instance, the original TCA clears and a new TCA is raised if a threshold is crossed.

Note: The default TCA notification type is summary event.

NBI and SNMP support

Any operations that can be performed in CLI are supported for NBI interfaces such as YANG and REST. All creation or deletion of TCA profiles generates a websocket notification and some of the modify operations generate a websocket notification.

The YANG data model and PM MIB includes this TCA information:

- TCA thresholds and active flag for each TCA supported for 15-min, 24-hr, untimed, and history bins
- PM instance TCA administrative state
- Global TCA administrative state
- Global TCA administrative state
- Global TCA notification type
- Global Untimed-Read-And-Clear administrative state, see [“Untimed read and clear”](#)
- TCA profile used by 15-min, 24-hr, and untimed bins
- Baselines for optical power and photonics PM profiles

Only SNMP get operations are supported. An SNMP trap is generated, when there is a TCA alarm state change. For example, when a TCA alarm is raised or cleared.

For information on configuring PM TCAs, see [Configuring Performance Monitoring Threshold Crossing Alerts \(PM TCA\)](#).

Untimed read and clear

This feature enables the user to clear the untimed bin after a get (read) request.

A global Untimed-Read-And-Clear admin state is introduced in the PM. The default state of Untimed-Read-And-Clear is “disable” and the values are fetched and continue to increment as appropriate.

To modify the state, run the following command accordingly:

```
pm untimed-read-and-clear enable
```

or

```
pm untimed-read-and-clear disable
```

Note: System does not support an individual stat query using CLI.

If “Untimed-Read-And-Clear” is enabled, then:

- 1 PM fetch request that does not include untimed bin (at pm root level or PM instance level) works as expected.
- 2 PM fetch request that includes an untimed bin, PM stat(s) are first fetched from data manager and then:

if the fetch request is at PM instance level (applicable to both REST/Netconf and CLI):

- PM manager clears all stats for that untimed bin
- PM manager updates untimed bin clear time to current time
- PM sends out a pm instance cleared event

or

if the fetch request is for individual stat(s) in the specified PM instance untimed bin (applicable to REST/Netconf):

- PM manager clears those individual stat(s)
- PM sends out an individual pm stat cleared event

High Correction Count Second (HCCS), Pre-FEC Signal Fail threshold, and Pre-FEC Signal Degrade threshold

This section provides information on High Correction Count Second (HCCS), Pre-FEC signal Fail and Pre-FEC Signal Degrade threshold for line ports. These topics are covered:

- [Configurable HCCS threshold](#)
- [HCCS and Pre-FEC SF/SD thresholds for line port PTP facilities](#)
- [Pre-FEC Signal Fail threshold](#)
- [Pre-FEC Signal Degrade threshold](#)

Configurable HCCS threshold

Waveserver 5 supports configuration of High Correction Count Second (HCCS) for line ports. HCCS is the count of seconds where the number of FEC corrections in one second exceeds a user-configurable reference value or a FEC overrun occurs. The default value for the user-configurable reference value or HCCS threshold is 0.00 dB and it can be customized to the value range of -1.00 to 4.00 dB (2 decimal digits). For information on configuring HCCS threshold, see [Procedure 142, “Configuring HCCS threshold”](#).

HCCS and Pre-FEC SF/SD thresholds for line port PTP facilities

Effective Signal-to-Noise Ratio (ESNR) is a convenient metric in dB that is used to estimate the present value of link noise (accounts for all noise seen by the modem).

For the line port PTP facilities, ESNR-ref is used for Pre-FEC SF/SD. ESNR-ref is the ESNR in dB, where the system can see significant post FEC errors and become unstable. ESNR-ref is dependent on the transmission mode.

Pre-FEC Signal Fail threshold

The Pre-FEC Signal Fail threshold is the difference in dB from ESNR-ref where the Pre-FEC Signal Fail alarm is raised.

The Pre-FEC Signal Fail alarm is raised when ESNR-actual is less than or equal to the sum of ESNR-ref and the Pre-FEC Signal Fail threshold, for 4 seconds or TFEF overflow for 4 ms. Therefore, if the Pre-FEC Signal Fail threshold is set to 0 dB (default), the Pre-FEC Signal Fail alarm is raised when ESNR-actual is less than or equal to the ESNR-ref point.

Pre-FEC Signal Degrade threshold

The Pre-FEC Signal Degrade threshold is the difference in dB from ESNR-ref where the Pre-FEC Signal Degrade alarm is raised.

The Pre-FEC Signal Degrade alarm is raised when ESNR-actual is less than or equal to the sum of ESNR-ref and the Pre-FEC SD Threshold, for 4 seconds.

Therefore, if the Pre-FEC SD threshold is set to 0.5 dB (default), the Pre-FEC Signal Degrade alarm is raised when ESNR-actual is less than or equal to the sum of ESNR-ref point and 0.5 dB.

The following table shows the HCCS threshold attribute relative to transmission mode and Pre-FEC BER values, and SD detection and clearing time values.

Table 239 HCCS values relative to transmission mode and Pre-FEC SD/SF set and clear thresholds for line port PTP

Transmission Mode (Note 2, Note 3)	Link Optimization Mode	ESNR ref (dB) (Note 1, Note 7)	Parameter	HCCS ref/SF/SD default threshold (dB) (Note 4)	HCCS/Pre-FEC SD/SF set (dB)	Pre-FEC SD/SF clear (dB) (Note 6)	
					ESNR (ESNR-ref + threshold) (Note 5)	Rx (not OOR) (set +0.25)	Rx (OOR) (set +0.75 dB)
82-400-E	SNR-Tolerant	8.35	HCCS/SF	0.0	8.35	8.6	9.10
			SD	0.5	8.85	9.10	9.60
85-300-E	Standard	5.77	HCCS/SF	0.0	5.77	6.02	6.52
			SD	0.5	6.27	6.52	7.02
85-400-E	Standard, SNR-Tolerant	8.05	HCCS/SF	0.0	8.05	8.3	8.80
			SD	0.5	8.55	8.80	9.30
85-600-E	Standard	12.24	HCCS/SF	0.0	12.24	12.49	12.99
			SD	0.5	12.74	12.99	13.49
90-400-E	SNR-Tolerant	7.54	HCCS/SF	0.0	7.54	7.79	8.29
			SD	0.5	8.04	8.29	8.79
90-600-E	Standard SNR-Tolerant	11.60	HCCS/SF	0.0	11.60	11.85	12.35
			SD	0.5	12.10	12.35	12.85
93.3-400-E	SNR-Tolerant	7.22	HCCS/SF	0.0	7.22	7.47	7.97
			SD	0.5	7.72	7.97	8.44
93.3-600-E	SNR-Tolerant	11.22	HCCS/SF	0.0	11.22	11.47	11.97
			SD	0.5	11.72	11.97	12.47

Transmission Mode (Note 2, Note 3)	Link Optimization Mode	ESNR ref (dB) (Note 1, Note 7)	Parameter	HCCS ref/SF/SD default threshold (dB) (Note 4)	HCCS/Pre-FEC SD/SF set (dB)	Pre-FEC SD/SF clear (dB) (Note 6)	
					ESNR (ESNR-ref + threshold) (Note 5)	Rx (not OOR) (set +0.25)	Rx (OOR) (set +0.75 dB)
95-200-O	Standard, Non-linear	2.70	HCCS/SF	0.0	2.70	2.95	3.45
			SD	0.5	3.20	3.45	3.95
95-250-O	Non-linear	4.20	HCCS/SF	0.0	4.20	4.45	4.95
			SD	0.5	4.70	4.95	5.45
	Standard	4.00	HCCS/SF	0.0	4.00	4.25	4.75
			SD	0.5	4.50	4.75	5.25
95-300-O	Non-linear	5.45	HCCS/SF	0.0	5.45	5.70	6.20
			SD	0.5	5.95	6.20	6.70
	Standard, SNR-Tolerant	5.25	HCCS/SF	0.0	5.25	5.50	6.00
			SD	0.5	5.75	6.00	6.50
95-350-O	Standard, SNR-Tolerant	6.34	HCCS/SF	0.0	6.34	6.59	7.09
			SD	0.5	6.84	7.09	7.59
	Non-linear	6.51	HCCS/SF	0.0	6.51	6.76	7.26
			SD	0.5	7.01	7.26	7.76
95-400-O	Standard, SNR-Tolerant, Fast-Tracking	7.41	HCCS/SF	0.0	7.41	7.66	8.16
			SD	0.5	7.91	8.16	8.66
	Non-linear	7.59	HCCS/SF	0.0	7.59	7.84	8.34
			SD	0.5	8.09	8.34	8.84
95-450-O	Standard, SNR-Tolerant	8.43	HCCS/SF	0.0	8.43	8.68	9.18
			SD	0.5	8.93	9.18	9.68
	Non-linear	8.58	HCCS/SF	0.0	8.58	8.83	9.33
			SD	0.5	9.08	9.33	9.83

Transmission Mode (Note 2, Note 3)	Link Optimization Mode	ESNR ref (dB) (Note 1, Note 7)	Parameter	HCCS ref/SF/SD default threshold (dB) (Note 4)	HCCS/Pre-FEC SD/SF set (dB)	Pre-FEC SD/SF clear (dB) (Note 6)	
					ESNR (ESNR-ref + threshold) (Note 5)	Rx (not OOR) (set +0.25)	Rx (OOR) (set +0.75 dB)
95-500-O	Standard, SNR-Tolerant	9.38	HCCS/SF	0.0	9.38	9.63	10.13
			SD	0.5	9.88	10.13	10.63
	Non-linear	9.56	HCCS/SF	0.0	9.56	9.81	10.31
			SD	0.5	10.06	10.31	10.81
95-550-O	Standard, SNR-Tolerant	10.56	HCCS/SF	0.0	10.56	10.81	11.31
			SD	0.5	11.06	11.31	11.81
95-600-O	Standard, SNR-Tolerant, Fast-tracking	11.44	HCCS/SF	0.0	11.44	11.69	12.19
			SD	0.5	11.94	12.19	12.69
95-650-O	Standard, SNR-Tolerant	12.33	HCCS/SF	0.0	12.33	12.58	13.08
			SD	0.5	12.83	13.08	13.58
95-650-E	Standard, SNR-Tolerant	11.88	HCCS/SF	0.0	11.88	12.13	12.63
			SD	0.5	12.38	12.63	13.13
95-700-O	Standard	13.24	HCCS/SF	0.0	13.24	13.49	13.99
			SD	0.5	13.74	13.99	14.49
95-800-O	Standard, SNR-Tolerant	15.03	HCCS/SF	0.0	15.03	15.28	15.78
			SD	0.5	15.53	15.78	16.28
95-200-E	Standard	2.70	HCCS/SF	0.0	2.70	2.95	3.45
			SD	0.5	3.20	3.45	3.95
95-250-E	Standard	3.76	HCCS/SF	0.0	3.76	4.01	4.51
			SD	0.5	4.26	4.51	5.01

Transmission Mode (Note 2, Note 3)	Link Optimization Mode	ESNR ref (dB) (Note 1, Note 7)	Parameter	HCCS ref/SF/SD default threshold (dB) (Note 4)	HCCS/Pre-FEC SD/SF set (dB)	Pre-FEC SD/SF clear (dB) (Note 6)	
					ESNR (ESNR-ref + threshold) (Note 5)	Rx (not OOR) (set +0.25)	Rx (OOR) (set +0.75 dB)
95-300-E	Standard	4.97	HCCS/SF	0.0	4.97	5.22	5.72
			SD	0.5	5.47	5.72	6.22
	Non-Linear	5.17	HCCS/SF	0.0	5.17	5.42	5.92
			SD	0.5	5.67	5.92	6.42
95-350-E	Standard, SNR-Tolerant	6.07	HCCS/SF	0.0	6.07	6.32	6.82
			SD	0.5	6.57	6.82	7.32
	Non-Linear	6.24	HCCS/SF	0.0	6.24	6.49	6.99
			SD	0.5	6.74	6.99	7.49
95-400-E	Standard SNR-Tolerant	7.07	HCCS/SF	0.0	7.07	7.32	7.82
			SD	0.5	7.57	7.82	8.32
	Non-linear	7.25	HCCS/SF	0.0	7.25	7.50	8.00
			SD	0.5	7.75	8.00	8.50
95-450-E	Standard, SNR-Tolerant	8.10	HCCS/SF	0.0	8.10	8.35	8.85
			SD	0.5	8.60	8.85	9.35
	Non-linear	8.26	HCCS/SF	0.0	8.26	8.51	9.01
			SD	0.5	8.76	9.01	9.51
95-500-E	Standard, SNR-Tolerant	8.99	HCCS/SF	0.0	8.99	9.24	9.74
			SD	0.5	9.49	9.74	10.24
	Non-linear	9.15	HCCS/SF	0.0	9.15	9.40	9.90
			SD	0.5	9.65	9.90	10.40
95-550-E	Standard, SNR-Tolerant	10.16	HCCS/SF	0.0	10.16	10.41	10.91
			SD	0.5	10.66	10.91	11.41

Transmission Mode (Note 2, Note 3)	Link Optimization Mode	ESNR ref (dB) (Note 1, Note 7)	Parameter	HCCS ref/SF/SD default threshold (dB) (Note 4)	HCCS/Pre-FEC SD/SF set (dB)	Pre-FEC SD/SF clear (dB) (Note 6)	
					ESNR (ESNR-ref + threshold) (Note 5)	Rx (not OOR) (set +0.25)	Rx (OOR) (set +0.75 dB)
95-600-E	Standard, SNR-Tolerant	11.41	HCCS/SF	0.0	11.41	11.66	12.16
			SD	0.5	11.91	12.16	12.66
95-650-E	Standard	11.88	HCCS/SF	0.0	11.88	12.13	12.63
			SD	0.5	12.38	12.63	13.13
95-700-E	Standard	12.75	HCCS/SF	0.0	12.75	13.00	13.50
			SD	0.5	13.25	13.50	14.00
95-800-E	Standard, SNR-Tolerant	14.47	HCCS/SF	0.0	14.47	14.72	15.22
			SD	0.5	14.97	15.22	15.72
91.6-200-E	Standard, Non-linear	2.70	HCCS/SF	0.0	2.70	2.95	3.45
			SD	0.5	3.20	3.45	3.95
91.6-300-E	Standard	5.23	HCCS/SF	0.0	5.23	5.48	5.98
			SD	0.5	5.73	5.98	6.48
	Non-linear	5.43	HCCS/SF	0.0	5.43	5.68	6.18
			SD	0.5	5.93	6.18	6.68
91.6-400-E	Standard, SNR-Tolerant	7.38	HCCS/SF	0.0	7.38	7.63	8.13
			SD	0.5	7.88	8.13	8.63
	Non-linear	7.57	HCCS/SF	0.0	7.57	7.82	8.32
			SD	0.5	8.07	8.32	8.82
91.6-450-E	SNR-Tolerant	8.40	HCCS/SF	0.0	8.40	8.65	9.15
			SD	0.5	8.90	9.15	9.65
91.6-500-E	Standard, SNR-Tolerant	9.35	HCCS/SF	0.0	9.35	9.60	10.10
			SD	0.5	9.85	10.10	10.60

Transmission Mode (Note 2, Note 3)	Link Optimization Mode	ESNR ref (dB) (Note 1, Note 7)	Parameter	HCCS ref/SF/SD default threshold (dB) (Note 4)	HCCS/Pre-FEC SD/SF set (dB)	Pre-FEC SD/SF clear (dB) (Note 6)	
					ESNR (ESNR-ref + threshold) (Note 5)	Rx (not OOR) (set +0.25)	Rx (OOR) (set +0.75 dB)
91.6-550-E	SNR-Tolerant	10.53	HCCS/SF	0.0	10.53	10.78	11.28
			SD	0.5	11.03	11.28	11.78
91.6-600-E	Standard, SNR-Tolerant	11.41	HCCS/SF	0.0	11.41	11.66	12.16
			SD	0.5	11.91	12.16	12.66
91.6-700-E	Standard	13.20	HCCS/SF	0.0	13.20	13.45	13.95
			SD	0.5	13.70	13.95	14.45
91.6-800-E	Standard, SNR-Tolerant	14.99	HCCS/SF	0.0	14.99	15.24	15.74
			SD	0.5	15.49	15.74	16.24
91.6-400-O	Standard	7.73	HCCS/SF	0.0	7.73	7.98	8.48
			SD	0.5	8.23	8.48	8.98
91.6-600-O	Standard	11.83	HCCS/SF	0.0	11.83	12.08	12.58
			SD	0.5	12.33	12.58	13.08

Note 1: The master reference for the ESNR values can be obtained by issuing the `xcvr show xcvr <slot/port>` command on a live system.

Note 2: A value of 0 dB represents an operating point where the system may see significant post FEC errors and become unstable.

Note 3: The default threshold setting is 0.00 dB for HCCS threshold and Pre-FEC SF (0.5 dB for Pre-FEC SD), but is provisionable if another operating point is required.

Note 4: SF threshold provisioning range is -1.0 to +1.0 dB in steps of 0.01.

Note 5: HCCS threshold provisioning range is -1.0 to +4.0 dB in steps of 0.01 and SD threshold provisioning range is -1.0 to 3.5 dB in steps of 0.01.

Note 6: The Pre-FEC signal degrade and Pre-FEC signal fail alarms are raised after a 2.5 second hold-off time in addition to the Pre-FEC signal degrade/fail detection time. The Pre-FEC signal degrade and Pre-FEC signal fail alarms are cleared after a 10 second hold-on time in addition to the Pre-FEC signal degrade/fail clearing time.

Note 7: ESNR is only accurate within 10 dB of the theoretical limit at a given transmission mode.

Procedure list

This section provides the following procedures:

- [Procedure 137, “Configuring performance monitoring”](#)
- [Procedure 138, “Clearing PM data”](#)
- [Procedure 139, “Retrieving port statistics”](#)
- [Procedure 140, “Retrieving port information and status”](#)
- [Procedure 141, “Configuring Performance Monitoring Threshold Crossing Alerts \(PM TCA\)”](#)
- [Procedure 142, “Configuring HCCS threshold”](#)

Procedure 137 Configuring performance monitoring

Use the commands in this procedure to configure performance monitoring on Waveserver 5.

Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

- 1 To globally disable or enable performance monitoring:

```
pm <disable | enable>
```

Note: Performance monitoring is enabled by default
- 2 To disable or enable one or more PM instances:

```
pm <disable | enable> port <slot>/<port> profile  
<Ethernet | OTU | ODU>
```

```
pm <disable | enable> ptp <slot>/<ptp> profile  
<OpticalPower | Modem >
```

```
pm <disable | enable> channel <slot>/<port>.<channel>  
profile <ODU>
```

Note: PM instances are enabled by default.
- 3 To retrieve summary information for performance monitoring:

```
pm show [auto-created] [port] [ptp] [channel]
```
- 4 To save the provisioned settings to the configuration file:

```
configuration save
```

Procedure 138 Clearing PM data

Use the commands in this procedure to clear data and profiles for performance-monitoring instances.

Requirements

You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Overview

Note: When clearing PM data, the current count is cleared but since data is being continuously collected, the current count may be updated again immediately.

Steps

- 1 To clear statistics for all ports

pm clear all

- 2 To clear statistics for one or more ports:

```
pm clear port <slot>/<port> [history] [profile] [bin-type]
```

```
pm clear port <portList> [history] [profile] [bin-type]
```

- 3 To clear statistics for one or more channels:

```
pm clear channel <slot>/<port>.<channel> [history] [profile] [bin-type]
```

```
pm clear channel <channelList> [history] [profile] [bin-type]
```

- 4 To clear statistics for one or more ptps:

```
pm clear ptp <slot>/<ptp> [history] [profile] [bin-type]
```

```
pm clear ptp <ptpList> [history] [profile] [bin-type]
```

Procedure 139 Retrieving port statistics

Use the commands in this procedure to monitor or display port statistics.

Requirements

- You must be logged in to Waveserver 5 using an account with at least limited access privileges.

Steps

- 1** To display the PM state and a summary of all auto-created monitoring instances:
`pm show`
- 2** To display a summary of Tx and Rx packets collected for one or more ports:
`pm show port <slot>/<port>`
`pm show port <portList>`
- 3** To display a summary of Tx and Rx packets collected for one or more channels:
`pm show channel <slot>/<port>.<channel>`
`pm show channel <channelList>`
- 4** To display a summary of Tx and Rx packets for one or more ptps:
`pm show ptp <slot>/<ptp>`
`pm show ptp <ptpList>`

Procedure 140 Retrieving port information and status

Use the commands in this procedure to display port status and diagnostic information.

Requirements

- You must be logged in to Waveserver 5 using an account that has a minimum access level of limited.

Steps

- 1 Use the following CLI command to display the port status:
`ptp show ptp <ptp> status`

Procedure 141 Configuring Performance Monitoring Threshold Crossing Alerts (PM TCA)

Use this procedure to configure Performance Monitoring Threshold Crossing Alerts (PM TCAs).

Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

Steps

1	If you want to	Then go to
	set the global administrative state for PM TCA	step 2
	set the administrative state for a specific slot, port, ptp or channel	step 3
	set the PM TCA notification type	step 4
	create a PM TCA profile and set its attributes	step 5
	copy a previously created PM TCA profile	step 7
	delete a PM TCA profile	step 8
	display PM TCA profile summary information	step 9
	set the PM TCA notification type to the default value	step 10
	set PM TCA baseline	step 11
	clear PM TCA baseline	step 12
2	Set the global administrative state (disabled or enabled) for PM TCA. <code>pm tca <disable enable></code> Go to step 13 .	
3	Set the administrative state (disabled or enabled) for a specific slot, port, ptp or channel. <code>pm tca <disable enable> [slot <NUMBER: 1..8>] [port <slot>/<port> profile <Ethernet ODU OTU>] [ptp <slot>/<ptp> profile <OpticalPower Modem>] [channel <slot>/<port>.<channel> profile <ODU>]</code> Go to step 13 .	
4	Set the PM TCA notification type. <code>pm tca set notification-type <alarm event summary-alarm summary-event></code> Go to step 13 .	

5 Create a PM TCA profile.

```
pm tca create profile-name <Name (no !%,?*)
String[1..32]> profile-type <Ethernet | Modem |
OpticalPower | ODU | OTU> bin-type <15-min | 24-hr |
untimed>
```

6 Set the PM TCA profile attributes based on the profile type created in [step 5](#).

For PM TCA profile-type Modem:

```
pm tca Modem set profile-name <profile name> [ptp <slot>/
<ptp>] [CD <NUMBER: 0..2000000>] [CSI <NUMBER:
0..1000000000>] [DGD <Threshold 0.000 .. NN.mmm>] [ESNR
<Threshold 0.000 .. NN.mmm>] [SNR-Ext <Threshold 0.000 ..
NN.mmm>] [PDL <Threshold 0.000 .. NN.mmm>] [Rx-FecQ
<Threshold 0.000 .. NN.mmm>] [Rx-High-Corr <NUMBER:
0..86399>] [ESNR-Delta <Threshold 0.000 .. NN.mmm>] [SNR-
ASE <Threshold 0.000 .. NN.mmm>] [SNR-TNLE <Threshold
0.000 .. NN.mmm>] [CFO <NUMBER:0.....32767>] [EVM
<Threshold 0.000 .. NN.mmm>]
```

For PM TCA profile-type ODU:

```
pm tca ODU set profile-name <profile name> [channel
<slot>/<port>.<channel>] [port <slot>/<port>] | [SES
<NUMBER: 0..86399>] [BBE <NUMBER: 0..4294967295>] [BBE-FE
<NUMBER: 0..4294967295>] [ES <NUMBER: 0..86399>] [ES-FE
<NUMBER: 0..86399>] [SES-FE <NUMBER: 0..86399>] [UAS
<NUMBER: 0..86399>] [UAS-FE <NUMBER: 0..86399>]
```

For PM TCA profile-type OpticalPower:

```
pm tca OpticalPower set profile-name <profile name> [ptp
<slot>/<ptp>] [Rx-Baseline-Offset <Threshold 0.000 ..
NN.mmm>] [Tx-Baseline-Offset <Threshold 0.000 .. NN.mmm>]
```

For PM TCA profile-type OTU:

```
pm tca OTU set profile-name <profile name> [port <slot>/
<port>] [BBE <NUMBER: 0..4294967295>] [BBE-FE <NUMBER:
0..4294967295>] [ES <NUMBER: 0..86399>] [ES-FE <NUMBER:
0..86399>] [SES <NUMBER: 0..86399>] [SES-FE <NUMBER:
0..86399>] [UAS <NUMBER: 0..86399>] [UAS-FE <NUMBER:
0..86399>]
```

For PM TCA profile-type Ethernet

```
pm tca Ethernet set profile-name <profile name> [port
<slot>/<port>] [Pcs-Es <NUMBER: 0...86399>] [Pcs-Ses
<NUMBER: 0...86399>] [Pcs-Uas <NUMBER: 0...86399>] [Rx-
Crc-Pkts <NUMBER: 0...4294967295>] [Rx-Frame-Err-Ratio
<Frame Error Ratio: X.XXE-YY. Ex: 1.00E-06, 1.01E-07,
0.09E-05>] [Rx-Link-Utilization <Threshold 0.000 ..
NN.mmm >] [Tx-Crc-Pkts <NUMBER: 0...4294967295>] [Tx-
Frame-Err-Ratio <Frame Error Ratio: X.XXE-YY. Ex: 1.00E-
06, 1.01E-07, 0.09E-05>] [Tx-Link-Utilization <Threshold
0.000 .. NN.mmm >]
```

- Go to [step 13](#).
- 7** Copy a previously created PM TCA profile.
- ```
pm tca copy from-profile-name <from profile name> to-
profile-name <to profile name>
```
- Go to [step 13](#).
- 8** Delete a PM TCA profile.
- ```
pm tca delete profile-name <profile-name>
```
- Note:** Only custom PM TCA profiles can be deleted.
- Go to [step 13](#).
- 9** Display PM TCA profile summary information.
- ```
pm tca show [all] [profile-name <profile name>] [profile-
type <Ethernet | Modem | ODU | OpticalPower | OTU>]
```
- Go to [step 13](#).
- 10** Set the PM TCA notification type to the default value.
- ```
pm tca unset notification-type
```
- Go to [step 13](#).
- 11** Set the PM TCA baseline for a chassis, slot, or ptp.
- ```
pm tca baseline set {chassis | slot <NUMBER: 1..8> | ptp
<slot>/<ptp> [direction <Rx | Tx>]}
```
- Note:** Optical power PM TCAs are only raised if the baseline is set.
- Go to [step 13](#).
- 12** Clear the PM TCA baseline for a chassis, slot or ptp.
- ```
pm tca baseline clear {chassis | slot <NUMBER: 1..8> | ptp  
<slot>/<ptp> [direction <Rx | Tx>]}
```
- 13** Save the provisioned settings to the configuration file.
- ```
configuration save
```

## Procedure 142 Configuring HCCS threshold

Use this procedure to configure the HCCS threshold for a line port.

### Requirements

You must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Steps

- |          |                                                                      |                                             |
|----------|----------------------------------------------------------------------|---------------------------------------------|
| <b>1</b> | <b>If you want to set the HCCS threshold to</b><br>a specified value | <b>Then go to</b><br><a href="#">step 2</a> |
|          | the default value                                                    | <a href="#">step 3</a>                      |
- 2** Set the HCCS threshold to a specified value:  

```
ptp set ptp <slot>/<ptp> hccs-threshold <HCCS Threshold in dB (N.MM): -1.00 .. 4.00>
```

Go to [step 4](#).
  - 3** Set the HCCS threshold to the default value of 0.00:  

```
ptp unset ptp <slot>/<ptp> hccs-threshold
```
  - 4** Optionally display the HCCS threshold configured values:  

```
ptp show ptp <slot>/<ptp>
```
  - 5** Save the provisioned settings to the configuration file:  

```
configuration save
```



## CHAPTER 28

# Alarms

### Overview

This section provides details on the alarms that are raised on Waveserver 5 and the associated alarm clearing procedures.

### Alarms

**Table 240** Alarm fields

| Field                 | Description                                                                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm Table ID (ATID) | The ID number of the alarm in the alarm catalog. To list the alarm catalog, use CLI command <code>alarm show defined-alarms</code> .                                                                                                                                                                            |
| Description           | The condition that caused the alarm to be raised.                                                                                                                                                                                                                                                               |
| Severity              | The alarm severity. Possible values: critical, major, minor, warning. All critical alarms are service-affecting. Alarms with any other severity (major, minor, warning) are non-service affecting.                                                                                                              |
| Instance              | The source of the alarm. Possible values: Chassis, Slot, XCVR, Port, PTP, Channel, License, and PM TCA.<br><br>The Instance field indicates the specific channel, port, slot, transceiver, or PTP. For example, a port may be specified as Port-3/2, a channel as Channel-3/2.1, or a transceiver as XCVR-1/10. |

The following table lists the supported alarms.

**Table 241** Alarms

| ATID | Description/Condition | Severity                                                                                         | Instance                                                    |
|------|-----------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1    | Over Temperature      | <ul style="list-style-type: none"> <li>• Critical</li> <li>• Major</li> <li>• Warning</li> </ul> | <ul style="list-style-type: none"> <li>• Chassis</li> </ul> |
| 3    | Disk Full             | <ul style="list-style-type: none"> <li>• Critical</li> <li>• Major</li> <li>• Warning</li> </ul> | <ul style="list-style-type: none"> <li>• Chassis</li> </ul> |

| ATID | Description/Condition             | Severity              | Instance                           |
|------|-----------------------------------|-----------------------|------------------------------------|
| 4    | Upgrade In Progress               | • Warning             | • Chassis                          |
| 5    | Upgrade Download Failed           | • Major               | • Chassis                          |
| 6    | Upgrade Installation Failed       | • Major               | • Chassis                          |
| 7    | Upgrade Activation Failed         | • Major               | • Chassis                          |
| 8    | Upgrade Commit Failed             | • Major               | • Chassis                          |
| 9    | Upgrade Cancel Failed             | • Major               | • Chassis                          |
| 10   | Software Library Missing          | • Warning             | • Chassis                          |
| 12   | Credentials Warning               | • Warning             | • Chassis                          |
| 13   | Adjacency Discovery Unreliable    | • Minor               | • Chassis                          |
| 16   | Certificate Not Yet Valid         | • Major               | • Chassis                          |
| 17   | Certificate About to Expire       | • Warning             | • Chassis                          |
| 18   | Certificate Expired               | • Major               | • Chassis                          |
| 19   | Software Audit Failed             | • Warning             | • Chassis                          |
| 21   | Equipment Missing                 | • Critical<br>• Major | • Module<br>• PSU<br>• Fan<br>• AP |
| 22   | Equipment Failed                  | • Critical            | • Module<br>• PSU<br>• CFU<br>• AP |
| 23   | Equipment Over Temperature        | • Major               | • Slot                             |
| 24   | Equipment Mismatch                | • Critical            | • Slot                             |
| 25   | Equipment Power Disabled          | • Warning             | • Slot                             |
| 26   | Equipment Communication Suspected | • Major               | • Slot                             |
| 27   | Equipment Unknown                 | • Major               | • Slot                             |
| 28   | Power Supply Feed Failed          | • Major               | • PSU                              |
| 29   | Filler Module Missing             | • Minor               | • Slot                             |
| 30   | Auto Upgrade in Progress          | • Warning             | • Slot                             |
| 31   | Auto Upgrade Failed               | • Major               | • Slot                             |

| ATID | Description/Condition           | Severity              | Instance        |
|------|---------------------------------|-----------------------|-----------------|
| 34   | Software Mismatch               | • Critical            | • Slot          |
| 35   | Configuration Database Mismatch | • Critical            | • Slot          |
| 36   | Auto-Config Backup Failed       | • Critical            | • CP            |
| 37   | Auto-Config Recovery Failed     | • Critical            | • CP            |
| 41   | XCVR Missing                    | • Critical            | • XCVR          |
| 42   | XCVR Failed                     | • Critical            | • XCVR          |
| 43   | XCVR Not Supported              | • Minor               | • XCVR          |
| 44   | Provisioning Mismatch           | • Major               | • XCVR<br>• PTP |
| 45   | XCVR Over Temperature           | • Major<br>• Warning  | • XCVR          |
| 46   | XCVR Under Temperature          | • Major<br>• Warning  | • XCVR          |
| 47   | XCVR Communications Suspect     | • Major               | • XCVR          |
| 48   | XCVR Unknown                    | • Minor               | • XCVR          |
| 51   | Rx Loss of Signal               | • Critical            | • PTP           |
| 52   | Rx Loss of Clock                | • Critical<br>• Minor | • PTP<br>• Port |
| 53   | Tx Loss of Clock                | • Critical<br>• Minor | • PTP           |
| 54   | Rx Power Low                    | • Major<br>• Warning  | • PTP           |
| 55   | Rx Power High                   | • Major<br>• Warning  | • PTP           |
| 56   | Tx Power Low                    | • Major<br>• Warning  | • PTP           |
| 57   | Tx Power High                   | • Major<br>• Warning  | • PTP           |
| 58   | Rx Loss of Channel              | • Critical            | • PTP           |
| 59   | Rx Channel Power Low            | • Major<br>• Warning  | • PTP           |



| ATID | Description/Condition                   | Severity              | Instance            |
|------|-----------------------------------------|-----------------------|---------------------|
| 60   | Rx Channel Power High                   | • Major<br>• Warning  | • PTP               |
| 61   | Tx Channel Contention                   | • Critical            | • PTP               |
| 62   | Tx Tuning in Progress                   | • Major               | • PTP               |
| 63   | Tx Provisioning Required                | • Major               | • PTP               |
| 64   | Tx Power Reduction                      | • Minor               | • PTP               |
| 65   | Frequency Provisioning Mismatch         | • Major               | • PTP               |
| 66   | Laser Frequency Out of Range            | • Major               | • PTP               |
| 67   | Rx Frequency Out of Range               | • Major               | • PTP<br>• Port     |
| 68   | Tx Frequency Out of Range               | • Major               | • PTP               |
| 71   | Modem Class Mismatch                    | • Critical            | • PTP               |
| 72   | Reach Violation                         | • Critical            | • PTP               |
| 73   | Remote Defect Indicator                 | • Minor               | • PTP               |
| 74   | Duplicate Adjacency Discovered          | • Minor               | • PTP               |
| 75   | Dispersion Scan In Progress             | • Warning             | • PTP               |
| 76   | Link Data Retrieval In Progress         | • Warning             | • PTP               |
| 80   | Dispersion Search Mode Scan In Progress | • Major               | • PTP               |
| 81   | Loss of Frame                           | • Critical            | • PTP<br>• Port     |
| 82   | Loss of Multiframe                      | • Critical            | • PTP<br>• Port     |
| 83   | Loss of Frame and Loss of Multiframe    | • Critical            | • Port<br>• Channel |
| 84   | OTU AIS                                 | • Critical            | • Port              |
| 85   | OTU BDI                                 | • Minor               | • Port              |
| 86   | OTU Signal Degrade                      | • Major               | • Port              |
| 87   | OTU Trace Identifier Mismatch           | • Critical<br>• Minor | • Port              |
| 88   | Pre-FEC Signal Degrade                  | • Major               | • PTP               |

| ATID | Description/Condition             | Severity   | Instance                                                     |
|------|-----------------------------------|------------|--------------------------------------------------------------|
| 89   | Pre-FEC Signal Fail               | • Critical | • PTP                                                        |
| 90   | ODU LCK                           | • Critical | • Port<br>• Channel                                          |
| 91   | ODU AIS                           | • Critical | • Port<br>• Channel                                          |
| 92   | ODU OCI                           | • Critical | • Port<br>• Channel                                          |
| 93   | ODU BDI                           | • Minor    | • Port<br>• Channel                                          |
| 94   | ODU Signal Degrade                | • Major    | • Port<br>• Channel                                          |
| 95   | ODU Trace Identifier Mismatch     | • Minor    | • Port<br>• Channel                                          |
| 96   | OPU Payload Type Mismatch         | • Critical | • Port                                                       |
| 97   | Loss of OPU Multiframe Identifier | • Critical | • Port                                                       |
| 98   | Loss of Lane Alignment            | • Critical | • Port                                                       |
| 101  | High Bit Error Rate               | • Major    | • Port                                                       |
| 102  | Rx Local Fault                    | • Critical | • Port                                                       |
| 103  | Rx Remote Fault                   | • Critical | • Port                                                       |
| 104  | PCS Loss of Alignment Marker      | • Critical | • Port                                                       |
| 105  | PCS Loss of Block Lock            | • Critical | • Port                                                       |
| 107  | Deskew Failed                     | • Critical | • Port                                                       |
| 108  | Skew Out of Range                 | • Critical | • Port                                                       |
| 109  | Link Down                         | • Major    | • DCN-1 to DCN-3<br>• ILAN-1 to ILAN-2<br>• ILAN-2<br>• IFP0 |
| 110  | Port Rx Loopback                  | • Minor    | • Port                                                       |
| 111  | Port Tx Loopback                  | • Minor    | • Port                                                       |
| 112  | Port Test Mode Active             | • Minor    | • Port                                                       |

| ATID | Description/Condition                                          | Severity             | Instance  |
|------|----------------------------------------------------------------|----------------------|-----------|
| 113  | Loss of Adjacency                                              | • Major              | • Port    |
| 114  | Remote Port Disabled                                           | • Major              | • Port    |
| 115  | Remote Port Down                                               | • Critical           | • Port    |
| 116  | Remote Port Unreachable                                        | • Critical           | • Port    |
| 117  | TCA Summary (Port)                                             | • Warning            | • Port    |
| 121  | Far End Client Signal Fail                                     | • Critical           | • Channel |
| 122  | Bandwidth Not Available                                        | • Minor              | • Channel |
| 126  | TCA Summary (Channel)                                          | • Warning            | • Channel |
| 131  | External License Server Comms Failed                           | • Major<br>• Warning | • Chassis |
| 132  | External License Server Time Out Of Sync                       | • Major              | • Chassis |
| 135  | License Missing                                                | • Major              | • License |
| 136  | License Expiration Warning                                     | • Minor<br>• Warning | • License |
| 137  | Entitlement Enforcement Disabled                               | • Warning            | • Chassis |
| 144  | Rx Dispersion Compensation Out of Range                        | • Major              | • PTP     |
| 145  | Laser off active                                               | • Major              | • PTP     |
| 146  | TCA Summary (PTP)                                              | • Warning            | • PTP     |
| 153  | TCA Q-Factor                                                   | • Warning            | • PM TCA  |
| 154  | TCA DGD                                                        | • Warning            | • PM TCA  |
| 155  | TCA PDL                                                        | • Warning            | • PM TCA  |
| 156  | TCA ESNR                                                       | • Warning            | • PM TCA  |
| 158  | TCA Chromatic Dispersion                                       | • Warning            | • PM TCA  |
| 159  | TCA High Correction Count Second                               | • Warning            | • PM TCA  |
| 160  | TCA RX Power High Baseline                                     | • Warning            | • PM TCA  |
| 161  | TCA RX Power Low Baseline                                      |                      |           |
| 162  | TCA TX Power High Baseline                                     |                      |           |
| 163  | TCA TX Power Low Baseline<br>TCA RX/TX Power High/Low Baseline |                      |           |

| ATID                     | Description/Condition                                                                                                                                   | Severity  | Instance |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| 164<br>165               | TCA RX Channel Power High Baseline<br>TCA RX Channel Power Low Baseline<br><a href="#">TCA RX Channel Power High/Low Baseline</a>                       | • Warning | • PM TCA |
| 166<br>167<br>168<br>169 | TCA ODU BBE<br>TCA ODU ES<br>TCA ODU SES<br>TCA ODU UAS<br><a href="#">TCA ODU BBE / TCA ODU ES / TCA ODU SES / TCA ODU UAS</a>                         | • Warning | • PM TCA |
| 170<br>171<br>172<br>173 | TCA ODU BBE FE<br>TCA ODU ES FE<br>TCA ODU SES FE<br>TCA ODU UAS FE<br><a href="#">TCA ODU BBE FE / TCA ODU ES FE / TCA ODU SES FE / TCA ODU UAS FE</a> | • Warning | • PM TCA |
| 174<br>175<br>176<br>177 | TCA OTU BBE<br>TCA OTU ES<br>TCA OTU SES<br>TCA OTU UAS<br><a href="#">TCA OTU BBE / TCA OTU ES / TCA OTU SES / TCA OTU UAS</a>                         | • Warning | • PM TCA |
| 178<br>179<br>180<br>181 | TCA OTU BBE FE<br>TCA OTU ES FE<br>TCA OTU SES FE<br>TCA OTU UAS FE<br><a href="#">TCA OTU BBE FE / TCA OTU ES FE / TCA OTU SES FE / TCA OTU UAS FE</a> | • Warning | • PM TCA |
| 182<br>183               | TCA RX Frame Error Ratio<br>TCA TX Frame Error Ratio<br><a href="#">TCA RX/TX Frame Error Ratio</a>                                                     | • Warning | • PM TCA |
| 184<br>185<br>186        | TCA PCS ES<br>TCA PCS SES<br>TCA PCS UAS<br><a href="#">TCA PCS ES / TCA PCS SES / TCA PCS UAS</a>                                                      | • Warning | • PM TCA |

| ATID       | Description/Condition                                                                               | Severity  | Instance  |
|------------|-----------------------------------------------------------------------------------------------------|-----------|-----------|
| 187<br>188 | TCA RX Link Utilization<br>TCA TX Link Utilization<br><a href="#">TCA RX/TX Link Utilization</a>    | • Warning | • PM TCA  |
| 189<br>190 | TCA RX CRC Error Packets<br>TCA TX CRC Error Packets<br><a href="#">TCA RX/TX CRC Error Packets</a> | • Warning | • PM TCA  |
| 194        | <a href="#">TCA SNR External</a>                                                                    | • Warning | • PM TCA  |
| 195        | <a href="#">TCA CSI</a>                                                                             | • Warning | • PM TCA  |
| 228        | <a href="#">Fiber Mismatch</a>                                                                      | • Major   | • PTP     |
| 229        | <a href="#">ZTP In Progress</a>                                                                     | • Warning | • Chassis |
| 233        | <a href="#">Power Supply Unit Disabled</a>                                                          | • Major   | • Slot    |
| 234        | <a href="#">Connection Validation Test In Progress</a>                                              | • Warning | • PTP     |

The following table describes possible alarm instances and provides examples.

**Table 242** Alarm instance descriptions

| Alarm Instance | Display                                                                                                                                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                         | Example                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Chassis        | • Chassis                                                                                                                                                                                                          | Identifies the location of the alarm as the Waveserver 5 chassis.                                                                                                                                                                                                                                                                                                                                                                   | • Chassis                                                                                                                      |
| Slot           | <ul style="list-style-type: none"> <li>• Slot-x, where x= 1 to 8</li> <li>• Fan-x, where x= 1/1, 1/2, 2/1, 2/2</li> <li>• PSU-x, where x = 1, 2</li> <li>• AP-x, where x = 1</li> <li>• CM-x, where x=1</li> </ul> | Identifies the physical location of the alarm. If the alarm occurs on one of the service modules, possible slot numbers are 1-8. If the alarm occurs on a fan module, possible instances are 1/1, 1/2, 2/1, or 2/2. If an alarm is raised on a power module, possible slot numbers are 1 or 2. If the alarm occurs on the access panel, the slot is number 1. If the alarm occurs on the control module (CM), the slot is number 1. | <ul style="list-style-type: none"> <li>• Slot-3</li> <li>• Fan-2/2</li> <li>• PSU-2</li> <li>• AP-1</li> <li>• CM-1</li> </ul> |
| XCVR           | • XCVR-<slot>/<xcvr>                                                                                                                                                                                               | Identifies the location of the alarm against an XCVR with the slot and XCVR numbers. XCVR numbers vary depending on the type of module where the pluggable is located.                                                                                                                                                                                                                                                              | • XCVR-1/3                                                                                                                     |

| Alarm Instance | Display                                                                                                                                                                                                                                                       | Description                                                                                                                                                                                                             | Example                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| PTP            | <ul style="list-style-type: none"> <li>• PTP-&lt;slot&gt;/&lt;ptp&gt;</li> <li>• PTP-&lt;slot&gt;/&lt;ptp&gt;.&lt;ptp section&gt;</li> </ul>                                                                                                                  | Identifies the location of an alarm against a PTP with slot and ptp numbers.                                                                                                                                            | <ul style="list-style-type: none"> <li>• PTP-3/5</li> <li>• PTP-3/5.1, PTP-3/5.2, PTP-3/5.3, PTP-3/5.4</li> </ul> |
| Port           | <ul style="list-style-type: none"> <li>• Port-&lt;slot&gt;/&lt;port&gt;</li> <li>• Port-&lt;slot&gt;/&lt;port&gt;.&lt;port section&gt;</li> </ul>                                                                                                             | Identifies the location of an alarm against a port with slot and port numbers.                                                                                                                                          | Port-3/5<br>Port-3/5.1, Port-3/5.2, Port-3/5.3, Port-3/5.4                                                        |
| Channel        | Channels under x00G line port 1:<br><ul style="list-style-type: none"> <li>• Channel-&lt;slot&gt;/&lt;1.1 - 1.4&gt;</li> </ul> Channels under x00G line port 2:<br><ul style="list-style-type: none"> <li>• Channel-&lt;slot&gt;/&lt;2.1 - 2.4&gt;</li> </ul> | Identifies the location of an alarm against a channel with slot, port, and channel numbers. Channels are available only under line ports. For x00G modes, there are four channels under each line port (Ports 1 and 2). | <ul style="list-style-type: none"> <li>• Channel-3/1.1</li> <li>• Channel-3/101.1</li> </ul>                      |

| Alarm Instance | Display                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Example                                                      |
|----------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| License        | <ul style="list-style-type: none"> <li>License-n, where n = 1, 17 - 35</li> </ul> | <p>Identifies the location of an alarm against a license number.</p> <p>Numbers correspond to the licenses as follows:</p> <ul style="list-style-type: none"> <li>License-01 — Waveserver-OSSWR</li> <li>License-17 — Waveserver-OSFRGNCLPLUG</li> <li>License-18 — Waveserver-OSFRGNLNPLUG</li> <li>License-19 — Waveserver-OSAPIS</li> <li>License-20 — Waveserver-OSITS</li> <li>License-21 — WaveLogic5e-TERADD100G</li> <li>License-22 — WaveLogic5e-TER250KM</li> <li>License-23 — WaveLogic5e-TER500KM</li> <li>License-24 — WaveLogic5e-TER1000KM</li> <li>License-25 — WaveLogic5e-TER2000KM</li> <li>License-26 — WaveLogic5e-TER2100KM</li> <li>License-27 — WaveLogic5e-TER3000KM</li> <li>License-28 — WaveLogic5e-TER4000KM</li> <li>License-29 — WaveLogic5e-TER5000KM</li> <li>License-30 — WaveLogic5e-TER8000KM</li> <li>License-31 — WaveLogic5e-SUBADD100G</li> <li>License-32 — WaveLogic5e-SUB3500KM</li> <li>License-33 — WaveLogic5e-SUB9000KM</li> <li>License-34 — WaveLogic5e-SUB14000KM</li> <li>License-35 — WaveLogic5e-BWADD100G</li> </ul> | <ul style="list-style-type: none"> <li>License-01</li> </ul> |

| Alarm Instance | Display                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Description                            | Example                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PM TCA         | <ul style="list-style-type: none"> <li>• PTP-&lt;slot&gt;/&lt;ptp&gt;-&lt;bin-type&gt;</li> <li>• PTP-&lt;slot&gt;/&lt;ptp&gt;.&lt;minor PTP&gt;-&lt;bin-type&gt;</li> <li>• Port-&lt;slot&gt;/&lt;Port&gt;-&lt;bin-type&gt;</li> <li>• Port-&lt;slot&gt;/&lt;port&gt;.&lt;minor Port&gt;-&lt;bin-type&gt;</li> <li>• Channel-&lt;slot&gt;/&lt;1.1 - 1.4&gt;-&lt;bin-type&gt;</li> <li>• Channel-&lt;slot&gt;/&lt;2.1 - 2.4&gt;-&lt;bin-type&gt;</li> <li>• Channel-&lt;slot&gt;/&lt;101.1 - 101.8&gt;-&lt;bin-type&gt;,<br/>where bin-type = 15m, 24h, or Untimed</li> </ul> | Identifies the location of the PM TCA. | <ul style="list-style-type: none"> <li>• PTP-2/2-15m</li> <li>• PTP-2/2.1-24h</li> <li>• Port-3/5-Untimed</li> <li>• Port-3/5.1-15m</li> <li>• Channel-3/1.1-24h</li> <li>• Channel-3/2.1-Untimed</li> <li>• Channel-3/101.1-15m</li> </ul> |



## Procedure 143 Over Temperature

### Requirements

You are logged in to Waveserver 5 using an account with at least limited access privileges.



#### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 243** Alarm details

|                |                                                                |
|----------------|----------------------------------------------------------------|
| Alarm TID      | 1                                                              |
| Probable Cause | Waveserver 5 chassis temperature is higher than specification. |
| Instance       | Chassis                                                        |
| Severity       | Warning                                                        |

### Steps

- 1 Check for and clear any fan-module-related alarms.
- 2 Check the alarm status:  
`alarm show`
- 3 If the alarm does not clear, check the airflow at the front and rear of the chassis and remove any obstructions.
- 4 Check the alarm status:  
`alarm show`
- 5 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 144 Disk Full

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 244** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 3                                                                                                                                                                                                                                                                                                                                                                 |
| Probable Cause | Disk space usage on the Waveserver 5 chassis is nearing or at capacity.                                                                                                                                                                                                                                                                                           |
| Instance       | Chassis                                                                                                                                                                                                                                                                                                                                                           |
| Severity       | <p>When a Disk Full condition is reached, some applications or operations are blocked. For example, the system blocks upgrades, provisioning, and initializations.</p> <ul style="list-style-type: none"> <li>• Major, non-service-affecting: 75% to 95% of disk space used</li> <li>• Critical, non-service-affecting: 95% to 100% of disk space used</li> </ul> |

### Steps

- 1 Retrieve a list of the software versions stored on the processor:

```
software show installed
+-----+-----+-----+-----+
| Version | Build | Tag | Build Time |
+-----+-----+-----+-----+
| 2.0.0 | 561 | GA | Wed Apr 29 01:05:49 2020 |
+-----+-----+-----+-----+
```

**Note:** Available software versions are listed under Installed Software.

- 2 Delete each software version that is not required:

```
software delete version <version>
```

For example:

```
software delete version waveserver-2.2.0.556-GA
```

**Note:** You cannot delete the current and backup software versions.

- 3 Confirm that the deleted software version is no longer stored on the processor:

```
software show installed
```

- 4 Check the alarm status:

```
alarm show
```

- 5 Delete any loads that you do not require on the disk.
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

---

## Procedure 145 Upgrade In Progress

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 245** Alarm details

|                |                                                                                                                                                                                                                                                                            |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 4                                                                                                                                                                                                                                                                          |
| Probable Cause | This alarm is raised when the system software is downloaded to Waveserver 5, and persists until the manual software upgrade process is complete (the software is activated). This alarm is also raised when you cancel the system software upgrade while it is in process. |
| Instance       | Chassis                                                                                                                                                                                                                                                                    |
| Severity       | Warning, non service-affecting                                                                                                                                                                                                                                             |

### Steps

- 1 Wait for the system software upgrade to complete. No action is required. The upgrade can take up to 10 minutes to complete.  
**Note:** If you want to clear the alarm after downloading the system software but prior to activating it, use the CLI command `software cancel`. This command does not delete the downloaded software package.
- 2 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 146 Upgrade Download Failed

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

**Table 246** Alarm details

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| Alarm TID      | 5                                                                        |
| Probable Cause | Failure occurred during download of the software from the remote server. |
| Instance       | Chassis                                                                  |
| Severity       | Major, non-service-affecting                                             |

### Steps

- Check the URL used for the download and attempt the installation again:

```
software install url sftp://<userid>@<ip address>:<path>
password <password>
```

Or, if the software release file was manually downloaded to Waveserver 5, use the syntax:

```
software install url file://<absolute path to release
file>
```

For example:

```
software install url file:///tmp/users/diag/waveserver-
2.2.0.10-GA.tar.gz.
```

**Note 1:** The password may contain special characters:

```
~!^&*() - _ : = + { } [] | \ ; ' < > . ,
```

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

**Note 2:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ( [https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 ( <https://tools.ietf.org/html/rfc3986>).
- Check the alarm status:

```
alarm show
```
- If the alarm does not clear, contact your next level of support or your Ciena support group.

---

## Procedure 147 Upgrade Installation Failed

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 247** Alarm details

|                |                                                       |
|----------------|-------------------------------------------------------|
| Alarm TID      | 6                                                     |
| Probable Cause | Failure occurred during the software-upgrade process. |
| Instance       | Chassis                                               |
| Severity       | Major, non service-affecting                          |

### Steps

- 1 Contact your next level of support or your Ciena support group.

## Procedure 148 Upgrade Activation Failed

### Requirements

You are logged in to Waveserver 5 using an account with at least super user access privileges.

### Overview

**Table 248** Alarm details

|                |                                                            |
|----------------|------------------------------------------------------------|
| Alarm TID      | 7                                                          |
| Probable Cause | Error occurred during software-upgrade activation process. |
| Instance       | Chassis                                                    |
| Severity       | Major, non-service-affecting                               |

### Steps

#### ***Remote host is available and state dump file can be transferred instantly***

- 1 Create a debug log file and transfer it from Waveserver 5 to a remote host:  

```
system state-dump file-name <String> [default-server |
default-sftp-server | default-scp-server | sftp-server
<IP address or host name> login-id <String: 1..32>
password <String: 1..128> secret <String: 1..256> | scp-
server <IP address or host name> login-id <String: 1..32>
password <String: 1..128> secret <String: 1..256>]
```

#### ***Remote host is not available***

- 2 Create a debug log file locally on the Waveserver 5 node.  

```
system state-dump file-name <String>
```

**Note:** The state dump process is an asynchronous activity that keeps running in the background without interrupting any other activity or action on the Waveserver 5 node.

#### ***Remote host is available again***

- 3 Check state dump process status and also the debug log file location:  

```
system show state-dump
```

*The typical location of the debug log file is /var/opt/stateDump/<filename>.txz*
- 4 If the state dump process is complete, transfer the generated debug log file from the Waveserver 5 node to your local PC or server. Use the following CLI command with appropriate protocol (xftp, sftp, or scp):  

```
system xftp putfile ... <path/filename>
```

For example:

```
system xftp putfile sftp-server <ip address> login-id
<login id> password <password> local-filename /var/opt/
stateDump/sample_debug_log.txz
```

See [Transferring a debug log file](#) for details.

- 5 Retrieve status information about the software upgrade and refer to the parameter Last Upgrade Operation:

```
software show upgrade-status
```

**Figure 147** Software upgrade status information

| UPGRADE STATUS INFORMATION |                                        |
|----------------------------|----------------------------------------|
| Parameter                  | Value                                  |
| Committed Release Version  | waveserver-2.1.0                       |
| Active Release Version     | waveserver-2.1.0                       |
| Upgrade To Version         |                                        |
| Upgrade State              | Idle                                   |
| Last Upgrade Operation     | Commit of waveserver-2.1.0 is complete |

- 6 If required, check the software log for additional information about the failed operation:  

```
software show log
```
- 7 Cancel the software upgrade process:  

```
software cancel
```
- 8 Resolve the issue that caused the upgrade activation failure, and reattempt the software upgrade procedure.
- 9 If the alarm does not clear, contact your next level of support or your Ciena support group.

---

## Procedure 149 Upgrade Commit Failed

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 249** Alarm details

|                |                                                        |
|----------------|--------------------------------------------------------|
| Alarm TID      | 8                                                      |
| Probable Cause | Error occurred during software-upgrade commit process. |
| Instance       | Chassis                                                |
| Severity       | Major, non-service-affecting                           |

### Steps

- 1 Attempt to commit the software upgrade again:  
`software commit`
- 2 Check the alarm status:  
`alarm show`
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.



---

## Procedure 150 Upgrade Cancel Failed

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 250** Alarm details

|                |                                                              |
|----------------|--------------------------------------------------------------|
| Alarm TID      | 9                                                            |
| Probable Cause | Error occurred during software-upgrade cancellation process. |
| Instance       | Chassis                                                      |
| Severity       | Major, non-service-affecting                                 |

### Steps

- 1 Attempt to cancel the software upgrade process again:  
`software cancel`
- 2 Check the alarm status:  
`alarm show`
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 151 Software Library Missing

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- Configuration settings are saved and a backup configuration file is available.

### Overview

**Table 251** Alarm details

|                |                                                                    |
|----------------|--------------------------------------------------------------------|
| Alarm TID      | 10                                                                 |
| Probable Cause | The software load library is either corrupted or has been deleted. |
| Instance       | Chassis                                                            |
| Severity       | Warning                                                            |

### Steps

- Download and install the software:  

```
software install url <String> [login-id <username>]
[password <password>]
```

For example:

```
software install url sftp://userid@10.10.10.10:/path/
waveserver-2.2.0.10-GA.tar.gz password my_password
```

The software is downloaded from the default software server and installed.

**Note 1:** The password may contain special characters:  
~ ! ^ & \* ( ) - \_ : = + { } [ ] | \ ; ' < > . ,  
% (use escape character %25 in the URL input)  
@ (use escape character %40 in the URL input)  
# (use escape character %23 in the URL input)  
? (use escape character %3F in the URL input)  
/ (use escape character %2F in the URL input)

**Note 2:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ( [https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 ( <https://tools.ietf.org/html/rfc3986>).
- Check the alarm status:  

```
alarm show
```
- If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 152 Credentials Warning

### Requirements

You are logged in to Waveserver 5 using an account with super user access privileges.

### Overview

**Table 252** Alarm details

|                |                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 12                                                                                                                                                                                                                                                         |
| Probable Cause | The factory-set empty default passwords of the super user su account and/or the Linux root account have not been changed.<br><b>Note:</b> Both the su and root accounts must have the password changed, or set the no password option to clear this alarm. |
| Instances      | Chassis                                                                                                                                                                                                                                                    |
| Severity       | Warning                                                                                                                                                                                                                                                    |

### Steps

- 1 To set the super user (su) account password:  

```
user set user su password <String[1..128]>
```

**Note:** You can also create another super user account and delete user su.
- 2 To set the Linux root account password:  

```
system environment root set password <String[1..128]>
```
- 3 Check the alarm status:  

```
alarm show
```
- 4 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 153 Adjacency Discovery Unreliable

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.

### Overview

**Table 253** Alarm details

|                |                                                      |
|----------------|------------------------------------------------------|
| Alarm TID      | 13                                                   |
| Probable Cause | Communication problems between Waveserver 5 and 6500 |
| Instance       | Chassis                                              |
| Severity       | Minor, non-service-affecting                         |

### Steps

- | 1 | If you                                                                                                                                                                                                                                                           | Then go to             |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|   | have recently changed the Waveserver 5/6500 connectivity                                                                                                                                                                                                         | <a href="#">step 2</a> |
|   | have not recently changed the Waveserver 5/6500 connectivity                                                                                                                                                                                                     | <a href="#">step 6</a> |
| 2 | If you have recently moved a Waveserver 5 connection and deleted the original SPLI provisioning on the 6500 for that Waveserver 5, the corresponding SPLI database entry on the Waveserver 5 will change to the unreliable state after approximately 20 minutes. |                        |
| 3 | Confirm that the Waveserver 5 connection you moved is the only one in the unreliable state:<br><code>spli database show</code>                                                                                                                                   |                        |
| 4 | Delete the unreliable entry in the SPLI database with:<br><code>spli database delete all-unreliable</code>                                                                                                                                                       |                        |
| 5 | Check the alarm status:<br><code>alarm show</code><br>If the alarm does not clear, go to <a href="#">step 6</a> .                                                                                                                                                |                        |
| 6 | Ensure that there are no general data center or LAN connectivity issues, if yes, fix those issue first before proceeding further. If no issues are present, go to <a href="#">step 7</a> .                                                                       |                        |
| 7 | If the 6500 is in reboot cycle, this may cause the “Adjacency Discovery Unreliable” alarm to be raised. Wait until reboot has finished and check the alarm status again. If the alarm does not clear, go to <a href="#">step 8</a> .                             |                        |

- 8 Determine which adjacency is unreliable using:  
`spli database show`
- 9 Confirm that the Ethernet cable on the DCN port of the 6500 and affected Waveserver 5 is in place and properly seated.
- 10 Check the alarm status. If the alarm does not clear, go to [step 11](#).
- 11 Check that the 6500 IPv4 address is visible to Waveserver 5. In the Waveserver 5 command line, type:  
`file ping <6500 IPv4 address>`
- 12 If the IPv4 address is not visible to Waveserver 5, check that the 6500 is not in Private-IP or Redundant NAT configuration.
- 13 If the IPv4 address is visible to Waveserver 5 and the 6500 is in the correct configuration, check that the data center LAN network does not block port 20040.
- 14 Check that the Waveserver 5 IPv4 address is visible to the 6500. In the 6500 command line, type:  
`ping <WS IPv4 address>`
- 15 Check the alarm status. If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 154 Certificate Not Yet Valid

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 254** Alarm details

|                |                                                                                                                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 16                                                                                                                                                                                                                            |
| Probable Cause | This alarm is raised when the date and time set on the Waveserver 5 is ahead of the “valid from” date of an enrolled certificate.<br><b>Note:</b> To display the system time, use CLI command <code>system show time</code> . |
| Instance       | Chassis                                                                                                                                                                                                                       |
| Severity       | Major                                                                                                                                                                                                                         |

### Steps

- 1 Display the date and time on enrolled certificates:  

```
certificates entity show [cert-name <Certificate name>]
certificates authorities show [ca-cert-hash <CA
certificate hash[1..8]>]
```
- 2 Install a valid certificate.  

```
certificates entity install cert-name <String: [1..32]>
{default-ftp-server | default-sftp-server | default-tftp-
server | default-server | default-scp-server | sftp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128] secret <String [1..128]> |
tftp-server <IP address or host name> | scp-server <IP
address or host name> login-id <String [1..32]> password
<String [1..128] secret <String [1..128]> | ftp-server
<IP address or host name> login-id <String [1..32]>
password <String [1..128] secret <String [1..128]> | url
<String [1..128]>} filename <String [1..127]> cert-
passphrase [certificate-only]

certificates authorities install {default-ftp-server |
default-sftp-server | default-tftpserver | default-server
| default-scp-server | sftp-server <IP address or host
name> login-id <String [1..32]> password <String [1..128]
secret <String [1..128] | tftp-server <IP address or host
name> | scp-server <IP address or host name> login-id
<String [1..32]> password <String [1..128] secret <String
[1..128] | ftp-server <IP address or host name> login-id
```

```
<String [1..32]> password <String [1..128] secret <String
[1..128] | url <String [1..128]} filename <String
[1..127]>
```

**3** Check the alarm status:

```
alarm show
```

**4** If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 155 Certificate About to Expire

### Requirements

You are logged in to Waveserver 5 using an account with at least super user access privileges.

### Overview

**Table 255** Alarm details

|                |                                                                                              |
|----------------|----------------------------------------------------------------------------------------------|
| Alarm TID      | 17                                                                                           |
| Probable Cause | This alarm is raised when an enrolled certificate on the chassis will expire within 90 days. |
| Instance       | Chassis                                                                                      |
| Severity       | Warning                                                                                      |

### Steps

- 1 Install a valid certificate.

```
certificates entity install cert-name <String: [1..32]>
{default-ftp-server | default-sftp-server | default-tftp-
server | default-server | default-scp-server | sftp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128]> secret <String [1..128]> |
tftp-server <IP address or host name> | scp-server <IP
address or host name> login-id <String [1..32]> password
<String [1..128]> secret <String [1..128]> | ftp-server
<IP address or host name> login-id <String [1..32]>
password <String [1..128]> secret <String [1..128]> | url
<String [1..128]>} filename <String [1..127]> cert-
passphrase [certificate-only]

certificates authorities install {default-ftp-server |
default-sftp-server | default-tftpserver | default-server
| default-scp-server | sftp-server <IP address or host
name> login-id <String [1..32]> password <String [1..128]
secret <String [1..128]> | tftp-server <IP address or host
name> | scp-server <IP address or host name> login-id
<String [1..32]> password <String [1..128] secret <String
[1..128] | ftp-server <IP address or host name> login-id
<String [1..32]> password <String [1..128] secret <String
[1..128] | url <String [1..128]>} filename <String
[1..127]>
```
- 2 Check the alarm status:

```
alarm show
```
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.



## Procedure 156 Certificate Expired

### Requirements

You are logged in to Waveserver 5 using an account with at least super user access privileges.

### Overview

**Table 256** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 18                                                                                                                                                                                                                                                                                                                                                      |
| Probable Cause | An enrolled certificate is expired.<br><b>Note:</b> The alarm is raised when Waveserver 5 compares the time and date on the certificate with the system time. To display the system time for Waveserver 5, use CLI command <code>system show time</code> . If the enrolled certificate expires, it may take up to one hour for this alarm to be raised. |
| Instance       | Chassis                                                                                                                                                                                                                                                                                                                                                 |
| Severity       | Major                                                                                                                                                                                                                                                                                                                                                   |

### Steps

- 1 Install a valid certificate.  

```
certificates entity install cert-name <String: [1..32]>
{default-ftp-server | default-sftp-server | default-tftp-
server | default-server | default-scp-server | sftp-
server <IP address or host name> login-id <String [1..32]>
password <String [1..128]> secret <String [1..128]> |
tftp-server <IP address or host name> | scp-server <IP
address or host name> login-id <String [1..32]> password
<String [1..128]> secret <String [1..128]> | ftp-server
<IP address or host name> login-id <String [1..32]>
password <String [1..128]> secret <String [1..128]> |
url <String [1..128]>} filename <String [1..127]> cert-
passphrase [certificate-only]

certificates authorities install {default-ftp-server |
default-sftp-server | default-tftpserver | default-server
| default-scp-server | sftp-server <IP address or host
name> login-id <String [1..32]> password <String
[1..128]> secret <String [1..128]> | tftp-server <IP
address or host name> | scp-server <IP address or host
name> login-id <String [1..32]> password <String
[1..128]> secret <String [1..128]> | ftp-server <IP
address or host name> login-id <String [1..32]> password
<String [1..128]> secret <String [1..128]> | url <String
[1..128]>} filename <String [1..127]>
```

- 2 Check the alarm status:  
`alarm show`
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 157 Software Audit Failed

### Requirements

You are logged in to Waveserver 5 using an account with at least super user access privileges.

### Overview

**Table 257** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Probable Cause | <p>This alarm is raised for multiple reasons:</p> <ul style="list-style-type: none"> <li>• The BOOT/ONIE images or disk image file is corrupt, missing or otherwise been tampered (like renamed, replaced, and so on) in the software repository directory.</li> <li>• The audit has determined that the installation of software on the disk does not match the release distribution – for example, someone tampered with a version of a file that was installed from one of the RPMs that make up the release.</li> </ul> |
| Instance       | Chassis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Severity       | Warning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### Steps

- 1**      **If** the probable cause is      **Then** go to

---

Bullet item **1** as mentioned in the [Alarm step 2 details](#) table

Bullet item **2** as mentioned in the [Alarm step 4 details](#) table
- 2**      Download the software upgrade file:

```
software download url <url> login-id <String> password <String>
```

For example,

```
software download url sftp://10.10.10.10/path/waveserver-2.2.0.10-GA.tar.gz login-id userid password my_password
```

**Note 1:** The target software upgrade file should end with `.tar.gz`

**Note 2:** You must specify the absolute path for the remote server.

**Note 3:** The password may contain special characters:

```
~ ! ^ & * () - _ : = + { } [] | \ ; ' < > . ,
```

% (use escape character %25 in the URL input)

@ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

For example, using a password admin?

```
software download url sftp://
[2222:111:dddd:5444:aced::293]/262/waveserver-2.2.0.10-
GA.tar.gz login-id waveserver password admin?
```

**Note 4:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

```
For example, software download url sftp://
[2222:111:dddd:5444:aced::293]/262/waveserver-
2.2.0.10.tar.gz login-id waveserver password ****
```

**3** Cancel the software upgrade:

```
software cancel
```

Waveserver 5 starts rebooting and re-install the correct version of the bad package.

Go to [step 5](#)

**4** No action is required. This alarm clears when a subsequent audit completes successfully, or after the recovery install runs on the next reboot. The recovery install runs automatically.

If the subsequent audit is not able to complete and recovery install is not executed, go to [step 2](#).

**5** If the alarm does not clear or is unexpected, contact your next level of support or your Ciena support group.

## Procedure 158 Equipment Missing

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 258** Alarm details

|                |                                                                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 21                                                                                                                                                                                                                                                    |
| Probable Cause | An equipment module was not installed or installed incorrectly in a provisioned slot, or has failed, making the equipment undetectable. A missing service module has a critical severity and all other field-replaceable units have a major severity. |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release.<br>PSU-<slot> where slot = 1 - 2<br>Fan-<slot> where slot = 1 - 2<br>AP-<slot> where slot = 1                                                                |
| Severity       | <ul style="list-style-type: none"> <li>Critical</li> <li>Major</li> </ul>                                                                                                                                                                             |

### Steps

- A generic equipment missing alarm is raised and the type of missing equipment (PSU, Fan, AP, and Module) is identified in the alarm instance name.
- Verify that the equipment is installed correctly. If necessary, install the missing equipment. See [Removing, replacing, or reseating equipment](#).
- If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 159 Equipment Failed

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- A replacement for the failed unit is available, if needed.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of damage to the chassis, modules and transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and modules or transceivers.

### Overview

**Table 259** Alarm details

|                |                                                                                                                                                                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 22                                                                                                                                                                                                                                                                                                 |
| Probable Cause | <p>This alarm is raised in the following situations:</p> <ul style="list-style-type: none"> <li>a module or pluggable transceiver fails</li> <li>the chassis or control subsystem detects a major failure on another unit</li> <li>a power failure on a fan module or power supply unit</li> </ul> |
| Instance       | <p>Slot-&lt;slot&gt; where slot = 1 - 8</p> <p><b>Note:</b> Only odd slots are used for this release.</p> <p>PSU-&lt;slot&gt; where slot = 1 - 2</p> <p>Fan-&lt;slot&gt; where slot = 1 - 2</p> <p>AP-&lt;slot&gt; where slot = 1</p>                                                              |
| Severity       | <ul style="list-style-type: none"> <li>Critical</li> <li>Major</li> </ul> <p>A failed service module has a critical severity and all other field-replaceable units have a major severity.</p>                                                                                                      |

**Steps**

- 1 Identify the module raising the alarm.  
**Note:** A generic Equipment Failed alarm is raised against different FRU types and the actual type (PSU, Fan, AP, and Module) is identified in the alarm instance name.
- 2 Verify that the module identified in the previous step is properly installed, and reseal the module if necessary. Refer to the procedure for the specific module in [Removing, replacing, or reseating equipment](#).
- 3 If the alarm persists after 10 minutes:
  - If necessary, perform a warm restart on the affected module.  
`module restart slot <Number: 1..8>`
  - If the problem persists, perform a warm restart on the chassis.  
`chassis restart`  
**Note:** This command is not traffic impacting.
  - If necessary, perform a cold restart on the chassis.  
`chassis restart-cold`  
**Note:** This command is traffic impacting.
- 4 If the alarm persists after 10 minutes, replace the module you identified in [step 1](#). Refer to the procedure for the specific module in [Removing, replacing, or reseating equipment](#).
- 5 If the alarm does not clear after 10 minutes, contact your next level of support or your Ciena support group.

## Procedure 160 Equipment Over Temperature

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



#### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

**Note:** It may take some time for a shelf to cool off and the alarm to clear after the cause is corrected.

### Overview

**Table 260** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Probable Cause | <p>This alarm is raised against a slot when the recorded temperature is too high. The high temperature can be due to:</p> <ul style="list-style-type: none"> <li>• The operating environment exceeds the temperature or altitude limits specified for the shelf.</li> <li>• One or more empty slots must be equipped with modules, circuit packs, or filler modules.</li> <li>• One or more chassis fan modules are not functioning correctly, are missing, or not fully inserted.</li> <li>• The chassis air exhaust or air intake is blocked, or is insufficiently ventilated.</li> <li>• Warm exhaust air from neighboring equipment is mixing with the shelf's intake air.</li> </ul> |
| Instance       | <p>Slot-&lt;slot&gt; where slot = 1 - 8</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Severity       | Major                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



## Precautions



### DANGER

#### Risk of eye injury

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.



### CAUTION

#### Risk of damage to circuit packs and modules

Do not attempt to cause this alarm to be raised by removing any cooling fan modules or filler cards or by blocking the shelf air inlet or exhaust ports as it will compromise shelf cooling. A shelf with a compromised cooling system may result in module failures prior to the assertion of an Equipment Over Temperature alarm.

## Steps

- 1 Make sure that the shelf operating environment is within the specified temperature and altitude limits.
- 2
 

|                                                         |                                                                 |
|---------------------------------------------------------|-----------------------------------------------------------------|
| <b>If</b> the operating temperature for the altitude is | <b>Then</b>                                                     |
| within the specified limits                             | go to <a href="#">step 3</a>                                    |
| not within the specified limits                         | correct the office temperature and go to <a href="#">step 7</a> |
- 3 Perform visual checks to ensure that air intake is not obstructed and that air exhausts correctly from each fan module.
  - Make sure that there is clearance between the back of the shelf or unit and any other shelf, wall, or obstruction.
  - Remove any accumulation of dust from the air exhaust of the fan modules, and the grill on the air intake.
- 4 Visually inspect the chassis to confirm that all slots are filled with modules or filler modules.
- 5
 

|                         |                        |
|-------------------------|------------------------|
| <b>If</b> the shelf has | <b>Then</b> go to      |
| all slots filled        | <a href="#">step 7</a> |
| empty slots             | <a href="#">step 6</a> |
- 6 Insert any missing modules or install filler modules into any empty slots. Refer to [Removing, replacing, or reseating equipment](#).
- 7 Monitor the temperatures on the modules. If the temperature does not decrease after ten minutes or if the system still reports a current temperature of 50 °C or greater, continue to [step 8](#).
- 8 Replace the cooling fan module(s) if necessary. Refer to [Removing, replacing, or reseating equipment](#).

- 9** Monitor the temperatures on the modules. If the temperature does not decrease after ten minutes or if the system still reports a current temperature of 50 °C or greater, go to [step 10](#).
- 10** If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 161 Equipment Mismatch

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity. If needed, you have a replacement pluggable transceiver.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 261** Alarm details

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| Alarm TID      | 24                                                                                       |
| Probable Cause | The provisioning for the slot is different from the installed module.                    |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Critical                                                                                 |

### Steps

- 1 Identify the transceiver or module raising the alarm.
- 2 Replace the transceiver or module identified in [step 1](#) with a supported transceiver or module.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 162 Equipment Power Disabled

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 262** Alarm details

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| Alarm TID      | 25                                                                                       |
| Probable Cause | A Waveserver 5 module has been manually or automatically powered down.                   |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Warning                                                                                  |

### Steps

- Restore power to the Waveserver 5 module that raised the alarm by using following options:
  - Check the provisioned power state of the module:  
`chassis slot show`
  - If the provisioned state is shutdown then set the chassis state to automatic:  
`chassis set slot <x> power automatic`
  - Check if the Power Supply Feed Failed alarm is active. If yes, try clearing this alarm. See [Procedure 165, "Power Supply Feed Failed"](#) for details.
- If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 163 Equipment Communication Suspected

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 263** Alarm details

|                |                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 26                                                                                                                                                                                                            |
| Probable Cause | This alarm is raised against the module when the control module is unable to communicate with the service module.<br><br>This alarm may also be raised when a module is performing an automatic warm restart. |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                      |
| Severity       | Major                                                                                                                                                                                                         |

### Steps

- 1 Identify the module raising the alarm.
- 2 Verify that the module identified in [step 1](#) is properly installed. Refer to [Removing, replacing, or reseating equipment](#).
- 3 If the alarm persists after 10 minutes:
  - If necessary, perform a warm restart on the affected module.  
`module restart slot <Number: 1..8>`  
**Note:** Only odd slots are used for this release.
  - If the problem persists, perform a warm restart on the chassis.  
`chassis restart`  
**Note:** This command is not traffic impacting.
- 4 If the alarm persists, delete and then re-provision and re-enable the module:  
`module disable slot <slot>`  
`module delete slot <slot>`  
`module create slot <slot> type <motr-400-2-16>`  
`module enable slot <slot>`
- 5 If the alarm persists, reseal the module. Holding the module by its handle, slide the latch to the right and pull the module straight out until the Status LED extinguishes. Then carefully insert it back into the slot until you hear a click, which indicates that the holding clip has engaged. Wait for the Status LED to light green.

- 6 If the alarm persists after 10 minutes, replace the module you identified in [step 1](#). Refer to [Reseating, removing, or replacing a service module](#).
- 7 If the alarm does not clear after 10 minutes, contact your next level of support or your Ciena support group.

---

## Procedure 164 Equipment Unknown

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 264** Alarm details

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| Alarm TID      | 27                                                                                       |
| Probable Cause | This alarm is raised when the device-id of the module is unknown.                        |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Major                                                                                    |

### Steps

- 1 Identify the module raising the alarm.
- 2 Replace the unknown module with a valid Waveserver 5 module, see for details on replacing modules.
- 3 Contact Ciena for further information about the unknown module.
- 4 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 165 Power Supply Feed Failed

### Requirements

You are logged in to Waveserver 5 using an account with at least limited access privileges.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 265** Alarm details

|                |                                                                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 28                                                                                                                                                                                            |
| Probable Cause | Power feed to a power module has failed.<br><b>Note:</b> This alarm is also raised while <a href="#">removing or replacing a fan module</a> because the corresponding power module shuts off. |
| Instance       | PSU-<slot> where slot = 1 - 2                                                                                                                                                                 |
| Severity       | Major, non-service-affecting                                                                                                                                                                  |

### Steps

- 1 Visually inspect the power supply feed to the module:
  - Reconnect the power cord to either the power module or the power source if it is disconnected.
  - Replace the power cord if it is damaged.

Refer to [Removing, replacing, or reseating equipment](#) for information about connecting or removing a power cable.
- 2 Check the alarm status:  
`alarm show`
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.



---

## Procedure 166 Filler Module Missing

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 266** Alarm details

|                |                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------|
| Alarm TID      | 29                                                                                                    |
| Probable Cause | This alarm is raised when a slot is not provisioned and a filler module is not installed in the slot. |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release.              |
| Severity       | Minor                                                                                                 |

### Steps

- 1 Log on to the desired chassis.
- 2 Install the filler module in the slot, see *Waveserver 5 Installation*, 323-3001-200 for details.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

---

## Procedure 167 Auto Upgrade in Progress

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 267** Alarm details

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| Alarm TID      | 30                                                                                       |
| Probable Cause | This alarm is raised when an automatic module software upgrade is in progress.           |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Warning                                                                                  |

### Steps

- 1 Wait for the automatic software upgrade to complete. The alarm clears when the module upgrade is complete.
- 2 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 168 Auto Upgrade Failed

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 268** Alarm details

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| Alarm TID      | 31                                                                                       |
| Probable Cause | This alarm is raised when a failure occurs during an automatic software upgrade.         |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Major                                                                                    |

### Steps

- 1 Log on to the desired chassis.
- 2 You can reattempt the software upgrade by performing a warm restart of the Waveserver 5 chassis, see [Performing a warm restart on the Waveserver 5 chassis](#).
- 3 Verify that the alarm cleared using the command `alarm show`.
  - If the alarm cleared, you have completed this procedure.
  - If the alarm did not clear, go to [step 4](#).
- 4 Install a new software release using the CLI command:  

```
software install url <URL> login-id <username> password <password>
```

For example:

```
software install url sftp://userid@10.10.10.10:/path/waveserver-2.2.0-10-GA.tar.gz password my_password
```

Or, if the software release file was manually downloaded to Waveserver 5, use the syntax:

```
software install url file://<absolute path to release file>
```

For example:

```
software install url file:///tmp/users/diag/waveserver-2.2.0.10-GA.tar.gz.
```

**Note 1:** The password may contain special characters:  
`~!^&*()-_:=+{ }[]|\'<>.,`  
 % (use escape character %25 in the URL input)  
 @ (use escape character %40 in the URL input)

# (use escape character %23 in the URL input)

? (use escape character %3F in the URL input)

/ (use escape character %2F in the URL input)

**Note 2:** When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT\_URL documentation ([https://curl.haxx.se/libcurl/c/CURLOPT\\_URL.html](https://curl.haxx.se/libcurl/c/CURLOPT_URL.html)) and RFC 3986 (<https://tools.ietf.org/html/rfc3986>).

- 5 Verify that the alarm cleared using the command.

```
alarm show
```

- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 169 Software Mismatch

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have access to the software load that is running on the modules in the chassis.
- You have network connectivity and you have resolved any remote management access issues.



### CAUTION

#### Risk of interruption to service

Before using CLI command `system auto-config-recovery override-config-mismatch`, **reroute** traffic if required. Default configuration will be applied and traffic will be lost.

### Overview

**Table 269** Alarm details

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 34                                                                                                                   |
| Probable Cause | This alarm is raised when there is a software load mismatch on the CP and the modules in the chassis.                |
| Instance       | Slot-<slot> where slot = 1 - 8<br><b>Note:</b> Only odd slots are used for this release.<br>CM-<slot> where slot = 1 |
| Severity       | Critical                                                                                                             |

### Steps

- 1 Verify the currently running software load on the CP.  
`software show`  
The software detail are available in the ACTIVE RELEASE INFORMATION section of the output.
- 2 Verify the software load on the module.  
`software show slot <slot 1...8>`  
The software detail are available in the Module SW Version row of the output.

- |   | If you want to                                                                                                                                                                                                                                                                                                                                                                 | Then go to                                                  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 3 | <p>upgrade the software load on the CP to match the software load running on the modules in the chassis</p> <p>override the configuration mismatch and apply the software load from the CP, and the default configuration to the modules in the chassis.</p> <p>Traffic will be lost.</p>                                                                                      | <p><a href="#">step 4</a></p> <p><a href="#">step 5</a></p> |
| 4 | <p>Upgrade the software load on the CP to match the software load running on the modules in the chassis.</p> <pre>software install url &lt;url&gt; login-id &lt;userid&gt; password &lt;password&gt; [delete-from-load]</pre> <p>For detailed steps to upgrade the software load see <a href="#">Performing the software upgrade</a>.</p> <p>Go to <a href="#">step 6</a>.</p> |                                                             |
| 5 | <p>Override the configuration mismatch and apply the software load, and default configuration to the modules in the chassis.</p> <pre>system auto-config-recovery override-config-mismatch</pre> <p>See <a href="#">Procedure 170, "Configuration Database Mismatch"</a> for detailed steps.</p>                                                                               |                                                             |
| 6 | <p>If the alarm cleared, you have completed this procedure. Otherwise, contact your next level of support or your Ciena support group.</p>                                                                                                                                                                                                                                     |                                                             |

## Procedure 170 Configuration Database Mismatch

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have the latest configuration file on a server that supports SFTP or SCP.
- You have network connectivity and you have resolved any remote management access issues.



### CAUTION

#### Risk of interruption to service

Before using CLI command `system auto-config-recovery override-config-mismatch`, reroute traffic if required. Default configuration will be applied and traffic will be lost.

### Overview

**Table 270** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 35                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Probable Cause | <p>This alarm is raised when the configuration on the replacement CP does not match the configuration data on the chassis.</p> <p><b>Note:</b> If you are replacing a CP that has release 2.0.2 or lower, verify that Config-Mismatch-Detection is ENABLED to ensure there is no traffic loss upon CP replacement and auto-recovery. This parameter is ENABLED by default. If the Config-Mismatch-Detection is DISABLED and a replacement CP running release 2.0.2 or less is inserted, traffic will be affected at the manual upgrade (since the CPs with older releases than 2.0.2 do not support auto-recovery) .</p> |
| Instance       | CM-<slot> where slot = 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Severity       | Critical                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### Steps

|   |                                                                                                                                                                                                                                 |                                                             |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| 1 | If you want to                                                                                                                                                                                                                  | Then go to                                                  |
|   | <p>restore the configuration that is running on the modules in the chassis</p> <p>override the configuration mismatch and apply the software load from the CP, and the default configuration to the modules in the chassis.</p> | <p><a href="#">step 2</a></p> <p><a href="#">step 4</a></p> |

- 2 Install latest configuration file from a server that supports SFTP or SCP.  

```
configuration install filename [default-server | default-sftp-server | default-scp-server | sftp-server <IP address or host name> login-id <String> {password <String> secret <String>} | scp-server <IP address or host name> login-id <String> {password <String> secret <String>}]
```

**Note 1:** Ensure that you specify the full path to the file in the command.

**Note 2:** You must specify [login-id] and [password] for SFTP and SCP servers.

**Note 3:** The CLI command `configuration install`, installs the specified configuration file to the local configuration directory.
- 3 Restore the configuration file.  

```
configuration reset-to-user-config filename <filename> [revert-timeout <Duration: number/time format N[yMwdhms]>]
```

**Or**

```
configuration restore filename <filename string>
```

Go to [step 5](#).

**Note 1:** You can use any one of the above commands to restore the configuration. It depends on which backup is available either config database or a config file.

**Note 2:** Restoring the configuration file can take several minutes.

**Note 3:** For [revert-timeout], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. If the specified time elapses before the installation is successfully completed, the configuration reverts to the currently running configuration file.
- 4 Override the configuration mismatch and apply the software load and default configuration to the modules in the chassis.  

```
system auto-config-recovery override-config-mismatch
```

**Note:** Use this CLI command with caution, as traffic will be lost.
- 5 If the alarm does not clear, contact your next level of support or your Ciena support group.



## Procedure 171 Auto-Config Backup Failed

### Requirements

- You are logged in to the Waveserver 5 using an account with at least admin access privileges.
- You have network connectivity and you have resolved any remote management access issues.

### Overview

**Table 271** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 36                                                                                                                                                                                                                                                                                                                                                                              |
| Probable Cause | <p>This alarm is raised when the Control Processor fails to create a configuration backup to the Access Panel. It happens because the changes to configuration file were made while the configuration file was being backed up.</p> <p>On an encryption enabled system, this alarm is also raised when passphrase is not present while system encryption backup is enabled.</p> |
| Instance       | Control Processor (CP)                                                                                                                                                                                                                                                                                                                                                          |
| Severity       | Critical                                                                                                                                                                                                                                                                                                                                                                        |

### Steps

- 1 Verify that you have the latest configuration running on the system.  
`configuration show`
- 2 Disable encryption backup if it is not required.  
`system encryption backup disable`
- 3 Set a passphrase if encryption backup is required.  
`system encryption set backup-passphrase <password String [14..128]>`
- 4 Perform a manual backup of the configuration.  
`system auto-config-recovery force-backup`
- 5 If the alarm does not clear, wait until the audit interval expires and a new backup is automatically attempted and completed. Alternatively, contact your next level of support or your Ciena support group.

## Procedure 172 Auto-Config Recovery Failed

### Requirements

- You are logged in to the Waveserver 5 using an account with at least admin access privileges.
- You have network connectivity and you have resolved any remote management access issues.

**Note 1:** If you are replacing a CP that has release 2.0.2 or lower, verify that Config-Mismatch-Detection is ENABLED to ensure there is no traffic loss upon CP replacement and auto-recovery. This parameter is ENABLED by default. If the Config-Mismatch-Detection is DISABLED and a replacement CP running release 2.0.2 or less is inserted, traffic will be affected at the manual upgrade (since the CPs with older releases than 2.0.2 do not support auto-recovery).

### Overview

**Table 272** Alarm details

|                |                                                                                        |
|----------------|----------------------------------------------------------------------------------------|
| Alarm TID      | 37                                                                                     |
| Probable Cause | This alarm is raised when CP failed to recover from the backup as available on the AP. |
| Instance       | Control Processor (CP)                                                                 |
| Severity       | Critical                                                                               |

### Steps

- 1 Restart the control processor to attempt recovery of the configuration again.  

```
chassis control-module restart
```
- 2 Verify that the original alarm cleared.
  - If the alarm cleared, you have completed this procedure.
  - Alternatively, you can manually clear the alarm and recover the configuration by upgrading the software to the expected release and restoring a known good configuration file.  
 Once the CP is confirmed to be running the expected node release, enable and then disable the admin state of auto-config recovery to clear the alarm first and then proceed with restoring the configuration.
  - If the alarm did not clear, contact your next level of support or your Ciena support group.

## Procedure 173 XCVR Missing

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 273** Alarm details

|                |                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 41                                                                                                                              |
| Probable Cause | Pluggable transceiver has been removed from a provisioned client port.                                                          |
| Instance       | XCVR- <b>&lt;slot&gt;/&lt;port&gt;</b> , slot = 1 - 8, port = 3 - 18.<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Critical, service-affecting when the administrative state of the transceiver is enabled                                         |

### Steps

- 1 Perform one of the following:
  - Disable the transceiver, then delete the provisioning data for the transceiver:
 

```
xcvr disable xcvr <slot>/<xcvr>
xcvr delete xcvr <slot>/<xcvr>
```
  - Reseat the transceiver. See [Replacing or reseating a pluggable transceiver](#).
- 2 Check the alarm status:
 

```
alarm show
```
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 174 XCVR Failed

### Requirements

- You are logged in to Waveserver 5 using an account with at least limited access privileges.
- A replacement pluggable transceiver (alarm raised against client port) or a service module (alarm raised against line port) is available.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 274** Alarm details

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 42                                                                                                                  |
| Probable Cause | This alarm is raised when a pluggable transceiver has failed on a client port or the modem (line port) has failed.  |
| Instance       | XCVR-<slot>/<port>, slot = 1 - 8, port = 1 - 18<br><b>Note:</b> Only odd slots are used for this release.           |
| Severity       | Critical, service-affecting when the administrative state of the pluggable transceiver or the line port is enabled. |

### Steps

- |          |                                                                       |                                                                       |
|----------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>1</b> | <b>If</b> the alarm is raised against the<br>client port<br>line port | <b>Then</b> go to<br><a href="#">step 2</a><br><a href="#">step 3</a> |
|----------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
- Replace the transceiver. See [Replacing or reseating a pluggable transceiver](#). Go to [step 4](#).
- Replace the service module. See [Reseating, removing, or replacing a service module](#).
- Check the alarm status:  
`alarm show`
- If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 175 XCVR Not Supported

### Requirements

You are logged in to Waveserver 5 using an account with at least limited access privileges.



#### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 275** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 43                                                                                                                                                                                                                                                                                                                                           |
| Probable Cause | <p>This alarm is raised if the pluggable transceiver is inserted and the system can identify the pluggable type and capabilities, but the type or capabilities are not supported by the module.</p> <p>This alarm is also raised when a QSFP-DD module is inserted in the supporting client port but it requires more than 14w of power.</p> |
| Instance       | <p>XCVR-&lt;slot&gt;/&lt;port&gt;, slot = 1 - 8, port = 3 - 18</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                                                                                                             |
| Severity       | Minor, non-service-affecting                                                                                                                                                                                                                                                                                                                 |

### Steps

- 1 Replace the transceiver. See [Replacing or reseating a pluggable transceiver](#).
- 2 Check the alarm status:  
`alarm show`
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 176 Provisioning Mismatch

### Requirements

You are logged in to Waveserver 5 using an account with at least limited access privileges.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 276** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 44                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Probable Cause | <p>XCVR provisioning mismatch alarm is raised on the line port when the provisioned XCVR mode is not supported by the current version of the modem hardware or it is not compatible with the currently provisioned link optimization mode.</p> <p>XCVR provisioning mismatch alarm is raised on the client port when a valid pluggable is inserted into a port but it does not match the existing provisioned protocol type.</p> <p>PTP provisioning mismatch alarm is raised on the line port when the provisioned frequency or link optimization mode is not supported by the current version of the hardware. This alarm is also raised if the provisioned link optimization mode is not compatible with the currently provisioned XCVR mode.</p> |
| Instance       | <p>XCVR-<code>&lt;slot&gt;/&lt;xcvr&gt;</code> , slot = 1 - 8, xcvr = 3 - 18</p> <p>PTP-<code>&lt;slot&gt;/&lt;ptp&gt;</code>, slot = 1 - 8, ptp = 1 - 2</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Severity       | <p>Critical</p> <p>Major, non service-affecting</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### Steps

- 1**

|                                                                                       |                                                                                                                           |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <p><b>If</b> the alarm is raised for</p> <p>an XCVR</p> <p>a module</p> <p>a port</p> | <p><b>Then</b> go to</p> <hr/> <p><a href="#">step 2</a></p> <p><a href="#">step 4</a></p> <p><a href="#">step 11</a></p> |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
- 2**

Check to see if the settings currently provisioned for the XCVR are supported by the hardware:

```
xcvr show
```

- 3 Replace the transceiver with a transceiver that matches the provisioned settings. See [Replacing or reseating a pluggable transceiver](#).  
Go to [step 13](#).
- 4 Check the settings currently provisioned on the module:  

```
ptp show ptp
```
- 5 Ensure that the Link Optimization Mode (LOM) is supported for the configured transmission mode. See [Facilities](#) for information on supported transmission modes and link optimization mode on different service modules.
- 6 If the configured LOM and transmission mode combination is not supported, change the LOM setting:  

```
ptp disable ptp <slot>/<port>
ptp set ptp <slot>/<port> link-optimization-mode
<Standard | Non-linear | SNR-Tolerant>
ptp enable ptp <slot>/<port>
```
- 7 Check the alarm status:  

```
alarm show
```
- 8 If the alarm does not clear, ensure that other settings configured on the module are supported by the hardware. Replace the module with a module type that supports the settings, if required. See [Reseating, removing, or replacing a service module](#).
- 9 Check the alarm status:  

```
alarm show
```
- 10 If the alarm does not clear, contact your next level of support or your Ciena support group.
- 11 Check the client rate of the near end and far end Waveserver 5:  

```
port show port <slot>/<port>
```
- 12 If the client rates do not match, change the client rates to match:  

```
xcvr disable xcvr <slot>/<xcvr>
xcvr set xcvr <slot>/<xcvr> mode <mode>
xcvr enable xcvr <slot>/<xcvr>
```
- 13 Check the alarm status:  

```
alarm show
```
- 14 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 177 XCVR Over Temperature

### Requirements

You are logged in to Waveserver 5 using an account with at least limited access privileges.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 277** Alarm details

|                |                                                                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 45                                                                                                                                                                                                       |
| Probable Cause | The operating temperature of a pluggable transceiver is higher than specification.                                                                                                                       |
| Instance       | XCVR-<slot>/<port>, slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                |
| Severity       | <ul style="list-style-type: none"> <li>Major, service-affecting when temperature has exceeded the threshold</li> <li>Warning, non service-affecting when temperature is nearing the threshold</li> </ul> |

### Steps

- 1 Retrieve diagnostic data for the transceiver, and determine whether the operating temperature of the transceiver is higher than specification:  
`xcvr show xcvr diagnostics`
- 2 If the operating temperature of the transceiver is higher than specification, check for and clear any fan-module-related alarms.
- 3 If a fan module is missing or defective, install or replace it immediately to ensure adequate air flow within the chassis. Refer to [Removing, replacing, or reseating equipment](#).
- 4 If the alarm does not clear, check for and clear any power-supply unit-related alarms.
- 5 If a PSU is missing or defective, install or replace it immediately to ensure adequate air flow within the chassis. Refer to [Removing, replacing, or reseating equipment](#).
- 6 If the alarm does not clear, reseal the transceiver. See [Replacing or reseating a pluggable transceiver](#).
- 7 If the alarm does not clear, replace the transceiver. See [Replacing or reseating a pluggable transceiver](#).
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.



## Procedure 178 XCVR Under Temperature

### Requirements

You are logged in to Waveserver 5 using an account with at least limited access privileges.



#### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 278** Alarm details

|                |                                                                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 46                                                                                                                                                                                                       |
| Probable Cause | Temperature of the transceiver is below specification.                                                                                                                                                   |
| Instance       | XCVR-<slot>/<port>, slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                |
| Severity       | <ul style="list-style-type: none"> <li>Major, service-affecting when temperature has exceeded the threshold</li> <li>Warning, non-service-affecting when temperature is nearing the threshold</li> </ul> |

### Steps

- 1 Retrieve diagnostic data for the transceiver, and determine whether the operating temperature of the transceiver is below specification:  
`xcvr show xcvr diagnostics`
- 2 If the operating temperature of the transceiver is lower than specification, check for and clear any fan-module-related alarms.
- 3 Check and adjust the ambient temperature so that it is within the normal operating range for the Waveserver 5. See [Environmental and electromagnetic specifications](#) for information about environmental specifications for the Waveserver 5.
- 4 Reseat the transceiver if the alarm does not clear. See [Replacing or reseating a pluggable transceiver](#).
- 5 Replace the transceiver if the alarm does not clear. See [Replacing or reseating a pluggable transceiver](#).
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 179 XCVR Communications Suspect

### Requirements

- You are logged in to Waveserver 5 using an account with at least limited access privileges.
- Replacement pluggable transceiver.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 279** Alarm details

|                |                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 47                                                                                                                                   |
| Probable Cause | Communications failure with a pluggable transceiver is suspected.                                                                    |
| Instance       | XCVR- <code>&lt;slot&gt;/&lt;port&gt;</code> , slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Major, non service-affecting                                                                                                         |

### Steps

- 1 Check the alarm status, as this alarm gets cleared automatically after sometime:  
`alarm show`
- 2 If the alarm does not clear, replace the pluggable transceiver. See [Replacing or reseating a pluggable transceiver](#).
- 3 Again, check the alarm status:  
`alarm show`
- 4 If the alarm does not clear, contact your next level of support or your Ciena support group.

# Procedure 180 XCVR Unknown

## Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

## Overview

**Table 280** Alarm details

|                |                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 48                                                                                                                                                                                                          |
| Probable Cause | A pluggable transceiver is inserted in a port but its inventory data cannot be identified.<br><b>Note:</b> This alarm is NOT raised for any third party pluggable transceiver inserted in the Waveserver 5. |
| Instance       | XCVR- <b>&lt;slot&gt;/&lt;port&gt;</b> , slot = 1 - 8, port 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                |
| Severity       | Minor                                                                                                                                                                                                       |

## Steps

- 1 Identify the client pluggable transceiver raising the alarm.
- 2 Replace the unknown client pluggable with a new version of the pluggable.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 181 Rx Loss of Signal

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 281** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 51                                                                                                                                                                                                                                                                                                                                 |
| Probable Cause | <p>This alarm is raised when one of the following conditions exists:</p> <ul style="list-style-type: none"> <li>• the remote equipment is faulty</li> <li>• the transmitter on the remote equipment stops transmitting</li> <li>• the Rx fiber is removed or disconnected</li> <li>• the Rx fiber is defective or dirty</li> </ul> |
| Instance       | <p>PTP-&lt;slot&gt;/&lt;ptp&gt;, slot = 1 - 8, ptp = 1 - 18</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                                                                                                      |
| Severity       | Critical                                                                                                                                                                                                                                                                                                                           |

### Steps

- 1 Identify the module raising the Rx Loss of Signal alarm.
- 2 Retrieve diagnostic data for the transceiver, and determine whether its Rx optical power is lower than specification:
 

```
xcvr show xcvr <slot>/<xcvr> diagnostics
```

- 3** If the optical power is lower than specification, inspect the fiber connected to the port:
  - Replace the fiber-optic cable if it is damaged.
  - Ensure that the cable connectors are clean and that the cable is properly connected.
- 4** Check the alarm status.  
`alarm show`
- 5** If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-insert it. Additionally, if required, replace the plug.
- 6** If the alarm does not clear, use a power meter to check the Rx power level coming from the equipment connected to the port.
- 7** If the power is lower than specification, contact your next level of support or your Ciena support group.

## Procedure 182 Rx Loss of Clock

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 282** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 52                                                                                                                                                                                                                                                                                                                                                                          |
| Probable Cause | This alarm is raised when there is a line mode mismatch at the two ends of the line. It is raised against a facility when the measured frequency of the input client clock exceeds the allowed frequency deviation outlined by the corresponding standard or when the provisioned line rate at the far-end module does not match the line port rate at the near-end module. |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 18<br>PORT-<slot>/<port>, slot = 1 - 8, port = 1 - 3<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                                                                                    |
| Severity       | <ul style="list-style-type: none"> <li>• Critical</li> <li>• Minor</li> </ul>                                                                                                                                                                                                                                                                                               |

### Steps

- 1 Identify the module and facility raising this alarm.
- 2 Review and adjust Dispersion Compensation Mode setting if required as follows:
  - For WL5e 2x MOTR 16x module: [Table 122, “Line port PTP parameters - chromatic dispersion compensation details”](#)
  - For WL5e 2x MOTR 12x module: [Table 138, “Line port PTP parameters - chromatic dispersion compensation details”](#)

- 3 Ensure the far-end equipment is working and providing an error-free signal that meets the allowed frequency deviation specifications.
- 4 Ensure that all the optical fibers between the subtending equipment Tx port and at the alarmed port are correctly connected.
- 5 Ensure that the provisioning (rate, wavelength, reach) on modules matches at both ends of the link.
- 6 Use an optical power meter to measure the received power level at the Rx port of the alarmed module. Verify if the Rx power level is within the operational threshold. Record the value.
- 7 Clean all the connectors between the far-end Tx port and the alarmed Rx port.
- 8 Record the operating power level after you clean each connector and compare it to the value recorded in [step 6](#). This will allow you to see if there is any improvement to the Rx power.
- 9
 

| If the original alarm has | Then                          |
|---------------------------|-------------------------------|
| cleared                   | the procedure is complete     |
| not cleared               | go to <a href="#">step 10</a> |
- 10 Warm restart the alarmed module. Refer to [Restarting Waveserver 5](#).
- 11 Check the alarm status.  
`alarm show`
- 12 If the alarm does not clear, inspect the fiber connected to the port:
  - Replace the fiber-optic cable if it is damaged.
  - Ensure that the cable connectors are clean and the cable is properly connected.
- 13 If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-insert it. Additionally, if required, replace the plug.
- 14 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 183 Tx Loss of Clock

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 283** Alarm details

|                |                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 53                                                                                                                    |
| Probable Cause | This alarm is raised when the client port Tx direction DPLL is out of lock or measured frequency exceeds +/- 120 ppm. |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                |
| Severity       | <ul style="list-style-type: none"> <li>• Critical</li> <li>• Minor</li> </ul>                                         |

### Steps

- 1 Identify the module and facility raising the Tx Loss of Clock alarm.
- 2 Ensure the far-end and near-end equipment is working and providing an error-free signal that meets the allowed frequency deviation specifications.
- 3 Ensure that all the optical fibers between the subtending equipment Rx port and at the alarmed port are connected correctly.
- 4 Ensure that the provisioning (rate, wavelength, reach) on the modules at both ends of the link are the same.
- 5 Use an optical power meter to measure the power level at the Tx port of the alarmed module. Verify that the Tx power level is within the operational threshold. Record the value.



- 6 Clean all the connectors between the far-end Rx port and the alarmed Tx port.
- 7 Record the operating power level after you clean each connector and compare it to the value recorded in [step 5](#). This allows you to see if there is any improvement to the Tx power.
- 8

| <b>If</b> the original alarm has | <b>Then</b>               |
|----------------------------------|---------------------------|
| cleared                          | the procedure is complete |
| not cleared                      | go to the next step       |
- 9 Warm restart the alarmed module. Refer to [Restarting Waveserver 5](#).
- 10 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 184 Rx Power Low

### Requirements

- A power meter is available.
- You are logged in to Waveserver 5 using an account with at least limited access privileges.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 284** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 54                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Probable Cause | Transceiver is indicating low optical Rx power.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Severity       | <ul style="list-style-type: none"> <li>• Major, service-affecting when Rx power level has exceeded low threshold</li> <li>• Warning, non-service-affecting when Rx power is nearing low threshold</li> </ul> <p><b>Note:</b> The warning severity is available only for third party pluggables. For detail, refer to the <a href="#">Commission third-party XCVR pluggables</a> section. For the Ciena branded plugs, no warning alarm is raised. However, warning alarm threshold, which is readable from EEPROM of the pluggable, and warning alarm flags indicating if the thresholds are crossed, are visible using the <code>ptp show ptp status</code> command.</p> |

### Steps

- 1 Retrieve diagnostic data for the transceiver, and determine whether its Rx optical power is lower than specification:  
`xcvr show xcvr <slot>/<xcvr> diagnostics`
- 2 If the optical power is lower than specification, inspect the fiber connected to the port:
  - Replace the fiber-optic cable if it is damaged.
  - Ensure that the cable connectors are clean and that the cable is properly connected.
- 3 Check the alarm status.  
`alarm show`

- 4 If the alarm does not clear, use a power meter to check the Rx power level coming from the equipment connected to the port.
- 5 If the power is lower than specification, contact your next level of support or your Ciena support group.

## Procedure 185 Rx Power High

### Requirements

- A power meter is available.
- You are logged in to Waveserver 5 using an account with at least limited access privileges.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 285** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Probable Cause | Transceiver is indicating high optical Rx power.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Severity       | <ul style="list-style-type: none"> <li>• Major, service-affecting when Rx power level has exceeded high threshold</li> <li>• Warning, non-service-affecting when Rx power is nearing high threshold</li> </ul> <b>Note:</b> The warning severity is available only for third party pluggables. For detail, refer to the <a href="#">Commission third-party XCVR pluggables</a> section. For the Ciena branded plugs, no warning alarm is raised. However, warning alarm threshold, which is readable from EEPROM of the pluggable, and warning alarm flags indicating if the thresholds are crossed, are visible using the <code>ptp show ptp status</code> command. |

### Steps

- 1 Retrieve diagnostic data for the transceiver, and determine whether its Rx optical power is higher than specification:  
`xcvr show xcvr <slot>/<xcvr> diagnostics`
- 2 Use a power meter to measure the incoming power from the fiber to the transceiver.
- 3 If the optical power is higher than specification, the signal may require padding. Contact your next level of support or your Ciena support group.

## Procedure 186 Tx Power Low

### Requirements

- A power meter is available.
- You are logged in to Waveserver 5 using an account with at least admin access privileges.



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 286** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Probable Cause | Transceiver is indicating low optical Tx power.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Severity       | <ul style="list-style-type: none"> <li>• Major, service-affecting when Tx power level has exceeded low threshold</li> <li>• Warning, non-service-affecting when Tx power is nearing low threshold</li> </ul> <p><b>Note:</b> The warning severity is available only for third party pluggables. For detail, refer to the <a href="#">Commission third-party XCVR pluggables</a> section. For the Ciena branded plugs, no warning alarm is raised. However, warning alarm threshold, which is readable from EEPROM of the pluggable, and warning alarm flags indicating if the thresholds are crossed, are visible using the <code>ptp show ptp status</code> command.</p> |

### Steps

- 1 Use a power meter to measure the transceiver power.
- 2 Retrieve diagnostic data for the transceiver, and determine whether its optical power is lower than specification:  
`xcvr show xcvr <slot>/<xcvr> diagnostics`
- 3 If the optical power is lower than specification, retrieve PTP data for the transceiver to view the status (enabled or disabled) of the transmitter state attribute:  
`ptp show ptp <slot>/<ptp>`

- 
- 4**      **If** the transmitter state for the PTP is      **Then** go to
- |          |                        |
|----------|------------------------|
| disabled | <a href="#">step 5</a> |
| enabled  | <a href="#">step 7</a> |
- 
- 5**      Enable the transmitter state for the PTP.
- ```
ptp disable ptp <ptpList>
ptp tx-state enable ptp <ptpList>
ptp enable ptp <ptpList>
```
- Go to [step 6](#).
- 6** Check the alarm status.
- ```
alarm show
```
- 7**      If the alarm does not clear, do one of the following:
- a.** For a modem transceiver, contact your next level of support or your Ciena support group.
  - b.** For a pluggable transceiver, go to [step 8](#).
- 8**      Reseat the pluggable transceiver. See [Replacing or reseating a pluggable transceiver](#).
- 9**      If the alarm does not clear, replace the transceiver.
- 10**     Check the alarm status.
- ```
alarm show
```
- 11** If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 187 Tx Power High

Requirements

- A power meter is available.
- You are logged in to Waveserver 5 using an account with at least limited access privileges.



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 287 Alarm details

Alarm TID	57
Probable Cause	Transceiver is indicating high optical Tx power.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 18 Note: Only odd slots are used for this release.
Severity	• Major, service-affecting when Tx power level has exceeded high threshold • Warning, non-service-affecting when Tx power is nearing high threshold Note: The warning severity is available only for third party pluggables. For detail, refer to the Commission third-party XCVR pluggables section. For the Ciena branded plugs, no warning alarm is raised. However, warning alarm threshold, which is readable from EEPROM of the pluggable, and warning alarm flags indicating if the thresholds are crossed, are visible using the <code>ptp show ptp status</code> command.

Steps

- 1 Use a power meter to measure the transceiver Tx power.
- 2 Retrieve diagnostic data for the transceiver, and determine whether its optical power is higher than specification:
`xcvr set xcvr <slot>/<xcvr> diagnostics`
- 3 If the optical Tx power is higher than specification, do the following:
 - a. For a modem transceiver, contact your next level of support or your Ciena support group.
 - b. For a pluggable transceiver, go to [step 4](#).
- 4 Reseat the pluggable transceiver. See [Replacing or reseating a pluggable transceiver](#).

- 5 Check the alarm status:
`alarm show`
- 6 If the alarm does not clear, replace the transceiver. See [Replacing or reseating a pluggable transceiver](#).
- 7 Check the alarm status.
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 188 Rx Loss of Channel

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 288 Alarm details

Alarm TID	58
Probable Cause	This alarm is raised when the optical power is below the Loss of channel threshold.
Instance	PTP-<slot>/<ptp> where slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the module and ptp raising the alarm.
- 2 Check the corresponding upstream module and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.
- 3 Retrieve the value of the **Rx Channel Actual Power (dBm) for the alarmed ptp**:

```
ptp show ptp <slot>/<port> status
```

If the received power is out of the power monitoring range for the ptp, the Rx Actual Power (dBm) column will display “OOR-HI” or “OOR-LO” instead of a numerical value.
- 4 Compare the **Rx Channel Actual Power (dBm)** value with the **Rx Channel minimum power (dBm)** and **Rx Channel maximum power (dBm)** values displayed.

- 5 Use the optical power meter to measure the optical power level at the far-end upstream monitored point. Also, verify the losses between the Tx/Rx and CMDs along the channel path are within the expected values.

If the value of the loss is Then

unexpected	troubleshoot the fiber or Photonic layer equipment
expected	go to step 6

- 6 **If the original alarm has Then**

cleared	the procedure is complete
not cleared	restart the module reporting the alarm. Refer to Restarting Waveserver 5 .

- 7 **If the original alarm has Then**

cleared	the procedure is complete
not cleared	reseat the module reporting the alarm. Refer to Restarting Waveserver 5 .

- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 189 Rx Channel Power Low

Requirements

- You are logged in to Waveserver 5 using an account with at least limited access privileges.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).

Overview

Table 289 Alarm details

Alarm TID	59
Probable Cause	This alarm is raised when the Rx channel power is below the Rx channel minimum power.
Instance	PTP-<slot>/<ptp> where slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	- Major, non-serving affecting - Warning, non-serving affecting

Steps

- Identify the module and PTP raising the alarm.
- If the “Rx Power Low” alarm is active, clear it first. Refer to [Rx Power Low](#).
- | | |
|--|------------------------------|
| If the “Rx Channel Power Low” alarm | Then |
| clears | the procedure is complete |
| does not clear | go to step 4 |
- Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.
- Check the Rx Channel Actual Power for the alarmed ptp:

```
ptp show ptp <slot/port> status
```

If the received power is below the power monitoring range for the ptp, the Rx channel Actual Power (dBm) column displays “OOR-LO” instead of a numerical value.
- Compare the **Rx Channel Actual Power (dBm)** value with the **Rx channel minimum power (dBm)** and **Rx channel maximum power (dBm)** values displayed.
- Use the optical power meter to measure the optical power level at the far-end upstream monitored point. Also, verify the losses between the Tx/Rx and CMDs along the channel path are within the expected values.

	If the value of the loss is	Then
	unexpected	troubleshoot the fiber or Photonic layer equipment
8	expected	go to step 8
	If the original alarm	Then
	clears	the procedure is complete
	does not clear	restart the circuit pack reporting the alarm. Refer to Restarting Waveserver 5
9	If the original alarm	Then
	clears	the procedure is complete
	does not clear	reseat the unit reporting the alarm. Refer to the appropriate reseating procedure in Removing, replacing, or reseating equipment .
10	If the alarm does not clear, contact your next level of support or your Ciena support group.	

Procedure 190 Rx Channel Power High

Requirements

- You are logged in to Waveserver 5 using an account with at least limited access privileges.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).

Overview

Table 290 Alarm details

Alarm TID	60
Probable Cause	This alarm is raised when the Rx channel power exceeds the Rx channel maximum power.
Instance	PTP-<slot>/<ptp> where slot = 1 - 8, ptp = 1 - 2
Severity	- Major, non-serving affecting - Warning, non-serving affecting

Steps

- Identify the unit and PTP raising the alarm.
- If the “Rx Power High” alarm is active, clear it first. Refer to [Rx Power High](#).
- | | |
|---|------------------------------|
| If the “Rx Channel Power High” alarm | Then |
| clears | the procedure is complete |
| does not clear | go to step 4 |
- Check the corresponding upstream circuit pack and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.
- Select the alarmed PTP and check the value in the **Rx channel Actual Power (dBm)** column using the command:

```
ptp show ptp <slot/port> status
```

If the received power is above the power monitoring range for the PTP, the Rx channel Actual Power (dBm) column displays “OOR-HI” instead of a numerical value.
- Compare the **Rx Channel Actual Power (dBm)** value with the **Rx channel minimum power (dBm)** and **Rx channel maximum power (dBm)** values displayed.
- Use the optical power meter to measure the optical power level at the far-end upstream monitored point. Also, verify the losses between the Tx/Rx and CMDs along the channel path are within the expected values.

	If the value of the loss is	Then
	unexpected	troubleshoot the fiber or Photonic layer equipment
8	expected	go to step 8
	If the original alarm	Then
	clears	the procedure is complete
	does not clear	restart the circuit pack reporting the alarm. Refer to Restarting Waveserver 5 .
9	If the original alarm	Then
	clears	the procedure is complete
	does not clear	reseat the unit reporting the alarm. Refer to the appropriate reseating procedure in Removing, replacing, or reseating equipment .
10	If the alarm does not clear, contact your next level of support or your Ciena support group.	

Procedure 191 Tx Channel Contention

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 291 Alarm details

Alarm TID	61
Probable Cause	A line port is provisioned to use a wavelength that is being used by another line port in the system.
Instance	PTP-<slot>/<ptp> where slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Identify the module in channel contention.
- Verify the provisioning information (rate, wavelength, reach) on the modules at both ends of the datapath is correct. Correct any inconsistencies with the far-end module.

If all the provisioning information is correct and the alarm is still active, then disable Channel Contention Detection.
- | If the alarm | Then |
|----------------|----------------------------------|
| clears | you have completed the procedure |
| does not clear | go to step 7 |
- Re-provision the facility wavelength to an available wavelength/frequency.

-
- | 5 | If the alarm | Then |
|---|---------------------|----------------------------------|
| | clears | you have completed the procedure |
| | does not clear | go to step 6 |
- 6** If the alarm does not clear and the facility wavelength is not already in use on the network, disable Contention Detection.
- 7** If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 192 Tx Tuning in Progress

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Table 292 Alarm details

Alarm TID	62
Probable Cause	A line port is in the process of tuning to the specified wavelength.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Major, non-service-affecting

Steps

- 1 This alarm clears when Tx wavelength tuning is completed successfully. No action is required.
- 2 If the alarm does not clear or is unexpected, contact your next level of support or your Ciena support group.

Procedure 193 Tx Provisioning Required

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

Note: If Waveserver 5 has not yet been commissioned, see [Commissioning](#) for information about setting the member ID attribute of a line port to provision its wavelength.

Overview

Table 293 Alarm details

Alarm TID	63
Probable Cause	This alarm is raised when the frequency setting for a line port is not set and it is raised on the client ports as well if the ZR plug is present.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Major, non-service-affecting

Steps

- 1 Specify a frequency for the line port:
`ptp set ptp <slot>/<ptp> frequency`
- 2 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 194 Tx Power Reduction

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Table 294 Alarm details

Alarm TID	64
Probable Cause	This alarm is raised when a line port is in a power reduction state. To clear this alarm, disable the tx-power-reduction-state on the specified line port.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Minor, generally non-service-affecting

Steps

- 1 Identify the line port raising the alarm.
- 2 Set the Tx power reduction mode on the port to disabled using the CLI commands:

```
ptp disable ptp <slot/port>
ptp set ptp <slot/port> tx-power-reduction-state disabled
ptp enable ptp <slot/port>
```
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 195 Frequency Provisioning Mismatch

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 295 Alarm details

Alarm TID	65
Probable Cause	This alarm is raised against a line port when the provisioned laser transmit frequency on both ends of the link do not match.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Major

Steps

- 1 Ensure that both ends are provisioned for the same lane spacing. If both ends are provisioned for gridless channel spacing, ensure that both ends are provisioned for the same frequency. If not, reprovision the transmit frequency on the PTP facility so both ends match.
- 2 If the alarm does not clear, replace the alarmed unit. Refer to the chapter [Removing, replacing, or reseating equipment](#).
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 196 Laser Frequency Out of Range

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 296 Alarm details

Alarm TID	66
Probable Cause	This alarm is raised when the difference between the nominal provisioned frequency and the actual frequency exceeds the provisioned threshold.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Major

Steps

- 1 If possible, clear all other transmission related alarms from the network.
- 2 Check the frequency specification at the line port at either end of the fiber span.
- 3 If the alarm does not clear, verify the subtending equipment to ensure that it meets the frequency specification. If the equipment does not meet the specification, change the client signal to use subtending equipment that is within specifications.
- 4 If the alarm persists, re-provision the thresholds of the line port. See [Commissioning](#).
- 5 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 197 Rx Frequency Out of Range

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 297 Alarm details

Alarm TID	67
Probable Cause	This alarm is raised when the network element detects an input signal it cannot lock onto because of a frequency difference.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 3 - 18 Port-<slot>/<port>, slot = 1 - 8, port = 1 - 3 Note: Only odd slots are used for this release.
Severity	Major. The impacted service cannot carry traffic.

Steps

- 1 Identify the source of the alarm. If possible, clear all signal-related alarms from the network.
- 2 Verify the far-end equipment and ensure it meets the specification. If the equipment does not meet the frequency specification, change the client signal to use another subtending equipment.
- 3 Contact your next level of support or your Ciena support group if the alarm does not clear.

Procedure 198 Tx Frequency Out of Range

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 298 Alarm details

Alarm TID	68
Probable Cause	This alarm is raised when the frequency of the transmitting signal is out of the normal frequency range.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 3 - 18 Note: Only odd slots are used for this release.
Severity	Major

Steps

- 1 Identify the source of the alarm. If possible, clear all Rx alarms on the associated line port and on the far-end client port.
- 2 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 199 Modem Class Mismatch

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Table 299 Alarm details

Alarm TID	71
Probable Cause	This alarm is raised on the PTP facility when the line port detects a mismatch between the modem class at either end of the datapath. Modem class allows you to provision the appropriate license type for Submarine or Terrestrial applications on a line port. The modem class license is released when the line port is deleted.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Verify that the modem class at each end of the datapath is the same.
- 2 If the modem class on either side is different, re-provision the line port PTP so that the modem class is the same at both ends of the link. Use these commands:


```
ptp disable ptp <slot>/<ptp>
ptp set ptp <slot>/<ptp> modem-class <terrestrial
|submarine>
ptp enable ptp <slot>/<ptp>
```


Procedure 200 Reach Violation

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Table 300 Alarm details

Alarm TID	72
Probable Cause	This alarm is raised on the PTP facility if the actual length of the link exceeds the provisioned reach class. Once the violation is detected, customer traffic is blocked by conditioning of the on ramp direction from the far-end of the link.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Verify the reach class of the link.

```
ptp show ptp <slot>/<port>
```
- 2 Re-provision the line port PTP of the module to make sure the actual length of the link does not exceed the provisioned reach class. Use these commands:

```
ptp disable ptp <slot>/<ptp>

ptp set ptp <slot>/<ptp> reach-limit <250 | 500 | 1000 |
2000 | 2100 | 3000 | 3500 | 4000 | 5000 | 8000 | 9000 |
14000>

ptp enable ptp <slot>/<ptp>
```

Procedure 201 Remote Defect Indicator

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 301 Alarm details

Alarm TID	73
Probable Cause	This alarm is raised against the PTP facilities when an incoming RDI from the far-end network element is detected.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Minor

Steps

- 1 Identify the module raising the alarm.
- 2 Verify that the module transmit path and receive path are working correctly.
- 3 Log into the network element from which the signal originates.
- 4 Retrieve all alarms from the far-end network element.

- | | If there are | Then |
|----------|---|--|
| 5 | no alarms for the facility | ensure that alarm reporting of related alarms is enabled.

Also ensure that the equipment and facility of the far-end module are in-service.

If there are masked alarms on this facility, clear them and go to step 6 . |
| | alarms for the facility | refer to the appropriate alarm clearing procedures. The Remote Defect Indication alarm may be expected if the local alarm is AIS, or Unequipped. |
| 6 | At the local network element, retrieve all alarms to determine if the original alarm has cleared. | |
| 7 | Contact your next level of support or your Ciena support group if the alarm does not clear. | |

Procedure 202 Duplicate Adjacency Discovered

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- For information on 6500, refer to the 6500 Technical Publications.

Overview

Table 302 Alarm details

Alarm TID	74
Probable Cause	Two or more 6500 expected far end addresses are set to the same Waveserver 5 node name, slot and port. Note: If the associated module has its PTP or SPLI disabled, this alarm will not be raised and it will be masked.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 3 Note: Only odd slots are used for this release.
Severity	Minor, non-service-affecting

Steps

- 1 Identify the Waveserver 5 that is raising this alarm and the port that the alarm is raised against:
`alarm show`
- 2 Examine the adjacency summary table for the Waveserver 5 units and note all matches found for all Waveserver 5 addresses.
`spli adjacency show`
- 3 On the 6500, delete the adj-tx provisioning pointing to the Waveserver 5 with the duplicate entry.
`adjacencies adjacency set ADJ-<shelf>-<slot>-<port> prov-fe-addr "" prov-addr-form NULL`
- 4 Determine which 6500 node has the correct far-end address and remove the duplicate entry from the other 6500 nodes.
Note: If you are deleting the duplicate SPLI adjacency using the 6500 CLI: Log into the 6500 CLI with admin or higher privilege level.
`adjacencies spli delete [aid] [index <Value>]`
Note: If you are deleting the duplicate SPLI adjacency using 6500 Site Manager, refer to the procedure on deleting SPLI entries in *Administration and Security*, 323-1851-301.

- 5 Check the Waveserver 5 alarm status.
`alarm show`
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 203 Dispersion Scan In Progress

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Table 303 Alarm details

Alarm TID	75
Probable Cause	This alarm is raised when a Q-based pre-compensation dispersion scan session is in progress.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Note: Only odd slots are used for this release.
Severity	Warning

Steps

- 1 No action is required. The alarm clears when the dispersion scan process is complete.
- 2 If the alarm does not clear or is unexpected, contact your next level of support or your Ciena support group.

Procedure 204 Link Data Retrieval In Progress

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Table 304 Alarm details

Alarm TID	76
Probable Cause	This alarm is raised when the link data collection process is being performed on the PTP.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, PTP = 1 - 2 Note: Only odd slots are used for this release.
Severity	Warning

Steps

- 1 No action is required. The alarm clears when the data collection process is complete.
- 2 If the alarm does not clear or is unexpected, contact your next level of support or your Ciena support group.

Procedure 205 Dispersion Search Mode Scan In Progress

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.

Overview

Table 305 Alarm details

Alarm TID	80
Probable Cause	This alarm is raised when the dispersion scan mode of line PTP is set to search and traffic is lost.
Instance	PTP-<slot>/<ptp>, slot = 1 - 3, ptp depends on the module type: - MOTR-400-1-8: Ports 1,3-10 - MOTR-100-2: Ports 1-10, 101, 201, 202 and 301 - MOTR-100-1: Ports 1, 3-6
Severity	Major

Steps

- Alarm is raised on line ports. The alarm clears when the dispersion scan process is complete.
- To set the dispersion scan mode, run the following CLI command:
ptp set ptp 2/1 tx-dispersion-compensation-mode search
- If the alarm is not cleared, follow the clear procedure used for [Rx Loss of Clock](#) or [Rx Loss of Channel](#).

Procedure 206 Loss of Frame

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 306 Alarm details

Alarm TID	81
Probable Cause	This alarm is raised when Waveserver 5 repeatedly receives errored optical transport unit (OTU) overhead bytes (OA1 or OA2) for three consecutive ms or more.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the module raising the Loss of Frame alarm.
- 2 Verify that no equipment alarms exist for the module raising the Loss of Frame alarm.
 - If an equipment alarm exists, clear the alarm for the alarmed equipment.
 - If an equipment alarm does not exist, go to [step 3](#).
- 3 Verify that the far-end module is the correct type (rate, wavelength, reach) and the provisioned optical rate (if applicable) is correct. Correct any inconsistencies with the far-end module.
 - If the alarm clears, you have completed this procedure.
 - If the alarm persists, go to [step 4](#).

- 4 Verify that the received light level is within specification.
 - If the received light level is within specification, go to [step 7](#).
 - If the received light level is not within specification, go to [step 5](#).
- 5 The low light level and resulting bit errors may be the result of a dirty or defective fiber or connector. Isolate and clean or repair the fiber or connector.
 - If a defective or dirty fiber or connector was cleaned or repaired, go to [step 6](#).
 - If no defective or dirty fibers or connectors were cleaned or repaired, go to [step 8](#).
- 6 Check the output of the far-end system and check/clean the transmit fiber and connectors from the output to the input of the alarmed port. Correct any problems you find. Reconnect fibers and if applicable, reinstall any pluggables.
 - If you isolated and corrected problems, go to [step 8](#).
- 7 Isolate/repair or refer any BER problem across line systems between the far-end module and the receive module.
 - If any problems were isolated and corrected, go to [step 8](#).
 - If no problems were isolated and corrected, go to [step 11](#).
- 8 Verify that the alarm cleared.
 - If the alarm cleared, you have completed this procedure.
 - If the alarm did not clear, replace the module, then go to [step 9](#).
- 9 If the alarm does not clear, inspect the fiber connected to the port:
 - Replace the fiber-optic cable if it is damaged.
 - Ensure that the cable connectors are clean and that the cable is properly connected.
- 10 If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-insert it. Additionally, if required, replace the plug.
- 11 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 207 Loss of Multiframe

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 307 Alarm details

Alarm TID	82
Probable Cause	This alarm is raised when errors in the multiframe alignment process or in the mapping of the optical channel payload (OPU) are detected.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the module raising the Loss of Multiframe alarm.
- 2 Verify that no equipment alarms exist for the module raising the Loss of Multiframe alarm.
 - If an equipment alarm exists, clear the alarm for the alarmed equipment.
 - If an equipment alarm does not exist, go to [step 3](#).
- 3 Verify that the far-end module is the correct type (rate, wavelength, reach) and the provisioned optical rate (if applicable) is also correct. If there are any inconsistencies between the far-end module and near-end module, correct them to match.
- 4 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 208 Loss of Frame and Loss of Multiframe

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 308 Alarm details

Alarm TID	83
Probable Cause	This alarm is raised on an OTU trail termination point (TTP) when a Loss Of Frame or Loss Of Multiframe is detected on any of the lanes.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 3 Channel-<slot>/<port>.<channel>, slot = 1 - 8, port = 1 - 3, channel = 1 - 4, 1 - 8 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Identify the source of the alarm, and identify the transmit and receive sites of the alarmed signal.
- Log into the remote network element.
- Retrieve alarms at the far-end module. Clear any alarms using the appropriate procedure.
- | | |
|----------------------------------|------------------------------|
| If the original alarm has | Then |
| cleared | the procedure is complete |
| not cleared | go to step 5 |
- Verify that the module provisioning (rate, wavelength, reach) on the far-end and near-end modules match. Correct any mismatches.

6	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 7
7	Verify that the FEC setting on the far-end and near-end modules match. Correct any mismatches.	
8	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 9
9	If a photonic line system is used to transport the signal of the unit raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms on the photonic line system associated with the alarmed signal.	
10	If the original alarm has cleared	Then the procedure is complete
	not cleared	go to step 11
11	Determine the optical power level into the Rx interface, and verify it is within technical specifications.	
12	If the measured optical power is below the specified receiver sensitivity	Then go to step 13
	above the specified receiver sensitivity	step 15
	within the specified receiver sensitivity range	step 21
13	Remove the Tx fiber from the far-end unit, and measure the transmit power at the far-end module.	
14	If the transmit power at the far-end module is above the launch power (minimum)	Then Possible causes are the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Determine the cause and correct the problem. Go to step 20 .
	below the launch power (minimum)	Replace the transmitting module. Refer to the procedures in Removing, replacing, or reseating equipment . Go to step 20 .
	within the launch power range specified for the module	Go to step 20 .
15	Clean all connections at both ends of the optical fiber link and re-attach the optical fibers.	

16	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	go to step 17

- 17 If a photonic line system is used to transport the signal, the high optical power can be caused by the photonic line system. Clear any signal-related alarms on the photonic line.

18	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	go to step 19

**CAUTION****Risk of traffic loss**

A cold restart is service-affecting and can cause traffic loss.

Once the optical path is verified to be correct, perform a cold restart on the module raising the alarm. See [Restarting Waveserver 5](#).

19	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	perform a warm restart on the module reporting the alarm. See Restarting Waveserver 5 .

20	If the original alarm has	Then
	cleared	the procedure is complete
	not cleared	the module is faulty. Replace the module reporting the alarm. Refer to the procedures in Removing, replacing, or reseating equipment .

- 21 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 209 OTU AIS

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 309 Alarm details

Alarm TID	84
Probable Cause	A generic AIS signal originating from the far-end network element was received by the near-end network element. Note: This alarm is not raised on 100GE client ports.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Log in to the reporting module, and identify the port that displays the OTU Alarm Indication Signal (AIS) alarm.
- Verify that no equipment alarms exist for the port that displays the OTU Alarm Indication Signal (AIS) alarm.
 - If equipment alarms exist, clear the alarms on the alarmed equipment before continuing with this procedure.
 - If equipment alarms do not exist, go to [step 3](#).
- At the local network element, retrieve all alarms to determine whether the original alarm has cleared.
- Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.
- Log into the remote network element and retrieve all alarms. Look for an alarm from the remote network element.

6	If there are	Then
	no alarms	ensure that the equipment and facility or the client and line facilities of the remote equipment are in-service and connected. Go to step 7 .
	additional alarms	refer to the appropriate alarm clearing procedures. Go to step 7 .
7	At the local network element, retrieve all alarms to determine whether the original alarm has cleared.	
8	If the alarm does not clear, contact your next level of support or your Ciena support group.	

Procedure 210 OTU BDI

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 310 Alarm details

Alarm TID	85
Probable Cause	A defect was detected on the far-end equipment by the near-end equipment. The OTU Backward Defect Indication (BDI) is a maintenance signal that informs the far-end chassis that the near-end chassis detected an AIS that originated from the far-end chassis. Note: This alarm is not raised on 100GE client ports.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18 Note: Only odd slots are used for this release.
Severity	Minor

Steps

- 1 Identify the module raising the alarm.
- 2 Determine whether there is additional port alarm information:
 - If an equipment alarm exists, resolve it before continuing with this procedure.
 - If an equipment alarm does not exist, go to [step 3](#).
- 3 Check the far-end module and identify the chassis generating this alarm. Resolve any alarms reported by the far-end network element.
- 4 Contact your next level of support or your Ciena support group if the alarm does not clear.

Procedure 211 OTU Signal Degrade

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 311 Alarm details

Alarm TID	86
Probable Cause	The quality of the optical signal exceeds the signal degrade threshold.
Instance	Port-<slot>/<port>, slot = 1 - 8, port 1 - 18 Note: Only odd slots are used for this release.
Severity	Major

Steps

- 1 Identify the module raising the OTU Signal Degrade alarm.
- 2 Verify that no equipment alarms exist for the module raising the OTU Signal Degrade alarm.
 - If an equipment alarm exists, clear the alarm for the alarmed equipment.
 - If an equipment alarm does not exist, go to [step 3](#).
- 3 Make sure that the module type and provisioning (rate, wavelength, power, reach) is consistent at both ends of the datapath.
- 4 Make sure the signal optical power level into the Rx interface is within specification.

	If the signal level is greater than the Rx maximum value less than the Rx minimum value within specifications	Then go to step 5 step 7 step 11
5	Confirm that the correct pluggables are installed and add attenuation pads as needed until the Rx power is within the reported specification.	
6	If the original alarm clears does not clear	Then procedure is complete go to step 11
7	The fiber may be dirty, pinched, folded, the fiber connectors may not be fully seated, or an incorrect pluggable type is used on the module reporting the alarm or on the subtending equipment. Confirm that the correct pluggables are installed, clean all fiber connectors and ensure they are not pinched or folded and are fully seated. If the alarm does not clear, replace the fiber.	
8	Recheck the Rx power value for the facility reporting the alarm.	
9	If the original alarm clears power is within specifications power is still below specifications	Then procedure is complete go to step 11 there is an issue with the link or source fault. Contact your next level of support or Ciena support group.
10	Replace the pluggable that supports the facility raising the alarm.	
11	If the original alarm has cleared not cleared	Then the procedure is complete go to step 12
12	Replace the far-end pluggable. Refer to Removing, replacing, or reseating equipment .	
13	If the alarm does not clear, contact your next level of support or your Ciena support group.	

Procedure 212 OTU Trace Identifier Mismatch

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 312 Alarm details

Alarm TID	87
Probable Cause	• A connectivity defect caused by a provisioning mismatch within the network. • Incorrect fiberling or incorrect provisioning of the TTI values. • Differences in the actual and expected (provisioned) values of these fields in the TTI string can cause this alarm: — Source Access Point Identifier (SAPI) only — Destination Access Point Identifier (DAPI) only — SAPI + DAPI — Operator specific information (default)
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18 Note: Only odd slots are used for this release.
Severity	Critical (Line port), Minor (Client port)

Steps

- 1 Identify the module raising the OTU TTI Mismatch alarm.
- 2 Verify that the optical fiber connections are correct on the network element that raised the alarm and on the far-end network element.
- 3 Retrieve the TTI values for the transmitting network element and the receiving (alarmed) network element.

- | | If the alarm | Then |
|----------|---|------------------------------|
| 4 | clears | this procedure is complete |
| | does not clear | go to step 5 |
| 5 | Compare the outgoing TTI value of the transmit signal at the transmit network element with the expected TTI value of the receive signal at the receive network element.
If the values are different, change the expected TTI value of the receive signal to match the outgoing TTI value of the transmit signal. | |
| 6 | If the alarm does not clear, contact your next level of support or your Ciena support group. | |

Procedure 213 Pre-FEC Signal Degrade

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 313 Alarm details

Alarm TID	88
Probable Cause	The number of corrected errors recorded by the FEC Error Correction process exceeds the threshold for consecutive one second intervals. The threshold value is dependent on the user-defined FEC error correction scheme.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Major

Steps

- 1 Identify the module raising the alarm.
- 2 Confirm that the FEC setting is the same at both ends of the datapath.
- 3 Verify that the Pre-FEC Signal Degrade Threshold setting is correct.
`ptp show ptp <ptpList>`
- 4 If the Pre-FEC Signal Degrade Threshold is not correctly set, set the value.
`ptp set ptp <ptpList> prefec-sd-threshold <Pre-Fec Threshold in dB (N.MM): -1.00..3.50>`

- | | | |
|-----------|---|--|
| 5 | If the alarm has
cleared
not cleared | Then
the procedure is complete
go to step 6 |
| 6 | Check the corresponding far-end module and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding. | |
| 7 | If the alarm has
cleared
not cleared | Then
the procedure is complete
go to step 8 |
| 8 | Clean the port receive fiber. | |
| 9 | Verify that the received light level is within specification. <ul style="list-style-type: none"> • If the received light level is within specification, go to step 12. • If the received light level is not within specification, go to step 10. | |
| 10 | The low light level and resulting bit errors may be the result of a dirty or defective fiber or connector. Isolate and clean or repair the fiber or connector. <ul style="list-style-type: none"> • If a defective or dirty fiber or connector was cleaned or repaired, go to step 11. • If no defective or dirty fibers or connectors were cleaned or repaired, go to step 19. | |
| 11 | Check the output of the far-end system and check/clean the transmit fiber and connectors from the output to the input of the alarmed port. Correct any problems that are found. <ul style="list-style-type: none"> • If any problems were isolated and corrected, go to step 15. | |
| 12 | Isolate/repair or refer any BER problem across line systems between modules at both ends of the datapath. <ul style="list-style-type: none"> • If any problems were isolated and corrected, go to step 15. • If no problems were isolated and corrected, go to step 19. | |
| 13 | Clean the port fiber connector. | |
| 14 | Check that the receive port is the correct type (rate, wavelength, and reach). | |
| 15 | Reconnect the receive fiber and if applicable, reinstall any pluggables. | |
| 16 | Verify that the alarm cleared. <ul style="list-style-type: none"> • If the alarm cleared, you have completed this procedure. • If the alarm did not clear, go to step 19. | |
| 17 | Replace the module. | |
| 18 | Reconnect the receive fiber and if applicable, reinstall the pluggables. | |
| 19 | If the alarm does not clear, contact your next level of support or your Ciena support group. | |

Procedure 214 Pre-FEC Signal Fail

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 314 Alarm details

Alarm TID	89
Probable Cause	This alarm is raised when the Pre-FEC Signal Fail provisionable threshold has been crossed.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2 Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the module raising the alarm.
- 2 Confirm that the FEC setting on the facility is the same at both ends of the datapath.
- 3 Check that the Pre-FEC Signal Fail Threshold value is correct.
`ptp show ptp <ptpList>`
- 4 If the Pre-FEC Signal Fail Threshold is not correctly set, set the value:
`ptp set ptp <ptpList> prefec-sf-threshold <Pre-Fec Threshold in dB(N.MM): -1.00..1.00>`

5	If the alarm has	Then
	cleared	the procedure is complete
	not cleared	go to step 6
6	Check the corresponding upstream module and photonic layer for failures or alarms. Troubleshoot these alarms/failures before proceeding.	
7	Clean the port receive fiber.	
8	Verify that the received light level is within specification.	
	- If the received light level is within specification, go to step 11. - If the received light level is not within specification, go to step 9.	
9	The low light level and resulting bit errors may be the result of a dirty or defective fiber or connector. Isolate and clean or repair the fiber or connector.	
	- If a defective or dirty fiber or connector was cleaned or repaired, go to step 10 - If no defective or dirty fibers or connectors were cleaned or repaired, go to step 18.	
10	Check the output of the far-end system and check/clean the transmit fiber and connectors from the output to the input of the alarmed port. Correct any problems that are found.	
	If any problems were isolated and corrected, go to step 17.	
11	Isolate/repair or refer any BER problem across line systems between modules at both ends of the datapath.	
	- If any problems were isolated and corrected, go to step 17. - If no problems were isolated and corrected, go to step 18.	
12	Clean the port fiber connector.	
13	Check that the receive port is the correct type (rate, wavelength, and reach).	
14	Reconnect the receive fiber and if applicable, reinstall any pluggables.	
15	Verify that the alarm cleared.	
	- If the alarm cleared, you have completed this procedure. - If the alarm did not clear, go to step 16.	
16	Replace the module.	
17	Reconnect the receive fiber and if applicable, reinstall the pluggables.	
18	If the alarm does not clear, contact your next level of support or your Ciena support group.	

Procedure 215 ODU LCK

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 315 Alarm details

Alarm TID	90
Probable Cause	The far-end TTP is administratively locked.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18 Channel-<slot>/<port>.<channel>, slot = 1 - 8, port = 1 - 18, channel 1 - 8 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the module raising the alarm.
`alarm show`
The module name is visible in the Instance column of the active alarm list.
- 2 Verify the administrative state of the PTP or facility that is locked, and set it to unlocked if necessary.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 216 ODU AIS

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 316 Alarm details

Alarm TID	91
Probable Cause	The far-end equipment has experienced a facility or equipment failure. Resolving the fault in the adjoining chassis clears the ODU Alarm Indication Signal (AIS) alarm in the reporting module. Note: This alarm is not raised on 100GE client ports.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 10 Channel-<slot>/<port>.<channel>, slot = 1 - 8, port = 1 - 18, channel = 1 - 8 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Log in to the reporting module, and identify the port that displays the ODU AIS alarm.
- Verify that no equipment alarms exist for the port that displays the ODU AIS alarm.
 - If equipment alarms exist, clear the alarms on the alarmed equipment before continuing with this procedure.
 - If equipment alarms do not exist, go to [step 3](#).
- At the local network element, retrieve all alarms to determine if the original alarm has cleared.
- Use the optical fiber/cable connection information to identify the transmit and receive sites of the alarmed signal.

-
- 5** Log into the remote network element and retrieve all alarms. Look for an alarm from the remote network element.
- 6**
- | If there are | Then |
|---------------------|--|
| no alarms | ensure that the equipment and facility or the client and line facilities of the remote equipment are in-service and connected.
Go to step 7 . |
| additional alarms | refer to the appropriate alarm clearing procedures.
Go to step 7 . |
- 7** At the local network element, retrieve all alarms to determine if the original alarm has cleared.
- 8** If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 217 ODU OCI

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 317 Alarm details

Alarm TID	92
Probable Cause	The far-end ODU4 channel is either not connected or the associated client port is not provisioned; or the OTL4.4 client port is receiving ODU Open Connection Indication (OCI) from the subtending device.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18 Channel-<slot>/<port>.<channel>, slot = 1 - 8, port = 1 - 18, channel 1 - 8 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Select your step:

If the alarm is raised	Then go to
on a line port channel on the far-end Waveserver 5	step 2
on an OTL4.4 client port on the near-end Waveserver 5	step 7
- Log into the far-end Waveserver 5 and display the line port channel raising the alarm. Determine the associated client port on the near-end Waveserver 5.
`alarm show`
- Trace the fiber to the associated client port on the near-end Waveserver 5. Alarms raised on line port 1 are associated with client ports 3 to 10. Alarms raised on line port 2 are associated with client ports 11 to 18.

- 4 Log into the near-end Waveserver 5 and check the associated client port.
`xcvr show xcvr <slot>/<xcvr>`
- 5 Provision the client port, if required.
`xcvr set xcvr <slot>/<xcvr>`
`port set port <slot>/<port>`
`ptp set ptp <slot>/<port>`
 For a list of supported parameters, see `xcvr set xcvr`, `port set port` and `ptp set ptp`.
- 6 Check the fiber connections and ensure that the associated client port is connected to the subtending equipment (optional).
 Go to [step 9](#).
- 7 Log into the near-end Waveserver 5 and display the OTL4.4 client port raising the alarm.
`port show port <slot>/<port>`
`xcvr show xcvr <slot>/<xcvr>`
- 8 Check the provisioning for any subtending equipment and provision as required.
Note: The ODU4 OCI signal is also conditioned to the far-end Waveserver 5, where it results in an ODU-OCI alarm on the far-end channel.
- 9 Check the alarm status.
`alarm show`
- 10 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 218 ODU BDI

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 318 Alarm details

Alarm TID	93
Probable Cause	A defect was detected on the far-end equipment by the near-end equipment. The ODU Backward Defect Indication (BDI) is a maintenance signal that informs the far-end chassis that the near-end chassis detected a failure. Note: This alarm is not raised on 100GE client ports.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18 Channel-<slot>/<port>.<channel> , slot = 1 - 8, port = 1 - 18, channel = 1 - 8 Note: Only odd slots are used for this release.
Severity	Minor

Steps

- 1 Identify the module raising the alarm.
- 2 Determine whether there is additional port alarm information available to troubleshoot the issue:
 - If an equipment alarm exists, resolve it before continuing with this procedure.
 - If an equipment alarm does not exist, proceed to the next step to troubleshoot further.
- 3 Check the far-end module and identify the chassis that may be generating this alarm. Resolve any alarms reported by the far-end network element.
- 4 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 219 ODU Signal Degrade

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 319 Alarm details

Alarm TID	94
Probable Cause	The quality of the optical signal of an ODU layer exceeds the signal degrade threshold.
Instance	Port- <code><slot>/<port></code> , slot = 1 - 8, port = 1 - 18 Channel- <code><slot>/<port>.<channel></code> , slot = 1 - 8, port = 1 - 18, channel = 1 - 8 Note: Only odd slots are used for this release.
Severity	Major

Steps

- Identify the module raising the ODU Signal Degrade alarm.
- Verify that no equipment alarms exist for the module raising the ODU Signal Degrade alarm.
 - If an equipment alarm exists, clear the alarm for the alarmed equipment.
 - If an equipment alarm does not exist, go to [step 3](#).
- Verify that the far-end module is the correct type (rate, wavelength, reach) and the provisioned optical rate (if applicable) is correct. Correct any inconsistencies with the far-end module.
 - If the alarm clears, you have completed this procedure.
 - If the alarm persists, go to [step 4](#).

- 4 Verify that the received light level is within specification.
 - If the received light level is within specification, go to [step 7](#).
 - If the received light level is not within specification, go to [step 5](#).
- 5 The low light level and resulting bit errors may be the result of a dirty or defective fiber or connector. Isolate and clean or repair the fiber or connector.
 - If a defective or dirty fiber or connector was cleaned or repaired, go to [step 6](#).
 - If no defective or dirty fibers or connectors were cleaned or repaired, go to [step 10](#).
- 6 Check the output of the far-end system and check/clean the transmit fiber and connectors from the output to the input of the alarmed port. Correct any problems you find. Reconnect fibers and if applicable, reinstall any pluggables.
 - If any problems were isolated and corrected, go to [step 8](#).
- 7 Isolate/repair or refer any BER problem across line systems between the far-end module and the receive module.
 - If any problems were isolated and corrected, go to [step 8](#).
 - If no problems were isolated and corrected, go to [step 10](#).
- 8 Verify that the alarm cleared.
 - If the alarm cleared, you have completed this procedure.
 - If the alarm did not clear, go to [step 10](#).
- 9 If necessary, reconnect fibers and if applicable, reinstall the pluggables.
- 10 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 220 ODU Trace Identifier Mismatch

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 320 Alarm details

Alarm TID	95
Probable Cause	• The trail trace identifier (TTI) string received by the facility differs from the expected provisioned TTI value. • Incorrect fiberling or incorrect provisioning of the TTI value. • Differences in the actual and expected (provisioned) values of these fields in the TTI string can cause this alarm: — Source Access Point Identifier (SAPI) only — Destination Access Point Identifier (DAPI) only — SAPI + DAPI — Operator specific information (default)
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 3 Channel-<slot/port.channel> slot = 1 - 8, port = 1 - 3, channel = 1 - 8 Note: Only odd slots are used for this release.
Severity	Minor

Steps

- 1 Identify the module raising the ODU TTI Mismatch alarm.
- 2 Verify that the optical fiber connections are correct on the network element that raised the alarm and on the far-end network element.
- 3 Retrieve the TTI values for the transmitting network element and the receiving (alarmed) network element.

- | | If the alarm | Then |
|----------|---|------------------------------|
| 4 | clears | this procedure is complete |
| | does not clear | go to step 5 |
| 5 | Compare the outgoing TTI value of the transmit signal at the transmit network element with the expected TTI value of the receive signal at the receive network element.
If the values are different, change the expected TTI value of the receive signal to match the outgoing TTI value of the transmit signal. | |
| 6 | If the alarm does not clear, contact your next level of support or your Ciena support group. | |

Procedure 221 OPU Payload Type Mismatch

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 321 Alarm details

Alarm TID	96
Probable Cause	An OPU Payload Type Mismatch alarm is raised when ports are connected incorrectly or the expected payload type provisioned on the OPU does not match the received payload type.
Instance	Port- <slot>/<port> , slot = 1 - 8, port = 1 - 3 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Identify the module raising the OPU Payload Type Mismatch alarm.
- Verify the payload type is correct.
channel show channel <slot>/<port>.<channel>
Note: Match Tx Payload Type should match the Rx Payload Type in the above command for the alarm to clear.
- Use the network connection information to check connectivity to far-end modules and client interfaces.
- Ensure the outgoing signal label message matches the incoming expected signal label message.
- Verify that the alarm has cleared.
- If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 222 Loss of OPU Multiframe Identifier

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have the optical fiber connection information (how the optical modules and network elements are connected to each other) and you have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 322 Alarm details

Alarm TID	97
Probable Cause	Loss of an OPU Multiframe Identifier (OMFI) for an OPUCn frame was detected.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 1 - 3 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the source of the alarm, and identify the transmit and receive sites of the alarmed signal.
- 2 Log into the remote network element.
- 3 Retrieve all alarms at the transmit end. Clear high-order alarms using the appropriate procedure.
- 4

If the original alarm has	Then
cleared	the procedure is complete
not cleared	go to step 5
- 5 Verify that the provisioned line rate on the far-end module and local module match. Correct any mismatches.

- | | | |
|--|---|--|
| 6 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
go to step 7 |
| 7 | | |
| Verify that the FEC setting on the far-end and local modules match. Correct any mismatches. | | |
| 8 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
go to step 9 |
| 9 | | |
| If a photonic line system is used to transport the signal of the unit raising the alarm, retrieve all alarms on the photonic line system. Clear any alarms on the photonic line system associated with the alarmed signal. | | |
| 10 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
go to step 11 |
| 11 | | |
| Determine the optical power level into the Rx interface, and verify it is within technical specifications. | | |
| 12 | If the measured optical power is
below the specified receiver sensitivity
above the specified receiver sensitivity
within the specified receiver sensitivity range | Then go to
step 13
step 15
step 22 |
| 13 | | |
| Remove the Tx fiber from the far-end unit, and measure the transmit power at the far-end module. | | |
| 14 | If the transmit power at the far-end module is
above the launch power (minimum)
below the launch power (minimum)
within the launch power range specified for the module | Then
the optical fiber attenuation is too high, the optical fiber connections are dirty, or the optical fiber is damaged. Determine the cause and correct the problem.
Go to step 21 .
Replace the transmitting module. Refer to the procedures in Removing, replacing, or reseating equipment .
Go to step 21 .
go to step 21 . |
| 15 | | |
| Clean all connections at both ends of the optical fiber link and re-attach the optical fibers. | | |

- | | | |
|--|---|--|
| 16 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
go to step 17 |
| 17 | If a photonic line system is used to transport the signal, the high optical power can be caused by the photonic line system. Clear any signal-related alarms on the photonic line. | |
| 18 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
go to step 19 |
| 19 | <div style="border: 1px solid black; padding: 10px;">  <p>CAUTION
 Risk of traffic loss
 A cold restart is service-affecting. On restart, only the saved configuration file is recovered (that is, the running configuration file is not recovered).</p> </div> | |
| Once the optical path is verified to be correct, perform a cold restart on the module raising the alarm. See Restarting Waveserver 5 . | | |
| 20 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
perform a warm restart on the module reporting the alarm. See Restarting Waveserver 5 . |
| 21 | If the original alarm has
cleared
not cleared | Then
the procedure is complete
the module is faulty. Replace the module reporting the alarm. Refer to the procedures in Removing, replacing, or reseating equipment . |
| 22 | If the alarm does not clear, contact your next level of support or your Ciena support group. | |

Procedure 223 Loss of Lane Alignment

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 323 Alarm details

Alarm TID	98
Probable Cause	In a mated configuration, the Waveserver 5 is unable to deskew due to the skew between the two line ports exceeding the maximum allowed tolerance. Note: This alarm masks the “Loss of Frame and Multiframe” and “Frequency Out of Range” alarms.
Instance	PTP-<slot>/<ptp>, slot = 1 - 8, PTP = 1 - 2 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the line port raising the alarm.
- 2 Verify if the length difference between the patch cord fibers for the line port 1 and line port 2 is less than 40 m. If not, replace them to meet this requirement.
- 3 If the original alarm does not clear, verify that the line port 1 and line port 2 go to the same CCMD. If not, ensure that the line port 1 and line port 2 go to the same CCMD/CMD.
- 4 If the original alarm does not clear, verify both line ports 1 and 2 have the appropriate wavelengths/frequencies provisioned based on [Engineering rules for mated line configurations for WL5e 2x MOTR 16x modules](#).
To verify wavelength/frequency:
`ptp show ptp <slot>/<ptp>`

To set the wavelength/frequency:

```
ptp disable ptp <slot>/<ptp>

ptp set ptp <slot>/<ptp> frequency <Frequency in
GHz (N.M): C-Band: 191250.0..196125.0> | <Frequency in
GHz (N.M): L-Band: 186087.5..190956.2>

ptp enable ptp <slot>/<ptp>
```

- 5 If the original alarm does not clear, ensure the provisioned transmission mode and link optimization mode on both line ports 1 and 2 are the same. If not, provision them to be the same.

To verify the transmission mode and link optimization mode:

```
ptp show ptp <slot>/<ptp>
```

To set transmission mode:

```
xcvr disable xcvr <slot 1..8>/<xcvr 1..2>

xcvr set xcvr <slot>/<xcvr 1..2> mode <95-200-O | 95-250-
O | 95-300-O | 95-350-O | 95-400-O | 95-450-O | 95-500-O
| 95-550-O | 95-600-O | 95-650-O | 95-700-O | 95-800-O |
95-300-E | 95-350-E | 95-400-E | 95-450-E | 95-500-E | 95-
550-E | 95-600-E | 95-650-E | 95-700-E | 95-800-E | 93.3-
600-E | 93.3-400-E | 91.6-300-E | 91.6-400-E | 91.6-450-E
| 91.6-500-E | 91.6-550-E | 91.6-600-E | 91.6-700-E |
91.6-800-E | 90-600-E | 90-400-E | 85-600-E | 85-400-E |
85-300-E | 82-400-E | 4x100GE-ZR>

xcvr enable xcvr <slot 1..8>/<xcvr 1..2>
```

To set link optimization mode:

```
ptp disable ptp <slot>/<ptp>

ptp set ptp <slot>/<ptp> link-optimization-mode <Standard
| Non-linear | SNR-Tolerant | Fast-Tracking>

ptp enable ptp <slot>/<ptp>
```

Note: Ensure the Link Optimization Mode (LOM) supports the configured transmission mode. See [Facilities](#) for information on supported transmission modes and link optimization mode on different service modules.

- 6 If alarm does not clear, replace the affected service module with the line port identified in [step 1](#). Refer to [Reseating, removing, or replacing a service module](#).
- 7 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 224 High Bit Error Rate

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.

Overview

Table 324 Alarm details

Alarm TID	101
Probable Cause	A high-bit error rate condition was detected.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Major

Steps

- 1 Log in to the affected chassis, and note the port that displays the High Bit Error Rate alarm.
- 2 Check the actual received optical power on the associated PTP and verify that the received optical power meets the optical interface specifications. If the received optical power is outside the specified operating range, troubleshoot or repair the link.
- 3 If the alarm does not clear, inspect the fiber connected to the port:
 - Replace the fiber-optic cable if it is damaged.
 - Ensure that the cable connectors are clean and that the cable is properly connected.
- 4 If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-insert it. Additionally, if required, replace the plug.
- 5 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 225 Rx Local Fault

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 325 Alarm details

Alarm TID	102
Probable Cause	This alarm is raised on the client-side interface. This alarm can indicate a problem at the far-end transmitting equipment.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Identify the module and facility raising the alarm.
- 2 Make sure the transmitting far-end equipment is functioning correctly and transmitting a valid signal.
- 3 Use an optical power meter to measure the transmit power on the far-end equipment. Ensure that the power of the transmitted signal is at the correct power level according to the manufacturer's specifications.

If the transmitting far-end equipment **Then**

is out of specification	adjust or replace the far-end equipment. Go to the next step.
is within specification	go to step 5 .

4	If the alarm	Then
	clears	you have completed this procedure.
	does not clear	go to the next step.
5	Reseat the module. Refer to Removing, replacing, or reseating equipment .	
6	If the alarm	Then
	clears	you have completed this procedure.
	does not clear	go to the next step.
7	Replace the module. Refer to Removing, replacing, or reseating equipment .	
8	If the alarm	Then
	clears	you have completed this procedure.
	does not clear	Reseat the module. Go to the next step.
9	If the alarm does not clear, contact your next level of support or your Ciena support group.	

Procedure 226 Rx Remote Fault

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 326 Alarm details

Alarm TID	103
Probable Cause	This alarm is raised when the client interface receives an indication that its transmitted signal is not being received at the far end.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Identify the module raising the alarm.
- Log into the remote network element at the transmit end. Retrieve all alarms from the remote network element at the transmit end.
- | If there are | Then |
|---------------------------------------|---|
| no alarms at the transmit end | ensure that the equipment or the client and line facilities of the remote chassis are in-service and connected. |
| additional alarms at the transmit end | refer to the appropriate alarm clearing procedures. |

Go to [step 4](#).
- At the local network element, retrieve all alarms to determine if the original alarm has cleared.
- If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 227 PCS Loss of Alignment Marker

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 327 Alarm details

Alarm TID	104
Probable Cause	A valid alignment marker was not received for several consecutive alignment markers for the PCS lane. For more details on when this alarm is raised/cleared, refer to IEEE 802.3 Clause 82.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- Log in to the affected chassis and note the port that displays the PCS Loss of Alignment Marker alarm.
- Confirm that the FEC settings for the near-end client ports, far-end client ports, and subtending equipment are matched. If the FEC settings are not matched, change them to match.


```
ptp show ptp <slot>/<ptp>
```

```
ptp set ptp <slot>/<ptp> forward-error-correction  
<enabled | disabled>
```
- Verify that the alarm cleared.
 - If the alarm cleared, you have completed this procedure.
 - If the alarm did not clear, go to the next step.

- 4 Confirm that the rates for the near-end client ports, far-end client ports, and subtending equipment are matched. If the rates do not match, change them to match.

```
port show port <slot>/<port>
```

```
xcvr set xcvr <slot>/<xcvr> mode <100GE | 400GE | OTL4.4>
```
- 5 Verify that the alarm cleared.
 - If the alarm cleared, you have completed this procedure.
 - If the alarm did not clear, go to the next step.
- 6 If the alarm does not clear, inspect the fiber connected to the port:
 - Replace the fiber-optic cable if it is damaged.
 - Ensure that the cable connectors are clean and the cable is properly connected.
- 7 Verify that the alarm cleared.
 - If the alarm cleared, you have completed this procedure.
 - If the alarm did not clear, go to the next step.
- 8 Inspect far-end client equipment and clear alarms on the far-end client equipment.
- 9 If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-attach it to the module. Additionally, if required, replace the plug.
- 10 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 228 PCS Loss of Block Lock

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You observe all the safety requirements described in [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 328 Alarm details

Alarm TID	105
Probable Cause	This alarm is raised when a loss of block lock occurs on a logical or an individual lane. For more details on when this alarm is raised/cleared, refer to IEEE 802.3 Clause 82.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical

Steps

- 1 Check the ptp status and verify that number of lanes must match at both ends:

```
ptp show ptp <slot/port> status
```

If the number of lanes doesn't match, change the plug at one end to make them compatible with each other.
- 2 Confirm that the FEC settings for the near-end, far-end client ports, and subtending equipment are matched. If the FEC settings are not matched, change them to match.

```
ptp show ptp <slot>/<ptp>
ptp set ptp <slot>/<ptp> forward-error-correction
<enabled | disabled>
```


- 3 Verify that the alarm cleared.
`alarm show`
- 4 If the alarm does not clear, confirm that the rates for the near-end client ports, far-end client ports, and subtending equipment are matched. If the rates do not match, change them to match.
`port show port <slot>/<port>`
`xcvr set xcvr <slot>/<xcvr> mode <100GE | 400GE | OTL4.4 | 4x100GE-ZR>`
- 5 Verify that the alarm cleared.
`alarm show`
- 6 If the alarm does not clear, clean all connections at both ends of the optical fiber link and re-attach the optical fibers.
- 7 Verify that the alarm cleared.
`alarm show`
- 8 If the alarm does not clear, inspect the fiber connected to the port:
 - Replace the fiber-optic cable if it is damaged.
 - Ensure that the cable connectors are clean and the cable is properly connected.
- 9 If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-insert it. Additionally, if required, replace the plug.
- 10 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 229 Deskew Failed

Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity. If needed, you have a replacement QSFP28.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 329 Alarm details

Alarm TID	107
Probable Cause	The skew exceeds the device capability or duplicate alignment markers were detected
Instance	Port-<slot>/<port> where slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical, service-affecting (C, SA) alarm Minor, non-service-affecting (m, NSA) alarm

Steps

- Determine the type of client service on the subtending equipment connected to the line port.
- Determine the facility provisioned on the port reporting the alarm.
- | | |
|--|------------------------|
| If the subtending equipment noted in step 1 is | Then go to |
| not provisioned with a service as required by the facility noted in step 2 | step 4 |
| provisioned with a service as required by the facility noted in step 2 | step 6 |
- If possible, provision the subtending equipment for client services, otherwise contact your next level of support or your Ciena support group.

- | | If the original alarm | Then |
|----------|------------------------------|------------------------------|
| 5 | has cleared | the procedure is complete |
| | has not cleared | go to step 6 |
- 6** Wear an antistatic wrist strap to protect the shelf from static damage. Connect the wrist strap to the ESD jack on the shelf.
- 7** If applicable, ensure that the optical transceiver is the correct type. If required, replace the optical transceiver. See [Removing, replacing, or reseating equipment](#).
- 8** If the alarm does not clear, inspect the fiber connected to the port:
- Replace the fiber-optic cable if it is damaged.
 - Ensure that the cable connectors are clean and the cable is properly connected.
- 9** If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-insert it. Additionally, if required, replace the plug.
- 10** If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 230 Skew Out of Range

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

Overview

Table 330 Alarm details

Alarm TID	108
Probable Cause	There is a difference in propagation time between two physically separate paths that are sharing traffic, so the data cannot be fully recovered at the far end.
Instance	Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18 Note: Only odd slots are used for this release.
Severity	Critical, service-affecting

Steps

- 1 Loopback the client port using a patch cord fiber.
- 2 Check the alarm status:
alarm show

If the alarm has	Then go to
cleared	step 3
not cleared	step 6
- 3 Restore the original configuration.
- 4 Check the alarm status:
alarm show

If the alarm has	Then
cleared	the procedure is complete
not cleared	go to step 5
- 5 The problem is caused by the subtending equipment. Troubleshoot the subtending equipment.
Go to [step 13](#).

6 Replace the pluggable transceiver. See [Removing, replacing, or reseating equipment](#).

7 Loopback the client port using a patch cord fiber.

8 Check the alarm status:

`alarm show`

If the alarm has	Then go to
cleared	step 9
not cleared	step 13

9 Restore the original configuration.

10 Check the alarm status:

`alarm show`

If the alarm has	Then
cleared	the procedure is complete
not cleared	go to step 13

11 If the alarm does not clear, inspect the fiber connected to the port:

- Replace the fiber-optic cable if it is damaged.
- Ensure that the cable connectors are clean and the cable is properly connected.

12 If the alarm does not clear and the port has a pluggable transceiver type then remove the plug, clean it, and re-insert it. Additionally, if required, replace the plug.

13 If the alarm does not clear, contact your next level of support or your Ciena support group.

Procedure 231 Link Down

Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.



CAUTION

Risk of damage to the chassis, modules and pluggable transceivers

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.



DANGER

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

Overview

Table 331 Alarm details

Alarm TID	109
Probable Cause	The link on a line or client port is lost, the link on the management network interface (DCN or ILAN) is lost. For the loss of the DCN or ILAN link, this alarm is raised when the administrative state of the management network is up, but the operating state is down. When the Link Down alarm is raised against the management network facility, the link cannot carry traffic. The Link Down alarm raised against a management network masks the Link Down alarm against an Ethernet facility. This alarm is raised on an IFP0 instance when the link is down for the GCC interface, caused by a disconnected or damaged fiber. Note: When the administrative state of a port is disabled, alarm reporting is inhibited on that port, and any link down alarms raised against that port will clear. Conversely, when the administrative state of a port is enabled, the link down alarm will be raised or cleared based on the port's operational state.
Instance	DCN-1, DCN-2, DCN-3, ILAN-1, ILAN-2, IFP0
Severity	Major

Steps

1	If the Link Down alarm is on a	Then go to
	Line port	step 2
	Client port	step 13
	DCN or ILAN port	step 21
	IFP0 port	step 25

Link Down (on line port)

- 2 At the local and remote ends, check the administrative state of the port and the PTP, and change the status to enabled if they are disabled:


```
port show port <slot>/<port>
port enable port <portList>
ptp show ptp <slot>/<ptp>
ptp enable ptp <ptpList>
```
- 3 Check the alarm status:


```
alarm show
```
- 4 If the alarm does not clear, use the following command to view status information for the PTP:


```
ptp show ptp <slot>/<ptp> status
```

If any optical defects are indicated, take appropriate action to clear them:

 - Rx LOS – ensure that the cables are properly connected, replace damaged cables. Check plugs and replace if defective.
 - Rx LOL – for future use
 - Tx LOS – for future use
 - Tx LOL – for future use
- 5 If the line system is colourless, use the following command to check that the channel power is within range:


```
xcvr show xcvr <slot>/<xcvr>
```

Note: Refer to [Line port specifications](#) for details about channel specifications.
- 6 If the alarm does not clear, check the performance monitoring statistics Errored Seconds - ODU, Severely Errored Seconds - ODU, and Unavailable Seconds - ODU for that port. If any of these performance monitoring counters increases, visually inspect the cable connected to the port:
 - Ensure that each end of the cable is properly connected.
 - Replace the cable if it is damaged.
- 7 If the alarm does not clear, observe the port LEDs. If the LEDs are not green, visually inspect the cable connected to the port:
 - Ensure that each end of the cable is properly connected.
 - Replace the cable if it is damaged.

- 8 For line ports 1 and 2 only, if the alarm does not clear use the following command to view diagnostic information for the PTP:
- ```
ptp show ptp <slot>/<ptp> status
```
- If any OTU4 defects (OTU4: LOC, Freq OOR, LOF, LOMF, Pre FEC SD, FEC SF) are indicated, check that the provisioned line rate and modulation at the far-end line port matches the line port rate and modulation at the near-end line port.
- Note:** If the OTU4 defect persists after taking corrective measures, or if any ODU4 defects are indicated, contact your next level of support or your Ciena support group.
- 9 If the alarm does not clear, use the following command to view diagnostic information for the PTP:
- ```
ptp show ptp <slot>/<ptp> status
```
- ```
xcvr show xcvr <slot>/<xcvr> diagnostics
```
- 10 If any Ethernet defects are indicated, take appropriate action to clear them:
- PMA:OOL - contact your next level of support or your Ciena support group.
  - FEC:LOA - contact your next level of support or your Ciena support group.
  - PCS:LOAM, LOBL, HIBER - contact your next level of support or your Ciena support group.
  - RS:LF - this defect is raised when the client Rx interface detects a IEEE 802.3 for the local fault. Check the subtending client equipment provisioning.
  - RS:RF - this defect is raised when the client Rx interface detects a IEEE 802.3 for the remote fault. Check the subtending client equipment provisioning.
- 11 If the alarm does not clear, check for and clear any remote-end-related alarms.
- 12 If the alarm does not clear, contact your next level of support or your Ciena support group.

***Link Down (on client port)***

- 13 At the local and remote ends, check the administrative state of the port, the PTP, and the transceiver, and change the status to enabled if they are disabled:
- ```
port show port <slot>/<port>
```
- ```
port enable port <portList>
```
- ```
ptp show ptp <slot>/<ptp>
```
- ```
ptp enable ptp <ptpList>
```
- ```
xcvr show
```
- ```
xcvr enable xcvr <slot>/<xcvr>
```



- 14** Check the alarm status:
- ```
alarm show
```
- 15** If the alarm does not clear, use the following command to view diagnostic information for the PTP:
- ```
ptp show ptp <slot>/<ptp> status
```
- ```
xcvr show xcvr <slot>/<xcvr> diagnostics
```
- If any optical defects are indicated, take appropriate action to clear the fault:
- Rx LOS, Rx LOL – ensure that the cables are properly connected, replace damaged cables. Check plugs and replace if defective.
 - Tx LOS, Tx LOL – Check plugs and replace if defective.
- 16** If the alarm does not clear, observe the port LEDs. If the LEDs are not green, visually inspect the cable connected to the port:
- Ensure that each end of the cable is properly connected.
 - Replace the cable if it is damaged.
- 17** If the alarm does not clear, use the following command to view diagnostic information for the PTP:
- ```
ptp show ptp <slot>/<ptp> status
```
- If any Ethernet defects are indicated, take appropriate action to clear them:
- PMA:OOL - verify the connectivity to the subtending equipment and the rate configuration for the subtending equipment (i.e. ensure you are connecting 10GE to 10GE or 100GE to 100GE).
  - FEC:LOA - ensure FEC is enabled on subtending equipment.
  - PCS:LOAM, LOBL: - verify the connectivity to the subtending equipment and the rate configuration for the subtending equipment (i.e. ensure you are connecting 10GE to 10GE or 100GE to 100GE).
  - PCS:HIBER - clean Rx optics.
  - RS:LF - this defect is raised when the client Rx interface detects a IEEE 802.3 for the local fault. Check the subtending client equipment provisioning.
  - RS:RF – the Waveserver 5 is conditioning LF towards subtending equipment or there is a problem in the Tx direction from Waveserver 5 to subtending equipment.
- 18** If the alarm does not clear, check for and clear any remote-end-related alarms.
- 19** If the alarm does not clear, verify that there are no faults on the line port where the service domain is attached.
- 20** If the alarm does not clear, contact your next level of support or your Ciena support group.

**Link Down on DCN or ILAN**

- 21 Check the administrative state of the management interface, and enable it if it is disabled:
- ```
chassis port show
chassis port enable port <dcn | ilan-1 | ilan-2>
```
- 22 Check the alarm status:
- ```
alarm show
```
- 23 If the alarm does not clear, observe the LEDs on the management interface. If the LEDs are off, visually inspect the cable connected to the interface:
- Ensure that the cable is properly connected.
  - Replace the cable if it is damaged.
- 24 If the alarm does not clear, contact your next level of support or your Ciena support group.

**Link Down on IFP0**

- 25 Clean the near-end and far-end fiber connectors, ensure they are not pinched, folded and are fully seated for the interface ( ifp0-<slot>-<port>) raising the alarm.
- 26 Check the alarm status:
- ```
alarm show
```
- 27 If the alarm does not clear, replace the fiber.
- 28 Check the alarm status:
- ```
alarm show
```
- 29 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 232 Port Rx Loopback

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.

### Overview

**Table 332** Alarm details

|                |                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Alarm TID      | 110                                                                                                       |
| Probable Cause | Rx loopback is operating on a client or line port.                                                        |
| Instance       | Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Minor                                                                                                     |

### Steps

- 1 If the alarm is raised during testing, no action is required. The alarm will clear once the testing is complete and the loopback is released.
- 2 Display summary information about all provisioned ports to determine where the loopback is happening:  
`port show`
- 3 Note the port number of the affected port.
- 4 Restore the affected port to the default disabled loopback state:  
`port unset port <portList> loopback`
- 5 Set the administrative state of the port to enabled to restore end-to-end traffic on the port:  
`port enable port <portList>`
- 6 Display summary information about all provisioned ports to confirm that the loopback has been removed:  
`port show`
- 7 Check the alarm status:  
`alarm show`
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 233 Port Tx Loopback

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.

### Overview

**Table 333** Alarm details

|                |                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Alarm TID      | 111                                                                                                       |
| Probable Cause | Tx loopback is operating on a client or line port.                                                        |
| Instance       | Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Minor                                                                                                     |

### Steps

- If the alarm is raised during testing, no action is required. The alarm clears once the testing is complete and the loopback is released.
- Display summary information about all provisioned ports to determine where the loopback is happening:  
`port show`
- Note the port number of the affected port.
- Restore the affected port to the default disabled loopback state:  
`port unset port <portList> loopback`
- Set the administrative state of the port to enabled to restore end-to-end traffic on the port:  
`port enable port <portList>`
- Display summary information about all provisioned ports to confirm that the loopback has been removed:  
`port show`
- Check the alarm status:  
`alarm show`
- If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 234 Port Test Mode Active

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.

### Overview

**Table 334** Alarm details

|                |                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Alarm TID      | 112                                                                                                       |
| Probable Cause | The Port Test Mode is active on the port.                                                                 |
| Instance       | Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Minor                                                                                                     |

### Steps

- If the alarm is raised during testing, no action is required. The alarm clears when the testing is complete and the Port Test Mode is disabled. If the alarm does not clear, go to [step 2](#).
- Display summary information about all provisioned ports to determine where the Port Test Mode is occurring:  
`application its show`
- Note the port number of the affected port.
- Restore the affected port to the default by disabling the Port Test Mode state:  
`application its disable port <portList>`

**Note 1:** See [Port Tx Loopback](#) if Port Tx Loopback is enabled.

**Note 2:** If the port is currently being monitored, you may want to stop monitoring and retrieve the monitor logs before disabling as follows:

```
application its stop port <portList>
application its upload-detailed-logs port <portList>
{scp-server remote-path <String[1..127]> suffix
<String[1..127]> login-id <String[1..32]> password
<Password String[1..128]> | url <String> suffix
<String[1..127]> login-id <String[1..32]> password
<Password String[1..128]>}
```

For more information on retrieving monitor logs, see [Retrieving and saving the test report and logs after performing a test using the Integrated Test Set](#).

- 5 Set the administrative state of the port to enabled to restore end-to-end traffic on the port:  
`port enable port <portList>`
- 6 Display summary information about all provisioned ports to confirm that the Port Test Mode has been removed:  
`application its show`
- 7 Check the alarm status:  
`alarm show`
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 235 Loss of Adjacency

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 335** Alarm details

|                |                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 113                                                                                                                                                                      |
| Probable Cause | This alarm is raised by a LOM mismatch on the line and a Service Profile mismatch. The far end adjacency cannot be established on the remote management interface (GCC). |
| Instance       | IF0-<slot>/<port>, where slot = 1-8, port = 1 - 3<br><b>Note:</b> Only odd slots are used for this release.                                                              |
| Severity       | Major                                                                                                                                                                    |

### Steps

- 1 Identify the associated line XCVR, PTP and port that is alarming.
- 2 Check alarm status and troubleshoot any alarms raised against the line XCVR, PTP or port. Use the alarm clearing procedures in this chapter.  
`alarm show`
- 3 Identify the near end remote management interface that is alarming and the associated far end remote management interface.  
`interface show interface if0-<slot>-<port>`
- 4 Confirm that the far end interface port and the far end remote management interface is enabled.  
`interface show`  
`interface port show`  
`interface show interface if0-<slot>-<port>`
- 5
 

|                                                                   |                                          |
|-------------------------------------------------------------------|------------------------------------------|
| <b>If</b> the far end interface port is administratively disabled | <b>Then</b> go to <a href="#">step 6</a> |
| enabled                                                           | <a href="#">step 7</a>                   |
- 6 Enable interface port.  
`interface port enable port ifp0-<slot>-<port>`
- 7
 

|                                                               |                                          |
|---------------------------------------------------------------|------------------------------------------|
| <b>If</b> the far end remote management interface is disabled | <b>Then</b> go to <a href="#">step 8</a> |
| enabled                                                       | <a href="#">step 9</a>                   |

- 8 Enable the far end remote management interface.  
`interface enable interface if0-<slot>-<port>`
- 9 Verify any missing configuration for remote management. For supported configurations, see [Remote Group Management](#).
- 10 If the alarm does not clear, contact your next level of support or your Ciena support group.



## Procedure 236 Remote Port Disabled

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 336** Alarm details

|                |                                                                                                                                                                                                                                                                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 114                                                                                                                                                                                                                                                                   |
| Probable Cause | This alarm is raised on the service module when the far end client port is disabled.<br><b>Note:</b> Ethernet Backward Defect Indication (BDI) for E-Line transcoding is not supported and therefore not returned on the far-end equipment when this alarm is raised. |
| Instance       | Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                             |
| Severity       | Major                                                                                                                                                                                                                                                                 |

### Steps

- 1 Identify the client port raising the alarm and the far end client port.
- 2 On the far end client port, check the administrative status of the port, PTP and transceiver and change the status to enabled if they are disabled.  
  

```
port show port <slot>/<port>
port enable port <slot>/<port>
ptp show ptp <slot>/<ptp>
ptp enable ptp <slot>/<ptp>
xcvr show xcvr <slot>/<xcvr>
xcvr enable xcvr <slot>/<xcvr>
```
- 3 Check alarm status.  
  

```
alarm show
```
- 4 If the alarm does not clear, contact your next level of support or your Ciena support group.

---

## Procedure 237 Remote Port Down

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 337** Alarm details

|                |                                                                                                                                                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 115                                                                                                                                                                                                                                                               |
| Probable Cause | This alarm is raised on the service module when the far end client port is down.<br><b>Note:</b> Ethernet Backward Defect Indication (BDI) for E-Line transcoding is not supported and therefore not returned on the far-end equipment when this alarm is raised. |
| Instance       | Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                         |
| Severity       | Critical                                                                                                                                                                                                                                                          |

### Steps

- 1 Identify the client port raising the alarm and the far end client port.
- 2 On the near and far end Waveserver 5, check the alarm status.  
`alarm show`
- 3 Troubleshoot the alarms accordingly, using the alarm clearing procedures in this chapter.
- 4 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 238 Remote Port Unreachable

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 338** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 116                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Probable Cause | This alarm is raised on the service module's client port when it does not receive the link state signaling from the corresponding far-end client port. If the corresponding near-end client port is not provisioned, the alarm is raised on the far-end client port.<br><b>Note:</b> Ethernet Backward Defect Indication (BDI) for E-Line transcoding is not supported and therefore not returned on the far-end equipment when this alarm is raised. |
| Instance       | Port-<slot>/<port>, slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                                                                                                                                                                                                             |
| Severity       | Critical                                                                                                                                                                                                                                                                                                                                                                                                                                              |

### Steps

- 1 Identify the client port raising the alarm and the far-end client port.
- 2 Verify the near-end and far-end client port fiber connections. Repair, clean, and reconnect the fibers as required.
- 3 On the near-end and far-end Waveserver 5, check alarm status.  
`alarm show`  

| If the alarm has | Then                         |
|------------------|------------------------------|
| cleared          | the procedure is complete    |
| not cleared      | go to <a href="#">step 4</a> |
- 4 Check the client port provisioning on the far-end client port. The XCVR mode should be set and the modes on the near-end and far-end client ports should match.  
`xcvr show xcvr <slot>/<xcvr> status`
- 5 On the near-end client port, set the XCVR mode to match the setting on the far-end client port.  
`xcvr set xcvr <slot>/<xcvr> mode <100GE | 400GE | OTL4.4>`
- 6 Check the alarm status.  
`alarm show`
- 7 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 239 TCA Summary (Port)

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 339** Alarm details

|                |                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Alarm TID      | 117                                                                                                       |
| Probable Cause | This alarm is raised when there is a PM threshold crossing alert on the Port.                             |
| Instance       | Port-<slot>/<port>, slot = 1 - 8, port = 1 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Warning                                                                                                   |

### Steps

- 1 Review the 15-min, 24-h, and Untimed PMs for the port raising the alarm and identify which PM raised the alarm.  
`pm show port <slot>/<port> bin-type <15-min | 24-hr | untimed>`
- 2 Use the PM TCA alarm clearing procedure in this chapter, for the PM identified in the previous step to clear this alarm.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 240 Far End Client Signal Fail

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.
- You have an optical power meter with the same optical connectors as the network element.
- You have the optical fiber connection information (how the optical modules and network elements are connected to each other) and have a fiber cleaning kit.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



#### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 340** Alarm details

|                |                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 121                                                                                                                                                                     |
| Probable Cause | This alarm is raised when the far-end port for the service is experiencing a client Rx signal fault.                                                                    |
| Instance       | Channel- <code>&lt;slot&gt;/&lt;port&gt;.&lt;channel&gt;</code> , slot = 1 - 8, port = 1 - 3, channel = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Critical, service-affecting (C, SA) alarm                                                                                                                               |

### Steps

- 1 Verify the fibers are connected correctly and not crossed, looped back, or misconnected at the corresponding far-end module.
- 2 Retrieve alarms from the corresponding far-end Waveserver 5.
- 3 Use the appropriate alarm clearing procedure to clear any alarms raised at the far-end module.
- 4 If the alarms do not clear, contact your next level of support or your Ciena support group.

## Procedure 241 Bandwidth Not Available

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine the SPLI connectivity.

### Overview

**Table 341** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 122                                                                                                                                                                                                                                                                                                                                                        |
| Probable Cause | This alarm is raised when the mode value at the line port is set to less than 800G. The alarm is raised for the channels that cannot be used. For example, at 600G, channels 7 and 8 cannot be used; at 500G, channels 6, 7, and 8 cannot be used; at 400G, channels 5, 6, 7, and 8 cannot be used; at 200G, channels 3, 4, 5, 6, 7, and 8 cannot be used. |
| Instance       | Channel- <code>&lt;slot&gt;/&lt;port&gt;.&lt;channel&gt;</code> , slot = 1 - 8, port = 1 - 3, channel = 1 - 8<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                                                    |
| Severity       | Minor                                                                                                                                                                                                                                                                                                                                                      |

### Steps

- 1 Confirm that the line port is set to less than 800G, and that it is the correct setting.  
`xcvr show xcvr <slot>/<xcvr> diagnostics`
- 2 Disable the unused channels on the client port.  
`channel disable channel <slot>/<port>.<channel>`
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

---

## Procedure 242 TCA Summary (Channel)

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 342** Alarm details

|                |                                                                                                                                        |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 126                                                                                                                                    |
| Probable Cause | This alarm is raised when a PM threshold crossing alert is raised on a channel.                                                        |
| Instance       | Channel-<slot>/<port>.<channel>, slot = 1 - 8, port = 1 - 3, channel = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Warning                                                                                                                                |

### Steps

- 1 Review the 15-min, 24-h, and Untimed PMs for the channel raising the alarm and identify which PM raised the alarm.  

```
pm show channel <slot>/<port>.<channel> bin-type <15-min | 24-hr | untimed>
```
- 2 Use the PM TCA alarm clearing procedure in this chapter, for the PM identified in the previous step to clear this alarm.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 243 External License Server Comms Failed

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 343** Alarm details

|                |                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 131                                                                                                        |
| Probable Cause | This alarm is raised after three successive attempts to communicate with the external license server fail. |
| Instance       | Chassis                                                                                                    |
| Severity       | Major                                                                                                      |

### Steps

- 1 Verify network connections and communication.
  - a. Verify that the Waveserver 5 and the external license server can communicate. Ping the Waveserver 5 from the external license server, and ping the external license server from the Waveserver 5.
  - b. Verify the communication protocol and its corresponding port number are aligned, by using the following command:
 

```
license client server show
```

For HTTPS the port should be 7071 and for HTTP it should be 7072.
  - c. Verify the physical connection of the Waveserver 5 to the network is correct.
  - d. Verify the physical connection of the external license server to the network is correct.
- 2 If the network connections and communication are working use the following command to perform an audit between Waveserver 5 and the external license server.
 

```
license client server sync-now
```
- 3 If the warning does not clear, perform a warm restart on the Waveserver 5. For more information, see [Restarting Waveserver 5](#).
- 4 If the failure persists, contact your next level of support or your Ciena support group.



---

## Procedure 244 External License Server Time Out Of Sync

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 344** Alarm details

|                |                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 132                                                                                                                                                 |
| Probable Cause | This alarm is raised when the difference between the Waveserver 5 system time and the external license server system time is greater than 24 hours. |
| Instance       | Chassis                                                                                                                                             |
| Severity       | Major                                                                                                                                               |

### Steps

- 1 Reset the system time of the Waveserver 5, the external license server, or both devices so that the time difference is within the 24-hour tolerance. For information on setting the system clock of the Waveserver 5, see [Network Time Protocol \(NTP\)](#).
- 2 If the problem persists, contact your next level of support or your Ciena support group.

## Procedure 245 License Missing

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 345** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 135                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Probable Cause | <p>This alarm is raised when:</p> <ul style="list-style-type: none"> <li>• a purchased software license is not installed on Waveserver 5.</li> <li>• you changed the Registration ID after you applied a permanent client-based license.</li> <li>• there are communication issues with external license server, if configured.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Instance       | <p>License-n, where n = 1, 17 - 35</p> <p>Numbers correspond to the licenses as follows:</p> <ul style="list-style-type: none"> <li>• License-01 — Waveserver-OSSWR</li> <li>• License-17 — Waveserver-OSFRGNCLPLUG</li> <li>• License-18 — Waveserver-OSFRGNLNPLUG</li> <li>• License-19 — Waveserver-OSAPIS</li> <li>• License-20 — Waveserver-OSITS</li> <li>• License-21 — WaveLogic5e-TERADD100G</li> <li>• License-22 — WaveLogic5e-TER250KM</li> <li>• License-23 — WaveLogic5e-TER500KM</li> <li>• License-24 — WaveLogic5e-TER1000KM</li> <li>• License-25 — WaveLogic5e-TER2000KM</li> <li>• License-26 — WaveLogic5e-TER2100KM</li> <li>• License-27 — WaveLogic5e-TER3000KM</li> <li>• License-28 — WaveLogic5e-TER4000KM</li> <li>• License-29 — WaveLogic5e-TER5000KM</li> <li>• License-30 — WaveLogic5e-TER8000KM</li> <li>• License-31 — WaveLogic5e-SUBADD100G</li> <li>• License-32 — WaveLogic5e-SUB3500KM</li> <li>• License-33 — WaveLogic5e-SUB9000KM</li> <li>• License-34 — WaveLogic5e-SUB14000KM</li> <li>• License-35 — WaveLogic5e-BWADD100G</li> </ul> |
| Severity       | Major                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**Steps**

- 1**      Verify that the software license is actually installed:  
`license client show inventory all`  
If the correct software license is not installed, acquire the license and install it.
- 2**      Verify that the Registration ID is correct:  
`license client show`  
If the Registration ID was changed, change it back to the original ID.  
`license client set registration-id <Registration ID>`
- 3**      If any Waveserver 5 in your network obtains licenses from an external license server, determine whether alarms have been raised against the license server, and clear them:  
`alarm show active`  
See [External License Server Comms Failed](#) and [External License Server Time Out Of Sync](#).
- 4**      To purchase or acquire a new Waveserver 5 software license, contact Ciena.  
For details on acquiring and installing software licenses, see [Licensing](#).

## Procedure 246 License Expiration Warning

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 346** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 136                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Probable Cause | The Waveserver 5 software license is about to expire.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Instance       | <p>License-n, where n = 1, 17 - 35</p> <p>Numbers correspond to the licenses as follows:</p> <ul style="list-style-type: none"> <li>• License-01 — Waveserver-OSSWR</li> <li>• License-17 — Waveserver-OSFRGNCLPLUG</li> <li>• License-18 — Waveserver-OSFRGNLNPLUG</li> <li>• License-19 — Waveserver-OSAPIS</li> <li>• License-20 — Waveserver-OSITS</li> <li>• License-21 — WaveLogic5e-TERADD100G</li> <li>• License-22 — WaveLogic5e-TER250KM</li> <li>• License-23 — WaveLogic5e-TER500KM</li> <li>• License-24 — WaveLogic5e-TER1000KM</li> <li>• License-25 — WaveLogic5e-TER2000KM</li> <li>• License-26 — WaveLogic5e-TER2100KM</li> <li>• License-27 — WaveLogic5e-TER3000KM</li> <li>• License-28 — WaveLogic5e-TER4000KM</li> <li>• License-29 — WaveLogic5e-TER5000KM</li> <li>• License-30 — WaveLogic5e-TER8000KM</li> <li>• License-31 — WaveLogic5e-SUBADD100G</li> <li>• License-32 — WaveLogic5e-SUB3500KM</li> <li>• License-33 — WaveLogic5e-SUB9000KM</li> <li>• License-34 — WaveLogic5e-SUB14000KM</li> <li>• License-35 — WaveLogic5e-BWADD100G</li> </ul> |
| Severity       | <ul style="list-style-type: none"> <li>• Minor: License will expire within 2 days (if license is sourced from external license server)</li> <li>• Warning: License will expire within 45 days</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### Steps

- 1 If you configured an external license server in your Waveserver 5 network, clear any alarms raised against the license server:  
`alarm show active`  
See [External License Server Comms Failed](#) and [External License Server Time Out Of Sync](#).
- 2 Contact Ciena to purchase a Waveserver 5 software license.

---

## Procedure 247 Entitlement Enforcement Disabled

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

This is a non service-affecting alarm.

**Table 347** Alarm details

|                |                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 137                                                                                                                                                                              |
| Probable Cause | This alarm is raised when no source of license information is provided. This alarm clears on activating a license file or configuring a license server on the Waveserver 5 node. |
| Instance       | Chassis                                                                                                                                                                          |
| Severity       | Warning                                                                                                                                                                          |

### Steps

- 1 Activate the appropriate licenses on the Waveserver 5 node. Refer to the following sections to activate licenses on the Waveserver 5 node:
  - a. [Procedure 6, “Installing and activating a software license”](#)
  - b. [Procedure 7, “Provisioning access to an external license server”](#)
- 2 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 248 Rx Dispersion Compensation Out of Range

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 348** Alarm details

|                |                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------|
| Alarm TID      | 144                                                                                                   |
| Probable Cause | The receive dispersion compensation is out of range.                                                  |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Major                                                                                                 |

### Steps

- On the far-end Waveserver 5, ensure the correct Tx PTP dispersion settings have been set. Change the Tx PTP dispersion settings, if required.  

```
ptp show ptp <slot>/<ptp>
ptp set ptp <slot>/<ptp> [tx-dispersion-compensation-mode
<automatic | fixed | search>] [tx-dispersion-
precompensation-value <NUMBER: -44500..1000000>] [tx-
dispersion-path-value <NUMBER: -89000..1044500>] [tx-
dispersion-search-mode-minimum <NUMBER: -
89000..1044500>] [tx-dispersion-search-mode-maximum
NUMBER: -89000..1044500]
```
- | If the original alarm is | Then                         |
|--------------------------|------------------------------|
| cleared                  | the procedure is complete    |
| not cleared              | go to <a href="#">step 3</a> |
- If the alarm did not clear, contact your next level of support or your Ciena support group.

---

## Procedure 249 Laser off active

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 349** Alarm details

|                |                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 145                                                                                                                                                    |
| Probable Cause | The laser off is active.                                                                                                                               |
| Instance       | PTP-<slot>/<ptp><br>slot = 1, 3, 5 and 7<br>ptp depends on the module type, for example, the WL5n DCO 3xQSFP-DD 12x module ptp numbers are 1,2, and 3. |
| Severity       | Major                                                                                                                                                  |

### Steps

- 1 Check on the far-end Waveserver 5 for faults and troubleshoot according to the alarms raised. Continue with the next step if the alarm does not clear.
- 2 If the alarm does not clear, contact your next level of support or your Ciena support group.



---

## Procedure 250 TCA Summary (PTP)

---

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 350** Alarm details

|                |                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------|
| Alarm TID      | 146                                                                                                    |
| Probable Cause | This alarm is raised when a PM threshold crossing alert is raised on a PTP.                            |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity       | Warning                                                                                                |

### Steps

- 1 Review the 15-min, 24-h, and Untimed PMs for the PTP raising the alarm and identify which PM raised the alarm.  

```
pm show ptp <slot>/<ptp> bin-type <15-min | 24-hr |
untimed>
```
- 2 Use the PM TCA alarm clearing procedure in this chapter, for the PM identified in the previous step to clear this alarm.
- 3 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 251 TCA Q-Factor

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a fiber patchcord, if required.
- You have a fiber cleaning kit, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 351** Alarm details

|                |                                                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 153                                                                                                                                                                                                 |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for the Q-Factor.<br><br>A PM threshold has been set incorrectly, a dirty fiber, a faulty fiber, or a misconnection in the fiber. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><br><b>Note:</b> Only odd slots are used for this release.                                             |
| Severity       | Warning                                                                                                                                                                                             |

**Steps**

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold, if required.  

```
pm tca show profile-type Modem
pm tca Modem set profile-name <profile name> Rx-FecQ
<Threshold 0.000 .. NN.mmm>
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4 Inspect and clean the alarmed line port fibers. Replace fiber patchcord, if required. Ensure fibers are seated correctly and not misconnected.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 252 TCA DGD

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of equipment damage**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 352** Alarm details

|                |                                                                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 154                                                                                                                                                                                                           |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for the Differential Group Delay.<br>A PM threshold has been set incorrectly or a line port is connected to an incorrect far end line port. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><b>Note:</b> Only odd slots are used for this release.                                                           |
| Severity       | Warning                                                                                                                                                                                                       |

**Steps**

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold, if required.  

```
pm tca show profile-type Modem
pm tca Modem set profile-name <profile name> DGD
<Threshold 0.000 .. NN.mmm>
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4 Verify line port is connected to the correct far end line port using the TTI feature. See [Assigning, displaying, and clearing TTI settings](#).
- 5 Connect to correct line port, if required.
- 6 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 7 If the alarm does not clear, contact your next level support or your Ciena support group.

## Procedure 253 TCA PDL

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have fiber patchcord, if required.
- You have a fiber cleaning kit, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 353** Alarm details

|                |                                                                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 155                                                                                                                                                                                                                    |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for the Polarization Dependent Loss.<br><br>A PM threshold has been set incorrectly, a dirty fiber, a faulty fiber, or a misconnection in the fiber. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><br><b>Note:</b> Only odd slots are used for this release.                                                                |
| Severity       | Warning                                                                                                                                                                                                                |

**Steps**

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold, if required.  

```
pm tca show profile-type Modem
pm tca Modem set profile-name <profile name> PDL
<Threshold 0.000 .. NN.mmm>
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4 Inspect and clean the alarmed line port. Replace the fiber patchcord, if required. Ensure fibers are seated correctly and not misconnected.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level support or your Ciena support group.

## Procedure 254 TCA ESNR

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have fiber patchcord, if required.
- You have a fiber cleaning kit, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 354** Alarm details

|                |                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 156                                                                                                                                                                                                                     |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for the Electrical Signal to Noise Ratio.<br>A PM threshold has been set incorrectly, a dirty fiber, a faulty fiber, or a misconnection in the fiber. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><b>Note:</b> Only odd slots are used for this release.                                                                     |
| Severity       | Warning                                                                                                                                                                                                                 |



**Steps**

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold, if required.  

```
pm tca show profile-type Modem
pm tca Modem set profile-name <profile name> ESNR
<Threshold 0.000 .. NN.mmm>
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4 Use an optical power meter to measure the power level on the alarmed line port and ensure power levels are as expected. Inspect and clean the alarmed line port fibers. Replace fiber patchcord, if required.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level support or your Ciena support group.

## Procedure 255 TCA Chromatic Dispersion

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 355** Alarm details

|                |                                                                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 158                                                                                                                                                                                                                      |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for the chromatic dispersion.<br>A PM threshold has been set incorrectly or the line port PTP parameters for chromatic dispersion are set incorrectly. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><b>Note:</b> Only odd slots are used for this release.                                                                      |
| Severity       | Warning                                                                                                                                                                                                                  |

### Steps

- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- Verify the appropriate PM threshold is set. Change the PM threshold, if required.  
  

```
pm tca show profile-type Modem
pm tca Modem set profile-name <profile name> CD <NUMBER: 0..2000000>
```
- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.

- 4 Verify that the line port PTP parameters for chromatic dispersion compensation are set correctly. Change the values, if required. See below:
  - For WL5e 2x MOTR 16x module: [Table 122, “Line port PTP parameters - chromatic dispersion compensation details”](#)
  - For WL5e 2x MOTR 12x module: [Table 138, “Line port PTP parameters - chromatic dispersion compensation details”](#)
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level support or your Ciena support group.

## Procedure 256 TCA High Correction Count Second

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a fiber cleaning kit, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 356** Alarm details

|                |                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 159                                                                                                                                                                                                                     |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for the high correction count second.<br><br>A PM threshold has been set incorrectly, a dirty fiber, a faulty fiber, or a misconnection in the fiber. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><br><b>Note:</b> Only odd slots are used for this release.                                                                 |
| Severity       | Warning                                                                                                                                                                                                                 |

**Steps**

- 1** Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2** Verify the appropriate PM threshold is set. Change the PM threshold, if required.  

```
pm tca show profile-type Modem
```

```
pm tca Modem set profile-name <profile name> Rx-High-Corr
<NUMBER: 0..86399>
```
- 3** Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4** Clear any alarms raised on the line port raising this alarm, using the alarm clearing procedures in this chapter.
- 5** Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6** Inspect and clean the alarmed line port fibers.
- 7** Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 8** If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 257 TCA RX/TX Power High/Low Baseline

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have attenuators, if required.
- You have a fiber cleaning kit, if required.
- You have a fiber patch cord, if required.
- You have the required pluggable transceiver, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 357** Alarm details

|                               |                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Alarm Description / Condition | TCA RX Power High Baseline<br>TCA RX Power Low Baseline<br>TCA TX Power High Baseline<br>TCA TX Power Low Baseline |
| Alarm TID                     | 160, 161, 162, 163                                                                                                 |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Probable Cause | <p>This alarm is raised when the Rx power is higher than the baseline by the provisioned offset value.</p> <p>This alarm is raised when the Rx power is lower than the baseline by the provisioned offset value.</p> <p>This alarm is raised when the Tx power is higher than the baseline by the provisioned offset value.</p> <p>This alarm is raised when the Tx power is lower than the baseline by the provisioned offset value.</p> <p>The default offset value is 3 dB.</p> <p>A PM threshold and/or baseline has been set incorrectly, a dirty fiber, a faulty fiber, misconnection in the fiber or a faulty pluggable transceiver.</p> <p><b>Note:</b> These alarms are only raised if the baseline has been set.</p> |
| Instance       | <p>PTP-&lt;slot/ptp&gt;-15m, PTP-&lt;slot/ptp&gt;-24h, PTP-&lt;slot/ptp&gt;-Untimed, slot = 1 - 8, ptp = 1 - 18</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Severity       | Warning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### Steps

- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- Verify the appropriate offset and baseline are set. Change the offset and/or baseline, if required.
 

```
pm tca show profile-type OpticalPower
pm tca OpticalPower set profile-name <profile name> {Rx-
Baseline-Offset <Threshold 0.000 .. NN.mmm > | Tx-
Baseline-Offset <Threshold 0.000 .. NN.mmm >}
pm tca baseline set ptp <slot>/<ptp> [direction <Rx | Tx>]
```
- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- Use an optical power meter to measure the power level on the alarmed port. Add attenuators, if the Rx power levels are too high. Inspect and clean fibers, if the Rx power levels are too low. Replace fibers, if required.

- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 Replace pluggable transceiver, if the alarmed port is a client port. See [Replacing or reseating a pluggable transceiver](#). Otherwise, continue to the next step.
- 7 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.



## Procedure 258 TCA RX Channel Power High/Low Baseline

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have attenuators, if required.
- You have fiber cleaning kit, if required.
- You have a fiber patchcord, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 358** Alarm details

|                               |                                                                         |
|-------------------------------|-------------------------------------------------------------------------|
| Alarm Description / Condition | TCA RX Channel Power High Baseline<br>TCA RX Channel Power Low Baseline |
| Alarm TID                     | 164, 165                                                                |

|                |                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Probable Cause | <p>This alarm is raised when the Rx channel power is higher/lower than the baseline by the provisioned offset value.</p> <p>The default offset value is 3 dB.</p> <p>A PM threshold and/or baseline has been set incorrectly, a dirty fiber, a faulty fiber, or a misconnection in the fiber.</p> <p><b>Note:</b> These alarms are only raised if the baseline has been set.</p> |
| Instance       | <p>PTP-&lt;slot/ptp&gt;-15m, PTP-&lt;slot/ptp&gt;-24h, PTP-&lt;slot/ptp&gt;-Untimed, slot = 1 - 8, ptp = 1 - 3</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                                                                                                 |
| Severity       | Warning                                                                                                                                                                                                                                                                                                                                                                          |

### Steps

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate offset and baseline are set. Change the offset and/or baseline, if required.
 

```
pm tca show profile-type OpticalPower
pm tca OpticalPower set profile-name <profile name> Rx-Baseline-Offset <Threshold 0.000 .. NN.mmm >
pm tca baseline set ptp <slot>/<ptp> [direction <Rx | Tx>]
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4 Use an optical power meter to measure the power level on the alarmed port. Add attenuators, if the Rx power levels are too high. Inspect and clean fibers, if the Rx power levels are too low. Replace fibers, if required.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 259 TCA ODU BBE / TCA ODU ES / TCA ODU SES / TCA ODU UAS

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have attenuators, if required.
- You have fiber cleaning kit, if required.
- You have a fiber patchcord, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 359** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 166, 167, 168, 169                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Probable Cause | <p>This alarm is raised when the provisioned PM threshold is crossed for ODU BBE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for ODU ES.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for ODU SES.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for ODU UAS.</p> <p>A PM threshold has been set incorrectly, an ODU related alarm is raised, a dirty or faulty fiber.</p> |

|          |                                                                                                                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instance | Port-<slot/port>-15m, Port-<slot/port>-24h, Port-<slot/port>-Untimed, slot = 1 - 8, port = 1 - 18<br>Channel-<slot/port.channel>-15m, Channel-<slot/port.channel>-24h, Channel-<slot/port.channel>-Untimed, slot = 1 - 8, port = 1 - 3, channel = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity | Warning                                                                                                                                                                                                                                                                                                           |

## Steps

- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:

  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type untimed
```

Verify if the alarm has cleared.
- Verify the appropriate PM threshold setting has been set based on the alarm. Change the PM threshold setting, if required.

```
pm tca show profile-type ODU
pm tca ODU set profile-name <profile name> [BBE <NUMBER: 0..4294967295> | ES <NUMBER: 0..86399> | SES <NUMBER: 0..86399> | UAS <NUMBER: 0..86399>]
```
- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:

  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type untimed
```

Verify if the alarm has cleared.
- Clear any ODU related alarms, such as ODU LCK, ODU AIS, ODU OCI, ODU BDI, ODU Signal Degrade, and ODU Trace Identifier Mismatch. Use the alarm clearing procedures in this chapter to clear these alarms.

- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type
untimed
```

Verify if the alarm has cleared.
- 6 Use an optical power meter to measure the power level on the alarmed port. Add attenuators, if the Rx power levels are too high. Inspect and clean fibers, if the Rx power levels are too low. Replace fibers, if required.
- 7 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type
untimed
```

Verify if the alarm has cleared.
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 260 TCA ODU BBE FE / TCA ODU ES FE / TCA ODU SES FE / TCA ODU UAS FE

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have attenuators, if required.
- You have fiber cleaning kit, if required.
- You have a fiber patchcord, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 360** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 170, 171, 172, 173                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Probable Cause | <p>This alarm is raised when the provisioned PM threshold is crossed for ODU BBE FE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for ODU ES FE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for ODU SES FE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for ODU UAS FE.</p> <p>A PM threshold has been set incorrectly, a far-end ODU related alarm is raised, a dirty or faulty fiber.</p> |

|          |                                                                                                                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instance | Port-<slot/port>-15m, Port-<slot/port>-24h, Port-<slot/port>-Untimed, slot = 1 - 8, port = 1 - 18<br>Channel-<slot/port.channel>-15m, Channel-<slot/port.channel>-24h, Channel-<slot/port.channel>-Untimed, slot = 1 - 8, port = 1 - 3, channel = 1 - 8<br><b>Note:</b> Only odd slots are used for this release. |
| Severity | Warning                                                                                                                                                                                                                                                                                                           |

### Steps

- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:

  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type untimed
```

Verify if the alarm has cleared.
- Verify the appropriate PM threshold setting has been set based on the alarm. Change the PM threshold setting, if required.

```
pm tca show profile-type ODU
pm tca ODU set profile-name <profile name> [BBE-FE <NUMBER: 0..4294967295> | ES-FE <NUMBER: 0..86399> | SES-FE <NUMBER: 0..86399> | UAS-FE <NUMBER: 0..86399>]
```
- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:

  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type untimed
```

Verify if the alarm has cleared.
- On the far-end Waveserver 5, clear any ODU related alarms, such as ODU LCK, ODU AIS, ODU OCI, ODU BDI, ODU Signal Degrade, ODU Trace Identifier Mismatch. Use the alarm clearing procedures in this chapter to clear these alarms.

- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type
untimed
```

Verify if the alarm has cleared.
- 6 On the far end Waveserver 5, use an optical power meter to measure the power level on the alarmed port. Add attenuators, if the Rx power levels are too high. Inspect and clean fibers, if the Rx power levels are too low. Replace fibers, if required.
- 7 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins, depending on the alarm being raised on a port or a channel, use CLI commands:

```
pm clear port <slot>/<port> bin-type untimed
pm clear channel <slot>/<port.channel> bin-type
untimed
```

Verify if the alarm has cleared.
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.



# **Procedure 261 TCA OTU BBE / TCA OTU ES / TCA OTU SES / TCA OTU UAS**

## **Requirements**

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have attenuators, if required.
- You have fiber cleaning kit, if required.
- You have a fiber patchcord, if required.



### **DANGER**

#### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



### **CAUTION**

#### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



### **CAUTION**

#### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

## **Overview**

**Table 361** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 174, 175, 176, 177                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Probable Cause | <p>This alarm is raised when the provisioned PM threshold is crossed for OTU BBE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for OTU ES.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for OTU SES.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for OTU UAS.</p> <p>A PM threshold has been set incorrectly, an OTU related alarm is raised, a dirty or faulty fiber.</p> |

|          |                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instance | Port-<slot/port>-15m, Port-<slot/port>-24h, Port-<slot/port>-Untimed, slot = 1 - 8, port = 1 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity | Warning                                                                                                                                                     |

### Steps

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold setting has been set based on the alarm. Change the PM threshold setting, if required.  

```
pm tca show profile-type OTU
pm tca OTU set profile-name <profile name> [BBE <NUMBER: 0..4294967295> | ES <NUMBER: 0..86399> | SES <NUMBER: 0..86399> | UAS <NUMBER: 0..86399>]
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 4 Clear any OTU related alarms, such as OTU AIS, OTU BDI, OTU Signal Degrade and OTU Trace Identifier Mismatch. Use the alarm clearing procedures in this chapter to clear these alarms.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 6 Use an optical power meter to measure the power level on the alarmed port. Add attenuators, if the Rx power levels are too high. Inspect and clean fibers, if the Rx power levels are too low. Replace fibers, if required.
- 7 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`.

- 8 Verify if the alarm has cleared.
- 9 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 262 TCA OTU BBE FE / TCA OTU ES FE / TCA OTU SES FE / TCA OTU UAS FE

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have attenuators, if required.
- You have fiber cleaning kit, if required.
- You have a fiber patchcord, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 362** Alarm details

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 178, 179, 180, 181                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Probable Cause | <p>This alarm is raised when the provisioned PM threshold is crossed for OTU BBE FE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for OTU ES FE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for OTU SES FE.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for OTU UAS FE.</p> <p>A PM threshold has been set incorrectly or a far end OTU related alarm is raised.</p> |

|          |                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instance | Port-<slot/port>-15m, Port-<slot/port>-24h, Port-<slot/port>-Untimed, slot = 1 - 8, port = 1 - 18<br><b>Note:</b> Only odd slots are used for this release. |
| Severity | Warning                                                                                                                                                     |

### Steps

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold setting has been set based on the alarm. Change the PM threshold setting, if required.  

```
pm tca show profile-type OTU
pm tca OTU set profile-name <profile name> [BBE-FE
<NUMBER: 0..4294967295> | ES-FE <NUMBER: 0..86399> | SES-
FE <NUMBER: 0..86399> | UAS-FE <NUMBER: 0..86399>]
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 4 On the far-end Waveserver 5, clear any OTU related alarms, such as OTU AIS, OTU BDI, OTU Signal Degrade and OTU Trace Identifier Mismatch. Use the alarm clearing procedures in this chapter to clear these alarms.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 6 On the far-end Waveserver 5, use an optical power meter to measure the power level on the alarmed port. Add attenuators, if the Rx power levels are too high. Inspect and clean fibers, if the Rx power levels are too low. Replace fibers, if required.

- 7 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 263 TCA RX/TX Frame Error Ratio

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a fiber patchcord, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 363** Alarm details

|                |                                                                                                                                                                                                                                                                                            |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 182, 183                                                                                                                                                                                                                                                                                   |
| Probable Cause | <p>This alarm is raised when the provisioned PM threshold is crossed for Rx frame error ratio.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for Tx frame error ratio.</p> <p>A PM threshold has been set incorrectly or there is a faulty fiber patchcord.</p> |
| Instance       | <p>Port-&lt;slot/port&gt;-15m, Port-&lt;slot/port&gt;-24h, Port-&lt;slot/port&gt;-Untimed, slot = 1 - 8, port = 3 - 18</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                   |
| Severity       | Warning                                                                                                                                                                                                                                                                                    |

## Steps

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold if required.
 

```
pm tca show profile-type Ethernet
pm tca Ethernet set profile-name <profile name> Rx-Frame-Err-Ratio <Frame Error Ratio: X.XXE-YY. Ex: 1.00E-06, 1.01E-07, 0.09E-05>
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 4 Replace the fiber patchcord on the port that is alarming.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.



## Procedure 264 TCA PCS ES / TCA PCS SES / TCA PCS UAS

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 364** Alarm details

|                |                                                                                                                                                                                                                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 184, 185, 186                                                                                                                                                                                                                                                                                                          |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for PCS ES.<br>This alarm is raised when the provisioned PM threshold is crossed for PCS SES.<br>This alarm is raised when the provisioned PM threshold is crossed for PCS UAS.<br>A PM threshold has been set incorrectly or a PCS alarm is raised. |
| Instance       | Port-<slot/port>-15m, Port-<slot/port>-24h, Port-<slot/port>-Untimed, slot = 1 - 8, port = 3 - 18<br><b>Note:</b> Only odd slots are used for this release.                                                                                                                                                            |
| Severity       | Warning                                                                                                                                                                                                                                                                                                                |

### Steps

- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- Verify the appropriate PM threshold is set based on the alarm (TCA PCS ES, TCA PCS SES or TCA PCS UAS) raised.  
Change the PM threshold if required.  

```
pm tca show profile-type Ethernet
pm tca Ethernet set profile-name <profile name> {Pcs-Es <NUMBER: 0..86399> | Pcs-Ses <NUMBER: 0..86399> | Pcs-Uas <NUMBER: 0..86399>}
```
- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.

- 4 Clear any PCS related alarms, such as PCS Loss of Alignment Marker and PCS Loss of Block Lock. Use the alarm clearing procedures in this chapter to clear these alarms.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 265 TCA RX/TX Link Utilization

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 365** Alarm details

|                |                                                                                                                                                                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 187, 188                                                                                                                                                                                                                                                                                                                 |
| Probable Cause | <p>This alarm is raised when the provisioned PM threshold is crossed for Rx link utilization.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for Tx link utilization.</p> <p>A PM threshold has been set incorrectly or an excessive amount of Ethernet traffic is being received or sent.</p> |
| Instance       | <p>Port-&lt;slot/port&gt;-15m, Port-&lt;slot/port&gt;-24h, Port-&lt;slot/port&gt;-Untimed, slot = 1 - 8, port = 3 - 18</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                                 |
| Severity       | Warning                                                                                                                                                                                                                                                                                                                  |

### Steps

- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:

  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- Verify the appropriate PM threshold is set. Change the PM threshold, if required.

```
pm tca show profile-type Ethernet
pm tca Ethernet set profile-name <profile name> {Rx-Link-Utilization <Threshold 0.000 .. NN.mmm> | Tx-Link-Utilization <Threshold 0.000 .. NN.mmm>}
```
- Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:

  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.

---

|   |                                                                                                                                                                                                                                                                                             |                        |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 4 | <b>If TCA</b>                                                                                                                                                                                                                                                                               | <b>Then go to</b>      |
|   | Rx Link Utilization is raised                                                                                                                                                                                                                                                               | <a href="#">step 5</a> |
|   | Tx Link Utilization is raised                                                                                                                                                                                                                                                               | <a href="#">step 6</a> |
| 5 | On the far-end Waveserver 5 client port, reduce the amount of Ethernet packets being sent.<br>Go to <a href="#">step 7</a> .                                                                                                                                                                |                        |
| 6 | On the near-end Waveserver 5 client port, reduce the amount of Ethernet packets being sent.                                                                                                                                                                                                 |                        |
| 7 | Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:                                                                                                                                                                                         |                        |
|   | <ul style="list-style-type: none"><li>For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.</li><li>For Untimed PM bins: Use CLI command <code>pm clear port &lt;slot&gt;/&lt;port&gt; bin-type untimed</code>. Verify if the alarm has cleared.</li></ul> |                        |
| 8 | If the alarm does not clear, contact your next level of support or your Ciena support group.                                                                                                                                                                                                |                        |

## Procedure 266 TCA RX/TX CRC Error Packets

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a fiber patchcord, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 366** Alarm details

|                |                                                                                                                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 189, 190                                                                                                                                                                                                                                                                                                |
| Probable Cause | <p>This alarm is raised when the provisioned PM threshold is crossed for Rx CRC error packets.</p> <p>This alarm is raised when the provisioned PM threshold is crossed for Tx CRC error packets.</p> <p>A PM threshold has been set incorrectly or there is a faulty patchcord on the client port.</p> |
| Instance       | <p>Port-&lt;slot/port&gt;-15m, Port-&lt;slot/port&gt;-24h, Port-&lt;slot/port&gt;-Untimed, slot = 1 - 8, port = 3 - 18</p> <p><b>Note:</b> Only odd slots are used for this release.</p>                                                                                                                |
| Severity       | Warning                                                                                                                                                                                                                                                                                                 |

## Steps

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold, if required.
 

```
pm tca show profile-type Ethernet
pm tca Ethernet set profile-name <profile name> {Rx-Crc-Pkts <NUMBER: 0..4294967295> | Tx-Crc-Pkts <NUMBER: 0..4294967295>}
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 4 Replace the fiber patchcord on the client port that is alarming.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear port <slot>/<port> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 267 TCA SNR External

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have fiber patchcord, if required.
- You have a fiber cleaning kit, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 367** Alarm details

|                |                                                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 194                                                                                                                                                                                                 |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for SNR external.<br><br>A PM threshold has been set incorrectly, a dirty fiber, a faulty fiber, or a misconnection in the fiber. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><br><b>Note:</b> Only odd slots are used for this release.                                             |
| Severity       | Warning                                                                                                                                                                                             |

## Steps

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold, if required.
 

```
pm tca show profile-type Modem
pm tca Modem set profile-name <profile name> SNR-Ext
<Threshold 0.000 .. NN.mmm>
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4 Use an optical power meter to measure the power level on the alarmed line port and ensure power levels are as expected. Inspect and clean the alarmed line port fibers. Replace fiber patchcord, if required.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm does not clear, contact your next level of support or your Ciena support group.



## Procedure 268 TCA CSI

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have an optical power meter with the same optical connectors as the network element.
- You have a fiber cleaning kit, if required.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Risk of interruption to service**

Before you disconnect fiber patchcord, make sure you do not affect traffic. Reroute traffic, if required.



#### **CAUTION**

##### **Risk of damage to the equipment**

Wear an appropriate ESD personal grounding device to dissipate electrostatic charges and to prevent damage to the equipment.

### Overview

**Table 368** Alarm details

|                |                                                                                                                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 195                                                                                                                                                                                                         |
| Probable Cause | This alarm is raised when the provisioned PM threshold is crossed for Cycle Slip Indicator.<br><br>A PM threshold has been set incorrectly, a dirty fiber, a faulty fiber, or a misconnection in the fiber. |
| Instance       | PTP-<slot/ptp>-15m, PTP-<slot/ptp>-24h, PTP-<slot/ptp>-Untimed, slot = 1 - 8, ptp = 1 - 2<br><b>Note:</b> Only odd slots are used for this release.                                                         |
| Severity       | Warning                                                                                                                                                                                                     |

## Steps

- 1 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 2 Verify the appropriate PM threshold is set. Change the PM threshold, if required.
 

```
pm tca show profile-type Modem
pm tca Modem set profile-name <profile name> CSI <NUMBER:
0..1000000000>
```
- 3 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 4 Use an optical power meter to measure the power level on the alarmed line PTP fibers. Inspect and clean fibers the alarmed port. Replace fiber patchcord, if required.
- 5 Do one of the following, depending on the PM bin (15m, 24h, or Untimed) instance raising the alarm:
  - For 15m or 24h PM bins: Wait for the PM bin rollover. Verify if the alarm has cleared.
  - For Untimed PM bins: Use CLI command `pm clear ptp <slot>/<ptp> bin-type untimed`. Verify if the alarm has cleared.
- 6 If the alarm did not clear, contact your next level of support or your Ciena support group.

---

## Procedure 269 Fiber Mismatch

---

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine how the line ports should be connected.

### Overview

**Table 369** Alarm details

|                |                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 228                                                                                                                     |
| Probable Cause | This alarm is raised when the Tx or Rx fiber for the near-end line port is connected to an incorrect far-end line port. |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 8, ptp = 1 - 2<br><b>Note:</b> Only odd slots are used for this release.                   |
| Severity       | Major                                                                                                                   |

### Steps

- 1 Check the network plan or document to determine if the Tx and/or Rx fiber for the near-end line port is connected to the correct far-end line port. If there is a discrepancy from the network plan or document, connect the fibers to match the network plan or document. The Tx and Rx fibers for the near-end line port must be connected to the same far-end line port.
- 2 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 270 ZTP In Progress

### Requirements

You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

**Table 370** Alarm details

|                |                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 229                                                                                                                                                |
| Probable Cause | This alarm is raised as soon as the ZTP process starts. The ZTP in Progress alarm clears once commissioning is complete and the ZTP process stops. |
| Instance       | Chassis                                                                                                                                            |
| Severity       | Warning, non service-affecting                                                                                                                     |

### Steps

- 1 Wait for the ZTP process to complete, it may take several minutes to complete.
- 2 If the alarm does not clear, use the following CLI command:  
`configuration ztp show`
- 3 Check value of the Oper State field in the output.
- 4

| If the Oper State field value is | Then go to             |
|----------------------------------|------------------------|
| Idle                             | <a href="#">step 5</a> |
| Completed                        | <a href="#">step 7</a> |
- 5 Check value of the Last ZTP field in the output.
- 6 If the Last ZTP field value is Failure, you can manually disable the ZTP by using the following command:  
`configuration ztp disable`
- 7 Check the alarm status:  
`alarm show`
- 8 If the alarm does not clear, contact your next level of support or your Ciena support group.

## Procedure 271 Power Supply Unit Disabled

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You observe all the safety requirements described in the chapter [Removing, replacing, or reseating equipment](#).



### CAUTION

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

### Overview

**Table 371** Alarm details

|                |                                                                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm TID      | 233                                                                                                                                                                                                                                                                                                        |
| Probable Cause | This alarm is raised when a fan module is missing from the power supply unit (PSU) and the PSU goes into a “safety shutdown” state. This alarm is also raised when a PSU is not seated properly. In both cases, this alarm indicates that the PSU is no longer supplying a redundant power to the chassis. |
| Instance       | Slot                                                                                                                                                                                                                                                                                                       |
| Severity       | Major                                                                                                                                                                                                                                                                                                      |

### Steps

- 1 Verify that there is no generic equipment missing alarm is raised for the Fan module. The type of missing equipment (PSU, Fan, AP, and Module) is identified in the alarm instance name.
- 2
 

|                                                                  |                        |
|------------------------------------------------------------------|------------------------|
| <b>If</b>                                                        | <b>Then go to</b>      |
| the generic equipment missing alarm is raised for the Fan module | <a href="#">step 3</a> |
| no generic equipment missing alarm is raised                     | <a href="#">step 4</a> |
- 3 Install the missing Fan module in the PSU. See [Removing, replacing, or reseating equipment](#).  
Go to [step 5](#)
- 4 Verify that all PSUs are installed or seated correctly. If necessary, install the PSU again. See [Removing, replacing, or reseating equipment](#).
- 5 If the alarm does not clear, contact your next level of support or your Ciena support group.

---

## Procedure 272 Connection Validation Test In Progress

---

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.
- You have a network plan or other documents that allow you to determine how the line ports should be connected.

### Overview

**Table 372** Alarm details

|                |                                                                                         |
|----------------|-----------------------------------------------------------------------------------------|
| Alarm TID      | 234                                                                                     |
| Probable Cause | This alarm is raised when SPLI sends a request to start the connection validation test. |
| Instance       | PTP-<slot>/<ptp>, slot = 1 - 3, ptp = 1 - 2                                             |
| Severity       | Warning                                                                                 |

### Steps

- 1 Alarm is raised on line ports and is not applied on the legacy MOTR-MCS cards.
- 2 No action is required. The alarm clears when the SPLI sends request to stop the connection validation test process.



---

## CHAPTER 29

# REST, NETCONF, gRPC, OpenConfig, and WebSocket

---

This section covers the following topics:

- [Overview](#)
  - [General NBI guidelines](#)
  - [YANG model support](#)
  - [REST](#)
  - [NETCONF](#)
  - [gRPC/gNMI](#)
  - [OpenConfig](#)
  - [WebSocket](#)
- [Procedure list](#)

**Note:** Software license S86-LIC-WOSAPIS is required when accessing the Waveserver 5 using NETCONF, REST, gRPC/gNMI, or WebSocket.

## Overview

This section provides guidelines for use of northbound interfaces (NBI) on Waveserver 5. Waveserver 5 supports REST, NETCONF, gRPC/gNMI, OpenConfig, and Websocket.



**ATTENTION**

Waveserver 5 performance issues can occur when multiple northbound interfaces (NBIs) are used simultaneously on Waveserver 5.

The use of multiple primary NBIs is not supported on Waveserver 5. Use one instance of a single primary NBI (REST, NETCONF, or SNMP) to extract data from Waveserver 5. If you are running a secondary interface for light monitoring, for example an SNMP trap monitor or Websocket, do not duplicate the data management activity on the secondary interface. Avoid overlapping requests for the same data over two interfaces on the same Waveserver 5.

Avoid bulk data retrievals on Waveserver 5. Retrieve the specific data whenever possible. APIs allow you to specify individual data elements or group data.

For more information, see [General NBI guidelines](#).

**General NBI guidelines**

The following section provides general NBI guidelines as well as more detailed recommendations for implementing data collection on Waveserver 5.

When collecting data from Waveserver 5 using an NBI, consider the following:

- Waveserver 5 supports one machine interface session at a time, which is used as the primary management interface (NETCONF, REST, or SNMP).
- All management interfaces are subject to a global inactivity timeout setting. If enabled, any management interface session that has been inactive for the duration of the configured timeout period will be closed. Management interfaces that use the Waveserver's webserver (Web user interface, REST, and WebSocket) are also subject to an HTTPS inactivity timeout setting. If enabled, the HTTPS inactivity timeout setting takes precedence over the global inactivity timeout setting.
- The Waveserver 5 RESTCONF interface is aligned with RFC-7951 with respect to namespace-qualified prefixes in `GET` responses. The YANG module name is returned in the top-level data node in the response. Note that on user-input (for example, `PATCH/PUT/POST` requests), the prefix is optional and either format is accepted. It is recommended that clients/scripts utilizing the REST API be able to accept/ignore prefix strings flexibly, to comply with both types of behavior laid out in RFC-7951.
- When accessing data elements, insert a brief pause (1 second) between element retrievals.

- Upon system initialization or after an object creation, when modifying a container or a list resource via NETCONF or RESTCONF for the first time, it is recommended to retrieve (GET) the object once prior to performing a modify (SET). Subsequent modifies can be performed without retrieving this object again.
- Certain types of data take a much longer time to retrieve and require more processing than other data. For example, PM data collection is a large body of data that is processing-intensive and requires adherence to engineering guidelines, see [Data retrieval rates for PM data](#) for more information.
- Static or slow-changing data should not be retrieved more frequently than necessary. For example, retrieve the data once after a configuration change or aligned to the rate of change expected for that data.
- When retrieving data that belongs to a list, bulk read requests at the top-level container should be avoided, and finer granularity reads should be used instead to reduce the processing strain on the system. For example, rather than reading all of /waveserver-pm, implement the data collection using /waveserver-pm/ethernet-performance-instances= {instance-name}/current-bin.

If you need to retrieve all of the data from a list, avoid looping through the list from the individual index level. For example, instead of looping through using this command:

```
/waveserver-pm/ethernet-performance-
instances={instance-name}
```

use instead:

```
/waveserver-pm/ethernet-performance-
instances={instance-name}/current-bin
```

- When performing a replace operation, either send the complete network element configuration in the replace command or send only the container being replaced. Incomplete provisioning specifications might delete the non-specified information.
- For performance reasons, the number of concurrent REST sessions is limited to five.

For sample GET requests, see below.

To retrieve an individual element in a list using the REST interface, the GET request must contain an index. For example:

```
https://123.123.123.123/restconf/data/waveserver-pm/
ethernet-performance-instances=1-3-Ethernet/current-
bin?content=nonconfig
```

or

```
https://123.123.123.123/restconf/data/waveserver-pm/
ethernet-performance-instances=1-3-Ethernet/
history?content=nonconfig
```

### Data retrieval rates for PM data

PM history bins should be retrieved within every 24 hours since the information is intentionally cached into 96 15-minute bins (or 24 hours of PM history information).

For REST and NETCONF you can set the query limit for PM history bins, using the CLI command `system management query-limits <netconf | rest | gnmi> set pm-history-bins <NUMBER: 0..96>`. To display the NETCONF and REST query limit summary information, use the CLI command `system management query-limits show`. For additional information, see [Configuring query limits for PM history bins](#).

Poll each type of bin separately, which will return those bins for every port on the Waveserver 5. The commands for the 15-minute bins can take a while to return when requested on a fully filled system, due to the quantity of data being requested.

For NETCONF client applications, if a timeout value can be specified for each GET request, Ciena recommends using a timeout value of 120 seconds. If a timeout value can be specified for a NETCONF connection session, Ciena recommends using a timeout value greater than 120 seconds to avoid the connection session timing out before a Get request is completed.

See the following examples of polling history bins for PM data.

### Example - Polling PM 15-minute History Bins (NETCONF)

#### Sign in to the NETCONF client and stay signed in for once every 4 hours

```
get /waveserver-pm/ethernet-performance-instances/
pm:history
```

**wait 1 second**

```
get /waveserver-pm/optical-power-instances/pm:history
```

**wait 1 second**

```
get /waveserver-pm/modem-performance-instances/
pm:history
```

**wait 1 second**

**repeat**

### **Example - Polling PM 24-hour History Bins (NETCONF)**

**Sign in to NETCONF client and stay signed in once every 24 hours**

```
get /waveserver-pm/ethernet-performance-instances/
pm:history-24-hour-bin
```

**wait 1 second**

```
get /waveserver-pm/optical-power-instances/pm:history-
24-hour-bin
```

**wait 1 second**

```
get /waveserver-pm/modem-performance-instances/
pm:history-24-hour-bin
```

**wait 1 second**

**repeat**

PM current bins can be retrieved every minute. If you want to retrieve the current and first historical 15-minute bin, they can be retrieved every 10 to 15 minutes.

See the following example of polling current PM bins with port stats.

### **Example - Polling current PM bins (NETCONF)**

**Sign in to NETCONF client and stay signed in once every minute**

```
get /waveserver-pm/ethernet-performance-instances/
pm:current-bin
```

**wait 1 second**

```
get /waveserver-pm/optical-power-instances/pm:current-
bin
```

**wait 1 second**

```
get /waveserver-pm/modem-performance-instances/
pm:current-bin
```

**wait 1 second**

**repeat**

**Note:** The same procedure can be used for 24-hour bins by changing `/current-bin` to `/current-24-hour-bin`, however it is recommended that frequent polling be done with 15-minute current bins.

## YANG model support

Each YANG-based interface (REST, NETCONF, gNMI) supports the Ciena YANG modules listed below and the OpenConfig modules listed in [OpenConfig](#).

A REST, NETCONF, or gNMI request can contain references to any supported data model. As OpenConfig does not provide full configuration support, OpenConfig users can supplement operation requests with native Ciena YANG models as required. You can send separate requests containing references to a single model or send a single request containing references to both OpenConfig and native Ciena models.

## Considerations when using different data models and interfaces

- Retrieve operations
  - The format of the data in the protocol-specific response matches the format used for specifying the element to be retrieved in the request.
  - Protocol-specific capability supporting retrieval of more than one element in a single request relies on the format used in the request to specify each element in the request. For example, a gNMI `GET` RPC allows the specification of a repeated set of path elements. The `GET` response message contains separate notification and update messages for each response corresponding to a path in the request. The format of data in each response is the format used in the request.
  - Root-level retrieval of state data returns the full current state of the Waveserver in both native Ciena YANG and OpenConfig YANG formats. The response therefore contains redundant data.
- Configuration operations
  - Avoid specifying redundant or intersecting paths in both OpenConfig and native Ciena YANG models in the request as the configuration request is declarative and the system processes all elements of the supplied input before proceeding. As a result, duplicated references with different values or implied behaviour may not be applied to the system in the way that was intended or may result in an unexpected rejection of the request if the combination of input creates an

incomplete or invalid configuration request. The system that creates the input should only include native Ciena YANG references for specific elements not available in OpenConfig YANG models.

- Subscription operations
  - The format of the data in the NETCONF or gNMI notification published to the subscriber uses the format used to subscribe to that element.
  - Protocol-specific capability supporting subscription to more than one element in a single request relies on the format used to specify each element in the subscription request. For example, the gNMI Subscribe RPC allows the specification of a subscription-list where individual subscriptions are specified with a unique path. The notifications published use the format of the path specified in the subscriptions.

### Ciena YANG modules

The following are the available Ciena YANG modules.

- Common version (ciena-common-version.html)
- Common YANG (ciena-common-yang.html)
- Streaming telemetry (ciena-pro-telemetry.html)
- Authentication (ciena-waveserver-aaa.html)
- Alarm (ciena-waveserver-alarm.html)
- Dispersion Scan (ciena-waveserver-application-dispersion-scan.html)
- Integrated Test Set (ciena-waveserver-application-its.html)
- Link Data Collection (ciena-waveserver-application-link-data-collection.html)
- Link Data Collection streaming (ciena-waveserver-link-data-telemetry)
- OTDR (ciena-waveserver-application-otdr.html)
- Chassis (ciena-waveserver-chassis.html)
- Configuration (ciena-waveserver-configuration.html)
- Encryption (ciena-waveserver-encryption.html)
- Interfaces (ciena-waveserver-interfaces.html)
- License (ciena-waveserver-license.html)
- LLDP (ciena-waveserver-lldp.html)
- Logging (ciena-waveserver-logging.html)
- Module (ciena-waveserver-module.html)
- NDP (ciena-waveserver-ndp.html)
- PKIX (ciena-waveserver-pkix.html)

- PM TCA (ciena-waveserver-pm-tca.html)
- PM (ciena-waveserver-pm.html)
- Port (ciena-waveserver-port.html)
- Protection (ciena-waveserver-protection.html)
- PTP (Physical Termination Point: ciena-waveserver-ntp.html)
- PTP Modem (Physical Termination Point: ciena-waveserver-ntp-modem.html)
- SNMP (ciena-waveserver-snmp.html)
- Software (ciena-waveserver-software.html)
- SPLI (ciena-waveserver-spli.html)
- Statistics (ciena-waveserver-statistics.html)
- System (ciena-waveserver-system.html)
- Typedefs (ciena-waveserver-typedefs.html)
- XCVR Modem (Transceiver: ciena-waveserver-xcvr-modem.html)
- XCVR Pluggables (Transceiver: ciena-waveserver-xcvr-pluggable.html)
- XCVR (Transceiver: ciena-waveserver-xcvr.html)
- IETF Yang types (ietf-yang-types.html)
- Definitions for NETCONF extensions (yuma-ncx.html)

## REST

You can use REST API to configure Waveserver 5 and view system information. REST systems typically communicate over the Secure Hypertext Transfer Protocol (HTTPS) using the same HTTP methods that web browsers use to retrieve and send data to remote servers.

REST services are enabled by default on Waveserver 5. The following HTTP methods are supported, as required by RFC 8040:

- GET—used to retrieve resources
- POST—used for remote procedure calls (RPC)
- PATCH—used to update resources without replacing the resource
- PUT—used to create resources
- DELETE—used to remove resources

RESTCONF uses a base path URI in the format: `https://<ip>/restconf/data/`

You can retrieve configuration and operational data in a single request: `?content=config|nonconfig|all`

List keys are specified as: `/path/to/list={key}`

REST API calls can be synchronous or asynchronous.

- Asynchronous calls return an immediate response, indicating that the call has been received and specifying whether the request has been accepted
- Synchronous calls return a pass or fail response after the operation completes

Responses to REST API calls are in JavaScript Object Notation (JSON) format.

If enabled, TACACS+ authorization and accounting apply to REST operations. For more information on TACACS+, refer to [TACACS+](#).

Waveserver 5 supports up to ten concurrent user sessions between the Web, REST, and WebSocket interfaces. Any attempt to connect an 11th user session without first closing an existing session will be rejected.

## REST APIs

Waveserver 5 provides API documentation using Swagger, a specification language that defines and documents RESTful APIs. A Swagger document is in a JSON schema-like format. The Swagger user interface (UI) is a browser-based interface for reading, exploring, and trying the REST APIs.

You can access Waveserver 5 REST APIs (YANG modules) from a Web browser pointed to Waveserver 5. You can also explore the APIs and execute API calls.

## REST client guidelines

The following REST API conditions and usage guidelines exist for the Waveserver 5 REST interface:

- For `GET` request, all YANG models are wrapped in a 'data' JSON object or XML element depending on the content type. It is recommended that you disable web browser caching when using Swagger in order to avoid inconsistent/stale results.
- Retrieval of port statistics operational data may have a long performance time.

If the system is fully loaded, or nearly fully loaded, querying all port statistic operational data can take up to a minute or longer. Avoid querying all operational data or all port operational data unless it is absolutely necessary. Instead, when querying port operational data, specify the exact port and data.



For example:

```
/data/waveserver-ports/ports={port-id}/
state?content=all
```

```
/data/waveserver-ports/ports={port-id}/
properties?content=all
```

**Note:** This performance issue does not affect port configuration data.

- REST sub-systems can take several minutes to fully initialize after boot up.

After a restart or power up, delay querying of port, ptp, xcvr, service, until the system has stabilized and the REST sub-system is fully initialized. This can be done by first querying URI: `/restconf/data/waveserver-modules/modules?content=nonconfig`

or

```
/restconf/data/waveserver-modules/modules={module-id}
```

To query the restart and Waveserver 5 uptime status use:

```
/restconf/data/waveserver-modules/
modules?content=nonconfig
```

- Leaf “uptime” indicates how long the system has been up since the last restart.

- Leaf “last-restart-reason” indicates the last restart reason.

- If the “last-restart-reason” is “User-Warm”, wait until “uptime” is at least 5 minutes before executing the full query:

```
/restconf/data/waveserver-modules/
modules?content=nonconfig
```

- If the “last-restart-reason” is anything other than “User-Warm”, wait until “uptime” is at least 10 minutes before executing the full query:

```
/restconf/data/waveserver-modules/
modules?content=nonconfig
```

- REST sub-systems can take several seconds to synchronize with a major configuration change.

After receiving a “websocket db-change” event on a configuration update, create, or delete, use the “data” portion of the db-change event message directly.

In the following sample db-change event message, the “path” field indicates the exact path that can be used to query the changed data, and the “data” portion contains the latest changed data. Note that both configuration and operational data are included in the “data” portion.

**Figure 148** Sample db-change event message

```

{
 "event": {
 "version": 1,
 "seq-number": 76,
 "timestamp": 1469712968,
 "db-change": {
 "action": "update",
 "path": "/ws-ports/ports/4/state",
 "data": {
 "state": {
 "admin-state": "disabled",
 "operational-state": "down",
 "operational-state-duration": "2d19h35m35s"
 }
 }
 }
 }
}

```

- Multi-threaded client is not recommended.

If a REST query is not yet complete, all other REST commands will be put in a queue, creating the possibility of timing out. Due to this, creating multi-threaded clients for the REST API is not recommended.

## NETCONF

NETCONF is an XML-based network management protocol specification developed by the Internet Engineering Task Force (IETF). You can use NETCONF to install, edit, and delete network device configurations, and to provision and activate network services.

YANG is a data modeling language, also developed by the IETF. YANG modules define the configuration data, state data, and the capabilities available on a Waveserver 5.

To communicate with a Waveserver 5 through the NETCONF interface you require a NETCONF client: either a purpose-built NETCONF browser or a network management tool with the appropriate plug-ins that enable it to act as NETCONF browser.

Communication between a NETCONF client and Waveserver 5 is session-based. A NETCONF client uses XML data encoding to exchange configuration data and protocol messages with the Waveserver 5.

The NETCONF client and Waveserver 5 communicate using Transmission Control Protocol (TCP), for reliability, and Secure Shell (SSH), for security. On Waveserver 5, NETCONF accesses TCP port 830.

NETCONF uses a simple remote procedure call (RPC) mechanism to facilitate communication between a client and server. A NETCONF session is the logical connection between a network administrator or network

configuration application and a network device. Global configuration attributes can be changed during any session and the effects are visible in all sessions. Session-specific attributes affect only the session in which they are changed.

If enabled, TACACS+ authorization and accounting apply to NETCONF operations. For more information on TACACS+, refer to [TACACS+](#).

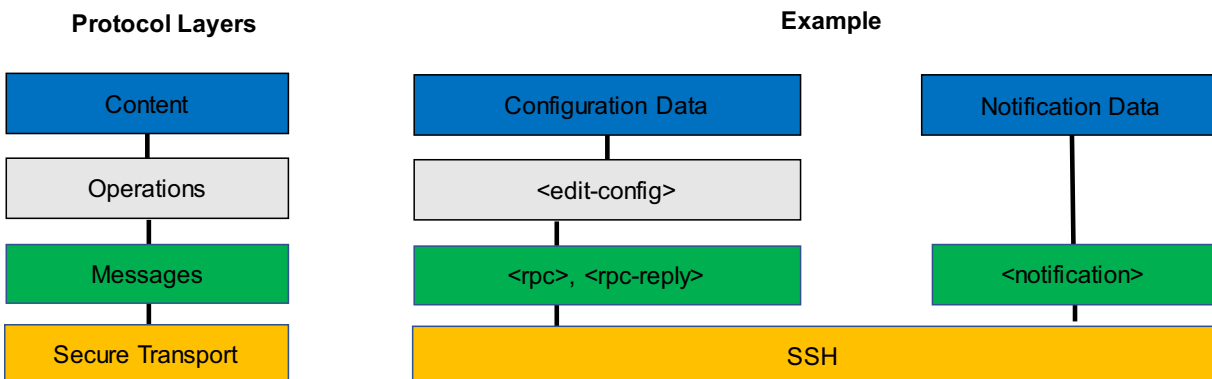
### Downloading YANG modules

To download YANG modules supported by Waveserver 5, use the NETCONF `<get-schema>` operation.

### NETCONF protocol layers

NETCONF can be conceptually partitioned into four layers, as shown below. The secure transport layer (Layer 1) provides a communication path between the client and server. The messages layer (Layer 2) provides a simple, transport-independent framing mechanism for encoding RPCs and notifications. The operations layer (Layer 3) defines a set of base protocol operations that are invoked as RPC methods with XML-encoded parameters. The content layer (Layer 4) consists of configuration and notification data. For a list of supported protocol operations, see [NETCONF protocol operations](#).

**Figure 149** NETCONF protocol layers



### NETCONF RPC model

NETCONF performs operations and capabilities through its own remote procedure call (RPC) mechanism, which is similar to the XML RPC mechanism. Requests from the NETCONF client to perform an action on Waveserver 5 must be well-formed XML documents; otherwise the requests fail.

When a client sends a series of RPC request messages, the server responds with a series of corresponding RPC reply messages. Four basic elements are used in these messages: `<rpc>`, `<rpc-reply>`, `<rpc-error>`, and `<ok>`. Each of the elements has additional attributes, some of which are mandatory.

The `<rpc>` element is used to enclose a NETCONF request sent from the client to Waveserver 5. The “message-id” attribute is mandatory. The name and parameters of an RPC are contained in the element. The following example of an `<rpc>` element shows a method called `<my-own-method>` and has two parameters, `<my-first-parameter>` (14) and `<another-parameter>` (fred).

```
<rpc message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
 <my-own-method xmlns="http://example.net/me/my-own/1.0">
 <my-first-parameter>14</my-first-parameter>
 <another-parameter>fred</another-parameter>
 </my-own-method>
</rpc>
```

The `<rpc-reply>` element is sent in response to an `<rpc>` message. The “message-id” attribute is mandatory and must correspond to the ID of the `<rpc>` message. The following example shows an `<rpc>` message and the corresponding `<rpc-reply>`.

```
<rpc message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0"
 xmlns:ex="http://example.net/content/I.0" ex:user-id="fred">
 <get/>
</rpc>

<rpc-reply message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0"
 xmlns:ex="http://example.net/content/I.0" ex:user-id="fred">
 <data xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
 <!-- contents here... -->
 </data xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
</rpc-reply>
```

The `<rpc-error>` element is sent in an `<rpc-reply>` message if an error occurs when the request is processed. Multiple `<rpc-error>` elements can be returned. The `<rpc-error>` element includes information such as the error location, the type and severity of the error, the path for the node reporting the error, and an error message (string). The following example shows an `<rpc-reply>` with an `<rpc-error>` where the MTU value is not within the specified range.

```
<rpc-reply message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0"
 xmlns:xc="urn:ietf:params:xml:ns:netconf:base:1.0">
 <rpc-error>
 <error-type>application</error-type>
 <error-tag>invalid-value</error-tag>
 <error-severity>error</error-severity>
 <error-path xmlns:t="http://example.com/schema/1.2/config">
 /t:top/t:interface[t:name="Ethernet0/0"]/t:mtu </error-path>
 <error-message xml:lang="en">
 MTU value 25000 is not within range 256..9192 </error-message>
 </rpc-error>
 <error-type>application</error-type>
```

The `<ok>` element is sent in `<rpc-reply>` messages if no errors or warnings occurred during the processing of the request and no data was returned. The following example shows an `<rpc-reply>` message containing the `<ok>` element.

```
<rpc-reply message-id="101"
 xmlns="urn:ietf:params:xrnl:ns:netconf:base:1.0">
 <ok/>
</rpc-reply>
```

### Example - `<get>` operation and the reply

```
<?xml version="1.0" encoding="utf-8"?>
<rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="13">
 <get>
 <filter type="subtree">
 <if:waveserver-interfaces xmlns:if="urn:ciena:params:xml:ns:yang:ciena-
ws:ciena-waveserver-interfaces">
 <if:logical-interface>
 <if:name>local</if:name>
 </if:logical-interface>
 </if:waveserver-interfaces>
 </filter>
 </get>
</rpc>

<?xml version="1.0" encoding="utf-8"?>
<rpc-reply xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="13">
 <data>
 <waveserver-interfaces xmlns="urn:ciena:params:xml:ns:yang:ciena-
ws:ciena-waveserver-interfaces">
 <logical-interface>
 <name>local</name>
 <id>
 <name>local</name>
 <ifindex>8</ifindex>
 <internal>false</internal>
 <bgp>false</bgp>
 <associated-physical-interface>ifp0-dcn-1</associated-physical-
interface>
 <associated-physical-interface>ifp0-ilan-1</associated-physical-
interface>
 <associated-physical-interface>ifp0-ilan-2</associated-physical-
interface>
 <associated-physical-interface>ifp0-dcn-2</associated-physical-
interface>
 <associated-physical-interface>ifp0-dcn-5</associated-physical-
interface>
 </id>
 <state>
 <admin-state>enabled</admin-state>
 <operational-state>up</operational-state>
 </state>
 <properties>
 <ipv4>
 <admin-state>enabled</admin-state>
 <user-ip-address>0.0.0.0/0</user-ip-address>
 <oper-ip-address>10.184.96.33/22</oper-ip-address>
 <user-gateway>0.0.0.0</user-gateway>
```

```

 <oper-gateway>10.184.96.1</oper-gateway>
 </ipv4>
 <ipv6>
 <oper-ip-address>
 <ip-address>fe80::2280:58ff:fe5b:aa10/64</ip-address>
 <source>local</source>
 <dad-failure>>false</dad-failure>
 </oper-ip-address>
 <oper-ip-address>
 <ip-address>2620:11b:d06d:f103:2280:58ff:fe5b:aa10/64</ip-
address>
 <source>slaac</source>
 <dad-failure>>false</dad-failure>
 </oper-ip-address>
 <oper-gateway>
 <ip-address>fe80::ce4e:24ff:fe6e:dc80</ip-address>
 <source>local</source>
 <metric>1024</metric>
 <interface-name>local</interface-name>
 </oper-gateway>
 <gateway-configuration>
 <ip-address>::/128</ip-address>
 <source>>manual</source>
 <operational-state>enabled</operational-state>
 </gateway-configuration>
 </ipv6>
</properties>
</logical-interface>
</waveserver-interfaces>
</data>
</rpc-reply>

```

### Example - <get-config> operation where the entire <user> subtree is requested

```

<?xml version="1.0" encoding="utf-8"?>
<rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="20">
 <get-config>
 <source>
 <running/>
 </source>
 <filter type="subtree">
 <aaa:waveserver-aaa xmlns:aaa="urn:ciena:params:xml:ns:yang:ciena-
ws:ciena-waveserver-aaa"
 xmlns:nc="urn:ietf:params:xml:ns:netconf:base:1.0">
 <aaa:user/>
 </aaa:waveserver-aaa>
 </filter>
 </get-config>
</rpc>

<?xml version="1.0" encoding="utf-8"?>
<rpc-reply xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="20">
 <data>
 <waveserver-aaa xmlns="urn:ciena:params:xml:ns:yang:ciena-
waveserver-aaa">
 <user>
 <user-name>su</user-name>
 <password>****</password>
 <secret/>
 </user>
 </waveserver-aaa>
 </data>
</rpc-reply>

```

```
 <access-level>super</access-level>
 </user>
 <user>
 <user-name>tester1</user-name>
 <password>****</password>

<secret>6srsmmmcH58KSxsFa$qJmKc19m1CPPRpXZsIregStsa0wAu6mMie1AfW1bdq3ARLI7
QRzmaN6mRXwl5l/14Ce01alz4zYDLVOaJ2A8u0</secret>
 <access-level>super</access-level>
 </user>
</waveserver-aaa>
</data>
</rpc-reply>
```

### Example - <edit-config> operation

The "password" and "access-level" for user "tester" is being modified.

```
<?xml version="1.0" encoding="utf-8"?>
<rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="34">
 <edit-config>
 <target>
 <running/>
 </target>
 <config>
 <waveserver-aaa xmlns="urn:ciena:params:xml:ns:yang:ciena-ws:ciena-
waveserver-aaa"
 xmlns:nc="urn:ietf:params:xml:ns:netconf:base:1.0">
 <user nc:operation="merge">
 <user-name>tester1</user-name>
 <password>waveserver</password>
 <access-level>limited</access-level>
 </user>
 </waveserver-aaa>
 </config>
 </edit-config>
 </rpc>

<?xml version="1.0" encoding="utf-8"?>
<rpc-reply xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="34">
 <ok/>
</rpc-reply>
```

### Example - <close-session> operation and the reply

```
<rpc message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
 <close-session/>
</rpc>

<rpc-reply message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
 <ok/>
</rpc-reply>
```

### Example - <kill-session> operation and the reply

```
<rpc message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
 <kill-session>
```

```

 <session-id>4</session-id>
 </kill-session>
</rpc>

<rpc-reply message-id="101"
 xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
 <ok/>
</rpc-reply>

```

### NETCONF protocol operations

NETCONF defines a set of base protocol operations. A protocol operation is a specific remote procedure call used within NETCONF. These operations allow you to manage device configurations and retrieve device state information. Additional operations may also be available, based on the capabilities advertised by a device. Waveserver 5 supports the protocol operations listed in the table below.

**Table 373** NETCONF protocol operations

Operation	Description	Parameters
<get>	Retrieves the running configuration datastore and device state information.  If Waveserver 5 is unable to satisfy the <get> request, an error is sent in the RPC reply.  <b>Note:</b> When the returned configuration datastore contains a password, the password is hidden and only asterisks are displayed.	<b>filter</b> - identifies the portion of the system configuration and state data to retrieve. If the filter parameter is not used, all of the device configuration and state information is returned.
<get-config>	Retrieves all or part of a configuration data store.  If Waveserver 5 is unable to satisfy the <get-config> request, an error is sent in the RPC reply.  <b>Note:</b> When the returned configuration datastore contains a password, the password is hidden and only asterisks are displayed.	<b>source</b> - name of the configuration datastore being queried, for example, &lt;running/&gt;     <b>filter</b> - identifies the portions of the device configuration datastore to retrieve. If the filter parameter is not used, the entire configuration is returned.



Operation	Description	Parameters
<edit-config>	Edits a configuration datastore by creating, deleting, merging, or replacing content.   <b>Note:</b> When multiple operations are given in a single command line, the order of execution is undefined. Ciena recommends that you give one command at a time.	• <b>target</b> - the name of the configuration datastore being edited, for example, &lt;running/&gt;     • <b>default-operation</b> - can have any of the following values:      — <b>merge</b>: the configuration data in the &lt;config&gt; parameter is merged with the configuration at the corresponding level in the target datastore. (Default value)     — <b>replace</b>: the configuration data in the &lt;config&gt; parameter completely replaces the configuration in the target datastore. This can be used to load a previously-saved configuration data.     — <b>none</b>: the target datastore is unaffected by the configuration in the &lt;config&gt; parameter, unless the incoming configuration data uses the <b>operation</b> attribute to request a different operation.           • <b>error-option</b> - can have the following value:      — <b>rollback-on-error</b>: if an error condition occurs that generates an error severity &lt;rpc-error&gt; element, the server stops processing the &lt;edit-config&gt; operation and restores the specified configuration to its complete state at the start of this &lt;edit-config&gt; operation.           • <b>config</b> - the portion of the configuration to edit.
<copy-config>	Creates or replaces an entire configuration datastore with the contents of another complete configuration datastore. The target datastore is overwritten if it exists; otherwise, a new datastore is created, if allowed.	• <b>target</b> - name of the configuration datastore to use as the destination of the operation     • <b>source</b> - name of the configuration datastore to use as the source of the operation
<delete-config>	Deletes a configuration previously created with <copy-config>.	—
<close-session>	Orderly termination of a NETCONF session.	—

Operation	Description	Parameters
<code>&lt;kill-session&gt;</code>	Forced termination of a NETCONF session. All operations currently in progress are aborted, any locks or resources associated with the session are released, and any associated connections are closed.	<code>session-id</code> - the session identifier of the NETCONF session to be terminated. If this value is equal to the current session ID, an <code>invalid-value</code> error is returned.
<code>&lt;lock&gt;</code>	Locks the entire configuration datastore of a device. Allows a NETCONF client to make changes without interfering with other NETCONF clients. Non-NETCONF clients (for example, CLI scripts) can still modify the running configuration when locked. Applies to the <code>&lt;running&gt;</code> and <code>&lt;candidate&gt;</code> datastores.	—
<code>&lt;unlock&gt;</code>	Releases a configuration previously locked with the <code>&lt;lock&gt;</code> operation. Applies to the <code>&lt;running&gt;</code> and <code>&lt;candidate&gt;</code> datastores.	—
<code>&lt;commit&gt;</code>	Commits a configuration to the device.	—
<code>&lt;confirm-timeout&gt;</code>	Timeout period for the confirmed commit. If a commit is not issued within the timeout period, the operation is reverted.	—
<code>&lt;cancel-commit&gt;</code>	Cancels a confirmed commit and reverts the changes.	—

Operation	Description	Parameters
<code>&lt;discard-changes&gt;</code>	Removes the configuration from the <code>&lt;candidate/&gt;</code> datastore when determined that the <code>&lt;candidate/&gt;</code> datastore will not be committed to the <code>&lt;running/&gt;</code> datastore. Enabled by the <code>:candidate:1.0</code> capability. <b>Note:</b> The <code>&lt;discard-changes&gt;</code> operation causes the <code>&lt;candidate/&gt;</code> datastore to revert to its initial state. If required, use <code>&lt;copy-config&gt;</code> to change the contents of the <code>&lt;candidate/&gt;</code> datastore to the current <code>&lt;running/&gt;</code> datastore.	—
<code>&lt;validate&gt;</code>	Checks the configuration for syntax errors before applying the configuration.	—

### Configuration datastores

A configuration datastore is defined as a complete set of configuration data that is required to configure a device from its initial default state into a desired operational state. NETCONF defines three configuration datastores:

- `<running>` — is the datastore defined in the base model. It is the configuration that is currently running on a Waveserver 5.
- `<candidate>` — is an optional datastore. It defines a configuration that is a candidate for future use. It contains the full configuration data set that can be edited without impacting the device's current configuration.
- `<startup>` — is an optional datastore. It defines a configuration that is used when a Waveserver 5 starts. Once the chassis starts, the startup configuration becomes the running configuration.

NETCONF supports editing of configuration datastores through the `<edit-config>` operation, including testing and validating a configuration before you implement it. [NETCONF protocol operations](#) describes the `<edit-config>` operation parameters. For an example of an `<edit-config>` operation, see [Example - `<edit-config>` operation](#).

### NETCONF declarative configurations

NETCONF supports declarative configurations that allow you to send a full device configuration to Waveserver 5. Waveserver 5 then determines the differences between its current running configuration and the candidate configuration and makes only the minimum number of changes required to achieve the new configuration.

Declarative configurations can be applied using standard NETCONF `<edit-config>` operations.

- For a full replace of the candidate or running configuration:
  - `target datastore = candidate or running`
  - `error-option = rollback-on-error` (other options not supported)
  - `default-operation = replace`
- For a partial merge of the candidate or running configuration:
  - `target datastore = candidate or running`
  - `error-option = rollback-on-error` (other options not supported)
  - `default-operation = merge` (default value)

The system validates the configuration as a valid end state. If the validation fails, the configuration is rejected and not applied.

### NETCONF capabilities

NETCONF capabilities are a set of functionalities that supplement the protocol operations and are identified by a uniform resource identifier (URI). Capabilities augment the base operations of the device and describe both additional operations and the content allowed inside those operations. When a NETCONF client establishes a session with Waveserver 5, each side advertises its capabilities to the other, in the form of the YANG modules. See following example.

#### Example - server advertises base NETCONF capability

```
<hello xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
 <capabilities>
 <capability>
 urn:ietf:params:netconf:base:1.1
 </capability>
 <capability>
 urn:ietf:params:netconf:capability:startup:1.0
 </capability>
 <capability>
 http://example.net/router/2.3/myfeature
 </capability>
 </capabilities>
 <session-id>4</session-id>
</hello>
```

When a NETCONF session is opened, each peer (both client and server) must simultaneously send a `<hello>` element containing a list of its capabilities. Each peer must send at least the base NETCONF capability. A peer may also include capabilities for previous NETCONF versions, to indicate that it supports multiple protocol versions.

A server sending a <hello> element to the client must include a <session-id> element containing the session ID for the NETCONF session. If the client does not receive a <session-id> element, it terminates the NETCONF session.

In the previous example, a server advertises the base NETCONF capability, with one NETCONF capability defined in the base NETCONF document, and one implementation-specific capability.

The following table lists supported NETCONF capabilities. Only the <running> configuration is present in the base NETCONF model.

**Table 374** Waveserver 5 NETCONF capabilities

NETCONF capability	Description
base:1.0	RFC 4741 protocol version
base:1.1	RFC 6241 protocol version
writable-running:1.0	Indicates that the device supports direct writes to the <running> configuration datastore; that is, the device supports <edit-config> and <copy-config> operations where the <running> configuration is the target.
url:1.0	URL parameter support. The “file” scheme is allowed in the <url> parameter to backup and restore named config files.
xpath:1.0	Full XPath 1.0 + YANG extensions for <get> and <get-config> operations.
rollback-on-error:1.0	Indicates that the device supports the <code>rollback-on-error</code> value in the &lt;error-option&gt; parameter to the &lt;edit-config&gt; operation.   If any error occurs during the requested edit operation, the target database (usually the running configuration) will be left unaffected. This provides an 'all-or-nothing' edit mode for a single &lt;edit-config&gt; request.   For shared configurations, this feature can cause other configuration changes (for example, through other NETCONF sessions) to be inadvertently altered or removed, unless the configuration locking feature is used (that is, the lock is obtained before the &lt;edit-config&gt; operation is started). Therefore, to use this feature with shared configuration datastores, it is recommended that configuration locking also be used.

NETCONF capability	Description
candidate:1.0	Indicates that the device supports a candidate configuration datastore, which is used to hold configuration data that can be edited without impacting the device's current configuration.   The candidate configuration is a full configuration data set that serves as a workplace for creating and editing configuration data.   Additions, deletions, and changes can be made to this data to construct the desired configuration data.   A &lt;commit&gt; operation may be performed at any time that causes the device's running configuration to be set to the value of the candidate configuration.   The &lt;commit&gt; operation effectively sets the running configuration to the current contents of the candidate configuration. The candidate configuration can be shared among multiple sessions.   Unless a client has specific information that the candidate configuration is not shared, it must assume that other sessions are able to modify the candidate configuration at the same time. A client must therefore lock the candidate configuration before modifying it.   The client can discard any uncommitted changes to the candidate configuration by executing the &lt;discard-changes&gt; operation. This operation reverts the contents of the candidate configuration to the contents of the running configuration.
interleave	Indicates that the device accepts &lt;rpc&gt; requests (besides &lt;close-session&gt;) while notification delivery is active. The :notification capability must also be present if this capability is advertised.
notification	Indicates that the device supports the basic notification delivery mechanisms defined in RFC 5277, that is, the &lt;create-subscription&gt; operation will be accepted by the server.   Unless the :interleave capability is also supported, only the &lt;close-session&gt; operation must be supported by the server while notification delivery is active.   The &lt;create-subscription&gt; operation is used to set up session-specific filtering and optional buffered notification delivery parameters. These parameters cannot be modified after notification delivery has started on a particular session. The only way to stop live notification delivery is to terminate the session, usually with the &lt;close-session&gt; operation.   The NETCONF notification definitions are divided into two namespaces, one for the top-level &lt;notification&gt; element, and the other for some initial notification content.   These definitions have been converted to YANG, named ncnotifications.yang and notifications.yang.

### NETCONF guidelines

Consider the following when using NETCONF:

- To use the NETCONF server on Waveserver 5, the server must be enabled. The NETCONF server is enabled by default.
- Users with super user account privileges can query current active NETCONF user sessions and terminate a logged-in NETCONF session.
- NETCONF user operations are recorded in the Waveserver 5 command logs. As well, NETCONF syslog security logs are recorded in the Waveserver 5 syslog and NETCONF debug log files are included in the Waveserver 5 system debug log file.
- To address security concerns, you can turn off the NETCONF server using CLI command `system server netconf disable`. NETCONF is enabled by default.
- A maximum of ten active NETCONF sessions are supported.

### OpenConfig support over NETCONF

Waveserver 5 supports OpenConfig management over NETCONF using TCP port 830.

Consider the following:

- You cannot combine Waveserver-YANG and OpenConfig YANG paths in the same NETCONF request. Use either OpenConfig paths or Waveserver YANG paths in a NETCONF request.
- OpenConfig over NETCONF can be used with other management interfaces for provisioning and monitoring, such as:
  - Streaming telemetry for monitoring
  - Waveserver YANG NETCONF/REST RPCs in conjunction with the OpenConfig management
  - gRPC with OpenConfig creates, edits and deletes
  - NETCONF with Waveserver YANG creates, edits and deletes

### OpenConfig support over RESTCONF

Waveserver 5 supports OpenConfig over RESTCONF for provisioning and information retrieval on all traffic modules.

### Streaming telemetry for NETCONF

NETCONF supports the YANG-Push based subscription model as defined in RFC 8639, RFC 8640, and RFC 8641. Currently, only dynamic subscription (dial-in) is supported. Using a dynamic subscription, the client application can subscribe to receive notifications on parts of the datastore; the notifications can be sampled and sent periodically at fixed intervals or only when changes to them are detected.

For a dynamic subscription, as described in RFC 8639, a subscriber initiates a subscription request via an `<establish-subscription>` RPC over an established NETCONF transport session. A `<selection-filter>` (as described in RFC 8641) is used to identify the targeted YANG datastore node/subtree. An update `<trigger>` (as described in RFC 8641) defines when to push the data back to the client. The trigger can be either `<periodic>` (based on the desired interval) or `<on-change>` (based on when the server detects changes to the selected datastore node/subtree).

If the server is able to support the request, a positive acknowledgement with the subscription ID is returned. Otherwise, an error response is returned.

Once a subscription has been established, data will be streamed over the same transport session on which the `<establish-subscription>` RPC was received. The streamed `<push-update>` or `<push-change-update>` message will be encoded within notification messages as defined in RFC 5277. A `<push-update>` is a complete update containing all subscribed data. A `<push-change-update>` is a partial update containing only the changed portions of the subscription.

You can globally enable or disable subscriptions (using the `ciena-pro-telemetry.yang` or the CLI). If disabled, all publishing for any existing subscription on the Waveserver is suspended on all interfaces. Once re-enabled, a synchronization to the current state of the subscription data is sent to the receiver as the first update, if synchronization at the start of the subscription is specified during subscription creation.

### Periodic sampling and notifications

An `<establish-subscription>` RPC can be used to sample the data at periodic intervals in order to receive updates on an ongoing basis.

The targeted portion of the datastores is specified using a `<selection-filter>`. In the following example, the `<selection-filter>` uses the `<datastore-subtree-filter>` format. The sampling interval period is set as 1000 centiseconds (10 seconds). `<ietf-yang-push.yang>` defines the `<yp:period>` field to be of type centiseconds.

### Example of periodic sampling `<establish-subscription>` request/reply

In this example, the client is subscribing for periodic updates to a subset of the performance monitoring data, sampled every 10 seconds.

```
<?xml version="1.0" encoding="utf-8"?>
<rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="7">
 <establish-subscription xmlns="urn:ietf:params:xml:ns:yang:ietf-
subscribed-notifications"
 xmlns:yp="urn:ietf:params:xml:ns:yang:ietf-yang-push">
 <yp:datastore xmlns:ds="urn:ietf:params:xml:ns:yang:ietf-datastores">
 ds:running
```



```
 </yp:datastore>
 <yp:datastore-subtree-filter
xmlns:pm="urn:ciena:params:xml:ns:yang:ciena-ws:ciena-waveserver-pm">
 <pm:waveserver-pm>
 <pm:otu-performance-instances>
 <pm:instance-name>5-1-OTU</pm:instance-name>
 <pm:current-bin>
 <pm:statistics>
 <pm:unavailable-seconds/>
 </pm:statistics>
 </pm:current-bin>
 </pm:otu-performance-instances>
 </pm:waveserver-pm>
 </yp:datastore-subtree-filter>
 <yp:periodic>
 <yp:period>1000</yp:period>
 </yp:periodic>
 </establish-subscription>
</rpc>
```

#### RPC reply:

```
<?xml version="1.0" encoding="utf-8"?>
<rpc-reply xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="7">
 <id xmlns="urn:ietf:params:xml:ns:yang:ietf-subscribed-notifications">5</
id>
</rpc-reply>
```

#### Example of a periodic notification:

```
<?xml version="1.0" encoding="utf-8"?>
<notification xmlns="urn:ietf:params:xml:ns:netconf:notification:1.0">
 <eventTime>2021-03-22T20:24:45.741607+00:00</eventTime>
 <push-update xmlns="urn:ietf:params:xml:ns:yang:ietf-yang-push">
 <id>5</id>
 <datastore-contents>
 <waveserver-pm xmlns="urn:ciena:params:xml:ns:yang:ciena-ws:ciena-
waveserver-pm">
 <otu-performance-instances>
 <instance-name>5-1-OTU</instance-name>
 <current-bin>
 <statistics>
 <unavailable-seconds>
 <value>0</value>
 <tca-threshold>10</tca-threshold>
 <invalid-data-flag>true</invalid-data-flag>
 <supported>true</supported>
 <tca-supported>true</tca-supported>
 <tca-active>false</tca-active>
 </unavailable-seconds>
 </statistics>
 </current-bin>
 </otu-performance-instances>
 </waveserver-pm>
 </datastore-contents>
 </push-update>
</notification>
```

### On-Change subscriptions

An <establish-subscription> RPC can be used to receive notifications when changes to targeted parts of the datastore occur.

For on-change subscriptions, an initial update containing all of the requested data is first provided. After the initial update, only the data elements that have changed are streamed.

### Example of an On-Change subscription request/reply

In this example, the client is subscribing for notifications whenever the system detects any changes to the `user-management` subtree of the datastore.

```
<?xml version="1.0" encoding="utf-8"?>
<rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="8">
 <establish-subscription xmlns="urn:ietf:params:xml:ns:yang:ietf-subscribed-notifications"
 xmlns:yp="urn:ietf:params:xml:ns:yang:ietf-yang-push">
 <yp:datastore xmlns:ds="urn:ietf:params:xml:ns:yang:ietf-datastores">
 ds:running
 </yp:datastore>
 <yp:datastore-subtree-filter xmlns:t="urn:ciena:params:xml:ns:yang:ciena-ws:ciena-waveserver-aaa">
 <t:waveserver-aaa>
 <t:user-management>
 <t:active-users/>
 </t:user-management>
 </t:waveserver-aaa>
 </yp:datastore-subtree-filter>
 <yp:on-change>
 <yp:dampening-period>100</yp:dampening-period>
 </yp:on-change>
 </establish-subscription>
</rpc>
```

### Positive reply:

```
<?xml version="1.0" encoding="utf-8"?>
<rpc-reply xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="8">
 <id xmlns="urn:ietf:params:xml:ns:yang:ietf-subscribed-notifications">6</id>
</rpc-reply>
```

### Initial snapshot containing all data for all the subscribed nodes:

```
<?xml version="1.0" encoding="utf-8"?>
<notification xmlns="urn:ietf:params:xml:ns:netconf:notification:1.0">
 <eventTime>2021-03-22T20:32:01.739546+00:00</eventTime>
 <push-update xmlns="urn:ietf:params:xml:ns:yang:ietf-yang-push">
 <id>6</id>
 <datastore-contents>
 <waveserver-aaa xmlns="urn:ciena:params:xml:ns:yang:ciena-ws:ciena-waveserver-aaa">
 <user-management>
 <active-users>
 <session-id>4</session-id>
 <username>su</username>
 <interface-type>cli</interface-type>
```

```
 <client>192.168.122.1:41270</client>
 </active-users>
</active-users>
 <session-id>5</session-id>
 <username>root</username>
 <interface-type>shell</interface-type>
 <client>Console</client>
</active-users>
</active-users>
 <session-id>7</session-id>
 <username>su</username>
 <interface-type>netconf</interface-type>
 <client>192.168.122.1:52836</client>
</active-users>
</active-users>
 <session-id>8</session-id>
 <username>root</username>
 <interface-type>shell</interface-type>
 <client>192.168.122.1:39032</client>
</active-users>
</active-users>
 <session-id>9</session-id>
 <username>su</username>
 <interface-type>cli</interface-type>
 <client>192.168.122.1:41390</client>
</active-users>
</user-management>
</waveserver-aaa>
</datastore-contents>
</push-update>
</notification>
```

Subsequent change update. Following is an example of an incremental update containing only the change since the last update.

```
<?xml version="1.0" encoding="utf-8"?>
<notification xmlns="urn:ietf:params:xml:ns:netconf:notification:1.0">
 <eventTime>2021-03-22T20:33:36.971527+00:00</eventTime>
 <push-change-update xmlns="urn:ietf:params:xml:ns:yang:ietf-yang-push">
 <id>6</id>
 <datastore-changes>
 <yang-patch>
 <patch-id>0</patch-id>
 <edit>
 <edit-id>0</edit-id>
 <operation>delete</operation>
 <target>/ciena-waveserver-aaa:waveserver-aaa/user-management/active-
users[session-id='4']</target>
 </edit>
 </yang-patch>
 </datastore-changes>
 </push-change-update>
</notification>
```

**Deleting/terminating an on-going subscription**

By default, terminating/closing the session automatically deletes all active subscriptions. In addition, subscriptions can be selectively deleted or killed using their subscription-identifier. The subscription-identifier is the ID value returned when a subscription is successfully established.

**<delete-subscription> RPC**

A <delete-subscription> RPC is sent to delete an active subscription. The <id> attribute is the value returned in the <rpc-reply> of the <establish-subscription> request.

Below is an example:

```
<?xml version="1.0" encoding="utf-8"?>
<rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="16">
 <delete-subscription xmlns="urn:ietf:params:xml:ns:yang:ietf-subscribed-
notifications"
 xmlns:t="urn:ciena:params:xml:ns:yang:ciena-ws:ciena-
waveserver-aaa"
 xmlns:yp="urn:ietf:params:xml:ns:yang:ietf-yang-push">
 <id>6</id>
 </delete-subscription>
</rpc>

<?xml version="1.0" encoding="utf-8"?>
<rpc-reply xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="16">
 <ok/>
</rpc-reply>
```

**<kill-subscription> RPC**

A <kill-subscription> RPC can also be sent to kill an active subscription. The <id> attribute is the value returned in the <rpc-reply> of an <establish-subscription> request. The <kill-subscription> does not need to be associated with the transport session associated with the RPC.

Below is an example:

```
<?xml version="1.0" encoding="utf-8"?>
<rpc xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="21">
 <kill-subscription xmlns="urn:ietf:params:xml:ns:yang:ietf-subscribed-
notifications"
 xmlns:t="urn:ciena:params:xml:ns:yang:ciena-ws:ciena-
waveserver-aaa"
 xmlns:yp="urn:ietf:params:xml:ns:yang:ietf-yang-push">
 <id>7</id>
 </kill-subscription>
</rpc>

<?xml version="1.0" encoding="utf-8"?>
<rpc-reply xmlns="urn:ietf:params:xml:ns:netconf:base:1.0" message-id="21">
 <ok/>
</rpc-reply>
```

## gRPC/gNMI

The gRPC Remote Procedure Call (gRPC) protocol is a communication protocol that allows application developers to use software and scripts to easily interact with and monitor Waveserver 5. gRPC is a communication protocol and framework built to standardize data transport over HTTP/2. It uses a client-server model to build a connected system and works across a variety of programming languages and platforms. A client application directly calls methods on a remote server application, in this case Waveserver 5, and receives a response.

The gRPC Network Management Interface (gNMI) specification outlines the protocol buffer definitions and services used by the gRPC server to perform management actions, or retrieve and/or modify configuration on Waveserver 5.

Waveserver 5 uses gRPC library 1.40.0, protobuf v3.14.0, and gNMI proto v0.7.0.

The gNMI specifications can be downloaded at <https://github.com/openconfig/reference/tree/master/rpc/gnmi>.

Waveserver 5 supports management of chassis configuration using the gNMI specification with the following YANG data models:

- OpenConfig YANG
- Waveserver-YANG
- Ciena-Pro-YANG

The OpenConfig YANG models can be retrieved at the following location: <https://github.com/openconfig/public>.

Waveserver 5 is based on the OpenConfig commit version: ae5ec3b9c09f8c3b963f4eaf3fa899f49698348c dated 13 Nov 2020.

A maximum of ten active gRPC sessions are supported.

For information on OpenConfig, see [OpenConfig](#).

## Streaming telemetry for gNMI

Streaming telemetry is a method of network monitoring that allows you to continuously stream data from network devices with efficient, incremental updates. Telemetry is sent based on subscriptions. Waveserver 5 supports the subscription services defined by the gNMI protocol buffer specifications.

Streaming telemetry notifications are generated according to the type of subscription and at the frequency requested by the client. Waveserver 5 supports the following subscription RPC modes:

- **ONCE** — returns one-off data
- **STREAM** — a long-lived subscription that streams data according to triggers specified within the subscription, for example, a trigger could be when the state of an object changes, or based on a sampling interval. The **ON\_CHANGE** trigger and **SAMPLE** interval trigger are supported.

For subscription list messages, you can specify a prefix that will be applied to all paths specified within the message.

You can create subscriptions on a set of paths defined by the following OpenConfig modules:

- openconfig-platform
- openconfig-platform-transceiver
- openconfig-platform-port
- openconfig-terminal-device
- openconfig-system
- openconfig-logging
- openconfig-interfaces
- openconfig-local-routing
- openconfig-license
- openconfig-telemetry
- openconfig-transport-line-common

You can also create subscriptions on a set of paths defined by the Ciena YANG modules.

You create subscriptions for a set of paths, and you cannot modify the paths while the subscription is active. The following sections describe the recommended paths within the modules that you can use to configure subscriptions.

**Note 1:** Subscriptions and their associated channels are not persistent. If the chassis undergoes a warm or cold restart, all subscription channels are closed and the previously configured subscriptions must be re-created.

**Note 2:** You can globally enable or disable subscriptions (using the `ciena-pro-telemetry.yang` or the CLI). If disabled, all publishing for any existing subscription on the Waveserver is suspended on all interfaces. Once re-

enabled, a synchronization to the current state of the subscription data is sent to the receiver as the first update, if synchronization at the start of the subscription is specified during subscription creation.

### Subscribe requests and responses

Use the gNMI SubscribeRequest to initiate subscriptions to a set of paths within the data tree of the device. The following SubscribeRequest parameters are supported:

- Subscription list – the subscription list contains the following fields:
  - Prefix – the prefix for the subscription paths
  - Stream mode – the type of stream to be initiated. Valid values are STREAM and ONCE.
  - Encoding – the type of encoding to be used in the subscribe response. Valid values are JSON and PROTO.
  - List of subscriptions. Each subscription contains:
    - Path. The path to the data tree of the subscription
    - Mode. The subscription mode. Valid values are ON\_CHANGE and SAMPLE.
    - Sample\_interval. Only applies if the subscription mode is SAMPLE. Only a sample mode of 10s is supported.
  - updates\_only. If set, the initial states of the subscription paths are not sent to the client. A sync message is sent followed by any subsequent updates to the current state.

The gNMI SubscribeResponse includes:

- A notification message associated with the paths specified in the subscribeRequest. The notification message has a prefix which applies to all path fields in the notification.
- The timestamp at which the notification message was generated by the device.

If PROTO encoding is specified in the subscribeRequest, the notification can contain a series of update messages for each data node under the subscribed path. Each update message contains:

- Path. The path to the data node.
- Typed Value. The scalar protobuf type and value of the data node based on the scalar type. Refer to the gNMI specification for a list of supported scalar protobuf types.

If JSON encoding is specified in the subscribeRequest, the notification contains a single update message which contains:

- **Typed Value.** A structured JSON data structure containing all data nodes under the subscribed path. The type of the data in the SubscribeResponse is the json\_val.

### **Streaming of link data collection information**

Waveserver 5 supports gNMI streaming telemetry of WL5e link data. Use the ciena-waveserver-link-data-telemetry YANG model for subscription and streaming of link data.

Link data collects State of Polarization (SOP) and Bit Error Rate (BER) measurements on the WL5e line port.

Waveserver 5 uses the following sampling values to stream link data:

- Sample interval = 50 ms
- Timestamp accuracy = 1 second
- Publishing rate = 1 second

Waveserver 5 transforms the 1-second raw sample data into a single gNMI notification message and sends the message to the subscriber/collector over the gRPC channel defined for the subscription.

Note the following:

- Use the ON\_CHANGE subscription mode to stream link data (refer to the following section for information about this mode)
- The path to the subscription is:  

```
/ciena-waveserver-link-data-telemetry/waveserver-link-data/high-res-sop-data[slot=x][port=y]
```

where *x* refers to the slot number and *y* refers to the port number
- Ciena recommends streaming link data on only one WL5e line port per chassis

### **ONCE subscriptions**

Any path in the supported OpenConfig module set is valid for subscription. You can use a ONCE subscription to obtain an initial view of the entire Waveserver 5 configuration and state by requesting updates for all nodes, rather than performing a Get request. This transmits a sync\_responseSubscribe Response message after it has transmitted an update for every object requested, and then closes the subscription channel.



## STREAM subscriptions

Stream subscriptions for ON\_CHANGE and SAMPLE modes are supported on a list of pre-defined paths in Waveserver 5. You can use the STREAM subscription mode to receive a long-lived data stream of updates for all requested data elements.

After Waveserver 5 transmits an initial set of updates, it sends a sync\_response message to indicate that the initial synchronization is complete and leaves the subscription channel open.

## ON\_CHANGE subscription mode

You can create a subscription that is defined as ON\_CHANGE to receive notifications about any state change in the Waveserver 5 chassis, modules, and services. Waveserver 5 supports paths that allow for the notification of events such as the installation or removal of equipment and changes in the operational state of managed entities defined in the platform and terminal-device data models. You can also monitor active alarms. You can monitor the following equipment inventory entities using the ON\_CHANGE subscription mode:

- Common equipment such as power modules, fan modules, the access panel, and the control module.
- Service modules and associated transceivers, including filler modules, Waveserver 5 MOTR modules, modem transceivers, and QSFP28 transceivers.
- Active alarms on the Waveserver 5.

You can also use the ON\_CHANGE stream subscription to monitor the state of the logical channel. See [Example ON\\_CHANGE OpenConfig subscription paths](#) for examples of subscription paths. Waveserver 5 supports wild cards in ON\_CHANGE subscription paths. For example, you can replace `Name = slot-1` with `name = *`. Waveserver 5 does not support partial wild cards in the current release. If using wild cards and subscribing to a path with multiple keys, all keys in the path should be wild cards. Waveserver 5 does not support a heartbeat with the ON\_CHANGE subscription mode in the current release.

**Note 1:** Waveserver 5 supports a maximum of five simultaneous STREAM channels.

**Note 2:** Waveserver 5 only supports SAMPLE subscriptions to performance monitoring and statistical data. A SAMPLE subscription will generate notifications every 10 seconds with the complete set of statistical data for the subscribed paths. No config or state data will be streamed.

**Note 3:** Waveserver 5 only supports ON\_CHANGE subscriptions to config and state data. An ON\_CHANGE subscription will only generate notifications when config or state data changes. No performance monitoring or statistical data will be streamed.

**Note 4:** The same path can be subscribed with both ON\_CHANGE and SAMPLE update triggers, as long as both are supported.

**Note 5:** Contact Ciena for the supported update triggers for all waveserver-yang and OpenConfig paths. Subscription requests to a path with an unsupported update trigger will be rejected with error.

The following table lists some OpenConfig example paths to be used for ON\_CHANGE subscriptions. For example, the `/components/component[name=XCVR-1-1]/state/oper-status` for a transceiver configures the subscription to notify you of changes in the XCVR state.

**Note:** Only odd slots are used for this release.

**Table 375** Example ON\_CHANGE OpenConfig subscription paths

Event	Path	Key Values/Response
Component state changes	<code>/components/ component[name=n]/state</code>	n = OR or { XCVR--, LM-1, LM-3, LM-5, LM-7, AP-1, FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, CM-1 Waveserver }
Software version	<code>/components/ component[name=n]/state/ software-version</code>	n = {CM-1}
Hardware version	<code>/components/ component[name=n]/state/ hardware-version</code>	n = {XCVR-<slot>-<port>, LM-1, LM-3, LM-5, LM-7, AP-1, FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, CM-1 Waveserver }
Module presence	<code>/components/ component[name=n]/state/ empty</code>	n = {LM-1, LM-3, LM-5, LM-7}
Client and line XCVR presence	<code>/components/ component[name=n]/state/ empty</code>	n = {XCVR-<slot>-<port>}
Module description, including filler modules	<code>/components/ component[name=n]/state/ description</code>	n = {SLOT-1..SLOT-8}

Event	Path	Key Values/Response
Removable components presence	/components/ component[name=n]/state/ empty	n = {FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, AP-1, CM-1} Response = {True, False}
Module operational status	/components/ component[name=n]/state/ oper-status	n = {LM-1, LM-3, LM-5, LM-7} Response = {Disabled, Active, Inactive}
Transceiver state changes	/components/ component[name=n]/ transceiver/state	n = OR or {XCVR-<slot>-<port>}
Transceiver operational status	/components/ component[name=XCVR-<s>-<p>]/state/oper-status	n = XCVR-<slot>-<port> Response = {Disabled, Active, Inactive}
Optical channel operational status	/components/ component[name=n]/state/ oper-status	n = OCH-<slot>-<port> Response = {Disabled, Active, Inactive}
Waveserver 5 components operational status	/components/ component[name=n]/state/ oper-status	n = {FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, CM-1, AP-1} Response = {Disabled, Active, Inactive}
Transceiver fault condition	/components/ component[name=n]/ transceiver/state/fault-condition	n = XCVR-<slot>-<port> value = {True, False}
Logical channel state changes	/terminal-device/logical-channels/channel[index=i]/state	i = * OR see the <a href="#">openconfig-terminal-device model logical-channel index definitions</a>
Logical channel link state	/terminal-device/logical-channels/channel[index=i]/state/link-state	See the <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
TTI received message changed	/terminal-device/logical-channels/channel[index=i]/otn/state/tti-msg-recv	See the <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
TTI transmitted message changed	/terminal-device/logical-channels/channel[index=i]/otn/state/tti-msg-transmit	See the <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Test signal configuration changed	/terminal-device/logical-channels/channel[index=i]/test-signal	See the <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .

Event	Path	Key Values/Response
Monitor active alarms	/system/alarms	N/A

The following table lists some waveserver-yang example paths to be used for ON\_CHANGE subscriptions.

**Table 376** Example ON\_CHANGE waveserver-yang subscription paths

Event	Path	Key values
Chassis component state changes	/waveserver-chassis	N/A
Software state changes	/waveserver-software	N/A
System alarm changes	/waveserver-alarms/ active	N/A
XCVR state changes	/waveserver-xcvrs/ xcvrs[xcvr-id=i]/state	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
XCVR vendor data	/waveserver-xcvrs/ xcvrs[xcvr-id=i]/ properties/vendor-data	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
PTP state changes	/waveserver-pts/ pts[ptp-id=i]/state	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
PTP wavelength changes	/waveserver-pts/ pts[ptp-id=i]/ properties/transmitter	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
Port state changes	/waveserver-ports/ ports[port-id=i]/state	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
Port loopback changes	/waveserver-ports/ ports[port-id=i]/ properties/loopback	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
Port TTI changes	/waveserver-ports/ ports[port-id=i]/ properties/otn/trace	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
Port channel state changes	/waveserver-ports/ ports[port-id=i]/ channels[channel-id=j]/ state	{i = * j = *} OR {i = <slot>-<port> and j = 1-8 where <slot> = {1,3,5,7} and <port> = 1-18 }

**SAMPLE subscription mode**

A subscription stream that is defined to be sampled provides performance and metric data updates at defined intervals. Waveserver 5 supports a fixed sample interval of 10 seconds in the current release.

The following table lists some example OpenConfig paths for SAMPLE subscriptions.

**Table 377** Example OpenConfig paths for SAMPLE subscriptions

Data element	Path(s)	Key Values
Optical channel statistics	/components/component[name=n]/optical-channel/state	n = * OR OCH-<slot>-<port>, where <port> = 1 or 2
Optical channel output power	/components/component[name=n]/optical-channel/state/output-power (Note)	n = OCH-<slot>-<port>, where <port> = 1 or 2
Optical channel input power	/components/component[name=n]/optical-channel/state/input-power (Note)	n = OCH-<slot>-<port>, where <port> = 1 or 2
Optical channel chromatic dispersion	/components/component[name=n]/optical-channel/state/chromatic-dispersion	n = OCH-<slot>-<port>, where <port> = 1 or 2
XCVR statistics	/components/component[name=n]/transceiver/state	n = * OR XCVR-<slot>-<port>
XCVR physical channel statistics	/components/component[name=n]/transceiver/physical-channels/channel[index=y]/state	{ n = * y = * } OR { n = XCVR-<slot>-<port> y = 1 to 4
Client XCVR physical channel output power	/components/component[name=n]/transceiver/physical-channels/channel[index=y]/state/output-power	n = XCVR-<slot>-<port>, where <port> = 3 to 18 y = 1 to 4 for QSFP28 lane
Client XCVR physical channel input power	/components/component[name=n]/transceiver/physical-channels/channel[index=y]/state/input-power	n = XCVR-<slot>-<port>, where <port> = 3 to 18 y = 1 to 4 for QSFP28 lane
Line XCVR physical channel output power	/components/component[name=n]/transceiver/physical-channels/channel[index=y]/state/output-power	n = XCVR-<slot>-<port>, where <port> = 1 or 2 y = 1
Line XCVR physical channel input power	/components/component[name=n]/transceiver/physical-channels/channel[index=y]/state/input-power	n = XCVR-<slot>-<port>, where <port> = 1 or 2 y=1

Data element	Path(s)	Key Values
Chassis level power consumption	/components/component[name=n]/state/used-power	n = {Waveserver}
Ethernet performance statistics	/terminal-device/logical-channels/channel[index=i]/ethernet/state	i = * OR See <a href="#">openconfig-terminal-device model logical-channel index definitions</a>
Ethernet Rx CRC Errored Packets	/terminal-device/logical-channels/channel[index=i]/ethernet/state/in-crc-errors	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Ethernet Tx CRC Errored Packets	/terminal-device/logical-channels/channel[index=i]/ethernet/state/out-crc-errors	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Ethernet Rx Block Error Count	/terminal-device/logical-channels/channel[index=i]/ethernet/state/in-block-errors	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Ethernet Rx PCS Lane BIP Error Count	/terminal-device/logical-channels/channel[index=i]/ethernet/state/in-pcs-bip-errors	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Ethernet PCS ES	/terminal-device/logical-channels/channel[index=i]/ethernet/state/in-pcs-errored-seconds	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Ethernet PCS SES	/terminal-device/logical-channels/channel[index=i]/ethernet/state/in-pcs-severely-errored-seconds	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Ethernet PCS UAS	/terminal-device/logical-channels/channel[index=i]/ethernet/state/in-pcs-unavailable-seconds	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .

Data element	Path(s)	Key Values
Ethernet Tx Block Error Count	/terminal-device/logical-channels/channel[index=i]/ethernet/state/out-block-errors	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
Ethernet Tx PCS Lane BIP Error Count	/terminal-device/logical-channels/channel[index=i]/ethernet/state/out-pcs-bip-errors	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
OTN performance statistics	/terminal-device/logical-channels/channel[index=i]/otn/state	i = * OR See <a href="#">openconfig-terminal-device model logical-channel index definitions</a>
OTN errored seconds	/terminal-device/logical-channel/channel[index=i]/otn/state/errored-seconds	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
OTN severely errored seconds	/terminal-device/logical-channel/channel[index=i]/otn/state/severely-errored-seconds	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
OTN unavailable seconds	/terminal-device/logical-channel/channel[index=i]/otn/state/unavailable-seconds	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
OTN background block errors	/terminal-device/logical-channel/channel[index=i]/otn/state/background-block-errors	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
OTN FEC uncorrected words	/terminal-device/logical-channel/channel[index=i]/otn/state/fec-uncorrected-words	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .

Data element	Path(s)	Key Values
OTN pre-FEC BER instant	/terminal-device/logical-channel/channel[index=i]/otn/state/pre-fec-ber/instant	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
OTN pre-FEC BER average	/terminal-device/logical-channel/channel[index=i]/otn/state/pre-fec-ber/avg	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
OTN pre-FEC BER maximum	/terminal-device/logical-channel/channel[index=i]/otn/state/pre-fec-ber/max	See the Logical Channel columns in <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .
<b>Note:</b> You can also use leaf nodes for the path.		

The following table lists some example waveserver-yang paths for SAMPLE subscriptions.

**Table 378** Example waveserver-yang paths for SAMPLE subscriptions

Event	Path	Key values
PTP Power statistics	/waveserver-pts/pts[ptp-id=i]/properties/lanes	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
PTP Coherent statistics	/waveserver-pts/pts[ptp-id=i]/statistics	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
Chassis power consumption	/waveserver-chassis/power	N/A
Chassis temperature statistics	/waveserver-chassis/statistics	N/A
AP temperature statistics	/waveserver-chassis/access-panel/statistics	N/A
CM temperature and resource statistics	/waveserver-chassis/control-module/statistics	N/A



Event	Path	Key values
LM temperature statistics	/waveserver-modules/ modules[module-id=i]/ statistics	i = * OR {1,3,5,7}
Port statistics (Ethernet, OTU, ODU)	/waveserver-ports/ ports[port-id=i]/ statistics	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18
ODU channel statistics	/waveserver-ports/ ports[port-id=i]/ channels/statistics	i = * OR <slot>-<port> where <slot> = {1,3,5,7} and <port> = 1-18

### gRPC guidelines

Consider the following when using gRPC:

- The gRPC server is supported on Waveserver 5.
- The gRPC server on Waveserver 5 is enabled by default. To view the status of the gRPC server, or to enable or disable the server, see [Enabling and disabling the gRPC server](#).
- gNMI/gNOI operations support TACACS+ authentication, authorization, and accounting. TACACS+ operations are supported for `GetRequest`, `SetRequest`, and `SubscribeRequest` services. gNMI capability requests are not covered under these TACACS+ operations.

### gRPC authentication

Waveserver 5 supports two different authentication mechanisms between the gRPC client and the gRPC server: one-way authentication and mutual (two-way) authentication. One-way authentication is the default.

One-way authentication uses the public certification from the SSL server port of the Waveserver 5 to negotiate a secure TLS connection between the gRPC client and the gRPC server. One way authentication requires a user ID and password for the client to authenticate against the server.

Mutual authentication requires a trusted certificate to be used by the gRPC client to authenticate mutually with the gRPC server. To gain access to the server using mutual authentication, upload a PKCS12 format RSA certificate to the Waveserver 5 device. Certificates should meet the following criteria:

- Key length: 2048, 4096
- Supported Ciphers: AES128, AES192, AES256, 3DES
- Hash: sha256, sha384, sha512

The certificate should be generated by a trusted certificate authority from the publicly available Mozilla roots.pem trusted store. See [Installing a new SSL certificate](#) for details.

The gRPC server must be restarted after the certificate is installed. The gRPC server does not support unsecured connections.

For more information about configuring gRPC authentication, see [Enabling or disabling mutual authentication on a gRPC server](#).

## gNOI

Waveserver 5 supports the gRPC Network Operations Interface (gNOI) protocol buffer specifications which define a set of RPC services used for invoking configuration and operational commands on Waveserver 5. These services are meant to be used in conjunction with gNMI for state management on network targets.

All gNOI RPC commands are logged in the command and event logs, indicating the session-authenticated user which issued the command. You can access logs through the CLI or Waveserver-YANG interfaces.

Waveserver 5 supports the following gNOI features:

- `gNOI:system:RebootRequest` with methods: WARM, COLD or RESET
  - Issuing a `gNOI:RebootRequest` with a WARM or COLD method results in a corresponding warm or cold reboot of Waveserver 5.
  - If no subcomponents are specified in the request (for example, the repeated `gnoi.Path` subcomponents array is empty), it is assumed to be a chassis restart request.
  - Line module restart requests are also supported on the following paths. If any paths in the array are invalid or unsupported, the entire RPC will fail without any subcomponents rebooting.
    - `/components/component [name=LM-1]`
    - `/components/component [name=LM-3]`
    - `/components/component [name=LM-5]`
    - `/components/component [name=LM-7]`
  - Issuing a `gNOI:RebootRequest` with method = RESET results in a chassis reset to factory default. All current configuration on the Waveserver 5 chassis is lost.
  - Restart delays are not supported. All restart requests occur immediately.
- `gNOI:system:FactoryReset`

- Issuing a `gNOI:system:FactoryReset` resets the shelf to the factory default.
- `gNOI:system:TimeRequest`
  - Issuing a `gNOI:TimeRequest` returns the current time on Waveserver 5.
- `gNOI/file: Put, Get, Stat, Remove`
  - `rpc Put`
    - You can only put file to valid working directories. Valid working directories are `/tmp/public/` or `/tmp/users/<userid>/`.
  - `rpc Get`
    - You can only retrieve files from valid working directories. Valid working directories are `/tmp/public/` or `/tmp/users/<userid>/`.
    - Any file within valid working directories can be retrieved.
  - `rpc Stat`
    - You can get file status information from files in the valid working directories. Valid working directories are `/tmp/public/` or `/tmp/users/<userid>/`.
    - Status data includes file size and permissions.
  - `rpc Remove`
    - You can only delete files from valid working directories. Valid working directories are `/tmp/public/` or `/tmp/users/<userid>/`.
    - Any file within the valid working directories can be deleted.
- `gNOI:System:SetPackage` (for software upgrade)

The `gNOI:System:SetPackage` can be used to initiate a software upgrade on the **Waveserver 5**. There are two modes in which the `SetPackage` can transfer the software load onto the Waveserver 5: `SetPackage` using remote download and `SetPackage` as byte stream. The following table provides information on these two modes.

**Table 379** SetPackage using remote download and SetPackage as byte stream

Commands	Description
SetPackage using remote download	The Waveserver 5 receives the gNOI command and initiates the software load transfer from a specified server to the Waveserver 5. The path information in the RemoteDownload provides information on where the software load is downloaded from. Supported protocols for the software load transfer are HTTP(s), SFTP and SCP. For HTTP(s), the path is a URL (for example, <code>foo.com/file.tbz2</code> ). For SFTP and SCP, the path is <code>path/to/file</code> (for example, <code>host.foo.com:/bar1/bar2</code> ).
SetPackage as byte stream	When a SetPackage request contains a filename and byte stream, the transfer of the software load is initiated from the local PC to the Waveserver 5. If the software load is larger than 64 Kb, it is broken up in 64 Kb chunks for the software load transfer. The path of the load file on the local PC is specified by <code>SetPackageRequest::Package::filename</code> .
SetPackageRequest::Package::activate is set to TRUE	The <b>Waveserver 5</b> is upgraded in a one-step process. The software load is downloaded from the remote server ( SetPackage using remote download) or local PC ( SetPackage as byte stream) to the Waveserver 5. Once the software is downloaded to the Waveserver 5, the software load is activated. A successful upgrade triggers a Waveserver 5 reboot. After the completion of the upgrade, the upgrade-from software load is automatically deleted from the Waveserver 5.
SetPackageRequest::Package::activate is set to FALSE	The software load is downloaded from the remote server ( SetPackage using remote download) or local PC ( SetPackage as byte stream) to the Waveserver 5. The upgrade can then be activated at a later time by sending a SetPackage request with <code>Package:activate</code> set to TRUE and the <code>Package:version</code> set to the software version you want to upgrade to. A successful upgrade triggers a system reboot. After the completion of the upgrade, the upgrade-from software load is automatically deleted from the system.

See the following example for `gNOI:System:SetPackage`.

**Figure 150** gNOI:System:SetPackage example

```
message SetPackageRequest {
 oneof request {
 Package package = 1;
 bytes contents = 2;
 types.HashType hash = 3; // Verification hash of data.
 }

 message Package {
 filename
 version
 activate
 RemoteDownload
 }

 message RemoteDownload {
 path
 protocol
 credentials
 }
}
```

- gNOI: cert.proto install (for license activation)
  - The gNOI: cert.proto install RPC can be used to upload and install licenses on Waveserver 5 and the gNOI cert.proto install (load\_certificate: LoadCertificateRequest) RPC can be used to activate the licenses on the using a one-step process.
  - To use gNOI to activate licenses on the Waveserver 5, the license registration ID must be configured using the OpenConfig declarative configuration request using the path /components/component[name=]/properties/property[name=license-registration-id].
  - The LoadCertificateRequest:certificate-id must be the same as the license-registration-id on the Waveserver 5. Also, the LoadCertificateRequest:certificate:type field must have a value of 600.

See the following example for gNOI: cert.proto install.

**Figure 151** gNOI: cert.proto install example

```

message LoadCertificateRequest {
 certificate
 certificate_id
}

message Certificate {
 type
 bytes
}

```

- gNOI:file:TransferToRemote (for state-dump)
  - Issuing a gNOI.file.TransferToRemote request can be used to initiate a collection of state-dump on the Waveserver 5 and the state-dump Waveserver 5 can be transferred from the to a specified location.
  - The TransferToRemote local-path should be set to Request:State-Dump, to initiate the collection of the state-dump on the Waveserver 5. To transfer the state-dump to a specified location, the RemoteDownload parameters must be set to indicate where to transfer the file and the protocol to use for the file transfer. Supported protocols for the file transfer include HTTP(s), SFTP and SCP.

See the following example for gNOI:file:TransferToRemote.

**Figure 152** gNOI:file:TransferToRemote example

```

message TransferToRemoteRequest {
 local_path
 gnoi.RemoteDownload
}

message RemoteDownload {
 path
 protocol
 credentials
}

```

### gNMI declarative configurations

Waveserver 5 supports intent-driven declarative configuration. Declarative configuration allows you to send a full device configuration to Waveserver 5. Waveserver 5 then determines the differences between its current running configuration and the candidate configuration and makes only the minimum number of changes required to achieve the new configuration.

To apply a customized declarative configuration to Waveserver 5, build the configuration off-line based on the OpenConfig view of the Waveserver 5. Use the minimal configurations specified in [OpenConfig read/write paths or](#)

[configurable paths](#) as a guideline to build the configuration. Declarative configuration changes are supported via the gNMI `SetRequest` `Update` message of the type `replace`, where the entire configuration is specified as a JSON string in a single `replace` message with the `TypedValue` of `json_val`. The following is an example of a `SetRequest` `Update` message in Python.

```
request = gnmi.SetRequest()

jsonOutput = config2Json()

replace = request.replace.add()

replace.val.json_val = jsonOutput.encode()
```

### Declarative configuration considerations

Consider the following when creating a declarative configuration:

- Waveserver 5 requires the declarative configuration to contain a full device configuration. Each object that requires configuration have mandatory configurable paths, as well as optional configurable paths indicated in the [OpenConfig read/write paths or configurable paths](#) table.
- You can re-use the declarative configuration file for multiple Waveserver 5 chassis.
- Waveserver 5 returns an error message and the declarative configuration fails if an invalid declarative configuration path is set.
- Waveserver 5 validates declarative configurations against the OpenConfig YANG model definitions. Model definitions validation include: range checks, enumerated values, regular expression (for example, IP addresses), string lengths, data types, and leafref pointer. The OpenConfig leaf that a leafref points to must exist in the declarative configuration request.
- Waveserver 5 validates declarative configurations against the Waveserver 5 device internal data models. Validation include: range checks, enumerated values, regular expression (for example, IP addresses), string lengths, and data types.
- Waveserver 5 validates the declarative configuration requests against its internal policies and constraints. For example, the declarative configuration request must contain at least one super user.

### gRPC set requests

Declarative configuration can be pushed to the **Waveserver 5** using the gNMI `SetRequest` service. Waveserver 5 replies to the `SetRequest` message with a `SetResponse` message.

A gNMI `SetRequest` supports update, delete, and replace operations. All operations specified in a `SetRequest` are applied as a single transaction. Either all of it is applied or none of it is applied to the device.

The `SetRequest` supports the following parameters:

- **Prefix.** A single prefix for all of the paths that are specified for each operation.
- **Delete.** The list of paths to be deleted in the data tree of the device.
- **Update.** The list of modifications to be performed on the data tree of the device. Each update message contains:
  - **Path.** The path to be modified.
  - **Typed Value.** The type and value of the data to be applied. The supported types and values include both scalar protobuf types and JSON data of type `json_val`.
- **Replace.** List of replace operations to be performed on the data tree of the device. Each update message contains:
  - **Path.** The path to be replaced.
  - **Typed Value.** The type and value of the data for the replace operation. The supported types and values include both scalar protobuf types and JSON data of type `json_val`.

The device processes the delete operations first, followed by paths to be replaced, and finally the paths to be updated/modified.

A gNMI `SetResponse` includes the status of each operation within the `SetRequest`. For each operation specified in the `SetRequest` message, an `UpdateResult` message is included in the response field of the `SetResponse`. The order in which the operations are applied to the device is maintained.

Use the following table [OpenConfig read/write paths or configurable paths](#) as a guideline to determine the paths and parameter values when creating a `SetRequest` to configure the components of Waveserver 5.

### **gRPC get requests and responses**

You can use the gNMI `GetRequest` service with OpenConfig to retrieve configuration data for any of the OpenConfig paths supported by Waveserver 5.

A gNMI `GetRequest` supports the following options:

- **Prefix.** Specifies the prefix to the paths specified in the request.
- **Paths.** The list of paths specified for the request.



- **Type.** The type of data to be returned by the device. Supported values are: ALL, CONFIG, STATE.
- **Encoding.** The format in which the response is to be sent by the device. Supported values are: JSON (default), PROTO.

**Note 1:** The GET request service is intended for retrieving relatively small sets of data that are not resource-intensive, for example, part of the configuration. As a result, Waveserver 5 supports two simultaneous GET requests to the lowest level container of a specific OpenConfig path only.

**Note 2:** The current release supports a single path in the `GetRequest`. If the root level path is specified, then only the OpenConfig data paths are returned in the `GetResponse`.

The gNMI `GetResponse` to a request includes:

- A notification message for each path specified in the `GetRequest`. The notification message has a prefix which is the prefix and path specified in the `getRequest`. The prefix applies to all path fields in the notification.
- The timestamp at which the notification message was generated by the device.

If PROTO encoding is specified in the `GetRequest`, the notification can contain a series of update messages for each data node under the requested path. Each update message contains:

- **Path.** The path to the data node.
- **Typed Value.** The scalar protobuf type and value of the data node based on the scalar type. Refer to the gNMI specification for a list of supported scalar protobuf types.

If JSON encoding is specified in the `GetRequest`, the notification contains a single update message which contains:

- **Typed Value.** A structured JSON data structure containing all data nodes under the requested path. The type of the data in the `getResponse` is the `json_val`.

For information on OpenConfig read-only paths, see [OpenConfig read-only paths](#).

## OpenConfig

OpenConfig is a collaborative effort by network operators to develop a common set of APIs for a dynamic, programmable method for configuring and managing multi-vendor networks. These vendor-neutral data models allow you to configure and manage a network, and define the configuration and operational state of network devices for common network protocols or services. The data models are written in YANG.

Waveserver 5 supports the following OpenConfig modules:

- openconfig-platform
- openconfig-terminal-device
- openconfig-alarms
- openconfig-system
- openconfig-interfaces
- openconfig-local-routing
- openconfig-aaa
- openconfig-aaa-tacacs
- openconfig-platform-linecard
- openconfig-system-logging
- openconfig-platform-transceiver
- openconfig-license
- openconfig-transport-line-common
- openconfig-platform-port

**Note 1:** OpenConfig is supported for IPv4 only on Waveserver 5.

**Note 2:** The following modules do not support OpenConfig:

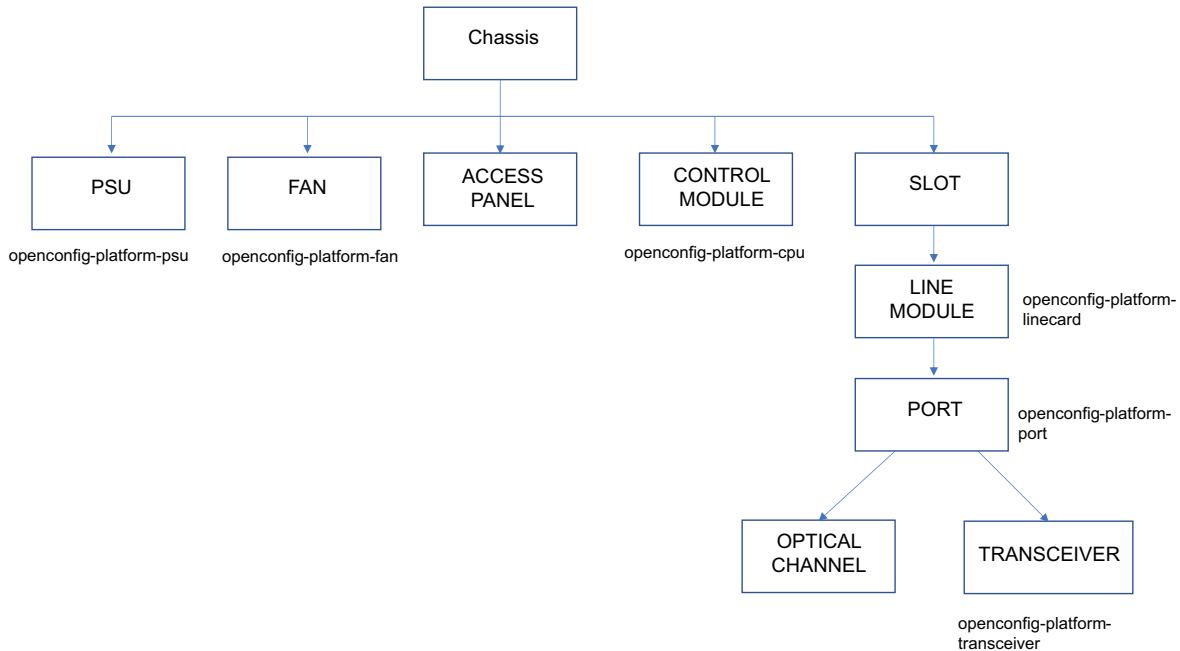
Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900)

Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module (186-3120-900)

Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module (186-3121-900)

#### **openconfig-platform module**

The openconfig-platform module defines a data model that allows you to capture the hardware and software inventory of a Waveserver 5. Each component is represented in the module. The following figure shows the structure of the openconfig-platform module.

**Figure 153** openconfig-platform model

The following components are supported on the openconfig-platform module for Waveserver 5:

- Chassis — `component [name=Waveserver]`
- Control Module — `component [name=CM-1]`
- Access Panel — `component [name=AP-1]`
- Power Modules — `component [name=PSU-<x>]` where `<x> = 1 to 2`
- Fan Units — `component [name=FAN-<x>-<y>]` where `<x> = 1 to 2`, `<y> = 1 to 2`
- Line MOTR modules — `component [name=LM-<slot>]` where `<slot> = 1, 3, 5, or 7`
- Slots — `component [name=SLOT-<slot>]` where `<slot> = 1 to 8`
- Management Ports — `component [name=<ILAN-1 | ILAN-2 | DCN-n | CONSOLE-1 | CONSOLE-2>]`, where `n = 1 to 5`
- Optical-channels — `component [name=OCH-<slot>-<port>]` where `<slot> = 1, 3, 5, or 7` and `<port> = 1 to 2`
- Transceivers — `component [name=XCVR-<slot>-<port>]` where `<slot> = 1, 3, 5, or 7` and `<port> = 1 to 18`
- Optical-port — `component [name=PORT-<slot>-<port>]` where `<slot> = 1, 3, 5, or 7` and `<port> = 1 to 18`

The following figure shows an example of the OpenConfig JSON configuration used to provision one line module in Slot 1.

**Figure 154** openconfig-platform - provisioning line module in slot 1

```
'components': {
 'component': [
 {
 'config': {
 'name': 'LM-1'
 },
 'name': 'LM-1',
 'properties': {
 'property': [
 {
 'config': {
 'name': 'module-type'
 'value': 'motr-400-2-16'
 },
 'name': 'module-type'
 }
]
 }
 }
]
}
```

Waveserver 5 allows you to set the administrative state (enabled or disabled) for the line XCVR and the client XCVR using OpenConfig. Use openconfigtransport-line-common:optical-port admin-state as shown in the following example:

**Figure 155** openconfig-platform - provisioning the administrative state of a port

```

{
 "openconfig-platform:component": [
 {
 "name": "PORT-1-1",
 "config": {
 "name": "PORT-1-1"
 },
 "port": {
 "openconfig-transport-line-common:optical-port": {
 "config": {
 "admin-state": "ENABLED"
 }
 }
 }
 }
]
}

```

Waveserver 5 allows you to enable or disable the administrative state of management ports (ILAN-1, ILAN-2, or DCN-n) using OpenConfig. The following figure shows an example of changing the ILAN-1 port admin-state.

**Figure 156** openconfig-platform - ILAN-1 management port admin-state configuration example

```

'components': {
 'component': [{
 'name': 'ILAN-1',
 'config': {
 'name': 'ILAN-1'
 },
 'properties': {
 'property': [{
 'name': 'admin-state',
 'config': {
 'name': 'admin-state',
 'value': 'ENABLED'
 }
 }]
 }
 }]
}

```

The openconfig-platform module paths are supported in declarative configuration requests, see [gNMI declarative configurations](#). For a list of supported paths, see [OpenConfig read/write paths or configurable paths](#).

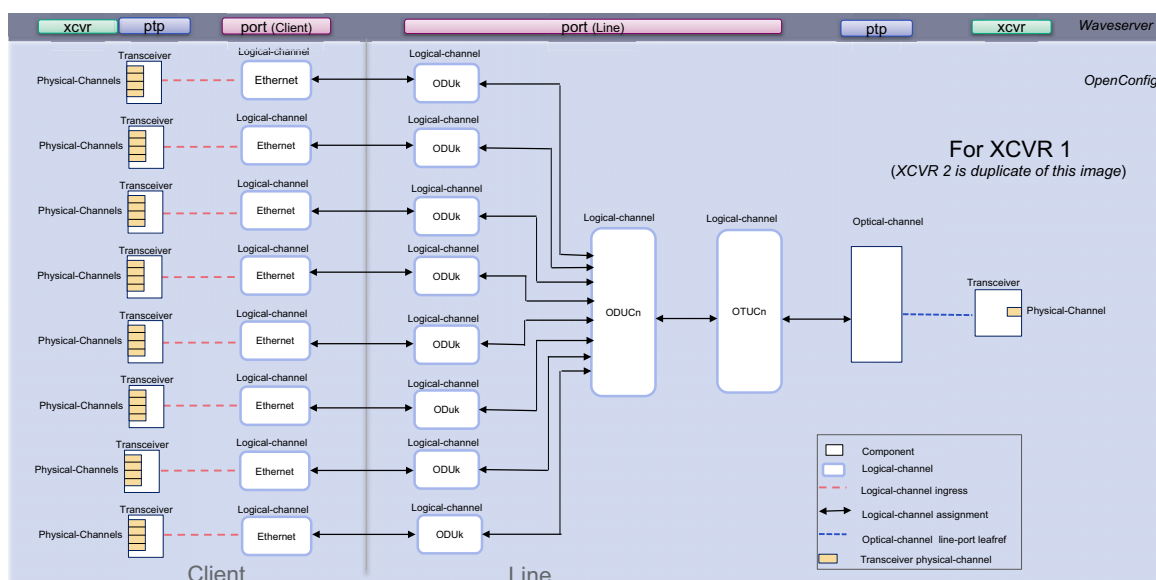
### openconfig-terminal-device module

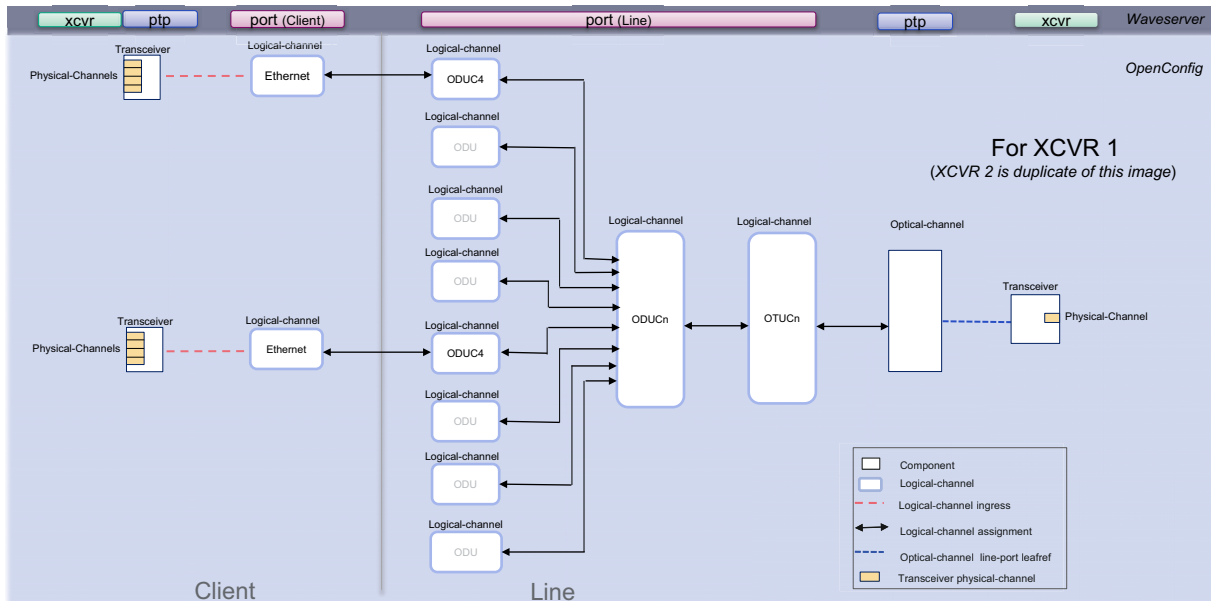
The openconfig-terminal-device module describes a terminal optics device model for managing the terminal systems (client and line side) in a DWDM transport network.

The following figures show examples of the modeling for standalone 100GE and 400GE clients over OTN transcoded lines.

Logical-channels model the ODU/OTU logical layers. Physical-channels model the physical ports, including PMs and power levels. Optical-channels model the optical carrier frequency/wavelengths and PMs.

**Figure 157** 100GE clients over OTN transcoded lines (standalone)



**Figure 158** 400GE clients over OTN transcoded lines (standalone)

The oc-terminal-device keys have two layers:

- **/terminal-device/logical-channels/channel** — represents the top level channel key representing the layers: {OTUCn | ODUc | ODUk | Client}
- **/terminal-device/logical-channels/logical-channel-assignments/assignment** — a list {keys: 1,2,3,4,5} of all mux channels in the OTN layer to which a top level key is assigned.

The following table shows a representation of the openconfig-terminal-device model index definitions.

**Table 380** openconfig-terminal-device model logical-channel index definitions

Logical Channel	Logical Channel	Logical Channel	Logical Channel	Optical Channel
Client Port <slot><port> <channel> where <slot> = 1, 3, 5, 7 <port> = 3 to 18 <channel> = 0 for 100G and 400G clients	ODUk Line Port Channels For standalone lines: <slot><port>1 <channel> For mated lines: <slot>98<port><c hannel> where <slot> = 1, 3, 5, 7 <port> = 1 to 2 <channel> = 1 to 8 for standalone lines, 1 to 16 for mated lines	ODUCn Line Ports For standalone lines: <slot><port>10 For mated lines: <slot>98<port>0 where <slot> = 1, 3, 5, 7 <port> = 1 to 2	OTUCn Line Ports For standalone lines: <slot><port>00 For mated lines: <slot>{98,99}00 where <slot> = 1, 3, 5, 7 <port> = 1 to 2	Optical Channel OCH-<slot>-<port>, where <slot>= 1, 3, 5, 7 <port>= 1 to 2

Openconfig-terminal-device on Waveserver 5 is modeled as follows:

**Table 381** openconfig-terminal-device model

openconfig-terminal-device object	Waveserver 5 object
Optical-port	WaveLogic 5 XCVR
Optical-channel	WaveLogic 5 PTP
Logical-channel rate-class = TRIB_RATE_x00G, where x is the rate of the logical channel 100G..800G for standalone lines, 100G..1600G for mated lines trib-protocol = PROT_OTUCN logical-channel-type = PROT_OTN	Line interface PORT object's OTUCn data
Logical-channel rate-class = TRIB_RATE_x00G, where x is the rate of the logical channel 100G..800G for standalone lines, 100G..1600G for mated lines trib-protocol= PROT_ODUCN logical-channel-type= PROT_OTN	Line interface PORT object's ODUCn data



openconfig-terminal-device object	Waveserver 5 object
Logical-channel rate-class = TRIB_RATE_100G trib-protocol = PROT_ODU4 logical-channel-type = PROT_OTN	Line interface CHANNEL object's ODUk data
For 100GE clients: rate-class = TRIB_RATE_100G trib-protocol = PROT_100GE logical-channel-type = PROT_ETHERNET For OTL4.4 clients: rate-class = TRIB_RATE_100G trib-protocol = PROT_OTU4 logical-channel-type = PROT_OTN or PROT_ODU4	100GE and OTL4.4 client interface PORT
For 400GE clients: rate-class = TRIB_RATE_400G trib-protocol = PROT_400GE logical-channel-type = PROT_ETHERNET	400GE client interface PORT

The operational-mode of the optical-channel specifies the modulation scheme or transmission mode of the line optical channel.

The following figure shows an example of the OpenConfig JSON configuration file used to provision the line optical channel on Slot 1, Port 1 with a line frequency of 193624.9 GHz and a target output power of 2.00 dBm.

**Figure 159** Example - Provisioning optical channel parameters

```

 'components': {
 'component': [
 {
 'config': {
 'name': 'OCH-1-1'
 },
 'name': 'OCH-1-1',
 'optical-channel': {
 'config': {
 'frequency': '193624900',
 'operational-mode': 2,
 'target-output-power': '2.00'
 }
 },
 'properties': {
 'property': [
 {
 'config': {
 'name': 'line-system-type',
 'value': 'colourless'
 },
 'name': 'line-system-type'
 }
]
 }
 }
]
 }
]
}

```

The following figure shows an example of an OpenConfig JSON configuration file used to provision Client Port 3 in Slot 1 to 100GE and Client Port 4 in Slot 1 to OTL4.4.

**Figure 160** openconfig-terminal-device - provisioning client ports example for setting client port 3 on slot 1 to 100GE and client port 8 on slot 1 to OTL4.4

```
"terminal-device": {
 "logical-channels": {
 "channel": [
 {
 "config": {
 "index": 10300,
 "logical-channel-type": "PROT_ETHERNET",
 "rate-class": "TRIB_RATE_100G",
 "trib-protocol": "PRTO_100GE"
 },
 "index": 10300
 },
 {
 "config": {
 "index": 10800,
 "logical-channel-type": "PROT_OTN",
 "rate-class": "TRIB_RATE_100G",
 "trib-protocol": "PRTO_100GE"
 },
 "index": 10800
 }
]
 }
}
```

#### Terminal device data model indexes

The following optical-channel and logical-channel objects in the openconfig-terminal-device data model are supported on Waveserver 5:

**Table 382** openconfig-terminal-device data models

Logical channel index type	Index	Location	Possible values	Example
OTUCn logical channel indexes for standalone lines	<s><p>00	slot-<s>, port-<p>	where s = 1, 3, 5, 7 p = 1 to 2	1100 for slot-1, port-1
OTUCn logical channel indexes for mated lines	<s>{98,99}00	slot-<s>	where s = 1, 3, 5, 7	19800 for slot 1
ODUCn logical channel indexes for standalone lines	<s><p>10	slot-<s>, port-<p>	where s = 1, 3, 5, 7, p = 1 to 2	1110 for slot-1, port -1
ODUCn logical channel indexes for mated lines	<s>98<p>0	slot-<s>, port-<p>	where s = 1, 3, 5, 7, p = 1 to 2	19810 for slot-1, port-1
ODUk logical channel indexes for standalone lines	<s><p>1<c>	slot-<s>, port-<p>, channel-<c>	where s = 1, 3, 5, 7, p = 1 to 2, c = 1 to 8	1111 for slot-1, port-1, channel-1
ODUk logical channel indexes for mated lines	<s>98<p><c>	slot-<s>, port-<p>, channel-<c>	where s = 1, 3, 5, 7, p = 1 to 2, c = 1 to 16	19811 for slot-1, port-1, channel-1
OTU4 logical channel on OTL4.4 clients	<s><pp>00	slot-<s>, port-<pp>	where s = 1, 3, 5, 7, pp = 03 - 18	10500 for slot 1 port 5 OTU4
ODU4 logical channel on OTL4.4 clients	<s><pp>10	slot-<s>, port-<pp>	where s = 1, 3, 5, 7, pp = 03 - 18	10510 for slot 1 port 5 ODU4
100GE and 400GE clients	<s><pp>00	slot-<s>, port-<pp>	where s = 1, 3, 5, 7, pp = 03 - 18	10500 for slot 1 port 5 100GE or 400GE

**Openconfig ethernet counters**

Waveserver 5 supports openconfig ethernet counters for ethernet clients on a 10 second bin level. See the following figure for supported leaves.

**Figure 161** Supported leaves for openconfig ethernet counters

```

"terminal-device":{
 "logical-channels":{
 "channel[index=<i>]":{
 "ethernet":{
 "state":{
 "in-crc-errors":
 "in-block-errors":
 "in-pcs-bip-errors":
 "in-pcs-errored-seconds":
 "in-pcs-severely-errored-seconds":
 "in-pcs-unavailable-seconds":
 "out-pcs-bip-errors":
 "out-crc-errors":
 "out-block-errors":
 }
 }
 }
 },
}

```

**OpenConfig test-signal**

Waveserver 5 supports a single leaf under the terminal-device logical channel to configure a test signal on the client 100GE port. The OpenConfig test-signal leaf is only supported on 100GE client ports and it controls CLI command `application its enable` and `application its disable`. See the following figure.

**Figure 162** Configuring test-signal

```

"terminal-device": {
 "logical-channels": {
 "channel": [
 {
 "config": {
 "index": 10300,
 "test-signal": true
 },
 "index": 10300
 }
]
 }
}

```

**Note:** For the Openconfig test-signal, default values are used: `test-signal-ber=0` and `test-signal-duration=continuous`. These attributes, including the test signal reporting and monitoring are performed via CLI.

The openconfig-terminal-device data module paths are supported in declarative configuration requests. For a list of supported paths, see [OpenConfig read/write paths or configurable paths](#).

### openconfig-system module

Waveserver 5 supports the openconfig-system module. The module allows you to provision the Waveserver 5 hostname, domain name, DNS servers, and NTP servers (see the following figure for an example OpenConfig JSON configuration file). For a list of supported paths, see [OpenConfig read/write paths or configurable paths](#).

**Figure 163** openconfig-system module example - provisioning hostname, domain name, and DNS servers

```
{
 "system": {
 "config": {
 "domain-name": "xyz.abcx.com",
 "hostname": "ABX1"
 }
 },
 "ntp": {
 "servers": {
 "server": [{
 "address": "10.100.10.100",
 "config": {
 "address": "10.100.10.100"
 }
 }]
 },
 "config": {
 "enabled": true
 }
 },
 "dns": {
 "servers": {
 "server": [{
 "address": "10.100.10.100",
 "config": {
 "address": "10.100.10.100"
 }
 }]
 }
 }
}
```

### Openconfig-system-logging module

The openconfig-logging module defines syslog server configuration on Waveserver 5. The openconfig-logging module paths are supported in declarative configuration requests, allowing you to define the host, the source address, remote port, the facility and the severity. For a list of supported paths, see [OpenConfig read/write paths or configurable paths](#).

The following figure shows an example of an openconfig JSON configuration file used to provision a syslog collector and supported severities on Waveserver 5.

**Figure 164** openconfig-logging module example

```

"system": {
 "logging": {
 "remote-servers": {
 "remote-server": [{
 "config": {
 "remote-port": 514,
 "host": "10.100.10.100"
 },
 "selectors": {
 "selector": [{
 "config": {
 "facility": "SYSTEM_DAEMON",
 "severity": "ALERT"
 },
 "facility": "SYSTEM_DAEMON",
 "severity": "ALERT"
 },
 {
 "config": {
 "facility": "SYSTEM_DAEMON",
 "severity": "DEBUG"
 },
 "facility": "SYSTEM_DAEMON",
 "severity": "DEBUG"
 },
 {
 "config": {
 "facility": "SYSTEM_DAEMON",
 "severity": "EMERGENCY"
 },
 "facility": "SYSTEM_DAEMON",
 "severity": "EMERGENCY"
 },
 {
 "config": {
 "facility": "SYSTEM_DAEMON",
 "severity": "ERROR"
 },
 "facility": "SYSTEM_DAEMON",
 "severity": "ERROR"
 },
 {
 "config": {
 "facility": "SYSTEM_DAEMON",
 "severity": "INFORMATIONAL"
 },
 "facility": "SYSTEM_DAEMON",
 "severity": "INFORMATIONAL"
 }
]
 }
]
 }
}

```



```
 {
 "config":{
 "facility": "SYSTEM_DAEMON",
 "severity": "NOTICE"
 },
 "facility": "SYSTEM_DAEMON",
 "severity": "NOTICE"
 },
 {
 "config":{
 "facility": "SYSTEM_DAEMON",
 "severity": "WARNING"
 },
 "facility": "SYSTEM_DAEMON",
 "severity": "WARNING"
 }
],
 "host": "10.100.10.100"
}
}
```

### Openconfig-interfaces module

The openconfig-interfaces module allows you to define and manage interfaces and sub-interfaces on Waveserver 5. You can use the openconfig-interface model to provision the IP address and subnet mask, or enable/disable DHCP. For a list of supported paths, see [OpenConfig read/write paths or configurable paths](#).

The following figure shows an example of disabling DHCP and provisioning a static IP address and subnet mask on Waveserver 5.

**Figure 165** openconfig-interfaces module example

```

{
 'interfaces': {
 'interface': [{
 'subinterfaces': {
 'subinterface': [{
 'index': 1,
 'config': {
 'index': 1
 },
 'ipv4': {
 'addresses': {
 'address': [{
 'config': {
 'prefix-length': 22,
 'ip': '10.100.10.100'
 },
 'ip': '10.100.10.100'
 }
],
 'config': {
 'dhcp-client': false
 }
 }
]
 },
 'config': {
 'name': 'local'
 },
 'name': 'local'
]
 }
}

```

**openconfig-local-routing module**

Waveserver 5 supports the openconfig-local-routing module. Waveserver 5 uses openconfig-local-routing to provision the default static route to the gateway NE. For a list of supported paths, see [OpenConfig read/write paths or configurable paths](#).

See the following figure for an example OpenConfig JSON configuration file using the openconfig-local-routing model to provision the default static route.

**Figure 166** openconfig-local-routing module configuration example

```

{
 "local-routes": {
 "static-routes": {
 "static": [
 {
 "config": {
 "prefix": "0.0.0.0/0"
 },
 "prefix": "0.0.0.0/0",
 "next-hops": [
 {
 "next-hop": [
 {
 "index": "1",
 "config": {
 "index": "1",
 "next-hop": "192.168.20.3"
 }
 }
]
 }
]
 }
]
 }
 }
}

```

**openconfig-aaa module**

The openconfig-aaa module allows you to provision local user accounts on Waveserver 5, manage the root password, and add TACACS servers for authentication.

The following figure shows an example OpenConfig JSON configuration file that is used to provision the root password and a local user account.

**Figure 167** openconfig-aaa module example

```

{
 "system": {
 "aaa": {
 "authentication": {
 "admin-user": {
 "config": {
 "admin-password": "NewPassword"
 }
 }
 },
 "users": {
 "user": [
 {
 "config": {
 "role": "super",
 "username": "suser1",
 "password": "NewPassword2"
 },
 "username": "suser1"
 }
]
 }
 }
 }
}

```

**openconfig-aaa-tacacs module**

The openconfig-aaa-tacacs module allows you to provision TACACS servers for authentication, authorization, and accounting.

**Note:** The remote authentication, authorization, and accounting server priority are mandatory. Since openconfig-aaa does not allow for a priority against a particular server, the Waveserver model uses the server name config attribute to specify the priority of the server.

The following shows an example OpenConfig JSON configuration file to provision TACACS servers for authentication, authorization, and accounting.

```
"system": {
 "aaa": {
 "server-groups": {
 "server-group[name=accounting-tacacs]": {
 "name": "accounting-tacacs",
 "config": {
 "name": "accounting-tacacs"
 }
 }
 }
 },
 "servers": {
 "server[address=<IP Address>]": {
 "address": <IP Address1>,
 "config": {
 "address": <IP Address1>,
 "timeout": 5
 "name": <server priority 1..8>
 }
 "tacacs": {
 "config": {
 "secret-key": "*****"
 }
 }
 }
 }
},
"server-group[name=authentication-tacacs]": {
 "name": "authentication-tacacs",
 "config": {
 "name": "authentication-tacacs"
 }
}
"servers": {
 "server[address=<IP Address>]": {
 "address": <IP Address2>,
 "config": {
 "address": <IP Address2>,
 "timeout": 5
 "name": <server priority 1..8>
 }
 "tacacs": {
 "config": {
 "secret-key": "*****"
 }
 }
 }
}
},
```

```
"server-group[name=authorization-tacacs]": {
 "name": "authorization-tacacs",
 "config": {
 "name": "authorization-tacacs"
 }
},
"servers": {
 "server[address=<IP Address>]": {
 "address": <IP Address3>,
 "config": {
 "address": <IP Address3>,
 "timeout": 5
 },
 "name": <server priority 1..8>
 },
 "tacacs": {
 "config": {
 "secret-key": "*****"
 }
 }
},
},
"server-group[name=global-tacacs]": {
 "name": "global-tacacs",
 "config": {
 "name": "global-tacacs"
 },
 "servers": {
 "server[address=<IP Address>]": {
 "address": <IP Address4>,
 "config": {
 "address": <IP Address4>,
 "timeout": 5
 },
 "name": <server priority 1..8>
 },
 "tacacs": {
 "config": {
 "secret-key": "*****"
 }
 }
 },
},
},
},
},
},
}
```

### **openconfig-platform-transceiver**

This model can be used to turn off/on the line and client lasers.

The following figure shows an example OpenConfig JSON configuration file that is used to turn off/on the line and client lasers.

**Figure 168** Openconfig-platform-transceiver example - turning the client laser off on XCVR-1-3

```

"components": {
 "component": [
 {
 "config": {
 "name": "XCVR-1-3"
 },
 "name": "XCVR-1-3",
 "transceiver": {
 "physical-channels": {
 "channel": [
 {
 "config": {
 "tx-laser": false
 },
 "index": 1
 }
]
 }
 }
 }
]
}

```

**openconfig-license**

This model can be used to upload and activate a license file.

**Note:** License server configuration is not available through OpenConfig.

The following figures show example OpenConfig JSON configuration files to upload and activate a license file and retrieve the configuration.

**Figure 169** openconfig-license example: license file upload/activate

```

{
 "openconfig-license:licenses": {
 "license": [
 {
 "license-id": "license-registration-id",
 "config": {
 "license-id": "license-registration-id",
 "license-key": "<<license registration id here>>>"
 }
 },
 {
 "license-id": "license-data1",
 "config": {
 "license-id": "license-data1",
 "license-data": "<< binary stream of data from license file>>>"
 }
 },
 {
 "license-id": "license-data2",
 "config": {
 "license-id": "license-data2",
 "license-data": "<< binary stream of data from license file>>>"
 }
 }
]
 }
}

```

**Figure 170** openconfig-license example: license file retrieval (only partial output shown)

```

{
 "openconfig-license:licenses": {
 "license": [
 {
 "license-id": "license-registration-id",
 "config": {
 "license-id": "license-registration-id",
 "license-key": "<<license registration id>>"
 },
 "state": {
 "license-id": "license-registration-id",
 "license-key": "<<license registration id>>",
 "active": true
 }
 },
 {
 "license-id": "License-01",
 "config": {
 "license-id": "License-01"
 },
 "state": {
 "license-id": "License-01",
 "description": "Software Release License",
 "active": true
 }
 }
]
 }
}

```

**openconfig-transport-line-common**

You can use the openconfig-transport-line-common model to manage transport line elements.

**openconfig-telemetry**

You can use the openconfig-telemetry model to manage subscriptions. The following leaves are not supported within the dynamic subscriptions container:

- destination-address
- destination-port
- heartbeat-interval
- suppress-redundant
- originated-qos-marking
- exclude-filter

**OpenConfig read/write paths or configurable paths**

The following table provides information on OpenConfig read/write paths or configurable paths.



**Table 383** OpenConfig read/write paths or configurable paths

Category	Path	Value	Additional information
Access panel	/components/component[name=AP-1]/config/name	AP-1	—
Access panel	/components/component[name=AP-1]/properties/property[name=admin-state]/config/value	true, false	Configures the access panel administrative state. Default: TRUE
Fan Unit 1	/components/component[name=FAN-1-1]/config/name  /components/component[name=FAN-1-2]/config/name	FAN-1-1 FAN-1-2	—
Fan Unit 1	/components/component[name=FAN-1-1]/properties/property[name=admin-state]/config/value  /components/component[name=FAN-1-2]/properties/property[name=admin-state]/config/value	true, false	Default: TRUE
Fan Unit 2	/components/component[name=FAN-2-1]/config/name  /components/component[name=FAN-2-2]/properties/property[name=admin-state]/config/value	FAN-2-1 FAN-2-2	—
Fan Unit 2	/components/component[name=FAN-2-1]/properties/property[name=admin-state]/config/value  /components/component[name=FAN-2-2]/properties/property[name=admin-state]/config/value	true, false	Default: TRUE

Category	Path	Value	Additional information
Power Supply 1	/components/ component[name=PSU-1]/config/ name	PSU-1	—
Power Supply 1	/components/ component[name=PSU-1]/ properties/ property[name=admin-state]/ config/value	true, false	Default: TRUE
Power Supply 2	/components/ component[name=PSU-2]/config/ name	PSU-2	—
Power Supply 2	/components/ component[name=PSU-2]/ properties/ property[name=admin-state]/ config/value	true, false	Default: TRUE
CM	/components/component[name=CM-1]/config/name	CM-1	—
CM	/components/component[name=CM-1]/properties/ property[name=admin-state]/ config/value	true, false	Default: TRUE
Management Ports	/components/component[name=n]/ config/name	ILAN-1, ILAN-2, DCN- n	—
Management Ports	/components/component[name=n]/ properties/ property[name=admin-state]/ config/name=admin-state	DISABLED, ENABLED	Default: TRUE
MOTR provisioning	/components/component[name=LM- <slot>]/config/name	LM-<slot>	If absent, the module and dependent objects are deleted.
MOTR provisioning	/components/component[name=LM- <slot>]/properties/ property[name=module-type]/ config/value	MOTR-400-2-16	Specifies the line module type as MOTR.

Category	Path	Value	Additional information
MOTR provisioning	/components/component[name=n]/properties/property[name=label]/config/value=<label>	Where n = LM-1, LM-3, LM-5, and LM-7  <label> = string, up to 128 characters in length	Specifies a label for the line module (optional).
MOTR provisioning	/components/component[name=n]/properties/property[name=admin-state]/config/value	DISABLED, ENABLED	Specifies the administrative state of the line module (optional)
MOTR provisioning	/components/component[name=n]/config/name	Where n = XCVR-x-y, x={1..8}, y={3..18}	—
MOTR provisioning	/components/component[name=LM-<slot>]/properties/property[name=mode]/config/value	NONE, MATED	NONE for x00G standalone lines, MATED for x50G or x00G mated lines

Category	Path	Value	Additional information
MOTR provisioning	/components/ component[name=OCH-<slot>-<port>]/optical-channel/ config/frequency <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 1 to 2	Frequency in GHz See <a href="#">Line port specifications</a> for details. <b>Note:</b> To maintain precision, the OpenConfig configuration request should be in multiples of 100 MHz. This is because OpenConfig frequency specifications are in GHz and Waveserver 5 supports frequency specifications in MHz to 1 decimal place.	Default: 0
MOTR provisioning	/components/ component[name=OCH-<slot>-<port>]/optical-channel/ config/target-output-power <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 1 to 2	See <a href="#">Line port specifications</a> for details.	Default: -9.00
MOTR provisioning	/components/ component[name=OCH-<slot>-<port>]/optical-channel/ config/operational-mode <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 1 to 2	The transmission mode of the transmitter.	—

Category	Path	Value	Additional information
MOTR provisioning	/terminal-device/logical-channels/ channel[index=<slot><port>]/ config/trib-protocol  <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 3 to 18	PROT_100GE, PROT_400GE, PROT_OTU4	—
MOTR provisioning	/terminal-device/logical-channels/ channel[index=<slot><port>]/ config/rate-class  <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 3 to 18	TRIB_RATE_100G, TRIB_RATE_400G	—
MOTR provisioning	/terminal-device/logical-channels/ channel[index=<slot><port>]/ config/logical-channel-type  <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 3 to 18	PROT_ETHERNET, PROT_OTN	—
MOTR provisioning	/terminal-device/logical-channels/ channel[index=<slot><port>]/ config/admin-state  <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 3 to 18	DISABLED, ENABLED,	Default: Enabled
MOTR provisioning	/terminal-device/logical-channels/ channel[index=<slot><port>]/ config/loopback-mode  <b>Note:</b> Where <slot> = 1, 3, 5, 7 and <port> = 3 to 18	NONE, TERMINAL, FACILITY	Default: NONE
Line transceivers	/components/component[name=n]/ port/optical-port/config/ admin-state  <b>Note:</b> n = PORT-x-y, x = {1..8}, y = {1, 2}	DISABLED, ENABLED	Specifies the administrative state of the line XCVR object. Default: ENABLED
Line transceivers	/components/component[name=n]/ properties/ property[name=service-profile]/config/value	100GE, 400GE, MIXED-ETHERNET, MIXED-OTN-ETHERNET	Specifies the service profile of the line XCVR object.

Category	Path	Value	Additional information
Client transceivers	/components/component[name=n]/properties/property[name=forward-error-correction]/config/value <b>Note:</b> n = XCVR-x-y, x = {1..8}, y = {3..18}	DISABLED, ENABLED	Enables or disables FEC on the client transceiver. Default: Based on XCVR type
Client transceivers	/components/component[name=n]/properties/property[name=conditioning-type]/config/value <b>Note 1:</b> n = XCVR-x-y, x = {1..8}, y = {3..18} <b>Note 2:</b> Applicable for client transceivers with an XCVR mode of 100GE only.	Ethernet, Laser-off, None	Specifies the conditioning type for the client transceiver. Default: Ethernet
Client transceivers	/components/component[name=n]/properties/property[name=conditioning-holdoff]/config/value <b>Note:</b> n = XCVR-x-y, x = {1..8}, y = {3..18}	Value = conditioning holdoff in ms: < 0   10   20   30   40   50   60   70   80   90   100   200   300   400   500   600   700   800   900   1000 >	Specifies the conditioning holdoff time. Default: 10 ms
Client transceivers	/components/component[name=n]/port/optical-port/config/admin-state <b>Note:</b> n = PORT-x-y, x = {1..8}, y = {3..18}	DISABLED, ENABLED	Enables or disables the administrative state of the client XCVR and client PTP objects.
Client/line transceivers	/components/component[name=XCVR-n]/transceiver/physical-channels/channel[index=n]/config/tx-laser	DISABLED, ENABLED	Enables or disables the transmit label for the channel.

Category	Path	Value	Additional information
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/ property[name=expected-restoration-time]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	1 - 300 seconds	Sets the maximum restoration time (in seconds)  See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=line-system-type]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	coloured, colourless, contentionless, cs-coloured, cscolourless	Sets the line system type.  See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/ property[name=modem-class]/ config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	Terrestrial, Submarine	Sets the modem class.  See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/ property[name=reach-limit]/ config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	reach-limit for terrestrial: 1000, 2000, 5000  reach-limit for submarine: 3500, 9000, 14000	Sets the reach limit.  See <a href="#">ptp set ptp</a> for default value.

Category	Path	Value	Additional information
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=odu-termination-mode]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	Terminate	Sets the odu-termination-mode. Only <code>Terminate</code> supported in OpenConfig.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=link-optimization-mode]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	standard, snr-tolerant, fast-tracking, spectral-efficiency, latency, foreign, narrow, non-linear	Sets the link optimization mode. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=laser-tuning-mode]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	normal, accelerated	Sets the laser tuning mode. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=laser-centering-mode]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	disabled, enabled, holdover	Sets the laser centering mode. See <a href="#">ptp set ptp</a> for default value.



Category	Path	Value	Additional information
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=laser-centering-range]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	Laser Centering Range in GHz (N.M): 0.0 .. 3.0	Sets the laser centering range. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=laser-shutdown-mode]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	normal, accelerated	Sets the shutdown mode. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=channel-contention-detection-avoidance]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	false, true	Sets the contention detection avoidance. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=tx-power-reduction-state]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	false, true	Sets the tx power reduction state. See <a href="#">ptp set ptp</a> for default value.

Category	Path	Value	Additional information
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=tx-dispersion-compensation-mode]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	automatic, fixed, search	Sets the tx dispersion compensation value. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=tx-dispersion-precompensation-value]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	-44500..1000000	Sets the tx dispersion precompensation value. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=tx-dispersion-search-mode-min]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	Units: ps/nm Range: -89000 to 1044500	Sets the tx dispersion search mode min. See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=tx-dispersion-search-mode-max]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	Units: ps/nm Range: -89000 to 1044500	Sets the tx dispersion search mode max. See <a href="#">ptp set ptp</a> for default value.

Category	Path	Value	Additional information
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=tx-dispersion-path-value]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	Units: ps/nm Range: -89000 ..1244500 ps/nm	Sets the tx dispersion path value See <a href="#">ptp set ptp</a> for default value.
Line optical-channels <b>Note:</b> See <a href="#">Physical Termination Point (PTP) commands</a> for additional details.	/components/component[name=n]/properties/property[name=admin-state]/config/value <b>Note:</b> n = OCH-x-y, x = {1..8}, y = {1..2}	DISABLED, ENABLED	Enables or disables the administrative state of the line PTP object. See <a href="#">ptp set ptp</a> for default value.
Line terminal-device logical-channels	/terminal-device/logical-channels/channel[index=i]/config/description <b>Note:</b> i = <slot><port>00 and refers to the line PORT object	String up to 128 characters	Sets the description (optional).
Line terminal-device logical-channels	/terminal-device/logical-channels/channel[index=i]/config/admin-state <b>Note:</b> i = <slot><port>00 and refers to the line PORT object	DISABLED, ENABLED	Enables or disables the administrative state of the line PORT object. Default: Enabled
Line terminal-device logical-channels	/terminal-device/logical-channels/channel[index=i]/config/loopback-mode <b>Note:</b> i = <slot><port>10 and refers to the line PORT object	none, terminal, facility	Sets the loopback mode. Default: None
Line terminal-device logical-channels	/terminal-device/logical-channels/channel[index=i]/config/description <b>Note:</b> i = <slot><port>1<channel> and refers to the line CHANNEL object	String up to 128 characters	Sets the description (optional).

Category	Path	Value	Additional information
Line terminal-device logical-channels	/terminal-device/logical-channels/channel[index=i]/config/admin-state <b>Note:</b> i = <slot><port>1<channel> and refers to the line CHANNEL object	DISABLED, ENABLED	Enables or disables the administrative state of the line CHANNEL object. Default: ENABLED
Client terminal-device channels	/terminal-device/logical-channels/channel[index=20]/config/description <b>Note:</b> i = <slot><port> and refers to the client PORT object	String up to 128 characters	Sets the description for the client port object (optional).
Client terminal-device channels	/terminal-device/logical-channels/channel[index=20]/config/admin-state <b>Note:</b> i = <slot><port> and refers to the client PORT object	DISABLED, ENABLED	Enables or disables the administrative state of the client PORT object. Default: ENABLED
Client terminal-device channels	/terminal-device/logical-channels/channel[index=n]/otn/config/tti-msg-transmit	String	Specifies the transmitted TTI message.
Client terminal-device channels	/terminal-device/logical-channels/channel[index=n]/otn/config/tti-msg-expected	String	Specifies the expected TTI message.
Client terminal-device channels	/terminal-device/logical-channels/channel[index=n]/otn/config/tti-msg-auto	TRUE, FALSE	Whether to automatically create the TTI message.
Client terminal-device channels	/terminal-device/logical-channels/channel[index=n]/config/test-signal	TRUE, FALSE	Enable or disable test signal generation.
Local User operations	/system/aaa/authentication/users/user[username=<name>]/config/username	See <a href="#">user create user</a> for values.	—
Local User operations	/system/aaa/authentication/users/user[username=<name>]/config/password	See <a href="#">user create user</a> for values.	—
Local User operations	/system/aaa/authentication/users/user[username=<name>]/config/role <string>	The access level of the user: super, admin, limited	—

Category	Path	Value	Additional information
Admin User operations	/system/aaa/authentication/admin-user/config/admin-password	See <a href="#">user create user</a> for values.	—
SSH Server management	/system/ssh-server/config/timeout	60...90000 seconds in multiples of 60	Rounding takes place and rounds down to the nearest minute.
TACACS Server management	/system/aaa/authentication/config/authentication-method	TACACS_ALL, LOCAL_ALL, RADIUS_ALL	—
TACACS Server management	/system/aaa/authentication/config/authorization-method	TACACS_ALL, LOCAL	—
TACACS Server management	/system/aaa/authentication/config/accounting-method	TACACS_ALL, LOCAL	—
TACACS Server management	/system/aaa/server-groups/server-group[name=servergroup-1]/servers/server[address=<ip address>]/config/address	A valid IP address	TACACS supports only one server-group, with a fixed name of “servergroup-1”.
TACACS Server management	/system/aaa/server-groups/server-group[name=servergroup-1]/servers/server[address=<ip address>]/config/timeout	—	TACACS supports only one timeout parameter and one secret-key that are used by all provisioned TACACS servers.   OpenConfig models the timeout and secret-key at a per-server level. This means that if you change the timeout and secret-key via OpenConfig for any of the TACACS servers, it will be changed for all provisioned TACACS servers.

Category	Path	Value	Additional information
TACACS Server management	/system/aaa/server-groups/server-group[name=servergroup-1]/servers/server[address=<ip address>]/tacacs/config/secret-key	—	The TACACS secret-key has a default minimum setting of 8 characters.
NTP Server Management	/system/ntp/config/enabled	false, true	NTP MD5 authentication keys are not supported via OpenConfig in OS 2.2.
NTP Server Management	/system/ntp/servers/server[address=<ip address>]/address	A valid IP address	NTP MD5 authentication keys are not supported via OpenConfig in OS 2.2.
DNS services	/system/config/domain-name	String 1..63	—
DNS services	/system/dns/servers/server[address=<ip address>]/config/address	A valid IP address	—
DNS services	/system/dns/servers/server[address=<ip address>]/config/port	—	—
Hostname Provisioning	/system/config/hostname	A valid host name	—
Syslog Provisioning	/system/logging/remote-servers/remote-server[host=<hostname>]/config/host	The host name for the remote syslog server (collector)	—
Syslog Provisioning	/system/logging/remote-servers/remote-server[host=<hostname>]/config/source-address	A valid IP address for the remote syslog server	—
Syslog Provisioning	/system/logging/remote-servers/remote-server[host=<hostname>]/config/remote-port	The remote port number for UDP messages sent to the server.	The default port number is 514.

Category	Path	Value	Additional information
Syslog Provisioning	/system/logging/remote-servers/remote-server[host=<hostname>]/selectors/selector[facility=<facility>][severity=<severity>]/config/facility	0..23	Default value is 3.
Syslog Provisioning	/system/logging/remote-servers/remote-server[host=<hostname>]/selectors/selector[facility=<facility>][severity=<severity>]/config/severity	Severities include: emergency, alert, error, warning, notice, info, all	—
Interfaces Provisioning	/interfaces/interface[name=local]/config/name	local	—
Interfaces Provisioning	/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/ipv4/addresses/address[ip=<Ip Address>]/config/ip=<Ip Address>	A valid IP address	—
Interfaces Provisioning	/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/ipv4/addresses/address[ip=<Ip Address>]/config/prefix-length	subnet mask	—
Interfaces Provisioning	/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/ipv4/config/dhcp-client	false, true	—
Interfaces Provisioning	/interfaces/interface[name=if0-n]/config/enabled	enabled, disabled	—

Category	Path	Value	Additional information
Local Routing	/local-routes/static-routes/ static[prefix=p]/config/prefix <b>Note:</b> p=<0.0.0.0/00>, i=1	IP address prefix	—
Local Routing	/local-routes/static-routes/ static[prefix=p]/next-hops/ next-hop[index=i]/config/index <b>Note:</b> p=<0.0.0.0/00>, i=1	IP address index	—
Local Routing	/local-routes/static-routes/ static[prefix=p]/next-hops/ next-hop[index=i]/config/next- hop <b>Note:</b> p=<0.0.0.0/00>, i=1	IP address of the gateway	—
License Activation	/components/ component[name=n1]/properties/ property[name=n2] <b>Note:</b> n1 = {Waveserver}	n2 = license- registration- id	—

### OpenConfig read-only paths

The following table provides information on OpenConfig read-only paths.

**Table 384** OpenConfig read-only paths

Path	Values
/interfaces/interface[name=local]/config/ enabled	false, true
/interfaces/interface[name=local]/config/ name	local
/interfaces/interface[name=local]/config/ type	ethernetCsmacd
/interfaces/interface[name=local]/state/ admin-status	DOWN, UP
/interfaces/interface[name=local1]/state/ enabled	false, true
/interfaces/interface[name=local]/state/name	local
/interfaces/interface[name=local]/state/ oper-status	DOWN, UP
/interfaces/interface[name=local]/state/type	ethernetCsmacd



Path	Values
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/config/enabled	false, true
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/config/index	1
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/ipv4/addresses/address/state/origin	DHCP, STATIC
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/ipv4/state/dhcp-client	false, true
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/state/admin-status	DOWN, UP
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/state/enabled	false, true
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/state/index	1
/interfaces/interface[name=local]/subinterfaces/subinterface[index=1]/state/oper-status	DOWN, UP
/components/component[name=n]/config/name <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	XCVR-x-y, x={1..8}, y={1..2}
/components/component[name=n]/config/name <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2	FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2
/components/component[name=n]/config/name <b>Note:</b> n = PSU-1, PSU-2	PSU-1, PSU-2
/components/component[name=AP-1]/config/name	AP-1
/components/component[name=CM-1]/config/name	CM-1
/components/component[name=n]/config/name <b>Note:</b> n = SLOT-1..SLOT-8	SLOT-1..SLOT-8
/components/component[name=Waveserver]/config/name	Waveserver

Path	Values
/components/component[name=n]/config/name <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	OCH-x->y, x={1..8}, y={1..2}
/components/component[name=n]/config/name <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	ILAN-1, ILAN-2, CONSOLE
/components/component[name=n]/linecard/ config/power-admin-state <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	POWER_DISABLED, POWER_ENABLED
/components/component[name=n]/linecard/ state/power-admin-state <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	POWER_DISABLED, POWER_ENABLED
/components/component[name=n]/linecard/ state/slot-id <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	1, 2, 3
/components/component[name=n]/optical- channel/config/line-port <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	XCVR-x-y, x={1..3}, y={1..2}
/components/component[name=n]/optical- channel/state/frequency <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	0.0.0.0, 186087.5..190956.2, 191281.0..196119.0
/components/component[name=n]/optical- channel/state/input-power/avg <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
/components/component[name=n]/optical- channel/state/input-power/instant <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-21474836.0..21474836.0
/components/component[name=n]/optical- channel/state/input-power/max <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
/components/component[name=n]/optical- channel/state/input-power/min <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
/components/component[name=n]/optical- channel/state/input-power/interval <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	10000000000

Path	Values
<code>/components/component[name=n]/optical-channel/state/line-port</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	XCVR-x-y, x={1..3}, y={1..2}
<code>/components/component[name=n]/optical-channel/state/output-power/avg</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
<code>/components/component[name=n]/optical-channel/state/output-power/instant</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-21474836.0..21474836.0
<code>/components/component[name=n]/optical-channel/state/output-power/max</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
<code>/components/component[name=n]/optical-channel/state/output-power/min</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
<code>/components/component[name=n]/optical-channel/state/output-power/interval</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	10000000000
<code>/components/component[name=n]/optical-channel/state/target-output-power</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-9.0..4.0
<code>/components/component[name=n]/optical-channel/state/chromatic-dispersion/interval</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	10000000000
<code>/components/component[name=n]/optical-channel/state/polarization-dependent-loss/interval</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	10000000000
<code>/components/component[name=n]/power-supply/state/enabled</code> <b>Note:</b> n = PSU-1, PSU-2	false, true
<code>/components/component[name=n]/properties/property[name=forward-error-correction]/config/value</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	DISABLED, ENABLED

Path	Values
/components/component[name=Waveserver]/properties/property[name=Power-Supply-Redundant]/config/value	True
/components/component[name=Waveserver]/properties/property[name=DC-Power-Supported]/config/value	True
/components/component[name=n]/properties/property[name=type]/config/value <b>Note:</b> n = PSU-1, PSU-2	AC, AC/240DC, DC, unequipped
/components/component[name=AP-1]/properties/property[name=type]/config/value	type-1, unequipped
/components/component[name=CM-1]/properties/property[name=type]/config/value	type-1, type-2, unequipped
/components/component[name=DCN]/properties/property[name=type]/config/value	ethernet, serial
/components/component[name=n]/properties/property[name=automatic-control]/config/value <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2	false, true
/components/component[name=n]/properties/property[name=mode]/config/value <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE	management, wayside-channel
/components/component[name=n]/properties/property[name=tunable]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	no, yes
/components/component[name=n]/properties/property[name=wavelength-min]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-21474836.0..21474836.0
/components/component[name=n]/properties/property[name=wavelength-max]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-21474836.0..21474836.0
/components/component[name=n]/properties/property[name=frequency-min]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	0..0, 186087.5..190956.2, 191281.0..196119.0
/components/component[name=n]/properties/property[name=frequency-max]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	0..0, 186087.5..190956.2, 191281.0..196119.0

Path	Values
/components/component[name=n]/properties/ property[name=edfa]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	no, yes
/components/component[name=n]/properties/ property[name=bit-rate-min]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	OTU2
/components/component[name=n]/properties/ property[name=bit-rate-max]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	OTU4
/components/component[name=n]/properties/ property[name=56GBaud-capable]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	true
/components/component[name=n]/properties/ property[name=35GBaud-capable]/config/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	true, false
/components/component[name=n]/properties/ property[name=last-restart-reason]/config/ value <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7, CM-1	power-on, system-cold, system-warm, unknown, user- cold, user-warm
/components/component[name=n]/properties/ property[name=speed]/config/value <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	100M, 10G, 10M, 13M, 1G, Unknown
/components/component[name=n]/properties/ property[name=duplex]/config/value <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	Unknown, full, half
/components/component[name=n]/properties/ property[name=spli-management-status]/ config/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	matched, mismatched, unavailable
/components/component[name=n]/properties/ property[name=admin-state]/state/ configurable <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, AP-1, CM-1, LM-1, LM-3, LM-5, LM-7, ILAN-1, ILAN-2, CONSOLE, DCN-n n = XCVR-x-y, x={1..8}, y={1..18}	true

Path	Values
/components/component[name=n]/properties/ property[name=forward-error-correction]/ state/configurable <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	false
/components/component[name=n]/properties/ property[name=forward-error-correction]/ state/configurable <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	true
/components/component[name=n]/properties/ property[name=model]/state/configurable <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	true
/components/component[name=n]/properties/ property[name=model]/state/configurable <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN	false
/components/component[name=n]/properties/ property[name=conditioning-type, conditioning-holdoff]/state/configurable <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	true
/components/component[name=Waveserver]/ properties/property[name=Max-Num-Control- Modules, Max-Num-Line-Modules, Max-Num-Fan- Modules, Max-Num-Power-Supplies, Max-Num- Access-Panels, Power-Supply-Redundant, DC- Power-Supported]/state/configurable	false
/components/component[name=n]/properties/ property[name=model]/state/configurable <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, AP-1, CM-1	false
/components/component[name=n]/properties/ property[name=manufacture-date]/state/ configurable <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, AP-1, CM-1, LM-1, LM-3, LM-5, LM-7	false
/components/component[name=n]/properties/ property[name=type]/state/configurable <b>Note:</b> n = PSU-1, PSU-2, AP-1, CM-1, ILAN-1, ILAN-2, CONSOLE, DCN-n	false

Path	Values
<code>/components/component[name=n]/properties/ property[name=automatic-control, number-of- fans]/state/configurable</code> <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2	false
<code>/components/component[name=n]/properties/ property[name=tunable, wavelength-min, wavelength-max, frequency-min, frequency-max, tx-dispersion-min, tx-dispersion-max, edfa, bit-rate-min, bit-rate-max, pre-fec-sd-ber, pre-fec-sf-ber]/state/configurable</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	false
<code>/components/component[name=n]/properties/ property[name=pre-fec-sd-dbg, pre-fec-sf- dbg]/state/configurable</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	true
<code>/components/component[name=n]/properties/ property[name=label]/state/configurable</code> <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	true
<code>/components/component[name=n]/properties/ property[name=module-type]/state/ configurable</code> <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	false
<code>/components/component[name=n]/properties/ property[name=label]/state/configurable</code> <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	false
<code>/components/component[name=n]/properties/ property[name=last-restart, last-restart- reason]/state/configurable</code> <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7, CM-1	false
<code>/components/component[name=n]/properties/ property[name=module-description]/state/ configurable</code> <b>Note:</b> n = SLOT-1..SLOT-8	false

Path	Values
<code>/components/component[name=n]/properties/ property[name=expected-restoration-time, line-system-type, modem-class, reach-limit, odu-termination-mode, link-optimization-mode, laser-tuning-mode, laser-centering-mode, laser-centering-range, laser-shutdown-mode, channel-contention-detection-avoidance, tx- power-reduction-state, tx-dispersion- compensation-mode, tx-dispersion- precompensation-value, tx-dispersion-path- value, tx-dispersion-search-mode-min, tx- dispersion-search-mode-max]/state/ configurable</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	true
<code>/components/component[name=n]/properties/ property[name=actual-tx-link-dispersion, actual-rx-link-dispersion, actual-tx- precompensation-dispersion, actual-rx-post- compensation-dispersion, estimated-round- trip-delay, estimated-unidirectional-latency, estimated-fiber-length]/state/configurable</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	false
<code>/components/component[name=n]/properties/ property[name=speed, duplex]/state/ configurable</code> <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	false
<code>/components/component[name=n]/properties/ property[name=spli-management]/state/ configurable</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	true
<code>/components/component[name=n]/properties/ property[name=spli-management-status]/state/ configurable</code> <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	false
<code>/components/component[name=Waveserver]/ properties/property[name=license- registration-id]/state/configurable</code>	true



Path	Values
/components/component[name=n]/properties/ property[name=admin-state]/state/value <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, AP-1, CM-1, LM-1, LM-3, LM-5, LM-7, ILAN-1, ILAN-2, CONSOLE, DCN-1, DCN-2, DCN-3, DCN-5, DCN-5 n = OCH-x-y, x={1..8}, y={1..2} n = XCVR-x-y, x={1..8}, y={1..18}	DISABLED, ENABLED
/components/component[name=n]/properties/ property[name=forward-error-correction]/ state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	DISABLED, ENABLED
/components/component[name=n]/properties/ property[name=conditioning-type]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	ethernet, laser-off, none, otn
/components/component[name=n]/properties/ property[name=conditioning-holdoff]/state/ value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	0 to 1000, increments of 10 from 0 to 100, increments of 100 from 100 to 1000
/components/component[name=Waveserver]/ properties/property[name=DC-Power- Supported]/state/value	True
/components/component[name=n]/properties/ property[name=type]/state/value <b>Note:</b> n = PSU-1, PSU-2	AC, AC/240VDC, DC, unequipped
/components/component[name=AP-1]/properties/ property[name=type]/state/value	type-1, unequipped
/components/component[name=CM-1]/properties/ property[name=type]/state/value	type-1, type-2, unequipped
/components/component[name=n]/properties/ property[name=type]/state/value <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	ethernet, serial
/components/component[name=n]/properties/ property[name=automatic-control]/state/value <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2	false, true

Path	Values
/components/component[name=n]/properties/ property[name=mode]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	100GE, 400GE, OTL4.4, 95-200-O, 95-250-O, 95-300-O, 95-350-O, 95-400-O, 95-450-O, 95-500-O, 95-550-O, 95-600-O, 95-650-O, 95-700-O, 95-800-O, 95-300-E, 95-350-E, 95-400-E, 95-450-E, 95-500-E, 95-550-E, 95-600-E, 95-650-E, 95-700-E, 95-800-E, 91.6-300-E, 91.6-400-E, 91.6-450-E, 91.6-500-E, 91.6-550-E, 91.6-600-E, 91.6-700-E, 91.6-800-E, 85-300-E, 85-400-E, 85-600-E, 85-300-O, 85-400-O, 85-600-O, 90-400-E, 90-600-E, 93.3-400-E, 93.3-600-E
/components/component[name=n]/properties/ property[name=mode]/state/value <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	management, wayside-channel
/components/component[name=n]/properties/ property[name=tunable]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	no, yes
/components/component[name=n]/properties/ property[name=wavelength-min]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-21474836.0..21474836.0
/components/component[name=n]/properties/ property[name=wavelength-max]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-21474836.0..21474836.0
/components/component[name=n]/properties/ property[name=frequency-min]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	0.0..0.0, 186087.5..190956.2, 191281.0..196119.0
/components/component[name=n]/properties/ property[name=frequency-max]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	0.0..0.0, 186087.5..190956.2, 191281.0..196119.0
/components/component[name=n]/properties/ property[name=edfa]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	no, yes

Path	Values
/components/component[name=n]/properties/ property[name=bit-rate-min]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	OTU2, OTU4
/components/component[name=n]/properties/ property[name=56GBaud-capable]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	true
/components/component[name=n]/properties/ property[name=35GBaud-capable]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	true, false
/components/component[name=n]/properties/ property[name=module-type]/state/value <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	motr-400-2-16
/components/component[name=n]/properties/ property[name=last-restart-reason]/state/ value <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7, CM-1	power-on, system-cold, system-warm, unknown, user- cold, user-warm
/components/component[name=n]/properties/ property[name=pre-fec-sd-dbq]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-1.0, 3.5
/components/component[name=n]/properties/ property[name=pre-fec-sf-dbq]/state/value <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-1.0, 1.0
/components/component[name=n]/properties/ property[name=expected-restoration-time]/ state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	0, 255
/components/component[name=n]/properties/ property[name=line-system-type]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	coloured, colourless, contentionless, cs-coloured, cs-colourless
/components/component[name=n]/properties/ property[name=modem-class]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	submarine, terrestrial, unavailable
/components/component[name=n]/properties/ property[name=odu-termination-mode]/state/ value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	passthrough, terminated

Path	Values
/components/component[name=n]/properties/ property[name=link-optimization-mode]/state/ value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	non-linear, snr-tolerant, standard
/components/component[name=n]/properties/ property[name=laser-tuning-mode]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	accelerated, fast, normal
/components/component[name=n]/properties/ property[name=laser-centering-mode]/state/ value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	disabled, enabled, holdover
/components/component[name=n]/properties/ property[name=laser-centering-range]/state/ value n = OCH-x-y, x={1..8}, y={1..2}	0.0..3.0
/components/component[name=n]/properties/ property[name=laser-shutdown-mode]/state/ value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	accelerated, normal
/components/component[name=n]/properties/ property[name=channel-contention-detection- avoidance]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	false, true
/components/component[name=n]/properties/ property[name=tx-power-reduction-state]/ state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	false, true
/components/component[name=n]/properties/ property[name=tx-dispersion-compensation- mode]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	automatic, fixed, search
/components/component[name=n]/properties/ property[name=tx-dispersion-precompensation- value]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-44500..1000000

Path	Values
/components/component[name=n]/properties/ property[name=tx-dispersion-path-value]/ state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-89000..104450
/components/component[name=n]/properties/ property[name=tx-dispersion-search-mode- min]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-89000..1044500
/components/component[name=n]/properties/ property[name=tx-dispersion-search-mode- max]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	-89000..1044500
/components/component[name=n]/properties/ property[name=speed]/state/value <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	100M, 10G, 10M, 13M, 1G, Unknown
/components/component[name=n]/properties/ property[name=duplex]/state/value <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	Unknown, full, half
/components/component[name=n]/properties/ property[name=spli-management]/state/value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	DISABLED, ENABLED
/components/component[name=n]/properties/ property[name=spli-management-status]/state/ value <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	matched, mismatched, unavailable
/components/component[name=n]/state/empty <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, AP-1, CM-1, LM-1, LM-3, LM-5, LM-7 n = XCVR-x-y, x={1..8}, y={3..18}	false, true
/components/component[name=n]/state/empty <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	false
/components/component[name=AP-1]/state/id	AP-1
/components/component[name=CM-1]/state/id	CM-1
/components/component[name=n]/state/id <b>Note:</b> n = SLOT-1..SLOT-8	SLOT-1..SLOT-8

Path	Values
/components/component[name=n]/state/id <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	OCH-x-y, x={1..8}, y={1..2}
/components/component[name=n]/state/id <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	ILAN-1, ILAN-2, CONSOLE, DCN-n
/components/component[name=n]/state/mfg-name <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, AP-1, CM-1, LM-1, LM-3, LM-5, LM-7, Waveserver n = XCVR-x-y, x={1..8}, y={1..18}	Ciena
/components/component[name=n]/state/name <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	XCVR-x-y, x={1..8}, y={1..18}
/components/component[name=AP-1]/state/name	AP-1
/components/component[name=CM-1]/state/name	CM-1
/components/component[name=CM-1]/state/memory/available	-
/components/component[name=CM-1]/state/memory/utilized	-
/components/component[name=n]/state/name <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	LM-1, LM-3, LM-5, LM-7
/components/component[name=n]/state/name <b>Note:</b> n = SLOT-1..SLOT-8	SLOT-1..SLOT-8
/components/component[name=Waveserver]/state/name	Waveserver
/components/component[name=n]/state/name <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	OCH-x-y, x={1..8}, y={1..2}
/components/component[name=n]/state/name <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	CONSOLE, ILAN-1, ILAN-2, DCN-n
/components/component[name=n]/state/removable <b>Note:</b> n = AP-1, FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2, PSU-1, PSU-2, LM-1, LM-3, LM-5, LM-7, CM-1 n = XCVR-x-y, x={1..8}, y={3..18}	true

Path	Values
/components/component[name=n]/state/removable <b>Note:</b> n = SLOT-1..SLOT-8, ILAN-1, ILAN-2, CONSOLE, DCN-n, Waveserver n = XCVR-x-y, x={1..8}, y={1..2} n = OCH-x-y, x={1..8}, y={1..2}	false
/components/component[name=n]/state/type <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	TRANSCEIVER
/components/component[name=n]/state/type <b>Note:</b> n = FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2	FAN
/components/component[name=n]/state/type <b>Note:</b> n = PSU-1, PSU-2	POWER_SUPPLY
/components/component[name=AP-1]/state/type	MODULE
/components/component[name=CM-1]/state/type	CPU
/components/component[name=CM-1]/cpu/utilization/instant	-
/components/component[name=CM-1]/cpu/utilization/min	-
/components/component[name=CM-1]/cpu/utilization/max	-
/components/component[name=CM-1]/cpu/utilization/avg	-
/components/component[name=n]/cpu/utilization/state/interval	3600000000000
/components/component[name=n]/state/type <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7 n = SLOT-1..SLOT-8	MODULE
/components/component[name=Waveserver]/state/type	CHASSIS
/components/component[name=n]/state/type <b>Note:</b> n = OCH-x-y, x={1..8}, y={1..2}	OPTICAL_CHANNEL
/components/component[name=n]/state/type <b>Note:</b> n = ILAN-1, ILAN-2, CONSOLE, DCN-n	PORT

Path	Values
<code>/components/component[name=n]/state/temperature/instant</code> <b>Note:</b> Supported on transceiver, control module, chassis, line module, and access panel components. For a transceiver, n = XCVR-x-y where x={1..8}, y={1..18}. For the control module, n = CM-1. For the chassis, n = Waveserver. For a line module, n = LM-1, LM-3, LM-5, LM-7. For the access panel, n = AP-1.	decimal64
<code>/components/component[name=n]/state/temperature/min</code> <b>Note:</b> Supported on control module, chassis, line module, and access panel components. For the control module, n = CM-1. For the chassis, n = Waveserver. For a line module, n = LM-1, LM-3, LM-5, LM-7. For the access panel, n = AP-1.	decimal64
<code>/components/component[name=n]/state/temperature/max</code> <b>Note:</b> Supported on control module, chassis, line module, and access panel components. For the control module, n = CM-1. For the chassis, n = Waveserver. For a line module, n = LM-1, LM-3, LM-5, LM-7. For the access panel, n = AP-1.	decimal64
<code>/components/component[name=n]/state/temperature/avg</code> <b>Note:</b> Supported on control module, chassis, line module, and access panel components. For the control module, n = CM-1. For the chassis, n = Waveserver. For a line module, n = LM-1, LM-3, LM-5, LM-7. For the access panel, n = AP-1.	decimal64
<code>/components/component[name=n]/state/temperature/interval</code> <b>Note:</b> Supported on transceiver, control module, chassis, line module, and access panel components. For a transceiver, n = XCVR-x-y where x={1..8}, y={1..18}. For the control module, n = CM-1. For the chassis, n = Waveserver. For a line module, n = LM-1, LM-3, LM-5, LM-7. For the access panel, n = AP-1.	3600000000000
<code>/components/component[name=n]/subcomponents/subcomponent[name=XCVR-1-1, XCVR-1-2, XCVR-2-1, XCVR-2-2, XCVR-3-1, XCVR-3-2, XCVR-4-1, XCVR-4-2, XCVR-5-1, XCVR-5-2, XCVR-6-1, XCVR-6-2, XCVR-7-1, XCVR-7-2, XCVR-8-1, XCVR-8-2]/config/name</code> <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	XCVR-1-1, XCVR-1-2, XCVR-2-1, XCVR-2-2, XCVR-3-1, XCVR-3-2, XCVR-4-1, XCVR-4-2, XCVR-5-1, XCVR-5-2, XCVR-6-1, XCVR-6-2, XCVR-7-1, XCVR-7-2, XCVR-8-1, XCVR-8-2



Path	Values
/components/component[name=n]/subcomponents/subcomponent[name=XCVR-1-3..XCVR-1-18, XCVR-2-3..XCVR-2-18, XCVR-3-3..XCVR-3-18, XCVR-4-3..XCVR-4-18, XCVR-5-3..XCVR-5-18, XCVR-6-3..XCVR-6-18, XCVR-7-3..XCVR-7-18, XCVR-8-3..XCVR-8-18]/config/name <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	XCVR-1-x..XCVR-8-x, where x = {3..18}
/components/component[name=Waveserver]/subcomponents/subcomponent[name=SLOT-1..SLOT-8]/config/name	SLOT-x, where x = {1..8}
/components/component[name=Waveserver]/subcomponents/subcomponent[name=FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2]/config/name	FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2
/components/component[name=Waveserver]/subcomponents/subcomponent[name=PSU-1, PSU-2]/config/name	PSU-1, PSU-2
/components/component[name=CM-1]/subcomponents/subcomponent[name=AP-1]/config/name	AP-1
/components/component[name=Waveserver]/subcomponents/subcomponent[name=CM-1]/config/name	CM-1
/components/component[name=LM-1, LM-3, LM-5, LM-7]/subcomponents/subcomponent[name=SLOT-1..SLOT-8]/config/name	SLOT-1..SLOT8
/components/component[name=AP-1]/subcomponents/subcomponent[name=CONSOLE, ILAN-1, ILAN-2, DCN-n]/config/name	CONSOLE, ILAN-1, ILAN-2, DCN-n
/components/component[name=n]/subcomponents/subcomponent[name=XCVR-1-1, XCVR-1-2, XCVR-2-1, XCVR-2-2, XCVR-3-1, XCVR-3-2, XCVR-4-1, XCVR-4-2, XCVR-5-1, XCVR-5-2, XCVR-6-1, XCVR-6-2, XCVR-7-1, XCVR-7-2, XCVR-8-1, XCVR-8-2]/state/name <b>Note:</b> n = LM-1, LM-3, LM-5, LM-7	XCVR-1-1, XCVR-1-2, XCVR-2-1, XCVR-2-2, XCVR-3-1, XCVR-3-2, XCVR-4-1, XCVR-4-2, XCVR-5-1, XCVR-5-2, XCVR-6-1, XCVR-6-2, XCVR-7-1, XCVR-7-2, XCVR-8-1, XCVR-8-2
/components/component[name=LM-1, LM-3, LM-5, LM-7]/subcomponents/subcomponent[name=XCVR-1-3..XCVR-1-18, XCVR-2-3..XCVR-2-18, XCVR-3-3..XCVR-3-18, XCVR-4-3..XCVR-4-18, XCVR-5-3..XCVR-5-18, XCVR-6-3..XCVR-6-18, XCVR-7-3..XCVR-7-18, XCVR-8-3..XCVR-8-18]/state/name	XCVR-1-x..XCVR-8-x, where x = 3 - 18

Path	Values
/components/component[name=Waveserver]/subcomponents/subcomponent[name=SLOT-1.., SLOT-8]/state/name	SLOT-1..SLOT-8
/components/component[name=Waveserver]/subcomponents/subcomponent[name=FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2]/state/name	FAN-1-1, FAN-1-2, FAN-2-1, FAN-2-2
/components/component[name=Waveserver]/subcomponents/subcomponent[name=PSU-1, PSU-2]/state/name	PSU-1, PSU-2
/components/component[name=CM-1]/subcomponents/subcomponent[name=AP-1]/state/name	AP-1
/components/component[name=Waveserver]/subcomponents/subcomponent[name=CM-1]/state/name	CM-1
/components/component[name=SLOT-1]/subcomponents/subcomponent[name=LM-1, LM-3, LM-5, LM-7]/state/name <b>Note:</b> n = SLOT-1..SLOT-8	LM-1, LM-3, LM-5, LM-7
/components/component[name=AP-1]/subcomponents/subcomponent[name=CONSOLE, ILAN-1, ILAN-2, DCN-n]/state/name	CONSOLE, ILAN-1, ILAN-2, DCN-n
/components/component/transceiver[name=n]/config/enabled <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	false, true
/components/component/transceiver[name=n]/physical-channels/channel[index=1]/config/index <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	1..4
/components/component/transceiver[name=n]/physical-channels/channel[index=i]/config/index <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	1..4
/components/component/transceiver[name=n]/physical-channels/channel[index=1]/state/input-power/avg <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0

Path	Values
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=i]/state/input-power/avg</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-2147483.0..2147483.0
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=1]/state/input-power/instant</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-21474836.0.. 21474836.0
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=i]/state/input-power/instant</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-21474836.0..21474836.0
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=1]/state/input-power/max</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=i]/state/input-power/max</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-2147483.0..2147483.0
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=1]/state/input-power/min</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=i]/state/input-power/min</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-2147483.0..2147483.0
<code>/components/component/transceiver[name=n]/physical-channels/channel[index=1]/state/output-power/avg</code> <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0

Path	Values
/components/component/transceiver[name=n]/ physical-channels/channel[index=i]/state/ output-power/avg <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-2147483.0..2147483.0
/components/component/transceiver[name=n]/ physical-channels/channel[index=1]/state/ output-power/instant <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-21474836.0..21474836.0
/components/component/transceiver[name=n]/ physical-channels/channel[index=i]/state/ output-power/instant <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-21474836.0..21474836.0
/components/component/transceiver[name=n]/ physical-channels/channel[index=1]/state/ output-power/max <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
/components/component/transceiver[name=n]/ physical-channels/channel[index=i]/state/ output-power/max <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-2147483.0..2147483.0
/components/component/transceiver[name=n]/ physical-channels/channel[index=1]/state/ output-power/min <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..2}	-2147483.0..2147483.0
/components/component/transceiver[name=n]/ physical-channels/channel[index=i]/state/ output-power/min <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	-2147483.0..2147483.0
/components/component[name=n]/transceiver/ state/output-power/interval <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	10000000000
<b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}	10000000000

Path	Values
/components/component[name=n]/transceiver/ physical-channels/channel[index=i]/state/ input-power/interval <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}, i = 1	10000000000
/components/component[name=n]/transceiver/ physical-channels/channel[index=i]/state/ output-power/interval <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18}, i = 1	10000000000
/components/component/transceiver[name=n]/ physical-channels/channel[index=i]/state/ laser-bias-current/instant <b>Note:</b> n = XCVR-x-y, x={1..8}, y={3..18} i = 1, 4	decimal64
/components/component/transceiver[name=n]/ state/connector-type <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	LC_CONNECTOR, MPO_CONNECTOR, SC_CONNECTOR
/components/component/transceiver[name=n]/ state/enabled <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	false, true
/components/component/transceiver[name=n]/ state/fault-condition <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	false, true
/components/component/transceiver[name=n]/ state/form-factor <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	NON_PLUGGABLE, OTHER, QSFP28, QSFP_PLUS
/components/component/transceiver[name=n]/ state/present <b>Note:</b> n = XCVR-x-y, x={1..8}, y={1..18}	NOT_PRESENT, PRESENT
/terminal-device/logical-channels/ channel[index=i]/config/index <b>Note:</b> i = Any Logical Channel index from <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	Any Logical Channel index from <a href="#">openconfig-terminal-device model logical-channel index definitions</a>
/terminal-device/logical-channels/ channel[index=i]/otn/state/q-value/instant <b>Note:</b> i = Any Logical Channel index from <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	-2147483.0, 2147483.0

Path	Values
/terminal-device/logical-channels/ channel[index=i]/otn/state/q-value/avg <b>Note:</b> i = OTUCn Logical Channels in <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	-2147483.0, 2147483.0
/terminal-device/logical-channels/ channel[index=i]/otn/state/q-value/max <b>Note:</b> i = OTUCn Logical Channels in <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	-2147483.0, 2147483.0
/terminal-device/logical-channels/ channel[index=i]/otn/state/q-value/min <b>Note:</b> i = OTUCn Logical Channels in <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	-2147483.0, 2147483.0
/terminal-device/logical-channels/ channel[index=i]/otn/state/q-value/interval <b>Note:</b> i = OTUCn Logical Channels in <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	10000000000
/terminal-device/logical-channels/ channel[index=i]/otn/state/tti-msg-auto <b>Note:</b> i = Any line port Logical Channel from <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	false, true
/terminal-device/logical-channels/ channel[index=i]/otn/state/pre-fec-ber/ interval <b>Note:</b> i = Any Logical Channel index from <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .	10000000000
/terminal-device/logical-channels/ channel[index=i]/otn/state/esnr/interval <b>Note:</b> i = Any Logical Channel index from <a href="#">openconfig-terminal-device model logical-channel index definitions</a> .	10000000000
/terminal-device/logical-channels/ channel[index=i]/state/index <b>Note:</b> i = Any Logical Channel Index from <a href="#">openconfig-terminal-device model logical-channel index definitions</a>	Any Logical Channel Index from <a href="#">openconfig-terminal-device model logical-channel index definitions</a>
/terminal-device/operational-modes/ mode[index=i]/state/description <b>Note:</b> i = 1 - 11	Line port XCVR transmission mode

Path	Values
/terminal-device/operational-modes/ mode[index=i]/state/mode-id <b>Note:</b> i = 1 - 11	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
/terminal-device/operational-modes/ mode[index=i]/state/vendor-id <b>Note:</b> i = 1 - 11	Ciena
/system/aaa/authentication/admin-user/state/ admin-username	root
/system/aaa/authentication/users/user/state/ role	super, admin, limited
/system/aaa/server-groups/server-group[ name=servergroup-1]/config/name	servergroup-1
/system/aaa/server-groups/server-group/ servers/server/state/timeout	1..30
/system/aaa/server-groups/server-group/ servers/server/tacacs/state/port	1..65535
/system/logging/remote-servers/remote- server/state/remote-port	1..65535
/system/ntp/ntp-keys/ntp-key/config/key-type	NTP_AUTH_MD5
/system/ntp/ntp-keys/ntp-key/state/key-type	NTP_AUTH_MD5
/system/ntp/servers/server/state/offset	-2147483.0..2147483.0
/system/ntp/servers/server/state/poll- interval	(16, 16), (32, 32), (64, 64), (128, 128), (256, 256), (512, 512), (1024, 1024), (2048, 2048), (4096, 4096), (8192, 8192), (16384, 16384), (32768, 32768), (65536, 65536)
/system/ntp/servers/server/state/root-delay	-2147483.0..2147483.0
/system/ntp/servers/server/state/root- dispersion	-2147483.0..2147483.0
/system/ntp/state/enable-ntp-auth	false, true
/system/ntp/state/enabled	false, true
/system/ssh-server/config/enable	false, true
/system/ssh-server/config/protocol-version	V2
/system/ssh-server/state/enable	false, true

Path	Values
/system/ssh-server/state/protocol-version	V2
/system/ssh-server/state/timeout	1, 1500

## WebSocket

You can use the WebSocket protocol, as defined in RFC 6455, to listen for Waveserver 5 autonomous messages. The WebSocket protocol enables communication between a client and server using the same security model used by Web browsers.

The WebSocket protocol supports ping and pong messages between the server and client. Ping messages from the server are sent every minute.

You can connect to WebSocket using a program or a browser. To connect to a WebSocket client using a browser, use the authentication token and HTTP cookie automatically generated for the current session in the response to a successful login request. The authentication token needs to be passed as a query parameter with parameter name `-xsrif-`.

The WebSocket connection will keep running until closed or the session times out.

Enter the following URL for the WebSocket interface:

**`https://<IP address>/notification/recv.html`**

**Note:** Waveserver 5 uses a self-signed browser security certificate. You can access Waveserver 5 by either adding the self-signed security certificate to your web browser, or by uploading and activating an RSA certificate. See [Installing a new SSL certificate](#).

You can display the Waveserver 5 Event Stream to monitor autonomous messages that capture events and database changes. Autonomous messages are retrieved in JSON format.

To access the Waveserver 5 Event Stream, click the Event Stream link on the Waveserver 5 homepage. See [Waveserver 5 Web User Interface](#).

Notifications contain only the differences instead of the full content of the current state.

Waveserver 5 supports up to ten concurrent user sessions between the Web, REST, and WebSocket interfaces. Any attempt to connect an 11th user session without first closing an existing session will be rejected.



### WebSocket subscribed paths

WebSocket subscribes to the following Ciena YANG model paths. Exceptions are listed in the following section.

- /waveserver-aaa
- /waveserver-alarms
- /waveserver-chassis
- /waveserver-configuration
- /waveserver-interfaces
- /waveserver-license
- /waveserver-ldp
- /waveserver-logging/syslog
- /waveserver-logging/syslog-tls
- /waveserver-modules
- /waveserver-ndp
- /waveserver-pkix
- /waveserver-pm/global-config
- /waveserver-ports
- /waveserver-ptps
- /waveserver-snmp
- /waveserver-software
- /waveserver-spli
- /waveserver-system
- /waveserver-topology
- /waveserver-xcvrs

### WebSocket exclusions

The following Waveserver YANG paths will not generate WebSocket notifications.

- /waveserver-aaa/authentication/authentication-method/statistics/called
- /waveserver-aaa/authentication/authentication-method/statistics/success
- /waveserver-aaa/authentication/authentication-method/statistics/failure
- /waveserver-aaa/authentication/authentication-method/statistics/skipped
- /waveserver-aaa/user-management/active-users/idle-time
- /waveserver-aaa/user-management/statistics/ssh/active-sessions
- /waveserver-aaa/user-management/statistics/ssh/closed-sessions

- /waveserver-aaa/user-management/statistics/console/active-sessions
- /waveserver-aaa/user-management/statistics/console/closed-sessions
- /waveserver-aaa/user-management/statistics/netconf/active-sessions
- /waveserver-aaa/user-management/statistics/netconf/closed-sessions
- /waveserver-aaa/user-management/statistics/webserver/active-sessions
- /waveserver-aaa/user-management/statistics/webserver/closed-sessions
- /waveserver-aaa/user-management/statistics/grpc/active-sessions
- /waveserver-aaa/user-management/statistics/grpc/closed-sessions
- /waveserver-alarms/statistics/index
- /waveserver-alarms/statistics/count
- /waveserver-alarms/statistics/cumulative
- /waveserver-alarms/statistics/type
- /waveserver-alarms/statistics-all/cumulative
- /waveserver-chassis/power/state/total-power
- /waveserver-license/licenses/licenses-list/time-remaining
- /waveserver-lldp/chassis/statistics/inserts
- /waveserver-lldp/chassis/statistics/deletes
- /waveserver-lldp/chassis/statistics/drops
- /waveserver-lldp/chassis/statistics/age-outs
- /waveserver-logging/syslog-tls/collector/operational/greylist-remaining
- /waveserver-logging/syslog-tls/collector/operational/statistics/connection-attempts
- /waveserver-modules/modules/state/uptime
- /waveserver-pm paths other than /waveserver-pm/global-config
- /waveserver-ports/ports/state/operational-state-duration
- /waveserver-snmp/global-config/engine-boots
- /waveserver-snmp/global-config/engine-time
- /waveserver-system/time-config/time
- /waveserver-system/time-config/date
- /waveserver-system/time-config/time-offset
- /waveserver-system/time-config/local-date-time
- /waveserver-system/time-config/coordinated-universal-time
- /waveserver-system/time-config/system-uptime
- /waveserver-system/client-config/dhcp/lease-time

- `/waveserver-system/client-config/dhcp/lease-remaining`
- `/waveserver-system/client-config/dhcp/renewal-time`
- `/waveserver-system/client-config/dhcp/renewal-remaining`
- `/waveserver-system/client-config/dhcp/rebinding-time`
- `/waveserver-system/client-config/dhcp/rebinding-remaining`
- `/waveserver-system/ntp/delay`
- `/waveserver-system/ntp/offset`
- `/waveserver-system/ntp/jitter`
- `/waveserver-system/ntp/server/offset`

## Procedure list

This section provides the following procedures:

- [Procedure 273, “Accessing the Waveserver 5 REST API”](#)
- [Procedure 274, “Enabling and disabling the NETCONF server”](#)
- [Procedure 275, “Configuring query limits for PM history bins”](#)
- [Procedure 276, “Enabling and disabling the gRPC server”](#)
- [Procedure 277, “Enabling or disabling mutual authentication on a gRPC server”](#)
- [Procedure 278, “Configuring telemetry subscriptions”](#)

---

## Procedure 273 Accessing the Waveserver 5 REST API

---

Use this procedure to access the Waveserver 5 REST API.

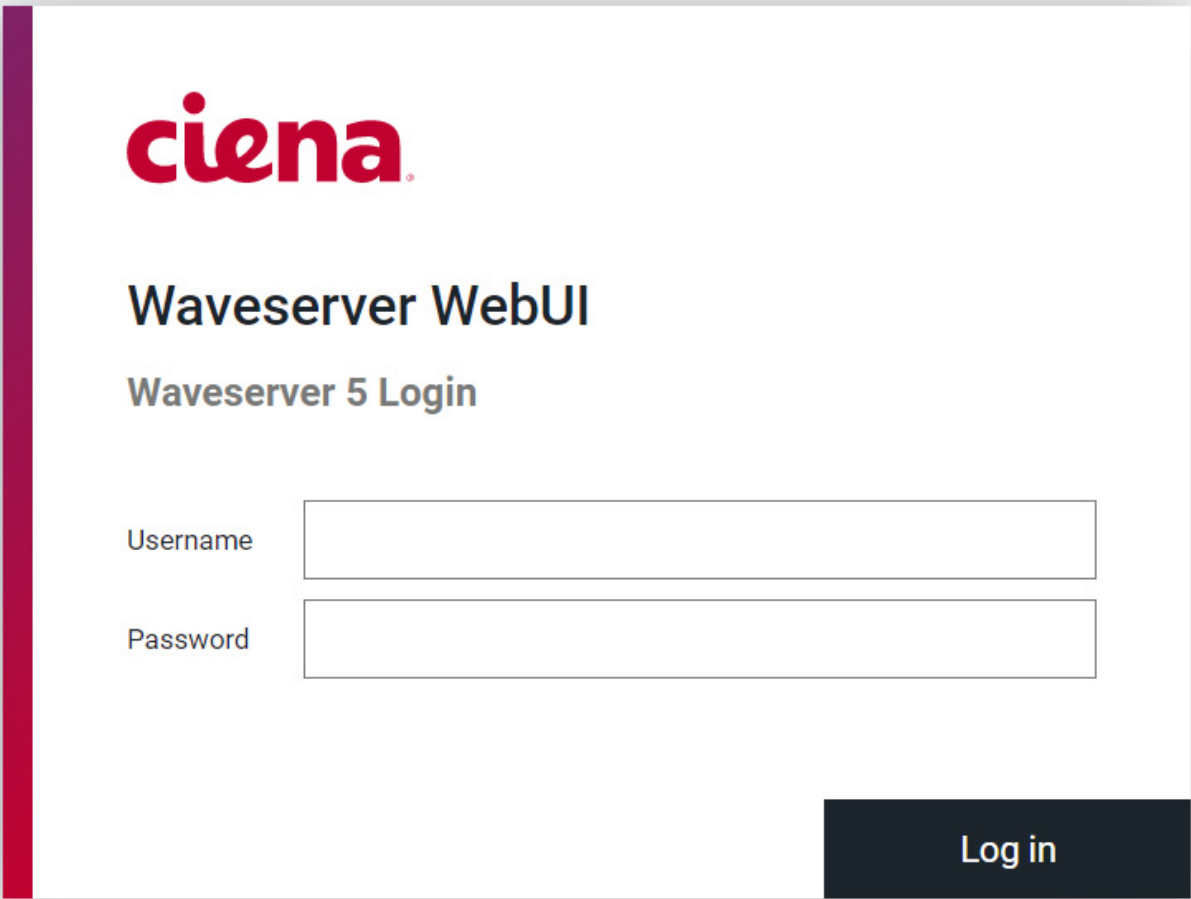
### Overview

**Note:** Waveserver supports up to ten concurrent user sessions between the Web, REST, and WebSocket interfaces. Any attempt to connect an 11th user session without first closing an existing session will be rejected.

### Steps

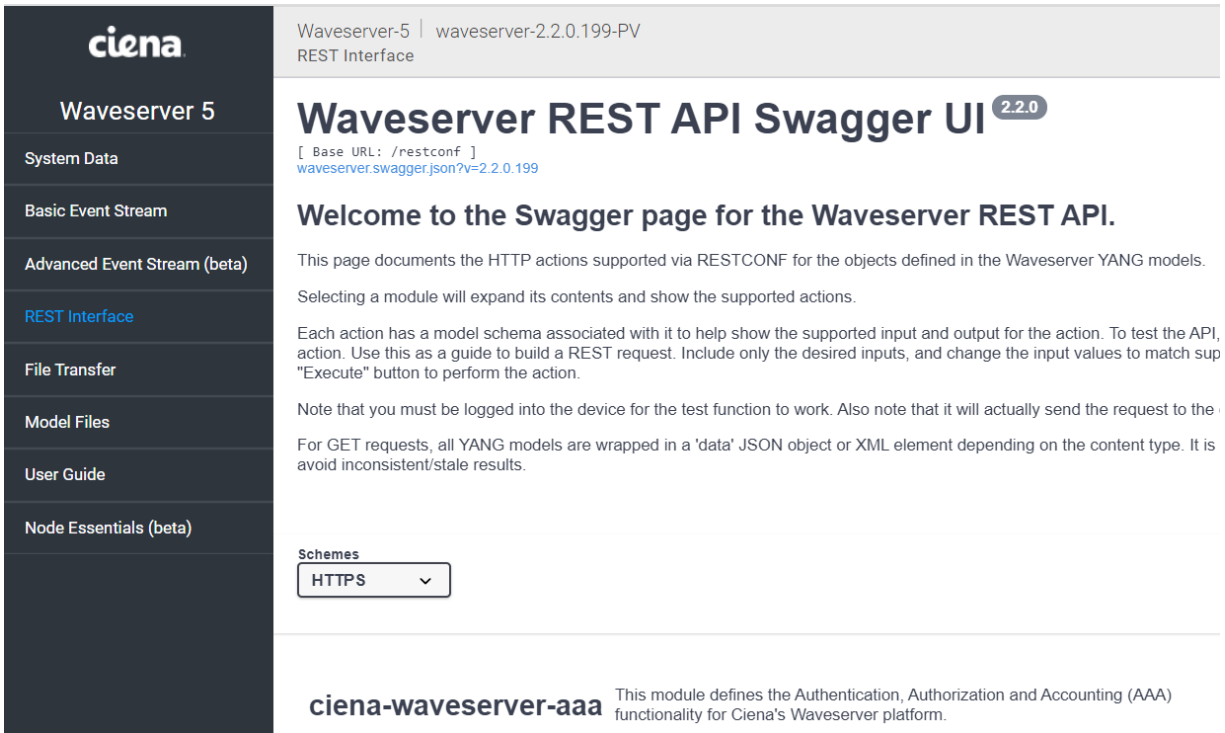
- 1 In a Web browser, enter the following:  
`https://<Waveserver IP address>:443/`  
where 443 is the secure HTTP port number  
**Note:** Waveserver 5 uses a self-signed browser security certificate. You can access Waveserver 5 by either adding the self-signed security certificate to your web browser, or by uploading and activating an RSA certificate on Waveserver 5. See [Installing a new SSL certificate](#).
- 2 On the Login screen, enter a user name and password.  
**Note:** When using the Swagger user interface (UI), Waveserver 5 verifies that the current login user has privileges to execute the REST requests.

**Figure 171** Waveserver 5 homepage login



The image shows the Waveserver 5 login page. It features the Ciena logo in red at the top left. Below the logo, the text "Waveserver WebUI" is displayed in a large, bold, black font. Underneath that, "Waveserver 5 Login" is written in a smaller, bold, black font. There are two input fields: "Username" and "Password", each with a corresponding text box. A dark blue "Log in" button is located at the bottom right of the form area.

- 3 On the Waveserver 5 screen, click the appropriate Swagger link under REST Interface to access the Swagger user interface (UI).

**Figure 172** Link to Swagger UI


Waveserver-5 | waveserver-2.2.0.199-PV  
REST Interface

## Waveserver REST API Swagger UI 2.2.0

[ Base URL: /restconf ]  
[waveserver.swagger.json?v=2.2.0.199](https://waveserver.swagger.json?v=2.2.0.199)

### Welcome to the Swagger page for the Waveserver REST API.

This page documents the HTTP actions supported via RESTCONF for the objects defined in the Waveserver YANG models.

Selecting a module will expand its contents and show the supported actions.

Each action has a model schema associated with it to help show the supported input and output for the action. To test the API, action. Use this as a guide to build a REST request. Include only the desired inputs, and change the input values to match sup "Execute" button to perform the action.

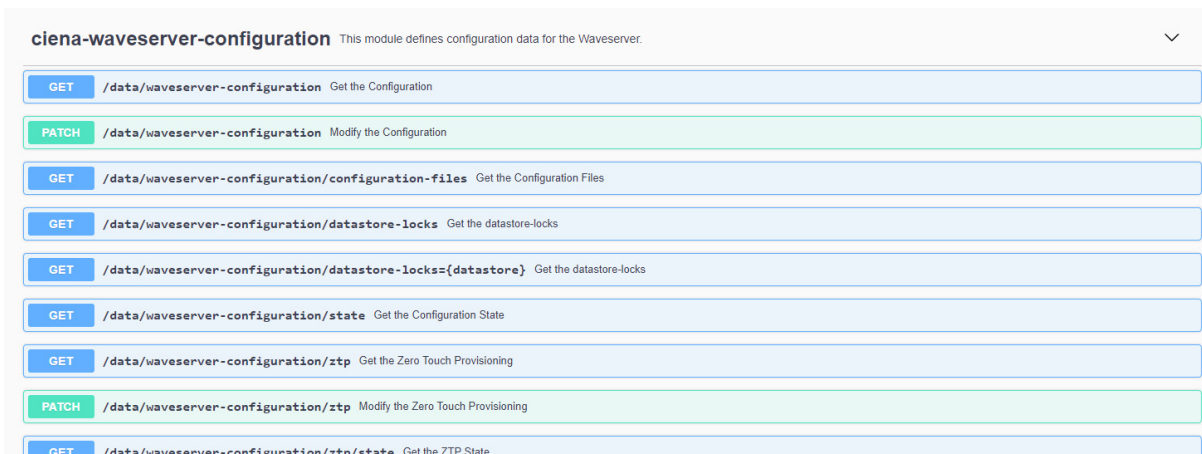
Note that you must be logged into the device for the test function to work. Also note that it will actually send the request to the

For GET requests, all YANG models are wrapped in a 'data' JSON object or XML element depending on the content type. It is avoid inconsistent/stale results.

Schemes  
HTTPS

**ciena-waveserver-aaa** This module defines the Authentication, Authorization and Accounting (AAA) functionality for Ciena's Waveserver platform.

- 4 Click a component to display a list of API operations available for that component.

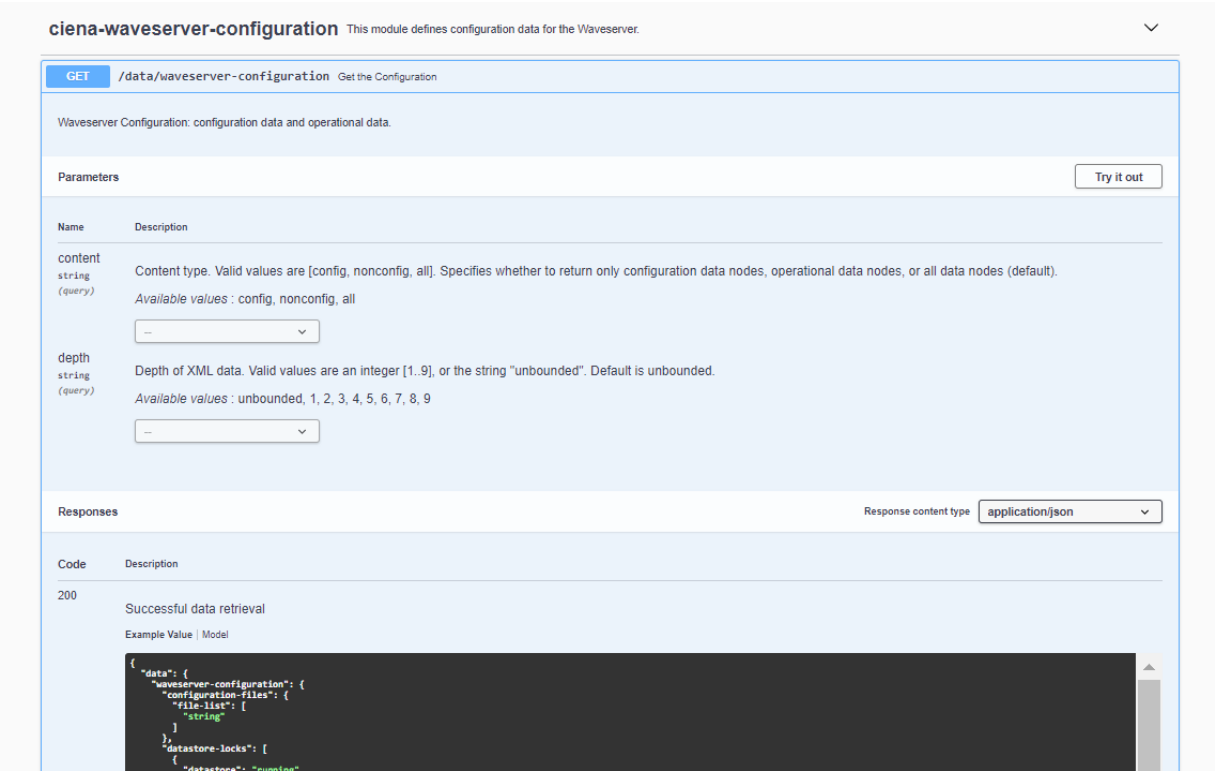
**Figure 173** Example of Waveserver 5 API operations


**ciena-waveserver-configuration** This module defines configuration data for the Waveserver.

Method	Path	Description
GET	/data/waveserver-configuration	Get the Configuration
PATCH	/data/waveserver-configuration	Modify the Configuration
GET	/data/waveserver-configuration/configuration-files	Get the Configuration Files
GET	/data/waveserver-configuration/datastore-locks	Get the datastore-locks
GET	/data/waveserver-configuration/datastore-locks={datastore}	Get the datastore-locks
GET	/data/waveserver-configuration/state	Get the Configuration State
GET	/data/waveserver-configuration/ztp	Get the Zero Touch Provisioning
PATCH	/data/waveserver-configuration/ztp	Modify the Zero Touch Provisioning
GET	/data/waveserver-configuration/ztp/ztp-state	Get the ZTP State

- 5 Click an operation to view the associated parameters, response types, and status codes.

Figure 174 API example



**Note:** For Swagger PATCH commands, ensure that the key used to identify the instance is specified in the request body. If the parameters do not match, but are both valid, the parameter specified in the Body will be used to apply the PATCH command.

## Procedure 274 Enabling and disabling the NETCONF server

Use this procedure to enable or disable the NETCONF server.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

NETCONF is enabled by default.

**Note:** Software license S86-LIC-WOSAPIS is required to access the Waveserver 5 using NETCONF.

### Steps

- Retrieve the current administrative state of the servers, if the state is unknown:  
`system server netconf show`
- | If you want to             | Then go to             |
|----------------------------|------------------------|
| disable the NETCONF server | <a href="#">step 3</a> |
| enable the NETCONF server  | <a href="#">step 4</a> |
- To disable the NETCONF server, use the command:  
`system server netconf disable`  
Go to [step 5](#).
- To enable the NETCONF server, use the command:  
`system server netconf enable`
- Retrieve the current administrative state of the servers, if you want to confirm the setting:  
`system server netconf show`
- Save the provisioned settings to the configuration file:  
`configuration save`



## Procedure 275 Configuring query limits for PM history bins

---

Use this procedure to configure NETCONF, REST, or gNMI query limits for PM history bins.

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Steps

- 1 Set the query limit for PM history bins.  
`system management query-limits <netconf | rest | gnmi> set pm-history-bins <NUMBER: 0..96>`
- 2 Display the query limit summary information.  
`system management query-limits show`
- 3 Save the provisioned settings to the configuration file.  
`configuration save`

## Procedure 276 Enabling and disabling the gRPC server

Use this procedure to enable or disable the gRPC server. The gRPC server is enabled by default.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Steps

- Optionally retrieve the current administrative state of the servers:  
`system server grpc show`
- |                         |                        |
|-------------------------|------------------------|
| <b>If you want to</b>   | <b>Then go to</b>      |
| enable the gRPC server  | <a href="#">step 3</a> |
| disable the gRPC server | <a href="#">step 4</a> |
- To enable the gRPC server, use the command:  
`system server grpc enable`  
Go to [step 6](#).
- To disable the gRPC server, use the command:  
`system server grpc disable`
- Save the provisioned settings to the configuration file:  
`configuration save`
- Optionally retrieve the current administrative state of the server to confirm the setting:  
`system server grpc show`

## Procedure 277 Enabling or disabling mutual authentication on a gRPC server

Use this procedure to enable or disable mutual (two-way) authentication on a gRPC server. Mutual authentication is set to enabled by default and can only be changed using the CLI. Changing the mutual authentication setting automatically triggers a gRPC server restart.

### Requirements

- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Steps

- Optionally retrieve the current authentication setting of the server(s):  
`system server grpc show`
- |                                                |                        |
|------------------------------------------------|------------------------|
| <b>If you want to</b>                          | <b>Then go to</b>      |
| disable mutual authentication on a gRPC server | <a href="#">step 3</a> |
| enable mutual authentication on a gRPC server  | <a href="#">step 5</a> |
- To disable mutual authentication on a gRPC server, use the command:  
`system server grpc mutual-authentication disable`
- Type Y to continue.  
The gRPC server restarts. The restart can take several minutes to complete.  
Go to [step 6](#).
- To enable mutual authentication on a gRPC server, use the command:  
`system server grpc mutual-authentication enable`
- Save the provisioned settings to the configuration file:  
`configuration save`
- Optionally retrieve the current status of the server(s) to confirm the setting:  
`system server grpc show`  
A gRPC server with disabled mutual-authentication displays an Authentication of "One-way" and a gRPC server with mutual-authentication enabled displays an Authentication of "Mutual".

---

## Procedure 278 Configuring telemetry subscriptions

---

Use this procedure to configure telemetry subscriptions.

### Requirements

- Use an account with a UPC level of 4 or higher.

### Steps

- 1 Globally enable or disable telemetry subscriptions:
  - `telemetry enable`
  - `telemetry disable`

**Note:** If disabled, all publishing for any existing subscription on the Waveserver is suspended on all interfaces.
- 2 Create a telemetry subscription with a subscription identifier and label:
  - `telemetry set subscription <String [1..32]> label <String [1..32]>`
- 3 Set the keep-alive message for subscriptions:
  - `telemetry set keep-alive <String [1..32]>`
- 4 Display telemetry settings:
  - `telemetry show [keep-alive]`
- 5 Display telemetry subscriptions:
  - `telemetry show subscriptions [subscription-id <String [1..32]>]`
- 6 Delete a subscription:
  - `telemetry delete subscription [subscription-id <String [1..32]>]`



## CHAPTER 30

# Removing, replacing, or reseating equipment

This section covers the following topics:

- [Overview](#)
  - [Safety](#)
  - [Auto configuration recovery](#)
- [Procedure list](#)

## Overview

This section provides information about safety, preparation, removing, replacing, and reseating equipment.

## Safety



### **DANGER**

Risk of electrical shock

- Before connecting the chassis to a power source, read all installation instructions.
- Do not work on the unit or connect or disconnect cables during periods of lightning activity.
- Before installing the unit, ensure that wiring at the installation area meets all local and national electrical codes.
- All electrical interfaces on the unit are intended for intra-building connections only. The interfaces are designed as type 2 and type 4 ports, as described in GR-1089-CORE, Issue 7, and require isolation from exposed outside plant cabling.



### **DANGER**

**Risk of personal injury**

Do not stack anything on top of the unit. The mounting brackets are designed to support only the weight of the unit. Bracket failure due to excess weight can result in bodily injury and equipment damage.



**DANGER**

**Risk of electrical shock**

Ensure that the input power cable is disconnected from the power module before removal. Removing the power module while it is connected to live power may result in personal injury.



**CAUTION**

**Risk of equipment damage**

Ensure that the input power cable is disconnected from the power module before removal. Removing the power module while it is connected to live power may result in damage to the module.



**CAUTION**

**Hazardous moving parts**

Fan and power modules contain hazardous moving parts. Keep fingers and other body parts away.



**CAUTION**

**Risk of damage to the chassis, pluggable modules and transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable modules or transceivers.



**DANGER**

**Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



**DANGER**

**Risk of electric shock**

Disconnect all power sources before servicing to avoid shock hazard.

## Auto configuration recovery

When enabled, the system maintains an encrypted copy of the committed software release and the full system configuration including the configuration databases, installed license file, and installed certificate files on the memory of the Access Panel module. On a Control Processor module replacement, during module initialization, the system automatically recovers the software release and configuration from the backup on the Access Panel. Automatic Configuration Recovery feature is optional and must be enabled if it is used.

When choosing to override the software or configuration mismatch alarms, the configuration on the new CP is deleted if it was not originally from this chassis serial number. Overriding the mismatch, forces the CP to initialize with no-configuration as it would after a reset-to-factory-defaults. If the current Control Processor configuration was made in this chassis, it is not deleted and initialization proceeds with this configuration. Previous system and traffic configuration can be affected by this process.

Configuration data will automatically be backed-up each time you use CLI command `configuration save`. You can manually force the CP to copy its backup information to the Access Panel (AP) using CLI command `system auto-config-recovery force-backup`.

You can also set an audit interval in which the CP and AP can check to ensure that they have the same backup data. The audit interval can be set using CLI command `system auto-config-recovery set audit-interval`. The backup on the AP is encrypted with an internal key. To further secure the backup on the AP, you can optionally provide your own passphrase. The passphrase is set using CLI command `system auto-config-recovery set passphrase`.

If a passphrase has been set, the passphrase must be entered using the serial console during a CP replacement, when prompted.

**Note:** Before configuration is restored for encryption modules, an authentication is required as an extra security measure. For more information and procedures, refer to [“User accounts, user authentication and security”](#) section in this document.

The second is a check that is performed during CP replacements to detect if the software load and/or the configuration on the replacement CP matches the software load and/or configurations running on the modules in the chassis. The default value for auto configuration mismatch detection is enabled.

To disable or enable auto configuration mismatch detection, use CLI command:



```
system auto-config-recovery config-mismatch-detection
<disable | enable>
```

**Note:** If you are replacing a CP that has release 2.0.2 or lower, verify that Config-Mismatch-Detection is ENABLED to ensure there is no traffic loss upon CP replacement and auto-recovery. This parameter is ENABLED by default. If the Config-Mismatch-Detection is DISABLED and a replacement CP running release 2.0.2 or less is inserted, traffic will be affected at the manual upgrade (since the CPs with older releases than 2.0.2 do not support auto-recovery).

When there is a mismatch between the software load and/or the configuration of the replacement CP and the modules in the chassis, alarms will be raised, and the CP will be blocked from sending the mismatched configuration to the modules to preserve running traffic. To recovery, you have these options:

- The Software Mismatch alarm is raised. To clear this alarm, you can upgrade the CP's software load to match the modules in the chassis. For additional details, see [Software Mismatch](#).
- The Configuration Database Mismatch alarm is raised. To clear this alarm, you can load the configuration that matches the configuration running on the chassis. For additional details, see [Configuration Database Mismatch](#).
- If you want the modules in the chassis to match the software load of the CP and a default configuration, use CLI command:

```
system auto-config-recovery override-config-mismatch
```

Use this CLI command with caution, traffic will be lost.

For more information on configuring auto configuration recovery, see [Configuring auto configuration recovery](#).

## Procedure list

This section provides the following procedures:

- [Procedure 279, "Configuring auto configuration recovery"](#)
- [Procedure 280, "Replacing a Control Processor"](#)
- [Procedure 281, "Replacing an Access Panel"](#)
- [Procedure 282, "Removing or replacing a power and fan module"](#)
- [Procedure 283, "Replacing a fan module"](#)
- [Procedure 284, "Replacing or reseating a pluggable transceiver"](#)
- [Procedure 285, "Replacing a QSFP28-QSFP28 or QSFP-DD - QSFP-DD Active Optical Cable"](#)
- [Procedure 286, "Reseating, removing, or replacing a service module"](#)

- [Procedure 287, “Removing a Waveserver 5 chassis from a 19-in, 23-in, or ETSI rack”](#)
- [Procedure 288, “Removing a Waveserver 5 chassis from a 19-in or 23-in data center rack”](#)
- [Procedure 289, “Replacing a Waveserver 5 chassis”](#)

## Procedure 279 Configuring auto configuration recovery

Use this procedure to configure auto configuration recovery. This feature is used when you are replacing a CP.

### Requirements

- You are logged in to Waveserver 5 using an account with at least admin privileges.

### Steps

- |   |                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                  |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p><b>If you want to</b></p> <p>set the administrative state for the auto configuration recovery mismatch detection</p> <p>override the configuration mismatch and apply the software load from the CP and a default configuration to the modules in the chassis.</p> <p>configure the automatic backup for the auto configuration recovery</p> <p>force the backup data for the CP to be copied to the AP</p> | <p><b>Then go to</b></p> <p><a href="#">step 2</a></p> <p><a href="#">step 3</a></p> <p><a href="#">step 4</a></p> <p><a href="#">step 6</a></p> |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
- 2 Set the administrative state for the auto configuration recovery mismatch detection.
- ```
system auto-config-recovery config-mismatch-detection
<enable | disable>
```
- Go to [step 7](#).
- Note:** If you set `config-mismatch-detection` to disabled, the CP software load and a default configuration will overwrite the configuration in the chassis. When `config-mismatch-detection` is set to disabled, traffic will be lost. You can determine if there is a traffic loss by checking for service-affecting alarms. The default for `config-mismatch-detection` is enabled and it is recommended to leave this attribute as enabled. When `config-mismatch-detection` is set to enabled, Software Mismatch and/or Configuration Database Mismatch alarms are raised when the software and/or the configuration file on the CP does not match the configuration data on the chassis.
- 3 Override the configuration mismatch and apply the software and the default configuration to the modules in the chassis.
- ```
system auto-config-recovery override-config-mismatch
```
- Go to [step 8](#).
- Note:** Use this CLI command with caution. Traffic may be lost, depending on the circumstances raising the mismatch alarm. Ideally, the CP database can be restored from an external server and merged with the config on the modules without traffic loss. Overriding the configuration mismatch clears Software Mismatch and Configuration Database Mismatch alarms.

- 4** Set the administrative state for the automatic backup.  
`system auto-config-recovery <disable | enable>`
- 5** Set the attributes for the automatic backup.  
`system auto-config-recovery set [passphrase <String:  
[8..64]>] [audit-interval <Minutes: 0..28800>]`  
Go to [step 7](#).
- 6** Force the CP to copy the software load, software configuration, certificate files, and license files to the AP.  
`configuration save`  
`system auto-config-recovery force-backup`  
Go to [step 7](#).
- 7** Display the auto configuration summary information.  
`system auto-config-recovery show`
- 8** Save the running configuration settings to the configuration file.  
`configuration save`

## Procedure 280 Replacing a Control Processor

---

### Requirements

Make sure that you

- have the appropriate personal grounding device to dissipate electrostatic charges.
- have a secure ladder for installations above shoulder level
- have console port access to Waveserver 5.
- observe all safety requirements described in *Waveserver 5 Installation*, 323-3001-200.
- have configured the auto configuration recovery, see [step 4](#) of the [Configuring auto configuration recovery](#) for detail.
- have access to or have backed up the following information to a secure location:
  - the upgrade package file (`tar.gz`) and the signature file (`tar.gz.sig`) that was used to install the current software load. See [Out-of-service software upgrades](#).
  - a configuration backup. See [Procedure 118, “Backing up the configuration datastores”](#).

**Note:** A configuration backup on the Access Panel is generated on a config save only after the DB has been modified, not after every config save.

- user-installed authority certificates, entity certificates, and license files. See [User accounts, user authentication and security](#).
- have comms configuration details

**Note:** CP ports support Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

### Overview

Use this procedure to replace the following Control Processors:

- Waveserver 5 Control Processor Subsystem (186-3010-900)

The Control Processor can be reseated or replaced while Waveserver 5 is in service

Replacing the Control Processor does not disrupt currently provisioned services, however, Ciena recommends performing this procedure during a maintenance window due to loss of comms and a CP cold restart.

The CP is located at the front of the Waveserver 5 chassis (see the following figure for an example).

**Figure 175** Example of Waveserver 5 chassis—front view, equipped



#### Legend

- 1 Control Processor

**Note:** Do not install or restart modules during the CP replacement.

#### Precautions



##### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the ends of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



##### **CAUTION**

##### **Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.



##### **CAUTION**

##### **Risk of module damage**

Handle modules with care at all times. Make sure that you do not drop any module. Otherwise, you will damage it. Also, do not remove the connector protectors until you are ready to install the module.

**CAUTION****Risk of module damage**

Do not force a module all the way to the back of its slot if it resists insertion. Before installing a module, make sure that you understand the detailed procedure for inserting modules.

**ATTENTION**

Do not replace the Control Processor (CP) and the Access Panel (AP) at the same time.

**ATTENTION**

When you remove the CP, logs of events, PMs and alarms are cached, and the Time-of-Day updates are stopped.

**Steps**

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack, or clip to a suitable ground point.
- 2 Retrieve all the active alarms and alarm details:
- 3 The following step is performed using CLI commands.  

```
show active-alarm <id> [name] [resource] [raise-time]
[raise-time] [cause] [condition-type] [direction]
[location] [severity] [service-impact] [additional-info]
[user-notes...]
```

The current alarms and alarm details will appear.
- 4 Record the current state of the alarms.
- 5 Verify that configuration mismatch detection is enabled. This ensures that Waveserver 5 notifies you in the event of a configuration mismatch between the replacement CP and the configuration used to provision the other modules on the chassis. Enter  

```
system auto-config-recovery show
```

If configuration mismatch detection is disabled, enable it.

```
system auto-config-recovery config-mismatch-detection
enable
```

Automatic backup is disabled by default. To enable generating a backup to be used during Auto Config Recovery, enter the following CLI command:

```
system auto-config-recovery enable
```

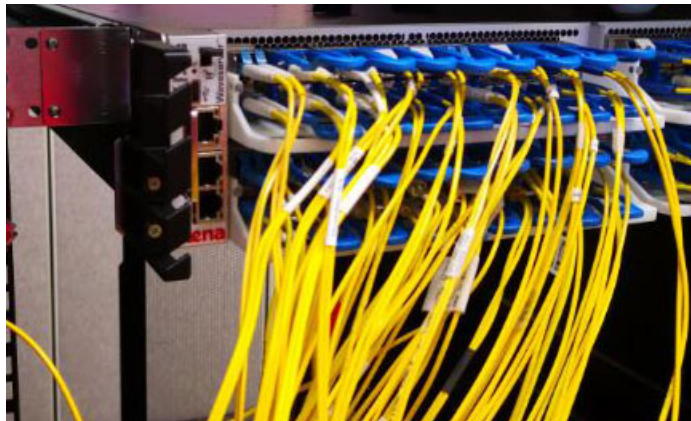
- 6 To enable the secure backup portion for encryption authentication data, enter the following CLI commands in the order specified (passphrase need to be more than 14 characters):

```
system encryption set backup-passphrase
```

```
system encryption backup enable
```

- 7 Label and disconnect any cables attached to the CP. If a large number of cables make accessing the CP difficult, remove the cables from the fiber manager one at a time and leave the cables hanging in front of the chassis (up to 18 pairs) until the CP is accessible.

**Figure 176** Removing cables for CP access



- 8 Grasp the module by its handle, depress the latch and slide the module out of the slot. Place the CP on a flat, stable surface.



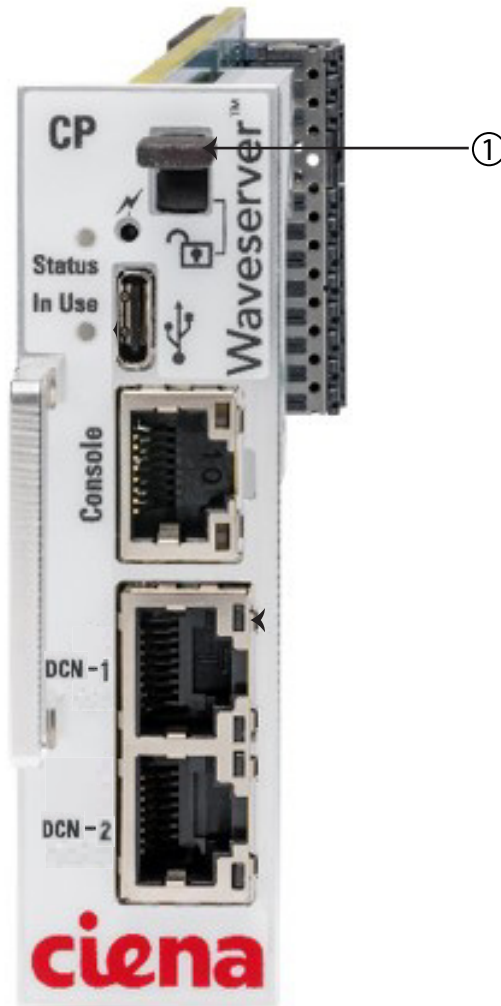
**WARNING**

**Hot Surface**

The module may be hot. Do not touch hot surfaces.



**Figure 177** CP latch



**Legend**

1 CP Latch

- 9 Position the replacement CP in the track of the slot at the front of the chassis.
- 10 Depress the CP latch and push firmly on the CP handle or faceplate until the module is fully engaged in the backplane.
- 11 Reconnect the cables that you disconnected in [step 7](#).
- 12 Route the cables through the fiber manager, if required.
- 13 Wait for the CP to boot up (the “In Use” and “Status” LEDs are solid green). The CP LED boot up sequence is as follows:
  - a. “Status” LED flashes red

- b. "In Use" and "Status" LEDs flash red
  - c. "Status" LED flashes green
  - d. "In Use" and "Status" LEDs turn off
  - e. "Status" LED is solid green
  - f. "In Use" and "Status" LEDs are solid green
- 14** If automatic backup is enabled and the optional passphrase is set, you must enter the passphrase at the serial port.
- 15** Login to the Waveserver 5 using a username and a password. Depending on whether the DB has been restored or not, you may need to login as super user (su) to configure comms and restore the DB. Refer to ["Restoring backed-up configuration datastores"](#) procedure.
- 16** Restore network connectivity to the chassis. See [Setting up management access using the console or USB port](#).
- 17** Retrieve all active alarms.
- If the automatic backup was enabled (before replacement) and config file was recovered successfully from the Access Panel (AP), no alarms is raised. This procedure is completed.
- Otherwise, the following mismatch alarms may be raised:
- Software Mismatch — indicates that the replacement CP software load does not match the software running on the other modules in the chassis. Refer to ["Software Mismatch"](#) alarm clearing procedure in this document.
  - Configuration Database Mismatch — indicates that the configuration on the replacement CP does not match the configuration data on the chassis. Refer to ["Configuration Database Mismatch"](#) alarm clearing procedure in this document.
- 18** Troubleshoot any other alarms using the "Alarm clearing" section of this publication. See [Alarms](#) for more information.

## Procedure 281 Replacing an Access Panel

Use this procedure to replace an Access Panel (AP). The Access Panel can be removed or replaced while Waveserver 5 is in service.

### Requirements

Make sure that you

- have the appropriate personal grounding device to dissipate electrostatic charges.
- have the replacement AP (186-3020-900) available
- observe all safety requirements described in *Waveserver 5 Installation*, 323-3001-200.
- have a secure ladder for installations above shoulder level

**Note:** For AP ports deployed at 1G, Waveserver 5 supports Category 5 enhanced Unshielded Twisted Pair (Cat 5e UTP) Ethernet cables.

### Overview

Replacing the AP does not disrupt currently provisioned services, however, Ciena recommends performing this procedure during a maintenance window due to loss of comms and a cold restart of the CP.

The AP is located at the rear of the Waveserver 5 chassis (see the following figure for an example).

**Figure 178** Example of Waveserver 5 chassis—rear view, equipped



### Precautions



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the ends of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

**CAUTION****Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic sensitive devices. Use anti-static protection to avoid damaging modules.

**CAUTION****Risk of equipment damage**

Make sure that your hands or tools do not come into contact with the circuitry of the access panel (AP) or backplane pins. Avoid contact until the shelf is grounded to the rack.

**CAUTION****Risk of equipment damage**

The shelf is suitable for connection to intrabuilding or unexposed wiring or cabling only.

**ATTENTION**

Do not replace the Control Processor (CP) and the Access Panel (AP) at the same time.

**ATTENTION**

While the AP is missing from the chassis, ILANs are down and comms fails between the CP and Service Modules, and between Service Modules.

**ATTENTION**

Replacing the AP results in a CP cold restart.

**Steps**

- 1 Verify the auto configuration recovery status. If auto configuration recovery is disabled, go to [step 3](#).
- 2 If auto configuration recovery is enabled, chassis information is backed up on the AP (after the AP replacement, chassis information is lost). After the replaced AP starts up, the Control Processor (CP) restores the backup files to the AP.
- 3 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack, or clip to a suitable ground point.
- 4 Label and disconnect any cables attached to the AP.



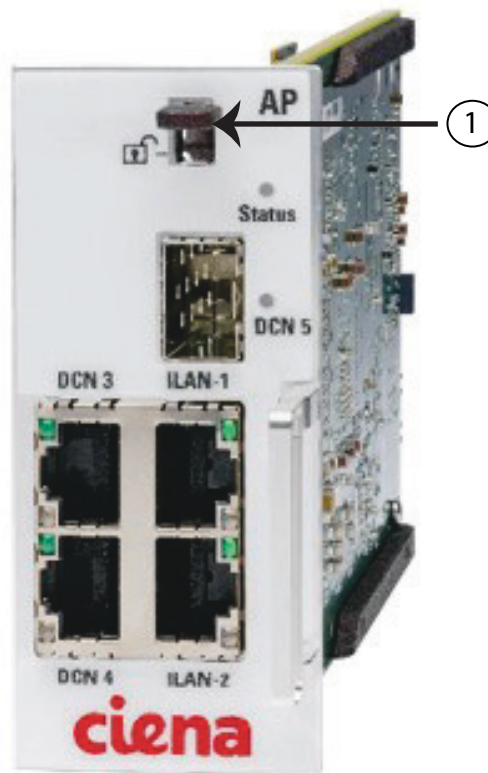
**WARNING**

**Hot Surface**

The module may be hot. Do not touch hot surfaces.

- 5 Grasp the AP by its handle, depress the latch and slide the module out of the slot. Place the AP on a flat, stable surface. See the following figure for the latch location.

**Figure 179** AP latch location



**Legend**

- 1 AP Latch

- 6 Position the replacement AP in the track of the slot at the rear of the chassis.
- 7 Depress the module latch and push firmly on the AP handle or faceplate until the AP is fully engaged in the backplane.

**Note:** When the new AP is installed, the CP boots up automatically.

- 8 Reconnect cables to the AP faceplate that you disconnected in [step 4](#), using the labels as guides.

- 9 If auto configuration recovery was enabled and the CP restored the backup files to the AP, ensure the restore is completed and the system status is as before the replacement.

## Procedure 282 Removing or replacing a power and fan module

---

Use this procedure to remove or replace a Waveserver 5 Power and Fan Module (Waveserver 5 100-277Vac/240Vdc Power and Fan Module (186-3040-900) or Waveserver 5 -48Vdc Power and Fan Module (186-3041-900)). The power and fan module is field-replaceable while the chassis is in service.

### Requirements

Make sure that you

- observe all the safety requirements described in the “Observing product and personnel safety guidelines” section of *Waveserver 5 Installation*, 323-3001-200.
- have a replacement power and fan module assembly with the correct Product Engineering Code (PEC)
- have Phillips screwdrivers and torque flat head screwdrivers suitable for at least 18 lb-in (2 N-m)
- if working with lug-based power connections, antioxidant paste and a metric or imperial torque driver/socket set suitable for at least 30 lb-in (3.4 N-m)
- have the Engineering Documentation Package (EDP), Installation Drawing Package (IDP), or equivalent site/network engineering plans
- have a secure ladder for installations above shoulder level
- have the appropriate personal grounding device to dissipate electrostatic charges

### Overview

See the following figures for examples of the power and fan modules installed at the back of the Waveserver 5 chassis.

**Figure 180** Waveserver 5 chassis with AC power module, fan and access panel - rear panel view





**Figure 181** Waveserver 5 chassis with DC power module, fan and access panel - rear panel view

### Precautions



#### **DANGER**

##### **Risk of electrical shock**

When working with the Waveserver 5 100-277Vac/240Vdc Power and Fan Module, disconnect from the power source before extracting the module, or partially extract the module and allow at least two seconds for hazardous voltage to dissipate before removing completely.

Removing a fan module from a Waveserver 5 100-277Vac/240Vdc Power and Fan Module exposes hazardous voltages in the empty slot of the module. Replacing a fan module must only be performed by skilled personnel familiar with the safety standards of working with live power.

Do not insert hands, fingers, or tools into a live power module. Power modules are safe to handle two seconds after the power has been disconnected or the power module has been partially extracted.





**DANGER**

**Risque de décharge électrique**

Lorsque vous travaillez avec le module d'alimentation 100-277 VAC, déconnectez-le de la source d'alimentation avant de l'extraire, ou extrayez-le partiellement et attendez deux secondes pour que la tension dangereuse se dissipe avant de le retirer complètement.

Le retrait d'un module de ventilation d'un module d'alimentation 100-277 VAC expose des tensions dangereuses dans l'emplacement vide du module. Le remplacement d'un module de ventilation ne doit être effectué que par du personnel qualifié connaissant les normes de sécurité du travail sous tension.

N'insérez pas les mains, les doigts ou les outils dans un module d'alimentation sous tension. Vous pouvez manipuler un module d'alimentation de façon sécuritaire deux secondes après sa mise hors tension ou son extraction partielle.



**DANGER**

**Risk of electrical shock**

Removing a fan module from a Waveserver 5 -48Vdc Power and Fan Module exposes 60V of live power in the empty slot of the PSU. Replacing a fan module must only be performed by qualified personnel familiar with the safety standards of working with live power.



**DANGER**

**Risk of electrical shock**

Ensure that the input power cable is disconnected from the power module before removal. Removing the power module while it is connected to live power may result in personal injury.



**CAUTION**

**Risk of equipment damage**

Ensure that the input power cable is disconnected from the power module before removal. Removing the power module while it is connected to live power may result in damage to the module.

**CAUTION****Risk of damage to the chassis, modules and pluggable transceivers**

Replace only one power module at a time. Install the replacement power module within 30 seconds of removing the failed power module.

**CAUTION****Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable transceivers.

**CAUTION****Risk of service interruption**

Do not remove a power module (AC or DC) from a live Waveserver 5 chassis unless the redundant power module is live and equipped with both fan modules. Disregarding this precaution will turn off power to the chassis and result in a loss of traffic.

**Steps**

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack or clip to a suitable ground point.
- 2 Verify that the equipment rack is grounded. Also, you must ground every shelf (and any peripheral trays as described in the applicable installation procedure). Grounding is mandatory.  
Follow the instructions in the precautionary messages listed above.
- 3 Retrieve all the active alarms and alarm details:
- 4 The following step is performed using CLI commands.  

```
show active-alarm <id> [name] [resource] [raise-time]
[raise-time] [cause] [condition-type] [direction]
[location] [severity] [service-impact] [additional-info]
[user-notes...]
```

  
The current alarms and alarm details will appear.
- 5 Record the current state of the alarms.

***Making sure that power to the system is off***

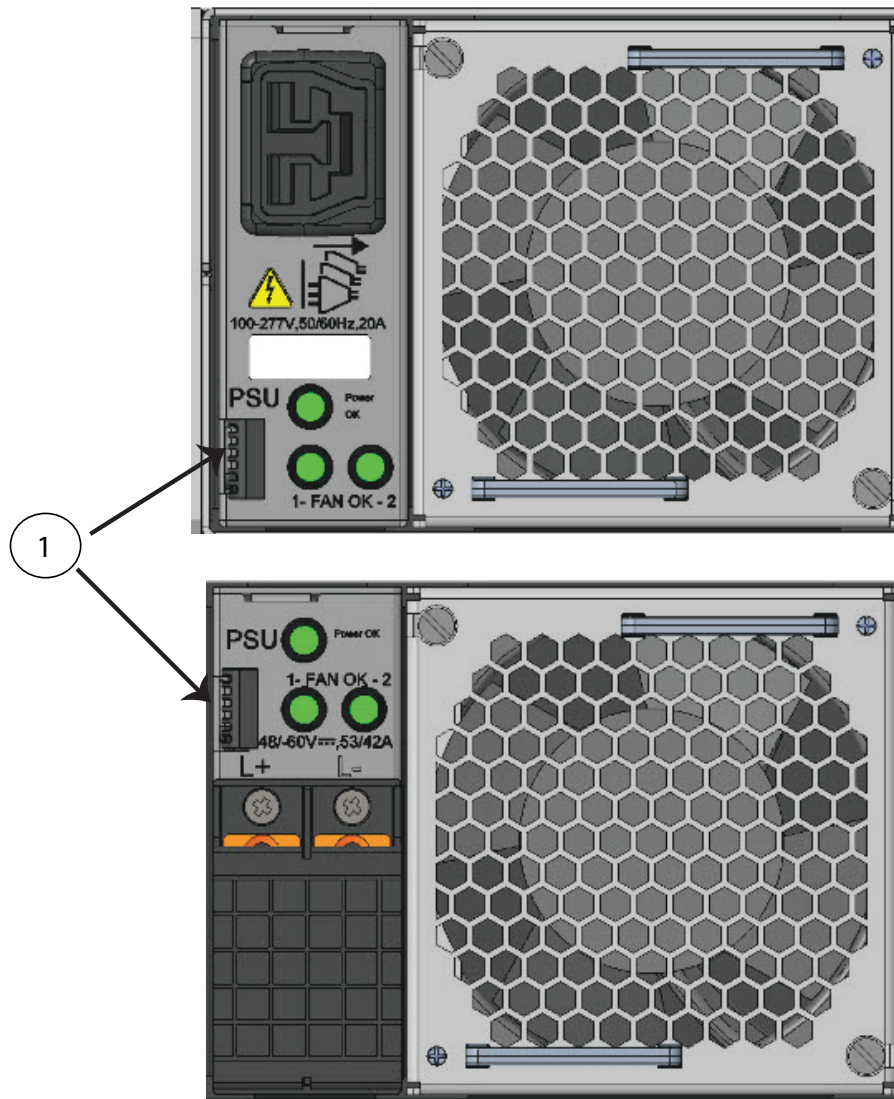
- 6 Locate the power to the power module that is going to be replaced. Verify the modules affected by the replacement. Make sure that power is off at the secondary power distribution panel (any upstream protector).

- 7 Turn off the power for the power module that is going to be replaced. Follow your company procedure for labeling and/or locking out each of the disconnected power sources in order to prevent any of the power sources from being energized before the procedure is completed.
- 8 Confirm that the green Power OK LED for the disconnected power module is off.  
A “Power Supply Feed Failed” alarm is raised against the affected power module.

***Removing the power and fan module***

- 9 Disconnect the AC or DC power cable from the power module as follows:
  - for DC power cables, remove the plastic safety cover from the terminals and remove the nut/washer assembly and nut on each of the power cables
  - for AC power cables, pull the AC power connector out of the power module receptacle
- 10 Holding the power and fan module by its handle, slide the latch towards the middle of the module until it disengages from the backplane. See the following figure for the location of the latch.

**Figure 182** Power and fan module latch



**Legend**

1 Latch

- 11 Using both hands, slide the module out of the slot and place it on a flat, stable surface.

**Installing the power and fan module**

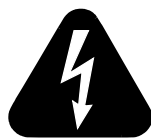
- 12 Hold the replacement power and fan module with both hands and insert it in its assigned slot.
- 13 Push the module into the slot by its handles or faceplate until it fully engages in the backplane, indicated by an audible click.

- 14 Select your next step.

If you are installing	Then go to
a DC power and fan module	<a href="#">step 15</a>
an AC power and fan module	<a href="#">step 26</a>

### ***Installing the DC power cable***

- 15 Prepare the appropriate DC power cable with the appropriate lug based on industry standards according to the EDP, IDP, or equivalent site/network engineering plans. See the “DC power modules” section of *Waveserver 5 Installation*, 323-3001-200 for a list of recommended lugs and power cable sizes.
- 16 Remove the plastic safety covers from the terminals.
- 17 Check that the nut/washer assemblies are not installed on the terminals. If they are installed, remove them.
- 18 Lightly coat with non-oxidizing compound as required by your company’s practices and connect the prepared DC power cable as applicable (based on the configuration) to the appropriate terminals on the power module.
- 19 Reinstall the nut/washer assembly and nut on the DC power cable. Verify that the torque is 20 lb-in. (2.26 N-m) using a torque driver. If required, lightly coat non-oxidizing compound on the terminals and the two-hole lugs.
- 20 After you attach the cable, align the holes on the plastic cover with the threaded studs (diagonal) of the power terminals and push the safety cover until it snaps into place.
- 21 Route and secure the DC power cable as follows:
- From the left module (facing the chassis rear), route the cable out and to the left. Secure the cables to the cable management bar.
  - From the right module (facing the chassis rear), route the cable out and to the right. Secure the cable to the cable management bar.



#### **DANGER**

##### **Risk of equipment damage and electrical shock**

When routing power cables over metallic edges, wrap and secure an insulating sheet fiber around the power cables to protect them against damage and to prevent personal injury from electrical shock.

- 22 Turn on the power for the DC power module.
- 23 Confirm that the green “Power OK” LED for the replacement power module is lit.
- 24 Retrieve all the active alarms and confirm that the “Power Supply Feed Failed” alarm has cleared.
- 25 If the alarm does not clear, refer to [Alarms](#).

If you are replacing a DC power and fan module, you have completed this procedure.

**Installing the AC power cable**

- 26 Connect the AC power cable to the power module. Route and secure the power cable as follows:
- For modules on the left side of the chassis, (facing the chassis rear), route the cable out and to the left. Secure the cable to the cable management bar.
  - From modules on the right side of the chassis (facing the chassis rear), route the cable out and to the right. Secure the cable to the cable management bar.

**DANGER****Risk of equipment damage and electrical shock**

When routing power cables over metallic edges, wrap and secure an insulating sheet fiber around the power cables to protect them against damage and to prevent personal injury from electrical shock.

- 27 Turn on the power for the AC power module.
- 28 Confirm that the green “Power OK” LED for the replacement power module is lit.
- 29 Retrieve all the active alarms and confirm that the “Power Supply Feed Failed” alarm has cleared.
- 30 If the alarm does not clear, refer to [Alarms](#) section for instructions to clear the alarm.

## Procedure 283 Replacing a fan module

---

Use this procedure to replace a Waveserver 5 fan module (186-3042-900). Fan modules are field-replaceable while the chassis is in service.

### Requirements

To perform this procedure, you must

- be qualified personnel who is familiar with the safety standards required for working with live power.
- have a replacement fan module (186-3042-900)
- have a flat head (slotted) screwdriver
- observe all safety requirements in the “Observing product and personnel safety guidelines” section of *Waveserver 5 Installation*, 323-3001-200
- have the appropriate personal grounding device to dissipate electrostatic charges

### Overview

Waveserver 5 is shipped with two power and fan module assemblies. Each assembly is equipped with two fan modules, for a total of four fan modules per Waveserver 5. Air is drawn in at the front of the chassis and exhausted out the rear. For proper operation, Waveserver 5 must be equipped with working fan modules in all fan module slots.

**Note:** When you remove a fan module, the associated power module shuts off, but the remaining fan draws 12 V of power from the other power module and continues to operate. This will cause the “Power Supply Feed Failed” alarm to be raised.



#### CAUTION

##### Risk of traffic loss

When one power supply is down on a live Waveserver 5, do not remove a fan module from the remaining good side. Otherwise, the power will drop potentially losing traffic to the entire node.

Do not operate Waveserver 5 with empty slots or empty pluggable transceiver ports/cages as it affects cooling. You must equip unused slots with appropriate filler modules and protect all unequipped pluggable module ports/cages with appropriate dust covers.

See the following figures for examples of the fan modules installed at the back of the Waveserver 5 chassis.



**Figure 183** Waveserver 5 chassis with AC PSU, fan and access panel - rear panel view



**Figure 184** Waveserver 5 chassis with DC PSU, fan and access panel - rear panel view



## Precautions



### **DANGER**

#### **Risk of electrical shock, energy hazard**

Removing a fan module from the AC and/or DC power and fan module exposes 500A of live power in the empty slot of the power module. Replacing a fan module must only be performed by qualified personnel familiar with the safety standards of working with live power.





### **DANGER**

#### **Risk of electrical shock**

When working with the 100-277Vac/240Vdc Power and Fan Module (186-3040-900), disconnect from the power source before extracting the module, or partially extract the module and allow at least two seconds for hazardous voltage to dissipate before removing completely.

Removing a fan module from a 100-277Vac/240Vdc Power and Fan Module exposes hazardous voltages in the empty slot of the module. Replacing a fan module must only be performed by skilled personnel familiar with the safety standards of working with live power.

Do not insert hands, fingers, or tools into a live power module. Power modules are safe to handle two seconds after the power has been disconnected or the power module has been partially extracted.



### **DANGER**

#### **Risque de décharge électrique**

Lorsque vous travaillez avec le module d'alimentation 100-277 VAC, déconnectez-le de la source d'alimentation avant de l'extraire, ou extrayez-le partiellement et attendez deux secondes pour que la tension dangereuse se dissipe avant de le retirer complètement.

Le retrait d'un module de ventilation d'un module d'alimentation 100-277 VAC expose des tensions dangereuses dans l'emplacement vide du module. Le remplacement d'un module de ventilation ne doit être effectué que par du personnel qualifié connaissant les normes de sécurité du travail sous tension.

N'insérez pas les mains, les doigts ou les outils dans un module d'alimentation sous tension. Vous pouvez manipuler un module d'alimentation de façon sécuritaire deux secondes après sa mise hors tension ou son extraction partielle.



### **DANGER**

#### **Risk of electrical shock**

Removing a fan module from a -48Vdc Power and Fan module (186-3041-900) exposes 60V of live power in the empty slot of the PSU. Replacing a fan module must only be performed by qualified personnel familiar with the safety standards of working with live power.



### **DANGER**

#### **Risk of eye injury**

Wear eye protection such as safety goggles or safety glasses with side guards when you work with fan modules or in proximity to the shelf air exhaust.



### **CAUTION**

#### **Hazardous moving parts**

Fan modules contain hazardous moving parts. Keep fingers and other body parts away.



### **CAUTION**

#### **Risk of damage to the chassis, modules and pluggable transceivers**

Replace only one fan module at a time. Insert the replacement fan module in the slot within 30 seconds of removing the failed fan module to avoid overheating of equipment due to insufficient airflow. Have a replacement fan module ready before removing the fan module.



### **CAUTION**

#### **Risk of equipment damage**

Replace any defective fan modules immediately. Operating Waveserver 5 with insufficient cooling can cause equipment to overheat and malfunction or fail.



### **CAUTION**

#### **Risk of equipment damage**

Electrostatic discharge (ESD) can damage electrostatic-sensitive devices. You must use anti-static protection while unpacking and installing all field-replaceable units (FRU) and cables.

## **Steps**

- 1 Retrieve all the active alarms and alarm details:
- 2 The following step is performed using CLI commands.  

```
show active-alarm <id> [name] [resource] [raise-time]
[raise-time] [cause] [condition-type] [direction]
[location] [severity] [service-impact] [additional-info]
[user-notes...]
```

The current alarms and alarm details will appear.

- 3 Ensure there are no unexpected alarms active and all alarms are accounted for and understood.
- 4 Wear an anti-static wrist strap or foot straps to protect the shelf from static damage. If you are using a wrist strap, connect the wrist strap to the ESD jack on the shelf.

***Removing the fan module***

- 5 Locate the fan module to be replaced.
- 6 Use a flat head screwdriver to unfasten the two screws on the fan module faceplate.
- 7 Gripping the pull handles, carefully remove the fan module from the PSU housing, being careful not to come into contact with the fan blades.
- 8 Place the fan module on a flat, stable surface.

***Installing the replacement fan module***

- 9 Firmly grip the replacement fan module on the left and right sides, being careful not to touch the fan blades. Slide the fan module into the slot.
- 10 Fasten the two screws on the fan module faceplate.  
The corresponding Fan LED on the power module faceplate lights green to indicate that the fan is operating normally.  
For more information about fan module LED behavior, see [Power and fan modules \(186-3040-900/186-3041-900/186-3042-900\)](#).
- 11 Retrieve all active alarms and alarm details.  
“Fan Missing” and “Power Supply Feed Failed” alarms must be cleared.
- 12 Troubleshoot any remaining alarms using the alarm clearing section of this publication. See [Alarms](#) for more information.

---

## Procedure 284 Replacing or reseating a pluggable transceiver

---

Use this procedure to replace or reseal a pluggable transceiver in Waveserver 5.

### Requirements

Make sure that

- you are logged in to Waveserver 5 using an account with at least admin privileges.
- you have a replacement transceiver (if required). See the [Pluggables and specifications](#) section for a list of supported pluggable transceivers.
- you observe all safety requirements in the “Observing product and personnel safety guidelines” section of *Waveserver 5 Installation*, 323-3001-200.

### Precautions



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Service-affecting procedure**

Replacing or reseating a pluggable transceiver is a service-impacting procedure.



#### **CAUTION**

##### **Risk of damage to the chassis, pluggable modules and transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable modules or transceivers.

### Steps

- 1 Reroute traffic for the pluggable you are replacing or reseating.
- 2 Select your first step.

If you want to	Then go to
reseat a pluggable transceiver	<a href="#">step 3</a>
replace a pluggable transceiver	<a href="#">step 9</a>

***Reseating a pluggable transceiver***

- 3 Remove the fiber-optic cables connected to the pluggable transceiver, and move them away from the Waveserver 5 chassis.
- 4 Carefully press or pull the transceiver's ejector lever and slide the transceiver out of the socket cage.
- 5 Clean the optical surfaces of the pluggable transceiver.
- 6 After at least 15 seconds, carefully insert the transceiver into the socket cage until it is fully seated.
- 7 Clean the ends of the fiber-optic cables and reconnect them to the transceiver.
- 8 Route traffic back to the pluggable you reseated.  
You have completed this procedure.

***Replacing a pluggable transceiver***

- 9 Change the admin state of the pluggable transceiver to "Disabled".
- 10 To place the pluggable transceiver to "Disabled". Enter the following command:  

```
xcvr disable xcvr <slot>/<xcvr>
```
- 11

If the replacement transceiver uses	Then go to
a different port mode than is currently provisioned	<a href="#">step 12</a>
the same port mode that is currently provisioned	<a href="#">step 13</a>
- 12 Delete the provisioning data of the pluggable transceiver from the configuration file on Waveserver 5:  

```
xcvr delete xcvr <slot>/<xcvr>
```
- 13 Remove the pluggable transceiver from the client interface:
  - a. Remove the fiber-optic cables connected to the pluggable transceiver, and move them away from the Waveserver 5 chassis.
  - b. Carefully press or pull the transceiver's ejector lever, and slide the transceiver out of the socket cage.
  - c. Place the pluggable transceiver in protective packaging.
- 14 If you will not be replacing the pluggable transceiver, install a dust cap on the empty socket cage to protect the interface from ambient dust.  
Continue with the next step if you are replacing the pluggable transceiver.

### ***Installing the replacement pluggable transceiver***

- 15 Remove the replacement pluggable transceiver from its packaging, and clean its optical surfaces. Ensure that the transceiver is supported in the client port (for traffic configuration options, see [Line ports and client ports in WL5e 2x MOTR 16x modules](#) or [Line ports and client ports in WL5n DCO 3xQSFP-DD 12x modules](#)).
- 16 Hold the transceiver by its sides and orient it so that the connector faces the socket cage on the chassis. Orient the transceiver as follows:
  - If you are installing the transceiver in the top row of the module, the plastic hinge (or tab) is facing up (above the fiber).
  - If you are installing the transceiver in the bottom row of the module, the plastic hinge (or tab) is facing down (below the fiber).
- 17 Carefully insert the transceiver into the socket cage until it is fully seated.
- 18 Clean the bulkhead of the transceiver and the ends of the fiber-optic cables.
- 19 Connect the fiber-optic cables to the pluggable transceiver. Refer to the [Cabling](#) section for a list of supported cables.
- 20

<b>If</b> the replacement transceiver uses	<b>Then</b> go to
a different port mode than is currently provisioned	<a href="#">step 21</a>
the same port mode as is currently provisioned	<a href="#">step 23</a>
- 21 Provision the pluggable transceiver as desired. See [“Provisioning client ports” on page 453](#) in this document.
- 22 Save the configuration.  
`configuration save`
- 23 Change the admin state of the pluggable transceiver to “Enabled”.
- 24 Repeat this procedure at the other end of the link if the transceiver type was changed.
- 25 Reroute traffic back to the pluggables you have replaced.

## Procedure 285 Replacing a QSFP28-QSFP28 or QSFP-DD - QSFP-DD Active Optical Cable

---

Use this procedure to replace the following AOCs that connects a Waveserver 5 client port to subtending equipment.

### Requirements

Make sure that you

- have the appropriate personal grounding device to dissipate electrostatic charges
- have the EDP, IDP, or equivalent site/network engineering plans
- have the required number of replacement QSFP28-QSFP28 Active Optical Cables (160-9460-xxx) or QSFP-DD - QSFP-DD Active Optical Cables (160-9660-xxx) of the correct length

### Precautions



#### **CAUTION**

**Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit, modules, and cables.

### Steps

- 1 Identify the cable at both ends to ensure that it can be removed.
- 2 Disable the Waveserver 5 client port where the AOC is installed.
- 3 On the Waveserver 5 client port, hold the AOC by its pull-tab and slide the cable plugs straight out of the ports. Repeat this action for the AOC cable plug that is installed on the subtending equipment.
- 4 Replace the dust caps on the ends of the cable and place the cable in protective packaging.
- 5 Remove the replacement AOC from its protective packaging and remove the dust caps from the ends of the cable.
- 6 Insert the cable plugs into the Waveserver 5 client port and into the port on the subtending equipment.
- 7 Re-enable the client port where the AOC is installed.

---

## Procedure 286 Reseating, removing, or replacing a service module

---

Use this procedure to reseal, remove and/or replace a service module. The Waveserver 5 can accommodate four double-width modules. Modules operate independently of one another. The service modules are field-replaceable while the chassis is in service.

### Requirements

- A replacement module of the same type is available for immediate installation, or a filler module is available if you are not replacing the Waveserver 5 module.
- You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

Waveserver 5 supports the following service modules:

- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-900)
- Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901)
- Waveserver WL5e 2x800G L-Band Submarine 16xQSFP28/QSFP-DD Module (186-3110-901)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-900)
- Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901)
- Waveserver WL5e 2x800G L-Band Premium 16xQSFP28/QSFP-DD Module (186-3111-901)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-900)
- Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901)
- Waveserver WL5e 2x800G L-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3112-901)
- Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28/QSFP-DD Module (186-3302-900)
- Waveserver WL5e 2x800G C-Band Submarine 12xQSFP28/QSFP-DD Module (186-3120-900)



- Waveserver WL5e 2x800G C-Band Premium 12xQSFP28/QSFP-DD Module (186-3121-900)

You can replace an older version (186-3100-900/186-3101-900/186-3102-900) of a service module with a newer version (186-3100-901/186-3101-901/186-3102-901) of the same service module, but you cannot replace a newer version of a service module with an older version of the same service module.

See the following examples:

- 1 Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901) can be used as a replacement for Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-900). You cannot replace Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-901) with Waveserver WL5e 2x800G C-Band Submarine 16xQSFP28/QSFP-DD Module (186-3100-900).
- 2 Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901) can be used as a replacement for Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-900). You cannot replace Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-901) with Waveserver WL5e 2x800G C-Band Premium 16xQSFP28/QSFP-DD Module (186-3101-900).
- 3 Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901) can be used as a replacement for Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-900). You cannot replace Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-901) with Waveserver WL5e 2x800G C-Band Enhanced 16xQSFP28/QSFP-DD Module (186-3102-900).

### Precautions



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **WARNING**

##### **Risk of burn**

Take care when removing or replacing Waveserver 5 modules. The heat sinks of the modules are exposed to touch and are very hot.

**CAUTION****Service-affecting procedure**

Reseating, replacing or removing a service module is service-impacting. The traffic on the respective module is lost even if the chassis remains in service.

**CAUTION****Risk of damage to the chassis, modules and pluggable transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and modules or transceivers.

**ATTENTION**

After installation, the Waveserver 5 module automatically upgrades to the software load installed on the CP, if the software versions are different. If the module type is unsupported by the CP software version, the module powers off and becomes unusable.

**Guidelines**

**Note 1:** For module replacements, ensure that the replacement module is of the same type as the module you are removing.

**Note 2:** For return merchandise authorization (RMA) of field replaceable units (FRUs), do not return the entire Waveserver 5 pick-to-order kit. Return only the FRU(s) to Ciena. FRUs include pluggable modules, AC/DC power modules with fans (186-3040-900 / 186-3041-900), access panels, and fan modules(186-3042-900).

**Steps**

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges.
- 2 Perform a Lamp Flash on the module to flag the module to be removed, reseated or replaced. See [“system lamps set”](#) in the [“CLI command reference”](#) section of this document.
- 3 Reroute traffic away from the module being replaced.
- 4
 

If you want to	Then go to
reseat a module	<a href="#">step 5</a>
remove a module	<a href="#">step 7</a>
replace a module	<a href="#">step 7</a>

**Reseating a module**

- 5 Holding the module by its handle, slide the latch to the right and pull the module straight out until the module's Status LED extinguishes.

- 6 Carefully re-insert the module back into the slot until you hear a click, which indicates that the holding clip has engaged. Wait for the Status LED to light green.

You have completed this procedure.

### ***Removing a module***

- 7 Disable the administrative state of the module.
- 8 If applicable, label and detach the cables that are attached to the module you are replacing. Place protective caps on the optical connectors and cables to protect them from contamination and damage.
- 9 If applicable, remove pluggable transceivers from the module, refer to [Removing, replacing, or reseating equipment](#).
- 10 Holding the module by its handle, slide the latch to the right and pull the module straight out until it clears the slot. Place it on a flat, stable surface.

If you are	Then go to
not replacing the module	<a href="#">step 11</a>
replacing the module	<a href="#">step 17</a>

- 11 Delete the provisioning for the module you are replacing:  
`module delete slot <slot>`
- 12 Remove the filler module from its packaging. Carefully insert it into the slot until you hear a click, which indicates that the holding clip has engaged.
- 13 Confirm that a filler module is installed:
- 14 The following step is performed using CLI commands.  
`module show`
- 15 The following step is performed using CLI commands.  
`show active-alarm <id> [name] [resource] [raise-time]  
[raise-time] [cause] [condition-type] [direction]  
[location] [severity] [service-impact] [additional-info]  
[user-notes...]`  
The current alarms and alarm details will appear.
- 16 Clear any alarms raised against the system. Refer to [Alarms](#) for information on alarm clearing procedures.  
You have completed the module removal procedure.

### ***Replacing a module***

- 17 Remove the replacement module from its packaging, and ensure that its optical surfaces are clean.
- 18 Lift the module by the side rail edges.
- 19 Carefully guide the back end of the module into the slot. Push the module towards the back of the shelf until it fully engages with the backplane. Push on the faceplate to fully seat it.

- 20 If applicable, re-install the pluggable transceivers. If you are installing new pluggable transceivers, refer to [Replacing or reseating a pluggable transceiver](#) or the relevant installation procedure in *Waveserver 5 Installation*, 323-3001-200.
- 21 If applicable, re-attach any fiber-optic cables removed in [step 8](#). Make sure that the cables are properly cleaned before reconnection. Refer to the “Cleaning connectors” section of *Waveserver 5 installation*, 323-3001-200.
- 22 Enable the administrative state of the module.
- 23 Disable or unset the Lamp Flash on the module. See [“system defaults unset”](#) in the [“CLI command reference”](#) section of this document.
- 24 Check the status of the module:  

```
module show <slot>
```

The module State is “Initializing” and then changes to “Up”.
- 25 If you have replaced the Waveserver 5 DCO 3x400G-QSFP-DD 12xQSFP28 / QSFP-DD module (186-3302-900), the CP LED boot up sequence is as follows:  
**Status LED**
  - a. Red flashing slow
  - b. Red flashing fast
  - c. Green flashing
  - d. LED is off
  - e. Green solid**Line LEDs, QSFP-DD LEDs, QSFP28 LEDs**
  - f. Hardware fault: solid red
  - g. Low power level/LOS: solid yellow
  - h. Admin state enabled: solid green
- 26 Route traffic back onto the module you replaced.
- 27 Retrieve active alarms and clear any alarms raised against the system. Refer to [Alarms](#) for information on alarm clearing procedures.

## Procedure 287 Removing a Waveserver 5 chassis from a 19-in, 23-in, or ETSI rack

---

Use this procedure to remove a Waveserver 5 chassis from a 19-in, 23-in, or ETSI rack.

### Requirements

Make sure that

- you have the following tools available:
  - #0 and #1 Phillips screwdrivers
  - 3/8-in nut driver or socket
  - tools to remove the rack mounting screws (specific to your installation)
- two people are available to perform this procedure.
- all traffic has been rerouted. If required, all configuration settings have been saved and a backup configuration file is available.

### Precautions



#### **DANGER**

##### **Risk of electric shock**

Disconnect all power sources before removing components and chassis to avoid shock hazard.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Hazardous moving parts**

Fan and power modules contain hazardous moving parts. Keep fingers and other body parts away.



#### **WARNING**

##### **Risk of burn**

Take care when removing or replacing modules. The heat sinks of the modules are exposed to touch and are very hot.

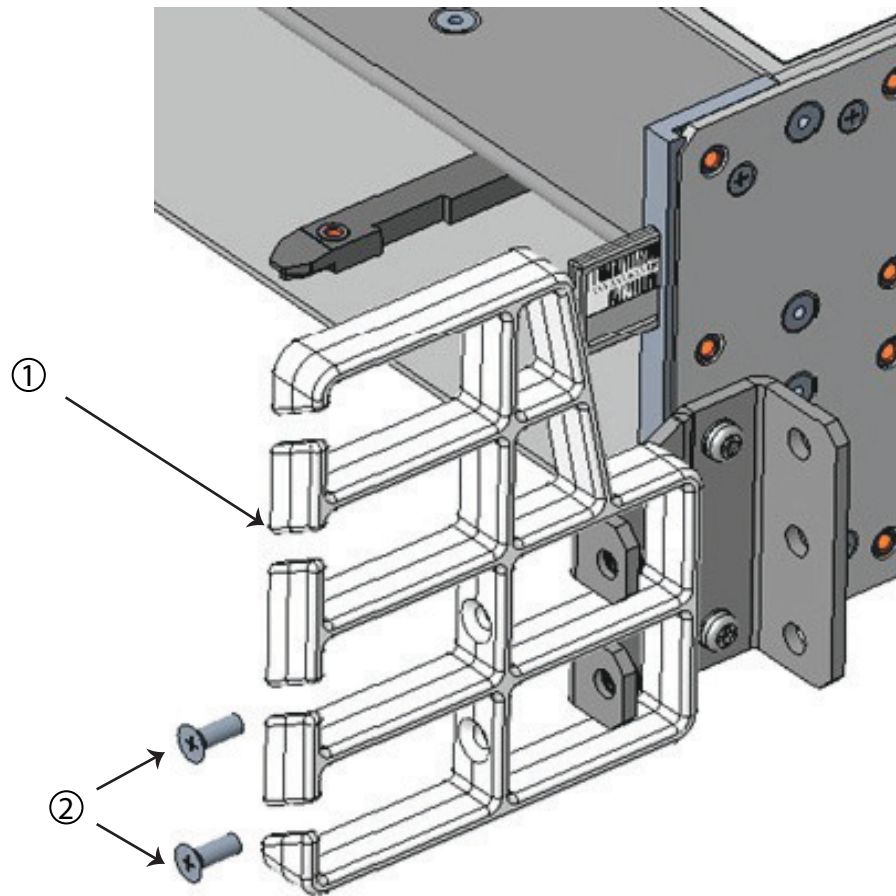
**CAUTION**

**Risk of damage to the chassis, pluggable modules and transceivers**

Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable modules or transceivers.

**Steps**

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack, or clip to a suitable ground point.
- 2 Ensure that all power is OFF before disconnecting the power connections from the chassis.
- 3 If you will be replacing the chassis, label and detach the cables that are attached to the modules. Place protective caps on the optical connectors and cables to protect them from contamination and damage.
- 4 Move all disconnected cables away from the chassis.
- 5 If you will be repackaging the Waveserver 5 for shipping, remove the fiber managers from each side of the chassis. Remove the screws securing the fiber managers using a Phillips screwdriver.

**Figure 185** Removing fiber managers and fasteners**Legend**

1 Fiber manager

2 Flat head screws

- 6** Remove the Control Processor (CP) from the chassis. Depress the module latch, grasp the CP by its handle and slide the module out of the slot. Place the CP on a flat, stable surface.

**Note:** If it is difficult to remove the CP, push it back into the slot and repeat this step.

- 7** Remove the Access Panel (AP). Depress the module latch, grasp the AP by its handle and slide the module out of the slot. Place the AP on a flat, stable surface.

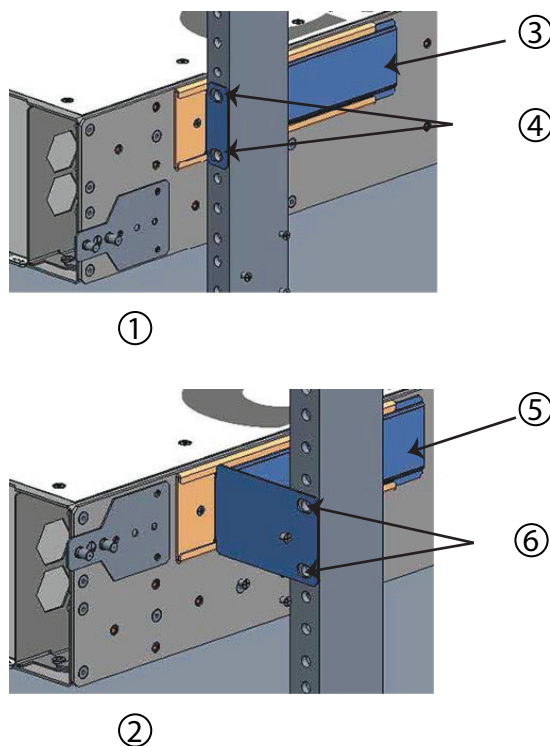
**Note:** If it is difficult to remove the AP, push it back into the slot and repeat this step.

- 8** Remove pluggable transceivers:

- a.** Carefully press or pull the transceiver's ejector lever, and slide the transceiver out of the socket cage.

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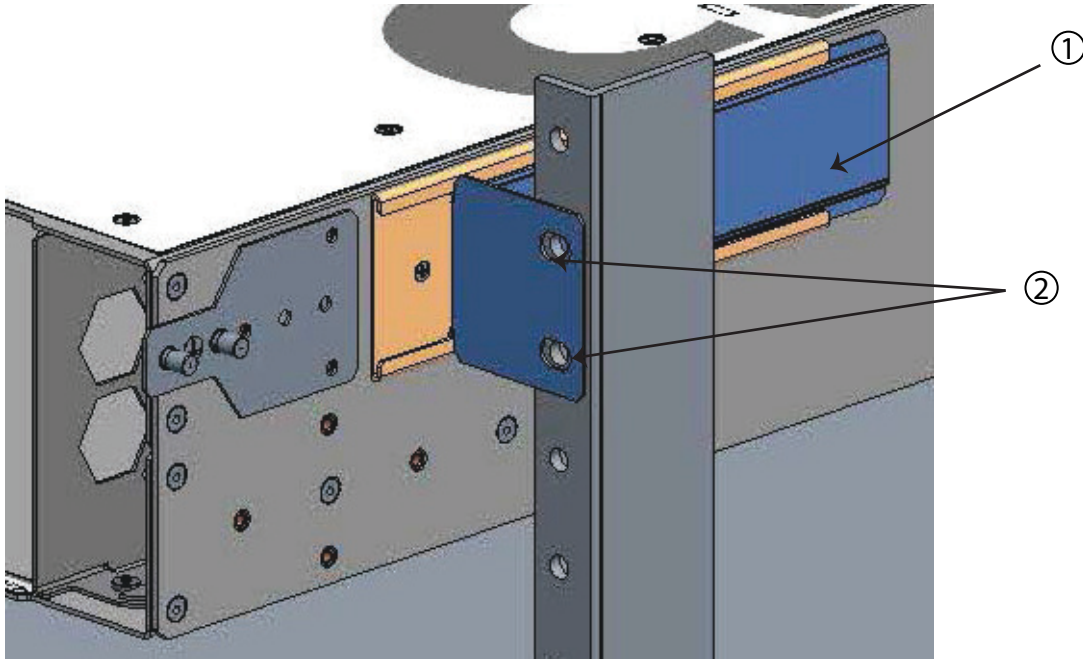




**Legend**

- |   |                                    |   |                                    |
|---|------------------------------------|---|------------------------------------|
| 1 | 19-in rack                         | 4 | Fastened with 2 rack screws        |
| 2 | 23-in rack                         | 5 | Rear mounting bracket 410-7036-001 |
| 3 | Rear mounting bracket 410-7036-001 | 6 | Fastened with 2 rack screws        |

**Figure 187** Rear mounting brackets — ETSI rack

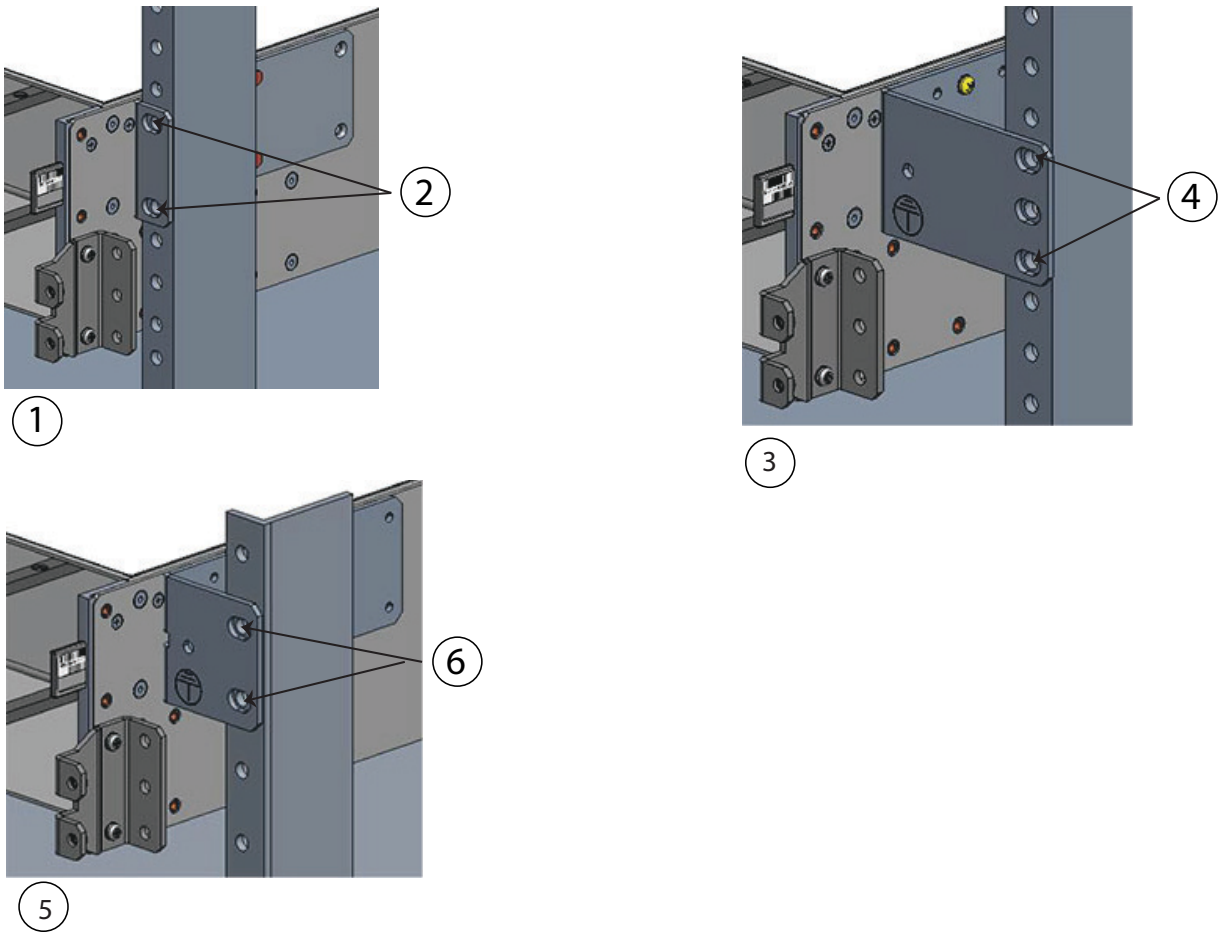


**Legend**

- |   |                                    |   |                             |
|---|------------------------------------|---|-----------------------------|
| 1 | Rear mounting bracket 410-7036-003 | 2 | Fastened with 2 rack screws |
|---|------------------------------------|---|-----------------------------|

- 13** Holding the chassis securely at the front and rear, remove the 2 screws per side that fasten the chassis to the rack at the front.

**Figure 188** Removing rack screws from the front mounting brackets

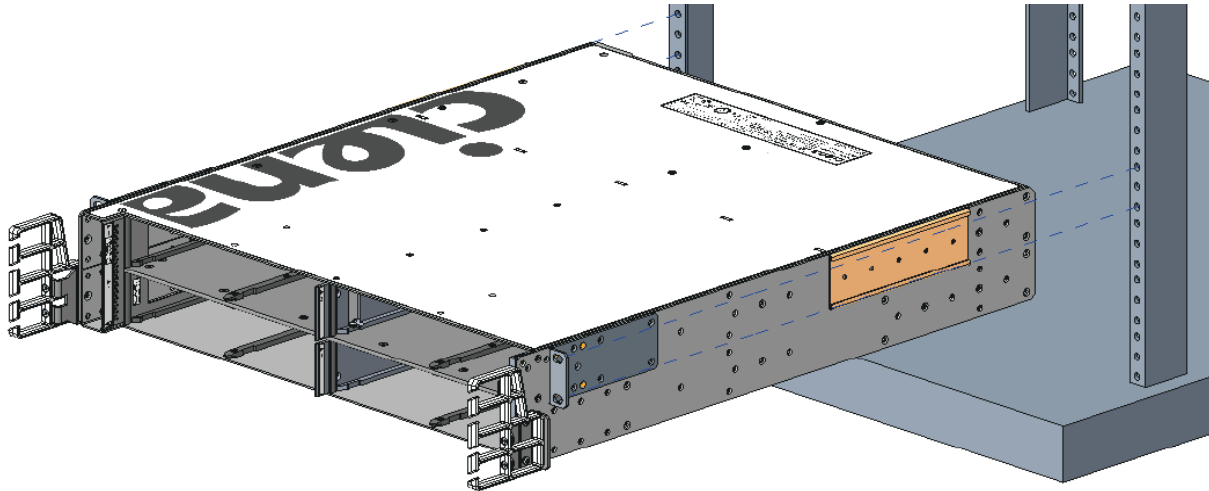


**Legend**

- |   |                             |   |                             |
|---|-----------------------------|---|-----------------------------|
| 1 | 19-in rack                  | 4 | Two rack screws per bracket |
| 2 | Two rack screws per bracket | 5 | ETSI rack                   |
| 3 | 23-in rack                  | 6 | Two rack screws per bracket |

- 14** Remove the Waveserver 5 chassis from the rack and place the chassis on a flat, stable surface.

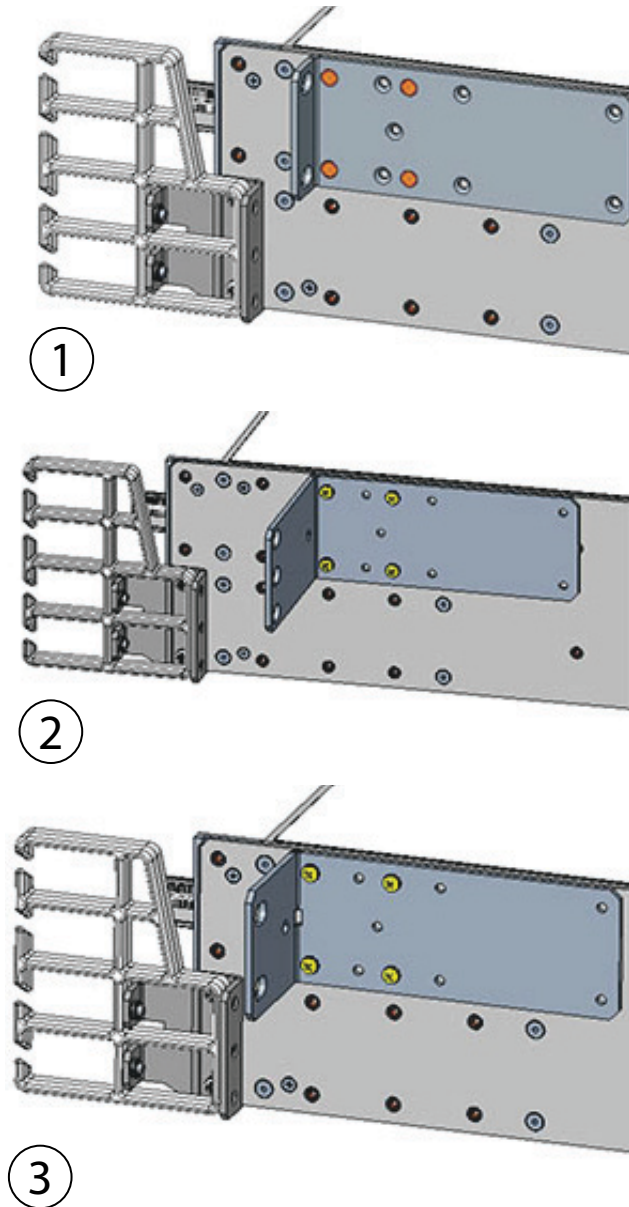
**Figure 189** Removing the Waveserver 5 chassis from the rack



- 15** Remove the rear bracket guide, front mounting brackets, and ground bracket from the Waveserver 5 chassis as follows:
- Remove the front brackets from the sides of the chassis. For a 19-in bracket kit, use a Phillips screwdriver to remove the four flat head screws that secure the bracket on each side. For a 23-in bracket kit or an ETSI bracket kit, use a Phillips screwdriver to remove the four pan head SEMS screws that secure the bracket on each side.
  - Remove the rear bracket guides from each side of the chassis. Unfasten the three wafer head screws using a Phillips screwdriver.
  - Remove the ground bracket from the chassis, if required. Unfasten the three wafer head screws securing the bracket to the side of the chassis using a Phillips screwdriver.

The following figures illustrate the locations of the brackets.

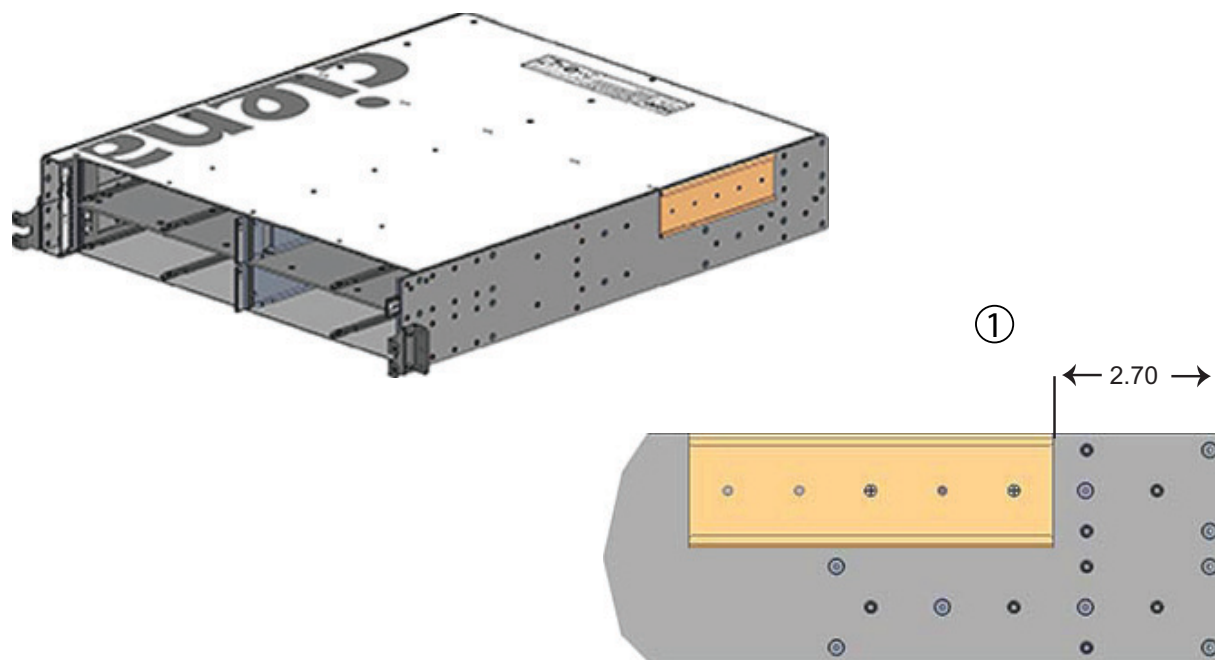
**Figure 190** Uninstall front mounting brackets



**Legend**

- |   |                                                                |   |                                                               |
|---|----------------------------------------------------------------|---|---------------------------------------------------------------|
| 1 | 19-in bracket kit                                              | 3 | ETSI bracket kit Front brackets fastened with pan head screws |
|   | Front brackets fastened with flat head screws                  |   |                                                               |
| 2 | 23-in bracket kit Front brackets fastened with pan head screws |   |                                                               |

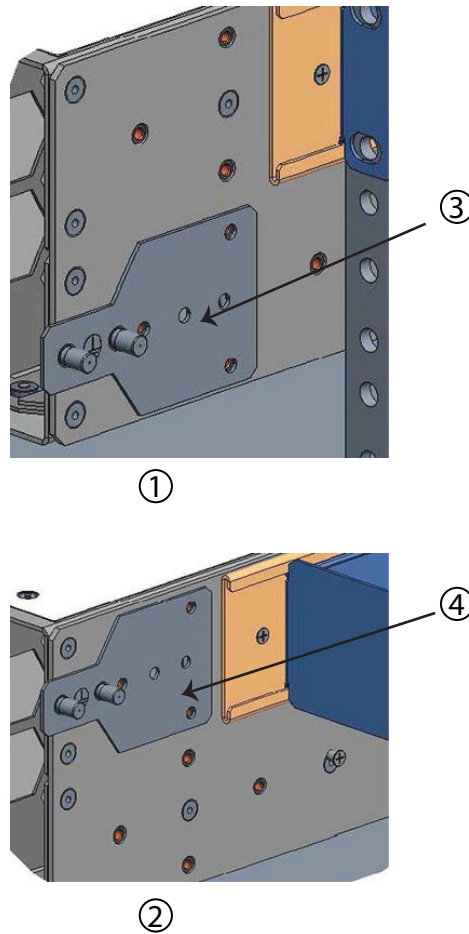
**Figure 191** Uninstall rear bracket guide (mid-mount position shown)



**Legend**

- 1 Recommended location for most racks and setbacks.

**Figure 192** Uninstall ground bracket (19-in and 23-in bracket kits shown)



### Legend

- |   |                                          |                             |
|---|------------------------------------------|-----------------------------|
| 1 | Ground bracket installed on 19-3 in rack | Ground bracket 410-6903-002 |
| 2 | Ground bracket installed on 23-4 in rack | Ground bracket 410-6903-002 |

**Note:** The mounting bracket, ground bracket and rear bracket guide positions may vary depending on the installation requirements, for example the type of rack used or the required setback.

- 16** Store all removed parts and fasteners for future re-installation.

## Procedure 288 Removing a Waveserver 5 chassis from a 19-in or 23-in data center rack

---

Use this procedure to remove a Waveserver 5 chassis from a 19-in or 23-in wide data center rack.

### Requirements

Make sure that

- you have the following tools available:
  - #0 and #1 Phillips screwdrivers
  - 3/8-in nut driver or socket
  - tools to remove the rack mounting screws (specific to your installation)
- All traffic has been rerouted. If required, all configuration settings have been saved and a backup configuration file is available.

### Precautions



#### **DANGER**

##### **Risk of electric shock**

Disconnect all power sources before removing components and chassis to avoid shock hazard.



#### **DANGER**

##### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



#### **CAUTION**

##### **Hazardous moving parts**

Fan and power modules contain hazardous moving parts. Keep fingers and other body parts away.



#### **WARNING**

##### **Risk of burn**

Take care when removing or replacing modules. The heat sinks of the modules are exposed to touch and are very hot.

**CAUTION**

**Risk of damage to the chassis, pluggable modules and transceivers**

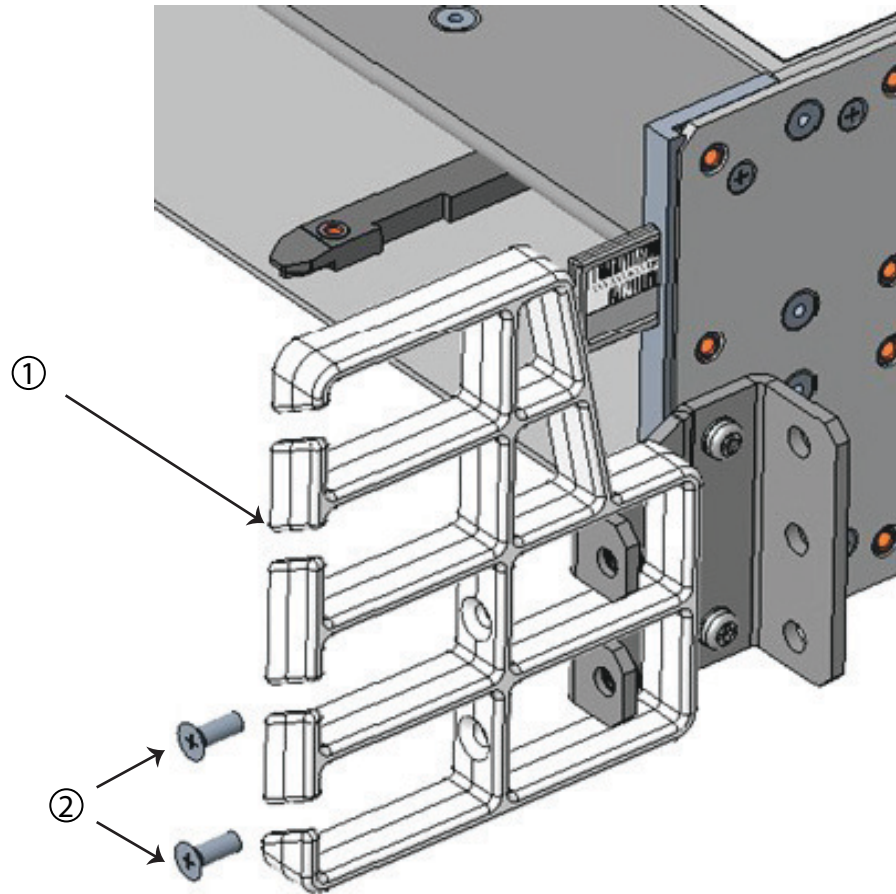
Wear a properly fitted electrostatic discharge (ESD) strap when handling the unit and pluggable modules or transceivers.

**Steps**

- 1 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges. If you are wearing an anti-static wrist strap, connect the cord to the ESD jack on the shelf installed and grounded in a grounded rack, or clip to a suitable ground point.
- 2 Ensure that all power is OFF before disconnecting the power connections from the chassis.
- 3 If you will be replacing the chassis, label and detach the cables that are attached to the modules. Place protective caps on the optical connectors and cables to protect them from contamination and damage.
- 4 Move all disconnected cables away from the chassis.
- 5 If you will be repackaging the Waveserver 5 for shipping, remove the fiber managers from each side of the chassis. Remove the screws securing the fiber managers using a Phillips screwdriver.



**Figure 193** Removing fiber managers and fasteners



**Legend**

- |   |               |   |                  |
|---|---------------|---|------------------|
| 1 | Fiber manager | 2 | Flat head screws |
|---|---------------|---|------------------|

- 6** Remove the Control Processor (CP) from the chassis. Depress the module latch, grasp the CP by its handle and slide the module out of the slot. Place the CP on a flat, stable surface.

**Note:** If it is difficult to remove the CP, push it back into the slot and repeat this step.

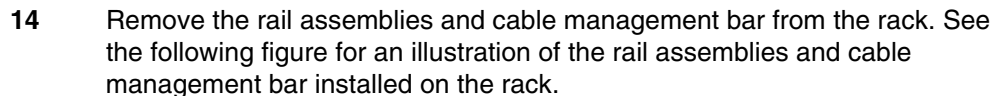
- 7** Remove the Access Panel (AP). Depress the module latch, grasp the AP by its handle and slide the module out of the slot. Place the AP on a flat, stable surface.

**Note:** If it is difficult to remove the AP, push it back into the slot and repeat this step.

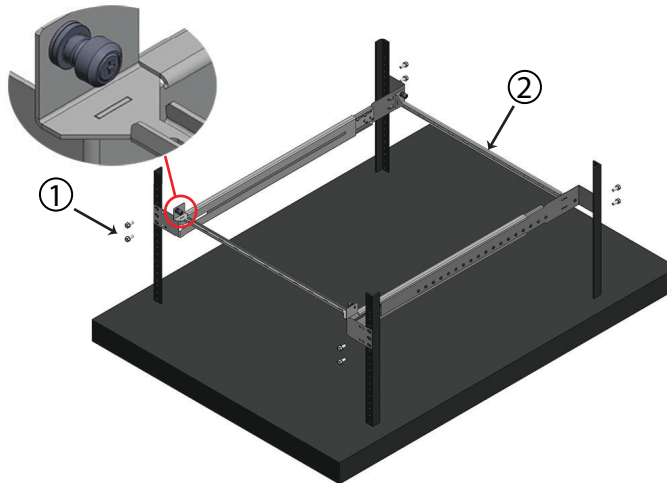
- 8** Remove pluggable transceivers:

- a. Carefully press or pull the transceiver's ejector lever, and slide the transceiver out of the socket cage.

- Figure 194** Removing the Waveserver 5 chassis from the rack



**Figure 195** Rail assemblies and cable management bar locations



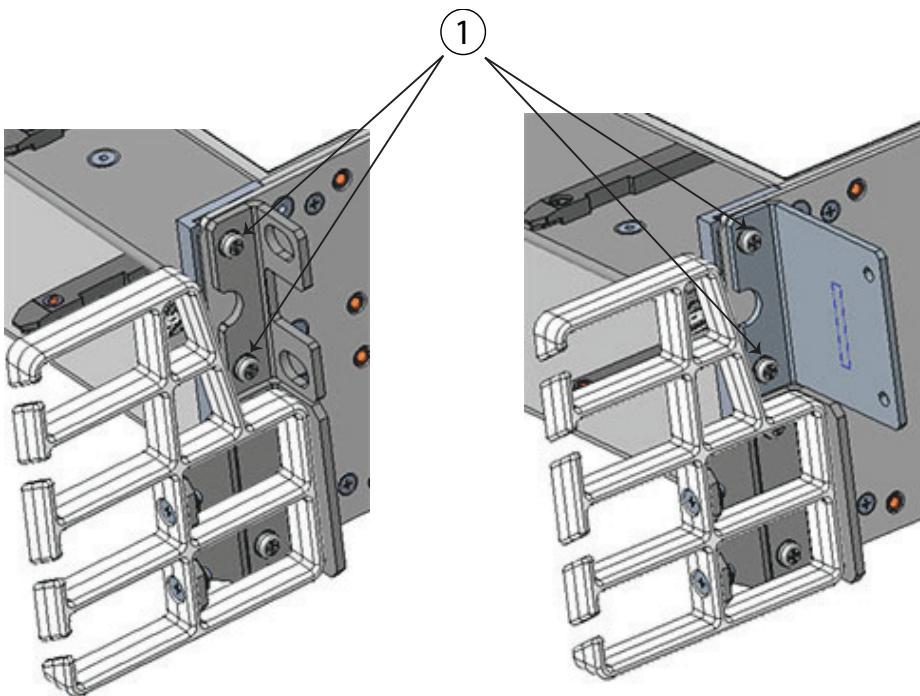
**Legend**

1 Rack manufacturer's recommended hardware

2 Rear Cable Management Bar (410-8191-001)

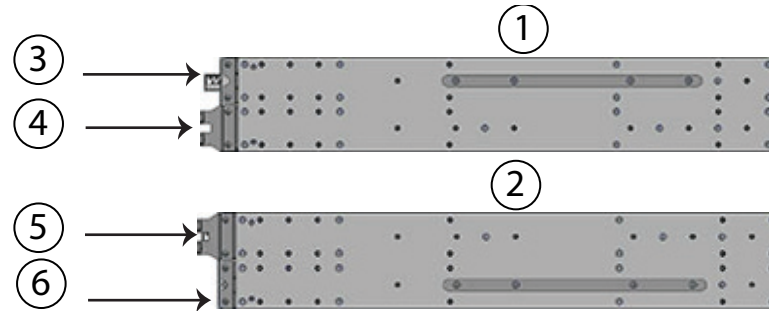
- 15 Remove the front mounting brackets and the T-Bars from the chassis using a Phillips screwdriver. The following figures illustrate the locations of the front mounting brackets and the T-Bars.

**Figure 196** Screws securing front mounting brackets



**Legend**

- |   |               |   |                  |
|---|---------------|---|------------------|
| 1 | Fiber manager | 2 | Flat head screws |
|---|---------------|---|------------------|

**Figure 197** T-Bar locations on chassis

**Legend**

- |   |                         |   |                    |
|---|-------------------------|---|--------------------|
| 1 | Upper mounting location | 4 | Utility bracket    |
| 2 | Lower mounting location | 5 | Utility bracket    |
| 3 | Rack mount bracket      | 6 | Rack mount bracket |

- 16** Store all removed parts and fasteners for future re-installation.

## Procedure 289 Replacing a Waveserver 5 chassis

---

Use this procedure to replace a Waveserver 5 chassis (186-3001-900) by removing all modules, replacing the chassis, and then re-installing the modules.

### Requirements

To perform this procedure you must

- reroute all traffic
- review all safety requirements described in “Observing product and personnel safety guidelines” section of *Waveserver 5 Installation*, 323-3001-200.
- be logged in to the Waveserver 5 using an account with at least admin access privileges.
- have backed up and saved the configuration file, if required. See [Procedure 118, “Backing up the configuration datastores”](#).
- be familiar with the manufacturer torque specifications for data center, 19-inch, 23-inch, or ETSI equipment frames, as required.
- have a replacement Waveserver 5 chassis that matches the PEC of the chassis you are replacing.
- have access to *Waveserver 5 Installation*, 323-3001-200.
- ensure that protective caps are available for unused ports on Waveserver 5.
- unpack and inspect the replacement chassis and place it on a flat, stable surface. If the replacement chassis is damaged, stop this procedure and contact your Ciena account representative.

### Overview

**Note:** This is an out-of-service procedure that contains references and links to other installation and replacement procedures; some of these references may contain steps that do not apply to this procedure or are duplicated in other procedures. Any steps in other procedures that are performed in-service, for example, querying, disabling, or enabling equipment, or steps pertaining to licensing, do not apply.

## Precautions



### **DANGER**

#### **Risk of laser radiation exposure**

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.



### **CAUTION**

#### **Risk of damage to the chassis, pluggable modules and transceivers**

Make sure you know how to handle electronic components correctly before you start the replacement procedure. Incorrect handling can cause damage to static-sensitive components.



### **CAUTION**

#### **Risk of network reliability**

Make sure that all connectors are cleaned before you make the connections (or re-connections) described in this procedure.

## Steps

- 1 Retrieve alarms and record the current state of the Waveserver 5 chassis. You can compare this information to the state of the new chassis after it is installed:
- 2 The following step is performed using CLI commands.  

```
show active-alarm <id> [name] [resource] [raise-time]
[raise-time] [cause] [condition-type] [direction]
[location] [severity] [service-impact] [additional-info]
[user-notes...]
```

The current alarms and alarm details will appear.
- 3 Wear an appropriate ESD personal grounding device to dissipate electrostatic charges.
- 4 Ensure that all power is OFF before disconnecting the power connections from the chassis.
- 5 Remove the Waveserver component such as CP, AP, SM, power and fan module from the chassis. Refer to [“Removing, replacing, or reseating equipment”](#) section for appropriate module replacement procedures in this document.
- 6 Remove the Waveserver 5 chassis that you want to replace from the rack.

- |  |                                                              |                                                                                                         |
|--|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|  | <b>If you want to replace a chassis that is installed in</b> | <b>Then go to</b>                                                                                       |
|  | a 19-in, 23-in, or ETSI rack                                 | <a href="#">Procedure 287, “Removing a Waveserver 5 chassis from a 19-in, 23-in, or ETSI rack”</a>      |
|  | a data center 19-inch or 23-inch wide, 24-36-inch deep rack  | <a href="#">Procedure 288, “Removing a Waveserver 5 chassis from a 19-in or 23-in data center rack”</a> |
- 7 Install the new chassis in the equipment frame. See *Waveserver 5 Installation*, 323-3001-200.
  - 8 Re-install the power and fan modules. See the “Installing the power and fan modules and connecting power cables to Waveserver 5” procedure in *Waveserver 5 Installation*, 323-3001-200.
  - 9 Test the power and fan modules. See the “Testing power to Waveserver 5” procedure in *Waveserver 5 Installation*, 323-3001-200.
  - 10 Remove the dust caps from the chassis ports.
  - 11 Re-install the Access Panel (AP) in the chassis. See the “Installing the Access Panel” procedure in *Waveserver 5 Installation*, 323-3001-200. Reconnect the AP cables.
  - 12 Re-install the Control Processor (CP) in the chassis. See the “Installing the Control Processor” procedure in *Waveserver 5 Installation*, 323-3001-200. Reconnect the CP cables.
  - 13 Re-install the SMs and/or filler modules, as required. See the “Installing or removing Service Modules (SMs)” and the “Installing or removing a filler module” procedures in *Waveserver 5 Installation*, 323-3001-200.
  - 14 Re-install the pluggable transceivers. See the “Installing or removing pluggable transceivers” procedure in *Waveserver 5 Installation*, 323-3001-200.
  - 15 If auto configuration mismatch detection is enabled, it detects whether the software load and/or the configuration file on the replacement CP matches the software load and/or configurations running on the modules in the chassis. For more information regarding the auto recovery, refer to [“Auto configuration recovery” on page 1106](#) of this document.  
  
**Note:** Database on CP contains the chassis ID. If it does not match, an empty database is used. Also the backup on AP is in a directory named with chassis ID, it will only restore if the backup is for the current chassis ID.
  - 16 Clean the fibers and reconnect to the pluggable transceivers. Use the fiber labels to guide you during reconnection.
  - 17 Route the fibers. See the “Routing cables from/to modules on the Waveserver 5” procedure in *Waveserver 5 Installation*, 323-3001-200.
  - 18 Repack and return the removed Waveserver OS chassis to Ciena.



---

## CHAPTER 31

# Restarting Waveserver 5

---

This section covers the following topics:

- [Overview](#)
  - [Impacts of restarting a service module](#)
  - [Impacts of restarting the Control Module \(CM\)](#)
  - [Impact of restarting the chassis](#)
- [Procedure list](#)

### Overview

Waveserver 5 supports warm and cold restarts on the individual Service Modules (SMs), Control Module (CM), and chassis.

Warm restarts should not affect traffic on the chassis, service modules, or the control module you are restarting.

Cold restarts are service affecting on the chassis or service module you are restarting. Under normal conditions, cold restarting the control module should not affect traffic.

**Note:** Performing a warm or cold restart on the control module and chassis clears the account lockout status. For additional information, see [User lockout](#).

### Impacts of restarting a service module

Warm restarting a service module has these consequences:

- Traffic is not affected.
- All unsaved and saved module provisioning recovers after the warm restart.
- Service modules can be warm restarted independently of one another.
- If you perform a warm restart on a service module that provides remote management access (via one of its interfaces) to a remote group of Waveserver 5 nodes, then management access to the remote group is lost



for up to three minutes. However, if at least one redundant interface is available to the remote group, then remote management access to the remote group is lost for up to 45 seconds.

- Performance monitoring statistics become invalid for all current 15-minute bins, current 24-hr bins, and untimed bins. The bins stop counting for the duration of the warm restart and once the module completes the restart the bins are marked as IDF (Invalid Data Flag).

Cold restarting a service module has the following consequences:

- Traffic is affected on the service module you are restarting.
- Traffic recovers after the service module cold restarts.
- All previously provisioned configurations are restored from a backed up configuration file after a cold restart.
- Service modules can be cold restarted independently of one another.
- If you perform a cold restart on a service module that provides remote management access (via one of its interfaces) to a remote group of Waveserver 5 nodes, then management access to that remote group is lost until the interface becomes operationally enabled. However, if at least one redundant interface is available to the remote group, then remote management access to the remote group is lost for up to 45 seconds.
- Performance monitoring statistics become invalid for all current 15-minute bins, current 24-hr bins, and untimed bins. The bins stop counting for the duration of the cold restart and once the module completes the restart the bins are marked as “IDF” (Invalid Data Flag).

### **Impacts of restarting the Control Module (CM)**

Warm restarting the control module has the following consequences:

- Traffic is not affected.
- All unsaved and saved module provisioning recovers after the warm restart.
- All current bins (15-minute, 24-hour, and untimed) are cached on the CM. Each time you restart the CM, the current bins are recreated on the CM, the counts in the bin are reset to zero and marked as “IDF” (Invalid Data Flag). The IDF flags on the timed bins (15-minute and 24-hour) clear when the bins roll over. The IDF flags on the untimed bin clear when you clear the untimed bin.

Cold restarting the control module has the following consequences:

- Under ideal conditions, traffic is not affected.
- The saved configuration is recovered and it may be different from the software and hardware configuration prior to the cold restart.

- All current bins (15-minute, 24-hour, and untimed) are cached on the CM. Each time you restart the CM, the current bins are recreated on the CM, the counts in the bin are reset to zero and marked as “IDF” (Invalid Data Flag). The IDF flags on the timed bins (15-minute and 24-hour) clear when the bins roll over. The IDF flags on the untimed bin clear when you clear the untimed bin.

### Impact of restarting the chassis

Warm restarting the chassis has these consequences:

- A warm restart of the chassis, including the control module and service modules.
- When a warm restart is performed on a Waveserver 5 that is connected in a linear ILAN configuration directly to the DCN, communications with that Waveserver 5 are lost during the restart.
- When a warm restart is performed on a Waveserver 5 that is in a linear ILAN configuration and not connected directly to the DCN, communications with that Waveserver 5 are lost during the restart.
- When a warm restart is performed on a Waveserver 5 in an ILAN ring configuration and the Waveserver 5 directly connected to the DCN, communications with that Waveserver 5 are lost during the restart.
- When a warm restart is performed on a Waveserver 5 in an ILAN ring configuration and the Waveserver 5 is not connected directly to a DCN, communication with that Waveserver 5 are lost during the restart.
- When a warm restart is performed on a Waveserver 5, which is providing management communication access to a remote Waveserver 5, the remote management access through the associated interfaces are be lost during the restart.
- Software and hardware configuration prior to the warm restart is maintained and recovered while the software reinitializes.
- Local (serial) and remote (SSH over DCN) connections to Waveserver 5 are automatically closed when warm restarts are performed.
- SPLI communications are automatically restored after warm restart.
- No log collection
- The collection of performance monitoring statistics is affected. A warm restart clears all current bins (15-min, 24-hr, and untimed), but historical bins persists.

Cold restarting the chassis has the following consequences:

- A cold restart of the chassis, including the control module and service modules.
- Traffic impact on all the service modules.

- When a cold restart is performed on a Waveserver 5 that is connected in a linear ILAN configuration directly to the DCN, communications with that Waveserver 5 will be lost during the restart. All interconnected Waveserver 5 chassis that connect to it using ILAN communications are lost during the restart.
- When a cold restart is performed on a Waveserver 5 that is in a linear ILAN configuration and not connected directly to the DCN, communications with that Waveserver 5 are lost during the restart. Waveserver 5 chassis interconnected to it using ILAN communications may be lost during the restart
- When a cold restart is performed on a Waveserver 5 in an ILAN ring configuration and that Waveserver 5 is directly connected to the DCN, all Waveserver 5 nodes in that ILAN ring are lost during the restart.
- When a cold restart is performed on a Waveserver 5 in an ILAN ring configuration and the Waveserver 5 is not directly connected to the DCN, communications with that Waveserver 5 are lost during the restart.
- The saved configuration is recovered and it may be different from the software and hardware configuration prior to the cold restart.
- Local (serial) and remote (SSH over DCN) connections to Waveserver 5 are automatically closed when cold restarts are performed.
- SPLI communications are automatically restored after a cold restart.
- The collection of performance monitoring statistics is affected. A cold restart clears all current bins (15-min, 24-hr, and untimed), but historical bins persists.

## Procedure list

This section provides the following procedures:

- [Procedure 290, “Performing a warm restart on a service module”](#)
- [Procedure 291, “Performing a cold restart on a service module”](#)
- [Procedure 292, “Performing a warm restart on the Control Module”](#)
- [Procedure 293, “Performing a cold restart on the Control Module”](#)
- [Procedure 294, “Performing a warm restart on the Waveserver 5 chassis”](#)
- [Procedure 295, “Performing a cold restart on the Waveserver 5 chassis”](#)

---

## Procedure 290 Performing a warm restart on a service module

---

Use this procedure to perform a warm restart on a service module. Warm restarts should not be traffic impacting.

### Requirements

- You are aware of the impact of a module warm restart and have taken appropriate precautions.
- You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

For the impact of this action, see [Impacts of restarting a service module](#).

### Steps

- 1 Perform a module warm restart of Waveserver 5:  

```
module restart slot <NUMBER 1..8>
```

A warning displays, and prompts you to confirm that you want to perform the warm restart.

WARNING:

You cannot abort the restart operation once it has started.

Enter [Y/y] to continue or [N/n] to abort
- 2 Type **Y** to perform the warm restart.  
The module restarts.  
**Note:** You can perform a module warm restart immediately, without the system prompting you for confirmation by typing this command:  

```
module restart slot <NUMBER 1..8> now
```
- 3 To confirm the restart status of the module, or to view the last restart time and reason, type the command:  

```
module show slot <NUMBER 1..8>
```

## Procedure 291 Performing a cold restart on a service module

---

Use this procedure to perform a cold restart on a service module. Cold restarts are traffic impacting.

### Requirements

- You are aware of the impact of a module cold restart and have taken appropriate precautions.
- You must be logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

For the impact of this action, see [Impacts of restarting a service module](#).

### Steps

- 1 If a backup configuration file is not available, create and save a backup configuration file. See [Configuration management](#).
- 2 Perform a module cold restart by typing:  

```
module restart-cold slot <NUMBER 1..8>
```

A warning displays, and prompts you to confirm that you want to perform the cold restart.

WARNING:

You cannot abort the restart operation once it has started.

Enter [Y/y] to continue or [N/n] to abort
- 3 Type **Y** to perform the module cold restart.  
The module restarts.  
**Note:** You can perform a module cold restart immediately, without the system prompting you for confirmation by typing this command:  

```
module restart-cold slot <NUMBER 1..8> now
```
- 4 To confirm the restart status of the module, or to view the last restart time and reason, type the command:  

```
module show slot <NUMBER 1..8>
```

---

## Procedure 292 Performing a warm restart on the Control Module

---

Use this procedure to perform a warm restart on the Control Module (CM).

### Requirements

- You are aware of the impact of a warm restart of the Control Module and have taken appropriate precautions.
- You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

For the impact of this action, see [Impacts of restarting the Control Module \(CM\)](#).

### Steps

- 1 Perform a warm restart on the Control Module.  

```
chassis control-module restart
```

A warning displays, and prompts you to confirm that you want to perform the cold restart.

WARNING:

You cannot abort the restart operation once it has started.

Enter [Y/y] to continue or [N/n] to abort
- 2 Type **Y** to perform the module cold restart.  
The module restarts.  
**Note:** You can perform a module warm restart immediately, without the system prompting you for confirmation by typing this command:  

```
chassis control-module restart now
```
- 3 Confirm the restart status of the Control Module, or view the last restart time and reason:  

```
chassis control-module show
```

## Procedure 293 Performing a cold restart on the Control Module

---

Use this procedure to perform a cold restart on the Control Module (CM).

### Requirements

- You are aware of the impact of a cold restart of the chassis and have taken appropriate precautions.
- You are logged in to Waveserver 5 using an account with at least admin access privileges.

### Overview

For impacts of this action, see [Impacts of restarting the Control Module \(CM\)](#).

### Steps

- 1 Create and save a configuration file, if required. See [Configuration management](#).
- 2 Perform a cold restart on the Control Module.  

```
chassis control-module restart-cold
```

A warning displays, and prompts you to confirm that you want to perform the cold restart.

WARNING:

You cannot abort the restart operation once it has started.

Enter [Y/y] to continue or [N/n] to abort
- 3 Type Y to perform the module cold restart.  
The Control Module restarts.  
**Note:** You can perform a module warm restart immediately, without the system prompting you for confirmation by typing this command:  

```
chassis control-module restart-cold now
```
- 4 Confirm the restart status of the Control Module, or view the last restart time and reason:  

```
chassis control-module show
```

# Procedure 294 Performing a warm restart on the Waveserver 5 chassis

Use this procedure to perform a warm restart on a Waveserver 5 chassis. A warm restart is not traffic impacting.

## Requirements

- You are aware of the impact of a warm restart and have taken appropriate precautions.
- You are logged in to Waveserver 5 using an account with at least admin access privileges.

## Overview

For impacts of this action, see [Impact of restarting the chassis](#).

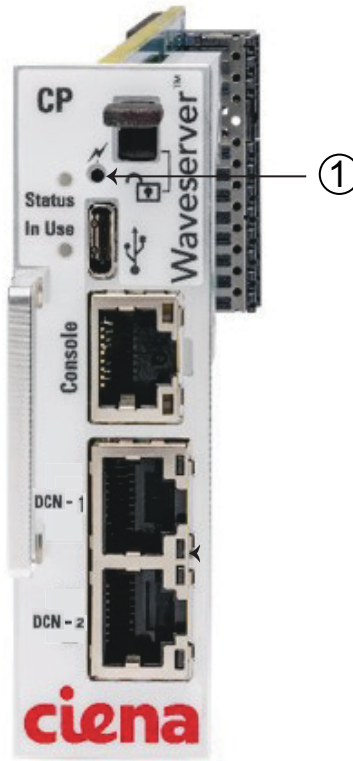
- Note 1:** After a warm restart, the performance monitoring (PM) profile statistics for all current bins (15-min, 24-hr, and untimed) are reset to zero. PM profile statistics for historical bins persist.
- Note 2:** You can perform a restart immediately, without the system prompting you for confirmation, by typing the `chassis restart now` command.

## Steps

- |   |                                                                                                                                                                                                                                                                                                                                                              |                                     |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1 | <div style="display: flex; justify-content: space-between;"><div>If you want to perform a warm restart using the software command</div><div>Then go to</div></div> <div>the reset button</div>                                                                                                                                                               | <div>step 2</div> <div>step 4</div> |
| 2 | <div>To perform a software warm restart, use the command:</div> <div><code>chassis restart</code></div> <div>A warning displays, and prompts you to confirm that you want to perform the warm restart.</div> <div>WARNING:</div> <div>You cannot abort the restart operation once it has started.</div> <div>Enter [Y/y] to continue or [N/n] to abort</div> |                                     |
| 3 | <div>Type Y to perform the warm restart.</div> <div>Waveserver 5 restarts.</div> <div>You have completed this procedure.</div>                                                                                                                                                                                                                               |                                     |
| 4 | <div>To perform a hardware warm restart operation, use a 1.3 mm (1/16 inch) hex key or other small, blunt object, then press and immediately release the pinhole reset button. This action triggers the warm restart.</div>                                                                                                                                  |                                     |



**Figure 198** Control Processor pinhole reset button



**Legend**

1 Pinhole reset button

# Procedure 295 Performing a cold restart on the Waveserver 5 chassis

Use this procedure to perform a cold restart on a Waveserver 5 chassis. Cold restarts are traffic impacting.

### Requirements

- You are aware of the impact of a cold restart and have taken appropriate precautions.
- For a software cold restart, you must be logged in to Waveserver 5 using an account with at least admin access privileges.
- For a Control Processor cold restart, you must have a 1.3 mm (1/16 inch) hex key or another small, blunt object to access the pinhole reset button.

### Overview

**Note:** After a cold restart, SPLI communications are automatically restored.

### Steps

- |   |                                                                                                                                              |                                          |
|---|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1 | <b>If</b> you want to perform a cold restart and the software is running                                                                     | <b>Then</b> go to <a href="#">step 2</a> |
|   | the server is unreachable                                                                                                                    | <a href="#">step 4</a>                   |
| 2 | If a backup configuration file is not available, create and save a backup configuration file. See <a href="#">Configuration management</a> . |                                          |



**WARNING**

A cold restart of Waveserver 5 is service-affecting. Before you perform the next step, ensure that configuration settings are saved and a backup configuration file is available. For more information, see [Configuration management](#).

- 3 Perform a software cold restart of Waveserver 5 by typing:
- ```
chassis restart-cold
```
- A warning displays, and prompts you to confirm that you want to perform the cold restart.
- WARNING:
- You cannot abort the restart operation once it has started.
- Enter [Y/y] to continue or [N/n] to abort
- Type Y to perform the cold restart.

Waveserver 5 restarts.

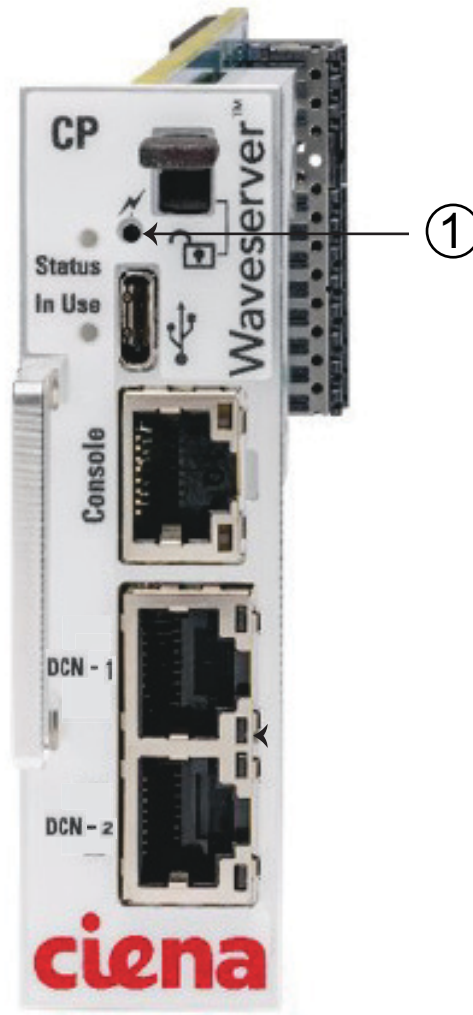
Note: You can perform a cold restart immediately, without the system prompting you for confirmation by typing this command:

```
chassis restart-cold now
```

You have completed this procedure.

- 4 To perform a hardware cold restart, use a 1.3 mm (1/16 inch) hex key or other small, blunt object to press and hold the pinhole reset button for 7.5 seconds.

Figure 199 Control Processor pinhole reset button



Legend

1 Pinhole reset button

Waveserver 5 Control Processor begins a cold restart.

CHAPTER 32

CLI command reference

Overview

This section is a reference to Waveserver 5 CLI commands.

CLI fundamentals

The Command Line Interface (CLI) is a text-based command interface used to configure Waveserver 5 and view system information.

CLI commands are available based on user access levels and installed licenses. Before entering a command, ensure that you have the proper access level. For more information, see [User account types](#).

Command completion

The CLI supports partial command recognition. This means that you can enter the first few characters that uniquely identify a command for the system to recognize it. For example, the command `port show` can be successfully recognized when entered as `po sh`.

If you do not enter enough characters to uniquely identify the command, a shell parser error will occur. For example, entering `p s` for the `port show` command, returns the message `SHELL PARSER FAILURE: 'p' - ambiguous input`.

In addition, pressing the Space bar and then the Tab key at the end of the command string (or partial command string) displays available actions, objects, or sub-objects. Also, after specifying the object, action, or instance, and even attributes, you can press the Space bar and then the Tab key to display additional parameters.

Command Confirmation

The CLI displays a command confirmation prompt before executing any command that impacts traffic or changes the running configuration. You need to confirm or decline the command execution by pressing the Y or N keys,

respectively. For example, while changing the frequency on a ptp, a confirmation message appears asking you to Enter Y/y or N/n to execute or abort the command.

```
ptp set ptp <slot>/<port> freq <Frequency in GHz (N.M)>
WARNING:
  This configuration change may be service impacting.
Enter [Y/y] to continue or [N/n] to abort Proceeding
```

Getting help

For more information about available commands, enter ? at a command prompt. Pressing the Space bar and entering ? or help at the end of a command string displays a list of valid parameters for the command.

Command response

Typically, error messages are generated only when a command is unsuccessful. A show command (for example, port show) that successfully retrieves data generates relevant output that is followed by the CLI prompt. Commands that do not generate output when they are successful, such as set and unset commands, return only the CLI prompt.

Command syntax

The following table summarizes the command syntax used in this document.

| Symbol | Description |
|--------|--|
| < > | Encloses a variable or literal value that must be specified. For example, alarm-instance-id <1...2147483647> |
| { } | Encloses either a single required argument or a list of optional arguments, at least one of which must be specified. For example:
{alarm-instance-id <1...2147483647>}
{[audible-severity <critical major minor>]
[warning-severity]} |
| [] | Encloses an optional value or a list of optional arguments. One or more values or arguments can be specified. For example:
port show [ethernet-config] [summary] [status] [port] |
| | Separates mutually exclusive options. For example:
search-method <cached priority> |

Searching and displaying CLI commands

Use the following commands to perform customized searches of the CLI:

- cli search keyword <String> [full] [object <String>] [persistent-only]

- `cli show [full] [object <String>] [persistent-only]`

Note: The results of each search depend on the access privileges of the account used to log in to the CLI.

CLI error response

Waveserver 5 generates CLI response messages in three severity levels: ERROR; WARNING; and INFO. The severity level precedes the message body separated by a colon, for example, `ERROR: Key already exists.`

The following list describes the possible CLI response message types:

- **ERROR** — indicates a failure in the completion of an operation.
- **WARNING** — indicates the operation succeeded and there is some additional system information related to the CLI command operation to take note, which may require action at some point in time. The details are in the text description.
- **INFO** — indicates the operation succeeded and there is some additional system information related to the CLI command operation to note. The details are in the text description.

Waveserver 5 also generates the following error messages during an interactive CLI session if you enter an incorrect CLI syntax or press the TAB key for command line completion for an incorrect partial command:

- **SHELL PARSER FAILURE** — an incorrect command was entered.
- **SHELL AUTOCOMPLETION FAILURE** — command auto-completion failed because an incorrect partial command was entered.

Waveserver 5 entities in CLI commands

The following three tables list the variables used to represent Waveserver 5 entities in CLI commands.

Table 385 Client entities in CLI commands

| Client Entity | Variable | Range of Values |
|---------------------------------|--|---|
| 100GE port, PTP, or transceiver | <code><slot>/<port></code> | <code><1..8>/<3..18></code> |
| OTL4.4 port | <code><slot>/<port></code> | <code><1..8>/<201 202></code>
Note: Logical ports 201 and 202 are present in x00G transmission modes. |

Table 386 Line entities in CLI commands

| Line Entity | Variable | Range of Values |
|-------------|--|---|
| Port | <code><slot>/<port></code> | <code><1..8>/<1 2 101></code> |

| Line Entity | Variable | Range of Values |
|-------------|-----------------------------|---|
| PTP | <slot>/<port> | <1..8>/<1 2> |
| Channel | <slot>/
<port>[.channel] | <1..8>/<1 2 101> <1.[1..4] 2.[1..4] 101.[1..8]> |

Entering port, PTP, channel, and transceiver entities in CLI commands

Execute CLI commands against individual entities only, not a range of entities. When using a command that includes a port, PTP, channel, or transceiver (xcvr), specify only one entity in the command.

Table 387 Entering port, PTP, XCVR, or PM-instance entities

| CLI reference | Single Slot / Port Entity | Range of Entities |
|---------------|---|--|
| portList | <slot>/<port>
Example: 1/1 | <slot>/<port>,...
Example: 1/1, 2/1 |
| ptpList | <slot>/<port>
Example: 2/1 | <slot>/<port>
Example: 1/3, 2/3, 3/3 |
| channelList | <slot>/<port>[.channel]
Example: 2/1.2 | <slot>/<port>[.channel],...
Example: 3/1.1, 3/1.2, 3/1.3, 3/1.4 |
| xcvrList | <slot>/<port>
Example: 1/2 | <slot>/<port>,...
Example: 1/4, 2/5, 3/7 |

IPv4, IPv6, and MAC addresses

The CLI accepts either IPv4 or IPv6 address format. For commands where only one or the other is accepted, the CLI returns an error if the incorrect IP address format is entered.

For MAC addresses, the CLI accepts the format `xx:xx:xx:xx:xx:xx`.

CLI commands

The following is a list of CLI commands menus:

- [Alarm commands](#)
- [Application commands](#)
- [Certificates commands](#)
- [Channel commands](#)
- [Chassis commands](#)
- [Command log commands](#)
- [Configuration commands](#)

- [Connection command](#)
- [DHCP client commands](#)
- [DNS client commands](#)
- [File commands](#)
- [Interface commands](#)
- [License commands](#)
- [Link Layer Discovery Protocol \(LLDP\) commands](#)
- [Logging commands](#)
- [Module commands](#)
- [NDP commands](#)
- [NTP \(Network Time Protocol\) MD5 authentication commands](#)
- [NTP client commands](#)
- [NTPv4 \(Network Time Protocol Version 4\) Autokey authentication commands](#)
- [Performance monitor \(PM\) commands](#)
- [Performance Monitoring \(PM\) TCA commands](#)
- [Physical Termination Point \(PTP\) commands](#)
- [Port commands](#)
- [RADIUS commands](#)
- [RADSec commands](#)
- [Service Photonic Layer Interoperations \(SPLI\) commands](#)
- [SNMP commands](#)
- [Software commands](#)
- [SSH commands](#)
- [SSL commands](#)
- [Syslog commands](#)
- [System commands](#)
- [TACACS+ commands](#)
- [Telemetry commands](#)
- [Transceiver \(xcvr\) commands](#)
- [User commands](#)

Alarm commands

alarm acknowledge

| | |
|--------------|--|
| Description | Acknowledges all, one, or multiple active alarms. |
| Access Level | admin |
| Syntax | alarm acknowledge [alarm-instance-id <Number: 1...2147483647>] |

alarm clear history

| | |
|--------------|---------------------------|
| Description | Clears the alarm history. |
| Access Level | admin |
| Syntax | alarm clear history |

alarm show

| | |
|------------------|---|
| Description | Displays summary information for active alarms. |
| Access Level | limited |
| Syntax | alarm show [statistics] [status] |
| Related Commands | <ul style="list-style-type: none"> • alarm show active-alarms • alarm show active-alarms last • alarm show active-alarms recent • alarm show active-alarms today • alarm show active-alarms yesterday • alarm show alarm-table-id • alarm show defined-alarms • alarm show history • alarm show status |

alarm show active-alarms

| | |
|--------------|---|
| Description | Displays active alarms. |
| Access Level | limited |
| Syntax | alarm show active-alarms [alarm-table-id <Number: 1...9999>] [from-date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <Time: hh:mm:ss or hh:mm>] [slot <1 3 5 7>] [to-date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <Time: hh:mm:ss or hh:mm>] [severity <critical major minor warning>] |

alarm show active-alarms last

| | |
|--------------|---|
| Description | Displays active alarms raised over a preceding span of time; for example, the last 4 hours (h), 10 minutes (m), and 3 seconds (s). |
| Access Level | limited |
| Syntax | alarm show active-alarms last <Duration: number/time format [yMwdhms]> [alarm-table-id <Number: 1...9999>] [severity <critical major minor warning>] [to-time <Time: hh:mm:ss or hh:mm>] [slot <1 3 5 7>] |

alarm show active-alarms recent

| | |
|--------------|--|
| Description | Displays active alarms raised over a recent span of time. |
| Access Level | limited |
| Syntax | alarm show active-alarms recent [alarm-table-id <Number: 1...9999>] [to-time <Time: hh:mm:ss or hh:mm>] [severity <critical major minor warning>] [slot <1 3 5 7>] |

alarm show active-alarms today

| | |
|--------------|---|
| Description | Displays active alarms raised today. |
| Access Level | limited |
| Syntax | alarm show active-alarms today [alarm-table-id <Number: 1...9999>] [from-time <Time: hh:mm:ss or hh:mm>] [to-time <Time: hh:mm:ss or hh:mm>] [severity <critical major minor warning>] [slot <1 3 5 7>] |

alarm show active-alarms yesterday

| | |
|--------------|---|
| Description | Displays active alarms raised yesterday. |
| Access Level | limited |
| Syntax | alarm show active-alarms yesterday [alarm-table-id <Number: 1...9999>] [from-time <Time: hh:mm:ss or hh:mm>] [to-time <Time: hh:mm:ss or hh:mm>] [severity <critical major minor warning>] [slot <1 3 5 7>] |

alarm show alarm-table-id

| | |
|--------------|--|
| Description | Displays summary information for one or more alarm tables. |
| Access Level | limited |
| Syntax | alarm show alarm-table-id <Number: 1...9999> |

alarm show defined-alarms

| | |
|-------------|---------------------------------------|
| Description | Displays a summary of defined alarms. |
|-------------|---------------------------------------|

| | |
|--------------|---|
| Access Level | limited |
| Syntax | alarm show defined-alarms [severity <critical major minor warning>] |

alarm show history

| | |
|------------------|---|
| Description | Displays the alarm history. |
| Access Level | limited |
| Syntax | alarm show history [alarm-table-id <Number 1...9999>] [from-date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <Time: hh:mm:ss or hh:mm>] [to-date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <Time: hh:mm:ss or hh:mm>] [severity <critical major minor warning info>] [slot <1 3 5 7>] [tail <Number: 1...4294967295>] |
| Related Commands | <ul style="list-style-type: none"> • alarm show history last • alarm show history recent • alarm show history since-activated • alarm show history tail • alarm show history today • alarm show history yesterday |

alarm show history last

| | |
|--------------|--|
| Description | Displays the alarm history for a preceding span of time, for example, the last 4 hours (h), 10 minutes (m), and 3 seconds (s). |
| Access Level | limited |
| Syntax | alarm show history last <Duration: number/time format [yMwdhms]> [alarm-table-id <Number: 1...9999>] [severity <critical major minor warning>] [tail <Number: 1...4294967295>] |

alarm show history recent

| | |
|--------------|---|
| Description | Displays the alarm history for a recent span of time. |
| Access Level | limited |
| Syntax | alarm show history recent [alarm-table-id <Number: 1...9999>] [severity <critical major minor warning>] [tail <Number: 1...4294967295>] [to-time <Time: hh:mm:ss or hh:mm>] [alarm-table-id <Number: 1...9999>] [slot <Fan-1 Fan-2 PSU-1 PSU-2 Slot-1 Slot-2 Slot-3>] |

alarm show history since-activated

| | |
|--------------|---|
| Description | Displays the alarm history since system start-up. |
| Access Level | limited |

| | |
|--------|---|
| Syntax | alarm show history since-activated [alarm-table-id <Number 1...9999>] [severity <critical major minor warning>] [from-time <Time: hh:mm:ss or hh:mm>] [to-date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <Time: hh:mm:ss or hh:mm>] [tail <Number: 1...4294967295>] [slot <Fan-1 Fan-2 PSU-1 PSU-2 Slot-1 Slot-2 Slot-3>] |
|--------|---|

alarm show history tail

| | |
|--------------|---|
| Description | Displays the specified number of the latest alarms in the alarm history, for example, the last five alarms or the last 10 alarms. |
| Access Level | limited |
| Syntax | alarm show history tail [alarm-table-id <Number: 1...9999>] [severity <critical major minor warning>] |

alarm show history today

| | |
|--------------|--|
| Description | Displays the alarm history for today. |
| Access Level | limited |
| Syntax | alarm show history today [alarm-table-id <Number: 1...9999>] [from-time <Time: hh:mm:ss or hh:mm>] [to-time <Time: hh:mm:ss or hh:mm>] [tail <Number: 1...4294967295>] [severity <critical major minor warning>] [slot <Fan-1 Fan-2 PSU-1 PSU-2 Slot-1 Slot-2 Slot-3>] |

alarm show history yesterday

| | |
|--------------|--|
| Description | Displays the alarm history for yesterday. |
| Access Level | limited |
| Syntax | alarm show history yesterday [alarm-table-id <Number: 1...9999>] [from-time <Time: hh:mm:ss or hh:mm>] [to-time <Time: hh:mm:ss or hh:mm>] [tail <Number: 1...4294967295>] [severity <critical major minor warning>] [slot <Fan-1 Fan-2 PSU-1 PSU-2 Slot-1 Slot-2 Slot-3>] |

alarm show status

| | |
|--------------|--|
| Description | Displays the alarm status and indicates if a Warning, Minor, Major, or Critical alarm is raised. |
| Access Level | limited |
| Syntax | alarm show status |

Application commands

application dispersion-scan show

| | |
|--------------|---|
| Description | Displays the dispersion scan results for a line port. The dispersion scan displays the pre-compensation value and Associated Q value for the specified line port. |
| Access Level | admin |
| Syntax | application dispersion-scan show ptp <slot>/<ptp> |

application dispersion-scan start

| | |
|--------------|---|
| Description | Starts a user-initiated Q-based pre-dispersion compensation scan on a line port. |
| Access Level | admin |
| Syntax | <p>application dispersion-scan start ptp <slot>/<ptp> [from <ps/nm (Range: -44500..1000000)>] [to <ps/nm (Range: -44500..1000000)>] [step <ps/nm (Range: 100 - 1000 in the increments of 100)>] [dwell <SECONDS (Range: 1 - 100 in the increments of 1)>]</p> <p>Note 1: Default values for the submarine PTP modem class are: From = -5000 ps/nm, To = 5000 ps/nm, Step = 200 ps/nm, Dwell time = 5 seconds</p> <p>Note 2: Default values for the terrestrial PTP modem class are: From = 0 ps/nm, To = 40000 ps/nm, Step = 1000 ps/nm, Dwell time = 5 seconds</p> |

application dispersion-scan stop

| | |
|--------------|---|
| Description | Stops the dispersion scan before it is complete. |
| Access Level | admin |
| Syntax | application dispersion-scan stop ptp <slot>/<ptp> |

application link-data-collector show

| | |
|--------------|---|
| Description | Displays status of the “application link-data-collector start” command, which is executed on the line port. |
| Access Level | admin |
| Syntax | application link-data-collector show ptp <slot>/<ptp> |

application link-data-collector start

| | |
|--------------|---|
| Description | Starts the optical link data collection on a line port. |
| Access Level | admin |
| Syntax | application link-data-collector start ptp <slot>/<port> interval <MILLISECONDS (Range: 10 - 60000)> numrecords <NumRecords count (Range: 100 - 50000)>. |

| | |
|----------------|--|
| Default values | <ul style="list-style-type: none"> Interval: 1000 ms NumRecords: 100 |
|----------------|--|

application link-data-collector stop

| | |
|--------------|--|
| Description | Stops the optical link data collection on a line port. |
| Access Level | admin |
| Syntax | application link-data-collector stop ptp <slot>/<ptp> |

application link-data-collector upload

| | |
|--------------|---|
| Description | Uploads the optical link statistics from the link data collector to an scp server. |
| Access Level | admin |
| Syntax | <p>application link-data-collector upload ptp <ptpList> {default-ftp-server default-scp-server default-server default-sftp-server default-tftp-server scp-server <IP address (w/optional port) or host name> file-name <String> login-id <String[1..32]> password <Password String[1..128]> url <String> login-id <String[1..32]> password <Password String[1..128]> secret <String[1..32]>}</p> <p>Note: When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (https://tools.ietf.org/html/rfc3986).</p> <p>For example, application link-data-collector upload ptp 1/1 url scp://user@10.20.30.40:/tmp/linkperformancedata.csv password *****</p> |

application its <disable | enable> port

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) of the integrated test set on one or more ports. |
| Access Level | admin |
| Syntax | <p>application its disable port <portList></p> <p>application its enable port <portList></p> |

application its set port

| | |
|---|---|
| Description | Sets the parameters of the integrated test set on the specified ports. |
| Access Level | admin |
| Syntax | application its set port <portList> [ber-threshold <NUMBER: 0 or 5 to 16>] [duration <number/time format N[yMwdhms]>] |
| Note: For [duration], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. | |

application its show

| | |
|--------------|---|
| Description | Displays the status of the integrated test set on provisioned ports in the chassis. |
| Access Level | limited |
| Syntax | application its show port <portList> [status] [log-summary] [report] |

application its <start | stop>

| | |
|--------------|---|
| Description | Starts or stops the integrated test set on the specified ports. |
| Access Level | limited |
| Syntax | application its start port <portList>
application its stop port <portList> |

application its unset port

| | |
|--------------|---|
| Description | Removes the parameters of the integrated test set on the specified ports. |
| Access Level | admin |
| Syntax | application its unset port <portList> [ber-threshold] [duration] |

application its upload-detailed-logs

| | |
|--------------|---|
| Description | Uploads to the specified target the detailed logs of the integrated test set for the specified ports. |
| Access Level | limited |
| Syntax | <p>application its upload-detailed-logs port <portList> {default-ftp-server default-scp-server default-server default-sftp-server default-tftp-server scp-server remote-path <String[1..127]> suffix <String[1..127]> login-id <String[1..32]> password <Password String[1..128]> url <String> suffix <String[1..127]> login-id <String[1..32]> password <Password String[1..128]> url <String> suffix <String[1..127]> secret <String[1..32]>}</p> <p>Note: When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (https://tools.ietf.org/html/rfc3986).</p> <p>For example, application its upload-detailed-logs port 2/3 url scp://user@[2222:111:dddd:5444:aced::293]:/tmp/itslogs.zip suffix ws53005 password *****</p> |

Certificates commands

certificates authorities install

| | |
|--------------|---|
| Description | Installs a CA certificate. You must assign a name to the CA certificate when you install it. |
| Access Level | super |
| Syntax | certificates authorities install cert-name <user_assigned_certificate_name> {default-ftp-server default-sftp-server default-tftp-server default-server default-scp-server sftp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] tftp-server <IP address or host name> scp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] ftp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] url <String [1..128]>} filename <String [1..127]> |

certificates authorities rename

| | |
|--------------|---|
| Description | Renames an installed CA certificate. |
| Access Level | super |
| Syntax | certificates authorities rename current-cert-name <current_certificate_name> new-cert-name <new_certificate_name> |

certificates authorities show

| | |
|--------------|---|
| Description | Displays summary information for CA certificates. |
| Access Level | limited |
| Syntax | certificates authorities show cert-name <user_assigned_certificate_name> ca-cert-hash <CA certificate hash[1..8]> |

certificates authorities uninstall

| | |
|--------------|--|
| Description | Uninstalls a CA certificate. |
| Access Level | super |
| Syntax | certificates authorities uninstall cert-name <user_assigned_certificate_name> ca-cert-hash <CA certificate hash[1..8]> |

certificates details show

| | |
|--------------|--|
| Description | Displays summary detail for CA certificates |
| Access Level | limited |
| Syntax | certificates details show cert-name <user_assigned_certificate_name> [hash <md5 sha256>] |

certificates entity csr

| | |
|--------------|--|
| Description | Generates a new CSR based on a user-provided .cnf file. |
| Access Level | super |
| Syntax | certificates entity csr generate cert-name <String: [1..32]> key-type <rsa2048 rsa3072 rsa4096 secp256r1 secp384r1 secp521r1 use-existing> filename <String [1..127]> {default-ftp-server default-sftp-server default-tftp-server default-server default-scp-server sftp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] tftp-server <IP address or host name> scp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] ftp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] url <String [1..128]>} |

certificates entity install

| | |
|--------------|---|
| Description | Installs the X.509 certificate and private key. |
| Access Level | super |
| Syntax | certificates entity install cert-name <String: [1..32]> cert-passphrase <cert-passphrase <Password String[1..128]> filename <String [1..127]> {default-ftp-server default-sftp-server default-tftp-server default-server default-scp-server sftp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] tftp-server <IP address or host name> scp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] ftp-server <IP address or host name> login-id <String [1..32]> password <String [1..128] secret <String [1..128] url <String [1..128]>} [certificate-only] |

certificates entity show

| | |
|--------------|--|
| Description | Displays summary information for all device certificates or a specific device certificate. |
| Access Level | limited |
| Syntax | certificates entity show [cert-name <Certificate name>] |

certificates entity uninstall

| | |
|--------------|---|
| Description | Uninstalls the X.509 certificate and private key. |
| Access Level | super |
| Syntax | certificates entity uninstall cert-name <String: [1..32]> |

certificates set support-RSA-2048 <disabled | enabled>

| | |
|----------------|---|
| Description | Sets the administrative state (disabled or enabled) for upgrade compatibility for certificates. |
| Access Level | super |
| Syntax | certificates set support-RSA-2048 disabled
certificates set support-RSA-2048 enabled |
| Default Values | Disabled |

certificates show

| | |
|--------------|--|
| Description | Displays the certificates upgrade compatibility state. |
| Access Level | limited |
| Syntax | certificates show |

Channel commands**channel <disable | enable>**

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of a channel.
Note: Disabling the channel using the <code>channel disable</code> command does not impact traffic. This command is used only for generating the necessary alarm. |
| Access Level | admin |
| Syntax | channel disable channel <channelList>
channel enable channel <channelList> |

channel set channel

| | |
|--------------|--|
| Description | Sets the label or ODU signal degrade threshold of a channel. |
| Access Level | admin |
| Syntax | channel set channel <channelList> [label <String[1..31]>] [odu-sd-threshold <BER String: XE-YY>] |

channel show

| | |
|--------------|--|
| Description | Displays summary information for the specified channels. |
| Access Level | limited |
| Syntax | channel show channel <channelList> |

channel trace set channel <tx-sapi | tx-dapi | tx-oper>

| | |
|--------------|---|
| Description | Transmits the SAPI, DAPI, or OPER string to the far-end chassis. |
| Access Level | admin |
| Syntax | channel trace set channel <slot>/<port>.<channel> {[tx-sapi <String: 0..15 char>]
[tx-dapi <String: 0..15>] [tx-oper <String: 0..32>]} |

channel trace set channel <exp-sapi | exp-dapi | exp-oper>

| | |
|--------------|--|
| Description | Sets the SAPI, DAPI, or OPER string expected to be returned from the far-end chassis. The expected string value is compared to the actual string value returned. |
| Access Level | admin |
| Syntax | channel trace set channel <slot>/<port>.<channel> {[exp-sapi <String: 0..15 char>]
[exp-dapi <String: 0..15>] [exp-oper <String: 0..32>]} |

channel trace set channel mismatch-mode

| | |
|--------------|--|
| Description | Specifies the string to be used to determine whether there is a mismatch. You cannot check for a mismatch on all three strings simultaneously. Default is operator-only. |
| Access Level | admin |
| Syntax | channel trace set channel <slot>/<port>.<channel> mismatch-mode <sapi dapi sapi-and-dapi operator-only> |

channel trace set channel mismatch-fail-mode

| | |
|--------------|---|
| Description | Specifies the action to take when a string mismatch is detected: do nothing, raise an alarm, or squelch traffic. Default is none. |
| Access Level | admin |
| Syntax | channel trace set channel <slot>/<port>.<channel> mismatch-fail-mode <none alarm-only squelch-traffic> |

channel trace set channel tx-oper-mode <automatic | manual>

| | |
|--------------|---|
| Description | Specifies how to manage the topology string. When tx-oper-mode is set to “automatic”, the OPER value transmitted is an automatically generated string. When the tx-oper-mode is set to “manual”, the OPER value transmitted is the value provisioned using the CLI command <code>channel trace set channel <tx-sapi tx-dapi tx-oper></code> . |
| Access Level | admin |
| Syntax | channel trace set channel <slot>/<port>.<channel> tx-oper-mode <automatic manual> |

| | |
|---------------|-----------|
| Default value | automatic |
|---------------|-----------|

channel trace set channel rx-to-exp

| | |
|--------------|--|
| Description | Establishes the currently received string (SAPI, DAPI, or OPER) as the expected string. The parameter allows you to select which receive/expect pair to use (SAPI, DAPI, or OPER). |
| Access Level | admin |
| Syntax | channel trace set channel <slot>/<port>.<channel> rx-to-exp <sapi dapi oper> |

channel trace show channel

| | |
|--------------|--|
| Description | Displays the TTI values for the channel. |
| Access Level | limited |
| Syntax | channel trace show channel <slot>/<port>.<channel> |

channel trace unset

| | |
|--------------|--|
| Description | Returns the TTI strings to their default values. |
| Access Level | admin |
| Syntax | channel trace unset channel <slot>/<port>.<channel> <tx-sapi exp-sapi tx-dapi exp-dapi tx-oper exp-oper mismatch-mode mismatch-fail-mode tx-oper-mode> |

channel unset channel

| | |
|--------------|--|
| Description | Clears the label and/or ODU signal degrade threshold of a channel. |
| Access Level | admin |
| Syntax | channel unset channel <slot>/<port>.<channel> [label] [odu-sd-threshold] |

Chassis commands

chassis access-panel <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) of the access panel. |
| Access Level | admin |
| Syntax | chassis access-panel disable
chassis access-panel enable |

chassis access-panel show

| | |
|-------------|--|
| Description | Displays summary information for the access panel. |
|-------------|--|

| | |
|--------------|---------------------------|
| Access Level | limited |
| Syntax | chassis access-panel show |

chassis access-panel show statistics

| | |
|--------------|--|
| Description | Displays temperature statistics of the access panel. |
| Access Level | limited |
| Syntax | chassis access-panel show statistics |

chassis control-module <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of the Control Module (CM). |
| Access Level | admin |
| Syntax | chassis control-module disable
chassis control-module enable |

chassis control-module restart

| | |
|--------------|--|
| Description | Warm restarts the Control Module (CM). |
| Access Level | admin |
| Syntax | chassis control-module restart [now] |

chassis control-module restart-cold

| | |
|--------------|---|
| Description | Cold restarts the Control Module (CM). |
| Access Level | admin |
| Syntax | chassis control-module restart-cold [now] |

chassis control-module show

| | |
|--------------|--|
| Description | Displays summary information for the Waveserver 5 Control Subsystem (CP) module. |
| Access Level | limited |
| Syntax | chassis control-module show |

chassis control-module show statistics

| | |
|--------------|--|
| Description | Displays temperature statistics and the utilization statistics for CPU, memory, and disk of the control processor. |
| Access Level | limited |

| | |
|--------|--|
| Syntax | chassis control-module show statistics |
|--------|--|

chassis device-id show

| | |
|--------------|---|
| Description | Displays summary information for the chassis, the Control Module (CM), the access panel, a fan module, a power module, or a slot. |
| Access Level | limited |
| Syntax | chassis device-id show [< chassis cm-1 ap-1 fan-1/1 fan-1/2 fan-2/1 fan-2/2 psu-1 psu-2 1 2 3 4 5 6 7 8>] |

chassis fan <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of a fan module. |
| Access Level | admin |
| Syntax | chassis fan disable slot <fan-1/1 fan-1/2 fan-2/1 fan-2/2>
chassis fan enable slot <fan-1/1 fan-1/2 fan-2/1 fan-2/2> |

chassis fan show

| | |
|--------------|---|
| Description | Displays summary information for the fan modules. |
| Access Level | limited |
| Syntax | chassis fan show |

chassis inventory show

| | |
|--------------|--|
| Description | Displays summary information for all equipment in the chassis. |
| Access Level | limited |
| Syntax | chassis inventory show |

chassis port <disable | enable> port

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of a communications interface. |
| Access Level | admin |
| Syntax | chassis port disable port <dcn-n ilan-1 ilan-2>
chassis port enable port <dcn-n ilan-1 ilan-2> |

chassis port set

| | |
|--------------|---|
| Description | Sets the port management mode for a DCN or ILAN port. |
| Access Level | admin |

| | |
|--------|---|
| Syntax | chassis port set port <dcn-n ilan-1 ilan-2> mode <management wayside-channel redundant-bgp> |
|--------|---|

chassis port show

| | |
|--------------|---|
| Description | Displays summary information for all communications interfaces or only the specified interface. |
| Access Level | limited |
| Syntax | chassis port show [port <dcn-n ilan-1 ilan-2>] |

chassis power <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of a power module. |
| Access Level | admin |
| Syntax | chassis power disable slot <psu-1 psu-2>
chassis power enable slot <psu-1 psu-2> |

chassis power show

| | |
|--------------|---|
| Description | Displays summary information for the power modules. |
| Access Level | limited |
| Syntax | chassis power show |

chassis restart

| | |
|--------------|---|
| Description | Performs a warm restart of Waveserver 5.
Note: A warm restart is non service-affecting. On restart, the running configuration file is recovered. |
| Access Level | admin |
| Syntax | chassis restart [now]

Note 1: A confirmation message appears when executing the <code>chassis restart</code> command, informing you that the restart of the system will affect traffic and asking you if you want to proceed.

Note 2: A confirmation message does not appear and the system restarts immediately if you use the <code>chassis restart now</code> command. |

chassis restart-cold

| | |
|-------------|---|
| Description | Performs a cold restart of Waveserver 5.
Note: A cold restart is service-affecting. On restart, only the saved configuration file is recovered (i.e., the running configuration file is not recovered). |
|-------------|---|

| | |
|--------------|--|
| Access Level | admin |
| Syntax | <p>chassis restart-cold [now]</p> <p>Note 1: A confirmation message appears when executing the <code>chassis restart-cold</code> command, informing you that the restart of the system will affect traffic and asking you if you want to proceed.</p> <p>Note 2: A confirmation message does not appear and the system performs a cold restart immediately if you use the <code>chassis restart-cold now</code> command.</p> |

chassis set

| | |
|--------------|---|
| Description | Sets the chassis slot power state attribute. |
| Access Level | admin |
| Prerequisite | <p>Before changing the power-state to shutdown on a service module, complete one of the following;</p> <ul style="list-style-type: none"> • Enable ptp tx-power-reduction-mode on all line ports of the service module, OR • Disable ptp tx-state on all line ports of the service module, OR • Set ptp frequency to 0 on all line ports of the service module, OR • Delete corresponding channels on 6500 photonics or RLS for all line ports of the service module. |
| Syntax | chassis set slot <1 2 3 4 5 6 7 8> power-state <automatic shutdown> |

chassis show

| | |
|--------------|---|
| Description | Displays all or specific summary information for the chassis. |
| Access Level | limited |
| Syntax | chassis show [capabilities] [mac] [restart] [secure-erase] |

chassis show restart

| | |
|--------------|---|
| Description | Displays the time and reason for the last restart of the Waveserver 5 chassis, CM-1, and modules. Also displays the type of restart (warm or cold). |
| Access Level | limited |
| Syntax | chassis show restart |

chassis show statistics

| | |
|--------------|---|
| Description | Displays temperature statistics of the chassis. |
| Access Level | limited |
| Syntax | chassis show statistics |

chassis slot show

| | |
|--------------|--|
| Description | Displays summary information for all module slots. |
| Access Level | limited |
| Syntax | chassis slot show |

Command log commands**command-log clear**

| | |
|--------------|---|
| Description | Clears the contents of the command log. |
| Access Level | super |
| Syntax | command-log clear |

command-log <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of command logging. |
| Access Level | super |
| Syntax | command-log disable
command-log enable |

command-log repeat

| | |
|--|--|
| Description | Repeats the specified command stored in the command log. |
| Access Level | limited
Note: Only commands that are available to the current user account can be repeated. |
| Syntax | command-log repeat [command <String>] [every <Duration: number/time format N[yMwdhms]] [times <Number: 1...4294967295>] |
| Note 1: The [command] string must be enclosed in quotation marks; for example, <code>command-log repeat command "alarm show"</code> . | |
| Note 2: For [every], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. | |

command-log repeat my-last-command

| | |
|--|--|
| Description | Repeats the last command stored in the command log for the current session. |
| Access Level | limited |
| Syntax | command-log repeat my-last-command [every <Duration: number/time format N[yMwdhms]] [times <Number: 1...4294967295>] |
| Note: For [every], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. | |

command-log repeat sequence-id

| | |
|--|--|
| Description | Repeats a sequence of one or more commands stored in the command log. The sequence is derived from the identifier assigned to each command in the log (see command-log show). |
| Access Level | limited
Note: Only commands that are available to the current user account can be repeated. |
| Syntax | command-log repeat sequence-id <Number: 1...4294967295 [...]> [every <Duration: number/time format N[yMwdhms]] [times <Number: 1...4294967295>] |
| Note: For [every], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. | |

command-log show

| | |
|---|--|
| Description | Displays the last 100 commands of the command log or the last 100 commands of the command log for a customized view of the command log contents. To see more than the last 100 commands for any given view, use the <code>history</code> argument. |
| Access Level | limited
Note: Only commands that are available to the current user account are displayed. |
| Syntax | command-log show [containing <String: <1...128>] [destination] [exit-status <Number: 0...4294967295>] [from-date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <Time: hh:mm:ss or hh:mm>] [last <Duration: number/time format N[yMwdhms]] [mine] [on-terminal <String: 1...32>] [recent] privilege <limited admin super> [sequence-id <Number: 1...4294967295[,...]>] [since-activated] [status] [tail <Number: 1...4294967295>] [today] [to-date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <Time: hh:mm:ss or hh:mm>] [user <String: 1...64>] [verbose] [yesterday] |
| Default Value | [destination] = flash |
| Note: For [last], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. | |

command-log show privilege

| | |
|--------------|--|
| Description | Displays the last 100 commands of the command log, customized for the specified access privilege. To see more than the last 100 commands, use the <code>history</code> argument. |
| Access Level | super |

| | |
|---|---|
| Syntax | command-log show privilege <limited admin super>] [containing <String: <1...128>] [destination] [exit-status <Number: 0...4294967295>] [from-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <time: hh:mm:ss or hh:mm>] [last <Duration: number/time format N[yMwdhms]>] [mine] [on-terminal <String: 1...32>] [recent] [sequence-id <NUMBER: 1...4294967295[,...]>] [since-activated] [status] [tail <NUMBER: 1..4294967295>] [today] [to-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <time: hh:mm:ss or hh:mm>] [user <String: 1...64>] [verbose] [yesterday] |
| Default Value | [destination] = flash |
| Note: For [last], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. | |

Configuration commands

cli candidate

| | |
|--------------|---|
| Description | Enables or disables candidate mode. When you enable candidate mode and make a configuration change in a CLI session, the change is only made to the Candidate datastore for that CLI session. Other, concurrent CLI sessions that reference the Running datastore are not affected. |
| Access Level | limited |
| Syntax | cli candidate < disable enable > |

configuration backup filename

| | |
|--------------|---|
| Description | Creates a backup archive of the datastores (running, startup, candidate) in native binary format, and logs and system files, and stores the archive in the config directory. The backup file can be restored using the <code>configuration restore</code> command.

Note: To ensure that the running and startup configuration datastores are the same when you back up the datastores, perform a <code>configuration save</code> operation just before you perform a full backup of the datastores. |
| Access Level | admin |
| Syntax | configuration backup filename <Filename> [passphrase <passphrase String[1..64]>] [no-passphrase]

Note: Waveserver 5 supports filenames containing alphanumeric characters, hyphens, and underscores. Spaces and special characters are not supported. |

configuration cancel-revert

| | |
|--------------|---|
| Description | Cancels the automatic configuration reversion when the auto-revert timer was set in the system. |
| Access Level | super |
| Syntax | configuration cancel-revert |

configuration check

| | |
|--------------|---|
| Description | Checks the specified configuration file for errors. This command is only applicable to a text-based configuration file consisting of CLI commands. |
| Access Level | super |
| Syntax | configuration check filename <Filename> [default-server] [default-ftp-server] [default-sftp-server] [default-scp-server] [default-tftp-server] [ftp-server <IP address or host name> {login-id <String> password <String> secret <String>}] [sftp-server <IP address or host name> {login-id <String> password <String> secret <String>}] [scp-server <IP address or host name> {login-id <String> password <String> secret <String>}] [tftp-server <IP address or host name> {login-id <String> password <String> secret <String>}] [url <remote_url>] [interactive] [verbose] [hide-progress]

Note: You must specify [login-id], [password], and [secret] for SFTP and SCP servers. |

configuration commit

| | |
|--------------|--|
| Description | Commits the candidate datastore to the running datastore. |
| Access Level | super |
| Syntax | configuration commit [revert-timeout <N[yMwdhms]>]

Note: The revert-timeout value sets a timer for execution of the commit command. If the specified time elapses before the commit completes, the commit operation is canceled. To specify 1 hour, 10 minutes and 3 seconds for example, the format is 1h10m3s. |

configuration copy

| | |
|--------------|---|
| Description | Creates a copy of a file in the config directory. |
| Access Level | admin |
| Syntax | configuration copy filename <filename> to-filename <filename> |

configuration datastore copy

| | |
|--------------|---|
| Description | Copies the contents of the specified source datastore to the specified destination datastore. |
| Access Level | super |
| Syntax | configuration datastore copy source <startup candidate running> destination <startup candidate running> |

configuration datastore delete

| | |
|-------------|--|
| Description | Deletes the contents of the specified datastore. |
|-------------|--|

| | |
|--------------|--|
| Access Level | super |
| Syntax | configuration datastore delete <startup candidate running> |

configuration datastore lock

| | |
|--------------|--|
| Description | Locks the specified datastore, preventing any changes to it. Typically, you lock the running datastore to prevent changes to the configuration settings. You can also lock the candidate datastore to limit access to it. When a lock exists for a particular datastore only the NBI user session that created the lock is allowed to update that datastore. |
| Access Level | admin |
| Syntax | configuration datastore lock <startup candidate running> |

configuration datastore show

| | |
|--------------|--|
| Description | Lists the contents of the specified datastore on screen. |
| Access Level | limited |
| Syntax | configuration datastore show datastore <startup candidate running> |

configuration datastore show differences

| | |
|--------------|--|
| Description | The datastore show differences command compares the contents of the two specified datastores then outputs a combined list of the datastore contents indicating which configuration settings were commands were added or removed. |
| Access Level | limited |
| Syntax | configuration datastore show differences source <startup candidate running> destination <startup candidate running> |

configuration datastore unlock

| | |
|--------------|--|
| Description | Releases a previously locked datastore. The force option forces the unlock operation on the specified datastore. |
| Access Level | admin, super (to use the [force] option) |
| Syntax | configuration datastore unlock <startup candidate running> [force] |

configuration datastore validate

| | |
|--------------|--|
| Description | Validates changes to the specified datastore. |
| Access Level | admin |
| Syntax | configuration datastore validate <startup candidate running> |

configuration delete

| | |
|--------------|---|
| Description | Deletes the specified configuration file. |
| Access Level | admin |
| Syntax | configuration delete filename <Filename> |

configuration install

| | |
|--------------|---|
| Description | Writes a copy of the configuration file (in its native database format or CLI commands in text format) to the \config directory. |
| Access Level | admin |
| Syntax | configuration install filename <Filename>
[verbose] [hide-progress] [default-server] [default-tftp-server] [default-ftp-server] [default-sftp-server] [default-scp-server] [tftp-server <ip-addr-or-hostname>] [ftp-server <ip-addr-or-hostname> login-id <username> password <password> secret <secret>] [sftp-server <ip-addr-or-hostname> login-id <username> password <password> secret <secret>] [scp-server <ip-addr-or-hostname> login-id <username> password <password> secret <secret>] [url <string>]
You must specify [login-id], [password], and [secret] for SFTP and SCP servers. |

configuration list

| | |
|--------------|---|
| Description | Displays a summary of the configuration files available locally (including backup configuration files). |
| Access Level | admin |
| Syntax | configuration list |

configuration rename filename

| | |
|--------------|--|
| Description | Changes the name of the specified configuration file. |
| Access Level | admin |
| Syntax | configuration rename filename <Filename> to-filename <Filename>
Note: Waveserver 5 supports filenames containing alphanumeric characters, hyphens, and underscores. Spaces and special characters are not supported. |

configuration reset-to-defaults

| | |
|--------------|---|
| Description | Restores default configuration values. This command resets configurable properties to their default values. This command deletes user-configured accounts and passwords. It does not reinstall the software load. |
| Access Level | super |

| | |
|--------|---|
| Syntax | <p>configuration reset-to-defaults</p> <p>Note: A confirmation message appears when you execute the <code>configuration reset-to-defaults</code> command to inform you that the operation will cause a restart of the system and affect traffic, and to prompt you to verify that you want to proceed.</p> |
|--------|---|

configuration reset-to-factory-defaults

| | |
|--------------|--|
| Description | <p>Restores factory-set configuration values. This command triggers a Reset to Factory Defaults and permanently deletes user-configured information such as the hostname, IP address, configuration file, logs, user accounts, and passwords (except for the default su account). The running software load is then reinstalled.</p> <p>For more information, see Reset to Factory Defaults (RTFD).</p> <p>Note: Executing the RTFD function on a chassis prior to its return to Ciena irrevocably deletes all provisioning information and troubleshooting logs which are typically required for root cause investigations or failure analysis. Please note that this could impact Ciena's ability to fully investigate a module or network failure event.</p> |
| Access Level | super |
| Syntax | <p>configuration reset-to-factory-defaults</p> <p>Note: A confirmation message appears when you execute the <code>configuration reset-to-factory-defaults</code> command to inform you that the operation will cause a restart of the system and affect traffic, and to prompt you to verify that you want to proceed.</p> |

configuration reset-to-user-config (for text-based CLI configuration)

| | |
|--|--|
| Description | Installs a local configuration file. |
| Access Level | super |
| Syntax | <p>configuration reset-to-user-config filename <Filename> [destination <datastore>] [revert-timeout <Duration: number/time format N[yMwdhms]>] [force]</p> |
| <p>Note: For [revert-timeout], to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. If the specified time elapses before the installation is successfully completed, the configuration reverts to the currently running configuration file.</p> | |

configuration restore

| | |
|-------------|--|
| Description | <p>Restores the specified configuration file to the datastore.</p> <p>Note 1: After saving a configuration to a file you can use the configuration reset-to-user-config (for text-based CLI configuration) command to apply the saved configuration.</p> <p>Note 2: When the restore operation is performed the CM is restarted.</p> |
|-------------|--|

| | |
|--------------|---|
| Access Level | super |
| Syntax | configuration restore filename <Filename> [passphrase <passphrase String[1..64]>] |

configuration save

| | |
|--------------|--|
| Description | <p>If you do not specify a filename or datastore, saves the currently active configuration settings (running datastore) to the startup datastore. (In this case, use <code>configuration restore</code> to restore saved settings to datastores).</p> <p>If you specify a filename only, this command saves the running datastore as a file of CLI commands under the specified filename. If you specify a filename and a datastore, this command saves the specified datastore as a file of CLI commands under the specified filename. (In these cases, use <code>configuration reset-to-user-config</code> to apply saved settings.)</p> |
| Access Level | admin |
| Syntax | <p>configuration save filename <Filename> datastore <startup candidate running></p> <p>By default the configuration is stored in the directory: <code>/var/opt/config</code>.</p> <p>Note: Waveserver 5 supports filenames containing alphanumeric characters, hyphens, and underscores. Waveserver 5 also allows the following special characters in filenames: <code>~ ' @ # \$ () + = { } . <></code> Spaces are not supported.</p> |

configuration show

| | |
|--------------|---|
| Description | Displays configuration settings in the active configuration file. |
| Access Level | limited |
| Syntax | configuration show |

configuration show differences-from-saved

| | |
|--------------|---|
| Description | Displays the differences between the running datastore and the startup datastore. |
| Access Level | limited |
| Syntax | configuration show differences-from-saved |

configuration show filename

| | |
|--------------|--|
| Description | Displays configuration settings in the specified configuration file. |
| Access Level | limited |
| Syntax | configuration show filename <Filename> |

configuration show status

| | |
|--------------|---|
| Description | Displays status information for the system configuration. |
| Access Level | limited |
| Syntax | configuration show status |

configuration ztp <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (enabled or disabled) for ZTP. |
| Access Level | super |
| Syntax | configuration ztp disable
configuration ztp enable |
| Default | enabled |

configuration ztp reset

| | |
|---|--|
| Description | Resets the name of the last configuration file or last command file (boot file) ZTP applied to the Waveserver. |
| Access Level | admin |
| Syntax | configuration ztp reset <last-config-file last-command-file> |
| <p>Note 1: By deleting the name of the last configuration file, ZTP will apply the currently specified configuration file, even if it is the one already installed.</p> <p>Note 2: By deleting the name of the last boot file, ZTP will apply the currently specified boot file, even if it is the one already installed.</p> | |

configuration ztp show

| | |
|--------------|---------------------------------|
| Description | Displays the status of the ZTP. |
| Access Level | limited |
| Syntax | configuration ztp show |

Connection command

connection show

| | |
|--------------|---|
| Description | Displays the connection information for the port and/or slot. |
| Access Level | limited |
| Syntax | connection show [port <slot>/<port>] [slot <slot>] |

DHCP client commands

dhcp client <disable | enable>

| | |
|--|---|
| Description | Sets the administrative state (enabled or disabled) of the DHCPv4 client. |
| Access Level | admin |
| Syntax | dhcp client disable
dhcp client enable |
| Note: Waveserver 5 does not clear the IP address when you disconnect it from the network. To clear the IP address and obtain a new one from the DHCP server, disable the DHCP client, then re-enable it. You must perform this operation through an IPv6 or console connection. | |

dhcp client lease renew

| | |
|--------------|-------------------------------|
| Description | Renews the DHCP client lease. |
| Access Level | admin |
| Syntax | dhcp client lease renew |

dhcp client options set

| | |
|----------------|---|
| Description | Changes writable DHCP client option settings. |
| Access Level | admin |
| Syntax | dhcp client options set {[boot-file <on off>] [dns <on off>] [domain-name <on off>] [host-name <on off>] [lease-time <on off>] [log-server <on off>] [ntp <on off>] [router <on off>] [subnet <on off>] [time-offset <on off>]} |
| Default Values | [boot-file] = on
[dns] = on
[domain-name] = on
[host-name] = on
[lease-time] = on
[log-server] = on
[ntp] = on
[router] = on
[subnet] = on
[time-offset] = on |

dhcp client set

| | |
|--------------|--|
| Description | Changes writable DHCP client settings. |
| Access Level | admin |

| | |
|----------------|---|
| Syntax | dhcp client set {[interface-type <active remote>] [discovery-interval <Seconds: 1...60>] [requested-lease-time <Seconds: 45...4294967294>]} |
| Default Values | <ul style="list-style-type: none"> • [discovery-interval] = 30 seconds • [requested-lease-time] = 3600 seconds |

dhcp client show

| | |
|--------------|---|
| Description | Displays summary information for the DHCPv4 client. |
| Access Level | admin |
| Syntax | dhcp client show [options] [state] |

dhcp relay show

| | |
|--------------|---|
| Description | Displays the administrative state of DHCP relay agent on the Waveserver 5 node. |
| Access Level | admin |
| Syntax | dhcp relay show |

dhcp relay <enable | disable>

| | |
|--------------|--|
| Description | Sets the administrative state of the DHCP relay agent. |
| Access Level | admin |
| Syntax | dhcp relay enable
dhcp relay disable |

DNS client commands**dns client <add | remove> server**

| | |
|--------------|---|
| Description | Assigns a server to or removes an assigned server from the DNS client. |
| Access Level | admin |
| Syntax | dns client add server <IP address>
dns client remove server <IP address> |

dns client <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (enabled or disabled) of either the DNS client or a server assigned to the DNS client. |
| Access Level | admin |
| Syntax | dns client disable [server <IP address>]
dns client enable [server <IP address>] |

dns client resolve ip

| | |
|--------------|---|
| Description | Resolves the IP address or host name of the DNS client. |
| Access Level | limited |
| Syntax | dns client resolve ip <IP address or host name> |

dns client show

| | |
|--------------|---|
| Description | Displays summary information for the DNS client and the servers assigned to it. |
| Access Level | limited |
| Syntax | dns client show [domain-name <Domain Name String: 1...63>] [servers] |

dns client set domain-name

| | |
|--------------|---|
| Description | Sets the domain name of the DNS client. |
| Access Level | admin |
| Syntax | dns client set domain-name <Domain Name String: 1...63> |

dns client set server

| | |
|--------------|---|
| Description | Sets the IP address of the DNS server. |
| Access Level | admin |
| Syntax | dns client set server <IP address> priority <Number: 1...3> |

dns client unset domain-name

| | |
|--------------|--|
| Description | Restores the domain name of the DNS client to the default configuration value. |
| Access Level | admin |
| Syntax | dns client unset domain-name |

File commands**file cat**

| | |
|--------------|--|
| Description | Copies a file to the terminal. |
| Access Level | limited |
| Syntax | file cat <Filename>
Note: Enter <code>file cat --help</code> for more information. |

file cd

| | |
|--------------|-------------------------------|
| Description | Changes the active directory. |
| Access Level | limited |
| Syntax | file cd <Directory name> |

file chmod

| | |
|--------------|--|
| Description | Changes the access mode of a file. |
| Access Level | limited |
| Syntax | file chmod mode <Mode> <Filename>
Note: Enter <code>file chmod --help</code> for more information. |

file cmp

| | |
|--------------|--|
| Description | Compares two files byte-by-byte. |
| Access Level | admin |
| Syntax | file cmp [-l] [-s] [-h] [file1 -] [file2] [pos1 [pos2]]
Note: Enter <code>file cmp --help</code> for more information. |

file cp

| | |
|--------------|--|
| Description | Copies one or more files within the local file system. |
| Access Level | admin |
| Syntax | file cp <Source> <Destination>
Note: Enter <code>file cp --help</code> for more information. |

file diff

| | |
|--------------|--|
| Description | Compares two files line-by-line. |
| Access Level | admin |
| Syntax | file diff <option> <file1 file2>
Note: Enter <code>file diff --help</code> for more information. |

file ls

| | |
|--------------|--|
| Description | Lists information about files in the active directory. |
| Access Level | limited |

| | |
|--------|--|
| Syntax | file ls [option]
Note: Enter <code>file ls --help</code> for more information. |
|--------|--|

file mkdir

| | |
|--------------|---|
| Description | Creates a directory. |
| Access Level | admin |
| Syntax | file mkdir [option] <Directory name>
Note: Enter <code>file mkdir --help</code> for more information. |

file mv

| | |
|--------------|---|
| Description | Moves a file to another directory. |
| Access Level | admin |
| Syntax | file mv [option] <Source> <Destination>
Note: Enter <code>file mv --help</code> for more information. |

file ping

| | |
|--------------|---|
| Description | Pings an IP address or host. |
| Access Level | limited |
| Syntax | file ping [option] <IP address or host name>
Note: Enter <code>file ping</code> to display available options. |

file ping6

| | |
|--------------|---|
| Description | Uses IPv6 to ping an IP address or host. |
| Access Level | limited |
| Syntax | file ping6 [option] <IPv6 address or host name>
Note: Enter <code>file ping6</code> to display available options. |

file pwd

| | |
|--------------|-------------------------------------|
| Description | Displays the active directory path. |
| Access Level | limited |
| Syntax | file pwd |

file ps

| | |
|-------------|-------------------------------|
| Description | Displays a list of processes. |
|-------------|-------------------------------|

| | |
|--------------|---|
| Access Level | admin |
| Syntax | file ps
Note: Enter <code>file ps --help</code> for more information. |

file rm

| | |
|--------------|---|
| Description | Deletes a file. |
| Access Level | admin |
| Syntax | file rm [option] <Filename>
Note: Enter <code>file rm --help</code> for more information. |

file rmdir

| | |
|--------------|---|
| Description | Deletes a directory. |
| Access Level | admin |
| Syntax | file rmdir [option] <Directory name>
Note: Enter <code>file rmdir --help</code> for more information. |

file rmtree

| | |
|--------------|---|
| Description | Deletes a directory path. |
| Access Level | admin |
| Syntax | file rmtree [option] <Path>
Note: Enter <code>file rmtree --help</code> for more information. |

file ssh

| | |
|--------------|--|
| Description | Establishes an SSH connection to an IP address or host. |
| Access Level | admin |
| Syntax | file ssh [option] <IP address or host name>
Note: Enter <code>file ssh --help</code> for more information. |

file scp

| | |
|--------------|---|
| Description | Uses SCP to copy file(s).
Note: Ensure the key exchange SSH algorithms match on the source and destination. |
| Access Level | admin |

| | |
|--------|---|
| Syntax | file scp [src] [dst]
Note: Enter <code>file scp --help</code> for more information. |
|--------|---|

file tget

| | |
|--------------|--|
| Description | Downloads a file from a remote server to the local server. |
| Access Level | admin |
| Syntax | file tget <IP address or host name> <Remote filename> [Local filename] |

file top

| | |
|--------------|--|
| Description | Displays the list of tasks managed by Linux on Waveserver 5. To close the display, press CTRL + C. |
| Access Level | limited |
| Syntax | file top [-n 1] [-b]
Where n = the number of times the display refreshes
and
b = batch mode, used for sending the output to another program or to save the list to a file |

file tput

| | |
|--------------|--|
| Description | Uploads a file to a remote server. |
| Access Level | limited |
| Syntax | file tput <IP address or host name> <Remote filename> [Local filename] |

file traceroute

| | |
|--------------|---|
| Description | Performs a traceroute to an IPv4 IP address or host. |
| Access Level | limited |
| Syntax | file traceroute [option] <IP address or host name>
Note: Enter <code>file traceroute</code> to display available options. |

file vols

| | |
|--------------|--|
| Description | Displays information about the file system on which each file resides, or all file systems by default. |
| Access Level | limited |
| Syntax | file vols [volume]
Note: Enter <code>file vols --help</code> to display available options. |

Interface commands

interface acl add

| | |
|--------------|--|
| Description | Sets the parameters for an ACL rule. |
| Access Level | super |
| Syntax | interface acl <ingress egress> <ipv4 ipv6> action <allow drop> destination-address <String> interface-name <String> protocol <tcp udp icmp ipv6-icmp all 0 to 255> sequence-id <1..50> source-address <String> |

interface acl delete

| | |
|--------------|--|
| Description | Deletes the specified ACL rule |
| Access Level | super |
| Syntax | interface acl delete <ingress egress> <ipv4 ipv6> sequence-id <#1..50> |

interface acl show

| | |
|--------------|--------------------------------|
| Description | Displays all of the ACL rules. |
| Access Level | limited |
| Syntax | interface acl show |

interface add gateway

| | |
|--------------|--|
| Description | Adds an IPv6 gateway for the Waveserver 5 on the management network. |
| Access Level | admin |
| Syntax | interface add gateway <IPv6 address> |

interface add interface

| | |
|--------------|--|
| Description | Adds an IPv6 address for the Waveserver 5 on the management network. |
| Access Level | admin |
| Syntax | interface add interface local ip <IP address /mask> |

interface <disable | enable>

| | |
|--------------|--|
| Description | Disables an interface. |
| Access Level | admin |
| Syntax | interface <disable enable> <if0-<slot>-<port> local> |

interface ipv6-stack <disable | enable>

| | |
|---------------|---|
| Description | Sets the administrative state (disabled or enabled) of the IPv6 stack. |
| Access Level | admin |
| Prerequisites | To disable IPv6:
interface ipv6-stack set stack-preference IPv4
interface ipv6-stack disable
To re-enable IPv6:
interface ipv6-stack enable
interface ipv6-stack set stack-preference IPv6 |
| Syntax | interface ipv6-stack <disable enable> |

interface ipv6-stack flush

| | |
|--------------|---|
| Description | Clears all IPv6 SLAAC addresses from the system or from only the specified interface. |
| Access Level | admin |
| Syntax | interface ipv6-stack flush [interface <interface name>] |

interface ipv6-stack set

| | |
|----------------|--|
| Description | Changes writable IPv6 stack configuration settings. |
| Access Level | admin |
| Syntax | interface ipv6-stack set [accept-router-advertisement <on off>] [max-slaac-addresses <Number: 1...32>] |
| Default Values | <ul style="list-style-type: none"> • [accept-router-advertisement]: on • [max-slaac-addresses]: 16 |

interface ipv6-stack set stack-preference

| | |
|---------------|---|
| Description | Sets the IP address type. |
| Access Level | admin |
| Syntax | interface ipv6-stack set stack-preference <IPv4 IPv6> |
| Default Value | IPv6 |

interface ipv6-stack show

| | |
|--------------|--|
| Description | Displays summary information for the IPv6 stack state. |
| Access Level | limited |

| | |
|--------|---------------------------|
| Syntax | interface ipv6-stack show |
|--------|---------------------------|

interface ipv6-stack unset

| | |
|--------------|--|
| Description | Restores IPv6 stack configuration settings to the default values. See interface ipv6-stack set and interface ipv6-stack set stack-preference . |
| Access Level | admin |
| Syntax | interface ipv6-stack unset [stack-preference] [accept-router-advertisement] [max-slaac-addresses] |

interface port <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state of the interface port. |
| Access Level | admin |
| Syntax | interface port <disable enable> ifp0-<slot>--<port> |

interface port show

| | |
|--------------|--|
| Description | Displays the summary information for all the interface ports or a specific interface port. |
| Access Level | limited |
| Syntax | interface port show [port ifp0-<slot>--<port>] |

interface protocol bgp clear

| | |
|--------------|--|
| Description | Performs a hard clear or soft clear on the BGP connection. |
| Access Level | admin |
| Syntax | interface protocol bgp clear [soft] |

interface protocol bgp <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state for BGP. |
| Access Level | admin |
| Syntax | interface protocol bgp <disable enable> |

interface protocol bgp set

| | |
|--------------|--|
| Description | Sets the BGP peer optional attributes. |
| Access Level | admin |

| | |
|--------|--|
| Syntax | interface protocol bgp set [autonomous-system <NUMBER: 1..4294967295>]
[keepalive <NUMBER: 1..21845>] [holdtime <NUMBER: 3..65535>] |
|--------|--|

interface protocol bgp set peer

| | |
|--------------|---|
| Description | Sets the BGP peers. |
| Access Level | admin |
| Syntax | interface protocol bgp set peer <peer-1 peer-2> autonomous-system <NUMBER: 1..4294967295> ip <IP Address> |

interface protocol bgp show

| | |
|--------------|--|
| Description | Displays the BGP summary information. |
| Access Level | limited |
| Syntax | interface protocol bgp show [peer-summary] [runtime-routes] [peer <peer-1 peer-2>] |

interface remove interface local ip

| | |
|--------------|--|
| Description | Removes a user-configured IPv6 IP address from the system. |
| Access Level | admin |
| Syntax | interface remove interface local ip <IPv6 address> |

interface remove gateway

| | |
|--------------|---|
| Description | Removes a user-specified IPv6 gateway. |
| Access Level | admin |
| Syntax | interface remove gateway <IPv6 gateway> |

interface route remove destination

| | |
|--------------|---|
| Description | Removes a route from the routing table. |
| Access Level | admin |
| Syntax | interface route remove destination <IP address with mask> |

interface route show

| | |
|--------------|-------------------------------------|
| Description | Displays routing table information. |
| Access Level | limited |
| Syntax | interface route show |

interface set gateway

| | |
|--------------|------------------------------------|
| Description | Sets an IPv4 gateway. |
| Access Level | admin |
| Syntax | interface set gateway <IP address> |

interface set interface

| | |
|----------------|---|
| Description | Sets an IPv4 IP address and gateway for Waveserver 5 on the management network. |
| Access Level | admin |
| Syntax | interface set interface local ip <IP address/mask> [gateway <IP address>] |
| Default Values | <ul style="list-style-type: none"> • <IP address> = 0.0.0.0/0 • [gateway] = 0.0.0.0 |

interface show

| | |
|--------------|--|
| Description | Displays summary information for the management interface. |
| Access Level | limited |
| Syntax | interface show [gateway] [interface <if0-<slot>-<port> local>] |

interface unset interface

| | |
|--------------|---|
| Description | Restores an IPv4 address and gateway to the configuration default values (see interface set interface).
Note: The interface unset interface local ip command cannot be used to unset IPv6 SLAAC addresses. |
| Access Level | admin |
| Syntax | interface unset interface local ip [gateway] |

License commands**license client activate filename <string>**

| | |
|--------------|--|
| Description | Activates a Waveserver 5 software license file that has been downloaded to the Waveserver 5. License files must be activated from the <code>/var/opt/license/download</code> location. |
| Access Level | admin |
| Syntax | license client activate filename <filename> |

license client set registration-id

| | |
|---------------|--|
| Description | Sets the customer-specified registration for a Waveserver 5 software license that can be used on more than one Waveserver 5. |
| Access Level | admin |
| Syntax | license client set registration-id <String: 1..64> |
| Default Value | <string> is empty |

license client server clear

| | |
|--------------|---|
| Description | Removes access to the external license server. Any licenses already directly acquired or installed are saved in the local trust store and are used. |
| Access Level | admin |
| Syntax | license client server clear |

license client server set

| | |
|--------------|---|
| Description | Configures access to the external license server. |
| Access Level | admin |
| Syntax | license client server set hostname <IP address> port <portNum> protocol <http https> sync-interval <15..1440> |

license client server show

| | |
|--------------|--|
| Description | Displays summary information for each license server the Waveserver 5 License client can access. |
| Access Level | limited |
| Syntax | license client server show |

license client server sync-now

| | |
|--------------|--|
| Description | Forces a synchronization of the licenses held on the Waveserver with the license inventory on the external license server. |
| Access Level | admin |
| Syntax | license client server sync-now |

license client show

| | |
|-------------|---|
| Description | Displays a summary of state information for the Waveserver 5 License client, as well as the Registration ID and the number of license servers the Waveserver 5 License client can access. |
|-------------|---|

| | |
|--------------|---------------------|
| Access Level | limited |
| Syntax | license client show |

license client show arrears

| | |
|--------------|---|
| Description | Displays a list of licenses in arrears. |
| Access Level | limited |
| Syntax | license client show arrears |

license client show inventory

| | |
|--------------|---|
| Description | Displays a customized view of license summary information. |
| Access Level | limited |
| Syntax | license client show inventory [all][served] [expired] [checked-out] [available] [preauth] |

license client show license

| | |
|--------------|---|
| Description | Displays summary and status information for all licenses installed on Waveserver 5. |
| Access Level | limited |
| Syntax | license client show license <license name> |

license client unset registration-id

| | |
|--------------|---|
| Description | Restores the license client registration ID to the default setting (see license client set registration-id). |
| Access Level | admin |
| Syntax | license client unset registration-id |

license file delete filename

| | |
|--------------|--|
| Description | Removes the specified license file from the Waveserver 5 License client. |
| Access Level | admin |
| Syntax | license file delete filename <string> |

license file download

| | |
|-------------|---|
| Description | Downloads the specified license file from the specified remote server.
Note: For SFTP and SCP license servers, ensure the key exchange SSH algorithms match on the Waveserver 5 and the license server. |
|-------------|---|

| | |
|--------------|--|
| Access Level | admin |
| Syntax | <p>license file download filename <full directory path and filename> [default-server] [default-sftp-server] [default-scp-server] [sftp-server <IP address or host name> login-id <String> password <String> secret <String>] [scp-server <IP address or host name> login-id <String> password <String> secret <String>]</p> <p>Note: You must specify [login-id] and [password] for SFTP and SCP servers.</p> |

license file download url

| | |
|--------------|---|
| Description | Downloads a license file from the specified URL. The supported transfer protocols are http, https, scp, and sftp. |
| Access Level | admin |
| Syntax | <p>license file download url <String> [login-id <username>] [password <password>]</p> <p><String> — use the format: <protocol>://<remote IP address>/<remote file path&name></p> <p><protocol> — use any one of the following protocols: http, https, scp, or sftp</p> <p><remote IP address> — use the remote server IP address or, if DNS is provisioned, use the domain name</p> <p><remote file path&name> — use the remote file path and file name</p> <p>Example:</p> <pre>license file download url https://123.123.123.123/waveserver/ license/ciena-license.lic login-id waveserver password *****</pre> <p>Note: When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (https://tools.ietf.org/html/rfc3986).</p> <p>For example, license file download url https://
[2222:111:dddd:5444:aced::293]/waveserver/license/ciena-
license.lic login-id waveserver password *****</p> |

license file generate request

| | |
|--------------|--|
| Description | Generates a Registration Request File for Waveserver 5 registration and licensing at the Ciena License portal. |
| Access Level | admin |
| Syntax | license file generate request |

license file list

| | |
|-------------|--|
| Description | Displays a customized view (downloaded, processed, and/or requested) of license files. |
|-------------|--|

| | |
|--------------|---|
| Access Level | limited |
| Syntax | license file list [downloaded] [processed] [requests] |

Link Layer Discovery Protocol (LLDP) commands

lldp clear interface-port statistics

| | |
|--------------|--|
| Description | Clears LLDP statistics for the DCN port. |
| Access Level | admin |
| Syntax | lldp clear interface-port <ifp0-dcn-1 ifp0-dcn-3> statistics |

lldp clear port statistics

| | |
|--------------|---|
| Description | Clears LLDP statistics for specified ports. |
| Access Level | admin |
| Syntax | lldp clear port <portList> statistics |

lldp clear statistics

| | |
|--------------|---|
| Description | Clears all LLDP bridge and port statistics. |
| Access Level | admin |
| Syntax | lldp clear statistics |

lldp <disable | enable>

| | |
|--------------|--|
| Description | Sets the global administrative status (disabled or enabled) of LLDP. |
| Access Level | admin |
| Syntax | lldp disable
lldp enable |

lldp set interface-port

| | |
|----------------|---|
| Description | Sets the LLDP mode (tx-rx or disabled) and the sets the SNMP notifications (on or off) on the DCN port. |
| Access Level | admin |
| Syntax | lldp set interface-port ifp0-dcn-n [mode <tx-rx disable>] [notification <on off>] |
| Default Values | <ul style="list-style-type: none"> • [mode] = disable • [notification] = off |

lldp set port

| | |
|--------------|--|
| Description | Sets the LLDP mode (snoop or disabled) for one or more ports.
Note: When the LLDP mode of the port is snoop, snooping is operational only while LLDP is globally enabled (see lldp <disable enable>) and the administrative state of the port is enabled (see port <disable enable> port). |
| Access Level | admin |
| Syntax | lldp set port <portList> [mode <snoop disable>] |

lldp show

| | |
|--------------|--|
| Description | Displays a summary of LLDP configuration attributes, bridge and port statistics. |
| Access Level | limited |
| Syntax | lldp show |

lldp show configuration

| | |
|--------------|---|
| Description | Displays a summary of global LLDP configuration settings. |
| Access Level | limited |
| Syntax | lldp show configuration |

lldp show interface-port

| | |
|--------------|--|
| Description | Displays LLDP summary information for the DCN port. |
| Access Level | limited |
| Syntax | lldp show interface-port ifp0-dcn-1 [configuration] [neighbors] [statistics] |

lldp show neighbors

| | |
|--------------|--|
| Description | Displays LLDP data for LLDP-enabled devices connected to Waveserver 5. |
| Access Level | limited |
| Syntax | lldp show neighbors |

lldp show port configuration

| | |
|--------------|---|
| Description | Displays a summary of LLDP configuration information for one or more ports. |
| Access Level | limited |
| Syntax | lldp show port <portList> configuration |

lldp show port statistics

| | |
|--------------|--|
| Description | Displays LLDP port statistics for one or more ports. |
| Access Level | limited |
| Syntax | lldp show port <portList> statistics |

lldp show statistics

| | |
|--------------|---|
| Description | Displays all LLDP bridge and port statistics. |
| Access Level | limited |
| Syntax | lldp show statistics |

Logging commands**logging clear**

| | |
|--------------|-----------------------------|
| Description | Clears logging information. |
| Access Level | super |
| Syntax | logging clear [history] |

logging disable

| | |
|--------------|-------------------|
| Description | Disables logging. |
| Access Level | super |
| Syntax | logging disable |

logging download

| | |
|--------------|--|
| Description | Downloads log files. |
| Access Level | super |
| Syntax | logging download {keyword <String>} {url <String>} [verbose] |

logging enable

| | |
|--------------|------------------|
| Description | Enables logging. |
| Access Level | super |
| Syntax | logging enable |

logging show

| | |
|-------------|------------------------------------|
| Description | Displays the event logging status. |
|-------------|------------------------------------|

| | |
|--------------|--------------|
| Access Level | limited |
| Syntax | logging show |

logging view

| | |
|--------------|--|
| Description | Displays the event logs. |
| Access Level | limited |
| Syntax | logging view history keyword <String>
logging view history verbose
logging view keyword <String>
logging view verbose |

Module commands

module create slot

| | |
|--------------|--|
| Description | Provisions a slot for the specified type of Waveserver 5 module. |
| Access Level | admin |
| Syntax | module create slot <NUMBER: 1..8> type <motr-400-2-16 motr-400-2-12 motr-dco-400-3-12> |

module delete slot

| | |
|--------------|--|
| Description | Deprovisions a slot for a Waveserver 5 module. |
| Access Level | admin |
| Syntax | module delete slot <NUMBER: 1..8> |

module <enable | disable> slot

| | |
|--------------|--|
| Description | Sets the administrative state (enabled or disabled) of the module in the specified slot in the Waveserver 5 chassis. |
| Access Level | admin |
| Syntax | module <disable enable> slot <NUMBER: 1..8> |

module restart-cold slot

| | |
|--------------|---|
| Description | Performs a cold restart on a Waveserver 5 module in the specified slot. |
| Access Level | admin |
| Syntax | module restart-cold slot <NUMBER: 1..8> [now] |

module restart slot

| | |
|--------------|---|
| Description | Restarts a Waveserver 5 module in the specified slot. |
| Access Level | admin |
| Syntax | module restart slot <NUMBER: 1..8> |

module set slot

| | |
|--------------|--|
| Description | Configures a label or mode for a module.
You must disable the module admin state before configuring the mode (see module <enable disable> slot). |
| Access Level | admin |
| Syntax | module set slot <slot> label <String [1..128]> mode <none mated> |

module show

| | |
|--------------|---|
| Description | Displays information about the modules installed in the Waveserver 5 chassis. |
| Access Level | limited |
| Syntax | module show |

module show slot

| | |
|--------------|---|
| Description | Displays information about the Waveserver 5 module in the specified slot. |
| Access Level | limited |
| Syntax | module show slot <NUMBER: 1..8> |

module show slot <slot> statistics

| | |
|--------------|---|
| Description | Displays temperature statistics of the individual module. |
| Access Level | limited |
| Syntax | module show slot <NUMBER: 1..8> statistics |

module unset slot

| | |
|--------------|---|
| Description | Deletes the label assigned to the specified Waveserver 5 module slot. |
| Access Level | admin |
| Syntax | module unset slot <NUMBER:1..8> [label] |

NDP commands**ndp delete ip**

| | |
|--------------|--------------------------------------|
| Description | Removes an entry from the NDP cache. |
| Access Level | admin |
| Syntax | ndp delete ip <IPv6 address> |

ndp flush

| | |
|--------------|--|
| Description | Clears all management interfaces or only the specified interface from the NDP cache. |
| Access Level | admin |
| Syntax | ndp flush [interface <local>] |

ndp set cachetime

| | |
|---------------|---|
| Description | Changes the NDP cache timeout value. |
| Access Level | admin |
| Syntax | ndp set cachetime <Seconds: 60 ...3600> |
| Default Value | 300 |

ndp show

| | |
|--------------|---|
| Description | Displays summary NDP information for all management interfaces or only the specified interface. |
| Access Level | limited |
| Syntax | ndp show [interface <local>] |

ndp unset cachetime

| | |
|--------------|--|
| Description | Restores the NDP cache timeout value to the default configuration setting. See ndp set cachetime . |
| Access Level | admin |
| Syntax | ndp unset cachetime |

NTP (Network Time Protocol) MD5 authentication commands**ntp md5-auth add**

| | |
|--------------|--------------------------------------|
| Description | Adds the NTP MD5 authentication key. |
| Access Level | admin |

| | |
|--------|---|
| Syntax | ntp md5-auth add {key-id <NTP MD5 key object number:1...65534> md5 <MD5 auth key string: 2...31>} |
|--------|---|

ntp md5-auth <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (enabled or disabled) for NTP MD5 authentication. |
| Access Level | admin |
| Syntax | ntp md5-auth disable
ntp md5-auth enable |

ntp md5-auth import

| | |
|--------------|--|
| Description | Imports NTP MD5 authentication keys from a file. |
| Access Level | admin |
| Syntax | ntp md5-auth import <filename> |

ntp md5-auth remove

| | |
|--------------|---|
| Description | Removes an NTP MD5 authentication key. |
| Access Level | admin |
| Syntax | ntp md5-auth remove {key-id <NTP MD5 key object number: 1...65534>} |

ntp md5-auth secret

| | |
|--------------|---|
| Description | Adds an NTP MD5 authentication key using a pre-encrypted string. |
| Access Level | admin |
| Syntax | ntp md5-auth secret {key-id <NTP MD5 key object number: 1...65534> md5 <Encrypted MD5 auth key string: 4...62>} |

ntp md5-auth show

| | |
|--------------|--|
| Description | Displays summary information for NTP MD5 authentication. |
| Access Level | limited |
| Syntax | ntp md5-auth show |

NTP client commands**ntp client <add | remove> multicast-server**

| | |
|-------------|--|
| Description | Assigns a multicast server to or removes an assigned multicast server from the NTP client. |
|-------------|--|

| | |
|--------------|---|
| Access Level | admin |
| Syntax | ntp client add multicast-server <IP address [1..63]>
ntp client remove multicast-server <IP address [1..63]> |

ntp client <add | remove> server

| | |
|--------------|---|
| Description | Assigns a server to or removes an assigned server from the NTP client. |
| Access Level | admin |
| Syntax | ntp client add server <IP address or host name: 1...63> [key-id <authentication key ID>] [autokey]
ntp client remove server <IP address or host name: 1...63>
Note: Up to 10 NTP client servers can be added for NTPv4 Autokey authentication. |

ntp client <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) for all servers assigned to the NTP client or for the specified server only. |
| Access Level | admin |
| Syntax | ntp client disable [server <IP address or host name>]
ntp client enable [server <IP address or host name>] |

ntp client set

| | |
|----------------|--|
| Description | Changes writable settings of the NTP client. |
| Access Level | admin |
| Syntax | ntp client set {[mode <polling broadcast multicast>] [polling-interval <Seconds: 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536>]} |
| Default Values | <ul style="list-style-type: none"> • [mode]: polling • [polling-interval]: 16 |

ntp client set server autokey

| | |
|--------------|--|
| Description | Changes a pre-added server to turn Autokey on or off. |
| Access Level | admin |
| Syntax | ntp client set server <IP address or host name> autokey <on off> |

ntp client set server key-id

| | |
|-------------|---|
| Description | Changes the MD5 authentication key for a server assigned to the NTP client. |
|-------------|---|

| | |
|---------------|---|
| Access Level | admin |
| Syntax | ntp client set server <IP address or host name> key-id <MD5 authentication key> |
| Default Value | [key-id]: blank |

ntp client show

| | |
|--------------|---------------------------------------|
| Description | Displays NTP summary information. |
| Access Level | limited |
| Syntax | ntp client show [oper] [state] [keys] |

ntp client show server

| | |
|--------------|--|
| Description | Displays summary information about an NTP client server. |
| Access Level | limited |
| Syntax | ntp client show server <IP address or host name> |

ntp show

| | |
|--------------|-----------------------------------|
| Description | Displays NTP summary information. |
| Access Level | limited |
| Syntax | ntp show |

NTPv4 (Network Time Protocol Version 4) Autokey authentication commands**ntp autokey-auth <disable | enable>**

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for NTPv4 Autokey authentication. |
| Access Level | super |
| Syntax | ntp autokey-auth disable
ntp autokey-auth enable |

ntp autokey-auth generate

| | |
|--------------|---|
| Description | Generates the certificate for NTPv4 Autokey authentication. |
| Access Level | super |
| Syntax | ntp autokey-auth generate |

ntp autokey-auth show

| | |
|--------------|--|
| Description | Displays NTPv4 Autokey authentication summary. |
| Access Level | limited |
| Syntax | ntp autokey-auth show |

Performance monitor (PM) commands**pm clear**

| | |
|--------------|--|
| Description | Clears the current PM statistics for the specified entity. When the [history] parameter is used, the 32 historical statistics are cleared and the current bin is not cleared. |
| Access Level | admin |
| Syntax | pm clear [port <portList> [profile <Ethernet ODU OTU>] [bin-type <15-min 24-hr untimed>] [history]] [ptp <ptpList> [profile <OpticalPower Modem Photonics>] [bin-type <15-min 24-hr untimed>] [history]] [channel <channelList> [profile <ODU>] [bin-type <15-min 24-hr untimed>] [history]] [slot <number> [history]] [all [history]] |

pm <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of the PM profile for the specified entity. |
| Access Level | admin |
| Syntax | pm <disable enable> [port <portList> [profile <Ethernet ODU OTU>]] [ptp <ptpList> [profile <OpticalPower Modem Photonics>]] [channel <channelList> [profile <ODU>]] |

pm show

| | |
|--------------|--|
| Description | Displays configuration information for performance monitoring instances. |
| Access Level | limited |
| Syntax | pm show [<auto-created port ptp channel>] |
| See also | <ul style="list-style-type: none"> • pm show port • pm show ptp • pm show channel |

pm show port

| | |
|--------------|--|
| Description | Displays configuration information for performance monitoring. |
| Access Level | limited |

| | |
|--------|---|
| Syntax | <p>pm show port <portList></p> <p>[active] [bin-number] [bin-type] [from-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <time: hh:mm:ss or hh:mm>] [history] [profile <Ethernet ODU OTU>] [scale <tera giga mega kilo none>] [to-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <time: hh:mm:ss or hh:mm>] [wide-view] <active profile scale></p> |
|--------|---|

pm show port detail

| | |
|--------------|--|
| Description | Displays detailed performance monitoring information for individual ports. |
| Access Level | limited |
| Syntax | <p>pm show port <portList> detail</p> <p>[active] [bin-number <1..96>] [bin-type <15-min 24-hr untimed>] [from-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <time: hh:mm:ss or hh:mm>] [history <active profile scale>] [profile <Ethernet ODU OTU>] [scale <tera giga mega kilo none>] [to-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <time: hh:mm:ss or hh:mm>] [wide-view] <active profile scale></p> |

pm show ptp

| | |
|--------------|--|
| Description | Displays configuration information for performance monitoring. |
| Access Level | limited |
| Syntax | <p>pm show ptp <ptpList></p> <p>[active] [bin-number] [bin-type <15-min 24-hr untimed>] [from-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <time: hh:mm:ss or hh:mm>] [history] [profile <OpticalPower Modem Photonics>] [scale <tera giga mega kilo none>] [to-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <time: hh:mm:ss or hh:mm>] [wide-view <active profile scale>]</p> |

pm show channel

| | |
|--------------|--|
| Description | Displays configuration information for performance monitoring. |
| Access Level | limited |
| Syntax | <p>pm show ptp <channellist></p> <p>[active] [bin-number] [bin-type <15-min 24-hr untimed>] [display <clear scroll>] [from-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [from-time <time: hh:mm:ss or hh:mm>] [history] [profile <ODU>] [scale <tera giga mega kilo none>] [to-date <date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [to-time <time: hh:mm:ss or hh:mm>] [wide-view <active profile scale>]</p> |

pm untimed-read-and-clear

| | |
|--------------|--|
| Description | Enables the user to clear the untimed bin after a get (read) request. By default untimed-read-and-clear is disabled. |
| Access Level | admin |
| Syntax | pm untimed-read-and-clear enable
or
pm untimed-read-and-clear disable |

Performance Monitoring (PM) TCA commands**pm tca baseline set**

| | |
|--------------|--|
| Description | Sets the PM TCA baseline. |
| Access Level | admin |
| Syntax | pm tca baseline set {chassis slot <NUMBER: 1..8> ptp <slot>/<ptp> [direction <Rx Tx>]} |

pm tca baseline clear

| | |
|--------------|--|
| Description | Clears the PM TCA baseline. |
| Access Level | admin |
| Syntax | pm tca baseline clear {chassis slot <NUMBER: 1..8> ptp <slot>/<ptp> [direction <Rx Tx>]} |

pm tca <create | delete>

| | |
|--------------|--|
| Description | Creates or deletes a PM TCA profile. |
| Access Level | admin |
| Syntax | pm tca create profile-name <Name (no !%,?*) String[1..32]> profile-type <Ethernet Modem ODU OpticalPower OTU> bin-type <15-min 24-hr untimed>
pm tca delete profile-name <profile name>
Note: Only custom PM TCA profiles can be deleted. |

pm tca copy

| | |
|--------------|---|
| Description | Creates a copy of a PM TCA profile. |
| Access Level | admin |
| Syntax | pm tca copy from-profile-name <from profile name> to-profile-name <to profile name> |

pm tca <disable | enable>

| | |
|---------------|---|
| Description | Sets the global administrative state (disabled or enabled) for PM TCA. Also sets the administrative state (disabled or enabled) for a specific channel, port, ptp or slot. |
| Access Level | admin |
| Syntax | pm tca enable [channel <slot>/<port>.<channel> profile <ODU>] [port <slot>/<port> profile <Ethernet ODU OTU>] [ptp <slot>/<ptp> profile <OpticalPower Modem>] [slot <NUMBER: 1..8>]

pm tca disable [channel <slot>/<port>.<channel> profile <ODU>] [port <slot>/<port> profile <Ethernet ODU OTU>] [ptp <slot>/<ptp> profile <OpticalPower Modem>] [slot <NUMBER: 1..8>] |
| Default Value | enabled |

pm tca Ethernet set

| | |
|----------------|---|
| Description | Sets the performance monitoring TCA Ethernet profile attributes. |
| Access Level | admin |
| Syntax | pm tca Ethernet set profile-name <profile name> [port <slot>/<port>] [Pcs-Es <NUMBER: 0..86399] [Pcs-Ses <NUMBER: 0..86399>] [Pcs-Uas <NUMBER: 0..86399>] [Rx-Crc-Pkts <NUMBER: 0..4294967295>] [Rx-Frame-Err-Ratio <Frame Error Ratio: X.XXE-YY. Ex: 1.00E-06, 1.01E-07, 0.09E-05>] [Rx-Link-Utilization <Threshold 0.000 .. NN.mmm>] [Tx-Crc-Pkts <NUMBER: 0..4294967295>] [Tx-Frame-Err-Ratio <Frame Error Ratio: X.XXE-YY. Ex: 1.00E-06, 1.01E-07, 0.09E-05>] [Tx-Link-Utilization <Threshold 0.000 .. NN.mmm>] |
| Default Values | See PMEthernetProfile . |

pm tca Modem unset

| | |
|--------------|--|
| Description | Sets the performance monitoring TCA Modem profile attributes to default settings. See PMModemProfile . |
| Access Level | admin |
| Syntax | pm tca Modem unset profile-name <profile name> [CD] [CSI] [DGD] [ESNR] [SNR-Ext] [PDL] [Rx-FecQ] [Rx-High-Corr] |

pm tca ODU set

| | |
|--------------|---|
| Description | Sets the performance monitoring TCA ODU profile attributes. |
| Access Level | admin |

| | |
|----------------|---|
| Syntax | pm tca ODU set profile-name <profile name> [channel <slot>/<port>.<channel>] [port <slot>/<port>] [SES <NUMBER: 0..86399>] [BBE <NUMBER: 0..4294967295>] [BBE-FE <NUMBER: 0..4294967295>] [ES <NUMBER: 0..86399>] [ES-FE <NUMBER: 0..86399>] [SES-FE <NUMBER: 0..86399>] [UAS <NUMBER: 0..86399>] [UAS-FE <NUMBER: 0..86399>] |
| Default Values | See PMBasicOduProfile . |

pm tca OpticalPower unset

| | |
|--------------|---|
| Description | Sets the performance monitoring TCA Optical Power profile attributes to default settings. See PMOpticalPowerProfile . |
| Access Level | admin |
| Syntax | pm tca OpticalPower unset profile-name <profile name> [Rx-Baseline-Offset] [Tx-Baseline-Offset] |

pm tca OTU unset

| | |
|--------------|---|
| Description | Sets the performance monitoring TCA OTU attributes to default settings. See PMBasicOtuProfile . |
| Access Level | admin |
| Syntax | pm tca OTU unset profile-name <profile name> [BBE] [BBE-FE] [ES] [ES-FE] [SES] [SES-FE] [UAS] [UAS-FE] |

pm tca set

| | |
|---------------|--|
| Description | Sets the PM TCA notification type. |
| Access Level | admin |
| Syntax | pm tca set notification-type <alarm event summary-alarm summary-event> |
| Default value | summary-event |

pm tca show

| | |
|--------------|--|
| Description | Displays the PM TCA profile summary information. |
| Access Level | limited |
| Syntax | pm tca show [all] [profile-name <profile name>] [profile-type <Ethernet Modem ODU OpticalPower OTU>] |

pm tca unset

| | |
|--------------|---|
| Description | Sets the PM TCA notification type to the default value. |
| Access Level | admin |

| | |
|--------|--------------------------------|
| Syntax | pm tca unset notification-type |
|--------|--------------------------------|

Physical Termination Point (PTP) commands

ptp clear ptp

| | |
|--------------|--|
| Description | Clears the current statistics for the specified PTP. |
| Access Level | admin |
| Syntax | ptp clear ptp <ptpList> statistics |

ptp <disable | enable> ptp

| | |
|---------------|---|
| Description | Sets the administrative state (disabled or enabled) of a PTP. |
| Access Level | admin |
| Prerequisite | For the PTP at a client port, a pluggable transceiver must be installed at the interface. |
| Syntax | ptp disable ptp <ptpList>
ptp enable ptp <ptpList> |
| Default Value | enabled |

ptp set ptp

| | |
|---------------|--|
| Description | Changes writable settings for the PTP at a line port.
Note: This command cannot be used to change the setting of certain attributes (Note 1) when the PTP is under SPLI control. |
| Access Level | admin |
| Prerequisites | The PTP must be disabled. See ptp <disable enable> ptp . |

| | |
|--------|---|
| Syntax | <pre>ptp set ptp <ptpList> {[adjacency-auto-provision <enabled disabled>] [adjacency- far-end-chassis <IP Address>] [adjacency-far-end-slot <String[1..31]>] [adjacency- far-end-port <String[1..31]>] [auto-span-loss-compensation <enabled disabled>] [auto-span-loss-compensation-offset <OTM Automatic Span Loss Compensation Tx Power Offset in dBm: -5.0 .. 5.0>] [channel-contention-detection-avoidance <enabled disabled>] [channel-rx-power-high-alarm-threshold <Power Threshold in dBm>] [channel-rx-power-low-alarm-threshold <Power Threshold in dBm>] [cmd-slot <NUMBER:0, 1..32, 83..96>] [forward-error-correction <enabled disabled>] [hccs-threshold <HCCS Threshold in dB (N.MM): -1.00 .. 4.00>] [prefec- sd-threshold <Pre-Fec Threshold in dB (N.MM): -1.00..3.50>] [prefec-sf-threshold <Pre-Fec Threshold in dB (N.MM): -1.00..1.00>] [frequency <Frequency in GHz(N.M): C-Band: 191250.0..196125.0 Frequency in GHz(N.M): L-Band: 186087.5..190956.2>] [laser-centering-mode <enabled disabled holdover>] [laser-centering-range <Laser Centering Range in GHz (N.M): 0.0 .. 3.0>] [laser- shutdown-mode <normal accelerated>] [laser-tuning-mode <normal accelerated fast>] [expected-restoration-time <Seconds: 0..255>] (Note 2) [line-system-type <coloured colourless contentionless cs-coloured cs-colourless>] [link- optimization-mode <Standard Non-linear SNR-Tolerant>] [manual-span-loss <OTM Manual Span Loss Tx Power boost in dB: 0.0 .. 34.0>] [modem-class < terrestrial submarine>] [rx-edfa <enabled disabled>] [rx-power-high-alarm- threshold <Power Threshold in dBm>] [rx-power-low-alarm-threshold <Power Threshold in dBm>] [spli <enabled disabled>] [tx-dispersion-compensation-mode <automatic fixed search>] (Note 2, Note 4, Note 3) [tx-dispersion-path-value <NUMBER>] (Note 2, Note 4) [tx-dispersion-precompensation-value <-44500 ..1000000 ps/nm>] (Note 2, Note 4) [tx-dispersion-search-mode-maximum <NUMBER: -89000..1044500>] (Note 3) [tx-dispersion-search-mode-minimum <NUMBER: -89000..1044500>] (Note 3) [tx-edfa <enabled disabled>] [tx-power <Transceiver Tx Power in dBm: -9.0 .. 4.0>] [tx-power-reduction-state <enabled disabled>] [tx-state <enabled disabled>]}</pre> |
|--------|---|

| | |
|----------------|--|
| Default Values | <ul style="list-style-type: none"> • [channel-contention-detection-avoidance]*: Disabled • [expected-restoration-time]: 10 seconds • [laser-centering-mode]: Enabled • [laser-centering-range]*: 2.5 GHz • [laser-shutdown-mode]: Accelerated • [laser-tuning-mode]: Normal • [link-optimization-mode]: Standard • [line-system-type]*: Coloured • [modem-class]: Terrestrial • [hccs-threshold]: 0.00 • [prefec-sd-threshold]: 0.50 • [prefec-sf-threshold]: 0.00 • [reach-limit]: 1000 • [tx-power]*: -10.0 dBm • [tx-power-reduction-state]*: Disabled • [tx-dispersion-compensation-mode]: Automatic • [tx-dispersion-search-mode-maximum]: 1044500 • [tx-dispersion-search-mode-minimum]: -89000 • [tx-dispersion-precompensation-value]: 0 ps/nm • [tx-dispersion-path-value]: 0 <p>Note 1: An asterisk (*) indicates that this attribute cannot be provisioned using the <code>ptp set ptp</code> command when the PTP is SPLI controlled.</p> <p>Note 2: The following attributes are under SPLI control only when 6500 is running Release 12.3 or higher: Tx Dispersion Compensation Mode, Tx Dispersion Pre-compensation Value, Tx Dispersion Path Value, and Expected Restoration Time. You can modify these attributes if 6500 is running pre-Release 12.3 software by disabling SPLI, modifying the attributes in the CLI, and then re-enabling SPLI.</p> <p>Note 3: If the tx-dispersion-compensation-mode is set to search, the values for tx-dispersion-search-mode-minimum and tx-dispersion-search-mode-maximum, constrains the dispersion sweep to this range.</p> <p>Note 4: For Waveserver 5 (all variants), the Tx-dispersion-compensation-mode is automatic or fixed and must be set based on the application; protected/unprotected terrestrial or submarine. For the required application settings, refer to the following tables:</p> <ul style="list-style-type: none"> • Table 122, “Line port PTP parameters - chromatic dispersion compensation details” • Table 138, “Line port PTP parameters - chromatic dispersion compensation details” |
|----------------|--|

ptp show

| | |
|-----------------|---|
| Description | Displays summary information for all PTPs. |
| Access Level | limited |
| Prerequisite | For the PTP at a client port, a pluggable transceiver must be installed at the interface. |
| Syntax | ptp show |
| Related Command | ptp show ptp |

ptp show ptp

| | |
|--------------|---|
| Description | <p>Displays status and diagnostic information for a PTP.</p> <p>For a SPLI-enabled PTP, displays SPLI-controlled configuration values for the following attributes:</p> <ul style="list-style-type: none"> • [wavelength] • [frequency] • [channel] • [tx-state] • [wavelength-spacing] • [tx-power] • [tuning-mode] • [tx-power-reduction] • [laser-centering-range] • [line-system-type] • [channel-contention-detection-avoidance] • [carrier-centering] • [expected-restoration-time] • [tx-dispersion-compensation-mode] • [tx-dispersion-precompensation-value] • [link-optimization-mode] • [tx-dispersion-path-value] <p>Note: An asterisk (*) indicates a SPLI-controlled attribute.</p> |
| Access Level | limited |
| Prerequisite | For the PTP at a client port, a pluggable transceiver must be installed at the interface. |
| Syntax | ptp show ptp <ptpList> |

ptp show ptp <slot/port> status

| | |
|--------------|--|
| Description | <p>Displays state, provision information, real-time instantaneous statistics, optical channel power, lane status and other related information for a PTP.</p> <ul style="list-style-type: none"> • Admin and Operational State • PTP information (PTP Type, XCVR Mode, Bit Rate (Gbps), Baud Rate (GBaud), and Number of Lanes) • Transmitter State • Tx Power Value (dBm) • Band Plan Channel Number • Actual Wavelength (nm) and Actual Frequency (GHz) • Minimum Spectral Width (GHz) • Supported 3/10 dBm Signal Bandwidth • Estimated Round Trip Delay (us), Estimated Unidirectional Latency (us), and Estimated Fiber Length (km) • The following instantaneous statistics related parameters: <ul style="list-style-type: none"> — Pre-FEC BER: Bit Error Rate before Forward Error Correction — Q-Factor: Q-PM representation of FEC Error Count — ESNR: Displays the Effective Signal to Noise Ratio (ESNR) — PDL: Polarization Dependent Loss — SNR/SNR-External: Signal To Noise Ratio External — Cycle Slip Indicator • Optical Channel Power (Rx/Tx) <ul style="list-style-type: none"> — Actual Optical Channel Power (dBm), Maximum power (dBm) and Time Recorded, Minimum power (dBm) and Time Recorded, and Loss of Optical Channel. • Actual Link Dispersion (ps/nm) • Compensation Dispersion (ps/nm) • Estimated Instance of DGD (ps) • Lane Status: displays following information with the corresponding lane number: <ul style="list-style-type: none"> — Actual Power (dBm) — Maximum Power (dBm) and corresponding recorded time — Minimum Power (dBm) and corresponding recorded time — High and Low thresholds for warning and alarms — Loss of Signal |
| Access Level | limited |
| Prerequisite | For the PTP at a client port, a pluggable transceiver must be installed at the interface. |

| | |
|--------|---------------------------------|
| Syntax | ptp show ptp <slot/port> status |
|--------|---------------------------------|

ptp show ptp <slot/port> ranges

| | |
|--------------|---|
| Description | <p>Displays the PTP ranges as Min/Max or High/Low based on the type of parameter.</p> <ul style="list-style-type: none"> • Frequency (GHz) as min/max • Wavelength (nm) as min/max • Tx Power - Provisioned (dBm) as min/max • Tx Pre-compensation Dispersion (ps/nm) • Tx Path Dispersion (ps/nm) • Tx Power Alarm Threshold (dBm) • Rx Power Alarm Threshold (dBm) • Tx Power Warning Threshold (dBm) • Rx Power Warning Threshold (dBm) • Rx Optical Channel Power Threshold (dBm) |
| Access Level | limited |
| Prerequisite | For the PTP at a client port, a pluggable transceiver must be installed at the interface. |
| Syntax | ptp show ptp <slot/port> ranges |

ptp show ptp <slot/port> statistics

| | |
|--------------|--|
| Description | <p>Displays real-time instantaneous statistics and other related information for a PTP.</p> <ul style="list-style-type: none"> • The following instantaneous statistics related parameters: <ul style="list-style-type: none"> — Pre-FEC BER: Bit Error Rate before Forward Error Correction — FEC Uncorrected Seconds: number of uncorrected seconds after applying Forward Error Correction. If this value is zero, it implies that packet transmission is error free. — Q-Factor: Q-PM representation of FEC Error Count — ESNR: Displays the Effective Signal to Noise Ratio (ESNR) — CFO: Carrier Frequency Offset. This parameter is not available WL5e type modules. — EVM: Error Vector Magnitude. This parameter is not available WL5e type modules — PDL: Polarization Dependent Loss. This parameter is not available WL5n type modules. — SNR/SNR-External: Signal To Noise Ratio External. This parameter is not available WL5n type modules. — Cycle Slip Indicator: This parameter is not available WL5n type modules. — Rx/Tx Actual Link Dispersion (ps/nm) — Rx/Tx Compensation Dispersion (ps/nm) — Rx/Tx Estimated Instance of DGD (ps): This parameter is not available WL5n type modules. <p>For more information about various parameters, refer to the Performance Monitoring chapter.</p> |
| Access Level | limited |
| Prerequisite | For the PTP at a client port, a pluggable transceiver must be installed at the interface. |
| Syntax | ptp show ptp <slot/port> statistics |

ptp unset ptp

| | |
|---------------|---|
| Description | Restores modified advanced attributes for a line PTP to their default values. See ptp set ptp . |
| Access Level | admin |
| Prerequisites | The PTP transmitter must be disabled. See ptp <disable enable> ptp . |

| | |
|--------|--|
| Syntax | <pre>ptp unset ptp <slot>/<ptp> [[frequency] [laser-centering-range] [rx-power-high-alarm-threshold] [tx-dispersion-precompensation-value] [link-optimization-mode] [laser-shutdown-mode] [rx-power-low-alarm-threshold] [tx-dispersion-path-value] [line-system-type] [spli] [tx-power] [modem-class] [cmd-slot] [forward-error-correction] [hccs-threshold] [prefec-sd-threshold] [prefec-sf-threshold] [channel-contention-detection-avoidance] [tx-state] [reach-limit] [laser-tuning-mode] [channel-rx-power-high-alarm-threshold] [tx-power-reduction-state] [laser-centering-mode] [channel-rx-power-low-alarm-threshold] [tx-dispersion-compensation-mode]]</pre> <p>Note: The [forward-error-correction] attribute applies only to a QSFP28 transceiver.</p> |
|--------|--|

Port commands

port <disable | enable> port

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of a provisioned port. |
| Access Level | admin |
| Prerequisite | For a client port, a pluggable transceiver must be installed at the interface. |
| Syntax | <pre>port disable port <portList> port enable port <portList></pre> |

port set port

| | |
|----------------|--|
| Description | Changes writable settings of a provisioned port. |
| Access Level | admin |
| Prerequisites | A port must be disabled before the [loopback] or [odu-termination-mode] attribute is changed (see port <disable enable> port). |
| Syntax | <pre>port set port <portList> [[label <String:[1...128]>] [conditioning-type <none laser-off ethernet otn>] [conditioning-holdoff <0ms 10ms 20ms 30ms 40ms 50ms 60ms 70ms 80ms 90ms 100ms 200ms 300ms 400ms 500ms 600ms 700ms 800ms 900ms 1000ms>] [loopback <disabled Rx Tx>] [connection-state <enabled disabled>][otu-sd-threshold <BER String: XE-YY>] [odu-sd-threshold <BER String: XE-YY>]]</pre> |
| Default Values | <ul style="list-style-type: none"> • [label]: blank • [conditioning-type]: Dependent on the client protocol. For 100GE client ports, the conditioning-type defaults to ethernet. For OTL4.4 client ports, the conditioning-type defaults to otn. • [conditioning-holdoff]: 10 ms • [loopback]: disabled • [odu-termination-mode]: terminate |

port show

| | |
|------------------|--|
| Description | Displays summary information about all provisioned ports. |
| Access Level | limited |
| Prerequisite | A pluggable transceiver must be provisioned at the client port, and at the line port, wherever a pluggable transceiver is supported. |
| Syntax | port show [ethernet-config] [status] |
| Related Commands | <ul style="list-style-type: none"> • port show port • port show summary |

port show port

| | |
|--------------|--|
| Description | Displays summary information about a provisioned port. |
| Access Level | limited |
| Prerequisite | For a client port, a pluggable transceiver must be provisioned at the interface. |
| Syntax | port show port <portList> |

port show summary

| | |
|--------------|--|
| Description | Displays summary information about all provisioned ports including, port name, speed, admin state, operational state, and description. |
| Access Level | limited |
| Prerequisite | For a client port, a pluggable transceiver must be provisioned at the interface. |
| Syntax | port show summary |

port show port <slot/port> statistics

| | |
|--------------|---|
| Description | Displays ODU, OTU, and Ethernet profile statistics of the port. For more information on various statistic parameters, refer to the ODU profile statistics , ODU profile statistics , and Ethernet profile statistics in the Performance Monitoring chapter. |
| Access Level | admin |
| Syntax | port show port <slot/port> statistics |

port trace path set port <tx-sapi | tx-dapi | tx-oper>

| | |
|--------------|--|
| Description | Transmits the SAPI, DAPI, or OPER string to the far-end chassis. |
| Access Level | admin |
| Syntax | port trace path set port <slot/port> {[tx-sapi <String: 0..15>] [tx-dapi <String: 0..15>] [tx-oper <String: 0..32>]} |

port trace path set port <exp-sapi | exp-dapi | exp-oper>

| | |
|--------------|--|
| Description | Sets the SAPI, DAPI, or OPER string expected to be returned from the far-end chassis. The expected string value is compared to the actual string value returned.
Note: Tx-oper cannot be provisioned on OTL4.4 client ports. |
| Access Level | admin |
| Syntax | port trace path set port <slot/port> {[exp-sapi <String: 0..15>] [exp-dapi <String: 0..15>] [exp-oper <String: 0..32>]} |

port trace path set port mismatch-mode

| | |
|--------------|--|
| Description | Specifies which string to use to determine whether there is a mismatch. You cannot check for a mismatch on all three strings simultaneously.
Note: Mismatch-mode cannot be provisioned on OTL4.4 client ports. |
| Access Level | admin |
| Syntax | port trace path set port <slot/port> mismatch-mode <sapi dapi sapi-and-dapi operator-only> |

port trace path set port mismatch-fail-mode

| | |
|--------------|--|
| Description | Specifies the action to take when a string mismatch is detected: do nothing, raise an alarm, or squelch traffic. Default is none (alarm-off).
Note: Mismatch-fail-mode cannot be provisioned on OTL4.4 client ports. |
| Access Level | admin |
| Syntax | port trace path set port <slot/port> mismatch-fail-mode <none alarm-on squelch-traffic> |

port trace path set port tx-oper-mode

| | |
|---------------|---|
| Description | Specifies how to manage the topology string. When tx-oper-mode is set to “automatic”, the OPER value transmitted is an automatically generated string. When the tx-oper-mode is set to “manual”, the OPER value transmitted is the value provisioned using the port trace path set port <tx-sapi tx-dapi tx-oper> command.
Note: Tx-oper-mode cannot be provisioned on OTL4.4 client ports. |
| Access Level | admin |
| Syntax | port trace path set port <slot/port> tx-oper-mode <automatic manual> |
| Default value | automatic |

port trace path set port rx-to-exp

| | |
|--------------|---|
| Description | Configures the currently received string (SAPI, DAPI, or OPER) to be the expected string. The parameter allows you to select which receive/expected pair to use (sapi, dapi, or operator). |
| Access Level | admin |
| Syntax | port trace path set port <slot/port> tx-oper-mode <automatic manual> |

port trace path show

| | |
|--------------|--|
| Description | Displays a summary of the path trace details for the port. |
| Access Level | limited |
| Syntax | port trace path show port <slot/port> |

port trace path unset

| | |
|--------------|---|
| Description | Returns the TTI strings to their default values. |
| Access Level | admin |
| Syntax | port trace path unset port <slot/port> <tx-sapi exp-sapi tx-dapi exp-dapi tx-oper exp-oper tx-oper-mode mismatch-fail-mode mismatch-mode> |

port trace section set port <tx-sapi | tx-dapi | tx-oper>

| | |
|-----------------|---|
| Description | Transmits the SAPI, DAPI, or OPER string to the far-end chassis. |
| Access Level | admin |
| Syntax | port trace section set port <slot/port> {[tx-sapi <String: 0..15>] [tx-dapi <String: 0..15>] [tx-oper <String: 0..32>]} |
| Related Command | port trace section set port <exp-sapi exp-dapi exp-oper> |

port trace section set port <exp-sapi | exp-dapi | exp-oper>

| | |
|-----------------|--|
| Description | Sets the SAPI, DAPI, or OPER string expected to be returned from the far-end chassis. The expected string value is compared to the actual string value returned. |
| Access Level | admin |
| Syntax | port trace section set port <slot/port> {[exp-sapi <String: 0..15>] [exp-dapi <String: 0..15>] [exp-oper <String: 0..32>]} |
| Related Command | port trace section set port <tx-sapi tx-dapi tx-oper> |

port trace section set port mismatch-mode

| | |
|--------------|--|
| Description | Specifies which string to use to determine whether there is a mismatch. You cannot check for a mismatch on all three strings simultaneously. |
| Access Level | admin |
| Syntax | port trace section set port <slot/port> mismatch-mode <sapi dapi sapi-and-dapi operator-only> |

port trace section set port mismatch-fail-mode

| | |
|--------------|---|
| Description | Specifies the action to take when a string mismatch is detected: do nothing, raise an alarm, or squelch traffic. Default is none (alarm-off). |
| Access Level | admin |
| Syntax | port trace section set port <slot/port> mismatch-fail-mode <none alarm-on squelch-traffic> |

port trace section set port tx-oper-mode <automatic | manual>

| | |
|---------------|---|
| Description | Specifies how to manage the topology string. When tx-oper-mode is set to “automatic”, the OPER value transmitted is an automatically generated string. When the tx-oper-mode is set to “manual”, the OPER value transmitted is the value provisioned using the port trace section set port <tx-sapi tx-dapi tx-oper> command. |
| Access Level | admin |
| Syntax | port trace section set port <slot/port> tx-oper-mode <automatic manual> |
| Default value | automatic |

port trace section set port rx-to-exp

| | |
|--------------|--|
| Description | Configures the currently received string (SAPI, DAPI, or OPER) to be the expected string. The parameter allows you to select which receive/expected pair to use (sapi, dapi, or operator). |
| Access Level | admin |
| Syntax | port trace section set port <slot/port> rx-to-exp <sapi dapi operator> |

port trace section port show

| | |
|--------------|---|
| Description | Displays a summary of the section trace details for the port. |
| Access Level | limited |
| Syntax | port trace section show port <slot/port> |

port trace section unset

| | |
|--------------|--|
| Description | Returns the TTI strings to their default values. |
| Access Level | admin |
| Syntax | port trace section unset port <slot/port> <tx-sapi exp-sapi tx-dapi exp-dapi tx-oper exp-oper tx-oper-mode mismatch-fail-mode mismatch-mode> |

port unset port

| | |
|--------------|--|
| Description | Restores modified port settings to their default values. See port set port . |
| Access Level | admin |
| Prerequisite | For a client port, a pluggable transceiver must be installed at the interface. |
| Syntax | port unset port <portList> {[label] [conditioning-type] [conditioning-holdoff] [loopback] [otu-sd-threshold] [odu-sd-threshold]} |

RADIUS commands**radius add server**

| | |
|---------------|---|
| Description | Adds a RADIUS server to the system.
Note: Waveserver 5 supports up to eight RADIUS servers. |
| Access Level | super |
| Syntax | radius add server <IP address or host name> [priority <Number:1...8>] [udp-port <Number:1...65535>] |
| Default Value | [udp-port] = 1812 |

radius <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) for all RADIUS servers or only a specific RADIUS server. |
| Access Level | super |
| Syntax | radius disable [server <IP address or host name>]
radius enable [server <IP address or host name>] |

radius remove server

| | |
|--------------|--|
| Description | Removes a RADIUS server from the system. |
| Access Level | super |
| Syntax | radius remove server <IP address or host name> |

radius set

| | |
|----------------|--|
| Description | Changes writable global RADIUS settings. |
| Access Level | super |
| Syntax | radius set {[key <Password String:8...64>] [retries <Number:0...3>] [secret <String:16...128>] [timeout <Seconds:1...30>] [search-method <cached priority>]} |
| Default Values | <ul style="list-style-type: none"> • [key]: blank • [retries]: 3 • [secret]: blank • [timeout]: 1 • [search-method]: priority |

radius set server

| | |
|---------------|---|
| Description | Changes writable settings for a RADIUS server. |
| Access Level | super |
| Syntax | radius set server <IP address or host name> [priority <Number:1...8>] [udp-port <Number:1...65535>] |
| Default Value | [udp-port]: 1812 |

radius show

| | |
|--------------|--|
| Description | Displays summary and statistic information for all RADIUS servers. |
| Access Level | limited |
| Syntax | radius show [statistics] |

radius unset

| | |
|--------------|--|
| Description | Restores RADIUS key and secret attributes settings to their default values. See radius set . |
| Access Level | super |
| Syntax | radius unset [key] [secret] |

RADSec commands**radsec <add | remove> server**

| | |
|-------------|--|
| Description | <p>Adds a RADSec server to or removes a RADSec server from the system.</p> <p>Note: Waveserver 5 supports up to eight RADSec servers. A newly added RADSec server is administratively enabled by default.</p> |
|-------------|--|

| | |
|--------------|---|
| Access Level | super |
| Syntax | radsec add server <IP address or host name> [fingerprint <a SHA256 certificate fingerprint[95]>] [priority <Number:1...8>] [port <Number:1...65535>] [trusted-dns <domain name>]
radsec remove server <IP address or host name>
Note: For the [priority] attribute, 1 is the highest priority. For the [trusted-dns] attribute, the value must be a fully qualified domain name that can accept a leading wildcard period. |

radsec <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) of all the RADSec servers. |
| Access Level | super |
| Syntax | radsec disable
radsec enable |

radsec <disable | enable> server

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for a specific RADSec server. |
| Access Level | super |
| Syntax | radsec disable server <IP address or host name>
radsec enable server <IP address or host name> |

radsec ocsdp <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) for the RADSec server Online Certificate Status Protocol (OCSP). |
| Access Level | super |
| Syntax | radsec ocsdp disable
radsec ocsdp enable |

radsec ocsdp set

| | |
|--------------|---|
| Description | Sets the RADSec server Online Certificate Status Protocol (OCSP) attributes. |
| Access Level | super |
| Syntax | radsec ocsdp set [default-responder <String[1..255]>] [nonce <on off>] [responder-preference <aia default-responder>] |

| | |
|----------------|--|
| Default Values | <ul style="list-style-type: none"> • [default-responder]: blank • [nonce]: on • [responder-preference]: aia |
|----------------|--|

radsec ocsd unset

| | |
|--------------|--|
| Description | Sets the RADSec server Online Certificate Status Protocol (OCSP) attributes to default values. See radsec ocsd set . |
| Access Level | super |
| Syntax | radsec ocsd unset [default-responder] [nonce] |

radsec set

| | |
|------------------|---|
| Description | Sets the global RADSec provisioning; this setting applies to all RADSec connections. |
| Access Level | super |
| Syntax | radsec set [cert-name <String: [1..32]>] [timeout <Seconds: 1..30>] [check-fingerprint <on off>] [check-ip-host <on off>] [search-method <cached priority>] |
| Default Values | <ul style="list-style-type: none"> • [cert-name] = blank • [timeout] = 6 • [check-fingerprint] = off • [check-ip-host] = on • [search-method] = priority |
| Related Commands | radsec set server |

radsec set server

| | |
|--------------|---|
| Description | Sets the RADSec server attributes for a specific RADSec server. |
| Access Level | super |
| Syntax | radsec set server <IP address or host name> [fingerprint <a SHA256 certificate fingerprint[95]>] [priority <Number:1...8>] [port <Number:1...65535>] [trusted-dns <domain name>]

Note: For the [priority] attribute, 1 is the highest priority. For the [trusted-dns] attribute, the value must be a fully qualified domain name that can accept a leading wildcard period. |

| | |
|----------------|---|
| Default Values | <ul style="list-style-type: none"> • [fingerprint] = blank • [priority] = 1 • [port] = 2083 • [trusted-dns] = blank |
|----------------|---|

radsec show

| | |
|-----------------|---|
| Description | Displays summary information for all configured RADSec servers.
Note: The RADSec server with the highest priority is displayed first. |
| Access Level | limited |
| Syntax | radsec show |
| Related Command | radsec show statistics |

radsec show statistics

| | |
|--------------|--|
| Description | Displays statistics information for all configured RADSec servers.
Note: The RADSec server with the highest priority is displayed first. |
| Access Level | limited |
| Syntax | radsec show statistics |

radsec unset

| | |
|-----------------|---|
| Description | Restores RADSec server cert-name and server settings (finger-print and trusted-dns) to default values. See radsec set and radsec set server . |
| Access Level | super |
| Syntax | radsec unset [cert-name] [server] |
| Related Command | radsec unset server |

radsec unset server

| | |
|--------------|--|
| Description | Restores a specific RADSec server setting to default values. See radsec set server . |
| Access Level | super |
| Syntax | radsec unset server <IP address or host name> [fingerprint] [trusted-dns] |

Service Photonic Layer Interoperations (SPLI) commands

spli adjacency show

| | |
|-------------|---|
| Description | Displays the SPLI adjacency summary table for all line ports. |
|-------------|---|

| | |
|--------------|---------------------|
| Access Level | limited |
| Syntax | spli adjacency show |

spli adjacency show ptp

| | |
|--------------|---|
| Description | Displays SPLI adjacency summary information for a particular line port. |
| Access Level | limited |
| Syntax | spli adjacency show ptp <slot>/<ptp> |

spli database delete all-unreliable

| | |
|--------------|--|
| Description | Deletes all unreliable entries in the SPLI database. |
| Access Level | admin |
| Syntax | spli database delete all-unreliable |

spli database show

| | |
|--------------|---|
| Description | Displays SPLI summary table for all line ports. |
| Access Level | limited |
| Syntax | spli database show |

spli database show ptp

| | |
|--------------|--|
| Description | Displays all SPLI database information for a particular line port. |
| Access Level | limited |
| Syntax | spli database show ptp <slot>/<ptp> |

spli <disable | enable>

| | |
|---------------|---|
| Description | Enables or disables chassis-level SPLI control. |
| Access Level | admin |
| Syntax | spli enable
spli disable |
| Default Value | enabled |

spli show

| | |
|--------------|--|
| Description | Displays chassis-level SPLI configuration information. |
| Access Level | limited |

| | |
|--------|-----------|
| Syntax | spli show |
|--------|-----------|

SNMP commands

snmp <disable | enable>

| | |
|---------------|--|
| Description | Sets the administrative state (disabled or enabled) of SNMP agent. |
| Access Level | admin |
| Syntax | snmp disable
snmp enable |
| Default Value | disabled |

snmp create access-entry

| | |
|--------------|--|
| Description | Creates an SNMP access entry. |
| Access Level | admin |
| Syntax | snmp create access-entry <String: 1...32> {sec-model <v1 v2c v3>} {sec-level <noAuthlauthNoPrivlauthWithPriv>} {read-view <SNMP View Tree: 1...32>} [write-view <SNMP View Tree: 1...32>] [notify-view <SNMP View Tree: 1...32>] |

snmp create community-index

| | |
|--------------|--|
| Description | Creates an SNMP community mapping. |
| Access Level | admin |
| Syntax | snmp create community-index <String: 1...32> {community <String: 1...64>} {sec-name <String: 1...32>} [transport-tag <String: 1...32>] [context-name <String: 1...32>] |

snmp create notify

| | |
|--------------|--|
| Description | Creates an SNMP notify entry. |
| Access Level | admin |
| Syntax | snmp create notify <String: 1...32> {notify-tag <String: 1...32>} [notify-type <trap>] |

snmp create target

| | |
|--------------|-------------------------------|
| Description | Creates an SNMP target entry. |
| Access Level | admin |

| | |
|---------------|--|
| Syntax | snmp create target <String: 1...32> {addr <IP address (with optional mask) or host name>} {param-name <String: 1...32>} [tag <String: 1...32>] [udp-port <Number: 0...65535>] [transport-domain <snmp-udp v4-udp v6-udp>] [retry-count <Number: 0...255>] [retry-timeout <Centi-seconds: 0...65535>] |
| Default Value | [tag]: TrapTag |

snmp create target-param

| | |
|--------------|--|
| Description | Creates an SNMP target parameter. |
| Access Level | admin |
| Syntax | snmp create target-param <String: 1...32> {sec-name <String>} [sec-model <v1 v2c v3>] [sec-level <noAuth authNoPriv authWithPriv>] |

snmp create user

| | |
|--------------|---|
| Description | Creates an SNMP user. |
| Access Level | admin |
| Syntax | snmp create user <String: 1...64> {auth-protocol <noAuth md5 sha>} [auth-password <String: 1...64>] [priv-protocol <aes-128 des 3des noPriv>] [priv-password <String: 1...64>] [auth-secret <String: 1...40>] [priv-secret <String: 1...64>] [engine-id <SNMP EngineID:XX:XX... (5-32 octets)>] |

snmp create viewtree

| | |
|--------------|--|
| Description | Creates an SNMP view tree. |
| Access Level | admin |
| Syntax | snmp create viewtree <String: 1...32> {sub-tree <String: 1...64>} mask <Octet String> {type <include exclude>} |

snmp delete access-entry

| | |
|--------------|--|
| Description | Deletes an SNMP access entry. |
| Access Level | admin |
| Syntax | snmp delete access-entry <String: 1...32> {sec-model <v1 v2c v3>} {sec-level <noAuth authNoPriv authWithPriv>} |

snmp delete community-index

| | |
|--------------|--|
| Description | Deletes an SNMP community mapping. |
| Access Level | admin |
| Syntax | snmp delete community-index <String: 1...32> |

snmp delete notify

| | |
|--------------|-------------------------------------|
| Description | Deletes an SNMP notify entry. |
| Access Level | admin |
| Syntax | snmp delete notify <String: 1...32> |

snmp delete target

| | |
|--------------|-------------------------------------|
| Description | Deletes an SMNP target. |
| Access Level | admin |
| Syntax | snmp delete target <String: 1...32> |

snmp delete target-param

| | |
|--------------|---|
| Description | Deletes an SNMP target parameter. |
| Access Level | admin |
| Syntax | snmp delete target-param <String: 1...32> |

snmp delete user

| | |
|---------------|---|
| Description | Deletes an SNMP user. |
| Access Level | admin |
| Syntax | snmp delete user <String: 1...64> [engine-id <SNMP Engine ID:XX:XX..(5-32 octets)>] |
| Default Value | [engine ID]: LocalSnmpEngineId |

snmp delete viewtree

| | |
|--------------|--|
| Description | Deletes an SNMP view tree. |
| Access Level | admin |
| Syntax | snmp delete viewtree <V12cView readview> sub-tree <String[1...64]> |

snmp security-to-group attach user

| | |
|--------------|---|
| Description | Attaches an SNMP user to a security group. |
| Access Level | admin |
| Syntax | snmp security-to-group attach user <String: 1...64> {sec-model <v1 v2c v3>} {group <String: 1...32>}
The current release supports SNMPv2c only. SNMPv1 and SNMPv3 are not supported. |

snmp security-to-group detach user

| | |
|--------------|--|
| Description | Detaches an SNMP user from a security group. |
| Access Level | admin |
| Syntax | snmp security-to-group detach user <String: 1...64> sec-model <v1 v2c v3>
The current release supports SNMPv2c only. SNMPv1 and SNMPv3 are not supported. |

snmp set contact

| | |
|---------------|---------------------------------------|
| Description | Sets SNMP system contact information. |
| Access Level | admin |
| Syntax | snmp set contact <String: 1...255> |
| Default Value | "Customer Support, Ciena" |

snmp set location

| | |
|---------------|--|
| Description | Sets SNMP system location information. |
| Access Level | admin |
| Syntax | snmp set location <String: 1...255> |
| Default Value | "Not Specified" |

snmp set notify

| | |
|--------------|--|
| Description | Changes writable settings for an SNMP notify entry. |
| Access Level | admin |
| Syntax | snmp set notify <String: 1...32> {notify-tag <String: 1...32>}[notify-type <trap>] |

snmp set user

| | |
|--------------|---|
| Description | Changes writable SNMP user settings. |
| Access Level | admin |
| Syntax | snmp set user <public private> {auth-protocol <noAuth md5 sha>} [auth-password <String: 1...64>] [priv-protocol <aes-128 des 3des noPriv>] [priv-password <String: 1...64>] [auth-secret <String: 1...40>] [priv-secret <String: 1...64>] engine-id <SNMP EngineID:XX:XX... (5-32 octets)>
Default:LocalSnmpEngineId |

snmp show

| | |
|-------------|--|
| Description | Displays global SNMP configuration settings. |
|-------------|--|

| | |
|--------------|--|
| Access Level | limited |
| Syntax | snmp show [access-entry] [community-map] [contact] [location] [notify] [security-to-group] [target] [target-params] [users] [viewtree] |

Software commands

software activate version

| | |
|--------------|---|
| Description | Installs a software upgrade version that was downloaded earlier and allows the software to be automatically committed. The <code>delete-from-load</code> option allows you to automatically delete the previous software load from the local machine after a new software release is installed. |
| Access Level | admin |
| Syntax | software activate version <version> [auto-commit] [delete-from-load] |

software cancel

| | |
|--------------|---|
| Description | Cancels the software upgrade in progress. |
| Access Level | admin |
| Syntax | software cancel |

software check

| | |
|--------------|---|
| Description | Checks the software version stored on either the processor (local) or on the default software server (remote) for errors. For a local software version, checks the file integrity. For a remote software version, checks the file extension and verifies that the URL is reachable. |
| Access Level | admin |

| | |
|--------|--|
| Syntax | <p>software check local version <version></p> <p>software check remote url <String> login-id <username> password <password></p> <p>Note 1: The password may contain special characters:
 ~ ! ^ & * () - _ : = + { } [] \ ; ' < > . ,
 % (use escape character %25 in the URL input)
 @ (use escape character %40 in the URL input)
 # (use escape character %23 in the URL input)
 ? (use escape character %3F in the URL input)
 / (use escape character %2F in the URL input)</p> <p>Note 2: When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (https://tools.ietf.org/html/rfc3986).</p> <p>For example, <code>software check remote url sftp://ciena@[2222:111:dddd:5444:aced::293]/262/waveserver-2.1.0.10-GA.tar.gz password ****</code></p> |
|--------|--|

software clear-log

| | |
|--------------|--|
| Description | Resets the log that is displayed by the software show log command. |
| Access Level | admin |
| Syntax | software clear-log |

software commit

| | |
|--------------|--|
| Description | Commits the activated software version. The <code>delete-from-load</code> option allows you to automatically delete the previous software load from the local machine after a new software release is installed. |
| Access Level | admin |
| Prerequisite | A software upgrade must be in progress |
| Syntax | software commit [delete-from-load] |

software delete version

| | |
|--------------|---|
| Description | Removes the local instance of the specified software version. |
| Access Level | admin |
| Syntax | software delete version <version> |

software download url

| | |
|-------------|---|
| Description | Downloads the specified software version from the from the specified URL. |
|-------------|---|

| | |
|--------------|--|
| Access Level | admin |
| Syntax | <p>software download url <url> login-id <String> password <String></p> <p>Note 1: The password may contain special characters:
 ~ ! ^ & * () - _ : = + { } [] \ ; ' < > . ,
 % (use escape character %25 in the URL input)
 @ (use escape character %40 in the URL input)
 # (use escape character %23 in the URL input)
 ? (use escape character %3F in the URL input)
 / (use escape character %2F in the URL input)</p> <p>Note 2: When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (https://tools.ietf.org/html/rfc3986).</p> <p>For example: software download url sftp://
 ciena@[2222:111:dddd:5444:aced::293]/262/waveserver-
 2.1.0.10-GA.tar.gz password ****</p> |

software install url

| | |
|--------------|--|
| Description | Downloads, installs, and activates the specified software version from the specified URL. If the software release file has been manually downloaded to a disk, Waveserver 5 supports URLs in the format: file://<absolute path to release file>. |
| Access Level | admin |

| | |
|--------|---|
| Syntax | <pre>software install url <String> [login-id <username>] [password <password>] [delete-from-load]</pre> <pre>software install url <String> [login-id <username>] [password <password>] [delete-from-load]</pre> <p>Example:</p> <pre>software install url sftp://userid@10.10.10.10/path/waveserver-2.1.0.10-GA.tar.gz password my_password</pre> <p>If the software release file was manually downloaded to Waveserver 5, use the syntax: <code>software install url file://<absolute path to release file></code></p> <p>Example:</p> <pre>software install url file:///tmp/users/diag/waveserver-2.1.0.10-GA.tar.gz.</pre> <p>Note 1: The password may contain special characters:
 <code>~!^&*()-_:=+{}[] \;'<>.,</code></p> <p>% (use escape character %25 in the URL input)
 @ (use escape character %40 in the URL input)
 # (use escape character %23 in the URL input)
 ? (use escape character %3F in the URL input)
 / (use escape character %2F in the URL input)</p> <p>Note 2: The [delete-from-load] option allows you to automatically delete the previous software load from the local machine after a new software release is installed.</p> <p>Note 3: When using an IPv6 address with the url parameter, the IPv6 address must be in square brackets as defined in the CURLOPT_URL documentation (https://curl.haxx.se/libcurl/c/CURLOPT_URL.html) and RFC 3986 (https://tools.ietf.org/html/rfc3986).</p> <p>For example, <code>software install url sftp://ciena@[2222:111:dddd:5444:aced::293]/262/waveserver-2.1.0.10-GA.tar.gz password ****</code></p> |
|--------|---|

software show

| | |
|------------------|---|
| Description | Displays software status information. |
| Access Level | limited |
| Syntax | <code>software show [slot <Number: 1..8>] [log]</code> |
| Related commands | <ul style="list-style-type: none"> software show installed software show upgrade-status |

software show installed

| | |
|-------------|---------------------------------------|
| Description | Displays software status information. |
|-------------|---------------------------------------|

| | |
|--------------|-------------------------|
| Access Level | limited |
| Syntax | software show installed |

software show upgrade-status

| | |
|--------------|---------------------------------------|
| Description | Displays software status information. |
| Access Level | limited |
| Syntax | software show upgrade-status |

SSH commands

ssh algorithm <disable | enable> encryption

| | |
|----------------|---|
| Description | Sets the administrative state (disabled or enabled) for the encryption SSH algorithm. |
| Access Level | super |
| Syntax | ssh algorithm disable encryption <aes256-gcm aes256-ctr aes128-gcm aes128-ctr>
ssh algorithm enable encryption <aes256-gcm aes256-ctr aes128-gcm aes128-ctr> |
| Default Values | <ul style="list-style-type: none"> • aes256-gcm: Enabled • aes256-ctr: Enabled • aes128-gcm: Disabled • aes128-ctr: Disabled |

ssh algorithm <disable | enable> kex

| | |
|----------------|---|
| Description | Sets the administrative state (disabled or enabled) for the key exchange SSH algorithm. |
| Access Level | super |
| Syntax | ssh algorithm disable kex <ecdh-sha2-nistp521 ecdh-sha2-nistp384 ecdh-sha2-nistp256 diffie-hellman-group14-sha1>
ssh algorithm enable kex <ecdh-sha2-nistp521 ecdh-sha2-nistp384 ecdh-sha2-nistp256 diffie-hellman-group14-sha1> |
| Default Values | <ul style="list-style-type: none"> • ecdh-sha2-nistp521: Enabled • ecdh-sha2-nistp384: Enabled • ecdh-sha2-nistp256: Enabled • diffie-hellman-group14-sha1: Enabled |

ssh algorithm <disable | enable> mac

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for the message authentication code SSH algorithm. |
| Access Level | super |
| Syntax | ssh algorithm disable mac <hmac-sha2-512 hmac-sha2-256>
ssh algorithm enable mac <hmac-sha2-512 hmac-sha2-256> |
| Syntax | shmac-sha2-512: Enabled
hmac-sha2-256: Enabled |

ssh algorithm <disable | enable> public-key-authentication

| | |
|----------------|---|
| Description | Sets the administrative state (disabled or enabled) for the public key authentication SSH algorithm. |
| Access Level | super |
| Syntax | ssh algorithm disable public-key-authentication <ssh-rsa ecdsa-sha2-nistp521 ecdsa-sha2-nistp384 ecdsa-sha2-nistp256>
ssh algorithm enable public-key-authentication <ssh-rsa ecdsa-sha2-nistp521 ecdsa-sha2-nistp384 ecdsa-sha2-nistp256> |
| Default Values | <ul style="list-style-type: none"> • ssh-rsa: Enabled • ecdsa-sha2-nistp521: Enabled • ecdsa-sha2-nistp384: Enabled • ecdsa-sha2-nistp256: Enabled |

ssh algorithm show

| | |
|--------------|---|
| Description | Displays summary information for the encryption, key exchange, message authentication code, and public key authentication SSH algorithms. |
| Access Level | limited |
| Syntax | ssh algorithm show <encryption kex mac public-key-authentication> |

ssh client connect

| | |
|--------------|---|
| Description | Establishes an SSH connection.
Note: Ensure the key exchange SSH algorithms match on the Waveserver 5 and the client you are connecting to. |
| Access Level | admin |
| Syntax | ssh client connect <IP address or host name> [user <String: 1...64>] [dss] |

ssh server <add | remove> client

| | |
|--------------|---|
| Description | Adds a valid client to, or removes a client from, the list of clients from which a connection is accepted.

Note: By default, all client IP addresses are accepted. However, after one client IP address is added, only that client, and other addresses subsequently added, are allowed to connect. |
| Access Level | super |
| Syntax | ssh server add client <IP address>
ssh server remove client <IP address> |

ssh server key activate

| | |
|--------------|---|
| Description | Activates the SSH server key file. |
| Access Level | super |
| Syntax | ssh server key activate host-key <path/filename> password <passphrase>

Note 1: The syntax “password” is used to indicate the “passphrase”.
Note 2: The passphrase must not begin with a space character.
Note 3: Waveserver 5 supports the OpenSSH format only for private authentication keys. |

ssh server key delete

| | |
|--------------|----------------------------------|
| Description | Deletes the SSH server key file. |
| Access Level | super |
| Syntax | ssh server key delete |

ssh server key delete user

| | |
|--------------|---|
| Description | Deletes the authentication key for a specific user. |
| Access Level | super |
| Syntax | ssh server key delete user <String: 1...64> |

ssh server key generate

| | |
|--------------|--|
| Description | Generates the public/private key pair for the SSH server and saves it to the specified xFTP server.

Note: The private key is encrypted and stored. |
| Access Level | super |

| | |
|---------------|--|
| Syntax | ssh server key generate key-type <rsa2048 rsa3072 rsa4096 ecdsa256 ecdsa384 ecdsa521> [force] <default-tftp-server default-ftp-server default-server default-sftp-server sftp-server <IP address or host name> login-id <String> password <String> secret <String> tftp-server <IP address or host name> ftp-server <IP address or host name> login-id <String> password <String> secret <String>>

Note 1: You must specify [login-id], [password], and [secret] for an SFTP or FTP server.

Note 2: Waveserver 5 supports the OpenSSH format only for private authentication keys.

Note 3: If the [force] option is used, the key will be overwritten. |
| Default value | [key-type]: rsa3072 |

ssh server key install user

| | |
|--------------|--|
| Description | Downloads the public authentication key file for a user from the specified xFTP server.

Note: The requested file is the specified user name string with the .pub or .pk2 extension. |
| Access Level | super |
| Syntax | ssh server key install user <String: 1...64> <default-tftp-server default-ftp-server default-server default-sftp-server sftp-server <IP address or host name> login-id <String> password <String> secret <String> tftp-server <IP address or host name> ftp-server <IP address or host name> login-id <String> password <String> secret <String>>

Note: You must specify [login-id], [password], and [secret] for FTP or SFTP servers. |

ssh server set

| | |
|----------------|--|
| Description | Changes writable SSH server settings. |
| Access Level | super |
| Syntax | ssh server set {[authentication-retries <Number: 1...3>] [listener-port <Number: 22...65535>]} |
| Default Values | <ul style="list-style-type: none"> • [authentication-retries]: 3 • [listener-port]: 22 |

ssh server show

| | |
|--------------|--|
| Description | Displays summary information about the SSH server. |
| Access Level | limited |
| Syntax | ssh server show |

ssh server show key

| | |
|--------------|--|
| Description | Displays status and fingerprint information for the SSH server key, as well as username information for the SSH public key authentication users. |
| Access Level | limited |
| Syntax | ssh server show key |

ssh server unset

| | |
|--------------|---|
| Description | Restores modified SSH server settings to their default values. See ssh server set . |
| Access Level | super |
| Syntax | ssh server unset {[authentication-retries] [listener-port]} |

SSL commands**ssl algorithm <enable | disabled> cipher-suite TLS <TLS_cipher_suite>**

| | |
|--------------|---|
| Description | Sets the administrative state (enabled or disabled) for the TLS cipher suite. |
| Access Level | super |

| | |
|--------|--|
| Syntax | <pre> ssl algorithm enable cipher-suite TLS <TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384 TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384 TLS_DHE_RSA_WITH_AES_256_CBC_SHA256 TLS_RSA_WITH_AES_256_GCM_SHA384 TLS_RSA_WITH_AES_256_CBC_SHA256 TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 TLS_DHE_RSA_WITH_AES_256_GCM_SHA384 TLS_DHE_RSA_WITH_AES_128_CBC_SHA256 TLS_RSA_WITH_AES_128_CBC_SHA256> ssl algorithm disable cipher-suite TLS <TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384 TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384 TLS_DHE_RSA_WITH_AES_256_CBC_SHA256 TLS_RSA_WITH_AES_256_GCM_SHA384 TLS_RSA_WITH_AES_256_CBC_SHA256 TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 TLS_DHE_RSA_WITH_AES_256_GCM_SHA384 TLS_DHE_RSA_WITH_AES_128_CBC_SHA256 TLS_RSA_WITH_AES_128_CBC_SHA256> </pre> |
|--------|--|

| | |
|----------------|---|
| Default Values | <ul style="list-style-type: none"> • TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384: Enabled • TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384: Enabled • TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384: Enabled • TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384: Enabled • TLS_DHE_RSA_WITH_AES_256_CBC_SHA256: Enabled • TLS_RSA_WITH_AES_256_GCM_SHA384: Enabled • TLS_RSA_WITH_AES_256_CBC_SHA256: Enabled • TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256: Disabled • TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256: Disabled • TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256: Disabled • TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256: Disabled • TLS_RSA_WITH_AES_128_CBC_SHA256: Disabled • TLS_DHE_RSA_WITH_AES_128_CBC_SHA256: Disabled • TLS_RSA_WITH_AES_128_CBC_SHA256: Disabled |
|----------------|---|

ssl algorithm show

| | |
|--------------|-------------------------------------|
| Description | Displays SSL algorithm information. |
| Access Level | limited |
| Syntax | ssl algorithm show |

ssl server rsa regenerate

| | |
|--------------|--|
| Description | <p>Regenerates a new, self-signed certificate and deletes the old certificate.</p> <p>Note: You cannot regenerate an SSL server key if a certificate is assigned to an HTTPS server. You can regenerate an SSL server key on default certificates only.</p> |
| Access Level | super |
| Syntax | <p>ssl server rsa regenerate</p> <p>Note: After successful completion of this command, all browser sessions and WebSocket sessions to the Waveserver 5 are terminated and reconnection is required.</p> |

Syslog commands

syslog add collector

| | |
|--------------|---|
| Description | Adds an available syslog collector for receiving syslog severity values. |
| Prerequisite | The collector must exist on the system (see syslog <create delete> collector). |
| Access Level | admin |

| | |
|--------|---|
| Syntax | syslog add collector <IP address or host name> severity < alert all critical debug emergency error info notice warning> |
|--------|---|

syslog add severity

| | |
|--------------|--|
| Description | Adds a value to, or removes a value from, the list of severity values that can egress to collectors. |
| Access Level | admin |
| Syntax | syslog add severity <emergency alert error warning notice info debug all> |

syslog <create | delete> collector

| | |
|--------------|---|
| Description | Creates or deletes a syslog collector that can receive syslog severity values. |
| Access Level | admin |
| Syntax | syslog create collector <IP address or host name> [custom-prefix <String: 1...15>] [facility <Number: 0...23>] [udp-port <Number: 514...65535>] [severity <emergency alert error warning notice info debug all>]
syslog delete collector <IP address or host name>
Note: Up to 16 collectors can be created. |

syslog <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative status (disabled or enabled) of syslog message logging. |
| Access Level | admin |
| Syntax | syslog disable
syslog enable |

syslog <disable | enable> collector

| | |
|--------------|---|
| Description | Sets the administrative status (disabled or enabled) for a specific syslog collector. |
| Access Level | admin |
| Syntax | syslog disable collector <IP address or host name>
syslog enable collector <IP address or host name> |

syslog remove collector

| | |
|--------------|---|
| Description | Removes a syslog collector. |
| Access Level | admin |
| Syntax | syslog remove collector <IP address or host name> |

syslog remove severity

| | |
|--------------|--|
| Description | Removes a severity from default severity list. Also, updates severity list for existing syslog collectors using the default severity list. |
| Access Level | admin |
| Syntax | syslog remove severity <emergency alert error warning notice info debug all> |

syslog send

| | |
|--------------|--|
| Description | Sends a debug syslog message. |
| Access Level | admin |
| Syntax | syslog send [event <Number: event ID> msg <String: 1...128> severity < alert critical debug emergency error info notice warning >] |

syslog set

| | |
|----------------|---|
| Description | Changes writable syslog settings for all collectors in the system or only a specific collector. |
| Access Level | admin |
| Syntax | syslog set [collector <IP address or host name>] {[custom-prefix <String: 1...15>] [facility <Number: 0...23>] [udp-port <Number: 1...65535>] [severity <all or String list: [...,...] emergency, alert, error, warning, notice, info, debug>]} |
| Default Values | <ul style="list-style-type: none">• [custom-prefix]: blank• [facility]: 3• [udp-port]: 514• [severity]: blank (none) |

syslog show

| | |
|--------------|--|
| Description | Displays summary information for the syslog configuration. |
| Access Level | limited |

| | |
|------------------|--|
| Syntax | <p>syslog show [default-attributes] [severity-mapping]</p> <p>Note: The [severity-mapping] option displays the mapping between the ITU Event Severity designations and the Waveserver 5 Syslog Event designations, as shown below:</p> <p>Event Severity, Syslog Severity</p> <ul style="list-style-type: none"> • cleared (1), notice (5) • indetermin (2), notice (5) • critical (3), emergency (0) • major (4), alert (1) • minor (5), error (3) • warning (6), warning (4) • config (7), notice (5) • info (8), info (6) • debug (9), debug (7) • n/a, critical (2) |
| Related Commands | syslog show collector |

syslog show collector

| | |
|--------------|--|
| Description | Displays summary information for a specific collector. |
| Access Level | limited |
| Syntax | syslog show collector <IP address or host name> |

syslog tls <add | remove> collector severity

| | |
|--------------|--|
| Description | Adds or removes a TLS syslog collector severity. |
| Prerequisite | The collector must exist on the system. See syslog tls <create delete> collector . |
| Access Level | admin |
| Syntax | <p>syslog tls add collector <IP address or host name> severity <emergency alert error warning notice info debug all></p> <p>syslog tls remove collector <IP address or host name> severity <emergency alert error warning notice info debug all></p> |

syslog tls clear statistics

| | |
|--------------|---|
| Description | Clears all TLS syslog statistics for a specified collector or all collectors. |
| Access Level | admin |
| Syntax | syslog tls clear [collector <IP address or host name>] statistics |

syslog tls <create | delete> collector

| | |
|--------------|--|
| Description | Creates or deletes a TLS syslog collector that can receive TLS syslog severity values. |
| Access Level | admin |
| Syntax | syslog tls create collector <IP address or host name> [custom-prefix <String: 1..15>] [fingerprint <fingerprint>] [facility <Number: 0..24>] [port <Number: 1..65535>] [severity <emergency alert error warning notice info debug all>] [trusted-dns]
syslog tls delete collector <IP address or host name> |

syslog tls <disable | enable>

| | |
|--------------|--|
| Description | Sets the global administrative state (disabled or enabled) for TLS syslog. |
| Access Level | admin |
| Syntax | syslog tls disable
syslog tls enable |

syslog tls <disable | enable> collector

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for a specific TLS syslog collector. |
| Access Level | admin |
| Syntax | syslog tls disable [collector <IP address or host name>]
syslog tls enable [collector <IP address or host name>] |

syslog tls ocsp <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) for the TLS syslog OCSP. |
| Access Level | super |
| Syntax | syslog tls ocsp disable
syslog tls ocsp enable |

syslog tls ocsp set

| | |
|--------------|--|
| Description | Sets the TLS syslog OCSP attributes. |
| Access Level | super |
| Syntax | syslog tls ocsp set [default-responder <String: 1..255>] [nonce <on off>] [responder-preference <aia default-responder>] |

| | |
|----------------|--|
| Default Values | <ul style="list-style-type: none"> • [default-responder]: blank • [nonce]: on • [responder-preference]: aia |
|----------------|--|

syslog tls ocap show

| | |
|--------------|--|
| Description | Displays the TLS syslog OCSP attributes. |
| Access Level | limited |
| Syntax | syslog tls ocap show |

syslog tls set

| | |
|------------------|---|
| Description | Sets the TLS syslog attributes for all collectors. |
| Access Level | super |
| Syntax | syslog tls set [cert-name <certificate name>] [check-fingerprint <on off>] [check-ip-host <on off>] [custom-prefix <String: 1..15>] [facility <Number: 0..23>] [port <Number: 1..65535>] [severity <emergency alert error warning notice info debug all>] [timeout <Number: 1..30>] |
| Default Values | <ul style="list-style-type: none"> • [cert-name]: blank • check-fingerprint: off • check-ip-host: on • [custom-prefix]: blank • [facility]: 3 • [port]: 6514 • [severity]: emergency, alert • timeout: 6 |
| Related Commands | syslog tls set collector |

syslog tls set collector

| | |
|--------------|---|
| Description | Sets the TLS syslog attributes for a specified collector. |
| Access Level | admin |
| Syntax | <p>syslog tls set collector <IP address or host name> [custom-prefix <String: 1..15>] [facility <Number: 0..23>] [fingerprint <fingerprint>] [port <Number: 1..65535>] [severity <emergency alert error warning notice info debug all>] [trusted-dns <trusteddns>]</p> <p>Note: The [trusted-dns] attribute must be a Fully Qualified Domain Name (FQDN), no wildcards(*) or quotes("").</p> |

| | |
|----------------|--|
| Default Values | <ul style="list-style-type: none">• [custom-prefix]: blank• [facility]: 3• [fingerprint]: blank• [port]: 6514• [severity]: emergency, alert• trusted-dns: blank |
|----------------|--|

syslog tls show

| | |
|------------------|---|
| Description | Displays the TLS syslog summary information all collectors. |
| Access Level | limited |
| Syntax | syslog tls show [default-attributes] [statistic] |
| Related Commands | syslog tls show collector |

syslog tls show collector

| | |
|--------------|--|
| Description | Displays the TLS syslog summary information for a specified collector. |
| Access Level | limited |
| Syntax | syslog tls show collector <IP address or host name> [statistic] |

syslog tls unset

| | |
|------------------|---|
| Description | Sets the TLS syslog attributes to default values for all collectors. See syslog tls set . |
| Access Level | super |
| Syntax | syslog tls unset [cert-name] [check-fingerprint] [check-ip-host] [custom-prefix] [facility] [port] [severity] [timeout] |
| Related Commands | syslog tls unset collector |

syslog tls unset collector

| | |
|--------------|---|
| Description | Sets the TLS syslog attributes to default values for a specified collector. See syslog tls set collector . |
| Access Level | admin |
| Syntax | syslog tls unset collector <IP Address or host name> [custom-prefix] [facility] [fingerprint] [port] [severity] [trusted-dns] |

syslog unset

| | |
|--------------|---|
| Description | Restores syslog settings for all collectors in the system to their default values. |
| Access Level | admin |
| Syntax | syslog unset [collector <IP address or host name>] {[custom-prefix] [facility] [udp-port] [severity]}
Note: The attribute [severity] is not applicable to collectors. |

System commands**system auto-config-recovery config-mismatch-detection <disable | enable>**

| | |
|--------------|--|
| Description | Sets the administrative state to be disabled or enabled for the auto configuration recovery configuration mismatch detection for CP replacement. |
| Access Level | super |
| Syntax | system auto-config-recovery config-mismatch-detection disable
system auto-config-recovery config-mismatch-detection enable |
| Default | Enabled |

system auto-config-recovery <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state to be disabled or enabled for the auto configuration recovery automatic backup of the CP and AP. |
| Access Level | super |
| Syntax | system auto-config-recovery disable
system auto-config-recovery enable |
| Default | Disabled |

system auto-config-recovery force-backup

| | |
|--------------|--|
| Description | Forces the CP to copy the backup information to the AP for the auto configuration recovery automatic backup. |
| Access Level | super |
| Syntax | system auto-config-recovery force-backup |

system auto-config-recovery override-config-mismatch

| | |
|--------------|---|
| Description | Sets the auto configuration recovery to override the configuration mismatch for CP replacement. |
| Access Level | super |

| | |
|--------|---|
| Syntax | system auto-config-recovery override-config-mismatch
Note: Use this CLI command with caution, as if the provisioned traffic configuration on the CP does not match the traffic configuration running on the modules in the chassis, traffic will be lost. |
|--------|---|

system auto-config-recovery set

| | |
|---------------|--|
| Description | Sets the attributes for the auto configuration recovery automatic backup. |
| Access Level | super |
| Syntax | system auto-config-recovery set [passphrase <String: [8..64]>] [audit-interval <Minutes: 0..28800>] |
| Default value | <ul style="list-style-type: none"> • [passphrase] = None • [audit-interval] = 10 minutes |

system auto-config-recovery show

| | |
|--------------|--|
| Description | Displays the auto configuration recovery summary information for CP replacement. |
| Access Level | limited |
| Syntax | system auto-config-recovery show |

system auto-config-recovery unset

| | |
|--------------|---|
| Description | Restores the auto configuration recovery automatic backup attributes to default values. See system auto-config-recovery set . |
| Access Level | super |
| Syntax | system auto-config-recovery unset [passphrase] [audit-interval] |

system defaults set

| | |
|--------------|---|
| Description | Sets chassis-wide port defaults for conditioning type and/or conditioning holdoff
Note: This command assigns default values to these attributes for all ports on a chassis, overwriting any previously set default values. The value of these attributes for newly created ports is taken from the system defaults. |
| Access Level | admin |
| Syntax | system defaults set [conditioning-type <laser-off none protocol-specific>]
[conditioning-holdoff <0ms 10ms 20ms 30ms 40ms 50ms 60ms 70ms 80ms 90ms 100ms 200ms 300ms 400ms 500ms 600ms 700ms 800ms 900ms 1000ms>] [terrestrial-reach-limit <250, 500, 1000, 2000, 3000, 4000, 5000, 8000>] |

| | |
|----------------|--|
| Default Values | <ul style="list-style-type: none"> • [conditioning-type] = protocol-specific • [conditioning-holdoff] = 10 ms • [terrestrial-reach-limit] = 1000 km |
|----------------|--|

system defaults unset

| | |
|--------------|---|
| Description | Sets the conditioning type and conditioning holdoff to factory defaults. |
| Access Level | admin |
| Syntax | system defaults unset conditioning-type
system defaults unset conditioning-holdoff |

system defaults show

| | |
|--------------|-------------------------------|
| Description | Displays the system defaults. |
| Access Level | limited |
| Syntax | system defaults show |

system environment diag set

| | |
|---------------|--|
| Description | Changes the SSH diag user shell type to CLI or system (Linux). |
| Access Level | super |
| Syntax | system environment diag set [shell <cli system>] [sudo <on off>] |
| Default Value | cli |

system environment diag unset

| | |
|--------------|--|
| Description | Restores the system environment diag user shell type to default value. See system environment diag set . |
| Access Level | super |
| Syntax | system environment diag unset shell |

system environment root set

| | |
|--------------|---|
| Description | Changes the Linux root account password. |
| Access Level | super |
| Syntax | system environment root set [password <String: 1..128>] [secret <String[1..256]>] |

system environment show

| | |
|-------------|---|
| Description | Displays summary information for the system environment parameters. |
|-------------|---|

| | |
|--------------|-------------------------|
| Access Level | limited |
| Syntax | system environment show |

system lamps enable

| | |
|--------------|---|
| Description | Causes all LEDs or a subset of LEDs on a Waveserver 5 to flash for identification purposes. |
| Access Level | admin |
| Syntax | system lamps enable |
| Defaults | System lamp-flash is disabled by default
[timeout]: 30 minutes |

system lamps disable

| | |
|--------------|--|
| Description | Disables the lamp-flash feature. |
| Access Level | admin |
| Syntax | system lamps disable |
| Defaults | System lamp-flash is disabled by default |

system lamps set

| | |
|--------------|--|
| Description | Configures the settings for the Waveserver 5 lamp flash feature. The lamp flash feature can be performed on slots provisioned with active modules. It can also be used on individual lamps per port or on an entire chassis. |
| Access Level | admin |
| Syntax | system lamps set target {[slot <NUMBER: 1..8>] [port <slot/port>] [chassis] [timeout <SECONDS: 0..4294967295>]}

Note: A zero-minute timeout setting will result in permanently flashing LEDs until <code>system lamps</code> is disabled or Waveserver 5 is restarted. |
| Default | System lamp-flash is disabled by default
[timeout]: 30 minutes |

system lamps show

| | |
|--------------|---|
| Description | Shows the Admin State, Lamp Mode, and Lamp Target settings for the Waveserver 5 lamp flash feature. |
| Access Level | limited |
| Syntax | system lamps show |

system lamps unset

| | |
|--------------|--|
| Description | Restores system lamps settings to their default values. See system lamps set . |
| Access Level | admin |
| Syntax | system lamps unset |

system management query-limits netconf set

| | |
|--------------|--|
| Description | Sets the NETCONF query limit for PM history bins. |
| Access Level | admin |
| Syntax | system management query-limits netconf set pm-history-bins <NUMBER: 0..96> |
| Default | 96 |

system management query-limits netconf unset

| | |
|--------------|---|
| Description | Sets the NETCONF query limit for PM history bins to default value. See system management query-limits netconf set . |
| Access Level | admin |
| Syntax | system management query-limits netconf unset pm-history-bins |

system management query-limits rest set

| | |
|--------------|---|
| Description | Sets the REST query limit for PM history bins. |
| Access Level | admin |
| Syntax | system management query-limits rest set pm-history-bins <NUMBER: 0..96> |
| Default | 96 |

system management query-limits rest unset

| | |
|--------------|--|
| Description | Sets the REST query limit for PM history bins to default values. See system management query-limits rest set . |
| Access Level | admin |
| Syntax | system management query-limits rest unset pm-history-bins |

system management query-limits gnmi set

| | |
|--------------|---|
| Description | Sets the gNMI query limit for PM history bins. |
| Access Level | admin |
| Syntax | system management query-limits gnmi set pm-history-bins <NUMBER: 0..96> |

| | |
|---------|----|
| Default | 96 |
|---------|----|

system management query-limits gnmi unset

| | |
|--------------|--|
| Description | Sets the gNMI query limit for PM history bins to default values. |
| Access Level | admin |
| Syntax | system management query-limits gnmi unset |

system management query-limits show

| | |
|--------------|--|
| Description | Displays the summary information for NETCONF and REST query limit for PM history bins. |
| Access Level | limited |
| Syntax | system management query-limits show |

system server grpc <enable | disable>

| | |
|---------------|---|
| Description | Sets the administrative state of the gRPC server. |
| Access Level | admin |
| Syntax | system server grpc enable
system server grpc disable |
| Default Value | enabled |

system server grpc show

| | |
|--------------|---|
| Description | Displays a list of gRPC servers available in the system and the administrative state (enabled or disabled) of each one. Also displays the mutual authentication status of the gRPC servers (One-way or Mutual). |
| Access Level | limited |
| Syntax | system server grpc show |

system server grpc mutual-authentication <enable | disable>

| | |
|--------------|---|
| Description | Sets the authentication state of a gRPC server (One-way or Mutual). |
| Access Level | super |
| Syntax | system server grpc mutual-authentication enable
system server grpc mutual-authentication disable |

system server grpc set cert-name

| | |
|--------------|---|
| Description | Assigns an installed certificate to a gRPC server. After the certificate is assigned, the gRPC server will no longer use the default certificate. |
| Access Level | super |
| Syntax | system server grpc set cert-name {String: 1..32} |

system server grpc unset cert-name

| | |
|--------------|---|
| Description | Removes the configured certificate from the gRPC server and reverts to the default certificate. |
| Access Level | super |
| Syntax | system server grpc unset cert-name |

system server https <enable | disable>

| | |
|---------------|---|
| Description | Sets the administrative state (enabled or disabled) of the HTTPs server. |
| Access Level | admin |
| Syntax | system server https enable
system server https disable |
| Default Value | enabled |

system server https mutual-authentication <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of the HTTPS server mutual authentication. |
| Access Level | super |
| Syntax | system server https mutual-authentication disable
system server https mutual-authentication enable |

system server https ocsp <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) of the HTTPS OCSP. |
| Access Level | super |
| Syntax | system server https ocsp disable
system server https ocsp enable |

system server https ocsp set

| | |
|-------------|---|
| Description | Sets the HTTPS server Online Certificate Status Protocol (OCSP) attributes. |
|-------------|---|

| | |
|----------------|---|
| Access Level | super |
| Syntax | <p>system server https ocsf set [default-responder <OCSP responder URL[1..127]>] [nonce <on off>] [responder-preference <aia default-responder>]</p> <p>Note 1: OCSP responder URL should be in URL format, for example, <code>http://1.2.3.4:2440</code>. Only http is supported.</p> <p>Note 2: When using an IPv6 address with the <code>default-responder <OCSP responder URL[1..127]></code>, the IPv6 address must be in square brackets as defined in RFC 3986 (https://tools.ietf.org/html/rfc3986).</p> <p>For example, <code>system server https ocsf set default-responder http://[2222:111:dddd:5444:aced::293]</code></p> <p>Note 3: If the [responder-preference] is not set to “default-responder”, the URL given will be used only if no responder URL is specified in the client certificate being verified.</p> <p>Note 4: If the [responder-preference] is set to “aia” the OCSP responder referred in the client certificate AIA extension will be used first during OCSP certificate validation. If it is set to default-responder, the configured default OCSP is used first.</p> |
| Default Values | <ul style="list-style-type: none"> • [default-responder]: blank • [nonce]: on • [responder-preference]: aia |

system server https ocsf unset

| | |
|--------------|--|
| Description | Sets the HTTPS server Online Certificate Status Protocol (OCSP) attributes to default values. See system server https ocsf set . |
| Access Level | super |
| Syntax | system server https ocsf unset [default-responder] [nonce] [responder-preference] |

system server https set cert-name

| | |
|--------------|---|
| Description | Assigns an installed RSA or ECC certificate to an HTTPS server. After the certificate is assigned, the HTTPS server will no longer use the default RSA certificate. |
| Access Level | super |
| Syntax | system server https set cert-name <String: [1..32]> |

system server https set trusted-dns

| | |
|--------------|---|
| Description | Sets the trusted DNS for the HTTPS server. When set, Waveserver 5 checks the HTTPS server trusted DNS name against CN/SAN of the peer certificate during mutual authentication. |
| Access Level | super |

| | |
|--------|---|
| Syntax | system server https set trusted-dns <domain name>
Note: For the [trusted-dns] attribute, the value must be a fully qualified domain name that can accept a leading wildcard period. |
|--------|---|

system server set global-inactivity timer

| | |
|---------------|--|
| Description | Sets the administrative state of the global inactivity timeout (on or off) for all management interfaces like CLI, Web UI, and RESTCONF. |
| Access Level | super |
| Syntax | system server set global-inactivity-timer <on off> |
| Default value | on |

system server set global-inactivity-timeout

| | |
|-----------------|--|
| Description | Sets the global inactivity timeout in minutes for all management interfaces like CLI, Web UI, and RESTCONF. |
| Access Level | super |
| Syntax | system server set global-inactivity-timeout <MINUTES: 1..1500>
Note: If you have configured the https inactivity-timeout than it will take precedence over global-inactivity-timeout for Web UI connections. |
| Default value | 10 (minutes) |
| Related Command | system server https set inactivity-timeout |

system server unset global-inactivity-timeout

| | |
|--------------|--|
| Description | Disables the global inactivity timeout for all management interfaces like CLI, Web UI, and RESTCONF. |
| Access Level | super |
| Syntax | system server unset global-inactivity-timeout |

system server https set inactivity-timeout

| | |
|--------------|---|
| Description | Sets the HTTPS server session inactivity timeout in minutes. It only applies to HTTPS sessions, so Web UI and RESTCONF sessions can have a different timeout value than the other management systems.
Note: If the global-inactivity-timer is disabled, it will take precedence, and Web UI sessions have an effectively infinite timeout/cookie expiration time. |
| Access Level | super |

| | |
|--------|--|
| Syntax | <p>system server https set inactivity-timeout <MINUTES: 1..1500></p> <p>Note: Waveserver 5 requires you to login again if the HTTPS session is inactive for a period of time that exceeds the HTTPS inactivity timeout setting. Depending on the browser you are using, Waveserver 5 may automatically renew your session using the credentials stored in the browser's cache and a login window will not be displayed. To confirm that your session was automatically renewed, view the login timestamps in the command log.</p> |
|--------|--|

system server https show

| | |
|--------------|---|
| Description | Displays a list of the HTTPS servers available in the system and the administrative state (enabled or disabled) of each one. |
| Access Level | limited |
| Syntax | system server https show |

system server https unset cert-name

| | |
|--------------|---|
| Description | Removes the configured certificate name from the HTTPS server and reverts the certificate to the default RSA certificate. |
| Access Level | super |
| Syntax | system server https unset cert-name |

system server https unset inactivity-timeout

| | |
|--------------|---|
| Description | Resets the HTTPS server session inactivity timeout value to 10 minutes. |
| Access Level | admin |
| Syntax | system server https unset inactivity-timeout |

system server https unset trusted-dns

| | |
|--------------|--|
| Description | Resets the HTTPS server trusted DNS name to the default value (blank). |
| Access Level | super |
| Syntax | system server https unset trusted-dns |

system server https web-ui-file-transfer enable | disable

| | |
|--------------|--|
| Description | Administratively enables or disables the file transfer feature on the Waveserver 5 homepage. |
| Access Level | admin |
| Syntax | <p>system server https web-ui-file-transfer enable</p> <p>system server https web-ui-file-transfer disable</p> |

system server https web-ui-file-transfer show

| | |
|--------------|---|
| Description | Displays the administrative state of the file transfer feature for the Waveserver 5 homepage. |
| Access Level | admin |
| Syntax | system server https web-ui-file-transfer show |

system server netconf <enable | disable>

| | |
|---------------|---|
| Description | Sets the administrative state (enabled or disabled) of the NETCONF server. |
| Access Level | admin |
| Syntax | system server netconf enable
system server netconf disable |
| Default Value | enabled |

system server netconf show

| | |
|--------------|---|
| Description | Displays a list of the NETCONF servers available in the system and the administrative state (enabled or disabled) of each one. |
| Access Level | limited |
| Syntax | system server netconf show |

system server scp <disable | enable>

| | |
|--------------|---|
| Description | Determines the administrative state (enabled or disabled) of the SCP server. |
| Access Level | admin |
| Syntax | system server scp disable
system server scp enable |

system server scp show

| | |
|--------------|---|
| Description | Displays the administrative state (enabled or disabled) of the SCP server. |
| Access Level | limited |
| Syntax | system server scp show |

system server sftp <disable | enable>

| | |
|--------------|--|
| Description | Determines the administrative state (enabled or disabled) of the SFTP server. |
| Access Level | admin |

| | |
|--------|---|
| Syntax | system server sftp disable
system server sftp enable |
|--------|---|

system server sftp show

| | |
|--------------|--|
| Description | Displays the administrative state (enabled or disabled) of the SFTP server. |
| Access Level | limited |
| Syntax | system server sftp show |

system server show

| | |
|--------------|---|
| Description | Displays a list of the servers available in the system and the administrative state (enabled or disabled) of each one.

It also displays the session settings configured on the system, which includes the administrative state of the global-inactivity-timer (disabled or enabled) and global-inactivity-timeout value (like 10 min). |
| Access Level | limited |
| Syntax | system server show |

system set

| | |
|--------------|---|
| Description | Changes writable system settings.

Ciena recommends that when you use this command to set the time you also set the date. When setting the date and time, do it in two steps: first set the date, wait for one or two minutes, then set the time. |
| Access Level | admin |
| Syntax | system set {[date <Date: yyyy-mm-dd or yy-mm-dd or mm-dd>] [host-name <String: 2...63>] [time <Time: hh:mm:ss or hh:mm>] [time-offset <Number: - 43200...50400>] [timestamp <local UTC>]}

Note: The host-name has the following rules:

- Must be between 2 to 63 characters. Host-name strings that exceed 63 characters raise a " String is too long" warning message.

- Must contain only alphanumeric characters, hyphens, periods, and underscores. Hyphens are not supported as the first or last character. Periods are not supported as the first character. Using unsupported characters raises an " Invalid Hostname" warning message. |

| | |
|----------------|---|
| Default Values | <ul style="list-style-type: none"> • [date]: blank • [host-name]: Waveserver 5 • [time]: blank • [time-offset]: 0 • [timestamp]: local |
|----------------|---|

system set group-id

| | |
|----------------|--|
| Description | Changes writable system group settings. |
| Access Level | admin |
| Syntax | system set group-id <Number: 1...99> [group-name <String: 1...32>] [group-description <String: 1...64>]
Note: A group name cannot contain ! % , ? * characters. Also, a string that contains one or more spaces must be enclosed in quotation marks. |
| Default Values | <ul style="list-style-type: none"> • [group-id]: 0 • [group-name]: blank • [group-description]: blank |

system set member-description

| | |
|--------------|---|
| Description | Sets the member description. |
| Access Level | admin |
| Syntax | system set member-description <String[1..64]> |

system set member-frame-id

| | |
|--------------|---|
| Description | Sets the member frame ID. |
| Access Level | admin |
| Syntax | system set member-frame-id <String[1..128]> |

system set member-id

| | |
|--------------|---------------------------------------|
| Description | Sets the member ID. |
| Access Level | admin |
| Syntax | system set member-id <NUMBER: 1..254> |

system set member-name

| | |
|-------------|-----------------------|
| Description | Sets the member name. |
|-------------|-----------------------|

| | |
|--------------|--|
| Access Level | admin |
| Syntax | system set member-name <Name (no !%,?*) String[1..32]> |

system set member-rack-unit-number

| | |
|--------------|--|
| Description | Sets the member rack unit number. |
| Access Level | admin |
| Syntax | system set member-rack-unit-number <NUMBER: 1..50> |

system set network-id

| | |
|--------------|---|
| Description | Sets the system network ID. |
| Access Level | admin |
| Syntax | system set network-id <Number: 1...254> |

system set network-name

| | |
|--------------|--|
| Description | Sets the system network name.
Note: Network name string cannot contain ! % , ? * characters. |
| Access Level | admin |
| Syntax | system set network-name <String[1...32]> |

system set network-description

| | |
|--------------|--|
| Description | Sets the system network description.
Note: If the network description string contains one or more spaces then it must be enclosed in quotation marks; for example, " My Test Network". |
| Access Level | admin |
| Syntax | system set network-description <String[1...64]> |

system set site-id

| | |
|--------------|--|
| Description | Changes writable system site settings. |
| Access Level | admin |

| | |
|----------------|---|
| Syntax | <p>system set site-id <Number: 1...65535> [site-name <String: 1...31>] [site-description <String: 1...64>] [site-latitude <Number: -90...90>] [site-longitude <Number: -180...180>] [site-address <String: 1...128>]</p> <p>Note 1: Site name, site description, and site address strings cannot contain ! % , ? * characters. Also, a string that contains one or more spaces must be enclosed in quotation marks; for example, " Main Street Data Center".</p> <p>Note 2: The value (in degrees) specified for the attribute site-latitude or site-longitude must contain at least five decimal places.</p> |
| Default Values | <ul style="list-style-type: none"> • [site-id]: 0 • [site-name]: blank • [site-description]: blank • [site-latitude]: 0.00000 • [site-longitude]: 0.00000 • [site-address]: blank |

system shell session set

| | |
|----------------|--|
| Description | Changes system shell settings for the current session. |
| Access Level | limited |
| Syntax | system shell session set {[more <on off>] [more-lines <Number: 0...999>] [tab-more <on off>] [window-height <Number: 0...999>] [window-width <Number: 0...999>]} |
| Default Values | <ul style="list-style-type: none"> • [more]: on • [more-lines]: 0 • [tab-more]: on • [window-height]: 0 • [window-width]: 0 |

system shell session show

| | |
|--------------|--|
| Description | Displays system shell information for the current session. |
| Access Level | limited |
| Syntax | system shell session show |

system shell session unset

| | |
|--------------|---|
| Description | Restores system shell settings for the current session to their default values (see system shell session set). |
| Access Level | limited |
| Syntax | system shell session unset [more] [more lines] [tab-more] |

system shell set

| | |
|----------------|---|
| Description | Changes global system shell settings. |
| Access Level | admin |
| Syntax | system shell set [login-authentication-message <on off>] [login-banner-file <String: 1...63>] [welcome-banner-file <String: 1...63>] [global-more <on off>] [global-more-lines <Number: 0...999>] |
| Default values | <ul style="list-style-type: none"> • [login-authentication-message]: on • [global-more]: off • [global-more-lines]: 0 |

system shell show

| | |
|--------------|--|
| Description | Displays global system shell information.
Note: This command does not display administrative state of global-inactivity-timer and the value of global-inactivity-timeout. For this you can execute the system server show command. |
| Access Level | limited |
| Syntax | system shell show |

system shell unset

| | |
|--------------|---|
| Description | Restores global system shell settings to their default values (see system shell set). |
| Access Level | admin |
| Syntax | system shell unset [login-authentication-message] [login-banner-file] [welcome-banner-file] [global-more] [global-more-lines] |

system show

| | |
|--------------|--|
| Description | Displays summary information for the system. |
| Access Level | limited |
| Syntax | system show [date] [configuration] [host-name] [time] [time-offset] [uptime] |

system sleep

| | |
|--------------|--|
| Description | Suspends interactive processing for the specified duration. Applies to the current session only.
Note: The system sleep command set is intended for use to test scripts or configuration files. This command set is not supported and must be used with caution. |
| Access Level | admin |

| | |
|---|---|
| Syntax | system sleep for <duration: number/time format N[yMwdhms] milliseconds: duration in milliseconds> [message <String: 1...128>] [verbose] |
| Note: For sleep duration, to specify 1 hour, 10 minutes and 3 seconds, the format is: 1h10m3s. | |

system state-dump

| | |
|---------------|---|
| Description | Dumps the system state to a file that can be saved to a specified location. |
| Access Level | super |
| Prerequisites | To access a default remote SFTP or SCP server, the server must be configured on the system (see system xftp set sftp-server or system xftp set scp-server). |
| Syntax | <p>system state-dump file-name <String> [default-server default-sftp-server default-scp-server sftp-server <IP address or host name> login-id <String: 1..32> password <String: 1..128> secret <String: 1..256> scp-server <IP address or host name> login-id <String: 1..32> password <String: 1..128> secret <String: 1..256> url <String>]</p> <p>Note 1: You must specify [login-id] with the corresponding [password] or [secret] for SCP or SFTP servers.</p> <p>Note 2: The file-name string supports alphanumeric characters (space not supported) including these special characters: ~ @ % ^ * _ = + { } [] : ?. However, some strings may not be allowed as valid filenames due to your management platform system's OS restrictions. For example, if your management platform system has Windows OS then you cannot use the ? (question mark) character in the file-name string, although it is supported by the Waveserver 5 and allowed on a Linux system.</p> |

system unset

| | |
|--------------|--|
| Description | Restores system settings to their default values. See system set . |
| Access Level | admin |
| Syntax | system unset {[host-name] [time-offset] [timestamp]} |

system unset group-id

| | |
|--------------|---|
| Description | Restores system group settings to their default values. See system set group-id . |
| Access Level | admin |
| Syntax | system unset group-id group-name group-description |

system unset site-id

| | |
|--------------|--|
| Description | Restores system site settings to their default values (see system set site-id). |
| Access Level | admin |

| | |
|--------|---|
| Syntax | system unset site-id <Number: 1...65535> [site-name] [site-description] [site-latitude] [site-longitude] [site-address] |
|--------|---|

system xftp getfile

| | |
|--------------|---|
| Description | Retrieves a file from a remote xFTP server. |
| Access Level | admin |
| Syntax | system xftp getfile remote-filename <String: full directory path and filename> [local-filename <String: 1..127>] [default-server] [default-sftp-server] [default-scp-server] [sftp-server <IP address or host name>] [scp-server <IP address or host name>] [login-id <String>] [password <String>]
Note: You must specify [login-id] and [password] for SCP or SFTP server. |

system xftp putfile

| | |
|--------------|---|
| Description | Saves a file to a remote xFTP server. |
| Access Level | admin |
| Syntax | system xftp putfile local-filename <String: full directory path and filename> [remote-filename <String: 1..127>] <default-server default-sftp-server default-scp-server sftp-server <IP address or host name> login-id <String> password <String> secret <String> scp-server <IP address or host name> login-id <String> password <String> secret <String>
Note: You must specify [login-id] and [password] for an SCP or SFTP server. |

system xftp set mode

| | |
|--------------|---|
| Description | Sets the global default xFTP server mode. |
| Access Level | admin |
| Syntax | system xftp set mode <sftp scp> |

system xftp set scp-server

| | |
|-----------------|--|
| Description | Sets the default SCP server that will be used when the system is set to use SCP mode. |
| Access Level | admin |
| Syntax | system xftp set scp-server <IP address or host name> login-id <String: 1...32> password <String: 1...128> secret <String: 1...256> |
| Related Command | system xftp set mode |

system xftp set sftp-server

| | |
|-----------------|---|
| Description | Sets the default SFTP server that will be used when the system is set to use SFTP mode. |
| Access Level | admin |
| Syntax | system xftp set sftp-server <IP address or host name> login-id <String: 1...32> password <String: 1...128> secret <String: 1...256> |
| Related Command | system xftp set mode |

system xftp show

| | |
|--------------|---|
| Description | Displays summary information for each default xFTP server configured on the system. |
| Access Level | limited |
| Syntax | system xftp show |

system xftp unset

| | |
|--------------|---|
| Description | Restores the default SFTP or SCP server to the default configuration value. |
| Access Level | admin |
| Syntax | system xftp unset <sftp-server scp-server> |

Telemetry commands**telemetry delete subscription**

| | |
|--------------|--|
| Description | Deletes a telemetry subscription. |
| Access Level | admin |
| Syntax | telemetry delete subscription <String [1..32]> |

telemetry disable

| | |
|--------------|--|
| Description | Disables telemetry for all supported interfaces. |
| Access Level | admin |
| Syntax | telemetry disable |

telemetry enable

| | |
|--------------|---|
| Description | Enables telemetry for all supported interfaces. |
| Access Level | admin |

| | |
|--------|------------------|
| Syntax | telemetry enable |
|--------|------------------|

telemetry set keep-alive

| | |
|--------------|--|
| Description | Sets the keep-alive message for telemetry subscriptions. |
| Access Level | admin |
| Syntax | telemetry set keep-alive <String [1..32]> |

telemetry set subscription

| | |
|--------------|--|
| Description | Creates a telemetry subscription. |
| Access Level | admin |
| Syntax | telemetry set subscription <String [1..32]> label <String [1..32]> |

telemetry show

| | |
|--------------|------------------------------|
| Description | Displays telemetry settings. |
| Access Level | limited |
| Syntax | telemetry show [keep-alive] |

telemetry show subscriptions

| | |
|--------------|---|
| Description | Displays telemetry subscriptions. |
| Access Level | limited |
| Syntax | telemetry show subscriptions [subscription-id <String [1..32]>] |

TACACS+ commands

tacacs accounting add server

| | |
|--------------|--|
| Description | Adds a TACACS+ accounting server to the system and, if required, specifies the following: a priority setting for the server, the TCP port it uses, a timeout value that specifies how long to wait for a response from the server before attempting to contact another accounting server, a clear-text shared key (key), and an encrypted shared key (secret) used to exchange information between Waveserver 5 and a TACACS+ accounting server. |
| Access Level | super |
| Syntax | tacacs accounting add server <IP address or host name> [priority <Number:1...8>] [tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]
Note: Ensure that the TCP port number specified is not already reserved. |

| | |
|---------------|-----------------|
| Default Value | [tcp-port] = 49 |
|---------------|-----------------|

tacacs accounting clear server statistics

| | |
|--------------|--|
| Description | Clears statistics information for the specified TACACS+ accounting server. |
| Access Level | super |
| Syntax | tacacs accounting clear server <IP address or host name> statistics |

tacacs accounting <disable | enable>

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) for TACACS+ accounting globally. |
| Access Level | super |
| Syntax | tacacs accounting disable
tacacs accounting enable |

tacacs accounting <disable | enable> server

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for the specified TACACS+ accounting server. |
| Access Level | super |
| Syntax | tacacs accounting disable server <IP address or host name>
tacacs accounting enable server <IP address or host name> |

tacacs accounting remove server

| | |
|--------------|--|
| Description | Removes the specified TACACS+ accounting server from the system. |
| Access Level | super |
| Syntax | tacacs accounting remove server <IP address or host name> |

tacacs accounting set

| | |
|---------------|--|
| Description | Sets the status (on or off) of command and/or session logging for TACACS+ accounting globally. |
| Access Level | super |
| Syntax | tacacs accounting set [command <on off>] [session <on off>] |
| Default Value | <ul style="list-style-type: none"> • [command] = on • [session] = on |

tacacs accounting set server

| | |
|---------------|---|
| Description | Sets the priority, TCP port, timeout value, shared clear-text key, or shared encrypted key (secret) for the specified TACACS+ accounting server. |
| Access Level | super |
| Syntax | tacacs accounting set server <IP address or host name> {[priority <Number:1...8>]
[tcp-port <Number:1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>]
[secret <String: 4...128>]}
Note: Ensure that the TCP port number specified is not already reserved. |
| Default Value | [tcp-port] = 49 |

tacacs accounting show

| | |
|--------------|--|
| Description | Displays summary or statistics information for TACACS+ accounting. |
| Access Level | limited |
| Syntax | tacacs accounting show [statistics] |

tacacs accounting unset server

| | |
|--------------|---|
| Description | Restores the TCP port for the specified TACACS+ accounting server to the default setting (see tacacs accounting set server). |
| Access Level | super |
| Syntax | tacacs accounting unset server <IP address or host name> tcp-port |

tacacs add server

| | |
|---------------|--|
| Description | Adds a server to the list of global TACACS+ servers available to Waveserver 5. If required, specifies a priority setting for the server, the TCP port it uses, a timeout value that specifies how long to wait for a response from the server before attempting to contact another server, a clear-text shared key (key), and an encrypted shared key (secret) for exchanging information between Waveserver 5 and a TACACS+ server. |
| Access Level | super |
| Syntax | tacacs add server <IP address or host name> {[priority <Number:1...8>] [tcp-port <Number:1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}
Note: Ensure that the TCP port number specified is not already reserved. |
| Default Value | [tcp-port] = 49 |

tacacs authentication add server

| | |
|---------------|--|
| Description | Adds a TACACS+ authentication server to the list of authentication servers available to Waveserver 5. If required, specifies a priority setting for the server, the TCP port it uses, a timeout value that specifies how long to wait for a response from the server before attempting to contact another server, a clear-text shared key (key), and an encrypted shared key (secret) for exchanging information between Waveserver 5 and a TACACS+ authentication server. |
| Access Level | super |
| Syntax | tacacs authentication add server <IP address or host name> {[priority <Number: 1...8>] [tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}
Note: Ensure that the TCP port number specified is not already reserved. |
| Default Value | [tcp-port] = 49 |

tacacs authentication clear server statistics

| | |
|--------------|--|
| Description | Clears statistics information for the specified TACACS+ authentication server. |
| Access Level | super |
| Syntax | tacacs authentication clear server <IP address or host name> statistics |

tacacs authentication <disable | enable>

| | |
|---------------|--|
| Description | Sets the administrative state (disabled or enabled) for TACACS+ authentication globally. |
| Access Level | super |
| Syntax | tacacs authentication disable
tacacs authentication enable |
| Default value | enabled |

tacacs authentication <disable | enable> server

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for the specified TACACS+ authentication server. |
| Access Level | super |
| Syntax | tacacs authentication disable server <IP address or host name>
tacacs authentication enable server <IP address or host name> |

tacacs authentication remove server

| | |
|-------------|--|
| Description | Removes the specified TACACS+ authentication server from the system. |
|-------------|--|

| | |
|--------------|---|
| Access Level | super |
| Syntax | tacacs authentication remove server <IP address or host name> |

tacacs authentication set server

| | |
|---------------|---|
| Description | Sets the priority and/or TCP port, timeout value, clear-text shared key (key), or encrypted shared key (secret) for the specified TACACS+ authentication server. |
| Access Level | super |
| Syntax | tacacs authentication set server <IP address or host name> {[priority <Number: 1...8>] [tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [password <String: 4...128>]}
Note: Ensure that the TCP port number specified is not already reserved. |
| Default Value | [tcp-port] = 49 |

tacacs authentication show

| | |
|--------------|--|
| Description | Displays summary information or statistics for TACACS+ authentication. |
| Access Level | limited |
| Syntax | tacacs authentication show [statistics] |

tacacs authentication unset server

| | |
|--------------|---|
| Description | Restores the TCP port for the specified TACACS+ authentication server to the default setting (see tacacs authentication set server). |
| Access Level | super |
| Syntax | tacacs authentication unset server <IP address or host name> tcp-port |

tacacs authorization add server

| | |
|--------------|--|
| Description | Adds the specified TACACS+ authorization server to the list of authorization servers available to Waveserver 5. If required, specifies a priority setting for the server, the TCP port it uses, a timeout value that specifies how long to wait for a response from the server before attempting to contact another server, a clear-text shared key (key), and an encrypted shared key (secret) for exchanging information between Waveserver 5 and a TACACS+ authentication server. |
| Access Level | super |
| Syntax | tacacs authorization add server <IP address or host name> {[priority <Number: 1...8>] [tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}
Note: Ensure that the TCP port number specified is not already reserved. |

| | |
|---------------|-----------------|
| Default Value | [tcp-port] = 49 |
|---------------|-----------------|

tacacs authorization clear server statistics

| | |
|--------------|---|
| Description | Clears statistics information for the specified TACACS+ authorization server. |
| Access Level | super |
| Syntax | tacacs authorization clear server <IP address or host name> statistics |

tacacs authorization <disable | enable>

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for TACACS+ authorization globally. |
| Access Level | super |
| Syntax | tacacs authorization disable
tacacs authorization enable |

tacacs authorization <disable | enable> server

| | |
|--------------|--|
| Description | Sets the administrative state (disabled or enabled) for the specified TACACS+ authorization server. |
| Access Level | super |
| Prerequisite | To enable TACACS+ authorization, either a TACACS+ authentication server must be present on the system (see tacacs authentication add server) or the global TACACS+ server must support TACACS+ authorization. |
| Syntax | tacacs authorization disable server <IP address or host name>
tacacs authorization enable server <IP address or host name> |

tacacs authorization remove server

| | |
|--------------|---|
| Description | Removes the specified TACACS+ authorization server from the system. |
| Prerequisite | If you are removing the only available authorization server, you must first set the administrative state of TACACS+ authorization to disabled (see tacacs authorization <disable enable>). Failing to disable authorization will prevent any further access to Waveserver 5. |
| Access Level | super |
| Syntax | tacacs authorization remove server <IP address or host name> |

tacacs authorization set server

| | |
|-------------|---|
| Description | Sets the priority, TCP port, timeout value, shared clear-text key, or shared encrypted key (secret) for the specified TACACS+ authorization server. |
|-------------|---|

| | |
|---------------|--|
| Access Level | super |
| Syntax | tacacs authorization set server <IP address or host name> {[priority <Number: 1...8>] [tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [key <String: 2...64>] [secret <String: 4...128>]}
Note: Ensure that the TCP port number specified is not already reserved. |
| Default Value | [tcp-port] = 49 |

tacacs authorization show

| | |
|--------------|---|
| Description | Displays summary information or statistics for TACACS+ authorization. |
| Access Level | limited |
| Syntax | tacacs authorization show [statistics] |

tacacs authorization unset server

| | |
|--------------|---|
| Description | Restores the TCP port for the specified TACACS+ authorization server to the default setting (see tacacs authorization set server). |
| Access Level | super |
| Syntax | tacacs authorization unset server <IP address or host name> tcp-port |

tacacs clear server statistics

| | |
|--------------|--|
| Description | Clears statistics information for the specified TACACS+ global server. |
| Access Level | super |
| Syntax | tacacs clear server <IP address or host name> statistics |

tacacs <disable | enable>

| | |
|---------------|--|
| Description | Sets the administrative state (disabled or enabled) for all TACACS+ servers. |
| Access Level | super |
| Syntax | tacacs disable
tacacs enable |
| Default value | enabled |

tacacs <disable | enable> server

| | |
|--------------|---|
| Description | Sets the administrative state (disabled or enabled) for the specified TACACS+ server. |
| Access Level | super |

| | |
|--------|---|
| Syntax | tacacs disable server <IP address or host name>
tacacs enable server <IP address or host name> |
|--------|---|

tacacs remove server

| | |
|--------------|---|
| Description | Removes the specified TACACS+ server from the list of global servers. |
| Access Level | super |
| Syntax | tacacs remove server <IP address or host name> |

tacacs set

| | |
|----------------|---|
| Description | Changes global attributes for TACACS+ servers. |
| Access Level | super |
| Syntax | tacacs set {[key <Password string>] [keyminlen <Number: 2...64>] [secret <String>] [timeout <Number of seconds:1...30>] [privilege-level-admin <Number: 1...13>] [privilege-level-super <Number: 2...15>] [global-servers <on off>] [search-method <priority cached>]} |
| Default Values | <ul style="list-style-type: none"> • [keyminlen] = 8 • [timeout] = 6 • [privilege-level-admin] = 2 • [privilege-level-super] = 10 • [global-servers] = on • [search-method] = priority |

tacacs set server

| | |
|---------------|---|
| Description | Sets the priority, the TCP port, timeout value, shared clear-text key, or shared encrypted key (secret) for the specified TACACS+ server. |
| Access Level | super |
| Syntax | tacacs set server <IP address or host name> {[priority <Number: 1...8>] [tcp-port <Number: 1...65535>] [timeout <Seconds: 1...30>] [secret <String: 2...64>] [password <String: 4...128>]}
Note: Ensure that the TCP port number specified is not already reserved. |
| Default Value | [tcp-port] = 49 |

tacacs show

| | |
|--------------|---|
| Description | Displays summary or statistics information for TACACS+. |
| Access Level | limited |

| | |
|--------|--------------------------|
| Syntax | tacacs show [statistics] |
|--------|--------------------------|

tacacs syslog <disable | enable>

| | |
|---------------|--|
| Description | Sets the administrative status (disabled or enabled) of TACACS+ message logging. |
| Access Level | super |
| Syntax | tacacs syslog disable
tacacs syslog enable |
| Default Value | disabled |

tacacs unset

| | |
|--------------|---|
| Description | Restores global settings for TACACS+ servers to their default values (see tacacs set). |
| Access Level | super |
| Syntax | tacacs unset {[key] [keyminlen] [secret] [timeout] [privilegeleveladmin] [privilegelevelrw] [global-servers] [search-method]} |

tacacs unset server

| | |
|--------------|---|
| Description | Restores the TCP port for the specified TACACS+ server to the default setting (see tacacs set server). |
| Access Level | super |
| Syntax | tacacs unset server <IP address or host name> tcp-port |

Transceiver (xcvr) commands

xcvr <disable | enable> xcvr

| | |
|--------------|---|
| Description | Sets the administrative state (enabled or disabled) of a transceiver. |
| Access Level | admin |
| Prerequisite | For a pluggable transceiver, the transceiver must be installed at the client interface. |
| Syntax | xcvr disable xcvr <xcvrList>
xcvr enable xcvr <xcvrList> |

xcvr delete xcvr

| | |
|---------------|--|
| Description | Removes all PTPs, ports, and services associated with a pluggable transceiver. |
| Access Level | admin |
| Prerequisites | The transceiver must be disabled. See xcvr <disable enable> xcvr . |

| | |
|--------|--------------------------------|
| Syntax | xcvr delete xcvr <slot>/<xcvr> |
|--------|--------------------------------|

xcvr profile apply

| | |
|--------------|---|
| Description | Applies an XCVR profile with a certain profile ID to all the matching XCVR pluggables or applies all XCVR profiles to all matching XCVR pluggables. |
| Access Level | admin |
| Syntax | xcvr profile apply [profile-id <XCVR Profile Index: 1..255>] [all] |

xcvr profile create

| | |
|--------------|--|
| Description | Creates an xcvr profile. |
| Access Level | admin |
| Syntax | <p>xcvr profile create profile-id <NUMBER: 1..255> from-xcvr <slot>/<xcvr></p> <p>Note: The above form of the command is used when a plug is already inserted. All parameter values are used from the inserted plug and you cannot override them.</p> <p>xcvr profile create profile-id <NUMBER: 1..255> vendor-revision <String[1..2]> vendor-partnum <String[1..16]> vendor-name <String[1..16]> vendor-oui <Organization Unique Identifier (xx:yy:zz)> [description <String[1..255]>] [rx-power-high-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-low-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-high-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-low-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-high-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-low-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-high-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-low-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-optical-power-monitor-override <off on>]</p> <p>Note: The above form of command is used to create an xcvr profile without inserting the plug. In this case, the vendor-name, vendor-oui, vendor-partnum, and vendor-rev parameters are mandatory.</p> |

xcvr profile delete

| | |
|--------------|---|
| Description | <p>Deletes an xcvr profile.</p> <p>Note: You can delete an xcvr profile even when it is applied on a currently provisioned third party plug. However, deleting a profile does not impact the provisioned plug.</p> |
| Access Level | admin |
| Syntax | xcvr profile delete profile-id <NUMBER: 1..255> |

xcvr profile show

| | |
|--------------|--|
| Description | Displays summary information for all the XCVR profiles or a specific XCVR profile. |
| Access Level | limited |
| Syntax | xcvr profile show [profile-id <XCVR Profile Index: 1..255>] |

xcvr profile set

| | |
|----------------|---|
| Description | Sets the profile attributes for a certain XCVR profile. |
| Access Level | admin |
| Syntax | xcvr profile set [vendor-name <String[1..16]>] [vendor-oui <Organization Unique Identifier (xx:yy:zz)>] [vendor-revision <String[1..2]>] [vendor-partnum <String[1..16]>] [description <String[1..255]>] [rx-power-high-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-low-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-high-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [rx-power-low-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-high-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-low-alarm-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-high-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-power-low-warning-threshold <Power Threshold in dBm (NN.MM): -40.00 .. 8.16>] [tx-optical-power-monitor-override <off on>] |
| Default values | <ul style="list-style-type: none"> [description] = blank <p>Note: Default values for all power thresholds are read from the XCVR pluggable's EEPROM.</p> |

xcvr profile unset

| | |
|--------------|--|
| Description | Sets the profile attributes to its default values. See xcvr profile set . |
| Access Level | admin |
| Syntax | xcvr profile unset profile-id <Number 1 ..255> [rx-power-high-alarm-threshold] [rx-power-low-alarm-threshold] [rx-power-high-warning-threshold] [rx-power-low-warning-threshold] [tx-power-high-alarm-threshold] [tx-power-low-alarm-threshold] [tx-power-high-warning-threshold] [tx-power-low-warning-threshold] [tx-optical-power-monitor-override] |

xcvr set xcvr

| | |
|---------------|---|
| Description | Sets writable attributes of the transceiver. |
| Access Level | admin |
| Prerequisites | Disable the transceiver before setting the attributes and then re-enable it. See xcvr <disable enable> xcvr . |

| | |
|--------|---|
| Syntax | <pre>xcvr set xcvr <xcvrList> [service-profile <100GE 4x100GE 400GE mixed-otn-ethernet mixed-ethernet>] [mode <100GE OTL4.4 95-200-O 95-250-O 95-300-O 95-350-O 95-400-O 95-450-O 95-500-O 95-550-O 95-600-O 95-650-O 95-700-O 95-800-O 95-300-E 95-350-E 95-400-E 95-450-E 95-500-E 95-550-E 95-600-E 95-650-E 95-700-E 95-800-E 93.3-600-E 93.3-400-E 91.6-300-E 91.6-400-E 91.6-450-E 91.6-500-E 91.6-550-E 91.6-600-E 91.6-700-E 91.6-800-E 90-600-E 90-400-E 85-600-E 85-400-E 85-300-E 82-400-E>]</pre> <p>Note 1: In Waveserver 5 CLI, “xcvr mode” indicates the transmission mode.</p> <p>Note 2: For additional information on service profiles, see Service profiles for WL5e 2x MOTR 16x modules in terminal applications.</p> |
|--------|---|

xcvr show

| | |
|--------------|---|
| Description | Displays summary information about all transceivers. |
| Access Level | limited |
| Prerequisite | For a pluggable transceiver, the transceiver must be installed at the client interface. |
| Syntax | xcvr show |

xcvr show xcvr

| | |
|--------------|---|
| Description | <p>Displays diagnostic, status, tunability, and vendor data for a transceiver. The XCVR Vendor Data table also displays license information.</p> <p>The XCVR information includes admin state, operational state, mode, service profile, and mate mode details.</p> |
| Access Level | limited |
| Prerequisite | For a pluggable transceiver, the transceiver must be installed at the client interface. |
| Syntax | <pre>xcvr show xcvr <slot>/<xcvr> [diagnostics] [status] [vendor]</pre> <p>Note: Pluggable transceivers 100G-SR4 and 100G-CWDM4 have FEC enabled by default. FEC is enabled when the value of the 100 GbE Compliance parameter is “w/FEC”.</p> |

xcvr upgrade xcvr

| | |
|--------------|--|
| Description | Upgrades firmware on pluggable transceivers. |
| Access Level | super |
| Syntax | <pre>xcvr upgrade xcvr <slot>/<port> file <filename> [restart <cold warm>]</pre> <p>Note: If you do not specify the restart argument, a warm restart occurs by default.</p> |

User commands

user auth clear

| | |
|--------------|---|
| Description | Clears authentication statistics for all authentication methods or only the specified method. |
| Access Level | super |
| Syntax | user auth clear [provider <local radius radsec tacacs>] |

user auth set method

| | |
|--------------|---|
| Description | Sets the authentication method to be used for logins using the specified source; i.e., Console port (serial) or DCN via SSH (remote). |
| Access Level | super |
| Syntax | user auth set method <local radius radsec tacacs> scope <all serial remote> |

user auth set order

| | |
|--------------|---|
| Description | Sets the authentication provider priority. |
| Access Level | super |
| Syntax | user auth set order <Value or a comma-separated list of values (value,value,value,value): local radius radsec tacacs> |

user auth show

| | |
|--------------|--|
| Description | Displays summary information for each authentication provider. |
| Access Level | limited |
| Syntax | user auth show |

user create user

| | |
|--------------|-------------------------|
| Description | Creates a user account. |
| Access Level | super |

| | |
|--------|---|
| Syntax | <p>user create user <String: 1...64> access-level <limited admin super> [password <String: 1...128>] [secret <1...256>]</p> <p>Note 1: The following restrictions apply to the user string:</p> <ul style="list-style-type: none"> — The user string must not contain only numbers; for example 583247. — The user string must not start with a number; for example, 21User. — The user string must start with an alpha character or an underscore (_). — The user string must not contain the characters ? # \$ & ' () + / : ; < > = @ [] ^ ` ~ { } ! % * , or a space. <p>Note 2: The password string may contain special characters ~ ! % ^ & * () - _ : = + { } [] \ ; ' < > . , @ # ? / \$.</p> <p>Note 3: For information on supported password characters and equivalent ASCII codes, see Creating local user accounts.</p> |
|--------|---|

user delete user

| | |
|--------------|---|
| Description | <p>Deletes a user account.</p> <p>Note: The last super user account cannot be deleted.</p> |
| Access Level | super |
| Syntax | user delete user <String: 1...64> |

user kill all-but-me

| | |
|-----------------|---|
| Description | Ends all user sessions but the current one. |
| Access Level | super |
| Syntax | user kill all-but-me |
| Related Command | user kill session |

user kill session

| | |
|--------------|---|
| Description | Ends the user session with the specified session identifier. See user who . |
| Access Level | super |
| Syntax | user kill session <identifier> |

user set

| | |
|--------------|---|
| Description | Sets limits for simultaneous user account sessions. |
| Access Level | super |
| Syntax | user set [max-login-attempt <Number: 2..10>] [lockout-interval <Number: <1..30>] [min-password-length <Number: 1..128>] |

| | |
|-----------------|--|
| Default Values | <ul style="list-style-type: none"> • [max-login-attempt]: 5 • [lockout-interval]: 10 • [min-password-length]: 1 |
| Related Command | user set user |

user set user

| | |
|--------------|--|
| Description | Sets attributes for a specific local user account. |
| Access Level | <ul style="list-style-type: none"> • limited users can change their own password, but not other user's passwords. • admin users can change their own password, but not other user's passwords. • super users can change the password for any account. |
| Syntax | <p>user set user <local user name> access-level <limited admin super> <password <String: 1...128> [secret <1...256>]</p> <p>Note 1: The password string may contain special characters ~ ! % ^ & * () - _ : = + { } [] \ ; ' < > . , @ # ? / \$</p> <p>Note 2: For information on supported password characters and equivalent ASCII codes, see Creating local user accounts.</p> |

user show

| | |
|--------------|---|
| Description | Displays user account information (account name, privilege, account status, minimum password length). |
| Access Level | limited |
| Syntax | user show |

user who

| | |
|--------------|---|
| Description | Displays user summary information (user name, idle time, session ID, and terminal). |
| Access Level | limited |
| Syntax | user who |

user whoami

| | |
|--------------|-------------------------|
| Description | Displays your username. |
| Access Level | limited |
| Syntax | user whoami |

user write

| | |
|--------------|---------------------------------------|
| Description | Sends a message to a user. |
| Access Level | admin |
| Syntax | user write user <user name> <message> |

Waveserver 5

User Guide

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